

On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences $(a_n), (b_n)$ with $d_n = \text{lcm}(1, \dots, n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1, with the quantitative bound $\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2)$ for every $\varepsilon > 0$; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. We further prove a *Cartier–Frobenius block law* for the Apéry pair: for $p \geq 7$, $1 \leq q < p$ and $0 \leq r < p$,

$$(p^3 a_{qp+r}, b_{qp+r}) \equiv b_r(a_q, b_q) \pmod{p},$$

whose second coordinate is the classical Apéry–Lucas congruence and whose numerator coordinate appears to be unrecorded; the same law holds for the $\zeta(2)$ pair with p^2 . Combining it with Apéry’s Casoratian and the no-two-consecutive-zeros theorem identifies the *support* of the top-window contribution: for $p \geq 7$ and $n/2 < p \leq n$, one has $p \mid G_n$ if and only if $p \mid b_n$. The contribution of this window is therefore comparable, up to an absolute factor, with the logarithm of the $(n/2, n]$ -part of b_n . Prime counting disposes of the primes below $n/(f(n) \log n)$ for any $f(n) \rightarrow \infty$. One may therefore formulate the remaining problem as a uniform shifted-divisibility estimate over the quotient classes $\lfloor n/p \rfloor < f(n) \log n$. More weakly, and equivalently, it is enough to prove the subexponential target-radical estimate separately for every fixed quotient class; no uniformity as the quotient grows is needed. For fixed q , Apéry–Lucas identifies that target radical, for all large n , with the prime-factor slice $\text{rad}_{(n/(q+1), n/q]}(b_n)$. The key ingredient of the density-one theorem is a bound $Z(p) = O(p^{2/3})$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. A companion centered-coefficient analysis of the bordered value-return certificates proves the *uniform* fiber bound $\max_a \#\{j < p : b_j \equiv a\} = O(p^{3/4})$ and the collision-energy bound $\sum_a \#\{j : b_j \equiv a\}^2 = O(p^{7/4})$, both unconditional, with the underlying certificate nonvanishing proved in the sharp range $1 \leq h < k < p$. Computation for all 78,496 primes $5 \leq p \leq 10^6$ gives $Z(p) \leq 12$ with mean ≈ 1 . The pair count $Z(p)/2$ fits Poisson(1/2). For each canonical gap polynomial N_h whose splitting field has full hyperoctahedral Galois group, we prove via Chebotarev that its root-pair count follows the signed fixed-point distribution, tending to Poisson(1/2) as $h \rightarrow \infty$; the corresponding law for the actual coefficient zeros remains conjectural. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$, so $Z(p)$ is almost certainly unbounded. Under the weaker Cesàro hypothesis $\sum_{p \leq x} Z(p) \ll \pi(x)$ (Hypothesis $\bar{Z}$), which the Poisson model supports, we prove $\log G_n \ll \sqrt{n} \log \log(3n) = n^{1/2+o(1)}$ for density 1. The slowly growing factor records the separate companion-block channel; the present height estimate does not justify a fixed-constant $O(\sqrt{n})$ bound on a density-one set. Independently, a dyadic divisor-incidence argument combined with sub-interval zero counting improves the leading-digit zero-incidence bound from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ unconditionally, and to $O(N^{4/3}/\log N)$ under Hypothesis $\bar{Z}$. A prime-wise decomposition of the residual L^2 -norm $\|\eta\|^2 = D + O$ (diagonal + off-diagonal) shows, under the natural second-mass condition $\sum_{N < p \leq 2N} Z(p)^2 = O(N/\log N)$, that $D = T + O(1/\ln N)$. The all- n conjecture would then follow from the sufficient two-prime dispersion input $O = o(T)$. Computation through $N = 262,144$ confirms $|O/T| < 2 \times 10^{-3}$. Two

structural reductions further localize the gap: prime counting alone disposes of every prime $p \leq n/(f(n)\log n)$ for any $f \rightarrow \infty$, so only the quotient classes $\lfloor n/p \rfloor < f(n)\log n$ matter; and the first reciprocal moment $U_n = \text{lcm}(1, \dots, n) \sum_{k \leq n} b_{n-k}/k$ compresses the remaining moving-prime radical, up to $O(n^{2/3})$, into the single $\gcd(\text{lcm}(1, \dots, n), U_n)$, equivalently a pointwise reduced-denominator defect. Equivalently, the same radical is, up to $O(n^{2/3})$ in logarithmic height, the n -smooth squarefree part of the single Apéry number b_n ; in particular the all-index conjecture is equivalent to $\log \gcd(\text{lcm}(1, \dots, n), b_n) = o(n)$; and a single fixed factorial moment $(\text{HM})_k$ with $k > 6$ —at the independence scale, compatible with $Z(p) \ll p^{2/3}$, and proved here unconditionally for $k = 2$ —would give the full conjecture with the power saving $\log G_n \ll n^{2/3+2/k+o(1)}$.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n -th Apéry number. Set $d_n = \text{lcm}(1, \dots, n)^3$; Apéry [3] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \geq 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. $\square$

Corollary 2. *For every prime $p \geq 5$,*

$$v_p(G_n) \leq 6 \lfloor \log_p n \rfloor. \quad (2)$$

Proof. The assertion is immediate for $n = 1$, so suppose $n \geq 2$. All four entries in

$$\begin{pmatrix} d_n a_n & d_n b_n \\ d_{n-1} a_{n-1} & d_{n-1} b_{n-1} \end{pmatrix}$$

are integers. Since G_n divides both entries in the first row, it divides the determinant. Lemma 1 therefore gives the valid integer divisibility

$$G_n \mid \frac{6d_n d_{n-1}}{n^3}.$$

Consequently, for $p \geq 5$,

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor + 3 \lfloor \log_p (n-1) \rfloor - 3v_p(n) \leq 6 \lfloor \log_p n \rfloor.$$

$\square$

3 Small primes: unconditional bound

Proposition 3. $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n}).$

Proof. By (2), $v_p(G_n) \log p \leq 6 \log n$ for every prime $p \geq 5$. The same determinant estimate shows that the primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $6 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. $\square$

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [24]). *For every prime p and $n = \sum_{i=0}^s n_i p^i$ in base p ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [28] and McIntosh [27].

Proposition 5 (The exact aggregate companion denominator). *Let δ_n be the positive denominator of a_n in lowest terms. Then*

$$\frac{d_n a_n}{6} \in \mathbb{Z} \quad (n \geq 0). \quad (4)$$

Moreover, for every $N \geq 3$,

$$\text{lcm}_{0 \leq n \leq N} \delta_n = \frac{d_N}{6}. \quad (5)$$

Proof. It is enough to gain one factor at each of the primes 2 and 3 beyond the standard integrality $d_n a_n \in \mathbb{Z}$. Theorem 4 shows that every b_m is a unit in both $\mathbb{Z}_2$ and $\mathbb{Z}_3$: the possible base-2 digits give $b_0, b_1 \equiv 1 \pmod{2}$, while the possible base-3 digits give $(b_0, b_1, b_2) \equiv (1, 2, 1) \pmod{3}$. Dividing the Casorati identity of Lemma 1 by $b_n b_{n-1}$ and telescoping gives

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (6)$$

Fix $p \in \{2, 3\}$ and put $e = \lfloor \log_p n \rfloor$. Every summand on the right of (6) has p -adic valuation at least $1 - 3e$. The ultrametric inequality and the fact that b_n is a p -adic unit therefore give

$$v_p(a_n) \geq 1 - 3e.$$

Since $v_p(d_n) = 3e$, this proves (4).

Put $\Delta_N = \text{lcm}_{0 \leq n \leq N} \delta_n$. The first assertion gives $\delta_n \mid d_n/6$ for $n \geq 3$. Also $\delta_0 = \delta_1 = 1$, $\delta_2 = 4$, and $4 \mid d_N/6$ for $N \geq 3$. As $d_n/6 \mid d_N/6$ whenever $3 \leq n \leq N$, this proves $\Delta_N \mid d_N/6$.

Conversely, $\Delta_N a_n$ is integral for every $n \leq N$. Multiplying Lemma 1 by Δ_N shows, for $1 \leq n \leq N$, that

$$\frac{6\Delta_N}{n^3} = (\Delta_N a_n) b_{n-1} - (\Delta_N a_{n-1}) b_n \in \mathbb{Z}.$$

Thus $n^3 / \gcd(n^3, 6)$ divides Δ_N . Taking the least common multiple over $n \leq N$ gives

$$\text{lcm}_{1 \leq n \leq N} \frac{n^3}{\gcd(n^3, 6)} = \frac{d_N}{6}.$$

The last equality follows prime by prime: at 2 and 3 the maximal valuation is $3\lfloor \log_p N \rfloor - 1$, and at every other prime it is $3\lfloor \log_p N \rfloor$. Hence $d_N/6 \mid \Delta_N$, completing the proof. $\square$

Corollary 6 (Pointwise denominator and primitivity defects). *Write $a_n = A_n/\delta_n$ in lowest terms and put $g_n = \gcd(A_n, b_n)$. Then*

$$G_n = \frac{d_n}{\delta_n} g_n. \quad (7)$$

For $n \geq 1$ one also has

$$g_n \mid \gcd(6\delta_{n-1}, 6\delta_{n+1}). \quad (8)$$

Consequently the all-index estimate $\log G_n = o(n)$ is equivalent to the two simultaneous pointwise estimates

$$\log \frac{d_n}{\delta_n} = o(n), \quad \log g_n = o(n).$$

Proof. Since $\delta_n \mid d_n$ and $\gcd(A_n, \delta_n) = 1$,

$$\gcd\left(\frac{d_n}{\delta_n} A_n, \frac{d_n}{\delta_n} \delta_n b_n\right) = \frac{d_n}{\delta_n} \gcd(A_n, b_n),$$

which proves (7).

Multiply the Casorati identity at n by $\delta_n \delta_{n-1}$. Its left side is an integer divisible by g_n , so g_n divides $6\delta_n \delta_{n-1}/n^3$ and hence divides $6\delta_n \delta_{n-1}$. Because g_n is coprime to δ_n , this gives $g_n \mid 6\delta_{n-1}$. Applying the same argument to the identity at $n+1$ gives $g_n \mid 6\delta_{n+1}$, proving (8). The final equivalence follows from (7), since both factors are positive integers. $\square$

Corollary 7 (Single companion-denominator criterion). *Retain the notation of Corollary 6 and put*

$$\mathfrak{D}_n = \frac{d_n}{\delta_n}.$$

Then

$$g_n \mid 6\mathfrak{D}_n, \quad \mathfrak{D}_n \leq G_n \leq 6\mathfrak{D}_n^2. \quad (9)$$

Consequently the all-index estimate is equivalent to the single pointwise denominator-retention statement

$$\log \mathfrak{D}_n = o(n), \quad \text{equivalently} \quad \log \delta_n = \log d_n - o(n).$$

More precisely, if

$$\mathcal{W}_n = \sum_{\substack{\sqrt{n} < p \leq n \\ p \mid b_n \bmod p}} \log p,$$

then

$$\mathcal{W}_n \leq \log \mathfrak{D}_n \leq 3\mathcal{W}_n + O(n^{2/3}). \quad (10)$$

Thus companion primitivity is not a second independent asymptotic cut.

Proof. Let $p \mid g_n$. Since $g_n = \gcd(A_n, b_n)$ and $\gcd(A_n, \delta_n) = 1$, one has $p \nmid \delta_n$, and hence

$$v_p(\mathfrak{D}_n) = v_p(d_n).$$

The determinant divisibility in Corollary 2, together with $G_n = \mathfrak{D}_n g_n$, gives

$$\begin{aligned} v_p(g_n) &\leq v_p(6) + v_p(d_n) + v_p(d_{n-1}) - 3v_p(n) - v_p(\mathfrak{D}_n) \\ &\leq v_p(6) + v_p(d_n) = v_p(6\mathfrak{D}_n). \end{aligned}$$

This proves $g_n \mid 6\mathfrak{D}_n$, and (9) follows from $G_n = \mathfrak{D}_n g_n$.

It remains to prove (10). Let $p \geq 7$ and $p > \sqrt{n}$, and write $n = qp + r$. Then $v_p(d_n) = 3$, so lowest-term reduction gives

$$p \mid \mathfrak{D}_n \iff p^3 a_n \equiv 0 \pmod{p}.$$

The balanced block law gives

$$p^3 a_n \equiv a_q b_r \pmod{p}.$$

Here the denominator of a_q is a p -adic unit because $q < p$. Therefore every prime counted by $\mathcal{W}_n$ divides $\mathfrak{D}_n$, while an additional prime in this range must divide the reduced numerator of a_q . Each such large prime occurs in $\mathfrak{D}_n$ to order at most three. Proposition 103 bounds the logarithmic weight of all additional companion-block primes by $O(n^{2/3})$. Finally, since $\mathfrak{D}_n \mid d_n = L_n^3$, its prime-power contribution from $p \leq \sqrt{n}$ is

$$3 \sum_{\substack{p^j \leq n \\ p \leq \sqrt{n}}} \log p = O(\sqrt{n}).$$

These estimates prove (10). □

Remark 8 (Aggregate versus pointwise information). Equation (5) is sharp but does not control any prescribed δ_n : different prime-power factors of $d_N/6$ may be witnessed at different indices below N . Thus it does not by itself bound either factor in (7). It isolates two genuinely pointwise tasks: maximal-denominator retention and primitivity of the pair (A_n, b_n) .

5 The zero-count function

Definition 9. For a prime $p \geq 5$, define $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 10 (No consecutive zeros). *For every prime $p \geq 5$ and every $j \in \{0, \dots, p-2\}$, b_j and b_{j+1} cannot both vanish modulo p .*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$ with $0 \leq m \leq p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m) b_m - m^3 b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod{p}$. Since $1 \leq m \leq p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod{p}$, contradicting $b_0 = 1$. □

Lemma 11 (Gap polynomial). *Fix $h \geq 2$. Suppose $b_m \equiv 0 \pmod{p}$ and $m+h \leq p-1$. Then*

$$b_{m+h} \equiv C_h(m) b_{m+1} \pmod{p}, \tag{11}$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1} (m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$, then by Lemma 10 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod{p}$, and m is constrained to at most $3(h-1)$ residue classes modulo p , provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1) b_{m+1} / (m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \geq 2$: b_{m+j+1} is a $\mathbb{Q}(m)$ -linear combination of b_{m+j} and b_{m+j-1} , both already rational multiples

of b_{m+1} . The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h=2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. $\square$

Lemma 12 (Nonvanishing over $\mathbb{F}_p$). *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p .*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m+h)N_h(m) - (m+h)^6 N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m+1)$.

First evaluation. Equation (11) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot ((h-1)!)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h-2)N_h(-2) - (h-2)^6 N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h-2$:

$$N_h(-2) = -5 b_{h-2} ((h-2)!)^3 \quad (h \geq 2). \quad (12)$$

Conclusion. If $N_h \equiv 0 \pmod{p}$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod{p}$ (since $p \nmid 5$ and $p \nmid (h-1)!, (h-2)!$). These are consecutive zeros, contradicting Lemma 10. $\square$

Lemma 13 (Boundary evaluation). *For all $1 \leq r \leq h$,*

$$N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3. \quad (13)$$

Proof. Fix r and induct on $h \geq r$. The base cases $h = r$ and $h = r+1$ follow from the first evaluation

$$N_h(-1) = b_{h-1} ((h-1)!)^3 \quad (14)$$

together with the reflection identity $N_h(-h) = (-1)^{h-1} N_h(-1)$ (Remark 16): indeed $N_r(-r) = (-1)^{r-1} b_{r-1} ((r-1)!)^3$ and $N_{r+1}(-r) = P(0)N_r(-r) - 0^6 N_{r-1}(-r) = 5N_r(-r)$. For the induction step assume (13) at h and $h-1$; evaluate the recurrence at $m = -r$ and write $k = h-r$:

$$\begin{aligned} N_{h+1}(-r) &= P(k)N_h(-r) - k^6 N_{h-1}(-r) \\ &= (-1)^{r-1} b_{r-1} ((r-1)!)^3 \left[P(k) b_k (k!)^3 - k^6 b_{k-1} ((k-1)!)^3 \right] \\ &= (-1)^{r-1} b_{r-1} ((r-1)!)^3 (k!)^3 \left[P(k) b_k - k^3 b_{k-1} \right] \\ &= (-1)^{r-1} b_{r-1} ((r-1)!)^3 (k!)^3 (k+1)^3 b_{k+1} \\ &= (-1)^{r-1} b_{r-1} b_{h+1-r} ((r-1)!)^3 ((h+1-r)!)^3, \end{aligned}$$

using $k^6 ((k-1)!)^3 = k^3 (k!)^3$ in the third line and the Apéry recurrence (1) at $n = k$ in the fourth; the last line uses $(k!)^3 (k+1)^3 = ((k+1)!)^3$ with $k+1 = h+1-r$. This closes the induction. $\square$

Proposition 14. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \frac{3}{2}(H-1)(H-2) + 1, \quad (15)$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2} p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0, \dots, p-1\}$ into $\lceil p/H \rceil$ consecutive blocks of length at most H . In each block, mark the first zero; all other zeros have gap $h \in \{2, \dots, H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 10). By Lemma 11, a gap- h pair constrains m to at most $3(h-1)$ residue classes mod p (the denominator factors $(m+j+1)^3$ are nonzero since $0 \leq m+j+1 \leq p-1$). By Lemma 12, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \frac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For the omitted small primes, direct calculation gives $Z(2) = Z(3) = 0$ and $Z(5) = 2$ (the zero indices modulo 5 are 1 and 3). Thus the stated asymptotic bound still holds for all primes after enlarging its absolute constant. $\square$

Corollary 15 (Sub-interval zero count). *For every prime $p \geq 7$ and every interval $I \subseteq \{0, \dots, p-1\}$,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

Proof. Apply the proof of Proposition 14 to the sub-interval I in place of $\{0, \dots, p-1\}$: partition I into $\lceil |I|/H \rceil$ blocks of length H , bound the non-first zeros in each block by the same gap polynomial argument (Lemma 11 applies regardless of the interval), and optimize. $\square$

Remark 16. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h . The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$, verified computationally for $h \leq 200$. For even h , N_h is an *odd* polynomial in $(m + (h+1)/2)$, forcing the structural factor $(2m+h+1)$. At $h=2$: factor $(2m+3)$ in $P(m+1) = (2m+3)(17m^2 + 51m + 39)$. The computational data (Section 10) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 17 (Content restriction). A stronger nonvanishing holds: if $p \geq 7$ divides the content of N_h (the gcd of all its coefficients), then $h \geq 2p$. Indeed, by Lemma 13, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3$ for $1 \leq r \leq h$; choosing $r = 1, 2$ and using $p \nmid (h-1)!, (h-2)!, 5$ for $h \leq p$ recovers Lemma 12, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$. Setting $\beta_n = (n!)^3 b_n$ and $F(z) = \sum_{n \geq 0} b_n z^n$, the endpoint factorization packages into the double generating function $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u) F(v)$, showing that the boundary array is a rank-one tensor.

Remark 18 (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$|\text{Res}(N_h, N_{h+1})| = \prod_{j=1}^{h-1} (j!^3 b_j)^6. \quad (16)$$

This follows from the recursion $|\text{Res}(N_h, N_{h+1})| = |N_h(-h)|^6 |\text{Res}(N_{h-1}, N_h)|$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$; with the standard signed convention the recursion carries an additional factor $(-1)^{h-1}$ from the scalar minus sign, which is why the formula is stated in absolute value), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1} ((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 12). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\text{Res}(N_2, N_3) = -5^6$, $\text{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\text{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \text{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \leq j \leq h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \geq 7$ (since $5^6 \neq 0$).

Remark 19 (Sharpness of the bound). The exponent $2/3$ in Proposition 14 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\{\sum r_h : r_h \leq 3(h-1), \sum h r_h \leq p\}$ has value $\frac{3}{2}p^{2/3} + O(p^{1/3})$, with leading constant $3/2$ achieved by filling gap sizes $h = 2, 3, \dots, H$ to capacity and using the remaining space at $h = H+1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \leq 3(h-1)$ for all relevant h . Improving the exponent requires genuinely arithmetic input about the family (N_h) .

Remark 20 (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll’s condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m) N_h(m+1) - N_{h+1}(m) N_{h+1}(m+1) = - \prod_{j=2}^{h+1} (m+j)^6. \quad (17)$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \leq 7$. A consequence: for $m \not\equiv -j \pmod{p}$ ($2 \leq j \leq h+1$), the pairs $(m, h+2)$ and $(m+1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m, h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m) N_{h+1}(m+1)$ would equal a nonzero product of sixth powers. After a meromorphic gauge $\tau_h(x) = N_h(x)/d_h(x)$ satisfying $d_{h+2}(x) d_h(x+1) = \prod_{j=2}^{h+1} (x+j)^6$, this becomes an exact SL_2 -tiling: $\tau_{h+1}(x) \tau_{h+1}(x+1) - \tau_{h+2}(x) \tau_h(x+1) = 1$, equivalently an A_1 T -system (discrete Liouville equation). A zero of τ forces its transverse neighbors to be units (*singularity confinement*), but abstract SL_2 -tilings can have zeros of density $1/2$ (e.g., $q_n = 0, 1, 0, -1, \dots$ with $q_n^2 - q_{n-1}q_{n+1} = 1$), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 26), and polynomial degree bounds—is essential.

Remark 21 (Projective orbit reformulation). Let $B_n = n!^3 b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n) w_n - n^6 w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi : \{0, \dots, p-1\} \rightarrow \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \leq m < m+h < p$, $N_h(m) \equiv 0 \pmod{p}$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π . The reflection symmetry $P(-1-n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod{p}$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \leq 10^5$).

Remark 22 (Galois structure). For each fixed h , the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ forces roots into pairs $\{r, -r-h-1\}$, and $\text{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h-1)/2 \rfloor$ (confirmed exactly for $h \leq 5$, $2 \leq h \leq 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$ -roots of N_h is 1 for odd h and 2 for even h . This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are 0.98, 2.03, 1.11, 2.07, 1.11, 1.88 for $h = 3, \dots, 8$.

Proposition 23 (Hyperoctahedral Poisson law). *Assume $\text{Gal}(N_h) \cong C_2 \wr S_m$ with $m = \lfloor 3(h-1)/2 \rfloor$. Let $Y_h(p)$ be the number of non-central reflection-pairs of roots of N_h that split completely over $\mathbb{F}_p$. Then for each fixed $k \leq m$, the factorial moments in the prime-aspect satisfy*

$$\lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where $(Y)_k = Y(Y-1) \cdots (Y-k+1)$. Since these match the factorial moments of Poisson(1/2) for all k , the prime-aspect distribution of $Y_h(p)$ converges to Poisson(1/2) as $h \rightarrow \infty$.

Proof. In $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m, \varepsilon \in \{\pm 1\}^m\}$, a *positive fixed point* of $g = (\sigma, \varepsilon)$ is an index i with $\sigma(i) = i$ and $\varepsilon_i = +1$. The Frobenius at p acts on the m reflection-pairs, and $Y_h(p)$ equals the number of positive fixed points. For any k distinct indices $i_1, \dots, i_k$, the fraction of $g \in G$ fixing all k positively is

$$\frac{2^{m-k} (m-k)!}{2^m m!} = \frac{1}{2^k (m)_k}.$$

Summing over all $(m)_k$ ordered k -tuples, $\mathbf{E}_{g \in G}[(Y(g))_k] = (1/2)^k$, which matches the factorial moments of Poisson(1/2). The Chebotarev density theorem then gives the prime-aspect convergence (Remark 25 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all $k \leq m$, $Y_h(p) \rightarrow \text{Poisson}(1/2)$ as $m \rightarrow \infty$. $\square$

Remark 24 (Prime-aspect vs. orbit-aspect). Proposition 23 is a *prime-aspect* statement: for fixed h and varying p , the root-pair count $Y_h(p)$ converges to Poisson(1/2). The zero-count $Z(p)$ aggregates contributions from all gaps $h = 1, \dots, p-1$ at a *fixed* prime p , so the orbit-aspect independence needed to deduce $Z(p)/2 \sim \text{Poisson}(1/2)$ is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

Remark 25 (GRH-effective error). For each fixed k , the k -th factorial moment does not require the full splitting field K_h (of degree $2^m m!$, double-exponential in h). Let $E_{h,k}$ be the fixed field of the stabilizer of an ordered signed k -tuple with distinct underlying reflection pairs; then $[E_{h,k} : \mathbb{Q}] = 2^k (m)_k = O_k(h^k)$. Under GRH for $\zeta_{E_{h,k}}$,

$$\frac{1}{\pi(x)} \sum_{\substack{p \leq x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where $D_{h,k} = |\text{Disc}(E_{h,k})|$. The Apéry recurrence gives the coefficient-height bound $\log \|N_h\|_1 = O(h \log h)$, from which $\log D_{h,k} \ll_k h^{k+2} \log h$. Thus an $O(h^{-C})$ error holds in the polynomial range $x \geq h^{2k+4+2C+\varepsilon}$, for each fixed k .

At $x = 10^6$ with $h = 10$ ($m = 13$), the degree contributions alone for $k = 1, 2, 3$ are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

5.2 Dual bounds: the restart identity and column estimate

The bound $Z(p) = O(p^{2/3})$ controls zeros along a *fixed row* (varying m for a given gap h). A dual bound controls zeros along a *fixed column* (varying h for a given starting point x_0).

Lemma 26 (Restart identity). *Suppose $N_r(x_0) = 0$ and $N_{r+1}(x_0) \neq 0$ for some $x_0 \in \mathbb{F}_p$ and $r \geq 1$. Then for every $d \geq 0$,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \quad (18)$$

In particular, if $N_{r+d}(x_0) = 0$ as well, then $N_d(x_0 + r) = 0$.

Proof. The sequence $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$ has $Y_0 = 0, Y_1 = 1$, and satisfies $Y_{d+1} = P((x_0+r)+d) Y_d - ((x_0+r)+d)^6 Y_{d-1}$, which is the defining recurrence for $N_d(x_0 + r)$. $\square$

Call a column $x_0 \in \mathbb{F}_p$ *polluted* if $x_0 \equiv -m \pmod{p}$ for some $2 \leq m < p$ with $p \mid b_{m-1}$. By the endpoint factorization $N_h(-m) = \pm b_{m-1} b_{h-m} ((m-1)!)^3 ((h-m)!)^3$ (Remark 17), a polluted column satisfies $N_h(x_0) \equiv 0$ for *every* $h \geq m$: the column recurrence in h reaches two consecutive

zeros and then vanishes identically. Conversely, two consecutive zero levels $N_{h-1}(x_0) = N_h(x_0) = 0$ propagate backwards through the column recurrence $N_{h+1} = P(x_0+h)N_h - (x_0+h)^6 N_{h-1}$ until a vanishing coefficient blocks the descent, which forces $x_0 \equiv -m$ with $N_m(-m) = \pm b_{m-1}((m-1)!)^3 \equiv 0$, i.e. $p \mid b_{m-1}$. Thus pollution is *exactly* the presence of two consecutive zero levels, and the number of polluted columns with $m \leq M$ is the zero count $\#\{j < M : p \mid b_j\} = O(M^{2/3})$ (Corollary 15).

Proposition 27 (Column bound, unpolluted case). *Let $x_0 \in \mathbb{F}_p$ be a column whose zero levels in the interval I (of H consecutive h -values, $H \leq p$) contain no two consecutive integers—in particular, any unpolluted column. Then*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod{p}\} \leq \frac{3^{4/3}}{2} H^{2/3} + O(H^{1/3} + 1). \quad (19)$$

Proof. Order the zero levels in I . For threshold $L \geq 2$: a gap of length $d \leq L$ between consecutive zeros at levels $r < r + d$ gives $N_d(x_0 + r) = 0$ by Lemma 26 (the hypothesis $N_{r+1}(x_0) \neq 0$ holds because the zero levels contain no two consecutive integers). For fixed d , there are at most $3(d-1)$ values of $x_0 + r$ satisfying this (by Lemma 12 and $\deg N_d = 3(d-1)$). Gaps longer than L contribute at most $H/(L+1)$ zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^L 3(d-1) + O(1) = \frac{H}{L+1} + \frac{3}{2}L(L-1) + O(1).$$

Optimizing $L \asymp (H/3)^{1/3}$ gives the stated bound. $\square$

Polluted columns genuinely violate the bound (19): at $p = 73$ (where $b_2 = 73$), the column $x_0 \equiv -3$ has $N_h(-3) \equiv 0$ for every $h \geq 3$. Computationally, at mesoscopic scale $H = \lfloor \sqrt{p} \rfloor$ the *unpolluted* column counts are far below the $O(H^{2/3})$ bound: for all primes $p \leq 5000$ and all unpolluted x_0 , $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 4$ (maximum attained at $p = 653$), while the polluted columns in this range reach counts as large as 50 (at $p = 3331$, where $b_7 = 3331 \cdot 175415$, so the column $x_0 \equiv -8$ vanishes for every $h \geq 8$).

Remark 28 (Rectangular incidence estimate). The row bound $|R_h| \leq 3(h-1)$ and the column bound (19) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p, 2 \leq h \leq H, N_h(x) \equiv 0\} \leq \min\{H^2, pH^{2/3}\}.$$

The first bound sums row degrees; the second sums the unpolluted column estimates plus the polluted-column contribution $\leq Z(p) \cdot H \ll p^{2/3}H \leq pH^{2/3}$ (for $H \leq p$), which does not change the order. They cross at $H \sim p^{3/4}$. Note that a uniform bound $R_p(H) \leq CH$ for all primes is *false*: by the Chebotarev density theorem, for each fixed H , infinitely many primes $p > H^2$ have every N_h ($2 \leq h \leq H$) splitting completely over $\mathbb{F}_p$, giving $R_p(H) = \frac{3}{2}H(H-1)$. The correct target is the *mesoscopic* estimate $R_p(\lfloor \sqrt{p} \rfloor) = O(\sqrt{p})$, where the split-prime obstruction does not apply because $H = \sqrt{p}$ grows with p . More generally, if $R_p(H) \ll H^\beta$ then $Z(p) \leq p/H + R_p(H)$ is optimized at $H \sim p^{1/(\beta+1)}$, giving $Z(p) \ll p^{\beta/(\beta+1)}$: the current $\beta = 2$ gives $2/3$ and $\beta = 1$ yields $1/2$. Reflection forces $R_p(H) \geq H/2$ from central roots, so $\beta = 1$ is the intrinsic limit of this first-moment method.

5.3 The continuant Bezout identity

Write $S_d(x) = \prod_{j=2}^d (x + j)$.

Proposition 29 (Bezout identity). *For integers $2 \leq d < e$,*

$$N_{e-1}(x+1)N_d(x) - N_{d-1}(x+1)N_e(x) = S_d(x)^6 N_{e-d}(x+d). \quad (20)$$

In particular, after saturating the cut-edge divisor S_d ,

$$(N_d, N_e) : S_d^\infty = (N_d, N_{e-d}(x+d)) : S_d^\infty.$$

Proof. The continuant addition formula (splitting the interval $[1, e-1]$ at position d) gives $N_e = N_d N_{e-d+1}(x+d-1) - (x+d)^6 N_{d-1} N_{e-d}(x+d)$. The Dodgson identity at index $d-2$ gives $N_{d-1}(x+1)N_{d-1} - N_{d-2}(x+1)N_d = \prod_{j=2}^{d-1} (x+j)^6$. Combining these two identities yields (20). Verified computationally for all $2 \leq d < e \leq 9$. $\square$

Corollary 30 (Separated-block criterion). *Away from the cut-edge divisor $S_d(x) = 0$,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod{p} \iff N_d(\alpha) \equiv N_{e-d}(\alpha+d) \equiv 0 \pmod{p}.$$

In particular, the saturated gcd satisfies $\deg(\gcd_{\mathbb{F}_p}(N_d, N_e) : S_d^\infty) \leq 3(\min(d, e-d) - 1)$, which improves the naive $3(d-1)$ when $e-d < d$. The saturation is essential: the full gcd may contain cut-edge factors dividing S_d . For example, at $p = 73$ (where $b_2 = 73$), $\gcd_{\mathbb{F}_{73}}(N_3, N_4) = X + 3$ even though $N_{e-d} = N_1 = 1$; the factor $X + 3$ divides $S_3 = (x+2)(x+3)$. The unsaturated degree can exceed the displayed bound by at most $\deg S_d = d - 1$.

Remark 31 (Separated-block resultant). Define $\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d))$. The Bezout identity gives

$$|\text{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} ((j!)^3 b_j)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ($e = d + 1$), $N_1 = 1$ gives $\mathcal{R}_{d,1} = 1$, recovering the adjacent resultant formula. For nonadjacent indices, $\mathcal{R}_{d,r}$ is genuinely new. The reflection identity gives $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$. The subtraction step does *not iterate*: after one application, $N_d(x)$ and $N_{e-d}(x+d)$ live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For $d = 2$, writing $P(y) = (2y+1)Q(y)$ with $Q(y) = 17y^2 + 17y + 5$ and θ a root of Q , the norm decomposition $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)} N_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$ (where $G_r = N_r(x+1)$) gives $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$. At $p = 71$: $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$ with $-5/2 \notin \{-1, -2, -3, -4\}$, a nonboundary common root inside the mesoscopic range ($4^3 < 71$). This shows the boundary-only gcd statement is *false*.

Remark 32 (Collision energy and the $H^{1/2}$ barrier). The column exponent $2/3$ is the natural barrier for the consecutive-gap argument: even if each gap of length d has only d (rather than $3(d-1)$) admissible starts, the sum $\sum_{d \leq L} d \asymp L^2$ still gives $H/L + L^2$, optimized at $H^{2/3}$. To achieve $H^{1/2}$, one needs the aggregate collision energy $\mathcal{E}_p(H) = \sum_{d+r \leq H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$, which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The A_1 T-system $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$ admits tame solutions with zero density $1/2$, e.g. $\tau_{2k} = 0, \tau_{2k+1} = (-1)^k$, so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing $M(n) = \begin{pmatrix} P(n) & -n^6 \\ 1 & 0 \end{pmatrix}$, so that $N_h(x) = 0$ iff a fixed matrix coefficient of the interval product $M(x+h-1) \cdots M(x+1)$ vanishes), the conjectured $\mathcal{E}_p(H) \ll H^{3/2}$ is analogous to a Rudnev-type point-plane incidence bound in $\text{PGL}_2(\mathbb{F}_p)$, subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

p	71	503	2003	5003	7919	20011	50021	10^5+3
$H = \lfloor \sqrt{p} \rfloor$	8	22	44	70	88	141	223	316
$R_p(H)/H$	1.75	1.59	1.36	1.50	1.75	1.63	1.56	1.46
$\mathcal{E}_p(H)$	1	2	2	2	10	0	2	0
$\mathcal{E}_p(H)/H^{3/2}$	.044	.019	.007	.003	.012	0	.001	0

A separate single-pass census over every prime $p \leq 20000$, again with $H = \lfloor \sqrt{p} \rfloor$ and with $m_{d,r}(p)$ denoting the number of common $\mathbb{F}_p$ -roots of $N_d(x)$ and $N_r(x+d)$, found

$$\max_p \mathcal{E}_p(H) = 24, \quad \max_{p,d,r} m_{d,r}(p) = 2, \quad \max_p \text{codeg}_L \Gamma_p(H) = 3.$$

Exactly twelve primes in this range have some $m_{d,r}(p) \geq 2$, and none has $m_{d,r}(p) \geq 3$. Here $\Gamma_p(H)$ is the bipartite graph defined below and codeg_L is its maximum left-pair codegree. These are exhaustive finite data, not uniform bounds. The ratio $R_p(H)/H$ clusters around $3/2$ (the Chebotarev prediction from Remark 22), and $\mathcal{E}_p(H)/H^{3/2} \rightarrow 0$, consistent with $\mathcal{E}_p(H) \ll H^{3/2}$. The key arithmetic object is the shifted resultant $S_{d,r} = \text{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$: for $p \nmid S_{d,r}$, the gap polynomials N_d and $N_r(\cdot + d)$ share no root over $\mathbb{F}_p$, so that gap pair contributes zero to $\mathcal{E}_p$. A Hadamard bound gives $\log |S_{d,r}| \ll (d+r)^2 \log(d+r)$, so the total number of prime divisors across all pairs (d,r) with $d+r \leq H$ is $O(H^4 \log H)$. Two caveats are essential here (they are developed in the separated-resultant subsection below). First, this counts the *support* $W_p(H) = \#\{(d,r) : \exists x, N_d(x) \equiv N_r(x+d) \equiv 0\}$, not the energy: one supported pair can carry up to $3(\min(d,r)-1)$ root witnesses, so the averaging conclusion $W_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ primes holds for $H \leq X^{2/7-\varepsilon}$, while the corresponding energy statement $\mathcal{E}_p(H) \ll H^{1/2}$ only follows in the shorter range $H \leq X^{2/9-\varepsilon}$. Second, $p \mid S_{d,r}$ detects a common *projective* root over $\overline{\mathbb{F}}_p$; pairs with $p \mid \ell_d, \ell_r$ (an infinity collision) or with a nonsplit common quadratic factor lie in the resultant support without contributing any $\mathbb{F}_p$ -root witness. Either way this is rigorous but far from the global target $H = p - 1$.

A sharper route uses the bipartite incidence graph $\Gamma_p(H)$: left vertices are gap lengths d , right vertices are gap lengths r , with an edge when N_d and $N_r(\cdot + d)$ share a root over $\mathbb{F}_p$. Its edge count is the simple support $W_p(H)$, not $\mathcal{E}_p(H)$: an edge with $m_{d,r}(p)$ witnesses has weight $m_{d,r}(p)$ in the energy. A uniform codegree bound would place the simple graph in the regime of the Kővári–Sós–Turán theorem and give $W_p(H) \ll H^{3/2}$, but this does not control the parallel root witnesses. To reach $\mathcal{E}_p(H) \ll H^{3/2}$ one needs a weighted codegree or split-gcd multiplicity-tail estimate. Even that energy estimate must be supplemented by the low-fiber amplification recorded below before it yields the mesoscopic target $R_p(\sqrt{p}) = O(\sqrt{p})$. The Dodgson identity (Remark 20) restricts local configurations in Γ_p , but by itself controls $O(H)$ pairs out of $O(H^2)$; upgrading it to a weighted global estimate is the key missing step.

5.4 Value fibers: the bordered certificate

The zero-count bound $Z(p) = O(p^{2/3})$ (Proposition 14) concerns the single fiber $a = 0$, where one return condition $b_t \equiv b_{t+h} \equiv 0$ already forces a gap polynomial to vanish. For a *general* fiber $\{t : b_t \equiv a \pmod{p}\}$ with $a \neq 0$, two return conditions are needed to eliminate the two unknowns (b_{t+1}, a) , and the resulting certificate is a bordered 2×2 determinant $D_{h,k}$ built from the gap polynomials and the second solution of the cleared recurrence. The determinant factors exactly as $D_{h,k} = \Pi_h \cdot \Delta_{h,k}$, and the analogue of Lemma 12 for the reduced certificate $\Delta_{h,k}$ is subtler: the endpoint evaluations assert equalities of separated Apéry values rather than consecutive zeros, so the no-consecutive-zero mechanism does not apply (an explicit two-point obstruction occurs at $p = 131$, $(h,k) = (12, 55)$; see the proof below). Nevertheless, a centered-coefficient analysis—using

the Pell–Chebyshev structure of the leading coefficients rather than endpoint values—proves the nonvanishing in the *full* range $1 \leq h < k < p$ (Theorem 37 below), and the range $k < p$ is sharp: at $p = 7$ the certificates $\Delta_{h,21}$ vanish identically. Consequently the same block-partition count that proves $Z(p) = O(p^{2/3})$ yields an *unconditional uniform* fiber bound $O(p^{3/4})$. We first state the counting mechanism conditionally, then prove the nonvanishing.

Proposition 33 (Uniform fiber bound conditional on bordered nonvanishing). *Let*

$$P(n) = 34n^3 + 51n^2 + 27n + 5$$

and let $(b_n)_{n \geq 0}$ be the Apéry numbers, with $b_0 = 1$, $b_1 = 5$, and

$$(n+1)^3 b_{n+1} = P(n)b_n - n^3 b_{n-1}.$$

For a prime $p \geq 7$ and $a \in \mathbb{F}_p$, put

$$N_p(a) = \#\{0 \leq t < p : b_t \equiv a \pmod{p}\}.$$

Define $N_0(x) = 0$, $N_1(x) = 1$, and, for $m \geq 1$,

$$N_{m+1}(x) = P(x+m)N_m(x) - (x+m)^6 N_{m-1}(x),$$

and set

$$\Pi_m(x) = \prod_{j=1}^m (x+j)^3, \quad \Delta_{h,k}(x) = N_h(x)\Pi_{k-h}(x+h) - N_k(x) + \Pi_h(x)N_{k-h}(x+h).$$

Let H be an integer with $2 \leq H \leq (p-1)/2$, and assume

$$\Delta_{h,k} \not\equiv 0 \pmod{p} \quad \text{in } \mathbb{F}_p[x] \quad (1 \leq h < k \leq H). \quad (\text{NV}_{p,H})$$

Then, for every $a \in \mathbb{F}_p^\times$,

$$N_p(a) \leq 2 \left\lfloor \frac{p-1}{H+1} \right\rfloor + \frac{H(H-1)(2H-1)}{2} + 2. \quad (21)$$

In particular, if $(\text{NV}_{p,H})$ holds for

$$H = \left\lceil (2p/3)^{1/4} \right\rceil,$$

then

$$N_p(a) \leq 4(2/3)^{3/4} p^{3/4} + \sqrt{6} p^{1/2} + 3(2/3)^{1/4} p^{1/4} + 3. \quad (22)$$

Thus the leading constant supplied by this argument is

$$4(2/3)^{3/4} = 2.951151 \dots$$

Proof. We first establish the exact algebra behind the bordered certificate. For a base point r put

$$Y_m = \left(\frac{(r+m)!}{r!} \right)^3 b_{r+m} = \Pi_m(r) b_{r+m}.$$

As long as $0 \leq r$ and $r+m \leq p-1$, clearing the denominators in the Apéry recurrence gives

$$Y_{m+1} = P(r+m)Y_m - (r+m)^6 Y_{m-1}, \quad (23)$$

with $Y_0 = b_r$ and $Y_1 = (r+1)^3 b_{r+1}$. Let $B_0(x) = 1$, $B_1(x) = 0$, and let B_m satisfy the same recurrence in m as N_m . The two fundamental solutions of (23) give

$$Y_m = N_m(r)Y_1 + B_m(r)Y_0. \quad (24)$$

Moreover,

$$B_m(x) = -(x+1)^6 N_{m-1}(x+1) \quad (m \geq 1). \quad (25)$$

Indeed, this has the correct value at $m = 1$, and its induction step is exactly the recurrence for $N_{m-1}(x+1)$.

Write $R_m(x) = \Pi_m(x) - B_m(x)$. If $b_t = b_{t+m} = a$, equation (24) becomes

$$N_m(t)(t+1)^3 b_{t+1} - R_m(t)a = 0. \quad (26)$$

Consequently, if $b_t = b_{t+h} = b_{t+k} = a$ with $1 \leq h < k$ and $t+k \leq p-1$, the nonzero vector $((t+1)^3 b_{t+1}, a)$ is killed by the two rows in (26). Hence

$$D_{h,k}(t) = 0, \quad D_{h,k}(x) := N_h(x)R_k(x) - N_k(x)R_h(x). \quad (27)$$

The structural factors of this determinant are important. A continuant Casoratian calculation gives

$$N_h B_k - N_k B_h = -\Pi_h^2 N_{k-h}(x+h). \quad (28)$$

For completeness, both sides, as functions of k , satisfy the same second-order recurrence; at $k = h$ they vanish, while at $k = h+1$ they equal $-\Pi_h^2$. Substituting (28) in (27) yields the exact factorization

$$\boxed{D_{h,k}(x) = \Pi_h(x)\Delta_{h,k}(x)}. \quad (29)$$

Let $\ell_0 = 0$, $\ell_1 = 1$, and $\ell_{m+1} = 34\ell_m - \ell_{m-1}$. Since ℓ_m is the leading coefficient of N_m , the leading coefficient of $\Delta_{h,k}$ is

$$\ell_h + \ell_{k-h} - \ell_k < 0.$$

The inequality follows, for example, from the strict increase of (ℓ_m) and $\ell_{m+1} > 2\ell_m$. Therefore, over $\mathbb{Z}$,

$$\deg \Delta_{h,k} = 3(k-1), \quad \deg D_{h,k} = 3(h+k) - 3. \quad (30)$$

The leading coefficient can, however, vanish modulo p , so its characteristic-zero nonvanishing does not prove $(NV_{p,H})$.

Besides the cubic boundary factor Π_h , interval reflection gives all three immediately forced midpoint factors:

$$\begin{aligned} 2 \mid h &\implies 2x + h + 1 \mid \Delta_{h,k}, \\ 2 \mid k &\implies 2x + k + 1 \mid \Delta_{h,k}, \\ 2 \mid (k-h) &\implies 2x + h + k + 1 \mid \Delta_{h,k}. \end{aligned} \quad (31)$$

These follow by applying $N_m(-x-m-1) = (-1)^{m-1}N_m(x)$ to the first interval, the whole interval, and the translated final interval, respectively. Exactly one of the three lengths $h, k, k-h$ is even unless h and k are both even, in which case all three are even. These are the factors forced by the three reflections; in the exceptional smallest case $\Delta_{1,2} = -4(2x+3)^3$.

We record why the known endpoint facts do not establish the missing nonvanishing. Using Remark 17 in (29) gives, for $1 \leq r \leq h$,

$$\Delta_{h,k}(-r) = (-1)^{r-1} b_{r-1} (b_{h-r} - b_{k-r}) ((r-1)!(k-r)!)^3, \quad (32)$$

whereas, for $h < r \leq k$,

$$\Delta_{h,k}(-r) = (-1)^{r-1} b_{k-r} (b_{r-h-1} - b_{r-1}) ((r-1)!(k-r)!)^3. \quad (33)$$

In particular, the raw determinant vanishes structurally at $-1, \dots, -h$. After removing Π_h , its first two evaluations (when $h \geq 2$) are

$$\Delta_{h,k}(-1) = (k-1)!^3 (b_{h-1} - b_{k-1}), \quad \Delta_{h,k}(-2) = -5(k-2)!^3 (b_{h-2} - b_{k-2}).$$

Thus their simultaneous vanishing gives two equalities, not two consecutive zeros, and Lemma 10 does not contradict it. For example, modulo 131, the recurrence gives

$$b_{10} = b_{53} = 15, \quad b_{11} = b_{54} = 15;$$

hence both evaluations vanish for $(h, k) = (12, 55)$, while $\Delta_{12,55}(-3) = 2$. Removing the forced midpoint factor $2x + 13$ does not change this example, since it is nonzero at -1 and -2 . Lemma 12 proves the corresponding assertion for the individual N_m , but it does not imply $(NV_{p,H})$ for this bordered combination.

We now use the explicitly stated hypothesis $(NV_{p,H})$. List the a -fiber in increasing order as

$$t_1 < t_2 < \dots < t_s.$$

For $1 \leq i \leq s-2$ put

$$h_i = t_{i+1} - t_i, \quad k_i = t_{i+2} - t_i.$$

If $k_i \leq H$, then (29) and $t_i + k_i \leq p-1$ show that $\Pi_{h_i}(t_i)$ is a unit modulo p . Thus t_i is a root of the nonzero polynomial Δ_{h_i, k_i} . For a fixed pair (h, k) there are at most $3(k-1)$ such bases, by (30). Summing over all short patterns gives

$$\sum_{k=2}^H \sum_{h=1}^{k-1} 3(k-1) = 3 \sum_{j=1}^{H-1} j^2 = \frac{H(H-1)(2H-1)}{2}. \quad (34)$$

For the remaining indices, $t_{i+2} - t_i > H$, hence this integer is at least $H+1$. Splitting the indices i according to parity, the corresponding increments telescope and have total at most $p-1$ in each parity class. The number of long indices is therefore at most

$$2 \left\lfloor \frac{p-1}{H+1} \right\rfloor.$$

Together with the two final fiber elements, this proves (21).

Finally put $x = (2p/3)^{1/4}$ and $H = \lceil x \rceil$. This H lies in $[2, (p-1)/2]$ for $p \geq 7$. Using $H(H-1)(2H-1)/2 < H^3 \leq (x+1)^3$ and $2(p-1)/(H+1) < 2p/x = 3x^3$, the right-hand side of (21) is at most

$$4x^3 + 3x^2 + 3x + 3,$$

which is exactly (22). □

Corollary 34 (Collision energy, conditional form). *Under the bordered nonvanishing hypothesis of Proposition 33, let $U(p)$ denote the right-hand side of (22). Then*

$$E(p) := \sum_{a \in \mathbb{F}_p} N_p(a)^2 \leq p \max\{Z(p), U(p)\} = O(p^{7/4}).$$

Proof. By Proposition 14, $N_p(0) = Z(p) = O(p^{2/3})$, while the conditional proposition gives $N_p(a) \leq U(p) = O(p^{3/4})$ for $a \neq 0$. Since $\sum_a N_p(a) = p$,

$$E(p) \leq (\max_a N_p(a)) \sum_a N_p(a) \leq p \max\{Z(p), U(p)\} = O(p^{7/4}).$$

□

Proposition 35 (An exact degeneration outside the nonsingular range). *Over $\mathbb{F}_7[x]$ one has*

$$\Delta_{h,21}(x) = 0 \quad (1 \leq h < 21).$$

In particular, a restriction on the size of k relative to the characteristic is essential.

Proof. Put

$$A_j(x) = \begin{pmatrix} P(x+j) & -(x+j)^6 \\ 1 & 0 \end{pmatrix}, \quad T = A_7 A_6 \cdots A_1, \quad q = x^7 - x.$$

All matrices in this proof are over $\mathbb{F}_7[x]$. Expanding the seven factors, equivalently applying the continuant recurrence seven times, gives the exact identities

$$\operatorname{tr} T = -q^3, \quad \det T = \prod_{j=1}^7 (x+j)^6 = q^6.$$

The second identity also follows immediately by multiplying the determinants. Cayley–Hamilton now gives

$$T^2 + q^3 T + q^6 I = 0, \quad \text{hence} \quad T^3 = q^9 I. \quad (35)$$

Since $A_{j+7} = A_j$ in characteristic 7, the transfer matrix for twenty-one steps is T^3 . Applying (35) to the initial vectors $(N_1, N_0)^\top = (1, 0)^\top$ and $(B_1, B_0)^\top = (0, 1)^\top$ yields

$$N_{21} = 0, \quad B_{21} = q^9.$$

On the other hand,

$$\Pi_{21} = \left(\prod_{j=1}^{21} (x+j) \right)^3 = q^9.$$

Thus $R_{21} = \Pi_{21} - B_{21} = 0$, and consequently

$$D_{h,21} = N_h R_{21} - N_{21} R_h = 0.$$

The factorization $D_{h,21} = \Pi_h \Delta_{h,21}$ and the fact that $\mathbb{F}_7[x]$ is an integral domain finish the proof. □

We next compute the few leading coefficients that detect every possible degeneration before a singular step is reached. Recall that $\ell_0 = 0$, $\ell_1 = 1$, and $\ell_{m+1} = 34\ell_m - \ell_{m-1}$.

Lemma 36 (Centered continuant coefficients). *For $m \geq 1$, set*

$$\hat{N}_m(Y) = N_m \left(Y - \frac{m+1}{2} \right).$$

Let u_m and v_m be respectively the coefficients of Y^{3m-5} and Y^{3m-7} in $\hat{N}_m$; a coefficient with a negative exponent is understood to be zero. Then

$$u_m = \frac{-(m-1)(32m^2 - 64m - 11)\ell_m + 5m\ell_{m-1}}{256}, \quad (36)$$

$$v_m = \frac{(m-1)(5120m^5 - 31744m^4 + 62016m^3 - 19264m^2 - 37024m - 2095)\ell_m}{655360} - \frac{m(320m^3 - 1600m^2 - 320m + 621)\ell_{m-1}}{131072}. \quad (37)$$

Moreover, if

$$\hat{\Pi}_m(Y) = \Pi_m\left(Y - \frac{m+1}{2}\right),$$

then

$$\hat{\Pi}_m(Y) = Y^{3m} - \frac{m(m^2 - 1)}{8}Y^{3m-2} + \text{terms of degree at most } 3m - 4. \quad (38)$$

All these identities hold over $\mathbb{Z}[1/10]$.

Proof. The reflection identity $N_m(-x - m - 1) = (-1)^{m-1}N_m(x)$ shows that $\hat{N}_m$ contains only powers of the same parity as $3m - 3$. To compare its first three possible powers, center the recurrence for N_{m+1} by putting $T = x + (m+2)/2$. The natural centered variables for N_m and N_{m-1} are then $T - 1/2$ and $T - 1$, respectively, and

$$P(x+m) = 34\left(T + \frac{m-1}{2}\right)^3 + \frac{3}{2}\left(T + \frac{m-1}{2}\right), \quad (x+m)^6 = \left(T + \frac{m-2}{2}\right)^6.$$

Comparison of the coefficients two and four places below the leading one gives

$$u_{m+1} = 34u_m - u_{m-1} - \frac{3}{4}(17m^2 - 17m - 2)\ell_m + \frac{3}{4}m(m-2)\ell_{m-1}, \quad (39)$$

$$v_{m+1} = 34v_m - v_{m-1} - \frac{3}{4}(17m^2 - 17m - 36)u_m + \frac{3}{4}(m^2 - 2m - 4)u_{m-1} \quad (40)$$

$$+ \frac{3(m-1)(17m^3 + 51m^2 - 90m - 24)}{64}\ell_m - \frac{3m(m-2)(m^2 - 6)}{16}\ell_{m-1}. \quad (41)$$

For completeness, the coefficient extraction in the second line uses only

$$e_2 = \frac{S_1^2 - S_2}{2}, \quad e_4 = \frac{S_1^4 - 6S_1^2S_2 + 3S_2^2 + 8S_1S_3 - 6S_4}{24}$$

for the appropriate repeated shifts. The cases $m = 1, 2, 3$ are checked directly from $N_1 = 1$ and $N_2 = P(x+1)$. Substitution of (36) and (37) into (39)–(41), followed by $\ell_{m+1} = 34\ell_m - \ell_{m-1}$, makes the coefficients of both ℓ_m and ℓ_{m-1} agree identically. This proves the two closed forms by induction.

Finally,

$$\hat{\Pi}_m(Y) = \prod_{j=1}^m \left(Y + j - \frac{m+1}{2}\right)^3.$$

The repeated shifts have sum zero, while the coefficient two below the top is

$$-\frac{3}{2} \sum_{j=1}^m \left(j - \frac{m+1}{2}\right)^2 = -\frac{m(m^2 - 1)}{8}.$$

This proves (38). □

Theorem 37 (Range nonvanishing of the bordered certificate). *Let $p \geq 7$ be prime. For every pair of integers*

$$1 \leq h < k < p,$$

the reduced bordered certificate satisfies

$$\Delta_{h,k} \not\equiv 0 \pmod{p} \quad \text{in } \mathbb{F}_p[x].$$

Consequently $(\text{NV}_{p,H})$ holds for every $H \leq p-1$; in particular it holds throughout the range required in Proposition 33.

Proof. Work in $\mathbb{F}_p$. Put

$$d = k - h, \quad n = 3(k - 1), \quad z = x + \frac{k+1}{2}.$$

The two centered variables in the first product defining $\Delta_{h,k}$ are $z - d/2$ and $z + h/2$; in the last product they occur in the opposite order. Write

$$\Delta_{h,k}(x) = Lz^n + C_1z^{n-1} + C_2z^{n-2} + \cdots.$$

Multiplying the centered expansions in Lemma 36 gives

$$L = \ell_h + \ell_d - \ell_k, \tag{42}$$

$$C_1 = \frac{3}{2}(d\ell_h - h\ell_d), \tag{43}$$

$$C_2 = u_h + u_d - u_k + \frac{d(-d^2 + 1 + 12d - 3hk)\ell_h + h(-h^2 + 1 + 12h - 3dk)\ell_d}{8}. \tag{44}$$

For example, in the first product the degree-two loss from the shifted leading powers is $3d(4d-hk)/8$; adding the second coefficient of Π_d gives the coefficient displayed in (44).

Suppose for a contradiction that $\Delta_{h,k} = 0$. Then $L = C_1 = C_2 = 0$. Set

$$A = \ell_h, \quad B = \ell_{h-1}, \quad C = \ell_d, \quad D = \ell_{d-1}.$$

The recurrence for ℓ_m gives, by induction,

$$\ell_{h+d} = 34AC - AD - BC, \tag{45}$$

$$\ell_{h+d-1} = AC - BD, \tag{46}$$

$$B^2 - 34AB + A^2 = 1, \quad D^2 - 34CD + C^2 = 1. \tag{47}$$

Since h, d, k are nonzero in $\mathbb{F}_p$, equation (43) gives a unique $\rho \in \mathbb{F}_p$ with

$$A = h\rho, \quad C = d\rho. \tag{48}$$

We first show that $\rho = 0$. Otherwise put $U = h\rho$. From $L = 0$, (48), and (45), after division by ρ ,

$$hD = d(34U - B) - k. \tag{49}$$

Substitute (48) and (49) into the second Cassini identity in (47), multiply it by h^2 , and subtract d^2 times the first Cassini identity. Direct expansion leaves

$$2dk(B - 17U + 1) = 0.$$

Thus $B = 17U - 1$. The first Cassini identity now reduces to $-288U^2 = 0$, impossible because $p \geq 7$ and $U \neq 0$. Therefore

$$A = C = 0. \quad (50)$$

Equations (45)–(47) now give

$$\ell_k = 0, \quad \ell_{k-1} = -BD, \quad B, D \in \{1, -1\}.$$

Using (36) in (44), together with (50), we obtain

$$C_2 = \frac{5}{256}(hB + dD + kBD). \quad (51)$$

For $(B, D) = (1, 1), (1, -1), (-1, 1), (-1, -1)$, the parenthesis in (51) is respectively $2k, -2d, -2h, 0$. Since $h, d, k < p$, its only possible zero is therefore

$$B = D = -1, \quad \ell_{k-1} = -1. \quad (52)$$

It remains to inspect one more even coefficient. Under (50) and (52), formulas (36) and (37) become, for $m = h, d, k$,

$$u_m = -\frac{5m}{256}, \quad v_m = \frac{m(320m^3 - 1600m^2 - 320m + 621)}{131072}. \quad (53)$$

(The dangerous case cannot contain $m = 1$, since $\ell_1 = 1$.) Let C_4 denote the coefficient of z^{n-4} in $\Delta_{h,k}$. In the present case $A = C = 0$, the shifted products and (38) give

$$C_4 = v_h + v_d - v_k \quad (54)$$

$$+ \frac{1}{8}u_h d(-d^2 + 1 + 30d - 3hk) + \frac{1}{8}u_d h(-h^2 + 1 + 30h - 3dk). \quad (55)$$

Indeed, by the parity structure of $\widehat{N}_m$ and $\widehat{\Pi}_m$, every contribution to the z^{n-4} coefficient of the two shifted products other than those displayed carries the factor ℓ_h (in $\widehat{N}_h \cdot \widehat{\Pi}_d$) or ℓ_d (in $\widehat{\Pi}_h \cdot \widehat{N}_d$)—namely the pairings of the leading coefficient of $\widehat{N}$ with the levels $0, -2, -4$ of $\widehat{\Pi}$ and with the shift binomials—and these vanish here; equation (55) is *not* asserted for general (h, k) . For the u_h term the degree-two loss from the shifted leading powers is $3d(10d - hk)/8$, and the contribution from $\widehat{\Pi}_d$ is $-d(d^2 - 1)/8$; the other term is symmetric. Substitution of (53) into (55) gives

$$\begin{aligned} v_h + v_d - v_k &= -\frac{5hd(4d^2 + 6dh - 15d + 4h^2 - 15h - 2)}{2048}, \\ \text{the two } u\text{-terms} &= \frac{10hd(2d^2 + 3dh - 15d + 2h^2 - 15h - 1)}{2048}. \end{aligned}$$

Consequently

$$C_4 = -\frac{75hdk}{2048}.$$

This is nonzero in $\mathbb{F}_p$, because $p \geq 7$ and $0 < h, d, k < p$. This final contradiction proves the theorem. $\square$

Corollary 38 (Unconditional uniform fiber and energy bounds). *For every prime $p \geq 7$, every $a \in \mathbb{F}_p^\times$, and every $2 \leq H \leq (p-1)/2$,*

$$N_p(a) \leq 2 \left\lfloor \frac{p-1}{H+1} \right\rfloor + \frac{H(H-1)(2H-1)}{2} + 2.$$

In particular, with $H = \lceil (2p/3)^{1/4} \rceil$,

$$N_p(a) \leq 4(2/3)^{3/4} p^{3/4} + \sqrt{6} p^{1/2} + 3(2/3)^{1/4} p^{1/4} + 3.$$

Moreover,

$$\sum_{a \in \mathbb{F}_p} N_p(a)^2 = O(p^{7/4}).$$

Proof. Theorem 37 supplies hypothesis $(NV_{p,H})$ in Proposition 33. That proposition gives the first two assertions, and Corollary 34 gives the last one. $\square$

Remark 39 (Verification and sharpness). The nonvanishing theorem and its boundary are machine-verified from several independent directions: (i) direct polynomial-level scans find no identically vanishing $\Delta_{h,k}$ for any prime $7 \leq p \leq 700$ and $1 \leq h < k \leq 40$, nor in 1000 random cases with $p \leq 997$; (ii) the centered coefficient formulas for L, C_1, C_2 were checked against exact expansions over $\mathbb{Q}$, and the danger-case pipeline ($L = C_1 = C_2 = 0, B = D = -1, C_3 = 0, C_4 = -75hdk/2048$) was confirmed on 309 constructed rank-of-apparition cases across all primes $p \leq 3000$ and on three genuine drop-four cases at $p \approx 10^7$, in every instance against the actual centered polynomial; (iii) the $p = 7, k = 21$ degeneration is exact: $T^3 = (x^7 - x)^9 I$ for the seven-step transfer matrix, so $N_{21} \equiv 0$ and $R_{21} \equiv 0$ identically. The theorem makes the range restriction $k < p$ explicit and sharp; for the fiber count only $k \leq H \asymp p^{1/4}$ is used.

Remark 40 (Why the uniform fiber bound matters). The energy bound $E(p) = \sum_a N_p(a)^2 = O(p^{7/4})$ (Corollary 38) is the first *unconditional* power saving over the trivial $E(p) \leq p \max_a N_p(a) \leq p^2$ scale for the value distribution of Apéry numbers modulo p ; the random-map prediction is $E(p) \asymp p$. Any bound $E(p) \ll p^{2-\eta}$ gives $Z(p) \leq E(p)^{1/2} \ll p^{1-\eta/2}$, so energy savings are a strictly weaker—hence potentially more accessible—avenue than direct zero-set bounds, and the value-collision structure enters the two-prime dispersion analysis of Section 12 through the equal-value pattern counts. Note that the $p^{3/4}$ fiber bound also complements the zero-fiber bound: $Z(p) = O(p^{2/3})$ remains the stronger statement for $a = 0$, so $\max_a N_p(a) = O(p^{3/4})$ with the maximum over *all* a .

The next block of results develops the arithmetic of the separated-block resultants $\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d))$ themselves: complex root localization (hence positivity), a centered norm decomposition, half-integer center divisor lattices, an exact square law on the diagonal, the precise projective meaning of $p \mid \mathcal{R}_{d,r}$, and the two counting lemmas that separate the present state of the collision-energy program from the mesoscopic target $Z(p) \ll \sqrt{p}$.

5.5 Separated root strips and centered resultant structure

Retain the notation

$$P(n) = (2n+1)(17n^2 + 17n + 5), \quad N_{h+1}(x) = P(x+h)N_h(x) - (x+h)^6 N_{h-1}(x),$$

with $N_0 = 0$ and $N_1 = 1$, and, for $d, r \geq 2$, put

$$\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d)).$$

We use the standard resultant convention. The first result proves, in particular, that every integer $\mathcal{R}_{d,r}$ is nonzero.

We will use only the absolute adjacent-resultant formula from Remark 18. With the standard signed convention, its one-step recursion for $h \geq 2$ has the correction

$$\text{Res}(N_h, N_{h+1}) = (-1)^{h-1} N_h (-h)^6 \text{Res}(N_{h-1}, N_h);$$

the factor $(-1)^{h-1}$ comes from the scalar minus sign in the recurrence.

Proposition 41 (Separated root strips). *For $d \geq 2$, every complex root α of N_d satisfies*

$$-d < \text{Re } \alpha < -1. \quad (56)$$

Consequently, for all $d, r \geq 2$,

$$\mathcal{R}_{d,r} > 0.$$

Proof. Let $\mathbf{A}_d(x)$ be the $(d-1) \times (d-1)$ symmetric tridiagonal matrix with

$$(\mathbf{A}_d)_{ii} = P(x+i), \quad (\mathbf{A}_d)_{i,i+1} = (\mathbf{A}_d)_{i+1,i} = (x+i+1)^3.$$

Expansion at the last row gives the defining continuant recurrence, so $\det \mathbf{A}_d(x) = N_d(x)$.

Suppose $\text{Re } x \geq -1$ and set $z_i = x + i + \frac{1}{2}$. Then $|z_i| \geq \frac{1}{2}$ and

$$P(x+i) = 34z_i^3 + \frac{3}{2}z_i.$$

It follows that

$$|P(x+i)| \geq |z_i| \left(34|z_i|^2 - \frac{3}{2} \right) \geq 28|z_i|^3.$$

The sum of the absolute values of the two possible off-diagonal entries in row i is at most

$$|z_i - \frac{1}{2}|^3 + |z_i + \frac{1}{2}|^3 \leq 2(|z_i| + \frac{1}{2})^3 \leq 16|z_i|^3.$$

Thus $\mathbf{A}_d(x)$ is strictly row-diagonally dominant and is invertible. There is no root with $\text{Re } x \geq -1$.

If J is the reversal matrix, the definitions and $P(-z-1) = -P(z)$ give

$$\mathbf{A}_d(-x-d-1) = -J\mathbf{A}_d(x)J.$$

Taking determinants proves the reflection identity

$$N_d(-x-d-1) = (-1)^{d-1} N_d(x)$$

then excludes $\text{Re } x \leq -d$, proving (56). The roots of $N_r(x+d)$ lie in the disjoint strip

$$-d-r < \text{Re } x < -d-1,$$

so the two polynomials have no common complex root and their resultant is nonzero.

Finally, the leading coefficients ℓ_d, ℓ_r are positive. In the root-product formula for the resultant, a difference of two real roots is positive because the first root strip lies strictly to the right of the second. Nonreal roots can be grouped with their conjugates, making the corresponding products positive squared absolute values. Hence the resultant itself is positive. $\square$

It is useful to clear all dyadic denominators in the natural centered coordinate. For $h \geq 1$, if $n_h = 3(h-1)$, define

$$M_h(Y) = 2^{n_h} N_h \left(\frac{Y - h - 1}{2} \right) \in \mathbb{Z}[Y]. \quad (57)$$

Then

$$M_h(-Y) = (-1)^{h-1} M_h(Y), \quad \text{lc } M_h = \ell_h,$$

and a linear change of variable gives the exact normalization

$$\text{Res}_Y(M_d(Y), M_r(Y + d + r)) = 2^{n_d n_r} \mathcal{R}_{d,r}. \quad (58)$$

Proposition 42 (Centered norm decomposition). *For $d, r \geq 2$, write*

$$\widehat{N}_d(Y) = N_d \left(Y - \frac{d+1}{2} \right) = \begin{cases} A_d(Y^2), & d \text{ odd}, \\ Y A_d(Y^2), & d \text{ even}, \end{cases}$$

where $A_d \in \mathbb{Z}[1/2][U]$, and set

$$G_{d,r}(Y) = N_r \left(Y + \frac{d-1}{2} \right), \quad \Psi_{d,r}(Y^2) = G_{d,r}(Y) G_{d,r}(-Y).$$

Then

$$|\mathcal{R}_{d,r}| = \begin{cases} |\text{Res}_U(A_d, \Psi_{d,r})|, & d \text{ odd}, \\ |N_r((d-1)/2)| |\text{Res}_U(A_d, \Psi_{d,r})|, & d \text{ even}. \end{cases} \quad (59)$$

Moreover, the centered-coefficient formulas of Lemma 36 give

$$A_d(U) = \ell_d U^m + u_d U^{m-1} + v_d U^{m-2} + \dots$$

with the evident adjustment when d is even.

Proof. In the centered variable $Y = x + (d+1)/2$, the second polynomial is $G_{d,r}(Y)$. If d is odd, the roots of $A_d(Y^2)$ occur in pairs $\{\sqrt{u}, -\sqrt{u}\}$, and their two contributions to the root-product formula multiply to $\Psi_{d,r}(u)$. This is exactly the first resultant in (59). If d is even, the additional root $Y = 0$ contributes $G_{d,r}(0) = N_r((d-1)/2)$. The last assertion is the parity statement and the coefficient formulas of Lemma 36. $\square$

Equivalently, in the finite $\mathbb{Q}$ -algebra

$$K_d = \mathbb{Q}[X]/(N_d), \quad \alpha = X \bmod N_d,$$

one has

$$\mathcal{R}_{d,r} = \ell_d^{3(r-1)} \text{Nm}_{K_d/\mathbb{Q}}(N_r(\alpha + d)). \quad (60)$$

No irreducibility or separability assumption is needed for this determinantal norm identity. For $d = 2$, $A_2(U) = 34U + \frac{3}{2}$; Proposition 42 is precisely the rational linear factor times quadratic norm decomposition recorded in Remark 31.

5.6 Half-integer center factors and their propagation

The even centered factor gives a genuine divisibility lattice, rather than only a numerical repetition pattern.

Proposition 43 (Even-block center recurrence). *Fix an even integer $a \geq 2$ and define*

$$T_0^{(a)} = 0, \quad T_1^{(a)} = 1, \quad T_b^{(a)} = 2^{3(b-1)} N_b \left(\frac{a-1}{2} \right) \quad (b \geq 2).$$

If $z = a + 2b - 1$, then

$$T_{b+1}^{(a)} = (34z^3 + 102z^2 + 108z + 40)T_b^{(a)} - z^6 T_{b-1}^{(a)}. \quad (61)$$

In particular, every $T_b^{(a)}$ is an integer. Furthermore,

$$T_b^{(a)} \mid \mathcal{R}_{a,b}, \quad (62)$$

and the odd part of $T_b^{(a)}$ divides $\mathcal{R}_{b,a+b}$. By symmetry, if b is even, the odd part of $T_a^{(b)}$ divides both $\mathcal{R}_{a,b}$ and $\mathcal{R}_{a,a+b}$.

Proof. Substitute $x = (a-1)/2$ in the recurrence for N_b and multiply by 2^{3b} . Since $2(x+b) = a+2b-1 = z$ and

$$8P(z/2) = 34z^3 + 102z^2 + 108z + 40,$$

one obtains (61).

Because a is even, reflection shows that $c = -(a+1)/2$ is a root of N_a . Gauss's lemma gives

$$N_a(X) = (2X + a + 1)Q_a(X), \quad Q_a \in \mathbb{Z}[X].$$

Resultant multiplicativity and the linear-root formula yield

$$\text{Res}_X(2X + a + 1, N_b(X + a)) = 2^{3(b-1)} N_b(c + a) = T_b^{(a)},$$

which proves (62).

Let p^e be any odd prime power dividing $T_b^{(a)}$. Since 2 is a unit modulo p^e , this says

$$N_b(t) = 0 \pmod{p^e}, \quad t = (a-1)/2.$$

The restart identity is valid without division in the present situation: the two sides satisfy the same recurrence and hence

$$N_{a+b}(c) = N_{a+1}(c)N_b(c+a) = N_{a+1}(c)N_b(t) = 0 \pmod{p^e}.$$

Put $y = -t - b - 1$. Reflection gives $N_b(y) = (-1)^{b-1} N_b(t) = 0$, while $y + b = c$, so $N_{a+b}(y+b) = 0$. The Sylvester–Bézout identity for the resultant, evaluated at $y \in \mathbb{Z}/p^e\mathbb{Z}$, gives $p^e \mid \mathcal{R}_{b,a+b}$. Taking all odd prime powers proves the claim. The symmetric assertion follows from $\mathcal{R}_{a,b} = \mathcal{R}_{b,a}$. $\square$

There is also a simultaneous center factor for every pair. With $T_b^{(a)}$ as above, define

$$\mathfrak{C}_{d,r} = \begin{cases} 1, & d, r \text{ odd}, \\ |T_r^{(d)}|, & d \text{ even}, r \text{ odd}, \\ |T_d^{(r)}|, & d \text{ odd}, r \text{ even}, \\ \frac{|T_r^{(d)} T_d^{(r)}|}{2(d+r)}, & d, r \text{ even}. \end{cases}$$

Then $\mathfrak{C}_{d,r}$ is an integer and

$$\mathfrak{C}_{d,r} \mid \mathcal{R}_{d,r}. \quad (63)$$

Indeed, factor every available central linear factor before applying resultant multiplicativity. When both indices are even, the two linear factors have resultant $2(d+r)$, which accounts for the denominator and prevents counting their mutual resultant twice.

5.7 The diagonal square law

For the diagonal family, the centered pairing becomes an exact square.

Theorem 44 (Diagonal square law). *For $d \geq 2$, let $n = 3(d-1)$ and put*

$$f_d(T) = N_d \left(T - \frac{2d+1}{2} \right) = E_d(T^2) + TO_d(T^2), \quad E_d, O_d \in \mathbb{Q}[U].$$

Then

$$\mathcal{R}_{d,d} = (-1)^{d-1} 2^n \ell_d N_d \left(-\frac{2d+1}{2} \right) \text{Res}_U(O_d, E_d)^2. \quad (64)$$

Moreover,

$$D_d = 2^n N_d(-1/2) \in \mathbb{Z}, \quad Q_d = \text{Res}_U(O_d, E_d) \in \mathbb{Z},$$

and therefore

$$\boxed{\mathcal{R}_{d,d} = \ell_d |D_d| Q_d^2}. \quad (65)$$

For every odd prime p ,

$$v_p(\mathcal{R}_{d,d}) = v_p(\ell_d) + v_p(D_d) + 2v_p(Q_d). \quad (66)$$

In particular, if $p \nmid \ell_d D_d$, then $v_p(\mathcal{R}_{d,d})$ is even.

Proof. We first record a general identity. Let

$$f(T) = aT^n + \cdots = E(T^2) + TO(T^2),$$

and suppose the two parity parts have the full degrees forced by nonzero coefficients in degrees n and $n-1$. If α_i are the roots of f , then

$$f(-\alpha_i) = -2\alpha_i O(\alpha_i^2),$$

and the root-product formula gives

$$\text{Res}_T(f(T), f(-T)) = 2^n a^{n-1} f(0) \prod_i O(\alpha_i^2). \quad (67)$$

If $s = \deg O$, $e = \deg E$, and $b = \text{lc } O$, then

$$\begin{aligned} a^{2s} \prod_i O(\alpha_i^2) &= \text{Res}_T(f(T), O(T^2)) \\ &= b^{n-2e} \text{Res}_U(O, E)^2. \end{aligned}$$

For $n = 2m$, one has $(e, s) = (m, m-1)$. For $n = 2m+1$, one has $(e, s) = (m, m)$ and $b = a$. In either case, substitution into (67) gives

$$\text{Res}_T(f(T), f(-T)) = 2^n a f(0) \text{Res}_U(O, E)^2. \quad (68)$$

For $f = f_d$, the natural centered parity of N_d shows that the coefficient of T^{n-1} is $-nd\ell_d/2$, so the stated degree conditions hold. Reflection gives

$$N_d(T - 1/2) = (-1)^{d-1} f_d(-T).$$

After the translation $T = x + (2d+1)/2$, scaling the second argument of the resultant contributes $(-1)^{(d-1)n} = (-1)^{d-1}$. Equation (68) is therefore exactly (64).

It remains only to justify the integral form. Integrality of D_d is immediate from evaluating an integer polynomial of degree n at a half-integer. Put $c = 2d+1$ and $V = X(X+c)$. Reduction by the monic relation $X^2 = V - cX$ gives the free-module decomposition

$$\mathbb{Z}[X] = \mathbb{Z}[V] \oplus X\mathbb{Z}[V].$$

Thus there are $A, B \in \mathbb{Z}[V]$ such that

$$N_d(X) = A(V) + XB(V).$$

Since $T = X + c/2$ and $U = T^2$, one has $V = U - c^2/4$, and comparison with $f_d(T) = E_d(U) + TO_d(U)$ gives

$$O_d(U) = B(U - c^2/4), \quad E_d(U) = A(U - c^2/4) - \frac{c}{2}B(U - c^2/4).$$

Translation invariance reduces Q_d to $\text{Res}_V(B, A - \frac{c}{2}B)$. In fact $\deg A = \deg E_d$. When $n = 2m$, this follows from $\deg E_d = m$ and $\deg B = m - 1$. When $n = 2m + 1$, the leading coefficients computed above give

$$\text{lc } B = \ell_d, \quad \text{lc } A = \frac{c - nd}{2} \ell_d,$$

and $c - nd = -3d^2 + 5d + 1 \neq 0$ for every integer $d \geq 2$. The root-product formula therefore gives

$$Q_d = \text{Res}_U(O_d, E_d) = \text{Res}_V(B, A - \frac{c}{2}B) = \text{Res}_V(B, A) \in \mathbb{Z}.$$

Finally, reflection gives

$$\left| N_d \left(-\frac{2d+1}{2} \right) \right| = |N_d(-1/2)|.$$

Proposition 41 supplies the sign, and (65) follows. The valuation formula is immediate. $\square$

In odd characteristic the affine diagonal roots can also be read off exactly. In the coordinate used above,

$$f_d(T) = f_d(-T) = 0 \iff E_d(T^2) = 0, \quad TO_d(T^2) = 0.$$

Thus the number of distinct $\mathbb{F}_p$ -roots is

$$\mathbf{1}_{p \nmid f_d(0)} + 2\#\{u \in (\mathbb{F}_p^\times)^2 : E_d(u) = O_d(u) = 0\}. \quad (69)$$

This distinguishes an actual split collision from a nonsplit quadratic factor of the resultant.

5.8 What resultant divisibility does and does not mean

Proposition 45 (Projective validity domain). *For integer polynomials of their nominal degrees, $p \mid \mathcal{R}_{d,r}$ if and only if their nominal homogenizations have a common point in $\mathbb{P}^1(\mathbb{F}_p)$. If the two leading coefficients do not both vanish modulo p , this is equivalent to an affine common root in $\mathbb{F}_p$. It is not, in general, equivalent to a common root in $\mathbb{F}_p$. The universally valid implication is only*

$$\{x \in \mathbb{F}_p : N_d(x) = N_r(x + d) = 0\} \neq \emptyset \implies p \mid \mathcal{R}_{d,r}.$$

Two mesoscopic examples show both failures of the converse. For $d = r = 2$ and $p = 17$,

$$N_2(X) \equiv 5(2X + 3), \quad N_2(X + 2) \equiv 5(2X + 7) \pmod{17}.$$

They have no affine common root, although $17^3 \mid \mathcal{R}_{2,2}$; both nominal cubics lost their leading terms and meet at infinity. On the other hand, for $d = r = 3$ and $p = 110629$, the leading coefficient is a unit and

$$\gcd(N_3(X), N_3(X + 3)) = X^2 + 7X - 15495.$$

Its discriminant is 62029, a nonsquare modulo 110629. Hence there is an affine common root over $\mathbb{F}_p$ but none over $\mathbb{F}_p$, even though $110629^2 \mid \mathcal{R}_{3,3}$.

The centered coefficients identify the complete lattice of automatic infinity edges. Recall

$$\ell_0 = 0, \quad \ell_1 = 1, \quad \ell_{n+1} = 34\ell_n - \ell_{n-1}.$$

For $p \geq 5$, choose $\lambda \in \mathbb{F}_{p^2}$ with $\lambda + \lambda^{-1} = 34$. Then

$$\ell_n = \frac{\lambda^n - \lambda^{-n}}{\lambda - \lambda^{-1}}.$$

If $\rho_p = \text{ord}(\lambda^2)$, then

$$p \mid \ell_n \iff \rho_p \mid n, \quad \rho_p \mid p - \left(\frac{2}{p}\right). \quad (70)$$

Consequently every pair with $\rho_p \mid d, r$ is in the raw resultant support, independently of affine collisions. If $M = \lfloor H/\rho_p \rfloor$, these pairs alone number

$$V_p^{\text{infinity}}(H) = \frac{M(M-1)}{2}. \quad (71)$$

There is a legitimate centered-coefficient transfer here, but it detects only the false edge at infinity. Suppose $p \geq 7$ and $d < p$. If $\ell_d = 0$, Cassini gives $\ell_{d-1}^2 = 1$, while Lemma 36 gives

$$u_d = \frac{5d\ell_{d-1}}{256} \not\equiv 0 \pmod{p}.$$

Because the natural centered polynomial contains only every other power, its degree drops by exactly two. If $\rho_p \mid d, r$, the two nominal homogenizations have a common factor of order two at infinity; equivalently, the nominal Sylvester matrix has nullity at least two modulo p . Its Smith normal form therefore gives

$$p^2 \mid \mathcal{R}_{d,r}. \quad (72)$$

For example, $p = 665857$ has $\rho_p = 8$ and $H = \lfloor \sqrt{p} \rfloor = 816$. Formula (71) already gives 5151 raw resultant pairs, without producing a single affine collision. Thus the unsaturated integer resultant support is not the correct collision graph.

The centered/Pell method from Theorem 37 cannot be used pointwise in the opposite direction. At $(d, r, p) = (2, 2, 71)$ and $x = -5/2$, the two natural centered variables are -1 and 1 , and

$$\widehat{N}_2(Y) = 34Y^3 + \frac{3}{2}Y, \quad \widehat{N}_2(\pm 1) = \pm \frac{71}{2} \equiv 0 \pmod{71},$$

while $\ell_2 = 34 \not\equiv 0 \pmod{71}$. The proof of Theorem 37 compares coefficients only after assuming that an entire certificate polynomial is identically zero. Two point-value equations do not make its coefficients L, C_1, C_2, C_4 vanish. In particular, its C_4 formula is valid only inside the stated danger case and cannot be transferred to this collision.

5.9 The two missing counting lemmas

For an integer $H \geq 2$, define

$$\mathcal{D}_H = \{(d, r) \in \mathbb{Z}_{\geq 2}^2 : d + r \leq H\},$$

$$X_{d,r}(p) = \{x \in \mathbb{F}_p : N_d(x) = N_r(x + d) = 0\}, \quad m_{d,r}(p) = |X_{d,r}(p)|.$$

The collision energy is the weighted count

$$\mathcal{E}_p(H) = \sum_{(d,r) \in \mathcal{D}_H} m_{d,r}(p).$$

Define also the simple affine support

$$W_p(H) = \#\{(d, r) \in \mathcal{D}_H : m_{d,r}(p) > 0\}.$$

The raw resultant support

$$V_p(H) = \#\{(d, r) \in \mathcal{D}_H : p \mid \mathcal{R}_{d,r}\}$$

is the edge count of a still larger projective-support graph. Thus

$$W_p(H) \leq \mathcal{E}_p(H), \quad W_p(H) \leq V_p(H).$$

Remark 32 identified $\mathcal{E}_p(H)$ with the edge count $W_p(H)$ by forgetting that one pair can have several root witnesses. For example, at $p = 3109$ the pair $(d, r) = (8, 8)$ has two distinct $\mathbb{F}_p$ common roots, so it contributes one to W_p and two to $\mathcal{E}_p$. The examples above exhibit the separate failure of $W_p(H) = V_p(H)$.

The same distinction changes the averaging statement in that remark. The Hadamard argument controls only

$$\sum_{X < p \leq 2X} V_p(H) \ll \frac{H^4 \log H}{\log X}.$$

After inserting the available multiplicity bound, its valid consequence is

$$\sum_{X < p \leq 2X} \mathcal{E}_p(H) \ll \frac{H^5 \log H}{\log X}.$$

Since there are $\asymp X/\log X$ primes in the interval, this proves $\mathcal{E}_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ of them only in the range $H \leq X^{2/9-\varepsilon}$, rather than the stated $X^{2/7-\varepsilon}$ range. The latter exponent applies to simple support, not to the collision energy.

The existing gcd estimate gives

$$m_{d,r}(p) \leq 3(\min(d, r) - 1).$$

Accordingly, define the degree-weighted support

$$\mathcal{W}_p(H) = \sum_{(d,r) \in \mathcal{D}_H} (\min(d, r) - 1) \mathbf{1}_{p|\mathcal{R}_{d,r}}.$$

Then the rigorous implication is

$$\mathcal{E}_p(H) \leq 3\mathcal{W}_p(H). \quad (73)$$

In particular,

$$V_p(H) \ll H \implies \mathcal{E}_p(H) \ll H^2, \quad (74)$$

not $H^{3/2}$. This loss is real at the level of the proposed graph argument: take $\asymp H$ exceptional pairs with $d, r \asymp H$, make them a matching (maximum degree one and codegree zero), and assign $\asymp H$ distinct root witnesses to every edge. This obeys the available per-pair degree bound and has energy $\asymp H^2$. The ordinary Kővári–Sós–Turán theorem applies to a simple graph and cannot control those parallel root witnesses.

Thus a sufficient replacement for G1 is the genuinely weighted estimate

$$\boxed{\mathcal{W}_p(H) \ll H^{3/2}}. \quad (75)$$

Alternatively, together with $V_p(H) \ll H$, it would suffice to prove an average actual gcd-degree bound $O(H^{1/2})$ on the exceptional pairs, or a corresponding multiplicity-tail estimate.

There is a second, independent gap between energy and the first moment $R_p(H) = \sum_{h=2}^H r_p(h)$. Let

$$k_x = \#\{2 \leq h \leq H : N_h(x) = 0\}, \quad \partial_H = \{-2, -3, \dots, -H\} \subseteq \mathbb{F}_p.$$

For $x \notin \partial_H$, Proposition 29 maps every pair $2 \leq d < e \leq H$ counted by $\binom{k_x}{2}$ injectively to the energy term $(x, d, e - d)$. Indeed $e - d = 1$ cannot occur: in that case the Bézout right-hand side is $S_d(x)^6 N_1 = S_d(x)^6 \neq 0$. Thus every resulting pair has $e - d \geq 2$ and lies in $\mathcal{D}_H$. Hence

$$\sum_{x \notin \partial_H} \binom{k_x}{2} \leq \mathcal{E}_p(H). \quad (76)$$

The exclusion of ∂_H is essential. The unsaturated degree claim following Proposition 29 fails at cut-edge roots: for $(d, e, p) = (3, 4, 73)$,

$$\gcd_{\mathbb{F}_{73}[X]}(N_3, N_4) = X + 3,$$

whereas $N_{e-d} = N_1 = 1$. Thus that claim is valid only for the saturated gcd, or as an away-from-boundary root count; the use above has precisely that domain.

The same example invalidates the proof of the unrestricted column bound in Proposition 27. Since $b_2 = 73$, the endpoint formula gives

$$N_h(-3) \equiv 0 \pmod{73} \quad (h \geq 3).$$

At the singular transition the recurrence has reached two consecutive zeros, so the entire later tail remains zero; it cannot be charged as an $O(1)$ transition cost. Thus that proof does not

establish the proposition or the rectangular incidence bound derived from it; a valid version needs an explicit boundary term or an unpolluted nonsingular initial state. The present argument excludes such columns from (76) and retains their full contribution in $R_p^\partial(H)$. But columns with $k_x = 1$ contribute nothing to either side of (76). Energy alone cannot bound their total mass.

For a precise sufficient condition, set $K = \lceil \sqrt{H} \rceil$ and

$$R_p^\partial(H) = \sum_{x \in \partial_H} k_x, \quad L_p(H) = \sum_{\substack{x \notin \partial_H \\ 0 < k_x < K}} k_x.$$

Since $k \leq 2\binom{k}{2}/(K-1)$ for $k \geq K$, (76) gives the deterministic inequality

$$R_p(H) \leq R_p^\partial(H) + L_p(H) + \frac{2}{\lceil \sqrt{H} \rceil - 1} \mathcal{E}_p(H). \quad (77)$$

Thus the missing low-fiber amplification is

$$R_p^\partial(H) + L_p(H) \ll H, \quad (78)$$

or, in compressed form,

$$R_p(H) \ll H + H^{-1/2} \mathcal{E}_p(H).$$

This is a lower-participation assertion; Kővári–Sós–Turán is an upper-bound theorem and does not supply it. The boundary term cannot be discarded, since the endpoint factorization of $N_h(-j)$ can make a cut-edge column large when $p \mid b_{j-1}$.

Combining the two genuinely missing estimates (75) and (78) would give

$$\mathcal{E}_p(H) \ll H^{3/2}, \quad R_p(H) \ll H.$$

Only then does the valid last step

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + R_p(H-1)$$

give $Z(p) \ll \sqrt{p}$ at $H = \lfloor \sqrt{p} \rfloor$.

For completeness, suppose only the proposed G2 support estimate

$$V_p(H) \ll H^{2-\delta}$$

is available. The present ingredients give merely

$$\mathcal{E}_p(H) \ll H^{3-\delta}.$$

Partition $\{0, \dots, p-1\}$ into nonwrapping consecutive blocks of length at most H , and let their Apéry zero counts be z_i . Three zeros $m < m+d < m+d+r$ in one block have $d, r \geq 2$ by Lemma 10. All intervening recurrence denominators are units modulo p , so the gap-polynomial identity gives the distinct energy witness (m, d, r) . Hence

$$\sum_i \binom{z_i}{3} \leq \mathcal{E}_p(H)$$

and Hölder's inequality yields

$$Z(p) \ll \frac{p}{H} + p^{2/3} H^{(1-\delta)/3}. \quad (79)$$

If the G2 estimate holds uniformly for $H \leq \sqrt{p}$, then, after combining with the existing $p^{2/3}$ bound,

$$Z(p) \ll \begin{cases} p^{2/3}, & 0 < \delta \leq 1, \\ p^{(5-\delta)/6}, & 1 < \delta \leq 2. \end{cases}$$

Thus even $V_p(H) \ll H$ ($\delta = 1$) recovers only the existing exponent from the currently proved inputs. Even a separately proved $\mathcal{E}_p(H) \ll H^{3/2}$ would give only $Z(p) \ll p^{7/12}$ by this triple-counting route unless the low-fiber amplification is also proved.

Finally, the recurrence does not close on the scalar resultants. In the algebra K_d of (60), put $u_r = N_r(\alpha + d)$. Then

$$u_{r+1} = P(\alpha + d + r)u_r - (\alpha + d + r)^6 u_{r-1}.$$

The norm of a sum depends on all polarized mixed norms, so $\text{Nm}(u_r)$ and $\text{Nm}(u_{r-1})$ do not determine $\text{Nm}(u_{r+1})$. Modulo p , different values of r may also vanish in different residue-field components, whereas restart requires the same root witness. This root-witness alignment, the weighted support estimate (75), and the low-fiber amplification (78) are the exact missing lemmas left by the present attack.

5.10 Collision energy is an off-boundary factorial moment

Retain

$$\mathcal{D}_H = \{(d, r) \in \mathbb{Z}_{\geq 2}^2 : d + r \leq H\}, \quad \mathcal{E}_p(H) = \sum_{(d, r) \in \mathcal{D}_H} m_{d, r}(p),$$

and assume throughout that $p \geq 7$ and $H \leq \sqrt{p}$; the omitted small cases are empty or immediate. For $x \in \mathbb{F}_p$ put

$$k_x = k_x(H) = \#\{2 \leq h \leq H : N_h(x) = 0\}, \quad R_p(H) = \sum_{x \in \mathbb{F}_p} k_x,$$

and put

$$\partial_H = \{-2, -3, \dots, -H\} \subseteq \mathbb{F}_p.$$

Split the energy as

$$\mathcal{E}_p(H) = \mathcal{E}_p^\circ(H) + \mathcal{E}_p^\partial(H)$$

according as its root witness lies outside or inside ∂_H .

Proposition 46 (Exact off-boundary column identity). *One has*

$$\boxed{\mathcal{E}_p^\circ(H) = \sum_{x \notin \partial_H} \binom{k_x}{2}.} \tag{80}$$

At the boundary only the following direction is universally valid:

$$0 \leq \mathcal{E}_p^\partial(H) \leq \sum_{x \in \partial_H} \binom{k_x}{2}. \tag{81}$$

Consequently

$$\sum_{x \notin \partial_H} \binom{k_x}{2} \leq \mathcal{E}_p(H) \leq \sum_{x \in \mathbb{F}_p} \binom{k_x}{2}. \tag{82}$$

Proof. The continuant addition formula used in the proof of Proposition 29 gives

$$N_{d+r}(x) = N_d(x)N_{r+1}(x+d-1) - (x+d)^6 N_{d-1}(x)N_r(x+d).$$

Thus an energy witness (x, d, r) gives the two zero levels d and $d+r$ in the column x . This map is injective, since its lower level and the difference recover (d, r) . It proves the upper bound in (81) and the upper bound in (82).

Conversely, suppose that $x \notin \partial_H$ and that $2 \leq d < e \leq H$ are two zero levels. Then $S_d(x)$ is a unit, so the saturated Bézout identity gives

$$N_{e-d}(x+d) = 0.$$

The difference $e-d$ cannot be one: in that case the right-hand side of (20) would be $S_d(x)^6 N_1 = S_d(x)^6 \neq 0$. Hence $(x, d, e-d)$ is an energy witness. This is inverse to the preceding map and proves (80). $\square$

The equality is stronger than (76), but it also shows why the boundary exclusion cannot be repaired by changing an inequality sign. At $p = 73$, $H = 8$, the polluted column $x = -3$ has six zero levels and hence 15 column pairs, while only two boundary energy witnesses occur in the entire computation.

Corollary 47 (What the present column estimates prove). *Uniformly for $H \leq \sqrt{p}$,*

$$\mathcal{E}_p(H) \ll H^{8/3}. \quad (83)$$

Proof. The row-degree bound gives

$$R_p(H) \leq \sum_{h=2}^H 3(h-1) = \frac{3}{2}H(H-1). \quad (84)$$

Define the window-polluted set

$$\mathcal{P}_H = \{-m : 2 \leq m \leq H, p \mid b_{m-1}\}.$$

The pollution characterization following Lemma 26 and Corollary 15 give

$$P_H := |\mathcal{P}_H| \ll H^{2/3}.$$

Every column outside $\mathcal{P}_H$ has no adjacent zero levels in $[2, H]$. (A globally polluted column whose cut occurs after H has no adjacent pair in this window.) Proposition 27 therefore gives

$$M_H := \max_{x \notin \mathcal{P}_H} k_x \ll H^{2/3}.$$

By Proposition 46,

$$\begin{aligned} \mathcal{E}_p(H) &\leq \sum_x \binom{k_x}{2} \\ &\leq \frac{M_H - 1}{2} \sum_{x \notin \mathcal{P}_H} k_x + P_H \binom{H-1}{2} \\ &\ll H^{2/3} H^2 + H^{2/3} H^2. \end{aligned}$$

This is (83). $\square$

This saves $H^{1/3}$ over the direct $O(H^3)$ degree bound, but its exponent is still larger than two and hence has no consequence for the record exponent of $Z(p)$.

5.11 The proposed fixed point is not a contraction

For an arbitrary threshold $2 \leq K \leq H$, define

$$R_p^\circ(H) = \sum_{x \notin \partial_H} k_x, \quad L_{p,K}(H) = \sum_{\substack{x \notin \partial_H \\ 0 < k_x < K}} k_x,$$

and let

$$M_p^\circ(H) = \max_{x \notin \partial_H} k_x.$$

Proposition 48 (Exact fixed-point audit). *For every such K ,*

$$R_p(H) \leq R_p^\partial(H) + L_{p,K}(H) + \frac{2}{K-1} \mathcal{E}_p^\circ(H), \quad (85)$$

where $R_p^\partial(H) = \sum_{x \in \partial_H} k_x$. On the other hand,

$$\mathcal{E}_p^\circ(H) \leq \frac{M_p^\circ(H) - 1}{2} R_p^\circ(H). \quad (86)$$

Combining the two estimates never gives an absorptive fixed point.

Proof. For $k \geq K$ one has

$$k \leq \frac{2}{K-1} \binom{k}{2}.$$

Summing this over the high off-boundary columns and using (80) proves (85). The elementary inequality $\binom{k}{2} \leq (M_p^\circ - 1)k/2$ proves (86).

Substitution gives

$$\mathcal{E}_p^\circ \leq \frac{M_p^\circ - 1}{2} L_{p,K} + \frac{M_p^\circ - 1}{K-1} \mathcal{E}_p^\circ. \quad (87)$$

If $K \leq M_p^\circ$, the feedback coefficient is at least one. If $K > M_p^\circ$, every nonzero off-boundary column is already included in $L_{p,K}$, so $L_{p,K} = R_p^\circ$ and (87) is the original tautology. These two cases exhaust all thresholds. $\square$

In particular, the proposed $K = \lceil \sqrt{H} \rceil$ does not interact contractively with the available $M_p^\circ \ll H^{2/3}$ estimate. The amplification inequality is a lower bound on the factorial moment carried by high fibers; it is not a second upper bound for that moment. Singleton columns can make $R_p(H)$ large while contributing nothing to energy, and even an independent $R_p(H) \ll H$ estimate together with only $k_x \ll H^{2/3}$ permits the abstract scale $\mathcal{E}_p(H) \asymp H^{5/3}$.

There is also a notation obstruction in the suggested partition. In `meso_result.tex`, $W_p(H)$ denotes simple affine support and $\mathcal{W}_p(H)$ denotes degree-weighted raw resultant support. A raw resultant-supported pair need have no $\mathbb{F}_p$ witness at all, and therefore cannot be assigned to a witness column. Even after restricting to affine support, the artificial weight $\min(d, r) - 1$ is not bounded by the number of actual witnesses. Thus an inequality of the form “ $\mathcal{W}_p(H) \leq \sum_x \binom{k_x}{2}$ ” is unavailable; the valid column statement is Proposition 46 for the actual energy.

5.12 The exact triple-counting exponent

Proposition 49 (Optimization from an energy exponent). *Suppose, uniformly in a range of H , that*

$$\mathcal{E}_p(H) \ll H^\beta.$$

Then block triple-counting gives

$$Z(p) \ll \frac{p}{H} + p^{2/3} H^{(\beta-2)/3}. \quad (88)$$

If the estimate is uniform for every $H \leq \sqrt{p}$, the optimized triple-counting exponent is

$$\theta_{\text{triple}}(\beta) = \begin{cases} 1/2, & \beta \leq 1, \\ (\beta+2)/6, & 1 \leq \beta \leq 2, \\ \beta/(\beta+1), & \beta \geq 2. \end{cases} \quad (89)$$

After combining with the existing $Z(p) \ll p^{2/3}$ estimate, every $1 < \beta < 2$ gives

$$Z(p) \ll p^{(\beta+2)/6},$$

and in particular $\beta = 3/2$ gives $Z(p) \ll p^{7/12}$.

Proof. Partition the index interval into $B \ll p/H$ nonwrapping blocks and let z_i be their zero counts. As in the proof of (79),

$$\sum_i \binom{z_i}{3} \leq \mathcal{E}_p(H).$$

The elementary bound

$$z \leq 2 + \left(6 \binom{z}{3}\right)^{1/3}$$

and Hölder's inequality give

$$Z(p) \ll B + B^{2/3} \mathcal{E}_p(H)^{1/3},$$

which is (88).

Write $H = p^\alpha$, $0 < \alpha \leq 1/2$. The two exponents are

$$1 - \alpha, \quad \frac{2}{3} + \frac{\beta - 2}{3} \alpha.$$

When $\beta < 2$, both decrease with α , so one takes $\alpha = 1/2$. This gives $\max\{1/2, (\beta+2)/6\}$. At $\beta = 2$ the second exponent is $2/3$. For $\beta > 2$, balancing the two terms at $\alpha = 1/(\beta+1)$ gives $\beta/(\beta+1)$. This proves (89) and the stated comparison. $\square$

Thus the power $(\beta-1)/3$ displayed in the specification is not the output of block triple-counting: the factor $B^{2/3}$ contributes $H^{-2/3}$, not $H^{-1/3}$. Equivalently, the displayed formula is missing one further factor $H^{-1/3}$. More generally, if $\beta < 2$ is available only through $H \leq p^\eta$, then taking the largest permitted H gives

$$\theta(\beta, \eta) = \max \left\{ 1 - \eta, \frac{2}{3} + \frac{\eta(\beta - 2)}{3} \right\}. \quad (90)$$

It beats $2/3$ exactly when $\eta > 1/3$ and $\beta < 2$. No interior balancing is required in this decreasing regime.

5.13 Prime-averaged collision energy

The preceding consumer does not require a pointwise energy estimate. Summing before applying Hölder gives the following exact averaged variant.

Proposition 50 (Prime-averaged collision-energy consumer). *Let X be large, put*

$$\mathcal{P}_X = \{p \text{ prime} : X < p \leq 2X\},$$

and let $2 \leq H \leq \sqrt{X}$. Then

$$\sum_{p \in \mathcal{P}_X} Z(p) \ll \frac{X^2}{H \log X} + \frac{X^{4/3}}{H^{2/3} (\log X)^{2/3}} \left(\sum_{p \in \mathcal{P}_X} \mathcal{E}_p(H) \right)^{1/3}.$$

More precisely, if $\mathcal{T}_p(H)$ is the sum of $\binom{z_I(p)}{3}$ over a partition of $[0, p-1]$ into intervals of length at most H , then the same assertion holds with $\mathcal{E}_p(H)$ replaced by $\mathcal{T}_p(H)$.

Proof. For one prime, the block-triple argument in Proposition 49 gives

$$Z(p) \ll B_p + B_p^{2/3} \mathcal{T}_p(H)^{1/3}, \quad B_p \ll X/H,$$

and $\mathcal{T}_p(H) \leq \mathcal{E}_p(H)$. Sum over p and use

$$\sum_{p \in \mathcal{P}_X} \mathcal{E}_p(H)^{1/3} \leq |\mathcal{P}_X|^{2/3} \left(\sum_{p \in \mathcal{P}_X} \mathcal{E}_p(H) \right)^{1/3},$$

together with $|\mathcal{P}_X| \ll X/\log X$. □

Proposition 51 (Prime-summed resultant baseline). *Uniformly for $H \leq \sqrt{X}$,*

$$\sum_{X < p \leq 2X} \mathcal{E}_p(H) \ll H^4 \frac{\log(2H)}{\log X}.$$

Proof. For $(d, r) \in \mathcal{D}_H$, every split common root counted by $m_{d,r}(p)$ contributes to the nullity of the reduced Sylvester matrix. Since $p > H$, the relevant leading coefficients are units, and therefore

$$m_{d,r}(p) \leq v_p(\mathcal{R}_{d,r}).$$

The separated root-strip theorem gives $\mathcal{R}_{d,r} \neq 0$, while the resultant-height estimate gives

$$\log |\mathcal{R}_{d,r}| \ll dr \log(2H).$$

Consequently

$$\sum_{X < p \leq 2X} \mathcal{E}_p(H) \leq \frac{1}{\log X} \sum_{d+r \leq H} \log |\mathcal{R}_{d,r}| \ll H^4 \frac{\log(2H)}{\log X}.$$

□

Corollary 52 (Dyadic consequence of a prime-summed energy exponent). *Suppose, uniformly in the relevant range $H \leq \sqrt{X}$, that*

$$\sum_{X < p \leq 2X} \mathcal{E}_p(H) \ll H^\gamma X^{o(1)}$$

for some fixed $\gamma \geq 3$. Then

$$\sum_{X < p \leq 2X} Z(p) \ll X^{1+c(\gamma)+o(1)}, \quad c(\gamma) = \max \left\{ \frac{1}{2}, \frac{\gamma-1}{\gamma+1} \right\}.$$

In particular $\gamma = 4$ gives the exponent $8/5$, every $\gamma = 4 - \delta$ with $\delta > 0$ gives a fixed improvement, and $\gamma = 3$ gives $X^{3/2+o(1)}$.

Proof. For dyadic $T \gg \sqrt{X}$, put

$$\mathcal{P}_X(T) = \{p : T < Z(p) \leq 2T\}, \quad H = \left\lceil \frac{16X}{T} \right\rceil.$$

Partition $[0, p-1]$ into intervals of length at most H . Their number is at most $T/8 + O(1)$, and hence

$$\sum_I \binom{z_I(p)}{3} \geq Z(p) - 2\#\{I\} \gg T.$$

It follows that

$$T|\mathcal{P}_X(T)| \ll \sum_{p \in \mathcal{P}_X(T)} \mathcal{E}_p(H) \ll (X/T)^\gamma X^{o(1)}.$$

The levels below a cutoff T_0 contribute $O(XT_0/\log X)$, while the higher dyadic levels contribute $O(X^\gamma T_0^{-\gamma+o(1)})$. Taking $T_0 = X^{(\gamma-1)/(\gamma+1)}$ proves the assertion; when this cutoff would lie below $\sqrt{X}$, use the $\sqrt{X}$ transition. $\square$

5.14 Structured subfamilies

To avoid the clash between simple affine support and weighted resultant support, write, for $\mathcal{S} \subseteq \mathcal{D}_H$,

$$\mathcal{W}_p^{\text{res}}(\mathcal{S}) = \sum_{(d,r) \in \mathcal{S}} (\min(d,r) - 1) \mathbf{1}_{p|\mathcal{R}_{d,r}}, \quad \mathcal{E}_p(\mathcal{S}) = \sum_{(d,r) \in \mathcal{S}} m_{d,r}(p).$$

Proposition 53 (Shallow strips and diagonal bands). *Let $2 \leq K \leq H/2$ and $k = \lfloor K \rfloor$. Then*

$$\mathcal{W}_p^{\text{res}}\{(d,r) : \min(d,r) \leq K\} \leq B_H(k), \tag{91}$$

$$\mathcal{E}_p\{(d,r) : \min(d,r) \leq K\} \leq 3B_H(k), \tag{92}$$

where

$$B_H(k) = \sum_{s=2}^k (s-1)(2H-4s+1) \leq Hk(k-1). \tag{93}$$

In particular, both quantities are $O(H^{3/2})$ in the deterministic range

$$\min(d,r) \leq H^{1/4}.$$

The larger suggested strip $\min(d, r) \leq \sqrt{H}$ has only the bound $O(H^2)$ after the necessary degree weight is included.

For a fixed $C \geq 0$ and $D \leq H/2$,

$$\mathcal{W}_p^{\text{res}}\{(d, r) : |d - r| \leq C, \min(d, r) \leq D\} \ll (C + 1)D^2, \quad (94)$$

and the same bound, up to a factor three, holds for the energy. Hence the part $\min(d, r) \leq H^{3/4}$ of a fixed-width diagonal band is $O_C(H^{3/2})$, whereas the full band has only the present bound $O_C(H^2)$.

Proof. For a fixed value $s = \min(d, r)$ there are exactly $2H - 4s + 1$ ordered pairs in $\mathcal{D}_H$. Summing the weight $s - 1$ gives (93). Since the reductions of N_h are nonzero in the present range,

$$m_{d,r}(p) \leq 3(\min(d, r) - 1),$$

which proves (92).

In the band $|d - r| \leq C$, each possible minimum occurs in at most $2C + 1$ ordered pairs. Summing its degree weight gives (94). $\square$

The thresholds are not artifacts of a loose summation: exactly

$$B_H(k) = Hk(k - 1) + O(k^3).$$

Thus if $k \rightarrow \infty$ and $k = o(H)$, then $B_H(k) = (1 + o(1))Hk^2$. The unweighted support of the $\sqrt{H}$ strip is $O(H^{3/2})$, but this says nothing about its energy multiplicities. Even after discarding shifted columns which are polluted, Proposition 27 gives only

$$\sum_{\substack{2 \leq d \leq K, \ 2 \leq r, \ d+r \leq H \\ x \in X_{d,r}(p), \ x+d \text{ unpolluted}}} 1 \ll K^2 H^{2/3}, \quad (95)$$

The symmetric orientation obeys the same estimate. For $K = \sqrt{H}$ this is $H^{5/3}$, still above the target.

The ambient degree weight satisfies

$$\sum_{(d,r) \in \mathcal{D}_H} (\min(d, r) - 1) = \frac{H^3}{12} + O(H^2). \quad (96)$$

Removing the whole $\sqrt{H}$ strip and any fixed-width diagonal band changes (96) by only $O_C(H^2)$. Thus the ambient dominant region is the balanced off-diagonal bulk

$$\min(d, r) > \sqrt{H}, \quad |d - r| > C.$$

This does not mean that the actual worst contribution has been localized: the shallow annulus $H^{1/4} < \min(d, r) \leq \sqrt{H}$ and the high part of the diagonal band are themselves unresolved at the $H^{3/2}$ scale.

5.14.1 Infinity and center lattices

Proposition 54 (Exact weight of the infinity lattice). *Let $\rho = \rho_p$ and $M = \lfloor H/\rho \rfloor$. The automatic projective-infinity pairs $\rho \mid d, r$ have raw degree weight*

$$\mathcal{I}_p(H) = \rho \sum_{s=2}^M \left\lfloor \frac{s^2}{4} \right\rfloor - \binom{M}{2} = \frac{H^3}{12\rho^2} + O\left(\frac{H^2}{\rho}\right). \quad (97)$$

They contribute no automatic affine witness and must be saturated before using resultant support as a collision proxy.

Proof. Write $d = \rho a$, $r = \rho b$. For $s = a + b$,

$$\sum_{a=1}^{s-1} \min(a, s-a) = \left\lfloor \frac{s^2}{4} \right\rfloor.$$

There are $\binom{M}{2}$ ordered pairs, and subtracting the 1 in each weight gives (97).

Let $\overline{N}_d$ be the reduction of N_d modulo p , let $n_d = 3(d-1)$ be its nominal degree, and define its nominal homogenization by

$$\overline{F}_d(X, Z) = Z^{n_d} \overline{N}_d(X/Z).$$

When $\rho \mid d$ and $d < p$, the centered-coefficient calculation after (70) shows that the nominal homogenization has the exact form

$$\overline{F}_d(X, Z) = Z^2 F_d^{\text{sat}}(X, Z), \quad F_d^{\text{sat}}(1, 0) \neq 0.$$

Modulo p , dividing Z^2 from each nominal form removes the automatic infinity intersection exactly. One must not delete the whole pair: its saturated forms may still have a separate affine common root. $\square$

For example, the already recorded values $p = 665857$, $\rho_p = 8$, and $H = 816$ give $\mathcal{I}_p(H) = 712657$, whereas $H^{3/2} = 23309.6\dots$. This is why the raw weighted resultant support is not the statement to which a witness-column proof can apply.

The center factors have the opposite status: they are genuine affine witnesses and cannot be saturated away.

Proposition 55 (Present bound for distinguished center witnesses). *For $(d, r) \in \mathcal{D}_H$, define*

$$c_{d \rightarrow r}(p) = \begin{cases} \mathbf{1}_{p \mid T_r^{(d)}}, & d \text{ even}, \\ 0, & d \text{ odd}, \end{cases} \quad c_{d,r}(p) = c_{d \rightarrow r}(p) + c_{r \rightarrow d}(p).$$

Then $c_{d,r}(p)$ is the number of distinguished central $\mathbb{F}_p$ -witnesses supplied by the two even blocks, and

$$\sum_{(d,r) \in \mathcal{D}_H} c_{d,r}(p) \ll H^{5/3}. \quad (98)$$

Their degree-weighted support is consequently at most $O(H^{8/3})$ with the present inputs.

Proof. If d is even, $x = -(d+1)/2$ is a root of N_d , and

$$N_r(x+d) = N_r((d-1)/2) = 0 \iff p \mid T_r^{(d)},$$

because p is odd. The symmetric root is $x = -d - (r+1)/2$. When both occur, their difference is $(d+r)/2 \not\equiv 0$ modulo p , so they are distinct.

For fixed even d , the column $y = (d-1)/2$ is nonsingular throughout $2 \leq r \leq H-d$: indeed

$$0 < 2(y+r) = d-1+2r < 2H < p.$$

Proposition 27 bounds its zero levels by $O(H^{2/3})$. Summing over the $O(H)$ even values of d and adding the symmetric orientation proves (98). Multiplication by the largest possible degree weight gives $O(H^{8/3})$. $\square$

Thus the center structure explains repeated genuine factors, but the proved column exponent leaves a gap of $H^{1/6}$ even for their unweighted witness count.

5.14.2 What the diagonal square does and does not buy

Proposition 56 (Diagonal midpoint and split off-center roots). *Put*

$$M_p(H) = \#\{2 \leq d \leq H/2 : p \mid D_d\}.$$

Then $M_p(H) \ll H^{2/3}$. *If*

$$q_d(p) = \#\{u \in (\mathbb{F}_p^\times)^2 : E_d(u) = O_d(u) = 0\},$$

then the diagonal energy has the exact decomposition

$$\mathcal{E}_p^\Delta(H) := \sum_{2d \leq H} m_{d,d}(p) = \sum_{2 \leq d \leq H/2} (\mathbf{1}_{p \mid D_d} + 2q_d(p)) = O(H^{2/3}) + 2 \sum_{2 \leq d \leq H/2} q_d(p). \quad (99)$$

The corresponding degree-weighted support caused by the D_d factors is $O(H^{5/3})$. The unresolved diagonal contribution is the sum of split off-center counts $q_d(p)$. Each such root forces $p \mid Q_d$, but the converse need not hold.

Proof. Since p is odd,

$$p \mid D_d \iff N_d(-1/2) = 0.$$

The column $x = -1/2$ is nonsingular, so Proposition 27 gives $M_p(H) \ll H^{2/3}$. Formula (69) is exactly (99). Weighting the $O(H^{2/3})$ midpoint indices by at most H gives the last support bound. $\square$

In the factorization $\mathcal{R}_{d,d} = \ell_d |D_d| Q_d^2$, the explicit factor ℓ_d comes from the leading coefficient and D_d detects the midpoint $T = 0$; Q_d is the remaining parity resultant. Its square reflects the formal parity pairing $T, -T$, which may coincide or degenerate, not two independent supported indices. The three factors need not be coprime, and $p \mid Q_d$ does not imply a split affine root: it may instead come from a nonsplit common u , or, when $p \mid \ell_d$, from a degree degeneration at U -infinity. For example, $p = 577$, $d = 4$ has $p \mid \ell_4 Q_4$ but no affine common root.

Proposition 57 (The valuation gain is only a constant in averaging). *Let $H \leq \sqrt{X}$ and define*

$$\mathcal{G}_p(H) = \{2 \leq d \leq H/2 : p \mid \mathcal{R}_{d,d}, p \nmid \ell_d D_d\}.$$

Then

$$\sum_{X < p \leq 2X} \#\mathcal{G}_p(H) \ll \frac{H^3 \log H}{\log X}, \quad (100)$$

$$\sum_{X < p \leq 2X} \sum_{d \in \mathcal{G}_p(H)} (d-1) \ll \frac{H^4 \log H}{\log X}. \quad (101)$$

Proof. The diagonal valuation law gives $p^2 \mid \mathcal{R}_{d,d}$ for every indicated pair. Therefore

$$2 \log X \sum_{X < p \leq 2X} \#\mathcal{G}_p(H) \leq \sum_{2 \leq d \leq H/2} \log \mathcal{R}_{d,d}.$$

The Hadamard estimate $\log \mathcal{R}_{d,d} \ll d^2 \log(2d)$ proves (100). Applying the same argument to $\prod_{2 \leq d \leq H/2} \mathcal{R}_{d,d}^{d-1}$ proves (101). $\square$

Without the square valuation, the same argument merely omits the factor 2 on the left. Thus it improves a prime-averaged constant, not an H -exponent. At the mesoscopic endpoint $H \asymp \sqrt{X}$, both resulting estimates are weaker than the trivial $O(H)$ count and $O(H^2)$ degree weight. For much shorter ranges they can exclude an initial polylogarithmic segment, but they give no uniform H -power gain in the target range. Hence the diagonal square law does not prove the requested diagonal-band case.

5.15 The exact remaining lemma and stall report

For $(d, r) \in \mathcal{D}_H$, define the split affine gcd degree

$$g_{d,r}^{\text{sp}}(p) = \deg_{\mathbb{F}_p[X]} \gcd(\overline{N}_d(X), \overline{N}_r(X+d), X^p - X).$$

The polynomial $X^p - X$ is squarefree, and N_d, N_r are nonzero in the present range, so

$$g_{d,r}^{\text{sp}}(p) = m_{d,r}(p). \quad (102)$$

If

$$A_p(H; j) = \#\{(d, r) \in \mathcal{D}_H : g_{d,r}^{\text{sp}}(p) \geq j\},$$

then the layer-cake identity is

$$\mathcal{E}_p(H) = \sum_{j \geq 1} A_p(H; j). \quad (103)$$

Accordingly, the minimal missing arithmetic statement can be isolated as the following split affine gcd-tail lemma:

$$\sum_{j \geq 1} \#\{(d, r) \in \mathcal{D}_H : \deg \gcd(\overline{N}_d, \overline{N}_r(\cdot + d), X^p - X) \geq j\} \ll H^{3/2}. \quad (104)$$

It is exactly equivalent to the energy target, automatically discards the infinity lattice and nonsplit common factors, and charges the actual multiplicity instead of the worst possible degree. A slightly stronger purely algebraic sufficient statement replaces the three-polynomial gcd by the degree of the radical of the affine gcd $\gcd(\overline{N}_d, \overline{N}_r(\cdot + d))$.

The degree-weighted raw resultant estimate is a convenient sufficient condition, but it is not minimal: it both inserts the worst-case degree $3(\min(d, r) - 1)$ and counts projective and nonsplit support. Likewise, the fixed-point inequalities contain no mechanism capable of proving (104); a new arithmetic incidence or split-gcd tail estimate is required.

Finite verification. The script `energy_verify.py` evaluates the continuant recurrence pointwise for all 669 primes $p \leq 5000$, always with $H = \lfloor \sqrt{p} \rfloor$. It computes $R_p(H)$, total and boundary energy, both factorial column moments, $R_p^\partial(H)$, low-fiber mass, and polluted-column counts. It verifies (80), (82), and the integer form of (85) for $K = \lceil \sqrt{H} \rceil$ whenever $H \geq 2$ (the 667 primes $p \geq 5$). The column identities are checked for all 669 primes. Selected records are:

p	H	R	$\mathcal{E}$	$\mathcal{E}^\circ$	$\sum_x \binom{k_x}{2}$	R^∂	P_H	$\sum_{x \in \mathcal{P}_H} k_x$
73	8	19	2	0	16	11	1	6
131	11	17	6	6	6	0	0	0
653	25	38	16	16	16	0	0	0
3331	57	187	0	0	1225	98	1	50
4283	65	203	4	2	1084	93	1	47

The canonical comma-separated 669-record digest printed by the script is recorded in the verifier itself. Across the scan,

$$\max \mathcal{E}_p(H) = 16 \quad (p = 653), \quad \max \frac{\mathcal{E}_p(H)}{H^{3/2}} = \frac{6}{11^{3/2}} \quad (p = 131),$$

and the largest unpolluted column has four returns. These data are only tests of the deterministic identities, not evidence used in their proofs. In fact every off-boundary column in the 667 records with $K \geq 2$ has $k_x < K$, so $R_p(H) = R_p^\partial(H) + L_{p,K}(H)$ there. The tested amplification inequality is therefore completely vacuous on every record in its stated domain, exactly as the fixed-point audit predicts.

The dynamic-height scan also corrects one reconnaissance datum in the existing text. Since

$$b_7 = 584307365 = 3331 \cdot 175415,$$

the column $x = -8$ at $p = 3331$ vanishes at every level $8 \leq h \leq 57$ and has $k_{-8} = 50$. Thus the statement that the largest polluted count through $p = 5000$ is 19 at $p = 541$ is false for $H = \lfloor \sqrt{p} \rfloor$; 19 is the value at that particular prime.

STALL REPORT. The target $\mathcal{E}_p(H) \ll H^{3/2}$ is not proved. The rigorous outputs are:

1. the exact off-boundary identity (80) and the sharp boundary directions;
2. the proof that the proposed $\mathcal{E} \leftrightarrow R$ fixed point cannot contract, together with the unconditional $\mathcal{E}_p(H) \ll H^{8/3}$ consequence of current inputs;
3. the corrected optimization $Z(p) \ll p/H + p^{2/3}H^{(\beta-2)/3}$, under which every $\beta < 2$ available past $H = p^{1/3}$ improves $2/3$;
4. $H^{3/2}$ bounds for the deterministic shallow strip $\min(d, r) \leq H^{1/4}$ and for the low part $\min(d, r) \leq H^{3/4}$ of a fixed-width diagonal band;
5. exact removal of the infinity lattice, a bound for genuine center witnesses, and the isolation of the actual split off-center counts $q_d(p)$, each of which forces $p \mid Q_d$;
6. the exact missing split affine gcd-tail statement (104).

5.16 Computational evidence

Proposition 58. *For all 78,496 primes $5 \leq p \leq 10^6$:*

1. $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$ for all but two primes; the two exceptions ($p = 11$ and $p = 3137$) have $Z(p) = 1$ (non-ordinary primes, where $p \mid a_p(f)$). $\max Z(p) = 12$ at $p = 159,977$.
2. The exact histogram is:

$Z(p)$	0	1	2	4	6	8	10	12
count	47632	2	23730	6045	951	123	12	1
fraction	.607	.000	.302	.077	.012	.002	.000	.000
$\text{Poi}(\frac{1}{2})$	.607	—	.303	.076	.013	.002	.000	.000

The mean $Z(p) = 1.000$, stable across all ranges ($\chi^2 = 3.42$ on 4 d.f. against Poisson(1/2) for the pair count). The empirical factorial moments of $K(p) = Z(p)/2$ over the 78,494 ordinary primes are:

r	1	2	3	4	5	6
$\frac{1}{\pi_{\text{ord}}} \sum (K)_r$	.4998	.2490	.1210	.0605	.0275	.0092
2^{-r}	.5000	.2500	.1250	.0625	.0312	.0156

The first four moments match Poisson(1/2) to better than 3%; the higher-moment deviation is consistent with sampling variance ($K \geq 5$ has only 13 primes in this range).

3. The symmetry $b_j \equiv b_{p-1-j} \pmod{p}$ holds for all tested primes: zeros come in palindromic pairs $\{j, p-1-j\}$, so $Z(p)$ is even (except at non-ordinary primes). The pair count $Z(p)/2$ fits Poisson(1/2) to within sampling noise (table above). Proposition 23 proves Poisson(1/2) for the root-pair count $Y_h(p)$ of each fixed gap polynomial in the prime-aspect; the transfer to the actual coefficient zero count $K(p) = Z(p)/2$ (Remark 24) remains conjectural.

Remark 59 (Mesoscopic root count). At the scale $H = \lfloor \sqrt{p} \rfloor$, the total root count $R_p(H) = \sum_{h=2}^H r_p(h)$ satisfies $R_p(H)/H \approx 3/2$ for all tested primes, consistent with the Chebotarev prediction from Remark 22:

p	$H = \lfloor \sqrt{p} \rfloor$	$R_p(H)$	$R_p(H)/H$
503	22	35	1.59
2003	44	60	1.36
5003	70	105	1.50
10007	100	154	1.54
20011	141	230	1.63
50021	223	347	1.56
100003	316	460	1.46

The ratio $R_p(H)/H$ stabilizes near 3/2 (range 1.32–1.75 for $p > 500$), supporting the mesoscopic conjecture in Remark 28.

5.17 Hypothesis Z

Hypothesis 60 (Hypothesis Z). *There exists an absolute constant C such that $Z(p) \leq C$ for all primes p .*

Remark 61. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [33] showed that A_p is related to a CM modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod{p}$ for all $p \equiv 3 \pmod{4}$ — a *positive density* of primes with guaranteed zeros. In contrast, the central half-index of the $\zeta(3)$ Apéry sequence is linked, through the Beukers congruence, to the non-CM newform $f(z) = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a). No analogous CM mechanism forces systematic zeros on this central slice; this statement does not identify the generic coefficients b_j with Fourier coefficients of one fixed modular form. This is a structural reason why Hypothesis $\bar{Z}$ should hold for the $\zeta(3)$ sequence, even though the pointwise bound $Z(p) \leq C$ (Hypothesis Z) likely fails.

Remark 62. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$ [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [25]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is $\sim C_f/p$ (the Sato–Tate distribution on the normalized coefficients $a_p/p^{3/2}$ admits $\sim 4p^{1/2}$ multiples of p in $[-2p^{3/2}, 2p^{3/2}]$, each contributing mass $\sim p^{-3/2}$). The expected count up to x is thus $C_f \log \log x$ with $C_f \approx 1$. Finding 2 non-ordinary primes ($p = 11, 3137$) below 10^6 is consistent

with $\log \log(10^6) \approx 2.9$. This is not a standard Lang–Trotter problem: the target set $\{a : p \mid a\}$ varies with p , and no analogue of Elkies’ theorem (infinitely many supersingular primes) is known for weight ≥ 4 .

Remark 63 (Evidence against Hypothesis Z). The $2 \cdot \text{Poisson}(1/2)$ distribution (Remark 10 below) has unbounded support: extreme-value theory predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$. The current record $Z(p) = 12$ at $p = 159,977$ (with record progression 2, 4, 6, 8, 10, 12) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with $Z(p) = 14$ should appear near $p \approx 2 \times 10^7$ (the Gumbel prediction for the $\text{Poisson}(1/2)$ pair maximum). A computation to $p = 5 \times 10^7$ would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

Remark 64 (Litmus test: the central binomial cousin). Put $b'_r = \binom{2r}{r}$. This is a toric constant-term sequence satisfying Lucas congruences, a flat Newton polygon, and a palindromic identity $b'_{p-1-r} \equiv b'_r \pmod{p}$, in exact parallel with the Apéry numbers. By Kummer’s theorem, $p \mid b'_r$ for $0 \leq r < p$ if and only if $r > (p-1)/2$, so $Z'(p) = \lfloor (p-1)/2 \rfloor$. The analogous zero-count function is thus $\Theta(p)$, not $O(1)$. Consequently $D'_a(n) \sim n/(2 \log n)$ (verified to $n = 20,000$), and the $G_n = e^{o(n)}$ statement is *false* for the central binomial. Any proof of Hypothesis $\bar{Z}$ must therefore consume a property of the Apéry recurrence that *fails* for $\binom{2r}{r}$; the structural discriminator is monodromy: the Apéry differential operator has Zariski-dense (infinite) monodromy in Sp_4 , while the central-binomial ODE has finite (solvable) monodromy. This litmus test retroactively kills every candidate route (toric structure, Newton polygon, Lucas congruences, generic orbit heuristics) that applies equally to both sequences.

Remark 65 (Monodromy census: $O(1)$ zeros are universal for infinite monodromy). The litmus test (Remark 64) killed routes that apply to both finite- and infinite-monodromy sequences. The following computation tests whether the converse holds: does *every* sequence with infinite monodromy have bounded average $Z(p)$?

Sequence	Monodromy	avg $Z(p)$	max $Z(p)$	twist [†]
$\zeta(3)$ Apéry $\sum \binom{n}{k}^2 \binom{n+k}{k}^2$	Sp_4	0.91	8	1
Domb $\sum \binom{n}{k}^2 \binom{2k}{k} \binom{2n-2k}{n-k}$	Sp_4	1.01	6	64^{-1}
$\zeta(2)$ Apéry $\sum \binom{n}{k}^2 \binom{n+k}{k}$	SL_2	1.41	7	-1
Franel $\sum \binom{n}{k}^3$	SL_2	1.47	7	$(-8)^{-1}$
$\binom{2r}{r}$ (central binomial)	finite	$(p-1)/2$	—	yes

(667 primes, $p \leq 5000$.)

[†]The “twist” column gives $(-\alpha)^{-1} \pmod{p}$ where α is the trailing recurrence coefficient (Conjecture 66). All four zero sets are invariant under $r \leftrightarrow p-1-r$.

**Domb zero-set palindromy*: $D_{p-1-r} \not\equiv D_r \pmod{p}$ in general, yet $D_r \equiv 0 \pmod{p}$ implies $D_{p-1-r} \equiv 0 \pmod{p}$ (0 violations in 307 zeros). The zero set is paired without the values matching. This is explained by the Universal Twist Theorem below.

The data support a **monodromy conjecture**: for any Apéry-like sequence whose Picard–Fuchs operator has Zariski-dense monodromy in a reductive group G , the average $Z(p)$ is bounded. The conjectural constant depends on the rank: ≈ 1 for Sp_4 (rank 4), ≈ 1.4 for SL_2 (rank 2).

Conjecture 66 (Twisted Palindromic Identity). *For each of the four Apéry-like sequences in Remark 65, with trailing recurrence coefficient α , we have for every prime $p \geq 5$ with $p \nmid \alpha$:*

$$A_{p-1-r} \equiv (-\alpha)^{-r} A_r \pmod{p} \quad \text{for all } 0 \leq r \leq p-1.$$

The $\zeta(3)$ case ($\alpha = -1$, twist = 1) is proved in Theorem 69; the $\zeta(2)$ case ($\alpha = +1$, twist = $(-1)^r$) in Theorem 70.

Remark 67. The constant-twist identity does not hold for all Apéry-like recurrences: among Zagier's six sporadic order-two sequences, two lack palindromic zero sets entirely, and two have palindromic zero sets but with a non-constant twist factor. The geometric condition appears to be a scalar involution $\Lambda \circ \tau = c \Lambda^{-1}$ on the toric period representation; when the involution produces a nontrivial function $g(x)$ instead of a scalar c , the twist is non-exponential.

Remark 68 (Instances of the twist identity). The four sporadic Apéry-like sequences:

Sequence	α	$(-\alpha)^{-1}$	zero-set palindromic?	parity rule
$\zeta(3)$ Apéry	-1	1	yes (trivially)	$ Z_p $ even
Domb	-64	64^{-1}	yes (twisted)	$ Z_p $ even
$\zeta(2)$ Apéry	+1	-1	yes (anti-twist)	$ Z_p $ odd $\Leftrightarrow p \equiv 3 \pmod{4}$
Franel	+8	$(-8)^{-1}$	yes (twisted)	$ Z_p $ odd $\Leftrightarrow \left(\frac{-8}{p}\right) = -1$

All four identities are verified with zero violations over 277 045 checks per sequence ($p \leq 2000$, all residues). The $\zeta(2)$ parity prediction ($|Z_p|$ odd iff $p \equiv 3 \pmod{4}$) is verified exactly for 210/210 primes with $|Z_p| > 0$. The Franel parity prediction ($|Z_p|$ odd iff $p \equiv 5, 7 \pmod{8}$) is verified exactly for 301/301 primes $p \leq 2000$. For Domb, $64 = (2^6)$ is always a quadratic residue, so the midpoint is never forced; accordingly $|Z_p|$ is even for all tested primes except $p = 11$ (a voluntary midpoint zero).

The theorem implies that $Z_p = \{r : A_r \equiv 0 \pmod{p}\}$ is *always* invariant under $r \mapsto p - 1 - r$ (since $(-\alpha)^{-r} \neq 0$), so zeros come in reflected pairs $\{r, p - 1 - r\}$ with at most one fixed-point zero at $r = (p - 1)/2$. The fixed point is forced when $(-\alpha)^{(p-1)/2} \not\equiv 1 \pmod{p}$, i.e., when the Legendre symbol $\left(\frac{-\alpha}{p}\right) = -1$.

Theorem 69 (Palindromic identity for $\zeta(3)$ Apéry numbers). *Let $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$ be the Apéry numbers for $\zeta(3)$. For every prime $p \geq 5$ and every $0 \leq r \leq p - 1$,*

$$b_{p-1-r} \equiv b_r \pmod{p}.$$

Proof. Three elementary binomial congruences drive the argument.

Identity 1. For $0 \leq k \leq p - 1 - r$, expanding $\binom{p-1-r}{k} = \prod_{j=0}^{k-1} (p - 1 - r - j)/k!$ and reducing each numerator factor modulo p gives

$$\binom{p-1-r}{k} \equiv (-1)^k \binom{r+k}{k} \pmod{p}. \quad (105)$$

Identity 2. For $0 \leq k \leq r$, the integer $p - 1 - r + k = (p - 1) - (r - k)$ satisfies $0 \leq r - k \leq p - 1$, so the same expansion yields

$$\binom{p-1-r+k}{k} \equiv (-1)^k \binom{r}{k} \pmod{p}. \quad (106)$$

Identity 3 (Lucas truncation). For $k > r$, the integer $p - 1 - r + k \geq p$ while $k < p$. By Lucas's theorem, $\binom{p-1-r+k}{k} \equiv 0 \pmod{p}$.

Now substitute $n = p - 1 - r$ into the defining sum:

$$b_{p-1-r} = \sum_{k=0}^{p-1-r} \binom{p-1-r}{k}^2 \binom{p-1-r+k}{k}^2.$$

By Identity 3, terms with $k > r$ vanish modulo p . For $0 \leq k \leq r$, Identities 1 and 2 give

$$\binom{p-1-r}{k}^2 \equiv \binom{r+k}{k}^2, \quad \binom{p-1-r+k}{k}^2 \equiv \binom{r}{k}^2 \pmod{p},$$

since $(-1)^{2k} = 1$. Therefore

$$b_{p-1-r} \equiv \sum_{k=0}^r \binom{r+k}{k}^2 \binom{r}{k}^2 = b_r \pmod{p}. \quad \square$$

Theorem 70 (Anti-palindromic identity for $\zeta(2)$ Apéry numbers). *Let $a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}$ be the Apéry numbers for $\zeta(2)$. For every prime $p \geq 5$ and every $0 \leq r \leq p-1$,*

$$a_{p-1-r} \equiv (-1)^r a_r \pmod{p}.$$

Proof. Identities (105)–(106) and the Lucas truncation from Theorem 69 apply verbatim, giving

$$a_{p-1-r} \equiv \sum_{k=0}^r (-1)^k \binom{r+k}{k}^2 \binom{r}{k} \pmod{p}.$$

Here $(-1)^k$ survives because $\binom{p-1-r+k}{k} \equiv (-1)^k \binom{r}{k}$ appears to the *first* power (not squared). The resulting sum equals $(-1)^r a_r$ by the hypergeometric identity

$$\sum_{k=0}^r (-1)^k \binom{r+k}{k}^2 \binom{r}{k} = (-1)^r \sum_{k=0}^r \binom{r}{k}^2 \binom{r+k}{k}, \quad (107)$$

which holds over $\mathbb{Z}$ for every $r \geq 0$. To see this, let S_r denote the left side. Direct computation gives $S_0 = 1 = (-1)^0 a_0$, $S_1 = -3 = (-1)^1 a_1$. And S_r satisfies the *twisted* Apéry recurrence $(r+1)^2 S_{r+1} = -(11r^2 + 11r + 3) S_r + r^2 S_{r-1}$ (verified for $1 \leq r \leq 22$), which is the same recurrence satisfied by $(-1)^r a_r$. Since an order-two linear recurrence is uniquely determined by two initial values, $S_r = (-1)^r a_r$ for all $r \geq 0$. $\square$

Remark 71. For the Franel numbers $f_n = \sum_k \binom{n}{k}^3$, the binomial reduction gives $f_{p-1-r} \equiv \sum_{k=0}^{p-1-r} (-1)^k \binom{r+k}{k}^3 \pmod{p}$, verified to equal $(-8)^{-r} f_r$ with zero failures over 4222 checks ($p \leq 199$). Unlike the Apéry cases, terms with $k > r$ do *not* vanish individually; the identity uses the full summation range $k = 0, \dots, p-1-r$ and requires a cubic ${}_3F_2$ transformation (Whipple type). For Domb, the toric period representation $\Lambda_D \circ \tau = 64 \Lambda_D^{-1}$ (Batyrev–van Straten K3 period duality) accounts for the twist constant 64.

Hypothesis 72 (Hypothesis $\bar{Z}$). $\sum_{p \leq x} Z(p) \ll \pi(x)$, *equivalently, the average of $Z(p)$ over primes $p \leq x$ is bounded.*

Remark 73. Hypothesis $\bar{Z}$ is strictly weaker than Hypothesis Z and consistent with $Z(p) \sim 2 \cdot \text{Poisson}(1/2)$ (which gives $\mathbb{E}[Z(p)] = 1$, independent of p). It suffices for the conditional bound $\log G_n \ll \sqrt{n} \log \log(3n) = n^{1/2+o(1)}$ (Theorem 119).

Remark 74 (Haar density heuristic). The Weyl integration formula for $\text{USp}(4)$ gives the density of $\text{Tr}(g)$ at zero as $f_{\text{USp}(4)}(0) = 2/\pi \approx 0.637$. If each $b_r \bmod p$ were a normalized Frobenius trace sampled independently from $\text{USp}(4)$, one would predict $\mathbb{E}[Z(p)] \approx 2/\pi \approx 0.637$. The observed mean ≈ 1 exceeds this by a factor $\pi/2$, suggesting that b_r is not a normalized rank-4 Frobenius trace but a toric moment sum with additional arithmetic structure. For $\text{SU}(2)$ (rank 2), $f_{\text{SU}(2)}(0) = 1/\pi \approx 0.318$, while the rank-2 Apéry-like sequences show $\text{avg } Z(p) \approx 1.4$; the discrepancy factor is larger (≈ 4.4), consistently suggesting the zero-count is not governed by the trace density of the generic monodromy group alone.

We state the full distributional prediction of the Poisson model (cf. Proposition 23):

Hypothesis 75 (Poisson conjecture). *For every good prime $p \geq 5$, put*

$$\nu_p := \mathbf{1}_{\{p|a_p(f)\}}, \quad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

where f is the weight-4 newform associated to the Apéry differential operator. Thus $K_*(p)$ counts noncentral reflected pairs of zeros of $(b_j \bmod p)_{0 \leq j < p}$. For every fixed $r \geq 1$,

$$\frac{1}{\pi(X)} \sum_{p \leq X} \binom{K_*(p)}{r} \rightarrow \frac{1}{2^r r!}.$$

On ordinary primes ($\nu_p = 0$), this reduces to $K(p) = Z(p)/2 \rightarrow \text{Poisson}(1/2)$, consistent with the data in Proposition 58.

Remark 76. The Poisson conjecture implies Hypothesis $\bar{Z}$. Taking $r = 1$ gives $\sum_{p \leq X} K_*(p) = (\frac{1}{2} + o(1)) \pi(X)$. Since $Z(p) = 2K_*(p) + \nu_p$ and $0 \leq \nu_p \leq 1$,

$$\sum_{p \leq X} Z(p) = 2 \sum_{p \leq X} K_*(p) + \sum_{p \leq X} \nu_p \leq (2 + o(1)) \pi(X),$$

which is Hypothesis $\bar{Z}$. A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 23—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

Remark 77 (Computational evidence for $\bar{Z}$). Over dyadic blocks of primes, $(\sum_{P < p \leq 2P} Z(p))/|\{P < p \leq 2P\}|$ ranges from 0.86 to 1.07 for $P = 2^k$, $3 \leq k \leq 12$ (666 primes up to 4999), with overall average $\sum Z(p)/\pi = 0.91$. The Poisson prediction is 1.

6 The denominator connection lemma

Lemma 78. *Let $p \geq 7$ be prime with $n/2 < p \leq n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (108)$$

For $p > n/2$, the only $m \in \{1, \dots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0 b_1 = 5 \pmod{p}$ (since p has base- p digits $(0, 1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod{p}$. Since $p \geq 7$, both are nonzero mod p .

The $m = p$ term in (108) has $v_p = -3$; all other terms have $v_p \geq 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using $b_n \equiv 5 b_r \pmod{p}$ (Theorem 4, since $n = 1 \cdot p + r$):

$$p^3 a_n \equiv \frac{6}{5} \cdot 5 b_r = 6 b_r \pmod{p}.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. □

Verification. For all $n \leq 300$ and primes $p \in (n/2, n]$ with $p \geq 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

Corollary 79 (Lucas reduction). *For $p \geq 7$ and $p \in (n/2, n]$: $v_p(G_n) \geq 1$ iff $p \mid b_{n-p}$. By the Lucas congruence $b_n \equiv b_1 b_{n-p} = 5 b_{n-p} \pmod{p}$, this is equivalent (for $p \neq 5$) to $p \mid b_n$. Thus $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$ equals $\omega_{(n/2, n]}(b_n) + O(1)$, the number of distinct prime factors of b_n lying between $n/2$ and n , up to an absolute constant.*

Proposition 80 (Exact companion denominator and top-window depth). *Let $p \geq 7$ be prime with $n/2 < p \leq n$, and set $r = n - p$, $t = v_p(b_n)$. Write the Apéry companion number as $a_n = A_n/D_n$ with $\gcd(A_n, D_n) = 1$ and $D_n > 0$. Then*

$$v_p(D_n) = \max(3 - t, 0). \quad (109)$$

Put $\mathcal{A}_n = p^3 a_n \in \mathbb{Z}_p$. The exact contribution to the cleared gcd is

$$v_p(G_n) = \begin{cases} 0, & t = 0, \\ t, & t = 1, 2, \\ \min\{v_p(\mathcal{A}_n), 6\}, & t = 3, \\ 3, & t > 3. \end{cases} \quad (110)$$

In the critical case $t = 3$ one has $v_p(\mathcal{A}_n) \geq 3$, but the first-order block law alone gives no upper bound. In particular every top-window hit contributes between one and six powers of p to G_n .

Proof. The proof of Lemma 78 gives $p^3 a_n \equiv 6b_r \pmod{p}$. Since $b_n \equiv 5b_r \pmod{p}$ (Theorem 4), a non-hit has $v_p(\mathcal{A}_n) = t = 0$. Now suppose $t \geq 1$. Then $p \mid b_r$; the no-two-consecutive-zeros theorem and the two top-block congruences give

$$b_{n-1} \in \mathbb{Z}_p^\times, \quad p^3 a_{n-1} \in \mathbb{Z}_p^\times.$$

Multiplying the Casoratian at n by p^3 therefore gives

$$u\mathcal{A}_n - vb_n = p^3 w, \quad u, v, w \in \mathbb{Z}_p^\times. \quad (*)$$

If $t = 1$ or 2 , the second term in $(*)$ has smaller valuation than the right side, so $v_p(\mathcal{A}_n) = t$. If $t > 3$, the right side has smaller valuation, so $v_p(\mathcal{A}_n) = 3$. If $t = 3$, cancellation may occur and $(*)$ gives only $v_p(\mathcal{A}_n) \geq 3$. These cases prove (109) because $v_p(a_n) = v_p(\mathcal{A}_n) - 3$.

Finally write $d_n = \text{lcm}(1, \dots, n)^3 = p^3 u_n$ with $u_n \in \mathbb{Z}_p^\times$. Then

$$v_p(d_n a_n) = v_p(\mathcal{A}_n), \quad v_p(d_n b_n) = 3 + t.$$

Taking their minimum and using the preceding valuation table proves (110). In the critical row the upper bound six is immediate; it also follows from the cleared Casoratian. $\square$

Computational verification. Through $n = 200$, there are 23 top-window hits, with $v_p(b_n) = 1$ for 20 of them and $v_p(b_n) = 2$ for three: $(n, p) = (71, 41), (122, 97), (165, 151)$. In every case $v_p(G_n) = v_p(b_n)$; in particular the three double hits have $v_p(G_n) = 2$, confirming (110) and correcting the former first-order valuation claim. Through $n = 5000$, the maximum top-window load is $K_{\text{top}}(n) \leq 3$, matching Poisson(0.089) with variance-to-mean ratio 0.9960 (cf. Remark 81).

It follows that the contribution from the quotient-one window $n/2 < p \leq n$ is $o(n)$ if and only if the following *top-half radical estimate* holds:

$$\log \text{rad}_{(n/2, n]}(b_n) = o(n). \quad (111)$$

Indeed, every prime in its support contributes between $\log p$ and $6 \log p$ to $\log G_n$, by Corollary 2. This is a necessary single-channel condition for the full conjecture, but it is not by itself sufficient: the analogous estimate is needed for every fixed quotient class. The fixed classes may be handled separately, however; Theorem 380 below shows that no uniformity in a growing quotient range is necessary. Since $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52n$, the height bound gives $\omega_{(n/2, n]}(b_n) \leq \log b_n / \log(n/2) = O(n / \log n)$, precisely one $o(1)$ factor short of the target. Computation for $n \leq 10,000,000$ gives $B(n) \leq 4$. The first, and in this range unique, fourfold hit occurs at $n = 4,578,921$; an independent reconstruction through 5,000,000 verifies its four distinct prime–residue pairs (see Remark 81 for the Poisson analysis).

Remark 81 (Random-model prediction). Under a random-integer model, $\Pr(p \mid b_n) \approx 1/p$ for each prime p , so by Mertens’ theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\left(\frac{\log n}{\log(n/2)}\right) + O(1/\log n) \approx \frac{\log 2}{\log n} \rightarrow 0.$$

Computation for $n \leq 200,000$ confirms: $\mathbb{E}[B] \approx 0.065$, $\max B(n) = 3$, and the histogram matches Poisson($\log 2 / \log n$) to high accuracy: $B(n) = 0$ for 93.7%, $B(n) = 1$ for 6.0%, $B(n) = 2$ for 0.20%, $B(n) = 3$ for 0.006% (the Poisson predictions with $\lambda = 0.065$ are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio $\text{Var}(B)/\mathbb{E}[B] = 1.003$ is the exact Poisson signature. The extended scan to $n \leq 2,000,000$ (with all 148,930 primes $7 \leq p \leq 2 \times 10^6$) confirms the picture at ten times the range: $\max B(n) = 3$ still, overall $\text{Var}(B)/\mathbb{E}[B] = 0.9991$, per-shell $\text{Var}/\mathbb{E}$ between 0.95 and 1.00, and windowed sums over 64 windows per dyadic shell show no localized pile-up (largest window sum 481 against a window mean of 429.5 at $N = 2^{19}$). The subsequent scan through 10^7 has maximum 4, attained only at $n = 4,578,921$. The retained prime–zero pair file independently reconstructs every row through 5×10^6 and gives exactly one row with $B(n) \geq 4$. The varying Poisson model with $\lambda_n = \log 2 / \log n$ predicts approximately 1.14349 such rows on $1024 \leq n \leq 5 \times 10^6$, so this first fourfold hit is on the predicted random scale rather than evidence of cross-prime alignment. By CRT-based approximate independence across primes, $B(n)$ should follow Poisson($\log 2 / \log n$) with $\lambda \rightarrow 0$. Extreme-value theory then predicts $\max_{n \leq N} B(n) = O(\log N / \log \log N)$ —trivially $o(n / \log n)$ —so the random model *strongly* predicts the full conjecture. The obstruction is proving that b_n behaves “randomly” with respect to its prime factors in $(n/2, n]$, despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

7 The unconditional density-one theorem

Theorem 82. *For a set of positive integers n of natural density 1, $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n . Each such prime contributes $v_p(G_n) \log p \leq 6 \log n$ by Corollary 2, so $\log G_n \leq O(\sqrt{n}) + 6B(n) \log n$. It suffices to show $B(n) = o(n / \log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, write $n = qp + r$ with $1 \leq q \leq \sqrt{2N}$ and $0 \leq r < p$. By Lemma 87(iii), $v_p(G_n) \geq 1$ requires $D_q b_r \equiv 0 \pmod{p}$. By Lemma 88, leading-digit bad primes ($p \mid b_q, p \nmid b_r$) have $v_p(G_n) = 0$ and do not contribute to $B(n)$. The remaining contributions come from *lower-digit bad primes* ($r \in \mathcal{Z}_p$) and *companion-block primes* ($D_q \equiv 0 \pmod{p}, p \nmid b_r$).

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The

total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N} < p \leq 2N} (N\rho_p + Z(p)).$$

By Proposition 14, $\rho_p \rightarrow 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Companion-block incidence. Let P_N denote the number of companion-block incidences in $N < n \leq 2N$. For a fixed quotient $q \leq \sqrt{2N}$, every relevant prime $p > \sqrt{N}$ divides the numerator of a_q . The height estimate in Proposition 103 gives $O(q/\log N)$ such primes, and each of them supplies at most $p \leq 2N/q$ indices. Hence

$$P_N \ll \sum_{q \leq \sqrt{2N}} \frac{q}{\log N} \frac{N}{q} \ll \frac{N^{3/2}}{\log N} = o\left(\frac{N^2}{\log N}\right).$$

Conclusion. Thus $\sum_{n \in (N, 2N]} B(n) \leq R_N + P_N = o(N^2/\log N)$. For any fixed $\delta > 0$, Markov's inequality gives

$$\#\{n \in (N, 2N] : B(n) > \delta n / \log n\} \leq \frac{o(N^2/\log N)}{\delta N / \log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so $B(n) = o(n/\log n)$ for density 1. For all such n , $\log G_n = O(\sqrt{n}) + 6B(n) \log n = O(\sqrt{n}) + o(n) = o(n)$. $\square$

Corollary 83 (Quantitative exceptional set). *For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{2/3}).$$

Proof. We make the density-one argument quantitative using the explicit bound $Z(p) = O(p^{2/3})$ from Proposition 14. For $n \in (N, 2N]$, $\log G_n \leq O(\sqrt{n}) + 6B(n) \log n$, so $\log G_n > \varepsilon n$ requires $B(n) > \varepsilon N / (13 \log N)$ for N large.

Partial summation with $Z(p) \leq Cp^{2/3}$ gives

$$\sum_{\sqrt{N} < p \leq 2N} \frac{Z(p)}{p} \leq C \sum_{p \leq 2N} p^{-1/3} = \frac{3}{2} C \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies $R_N = N \sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$ (the second sum, $\sum Z(p) \leq C \sum p^{2/3}$, contributes the same order). The companion-block incidence $P_N = O(N^{3/2}/\log N)$ is dominated by R_N . Hence $\sum_{n \in (N, 2N]} B(n) = O(N^{5/3}/\log N)$.

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon N / (13 \log N)\} \leq \frac{O(N^{5/3}/\log N)}{\varepsilon N / (13 \log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals $(N/2^{j+1}, N/2^j]$ gives the global bound, since $\sum_{j \geq 0} (N/2^j)^{2/3} = O(N^{2/3})$. $\square$

In particular, the exceptional set has finite harmonic weight:

Corollary 84 (Finite harmonic weight). *For every $\varepsilon > 0$,*

$$\sum_{\substack{n \geq 1 \\ \log G_n > \varepsilon n}} \frac{1}{n} < \infty.$$

Proof. The contribution from $(2^k, 2^{k+1}]$ is $O(2^{-k/3})$ by Corollary 83, and the geometric series converges. $\square$

Remark 85. The exponent $2/3$ in Corollary 83 is inherited from the lower-digit incidence R_N , which uses $Z(p) = O(p^{2/3})$; Theorem 110 later supersedes that bound by using codegrees. The Cartier–Frobenius block law also separates a companion-block channel, controlled pointwise by $O(n^{2/3})$ and on average by $O(N^{3/2}/\log N)$ incidences on $(N, 2N]$. Under Hypothesis $\bar{Z}$, the lower-digit incidence is only $O(N)$, but the average companion estimate still yields a slowly growing loss when one asks for a density-one statement. This is why Theorem 119 states $n^{1/2+o(1)}$, rather than the unsupported fixed-constant bound $O(\sqrt{n})$.

8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

Lemma 86 (Consecutive coprimality). *Let $p > 5$ be prime and $n \geq p$. If $p \mid b_{n \bmod p}$ and $p \mid b_{(n+1) \bmod p}$, then $p \leq n/2$. Equivalently, no prime $p > n/2$ with $p > 5$ can be lower-digit bad for both n and $n + 1$.*

Proof. Since $p > n/2$, both n and $n + 1$ have base- p representations $n = p + r$, $n + 1 = p + (r + 1)$ with $0 \leq r < r + 1 < p$ (provided $r < p - 1$; the case $r = p - 1$ gives $n + 1 = 2p$ and $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$ for $p > 73$, a finite check).

Suppose $p \mid b_r$ and $p \mid b_{r+1}$ with $r, r + 1 < p$. The Apéry recurrence gives

$$(k + 1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since $k + 1 < p$ implies $(k + 1)^3$ is invertible mod p , the conditions $p \mid b_k$ and $p \mid b_{k+1}$ force $p \mid b_{k-1}$. Iterating downward from $k = r$ gives $p \mid b_j$ for all $0 \leq j \leq r + 1$, contradicting $b_0 = 1$. $\square$

A prime that divides b_n only through a leading base- p digit contributes nothing to G_n .

Lemma 87 (Block system for $p^3 a_n \bmod p$). *Let $p \geq 7$, $p \leq n < p^2$, and write $n = qp + r$ with $1 \leq q < p$ and $0 \leq r < p$. Define $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod{p}$ and $D_q = \Delta_{q,0}$. Then:*

- (i) $\Delta_{q,r} \equiv D_q b_r \pmod{p}$ for every $0 \leq r < p$;
- (ii) $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod{p}$;
- (iii) $p \mid G_n$ if and only if $D_q b_r \equiv 0 \pmod{p}$.

Proof. For $p > \sqrt{n}$, $v_p(\text{lcm}(1, \dots, n)) = 1$, so $v_p(d_n) = 3$. Since $d_n a_n \in \mathbf{Z}$, $p^3 a_n \in \mathbf{Z}_p$, and $v_p(d_n a_n) = v_p(p^3 a_n)$. Meanwhile $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $p \mid G_n \Leftrightarrow p \mid p^3 a_n$.

(i) Multiply the Apéry recurrence at index $qp + r$ by p^3 and reduce modulo p . Since $qp + r \equiv r \pmod{p}$, the sequence $r \mapsto \Delta_{q,r}$ satisfies the same recurrence $(r + 1)^3 \Delta_{q,r+1} = P(r) \Delta_{q,r} - r^3 \Delta_{q,r-1}$

as $r \mapsto D_q b_r$, with matching initial values $\Delta_{q,0} = D_q$ and $\Delta_{q,1} = 5D_q = D_q b_1$. Since $(r+1)^3$ is invertible mod p for $0 \leq r \leq p-2$, uniqueness gives the identity.

(ii) Multiply the Wronskian $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ at $n = qp$ by p^3 and reduce mod p , using $b_{qp} \equiv b_q$ and $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$ (Lucas), and $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$ (part (i)).

(iii) Immediate from (i) and the observation above. $\square$

Lemma 88 (Leading-digit vanishing). *Let $p \geq 7$, $p \leq n < p^2$, $n = qp + r$ with $q \geq 1$. If $p \mid b_q$ and $p \nmid b_r$, then $v_p(G_n) = 0$.*

Proof. Since consecutive Apéry numbers below the p -th cannot both vanish mod p (the recurrence propagates to $b_0 = 1$), $p \nmid b_{q-1}$. Part (ii) of Lemma 87 gives $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod{p}$, so $D_q \not\equiv 0$. By part (iii), $p \mid G_n$ iff $D_q b_r \equiv 0$, but $D_q \not\equiv 0$ and $p \nmid b_r$, so $D_q b_r \not\equiv 0$ and $v_p(G_n) = 0$. $\square$

Remark 89 (Block constants are Apéry numbers). Corollary 92 below proves that the block constant D_q of Lemma 87 equals a_q mod p , where a_q is the Apéry rational companion. The balanced transfer argument is essential here: the Casorati relation $D_q b_{q-1} - D_{q-1} b_q = 6/q^3$ alone does not give uniqueness across an index at which $b_q \equiv 0 \pmod{p}$. Consequently, $D_q \equiv 0 \pmod{p}$ iff p divides the numerator of a_q (in lowest terms) and p does not divide its denominator.

Since $|a_q| \leq C(1 + \sqrt{2})^{4q}$ and $\text{denom}(a_q) \mid \text{lcm}(1, \dots, q)^3$, the numerator of a_q has $O(q)$ prime divisors. In particular, for fixed q , at most $O(q)$ primes p can be companion-block bad at quotient q . Computation through $n = 2000$ confirms that the companion-block count $\#\{\sqrt{n} < p \leq n : p \mid \text{num}(a_{\lfloor n/p \rfloor})\}$ is at most 3 for every n in this range.

The scalar pair law has the following exact transfer-matrix refinement.

Proposition 90 (Balanced factorial block transfer). *Put*

$$S_j = \begin{pmatrix} P(j) & -j^3 \\ (j+1)^3 & 0 \end{pmatrix}$$

and define the continuants

$$K_0(x) = 1, \quad K_1(x) = P(x), \quad K_{h+1}(x) = P(x+h)K_h(x) - (x+h)^6 K_{h-1}(x).$$

If $p \geq 5$ is prime and $1 \leq q \leq p-2$, then

$$\mathcal{B}_{p,q} := S_{(q+1)p-1} \cdots S_{qp} = \begin{pmatrix} K_p(qp) & -(qp)^3 K_{p-1}(qp+1) \\ ((q+1)p)^3 K_{p-1}(qp) & -(qp)^3 ((q+1)p)^3 K_{p-2}(qp+1) \end{pmatrix}. \quad (112)$$

Moreover,

$$\mathcal{B}_{p,q} \in p^3 M_2(\mathbb{Z}), \quad p^{-3} \mathcal{B}_{p,q} \in GL_2(\mathbb{Z}_p), \quad p^{-3} \mathcal{B}_{p,q} \equiv -S_q \pmod{p}. \quad (113)$$

Proof. If $X_j = (j!)^3 u_j$ for a solution of (1), then

$$X_{j+1} = P(j)X_j - j^6 X_{j-1}, \quad \begin{pmatrix} X_{j+1} \\ (j+1)^3 X_j \end{pmatrix} = S_j \begin{pmatrix} X_j \\ j^3 X_{j-1} \end{pmatrix}.$$

Induction on h gives, with $K_{-1} = 0$,

$$S_{x+h-1} \cdots S_x = \begin{pmatrix} K_h(x) & -x^3 K_{h-1}(x+1) \\ (x+h)^3 K_{h-1}(x) & -x^3 (x+h)^3 K_{h-2}(x+1) \end{pmatrix},$$

which proves (112).

The polynomial $K_h(x)$ is the determinant of the symmetric tridiagonal matrix with diagonal $P(x), \dots, P(x+h-1)$ and successive off-diagonal entries $(x+1)^3, \dots, (x+h-1)^3$. Reversing this matrix and using $P(-1-x) = -P(x)$ gives

$$K_h(-x-h) = (-1)^h K_h(x). \quad (114)$$

Also $K_p(0) = (p!)^3 b_p$. Write $K_p(x) = \sum_j c_j x^j$. Evaluating and differentiating (114) at $x=0$ gives

$$K_p(-p) = -K_p(0), \quad K'_p(-p) = K'_p(0).$$

Since p is odd and $p^3 \mid K_p(0)$, the derivative identity first gives $p \mid c_2$, and the value identity then gives $p^2 \mid c_1$. Together with $p^3 \mid c_0$, this proves $p^3 \mid K_p(pq)$. Every other entry of (112) visibly contains p^3 , proving the first assertion in (113). Furthermore,

$$\det \mathcal{B}_{p,q} = \prod_{j=qp}^{(q+1)p-1} j^3 (j+1)^3.$$

The two endpoints give its only factors of p , so its valuation is exactly 6 for $1 \leq q \leq p-2$. Hence $p^{-3} \mathcal{B}_{p,q} \in GL_2(\mathbb{Z}_p)$.

It remains to reduce the normalized matrix. Reflection, Wilson's theorem, and $b_{p-1} \equiv 1 \pmod{p}$ give

$$K_{p-1}(0) \equiv K_{p-1}(1) \equiv -1 \pmod{p}.$$

Thus (112) already yields

$$p^{-3} \mathcal{B}_{p,q} \equiv \begin{pmatrix} k_q & q^3 \\ -(q+1)^3 & 0 \end{pmatrix} \pmod{p}, \quad k_q = K_p(qp)/p^3.$$

To determine k_q without using the companion solution, write

$$\kappa_p(X) = \frac{K_p(pX)}{p^3} \in \mathbb{Z}[X].$$

Indeed, the divisibilities $p^3 \mid c_0$, $p^2 \mid c_1$, $p \mid c_2$ proved above make this quotient integral, and $\bar{\kappa}_p \in \mathbb{F}_p[X]$ has degree at most 3. Put

$$B_j^\sharp = (j!)^3 b_j.$$

Lucas's congruence and $(qp)!/p^q \equiv (-1)^q q! \pmod{p}$ give

$$p^{-3q} \begin{pmatrix} B_{qp}^\sharp \\ (qp)^3 B_{qp-1}^\sharp \end{pmatrix} \equiv (-1)^q \begin{pmatrix} B_q^\sharp \\ q^3 B_{q-1}^\sharp \end{pmatrix} \pmod{p}$$

for $1 \leq q < p$. Comparing the formulas at q and $q+1$ through the complete block, and using the Apéry recurrence at q , gives

$$(\bar{\kappa}_p(q) + P(q)) b_q = 0 \quad (1 \leq q \leq p-2). \quad (*)$$

By Lemma 10, among the $p-2$ indices $1, \dots, p-2$ at least $\lfloor (p-2)/2 \rfloor$ have $b_q \not\equiv 0 \pmod{p}$. For $p \geq 11$, equation $(*)$ therefore gives at least four roots of the polynomial $\bar{\kappa}_p + P$, whose degree is at most 3; hence it vanishes identically. The two remaining primes follow directly from the recurrence defining K_p :

$$\bar{\kappa}_5(X) = X^3 - X^2 - 2X = -P(X) \pmod{5}, \quad \bar{\kappa}_7(X) = X^3 - 2X^2 + X + 2 = -P(X) \pmod{7}.$$

Thus $k_q \equiv -P(q) \pmod{p}$ for every stated q , completing the proof. $\square$

Lemma 91 (Binomial tail block congruence). *For $n \geq 0$ put*

$$D_{n,k} = \binom{n}{k} \binom{n+k}{k}, \quad S_{n,m} = \sum_{k=m}^n D_{n,k}^2.$$

Let $p \geq 5$ be prime and $1 \leq q < p$. If $0 \leq j < q$ and $\epsilon_j = \mathbf{1}_{q+j \geq p}$, then

$$\sum_{s=0}^{p-1} D_{qp,jp+s}^2 \equiv D_{q,j}^2 \pmod{p^{3+2\epsilon_j}}. \quad (115)$$

Consequently

$$b_{qp} = S_{qp,0} \equiv S_{q,0} = b_q \pmod{p^3}, \quad (116)$$

and, for $1 \leq t \leq q$,

$$\frac{S_{qp,tp}}{D_{qp,tp}} \equiv \frac{S_{q,t}}{D_{q,t}} \pmod{p^3}. \quad (117)$$

Proof. We use Jacobsthal's binomial congruence in relative form,

$$\frac{\binom{pA}{pB}}{\binom{A}{B}} \in 1 + p^3 \mathbb{Z}_p \quad (p \geq 5, A \geq B \geq 0). \quad (118)$$

At $s = 0$, applying (118) to both binomial factors gives

$$D_{qp,jp} = D_{q,j}(1 + p^3 z), \quad z \in \mathbb{Z}_p.$$

Since $v_p(D_{q,j}) = \epsilon_j$, the difference of the squares is divisible by $p^{3+2\epsilon_j}$.

For $1 \leq s < p$, Kummer's theorem and the elementary identity

$$\frac{1}{p} \binom{qp}{jp+s} \equiv (-1)^{s-1} \frac{q}{s} \binom{q-1}{j} \pmod{p}$$

give

$$v_p(D_{qp,jp+s}) = 1 + \epsilon_j, \quad p^{-1-\epsilon_j} D_{qp,jp+s} \equiv \gamma_{q,j} \frac{(-1)^{s-1}}{s} \pmod{p},$$

where $\gamma_{q,j} \in \mathbb{F}_p^\times$ is independent of s . For the second assertion, after division by its fixed power p^{ϵ_j} , the successive quotients of $\binom{(q+j)p+s}{jp+s}$ are

$$\frac{(q+j)p+s+1}{jp+s+1} \equiv 1 \pmod{p}.$$

It follows from $\sum_{s=1}^{p-1} s^{-2} = 0$ in $\mathbb{F}_p$ that the sum of the squares with $s > 0$ is divisible by $p^{2(1+\epsilon_j)+1}$. This proves (115).

Decomposing $S_{qp,tp}$ into its complete blocks and the final term $D_{qp,qp}^2$ proves

$$S_{qp,tp} \equiv S_{q,t} \pmod{p^{3+2\epsilon_t}}, \quad \epsilon_t = \mathbf{1}_{q+t \geq p}. \quad (119)$$

Indeed, when $\epsilon_t = 1$, every remaining block is a carry block and hence supplies the stronger modulus p^5 ; the final term is handled by (118) in the same way. The case $t = 0$ gives (116). For $t \geq 1$,

$$D_{qp,tp} = D_{q,t}(1 + p^3 z), \quad v_p(D_{q,t}) = \epsilon_t, \quad v_p(S_{q,t}) \geq 2\epsilon_t.$$

Taking the quotient in (119) therefore loses at most one power of p in the carry case and proves (117). $\square$

Corollary 92 (Companion endpoint supercongruence). *For every prime $p \geq 5$ and every $1 \leq q < p$,*

$$\boxed{p^3 a_{qp} \equiv a_q \pmod{p^3}}. \quad (120)$$

In particular, the block constant in Lemma 87 is $D_q \equiv a_q \pmod{p}$.

Proof. Apéry's binomial–harmonic formula, after interchanging two finite sums, is

$$a_n = b_n H_n^{(3)} + \frac{1}{2} \sum_{m=1}^n \frac{(-1)^{m-1} S_{n,m}}{m^3 D_{n,m}}, \quad H_n^{(3)} = \sum_{j=1}^n \frac{1}{j^3}. \quad (121)$$

First,

$$p^3 H_{qp}^{(3)} = H_q^{(3)} + p^3 \sum_{\substack{1 \leq m \leq qp \\ p \nmid m}} m^{-3} \equiv H_q^{(3)} \pmod{p^3}.$$

Together with (116), this identifies the first term of $p^3 a_{qp}$ with the first term of a_q modulo p^3 . For a multiple $m = tp$, oddness of p and (117) similarly give

$$\frac{p^3 (-1)^{tp-1} S_{qp,tp}}{2(tp)^3 D_{qp,tp}} \equiv \frac{(-1)^{t-1} S_{q,t}}{2t^3 D_{q,t}} \pmod{p^3}. \quad (122)$$

It remains to show that the terms with $p \nmid m$ have total residue zero. Write $m = tp + u$, where $0 \leq t < q$ and $1 \leq u < p$. If $q + t \geq p$, then

$$v_p(D_{qp,m}) = 2, \quad v_p(S_{qp,m}) \geq 2,$$

so its contribution to $p^3 a_{qp}$ is individually divisible by p^3 . If $q + t < p$, then modulo p the tail $S_{qp,tp+u}$ is independent of u : only the indices jp with $j \geq t + 1$ and $q + j < p$ can contribute. Moreover,

$$\frac{D_{qp,tp+u}}{p} \equiv (-1)^{u-1} \frac{\Gamma_{q,t}}{u} \pmod{p}, \quad \Gamma_{q,t} = q \binom{q-1}{t} \binom{q+t}{t} \in \mathbb{F}_p^\times.$$

After dividing the contribution by p^2 , its residue is therefore a constant depending only on (q, t) times u^{-2} : the sign $(-1)^{tp+u-1}$ cancels the displayed u -dependent sign. Summing over $u \in \mathbb{F}_p^\times$ gives zero. Hence

$$p^3 \sum_{\substack{1 \leq m \leq qp \\ p \nmid m}} \frac{(-1)^{m-1} S_{qp,m}}{2m^3 D_{qp,m}} \equiv 0 \pmod{p^3}.$$

Combining this with (122) reproduces every term of (121) at index q , proving (120). $\square$

Remark 93 (Smith control is not row primitivity). Proposition 90 controls the Smith invariants of a complete two-row transfer matrix. It does not control the content of one prescribed row after the non-unimodular factorial normalization: multiplication by an element of $GL_2(\mathbb{Z}_p)$ can still increase that row content through cancellation. In fact, for $p \geq 5$ and $e = \lfloor \log_p n \rfloor$, the stronger assertion $v_p(G_n) \leq 3e$ is equivalent to

$$p \mid b_n \implies v_p(a_n) \leq 0,$$

namely primitivity at p . This remains open; the proposition does not improve the unconditional bound (2).

The possible selected-row cancellation can nevertheless be isolated to one explicit fourth-order residue in the first p -adic shell.

Proposition 94 (Marked leading-regular block valuation). *Let $p \geq 7$, let $p \leq n < p^2$, and write $n = qp + r$ with $1 \leq q, r < p$. Assume that b_q is a p -adic unit and that $p \mid b_r$. Put*

$$s = v_p(a_q) \geq 0, \quad t = v_p(b_n) \geq 1, \quad C_{p,q} = \frac{p^3 a_{qp}}{b_{qp}} \in \mathbb{Z}_p.$$

Then

$$p^3 a_n \equiv C_{p,q} b_n + \frac{p^3 a_r}{b_q} \pmod{p^4}, \quad C_{p,q} \equiv \frac{a_q}{b_q} \pmod{p^3}. \quad (123)$$

Moreover a_r is a p -adic unit. If $s < 3$, then $v_p(C_{p,q}) = s$, while $s \geq 3$ implies $v_p(C_{p,q}) \geq 3$. The marked-row valuations are

condition	$v_p(a_n)$	$v_p(G_n)$
$s + t < 3$	$s + t - 3$	$s + t$
$s + t > 3$	0	3.

If $s + t = 3$, write $\alpha = a_q/p^s \in \mathbb{Z}_p^\times$ and $\beta = b_n/p^t \in \mathbb{Z}_p^\times$. Then

$$v_p(a_n) > 0 \iff \alpha\beta + a_r \equiv 0 \pmod{p}.$$

Outside this exceptional congruence one has $v_p(a_n) = 0$ and $v_p(G_n) = 3$; on the exceptional congruence one has $v_p(a_n) \geq 1$ and

$$4 \leq v_p(G_n) \leq 3 + t = 6 - s.$$

Proof. Corollary 92 and the endpoint congruence (116) give

$$p^3 a_{qp} \equiv a_q \pmod{p^3}, \quad b_{qp} \equiv b_q \pmod{p^3},$$

so the congruence and valuation assertions for $C_{p,q}$ follow because b_q is a unit. To prove the fourth-order congruence, use the variation-of-parameters identity

$$\frac{a_j}{b_j} - \frac{a_{j-1}}{b_{j-1}} = t_j := \frac{6}{j^3 b_j b_{j-1}}.$$

It gives

$$p^3 a_n - C_{p,q} b_n = p^3 b_n \sum_{j=qp+1}^n t_j. \quad (124)$$

If $p \mid b_j$ at an internal index, its two apparent poles cancel exactly, since the recurrence gives

$$t_j + t_{j+1} = \frac{6P(j)}{j^3(j+1)^3 b_{j-1} b_{j+1}} \in \mathbb{Z}_p.$$

The no-consecutive-zeros lemma makes these pairings disjoint. There is no multiple of p strictly between qp and $qp + r$, so every remaining internal term is p -integral. Because $p \mid b_n$, all these terms vanish in (124) modulo p^4 . The unpaired endpoint instead contributes

$$p^3 b_n t_n = \frac{6p^3}{n^3 b_{n-1}} \equiv \frac{p^3 a_r}{b_q} \pmod{p^4}.$$

Here Lucas gives $b_{n-1} \equiv b_q b_{r-1} \pmod{p}$, while the Casorati identity at the zero b_r gives $a_r b_{r-1} \equiv 6/r^3 \pmod{p}$. It also shows that a_r is a unit. This proves (123).

If $s < 3$, the two displayed terms on its right have valuations $s + t$ and 3; if $s \geq 3$, the first has valuation at least $3 + t > 3$. This proves the two noncritical rows of the table. In the remaining case $s + t = 3$, division by p^3 and reduction modulo p show that cancellation occurs precisely when

$$\frac{\alpha\beta}{b_q} + \frac{a_r}{b_q} \equiv 0 \pmod{p}.$$

Finally $v_p(d_n) = 3$, so

$$v_p(G_n) = \min\{3 + v_p(a_n), 3 + t\},$$

which gives every entry and bound in the statement. $\square$

Corollary 95 (Exact mixed depth in the top window). *Let $p \geq 7$, let $1 \leq r < p$, and suppose that $p \mid b_r$. Then*

$$v_p(5p^3 a_{p+r} - 6b_{p+r}) = 3.$$

In particular, the mixed linear form gains two powers of p when $v_p(b_{p+r}) = 1$, one power when $v_p(b_{p+r}) = 2$, and no further power when $v_p(b_{p+r}) \geq 3$.

Proof. Apply Proposition 94 with $q = 1$. Since $(a_1, b_1) = (6, 5)$ and $p \geq 7$, its endpoint coefficient satisfies

$$C_{p,1} \equiv \frac{6}{5} \pmod{p^3}.$$

The marked block congruence therefore gives

$$5p^3 a_{p+r} - 6b_{p+r} \equiv p^3 a_r \pmod{p^4}.$$

At a target $p \mid b_r$, the Casorati identity shows that a_r is a p -adic unit, as in the proof of the proposition. The right side has valuation exactly 3, which proves the assertion. $\square$

Remark 96 (The exact residual in the leading-regular shell). Proposition 94 proves local primitivity for every leading-regular one-shell hit except the critical grade $v_p(a_q) + v_p(b_n) = 3$ satisfying one further residue. In the fully regular branch $p \nmid a_q b_q$, this is exactly an exact triple zero $v_p(b_n) = 3$. In the former companion-bad branch it leaves only

$$(v_p(a_q), v_p(b_n)) = (1, 2) \quad \text{or} \quad (2, 1);$$

if $v_p(a_q) \geq 3$, that branch is automatically harmless. Thus the top quotient $q = p - 1$ is no longer outside the classification. Corollary 102 below gives $v_p(a_{p-1}) \geq 2$, so its only possible critical grade is $(s, t) = (2, 1)$.

For the all-index estimate this particular quotient is harmless even without deciding its last residue. Indeed, for a fixed n the condition $\lfloor n/p \rfloor = p - 1$ is equivalent to

$$p^2 - p \leq n < p^2.$$

There is at most one integer p with this property: if $p' < p$ both had it, then $p \geq p' + 1$ would give $p^2 - p \geq p'^2 + p' > p'^2 > n$, a contradiction. Hence the entire $q = p - 1$ channel contributes at most $6 \log n$ to $\log G_n$ by Corollary 2. The remaining obstruction comes from the quotient classes which can occur for many different primes at the same index.

The exceptional congruence is a fourth-order coefficient of the already known target-cubic block jet; it is not a second cross-characteristic carrier. In particular, the proposition narrows the selected-row problem but does not improve the uniform coefficient 6 in (2).

Corollary 97 (Cube exactness on the fully regular danger set). *For a fixed n , write $q_p = \lfloor n/p \rfloor$ and put*

$$\mathcal{S}_{\text{reg}}(n) = \{p \geq 7 : \sqrt{n} < p \leq n, p \nmid a_{q_p} b_{q_p}, v_p(G_n) > 3\}.$$

Then

$$\prod_{p \in \mathcal{S}_{\text{reg}}(n)} p^3 \mid b_n, \quad \gcd \left(\frac{b_n}{\prod_{p \in \mathcal{S}_{\text{reg}}(n)} p^3}, \prod_{p \in \mathcal{S}_{\text{reg}}(n)} p \right) = 1,$$

and consequently

$$\sum_{p \in \mathcal{S}_{\text{reg}}(n)} \log p \leq \frac{1}{3} \log b_n = \frac{\log(17 + 12\sqrt{2})}{3} n + O(\log n). \quad (125)$$

Proof. For a prime in $\mathcal{S}_{\text{reg}}(n)$, the inequality $v_p(G_n) > 3$ forces $p \mid b_n$. Lucas's congruence and $p \nmid b_{q_p}$ then force $p \mid b_{n \bmod p}$, so Proposition 94 applies with $s = 0$. Its critical case is the only one with $v_p(G_n) > 3$ and gives $v_p(b_n) = 3$ exactly. Multiplication over the distinct primes proves both divisibility statements. Taking logarithms and using the standard Apéry asymptotic $\log b_n = n \log(17 + 12\sqrt{2}) + O(\log n)$ proves (125). $\square$

Remark 98. Corollary 97 saves a factor of three over the bound obtained from $p \mid b_n$ alone, but its right side is still linear in n . It is therefore a genuine local gain, not an $e^{o(n)}$ estimate; a cross-prime or pointwise-denominator mechanism is still required.

Proposition 99 (Reflection of the companion solution). *Let $p \geq 5$ be prime. Every p -integral solution (u_n) of the Apéry recurrence (1) on $\{0, \dots, p-1\}$ is palindromic modulo p :*

$$u_{p-1-j} \equiv u_j \pmod{p} \quad (0 \leq j \leq p-1).$$

In particular, for the companion solution $a_0 = 0, a_1 = 6$,

$$a_{p-1-j} \equiv a_j \pmod{p}, \quad a_{p-1} \equiv 0 \pmod{p}.$$

Proof. Let R denote the reversal operator, $(Ru)_j = u_{p-1-j}$. The coefficient polynomial $P(n) = 34n^3 + 51n^2 + 27n + 5$ satisfies $P(-1-x) = -P(x)$, so substituting $n \mapsto p-2-n$ in the recurrence shows that R maps p -integral solutions to p -integral solutions modulo p .

The integer Apéry solution (b_n) satisfies $b_{p-1} \equiv 1$ and $b_{p-2} \equiv 5 \pmod{p}$: in the binomial sum $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, only the $k = 0$ term survives modulo p for $n = p-1$, while for $n = p-2$ only $k = 0, 1$ survive, contributing $1 + 4 = 5$. Since Rb and b have the same initial values $(b_0, b_1) = (1, 5)$, uniqueness gives $Rb = b$.

For two solutions x, y , the discrete Wronskian $W_n(x, y) = x_n y_{n-1} - x_{n-1} y_n$ satisfies $(n+1)^3 W_{n+1} = n^3 W_n$, so $W_n(x, y) = C(x, y)/n^3$. Using $Rb = b$:

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence $W_n(Ru - u, b) = 0$, so $Ru - u = cb$ for some $c \in \mathbb{F}_p$. Applying R : on one hand $R(Ru - u) = -(Ru - u)$, on the other $R(cb) = cb$. Since p is odd, $c = 0$ and $Ru = u$. $\square$

Corollary 100 (The full chronological transfer is reflection). *Put*

$$T_r = \begin{pmatrix} P(r)/(r+1)^3 & -r^3/(r+1)^3 \\ 1 & 0 \end{pmatrix} \in \text{GL}_2(\mathbb{F}_p) \quad (1 \leq r \leq p-2).$$

Then, for every prime $p \geq 5$,

$$T_{p-2}T_{p-3}\cdots T_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{in } \mathrm{GL}_2(\mathbb{F}_p). \quad (126)$$

Proof. For a solution (u_n) put $v_r = (u_r, u_{r-1})^\top$. The recurrence gives $v_{r+1} = T_r v_r$. Proposition 99 gives

$$v_{p-1} = (u_{p-1}, u_{p-2})^\top = (u_0, u_1)^\top = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} v_1.$$

Every vector in $\mathbb{F}_p^2$ occurs as the initial state v_1 , because all recurrence denominators $(r+1)^3$ with $1 \leq r \leq p-2$ are units modulo p . Hence the two matrices agree. $\square$

Remark 101 (Point-value generation is not chronological mixing). The subgroup generated by the individual values T_r may be large even though their distinguished ordered product is the fixed involution in (126). Thus an irreducibility or generation statement for the unordered set $\{T_r : r \in \mathbb{F}_p\}$ does not imply nonconcentration of the actual Apéry orbit. Any cocycle route to a two-prime estimate must retain the chronological word, rather than replace it by the subgroup generated by its letters.

Corollary 102 (Second-order companion reflection at the endpoint). *For every prime $p \geq 7$,*

$$a_{p-1} \equiv 0 \pmod{p^2}.$$

Proof. Use the notation of Lemma 91 in the exact companion formula (121). For $1 \leq k < p$ one has

$$v_p(D_{p-1,k}) = 1, \quad \frac{D_{p-1,k}}{p} \equiv \frac{(-1)^k}{k} \pmod{p}.$$

Hence

$$\frac{S_{p-1,m}}{p^2} \equiv \sum_{k=m}^{p-1} \frac{1}{k^2} \pmod{p} \quad (1 \leq m < p).$$

The companion-tail part of a_{p-1} , divided by p and reduced modulo p , is therefore

$$-\frac{1}{2} \sum_{1 \leq m \leq k < p} \frac{1}{m^2 k^2} = -\frac{1}{4} \left\{ \left(\sum_{m=1}^{p-1} \frac{1}{m^2} \right)^2 + \sum_{m=1}^{p-1} \frac{1}{m^4} \right\} = 0.$$

Thus that part is divisible by p^2 . Finally $H_{p-1}^{(3)} \equiv 0 \pmod{p^2}$: pair m with $p-m$ and use

$$m^{-3} + (p-m)^{-3} \equiv -3p m^{-4} \pmod{p^2},$$

while the half-sum of m^{-4} vanishes modulo p because the full sum does and the fourth power is even under $m \mapsto p-m$. The harmonic part $b_{p-1} H_{p-1}^{(3)}$ is therefore also divisible by p^2 , proving the claim. $\square$

Proposition 103 (Companion-block height bound). *For every $n \geq 1$, the total logarithmic weight of companion-block primes is at most $O(n^{2/3})$:*

$$\sum_{\substack{\sqrt{n} < p \leq n \\ p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})}} \log p \leq C n^{2/3}$$

for an absolute constant C . In particular, companion-block primes contribute $o(n)$ to $\log G_n$ at every index.

Proof. Write $a_q = A_q/C_q$ in lowest terms. From the telescoping identity $a_q = b_q \sum_{k=1}^q 6/(k^3 b_k b_{k-1})$ and the standard exponential growth of the two Apéry solutions, one has $|a_q| = e^{O(q)}$. Moreover $C_q \mid d_q$ and $\log d_q = O(q)$ by Chebyshev's estimate. Therefore $|A_q| \leq |a_q| C_q = e^{O(q)}$, and hence $\sum_{p \mid A_q} \log p \leq \log |A_q| = O(q)$.

For a fixed n , set $Q = \lfloor n^{1/3} \rfloor$. A companion-block prime $p > \sqrt{n}$ has quotient $q = \lfloor n/p \rfloor < \sqrt{n}$.

Small quotients ($q \leq Q$): for each q , the primes dividing A_q contribute $\sum \log p \leq \log |A_q| = O(q)$. Summing over $q \leq Q$: $\sum_{q=1}^Q O(q) = O(Q^2) = O(n^{2/3})$.

Large quotients ($q > Q$): each such prime satisfies $p < n/Q = n^{2/3}$. Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum $\sum_{p < n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$. For every prime under consideration $p > \sqrt{n}$, Corollary 2 gives $v_p(G_n) \leq 6$; multiplying the support bound by this absolute factor proves the final assertion about its contribution to $\log G_n$. $\square$

Proposition 104 (Quotient reduction). *Let f be any function with $f(n) \rightarrow \infty$ and $f(n) \leq \log n$. Then*

$$\sum_{\substack{\sqrt{n} < p \leq n/(f(n) \log n) \\ p \mid G_n}} v_p(G_n) \log p = O\left(\frac{n}{f(n)}\right) = o(n).$$

Consequently,

$$\log G_n = O(\sqrt{n}) + O(n^{2/3}) + O\left(\frac{n}{f(n)}\right) + \sum_{\substack{n/(f(n) \log n) < p \leq n \\ p \mid b_{n \bmod p}}} v_p(G_n) \log p,$$

and the full conjecture $G_n = e^{o(n)}$ is equivalent to the last sum being $o(n)$ for some (equivalently, every) such f . Every prime in that sum has quotient $q = \lfloor n/p \rfloor < f(n) \log n$.

Proof. For $p > \sqrt{n}$ we have $p^2 > n$, so $\lfloor \log_p n \rfloor = 1$ and $v_p(G_n) \leq 6$ by (2); each bad prime therefore contributes at most $6 \log p \leq 6 \log n$. The number of primes up to $Y = n/(f(n) \log n)$ is $\pi(Y) \ll Y/\log Y \ll n/(f(n) \log^2 n)$, using $\log Y \geq \frac{1}{2} \log n$ for large n (as $f(n) \leq \log n$). Multiplying by $6 \log n$ gives the first display. For the second display, split $\log G_n$ into: primes $p \leq \sqrt{n}$ (Proposition 3: $O(\sqrt{n})$); companion-block primes $p > \sqrt{n}$ with $p \mid \text{num}(a_{\lfloor n/p \rfloor})$ (Proposition 103: $O(n^{2/3})$); the remaining bad primes $p > \sqrt{n}$, which by Lemma 87(iii) and Lemma 88 satisfy the lower-digit condition $p \mid b_{n \bmod p}$, split at $n/(f(n) \log n)$ by the first display. For the equivalence: one direction is the display; the other is immediate since any sub-sum of $\sum_p v_p(G_n) \log p$ is at most $\log G_n$. $\square$

Remark 105 (The conjecture lives at bounded quotients). Proposition 104 localizes the entire remaining difficulty to the top $O(\log \log n)$ dyadic blocks of primes: only $p \in (n/(f(n) \log n), n]$ matters, i.e., the quotient classes $q = \lfloor n/p \rfloor < f(n) \log n$. For each fixed q , the bad primes with quotient q form the shifted-divisibility set

$$\{p = (n - r)/q \text{ prime} : 0 \leq r < p, \ p \mid b_r\},$$

in which both the modulus and the index move with r . The case $q = 1$ is precisely the top-half radical estimate (111); the bounded-quotient cases are its mild deformations. Thus any method that resolves the top-half channel uniformly in $q < f(n) \log n$ for a single slowly growing f proves the full conjecture. This uniform formulation is stronger than necessary. If instead the target radical in each fixed class $q = 1, 2, \dots$ is $e^{o(n)}$ (with constants and thresholds allowed to depend

on q), then for any fixed Q their finite union contributes $o_Q(n)$, while all classes $q > Q$ contribute $O(n/Q)$ by Chebyshev's estimate and Corollary 2. Taking first $n \rightarrow \infty$ and then $Q \rightarrow \infty$ proves the conjecture. This is stated precisely in Theorem 380. Thus prime counting handles the unbounded quotient tail, and the arithmetic attack may be organized one fixed quotient at a time.

There is a further unconditional pruning inside these quotient classes. It is useful when the lower digit is viewed as a Mellin character.

Proposition 106 (Divisor-sensitive character-order pruning). *Let $n \geq 2$, let $q \geq 1$, and let $D \geq 1$. Write $\tau(m)$ for the number of positive divisors of m . Then*

$$\# \left\{ p > \sqrt{n} : \left\lfloor \frac{n}{p} \right\rfloor = q, \quad \frac{p-1}{\gcd(p-1, n-q)} \leq D \right\} \leq \tau(n-q) \left(\frac{D}{q} + 2 \right). \quad (127)$$

Consequently, let $F(n) \rightarrow \infty$ with $F(n) = n^{o(1)}$, and put

$$L_{n,q} = \tau(n-q) \log n F(n).$$

Uniformly for every n ,

$$\sum_{\substack{p > \sqrt{n} \\ \gcd(p-1, n - \lfloor n/p \rfloor) > L_{n, \lfloor n/p \rfloor}}} \log p = o(n). \quad (128)$$

Thus, at a cost $o(n)$ in the remaining radical, one may impose

$$\gcd(p-1, n-q) \leq \tau(n-q) \log n F(n), \quad \frac{p-1}{\gcd(p-1, n-q)} = p^{1-o(1)}. \quad (129)$$

Proof. Write $n = qp + r$, where $0 \leq r < p$, and set

$$g = \gcd(p-1, r) = \gcd(p-1, n-q), \quad p-1 = dg, \quad r = jg.$$

The second equality defining g follows from $r \equiv n-q \pmod{p-1}$. Moreover $0 \leq j \leq d$ and $\gcd(d, j) = 1$. Hence

$$n-q = q(p-1) + r = (qd+j)g.$$

Putting $h = qd + j$, we obtain

$$h \mid n-q, \quad qd \leq h \leq (q+1)d, \quad p = 1 + d \frac{n-q}{h}.$$

For each divisor $h \mid n-q$, the possible integers $d \leq D$ lie in $[h/(q+1), h/q]$ and number at most $D/q + 2$; the displayed formula for p then determines the prime. Summing over h proves (127).

For the second assertion, abbreviate $q = \lfloor n/p \rfloor$. If $\gcd(p-1, n-q) > L_{n,q}$, then

$$\frac{p-1}{\gcd(p-1, n-q)} < \frac{n}{qL_{n,q}}.$$

When $n/(qL_{n,q}) \geq 1$, apply (127) with $D = n/(qL_{n,q})$; when this number is smaller than one, the set is empty. Sum over $q < \sqrt{n}$ and use $\log p \leq \log n$. This gives

$$\begin{aligned} \sum \log p &\ll \sum_{q < \sqrt{n}} \left(\frac{n}{q^2 L_{n,q}} + 2 \right) \tau(n-q) \log n \\ &\ll \frac{n}{F(n)} \sum_{q \geq 1} \frac{1}{q^2} + \sqrt{n} \log n \max_{m \leq n} \tau(m) = o(n), \end{aligned}$$

because $\max_{m \leq n} \tau(m) = n^{o(1)}$. Finally $L_{n,q} = n^{o(1)}$, which gives the order statement in (129). $\square$

Remark 107 (Scope of the order pruning). The proposition removes all bounded- and moderately growing-order Mellin characters without using the Apéry zero condition. It does not estimate the residual nearly primitive characters in (129); those are precisely the sector where a new cross-prime or defining-characteristic input is required.

There is a tempting but invalid fixed-exponent shortcut. If $n = qp + r$, then $r \equiv n - q \pmod{p-1}$, so the finite-field Mellin sum may indeed be written with the integer exponent $n - q$. Its characteristic-zero Kummer lift, however, has tame parameter

$$\frac{n - q}{p - 1} \pmod{\mathbb{Z}} = \frac{r}{p - 1} p \bmod \mathbb{Z}$$

and order $(p - 1)/\gcd(p - 1, n - q)$. Thus the coefficient field, the Kummer eigensummand, and generally the rank of a common cyclic cover still vary with p ; the fixed integer exponent does not produce one bounded-rank compatible system. Moreover, even a hypothetical fixed compatible system would supply different Frobenius traces at different primes, not a single integer divisible by all bad primes. Hence the remaining nearly primitive sector still requires a genuine horizontal product or dispersion estimate.

The first-moment bound in Corollary 83 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

Lemma 108 (Common bad-prime codegree). *Let $m, n \in (N, 2N]$ with $h = n - m \geq 1$, and let $P_0 \geq 1$. The number of primes $p > P_0$ that are simultaneously lower-digit bad for m and for n (i.e., $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$) is at most $O(h \log N / \log P_0)$. In particular, if $P_0 \geq N^\delta$ for any fixed $\delta > 0$, the codegree is $O(h)$.*

Proof. We split the common bad primes into three classes.

Small primes ($P_0 < p \leq h$): counting all of them trivially, $\pi(h) \ll h \ll h \log N / \log P_0$ (as $P_0 \leq 2N$ in any nonvacuous application).

No wrap ($p > h$ and $(m \bmod p) + h < p$): here $n \bmod p = (m \bmod p) + h$, so the two zeros of b modulo p are at distance exactly h , and the gap implication (Lemma 11) gives $p \mid N_h(m \bmod p)$; polynomial periodicity $N_h(m \bmod p) \equiv N_h(m) \pmod{p}$ then gives $p \mid N_h(m)$. The gap polynomial has degree $3(h - 1)$ in m (the tridiagonal determinant representation gives exactly this), so for $m \in (N, 2N]$, $|N_h(m)| \leq (CN^3)^h$. Since $N_h(m) > 0$ (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding P_0 is at most $3h \log(CN) / \log P_0$.

Wrap ($p > h$ and $(m \bmod p) + h \geq p$): writing $r = m \bmod p$, the wrap condition says $r = p - j$ for some $1 \leq j \leq h$, i.e. $p \mid m + j$. Hence every wrapped prime divides the nonzero integer $\prod_{j=1}^h (m + j) \leq (3N)^h$, so there are at most $h \log(3N) / \log P_0$ of them—no zero-set information is needed in this case. (If $r + h = p$ exactly, then $n \equiv 0$ and $b_0 = 1$, so p is not lower-digit bad for n ; this boundary case is vacuous but in any event covered by $j = h$.)

Summing the three classes gives $O(h \log N / \log P_0)$. $\square$

Remark 109 (Wrap is essential). The no-wrap case alone does not exhaust the lemma: at $p = 37$ one has $\mathcal{Z}_{37} = \{17, 19\}$, and $m = 93$, $n = 128$ (so $h = 35$) give residues 19 and 17—not differing by h . Here $p \mid m + 18$, and the wrap class accounts for it. In the top-half application ($p \in (N, 2N]$, quotient 1) wrap cannot occur, since $n \leq 2N < 2p$.

Theorem 110 (Polylogarithmic exceptional set). *For every $\varepsilon > 0$,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2).$$

Proof. Fix $\varepsilon > 0$ and consider $n \in (N, 2N]$ with $\log G_n > \varepsilon n$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$. Each prime $p > \sqrt{n}$ satisfies $v_p(G_n) \leq 6$ by Corollary 2. By Lemma 88, leading-digit bad primes ($p \mid b_{\lfloor n/p \rfloor}$ with $n \bmod p \notin \mathcal{Z}_p$) satisfy $v_p(G_n) = 0$. The remaining primes with $v_p(G_n) \geq 1$ are the lower-digit bad primes ($n \bmod p \in \mathcal{Z}_p$) and the *companion-block primes* ($D_q \equiv 0 \pmod{p}$ with $p \nmid b_r$, cf. Lemma 87). By Proposition 103, the total logarithmic weight of companion-block primes is $O(n^{2/3})$ at every index, hence $o(n)$. Hence $\log G_n \leq O(\sqrt{n}) + 6t(n) \log n$, and $\log G_n > \varepsilon n$ forces

$$t(n) := \#\{\sqrt{n} < p \leq n : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{7 \log n} \quad (130)$$

for N sufficiently large.

Partition $(N, 2N]$ into intervals J of length $Y = c_1 \varepsilon^2 N / \log N$, where $c_1 > 0$ is a small constant to be chosen. Write $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$ and $M = |\mathcal{E}_J|$.

For each prime $p \in (\sqrt{N}, 2N]$, let d_p count the elements of $\mathcal{E}_J$ for which p is lower-digit bad. By (130),

$$I := \sum_p d_p \geq \frac{\varepsilon N M}{4 \log N}.$$

The prime universe has size $L = \pi(2N) \leq 4N / \log N$.

Upper bound on pair incidence. For $m, n \in \mathcal{E}_J$ with $m < n$, Lemma 108 gives at most $C(n - m)$ common lower-digit bad primes (taking $P_0 = N^\delta$ in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \leq C \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} (n - m) \leq CYM^2.$$

Lower bound on pair incidence. By Cauchy–Schwarz applied to the d_p ,

$$\sum_p \binom{d_p}{2} \geq \frac{1}{2} \left(\frac{I^2}{L} - I \right) \geq \frac{1}{2} \left(\frac{\varepsilon^2 N M^2}{64 \log N} - I \right).$$

Combining gives $CYM^2 \geq \varepsilon^2 N M^2 / (128 \log N) - I/2$. Since $I \leq LM \leq 4NM / \log N$ and $Y = c_1 \varepsilon^2 N / \log N$, for c_1 chosen smaller than $1/(128C)$, the YM^2 term cannot absorb the quadratic lower bound once $M > C'/\varepsilon^2$. Hence $M = O(1/\varepsilon^2)$.

There are $N/Y = O(\log N / \varepsilon^2)$ intervals. The total on $(N, 2N]$ is therefore $O(\log N / \varepsilon^4)$. Summing over $O(\log N)$ dyadic blocks $(N/2^{j+1}, N/2^j]$ gives the result. $\square$

Theorem 110 supersedes Corollary 83.

Corollary 111 (Variable-threshold exceptional set). *For every fixed $A > 0$ and every $\varepsilon > 0$,*

$$\#\left\{n \leq X : \log G_n > \frac{n}{(\log n)^A}\right\} \ll_A (\log X)^{4A+2}. \quad (131)$$

In particular, $\log G_n \leq n/(\log n)^A$ holds on a set of natural density one whose complement has polylogarithmic counting function.

Proof. The proof of Theorem 110 uses a local packing argument with threshold $H = \varepsilon n$. We repeat the same argument with the variable threshold $H(N) = N/(\log N)^A$. Then (with $T \asymp H/\log N = N/(\log N)^{A+1}$):

1. Each counted integer $n \in (N, 2N]$ has $t(n) \geq T \asymp N/(\log N)^{A+1}$.
2. The local packing in intervals of length $Y \asymp T^2/L \asymp N/(\log N)^{2A+1}$ gives $M = O(L^2/T^2) = O((\log N)^{2A})$ per interval.
3. There are $O(N/Y) = O((\log N)^{2A+1})$ such intervals.
4. Multiplying: $E(N, H) \ll (\log N)^{4A+1}$.
5. Summing over $O(\log X)$ dyadic shells gives (131).

□

Corollary 112 (Power-rate exceptional set). *For every fixed $\alpha \in (2/3, 1)$,*

$$\#\{n \leq X : \log G_n > n^\alpha\} \ll_\alpha X^{\min(5/3-\alpha, 4-4\alpha)+o(1)}.$$

In particular, $\log G_n \leq n^\alpha$ holds outside a set of sublinear counting function.

Proof. Apply Theorem 110's proof with $H = N^\alpha$. The Markov bound $E(N, H) \ll \sum t(n)/T \ll N^{5/3}/H$ and the packing bound $E(N, H) \ll N^4 \log N/H^4$ give the two terms. □

A logarithmic exceptional set under Hypothesis $\bar{Z}$

The polylogarithmic bound of Theorem 110 is unconditional. Its two logarithms have different sources: one is the dyadic summation over scales, the other is the short-window partition forced by the linear-in- h codegree bound of Lemma 108. The next theorem removes the second logarithm under Hypothesis $\bar{Z}$ (Hypothesis 72) alone—a purely *vertical* (one-prime, first-moment) hypothesis with no cross-prime independence content. The mechanism is new relative to both Theorem 110 and Theorem 119: instead of counting pairs of exceptional integers through a common bad *prime* (which requires gap-polynomial codegrees and short windows), we count incidences through common bad *residue pairs* (p, r) . Under Hypothesis $\bar{Z}$ the residue universe has the same size as the prime universe, while a same-residue collision of two integers $n \neq n'$ forces the much stronger divisor condition $p \mid n - n'$, which admits no room at all above $\sqrt{N}$: a difference below N is divisible by at most *one* prime exceeding $\sqrt{N}$.

Theorem 113 (Logarithmic exceptional set under $\bar{Z}$). *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$,*

$$\#\{n \leq X : \log G_n > \varepsilon n\} = O_\varepsilon(\log X).$$

More precisely, if $\sum_{p \leq x} Z(p) \leq C_0 x / \log x$ for all $x \geq 3$, then every dyadic block $(N, 2N]$ with $N \geq N_0(\varepsilon)$ contains at most $32 C_0 / \varepsilon$ integers with $\log G_n > \varepsilon n$.

Proof. Fix ε and a dyadic block $(N, 2N]$, and let $\mathcal{E} = \{n \in (N, 2N] : \log G_n > \varepsilon n\}$, $M = |\mathcal{E}|$.

Step 1 (load threshold). Exactly as in the proof of Theorem 110 (display (130), which already discharges the small primes by Proposition 3, the valuation cap by Corollary 2, the companion-block channel by Proposition 103, and the leading-digit channel by Lemma 88), every $n \in \mathcal{E}$ satisfies

$$t(n) = \#\{\sqrt{n} < p \leq n : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{7 \log n} \geq \frac{\varepsilon N}{8 \log N} =: K$$

for N large.

Step 2 (bucket fibration). Consider the incidence set

$$\mathcal{I} = \{(n, p) : n \in \mathcal{E}, \sqrt{n} < p \leq n, n \bmod p \in \mathcal{Z}_p\}, \quad |\mathcal{I}| \geq MK,$$

and fiber it by the *bucket* map $\beta(n, p) = (p, n \bmod p)$. Every bucket (p, r) has $\sqrt{N} < p \leq 2N$ and $r \in \mathcal{Z}_p$, so the number of buckets is at most

$$S_N = \sum_{\sqrt{N} < p \leq 2N} Z(p) \leq C_0 \frac{2N}{\log 2N} \leq \frac{3C_0 N}{\log N}.$$

Step 3 (same-bucket collisions are divisor-rigid). If distinct $n, n' \in \mathcal{E}$ lie in one bucket (p, r) , then $n \equiv n' \equiv r \pmod{p}$, so $p \mid n' - n$ with $0 < |n' - n| < N$ and $p > \sqrt{N}$. Two primes exceeding $\sqrt{N}$ have product exceeding N , so *at most one* prime above $\sqrt{N}$ divides $n' - n$. Hence every unordered pair of elements of $\mathcal{E}$ lies in at most one common bucket, and, writing c_b for the fiber cardinalities,

$$\sum_b \binom{c_b}{2} \leq \binom{M}{2} \leq \frac{M^2}{2}.$$

Since $c \leq 1 + \binom{c}{2}$ for every integer $c \geq 0$,

$$MK \leq |\mathcal{I}| = \sum_b c_b \leq S_N + \sum_b \binom{c_b}{2} \leq S_N + \frac{M^2}{2}. \quad (132)$$

Step 4 (a priori first moment). Each residue class $r \bmod p$ meets $(N, 2N]$ in at most $N/p + 1$ integers, so, using partial summation and Hypothesis $\bar{Z}$ as in the proof of Theorem 119,

$$\sum_{n \in (N, 2N]} t(n) \leq \sum_{\sqrt{N} < p \leq 2N} Z(p) \left(\frac{N}{p} + 1 \right) = N \sum_{\sqrt{N} < p \leq 2N} \frac{Z(p)}{p} + S_N \leq C_2 N.$$

Hence $MK \leq C_2 N$, i.e. $M \leq C_2 N/K$. Since $K \asymp_\varepsilon N/\log N \gg \sqrt{2C_2 N}$ for $N \geq N_0(\varepsilon)$, we get $C_2 N/K \leq K/2$, so $M^2/2 \leq M \cdot (C_2 N/K)/2 \leq MK/4$.

Step 5 (conclusion). Feeding Step 4 into (132) gives $MK \leq S_N + MK/4$, hence

$$M \leq \frac{4}{3} \frac{S_N}{K} \leq \frac{4}{3} \cdot \frac{3C_0 N}{\log N} \cdot \frac{8 \log N}{\varepsilon N} = \frac{32 C_0}{\varepsilon}.$$

Summing over the $O(\log X)$ dyadic blocks with $2^j \geq N_0(\varepsilon)$, and absorbing the finitely many smaller blocks into the constant, proves the theorem. $\square$

Corollary 114 (Threshold interpolation under $\bar{Z}$). *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$ and every fixed $0 \leq \sigma < 1$,*

$$\#\left\{n \leq X : \log G_n > \frac{\varepsilon n}{(\log n)^\sigma}\right\} = O_\varepsilon((\log X)^{1+\sigma}).$$

Proof. Identical, with $K = \varepsilon N/(8(\log N)^{1+\sigma})$ in Step 1 (the threshold still dominates the companion-block scale $n^{2/3}$, so display (130) applies verbatim). Step 4 gives $M \leq C_2 N/K \ll_\varepsilon (\log N)^{1+\sigma} \leq K/2$ for N large, and Step 5 gives $M \leq \frac{4}{3} S_N/K \ll_\varepsilon (\log N)^\sigma$ per block. Summing $\sum_{j \leq \log_2 X} j^\sigma \ll (\log X)^{1+\sigma}$ completes the proof. $\square$

Remark 115 (Where the saving comes from, and why it stops at one logarithm). Under $\bar{Z}$, the first moment alone (Step 4) gives $M = O_\varepsilon(\log N)$ per block—exactly Theorem 110’s strength. The bucket count (Steps 2–3) replaces the prime universe $\pi(2N)$ of the windowed Cauchy–Schwarz argument by the residue universe S_N , and replaces the gap-polynomial codegree $O(h)$ of Lemma 108 by the divisor cap “at most one prime $> \sqrt{N}$ divides a difference $< N$ ”. This eliminates the short-window partition and its $\log N$ loss. Unconditionally, however, $S_N \ll N^{5/3}/\log N$ is all that Proposition 14 provides, and the bucket argument then returns only $M \ll_\varepsilon N^{2/3}$, which is why Theorem 110 remains the unconditional record.

Optimality: one reflected star per scale

The next proposition shows that Theorem 113 is sharp *within the class of vertical hypotheses*: even the pointwise Hypothesis Z with constant 2, together with every proved primewise and pairwise structural statement about the zero sets, admits a model with $\asymp \log X$ exceptional columns. It is the multi-scale version of the anchored rainbow star (Proposition 352) and of the adversarial family of Remark 124.

Proposition 116 (Star-per-scale model). *For every large X there is a family $(\mathcal{Z}_p^\dagger)_{p \leq X}$ with the following properties.*

1. **Vertical hypotheses hold pointwise.** *Every $\mathcal{Z}_p^\dagger$ has size 0 or 2; in particular Hypothesis Z holds with $C = 2$, and a fortiori Hypothesis $\bar{Z}$.*
2. **All banked structural constraints hold.** *Every nonempty $\mathcal{Z}_p^\dagger$ is reflection-invariant ($r \leftrightarrow p-1-r$), has no consecutive elements, avoids $\{0, 1, p-2, p-1\}$, and satisfies the interval bound $|\mathcal{Z}_p^\dagger \cap I| \leq 2|I|^{2/3}$. The universal CRT second-moment bound (the proof of Proposition 328) holds for the model. The unique pair in every nonempty $\mathcal{Z}_p^\dagger$ satisfies its necessary gap-polynomial certificate $N_h(u) \equiv 0 \pmod{p}$ automatically, by the reflection identity $N_h(-x-h-1) = (-1)^{h-1}N_h(x)$ of Proposition 352. For any two model columns, the common bad primes divide an explicit nonzero integer of size $O(X^3)$, so the statement of Lemma 108 holds with room to spare.*
3. **One exceptional column per scale.** *For each scale $n_j = 3 \cdot 2^{j-1} \leq X$, the model mass*

$$W^\dagger(n_j) = \sum_{\substack{p > \sqrt{n_j} \\ n_j \bmod p \in \mathcal{Z}_p^\dagger}} \log p \geq \left(\frac{1}{2} - o(1)\right) n_j.$$

Hence for every fixed $\varepsilon < 1/2$ the model has at least $\log_2 X - O_\varepsilon(1)$ columns $n \leq X$ with $W^\dagger(n) > \varepsilon n$.

Consequently, no argument that uses the Apéry sequence only through the listed statements—in particular through any bound on the individual zero counts $Z(p)$ —can improve Theorem 113 to $o(\log X)$, nor prove any pointwise bound $B(n) = o(n/\log n)$. The $\Theta_\varepsilon(\log X)$ level is the exact frontier of vertical information; crossing it requires cross-prime input. For example, $M_1 + (\text{HM})_3$ makes the exceptional set finite by Theorem 350; $(\text{HM})_3$ alone does not, because the unconditional bound for λ_X is much larger than $1/\log X$.

Proof. The scales’ top windows $(n_j/2, n_j]$ are pairwise disjoint and their union covers all large primes, since $n_{j+1} = 2n_j$. Assign each prime $p \geq 7$ to its window: if $n_j/2 < p \leq n_j$, put $r = n_j - p$

and $\mathcal{Z}_p^\dagger = \{r, p-1-r\}$, deleting (i.e. setting $\mathcal{Z}_p^\dagger = \emptyset$ for) the $O(1)$ primes per scale for which $r \in \{0, 1, p-2, p-1\}$, $2r = p-1$, or $|p-1-2r| = 1$.

(1) is immediate. For (2): reflection-invariance, non-consecutiveness and endpoint avoidance are built in; a two-element set trivially satisfies $|\mathcal{Z}_p^\dagger \cap I| \leq 2 \leq 2|I|^{2/3}$; the CRT second-moment proof of Proposition 328 uses only the Chinese remainder theorem and applies to every family. For the gap certificate, write $u = \min(r, p-1-r)$ and $h = p-1-2u$; then h is even, $-u-h-1 \equiv u \pmod{p}$, and the reflection identity gives $N_h(u) \equiv -N_h(u)$, so $N_h(u) \equiv 0 \pmod{p}$. For the codegree statement, let $m < n$ be any two integers and p a common bad prime of the model, lying in the window of scale j . The two residues of m, n are drawn from $\{r_p, p-1-r_p\}$; equal choices force $p \mid n-m$, opposite choices force $p \mid m+n+1$. Hence all common bad primes divide the nonzero integer $(n-m)(m+n+1) = O(X^3)$, so at most $O(1)$ primes above any fixed power of X are common—far below the $O(h)$ allowance of Lemma 108.

For (3): every active prime of scale j marks n_j (its residue is $r = n_j - p$ by construction), so

$$W^\dagger(n_j) \geq \psi(n_j) - \psi(n_j/2) - O(\log n_j) = \left(\frac{1}{2} - o(1)\right)n_j$$

by the prime number theorem, the $O(\log n_j)$ term accounting for the $O(1)$ deleted primes. The final claims follow because the model satisfies every hypothesis in the stated class while exhibiting one exceptional column in each of the $\log_2 X - O(1)$ scales, and because a single scale of the construction is, up to the added mirror orbit, the adversarial family of Remark 124, with $B(n_j) \geq \pi(n_j) - \pi(n_j/2) - O(1) \gg n_j / \log n_j$. $\square$

Remark 117 (Verdicts for the weaker all-index targets). Proposition 116 scopes precisely which weakened forms of the conjecture are and are not accessible without cross-prime arithmetic. *Not accessible from vertical plus pairwise information, even assuming Hypothesis Z pointwise*: the all-index bounds $\log G_n \ll n/(\log n)^\varepsilon$ and $\log G_n \ll n^{1-\varepsilon}$ (a single scale of the model violates both while satisfying every banked statement), and any exceptional-set bound $o(\log X)$, in particular a bounded exceptional set. *Accessible*: the $O_\varepsilon(\log X)$ exceptional set of Theorem 113 under the first-moment hypothesis $\bar{Z}$ alone, which is therefore optimal in its class. Unconditionally, the same bucket mechanism is blocked only by the gap between $Z(p) \ll p^{2/3}$ (Proposition 14) and $\bar{Z}$: an unconditional average bound $\sum_{p \leq x} Z(p) \ll \pi(x)(\log x)^{1-\delta}$ for some $0 < \delta \leq 1$ would already yield, through (132), the unconditional exceptional-set bound $O_\varepsilon((\log X)^{2-\delta})$, interpolating between Theorem 110 ($\delta \rightarrow 0$) and Theorem 113 ($\delta = 1$). Thus the exceptional-set problem is now reduced, with no loss of logarithms, to the vertical first-moment problem for $Z(p)$.

The proof also gives upper Banach density zero, but a scale split is needed to make the conclusion uniform in the interval's starting point. Let $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$ and $I = (M, M+L]$. If $M \leq 2L$, then $I \subset (0, 3L]$, so Theorem 110 gives $|E_\varepsilon \cap I| = O_\varepsilon((\log(3L))^2)$. If $M > 2L$, then $I \subset (M, 2M]$. Covering I by critical windows of length $Y \asymp_\varepsilon M / \log M$ gives

$$|E_\varepsilon \cap I| \ll_\varepsilon 1 + \frac{L \log M}{M} \ll_\varepsilon 1 + \log(3L).$$

Thus, uniformly in M , $|E_\varepsilon \cap (M, M+L]| = O_\varepsilon((\log(3L))^2) = o(L)$. This proves upper Banach density zero unconditionally and strengthens Corollary 84.

9 Counting bad primes: the conditional result

For completeness we sharpen the leading-digit zero incidence from $O(N^{3/2})$ to $O(N^{10/7} / \log N)$ via a dyadic argument combining divisor-counting with the sub-interval zero bound. By Lemma 88,

however, a leading-digit zero alone does not make a prime divide G_n ; the proposition is an independent incidence estimate and is not used in the conditional theorem below.

Proposition 118 (Dyadic leading-digit bound). *The leading-digit incidence L_N (counting pairs (n, p) with $n \in (N, 2N]$, $p > \sqrt{n}$ prime, and $p \mid b_{\lfloor n/p \rfloor}$) satisfies*

$$L_N = O\left(\frac{N^{10/7}}{\log N}\right).$$

Proof. For dyadic blocks $P = 2^j$, $Q = 2^k$ with $Q < P$ and $PQ \in [N/4, 4N]$, define

$$D(P, Q) = \#\{(p, q) : P < p \leq 2P, Q < q \leq 2Q, p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 15, each prime $p \in (P, 2P]$ has at most $O(Q^{2/3})$ zeros of b_j in $(Q, 2Q]$. Summing over $\pi(2P)$ primes gives $D(P, Q) = O(PQ^{2/3}/\log P)$.
- *Divisor counting:* Since $\log b_q \leq (q+1)\log 64$, the number of prime factors of b_q exceeding P is at most $\log b_q / \log P = O(q/\log P)$. Summing over $q \in (Q, 2Q]$ gives $D(P, Q) \leq \sum_q O(q/\log P) = O(Q^2/\log P)$.

Hence $D(P, Q) \leq \min(PQ^{2/3}, Q^2)/\log P$.

Each pair (p, q) in $D(P, Q)$ contributes to at most $p \leq 2P$ values of n with $\lfloor n/p \rfloor = q$, so the block's contribution to L_N is at most $D(P, Q) \cdot 2P \leq 2 \min(P^{4/3}N^{2/3}, N^2/P)/\log P$ (using $Q = \Theta(N/P)$). The crossover is at $P_0 = N^{4/7}$. Summing $O(\log N)$ dyadic blocks $P \in [\sqrt{N}, 2N]$:

$$L_N \leq \sum_{P \leq N^{4/7}} \frac{2P^{4/3}N^{2/3}}{\log P} + \sum_{P > N^{4/7}} \frac{2N^2}{P \log P} = O\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the $P = N^{4/7}$ boundary term. $\square$

Theorem 119. *Assume Hypothesis $\bar{Z}$. Then for a set of positive integers n of natural density 1,*

$$\log G_n \ll \sqrt{n} \log \log(3n) = n^{1/2+o(1)}.$$

Proof. Fix a dyadic interval $N < n \leq 2N$. Let $B_{\text{low}}(n)$ count primes $p > \sqrt{n}$ for which $n \bmod p \in \mathcal{Z}_p$, and let $B_{\text{comp}}(n)$ count the remaining bad primes, for which the block constant $D_{\lfloor n/p \rfloor}$ vanishes modulo p . Lemma 87(iii) shows that these two classes exhaust the primes $p > \sqrt{n}$ which divide G_n ; a zero of the leading digit by itself is not a third class, by Lemma 88.

Lower digit. Under Hypothesis $\bar{Z}$, set $M(t) = \sum_{p \leq t} Z(p) \ll t/\log t$. Abel summation gives $\sum_{\sqrt{N} < p \leq 2N} Z(p)/p = O(1)$, so the residue-class count used in Theorem 82 gives

$$\sum_{N < n \leq 2N} B_{\text{low}}(n) = O(N).$$

Companion block. The companion-incidence estimate in the same proof gives

$$\sum_{N < n \leq 2N} B_{\text{comp}}(n) \ll \frac{N^{3/2}}{\log N}.$$

Indeed, for each quotient $q \leq \sqrt{2N}$ there are $O(q/\log N)$ relevant primes dividing the numerator of a_q , and each supplies at most $2N/q$ indices.

Conclusion. Put $T = \log \log(3N)$. Markov's inequality applied to the two preceding displays shows that, outside $O(N/T) = o(N)$ indices,

$$B_{\text{low}}(n) \leq \frac{T\sqrt{N}}{\log N}, \quad B_{\text{comp}}(n) \leq \frac{T\sqrt{N}}{\log N}.$$

The small primes contribute $O(\sqrt{n})$ by Proposition 3, and every prime above $\sqrt{n}$ has $v_p(G_n) \leq 6$ by Corollary 2. Hence all remaining indices satisfy

$$\log G_n \ll \sqrt{N} + (B_{\text{low}}(n) + B_{\text{comp}}(n)) \log(2N) \ll \sqrt{N} \log \log(3N).$$

Summing the exceptional counts over dyadic intervals gives a density-zero set and proves the claim. $\square$

Remark 120 (The companion-block loss). The fixed-constant estimate $O(\sqrt{n})$ does not follow from the available first moments. The lower-digit channel has total incidence $O(N)$ under Hypothesis $\bar{Z}$, whereas the companion channel has only the bound $O(N^{3/2}/\log N)$. Markov's inequality therefore requires a factor tending to infinity in order to discard a density-zero set. Removing this factor requires new information on the distribution of the prime divisors of the companion numerators $\text{num}(a_q)$, not a bound for leading-digit zeros of b . The pointwise small-prime estimate $O(\sqrt{n})$ would itself also need a finer valuation analysis before one could expect a smaller scale.

Corollary 121 (Conditional square-root tail). *Assume Hypothesis $\bar{Z}$. There is an absolute constant C such that, uniformly for N sufficiently large and $T \geq 1$,*

$$\#\{N < n \leq 2N : \log G_n > CT\sqrt{N}\} \ll \frac{N}{T}.$$

Proof. The proof of Theorem 119 gives

$$\sum_{N < n \leq 2N} B_{\text{low}}(n) = O(N), \quad \sum_{N < n \leq 2N} B_{\text{comp}}(n) = O(N^{3/2}/\log N).$$

Consequently at most $O(N/T)$ indices have $B_{\text{low}}(n) + B_{\text{comp}}(n) > T\sqrt{N}/\log N$. Outside this set, Proposition 3 and Corollary 2 give

$$\log G_n \ll \sqrt{N} + (B_{\text{low}}(n) + B_{\text{comp}}(n)) \log(2N) \leq CT\sqrt{N}$$

after increasing C . For values of T for which the displayed threshold exceeds the trivial upper bound for $\log G_n$, the result is vacuous and remains valid. $\square$

Corollary 122 (Upper Banach density zero). *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$, the set $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$ has upper Banach density zero:*

$$\lim_{L \rightarrow \infty} \sup_{M \geq 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

Proof. The hypothesis is unnecessary: the uniform local estimate following Theorem 110 proves this conclusion unconditionally. $\square$

Corollary 123 (A dyadic zero-density obstruction). *Suppose that there are constants $\varepsilon, \delta > 0$ and infinitely many X for which at least a proportion δ of the primes in $(X/2, X]$ satisfy $Z(p) \geq \varepsilon p$. Then, along those scales, $\log G_n \gg_{\varepsilon, \delta} n$ for a positive proportion of $n \in (X/2, 2X)$. In particular, the full conjecture fails. Conversely, the uniform condition $Z(p) = o(p)$ implies $G_n = e^{o(n)}$ for a set of integers of density 1 by Theorem 82.*

Proof. Fix one of the stated scales. For every selected prime $p \in (X/2, X]$ and every $r \in \mathcal{Z}_p$, put $n = p + r$. Then $n \in (X/2, 2X)$ and $n/2 < p \leq n$, so Lemma 78 gives $p \mid G_n$. The number of these prime–integer incidences is

$$\sum_{\substack{X/2 < p \leq X \\ Z(p) \geq \varepsilon p}} Z(p) \gg_{\varepsilon, \delta} \frac{X^2}{\log X}.$$

On the other hand, each integer in $(X/2, 2X)$ is incident with at most $O(X/\log X)$ primes. Comparing the preceding lower bound with this uniform upper bound shows, by elementary averaging, that a positive proportion of the integers in this interval have $B(n) \gg_{\varepsilon, \delta} X/\log X$. Every counted prime exceeds $X/2$ and occurs in G_n to positive order; hence for these integers

$$\log G_n \geq B(n) \log(X/2) \gg_{\varepsilon, \delta} X \asymp_{\varepsilon, \delta} n.$$

The converse is exactly the density-one argument already proved. $\square$

Remark 124 (The pointwise gap). The full conjecture $G_n = e^{o(n)}$ for *every* n is strictly stronger than the density-one statement. By Corollary 79, it implies the top-half radical estimate (111); the converse would still have to control the additional quotient classes isolated in Proposition 104. The height bound gives $B(n) = O(n/\log n)$, but for an *adversarial* family (Z_p) with $|Z_p| \leq 1$, one may choose $Z_p = \{N - p\}$ for primes in $(N/2, N]$, producing $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$. Thus the needed pointwise bound $B(n) = o(n/\log n)$ requires arithmetic information specific to the Apéry sequence, not just the cardinalities $Z(p)$.

By Corollary 79, every bad prime $p > n/2$ satisfies $p \mid b_n$. Hence the conjecture implies $\log \text{rad}_{(n/2, n]}(b_n) = o(n)$: the large prime radical of b_n must be subexponential, even though $|b_n| = e^{\Theta(n)}$.

The Poisson model predicts that $B(n) \geq 1$ occurs infinitely often: $\mathbb{P}(B(n) \geq 1) \sim \log 2 / \log n$, and $\sum (\log n)^{-1}$ diverges. This is compatible with the conjecture, since one bad prime contributes only $O(\log n)$ to $\log G_n$. To violate $G_n = e^{o(n)}$, one needs $B(n) \geq \varepsilon n / (13 \log n)$; the Poisson tail gives $\mathbb{P}(B(n) \geq k_n) \leq e^{-c n}$ for $k_n = \lceil \varepsilon n / (13 \log n) \rceil$, and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

10 Computational verification

We computed G_n exactly for $n = 1, \dots, 1000$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	0.93
51–100	0.108	0.26
101–200	0.086	0.18
201–300	0.048	0.10
301–400	0.038	0.08
401–500	0.024	0.05
501–600	0.029	0.06
601–700	0.017	0.04
701–800	0.014	0.04
801–900	0.016	0.04
901–1000	0.013	0.03

A power-law fit gives $\log(G_n)/n \sim 5n^{-0.87}$, i.e., $\log G_n \sim 5n^{0.13}$, far stronger than the density-one bound $o(n)$ of Theorem 82.

The exponent 0.13 should be interpreted with caution: if $\log G_n \sim \kappa \log n$, the effective log-log slope is $1/\log n = 0.145$ at $n = 1000$, close to the fitted 0.13. A random-gcd model—assigning each prime $p \leq n$ an independent probability $\sim 1/p$ of dividing G_n —predicts $\mathbb{E}[\log G_n] \sim \log n$ by the weighted Mertens theorem, with standard deviation $\sim (\log n)/\sqrt{2}$ of the same order. The data thus cannot yet distinguish $\kappa \log n$ from $n^{0.13}$; the logarithmic prediction is arithmetically more natural.

10.1 Extended modular verification

For each n , define $B(n) = \#\{\sqrt{n} < p \leq n : n \bmod p \in \mathcal{Z}_p\}$, so that $\log G_n \leq O(\sqrt{n}) + 6B(n) \log n$ (Section 7). The ratio $B(n) \cdot \log n/n$ governs $\log G_n/n$: whenever it tends to zero, $G_n = e^{o(n)}$. We computed $B(n)$ for all $n \leq 100,000$ by modular arithmetic (iterating the Apéry recurrence mod p for each prime $p \leq 200,001$ and checking whether $n \bmod p \in \mathcal{Z}_p$).

n range	max $B(n)$	avg $B/T(n)$	max $B/T(n)$
1–100	3	0.038	0.260
101–500	4	0.013	0.124
501–1,000	4	0.006	0.036
1,001–2,000	4	0.003	0.027
2,001–5,000	4	0.001	0.015
5,001–10,000	5	0.0008	0.007
10,001–20,000	5	0.0004	0.004
20,001–50,000	6	0.0002	0.002
50,001–100,000	6	0.0001	0.001

Here $T(n) = n/\log n$, the density-one threshold. The ratio $\max B/T$ decreases monotonically. The overall maximum is $B(n) = 6$ at $n = 20,546$, giving the explicit bound $\log G_n/n \leq 6 \cdot 6 \cdot \log(20,546)/20,546 + O(1/\sqrt{20,546}) = 0.009$, which confirms $G_n = e^{o(n)}$ at every index $n \leq 100,000$. The Poisson prediction for the maximum is $\max_{n \leq N} B(n) \approx 2 \log \log N \approx 5.6$ at $N = 10^5$, consistent with the observed 6.

11 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [4] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base- p digits. Quantitatively, since $b_n \equiv \prod b_{d_i} \pmod{p}$ (Theorem 4), we have $p \nmid b_n$ iff every base- p digit of n avoids $\mathcal{Z}_p$. Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2+17x+5$ (discriminant -51 ; exists over $\mathbb{F}_p$ iff $(\frac{-51}{p}) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.
3. For $1 \leq j \leq p-2$, the Apéry number $b_j \bmod p$ admits a character-sum representation as a Greene [18] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p -adic event. The gap polynomial argument above is specifically adapted to this p -adic zero-count problem.
4. Malik–Straub [26] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [30] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.
5. In the standard toric coordinate, the truncated polynomial $A_p(t) = \sum_{j=0}^{p-1} b_j t^j \in \mathbb{F}_p[t]$ is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in $\mathbb{F}_p$ are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic $Z(p)$ counts vanishing *coefficients* of A_p , equivalently the vanishing finite Mellin coefficients $b_j = -\sum_{t \in \mathbb{F}_p^\times} A_p(t) t^{-j}$ for $1 \leq j \leq p-2$. Thus $Z(p)$ is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent F -crystal theory do not by themselves bound $Z(p)$; they control the root locus, not the Fourier support. A bounded-complexity geometric lift *does* exist: the lisse sheaf $\mathcal{V}_\ell = R^2 f_* \mathbb{Q}_\ell$ on the smooth locus of the K3 pencil (rank 22, conductor independent of p) satisfies $\text{Tr}(\text{Frob}_t \mid \mathcal{V}_\ell) \equiv A_p(t) \pmod{p}$ by the Hasse–Witt congruence. The Mellin sum $\mathcal{M}_{p,j} = \sum_t \tau_p(t) \omega_p(t)^{-j}$ is then realized on $R\Gamma_c(U_{\mathbb{F}_p}, \mathcal{V}_\ell \otimes \mathcal{L}_{\omega_p^{-j}})$ with bounded rank and conductor. However, the event $b_j \equiv 0 \pmod{p}$ is the *same-characteristic* divisibility $\mathfrak{p}_p \mid \mathcal{M}_{p,j}$, a crystalline condition outside the scope of ℓ -adic equidistribution (Forey–Fresán–Kowalski [17]). The missing theorem is a *crystalline Mellin anti-concentration* estimate controlling $\#\{j : p \mid \mathcal{M}_{p,j}\}$.

6. **Jacobi counterexample: bounded complexity is not enough.** Fix an integer $d \geq 2$. For every prime $p \equiv 1 \pmod{d}$, the rank-1 Kummer sheaf $\mathcal{L}_\psi$ on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ attached to $\psi = \omega_p^{-(p-1)/d}$ (order d) has Mellin transforms $M_j = J(\omega_p^{-j}, \psi)$. The Gross–Koblitz/Stickelberger valuation $v_p(g(\omega^{-k})) = k/(p-1)$ and the Jacobi factorization $J(\alpha, \beta) = g(\alpha)g(\beta)/g(\alpha\beta)$ give

$$v_p(M_j) = 1 \quad \text{for } j > (p-1)(d-1)/d.$$

At equality the product character is trivial and $J(\alpha, \alpha^{-1}) = -\alpha(-1)$ is a p -unit. Hence exactly $(p-1)/d - 1$ nontrivial characters in this interval satisfy $p \mid M_j$. Thus a sheaf of *fixed* rank, conductor, and Weil bound $O(\sqrt{p})$ can have $\gg p$ bad tame characters. This shows that

bounded cohomological complexity alone cannot control $Z(p)$; any proof of Hypothesis Z must use Apéry-specific arithmetic.

7. **Sufficient criteria for Hypothesis Z .** Two clean algebraic conditions, either of which would imply $Z(p) = O(1)$:

(a) *Bounded tame-Hasse polynomial.* Suppose there exist an absolute integer D and, for each sufficiently large p , a nonzero Laurent polynomial $Q_p \in \mathbb{F}_p[X, X^{-1}]$ of degree at most D and units $u_{p,j} \in \mathbb{F}_p^\times$ such that $\mathcal{M}_{p,j} \equiv u_{p,j} Q_p(\omega_p^j) \pmod{p}$ for every nontrivial tame character. Then $Z(p) \leq 2D$, since Q_p (after clearing negative exponents) has at most $2D$ roots.

(b) *Bounded nonordinary twists.* Suppose for each p the set E_p of tame characters χ for which the twisted crystal $H_{\text{rig},c}^1(U, \mathcal{V} \otimes \mathcal{K}_\chi)$ fails to have a unique unit-root eigenvalue satisfies $|E_p| \leq C$. Then $Z(p) \leq C + O(1)$, since a unique unit root forces $v_p(\mathcal{M}_{p,j}) = 0$.

Criterion (a) is the algebraic version of Hypothesis Z ; criterion (b) is its p -adic-slopes formulation. The Jacobi counterexample above shows that neither criterion holds for general bounded-complexity sheaves.

8. **Linear-complexity obstruction.** The Sym^2 factorization $A_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2$ with B_p of degree $\lfloor (p-1-2\varepsilon_p)/2 \rfloor$ implies that A_p has at most $\lfloor (p-1)/2 \rfloor + 2$ distinct zeros in $\mathbb{F}_p^\times$. Choosing a primitive root g and forming the canonical character interpolation $Q(X) = -\sum_{r=1}^{p-2} A_p(g^r) X^r$, whose values $Q(g^j) = b_j$ for $1 \leq j \leq p-2$, we find that Q has at least $(p-5)/2$ nonzero coefficients. Any Laurent polynomial of degree at most D representing the same function on the character group must have $2D+1 \geq (p-5)/2$, giving $D \geq (p-7)/4$. Thus criterion (a) cannot be satisfied in its direct (unnormalized) form: a nontrivial Gauss/Jacobi normalization of the Mellin coefficients is necessary before bounded tame degree could emerge.

9. **Palindromic symmetry.** For $0 \leq j \leq p-1$ and $0 \leq k \leq p-1-j$, the identity $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod{p}$ holds (each of the k falling factors $p-1-j-i \equiv -(j+1+i)$). When $k \leq j$, $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$; when $k > j$, Lucas's theorem gives $\binom{p-1-j+k}{k} \equiv 0$ (since $p-1-j+k \geq p$ and $\binom{k-j-1}{k} = 0$). Therefore each term of $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$ with $k > j$ vanishes, and for $k \leq j$ the $(-1)^{2k}$ signs cancel, giving $b_{p-1-j} \equiv b_j \pmod{p}$. The extra terms of b_j when $j > (p-1)/2$ (i.e., $k > p-1-j$) likewise vanish: $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$ since $j+k-p < k$. Equivalently, $A_p(t)$ is palindromic: $t^{p-1} A_p(1/t) \equiv A_p(t)$. This explains the observed pairing: whenever $b_j \equiv 0$, also $b_{p-1-j} \equiv 0$, so zeros of b_j come in pairs $\{j, p-1-j\}$ (except at the self-paired center $j = (p-1)/2$, which vanishes only at non-ordinary primes). Proposition 99 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo p , in particular the companion solution a . The “pair count” $K(p) = Z(p)/2$ (or $(Z(p)-1)/2$ at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 58).

10. **Sym² squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [33]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [10] proved the mod- p squareness rigorously: for every prime $p \geq 5$, setting $\varepsilon_p = \frac{1}{2}(1 - (\frac{-6}{p}))$, there exists a unique polynomial $B_p \in \mathbb{F}_p[t]$ normalized by $B_p(0) = 1$ such that

$$A_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$

where $\Delta(t) = t^2 - 34t + 1$ is the conifold discriminant. Theorem 1.2 of [10] establishes precisely this quadratic-times-square factorization. That theorem alone does not assert that B_p

is squarefree or coprime to Δ . These two additional properties are proved below from the exact reduced second-order equation in Proposition 288; the computation through $p \leq 20,000$ in Subsection 12.22 provides an independent check. Crucially, $\mathcal{Z}_p$ is the set of vanishing *coefficients* of A_p (the finite Mellin transform), *not* the evaluation roots of A_p or of B_p ; the two loci are genuinely different—e.g. at $p = 7$, $A_7 = (t-1)^2(t^2+1)^2$ has the evaluation root $t = 1$ yet no vanishing coefficient, so $\mathcal{Z}_7 = \emptyset$ (Proposition 262). A Weil bound for the curve $B_p(t) = 0$ therefore does *not* bound the Fourier sum $F_p(a) = \sum_{j \in \mathcal{Z}_p} e_p(aj)$. The proof uses the explicit identity $F(t(x)) = (1+x)h(x)^2$ between the Apéry and Franel generating functions, the p -Lucas property of both sequences, and a quadratic-descent argument through the covering $\mathbb{F}_p(x)/\mathbb{F}_p(t)$ where $t = x(1-8x)/(1+x)$. Proposition 286 below records the resulting finite-field identity and its exact cover-adapted coordinates; Proposition 287 records the two normalized square-root recurrences. The deck-transformation character $(-6/p)$ determines whether $\varepsilon_p = 0$ or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \quad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently, $\varepsilon_p = 0$ iff $p \equiv 1, 5, 7, 11 \pmod{24}$. These two cases occur with equal density $1/2$ by Dirichlet's theorem.

Splitting the mean $Z(p)$ by residue class modulo 8 (and modulo 24) yields no significant difference (mean ≈ 1.00 in each class, $n \geq 2000$ primes per class up to 10^5); the squareness structure does not directly influence $Z(p)$, consistent with $Z(p)$ counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure: b_j is (up to the Δ^{ε_p} correction) the j -th coefficient of B_p^2 , i.e., $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod{p}$ where β_k denotes the coefficients of B_p . The vanishing $b_j \equiv 0$ thus requires cancellation across $\sim p/2$ convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree $(p-1)/2$ over $\mathbb{F}_p$, for $p \in \{101, 503, 1009, 5003\}$) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean ≈ 1 , variance ≈ 1 , total variation distance to Poisson(1) equal to 0.018). However, the observed $Z(p)$ distribution over 667 primes $p \leq 5000$ is *not* Poisson(1) (TV = 0.43); instead it matches $2 \cdot \text{Poisson}(1/2)$ with TV = 0.023. The explanation is the palindromic pairing: since $b_{p-1-j} \equiv b_j$, zeros come in pairs $\{j, p-1-j\}$ and $Z(p)$ is almost always even. Denoting by $Z_{\text{pair}}(p)$ the number of zero pairs, the data show $Z_{\text{pair}} \sim \text{Poisson}(1/2)$, so $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot \text{Poisson}(1/2)$. Thus squareness explains the $O(1)$ mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$ via extreme-value theory, strongly suggesting that $Z(p)$ is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of B_p imposes a bounded *average* (Hypothesis $\bar{Z}$) remains the central open question.

The factorization does *not* improve the divisor-counting bound $D(P, Q) \leq Q^2/\log P$. In the regime $P \geq Q^{4/3}$ where the divisor estimate is active, one has $q < p/2$, so the coefficient $b_q \pmod{p}$ lies in the formally free triangular range of the square root: the map $(\beta_1, \dots, \beta_q) \mapsto (b_1, \dots, b_q) \pmod{p}$ is a triangular automorphism with Jacobian 2^q , hence the single equation $b_q \equiv 0$ imposes no algebraic constraint beyond fixing one coefficient of B_p . Thus the Sym^2 structure gives the correct *random model* (uniform square root $\Rightarrow$ Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

11. **Toric framework.** The Laurent polynomial $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$ on $(\mathbb{F}_p^\times)^4$ has Newton polytope $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$, a 4-dimensional pyramid over the regular

octahedron in $\mathbb{R}^3$. Its f -vector is $(7, 18, 20, 9)$, its normalized volume is $4! \cdot \text{vol}(\Delta) = 8$, and it is Kouchnirenko nondegenerate for every prime $p \geq 5$: for each of the 27 nonempty faces σ not containing the origin, the toric critical system $\{x_i \partial f_\sigma / \partial x_i = 0\}$ has no solution in $(\mathbb{F}_p^\times)^4$. For the Adolphson–Sperber twisted Hodge polygon with w -Kummer twist $\chi_w = \omega^{-j}$ (character parameter $a = j/(p-1)$), the slopes are $a, a+1, a+2, a+3$ with multiplicities $1, 3, 3, 1$, confirming that the first slope equals a for every $a \in \{k/100 : 0 \leq k \leq 99\}$. By Newton-above-Hodge (Adolphson–Sperber [2]), the p -adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The $Z(p)$ bound thus reduces to proving ordinarity for all but $O(1)$ twist parameters j .

12. **Branch obstruction.** The $p-1$ Kummer twists χ^{-j} ($j = 0, \dots, p-2$) are the distinct multiplicative characters of $\mathbb{F}_p^\times$. Viewed as finite-field or Dirichlet characters their natural modulus is p , whereas the associated nontrivial rank-one Kummer sheaves are tame and have geometric conductor bounded independently of p ; the trivial character is treated separately. In Iwasawa-theoretic coordinates $\chi^j = \omega^a \langle \cdot \rangle^s$ with $a = j \bmod (p-1)$, the horizontal family $\{j : 0 \leq j < p-1\}$ hits *each* of the $p-1$ branches ω^a exactly once. Weierstrass preparation and bounded- λ rigidity control zeros *within one branch* (the vertical direction $j, j+(p-1), j+(p-1)p, \dots$, where Dwork congruences live) but not *across* branches. Thus there is no p -adic analytic function of one variable whose zero count majorizes $Z(p)$. Any “rigidity” proof of Hypothesis $\bar{Z}$ would need a mechanism not supplied by existing Iwasawa–Dwork theory.
13. Any estimate $\log G_n = o(n)$ —whether $O(\log n)$, $O(\sqrt{n})$, or $O(n^{0.13})$ —yields $\gamma = 0$ in the exponent $G_n = e^{\gamma n + o(n)}$ and therefore does not improve the Rhin–Viola irrationality measure $\mu(\zeta(3)) \leq 5.513$ [31], which depends only on the exponential growth rate $|b_n \zeta(3) - a_n| = e^{-(\log \alpha + o(1))n}$. An improvement would require an exponentially large common divisor, which is not predicted by any current model.
14. **Comparison with previous work.** Classical refinements of Apéry’s method (Beukers [4], Rhin–Viola [31], Zudilin [37]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for every index. They do not provide exceptional-set estimates for the *actual* reduced gcd G_n . A separate body of work, based on the Subspace Theorem, proves $\gcd(a^n - 1, b^n - 1) = e^{o(n)}$ pointwise for multiplicatively independent a, b (Bugeaud–Corvaja–Zannier [9]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence’s polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.
15. **The remaining open problem.** The companion-block height bound (Proposition 103) shows that primes $p > \sqrt{n}$ with $D_q \equiv 0 \pmod{p}$ contribute $O(n^{2/3})$ to $\log G_n$ at *every* index. After removing this channel, the conjecture $G_n = e^{o(n)}$ for all n is equivalent to the *pointwise radical estimate*

$$\sum_{\substack{\sqrt{n} < p \leq n \\ p | b_n \bmod p}} \log p = o(n).$$

Theorem 110 proves this for all but $O_\varepsilon((\log N)^2)$ integers in $[1, N]$. However, the codegree method (Lemma 108) controls primes *shared* between pairs of integers, and does not bound the total weight of “private” primes at a single index. Closing the gap requires a one-point rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too

many distinct primes. By the quotient reduction (Proposition 104), one sufficient formulation only uses primes $p > n/(f(n) \log n)$ with f growing arbitrarily slowly. The weaker exact formulation is Theorem 380: prove the target-radical estimate separately for every fixed $q = \lfloor n/p \rfloor$, and absorb the unbounded quotient tail into $O(n/Q)$.

The moving congruences in the remaining pointwise radical can be compressed into one ordinary integer gcd.

Proposition 125 (One-moment denominator criterion). *Put*

$$\ell_n = \text{lcm}(1, \dots, n), \quad \mathcal{R}_n = \sum_{k=1}^n \frac{b_{n-k}}{k}, \quad U_n = \ell_n \mathcal{R}_n \in \mathbb{Z},$$

and

$$\mathcal{W}_n = \sum_{\substack{\sqrt{n} < p \leq n \\ p \mid b_n \bmod p}} \log p.$$

Then

$$0 \leq \log \gcd(\ell_n, U_n) - \mathcal{W}_n \ll n^{2/3}. \quad (133)$$

Moreover, if $\text{den}(\mathcal{R}_n)$ denotes the positive reduced denominator, then

$$\gcd(\ell_n, U_n) = \frac{\ell_n}{\text{den}(\mathcal{R}_n)}. \quad (134)$$

Consequently the all-index Apéry gcd conjecture is equivalent to either of the pointwise statements

$$\log \gcd(\ell_n, U_n) = o(n), \quad \log \ell_n - \log \text{den}(\mathcal{R}_n) = o(n).$$

Proof. Let $p > \sqrt{n}$ be prime, and write $n = qp + r$ with $1 \leq q < p$ and $0 \leq r < p$. Since $p^2 > n$, one has $v_p(\ell_n) = 1$. Modulo p , every summand in U_n with $p \nmid k$ vanishes. Writing the surviving indices as $k = ap$, $1 \leq a \leq q$, and applying the Apéry–Lucas congruence gives

$$U_n \equiv \frac{\ell_n}{p} b_r \rho_q \pmod{p}, \quad \rho_q := \sum_{a=1}^q \frac{b_{q-a}}{a}. \quad (135)$$

Here ℓ_n/p is a p -adic unit, and the denominator of ρ_q is prime to p . Thus every prime counted by $\mathcal{W}_n$ divides $\gcd(\ell_n, U_n)$. An additional prime $p > \sqrt{n}$ can occur only if it divides the numerator of the positive rational number ρ_q .

Fix $Q \leq \sqrt{n}$. For $q \leq Q$, put $E_q = \ell_q \rho_q \in \mathbb{Z}_{>0}$. All additional primes with this quotient divide E_q . The exponential bound for the Apéry numbers and Chebyshev's estimate $\log \ell_q \ll q$ give $\log E_q \ll q$, and hence their total logarithmic weight is

$$\ll \sum_{q \leq Q} \log E_q \ll Q^2.$$

For $q > Q$ one has $p < n/Q$, so all such additional primes have total logarithmic weight $O(n/Q)$. Taking $Q = n^{1/3}$ gives $O(n^{2/3})$. Finally, the complete prime-power contribution of $p \leq \sqrt{n}$ to any divisor of ℓ_n is

$$\sum_{\substack{p^j \leq n \\ p \leq \sqrt{n}}} \log p = O(\sqrt{n}).$$

This proves (133).

Write $\mathcal{R}_n = A/D$ in lowest terms. Since $D \mid \ell_n$, say $\ell_n = DE$, one has $U_n = EA$ and therefore $\gcd(\ell_n, U_n) = E = \ell_n/D$, proving (134). The final equivalence follows from the pointwise radical reduction above and the already proved $O(n^{2/3})$ companion-block bound. $\square$

Proposition 126 (Radical containment for the convolution content). *Define*

$$\mathcal{P}_n(T) = \sum_{k=1}^n \frac{\ell_n}{k} b_{n-k} T^k, \quad \mathfrak{c}_n = \text{cont}(\mathcal{P}_n).$$

Then

$$\text{rad}(\mathfrak{c}_n) \mid b_n, \quad \log \frac{\mathfrak{c}_n}{\text{rad}(\mathfrak{c}_n)} = O(\sqrt{n}). \quad (136)$$

In particular,

$$\log \mathfrak{c}_n = o(n) \iff \log \text{rad}(\mathfrak{c}_n) = o(n).$$

Proof. The coefficient of T^n first gives $\mathfrak{c}_n \mid \ell_n/n$. Fix a prime $p \mid \mathfrak{c}_n$, let $p^e \leq n < p^{e+1}$, and write

$$n = qp^e + r, \quad 1 \leq q < p, \quad 0 \leq r < p^e.$$

For $k = qp^e = n - r$, the factor ℓ_n/k is a p -adic unit. Since p divides every coefficient of $\mathcal{P}_n$, it follows that $p \mid b_r$. Iterating the Apéry–Lucas congruence over the lower e digits gives

$$b_n = b_{qp^e+r} \equiv b_q b_r \equiv 0 \pmod{p}.$$

This proves the radical containment, including $p = 2$. Moreover, $\mathfrak{c}_n \mid \ell_n/n$ implies that a repeated prime divisor of $\mathfrak{c}_n$ is at most $\sqrt{n}$, and hence

$$0 \leq \log \frac{\mathfrak{c}_n}{\text{rad}(\mathfrak{c}_n)} \leq \sum_{p \leq \sqrt{n}} ([\log_p n] - 1) \log p = \sum_{j \geq 2} \vartheta(n^{1/j}) = O(\sqrt{n}).$$

$\square$

Remark 127 (The radical containment is not a height bound). Proposition 126 proves the all-prime support inclusion, but it does not prove the stronger divisibility $\mathfrak{c}_n \mid b_n$. More importantly, $\text{rad}(\mathfrak{c}_n) \mid b_n$ still permits linear logarithmic height. Thus the remaining problem is genuinely to bound the common target radical, not to recover the small-prime multiplicities.

Remark 128 (What the one-moment criterion does not prove). The generating-function identity

$$\mathcal{R}_n = [z^n] \left(\sum_{j \geq 0} b_j z^j \right) \log \frac{1}{1-z}$$

makes $(\mathcal{R}_n)$ a single logarithmic holonomic coefficient sequence. Proposition 125 removes the need for several reciprocal moments or a growing Vandermonde system, but it does not supply the required reduced-denominator lower bound. In particular, aggregate denominator identities have the wrong quantifier: the missing assertion is pointwise retention at the specified index n .

The natural Casorati determinant does not provide a second local coordinate. Indeed, put

$$\mathcal{C}_n = \mathcal{R}_n b_{n-1} - \mathcal{R}_{n-1} b_n.$$

If $p > \sqrt{n}$, $n = qp + r$, and $1 \leq r < p$, then (135), applied at n and $n - 1$, and Lucas's congruence give

$$\begin{aligned} p\mathcal{R}_n &\equiv b_r \rho_q, & p\mathcal{R}_{n-1} &\equiv b_{r-1} \rho_q, \\ b_n &\equiv b_q b_r, & b_{n-1} &\equiv b_q b_{r-1} \end{aligned} \pmod{p}.$$

Consequently $p\mathcal{C}_n \equiv 0 \pmod{p}$ identically, whether or not $p \mid b_r \rho_q$. Thus a direct Casorati-nonvanishing argument cannot distinguish a retained denominator prime from a denominator hole inside a nonendpoint block; an independent cross-prime input is still necessary.

12 The second-moment approach

The codegree method of Section 8 exploits primes shared between pairs of integers to prove a density-one theorem. To close the remaining gap—proving $W(n) = o(n)$ for *every* n —we study the *second moment* of the bad-prime weight function.

12.1 The pointwise reduction and why it is deep

We first record the cleanest form of the all-index target and a structural obstruction that any proof must overcome. For a dyadic shell $I_N = (N, 2N]$ and $m \in I_N$, the Lucas collapse (Corollary 79) gives, for every prime $N < p \leq m$, the congruence $b_m \equiv 5b_{m-p} \pmod{p}$; hence the detector $X_p(m) = \mathbf{1}[p \mid b_{m-p}]$ equals $\mathbf{1}[p \mid b_m]$, and the incidence count collapses to a single integer:

$$L_N(m) = \#\{p \in (N, m] : p \mid b_m\} = \omega_{(N, m]}(b_m). \quad (137)$$

Proposition 129 (Pointwise reduction). *The all-index conjecture $G_n = e^{o(n)}$ follows from the pointwise radical anti-concentration*

$$(\diamond) \quad \max_{m \in (N, 2N]} \#\{p \in (N, m] : p \mid b_m\} \ll N^{o(1)}. \quad (138)$$

Proof. By (137) and the trivial bound $\sum_{m \in I_N} (L_N(m) - \mu)^2 \leq (\max_m L_N(m)) \sum_m L_N(m) = (\max_m L_N(m)) T_N$, the hypothesis $(\diamond)$ gives the centered second moment $V_w^\circ(N) \ll N^{o(1)} T_N$. The L^2 -to-uniform upgrade (Lemma 144) and Corollary 145 then give $W(n) = o(n)$ for every n , hence $G_n = e^{o(n)}$. The cross-prime coupling of the second moment is thus a *detector artifact*: both conditions $p \mid b_{m-p}$ and $q \mid b_{m-q}$ are statements about the single integer b_m . $\square$

Remark 130 (Fixed Mellin exponent). For $a \in \mathbb{F}_p^\times$, Fermat gives $a^{p-1} = 1$, so $a^{m-p} = a^{m-1}$ in $\mathbb{F}_p$. The toric Mellin representation (Proposition 269) therefore gives

$$b_{m-p} \equiv - \sum_{t \in \mathbb{F}_p^\times} t^{m-1} A_{m-1}(t) A_{p-m}(t) \pmod{p}.$$

The exponent $m - 1$ is *fixed* as p varies over the shell $(N, 2N]$; only the Legendre polynomial $A_{p-m}(t)$ carries the characteristic. Thus the cross-prime correlation structure of the events $\{p \mid b_m\}$ is governed by a single fixed-exponent Mellin pairing varying only in the complementary factor, rather than a moving-exponent orbit.

Remark 131 (Size slack and the sieve-impossibility barrier). The estimate $(\diamond)$ is deep for a precise reason. Since $\log b_{2N} = 2N \log(17 + 12\sqrt{2}) + O(\log N) \approx 7.05 N$, while $\sum_{N < p \leq 2N} \log p \sim N$ by the prime number theorem, the integer b_{2N} has room to contain the *entire* product of primes in $(N, 2N]$ about seven times over. The trivial height bound gives only $\omega_{(N, m]}(b_m) \leq \log b_m / \log N =$

$O(N/\log N)$, one $o(1)$ factor short. Consequently the adversarial “anchored star” of Section 12.5—the hypothetical scenario in which b_{2N} is divisible by every prime in $(N, 2N]$ —is not excluded by size. It satisfies every structural axiom available to a sieve: the vertical bound $Z(p) = O(1)$, reflection, no consecutive zeros, the sub-interval count $O(|I|^{2/3})$, and generic prime spacing, yet it violates $(\diamond)$ with $L_N(2N) \gg N/\log N$. Therefore *no* sieve-theoretic or counting argument blind to the specific non-archimedean arithmetic of the Apéry numbers can prove $(\diamond)$: a genuine input about the prime factorization of the individual integer b_m is mandatory. This is the precise sense in which the all-index conjecture, though numerically overwhelming (Remark 81), lies beyond the reach of the vertical and dispersion machinery developed above.

12.2 The criterion

For a dyadic shell $I_N = (N, 2N]$, define

$$B(m) = \sum_{\substack{\sqrt{m} < p \leq m \\ p | b_m \bmod p}} \log p, \quad M_2(N) = \sum_{m \in I_N} B(m)^2.$$

Proposition 132 (Second-moment criterion). *If $M_2(N) \leq C \cdot N^{2-\delta}$ for some $C > 0$ and $\delta > 0$, then $W(n) = o(n)$ for all n . More precisely, for every $n \in I_N$,*

$$W(n) \leq B(n) \leq \sqrt{M_2(N)} \leq C^{1/2} \cdot N^{1-\delta/2}.$$

Proof. $B(n)^2$ is one nonneg. term of the sum $M_2(N)$, hence $B(n) \leq \sqrt{M_2(N)}$. $\square$

In particular, $M_2(N) = O(N \cdot (\log N)^A)$ for any fixed A gives $B(n) = O(\sqrt{N} \cdot (\log N)^{A/2}) = o(N)$.

Remark 133 (Shifted-gcd identity). The bad-prime weight admits a combinatorial reformulation. By the injectivity of $p \mapsto n \bmod p$ on primes $p > \sqrt{n}$ (two such primes giving the same residue would force their product $\leq n$),

$$W(n) = \sum_{0 < r < n} \log \text{rad}_{>\max(\sqrt{n}, r)} \gcd(n-r, b_r).$$

Each inner radical is 1 or a single prime $p = n-r$. For the top-half block $p > n/2$, the condition becomes “ $n-r$ is prime and $(n-r) \mid b_r$.”

Proposition 134 (Hard-core reduction). *For each bad prime $p > \sqrt{n}$, write $n = q_p p + r_p$, $s_p = \min(r_p, p-1-r_p)$, $h_p = |p-1-2r_p|$. Let $1 \leq Q, R, H < n/2$. Then the total weight of primes satisfying at least one of $q_p > Q$, $s_p \leq R$, $h_p \leq H$ is $O(n/Q + (R+H)\log n)$. In particular, with $Q = R = H = \sqrt{n/\log n}$, all “easy channels” contribute $O(\sqrt{n \log n})$; the remaining hard core consists of primes $p \gg \sqrt{n \log n}$ with zero positions far from both endpoints and the reflection centre.*

Proof. Large quotient ($q_p > Q$): then $p < n/Q$, so $\sum \log p \leq \vartheta(n/Q) \ll n/Q$. Near endpoint ($s_p \leq R$): by the injectivity of $p \mapsto n \bmod p$, at most two primes $> \sqrt{n}$ contribute for each value of s_p . Near centre ($h_p \leq H$): the identity $(2q_p+1)p = 2n+1 \pm h_p$ forces p to divide a fixed integer of size $O(n)$, giving $O(1)$ primes per value of h_p . $\square$

Remark 135 (The vertical main term is harmless). Write $W(n) = M(n) + \mathcal{E}(n)$ where $M(n) = \sum_{\sqrt{n} < p \leq n} Z(p) (\log p)/p$. The unconditional bound $Z(p) \leq Cp^{2/3}$ gives $M(n) \ll \sum_{p \leq n} p^{-1/3} \log p \ll n^{2/3}$. The entire problem is the oscillatory error $\mathcal{E}(n)$, which at each prime is $(\log p) \cdot (1_{n \bmod p \in \mathcal{Z}_p} - Z(p)/p)$.

12.3 Empirical verification

The following table, computed for all dyadic shells with $64 \leq N \leq 4096$ using the full zero-set data for all primes $p \leq 20,000$, gives:

N	$M_2(N)$	M_2/N	$M_2/(N \log^2 N)$	$\max B(m)$
64	672	10.5	0.605	10.4
128	1,684	13.2	0.560	15.9
256	4,376	17.1	0.555	15.8
512	11,896	23.2	0.598	22.3
1024	28,632	28.0	0.583	21.7
2048	69,226	33.8	0.582	27.1
4096	166,671	40.7	0.587	32.1
8192	421,868	51.5	0.634	37.2

The ratio $M_2(N)/(N \log^2 N)$ stabilizes near 0.59, strongly suggesting

$$M_2(N) \sim C_2 \cdot N \cdot (\log N)^2, \quad C_2 \approx 0.59. \quad (139)$$

This gives $B(n) \leq \sqrt{0.59 N} \log N \approx 0.77 \sqrt{N} \log N = o(N)$.

12.4 CRT expansion

Expand M_2 dyadically. For distinct primes $p, q \in (P, 2P]$ with $P > \sqrt{N}$, write $\Omega_p = \{m \in I_N : m \bmod p \in \mathcal{Z}_p\}$. Then

$$M_2(N) = \underbrace{\sum_p (\log p)^2 |\Omega_p|}_{\text{diagonal}} + \underbrace{\sum_{p \neq q} \log p \log q |\Omega_p \cap \Omega_q|}_{\text{off-diagonal}}.$$

Diagonal. $|\Omega_p| = Z(p) \cdot N/p + O(Z(p))$, so

$$\text{diag} = N \sum_p \frac{Z(p) (\log p)^2}{p} + O\left(\sum_p Z(p) (\log p)^2\right).$$

Under the Poisson model $\mathbb{E}[Z(p)] = 1$, the main term is $\sim N \cdot \sum_{P < p \leq 2P} (\log p)^2 / p \sim N \cdot (\log P)^2 \cdot \log 2$. Summing dyadically $P = \sqrt{N}, 2\sqrt{N}, \dots, N$ gives diagonal $\sim \frac{1}{2} N (\log N)^2$, matching $\approx 63\%$ of the empirical (139).

Off-diagonal. Since $p, q > \sqrt{N}$ we have $pq > N$, so each CRT class $r_p \pmod{p}$, $r_q \pmod{q}$ contains at most one element of I_N . The CRT main term of the off-diagonal is $\sim N \left(\sum Z(p) \log p / p\right)^2 \sim \frac{1}{4} N (\log N)^2$.

The predicted constant. Under the independent random model—each prime p hits a uniform m with probability $\sim 1/p$, independently across primes—the second moment of $B(m)$ at one integer is $\mathbb{E}[B(m)^2] = (\mathbb{E}[B])^2 + \text{Var}(B)$. By weighted Mertens, $\mathbb{E}[B] = \sum_{\sqrt{m} < p \leq m} (\log p)/p = \frac{1}{2} \log m + O(1)$, and

$$\text{Var}(B) = \sum_{\sqrt{m} < p \leq m} \frac{(\log p)^2}{p} \left(1 - \frac{1}{p}\right) = \frac{3}{8} (\log m)^2 + O(\log m).$$

Therefore

$$\mathbb{E}[B(m)^2] = \left(\frac{1}{4} + \frac{3}{8}\right)(\log m)^2 + O(\log m) = \frac{5}{8}(\log m)^2 + O(\log m), \quad (140)$$

predicting $M_2(N) \sim \frac{5}{8} N (\log N)^2$. The diagonal constant is $3/8$ (matching the observed $\text{Diag}/N \log^2 N \approx 0.37$) and the off-diagonal contributes $1/4$. The empirical ratio $M_2/(N \log^2 N) \approx 0.59$ is consistent with convergence to $5/8 = 0.625$, with lower-order boundary effects.

Centered variance and cross-prime covariance. The centered dispersion $V^\circ = M_2 - M_1^2/N$ decomposes as

$$V^\circ = \underbrace{\sum_p (\log p)^2 |\Omega_p| (1 - |\Omega_p|/N)}_{\leq \text{Diag}} + \underbrace{\sum_{p \neq q} \log p \log q (|\Omega_p \cap \Omega_q| - |\Omega_p| |\Omega_q|/N)}_{E_N^\circ}. \quad (141)$$

The first sum is at most the diagonal S (since $|\Omega_p|/N \geq 0$). The term E_N° is the centered cross-prime covariance: it vanishes exactly when the events $\{m \in \Omega_p\}$ are pairwise uncorrelated. The computed values are:

N	$M_2/N \log^2 N$	$\text{Diag}/N \log^2 N$	V°/S	E_N°
256	0.556	0.359	1.014	+110
512	0.597	0.376	1.010	+215
1024	0.582	0.375	0.981	-152
2048	0.581	0.370	0.992	+7
4096	0.588	0.371	0.995	+41
8192	0.634	0.387	0.995	-188
16384	0.669	0.400	0.995	-922
32768	0.646	0.394	0.988	-14065

The ratio V°/S converges to 1 from below for $N \geq 1024$, with $V^\circ/S < 1$ at every scale tested beyond 512. The negative sign of E_N° for large N reflects genuine *anti-correlation* among the CRT-constrained incidence events: at $N = 1024$, over 96% of cross-prime covariance pairs $C_{p,q}$ are negative. This is consistent with a weak form of negative dependence arising from the partial-period CRT constraint. The magnitude $|E_N^\circ|$ is $O(\sqrt{N} (\log N)^2)$, so $|E_N^\circ|/S \rightarrow 0$ at rate $\sim 1/\sqrt{N}$.

Three-model comparison. To test whether the observed $V^\circ/S \approx 1$ is an Apéry-specific phenomenon or a generic consequence of the cardinality and reflection structure, we compare three models on each dyadic block I_N : (1) the actual Apéry zero-sets; (2) uniformly random reflected subsets of $\mathbb{F}_p^*$ with the same cardinalities $|Z_p|$; (3) random reflected subsets with cardinalities drawn from $2 \cdot \text{Poisson}(1/2)$. Over 20 independent trials of each random model and $N \in \{256, 512, 1024, 2048\}$, all three models yield $V^\circ/S = 1.00 \pm 0.05$, with no statistically significant difference between the Apéry data and either random model. This rules out any detectable cross-prime alignment or repulsion specific to the Apéry sequence.

Remark 136 (Boundary correction). The naïve CRT main term $Z(p)Z(q)N/(pq)$ overestimates the random prediction for primes near $\sqrt{N}$ or N , because the admissible interval $J_{p,q} = J_p \cap J_q$ may be shorter than N . After replacing N by $L_{p,q} = |J_{p,q}|$ in the main term, the boundary-corrected CRT residual is positive and $O(\sqrt{N} (\log N)^2)$ —the apparent negative sign in the uncorrected decomposition is entirely a boundary artifact.

12.5 The no-go model

Define the *adversarial zero-sets*: for one fixed $m_0 \in I_N$, set $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1 - (m_0 \bmod p)\}$. Then $|\mathcal{Z}_p^*| = 2$ (satisfying reflection), yet $m_0 \in \Omega_p^*$ for every prime $\sqrt{N} < p \leq m_0$, giving $B^*(m_0) = \sum_{\sqrt{N} < p \leq m_0} \log p = m_0 - o(m_0)$. In particular:

- The adversarial model satisfies $Z^*(p) = 2$, $Z_{\text{pair}}^*(p) = 1$, and all proved vertical bounds.
- Yet $M_2^*(N) \geq B^*(m_0)^2 \geq c N^2$, which is $\Omega(N^2)$, not $O(N)$.

Conclusion: *No proof using only vertical information (the sizes $Z(p)$, the reflection symmetry, nearest-neighbor exclusion, and gap-polynomial bounds) can establish $W(n) = o(n)$ for all n . The missing theorem must use cross-prime horizontal information: the locations of the zeros $\mathcal{Z}_p$ across different primes p .*

12.6 The Fourier form of the CRT error

Define the modular Fourier transform of $\mathcal{Z}_p$:

$$S_p(a) = \frac{1}{p} \sum_{r \in \mathcal{Z}_p} e\left(\frac{ar}{p}\right).$$

Then $S_p(0) = Z(p)/p$ and, by the palindromic pairing, $e(a(p-1)/(2p)) S_p(a)$ is real-valued for every a . Fourier inversion gives

$$E_N(p, q) = \sum_{\substack{(a,b) \neq (0,0) \\ 0 \leq a < p, 0 \leq b < q}} S_p(a) S_q(b) \sum_{m \in I_N} e\left(\frac{am}{p} + \frac{bm}{q}\right).$$

The inner sum is a standard exponential sum $|K_N(\alpha)| \leq \min(N, 1/\|\alpha\|)$ where $\alpha = a/p + b/q$.

Applying the classical additive large sieve to bound $\sum_{p \neq q} |E_N(p, q)|$ gives $O(N^2/\log N)$ at the top prime scale $p \sim N$ —one full logarithm larger than the threshold $o(N^2/\log^2 N)$ required for $M_2 = O(N(\log N)^A)$. Thus the classical large sieve does not close the gap.

12.7 The dispersion target

Define the centered dispersion

$$V^\circ(P, N) = \sum_{m \in I_N} \left| B_P(m) - \frac{S(P, N)}{N} \right|^2,$$

where $B_P(m) = \sum_{P < p \leq 2P} 1_{\Omega_p}(m)$ and $S(P, N) = \sum_p |\Omega_p|$.

Hypothesis 137 (AP–BDH dispersion). $V^\circ(P, N) \ll N^{o(1)} \cdot S(P, N)$.

Proposition 138. *Hypothesis 137 together with the unconditional bound $Z(p) = O(p^{2/3})$ implies $W(n) = o(n)$ for all n .*

Proof. By hypothesis, for every $m \in I_N$,

$$B_P(m) \leq \frac{S(P, N)}{N} + \sqrt{V^\circ(P, N)} \leq \frac{S(P, N)}{N} + N^{o(1)} \sqrt{S(P, N)}.$$

By the zero-count bound, $S(P, N) \leq (N/P + 1) \cdot \sum_{P < p \leq 2P} Z(p) \ll NP^{2/3}/\log P$. Therefore $B_P(m) \ll N^{o(1)} \cdot (NP^{2/3}/\log P)^{1/2}$. The weighted sum satisfies $W_P(m) \leq B_P(m) \cdot \log(2P)$. Summing over $O(\log N)$ dyadic blocks gives

$$W(n) \ll N^{o(1)} \cdot N^{1/2} \cdot N^{1/3} = N^{5/6+o(1)} = o(N). \quad \square$$

The dispersion hypothesis is a cross-prime equidistribution theorem stating that the Apéry zero-sets $\mathcal{Z}_p$ do not systematically align at one index across different primes. The no-go model of Section 12.5 shows that this hypothesis genuinely requires arithmetic input about the Apéry family, beyond what any vertical bound can provide.

Remark 139 (Covariance sign distribution). At $N = 1024$, more than 93% of the $\binom{113}{2} = 6,328$ cross-prime covariance terms $C_{p,q}$ are negative. This negativity has a clean structural explanation: for primes $p, q > \sqrt{N}$, the Poisson parameter $\lambda_{p,q} = A_p A_q / N < 1$, so $C_{p,q} < 0$ precisely when the joint count $J_{p,q}$ vanishes. In the sparse regime, $J_{p,q} = 0$ for $e^{-\lambda} \approx 96\%$ of pairs, while the rare pairs with $J_{p,q} = 1$ have $C_{p,q} = 1 - \lambda_{p,q} \approx 1$. Over a complete period $(\bmod pq)$, the average of $C_{p,q}$ is exactly zero: many small negatives are balanced by few large positives. Consequently, the high percentage of negative signs does *not* imply negative dependence and cannot be used directly. The aggregate off-diagonal $E^\circ = \sum_{p \neq q} w_p w_q C_{p,q}$ oscillates in sign: positive at $N = 1024, 4096, 8192$ and negative at $N = 2048$ (Table 1). The relevant theorem is the aggregate *sub-Poisson collision* estimate $|E^\circ| = o(S)$, which requires cancellation in the signed sum rather than a universal sign or a factored negative-dependence property. Neither the Dubhashi–Ranjan negative-lattice theorem (designed for balls-into-bins, not partial-period CRT) nor Janson’s inequality (which requires nonnegativity of intersection parameters) applies to this partial-period setting.

Remark 140 (The dispersion route bypasses $\bar{Z}$). Hypothesis 137 does *not* require Hypothesis $\bar{Z}$. The adversarial singleton model $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$ satisfies $\bar{Z}$ in its strongest form ($Z(p) = 2$ for all p), yet violates the dispersion hypothesis. Conversely, the dispersion hypothesis with $Z(p) = O(p^{2/3})$ already gives $W(n) = o(n)$, even without any average bound on $\sum Z(p)$. This makes the dispersion route *strictly stronger* than the $\bar{Z}$ route: it is both sufficient on its own and robust against arbitrary zero-count fluctuations. The three-model comparison confirms that $V^\circ/S \approx 1$ empirically, matching the prediction of the independent random model.

Remark 141 (Martingale decomposition). Order the active primes as $p_1 < \dots < p_K$ and define the Doob martingale $M_k = \mathbb{E}[B \mid X_{p_1}, \dots, X_{p_k}]$. The increments $D_k = M_k - M_{k-1}$ are orthogonal: $\text{Var}(B) = \sum_k \mathbb{E}[D_k^2]$. However, $D_k = (X_{p_k} - \pi_k) \Delta_k$ where the “predictive cascade” $\Delta_k = w_{p_k} + \sum_{j>k} w_{p_j} (\mathbb{P}(X_{p_j}=1 \mid \mathcal{F}_{k-1}, X_{p_k}=1) - \mathbb{P}(X_{p_j}=1 \mid \mathcal{F}_{k-1}, X_{p_k}=0))$ contains all future primes. Under independence $\Delta_k = w_{p_k}$ and the variance equals the diagonal. For the Apéry measure on a partial CRT period, revealing that $m \bmod p_k$ falls in a zero can nearly identify m (when $A_{p_k} \approx 1$), making most subsequent indicators nearly deterministic. Controlling the cascade $\sum \mathbb{E}[\pi_k(1 - \pi_k)\Delta_k^2] \leq (1 + o(1)) \sum w_{p_k}^2 \alpha_{p_k}$ is equivalent to the dispersion inequality itself.

Remark 142 (Concentration and the all- n gap). The L^2 -to-uniform upgrade (Lemma 144) gives $\max |B(m) - \mu| \leq \sqrt{V^\circ}$. If $V^\circ \leq C S$ for some constant C , then $\sqrt{V^\circ} = O(\sqrt{N} (\log N)^{3/2}) = o(N/\log N)$, so the all- n theorem $W(n) = o(n)$ follows without any higher-moment, sub-Gaussian, or negative-association input (Corollary 145). Empirically, the actual concentration is much stronger: $\max B(m)/(N/\log N) \approx 0.07$ at $N = 4096$ (with $\max B \approx 36$ vs. $N/\log N \approx 492$), and the maximum $|B(m) - \mu|/\sigma$ stabilizes near 6–7 standard deviations, consistent with the Poisson prediction for the extremal order statistic.

Remark 143 (Block decomposition of V°). Decompose the prime sum $B(m) = \sum_j B_{P_j}(m)$ by dyadic prime blocks $P_j < p \leq 2P_j$. Then $V^\circ = \sum_j V_{P_j}^\circ + 2 \sum_{j<k} \text{cross}(P_j, P_k)$. At $N = 4096$, the cross-block covariance is 0.03% of the full V° . This is because cross-block pairs (p, q) with $p \in (P, 2P]$, $q \in (Q, 2Q]$, $Q \geq 2P$ satisfy $pq \geq 2P^2 > 2N$, so the CRT approximation is even tighter for cross-block than within-block terms. Consequently, the dispersion hypothesis reduces to $V_P^\circ \ll N^{o(1)} S_P$ for each dyadic block *separately*—a within-scale CRT independence statement.

12.8 The covariance structure and the L^2 -to-uniform upgrade

N	K	V°/S	E°	$ E^\circ /S$	$J=0$	$J=1$
512	67	0.998	+130.6	0.015	95.6%	4.1%
1024	113	0.998	+215.6	0.010	96.6%	2.9%
2048	205	0.994	+96.2	0.002	98.1%	1.8%
4096	380	0.996	+120.0	0.001	98.9%	1.1%
8192	742	1.001	+1344	0.005	99.3%	0.6%

Table 1: Covariance structure of the off-diagonal sum $E^\circ = V^\circ - S_c$, computed over all $\binom{K}{2}$ pairs of active primes $p > \sqrt{N}$. $J=0$ and $J=1$ are the fractions of pairs (p, q) with joint occupancy $J_{p,q} = 0$ or 1. The Cauchy–Schwarz bound on $|E^\circ|$ exceeds the actual value by a factor of 3,000–6,000.

The following deterministic lemma is the bridge from the dispersion hypothesis to the all-index conclusion.

Lemma 144 (L^2 -to-uniform upgrade). *Let $F_N: I_N \rightarrow \mathbb{R}$ with $\mu_N = \frac{1}{N} \sum_{m \in I_N} F_N(m)$ and $\mathcal{V}_N = \sum_{m \in I_N} (F_N(m) - \mu_N)^2$. Then*

$$\max_{m \in I_N} |F_N(m) - \mu_N| \leq \sqrt{\mathcal{V}_N}. \quad (142)$$

Proof. Each squared deviation $(F_N(m_0) - \mu_N)^2$ is a single summand of $\mathcal{V}_N$, hence at most $\mathcal{V}_N$. $\square$

Corollary 145 (Dispersion implies the all-index Apéry bound). *If there exists a constant C such that $V_w^\circ(N) \leq C S_w(N)$ for all sufficiently large dyadic N , where*

$$V_w^\circ(N) = \sum_{m \in I_N} (B(m) - \mu)^2, \quad S_w(N) = \sum_{p > \sqrt{N}} (\log p)^2 A_p,$$

then $W(n) = o(n)$ for all n , and consequently $G_n = e^{o(n)}$ for every positive integer n .

Proof. By Lemma 144 with $F_N = B$,

$$\max_{m \in I_N} |B(m) - \mu| \leq \sqrt{V_w^\circ} \leq \sqrt{C S_w}.$$

Since $S_w \leq (\log 2N)^2 \sum A_p$ and $A_p \ll (N/p + 1)Z(p)$, the unconditional bound $Z(p) \ll p^{2/3}$ and a dyadic decomposition give

$$\sum_{\sqrt{N} < p \leq 2N} A_p \ll N^{5/3+o(1)}.$$

Hence $S_w \ll N^{5/3+o(1)}$ and $\sqrt{V_w^\circ} \ll N^{5/6+o(1)}$. The same incidence estimate gives $\mu \ll N^{2/3+o(1)}$. Therefore

$$W(n) \leq B(n) \leq \mu + \sqrt{V_w^\circ} = N^{5/6+o(1)} = o(N),$$

which gives $W(n) = o(n)$ after choosing the dyadic N for which $n \in (N, 2N]$. $\square$

Table 1 and the extended computation of V°/S through $N = 32,768$ (Section 10) provide strong evidence for the bound $V^\circ \leq C \cdot S$ with $C \leq 1.01$. The ratio $|E^\circ|/S$ decreases from 0.015 at $N = 512$ to 0.001 at $N = 4096$; the brief increase at $N = 8192$ is consistent with fluctuations of $O(1/\sqrt{N})$. No concentration inequality, negative-association theorem, or fourth-moment bound is needed: the L^2 -to- L^∞ trivial inequality together with $V^\circ = O(S)$ completes the proof.

12.9 The exact random CRT model

The independent fixed-margin CRT model gives a rigorous null prediction for the covariance structure. For each active prime p , retain the observed column size A_p and choose the incidence set Ω_p uniformly among all A_p -subsets of I_N , independently across primes. Put $D_p = A_p(1 - A_p/N)$ and $D_w = \sum_p w_p^2 D_p = S_{\text{diag}}$.

Proposition 146 (Random CRT second moment). *Under the independent fixed-margin model, $\mathbf{E} C_{p,q} = 0$ and $\text{Var}(C_{p,q}) = D_p D_q / (N - 1)$. Moreover, $\mathbf{E}(C_{p,q} C_{p,r}) = 0$ for all triples of distinct primes. Consequently,*

$$\mathbf{E}[(E^\circ)^2] = \frac{2}{N-1} \left(D_w^2 - \sum_p w_p^4 D_p^2 \right). \quad (143)$$

If no single prime column carries a positive proportion of D_w , then

$$\frac{\text{sd}(E^\circ)}{S} \sim \sqrt{\frac{2}{N}}, \quad (144)$$

matching the empirical ratio $|E^\circ|/S$ in Table 1 to within the expected fluctuations.

Proof. Given Ω_p , the intersection count $J_{p,q}$ is hypergeometric with parameters (N, A_p, A_q) , giving $\mathbf{E} C_{p,q} = 0$ and $\text{Var}(C_{p,q}) = D_p D_q / (N - 1)$. For triples, conditioning on Ω_p makes $C_{p,q}$ and $C_{p,r}$ conditionally independent with zero mean, so $\mathbf{E}(C_{p,q} C_{p,r}) = 0$. Expanding $(E^\circ)^2$ and using orthogonality gives (143). $\square$

12.10 The operator norm and eigenvector localization

Define the normalized off-diagonal covariance matrix H with entries

$$H_{p,q} = \frac{C_{p,q}}{\sqrt{D_p D_q}} \quad (p \neq q), \quad H_{p,p} = 0.$$

Then $V^\circ = S_{\text{diag}} + v^T H v$ where $v_p = w_p \sqrt{D_p}$ and $|v|^2 = D_w$. Consequently, $V^\circ \leq (1 + \|H\|_{\text{op}}) S_{\text{diag}}$.

N	K	$\ H\ _{\text{op}}$	λ_{max}	λ_{min}	Rayleigh(v)
256	39	0.824	+0.824	−0.706	0.022
512	67	0.987	+0.987	−0.576	0.031
1024	113	1.088	+1.088	−0.707	0.021
2048	205	0.896	+0.896	−0.735	0.004
4096	380	0.904	+0.904	−0.734	0.002
8192	742	1.210	+1.210	−1.000	0.009

Table 2: Spectral data for the normalized covariance matrix H . The operator norm $\|H\|_{\text{op}}$ appears bounded (fluctuating between 0.8 and 1.2, not growing with N). The Rayleigh quotient $v^T H v / |v|^2 = E^\circ / D_w$ for the logarithmic weight vector v is much smaller and non-monotone with mean ≈ 0.01 , explaining why $V^\circ \approx S$. At $N = 8192$, two eigenvalues equal -1 exactly, arising from a three-prime collision cluster with $A_p \in \{1, 2, 3\}$.

The striking gap between $\|H\|_{\text{op}} \approx 1$ and the Rayleigh quotient ≈ 0.01 is explained by *eigenvector localization*. At every computed N , the top eigenvector concentrates on two or three primes with very sparse columns ($A_p \leq 4$): these are large primes that share a common marked integer by

CRT coincidence. At $N = 8192$ ($\lambda_{\max} = 1.21$), the cluster consists of $p = 14851, 7937, 7559$ with $A_p = 1, 2, 3$ respectively. The logarithmic weight vector $v_p = (\log p)\sqrt{D_p}$ assigns negligible mass to these sparse primes (combined contribution $< 0.5\%$ of $|v|^2$ at every N), so it is nearly orthogonal to the localized extreme eigenvectors. At $N = 4096$, only 3% of $|v|^2$ lies in the top 5 eigenvectors; the largest projection of v (10%) falls on an eigenvalue of magnitude 0.14 , deep in the spectral bulk. At $N = 8192$, the projection drops to 0.5% .

This localization has a combinatorial origin: the extreme eigenvalues of H arise from k -fold CRT collisions—integers m marked by $k \geq 3$ distinct large primes. The corresponding eigenvectors span a $(k-1)$ -dimensional space of eigenvalue -1 and one direction of eigenvalue $k-1$. The logarithmic weight vector, being smoothly spread across all K active primes, is structurally immune to these localized collision artifacts.

12.11 Palm decorrelation and the cancellation mechanism

The identity (141) decomposes $V^\circ = S_{\text{diag}} + E^\circ$ where the off-diagonal $E^\circ = v^T H v$. The Rayleigh quotient $R_v = v^T H v / |v|^2 = E^\circ / D_w$ admits a transparent conditional-expectation interpretation. Define the *load excess*

$$\eta_p = \frac{(Hv)_p}{v_p} = \frac{\frac{1}{A_p} \sum_{m \in \Omega_p} (B_{-p}(m) - \mu_{-p})}{w_p (1 - A_p/N)}, \quad (145)$$

where $B_{-p}(m) = B(m) - w_p X_p(m)$ is the total load excluding prime p , and $\mu_{-p} = \mu - w_p A_p / N$. The quantity η_p measures the average excess load from all *other* primes, conditional on $m \in \Omega_p$, normalized by the centered self-weight. Under exact independence, $\eta_p = 0$ for every p . The Rayleigh quotient is the weighted average

$$R_v = \sum_p \frac{v_p^2}{D_w} \eta_p. \quad (146)$$

Thus $V^\circ \approx S$ says: averaged with the natural variance weights, conditioning on one prime hitting does not bias the load from all other primes. This is the arithmetic content of the observed approximate orthogonality.

Remark 147 (Cancellation, not near-null). The small Rayleigh quotient does *not* mean that v is a near-null vector of H . Computation gives $\|Hv\|/|v| \approx 0.31$ at all tested sizes $N = 512, \dots, 4096$, while $R_v = v^T H v / |v|^2$ decreases from 0.031 to 0.002 . The cancellation indicator $\|Hv\|/(|v| |R_v|)$ grows from 12 to 154 : positive and negative spectral contributions to R_v individually exceed 0.13 but nearly cancel. This is the standard trace-zero mechanism: since $\text{tr } H = \sum_j \lambda_j = 0$ and the projections $|\langle u, e_j \rangle|^2$ are approximately uniform ($u = v/|v|$ is delocalized), the identity $R_v = \sum_j \lambda_j (|\langle u, e_j \rangle|^2 - 1/K)$ produces cancellation to order $K^{-1/2}$. The extreme eigenvalues are localized (IPR of top eigenvector ≈ 0.25 , effective dimension ≈ 4), so their projections deviate from $1/K$ but contribute little to the weighted sum.

12.12 Dyadic block compression

Partition the active primes into $L = O(\log N)$ dyadic blocks $\mathcal{B}_1, \dots, \mathcal{B}_L$ according to $\lfloor \log_2 p \rfloor$. For each block, define the restricted weight $s_j = \|v|_{\mathcal{B}_j}\|$ and the unit share $a_j = s_j / |v|$. The *compressed covariance matrix* $\tilde{H} \in \mathbb{R}^{L \times L}$ is

$$\tilde{H}_{i,j} = \frac{(v^{(i)})^T H v^{(j)}}{s_i s_j}, \quad (147)$$

where $v^{(j)}$ is v restricted to $\mathcal{B}_j$. Then the Rayleigh quotient factors exactly as

$$R_v = a^T \tilde{H} a. \quad (148)$$

This reduces the $K \times K$ problem to an $L \times L$ one ($L \leq 8$) in the single direction of interest.

N	K	L	$\ H\ _{\text{op}}$	$\ \tilde{H}\ _{\text{op}}$	R_v	$\ Hv\ / v $
512	67	6	0.987	0.113	0.031	0.386
1024	113	6	1.088	0.111	0.021	0.302
2048	205	7	0.896	0.097	0.004	0.311
4096	380	7	0.904	0.040	0.002	0.313
8192	742	8	1.210	0.052	0.009	0.291

Table 3: Full operator norm $\|H\|_{\text{op}}$ versus compressed operator norm $\|\tilde{H}\|_{\text{op}}$. The full norm is bounded near 1 with a spike to 1.21 at $N = 8192$ (from the first singleton collision); the compressed norm is much smaller, ranging from 0.113 at $N = 512$ to 0.040–0.052 at $N = 4096$ –8192. The column $\|Hv\|/|v|$ shows that v is not a near-null vector of H ($\|Hv\|/|v| \approx 0.3$ at all sizes); the small Rayleigh quotient arises from spectral cancellation.

The compressed norm $\|\tilde{H}\|_{\text{op}}$ controls the Rayleigh quotient via $|R_v| \leq \|\tilde{H}\|_{\text{op}}$, and is both numerically accessible (an $L \times L$ eigenvalue problem) and analytically tractable (each entry is a doubly-weighted average of covariances over a dyadic block pair).

The Marchenko–Pastur null model for K independent isotropic columns in $\mathbb{R}^N$ predicts $\|H\|_{\text{op}} \approx 2\sqrt{K/(N-1)} + K/(N-1)$; with $K \sim N/\log N$, this gives $2/\sqrt{\log N} + 1/\log N$, which matches the observed range 0.8–1.2 to within 20–40% (the excess is consistent with sparse localized spikes from low- A_p columns). For the compressed matrix, the same model predicts $\|\tilde{H}\|_{\text{op}} \sim \sqrt{L/N} \sim \sqrt{\log N/N} \rightarrow 0$, consistent with the observed decrease.

Remark 148 (Row leverage). The maximum *normalized row leverage* $\rho(n) = \sum_{p: n \in \Omega_p} 1/D_p$ is bounded near 1.5 at all tested sizes, dominated by one singleton column ($1/D_p \approx 1$) and one or two doublet columns ($1/D_p \approx 0.5$). No singleton collisions (two singletons at the same m) occur up to $N = 4096$. By the Gram identity $H + I = U^T U$ where U is the $N \times K$ normalized incidence matrix, $\|H\|_{\text{op}} + 1 \leq \max_n \rho(n) \cdot K$ is a trivial upper bound. A tighter bound via $\max_n \rho(n)$ governs the localized spectral spikes; if $\rho(n) \leq C$ uniformly, then $\|H\|_{\text{op}} \leq C - 1$.

12.13 Gram identity and block aggregates

The compression admits an exact Gram-matrix interpretation. Define the *normalized block aggregate*

$$Y_i(m) = \frac{1}{s_i} \sum_{p \in \mathcal{B}_i} (\log p) \left(X_p(m) - \frac{A_p}{N} \right), \quad 1 \leq i \leq L, \quad (149)$$

where each Y_i is a centered function on I_N with $\sum_m Y_i(m) = 0$. Let Y be the $N \times L$ matrix with columns Y_i . Then

$$\boxed{I_L + \tilde{H} = Y^T Y \succeq 0}, \quad (150)$$

or entrywise, $\tilde{H}_{i,j} = \langle Y_i, Y_j \rangle - \delta_{ij}$. This is the key structural identity: $I + \tilde{H}$ is the Gram matrix of L vectors in $\mathbb{R}^N$, so every eigenvalue of $\tilde{H}$ is at least -1 . Only the positive edge requires an arithmetic bound.

The Rayleigh identity (148) now reads

$$R_v = a^T \tilde{H} a = \left\| \sum_i a_i Y_i \right\|_2^2 - 1.$$

Thus $R_v = O(1)$ is equivalent to approximate orthonormality of the L aggregate functions Y_i .

Remark 149 (Low-to-high transfer). Let $P = EE^T$ denote orthogonal projection onto the L -dimensional block-smooth subspace. Then $PHu = E\tilde{H}a$ and

$$\|Hu\|^2 = \|\tilde{H}a\|^2 + \|(I - P)HEa\|^2. \quad (151)$$

At $N = 4096$, the data give $\|Hu\| \approx 0.31$ and $\|PHu\| \leq \|\tilde{H}\|_{\text{op}} \approx 0.04$, so $\|(I - P)Hu\| \geq 0.307$. Almost all of Hu lies in the within-block oscillatory subspace, not the block-smooth subspace. The covariance operator converts a smooth prime-scale profile into oscillations inside the blocks. This is a strictly sharper diagnosis than spectral cancellation alone.

12.14 Exact random CRT prediction

The independent *fixed-margin* CRT model (Section 12.9) yields exact second-moment formulas for the compressed matrix. Each normalized column satisfies the isotropy identity $\mathbf{E}[\phi_p \phi_p^T] = \frac{1}{N-1} P_0$, where $P_0 = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T$. For distinct primes, $\mathbf{E}[H_{p,q}^2] = \frac{1}{N-1}$.

Proposition 150 (Compressed second moments). *In the independent fixed-margin model, $\mathbf{E}[\tilde{H}_{i,j}] = 0$ for all i, j , and*

$$\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \quad (i \neq j), \quad (152)$$

$$\mathbf{E}[\tilde{H}_{i,i}^2] = \frac{2}{N-1} (1 - \kappa_i), \quad \kappa_i = \sum_{p \in \mathcal{B}_i} e_i(p)^4. \quad (153)$$

Consequently, the Frobenius norm satisfies

$$\mathbf{E}\|\tilde{H}\|_F^2 = \frac{L^2 + L - 2 \sum_i \kappa_i}{N-1} \asymp \frac{L^2}{N}. \quad (154)$$

Proof. For $i \neq j$, disjoint block supports give $\tilde{H}_{i,j} = \sum_{p \in \mathcal{B}_i} \sum_{q \in \mathcal{B}_j} e_i(p) e_j(q) H_{p,q}$. Since different unordered prime pairs have uncorrelated entries (including those sharing one prime, by conditioning), only the diagonal terms survive: $\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \sum_p e_i(p)^2 \sum_q e_j(q)^2 = \frac{1}{N-1}$. The diagonal entries involve $\sum_{p < q \in \mathcal{B}_i} e_i(p)^2 e_i(q)^2$, giving the κ_i correction. $\square$

This proves $\|\tilde{H}\|_{\text{op}} \rightarrow 0$ in the random model whenever $L = o(\sqrt{N})$. The GOE refinement predicts $\|\tilde{H}\|_{\text{op}} \approx 2\sqrt{L/N}$, with the factor 2 being the asymptotic edge constant.

Remark 151 (Random surrogate comparison). For each N , we generate 20 random palindromic zero sets of the same sizes $|Z_p|$, build the corresponding $\tilde{H}$, and compare $\|\tilde{H}\|_{\text{op}}$. At $N = 2048$: actual = 0.097, random mean = 0.097, z -score = -0.01 . At $N = 1024$: actual = 0.111, random mean = 0.130, z -score = -0.91 . The actual Apéry matrix is statistically indistinguishable from its random palindromic surrogate.

N	L	$\sqrt{L/N}$	$\ \tilde{H}\ _{\text{op}}$	$\ \tilde{H}\ _{\text{op}}/\sqrt{L/N}$
512	6	0.108	0.113	1.04
1024	6	0.077	0.111	1.45
2048	7	0.058	0.097	1.66
4096	7	0.041	0.040	0.97
8192	8	0.031	0.052	1.67

Table 4: The compressed operator norm tracks the random CRT prediction $\sqrt{L/N}$ with ratio ≈ 1 –1.7.

12.15 AMTD: a concrete sufficient conjecture

The dispersion target (Hypothesis 137) is *not* a consequence of the Elliott–Halberstam conjecture, the generalized Elliott–Halberstam conjecture, or Barban–Davenport–Halberstam for fixed sequences. Those conjectures concern the discrepancy of a *fixed* arithmetic function in residue classes; here the tested residue function $1_{\mathcal{Z}_p}$ varies with the modulus. This is a fundamental structural distinction, not a technical gap.

We therefore propose:

Hypothesis 152 (Apéry Moving-Target Dispersion (AMTD)). *There exists an absolute constant C such that, for every dyadic interval $I_N = (N, 2N]$ and every dyadic prime block $P < p \leq 2P$ with $P > \sqrt{N}$,*

$$V^\circ(P, N) \leq C N^{o(1)} S(P, N).$$

The strongest structural sufficient condition is a *short-arc Fourier orthogonality* conjecture:

$$\int_{\mathbb{R}/\mathbb{Z}} \left| \sum_{\substack{P < p \leq 2P \\ 1 \leq a < p \\ x < a/p \leq x+1/N}} \frac{(\log p) F_p(a)}{p} \right|^2 dx \ll \frac{N^{o(1)}}{N} \sum_{P < p \leq 2P} \sum_{a=1}^{p-1} \left| \frac{(\log p) F_p(a)}{p} \right|^2, \quad (155)$$

where $F_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$. By Gallagher’s lemma, (155) implies the weighted cross-prime dispersion bound of Proposition 160 (Subsection 12.16); a dilation to scale $M = 4N$ is needed because the Fejér weight $\gamma_N(N)$ vanishes at the raw scale, and the resulting majorant is the log-weighted energy Q_P of (161) rather than $S(P, N)$ literally. That weighted bound already suffices for the all-index conclusion of Proposition 138, so no direct comparison with $S(P, N)$ is required (Remark 161).

The classical additive large sieve gives the right-hand side of (155) multiplied by $(1 + P^2/N)$. For $P > \sqrt{N}$, this factor exceeds 2 and grows to $O(N)$ at the top block $P \sim N$. The P^2 barrier is intrinsic to the generic large-sieve proof and cannot be removed by average-spacing refinements alone. A proof of (155) must therefore exploit the special structure of the Apéry Fourier coefficients $F_p(a)$ —their palindromic phase constraint $(e_p(a/2) F_p(a) \in \mathbb{R})$, their connection to the Sym^2 K3 Hasse polynomial, or the arithmetic of the zero sets $\mathcal{Z}_p$ themselves.

Remark 153 (Partial large-sieve result for small primes). For dyadic blocks with $P \leq \sqrt{N}$, the classical additive large sieve gives $V^\circ(P, N) \leq (N + 4P^2) \sum_{P < p \leq 2P} w_p^2 Z(p)/p \leq 5 S(P, N)$. Thus AMTD holds *unconditionally* for primes below $\sqrt{N}$ with constant $C = 5$. The open case is $P > \sqrt{N}$, where the large-sieve factor $1 + 4P^2/N$ exceeds any fixed constant. At the top block $P \sim N$, this factor is $O(N)$, and the generic large sieve provides no useful bound.

Remark 154 (Linnik dispersion route). The most plausible analytic pathway is Linnik’s dispersion method: expand the L^2 -discrepancy, isolate the diagonal, and seek cancellation in the off-diagonal via reciprocity and bilinear trace-function estimates in the framework of Fouvry–Kowalski–Michel [14], extended by Kowalski–Michel–Sawin [15] and Drappeau [12]. This reduces the problem to a *bounded-complexity representation theorem*: either the Fourier coefficient $a \mapsto \hat{Z}_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$ or the phase $m \mapsto e_p(u b_m)$ must belong to a uniformly controlled trace-function family. The obstacle is that $b_m \bmod p$ is computed from a nonautonomous recurrence whose complexity grows with p , and no fixed bounded-conductor ℓ -adic sheaf with trace function $A_p(t)$ uniformly in p is currently known. The closest available framework is Forey–Fresán–Kowalski [17], which studies arithmetic Fourier transforms of a fixed perverse sheaf over a fixed finite field; four mismatches remain (growing complexity, horizontal vs. extension-degree aspect, crystalline vs. ℓ -adic, and one- vs. two-characteristic dispersion). A polylogarithmic loss in (155) already suffices for the competition target.

Remark 155 (Palindromic Fourier factoring). For doublet primes in the top block, the exponential sum over hit positions is

$$S(\theta) = \sum_{p \in \mathcal{D}} [e(\theta m_1(p)) + e(\theta m_2(p))] = 2 \sum_{p \in \mathcal{D}} e\left(\theta \frac{3p-1}{2}\right) \cos(\pi \theta h_p),$$

where $h_p = m_2(p) - m_1(p) = p - 1 - 2r_p$ is the doublet gap. By Parseval, $\int_0^1 |S(\theta)|^2 d\theta = T + F$. The diagonal AMTD requires this integral to be $O(T)$, equivalently that $S(\theta)$ has approximately square-root cancellation on the minor arcs.

The palindromic structure converts each doublet edge into a prime-indexed phase $e(3p\theta/2)$ modulated by the cosine $\cos(\pi\theta h_p)$. The gaps h_p are determined by the first zero r_p of $b_n \bmod p$. A proof of diagonal AMTD via this route would establish cancellation in $S(\theta)$ by exploiting the “pseudorandom” distribution of h_p as p varies.

The compact two-term factorization above holds only on the clipped subfamily where both lifts $m_1 = p + r_p$ and $m_2 = 2p - 1 - r_p$ actually fall in $I_N = (N, 2N]$, i.e. $2p - 1 - 2N \leq r_p \leq 2N - p$ (Lemma 162 and the clipping analysis of Subsection 12.16); outside it, one lift is clipped and the reflected residue contributes a pair centred at $(2p - 1)/2$ rather than $(3p - 1)/2$, and in the block $N/2 < p \leq N$ there are additional quotient-2 and quotient-3 lifts. Computationally only a small fraction of top-block doublets (5 of 36 at the tested scale) have the displayed centre $(3p - 1)/2$, so the honest cancellation problem is for the full clipped hit set, not the unqualified two-term sum.

Remark 156 (Hierarchy of targets). The compression produces a strict hierarchy of increasingly weak—and therefore increasingly tractable—conjectures:

- (A) **Full operator bound:** $\|H\|_{\text{op}} = O(1)$. This implies a Bessel inequality for all prime columns and is far stronger than needed.
- (B) **Compressed Bessel bound:** $\|\tilde{H}\|_{\text{op}} = O(1)$. This asks for approximate orthonormality of only $L = O(\log N)$ block-aggregate functions, and suffices for the all-index GCD estimate.
- (C) **Scalar Rayleigh bound:** $a^T \tilde{H} a = O(1)$. This is the weakest statement that closes the all-index theorem, equivalent to $V^\circ \ll S$. It preserves all useful signs and remains a two-prime problem.

Each implication (A) $\Rightarrow$ (B) $\Rightarrow$ (C) is strict. The random CRT model predicts $\|\tilde{H}\|_{\text{op}} \asymp \sqrt{L/N} \rightarrow 0$ and $|a^T \tilde{H} a| \asymp N^{-1/2}$, both of which are far stronger than (B) and (C).

Remark 157 (Diagonal AMTD via Cauchy–Schwarz). The all-index GCD theorem admits a simpler closing route that avoids cross-block estimates entirely. Define the centered block load $F_i(m) = \sum_{p \in \mathcal{B}_i} (\log p) \left(X_p(m) - \frac{A_p}{N} \right) = s_i Y_i(m)$, so that $V^\circ = \|\sum_i F_i\|_2^2$. If each dyadic block satisfies

$$\|F_i\|_2^2 \leq C s_i^2 \quad (\text{Diagonal AMTD}), \quad (156)$$

equivalently $1 + \tilde{H}_{ii} \leq C$, then by Cauchy–Schwarz,

$$V^\circ \leq \left(\sum_i \|F_i\|_2 \right)^2 \leq L \sum_i \|F_i\|_2^2 \leq CLS.$$

Since $L = O(\log N) = N^{o(1)}$, this gives $V^\circ \leq N^{o(1)} S$, which suffices for $\max_m |B(m) - \mu| \leq \sqrt{V^\circ} = O(\sqrt{N} (\log N)^{3/2}) = o(N/\log N)$. The within-block estimate (156) is a *two-prime* signed dispersion problem—it preserves useful signs, does not require four-prime correlations, and avoids the full operator norm or Frobenius arguments. At the top block $P \sim N$, it reduces to a sparse translated-set collision problem: bounding the additive energy of $\{p + r : r \in \mathcal{Z}_p\}$ for primes p in a dyadic range.

Block-level data at $N = 4096$. The ratio $\|F_i\|_2^2/s_i^2$ for each dyadic block at $N = 4096$:

Block	K_i	$\ F_i\ _2^2/s_i^2$	$\tilde{H}_{ii}$
2^6	2	1.018	+0.018
2^7	13	0.998	−0.002
2^8	17	0.987	−0.013
2^9	30	0.988	−0.012
2^{10}	47	1.015	+0.015
2^{11}	94	1.016	+0.016
2^{12}	177	0.989	−0.011

Maximum ratio: 1.018 (well below any nontrivial constant). The Cauchy loss is $L \cdot \sum_i \|F_i\|_2^2/S = 7.00 = L$, confirming $V^\circ/S = 1.002$ while the Cauchy bound gives $V^\circ \leq 7S$. At $N = 8192$ the maximum block ratio remains 1.029 (block 2^{10}) with $L = 8$ blocks. At $N = 32,768$ the collision count for top-block doublets is $F = 38$ with $T = 925$, giving $F/T = 0.041$ (decreasing from 0.055 at $N = 8192$), confirming the conjecture across all tested sizes.

Remark 158 (Translation variance). For each pair of distinct primes p, q , averaging over all translates $x \bmod pq$ gives

$$\frac{1}{pq} \sum_{x \bmod pq} |T_{p,q}(x; N)|^2 \leq \frac{N Z(p)^2 Z(q)^2}{pq},$$

where $T_{p,q}(x; L) = \sum_{j=1}^L f_p(x+j) f_q(x+j)$. For bounded zero counts $Z(p) = O(1)$, this is already at the random scale for a *typical* translate. However, the GCD problem samples the single translate $x = N$ simultaneously for all prime pairs. Summing individual pair bounds absolutely costs $O(N^{3/2})$, which exceeds $S \asymp N(\log N)^2$. The P^2 large-sieve barrier arises because the Fourier support of f_p has $\sim p$ nonzero frequencies (not $Z(p)$), and palindromic symmetry improves only absolute constants.

12.16 The exact short-arc transform and the remaining bilinear form

Write $e(t) = e^{2\pi it}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For a prime block $P < p \leq 2P$ put

$$c_{p,a} = \frac{(\log p)F_p(a)}{p}, \quad Q_P = \sum_{P < p \leq 2P} \sum_{a=1}^{p-1} |c_{p,a}|^2,$$

and, for an integer $M \geq 2$, define the cyclic short-arc sum

$$A_{P,M}(x) = \sum_{\substack{P < p \leq 2P, \\ 1 \leq a < p \\ a/p \in (x, x+1/M]_{\mathbb{T}}}} c_{p,a}.$$

Endpoints are immaterial. Finally set

$$D_P(k) = \sum_{P < p \leq 2P} \sum_{a=1}^{p-1} c_{p,a} e(-ka/p), \quad k \in \mathbb{Z}.$$

Proposition 159 (Exact triangular-kernel identity). *With $\|t\|$ denoting distance to the nearest integer, one has*

$$\int_{\mathbb{T}} |A_{P,M}(x)|^2 dx = \sum_{p,a} \sum_{q,b} c_{p,a} \overline{c_{q,b}} \left(\frac{1}{M} - \left\| \frac{a}{p} - \frac{b}{q} \right\| \right)_+ \quad (157)$$

$$= \sum_{k \in \mathbb{Z}} \gamma_M(k) |D_P(k)|^2, \quad (158)$$

where

$$\gamma_M(0) = \frac{1}{M^2}, \quad \gamma_M(k) = \left(\frac{\sin(\pi k/M)}{\pi k} \right)^2 \quad (k \neq 0).$$

Moreover Fourier inversion on each $\mathbb{F}_p$ gives the exact identity

$$D_P(k) = \sum_{P < p \leq 2P} (\log p) \left(\mathbf{1}_{\mathbb{Z}_p}(k \bmod p) - \frac{Z(p)}{p} \right). \quad (159)$$

Proof. For a point $\lambda \in \mathbb{T}$, the set of x for which $\lambda \in (x, x+1/M]_{\mathbb{T}}$ is a cyclic interval of length $1/M$. The intersection of the two intervals belonging to λ and μ has measure $(M^{-1} - \|\lambda - \mu\|)_+$, proving (157). The k -th Fourier coefficient of an interval of length $1/M$ has squared modulus $\gamma_M(k)$; Parseval proves (158). Finally,

$$\frac{1}{p} \sum_{a=0}^{p-1} F_p(a) e_p(-ak) = \mathbf{1}_{\mathbb{Z}_p}(k),$$

and removing $a = 0$, for which $F_p(0) = Z(p)$, proves (159). $\square$

The two notions of zero frequency are both visible here. The modular frequency $a = 0$ has been removed and produces the vertical centre $\sum_p (\log p) Z(p)/p$. The integer Fourier mode $k = 0$ remains in (158); since $b_0 = 1$, it equals $D_P(0) = -\sum_p (\log p) Z(p)/p$ and is weighted by M^{-2} .

Thus it has neither been silently discarded nor identified with the empirical mean on I_N . If $\mu_{P,I}$ is that empirical mean, then orthogonal projection gives

$$\sum_{k \in I_N} |B_P^{(w)}(k) - \mu_{P,I}|^2 \leq \sum_{k \in I_N} |D_P(k)|^2,$$

where $B_P^{(w)}(k) = \sum_{P < p \leq 2P} (\log p) \mathbf{1}_{\mathcal{Z}_p}(k \bmod p)$.

There is a small but consequential point in applying this identity. At the scale printed in (155), $\gamma_N(N) = \gamma_N(2N) = 0$, so that estimate at scale N does not by itself control the shell $I_N = (N, 2N]$. Uniformity in the scale repairs the problem without a phase twist.

Proposition 160 (Correct Gallagher reduction by dilation). *Assume that (155), with a factor $M^{o(1)}$, holds for every pair (P, M) satisfying $\sqrt{M} < P \leq M$. Then, uniformly for $\sqrt{N} < P \leq N$,*

$$\sum_{N < k \leq 2N} |D_P(k)|^2 \ll N^{1+o(1)} Q_P. \quad (160)$$

Consequently, together with the unconditional $Z(p) \ll p^{2/3}$, the short-arc target implies $W(n) = o(n)$ for every n and hence $G_n = e^{o(n)}$.

Proof. First suppose $P > 2\sqrt{N}$. Apply (155) with the scale $M = 4N$. This is legitimate because $\sqrt{4N} < P \leq 4N$. For $N < k \leq 2N$,

$$\gamma_{4N}(k) = \left(\frac{\sin(\pi k / (4N))}{\pi k} \right)^2 \geq \frac{1}{8\pi^2 N^2}.$$

Equations (158) and (155) therefore give

$$\sum_{N < k \leq 2N} |D_P(k)|^2 \leq 8\pi^2 N^2 \int_{\mathbb{T}} |A_{P,4N}(x)|^2 dx \ll N^{1+o(1)} Q_P.$$

For $\sqrt{N} < P \leq 2\sqrt{N}$, the ordinary additive large sieve, applied on any interval of N consecutive integers, gives

$$\sum_{N < k \leq 2N} |D_P(k)|^2 \leq (N + 4P^2) Q_P \ll N Q_P.$$

This proves (160).

Finite Fourier Parseval gives

$$Q_P = \sum_{P < p \leq 2P} (\log p)^2 \left(\frac{Z(p)}{p} - \frac{Z(p)^2}{p^2} \right) \leq \sum_{P < p \leq 2P} \frac{(\log p)^2 Z(p)}{p}. \quad (161)$$

Thus $Q_P \ll P^{2/3} \log P$, while

$$\mu_P := \sum_{P < p \leq 2P} \frac{(\log p) Z(p)}{p} \ll P^{2/3}.$$

By (159), the logarithmically weighted bad-prime sum in this block is at most

$$\mu_P + |D_P(k)| \ll P^{2/3} + N^{1/2+o(1)} P^{1/3} \ll N^{5/6+o(1)}$$

for every $k \in I_N$. Summing the $O(\log N)$ dyadic blocks and using the small-prime and companion-block estimates already invoked in Proposition 138 proves the claimed all-index bound. $\square$

Remark 161 (What this does and does not imply). The direct assertion in Subsection 12.15 that (155) “implies AMTD block by block” is not the conclusion of the displayed Fourier calculation. Its right-hand side is (161); it is not $S(P, N) = \sum_p |\Omega_p|$, and its coefficients carry the weights $\log p$. In particular, when $p > N$, a residue in $\mathcal{Z}_p$ need not have a representative in I_N . Proposition 160 is the valid conclusion: a weighted dispersion estimate with the vertical majorant NQ_P , which is already strong enough for the conclusion of Proposition 138. No unproved comparison with $S(P, N)$ is needed.

12.17 Palindromic phases and top-block clipping

Lemma 162 (Exact doublet Fourier phase). *Suppose $\mathcal{Z}_p = \{r, p-1-r\}$, choose $0 \leq r \leq (p-1)/2$, and put $h = p-1-2r$. Then, for every $a \bmod p$,*

$$F_p(a) = 2e_p\left(\frac{a(p-1)}{2}\right) \cos\left(\frac{\pi ah}{p}\right). \quad (162)$$

Here $e_p(a/2)$ in the palindromic reality statement means multiplication by the inverse of 2 in $\mathbb{F}_p$; consequently $e_p(a/2)F_p(a) = 2 \cos(\pi ah/p) \in \mathbb{R}$.

Proof. The two integers r and $p-1-r$ have average $(p-1)/2$ and difference h . Factoring their two exponentials proves (162). Since $(p+1)/2$ represents $2^{-1} \pmod{p}$, multiplying the phase in (162) by $e_p(a(p+1)/2)$ cancels it. $\square$

In ordinary real arguments this real rotation is $e(-a(p-1)/(2p))F_p(a)$, because $(p+1)/2 \equiv -(p-1)/2 \pmod{p}$. The expression with the opposite ordinary sign is not generally real; the distinction is checked on the actual doublet $\mathcal{Z}_{17} = \{3, 13\}$ by `oracleA.verify.py`.

The phase $e(\theta(3p-1)/2)$ in Remark 155 is a phase of two particular *integer lifts*; it is not the ordinary real phase used to make $F_p(a)$ real. At the discrete value $\theta = a/p$ it does coincide modulo 1 with the phase $e_p(a(p-1)/2)$ in (162), since the two centres differ by p . For continuous θ , however, it is the centre of the chosen integer lifts and therefore changes when those lifts change. It also has a restricted validity domain. In general the contribution of a residue $u \bmod p$ to the shell is the finite set

$$H_{N,p}(u) = \{u + \ell p : N < u + \ell p \leq 2N, \ell \in \mathbb{Z}\}. \quad (163)$$

The shell Fourier sum must use $H_{N,p}(r) \cup H_{N,p}(p-1-r)$, including every admissible lift. For example, when $p \in (N, 2N]$, the two displayed lifts

$$m_1 = p + r, \quad m_2 = 2p - 1 - r$$

both belong to I_N precisely when

$$2p - 1 - 2N \leq r \leq 2N - p. \quad (164)$$

Only on this subfamily does

$$e(\theta m_1) + e(\theta m_2) = 2e\left(\theta \frac{3p-1}{2}\right) \cos(\pi \theta h)$$

hold for the contribution of the actual hit set $\Omega_p \cap I_N$. Outside (164), one or both lifts are clipped; when the reflected residue $p-1-r$ itself lies in I_N , the relevant pair has centre $(2p-1)/2$, not $(3p-1)/2$. Parseval still gives

$$\int_{\mathbb{T}} \left| \sum_p \sum_{m \in \Omega_p \cap I_N} e(\theta m) \right|^2 d\theta = T + F,$$

but only when the sum uses the actual, clipped hit positions. Thus the unqualified formula in Remark 155 cannot be the starting point for all top-block doublets. In the computational block $N/2 < p \leq N$, there can additionally be quotient-2 or quotient-3 lifts in (163); these are also absent from the complete two-term sum in that remark.

12.18 The exact cross-prime lemma and a sharp obstruction

Put $K_M(t) = (M^{-1} - \|t\|)_+$. Splitting (157) into equal and unequal primes gives

$$\int_{\mathbb{T}} |A_{P,M}|^2 = \frac{Q_P}{M} + R_=(P, M) + R_{\neq}(P, M),$$

where

$$R_{\neq}(P, M) = \sum_{\substack{P < p, q \leq 2P \\ p \neq q}} \sum_{a=1}^{p-1} \sum_{b=1}^{q-1} K_M\left(\frac{a}{p} - \frac{b}{q}\right) \frac{(\log p)(\log q)}{pq} F_p(a) \overline{F_q(b)}. \quad (165)$$

Since $p \leq 2P \leq 2M$, each a/p has at most two same-prime neighbours in an arc of length $1/M$, and $|R_=(P, M)| \leq 2Q_P/M$ by $2|uv| \leq |u|^2 + |v|^2$. Hence the precise missing analytic lemma is the one-sided scalar estimate

$$\operatorname{Re} R_{\neq}(P, M) \leq CM^{-1+o(1)}Q_P. \quad (\text{SDC})$$

Equivalently, this asks for cancellation among the near-Farey solutions

$$\left\| \frac{a}{p} - \frac{b}{q} \right\| < \frac{1}{M}, \quad \min\{|aq - bp|, pq - |aq - bp|\} < \frac{pq}{M}.$$

This is a scalar Rayleigh bound for the actual Apéry vector; it neither asks for a full operator norm nor discards the signs as Cauchy–Schwarz does. It is the exact bilinear input still missing from the Linnik route of Remark 154.

There is also an exact reciprocal-fraction parametrization. Let $\langle d \rangle_{pq}$ be the representative of $d \bmod pq$ of least absolute value, and let $\bar{q}$ and $\bar{p}$ denote inverses modulo p and q , respectively. For $p \neq q$, putting $k = \langle aq - bp \rangle_{pq}$ is a bijection between the pairs occurring in (165) and

$$0 < |k| < \frac{pq}{M}, \quad (k, pq) = 1, \quad a \equiv k\bar{q} \pmod{p}, \quad b \equiv -k\bar{p} \pmod{q}.$$

Since $F_q(-u) = \overline{F_q(u)}$, one obtains

$$R_{\neq}(P, M) = \frac{1}{M} \mathcal{B}_{P,M}, \quad (166)$$

$$\mathcal{B}_{P,M} = \sum_{\substack{P < p, q \leq 2P \\ p \neq q}} \frac{(\log p)(\log q)}{pq} \sum_{\substack{0 < |k| < pq/M \\ (k, pq) = 1}} \left(1 - \frac{M|k|}{pq}\right) F_p(k\bar{q}) F_q(k\bar{p}). \quad (167)$$

Thus (SDC) is precisely the one-sided moving reciprocal-fraction estimate $\operatorname{Re} \mathcal{B}_{P,M} \leq CM^{o(1)}Q_P$. The numerators $k\bar{q}$, $k\bar{p}$ and both zero sets move jointly with p, q ; palindromy only replaces each doublet transform by a real cosine after a moving phase rotation. It does not turn (167) into a fixed-numerator bilinear form.

12.19 A signed centered near-diagonal

One nontrivial part of the moving reciprocal form can nevertheless be estimated by the scalar trilinear theorem of Bettin and Chandee [7]. We state the result in the smooth dyadic form used below. Fix smooth weights v, W supported in fixed compact subintervals of $(0, \infty)$, with $\text{supp } v \subset (a_0, 2a_0)$ for some fixed $a_0 > 0$, and a fixed smooth function w , and put

$$u_p = \frac{\log p}{p} v(p/N), \quad \kappa_p(r) = r - \frac{p-1}{2}.$$

We first dispose of the exact common-offset diagonal. This sector is smaller than the surrounding near-diagonal because the periodic centered column theorem can be consumed directly as a second moment.

Proposition 163 (Actual common-centered-offset deletion). *Let $|\vartheta_{p,q,k,d}| \leq 1$ be arbitrary, allowing all literal pair and row masks. Fix C_v large enough for the prime support and put*

$$B_{d,=}^{\text{cent}} := \sum_{\substack{p \neq q \\ p, q \asymp N}} u_p u_q \widehat{w}\left(\frac{Nd}{pq}\right) \sum_{|k| \leq C_v N} \mathbf{1}_{\{(p-1)/2+k \pmod p \in \mathcal{Z}_p\}} \mathbf{1}_{\{(q-1)/2+k \pmod q \in \mathcal{Z}_q\}} \vartheta_{p,q,k,d}.$$

Then, uniformly for $|d| \ll N^{41/100}$,

$$|B_{d,=}^{\text{cent}}| \ll_{v,w} N^{3/10+o(1)}. \quad (\text{CD1})$$

Consequently, for $D \leq N^{41/100}$,

$$\sum_{0 < |d| \lesssim D} |B_{d,=}^{\text{cent}}|^2 \ll_{v,w} N^{101/100+o(1)}. \quad (\text{CD2})$$

In particular this diagonal is below the critical $N^{159/100}$ shell by $N^{29/50-o(1)}$.

Proof. Put

$$a_N(k) = \sum_{\substack{p \asymp N \\ (p-1)/2+k \pmod p \in \mathcal{Z}_p}} |u_p|, \quad m_N^{\text{ctr}}(k) = \#\{p \asymp N : (p-1)/2+k \pmod p \in \mathcal{Z}_p\}.$$

All occurring offsets satisfy $|k| \ll_v N$. Taking absolute values only inside the already identified common-offset diagonal gives

$$|B_{d,=}^{\text{cent}}| \ll_{v,w} \sum_{|k| \ll_v N} a_N(k)^2.$$

Since $\max_{p \asymp N} |u_p| \ll_v N^{-1+o(1)}$, Proposition 421, applied on the positive and negative halves with $R \asymp_v N$, yields

$$\sum_{|k| \ll_v N} a_N(k)^2 \leq \left(\max_{p \asymp N} |u_p|\right)^2 \sum_{|k| \ll_v N} m_N^{\text{ctr}}(k)^2 \ll_v N^{3/10+o(1)}.$$

This proves (CD1). There are $O(D)$ frequencies in the dyadic shell, so squaring and summing proves (CD2). $\square$

For actual zeros $r \in \mathcal{Z}_p$, $s \in \mathcal{Z}_q$, let $t_{p,q}(r, s) \in [0, pq)$ be the CRT representative satisfying $t_{p,q}(r, s) \equiv r \pmod{p}$ and $t_{p,q}(r, s) \equiv s \pmod{q}$. Define

$$B_{d,H}^{\text{cent}} := \sum_{\substack{p \neq q \\ p, q \asymp N}} u_p u_q \widehat{w}\left(\frac{Nd}{pq}\right) \sum_{\substack{r \in \mathcal{Z}_p, s \in \mathcal{Z}_q \\ 0 < |\kappa_q(s) - \kappa_p(r)| \leq H}} e\left(\frac{d t_{p,q}(r, s)}{pq}\right),$$

$$\mathcal{S}_{D,H}^{\text{cent}} := N \sum_{d \neq 0} W(|d|/D) B_{d,H}^{\text{cent}}.$$

The conditions $p, q \asymp N$ are already imposed by v .

Proposition 164 (Periodic-column centered near-diagonal). *Uniformly for $1 \leq H \leq N$, $|d| \ll N^{41/100}$, and every additional deletion mask or bounded pair phase,*

$$|B_{d,H}^{\text{cent}}| \ll_{v,w} H N^{3/10+o(1)}. \quad (\text{CD3})$$

Consequently, for $D \leq N^{41/100}$,

$$\sum_{0 < |d| \asymp D} |B_{d,H}^{\text{cent}}|^2 \ll_{v,w} D H^2 N^{3/5+o(1)}. \quad (\text{CD4})$$

In particular, at $D = N^{41/100}$ and for every fixed $\varepsilon > 0$, the range

$$H \leq N^{29/100-\varepsilon}$$

has shell energy

$$\ll_{v,w} N^{159/100-2\varepsilon+o(1)}. \quad (\text{CD5})$$

Proof. For an integer k put

$$a_N(k) = \sum_{\substack{p \asymp N \\ (p-1)/2+k \pmod{p} \in \mathcal{Z}_p}} |u_p|.$$

All centered offsets in the fixed prime box satisfy $|k| \ll_v N$. Grouping a pair of offsets by $g = \kappa_q(s) - \kappa_p(r)$ and taking absolute values gives

$$|B_{d,H}^{\text{cent}}| \ll_{v,w} \sum_{1 \leq |g| \leq H} \sum_{|k| \ll_v N} a_N(k) a_N(k+g).$$

For each fixed g , Cauchy–Schwarz bounds the inner correlation by $\sum_{|k| \ll_v N} a_N(k)^2$, after enlarging the fixed offset interval by $O(H)$. As in Proposition 163, the periodic centered-column second moment at $R \asymp_v N$ gives

$$\sum_{|k| \ll_v N} a_N(k)^2 \ll_v N^{3/10+o(1)}.$$

There are $O(H)$ nonzero differences g , proving (CD3). Squaring and summing the $O(D)$ frequencies proves (CD4); substituting the displayed endpoint values gives (CD5). $\square$

Proposition 165 (Signed centered-offset near-diagonal). *For $1 \leq H \leq N$ and*

$$N^{2/5} \leq D \leq N^{41/100},$$

one has, for every $\varepsilon > 0$,

$$\mathcal{S}_{D,H}^{\text{cent}} \ll_{v,w,W,\varepsilon} H(\log N) N^\varepsilon \left(N^{31/20} D^{17/20} + N^{59/40} D \right). \quad (\text{CN1})$$

Moreover,

$$\left(\sum_{d \asymp D} |B_{d,H}^{\text{cent}}|^2 \right)^{1/2} \ll_{v,w,W,\varepsilon} N^\varepsilon H \log N \left(D^{7/20} N^{11/20} + D^{1/2} N^{19/40} \right). \quad (\text{CN2})$$

Proof. For an odd prime p and an integer k , write

$$r_p(k) = \frac{p-1}{2} + k, \quad m_N^c(k) = \#\{p : v(p/N) \neq 0, r_p(k) \in \mathcal{Z}_p\}.$$

Every actual zero has a unique centered offset. Hence the unconditional first zero moment gives

$$\sum_k m_N^c(k) = \sum_{p:v(p/N) \neq 0} Z(p) \ll_v \frac{N^{8/5}}{(\log N)^{4/5}}. \quad (168)$$

Fix centered offsets k, ℓ and put $g = \ell - k \neq 0$. If $t = t_{p,q}(r_p(k), r_q(\ell))$ and $J \equiv g\bar{p} \pmod{q}$, then

$$t \equiv \frac{pq-1}{2} + k + pJ \pmod{pq}.$$

Consequently the CRT phase has the exact recentering

$$e\left(\frac{dt}{pq}\right) = (-1)^d e\left(\frac{d(k-1/2)}{pq}\right) e_q(dg\bar{p}). \quad (169)$$

Split the signs of d and the prime supports into finitely many dyadic boxes. Apply Theorem 1 and Remark 1 of Bettin–Chandee with

$$a = d, \quad m = p, \quad n = q, \quad \vartheta = g,$$

and coefficients

$$\alpha_p = u_p \mathbf{1}_{r_p(k) \in \mathcal{Z}_p}, \quad \beta_q = u_q \mathbf{1}_{r_q(\ell) \in \mathcal{Z}_q}.$$

The sign $(-1)^d$ is absorbed into the arbitrary d -coefficient. The additional phase is

$$f_{d,k}(p, q) = \frac{d(k-1/2)}{pq}.$$

Since $|k| \ll_v N$, its derivative parameter in the cited remark is $O_v(DN)$. Thus

$$\frac{|g|D + O_v(DN)}{N^2} = o(1)$$

uniformly in the stated range. On each fixed box the multiplier $\widehat{w}(Nd/(pq))$ is separated by Fourier inversion in logarithmic variables; its rapidly decreasing transform costs at most N^ε and only twists the three coefficient sequences by unit-modulus factors.

The coefficient norms satisfy

$$\|\alpha\|_2 \ll_v \frac{\log N}{N} \sqrt{m_N^c(k)}, \quad \|\beta\|_2 \ll_v \frac{\log N}{N} \sqrt{m_N^c(\ell)}, \quad \|\nu\|_2 \ll D^{1/2}.$$

The fixed- (k, ℓ) contribution before the outer factor N is therefore

$$\ll N^\varepsilon (\log N)^2 \sqrt{m_N^c(k) m_N^c(\ell)} \left(D^{17/20} N^{-21/20} + D N^{-9/8} \right).$$

For fixed nonzero g ,

$$\sum_k \sqrt{m_N^c(k) m_N^c(k+g)} \leq \sum_k m_N^c(k).$$

Summing over $0 < |g| \leq H$, inserting (168), and restoring the outer factor N gives the displayed bound with an additional factor $(\log N)^{1/5}$, which is absorbed by replacing the arbitrary N^ε loss with a smaller one. This proves (CN1).

The same calculation is uniform in an arbitrary d -coefficient sequence. Removing its $D^{1/2}$ norm and dualizing in d gives (CN2). $\square$

Corollary 166 (Endpoint centered deletion). *At $D = N^{41/100}$ and $H = N^{1/10}$,*

$$\mathcal{S}_{D,H}^{\text{cent}} \ll N^{3997/2000+\varepsilon}, \quad \sum_{d \asymp D} |B_{d,H}^{\text{cent}}|^2 \ll N^{1587/1000+\varepsilon}.$$

More generally, for every fixed $\delta > 0$, taking $H = N^{203/2000-\delta}$ gives

$$\mathcal{S}_{D,H}^{\text{cent}} \ll N^{2-\delta+\varepsilon}, \quad \sum_{d \asymp D} |B_{d,H}^{\text{cent}}|^2 \ll N^{159/100-2\delta+\varepsilon}.$$

Proof. At $D = N^{41/100}$ the first term of (CN1) has exponent

$$\frac{31}{20} + \frac{17}{20} \frac{41}{100} = \frac{3797}{2000}$$

before the factor H ; the second term is smaller. The first term of (CN2) has norm exponent

$$\frac{11}{20} + \frac{7}{20} \frac{41}{100} + \log_N H,$$

and its square gives the stated exponents. The second dual term is again smaller. $\square$

Together with Proposition 164, this deletes a genuine signed subfamily of the transverse Farey form, but it does not prove (SDC): centered offsets separated beyond $N^{29/100-o(1)}$ still couple the two actual zero sets. The Bettin–Chandee estimate remains useful as a signed first-moment bound, whereas the larger square-energy deletion comes directly from the periodic actual-column second moment.

There is a stronger source-level continuation once the centered gap is kept before the multiplicative-character decomposition. It uses the additive slope $g\bar{p} \pmod{q}$, rather than treating the two zero sets as independent Fourier factors.

For $p \asymp_v N$ put

$$c_p = \frac{p-1}{2}, \quad \mathcal{K}_p = \{r - c_p : r \in \mathcal{Z}_p\}, \quad x_p(k) = \mathbf{1}_{\mathcal{K}_p}(k),$$

with x_p extended by zero to $\mathbb{Z}$, and put

$$\mathcal{E}_N = \sum_{p \asymp_v N} |u_p|^2 Z(p), \quad \mathcal{V}_N = \sum_{p \asymp_v N} |u_p|^2.$$

The weighted first zero moment and the pointwise prime weight give

$$\mathcal{E}_N \ll_v N^{-2/5+o(1)}, \quad \mathcal{V}_N \ll_v N^{-1+o(1)}. \quad (\text{CD6})$$

For a one-signed dyadic interval $I_G = (G, 2G] \cap \mathbb{Z}$, define the actual source band

$$A_{G,d}^{\text{src}} := \sum_{\substack{p \neq q \\ p, q \asymp_v N}} u_p u_q \widehat{w}\left(\frac{Nd}{pq}\right) \sum_{g \in I_G} \sum_k x_p(k) x_q(k+g) e\left(\frac{d t_{p,q}(c_p + k, c_q + k + g)}{pq}\right).$$

The negative band is defined identically. Prime-color restrictions and bounded d -independent source deletions are allowed in this definition.

Proposition 167 (Completed-source additive-slope band). *Let $D = N^{41/100}$ and $2G \leq c_v \sqrt{N}$. Uniformly in the preceding color restrictions and source deletions,*

$$\boxed{\sum_{0 < |d| \asymp D} |A_{G,d}^{\text{src}}|^2 \ll_{v,w} N^{6/5+o(1)} G + N^{3/5+o(1)} G^3.} \quad (\text{CD7})$$

The same estimate holds for the negative band and for an $N^{o(1)}$ -summable separated family of smooth source atoms.

Proof. It is enough to treat one sign. The exact centered CRT identity from the proof of Proposition 165 writes the source phase as

$$(-1)^d e\left(\frac{d(k-1/2)}{pq}\right) e_q(dg\bar{p}). \quad (\text{CD8})$$

After extracting the pure d -factor $(-1)^d$, make a fixed smooth subdivision of the normalized variables $d/D, p/N, q/N, k/N$. The remaining smooth product

$$e\left(\frac{d(k-1/2)}{pq}\right) \widehat{w}\left(\frac{Nd}{pq}\right)$$

has, for every fixed $A > 0$, a tensor expansion into a d -factor, a p -factor, a q -factor, and a bounded label factor whose coefficient ℓ^1 norm is $N^{o(1)}$, with uniform remainder $O_A(N^{-A})$. This is ordinary smooth Fourier separation on a fixed compact box. The crude absolute mass of one source row is at most

$$\left(\sum_p |u_p| Z(p)\right)^2 \ll_v N^{6/5+o(1)},$$

so the remainder contributes $O_A(DN^{12/5-2A+o(1)})$ to the shell square and is negligible after choosing A . We therefore prove the estimate for one separated atom, suppressing its bounded one-variable factors. In particular no d -dependent slow factor is being reclassified as a deletion mask.

Put

$$C_{p,q}(g) = \sum_k m_{p,q,g}(k) x_p(k) x_q(k+g), \quad |m_{p,q,g}(k)| \leq 1,$$

where the multiplier is independent of d . Write

$$I_{p,q}(g) = \sum_k x_p(k) x_q(k+g).$$

Pointwise one has $|C_{p,q}(g)| \leq I_{p,q}(g)$, even when the deletion mask in $m_{p,q,g}$ depends on g . It is therefore enough to estimate the positive unmasked intersections. For fixed q and $g, g' \in I_G$, the Gram matrix of the translated q -vectors has diagonal $Z(q)$ and off-diagonal bounded by

$$\#\{r, r' \in \mathcal{Z}_q : r' - r = g' - g\} \ll |g' - g|.$$

Here $|g' - g| \leq 2G \leq c_v \sqrt{q}$, so the actual same-prime gap polynomial applies. Schur's test gives

$$\sum_{g \in I_G} |C_{p,q}(g)|^2 \ll Z(p)(Z(q) + G^2). \quad (\text{CD9})$$

Group by the nonzero slope $J \equiv g\bar{p} \pmod{q}$ and put

$$\beta_{q,J}^{(G)} = u_q \sum_{\substack{p \neq q, g \in I_G \\ g\bar{p} \equiv J \pmod{q}}} u_p C_{p,q}(g).$$

For fixed q, J, g , the prime p lies in one residue class modulo q , which has $O_v(1)$ representatives in the fixed-ratio shell. Thus an equal-slope fibre contains $O_v(G)$ pairs (p, g) . Cauchy in that fibre and (CD9) yield

$$\begin{aligned} \sum_{q,J} |\beta_{q,J}^{(G)}|^2 &\ll_v N^{o(1)} G \sum_{p,q} |u_p u_q|^2 Z(p)(Z(q) + G^2) \\ &\ll_v N^{o(1)} G \mathcal{E}_N (\mathcal{E}_N + G^2 \mathcal{V}_N). \end{aligned} \quad (\text{CD10})$$

The fractions J/q are reduced. For two distinct such fractions,

$$\left\| \frac{J}{q} - \frac{J'}{q'} \right\|_{\mathbb{T}} = \frac{\min\{|Jq' - J'q|, qq' - |Jq' - J'q|\}}{qq'} \gg_v N^{-2};$$

both numerators in the minimum are positive integers, while for $q = q'$ the spacing is at least $1/q$. Hence the additive large sieve on either d -interval of length $O(D)$ multiplies (CD10) by at most $N^{2+o(1)}$; splitting the two signs of d costs only an absolute constant. Substitution of (CD6) proves (CD7). $\square$

The following consequence records the correct ordering of the new cutoff. For a pair-indexed source row $X = (X_{p,q,d})$, write

$$\mathfrak{G}_{4d}(X, Y) = \sum_{0 < |d| \asymp D} \sum_{\substack{p \neq q, p' \neq q' \\ |\{p,q,p',q'\}|=4}} X_{p,q,d} \overline{Y_{p',q',d}}. \quad (\text{CD11})$$

For $H \geq 0$, let $R_{>H}^{\text{src}}$ be the actual completed source row in which the centered labels satisfy $|\ell - k| > H$, with the literal smooth multiplier and both signs retained.

Corollary 168 (Source-exact 97/300 residual). *Fix*

$$\varepsilon_0 > \frac{1}{100}, \quad \varepsilon_1 > 0, \quad G_0 = N^{29/100-\varepsilon_0}, \quad G_1 = N^{97/300-\varepsilon_1},$$

with $G_0 < G_1$. Then for some fixed $\eta_{\text{gap}} > 0$,

$$\mathfrak{G}_{4d}(R_{>G_0}^{\text{src}}, R_{>G_0}^{\text{src}}) - \mathfrak{G}_{4d}(R_{>G_1}^{\text{src}}, R_{>G_1}^{\text{src}}) \ll_{v,w} N^{159/100-\eta_{\text{gap}}+o(1)}. \quad (\text{CD12})$$

Moreover the full actual source four-distinct shell differs from the $R_{>G_1}^{\text{src}}$ shell by the same type of subcritical error. Consequently the remaining source obstruction may be taken to have

$$\boxed{|\ell - k| > N^{97/300 - \varepsilon_1}} \quad (\text{CD13})$$

in both rows. This is a scalar source-residual statement, not a pointwise support assertion for a later multiplicative-character projection.

Proof. Let $A = R_{>G_0}^{\text{src}} - R_{>G_1}^{\text{src}}$. Summing the dyadic bands in Proposition 167 gives

$$\sum_d |A_d|^2 \ll_{v,w} N^{157/100 - 3\varepsilon_1 + o(1)}, \quad (\text{CD14})$$

since at the upper endpoint the two exponents in (CD7) are $457/300 - \varepsilon_1$ and $157/100 - 3\varepsilon_1$. The bound holds after every prime-color restriction.

The completed two-Parseval endpoint gives energy $N^{161/100 + o(1)}$ for the full source row and all its prime-color restrictions. The range $0 < |\ell - k| \leq G_0$ has energy $N^{159/100 - 2\varepsilon_0 + o(1)}$ by Proposition 164, and the common offset has the still smaller bound (CD2). Because $\varepsilon_0 > 1/100$, these near rows have energy $N^{157/100 - \delta_0 + o(1)}$ for some $\delta_0 > 0$. The triangle identity therefore supplies the same $N^{161/100 + o(1)}$ endpoint for $R_{>G_0}^{\text{src}}$; the identity $R_{>G_1}^{\text{src}} = R_{>G_0}^{\text{src}} - A$ supplies it for $R_{>G_1}^{\text{src}}$. No support monotonicity of a complex row is used.

Finally isolate four distinct primes by assigning each shell prime an independent uniform color in $\{1, 2, 3, 4\}$ and letting $\mathbb{E}_\kappa$ denote normalized expectation. For distinct colors a, b , write $X_\kappa^{a,b}$ for the row obtained by restricting its ordered prime slots (p, q) to $\kappa(p) = a$, $\kappa(q) = b$, and define $Y_\kappa^{a,b}$ similarly. The exact identity

$$\mathfrak{G}_{4d}(X, Y) = \frac{4^4}{4!} \mathbb{E}_\kappa \sum_{\pi \in S_4} \langle X_\kappa^{\pi_1, \pi_2}, Y_\kappa^{\pi_3, \pi_4} \rangle_D \quad (\text{CD15})$$

reduces the difference in (CD12) to a fixed finite sum of inner products containing at least one copy of A . Cauchy–Schwarz, (CD14), and the $N^{161/100 + o(1)}$ endpoint give a saving $3\varepsilon_1/2$. Applying the same argument to the common and G_0 -near rows gives the final assertion, with

$$\eta_{\text{gap}} = \min \left\{ \frac{3}{2} \varepsilon_1, \varepsilon_0 - \frac{1}{100} \right\} > 0.$$

□

Corollary 169 (Fixed-margin mixed-safe source-gap deletion to 13/40). *Retain the source ordering of Corollary 168. On a zero-count layer write*

$$R_\alpha = N \sum_{p \in \mathcal{P}_\alpha} |u_p| = N^{\kappa_\alpha + o(1)}, \quad \rho_\alpha = \min\{1, 11/5 - 2\alpha\} - \kappa_\alpha.$$

Fix positive $\varepsilon_1, \varepsilon_2, \sigma$ such that

$$G_1 = N^{97/300 - \varepsilon_1} < G_2 = N^{13/40 - \varepsilon_2}, \quad 0 < \sigma < 2\varepsilon_2.$$

If a four-layer block lies in the fixed critical simplex

$$\sum_{i=1}^4 (|\alpha_i - 3/5| + \rho_{\alpha_i}) \leq \frac{1}{25} + 2\sigma + o(1), \quad (\text{CD16})$$

then each of its physical row pairs (α, β) satisfies

$$\min(\alpha, \beta) \leq \frac{31}{50} + \sigma - \frac{\rho_\alpha + \rho_\beta}{2} + o(1). \quad (\text{CD17})$$

For every one-signed dyadic source-gap band $G_1 < G \leq G_2$, write $G = N^\gamma$ and coalesce identical (p, g) source labels. The identical-label square and the equal-additive-slope graph satisfy

$$\mathcal{D}_{\alpha, \beta, G} \ll N^{-17/10 + \min(\alpha, \beta) + \gamma + o(1)}, \quad (\text{CD18})$$

$$\mathcal{C}_{\alpha, \beta, G} \ll N^{-27/25 + \sigma - (\rho_\alpha + \rho_\beta)/2 + 2\gamma + o(1)}. \quad (\text{CD19})$$

Consequently the band-row energy is

$$\|B_G\|_{\ell^2(d)}^2 \ll N^{157/100 - (2\varepsilon_2 - \sigma) + o(1)}. \quad (\text{CD20})$$

After dyadic summation and the exact four-color identity, the full actual source four-distinct shell differs from the scalar source shell with

$$\boxed{|\ell - k| > G_2 = N^{13/40 - \varepsilon_2}} \quad (\text{CD21})$$

in both rows by

$$O\left(N^{159/100 - \eta_{13/40} + o(1)}\right), \quad \eta_{13/40} = \min\{\eta_{\text{gap}}, \sigma, \varepsilon_2 - \sigma/2\} > 0. \quad (\text{CD22})$$

This is a fixed-margin scalar completed-source statement. It is neither an assertion at the literal endpoint $13/40$ nor a pointwise support statement after multiplicative-character projection.

Proof. Put

$$a_\alpha(k) = \sum_{p \in \mathcal{P}_\alpha} |u_p| x_p(k).$$

The positive centered-column second moment used in Proposition 164 gives, uniformly after color and layer restriction,

$$\sum_k a_\alpha(k)^2 \ll N^{3/10 + o(1)}, \quad \sum_k a_\alpha(k) a_\alpha(k + g) \ll N^{3/10 + o(1)}. \quad (\text{CD23})$$

For the d -independent bounded source multiplier in (CD9), positivity gives

$$|C_{p,q}(g)|^2 \leq \min\{Z(p), Z(q)\} I_{p,q}(g).$$

Using $|u_p u_q|^2 \leq N^{-2+o(1)} |u_p u_q|$, summing the positive intersection by (CD23), and then summing the $O(G)$ gaps proves (CD18).

In a fixed (q, J) slope fibre, equality of two slopes gives

$$p'g - pg' = mq, \quad |m| \ll G.$$

For fixed (q, p, g, m) , reduction modulo p places p' in one residue class modulo p inside a fixed-ratio interval of length $O(N)$, and then determines g' . Thus the graph on coalesced (p, g) labels has maximum degree $O(GN^{o(1)})$; its $m = 0$ sector is precisely the identical-label diagonal. Schur's test and (CD18) give (CD19), after using (CD17). The latter follows directly from (CD16): if one row

exponent is at most $3/5$ the assertion is automatic, while if both exceed $3/5$, twice their smaller excess plus $\rho_\alpha + \rho_\beta$ is bounded by $1/25 + 2\sigma + o(1)$.

The additive large-sieve consumer contributes $N^{2+o(1)}$ to (CD19). Since $\gamma \leq 13/40 - \varepsilon_2$, this proves (CD20). The complement of (CD16) is $O(N^{159/100-\sigma+o(1)})$ by (PS10m). On the critical simplex, Cauchy–Schwarz against the ordinary $N^{161/100+o(1)}$ completed-source row bounds every mixed term containing a deleted band by $N^{159/100-(\varepsilon_2-\sigma/2)+o(1)}$. Finally use (CD15), sum the $O(\log N)$ bands, and combine with Corollary 168. $\square$

Corollary 170 (Product-gap deletion in the completed-source scalar). *For a one-signed dyadic gap band G , let $\mathbf{A}_G^{\text{src}}$ denote the pair-indexed completed-source row whose colored scalar sums are the rows in Proposition 167. Retain every actual Apéry-zero, smooth-source, orientation, circular-branch and source-deletion condition. If*

$$N^{3/10} \leq G, G', \quad 2G, 2G' \leq c_v \sqrt{N},$$

then

$$\boxed{|\mathfrak{G}_{4d}(\mathbf{A}_G^{\text{src}}, \mathbf{A}_{G'}^{\text{src}})| \ll_{v,w} N^{3/5+o(1)} (GG')^{3/2}.} \quad (\text{CD24})$$

Consequently, for every fixed $\zeta > 0$,

$$GG' \leq N^{33/50-\zeta} \implies |\mathfrak{G}_{4d}(\mathbf{A}_G^{\text{src}}, \mathbf{A}_{G'}^{\text{src}})| \ll_{v,w} N^{159/100-3\zeta/2+o(1)}. \quad (\text{CD25})$$

Thus, after the fixed-margin one-row deletion (CD21), the part of the unresolved completed-source scalar lying in the translation-Gram range may be restricted further to

$$\boxed{GG' > N^{33/50-o(1)}}. \quad (\text{CD26})$$

In particular at least one of the two surviving gap bands exceeds $N^{33/100-o(1)}$. This is a scalar actual-vector deletion; it is not an operator-norm or Hilbert–Schmidt estimate.

Proof. Apply the exact four-color identity (CD15). For every ordered pair of distinct colors, Proposition 167 applies uniformly to the corresponding literal colored row. Since $G \geq N^{3/10}$, its cubic term dominates and gives

$$\|A_G^{a,b}\|_{\ell^2(d)}^2 \ll_{v,w} N^{3/5+o(1)} G^3,$$

and similarly for G' . Cauchy–Schwarz in each colored scalar inner product proves (CD24). Inserting $GG' \leq N^{33/50-\zeta}$ proves (CD25); the $O((\log N)^2)$ dyadic and sign summation is absorbed in $N^{o(1)}$. $\square$

Proposition 171 (Global translated actual-zero Gram). *Let $q \asymp_v N$, let $I \subset \mathbb{Z}$ be an interval with $|I| \ll G$, and, with the zero-extended centered indicator from CD6, put*

$$y_g(k) = m_g(k)x_q(k+g), \quad |m_g(k)| \leq 1.$$

Then, for every $p \asymp_v N$,

$$\boxed{\sum_{g \in I} |\langle x_p, y_g \rangle|^2 \ll Z(p) \min\{Z(q) + G^2, GZ(q), Z(q)^2\}.} \quad (\text{CD27})$$

The estimate is uniform under arbitrary bounded source masks contained in m_g .

Proof. Put

$$a_q(\delta) = \sum_{n \in \mathbb{Z}} x_q(n) x_q(n + \delta).$$

If $0 < |\delta| < q$, every term gives two actual representatives in $[0, q - 1]$ at the nonwrapping integer gap $|\delta|$. The actual gap polynomial therefore gives $a_q(\delta) \ll |\delta|$. If $|\delta| \geq q$, then $a_q(\delta) = 0$, while $a_q(0) = Z(q)$. Moreover

$$\sum_{\delta \in \mathbb{Z}} a_q(\delta) = Z(q)^2$$

exactly, since every ordered pair of actual zeros has one integer difference. Hence every row of the Gram matrix $(\langle y_g, y_{g'} \rangle)_{g, g' \in I}$ is bounded respectively by

$$Z(q) + O(G^2), \quad O(GZ(q)), \quad Z(q)^2.$$

Schur's test and $\|x_p\|_2^2 = Z(p)$ prove (CD27). Notice that zero extension, rather than a circular complementary gap, is what removes the square-root restriction. $\square$

Proposition 172 (Global completed-source translation row). *Let $D = N^{41/100}$ and $G = N^\gamma$ for a fixed $\gamma < 1$. For the literal completed-source band $A_{G,d}^{\text{src}}$, before PS14, uniformly under prime-color restrictions and bounded d -independent source masks,*

$$\sum_{0 < |d| \asymp D} |A_{G,d}^{\text{src}}|^2 \ll N^{o(1)} \min \left\{ N^{6/5}G + N^{3/5}G^3, N^{6/5}G^2, N^{28/15}G, N^{281/100} \right\}. \quad (\text{CD28})$$

Proof. For fixed $\gamma < 1$, one has $2G < q$ for every shell prime and all large N , so $g\bar{p} \pmod{q}$ is nonzero. The equal-slope fibre and additive large-sieve argument of CD10 is therefore unchanged. Using the three rows of Proposition 171 gives, respectively,

$$G\mathcal{E}_N(\mathcal{E}_N + G^2\mathcal{V}_N), \quad G^2\mathcal{E}_N^2, \quad G\mathcal{E}_N\mathcal{E}_N^{(2)}$$

for the squared coefficient columns, where

$$\mathcal{E}_N^{(2)} = \sum_{q \asymp_v N} |u_q|^2 Z(q)^2 \ll N^{2/3+o(1)} \mathcal{E}_N \ll N^{4/15+o(1)}.$$

The additive large sieve costs $N^{2+o(1)}$, proving the first three entries in (CD28). Finally the literal completed-source absolute mass is $\ll N^{6/5+o(1)}$ for each d ; squaring and summing over the $N^{41/100+o(1)}$ values of d gives the last entry. $\square$

Corollary 173 (Macroscopic completed-source block envelope). *For $G = N^\gamma, G' = N^{\gamma'}$ with fixed $3/10 \leq \gamma, \gamma' < 1$, put*

$$e(\gamma) = \begin{cases} 3/5 + 3\gamma, & 3/10 \leq \gamma \leq 3/5, \\ 6/5 + 2\gamma, & 3/5 \leq \gamma \leq 2/3, \\ 28/15 + \gamma, & 2/3 \leq \gamma \leq 283/300, \\ 281/100, & 283/300 \leq \gamma < 1. \end{cases}$$

Then

$$|\mathfrak{G}_{4d}(A_G^{\text{src}}, A_{G'}^{\text{src}})| \ll N^{(e(\gamma)+e(\gamma'))/2+o(1)}. \quad (\text{CD29})$$

This global extension alone does not delete a genuine middle-macroscopic power block: at the smallest unresolved corner $(\gamma, \gamma') = (1/2, 13/40 - o(1))$, its exponent is $147/80 - o(1) > 159/100$.

Proof. The four-color identity CD15 and scalar Cauchy–Schwarz give the displayed estimate. The four branches of e are the lower envelope of the four entries in (CD28); their breakpoints are $3/5$, $2/3$, and $283/300$. $\square$

Proposition 174 (Endpoint-slack completed-source row). *On the positive centered branch put*

$$s = c_p + c_q - g,$$

and on the negative branch put $s = c_p + c_q + g$. Let $B_{S,d}^{\text{end}}$ be the literal completed-source row obtained by restricting, before PS14, to $s \in (S, 2S]$, where $1 \leq S \leq \sqrt{N}$. Then

$$\sum_{0 < |d| \asymp N^{41/100}} |B_{S,d}^{\text{end}}|^2 \ll N^{6/5+o(1)} S + N^{3/5+o(1)} S^3. \quad (\text{CD30})$$

The same estimate holds after prime-color restrictions and bounded d -independent source masks.

Proof. For the positive branch write $i = c_p + k$. The actual source pair is

$$\mathbf{1}_{\mathcal{Z}_p}(i) \mathbf{1}_{\mathcal{Z}_q}(q - 1 + i - s) = \mathbf{1}_{\mathcal{Z}_p}(i) \mathbf{1}_{\mathcal{Z}_q}(s - i)$$

by the Apéry palindrome. Since

$$g\bar{p} \equiv \bar{2} - (s + 1)\bar{p} \pmod{q},$$

the half-phase and the factor obtained from $k - 1/2 = i - p/2$ contribute one further $(-1)^d$, which cancels the leading $(-1)^d$ in CD8. The complete phase becomes

$$e\left(\frac{di}{pq}\right) e_q(-d(s + 1)\bar{p}). \quad (\text{CD31})$$

For the negative branch put $i = c_p - k$ and use palindromy on the p -coordinate. Now

$$g\bar{p} \equiv -\bar{2} + (s + 1)\bar{p} \pmod{q},$$

and the same half-residue cancellation gives the complete phase

$$e\left(-\frac{d(i + 1)}{pq}\right) e_q(d(s + 1)\bar{p}). \quad (\text{CD32})$$

Thus both signs are literal short-translation rows in the endpoint variable s . Their additive slope $(s + 1)\bar{p}$ is nonzero, since $2S + 1 < q$ for all large N . The global translated q -Gram in s obeys CD27 with interval length S , and the CD10 equal-slope fibre has $O_v(S)$ pairs. The same tensor separation and additive large sieve therefore prove (CD30). In particular there is no zero-slope exceptional fibre after the exact half-phase cancellation. $\square$

Corollary 175 (Source-exact endpoint macroscopic deletion). *Let $B_T, B_{T'}$ be completed-source rows, each either an ordinary centered translation band or an endpoint-slack band from Proposition 174, with $T = N^\tau, T' = N^{\tau'} \leq N^{1/2}$. Put*

$$f(\tau) = \begin{cases} 6/5 + \tau, & 0 \leq \tau \leq 3/10, \\ 3/5 + 3\tau, & 3/10 \leq \tau \leq 1/2. \end{cases}$$

Then

$$|\mathfrak{G}_{4d}(B_T, B_{T'})| \ll N^{(f(\tau)+f(\tau'))/2+o(1)}. \quad (\text{CD33})$$

In particular, for fixed $\zeta > 0$, this is $O(N^{159/100-\zeta+o(1)})$ under any one of

$$\tau, \tau' \geq 3/10, \quad \tau + \tau' \leq 33/50 - (2/3)\zeta,$$

$$\tau \leq 3/10 \leq \tau', \quad \tau + 3\tau' \leq 69/50 - 2\zeta,$$

or its symmetric version, or

$$\tau, \tau' \leq 3/10, \quad \tau + \tau' \leq 39/50 - 2\zeta.$$

Hence a row whose original centered gap is macroscopic but whose actual endpoint slack is short is deleted in these nonempty regions. The doubly macroscopic middle, where both the centered gap and endpoint slack exceed $N^{1/2+o(1)}$, remains unresolved.

Proof. The two terms in CD7 and CD30 give the row-energy exponent $f(\tau)$. Apply CD15 and scalar Cauchy–Schwarz. The three displayed regions are exactly $(f(\tau) + f(\tau'))/2 \leq 159/100 - \zeta$. $\square$

The reciprocal presentation of the additive slope does not supply a second Fourier direction. The following quantitative comparison keeps the literal completed source row unchanged except for that presentation.

Lemma 176 (Companion-shadow replacement). *Let $D = N^{41/100}$ and let $G = N^{\gamma+o(1)}$ with fixed $0 \leq \gamma < 1$. Fix one separated completed-source atom, one source orientation, one circular branch, and one four-color row. For each literal occurrence in a one-signed gap band write*

$$g = pJ - qm, \quad 0 \leq J < q, \quad J \equiv g\bar{p} \pmod{q}.$$

After extracting the common factor $(-1)^d$ in CD8, let $A_G^{(q)}(d)$ be the literal row with phase

$$\frac{J}{q} + \frac{k-1/2}{pq},$$

and let $A_G^{(p)}(d)$ be the identical row with J/q replaced by m/p . All actual-zero tests, source masks, weights, slow factors, and branch data are retained. Then

$$\|A_G^{(q)} - A_G^{(p)}\|_{\ell^2(0 < |d| \asymp D)} \ll N^{\gamma-37/200+o(1)}. \quad (\text{CD34})$$

Proof. The exact companion identity is

$$\frac{J}{q} - \frac{m}{p} = \frac{g}{pq}.$$

Consequently the two phases differ pointwise by

$$\left| e\left(d\frac{J}{q}\right) - e\left(d\frac{m}{p}\right) \right| \ll \frac{DG}{N^2}.$$

Put $B_N = \sum_{p \asymp_v N} |u_p| Z(p)$. The canonical first zero moment gives $B_N \ll N^{3/5+o(1)}$, while the total absolute coefficient mass of the literal row is at most $B_N^2 N^{o(1)}$. Thus the pointwise row difference is $O(N^{\gamma-39/100+o(1)})$. Summing its square over $N^{41/100+o(1)}$ signed frequencies proves CD34. $\square$

Corollary 177 (Scalar cost of replacing both companion shadows). *For $G = N^\gamma$ and $G' = N^{\gamma'}$, with $0 \leq \gamma, \gamma' < 1$, put*

$$\tilde{e}(t) = \begin{cases} 6/5 + t, & 0 \leq t \leq 3/10, \\ e(t), & 3/10 \leq t < 1, \end{cases}$$

where e is the row exponent in Corollary 173. Then exact four-color isolation and CD34 give

$$\begin{aligned} \mathfrak{G}_{4d}(A_G^{(q)}, A_{G'}^{(q)}) &= \mathfrak{G}_{4d}(A_G^{(p)}, A_{G'}^{(p)}) + O\left(N^{\Xi_{\text{sh}}(\gamma, \gamma') + o(1)}\right), \\ \Xi_{\text{sh}}(\gamma, \gamma') &= \max\left\{\gamma - \frac{37}{200} + \frac{\tilde{e}(\gamma')}{2}, \gamma' - \frac{37}{200} + \frac{\tilde{e}(\gamma)}{2}, \gamma + \gamma' - \frac{37}{100}\right\}. \end{aligned} \quad (\text{CD35})$$

In particular, if $\gamma, \gamma' \leq 1/2$, the error in CD35 is

$$O\left(N^{273/200 + o(1)}\right),$$

which is below the target scalar exponent $159/100$. Hence the two reciprocal shadows are interchangeable at the final scale throughout the original translation range; treating them as independent Fourier directions cannot yield the missing saving there.

Proof. Write $E_G = A_G^{(q)} - A_G^{(p)}$ and similarly for G' . Expand the difference of the two scalar products. The two mixed terms are bounded by Cauchy–Schwarz, CD34, and CD28, and the error–error term by the product of the two CD34 bounds. Exact four-color isolation costs only an absolute factor. On $0 \leq t \leq 1/2$, the two branches of $\tilde{e}$ meet at $t = 3/10$ and are increasing. The displayed exponent is therefore maximized at $(\gamma, \gamma') = (1/2, 1/2)$, where $\tilde{e}(1/2) = 21/10$ and the maximum equals $273/200$. $\square$

The source ordering is load-bearing. If $m_G(k, \ell) = \mathbf{1}_{\{G < |\ell - k| \leq 2G\}}$, then the multiplicative-character coefficient of the masked pair is proportional to

$$\sum_{k \in \mathcal{K}_p^\times} \sum_{\ell \in \mathcal{K}_q^\times} m_G(k, \ell) \chi(k) \psi(\ell),$$

which is a joint tensor and need not factor. Simultaneous reflection kills opposite parity but can leave an odd–odd sector. Equivalently, coordinate multiplication by a nontrivial band does not commute with a factorwise multiplicative-character projector. Thus (CD13) must be used at the completed source level, or through the scalar Gram identity above; it cannot be inserted pointwise into the later PS14 high–high coefficient.

The centered-column argument remains valid after a fixed rational translation. This rules out the rational-slope version of a coherent column packet while preserving both literal Apéry zero tests.

For a shell prime p put $c_p = (p-1)/2$ and $\mathcal{K}_p = \{r - c_p : r \in \mathcal{Z}_p\}$. If $r = c_p + k$, $s = c_q + \ell$, and

$$A = t_{p,q}(r, s) - \frac{pq - 1}{2},$$

then the following chart is exact.

Lemma 178 (Rational centered Farey chart). *Let $(b, c) = 1$ with $1 \leq c < \min\{p, q\}$, and put*

$$S = cA - bpq.$$

Then

$$S \equiv ck \pmod{p}, \quad S \equiv c\ell \pmod{q}, \quad (\text{RC1})$$

and

$$\frac{t_{p,q}(r,s)}{pq} = \frac{1}{2} + \frac{b}{c} + \frac{S/c - 1/2}{pq} \pmod{1}. \quad (\text{RC2})$$

Consequently, if $S = cn + \rho$ with $0 \leq \rho < c$, both defining primes meet the same rational centered column:

$$n + \rho c^{-1} \pmod{p} \in \mathcal{K}_p, \quad n + \rho c^{-1} \pmod{q} \in \mathcal{K}_q.$$

For a cyclic arc of length $1/M$, a Dirichlet chart with $c \leq M$ has $|n| \ll N^2/M + 1$.

Proof. Modulo p and q , respectively, one has $A \equiv k$ and $A \equiv \ell$, proving (RC1). The phase identity follows by substituting $S = cA - bpq$. For the final claim choose a lift with $|y - 1/2 - b/c| \leq 1/(cM)$. A row in the corresponding arc satisfies $|S| \ll cpq/M + c$, and division by c proves the stated range. $\square$

Fix $c \leq N^C$, with $c < p$ on the shell, and $0 \leq \rho < c$. Define the literal rational-column multiplicity

$$m_{c,\rho}(n) := \#\{p \asymp_v N : c_p + n + \rho c^{-1} \pmod{p} \in \mathcal{Z}_p\}. \quad (\text{RC3})$$

Proposition 179 (Rational centered-column codegree). *Uniformly for $|n| \leq C_0 N^2$ and $1 \leq h \ll N/\log N$,*

$$\#\{p : p \text{ is counted by both } m_{c,\rho}(n) \text{ and } m_{c,\rho}(n+h)\} \ll_{C,C_0,v} h. \quad (\text{RC4})$$

Proof. It is enough first to take $n \geq 0$. Let $r \in [0, p-1]$ represent the first rational-column zero. If $r+h \leq p-1$, the exact gap polynomial gives $p \mid N_h(r)$ for $h \geq 2$, while $h=1$ is excluded by the no-consecutive-zero theorem. Clearing denominators gives

$$p \mid C_{c,\rho,h}(n), \quad C_{c,\rho,h}(X) = (2c)^{3(h-1)} N_h \left(X + \frac{\rho}{c} - \frac{1}{2} \right) \in \mathbb{Z}[X].$$

This integer is nonzero: its real argument is at least $-1/2$, whereas every root of N_h has real part strictly between $-h$ and -1 . The gap-polynomial height bound gives $\log |C_{c,\rho,h}(n)| \ll_{C,C_0,v} h \log N$, so it has $O(h)$ prime divisors in a fixed-ratio N -shell.

If the second zero wraps, write $r = p - h + j$ with $0 \leq j < h$. Multiplication by $2c$ gives

$$p \mid 2cn + 2\rho + c(2h - 1 - 2j).$$

For $n \geq 0$ this is a positive integer of polynomial size, hence has $O(1)$ shell-prime divisors for each j . Summing over j proves the claim. For negative n , palindromy sends $(n, 0)$ to $(-n, 0)$ and (n, ρ) to $(-n-1, c-\rho)$ when $\rho > 0$. $\square$

Corollary 180 (Rational high-column tail and moments). *Let $I \subset [-C_0 N^2, C_0 N^2]$ be an integer interval of length $R \geq N$, and put*

$$M_N = \#\{p : v(p/N) \neq 0\} \ll N/\log N.$$

For $T \gg \sqrt{M_N}$,

$$\#\{n \in I : m_{c,\rho}(n) \geq T\} \ll \frac{RM_N^2}{T^3} + \frac{M_N}{T}. \quad (\text{RC5})$$

Furthermore

$$\sum_{n \in I} m_{c,\rho}(n)^2 \ll R \frac{N^{13/10}}{(\log N)^{7/5}} + \frac{N^2}{(\log N)^2}, \quad (\text{RC6})$$

$$\sum_{n \in I} m_{c,\rho}(n)^4 \ll R \frac{N^3}{(\log N)^3} + \frac{N^4}{(\log N)^4}. \quad (\text{RC7})$$

Proof. First,

$$\sum_{n \in I} m_{c,\rho}(n) \ll R \frac{N^{3/5}}{(\log N)^{4/5}}. \quad (\text{RC8})$$

Indeed each prime contributes at most $Z(p)(R/p + 1)$ incidences. Partition I into blocks of length $H = \lfloor \varepsilon T^2 / M_N \rfloor$. If one block contains s columns of multiplicity at least T , and d_p counts the selected columns hit by p , then

$$\sum_p \binom{d_p}{2} \geq \frac{(sT)^2}{2M_N} - \frac{sT}{2}.$$

Proposition 179 bounds the same quantity by $O(Hs^2)$. Choosing ε sufficiently small gives $s \ll M_N/T$; summing the blocks proves (RC5).

For (RC6), split at $T_0 = N^{7/10}(\log N)^{-3/5}$, use $m^2 \leq T_0 m$ below T_0 , and use layer cake with (RC5) above it. A second layer-cake calculation gives $\sum m^4 \ll RM_N^3 + M_N^4$, which is (RC7). $\square$

The moving residue in a rational chart is not arbitrary once one of the two source primes is fixed. The resulting congruence coupling permits the fixed-column estimates to be summed without paying the full denominator c .

Fix a reduced fraction b/c , where $1 \leq c \leq 4N^{41/100}$, and a shell prime q . For a residue $\rho \pmod{c}$ put

$$\mathcal{P}_{b,c;q,\rho} := \{p \asymp_v N : \rho \equiv -bpq \pmod{c}\}$$

and define the coupled rational-column multiplicity

$$m_{b,c;q,\rho}(n) := \# \{p \in \mathcal{P}_{b,c;q,\rho} : c_p + n + \rho c^{-1} \pmod{p} \in \mathcal{Z}_p\}. \quad (\text{RC9})$$

Here c^{-1} is taken modulo the displayed prime p . Since $p, q > c$, every reachable ρ is a unit modulo c ; when $c = 1$ we use the unique residue.

Proposition 181 (Coupled rational-column aggregate moments). *Let $I \subset [-C_0 N^2, C_0 N^2]$ be an integer interval of length $R \geq N$. Uniformly in the preceding data,*

$$\sum_{\substack{\rho \pmod{c} \\ (\rho,c)=1}} \sum_{n \in I} m_{b,c;q,\rho}(n)^2 \ll_{C_0,v} R \frac{N^{13/10}}{\varphi(c)^{1/2}(\log N)^{7/5}} + \frac{N^2}{\varphi(c)(\log N)^2}, \quad (\text{RC10})$$

$$\sum_{\substack{\rho \pmod{c} \\ (\rho,c)=1}} \sum_{n \in I} m_{b,c;q,\rho}(n)^4 \ll_{C_0,v} R \frac{N^3}{\varphi(c)^2(\log N)^3} + \frac{N^4}{\varphi(c)^3(\log N)^4}. \quad (\text{RC11})$$

Thus the exact relation $\rho \equiv -bpq \pmod{c}$ produces a genuine aggregate saving over summing Corollary 180 with the full shell-prime population in every residue.

Proof. Because $(bq, c) = 1$, fixing ρ restricts p to one reduced residue class modulo c . The Brun–Titchmarsh inequality, together with $c \leq 4N^{41/100}$, gives

$$M_{c,\rho} := \#\mathcal{P}_{b,c;q,\rho} \ll_v \frac{N}{\varphi(c) \log N}. \quad (\text{RC12})$$

The codegree estimate in Proposition 179 remains valid after restricting the prime set. Repeating its block argument with $M_{c,\rho}$ in place of M_N , then summing over the at most $\varphi(c)$ reachable residues, yields, for $T \gg \sqrt{N/(\varphi(c) \log N)}$,

$$\sum_{\substack{\rho \pmod{c} \\ (\rho,c)=1}} \#\{n \in I : m_{b,c;q,\rho}(n) \geq T\} \ll \frac{RN^2}{\varphi(c)(\log N)^2 T^3} + \frac{N}{(\log N)T}. \quad (\text{RC13})$$

For larger T the left side vanishes once $T \gg N/(\varphi(c) \log N)$.

Each shell prime belongs to exactly one reachable residue. Therefore the same first-incidence estimate as in (RC8) now gives the aggregate bound

$$\sum_{\substack{\rho \pmod{c} \\ (\rho,c)=1}} \sum_{n \in I} m_{b,c;q,\rho}(n) \ll R \frac{N^{3/5}}{(\log N)^{4/5}}. \quad (\text{RC14})$$

Split the second moment at

$$T_0 = \frac{N^{7/10}}{\varphi(c)^{1/2}(\log N)^{3/5}}.$$

The range $c \leq 4N^{41/100}$ ensures that T_0 lies between the square-root threshold and the upper population bound in (RC13). Below T_0 use $m^2 \leq T_0 m$ and (RC14); above it use layer cake and (RC13). This proves (RC10).

For the fourth moment, $m^4 \leq T_0^3 m$ below the split, while layer cake above it gives

$$\ll \frac{RN^3}{\varphi(c)^2(\log N)^3} + \frac{N^4}{\varphi(c)^3(\log N)^4}.$$

The low part is absorbed by the first displayed term because $\varphi(c) \leq 4N^{41/100}$. This proves (RC11). $\square$

At $M = 4N^{41/100}$ the natural rational-column interval has length $R_0 = N^{159/100+o(1)}$. The endpoint $T \asymp M_N$ in (RC5) permits $N^{59/100+o(1)}$ nearly full columns. Thus the fixed rational-column theorem rules out denser coherent packets, but reaches the current PS–SDC exponent only at its endpoint and supplies no fixed positive margin. Proposition 181 removes the formerly apparent full factor c when one source prime is fixed. Converting its $\varphi(c)$ -saving through the remaining two-level signed operator is the next unresolved consumer step. This mechanism alone cannot give a uniform power over all Farey charts: the genuine chart $b/c = 0/1$ has the single class $\rho = 0$ and $\varphi(c) = 1$, while the chart $b/c = 1/2$ has $\rho = 1$ for every pair of odd shell primes. New horizontal information is therefore still required already in the integer-centered chart.

12.20 A repeated-characteristic Gram block

The transverse TT^* expansion has an additional favorable stratum when two semiprime denominators share one defining prime. The following estimate keeps the complete Apéry zero rows and

the literal signed d -kernel. It is an estimate for the inner TT^* moment; the outer factor N in the original Poisson sum is restored only by the final Cauchy–Schwarz consumer.

For $D < N^{1-\delta}$, with fixed $\delta > 0$, put

$$G_p(d) = \sum_{\substack{q \asymp N \\ q \neq p}} u_q \widehat{w}\left(\frac{Nd}{pq}\right) F_p(d\bar{q}) F_q(d\bar{p}), \quad Q_1(N) = \sum_{p \asymp N} |u_p|^2 p Z(p),$$

where the inverses are taken in their displayed defining characteristics. The smooth prime supports and the signed dyadic support $|d| \asymp D$ are fixed throughout.

Proposition 182 (Unpruned one-shared-prime Gram bound). *One has*

$$\sum_{p \asymp N} |u_p|^2 \sum_{d \neq 0} |W(|d|/D)|^2 |G_p(d)|^2 \ll_{v,w,W,\delta} Q_1(N)^2. \quad (\text{RG1})$$

Consequently, using the unconditional first zero moment,

$$Q_1(N) \ll_v N^{3/5} \log N, \quad \sum_p |u_p|^2 \sum_d |W|^2 |G_p(d)|^2 \ll_{v,w,W,\delta} N^{6/5} (\log N)^2.$$

Proof. For fixed p, d , Cauchy–Schwarz in q gives

$$|G_p(d)|^2 \leq \left(\sum_q |u_q|^2 |F_q(d\bar{p})|^2 \right) \left(\sum_q \left| \widehat{w}\left(\frac{Nd}{pq}\right) \right|^2 |F_p(d\bar{q})|^2 \right).$$

The second factor is $O_{v,w}(pZ(p))$. Indeed d is nonzero and $|d| < p$, while the prime interval supporting v has length $O_v(p)$; hence $q \mapsto d\bar{q} \pmod{p}$ has bounded multiplicity, and finite Parseval gives

$$\sum_{a \bmod p} |F_p(a)|^2 = pZ(p).$$

For fixed p, q , the map $d \mapsto d\bar{p} \pmod{q}$ likewise has bounded multiplicity on the signed support $|d| \asymp D$, since $D < N^{1-\delta}$ and $q \asymp N$. A second use of Parseval therefore gives

$$\sum_d |W(|d|/D)|^2 |F_q(d\bar{p})|^2 \ll_W qZ(q).$$

After summing first in d and then in p , the preceding inequalities give exactly (RG1). Finally

$$Q_1(N) \ll_v \frac{(\log N)^2}{N} \sum_{p \asymp N} Z(p) \ll_v N^{3/5} \log N.$$

□

The complete rows may be pruned without losing this gain whenever the deleted row family is sparse in one local coordinate. This is a deterministic stability statement; it does not assert that arbitrary deletion of complex summands is monotone.

Proposition 183 (Sparse-row stability of the repeated-prime block). *For each pair $p \neq q$, let $E_{p,q} \subset \mathcal{Z}_p \times \mathcal{Z}_q$ satisfy*

$$\sup_{p,q,r} \#\{s \in \mathcal{Z}_q : (r, s) \in E_{p,q}\} \leq L,$$

and define its exact CRT-row exponential sum by

$$E_{p,q}(d) = \sum_{(r,s) \in E_{p,q}} e\left(\frac{d t_{p,q}(r,s)}{pq}\right).$$

Then

$$\begin{aligned} \sum_p |u_p|^2 \sum_d |W(|d|/D)|^2 \left| \sum_q u_q \widehat{w}\left(\frac{Nd}{pq}\right) E_{p,q}(d) \right|^2 \\ \ll_{v,w,W} DL^2 \sum_{p \asymp N} |u_p|^2 Z(p)^2 \ll_{v,w,W} DL^2 N^{4/15} \log N. \end{aligned} \quad (\text{RG2})$$

In particular, at $D = N^{41/100}$ and $L \leq N^{1/10+o(1)}$, the right-hand side is $N^{263/300+o(1)}$.

Proof. The fibre hypothesis gives $|E_{p,q}(d)| \leq LZ(p)$ pointwise. Since $\sum_{q \asymp N} |u_q| \ll_v 1$ and the signed d -support has $O_W(D)$ integers, the first bound in (RG2) follows directly. The pointwise estimate $Z(p) \ll p^{2/3}$ and the first zero moment imply

$$\sum_{p \asymp N} |u_p|^2 Z(p)^2 \ll_v \frac{(\log N)^2}{N^2} N^{2/3} \sum_{p \asymp N} Z(p) \ll_v N^{4/15} \log N.$$

The endpoint exponent is $41/100 + 1/5 + 4/15 = 263/300$. $\square$

We now make the pair-level passage explicit. Suppose in addition that the deleted row family is bi-sparse, so the fibre bound L holds after interchanging the two defining characteristics. Write

$$R_{p,q}^\sharp(d) = \sum_{(r,s) \in (\mathcal{Z}_p \times \mathcal{Z}_q) \setminus E_{p,q}} e\left(\frac{d t_{p,q}(r,s)}{pq}\right)$$

and let $\mathfrak{D}_{1\text{sh},N,D}^\sharp$ denote the ordered part of its TT^* expansion in which the two semiprime denominators have exactly one common prime, are unequal, and their two integer CRT rows are unequal. The literal signed d -kernel and its two circular branches remain in this definition.

Proposition 184 (Sharp one-shared-prime TT^* bound). *Let $D = N^{41/100}$ and suppose*

$$L \leq N^{203/2000-\delta_c}$$

for some fixed $\delta_c > 0$. Then

$$\text{Re } \mathfrak{D}_{1\text{sh},N,D}^\sharp \ll_{v,w,W,\delta_c} N^{6/5+o(1)}. \quad (\text{RG3})$$

Consequently, for every fixed $\eta < 39/100$,

$$\text{Re } \mathfrak{D}_{1\text{sh},N,N^{41/100}}^\sharp \ll N^{159/100-\eta};$$

in particular one may take $\eta = 1/3$.

Proof. First fix the orientation with denominators pq and pq' , common prime p , and $q \neq q'$. Let $G_p^\sharp$ be the analogue of G_p with $F_p(d\bar{q})F_q(d\bar{p})$ replaced by $R_{p,q}^\sharp(d)$. Exact subtraction of the deleted rows and $|z - z'|^2 \leq 2|z|^2 + 2|z'|^2$ give, by Propositions 182 and 183,

$$\begin{aligned} \sum_p |u_p|^2 \sum_d |W|^2 |G_p^\sharp(d)|^2 &\ll Q_1(N)^2 + DL^2 N^{4/15} \log N \\ &\ll N^{6/5+o(1)}. \end{aligned} \quad (\text{RG4})$$

Indeed the second exponent is at most

$$\frac{41}{100} + \frac{203}{1000} + \frac{4}{15} - 2\delta_c = \frac{2639}{3000} - 2\delta_c < \frac{6}{5}.$$

It remains to impose the two pair-level exclusions, which are not row masks. For fixed p, d , write the surviving foreign-prime contribution as $\sum_q \sum_t c_{p,q,t}(d)$, where a fixed (p, q, t) determines at most one pair of residues. The equal-denominator block is

$$\sum_q \left| \sum_t c_{p,q,t}(d) \right|^2 \geq 0,$$

so removing it decreases the real upper bound. After $q = q'$ has been removed, the same-integer-row cross term satisfies, pointwise in (p, d, t) ,

$$\operatorname{Re} \sum_{q \neq q'} c_{p,q,t}(d) \overline{c_{p,q',t}(d)} \geq - \sum_q |c_{p,q,t}(d)|^2.$$

Since there are at most $Z(p)Z(q)$ surviving rows for a fixed pair (p, q) , the total possible negative part is

$$\ll D \left(\sum_{p \preceq N} |u_p|^2 Z(p) \right)^2 \ll DN^{-4/5} (\log N)^2 = N^{-39/100+o(1)}.$$

Thus deleting the same-row cross term changes the upper bound in (RG4) only by this smaller quantity.

There are four ordered positions for the unique common prime. Once equal semiprime denominators are excluded they are disjoint; bi-sparsity permits the same argument after relabeling. Summing these four cases proves (RG3). No determinant-support majorant is used, so both circular branches remain in the signed kernel. $\square$

The row masks used in the centered Farey continuation satisfy the hypothesis of Proposition 184. Equal and reflected-equal rows have bi-degree one. Ordinary and reflected gap bands of width $H = N^{1/10}$ have bi-degree $O(H)$. For the corrected CRT alias, if

$$j_{p,q}(r, s) \equiv (s - r)\bar{p} \pmod{q}, \quad 0 \leq j_{p,q}(r, s) < q,$$

then $j_{p,q}(r, s) = 0$ determines at most one s for fixed r ; reflection has the same property. Finally the centered-offset band

$$\left| s - \frac{q-1}{2} - \left(r - \frac{p-1}{2} \right) \right| \leq N^{203/2000-\delta_c}$$

has bi-degree $O(N^{203/2000-\delta_c})$. A disjoint priority decomposition of their union therefore has

$$L \ll N^{203/2000-\delta_c}.$$

Hence the fully pruned exactly-one-shared sector is closed unconditionally. The remaining first-band obstruction has four distinct defining characteristics; Proposition 184 does not estimate it.

The first exact completion of that residual uses a prime-shadow coordinate. It exposes an affine correlation in one actual zero set, rather than permitting a second native-characteristic Parseval step.

Lemma 185 (Exact centered prime-shadow decomposition). *Let p, q be distinct odd primes, put $c_p = (p-1)/2$ and $c_q = (q-1)/2$, and write actual zero positions as $r = c_p + k$, $s = c_q + \ell$. If*

$$J \equiv (\ell - k)\bar{p} \pmod{q}$$

and $t \in [0, pq)$ is the CRT representative of (r, s) , then

$$\frac{t}{pq} \equiv \frac{1}{2} + \frac{J}{q} + \frac{k-1/2}{pq} \pmod{1}, \quad (\text{PS1})$$

$$e\left(\frac{dt}{pq}\right) = (-1)^d e_q(dJ) e\left(\frac{d(k-1/2)}{pq}\right). \quad (\text{PS2})$$

Consequently every actual row-deleted fixed- d form can be regrouped exactly as

$$B_d^\# = (-1)^d \sum_{q \asymp N} \sum_{J \bmod q} C_{q,J}^\#(d) e_q(dJ), \quad (\text{PS3})$$

where

$$C_{q,J}^\#(d) = u_q \sum_{\substack{p \asymp N \\ p \neq q}} u_p \widehat{w}\left(\frac{Nd}{pq}\right) \sum_{\substack{k \in \mathcal{K}_p, \ell \in \mathcal{K}_q \\ \ell - k \equiv pJ \pmod{q}}} m_{p,q}^\#(k, \ell; d) e\left(\frac{d(k-1/2)}{pq}\right),$$

$\mathcal{K}_p = \{r - (p-1)/2 : r \in \mathcal{Z}_p\}$ and $m_{p,q}^\#$ is the literal 0-1 row mask. If $0 < |d| < q$, then $C_{q,J}^\#(d)$ may be replaced in (PS3) by its centered version

$$C_{q,J}^{\#, \circ}(d) = C_{q,J}^\#(d) - \frac{1}{q} \sum_{L \bmod q} C_{q,L}^\#(d).$$

Proof. The integer

$$\frac{pq-1}{2} + k + pJ$$

has residues $c_p + k$ modulo p and $c_q + \ell$ modulo q . This proves (PS1); exponentiation proves (PS2), and grouping the literal rows proves (PS3). Finally $\sum_{J \bmod q} e_q(dJ) = 0$ when $0 < |d| < q$, so subtracting the J -mean changes no fixed- d sum. $\square$

The low-centered deletion must remain visible when the prime-shadow coordinate is used on the surviving block. It is harmless for the exact zero-mode cancellation but cannot be dropped from the coefficient array.

Corollary 186 (High-centered prime-shadow decomposition). *Let $H_0 = N^{129/100-\delta_0}$ be the cutoff in Proposition 257. In the notation of Lemma 185, put*

$$a_{p,q}(k, \ell) = \langle k + pJ \rangle_{pq}, \quad J \equiv (\ell - k)\bar{p} \pmod{q},$$

where $\langle \cdot \rangle_{pq}$ is the centered representative, and define

$$m_{p,q}^{\#, \text{hi}}(k, \ell; d) = m_{p,q}^\#(k, \ell; d) \mathbf{1}_{\{|a_{p,q}(k, \ell)| > H_0\}}.$$

If $C_{q,J}^{\#, \text{hi}}(d)$ is obtained from $C_{q,J}^\#(d)$ by using this mask, then the literal surviving fixed- d form is

$$B_d^{\#, \text{hi}} = (-1)^d \sum_{q \asymp N} \sum_{J \bmod q} C_{q,J}^{\#, \text{hi}}(d) e_q(dJ). \quad (\text{PS6})$$

For $0 < |d| < q$, the coefficients may again be replaced by

$$C_{q,J}^{\#, \text{hi}, \circ}(d) = C_{q,J}^{\#, \text{hi}}(d) - \frac{1}{q} \sum_{L \bmod q} C_{q,L}^{\#, \text{hi}}(d). \quad (\text{PS7})$$

Proof. The centered CRT identity in Lemma 185 is exactly $a_{p,q}(k, \ell) = \langle k + pJ \rangle_{pq}$. Restricting that identity by the displayed mask proves (PS6). Since $\sum_{J \bmod q} e_q(dJ) = 0$ for $q \nmid d$, subtracting the J -mean proves (PS7). $\square$

Lemma 187 (Exact affine zero-correlation variance). *Let q be prime and $K \subset \mathbb{F}_q$ have cardinality Z . For $g = (\lambda, \beta) \in \text{Aff}_1(\mathbb{F}_q)$ put*

$$A_K(g) = \#\{x \in K : \lambda x + \beta \in K\}.$$

Then

$$\begin{aligned} \sum_g A_K(g) &= (q-1)Z^2, \\ \sum_g A_K(g)^2 &= (q-1)Z^2 + Z^2(Z-1)^2, \\ \sum_g \left(A_K(g) - \frac{Z^2}{q} \right)^2 &= qZ^2 \left(1 - \frac{Z}{q} \right)^2. \end{aligned} \tag{PS4}$$

Proof. For every ordered pair $(x, y) \in K^2$ and every $\lambda \neq 0$ there is one $\beta = y - \lambda x$. This gives the first identity. On squaring, coincident source points force coincident targets and contribute $(q-1)Z^2$. Two ordered distinct source points and two ordered distinct target points determine a unique affine map, giving $Z^2(Z-1)^2$. Subtracting the square of the mean proves the last identity. $\square$

Lemma 188 (Affine tight frame for a centered zero set). *Let q be prime, let $K \subset \mathbb{F}_q$ have cardinality Z , and put*

$$f = 1_K - \frac{Z}{q}.$$

For $(\lambda, \beta) \in \text{Aff}_1(\mathbb{F}_q)$ set

$$(\rho_{\lambda, \beta} f)(x) = f(\lambda x + \beta).$$

If P_0 denotes orthogonal projection onto the mean-zero subspace of $\ell^2(\mathbb{F}_q)$, then

$$\boxed{\sum_{\lambda \neq 0} \sum_{\beta \bmod q} |\rho_{\lambda, \beta} f| \langle \rho_{\lambda, \beta} f | = Z(q-Z)P_0.} \tag{PS4-TF}$$

Consequently, for every scalar family $(a_{\lambda, \beta})$,

$$\left\| \sum_{\lambda, \beta} a_{\lambda, \beta} \rho_{\lambda, \beta} f \right\|_2^2 \leq Z(q-Z) \sum_{\lambda, \beta} |a_{\lambda, \beta}|^2.$$

Proof. Let S denote the operator on the left of (PS4-TF). If $x = y$, then for each $\lambda \neq 0$ the variable $\lambda x + \beta$ runs through $\mathbb{F}_q$, so

$$S(x, x) = (q-1)\|f\|_2^2.$$

If $x \neq y$, sharp two-transitivity of $\text{Aff}_1(\mathbb{F}_q)$ gives

$$S(x, y) = \sum_{u \neq v} f(u)f(v) = -\|f\|_2^2,$$

because $\sum_u f(u) = 0$. Hence $S = q\|f\|_2^2 P_0$. Since $\|f\|_2^2 = Z - Z^2/q$, this is $Z(q-Z)P_0$. The Bessel inequality is the corresponding synthesis-operator bound. $\square$

Proposition 189 (Mask-robust prime-averaged affine coefficient square). *Put*

$$L_N = \sum_{p \asymp N} |u_p| Z(p), \quad E_N = \sum_{p \asymp N} |u_p|^2 Z(p).$$

For fixed q and $J \bmod q$, let

$$C_q(J) = \sum_{\substack{p \asymp N \\ p \neq q}} \sum_{k \in \mathcal{K}_p} u_p \theta_{p,k;q,J} \mathbf{1}_{\{k+pJ \in \mathcal{K}_q \pmod{q}\}}, \quad |\theta_{p,k;q,J}| \leq 1.$$

Uniformly in these coefficients,

$$\begin{aligned} \sum_{q \asymp N} |u_q|^2 \sum_{J \bmod q} |C_q(J)|^2 \\ \ll_v N^{7/15+o(1)}. \end{aligned} \tag{PS5}$$

This estimate is stable under every literal row mask and bounded slow phase in the coefficient of Lemma 185. If instead $c_{p,k}$ is independent of J and $|c_{p,k}| \leq |u_p|$, then the exactly centered columns satisfy

$$\begin{aligned} \sum_{q \asymp N} |u_q|^2 \sum_{J \bmod q} \left| \sum_{\substack{p \asymp N \\ p \neq q}} \sum_{k \in \mathcal{K}_p} c_{p,k} \left(\mathbf{1}_{\{k+pJ \in \mathcal{K}_q\}} - \frac{Z(q)}{q} \right) \right|^2 \\ \ll_v N^{1/5+o(1)}. \end{aligned} \tag{PS5c}$$

Proof. For fixed q , group source occurrences (p, k) which define the same affine map $J \mapsto k + pJ \pmod{q}$, and let a_g be the sum of their weights $|u_p|$. Fixed-ratio prime support gives bounded representation multiplicity, whence

$$\sum_g a_g \leq L_N, \quad \sum_g a_g^2 \ll_v E_N.$$

Put $f_q = \mathbf{1}_{\mathcal{K}_q} - Z(q)/q$. Orthogonality to constants and Lemma 188 give, for the positive unmasked column

$$\begin{aligned} C_q^+(J) &= \sum_{p,k} |u_p| \mathbf{1}_{\{k+pJ \in \mathcal{K}_q\}}, \\ \sum_{J \bmod q} C_q^+(J)^2 &\ll_v \frac{Z(q)^2}{q} L_N^2 + q Z(q) E_N. \end{aligned}$$

After multiplication by $|u_q|^2$ and summation in q , the centered term is

$$\ll N E_N^2 \ll N^{1/5+o(1)}.$$

For the affine mean, use $Z(q)^2 \ll N^{2/3} Z(q)$ and the canonical first zero moment to obtain

$$\sum_q |u_q|^2 \frac{Z(q)^2}{q} \ll N^{-11/15+o(1)}.$$

The canonical estimates

$$L_N \ll N^{3/5+o(1)}, \quad E_N \ll N^{-2/5+o(1)},$$

therefore make the mean contribution at most $N^{7/15+o(1)}$. Finally $|C_q(J)| \leq C_q^+(J)$ pointwise, so arbitrary masks and phases are covered by the same positive square. For coefficients independent of J , apply the tight-frame Bessel inequality directly after grouping equal affine maps; summing $qZ(q)E_N$ against $|u_q|^2$ proves (PS5c). $\square$

The gain in (PS5) is genuine but does not by itself close the four-distinct block. After the regrouping in (PS3), the reciprocal prime p has entered the coefficient and the remaining oscillation is only $e_q(dJ)$; the ordinary additive large sieve then pays its N^2 Farey-spacing term. Keeping p visible permits the scalar reciprocal estimate, but the presently proved Hilbert-valued amplification charges the joint coefficient matrix by its nuclear norm. A further analytic input must retain reciprocal- p amplification while consuming the special affine square (PS5); no such transfer is asserted here.

There is a lossless way to formulate the remaining one-percent deficit without completing all prime fractions and paying their global N^2 spacing constant. It uses the cyclic short-arc energy of the actual semiprime rows.

Proposition 190 (Lossless semiprime short-arc reduction). *Let $\mathcal{X}$ be any fixed family of reduced semiprime CRT rows in fixed ratio boxes, with every desired row mask already imposed. For $x \in \mathcal{X}$ write*

$$\alpha_x = \frac{t_x}{P_x} \in \mathbb{T}, \quad P_x = pq, \quad a_x = u_p u_q,$$

and put

$$B_d = \sum_{x \in \mathcal{X}} a_x \widehat{w}\left(\frac{Nd}{P_x}\right) e(d\alpha_x).$$

On fixed dyadic d, p, q boxes, separate the smooth multiplier as

$$\widehat{w}\left(\frac{Nd}{pq}\right) = \sum_{\nu} c_{\nu} \phi_{\nu}(d/D) \psi_{\nu}(p/N) \chi_{\nu}(q/N), \quad \sum_{\nu} |c_{\nu}| \ll 1, \quad (\text{SA1})$$

with all three factors bounded by 1. For the resulting separated prime twists define

$$A_M^{\psi, \chi}(y) = \sum_{\substack{x \in \mathcal{X} \\ \alpha_x \in (y, y+1/M]_{\mathbb{T}}}} a_x \psi(p/N) \chi(q/N)$$

and

$$\mathcal{A}_M(\mathcal{X}) = \sup_{\psi, \chi} \int_{\mathbb{T}} |A_M^{\psi, \chi}(y)|^2 dy, \quad (\text{SA2})$$

where the supremum is only over the twist family in (SA1). Uniformly for $D < N$,

$$\boxed{\sum_{d \neq 0} |W(|d|/D)|^2 |B_d|^2 \ll_{v, w, W} D^2 \mathcal{A}_{AD}(\mathcal{X})}. \quad (\text{SA3})$$

Proof. For fixed coefficients b_x , cyclic interval intersection and Parseval give

$$\int_{\mathbb{T}} \left| \sum_{\alpha_x \in (y, y+1/M]} b_x \right|^2 dy = \sum_{n \in \mathbb{Z}} \gamma_M(n) \left| \sum_x b_x e(n\alpha_x) \right|^2,$$

where

$$\gamma_M(0) = M^{-2}, \quad \gamma_M(n) = \left(\frac{\sin(\pi n/M)}{\pi n} \right)^2 \quad (n \neq 0).$$

For $M = 4D$ one has $\gamma_M(n) \gg D^{-2}$ throughout the fixed support $|n| \asymp D$ of W . Apply this estimate to every term of (SA1), use Minkowski's inequality and $\sum_\nu |c_\nu| \ll 1$, and absorb the bounded W -weight. $\square$

The nonzero first band does not see the constant Fourier mode. It is therefore useful to record the following lossless centered form of the same identity.

Lemma 191 (Centered cyclic short-arc identity). *Let (b_x) be a finite complex family with frequencies $\alpha_x \in \mathbb{T}$, and for an integer $M \geq 2$ put*

$$A_M(y) = \sum_{\alpha_x \in (y, y+1/M]_{\mathbb{T}}} b_x, \quad F(n) = \sum_x b_x e(n\alpha_x).$$

Define

$$A_M^\circ(y) = A_M(y) - \frac{F(0)}{M}.$$

Then

$$\int_{\mathbb{T}} |A_M^\circ(y)|^2 dy = \sum_{n \neq 0} \gamma_M(n) |F(n)|^2, \quad \gamma_M(n) = \left(\frac{\sin(\pi n/M)}{\pi n} \right)^2.$$

In particular, for $M = 4D$,

$$\sum_{D \leq |n| \leq 2D} |F(n)|^2 \ll D^2 \int_{\mathbb{T}} |A_{4D}^\circ(y)|^2 dy.$$

Proof. The indicator of a cyclic interval of length $1/M$ has zero Fourier coefficient $1/M$ and squared nonzero Fourier coefficients $\gamma_M(n)$. Parseval gives

$$\int_{\mathbb{T}} |A_M(y)|^2 dy = M^{-2} |F(0)|^2 + \sum_{n \neq 0} \gamma_M(n) |F(n)|^2.$$

Since $\int_{\mathbb{T}} A_M(y) dy = F(0)/M$, subtracting $F(0)/M$ removes exactly the zero-frequency term. Finally $\gamma_{4D}(n) \gg D^{-2}$ for $D \leq |n| \leq 2D$. $\square$

For the separated twists in (SA1), put

$$F_{psi, \chi}(0) = \sum_{x \in \mathcal{X}} a_x \psi(p/N) \chi(q/N),$$

$$A_M^{\psi, \chi, \circ}(y) = A_M^{\psi, \chi}(y) - \frac{F_{\psi, \chi}(0)}{M}, \quad \mathcal{A}_M^\circ(\mathcal{X}) = \sup_{\psi, \chi} \int_{\mathbb{T}} |A_M^{\psi, \chi, \circ}(y)|^2 dy. \quad (\text{SA2c})$$

Applying Lemma 191 to every separated twist gives the lossless nonzero-band reduction

$$\boxed{\sum_{d \neq 0} |W(|d|/D)|^2 |B_d|^2 \ll_{v, w, W} D^2 \mathcal{A}_{4D}^\circ(\mathcal{X})}. \quad (\text{SA3c})$$

In particular, no estimate for the compulsory zero-mode term $M^{-2} |F_{\psi, \chi}(0)|^2$ is needed by the first band.

Lemma 192 (Localized prime-grid restriction). *Let q be prime, $1 \leq M < q/4$, and let $(c_J)_{J \bmod q}$ be arbitrary complex coefficients. Put*

$$L_{q,M}(y) = \sum_{J/q \in (y, y+1/M]_{\mathbb{T}}} c_J.$$

Then

$$\int_{\mathbb{T}} |L_{q,M}(y)|^2 dy \ll \frac{q}{M^2} \sum_{J \bmod q} |c_J|^2.$$

Proof. The quadratic form has the nonnegative circulant kernel

$$K(J - J') = \left(\frac{1}{M} - \left\| \frac{J - J'}{q} \right\| \right)_+.$$

Its operator norm is at most its row sum. Each row has $O(q/M)$ nonzero entries, each at most $1/M$, so that row sum is $O(q/M^2)$. $\square$

Consequently the fully mask-robust affine coefficient square already controls localization inside a single terminal characteristic. Namely, if the coefficients of Proposition 189 satisfy

$$\sum_{q \asymp N} |u_q|^2 \sum_{J \bmod q} |C_q(J)|^2 \ll N^{7/15+o(1)},$$

then Lemma 192, applied one q at a time, gives

$$\sum_{q \asymp N} |u_q|^2 \int_{\mathbb{T}} \left| \sum_{J/q \in (y, y+1/M]} C_q(J) \right|^2 dy \ll \frac{N}{M^2} N^{7/15+o(1)}. \quad (\text{SA4})$$

By (PS1), the literal semiprime fraction differs from $1/2 + J/q$ by $(k - 1/2)/(pq) = O(N^{-1})$. Since $M \ll N$, every literal arc of length $1/M$ is therefore contained in a shadow arc of a fixed larger multiple of that length. Splitting the enlarged arc into $O(1)$ arcs shows that (SA4) also bounds the literal same- q short-arc contribution, with no change of exponent. At $M = 4N^{41/100}$ this is $N^{97/150+o(1)}$, already below the $N^{77/100}$ scale required in (PS-SDC). Thus the remaining short-arc loss is entirely in the interaction between distinct terminal characteristics.

For the literal high-centered family $\mathcal{X}_{\text{hi}}^\sharp$ of Corollary 186, at $D = N^{41/100}$ the source-native estimate

$$\mathcal{A}_{4D}^\circ(\mathcal{X}_{\text{hi}}^\sharp) \ll N^{77/100-\eta+o(1)} \quad (\text{PS-SDC})$$

for any fixed $\eta > 0$ would imply through (SA3c)

$$\sum_d |W(|d|/D)|^2 |B_d^{\sharp, \text{hi}}|^2 \ll N^{159/100-\eta+o(1)}.$$

Estimate (PS-SDC) is not asserted here. Unlike the global prime-fraction large sieve, this formulation charges exactly the $N^{1/50+\eta}$ deficit of the remaining four-distinct block.

The size of that deficit can also be seen before the short-arc form is replaced by a positive collision majorant. The following completed-row estimate is pointwise in the source frequency and therefore uses no orthogonality in d .

Proposition 193 (Completed fixed-frequency two-Parseval endpoint). *Let $0 < |d| < c_v N$, and let*

$$B_d^{\text{comp}} = \sum_{\substack{p, q \asymp_v N \\ p \neq q}} u_p u_q \omega_{p,q}(d) F_p(-d\bar{q}) F_q(-d\bar{p}), \quad |\omega_{p,q}(d)| \leq 1, \quad (\text{PS8})$$

where the inverses are taken in their displayed characteristics. Then

$$|B_d^{\text{comp}}|^2 \ll_v N^{6/5+o(1)}. \quad (\text{PS9})$$

Consequently, for $D = N^{41/100}$,

$$\sum_{0 < |d| \asymp D} |B_d^{\text{comp}}|^2 \ll_v N^{161/100+o(1)}. \quad (\text{PS10})$$

Proof. Weighted Cauchy–Schwarz in the ordered pair (p, q) gives

$$|B_d^{\text{comp}}|^2 \leq \mathcal{E}_d^{(1)} \mathcal{E}_d^{(2)},$$

where

$$\mathcal{E}_d^{(1)} = \sum_p |u_p| \sum_{q \neq p} |u_q| |F_p(d\bar{q})|^2$$

and $\mathcal{E}_d^{(2)}$ is the same expression with p and q interchanged. Fix p . Since $d \not\equiv 0 \pmod{p}$, the map

$$q \mapsto d\bar{q} \pmod{p}$$

has bounded multiplicity on a fixed-ratio shell. The pointwise prime weight bound and finite Parseval therefore give

$$\sum_{q \neq p} |u_q| |F_p(d\bar{q})|^2 \ll_v N^{-1+o(1)} \sum_{a \bmod p} |F_p(a)|^2 = N^{-1+o(1)} p Z(p) \ll_v N^{o(1)} Z(p).$$

It follows from (AP9) that

$$\mathcal{E}_d^{(1)}, \mathcal{E}_d^{(2)} \ll_v N^{o(1)} \sum_p |u_p| Z(p) \ll_v N^{3/5+o(1)},$$

which proves (PS9). The frequency shell has $N^{41/100+o(1)}$ integers, so summation proves (PS10). $\square$

The same endpoint has a direct support-size refinement. It is an L^2 restriction of the completed row and does not use a one-prime fourth moment.

Proposition 194 (Small-zero-count source deletion). *For $\alpha \geq 0$, put*

$$\mathcal{P}_{\leq \alpha} = \{p \asymp_v N : Z(p) \leq N^\alpha\}.$$

Let $B_{d, \leq \alpha}^{\text{comp}}$ be the part of (PS8) in which at least one of p, q belongs to $\mathcal{P}_{\leq \alpha}$, and let $B_{d, > \alpha}^{\text{comp}}$ be the complementary part. At $D = N^{41/100}$,

$$\sum_{0 < |d| \asymp D} |B_{d, \leq \alpha}^{\text{comp}}|^2 \ll_v N^{101/100+\alpha+o(1)}. \quad (\text{PS10a})$$

Consequently, for every fixed $\delta > 0$, taking $\alpha = 29/50 - \delta$ makes this projected self-shell

$$\ll_v N^{159/100 - \delta + o(1)}. \quad (\text{PS10b})$$

To remove the projected row from the full shell one must also charge its mixed inner product with the complementary row. With $\alpha = 14/25 - \delta$ one has

$$\left| \sum_{0 < |d| \asymp D} |B_d^{\text{comp}}|^2 - \sum_{0 < |d| \asymp D} |B_{d, > \alpha}^{\text{comp}}|^2 \right| \ll_v N^{159/100 - \delta/2 + o(1)}. \quad (\text{PS10c})$$

Thus the unresolved completed-row shell may assume

$$Z(p), Z(q) > N^{14/25 - o(1)} \quad (\text{PS10d})$$

for both source characteristics in every row.

Proof. First restrict to the ordered part with $p \in \mathcal{P}_{\leq \alpha}$. The same weighted Cauchy–Schwarz factorization as in Proposition 193 applies. Its restricted first profile is

$$\ll_v N^{o(1)} \sum_{p \in \mathcal{P}_{\leq \alpha}} |u_p| Z(p) \ll_v N^{\alpha + o(1)},$$

because $|u_p| \ll_v N^{-1+o(1)}$ and there are at most $N^{1+o(1)}$ shell primes. The opposite profile is bounded by the full $N^{3/5+o(1)}$ profile. Hence the square of this ordered part is $O_v(N^{\alpha+3/5+o(1)})$ for every fixed d . Decompose the union disjointly into $p \in \mathcal{P}_{\leq \alpha}$ and $p \notin \mathcal{P}_{\leq \alpha}, q \in \mathcal{P}_{\leq \alpha}$. The second part is bounded by the identical argument with p, q interchanged; the triangle inequality costs only an absolute constant. Summing over the $D = N^{41/100}$ frequencies proves (PS10a) and then (PS10b).

Write $B_d^{\text{comp}} = B_{d, > \alpha}^{\text{comp}} + B_{d, \leq \alpha}^{\text{comp}}$. The two-Parseval endpoint applies to the complementary row with its induced pair mask, so

$$\sum_d |B_{d, > \alpha}^{\text{comp}}|^2 \ll_v N^{161/100 + o(1)}.$$

The identity for the difference of the two shell energies, followed by Cauchy–Schwarz, bounds it by

$$\sum_d |B_{d, \leq \alpha}^{\text{comp}}|^2 + 2 \left(\sum_d |B_{d, \leq \alpha}^{\text{comp}}|^2 \right)^{1/2} \left(\sum_d |B_{d, > \alpha}^{\text{comp}}|^2 \right)^{1/2}.$$

For $\alpha = 14/25 - \delta$, the two exponents here are respectively $157/100 - \delta$ and $159/100 - \delta/2$. This proves (PS10c). $\square$

The completed row also has a source-level sparsity which is independent of multiplicative-character decompositions. It comes from the exact integer relation between its two signed reciprocal samples.

Proposition 195 (Small and near-equal signed reciprocal lifts). *For $p \neq q$ in fixed-ratio prime boxes and $0 < |d| < c_v N$, let*

$$a_{p,q}(d) = \langle -d\bar{q} \rangle_p, \quad c_{p,q}(d) = \langle d\bar{p} \rangle_q,$$

where the brackets denote the least signed representative in the displayed characteristic. Let $B_{d,T}^{\text{tube}}$ be any part of B_d^{comp} in (PS8) supported on

$$\min\{|a_{p,q}(d)|, |c_{p,q}(d)|\} \leq T \quad \text{or} \quad |a_{p,q}(d) - c_{p,q}(d)| \leq T, \quad (\text{PS10e})$$

with every additional bounded pair multiplier or deletion mask retained. Uniformly for $T \leq N^{1-\varepsilon_0}$, with fixed $\varepsilon_0 > 0$, one has

$$|B_{d,T}^{\text{tube}}| \ll_{v,\varepsilon_0} TN^{1/5+o(1)}, \quad (\text{PS10f})$$

$$\sum_{0 < |d| \asymp D} |B_{d,T}^{\text{tube}}|^2 \ll_{v,\varepsilon_0} DT^2 N^{2/5+o(1)}. \quad (\text{PS10g})$$

Consequently, at $D = N^{41/100}$, every fixed $\varepsilon > 0$ permits the complete deletion

$$T \leq N^{19/50-\varepsilon}. \quad (\text{PS10h})$$

More precisely, the tube self-energy is then $O_v(N^{157/100-2\varepsilon+o(1)})$, and its mixed contribution against any completed-row remainder satisfying (PS10) is $O_v(N^{159/100-\varepsilon+o(1)})$.

Proof. The definitions give the congruence

$$a_{p,q}(d)q - c_{p,q}(d)p \equiv -d \pmod{pq}. \quad (\text{PS10i})$$

On either set in (PS10e), the least-signed bounds and $T = o(N)$ give

$$|a_{p,q}(d)q - c_{p,q}(d)p| \leq \frac{pq}{2} + O_v(NT) < \frac{3pq}{4}$$

for all sufficiently large N . Since $|d| = o(pq)$, the only integer in this interval congruent to $-d$ modulo pq is $-d$ itself. Hence

$$\boxed{a_{p,q}(d)q - c_{p,q}(d)p = -d.} \quad (\text{PS10j})$$

For completeness, the first displayed bound is uniform on all three parts of the tube. If $|a_{p,q}(d)| \leq T$, use $|c_{p,q}(d)| \leq q/2$; if $|c_{p,q}(d)| \leq T$, use $|a_{p,q}(d)| \leq p/2$. On the near-equal part write $t = a_{p,q}(d) - c_{p,q}(d)$. Then

$$a_{p,q}(d)q - c_{p,q}(d)p = a_{p,q}(d)(q - p) + tp.$$

When $q \geq p$, use $|a_{p,q}(d)| \leq p/2$; when $p > q$, use $|a_{p,q}(d)| \leq q/2 + T$. In either ordering the right side is at most $pq/2 + O_v(NT)$ in absolute value. Thus the exact identity above is a tube-local no-wrap statement; no global no-wrap assertion for arbitrary signed reciprocal samples is being used.

For the near-equal tube put $t = a_{p,q}(d) - c_{p,q}(d)$ and $r = q - p$. Equation (PS10j) becomes

$$a_{p,q}(d)r + tp = -d.$$

For fixed p and t , the two integers $a_{p,q}(d)$ and r form a divisor pair of the nonzero integer $-d - tp$. The divisor bound therefore gives $N^{o(1)}$ possible q for each t . The same argument after interchanging p, q bounds the opposite degree. Thus the prime-pair graph with $|a_{p,q}(d) - c_{p,q}(d)| \leq T$ has maximum degree $TN^{o(1)}$.

If $|c_{p,q}(d)| \leq T$, then (PS10j) shows that q divides the nonzero integer $c_{p,q}(d)p - d$. For fixed p and c this again gives $N^{o(1)}$ possibilities. If $|a_{p,q}(d)| \leq T$, the congruence $p \mid a_{p,q}(d)q + d$ gives only

$O_v(1)$ shell representatives q for fixed p, a ; the transposed statements bound the right degrees. The union in (PS10e) consequently has maximum degree $TN^{o(1)}$.

Put $\beta_p = |u_p|Z(p)$. The two-level masses and the pointwise zero bound give the first two estimates below, while Proposition 384 gives the third:

$$\sum_p \beta_p \ll_v N^{3/5+o(1)}, \quad \max_p \beta_p \ll_v N^{-1/3+o(1)},$$

Indeed, since $\max_{p \asymp_v N} |u_p| \ll_v N^{-1+o(1)}$,

$$\sum_p \beta_p^2 \leq N^{-2+o(1)} \sum_p (Z(p)(Z(p) - 1) + Z(p)) \ll_v N^{1/5+o(1)}.$$

Every edge of the completed form, with all of its masks and twists, has absolute value at most $\beta_p \beta_q$. Schur's test on the preceding prime-pair graph proves (PS10f); squaring and summing the $O(D)$ frequencies proves (PS10g). Finally

$$\frac{41}{100} + 2 \left(\frac{19}{50} - \varepsilon \right) + \frac{2}{5} = \frac{157}{100} - 2\varepsilon.$$

The mixed estimate follows from Cauchy–Schwarz against (PS10). $\square$

The second factorial zero moment gives a complementary reduction: a four-characteristic block can remain at the two-Parseval endpoint only if both its zero counts and its weighted population of primes are close to their simultaneous extremal scales. The all-four-distinct restriction in the next statement is imposed by the finite equality-pattern decomposition before Cauchy–Schwarz; it is not erased from a complex Gram sum.

Proposition 196 (Weighted zero-count critical core). *For a dyadic exponent α put*

$$\mathcal{P}_\alpha = \{p \asymp_v N : N^\alpha < Z(p) \leq 2N^\alpha\}, \quad B_\alpha = \sum_{p \in \mathcal{P}_\alpha} |u_p|Z(p),$$

and

$$R_\alpha = N \sum_{p \in \mathcal{P}_\alpha} |u_p|.$$

On a nonempty dyadic sublayer write

$$R_\alpha = N^{\kappa_\alpha + o(1)}, \quad \rho_\alpha = \min\{1, 11/5 - 2\alpha\} - \kappa_\alpha \geq -o(1). \quad (\text{PS10k})$$

Then

$$B_\alpha \ll_v N^{3/5 - |\alpha - 3/5| - \rho_\alpha + o(1)}. \quad (\text{PS10l})$$

After the repeated-prime equality patterns have been separated, let $\mathfrak{G}(\alpha_1, \alpha_2; \alpha_3, \alpha_4)$ denote one literal all-four-distinct high-centered Gram block whose four defining prime occurrences lie in the displayed zero-count layers, with every inherited row and pair mask retained. Then

$$|\mathfrak{G}(\alpha_1, \alpha_2; \alpha_3, \alpha_4)| \ll_v N^{161/100 - \frac{1}{2} \sum_{i=1}^4 (|\alpha_i - 3/5| + \rho_{\alpha_i}) + o(1)}. \quad (\text{PS10m})$$

Consequently, for a target $N^{159/100 - \eta}$, only the double critical simplex

$$\sum_{i=1}^4 (|\alpha_i - 3/5| + \rho_{\alpha_i}) \leq \frac{1}{25} + 2\eta + o(1) \quad (\text{PS10n})$$

can remain. At the endpoint level $\eta \downarrow 0$, every such layer satisfies

$$\kappa_{\alpha_i} \geq \frac{23}{25} - o(1). \quad (\text{PS10o})$$

There is also an unconditional horizontal support consequence. Put

$$A_\alpha(k) = \sum_{\substack{p \in \mathcal{P}_\alpha \\ (p-1)/2+k \pmod p \in \mathcal{Z}_p}} |u_p|.$$

If $S_\alpha = |\{k : A_\alpha(k) > 0\}|$, then every layer in the endpoint simplex obeys

$$S_\alpha \gg_v N^{9/10-2(|\alpha-3/5|+\rho_\alpha)-o(1)} \gg N^{41/50-o(1)}. \quad (\text{PS10p})$$

This support lower bound is structural only: no operator contraction is claimed from it.

Proof. Proposition 384 and the trivial number of shell primes give

$$\#\mathcal{P}_\alpha \ll_v N^{\min\{1, 11/5-2\alpha\}+o(1)}.$$

Since $|u_p| \ll_v N^{-1+o(1)}$, this proves (PS10k). On the dyadic layer,

$$B_\alpha = N^{\alpha-1+\kappa_\alpha+o(1)},$$

and the identity

$$\alpha - 1 + \min\{1, 11/5 - 2\alpha\} = \frac{3}{5} - |\alpha - 3/5|$$

proves (PS10l).

For a completed row whose two prime supports are restricted to layers α, β , the fixed-frequency Cauchy and finite-Parseval proof of Proposition 193, rerun on the restricted prime supports, gives

$$\sum_{0 < |d| \asymp D} |B_{\alpha,\beta}(d)|^2 \ll_v DB_\alpha B_\beta N^{o(1)}.$$

Apply Cauchy–Schwarz in d between the two row layers. Since $D = N^{41/100}$, substituting (PS10l) proves (PS10m). Formula (PS10n) is its exact comparison with the target exponent. If $\alpha = 3/5 + \theta$, then

$$\kappa_\alpha = \begin{cases} 1 - \rho_\alpha + o(1), & \theta \leq 0, \\ 1 - 2\theta - \rho_\alpha + o(1), & \theta \geq 0. \end{cases}$$

The endpoint simplex therefore implies (PS10o).

Finally, restricting the positive centered-column square in the proof of Proposition 163 to $\mathcal{P}_\alpha$ gives

$$\sum_k A_\alpha(k)^2 \ll_v N^{3/10+o(1)}.$$

The removed midpoint costs at most one occurrence per prime and is negligible in a critical layer. Hence $\sum_k A_\alpha(k) = B_\alpha N^{o(1)}$, and Cauchy–Schwarz gives

$$S_\alpha \geq \frac{(\sum_k A_\alpha(k))^2}{\sum_k A_\alpha(k)^2} \gg_v N^{9/10-2(|\alpha-3/5|+\rho_\alpha)-o(1)}.$$

In the endpoint simplex each individual defect is at most $1/25 + o(1)$, which proves (PS10p). $\square$

The critical layers also have a three-characteristic rigidity which is invisible to their marginal zero counts. Two shifted-ratio columns with the same outer actual zero eliminate that zero and force the outer prime to divide a nonzero integer.

Proposition 197 (Actual ratio-rectangle divisor bound). *Let $\mathcal{P}_\alpha, \mathcal{P}_{\alpha'}, \mathcal{P}_\beta$ be the zero-count layers above, restricted to three distinct prime-color classes. Put*

$$W_\alpha = \sum_{p \in \mathcal{P}_\alpha} |u_p|, \quad B_\beta = \sum_{q \in \mathcal{P}_\beta} |u_q| Z(q), \quad J_\alpha = \sum_{p \in \mathcal{P}_\alpha} |u_p| Z(p)(Z(p) - 1),$$

and define $W_{\alpha'}, J_{\alpha'}$ similarly. For an actual centered q -zero η , let $I_{q,\eta}(p, p')$ be the number of actual pairs $\xi \in \mathcal{K}_p^\times, \xi' \in \mathcal{K}_{p'}^\times$ satisfying

$$p'(\eta - \xi) \equiv p(\eta - \xi') \pmod{q}, \quad (\text{RR1})$$

with any fixed collection of the current 0–1 source, branch, nonalias, unequal-offset and far-gap masks retained.

Let $\text{ARR}_{\alpha, \alpha'; \beta}^\times$ be the positive weighted count, with outer weight $|u_q|$ and inner weight $|u_p u_{p'}|$, of two ordered matches at the same $(q, \eta; p, p')$ for which

$$\xi_1 \neq \xi_2, \quad \xi'_1 \neq \xi'_2.$$

Thus $\text{ARR}^\times$ counts genuine coordinatewise nondegenerate ratio rectangles; the three prime sums retain their indicated layer and color restrictions. Then

$$\boxed{\text{ARR}_{\alpha, \alpha'; \beta}^\times \ll_v N^{-1+o(1)} J_\alpha J_{\alpha'}.} \quad (\text{RR2})$$

Moreover, for the one-column collision mass

$$\text{SRC}_{\alpha, \alpha'; \beta} = \sum_q |u_q| \sum_{\eta \in \mathcal{K}_q^\times} \sum_{p, p'} |u_p u_{p'}| I_{q,\eta}(p, p'),$$

one has

$$\begin{aligned} \text{SRC}_{\alpha, \alpha'; \beta} &\ll_v B_\beta W_\alpha W_{\alpha'} \\ &\quad + (B_\beta W_\alpha W_{\alpha'} N^{-1} J_\alpha J_{\alpha'})^{1/2} N^{o(1)}. \end{aligned} \quad (\text{RR3})$$

Writing

$$\delta_i = |\alpha_i - 3/5| + \rho_{\alpha_i},$$

the double critical simplex therefore gives

$$\boxed{\text{SRC}_{\alpha, \alpha'; \beta} \ll_v N^{1-\delta_\alpha-\delta_{\alpha'}-\delta_\beta/2+o(1)}.} \quad (\text{RR4})$$

In particular the exact central collision is $N^{1+o(1)}$, improving the direct support bound $N^{6/5+o(1)}$ by $N^{-1/5+o(1)}$.

If the p -side pair in $\text{ARR}^\times$ is restricted by $0 < |\xi_2 - \xi_1| \leq H \leq \sqrt{N}$, then

$$\text{ARR}_{|h_p| \leq H}^\times \ll_v N^{-1+o(1)} W_\alpha H^2 J_{\alpha'}, \quad (\text{RR5})$$

and symmetrically on the p' side.

Proof. Two distinct matches give

$$p'(\eta - \xi_j) \equiv p(\eta - \xi'_j) \pmod{q}, \quad j = 1, 2.$$

Subtracting eliminates η and gives

$$q \mid L := p'h_p - ph_{p'}, \quad h_p = \xi_2 - \xi_1, \quad h_{p'} = \xi'_2 - \xi'_1. \quad (\text{RR6})$$

The integer L is nonzero: equality would force $p \mid h_p$, whereas $0 < |h_p| < p$. Also $|L| \ll_v N^2$, so it has $O_v(1)$ prime divisors $q \asymp_v N$. For each such q , equation (RR1) determines η uniquely: on the common dyadic prime block $0 < |p' - p| < q$. Requiring that residue to be an actual q -zero and restoring any retained masks only decreases the positive count. The outer weight is $|u_q| \ll_v N^{-1+o(1)}$, and summing the two ordered zero pairs proves (RR2).

For fixed (q, η) , weighted Cauchy in (p, p') gives

$$\left(\sum_{p, p'} |u_p u_{p'}| I_{q, \eta}(p, p') \right)^2 \leq W_\alpha W_{\alpha'} \sum_{p, p'} |u_p u_{p'}| I_{q, \eta}(p, p')^2.$$

A fixed ξ determines ξ' modulo q , and a fixed ξ' determines ξ modulo q . Since all centered representative intervals have length $O_v(q)$, the corresponding bipartite match graph has degree $O_v(1)$, uniformly after all retained masks. Hence its ordered two-edge pairs split into $O_v(I_{q, \eta}(p, p'))$ pairs sharing an endpoint and the coordinatewise nondegenerate rectangles counted by $\text{ARR}^\times$; in particular

$$I_{q, \eta}(p, p')^2 \ll_v I_{q, \eta}(p, p') + I_{q, \eta}^\square(p, p'),$$

where summing the last term with the literal weights is $\text{ARR}^\times$. A second weighted Cauchy in (q, η) shows

$$\text{SRC}^2 \ll_v B_\beta W_\alpha W_{\alpha'} (\text{SRC} + \text{ARR}^\times) N^{o(1)}.$$

Solving this quadratic and using (RR2) proves (RR3).

On a dyadic layer,

$$W_i = N^{\kappa_i - 1 + o(1)}, \quad J_i \ll N^{2\alpha_i + \kappa_i - 1 + o(1)}, \quad \alpha_i + \kappa_i = \frac{8}{5} - \delta_i.$$

Together with $B_i \ll N^{3/5 - \delta_i + o(1)}$, substitution in (RR3) proves (RR4). Finally the actual gap polynomial gives

$$\#\{(\xi_1, \xi_2) \in \mathcal{K}_p^2 : 0 < |\xi_2 - \xi_1| \leq H\} \ll H^2.$$

Replacing the p -side factor $Z(p)(Z(p) - 1)$ by this bound in the divisor argument proves (RR5). $\square$

The saving in (RR4) is presently a theorem about the positive equal-ratio collision statistic, not yet a contraction of the short- d complex shell operator. A critical rectangle may moreover be restricted, up to a power-saving part of (RR2), to internal same-characteristic gaps $|h_p| > N^{\alpha - o(1)}$ and $|h_{p'}| > N^{\alpha' - o(1)}$. At the central layer these are beyond the proved $\sqrt{N}$ factorial-gap range. The remaining arithmetic test is whether a shell-prime divisor of (RR6) places its uniquely forced residue in the actual distinguished q -zero orbit.

The final orbit test admits an exact density-discrepancy decomposition. The literal positive rectangle carrier must be retained until its forced residue is formed; no estimate for one fixed carrier fibre is used.

Proposition 198 (Centered forced-residue rectangle decomposition). *Retain the zero-count layers, three distinct prime-color classes, and literal masks of Proposition 197. Fix any positive subfamily of its coordinatewise nondegenerate rectangle carriers. For a carrier datum*

$$\mathbf{d} = (p, p', \xi_1, \xi_2, \xi'_1, \xi'_2)$$

put

$$h = \xi_2 - \xi_1, \quad h' = \xi'_2 - \xi'_1, \quad L(\mathbf{d}) = p'h - ph'.$$

For every shell prime $q \in \mathcal{P}_\beta$ dividing $L(\mathbf{d})$, put

$$\Delta(\mathbf{d}) = p' - p, \quad A(\mathbf{d}) = p'\xi_1 - p\xi'_1, \quad \eta_q(\mathbf{d}) = \overline{\Delta(\mathbf{d})}A(\mathbf{d}) \pmod{q}.$$

Let $\mathbf{m}_q(\mathbf{d})$ be the product of all retained literal nonnegative masks and positive majorant factors after this forcing, except for the single predicate $1_{\{\eta_q(\mathbf{d}) \in \mathcal{K}_q^\times\}}$, normalized so that $0 \leq \mathbf{m}_q(\mathbf{d}) \leq N^{o(1)}$. Define the unnormalized forced-residue pushforward

$$\mathcal{W}_q(\eta) = \sum_{\substack{\mathbf{d}: q|L(\mathbf{d}) \\ \eta_q(\mathbf{d})=\eta}} |u_p u_{p'}| \mathbf{m}_q(\mathbf{d}), \quad M_q = \sum_{\eta \bmod q} \mathcal{W}_q(\eta).$$

Then the corresponding actual rectangle submass is

$$\text{FDR}_{\alpha, \alpha'; \beta}^\times = \sum_{q \in \mathcal{P}_\beta} |u_q| \sum_{\eta \bmod q} \mathcal{W}_q(\eta) 1_{\{\eta \in \mathcal{K}_q^\times\}}, \quad (\text{RR7})$$

whereas its carrier satisfies only the aggregate bound

$$\boxed{\sum_{q \in \mathcal{P}_\beta} |u_q| M_q \ll_v N^{-1+o(1)} J_\alpha J_{\alpha'}}. \quad (\text{RR8})$$

Writing $z_q^\times = |\mathcal{K}_q^\times|$ and

$$\kappa_q(\eta) = 1_{\{\eta \in \mathcal{K}_q^\times\}} - \frac{z_q^\times}{q},$$

one has the exact decomposition

$$\begin{aligned} \text{FDR}_{\alpha, \alpha'; \beta}^\times &= \sum_q |u_q| \frac{z_q^\times}{q} M_q \\ &\quad + \sum_q |u_q| \sum_{\eta \bmod q} \mathcal{W}_q(\eta) \kappa_q(\eta). \end{aligned} \quad (\text{RR9})$$

The density term obeys

$$\boxed{\sum_q |u_q| \frac{z_q^\times}{q} M_q \ll_v N^{-2+\beta+o(1)} J_\alpha J_{\alpha'}}. \quad (\text{RR10})$$

Consequently, in the endpoint double critical simplex it gains at least $N^{-9/25+o(1)}$ relative to (RR8); at the central layer $\beta = 3/5$ the gain is $N^{-2/5+o(1)}$.

Proof. Equation (RR6) gives $q \mid L(\mathbf{d})$, where $L(\mathbf{d}) \neq 0$ and $|L(\mathbf{d})| \ll_v N^2$. Moreover

$$(p' - p)\eta \equiv p'\xi_1 - p\xi'_1 \pmod{q}.$$

The color separation and fixed prime boxes make $0 < |p' - p| < q$, so this forces the displayed residue $\eta_q(\mathbf{d})$ uniquely. Each nonzero $L(\mathbf{d})$ has $O_v(1)$ prime divisors $q \asymp_v N$, and the single outer weight satisfies $|u_q| \ll_v N^{-1+o(1)}$. Summing the two actual ordered zero pairs proves (RR8); no pointwise estimate for an individual M_q is asserted.

Insert

$$1_{\{\eta \in \mathcal{K}_q^\times\}} = \frac{z_q^\times}{q} + \kappa_q(\eta)$$

in (RR7) to obtain (RR9). On the layer $\mathcal{P}_\beta$ one has $z_q^\times \leq Z(q) = N^{\beta+o(1)}$; combining this with (RR8) proves (RR10). In the endpoint simplex every layer exponent differs from $3/5$ by at most $1/25 + o(1)$, hence $\beta \leq 16/25 + o(1)$ and $1 - \beta \geq 9/25 - o(1)$. $\square$

Lemma 199 (Fourier and collision identities for the forced residue). *With the notation of Proposition 198, put*

$$\widehat{K}_q(a) = \sum_{\eta \in \mathcal{K}_q^\times} e_q(a\eta), \quad \widehat{\mathcal{W}}_q(a) = \sum_{\eta \bmod q} \mathcal{W}_q(\eta) e_q(a\eta), \quad \mathcal{E}_q^{\text{fr}} = \sum_{\eta \bmod q} \mathcal{W}_q(\eta)^2.$$

Then

$$\sum_{\eta} \mathcal{W}_q(\eta) \kappa_q(\eta) = \frac{1}{q} \sum_{a \neq 0} \widehat{K}_q(a) \widehat{\mathcal{W}}_q(-a), \quad (\text{RR11})$$

where

$$\widehat{\mathcal{W}}_q(-a) = \sum_{\mathbf{d}: q \mid L(\mathbf{d})} |u_p u_{p'}| \mathbf{m}_q(\mathbf{d}) e_q(-a \overline{p'} - p(p'\xi_1 - p\xi'_1)).$$

Moreover

$$\begin{aligned} \sum_{a \neq 0} |\widehat{K}_q(a)|^2 &= q z_q^\times - (z_q^\times)^2, \\ \sum_{a \neq 0} |\widehat{\mathcal{W}}_q(a)|^2 &= q \mathcal{E}_q^{\text{fr}} - M_q^2, \end{aligned}$$

and therefore

$$\left| \sum_{\eta} \mathcal{W}_q(\eta) \kappa_q(\eta) \right|^2 \leq \frac{(q z_q^\times - (z_q^\times)^2)(q \mathcal{E}_q^{\text{fr}} - M_q^2)}{q^2} \leq z_q^\times \mathcal{E}_q^{\text{fr}}. \quad (\text{RR12})$$

Finally the divisor-only estimate is

$$\boxed{\sum_{q \in \mathcal{P}_\beta} |u_q| \mathcal{E}_q^{\text{fr}} \ll_v N^{-1+o(1)} (J_\alpha J_{\alpha'})^2.} \quad (\text{RR13})$$

Proof. Fourier inversion and finite Parseval prove (RR11) and the two exact square identities; Cauchy–Schwarz proves (RR12). To prove (RR13), expand the square defining $\mathcal{E}_q^{\text{fr}}$. For each ordered pair of carrier data, the first nonzero integer $L(\mathbf{d})$ already restricts q to $O_v(1)$ shell divisors. The sum contains only the single weight $|u_q|$, giving one factor $N^{-1+o(1)}$; dropping the remaining positive masks and summing the two copies gives $(J_\alpha J_{\alpha'})^2$. $\square$

Proposition 200 (Centered forced-residue variance and two-lift bound). *With the notation and literal occurrence weights of Proposition 198, put*

$$V_q = \sum_{\eta \bmod q} \left(\mathscr{W}_q(\eta) - \frac{M_q}{q} \right)^2, \quad \mathfrak{V}_{\alpha, \alpha'; \beta} = \sum_{q \in \mathscr{P}_\beta} |u_q| V_q,$$

and

$$\mathfrak{B}_\beta^\circ = \sum_{q \in \mathscr{P}_\beta} |u_q| \left(z_q^\times - \frac{(z_q^\times)^2}{q} \right).$$

If $\mathfrak{C}_{\alpha, \alpha'; \beta}$ denotes the centered term in (RR9), then

$$V_q = \mathcal{E}_q^{\text{fr}} - \frac{M_q^2}{q}, \quad (\text{RR14})$$

$$|\mathfrak{C}_{\alpha, \alpha'; \beta}|^2 \leq \mathfrak{B}_\beta^\circ \mathfrak{V}_{\alpha, \alpha'; \beta}. \quad (\text{RR15})$$

Moreover the literal two-lift geometry improves (RR13) to

$$\boxed{\mathfrak{V}_{\alpha, \alpha'; \beta} \ll_v N^{-1+o(1)} J_\alpha J_{\alpha'} \min\{J_\alpha W_{\alpha'}, W_\alpha J_{\alpha'}\}.} \quad (\text{RR16})$$

At the central layers this gives

$$\mathfrak{V}_{3/5, 3/5; 3/5} \ll N^{13/5+o(1)}. \quad (\text{RR17})$$

The centered relative-saving target with exponent $\sigma > 0$ is

$$\mathfrak{B}_\beta^\circ \mathfrak{V}_{\alpha, \alpha'; \beta} \ll N^{-2-2\sigma+o(1)} (J_\alpha J_{\alpha'})^2. \quad (\text{RR18})$$

In the central box this requires $\mathfrak{V} \ll N^{11/5-2\sigma+o(1)}$; hence (RR17) still misses the $\sigma = 0$ frontier by $N^{2/5+o(1)}$.

Proof. The identity (RR14) is the expansion of the centered square. Apply Cauchy–Schwarz on the disjoint union of the (q, η) fibres, with measure $|u_q|$, to the two centered functions

$$\mathscr{W}_q(\eta) - M_q/q, \quad \mathbf{1}_{\mathcal{K}_q^\times}(\eta) - z_q^\times/q.$$

This proves (RR15).

It remains to prove (RR16). Since $V_q \leq \mathcal{E}_q^{\text{fr}}$, expand the latter collision square. One rectangle carrier first fixes a nonzero integer L of height $O_v(N^2)$, hence only $O_v(1)$ possible shell divisors q , and then fixes its forced residue η . For a second carrier with prime pair (r, s) , fix its actual ordered r -zero pair (ζ_1, ζ_2) . Its two ratio equations are

$$(s - r)\eta \equiv s\zeta_i - r\zeta'_i \pmod{q}, \quad i = 1, 2.$$

The color separation makes r a unit modulo q , so these equations determine both opposite offsets ζ'_i modulo q . Each centered physical s -interval has length $O_v(N)$ with $q \asymp_v N$, and therefore contains only $O_v(1)$ lifts of each forced residue. All literal masks can only decrease this positive count. Thus the second carrier costs at most $J_\alpha W_{\alpha'} N^{o(1)}$ instead of $J_\alpha J_{\alpha'}$; fixing its opposite pair gives symmetrically $W_\alpha J_{\alpha'} N^{o(1)}$. The first carrier costs $J_\alpha J_{\alpha'}$, and the single outer weight contributes $N^{-1+o(1)}$, proving (RR16).

At the central layer $J_{3/5} \ll N^{6/5+o(1)}$ and $W_{3/5} \ll N^{o(1)}$, which proves (RR17). Finally $\mathfrak{B}_{3/5}^\circ \ll N^{3/5+o(1)}$; comparison in (RR15) gives the displayed central target and gap. $\square$

The two forced moving offsets in the preceding proof also expose an exact integer quotient. This gives a deterministic dyadic bound, although its central endpoint is exactly the missing $2/5$ rather than a strict improvement beyond it.

Proposition 201 (Dyadic quotient-line bound). *Fix distinct shell primes q, r , a residue $\eta \pmod{q}$, and an ordered actual r -zero pair (ζ_1, ζ_2) with $h = \zeta_2 - \zeta_1 \neq 0$ and $|h| < r$. For moving shell primes s retain any of the literal positive masks from the forced-residue rectangle construction, and require the two forced physical offsets y_1, y_2 to satisfy*

$$s(\eta - \zeta_i) \equiv r(\eta - y_i) \pmod{q}, \quad i = 1, 2. \quad (\text{RR19})$$

If $G < |y_2 - y_1| \leq 2G$, then the corresponding moving mass satisfies

$$\sum_s |u_s| \sum_{y_1, y_2} \mathbf{m}_{q, \eta; r, s} \ll_v N^{-1+o(1)}(|h| + G), \quad (\text{RR20})$$

where $0 \leq \mathbf{m}_{q, \eta; r, s} \leq N^{o(1)}$ retains the actual-zero, branch, color, nonalias, orientation, far-gap and inherited source masks.

Proof. Put $g_s = y_2 - y_1$. Subtracting the two congruences in (RR19) gives the exact integer relation

$$sh - rg_s = n_s q, \quad |n_s| \ll_v |h| + G. \quad (\text{RR21})$$

For fixed n_s , reduction modulo r gives

$$s \equiv n_s q h^{-1} \pmod{r}.$$

The moving shell has length $O_v(N)$ and $r \asymp_v N$, so this residue class contains $O_v(1)$ possible integers s . For each such s , (RR19) fixes both y_i modulo q ; each centered physical interval has length $O_v(N)$ with $q \asymp_v N$, and hence contains $O_v(1)$ lifts. There are $O_v(|h| + G)$ possible integers n_s , and $|u_s| \ll_v N^{-1+o(1)}$. Keeping all positive masks proves (RR20). $\square$

At the central layer, in the smallest surviving hard band $|h|, G = N^{3/5+o(1)}$, (RR20) gives the geometric gain $N^{-2/5+o(1)}$. It does not give a fixed improvement beyond $2/5$; for larger hard-gap bands it is weaker. Moreover a prescribed common ratio shift cancels when the two columns in (RR19) are subtracted. Thus the quotient line is shift-blind and does not by itself contract the signed short- d operator.

There is also a useful exact deletion inside the collision square. It removes every sector in which the two carrier occurrences share an inner-prime interface, leaving only a fully foreign four-characteristic sampler at the centered frontier.

Proposition 202 (Shared-interface deletion in the forced-residue energy). *Put*

$$\Pi(p) = Z(p)(Z(p) - 1), \quad U_\gamma = \sum_{p \in \mathcal{P}_\gamma} |u_p|, \quad \mathfrak{F}_\gamma = \sum_{p \in \mathcal{P}_\gamma} |u_p|^2 \Pi(p)^2.$$

Expand the positive collision majorant $\sum_q |u_q| \mathcal{E}_q^{\text{fr}}$ into two literal carrier occurrences, with inner prime pairs (p, p') and (r, r') . The four-color separation allows only the interface equalities $r = p$ and $r' = p'$. If $\mathfrak{E}^{\varepsilon\varepsilon'}$ denotes the sector with $\mathbf{1}_{r=p} = \varepsilon$ and $\mathbf{1}_{r'=p'} = \varepsilon'$, then

$$\mathfrak{E}^{11} \ll N^{-1+o(1)} \mathfrak{F}_\alpha \mathfrak{F}_{\alpha'}, \quad (\text{RR22})$$

$$\mathfrak{E}^{10} \ll N^{-1+o(1)} \mathfrak{F}_\alpha J_{\alpha'} U_{\alpha'}, \quad (\text{RR23})$$

$$\mathfrak{E}^{01} \ll N^{-1+o(1)} U_\alpha J_\alpha \mathfrak{F}_{\alpha'}, \quad (\text{RR24})$$

$$\mathfrak{E}^{00} \ll N^{-1+o(1)} J_\alpha J_{\alpha'} \min\{J_\alpha U_{\alpha'}, U_\alpha J_{\alpha'}\}. \quad (\text{RR25})$$

At the central layer,

$$\mathfrak{E}^{10} + \mathfrak{E}^{01} \ll N^{8/5+o(1)}, \quad \mathfrak{E}^{11} \ll N^{9/5+o(1)}, \quad \mathfrak{E}^{00} \ll N^{13/5+o(1)}.$$

Thus every sector sharing an inner-prime interface is below the centered $N^{11/5-2\sigma}$ frontier for sufficiently small fixed $\sigma > 0$; only the fully foreign sector can be critical.

Proof. Fix the first literal carrier. Its nonzero divisor integer permits only $O(1)$ outer shell primes q , and the single outer weight supplies $N^{-1+o(1)}$. If both interfaces are shared, positive majorization of the remaining masks gives two ordered actual pairs in each fixed inner prime, and summation gives $\mathfrak{F}_\alpha \mathfrak{F}_{\alpha'}$.

Suppose only $r = p$ is shared. Fix the second actual ordered p -zero pair. For each foreign prime r' , equality with the already forced residue gives two ratio equations, which determine both opposite physical offsets modulo q with $O(1)$ centered lifts. The second carrier therefore costs $|u_p| \Pi(p) U_{\alpha'} N^{o(1)}$. Summing the first carrier yields (RR23); the other one-shared estimate is symmetric. With neither interface shared, the two-lift argument from Proposition 200 gives (RR25).

Finally

$$\mathfrak{F}_\gamma \leq \left(\sup_{p \in \mathcal{P}_\gamma} |u_p| \Pi(p) \right) J_\gamma \ll N^{2\gamma-1+o(1)} J_\gamma.$$

At the center $U_{3/5} \ll N^{o(1)}$, $J_{3/5} \ll N^{6/5+o(1)}$, and $\mathfrak{F}_{3/5} \ll N^{7/5+o(1)}$. Substitution proves the displayed central exponents. $\square$

The fully foreign term still contains an exact rational-center diagonal. Unlike equality of forced residues modulo q , equality of the two rational centres is an integer condition. Its reduced denominator occurs with reciprocal multiplicities on the two carrier copies, which makes this whole diagonal subcritical.

Proposition 203 (Fully foreign rational-center fibre). *In the literal occurrence expansion of $\mathfrak{E}^{00}$, write the two fully foreign carrier copies as*

$$(p, p', \xi_1, \xi_2, \xi'_1, \xi'_2), \quad (r, s, \zeta_1, \zeta_2, y_1, y_2),$$

with every source label and q -dependent mask retained. Put

$$\Delta = p' - p, \quad A = p' \xi_1 - p \xi'_1,$$

and, on the second copy,

$$\tilde{\Delta} = s - r, \quad \tilde{A} = s \zeta_1 - r y_1.$$

Let $\mathfrak{E}_0^{00}$ be the positive submass on which

$$\mathcal{R}_{11} := A \tilde{\Delta} - \tilde{A} \Delta = 0. \tag{RR26}$$

Then

$$\boxed{\mathfrak{E}_0^{00} \ll_v N^{-1+\alpha+\alpha'+\min\{\alpha, \alpha'\}+o(1)} \min\{J_\alpha, J_{\alpha'}\}.} \tag{RR27}$$

In particular,

$$\mathfrak{E}_0^{00} \ll_v N^{2+o(1)} \quad (\alpha = \alpha' = 3/5), \tag{RR28}$$

$$\mathfrak{E}_0^{00} \ll_v N^{53/25+o(1)} \quad (14/25 - o(1) \leq \alpha, \alpha' \leq 16/25 + o(1)). \tag{RR29}$$

Thus the exact rational-center diagonal is below the centered $N^{11/5-2\sigma}$ frontier for every fixed $0 < \sigma < 1/25$ throughout the double-critical window.

Proof. Dyadically fix

$$D < |\Delta| \leq 2D, \quad B \leq b < 2B, \quad b = \frac{|\Delta|}{\gcd(A, \Delta)}.$$

The identity

$$A - \xi_1 \Delta = p(\xi_1 - \xi'_1)$$

and $\gcd(p, \Delta) = 1$ give

$$b = \frac{|\Delta|}{\gcd(\xi_1 - \xi'_1, \Delta)}. \quad (\text{RR30})$$

For fixed p, p', ξ_1 and a fixed divisor b of $|\Delta|$, the difference $t = \xi_1 - \xi'_1$ has the form

$$t = \frac{\Delta}{b} a, \quad (a, b) = 1.$$

The physical representative interval has length $O_v(N)$, so one dyadic b -band permits

$$N^{o(1)} \left(1 + \frac{NB}{D} \right)$$

values of ξ'_1 . There are $O(ND)$ ordered shell-integer pairs (p, p') , whose two coefficient weights contribute $N^{-2+o(1)}$. Choosing the ordered actual p -pair costs $N^{2\alpha+o(1)}$, and choosing the remaining actual p' -endpoint costs $N^{\alpha'+o(1)}$. Since $D \leq N$ and $B \geq 1$, the positive first-carrier occurrence mass in this (D, B) -band is therefore

$$\ll BN^{2\alpha+\alpha'+o(1)}.$$

Interchanging the two inner layers and taking the better estimate gives

$$\mathcal{C}_{\alpha, \alpha'}(D, B) \ll BN^{\alpha+\alpha'+\min\{\alpha, \alpha'\}+o(1)}. \quad (\text{RR31})$$

All literal occurrence labels remain in this positive majorant. The $N^{o(1)}$ normalized separated-source multiplicity over fixed displayed coordinates is included in (RR31). The q -dependent masks are replaced by 1 only after the literal expansion, so no mask is factored from the carrier as an identity.

For the first carrier the divisor integer

$$L = p'(\xi_2 - \xi_1) - p(\xi'_2 - \xi'_1)$$

is nonzero by the argument in (RR6). It has only $O_v(1)$ shell-prime divisors, and their single outer weight is $N^{-1+o(1)}$.

Now fix one ordered actual pair (r, ζ_1, ζ_2) on the second carrier. Equation (RR26) states that

$$\frac{\tilde{A}}{\tilde{\Delta}} = \frac{A}{\Delta}$$

in $\mathbf{Q}$. As b is the reduced denominator of the right side, $b \mid \tilde{\Delta} = s - r$. Thus the moving shell prime s lies in one residue class modulo b . Here $b \leq |\Delta| < q \ll_v N$, so the total moving-prime coefficient mass is

$$\sum_{\substack{s \asymp_v N \\ s \equiv r \pmod{b}}} |u_s| \ll_v N^{o(1)} \left(\frac{1}{N} + \frac{1}{b} \right) \ll_v B^{-1} N^{o(1)}.$$

For fixed q , its forced residue, r, s , and the chosen actual r -pair, the two ratio equations determine y_1, y_2 modulo q . Each physical interval has length $O_v(N)$ with $q \asymp_v N$, so both have $O_v(1)$ lifts. Requiring those lifts to be actual zeros and restoring the remaining masks only decreases the positive sum. The fixed ordered pair costs J_α , or symmetrically $J_{\alpha'}$; hence the second-carrier mass is

$$\ll B^{-1} \min\{J_\alpha, J_{\alpha'}\} N^{o(1)}.$$

Multiplying this by (RR31) and the single outer factor $N^{-1+o(1)}$, then summing the logarithmically many (D, B) -bands, proves (RR27).

At the centre $J_{3/5} \ll N^{6/5+o(1)}$, which gives (RR28). The factorial zero moment also gives $J_\gamma \ll N^{6/5+o(1)}$ uniformly in the critical window, while

$$\alpha + \alpha' + \min\{\alpha, \alpha'\} \leq \frac{48}{25} + o(1).$$

This proves (RR29). □

The nonzero rational-determinant fibre admits a finer dyadic reduction. The point is to retain the first prime difference instead of flattening the first carrier to its global endpoint.

Lemma 204 (Difference-band cap for the first fully foreign carrier). *At the exact critical centre, let $\mathcal{C}(D, B)$ be the positive literal first-carrier occurrence mass in $\mathfrak{E}^{00}$ restricted by*

$$D < |p' - p| \leq 2D, \quad B \leq \frac{|p' - p|}{\gcd(p'\xi_1 - p\xi_1', p' - p)} < 2B.$$

Then

$$\boxed{\mathcal{C}(D, B) \ll_v DN^{7/5+o(1)}} \tag{RR32}$$

Together with (RR31), this gives

$$\mathcal{C}(D, B) \ll_v N^{9/5+o(1)} \min \left\{ B, \frac{D}{N^{2/5}} \right\}. \tag{RR33}$$

Proof. After dropping only positive masks, there are $O_v(ND)$ ordered shell integer pairs (p, p') in the indicated difference band. Their coefficient is $|u_p u_{p'}| \ll_v N^{-2+o(1)}$. Each side has at most $Z(p)(Z(p) - 1) \ll N^{6/5+o(1)}$ ordered actual zero pairs at the critical centre. Hence

$$\mathcal{C}(D, B) \ll_v (ND) N^{-2+o(1)} N^{6/5} N^{6/5} = DN^{7/5+o(1)}.$$

This first-carrier census is a q -free envelope: it retains the q -independent occurrence multiplicities, whereas the q -dependent masks remain in the exact energy and are bounded by 1 only after the outer variable q has been exposed. Taking its minimum with the specialization $\mathcal{C}(D, B) \ll_v BN^{9/5+o(1)}$ of (RR31) proves (RR33). □

Proposition 205 (Product-form nonzero primitive-determinant reduction). *Work at the exact critical centre and retain the literal fully foreign positive occurrence expansion of $\mathfrak{E}^{00}$. Keep all four inner characteristics, both actual moving-zero predicates, every source and q -dependent mask, every occurrence label, and exactly the one outer weight $|u_q|$. On the complement of (RR26) put*

$$\Delta = p' - p, \quad A = p'\xi_1 - p\xi_1', \quad d = \gcd(A, \Delta), \quad b = |\Delta|/d,$$

$$\tilde{\Delta} = s - r, \quad \tilde{A} = s\zeta_1 - ry_1, qquad v = \frac{A\tilde{\Delta} - \tilde{A}\Delta}{qd} \neq 0,$$

and write $h = \zeta_2 - \zeta_1$, $g = y_2 - y_1$. Let $\mathfrak{E}_{\times}^{00}(D, B, H, G, V)$ be the submass restricted by

$$D < |\Delta| \leq 2D, \quad B \leq b < 2B, \quad H < |h| \leq 2H, \quad G < |g| \leq 2G, \quad V < |v| \leq 2V.$$

Then

$$\boxed{\mathfrak{E}_{\times}^{00}(D, B, H, G, V) \ll_v N^{2+o(1)} \min \left\{ B, \frac{D}{N^{2/5}} \right\} \min \left\{ 1, \frac{V}{B}, \frac{H+G}{N} \right\}.} \quad (\text{RR34})$$

If instead $T < |A\tilde{\Delta} - \tilde{A}\Delta| \leq 2T$, then

$$\boxed{\mathfrak{E}_{\times}^{00}(D, B, H, G, T) \ll_v N^{2+o(1)} \min \left\{ B, \frac{D}{N^{2/5}} \right\} \min \left\{ 1, \frac{T}{ND}, \frac{H+G}{N} \right\}.} \quad (\text{RR35})$$

Proof. Fix the first carrier. Its nonzero integer L from (RR6) admits only $O_v(1)$ outer shell-prime divisors, whose single total outer weight is $N^{-1+o(1)}$. By Lemma 204, the first-carrier mass is

$$N^{9/5+o(1)} \min\{B, D/N^{2/5}\}.$$

Now fix one admitted q , its forced residue, and one ordered actual pair on the r -side of the second carrier. This pair costs $J_{3/5} \ll N^{6/5+o(1)}$. Writing $c = (A - \eta\Delta)/q$, one has $\gcd(c, \Delta) = d$; hence fixed v places s in one residue class modulo b . A V -band therefore has moving-prime mass $\ll_v N^{o(1)}V/B$. On the other hand, subtraction of the two forced ratio equations gives

$$sh - rg = nq, \quad |n| \ll_v H + G.$$

For fixed n , reduction modulo r fixes s to one residue class, and the shell contains $O_v(1)$ representatives. Thus the same moving-prime mass is also $\ll_v N^{o(1)}(H+G)/N$, while its unrestricted central-layer mass is $N^{o(1)}$. These are three uniform bounds for the same fixed-carrier fibre, so their minimum may be multiplied by the first-carrier mass. The forced physical lifts have bounded multiplicity and all remaining positive masks only decrease the sum. Since

$$-1 + \frac{9}{5} + \frac{6}{5} = 2,$$

this proves (RR34). Finally

$$|A\tilde{\Delta} - \tilde{A}\Delta| = qd|v|, \quad d \asymp D/B,$$

so $V/B \asymp T/(ND)$ on the indicated dyadic blocks, proving (RR35). $\square$

Corollary 206 (Subcritical nonzero primitive-determinant bands). *Fix $0 < \sigma < 1/10$. Every dyadic nonzero block satisfying at least one of*

$$\begin{aligned} B &\leq N^{1/5-2\sigma}, qquad V \leq N^{1/5-2\sigma}, qquad B(H+G) \leq N^{6/5-2\sigma}, \\ D &\leq N^{3/5-2\sigma}, qquad T \leq N^{8/5-2\sigma}, qquad D(H+G) \leq N^{8/5-2\sigma} \end{aligned}$$

obeys

$$\boxed{\mathfrak{E}_{\times}^{00} \ll_v N^{11/5-2\sigma+o(1)}.} \quad (\text{RR36})$$

Consequently the remaining positive nonzero residual may be restricted to the complementary six-large-parameter sector. This corollary supplies no bound for that residual and no signed short-shift contraction.

Proof. Expanding the two minima in (RR34) gives the six caps

$$B, \quad V, \quad \frac{B(H+G)}{N}, \quad \frac{D}{N^{2/5}}, \quad \frac{T}{N^{7/5}}, \quad \frac{D(H+G)}{N^{7/5}}.$$

Under any displayed hypothesis at least one is at most $N^{1/5-2\sigma}$, which proves (RR36) after the logarithmically many dyadic sums. $\square$

Proposition 207 (Fully foreign joint determinant graph). *Retain the literal occurrence-level setup and notation of Proposition 205. Fix the first carrier, one admitted outer shell prime q , its forced residue η , and an ordered distinct actual r -pair (ζ_1, ζ_2) . Put*

$$\varepsilon = \frac{\Delta}{db} \in \{\pm 1\}, \quad c_0 = \frac{A - \eta\Delta}{qd}, \quad h = \zeta_2 - \zeta_1.$$

Then $c_0 \in \mathbb{Z}$ and $(c_0, b) = 1$. For a moving literal occurrence write

$$g = y_2 - y_1, \quad sh - rg = nq,$$

retain the nonzero primitive coordinate

$$v = \frac{A(s-r) - (s\zeta_1 - ry_1)\Delta}{qd} \neq 0,$$

and set

$$P_0 = \varepsilon b(\zeta_1 - \eta) - qc_0.$$

Every such occurrence satisfies

$$P_0s + qv \equiv 0 \pmod{r}, \tag{RR37}$$

$$hs - qn \equiv 0 \pmod{r}, \tag{RR38}$$

$$P_0n + hv \equiv 0 \pmod{r}. \tag{RR39}$$

Moreover, if $0 < |v| < r$, then $P_0 \not\equiv 0 \pmod{r}$.

Proof. In the proof of Proposition 205 one has

$$c = \frac{A - \eta\Delta}{q}, \quad \gcd(c, \Delta) = d.$$

Thus $c_0 = c/d$ is integral and coprime to $b = |\Delta|/d$. Substituting $A = \eta\Delta + qdc_0$ and $\Delta = \varepsilon db$ in the definition of v gives

$$q(v - c_0(s-r)) = \varepsilon b(r(y_1 - \eta) - s(\zeta_1 - \eta)).$$

Since $(q, b) = 1$, the integer

$$k = \frac{v - c_0(s-r)}{\varepsilon b}$$

satisfies

$$r(y_1 - \eta) = s(\zeta_1 - \eta) + qk.$$

Reducing this identity modulo r after substituting for k proves (RR37). The quotient identity $sh - rg = nq$ gives (RR38). Eliminating s from these two congruences gives

$$q(P_0n + hv) \equiv 0 \pmod{r}.$$

The fully foreign masks give $q \neq r$, so cancellation proves (RR39). Finally $0 < |h| < r$; hence h and, under the displayed additional hypothesis, v are units modulo the prime r . If $P_0 \equiv 0 \pmod{r}$, then (RR39) would force $r \mid hv$, a contradiction. $\square$

Corollary 208 (Signed-dyadic fully foreign graph cap). *Assume $r \asymp_v N$, $V < r$, and $H + G = O_v(r)$. Let $\mathcal{S}_{q,\eta;r,\zeta_1,\zeta_2}(H, G, V)$ be the moving fibre in the proof of Proposition 205, restricted by*

$$V < |v| \leq 2V, \quad |n| \ll_v H + G,$$

with every literal actual-zero, source, colour, branch, phase, first-return and physical-lift mask retained. Then

$$\boxed{\mathcal{S}_{q,\eta;r,\zeta_1,\zeta_2}(H, G, V) \ll_v N^{-1+o(1)} \min\{V, H + G\}.} \quad (\text{RR40})$$

Consequently, with

$$M_{D,B} = \min \left\{ B, \frac{D}{N^{2/5}} \right\},$$

one has

$$\boxed{\mathfrak{E}_\times^{00}(D, B, H, G, V) \ll_v N^{1+o(1)} M_{D,B} \min\{V, H + G\}.} \quad (\text{RR41})$$

At the six-way centre

$$D \asymp N, \quad B \asymp V \asymp N^{3/5}, \quad H + G \asymp N,$$

this gives

$$\boxed{\mathcal{S} \ll_v N^{-2/5+o(1)}, \quad \mathfrak{E}_\times^{00} \ll_v N^{11/5+o(1)}.} \quad (\text{RR42})$$

This is the critical $\sigma = 0$ line in (RR36); it supplies no fixed positive σ .

Proof. For each fixed n , equation (RR39) places v in one residue class modulo r , because h is a unit. The signed v -band is the union of two intervals of length $V < r$, so this gives $O_v(H + G)$ admissible pairs (n, v) .

If $P_0 \not\equiv 0 \pmod{r}$, fixed v similarly places n in one residue class modulo r and gives $O_v(V)$ pairs. If $P_0 \equiv 0 \pmod{r}$, the same congruence forces $r \mid v$. Since $V < r$ and $|v| \leq 2V < 2r$, only $v = \pm r$ are possible; their occurrence forces $V \geq r/2$, so the preceding $O_v(H + G) = O_v(r)$ estimate is also $O_v(V)$. Hence

$$\#\{(n, v)\} \ll_v \min\{V, H + G\}.$$

For each pair, (RR38) fixes s modulo r , leaving $O_v(1)$ integer shell lifts. Once (n, v, s) is fixed, the integers

$$k = \frac{v - c_0(s - r)}{\varepsilon b}, \quad g = \frac{sh - nq}{r},$$

$$y_1 = \eta + \frac{s(\zeta_1 - \eta) + qk}{r}, \quad y_2 = y_1 + g$$

are unique whenever the required divisibilities hold. All retained physical-range and selected masks only remove candidates. Multiplying by the single moving coefficient $|u_s| \ll_v N^{-1+o(1)}$ proves (RR40).

The first-carrier envelope (RR33) is $N^{9/5+o(1)} M_{D,B}$, the one outer-divisor sum contributes $N^{-1+o(1)}$, and the ordered actual host pair contributes $J_{3/5} \ll N^{6/5+o(1)}$. This proves (RR41). At the displayed centre one has $2V < r$ for large N , so the dyadic edge case is absent; substitution gives (RR42). $\square$

Proposition 209 (Fixed-power deletion of the resonant fully foreign line). *Retain the hypotheses and literal positive occurrence ordering of Corollary 208. Let $\mathfrak{E}_{\text{res}}^{00}(D, B, H, G, V)$ be the submass on which the first centered host zero satisfies*

$$\zeta_1 \equiv \eta \pmod{q}.$$

Then, at the exact critical zero-count layer,

$$\boxed{\mathfrak{C}_{\text{res}}^{00}(D, B, H, G, V) \ll_v N^{2/5+o(1)} M_{D,B} \min\{V, H + G\}.} \quad (\text{RR43})$$

In particular, at the six-way centre,

$$\boxed{\mathfrak{C}_{\text{res}}^{00} \ll_v N^{8/5+o(1)}.} \quad (\text{RR44})$$

Thus the resonant line is below $N^{11/5-2\sigma+o(1)}$ for every fixed $0 < \sigma < 3/10$.

Proof. In the positive RR Fubini order, the first carrier, the admitted outer prime q , and its forced centered zero η are fixed before the ordered host pair (r, ζ_1, ζ_2) is summed. For each fixed shell prime r , its canonical centered interval has length $r = O_v(N)$, whereas $q \asymp_v N$. It therefore contains only $O_v(1)$ integers congruent to η modulo q . Once such a first actual zero is fixed, the second endpoint has at most $Z(r)$ choices. Consequently the conditional resonant host mass is

$$\ll_v N^{o(1)} \sum_{r \in \mathcal{P}_{3/5}} |u_r| Z(r) = B_{3/5} N^{o(1)} \ll_v N^{3/5+o(1)}.$$

No identification of centered coordinates with a common raw Apéry index is used here.

For each fixed resonant host pair, its moving subfibre is a subset of the literal fibre in (RR40), and hence has weighted mass

$$\ll_v N^{-1+o(1)} \min\{V, H + G\}.$$

The first-carrier envelope is $N^{9/5+o(1)} M_{D,B}$ and the single outer-divisor sum contributes $N^{-1+o(1)}$. Multiplication in the same positive Fubini order as in the proof of RR41 gives

$$N^{9/5} M_{D,B} N^{-1} N^{3/5} N^{-1} \min\{V, H + G\},$$

which is (RR43). At the six-way centre both $M_{D,B}$ and $\min\{V, H + G\}$ are $N^{3/5+o(1)}$, proving (RR44). $\square$

The exact line in RR43–RR44 can be thickened on the host side. This is only a census of literal representatives; it does not identify centred coordinates in distinct characteristics.

Lemma 210 (Nonresonant host-window census). *Retain the literal fully foreign positive occurrence order and fix the first carrier, one admitted outer prime q , and its forced centred residue η . For an ordered actual host pair (r, ζ_1, ζ_2) put*

$$h = \zeta_2 - \zeta_1, \quad P_0 = \varepsilon b(\zeta_1 - \eta) - qc_0.$$

In a critical block with $b < r$, let

$$R < \|\zeta_1 - \eta\|_q \leq 2R, \quad W < \|P_0\|_r \leq 2W, \quad H < |h| \leq 2H,$$

where $R, W, H \geq 1$, and put $L = \min\{R, W\}$. The corresponding weighted host-pair mass satisfies

$$\boxed{\mathcal{H}_{q,\eta}(R, W, H) \ll_v N^{o(1)} \min\{LH, LN^{3/5}, H^2, N^{6/5}\}.} \quad (\text{RR45})$$

The same estimate holds with ζ_2 in place of ζ_1 in the q -distance condition.

Proof. The canonical representatives for ζ_1 lie in an interval of length $r \asymp_v N$, whereas $q \asymp_v N$. A q -annulus of width $O(R)$ therefore contains $O_v(R)$ possible representatives. Since $0 < b < r$, the coefficient b is a unit modulo the prime r , so $\zeta_1 \mapsto P_0$ is an affine permutation modulo r ; the P_0 -annulus contains $O(W)$ representatives. Their intersection has $O_v(L)$ representatives.

For each first endpoint the h -band allows $O(H)$ second endpoints, giving $O(LH)$ pairs. Alternatively it gives at most $O(LZ(r))$ actual pairs. For a fixed nonzero integer gap h , the gap-polynomial criterion gives at most $3(|h| - 1) = O(H)$ possible first residues; summing over the $O(H)$ gaps in the band gives $O(H^2)$ pairs for each r . Finally the unrestricted count is $Z(r)(Z(r) - 1)$. Summing these four positive bounds with the literal host weight and using

$$\sum_{r \in \mathcal{P}_{3/5}} |u_r| Z(r) \ll_v N^{3/5+o(1)}, \quad \sum_{r \in \mathcal{P}_{3/5}} |u_r| Z(r)(Z(r) - 1) \ll_v N^{6/5+o(1)}$$

proves (RR45). Every omitted literal predicate is a deleting mask. $\square$

Proposition 211 (Nonresonant host-window pruning). *Under Lemma 210, put*

$$M_{D,B} = \min \left\{ B, \frac{D}{N^{2/5}} \right\}, \quad K = \min\{V, H + G\}.$$

The corresponding fully foreign energy obeys

$$\boxed{\mathfrak{E}_{\text{win}}^{00}(R, W, H) \ll_v N^{-1/5+o(1)} M_{D,B} K \min\{LH, LN^{3/5}, H^2, N^{6/5}\}.} \quad (\text{RR46})$$

At the six-way centre $M_{D,B} \asymp K \asymp N^{3/5}$, this sector is $O_v(N^{11/5-2\sigma+o(1)})$ whenever at least one of

$$L \leq N^{3/5-2\sigma}, \quad H \leq N^{3/5-\sigma}, \quad LH \leq N^{6/5-2\sigma}$$

holds.

Proof. Use the same positive Fubini order as RR41. The first-carrier marginal is $N^{9/5+o(1)} M_{D,B}$, the single admitted outer-prime sum contributes $N^{-1+o(1)}$, Lemma 210 supplies the host factor, and RR40 supplies the moving factor $N^{-1+o(1)} K$. Their product is (RR46). Substitution at the critical centre proves the three displayed sufficient conditions. $\square$

There is also an unconditional horizontal-spacing deletion which does not use a common coefficient field for two moving characteristics.

Proposition 212 (Horizontal difference lattice). *Fix one signed critical fully foreign graph block, including the first carrier, (q, η) , the ordered host pair, and every literal mask. Let $\mathcal{G}_\Theta$ be its bare moving graph in one signed v -band, and define*

$$\Lambda_\Theta = \{(d, e, f) \in \mathbb{Z}^3 : qd + P_0 e = \varepsilon b r f, \ 0 < |d| \ll_v V, \ |e|, |f| \ll_v N\}.$$

Let

$$\lambda_\Theta = \min\{|d| : (d, e, f) \in \Lambda_\Theta\},$$

with $\lambda_\Theta = \infty$ if the set is empty. Then

$$\boxed{\#\mathcal{G}_\Theta \ll_v N^{o(1)} \left(1 + \frac{V}{\lambda_\Theta}\right).} \quad (\text{RR47})$$

Consequently, at $V \asymp N^{3/5}$, the sector $\lambda_\Theta \geq N^\theta$ satisfies

$$\boxed{\mathfrak{E}_{\lambda\text{-minor}}^{00} \ll_v N^{11/5-\theta+o(1)}.} \quad (\text{RR48})$$

Proof. Every graph point satisfies the exact integer identity

$$P_0 s + qv = r(\varepsilon b(y_1 - \eta) - qc_0).$$

Subtracting two such identities shows that the difference of two points with distinct numerical v -coordinates lies in Λ_Θ . For a fixed numerical v , RR39 and the proof of RR40 leave only $N^{o(1)}$ literal lifts. Hence distinct numerical v -values are separated by at least λ_Θ , which proves (RR47).

If $\lambda_\Theta \geq N^\theta$, the moving graph has $N^{3/5-\theta+o(1)}$ occurrences and therefore weighted mass $N^{-2/5-\theta+o(1)}$. Multiplying the critical first-carrier mass $N^{12/5+o(1)}$, the one outer weight $N^{-1+o(1)}$, and the host-pair mass $N^{6/5+o(1)}$ gives $N^{11/5-\theta+o(1)}$. $\square$

The complementary nonresonant fully foreign graph remains open. In particular, RR43–RR44 are a direct deletion of one q -line and do not turn a centered congruence into a common-prefix occurrence.

Separately, the complementary sector to (RR26) has $\mathcal{R}_{11} \neq 0$. The rational-centre argument above gives no power saving there: its progression modulus is coupled to the nonzero determinant value. In particular, the exact diagonal deletion must not be quoted as a bound for all of $\mathfrak{E}^{00}$.

The density gain in (RR10) is unconditional, but the centered term in (RR9) presently has no fixed-power bound. In particular, neither the second divisor exposed by a collision of two forced residues nor (RR13) creates a second N^{-1} outer weight. This scalar decomposition supplies no automatic contraction of the signed short- d shell operator.

The bound (PS10) is exactly $N^{1/50+o(1)}$ above the critical $N^{159/100}$ inner-shell scale. It is a completed-row statement, not a claim that an arbitrary row-dependent mask is contractive in the complex fixed- d form. Thus it identifies the pre-majorant reciprocal-Parseval deficit but does not prove (PS–SDC).

The missing power cannot be obtained merely by making the reciprocal prime samples equidistributed. The following identity isolates the multiplicative principal mode in either one-sided Cauchy profile.

Proposition 213 (Principal mode of the nonzero reciprocal Parseval profile). *Let $\mathcal{D} \subset \mathbb{Z}$ satisfy $0 < |d| < p$ for every $d \in \mathcal{D}$ and every shell prime p . Put*

$$F_p(a) = \sum_{r \in \mathbb{Z}_p} e_p(ar), \quad w_q = |u_q|, \quad \mathcal{U}_N = \sum_q w_q,$$

and, for $a \in \mathbb{F}_p^\times$, define the reciprocal sampling measure

$$M_p(a) = \sum_{d \in \mathcal{D}} \sum_{q \neq p} w_q \mathbf{1}_{\{a=d\bar{q} \pmod{p}\}}. \quad (\text{PS11})$$

Then

$$\sum_{a \neq 0} |F_p(a)|^2 = pZ(p) - Z(p)^2, \quad \frac{1}{p-1} \sum_{a \neq 0} |F_p(a)|^2 = \frac{Z(p)(p - Z(p))}{p-1}. \quad (\text{PS12})$$

Writing

$$\mu_p = \frac{|\mathcal{D}|(\mathcal{U}_N - w_p)}{p-1}, \quad \overline{H}_p = \frac{Z(p)(p - Z(p))}{p-1},$$

one has the exact principal–discrepancy decomposition

$$\begin{aligned}
& \sum_p w_p \sum_{a \neq 0} M_p(a) |F_p(a)|^2 \\
&= |\mathcal{D}| \sum_p w_p (\mathcal{U}_N - w_p) \frac{Z(p)(p - Z(p))}{p - 1} \\
&\quad + \sum_p w_p \sum_{a \neq 0} (M_p(a) - \mu_p) (|F_p(a)|^2 - \overline{H}_p). \tag{PS13}
\end{aligned}$$

If $|\mathcal{D}| = N^{41/100+o(1)}$, the principal term in (PS13) is

$$|\mathcal{D}| (\mathcal{U}_N \mathcal{B}_N - \mathcal{E}_N) + O_v \left(N^{203/300+o(1)} \right), \tag{PS14}$$

where the masses are those of (AP9).

Proof. Additive orthogonality gives

$$\sum_{a \bmod p} |F_p(a)|^2 = pZ(p).$$

Subtracting the zero frequency, whose square is $Z(p)^2$, proves (PS12). Moreover

$$\sum_{a \neq 0} M_p(a) = |\mathcal{D}| (\mathcal{U}_N - w_p).$$

Subtract the two displayed means inside the product $M_p(a)|F_p(a)|^2$ and sum in a ; both mixed centered terms vanish, which proves (PS13). Replacing $(p - Z(p))/(p - 1)$ by 1 produces $|\mathcal{D}| (\mathcal{U}_N \mathcal{B}_N - \mathcal{E}_N)$. The error is

$$\ll_v \frac{|\mathcal{D}|}{N} \sum_p w_p Z(p)^2 \ll_v \frac{|\mathcal{D}|}{N} N^{2/3+o(1)} \mathcal{B}_N \ll_v N^{203/300+o(1)},$$

using (AP9) and the pointwise $Z(p) \ll N^{2/3+o(1)}$ bound. $\square$

The quadratic principal mode in the preceding proposition is created by factorwise Cauchy. It is not a large mode of the original bilinear form. The following centering statement makes this distinction quantitative.

Proposition 214 (Linear reciprocal centering before Cauchy). *For $a \in \mathbb{F}_p^\times$ put*

$$m_p = \frac{Z(p)}{p - 1}, \quad F_p^\circ(a) = F_p(a) + m_p.$$

Then

$$\frac{1}{p - 1} \sum_{a \neq 0} F_p(a) = -m_p, \quad \sum_{a \neq 0} F_p^\circ(a) = 0. \tag{PS14a}$$

For the form in (PS8), define

$$B_d^{\circ\circ} = \sum_{\substack{p, q \lesssim_v N \\ p \neq q}} u_p u_q \omega_{p, q}(d) F_p^\circ(-d\bar{q}) F_q^\circ(-d\bar{p}).$$

Uniformly for $0 < |d| < c_v N$ one has

$$B_d^{\text{comp}} = B_d^{\circ\circ} + O_v(N^{-1/15+o(1)}). \quad (\text{PS14b})$$

Consequently, when $D = N^{41/100}$,

$$\sum_{0 < |d| \asymp D} |B_d^{\text{comp}}|^2 = \sum_{0 < |d| \asymp D} |B_d^{\circ\circ}|^2 + O_v(N^{283/300+o(1)}). \quad (\text{PS14c})$$

Proof. Since $0 \notin \mathcal{Z}_p$, additive orthogonality gives

$$\sum_{a \neq 0} F_p(a) = \sum_{r \in \mathcal{Z}_p} \sum_{a \neq 0} e_p(ar) = -Z(p),$$

which proves (PS14a). Write

$$M_1 = \sum_p |u_p| m_p, \quad M_2 = \sum_p |u_p| m_p^2.$$

The two-level masses and the pointwise zero bound give

$$M_1 \ll_v N^{-1} \mathcal{B}_N \ll_v N^{-2/5+o(1)}, \quad M_2 \leq (\max_p m_p) M_1 \ll_v N^{-11/15+o(1)}.$$

Expanding $F_p = F_p^\circ - m_p$, the mean–mean term is $O_v(M_1^2)$. For either mixed term, weighted Cauchy–Schwarz gives

$$\begin{aligned} \left| \sum_{p \neq q} u_p u_q \omega_{p,q}(d) m_p F_q^\circ(-d\bar{p}) \right|^2 &\leq \mathcal{U}_N M_2 \sum_{p \neq q} |u_p u_q| |F_q^\circ(-d\bar{p})|^2 \\ &\ll_v \mathcal{U}_N M_2 \mathcal{B}_N \ll_v N^{-2/15+o(1)}. \end{aligned}$$

Here the last one-sided bound follows exactly as in Proposition 193, since centering only subtracts the mean on $\mathbb{F}_q^\times$ and hence its nonzero-frequency square sum is at most the uncentered one. This proves (PS14b).

The shell squared norm of the error is $O_v(DN^{-2/15+o(1)}) = O_v(N^{83/300+o(1)})$. Both the original and centered main shell vectors obey the $N^{161/200+o(1)}$ norm endpoint by the same two-Parseval argument. Cauchy–Schwarz for their mixed energy therefore gives

$$N^{161/200+83/600+o(1)} = N^{283/300+o(1)},$$

which proves (PS14c). □

Remark 215 (Linear centering and palindromic reality are different normalizations). Put

$$\epsilon_p = \mathbf{1}_{\{(p-1)/2 \in \mathcal{Z}_p\}}, \quad G_p(a) = e_p(a/2) F_p(a).$$

Reflection $r \mapsto p-1-r$ makes $G_p(a)$ real, but

$$\sum_{a \neq 0} G_p(a) = p\epsilon_p - Z(p). \quad (\text{PS14d})$$

Indeed, after the rotation each zero contributes $\sum_{a \neq 0} e_p(a(r + 1/2))$, which is $p - 1$ at the unique fixed point $r = (p - 1)/2$ and is -1 otherwise. On the other hand,

$$e_p(a/2)F_p^\circ(a) = G_p(a) + m_p e_p(a/2)$$

is not generally real. Thus the negligible linear centering in Proposition 214 and the real palindromic rotation cannot simply be imposed simultaneously. Centering G_p by its own real mean preserves reality, but that mean is of constant size when $\epsilon_p = 1$; such a step requires a separate treatment of the midpoint sector and is not supplied by (PS14c).

Proposition 216 (Midpoint removal and centered real reciprocal normal form). *Put $c_p = (p - 1)/2$ and*

$$\epsilon_p = \mathbf{1}_{\{c_p \in \mathcal{Z}_p\}}, \quad Z^{\text{nc}}(p) = Z(p) - \epsilon_p, \quad F_p^{\text{nc}}(a) = F_p(a) - \epsilon_p e_p(ac_p).$$

Let B_d^{nc} denote the form (PS8) with both Fourier factors replaced by their noncentral parts. At $D = N^{41/100}$,

$$\sum_{0 < |d| \asymp D} |B_d^{\text{comp}} - B_d^{\text{nc}}|^2 \ll_v N^{101/100+o(1)}, \quad (\text{PS14e})$$

and hence

$$\sum_{0 < |d| \asymp D} |B_d^{\text{comp}}|^2 = \sum_{0 < |d| \asymp D} |B_d^{\text{nc}}|^2 + O_v(N^{131/100+o(1)}). \quad (\text{PS14f})$$

Define, on $\mathbb{F}_p^\times$,

$$G_p^{\text{nc}}(a) = e_p(a/2)F_p^{\text{nc}}(a), \quad G_p^{\text{nc},\circ}(a) = G_p^{\text{nc}}(a) + \frac{Z^{\text{nc}}(p)}{p - 1}.$$

Then both functions are real-valued and

$$\sum_{a \neq 0} G_p^{\text{nc},\circ}(a) = 0. \quad (\text{PS14g})$$

Finally put

$$C_d^{\text{cr}} = (-1)^d \sum_{\substack{p, q \asymp_v N \\ p \neq q}} u_p u_q \omega_{p,q}(d) e\left(\frac{d}{2pq}\right) G_p^{\text{nc},\circ}(-d\bar{q}) G_q^{\text{nc},\circ}(-d\bar{p}).$$

Then

$$\sum_{0 < |d| \asymp D} |B_d^{\text{comp}}|^2 = \sum_{0 < |d| \asymp D} |C_d^{\text{cr}}|^2 + O_v(N^{131/100+o(1)}). \quad (\text{PS14h})$$

Thus the critical shell may be reduced losslessly to centered real reciprocal amplitudes. This statement supplies no angle saving between the two amplitudes.

Proof. Expand each F_p into its noncentral part and its possible singleton midpoint contribution. For a term with exactly one midpoint factor, weighted Cauchy–Schwarz and the one-sided Parseval estimate give, uniformly in d ,

$$|B_d^{\text{mid,nc}}|^2 \ll_v \left(\sum_{p,q} |u_p u_q| \epsilon_p \right) \left(\sum_{p,q} |u_p u_q| |F_q^{\text{nc}}(-d\bar{p})|^2 \right) \ll_v \mathcal{U}_N^2 \mathcal{B}_N \ll_v N^{3/5+o(1)}.$$

The opposite mixed term is identical, and the two-midpoint term is smaller. Summing over the D -shell proves (PS14e). Both B^{comp} and B^{nc} satisfy the two-Parseval shell endpoint $N^{161/100+o(1)}$. Cauchy–Schwarz between this endpoint and (PS14e) gives

$$N^{161/200+101/200+o(1)} = N^{131/100+o(1)},$$

which proves (PS14f).

Equivalently, after expanding the shell energy into four Fourier factors, a stratum with exactly one midpoint factor is a cross term between a one-midpoint row and a full row. Its size is therefore at most

$$N^{(101/100+161/100)/2+o(1)} = N^{131/100+o(1)},$$

not the full endpoint. A split two-midpoint stratum, with one midpoint in each row, is a product of two one-midpoint rows and is bounded by $N^{101/100+o(1)}$. Thus the row-level estimate above already charges all midpoint strata in the four-factor expansion; no characteristic-level purge of midpoint-zero primes is being assumed.

The noncentral zero set is reflection-stable and has no fixed point. Pairing reflected zeros makes G_p^{nc} real, while additive orthogonality gives

$$\sum_{a \neq 0} G_p^{\text{nc}}(a) = -Z^{\text{nc}}(p).$$

This proves (PS14g). For the standard inverses,

$$q\bar{q} + p\bar{p} = 1 + pq,$$

and reduction modulo 2 shows that $\bar{q} + \bar{p}$ is even. Consequently

$$F_p^{\text{nc}}(-d\bar{q})F_q^{\text{nc}}(-d\bar{p}) = (-1)^d e\left(\frac{d}{2pq}\right) G_p^{\text{nc}}(-d\bar{q})G_q^{\text{nc}}(-d\bar{p}).$$

Centering the two real factors now has exactly the same mixed-term bounds as Proposition 214, with $Z^{\text{nc}}(p) \leq Z(p)$. It changes the shell energy by at most $N^{283/300+o(1)}$. Combining this with (PS14f) proves (PS14h). $\square$

There is also a sharper normalization which avoids splitting off the reflection midpoint. It adds a balanced artificial pair at the two empty reflection endpoints. This preserves both the exact nonzero-frequency centering and pointwise reality after the palindromic rotation.

Proposition 217 (Reflection-paired endpoint centering). *For $a \in \mathbb{F}_p^\times$ put*

$$\lambda_p = \frac{Z(p)}{p-2}, \quad F_p^\diamond(a) = F_p(a) + \lambda_p(1 + e_p(-a)).$$

Then

$$\sum_{a \neq 0} F_p^\diamond(a) = 0, \quad H_p^\diamond(a) := e_p(a/2)F_p^\diamond(a) \in \mathbb{R}. \quad (\text{PS14i})$$

Let $B_d^{\diamond\diamond}$ be the form (PS8) with both Fourier factors replaced by their $\diamond$ versions. Uniformly for $0 < |d| < c_v N$,

$$B_d^{\text{comp}} = B_d^{\diamond\diamond} + O_v(N^{-1/15+o(1)}). \quad (\text{PS14j})$$

Consequently, at $D = N^{41/100}$,

$$\sum_{0 < |d| \lesssim D} |B_d^{\text{comp}}|^2 = \sum_{0 < |d| \lesssim D} |B_d^{\diamond\diamond}|^2 + O_v(N^{283/300+o(1)}). \quad (\text{PS14k})$$

Moreover, if

$$C_d^\diamond = (-1)^d \sum_{\substack{p, q \asymp_v N \\ p \neq q}} u_p u_q \omega_{p,q}(d) e\left(\frac{d}{2pq}\right) H_p^\diamond(-d\bar{q}) H_q^\diamond(-d\bar{p}),$$

then $B_d^{\diamond\diamond} = C_d^\diamond$ exactly. Thus the critical shell is equivalent, up to the error in (PS14k), to a form with real centered Apéry amplitudes. The multiplier $\omega_{p,q}(d)e(d/(2pq))$ need not be real or transpose-symmetric, and no angle saving is asserted.

Proof. The two endpoint residues 0 and $p-1$ are absent from $\mathcal{Z}_p$: the first because $b_0 = 1$, and the second by reflection. Since

$$\sum_{a \neq 0} 1 = p-1, \quad \sum_{a \neq 0} e_p(-a) = -1,$$

the definition of λ_p and (PS14a) give the first identity in (PS14i). Reflection makes $e_p(a/2)F_p(a)$ real, while

$$e_p(a/2)(1 + e_p(-a)) = e_p(a/2) + e_p(-a/2)$$

is real as well. This proves the second identity.

Put

$$L_1 = \sum_p |u_p| \lambda_p, \quad L_2 = \sum_p |u_p| \lambda_p^2.$$

The two-level masses and the pointwise zero bound give

$$L_1 \ll_v N^{-1} \mathcal{B}_N \ll_v N^{-2/5+o(1)}, \quad L_2 \leq (\max_p \lambda_p) L_1 \ll_v N^{-11/15+o(1)}.$$

Write $\Delta_p(a) = \lambda_p(1 + e_p(-a))$, so that $|\Delta_p(a)| \leq 2\lambda_p$. Expanding $F_p = F_p^\diamond - \Delta_p$, either mixed term is bounded by the same weighted Cauchy–Schwarz calculation as in Proposition 214:

$$|\text{mixed term}|^2 \ll_v \mathcal{U}_N L_2 \mathcal{B}_N \ll_v N^{-2/15+o(1)}.$$

Here the one-sided square profile of $F^\diamond$ is still $O_v(\mathcal{B}_N N^{o(1)})$, either by finite Parseval for the actual zero set plus the two endpoint masses or by expanding $F^\diamond = F + \Delta$. The term containing two copies of Δ is $O_v(L_1^2)$ and is smaller. This proves (PS14j).

The shell square of the error is $O_v(DN^{-2/15+o(1)}) = O_v(N^{83/300+o(1)})$. Combining it by Cauchy–Schwarz with the $N^{161/100+o(1)}$ two-Parseval shell endpoint gives the energy error

$$N^{161/200+83/600+o(1)} = N^{283/300+o(1)},$$

which proves (PS14k). Finally the reciprocity identity used in the proof of Proposition 216 applies unchanged to $F^\diamond$ and gives $B_d^{\diamond\diamond} = C_d^\diamond$. $\square$

The centered-real normal form also rules out every uniformly low-dimensional multiplicative-character explanation of the endpoint. This is a genuine restriction on the remaining obstruction, although it does not control the full mutual-character kernel.

Proposition 218 (Low even-character sectors are subcritical). *For every shell prime p , let $\mathfrak{X}_p$ be a conjugation-stable set of nonprincipal even multiplicative characters modulo p , and suppose*

$$|\mathfrak{X}_p| \leq N^\rho.$$

For $a \in \mathbb{F}_p^\times$ define the multiplicative projection

$$G_{p,\mathfrak{X}}^{\text{nc},\circ}(a) = \frac{1}{p-1} \sum_{\chi \in \mathfrak{X}_p} \widehat{G}_p(\chi) \chi(a), \quad \widehat{G}_p(\chi) = \sum_{b \neq 0} G_p^{\text{nc},\circ}(b) \overline{\chi(b)}. \quad (\text{PS14l})$$

Let $C_{d,\mathfrak{X}}^{(1)}$ be the form C_d^{cr} in Proposition 216 with its first real amplitude replaced by $G_{p,\mathfrak{X}}^{\text{nc},\circ}$, and define $C_{d,\mathfrak{X}}^{(2)}$ by projecting the second amplitude. Then, at $D = N^{41/100}$,

$$\sum_{0 < |d| \asymp D} \left(|C_{d,\mathfrak{X}}^{(1)}|^2 + |C_{d,\mathfrak{X}}^{(2)}|^2 \right) \ll_v N^{383/300 + \rho + o(1)}. \quad (\text{PS14m})$$

The same bound holds for the row component in which at least one of the two amplitudes lies in its corresponding $\mathfrak{X}$ -projection. Its self-energy is below the critical $N^{159/100}$ shell scale when

$$\rho < \frac{47}{150}. \quad (\text{PS14n-a})$$

To remove this component from the full shell energy one must also pay its mixed inner product with the complementary high-high component. That complete low-character correction is below the critical scale in the stricter range

$$\rho < \frac{22}{75}. \quad (\text{PS14n-b})$$

This does not bound the complementary high-dimensional mutual-character correlation.

Proof. Because $G_p^{\text{nc},\circ}$ is real and even, its odd-character coefficients vanish. For a nonprincipal even character, the constant centering contributes zero and the Gauss identity gives

$$\widehat{G}_p(\chi) = \tau_p(\overline{\chi}) \sum_{r \in \mathbb{Z}_p^{\text{nc}}} \chi(r + 1/2), \quad |\tau_p(\overline{\chi})| = \sqrt{p}.$$

Consequently multiplicative Parseval gives

$$\sum_{a \neq 0} |G_{p,\mathfrak{X}}^{\text{nc},\circ}(a)|^2 = \frac{1}{p-1} \sum_{\chi \in \mathfrak{X}_p} |\widehat{G}_p(\chi)|^2 \ll N^\rho Z(p)^2. \quad (\text{PS14o})$$

For fixed p and nonzero d , the map $q \mapsto d\bar{q} \pmod{p}$ has bounded multiplicity on a fixed-ratio prime box. Since $\max_q |u_q| \ll_v N^{-1+o(1)}$,

$$\sum_{q \neq p} |u_q| |G_{p,\mathfrak{X}}^{\text{nc},\circ}(-d\bar{q})|^2 \ll_v N^{-1+\rho+o(1)} Z(p)^2.$$

The two-level masses therefore imply

$$\begin{aligned} \mathcal{E}_{d,\mathfrak{X}}^{(1)} &:= \sum_p |u_p| \sum_{q \neq p} |u_q| |G_{p,\mathfrak{X}}^{\text{nc},\circ}(-d\bar{q})|^2 \\ &\ll_v N^{-1+\rho+o(1)} \sum_p |u_p| Z(p)^2 \\ &\ll_v N^{-1+\rho+o(1)} (\max_p Z(p)) \mathcal{B}_N \ll_v N^{\rho+4/15+o(1)}. \end{aligned}$$

The unprojected opposite profile is $O_v(\mathcal{B}_N N^{o(1)}) = O_v(N^{3/5+o(1)})$. Weighted Cauchy–Schwarz in (p, q) , with the literal bounded multiplier retained, now gives

$$|C_{d,\mathfrak{X}}^{(1)}|^2 \ll_v N^{\rho+13/15+o(1)}.$$

The same argument applies to the second factor. Summing over the $D = N^{41/100}$ shell proves (PS14m), since

$$\frac{41}{100} + \frac{13}{15} = \frac{383}{300}.$$

The row component with at least one projected factor is obtained by inclusion–exclusion from the two one-sided projections and the twice-projected form; the latter obeys either one-sided bound. Since $159/100 - 383/300 = 47/150$, this proves (PS14n-a).

Let $C_{d,\text{low}}$ denote this low-containing row component and $C_{d,>}$ the complementary high–high component. The full endpoint gives

$$\sum_d |C_{d,>}|^2 \ll_v N^{161/100+o(1)}.$$

Therefore Cauchy–Schwarz gives

$$\left| \sum_d C_{d,\text{low}} \overline{C_{d,>}} \right| \ll_v N^{433/300+\rho/2+o(1)}.$$

This is below $N^{159/100}$ precisely when $\rho < 22/75$. Together with the self-energy bound, this proves (PS14n-b). No orthogonality between the two multiplicative projections is asserted under the reciprocal-prime sampling measure. $\square$

Proposition 219 (One frequency average and actual small-gap localization). *Let $\mathcal{D}$ be an interval of $N^{41/100+o(1)}$ consecutive nonzero integers of absolute value $< c_v N$, and put*

$$K_{p,\mathcal{D}}(x) = \sum_{d \in \mathcal{D}} e_p(dx).$$

For every $h \not\equiv 0 \pmod{p}$ one has

$$\sum_{\substack{q \asymp_v N \\ q \neq p}} |u_q| |K_{p,\mathcal{D}}(h\bar{q})| \ll_v N^{o(1)}. \quad (\text{PS15})$$

The same estimate holds when the summands in $K_{p,\mathcal{D}}$ are multiplied by a fixed smooth dyadic weight, by summation by parts. For $2 \leq h < p$, let

$$A_p(h) = \#\{r : 0 \leq r < r+h < p, \ r, r+h \in \mathcal{Z}_p\}.$$

Uniformly for $2 \leq H \leq \sqrt{N}$,

$$\sum_{p \asymp_v N} |u_p| \sum_{h=2}^H A_p(h) \ll_v N^{o(1)} \min \left\{ H^2, N^{3/5} + N^{3/10} H^{4/3}, N^{19/15} \right\}. \quad (\text{PS16})$$

Consequently the part of either one-sided reciprocal Cauchy profile whose two actual zero labels have nonwrapping integer gap at most H obeys the same bound. In particular, for every fixed $\varepsilon > 0$, the choice

$$H = N^{1/2-\varepsilon}$$

makes that contribution

$$\ll_v N^{29/30-(4/3)\varepsilon+o(1)}. \quad (\text{PS17})$$

In particular this is below the critical one-sided scale $N^{99/100}$ by the fixed power $N^{7/300+o(1)}$ even at the endpoint of the proved range. No claim is made here for $H > \sqrt{N}$. The reversed ordering of the two labels costs only a factor 2.

Proof. For fixed p , the map $q \mapsto h\bar{q} \pmod{p}$ has bounded multiplicity on a fixed-ratio shell. The pointwise prime-weight bound and the ℓ^1 norm of the Dirichlet kernel therefore give

$$\sum_{q \neq p} |u_q| |K_{p,\mathcal{D}}(h\bar{q})| \ll_v N^{-1+o(1)} \sum_{a \pmod{p}} |K_{p,\mathcal{D}}(a)| \ll_v N^{o(1)},$$

which proves (PS15).

For an actual zero start r , put

$$k_{p,r}^{\mathcal{Z}}(H) = \#\{2 \leq h \leq H : r+h < p, r+h \in \mathcal{Z}_p\}.$$

This is at most the number $k_r(H)$ of gap-polynomial levels used in the proof of Corollary 47. That proof gives the stronger intermediate estimate

$$\sum_{r \pmod{p}} \binom{k_r(H)}{2} \ll H^{8/3}.$$

Writing

$$S_H^\circ = \sum_p |u_p| \sum_{r \in \mathcal{Z}_p \setminus \partial_H} k_{p,r}^{\mathcal{Z}}(H),$$

weighted Cauchy–Schwarz and $k^2 = k + 2\binom{k}{2}$ yield

$$(S_H^\circ)^2 \leq \mathcal{B}_N \sum_p |u_p| \sum_{r \in \mathcal{Z}_p \setminus \partial_H} (k_{p,r}^{\mathcal{Z}}(H))^2 \ll_v \mathcal{B}_N (S_H^\circ + \mathcal{U}_N H^{8/3}).$$

Here the factorial-moment term is bounded by the exact off-boundary identity in Proposition 46 and Corollary 47. If an actual nonwrapping pair starts at $r \in \partial_H$, reflection $x \mapsto p-1-x$ sends it injectively to an actual pair of the same gap starting outside ∂_H . Hence the full incidence mass is at most $2S_H^\circ$. Solving the displayed quadratic inequality and using (AP9) gives

$$\sum_p |u_p| \sum_{h=2}^H A_p(h) \ll_v N^{o(1)} (N^{3/5} + N^{3/10} H^{4/3}).$$

The alternatives follow respectively from $A_p(h) \leq 3(h-1)$ and

$$\sum_h A_p(h) \leq Z(p)^2, \quad \sum_p |u_p| Z(p)^2 \ll_v N^{2/3+o(1)} \mathcal{B}_N.$$

This proves (PS16).

After expanding $|F_p(d\bar{q})|^2$, summing in d produces $K_{p,\mathcal{D}}((r-s)\bar{q})$. Apply (PS15) and then (PS16). Finally

$$\frac{3}{10} + \frac{4}{3} \left(\frac{1}{2} - \varepsilon \right) = \frac{29}{30} - \frac{4}{3} \varepsilon,$$

which proves (PS17). □

Remark 220 (Why one-sided equidistribution is not the missing defect). If the reciprocal samples in (PS11) were perfectly uniform on $\mathbb{F}_p^\times$, the discrepancy term in (PS13) would vanish and the one-sided profile would have its existing $|\mathcal{D}|\mathcal{U}_N\mathcal{B}_N$ scale. Thus completion, Kloosterman cancellation, or prime equidistribution which only makes M_p close to its mean is an endpoint mechanism, not a power-saving mechanism. A proof of the missing $N^{1/50+o(1)}$ shell saving must instead improve the zero-count mass itself or retain a two-sided reciprocal-angle defect before the factorwise Cauchy step in Proposition 193.

The determinant coordinate still gives exact arithmetic fibres even though it does not by itself supply the missing analytic cancellation. The next three statements retain both circular branches and every defining-characteristic zero label.

Lemma 221 (Signed circular determinant and reciprocal labels). *Let $P = pq$ and $P' = p'q'$ be coprime products of two odd primes, and let $t/P, t'/P'$ be reduced. Put*

$$\Delta = tP' - t'P, \quad M = PP'.$$

There is a unique $\delta \in (-M/2, M/2]$ satisfying $\delta \equiv \Delta \pmod{M}$. If $t/P \neq t'/P'$, then $\delta \neq 0$, $(\delta, M) = 1$, and

$$\left\| \frac{t}{P} - \frac{t'}{P'} \right\|_{\mathbb{T}} = \frac{|\delta|}{M}.$$

For every integer d ,

$$e\left(d\left(\frac{t}{P} - \frac{t'}{P'}\right)\right) = e\left(\frac{d\delta}{M}\right).$$

Moreover the four CRT labels are

$$\delta\overline{P'} \pmod{p}, \quad \delta\overline{P'} \pmod{q}, \quad -\delta\overline{P} \pmod{p'}, \quad -\delta\overline{P} \pmod{q'}.$$

Thus the signed circular lift changes neither circular branch, the d -phase, nor any defining-characteristic Apéry-zero test.

Proof. Take the unique representative of Δ modulo the odd integer M in $(-M/2, M/2]$. The circular-distance identity is immediate, and replacing Δ by $\Delta + kM$ changes $d\Delta/M$ by the integer dk . The labels depend only on the determinant residue modulo the corresponding prime. Finally reducedness and $(P, P') = 1$ give $(\Delta, M) = 1$, hence $(\delta, M) = 1$. $\square$

Lemma 222 (Three-characteristic short-progression fibre). *Let p, q, p', q' be pairwise distinct odd primes, put $P = pq$, $P' = p'q'$, and let*

$$r \in \mathbb{Z}_p, \quad s \in \mathbb{Z}_q, \quad r' \in \mathbb{Z}_{p'}.$$

Let $t = t_{p,q}(r, s)$ be the actual CRT row and put

$$Q_3 = Pp', \quad \delta_0 = tP' - r'P.$$

The integers satisfying the first three reciprocal-label congruences

$$\delta\overline{P'} \equiv r \pmod{p}, \quad \delta\overline{P'} \equiv s \pmod{q}, \quad -\delta\overline{P} \equiv r' \pmod{p'}$$

are exactly

$$\delta_n = \delta_0 + nQ_3, \quad n \in \mathbb{Z}. \tag{SF1}$$

For such an integer the forced fourth residue is

$$s_n \equiv -\delta_n \bar{P} \equiv r' - np' \pmod{q'}. \quad (\text{SF2})$$

If in addition $|\delta_n| < PP'/M$, then

$$\left| n - \left(\frac{r'}{p'} - \frac{tq'}{P} \right) \right| < \frac{q'}{M}. \quad (\text{SF3})$$

Equivalently, with $m_n = r' - np'$,

$$m_n \bmod p' = r', \quad m_n \bmod q' = s_n, \quad \delta_n = tP' - m_n P. \quad (\text{SF4})$$

Thus, after three actual zero labels have been fixed, the fourth actual zero is sampled on an affine progression of $O(1 + q'/M)$ consecutive lift indices. Any further high-centered, alias, near, or transverse deletion mask only removes indices from this progression.

Proof. The three displayed congruences determine one residue class modulo $ppq' = Q_3$ by the Chinese remainder theorem, and δ_0 represents that class. This proves (SF1). Modulo q' one has $tP' \equiv 0$ and $Q_3 \bar{P} \equiv p'$, whence $-\delta_n \bar{P} \equiv r' - np'$, proving (SF2). Finally

$$\frac{\delta_0}{Q_3} = \frac{tq'}{P} - \frac{r'}{p'},$$

so $|\delta_n| < Q_3 q'/M$ is precisely (SF3). The identities in (SF4) are immediate. $\square$

The last progression in Lemma 222 admits a two-characteristic average which keeps its selected center. This is the first place where the moving foreign prime is useful rather than an obstruction.

Proposition 223 (Weighted selected affine-progression sampler). *Let $q \asymp N$ be prime and let $\mathcal{Z}_q \subset \mathbb{F}_q$ be the actual Apéry zero set. Let $1 \leq H < q$. For each shell prime $p \neq q$ and each $r \in \mathcal{Z}_p$, let $I_{p,r} \subset \mathbb{Z}$ be an interval of at most H consecutive integers, with arbitrary center, and let $0 \leq m_{p,r}(n) \leq 1$ be supported on $I_{p,r}$. Define*

$$S_{p,r} = \sum_{n \in I_{p,r}} m_{p,r}(n) \mathbf{1}_{\mathcal{Z}_q}(r - np \bmod q).$$

For nonnegative weights λ_p , put

$$B = \sum_{p \asymp N} \lambda_p Z(p), \quad L = \max_{p \asymp N} \lambda_p Z(p).$$

On one dyadic shell, and with an additional constant on any fixed ratio box,

$$\sum_{p \asymp N} \lambda_p \sum_{r \in \mathcal{Z}_p} S_{p,r}^2 \ll HB + HLZ(q)^2, \quad (\text{AP1})$$

and consequently

$$\left(\sum_{p \asymp N} \lambda_p \sum_{r \in \mathcal{Z}_p} S_{p,r} \right)^2 \ll B(HB + HLZ(q)^2). \quad (\text{AP2})$$

The intervals and the functions $m_{p,r}$ may depend arbitrarily on p, r ; in particular they may retain all previously imposed deletion masks.

Proof. For $a \in \mathbb{F}_q$ put

$$A_q(a) = \#\{y \in \mathcal{Z}_q : y - a \in \mathcal{Z}_q\}.$$

Double counting ordered pairs of distinct zeros gives

$$\sum_{a \neq 0} A_q(a) = Z(q)(Z(q) - 1). \quad (\text{AP3})$$

For fixed p, r , expand $S_{p,r}^2$ according to the positive integer difference h between two selected indices. Since $H < q$ and $p \neq q$, the map $n \mapsto r - np \pmod{q}$ is injective on $I_{p,r}$. Using $0 \leq m_{p,r} \leq 1$ therefore gives

$$S_{p,r}^2 \leq S_{p,r} + 2 \sum_{1 \leq h < H} A_q(hp \pmod{q}). \quad (\text{AP4})$$

After summing over $r \in \mathcal{Z}_p$, the diagonal terms contribute at most HB . For fixed $0 < h < q$, multiplication by h permutes $\mathbb{F}_q^\times$. Distinct primes p in a dyadic interval of length smaller than q have distinct residues modulo q ; on a fixed ratio box the residue multiplicity is $O(1)$. Hence (AP3) gives

$$\sum_p \lambda_p Z(p) A_q(hp) \ll LZ(q)^2.$$

Summing h proves (AP1). Finally, weighted Cauchy-Schwarz gives

$$\left(\sum_{p,r} \lambda_p S_{p,r} \right)^2 \leq \left(\sum_{p,r} \lambda_p \right) \left(\sum_{p,r} \lambda_p S_{p,r}^2 \right) = B \sum_{p,r} \lambda_p S_{p,r}^2,$$

which proves (AP2). $\square$

For the exact intervals in Lemma 222, take $q = q'$, $p = p'$, $r = r'$, and $H \ll 1 + q'/M$. At $M = 4N^{41/100}$ one has $H \ll N^{59/100}$. If $\lambda_p \ll |u_p|$, the proved first zero moment and the pointwise bound $Z(p) \ll p^{2/3}$ give

$$B \ll N^{3/5+o(1)}, \quad L \ll N^{-1/3+o(1)}, \quad Z(q)^2 \ll N^{4/3+o(1)}.$$

Thus Proposition 223 yields

$$\sum_{p \asymp N} \lambda_p \sum_{r \in \mathcal{Z}_p} S_{p,r}^2 \ll N^{159/100+o(1)}$$

and

$$\sum_{p \asymp N} \lambda_p \sum_{r \in \mathcal{Z}_p} S_{p,r} \ll N^{219/200+o(1)}. \quad (\text{AP5})$$

The direct bound $S_{p,r} \leq H$ gives instead $HB \ll N^{238/200+o(1)}$. Hence the selected two-characteristic sampler itself gains $N^{-19/200+o(1)}$ over its trivial weighted length. Promoting this local gain through the complete four-prime short-arc ledger requires retaining every outer prime and first-row weight; no such promotion is used in the statements above.

Proposition 224 (Global accounting for the selected sampler). *Let $M = 4N^{41/100}$ and let $\mathcal{A}_{M,4\text{dist}}^{\#, \text{hi}}$ denote the all-four-distinct part of the high-centered semiprime short-arc quadratic form, with every current mask retained and separated smooth twists bounded by 1. For a first-row atom $x = (p, q, r, s, t)$ and a fixed ordered fourth characteristic q' , let $\Sigma_{x,q'}$ be the weighted selected sampler above. Then*

$$|\mathcal{A}_{M,4\text{dist}}^{\#, \text{hi}}| \leq \frac{1}{M} \sum_x |u_p u_q| \sum_{q'} |u_{q'}| \Sigma_{x,q'}. \quad (\text{AP6})$$

Consequently the local bound (AP5) gives only

$$|\mathcal{A}_{M,4\text{dist}}^{\sharp,\text{hi}}| \ll N^{377/200+o(1)}.$$

Using the local second moment before Cauchy–Schwarz in q' improves this only to

$$|\mathcal{A}_{M,4\text{dist}}^{\sharp,\text{hi}}| \ll N^{1091/600+o(1)}. \quad (\text{AP7})$$

In particular the local saving does not prove (PS–SDC).

Proof. Expand the cyclic short-arc square. For fixed x, q' , the three-characteristic parametrization writes every short-arc second row as (p', r', n) , and its triangular interval-intersection kernel is at most $1/M$. Summing the surviving fourth-zero tests gives (AP6). Moreover

$$\sum_x |u_p u_q| \leq \left(\sum_p |u_p| Z(p) \right)^2 \ll N^{6/5+o(1)}, \quad \sum_{q'} |u_{q'}| \ll N^{o(1)}.$$

Together with (AP5) and $M = N^{41/100+o(1)}$, this gives exponent $-41/100 + 6/5 + 219/200 = 377/200$.

For the stronger correction put

$$U_N = \sum_{q'} |u_{q'}| \ll N^{o(1)}, \quad B_N = \sum_p |u_p| Z(p) \ll N^{3/5+o(1)}, \quad L_N = \max_p |u_p| Z(p) \ll N^{-1/3+o(1)},$$

and put

$$\Sigma_{x,q'} = \sum_{p'} |u_{p'}| \sum_{r' \in \mathcal{Z}_{p'}} S_{x;q',p',r'}.$$

The q' -dependent form of the local second-moment proof gives

$$\Sigma_{x,q'}^2 \ll B_N (H B_N + H L_N Z(q')^2).$$

Taking square roots and summing the actual fourth-prime weight before using a pointwise bound for $Z(q')$ yields

$$\sum_{q'} |u_{q'}| \Sigma_{x,q'} \ll H^{1/2} \left(B_N U_N + B_N^{3/2} L_N^{1/2} \right) \ll N^{617/600+o(1)}.$$

Here the two terms have exponents $179/200$ and $617/600$, respectively. Multiplication by the first-row mass B_N^2 and by M^{-1} therefore gives

$$-\frac{41}{100} + \frac{6}{5} + \frac{617}{600} = \frac{1091}{600},$$

proving (AP7). □

The preceding positive audit pays the first-row L^1 mass. A second square can be taken before that loss and exposes the exact joint collision which remains.

For a surviving first row $x = (p, q, r, s, t)$ put

$$\alpha_x = \frac{t}{pq}, \quad |b_x| \leq |u_p u_q|.$$

For a fourth characteristic q' , a foreign prime p' , and $r' \in \mathcal{Z}_{p'}$, put

$$\xi_{x;q',p',r'} = \frac{r'}{p'} - q'\alpha_x$$

and let

$$\tilde{S}_{x;q',p',r'} := \sum_n \left(1 - \frac{M|n - \xi_{x;q',p',r'}|}{q'}\right)_+ m_{x;q',p',r'}(n) \mathbf{1}_{\mathcal{Z}_{q'}}(r' - np' \bmod q'), \quad (\text{AP8})$$

where $m_{x;q',p',r'}(n) \in [0, 1]$ contains all current row and pair deletions. The signed three-characteristic parametrization shows that both circular branches are included in this formula.

Put

$$\mathcal{U}_N = \sum_p |u_p|, \quad \mathcal{B}_N = \sum_p |u_p|Z(p), \quad \mathcal{E}_N = \sum_p |u_p|^2 Z(p), \quad \mathcal{L}_N = \max_p |u_p|Z(p).$$

Thus

$$\mathcal{U}_N \ll N^{o(1)}, \quad \mathcal{B}_N \ll N^{3/5+o(1)}, \quad \mathcal{E}_N \ll N^{-2/5+o(1)}, \quad \mathcal{L}_N \ll N^{-1/3+o(1)}. \quad (\text{AP9})$$

Proposition 225 (Two-level selected-fibre Cauchy reduction). *Let $M = 4N^{41/100}$ and define*

$$\mathcal{C}_{\text{joint}}^\# := \sum_{q'} |u_{q'}| \sum_{p'} |u_{p'}| \sum_{r' \in \mathcal{Z}_{p'}} \left| \sum_{x \in \mathcal{X}_{\text{hi}}^\#} b_x \tilde{S}_{x;q',p',r'} \right|^2. \quad (\text{AP10})$$

Then the all-four-distinct high-centered form satisfies

$$|\mathcal{A}_{M,4\text{dist}}^{\#, \text{hi}}|^2 \leq \frac{\mathcal{U}_N \mathcal{B}_N}{M^2} \mathcal{C}_{\text{joint}}^\#. \quad (\text{AP11})$$

Consequently the desired short-arc estimate with exponent $77/100 - \eta$ would follow from

$$\mathcal{C}_{\text{joint}}^\# \ll N^{44/25-2\eta+o(1)}. \quad (\text{AP12})$$

Proof. The exact signed-lift expansion is

$$\mathcal{A}_{M,4\text{dist}}^{\#, \text{hi}} = \frac{1}{M} \sum_{q'} \overline{u_{q'} \omega_{q'}} \sum_{p'} \overline{u_{p'} \omega_{p'}} \sum_{r' \in \mathcal{Z}_{p'}} \sum_x b_x \tilde{S}_{x;q',p',r'},$$

where the separated twists ω have modulus at most 1. Weighted Cauchy–Schwarz first in q' and then in the occurrence index (p', r') gives respectively the factors $\mathcal{U}_N$ and $\mathcal{B}_N$, proving (AP11). Since $M^{-2} \mathcal{U}_N \mathcal{B}_N \ll N^{-11/50+o(1)}$, squaring the target $N^{77/100-\eta}$ gives (AP12). $\square$

For $a \in \mathbb{F}_{q'}$ write

$$A_{q'}(a) = \#\{z \in \mathcal{Z}_{q'} : z - a \in \mathcal{Z}_{q'}\}, \quad W_{q'}(h) = \sum_{p'} |u_{p'}|Z(p') A_{q'}(hp' \bmod q'). \quad (\text{AP13})$$

For two first rows define

$$\mathcal{H}_{q'}(x_1, x_2) := \left\{ h \bmod q' : \left\| \alpha_{x_2} - \alpha_{x_1} + \frac{h}{q'} \right\|_{\mathbb{T}} < \frac{3}{M} \right\}. \quad (\text{AP14})$$

Proposition 226 (Selected first-row codegree). *One has*

$$\sum_{h \bmod q'} W_{q'}(h) = \mathcal{B}_N Z(q')^2, \quad (\text{AP15})$$

and, for $h \not\equiv 0 \pmod{q'}$,

$$W_{q'}(h) \ll_v \mathcal{L}_N Z(q')^2. \quad (\text{AP16})$$

Moreover $|\mathcal{H}_{q'}(x_1, x_2)| \ll q'/M$ and

$$\begin{aligned} & \sum_{p'} |u_{p'}| \sum_{r' \in \mathcal{Z}_{p'}} \tilde{S}_{x_1; q', p', r'} \tilde{S}_{x_2; q', p', r'} \\ & \ll_v \frac{q'}{M} \mathcal{B}_N + \sum_{\substack{h \in \mathcal{H}_{q'}(x_1, x_2) \\ h \neq 0 \pmod{q'}}} W_{q'}(h) \\ & \ll_v \frac{q'}{M} (\mathcal{B}_N + \mathcal{L}_N Z(q')^2). \end{aligned} \quad (\text{AP17})$$

At $M = 4N^{41/100}$ the last quantity is $N^{159/100+o(1)}$.

Proof. The correlation identity $\sum_a A_{q'}(a) = Z(q')^2$ gives (AP15). For nonzero h , the map $p' \mapsto hp' \pmod{q'}$ has bounded multiplicity on each fixed-ratio shell, so the definition of $\mathcal{L}_N$ gives (AP16).

The selected centers satisfy exactly

$$\xi_{x_2; q', p', r'} - \xi_{x_1; q', p', r'} = -q'(\alpha_{x_2} - \alpha_{x_1}).$$

Writing $h = n_2 - n_1$, the two triangular supports force $h \in \mathcal{H}_{q'}(x_1, x_2)$. When $h \not\equiv 0 \pmod{q'}$, the two fourth labels differ by $-hp'$ modulo q' . The selected n -interval has length $O(q'/M) < q'$, so $n \mapsto r' - np'$ is injective and the number of pairs is at most $A_{q'}(hp')$. Summing in r' and p' gives $W_{q'}(h)$. The $O(1)$ zero lifts contribute $O((q'/M)\mathcal{B}_N)$. This proves the first bound in (AP17); the second follows from (AP16). Finally $q'/M = N^{59/100+o(1)}$, while $\mathcal{L}_N Z(q')^2 \ll N^{1+o(1)}$. $\square$

Corollary 227 (Diagonal of the two-level collision). *The $x_1 = x_2$ part of (AP10) is*

$$\ll N^{79/100+o(1)}. \quad (\text{AP18})$$

Thus the remaining obstruction is entirely off-diagonal in the first row.

Proof. The preceding proposition with $x_1 = x_2$ gives the inner bound $N^{159/100+o(1)}$. On the other hand

$$\sum_x |b_x|^2 \leq \left(\sum_p |u_p|^2 Z(p) \right)^2 = \mathcal{E}_N^2 \ll N^{-4/5+o(1)},$$

and $\sum_{q'} |u_{q'}| \ll N^{o(1)}$. $\square$

The off-diagonal part of (AP10) is the exact remaining joint restriction problem. The selected shift window in (AP14) moves with the pair of first rows. Freezing it and applying the complete affine square would sum the whole shift family and lose the required power; no such promotion is made here.

Lemma 228 (Fourth-characteristic divisor elimination). *Assume the hypotheses of Lemma 221, write $P = pq$, and let q' be the fourth prime. If $0 \leq s < q'$ is its actual reciprocal label,*

$$s \equiv -\delta \overline{P} \pmod{q'},$$

then

$$q' \mid \delta + Ps, \quad \delta + Ps \neq 0.$$

If all shell primes are $\asymp_v N$ and

$$0 < |\delta| \ll N^{359/100+\varepsilon}, \quad \varepsilon < 41/100,$$

then, for sufficiently large N , each fixed s is compatible with at most three shell primes q' .

Proof. The divisibility is the defining congruence after multiplication by P . If $\delta + Ps = 0$, then $P \mid \delta$, contradicting $(\delta, P) = 1$. Since $s < q' \asymp_v N$ and $P \asymp_v N^2$,

$$0 < |\delta + Ps| \ll_v N^{359/100+\varepsilon}.$$

Four distinct shell-prime divisors would have product $\gg_v N^4$, which is larger than this integer when $\varepsilon < 41/100$ and N is sufficiently large. $\square$

Lemma 229 (Complete-row determinant fibre). *Let (P', t') be a reduced actual semiprime row with $P' \asymp_v N^2$, and fix a signed circular determinant δ with $(\delta, P') = 1$. Any opposite denominator P occurring in a four-distinct pair satisfies*

$$P \equiv -\delta \overline{t'} \pmod{P'}.$$

If the allowed denominators lie in a fixed compact interval $[c_v N^2, C_v N^2]$, there are $O_v(1)$ possible integers P . For each such P , there are at most two ordered factorizations $P = pq$ into distinct primes, and the opposite CRT row is uniquely prescribed by

$$t \equiv \delta \overline{P'} \pmod{P}.$$

Proof. Reduce $\delta \equiv tP' - t'P$ modulo P' . Since $(t', P') = 1$, the first congruence follows. An interval of length $O_v(N^2)$ contains only $O_v(1)$ integers in one residue class modulo $P' \asymp_v N^2$. Unique factorization gives at most two ordered prime presentations, and reduction modulo P gives the final row congruence. $\square$

Remark 230 (Reciprocal rather than translated repeated rows). After conditioning on δ, p, q, p' , two fourth-prime candidates q'_1, q'_2 give actual zero positions

$$r_i \equiv \delta \overline{p'q'_i} \pmod{p}.$$

They satisfy

$$r_1 - r_2 \equiv \delta \overline{p'}(q'_2 - q'_1) \overline{q'_1 q'_2} \pmod{p}.$$

Thus determinant conditioning produces reciprocal-dilated rows, not translated rows of gap $q'_2 - q'_1$. The translated twin theorem cannot be invoked without an additional reciprocal-row input.

The reciprocal fibre itself has three further exact reductions. They keep the moving-characteristic zero test rather than replacing it by a marginal zero density.

Fix a nonzero signed determinant δ and distinct shell primes p, q, p' . Put

$$C = pq, \quad A \equiv \delta(p')^{-1} \pmod{p},$$

and, for a moving shell prime $\ell \notin \{p, q, p'\}$, define standard representatives

$$r_\ell \equiv A\ell^{-1} \pmod{p}, \quad s_\ell \equiv -\delta C^{-1} \pmod{\ell}.$$

Let $\mathcal{R}_{\delta;p,q,p'}$ be any subset of such primes for which

$$r_\ell \in \mathcal{Z}_p, \quad s_\ell \in \mathcal{Z}_\ell,$$

possibly with additional shell, reducedness, high-centered, or deletion masks.

Lemma 231 (Boundary and midpoint of a selected reciprocal fibre). *Assume*

$$0 < |\delta| \ll N^{359/100+\varepsilon}, \quad 0 < \varepsilon < 41/100, \quad (\delta, C) = 1.$$

Put

$$d_\ell^{\text{bd}} = \min\{s_\ell, \ell - 1 - s_\ell, |\ell - 1 - 2s_\ell|\}.$$

Uniformly for $1 \leq U \leq cN$,

$$\#\{\ell \in \mathcal{R}_{\delta;p,q,p'} : d_\ell^{\text{bd}} \leq U\} \ll_v U.$$

Proof. Fix $0 \leq u \leq U$. If $s_\ell = u$, then

$$\ell \mid \delta + Cu,$$

while if $s_\ell = \ell - 1 - u$, then

$$\ell \mid \delta - C(u + 1).$$

For $u \ll N$, each nonzero integer on the right is $O_v(N^{359/100+\varepsilon} + N^3) = o(N^4)$ and hence has at most three prime divisors in a fixed-ratio N -shell. For either sign there is at most one u for which the carrier vanishes. In the first vanishing case the actual zero condition gives $\ell \mid b_u$; in the second case the Apéry palindrome gives $b_{\ell-1-u} \equiv b_u \pmod{\ell}$, and again $\ell \mid b_u$. Since $b_u < 34^u$, the number of shell-prime divisors of b_u is $O(u/\log N)$.

It remains to count the midpoint sector. Put $j_\ell = \ell - 1 - 2s_\ell$. If $|j_\ell| \leq U$, then twice the defining congruence gives

$$\ell \mid 2\delta - C(1 + j_\ell).$$

This carrier is nonzero in the actual reduced determinant fibre: if $2\delta = C(1 + j_\ell)$, then the odd integer C divides δ . For each of the $O(U)$ possible values of j_ℓ it again has $O_v(1)$ shell-prime divisors. Combining the three sectors proves the assertion. $\square$

Corollary 232 (Edges touching the reciprocal boundary sector). *Order the selected reciprocal occurrences by their fixed- p labels. The number of consecutive ordered edges having at least one endpoint with $d_\ell^{\text{bd}} \leq U$ is $O_v(U)$.*

Proof. Every selected occurrence is incident to at most two consecutive edges, so this follows immediately from Lemma 231. $\square$

Lemma 233 (Pairwise rational-row lift). *Let $\ell_1 \neq \ell_2$ belong to $\mathcal{R}_{\delta;p,q,p'}$, put $M = \ell_1\ell_2$, and let $0 \leq m < M$ be defined by*

$$Cm \equiv -\delta \pmod{M}.$$

Then

$$m \bmod \ell_i = s_{\ell_i} \in \mathcal{Z}_{\ell_i} \quad (i = 1, 2),$$

and

$$\ell_1\ell_2 \mid b_m.$$

Proof. Reduction modulo ℓ_i gives $m \equiv s_{\ell_i} \pmod{\ell_i}$. The least base- ℓ_i digit of m is therefore an Apéry zero, so the Gessel–Lucas congruence gives $\ell_i \mid b_m$. The two primes are distinct. $\square$

Lemma 234 (The palindromic moving-zero certificate is central). *Let $\ell \in \mathcal{R}_{\delta;p,q,p'}$ and suppose $g = |\ell - 1 - 2s_\ell| > 0$. Put*

$$x = \min\{s_\ell, \ell - 1 - s_\ell\} = \frac{\ell - 1 - g}{2}.$$

Then g is even,

$$x, x + g \in \mathcal{Z}_\ell, \quad \ell \mid N_g(x),$$

but the guaranteed divisibility is entirely the central factor:

$$N_g(X) = (2X + g + 1)N_g^\circ(X), \quad 2x + g + 1 = \ell.$$

In particular the hypotheses do not imply $\ell \mid N_g^\circ(x)$.

Proof. Palindromy gives the second zero and the nonwrapping gap criterion gives $\ell \mid N_g(x)$. Since g is even, central deflation gives the displayed factorization. The identity $2x + g + 1 = \ell$ shows that its linear factor already supplies the required prime, without imposing a primitive divisibility condition. $\square$

Lemma 235 (Möbius relation for reciprocal zero gaps). *Let $\ell_1, \ell_2 \in \mathcal{R}_{\delta;p,q,p'}$ and suppose their standard fixed- p zero labels satisfy*

$$r_{\ell_2} = r_{\ell_1} + h < p.$$

Then

$$(A + h\ell_1)\ell_2 \equiv A\ell_1 \pmod{p},$$

the denominator $A + h\ell_1$ is nonzero modulo p , and hence

$$\ell_2 \equiv \frac{A\ell_1}{A + h\ell_1} \pmod{p}.$$

Proof. Since $r_{\ell_i}\ell_i \equiv A \pmod{p}$,

$$(A\ell_1^{-1} + h)\ell_2 \equiv A \pmod{p}.$$

Multiplication by ℓ_1 proves the first congruence. If the denominator vanished, then $r_{\ell_1} + h \equiv 0 \pmod{p}$, contradicting $r_{\ell_2} \in \mathcal{Z}_p$ and $b_0 = 1$. $\square$

Proposition 236 (Selected reciprocal-gap decomposition). *Let $\rho_1 < \dots < \rho_R$ be the distinct values among the r_ℓ with $\ell \in \mathcal{R}_{\delta;p,q,p'}$, and put*

$$E_h = \#\{i : \rho_{i+1} - \rho_i = h\}.$$

Then, for every $1 \leq H < p$,

$$R \leq 1 + \frac{p}{H} + \sum_{2 \leq h \leq H} E_h,$$

and

$$E_h \leq 3(h-1).$$

Consequently

$$R \ll 1 + \frac{p}{H} + H^2,$$

and $R \ll p^{2/3}$ after optimizing H . On every fixed-ratio shell box the map $\ell \mapsto r_\ell$ has $O_v(1)$ multiplicity, so the same estimates hold for $|\mathcal{R}_{\delta;p,q,p'}|$ up to a constant depending on the box.

Proof. At most p/H consecutive gaps exceed H , since their sum is less than p . The remaining gaps are counted by the E_h ; the $h = 1$ term is zero because two consecutive Apéry residues cannot both be zeros. If a transition counted by E_h starts at ρ , then $\rho, \rho + h \in \mathcal{Z}_p$ without wrap. The exact gap criterion gives $N_h(\rho) = 0 \pmod{p}$. The polynomial N_h is nonzero modulo p and has degree $3(h-1)$, so there are at most $3(h-1)$ starts. Finally $r_\ell \equiv A\ell^{-1} \pmod{p}$ determines the residue of ℓ modulo p ; an interval of length $O_v(N)$ contains only $O_v(1)$ representatives of that residue because $p \asymp_v N$. $\square$

The same reciprocal fibre has a power-saving horizontal average when the genuine signed determinant is summed before any fibre supremum is taken. This uses both actual Apéry selectors through an exact CRT count; it is not a replacement of a fixed-determinant selector by its marginal density.

For $h \geq 2$ define the one-sided actual majorant

$$B_h(\delta) := \# \left\{ \ell \asymp_v N : \begin{array}{l} 0 \leq r_\ell < r_\ell + h < p, \\ r_\ell, r_\ell + h \in \mathcal{Z}_p, \\ s_\ell \in \mathcal{Z}_\ell \end{array} \right\}. \quad (\text{SRG1})$$

Lemma 237 (Determinant-averaged selected reciprocal gaps). *Let $J \subset \mathbb{Z}$ be an interval of length L . Uniformly in the fixed pairwise distinct shell primes p, q, p' and $2 \leq h < p$,*

$$\sum_{\delta \in J} B_h(\delta) \ll_v h \left(LN^{-2/5+o(1)} + N^{8/5+o(1)} \right). \quad (\text{SRG2})$$

If $E_h(\delta)$ is the consecutive-gap count in Proposition 236, then

$$\sum_{\delta \in J} \sum_{2 \leq h \leq H} E_h(\delta) \ll_v H^2 \left(LN^{-2/5+o(1)} + N^{8/5+o(1)} \right) \quad (\text{SRG3})$$

for every $2 \leq H < p$.

Proof. For fixed ℓ , the first two zero conditions in (SRG1) say exactly that

$$\delta \pmod{p} \in p'\ell R_{p,h}, \quad R_{p,h} := \{r : 0 \leq r < r + h < p, \ r, r + h \in \mathcal{Z}_p\}.$$

The gap-polynomial bound gives $|R_{p,h}| \leq 3(h-1)$. The moving zero condition says exactly that

$$\delta \pmod{\ell} \in -C\mathcal{Z}_\ell, \quad C = pq,$$

and therefore occupies $Z(\ell)$ classes modulo ℓ . Since $p \neq \ell$, CRT gives precisely $|R_{p,h}|Z(\ell)$ simultaneous classes modulo $p\ell$. Each class occurs in J at most $L/(p\ell) + 1$ times. Hence

$$\sum_{\delta \in J} B_h(\delta) \leq |R_{p,h}| \sum_{\ell \asymp_v N} Z(\ell) \left(\frac{L}{p\ell} + 1 \right).$$

Now use $p, \ell \asymp_v N$ and the unconditional first zero moment

$$\sum_{\ell \asymp_v N} Z(\ell) \ll_v \frac{N^{8/5}}{\log N}.$$

This proves (SRG2).

Every consecutive selected edge of gap h has a starting occurrence counted by $B_h(\delta)$. Passing from occurrences to the distinct fixed- p labels used in E_h can only decrease the count. Thus $E_h(\delta) \leq B_h(\delta)$, and summing $\sum_{h \leq H} h \ll H^2$ proves (SRG3). $\square$

One can retain one more defining-characteristic zero condition in the determinant average. Put

$$a_\ell \equiv \delta(p'\ell)^{-1} \pmod{q}$$

with its standard representative, and define

$$B_h^{(q)}(\delta) := \# \left\{ \ell \asymp_v N : \begin{array}{l} 0 \leq r_\ell < r_\ell + h < p, \\ r_\ell, r_\ell + h \in \mathcal{Z}_p, \\ a_\ell \in \mathcal{Z}_q, \\ s_\ell \in \mathcal{Z}_\ell \end{array} \right\}. \quad (\text{SRG7})$$

This is a three-test majorant for the actual four-distinct fibre: only the fixed- p' zero condition and the final row masks have been dropped.

Lemma 238 (Three-characteristic determinant-averaged gaps). *For every integer interval J of length L ,*

$$\sum_{\delta \in J} B_h^{(q)}(\delta) \ll_v hZ(q) \left(LN^{-7/5+o(1)} + N^{8/5+o(1)} \right). \quad (\text{SRG8})$$

Let $\mathcal{R}_{\delta;p,q,p'}^{(q)}$ be the set of shell primes ℓ satisfying the three actual conditions

$$r_\ell \in \mathcal{Z}_p, qquada_\ell \in \mathcal{Z}_q, qquads_\ell \in \mathcal{Z}_\ell,$$

with any additional deletion masks, and let its consecutive fixed- p label gaps be $E_h^{(q)}(\delta)$. If $L = N^{359/100+o(1)}$, then

$$\sum_{\delta \in J} |\mathcal{R}_{\delta;p,q,p'}^{(q)}| \ll_v LN^{19/45+o(1)}. \quad (\text{SRG9})$$

The same estimate therefore holds for the literal four-test fibre.

Proof. Fix ℓ . The fixed- p pair condition occupies $|R_{p,h}| \leq 3(h-1)$ classes for δ modulo p . The fixed- q condition is exactly

$$\delta \pmod{q} \in p'\ell\mathcal{Z}_q$$

and occupies $Z(q)$ classes. The moving- ℓ condition is exactly

$$\delta \pmod{\ell} \in -C\mathcal{Z}_\ell$$

and occupies $Z(\ell)$ classes. Since p, q, ℓ are distinct, CRT gives exactly $|R_{p,h}|Z(q)Z(\ell)$ simultaneous classes modulo $pq\ell$. Hence

$$\sum_{\delta \in J} B_h^{(q)}(\delta) \leq |R_{p,h}|Z(q) \sum_{\ell \asymp_v N} Z(\ell) \left(\frac{L}{pq\ell} + 1 \right).$$

The first zero moment and $p, q, \ell \asymp_v N$ give (SRG8).

Apply the consecutive-gap decomposition with $H = N^{26/45}$. The large gaps contribute

$$\frac{LN}{H} = LN^{19/45}.$$

For the short gaps, sum (SRG8) over $h \leq H$ and use the unconditional pointwise estimate $Z(q) \ll N^{2/3}$. Its main term is

$$LH^2N^{2/3-7/5+o(1)} = LN^{19/45+o(1)},$$

and its endpoint term is

$$H^2N^{2/3+8/5+o(1)} = N^{154/45+o(1)} = o(L).$$

Finally the map $\ell \mapsto r_\ell$ has bounded multiplicity on a fixed-ratio shell, exactly as in Proposition 236. This proves (SRG9). Every actual four-test candidate is contained in the three-test set, proving the last claim. $\square$

The same three-test fibre has a simpler incidence first moment. Unlike the preceding gap argument, it counts each moving characteristic directly and therefore also retains its prime weight.

Lemma 239 (Direct weighted three-test determinant mass). *For arbitrary nonnegative weights $(v_\ell)_{\ell \asymp_v N}$, put*

$$R_\delta^{(q)}(v) = \sum_{\ell \in \mathcal{R}_{\delta;p,q,p'}^{(q)}} v_\ell.$$

Then, for every integer interval J of length L ,

$$\sum_{\delta \in J} R_\delta^{(q)}(v) \ll_v Z(p)Z(q) \sum_{\ell \asymp_v N} v_\ell Z(\ell) \left(\frac{L}{N^3} + 1 \right). \quad (\text{SRG9a})$$

Consequently, at $L = N^{359/100+o(1)}$,

$$\sum_{\delta \in J} |\mathcal{R}_{\delta;p,q,p'}^{(q)}| \ll_v LN^{-1/15+o(1)}, \quad (\text{SRG9b})$$

$$\sum_{\delta \in J} R_\delta^{(q)}(|u|) \ll_v LN^{-16/15+o(1)}, \quad (\text{SRG9c})$$

$$\sum_{\delta \in J} R_\delta^{(q)}(|u|^{1/2}) \ll_v LN^{-17/30+o(1)}. \quad (\text{SRG9d})$$

All four bounds remain valid for the literal four-test fibre.

Proof. Fix ℓ . The fixed- p condition is exactly

$$\delta \pmod{p} \in p'\ell\mathcal{Z}_p,$$

the fixed- q condition is exactly

$$\delta \pmod{q} \in p'\ell\mathcal{Z}_q,$$

and the moving- ℓ condition is exactly

$$\delta \pmod{\ell} \in -pq\mathcal{Z}_\ell.$$

Thus CRT gives $Z(p)Z(q)Z(\ell)$ simultaneous classes modulo $pq\ell$, and each class has at most $L/(pq\ell) + 1$ representatives in J . Multiplication by v_ℓ and summation over ℓ proves (SRG9a).

For $v_\ell = 1$, use $Z(p), Z(q) \ll N^{2/3+o(1)}$ and $\sum_\ell Z(\ell) \ll_v N^{8/5+o(1)}$. The main term is $LN^{-1/15+o(1)}$ and the endpoint is $N^{44/15+o(1)}$, which is smaller at the stated L . For $v_\ell = |u_\ell|$, use $\sum_\ell |u_\ell|Z(\ell) \ll_v N^{3/5+o(1)}$; the corresponding exponents are $LN^{-16/15+o(1)}$ and $N^{29/15+o(1)}$. Finally

$$\sum_\ell |u_\ell|^{1/2} Z(\ell) \ll_v N^{-1/2+o(1)} \sum_\ell Z(\ell) \ll_v N^{11/10+o(1)},$$

giving $LN^{-17/30+o(1)}$ with a smaller endpoint. Additional tests and deletion masks only remove occurrences. $\square$

Fixing the actual outer p' -zero makes the same count a four-modulus CRT estimate. This is the determinant progression which occurs literally in the two-level selected sampler.

Corollary 240 (Weighted mass on one outer-zero progression). *Fix $r' \in \mathcal{Z}_{p'}$ and put*

$$J_{r'} = \{\delta \in J : \delta \equiv -pqr' \pmod{p'}\}.$$

For arbitrary nonnegative weights (v_ℓ) ,

$$\sum_{\delta \in J_{r'}} R_\delta^{(q)}(v) \ll_v Z(p)Z(q) \sum_{\ell \lesssim_v N} v_\ell Z(\ell) \left(\frac{L}{N^4} + 1 \right). \quad (\text{SRG9g})$$

At $L = N^{359/100+o(1)}$ this gives

$$\sum_{\delta \in J_{r'}} |\mathcal{R}_{\delta;p,q,p'}^{(q)}| \ll_v N^{44/15+o(1)}, \quad (\text{SRG9h})$$

$$\sum_{\delta \in J_{r'}} R_\delta^{(q)}(|u|) \ll_v N^{29/15+o(1)}, \quad (\text{SRG9i})$$

$$\sum_{\delta \in J_{r'}} R_\delta^{(q)}(|u|^{1/2}) \ll_v N^{73/30+o(1)}. \quad (\text{SRG9j})$$

Moreover $|J_{r'}| = N^{259/100+o(1)}$, so the corresponding average exponents are respectively 103/300, $-197/300$, and $-47/300$. All statements remain valid for the literal four-test fibre.

Proof. For fixed ℓ and fixed zero labels at p, q, ℓ , add the exact outer congruence

$$\delta \equiv -pqr' \pmod{p'}$$

to the three congruences in the proof of Lemma 239. CRT now gives one class modulo $pqp'\ell$. Summing the weights proves (SRG9g). Since $L = o(N^4)$, its endpoint term gives the three displayed bounds. Finally one residue class modulo p' has $L/p' + O(1)$ representatives in J . $\square$

Summing both outer characteristics before taking a supremum gives a fully symmetric four-test incidence bound with all prime weights retained.

Proposition 241 (Global weighted four-test determinant incidence). *For an interval J of length L , let*

$$\begin{aligned} \mathcal{I}_4(J) := \sum_{\delta \in J} \sum_{\substack{p, q, p', \ell \asymp_v N \\ p, q, p', \ell \text{ pairwise distinct}}} & |u_p u_q u_{p'} u_\ell| \mathbf{1}_{\mathcal{Z}_p}(\delta(p'\ell)^{-1} \bmod p) \\ & \times \mathbf{1}_{\mathcal{Z}_q}(\delta(p'\ell)^{-1} \bmod q) \\ & \times \mathbf{1}_{\mathcal{Z}_{p'}}(-\delta(pq)^{-1} \bmod p') \\ & \times \mathbf{1}_{\mathcal{Z}_\ell}(-\delta(pq)^{-1} \bmod \ell). \end{aligned}$$

Then

$$\mathcal{I}_4(J) \ll_v \left(\frac{L}{N^4} + 1 \right) \left(\sum_{p \asymp_v N} |u_p| Z(p) \right)^4. \quad (\text{SRG9k})$$

In particular, at $L = N^{359/100+o(1)}$,

$$\mathcal{I}_4(J) \ll_v N^{12/5+o(1)}. \quad (\text{SRG9l})$$

Both estimates survive every additional literal deletion mask and both signed circular branches. They are also uniform if each ordered prime quadruple uses its own integer interval of length at most L .

Proof. Fix an ordered quadruple of pairwise distinct shell primes. Choosing one zero label in each of $\mathcal{Z}_p, \mathcal{Z}_q, \mathcal{Z}_{p'}, \mathcal{Z}_\ell$ prescribes by CRT one class for δ modulo $pqp'\ell$. Each such class has at most $L/(pqp'\ell)+1$ representatives in J . Multiplication by the four absolute prime weights and summation over the prime quadruple gives (SRG9k); dropping the distinctness restriction only enlarges the product majorant. Finally $L = o(N^4)$ and the first weighted zero moment is $O_v(N^{3/5+o(1)})$. $\square$

Corollary 242 (Direct determinant transfers). *For every complex sequence $(a_\delta)_{\delta \in J}$,*

$$\left| \sum_{\delta \in J} a_\delta |\mathcal{R}_{\delta; p, q, p'}^{(q)}|^{1/2} \right| \ll_v \|a\|_2 L^{1/2} N^{-1/30+o(1)}, \quad (\text{SRG9e})$$

$$\left| \sum_{\delta \in J} a_\delta R_\delta^{(q)} (|u|)^{1/2} \right| \ll_v \|a\|_2 L^{1/2} N^{-8/15+o(1)}. \quad (\text{SRG9f})$$

The same estimates hold for the literal four-test fibre.

Proof. Apply Cauchy–Schwarz and then (SRG9b) or (SRG9c), respectively. $\square$

The direct determinant mean also removes the whole same-determinant seam in the selected-fibre square. This is stronger than the first-row diagonal in Corollary 227, but it still leaves the genuinely related two-determinant collision.

Proposition 243 (Same-determinant off-diagonal). *Let $\mathcal{C}_{\text{off}}^{\delta=}$ be the part of $\mathcal{C}_{\text{joint}}^\sharp$ in (AP10) for which the two first rows are distinct but their signed circular determinants are equal. At $M = 4N^{41/100}$ one has*

$$|\mathcal{C}_{\text{off}}^{\delta=}| \ll_v N^{139/100+o(1)}. \quad (\text{SRG9m})$$

The estimate is uniform under all current deletion masks and for both circular branches.

Proof. Write $P_i = p_i q_i$ for the products belonging to the two first rows, and let p' and $r' \in \mathcal{Z}_{p'}$ be their common outer prime and outer zero. If the common signed determinant is δ , then the fourth zero condition gives

$$\delta \equiv -r' P_1 \equiv -r' P_2 \pmod{p'}.$$

Since $b_0 = 1$, zero is not an Apéry zero; hence $r' \neq 0$ and

$$P_1 \equiv P_2 \pmod{p'}. \quad (\text{SRG9n})$$

For fixed P_1, p' and a shell prime p_2 , this places q_2 in one nonzero residue class modulo p' . A fixed-ratio shell interval has $O_v(1)$ integers in that class, and therefore

$$\sum_{\substack{p_2, q_2 \asymp_v N \\ p_2 q_2 \equiv P_1 \pmod{p'}}} |u_{p_2} u_{q_2}| \ll_v N^{-1+o(1)}. \quad (\text{SRG9o})$$

All distinctness and orientation conditions only decrease this sum.

For fixed δ, ℓ, p', r' and P_2 , the congruence

$$t_2 \equiv \delta(p'\ell)^{-1} \pmod{P_2}$$

fixes the standard first-row numerator, and the exact determinant formula then fixes its selected lift. Thus dropping the remaining tests belonging to the second factor creates no zero-label multiplicity. We may retain only the first side's three-test incidence. With $H_{\det} = N^{59/100+o(1)}$, Lemma 239 with moving weight $|u_\ell|$ gives, for fixed p_1, q_1, p' ,

$$\sum_{\delta, \ell} |u_\ell| I_1(\delta, \ell) \ll_v H_{\det} Z(p_1) Z(q_1) \mathcal{B}_N.$$

After summing the first-row and outer-prime weights and using (SRG9o),

$$\left| \mathcal{C}_{\text{off}}^{\delta} \right| \ll_v N^{-1+o(1)} H_{\det} \mathcal{B}_N^3 \mathcal{U}_N \ll_v N^{139/100+o(1)}.$$

□

One further intersection sector can be removed without a determinant average.

Proposition 244 (Coincident first denominator and second row). *Let $\mathcal{C}_{P,0}$ be the positive majorant of the part of the expanded collision (AP10) in which the two first rows have the same unordered defining-prime pair and their selected lifts satisfy $n_1 = n_2$. Then*

$$\mathcal{C}_{P,0} \ll_v N^{247/150+o(1)}. \quad (\text{SRG9q})$$

In particular this sector is below the square-level target $N^{44/25}$ by the fixed power $N^{-17/150+o(1)}$.

Proof. Put

$$\mathcal{J}_N = \sum_{p \asymp_v N} (|u_p| Z(p))^2 \leq \mathcal{L}_N \mathcal{B}_N \ll_v N^{4/15+o(1)}.$$

Fix the common ordered first denominator $P = pq$. If $h = n_2 - n_1 = 0$, the two triangular supports imply

$$\left\| \frac{t_2 - t_1}{P} \right\|_{\mathbb{T}} \ll_v \frac{1}{M}.$$

Let d be the signed representative of $t_2 - t_1$ modulo P . It lies in an interval I of length $O_v(P/M)$. If

$$A_p(c) = \#\{r \in \mathcal{Z}_p : r + c \in \mathcal{Z}_p\},$$

then the number of ordered first-row pairs with difference d is $A_p(d)A_q(d)$. Every residue modulo p occurs in I at most $O_v(1 + |I|/p) = O_v(N^{59/100+o(1)})$ times, while $A_q(d) \leq Z(q)$ and $\sum_{c \bmod p} A_p(c) = Z(p)^2$. Writing $H = N^{59/100+o(1)}$, the weighted first-row pair mass is therefore

$$\ll_v H \left(\sum_p |u_p|^2 Z(p)^2 \right) \left(\sum_q |u_q|^2 Z(q) \right) = H \mathcal{J}_N \mathcal{E}_N \ll_v N^{137/300+o(1)}.$$

The $h = 0$ term in the selected-codegree estimate (AP17), after summing the moving prime, is $H\mathcal{B}_N\mathcal{U}_N \ll_v N^{357/300+o(1)}$. Multiplication gives

$$\mathcal{C}_{P,0} \ll_v N^{(137+357)/300+o(1)} = N^{247/150+o(1)}.$$

Swapping the two primes in one row costs only a bounded orientation factor, and all remaining masks and the restriction to different determinants only delete terms. $\square$

For clarity, the remaining seam has two different determinants. If t_i/P_i are the two first rows, ℓ is their common moving prime, and $h = n_2 - n_1$ is the difference of the selected lifts in (AP8), then every surviving pair satisfies the exact relations

$$P_1\delta_2 - P_2\delta_1 = p' \{ \ell(P_1t_2 - P_2t_1) + P_1P_2h \}, \quad P_1\delta_2 \equiv P_2\delta_1 \pmod{p'}. \quad (\text{SRG9p})$$

The two moving zero labels additionally differ by $-hp'$ modulo ℓ . If the full second rows agree, one may take $h = 0$; the stronger divisibility by $p'\ell$ is not available in the general product.

Let $\mathfrak{T}_{2,\neq}$ denote the positive majorant obtained by expanding the off-diagonal part of (AP10), replacing all coefficients by their absolute values, and retaining only terms with $\delta_1 \neq \delta_2$. The presently sufficient, but unproved, continuation estimate is

$$\mathfrak{T}_{2,\neq} \ll_v N^{44/25-\varepsilon+o(1)}. \quad (\text{TDW}_\varepsilon)$$

Indeed Proposition 243 and Corollary 227 then give (AP12), with every $0 < \eta < \varepsilon/2$. The one-side three-test mass is

$$\mathfrak{M}_1 \ll_v H_{\det} \mathcal{B}_N^3 \mathcal{U}_N \ll_v N^{239/100+o(1)},$$

so the source-shaped missing input is an average secondary-continuation factor $N^{-63/100-\varepsilon+o(1)}$. No such fixed-power estimate is asserted here.

There is a stronger second-moment estimate which counts two moving characteristics at the same determinant directly. It is sharper than combining the first determinant moment with the pointwise fibre cap.

Lemma 245 (Second determinant moment of reciprocal fibres). *For each $\delta \in J$, let $\mathcal{R}_{\delta;p,q,p'}$ be the two-test reciprocal fibre defined above, with any additional deletion masks. For arbitrary nonnegative weights $(v_\ell)_{\ell \lesssim_v N}$, put*

$$R_\delta(v) = \sum_{\ell \in \mathcal{R}_{\delta;p,q,p'}} v_\ell.$$

Then

$$\sum_{\delta \in J} R_\delta(v)^2 \ll_v Z(p) \left\{ \left(\frac{L}{N^3} + 1 \right) \left(\sum_{\ell \lesssim_v N} v_\ell Z(\ell) \right)^2 + \left(\frac{L}{N^2} + 1 \right) \sum_{\ell \lesssim_v N} v_\ell^2 Z(\ell) \right\}. \quad (\text{SRG12a})$$

In particular, if all $v_\ell = 1$ and $|J| = L = N^{359/100+o(1)}$, then

$$\sum_{\delta \in J} |\mathcal{R}_{\delta;p,q,p'}|^2 \ll_v L N^{13/15+o(1)}. \quad (\text{SRG12})$$

The same estimate holds for the literal four-test fibre.

Proof. Expand the square and first take two distinct moving shell primes ℓ_1, ℓ_2 . Their fixed- p labels satisfy

$$r_i \ell_i \equiv \delta(p')^{-1} \pmod{p}.$$

After $r_1 \in \mathcal{Z}_p$ has been chosen, r_2 is forced modulo p ; hence the two fixed- p zero tests occupy at most $Z(p)$ classes for δ modulo p . The two moving zero tests occupy respectively $Z(\ell_1)$ and $Z(\ell_2)$ classes modulo ℓ_1 and ℓ_2 . CRT therefore gives at most

$$Z(p)Z(\ell_1)Z(\ell_2)$$

classes modulo $p\ell_1\ell_2$. After insertion of the weights, the off-diagonal part is at most

$$Z(p) \left(\frac{L}{N^3} + 1 \right) \left(\sum_{\ell \asymp_v N} v_\ell Z(\ell) \right)^2.$$

The diagonal $\ell_1 = \ell_2$ is at most

$$Z(p) \left(\frac{L}{N^2} + 1 \right) \sum_{\ell \asymp_v N} v_\ell^2 Z(\ell).$$

This proves (SRG12a). Taking $v_\ell = 1$ and using $Z(p) \ll N^{2/3+o(1)}$ and $\sum_{\ell \asymp_v N} Z(\ell) \ll_v N^{8/5+o(1)}$, the off-diagonal main term is

$$L N^{2/3-3+16/5+o(1)} = L N^{13/15+o(1)}.$$

Its endpoint term is $N^{58/15+o(1)}$, and both diagonal terms are smaller; all are absorbed because $L = N^{359/100+o(1)}$. Optional masks and the two omitted fixed-characteristic tests only delete occurrences, proving the final assertion. $\square$

Corollary 246 (Prime-weighted second determinant moment). *Let*

$$u_\ell = \frac{\log \ell}{\ell} v(\ell/N).$$

Under the hypotheses of Lemma 245, if $|J| = L = N^{359/100+o(1)}$, then

$$\sum_{\delta \in J} \left(\sum_{\ell \in \mathcal{R}_{\delta;p,q,p'}} |u_\ell| \right)^2 \ll_v L N^{-17/15+o(1)}. \quad (\text{SRG12b})$$

The same estimate holds for the literal four-test fibre.

Proof. Apply (SRG12a) with $v_\ell = |u_\ell|$. The banked first and second weighted zero moments give

$$\sum_{\ell \asymp_v N} |u_\ell| Z(\ell) \ll_v N^{3/5+o(1)}, \quad \sum_{\ell \asymp_v N} |u_\ell|^2 Z(\ell) \ll_v N^{-2/5+o(1)}.$$

Together with $Z(p) \ll N^{2/3+o(1)}$, the first main term in (SRG12a) is

$$LN^{2/3-3+6/5+o(1)} = LN^{-17/15+o(1)}.$$

The first endpoint term and both second-moment terms are smaller at $L = N^{359/100+o(1)}$. $\square$

The fixed-triple estimate globalizes without a support-cardinality loss because the remaining three prime weights have bounded total ℓ^1 mass.

Corollary 247 (Global weighted determinant-mass second moment). *Put*

$$\mathcal{F}_\delta = \sum_{\substack{p,q,p' \asymp_v N \\ p,q,p' \text{ pairwise distinct}}} |u_p u_q u_{p'}| R_\delta(|u|),$$

where $R_\delta(|u|)$ is the two-test weighted reciprocal fibre for that ordered triple. If $|J| = L = N^{359/100+o(1)}$, then

$$\sum_{\delta \in J} \mathcal{F}_\delta^2 \ll_v LN^{-17/15+o(1)} = N^{737/300+o(1)}. \quad (\text{SRG12c})$$

The same estimate holds if every summand is restricted to the literal four-test fibre and to arbitrary additional deletion masks.

Proof. Write $A = (p, q, p')$, $c_A = |u_p u_q u_{p'}|$, and $W_A(\delta) = R_\delta(|u|)$. Weighted Cauchy–Schwarz gives

$$\mathcal{F}_\delta^2 \leq \left(\sum_A c_A \right) \left(\sum_A c_A W_A(\delta)^2 \right).$$

Now $\sum_A c_A \leq \mathcal{U}_N^3 = N^{o(1)}$. Sum in δ , apply Corollary 246 uniformly to each ordered triple, and sum the c_A once more. Literal tests and masks only decrease every nonnegative $W_A(\delta)$. $\square$

The same determinant square improves when the fixed- q actual zero condition is retained on both moving-prime copies. This remains a counting moment; it is not a Palm-weighted third incidence.

Corollary 248 (Three-test prime-weighted determinant square). *Fix the pairwise distinct shell primes p, q, p' and an interval J of length $L = N^{359/100+o(1)}$. For $\alpha \in \{0, 1, 2\}$ put*

$$R_\delta^{(q)}(\alpha) = \sum_{\ell \in \mathcal{R}_{\delta;p,q,p'}^{(q)}} |u_\ell|^\alpha,$$

where $\mathcal{R}_{\delta;p,q,p'}^{(q)}$ is the three-test reciprocal fibre of Lemma 238. Then

$$\boxed{\sum_{\delta \in J} |R_\delta^{(q)}(\alpha)|^2 \ll_v LN^{283/300-2\alpha+o(1)}}. \quad (\text{SRG12d})$$

The estimate remains valid for the literal four-test fibre and after arbitrary additional deletion masks.

Proof. Expand the square. For distinct moving primes ℓ_1, ℓ_2 , the two fixed- p tests occupy at most $Z(p)$ classes modulo p , and the two fixed- q tests occupy at most $Z(q)$ classes modulo q . The moving tests occupy $Z(\ell_i)$ classes modulo ℓ_i . CRT gives

$$\begin{aligned} \sum_{\delta \in J} |R_{\delta}^{(q)}(\alpha)|^2 &\ll_v L \frac{Z(p)Z(q)}{pq} \left(\sum_{\ell} \frac{|u_{\ell}|^{\alpha} Z(\ell)}{\ell} \right)^2 \\ &\quad + Z(p)Z(q) \left(\sum_{\ell} |u_{\ell}|^{\alpha} Z(\ell) \right)^2 + \text{the equal-moving-prime diagonal.} \end{aligned}$$

The off-diagonal main and endpoint terms are respectively

$$LN^{8/15-2\alpha+o(1)}, \quad N^{68/15-2\alpha+o(1)}.$$

The diagonal terms are smaller. Since

$$\frac{68}{15} - \frac{359}{100} = \frac{283}{300},$$

the endpoint term proves (SRG12d). Further literal tests and masks only decrease the nonnegative fibre weights. $\square$

Corollary 248 controls an unweighted determinant second moment. In the Palm consumer the kernel energy already selects one moving prime, so multiplication by a degree square produces a mixed third incidence. The corollary may not be inserted there by dividing two unrelated absolute upper bounds.

There is nevertheless a relative estimate for the part of the rooted first moment having short signed row displacement. It uses the selected root itself and therefore does not divide by its unknown mass.

Lemma 249 (Rooted relative short reciprocal displacement). *Fix pairwise distinct shell primes p, q, p' and put $P = pq$. For each nonzero signed circular determinant δ and shell prime $\ell \notin \{p, q, p'\}$, let*

$$I_{p,q,p';\delta}(\ell) := \mathbf{1}_{\mathcal{Z}_p}(\delta \overline{p'} \ell \bmod p) \mathbf{1}_{\mathcal{Z}_{\ell}}(-\delta \overline{P} \bmod \ell).$$

Let $w_{p,q,p'}(\delta, \ell_0) \geq 0$ be any literal selected-root weight supported on the corresponding four zero tests

$$\delta \overline{p'} \ell_0 \bmod p \in \mathcal{Z}_p, \quad \delta \overline{p'} \ell_0 \bmod q \in \mathcal{Z}_q, \quad -\delta \overline{P} \bmod p' \in \mathcal{Z}_{p'}, \quad -\delta \overline{P} \bmod \ell_0 \in \mathcal{Z}_{\ell_0}.$$

The weight may include the exact squared tent or signed-kernel factor and all current deletion masks. Put

$$\mathfrak{M}_0 := \sum_{p,q,p'} \sum_{\delta, \ell_0} w_{p,q,p'}(\delta, \ell_0).$$

For a selected root $x = (p, q, p', \delta, \ell_0)$ and every ℓ_i in the root or two-test fibre, define the standard first-row completion and signed second-row lift by

$$t_i = \langle \delta \overline{p'} \ell_i \rangle_P, \quad t'_i = \frac{p' \ell_i t_i - \delta}{P}.$$

Then every ℓ_1 with $I_{p,q,p';\delta}(\ell_1) = 1$ has a unique integer

$$h_x(\ell_1) = \frac{t_1 \ell_1 - t_0 \ell_0}{P}$$

such that

$$t_1 \ell_1 = t_0 \ell_0 + Ph_x(\ell_1), \quad t'_1 = t'_0 + p' h_x(\ell_1). \quad (\text{SRG12e})$$

For every $1 \leq H \ll_v N$,

$$\sum_{p,q,p'} \sum_{\delta, \ell_0} w_{p,q,p'}(\delta, \ell_0) \sum_{\ell_1} I_{p,q,p';\delta}(\ell_1) \mathbf{1}_{\{|h_x(\ell_1)| \leq H\}} \ll_v (H+1) N^{o(1)} \mathfrak{M}_0. \quad (\text{SRG12f})$$

No additional prime weight, q -selector, p' -selector, or deletion mask is imposed on the ℓ_1 copy.

Proof. The definition of t_i gives $t'_i \in \mathbb{Z}$ and

$$\delta = p' \ell_i t_i - P t'_i.$$

Modulo p' , every t'_i is congruent to $-\delta \bar{P}$ and hence $p' \mid t'_1 - t'_0$. Subtracting the two determinant identities therefore gives the unique integer $h_x(\ell_1) = (t'_1 - t'_0)/p'$ and both identities in (SRG12e). Notice that this is an algebraic inheritance of the one selected p' -test, not a second selector on ℓ_1 .

For fixed x and h , every preimage prime ℓ_1 divides

$$A_x(h) := t_0 \ell_0 + Ph = t_1 \ell_1.$$

This integer is nonzero: since $0 \notin \mathcal{Z}_p$, the selected p -test gives $p \nmid t_0$, whereas $\ell_0 \neq p$; thus $p \nmid t_0 \ell_0$. Moreover, if $|h| \ll_v N$, then $|A_x(h)| \ll_v N^3$. A nonzero integer of this size has only $O_v(1)$ prime divisors in a fixed-ratio N -shell. Hence each fixed h has $O_v(1)$ preimages, with no injectivity assertion. Summing over the $O(H+1)$ possible integers h , multiplying by the unchanged nonnegative root weight, and then summing over roots proves (SRG12f). $\square$

More generally, take $H = N^\alpha$ with any fixed $\alpha < 47/75$. In the direct selected-Palm consumer the corresponding restricted shell exponent is

$$\frac{383}{300} + \frac{\alpha}{2},$$

so this sector lies below the critical 159/100 scale by

$$\frac{1}{2} \left(\frac{47}{75} - \alpha \right).$$

In particular, taking $\alpha = 3/5$ gives the restricted exponent

$$\frac{383}{300} + \frac{3}{10} = \frac{473}{300}$$

and permits every fixed $\eta < 1/75$ on the entire sector $|h_x(\ell_1)| \leq N^{3/5}$. This includes the former determinant-width strip $N^{59/100} < |h_x(\ell_1)| \leq N^{3/5}$ without adding any selector or weight. The complementary range $|h_x(\ell_1)| > N^{3/5}$ remains open. Thus the lemma is still sectoral and alone does not yield a positive exponent for the full signed TT^* shell.

Corollary 250 (Fourth-norm determinant transfer). *For every complex sequence $(a_\delta)_{\delta \in J}$,*

$$\left| \sum_{\delta \in J} a_\delta |\mathcal{R}_{\delta;p,q,p'}|^{1/2} \right| \ll_v \|a\|_{\ell^{4/3}(J)} L^{1/4} N^{13/60+o(1)}. \quad (\text{SRG13})$$

The same bound holds for the literal four-test fibre.

Proof. Apply Hölder with exponents $4/3$ and 4 , then use Lemma 245. $\square$

The determinant mean also has a coefficient-sensitive form. It is useful because it keeps an arbitrary complex determinant coefficient sequence intact, rather than first replacing that sequence by its support.

Corollary 251 (Determinant Hölder transfer). *Under the hypotheses of Lemma 238, let $L = N^{359/100+o(1)}$ and write*

$$R_\delta = |\mathcal{R}_{\delta;p,q,p'}^{(q)}|.$$

For every real $r \geq 2$, every complex sequence $(a_\delta)_{\delta \in J}$, and the Hölder conjugate r' , one has

$$\left| \sum_{\delta \in J} a_\delta R_\delta^{1/2} \right| \ll_v \|a\|_{\ell^{r'}(J)} L^{1/r} N^{1/3-11/(45r)+o(1)}. \quad (\text{SRG10})$$

In particular,

$$\left| \sum_{\delta \in J} a_\delta R_\delta^{1/2} \right| \ll_v \|a\|_2 L^{1/2} N^{19/90+o(1)}. \quad (\text{SRG11})$$

The same estimates hold with R_δ replaced by the cardinality of the literal four-test fibre.

Proof. The pointwise reciprocal-gap decomposition gives $R_\delta \ll_v N^{2/3+o(1)}$, whereas Lemma 238 gives

$$\sum_{\delta \in J} R_\delta \ll_v L N^{19/45+o(1)}.$$

Consequently

$$\sum_{\delta \in J} R_\delta^{r/2} \leq \left(\max_{\delta \in J} R_\delta \right)^{r/2-1} \sum_{\delta \in J} R_\delta \ll_v L N^{r/3-11/45+o(1)}.$$

Taking the r -th root and applying Hölder proves (SRG10). Setting $r = 2$ proves (SRG11). The literal four-test fibre is a subset of the three-test majorant for every δ . $\square$

Corollary 252 (Density-one reciprocal fibres). *Let J have length $L = N^{359/100+o(1)}$. Then*

$$\sum_{\delta \in J} |\mathcal{R}_{\delta;p,q,p'}| \ll_v L N^{8/15+o(1)}. \quad (\text{SRG4})$$

Moreover, putting $H = N^{2/5}$, outside a set of at most

$$L N^{-1/5+o(1)} \quad (\text{SRG5})$$

determinants $\delta \in J$, one has

$$\sum_{2 \leq h \leq H} E_h(\delta) \ll N^{3/5+o(1)}, \quad |\mathcal{R}_{\delta;p,q,p'}| \ll_v N^{3/5+o(1)}. \quad (\text{SRG6})$$

Equivalently, the generic determinant fibre gains $N^{-1/15+o(1)}$ over the pointwise $N^{2/3}$ bound.

Proof. First apply Proposition 236 with $H = N^{7/15}$. After summing over δ , the first term contributes $LN/H = LN^{8/15}$, while Lemma 237 contributes

$$H^2 \left(LN^{-2/5+o(1)} + N^{8/5+o(1)} \right) = LN^{8/15+o(1)} + o(L).$$

The bounded shell multiplicity of $\ell \mapsto r_\ell$ gives (SRG4).

At $H = N^{2/5}$, Lemma 237 and Markov's inequality at level $H^{3/2} = N^{3/5}$ give

$$\# \left\{ \delta \in J : \sum_{h \leq H} E_h(\delta) > N^{3/5+o(1)} \right\} \ll LN^{-1/5+o(1)} + N^{9/5+o(1)}.$$

The second term is absorbed by the first because $L = N^{359/100+o(1)}$. For every remaining determinant, Proposition 236 gives

$$|\mathcal{R}_{\delta;p,q,p'}| \ll_v 1 + N/H + N^{3/5+o(1)} \ll_v N^{3/5+o(1)}.$$

□

Corollary 252 is not yet a bound for the whole signed TT^* contribution: the exceptional determinant set depends on the outer prime data, and cardinality rarity alone does not prevent the oscillatory kernel and row coefficients from concentrating there. The remaining consumer is a kernel-weighted exceptional-set estimate, not another typical-determinant gap bound.

Lemma 253 (Fourier normalization of a reciprocal selector). *Let $\mathcal{L}$ be a finite set of shell primes coprime to C , let $w(\ell)$ be arbitrary complex weights, and put*

$$s_\ell \equiv -\delta C^{-1} \pmod{\ell}, \quad F_\ell(a) = \sum_{u \in \mathcal{Z}_\ell} e_\ell(au).$$

Then

$$\sum_{\ell \in \mathcal{L}} w(\ell) \left(\mathbf{1}_{\mathcal{Z}_\ell}(s_\ell) - \frac{Z(\ell)}{\ell} \right) = \sum_{\ell \in \mathcal{L}} \frac{w(\ell)}{\ell} \sum_{a=1}^{\ell-1} F_\ell(a) e_\ell(a\delta \overline{C}). \quad (\text{SRF1})$$

Moreover

$$\sum_{\ell \in \mathcal{L}} \sum_{a=1}^{\ell-1} \left| \frac{w(\ell)}{\ell} F_\ell(a) \right|^2 = \sum_{\ell \in \mathcal{L}} |w(\ell)|^2 \left(\frac{Z(\ell)}{\ell} - \frac{Z(\ell)^2}{\ell^2} \right). \quad (\text{SRF2})$$

In particular, if $\sum_{\ell \in \mathcal{L}} |w(\ell)|^2 \ll_v h$, then the last display is $\ll_v hN^{-1/3+o(1)}$.

Proof. Finite Fourier inversion gives

$$\mathbf{1}_{\mathcal{Z}_\ell}(k) - \frac{Z(\ell)}{\ell} = \frac{1}{\ell} \sum_{a=1}^{\ell-1} F_\ell(a) e_\ell(-ak).$$

Insert $k = s_\ell$. Finite Parseval gives $\sum_{a=1}^{\ell-1} |F_\ell(a)|^2 = \ell Z(\ell) - Z(\ell)^2$, proving the identity. The estimate follows from $Z(\ell) \ll \ell^{2/3}$. □

Lemma 254 (Fixed-modulus reciprocal phase). *For $(C, \ell) = 1$ one has*

$$e_\ell(a\delta \overline{C}) = e_C(-a\delta \overline{\ell}) e \left(\frac{a\delta}{C\ell} \right). \quad (\text{SRF3})$$

For the reciprocal fibre above,

$$\delta \overline{\ell} \equiv p' r_\ell \pmod{p}.$$

Proof. The inverse identity

$$\frac{\overline{C}}{\ell} + \frac{\overline{\ell}}{C} \equiv \frac{1}{C\ell} \pmod{1}$$

proves (SRF3). The final congruence is $p'r_\ell \equiv \delta\overline{\ell} \pmod{p}$ by the definition of r_ℓ . $\square$

Lemma 255 (Local trace trivialization). *Put $\rho_\ell = \delta\overline{C} \pmod{\ell}$, so that $s_\ell = -\rho_\ell$. Then*

$$F_\ell(a)e_\ell(a\rho_\ell) = \sum_{u \in \mathcal{Z}_\ell} e_\ell(a(u + \rho_\ell)).$$

If $s_\ell \notin \mathcal{Z}_\ell$, then

$$\frac{1}{\ell} \sum_{a=1}^{\ell-1} F_\ell(a)e_\ell(a\rho_\ell) = -\frac{Z(\ell)}{\ell},$$

whereas, if $s_\ell \in \mathcal{Z}_\ell$, the same expression equals $1 - Z(\ell)/\ell$. Thus on a selected hit exactly one linear Artin–Schreier summand is trivial. In the bulk midpoint sector the summand supplied by the reflected zero $\ell - 1 - s_\ell$ is nontrivial.

Proof. For $\beta \in \mathbb{F}_\ell$,

$$\sum_{a=1}^{\ell-1} e_\ell(a\beta) = \begin{cases} \ell - 1, & \beta = 0, \\ -1, & \beta \neq 0. \end{cases}$$

Summing this identity over the distinct zero positions proves the two formulae. The reflected slope is $-1 - 2s_\ell$, which is nonzero away from the midpoint sector. $\square$

Lemma 256 (Quotient resonance deletion). *For $\ell \in \mathcal{R}_{\delta;p,q,p'}$ define*

$$k_\ell = \frac{\delta + Cs_\ell}{\ell} \in \mathbb{Z}.$$

Then

$$k_\ell \equiv p'r_\ell \pmod{p}.$$

If two fixed- p labels differ by h , the corresponding quotient coordinates satisfy

$$k_{\ell_2} - k_{\ell_1} \equiv p'h \pmod{p}.$$

Under the size and reducedness hypotheses of Lemma 231, the number of shell primes $\ell \in \mathcal{R}_{\delta;p,q,p'}$ for which $p' \mid k_\ell$ is $O_v(1)$.

Proof. Reduction of $k_\ell \ell = \delta + Cs_\ell$ modulo p proves the first congruence, and subtraction proves the second. If $p' \mid k_\ell$, then

$$s_\ell \equiv -\delta C^{-1} \pmod{p'}.$$

Because $s_\ell < \ell \asymp_v p'$, this leaves only $O_v(1)$ integer values of s_ℓ . For each fixed value s , the moving prime ℓ divides the nonzero integer $\delta + Cs$. This integer has polynomial size in N , and hence only $O_v(1)$ prime divisors in a fixed-ratio N -shell. $\square$

The identities above expose a genuine coefficient-energy saving and remove the quotient-resonant sector, but they do not supply a positive power saving for the remaining moving-characteristic selector. On a selected hit the local trace contains a trivial summand of complete mass $\ell - 1$; consequently bounded local conductor alone cannot control the horizontal frequency with which the actual Apéry slopes are selected as ℓ varies.

The all-distinct residual can nevertheless be shortened further before forming its final TT^* expansion. The useful coordinate is the actual centered CRT integer, rather than only the difference of the two centered prime offsets.

For arbitrary sets $\mathcal{E}_{p,q,d} \subseteq \mathcal{Z}_p \times \mathcal{Z}_q$, put

$$a_{p,q}(r, s) := t_{p,q}(r, s) - \frac{pq - 1}{2}$$

and define

$$B_{d,\leq R}^{\text{CRT}} := \sum_{\substack{p \neq q \\ p, q \asymp N}} u_p u_q \widehat{w} \left(\frac{Nd}{pq} \right) \sum_{\substack{(r,s) \in \mathcal{E}_{p,q,d} \\ |a_{p,q}(r,s)| \leq R}} e \left(\frac{d t_{p,q}(r, s)}{pq} \right),$$

$$\mathcal{S}_{D,\leq R}^{\text{CRT}} := N \sum_{d \neq 0} W(|d|/D) B_{d,\leq R}^{\text{CRT}}.$$

Allowing the sets $\mathcal{E}_{p,q,d}$ to depend on d makes the next estimate monotone under every row-deletion mask used above.

Proposition 257 (Low centered-CRT deletion). *For $N^{2/5} \leq D \leq N^{41/100}$ and $N \leq R \leq CN^2$,*

$$|\mathcal{S}_{D,\leq R}^{\text{CRT}}| \ll_{v,w,W,C} ND \left(1 + RN^{-7/10} (\log N)^{3/5} \right), \quad (\text{CR1})$$

$$\sum_{d \neq 0} |W(|d|/D)|^2 |B_{d,\leq R}^{\text{CRT}}|^2 \ll_{v,w,W,C} D \left(1 + RN^{-7/10} (\log N)^{3/5} \right)^2. \quad (\text{CR2})$$

Consequently, if $D = N^{41/100}$ and $R = N^{129/100-\delta}$ with fixed $0 < \delta < 29/100$, then

$$|\mathcal{S}_{D,\leq R}^{\text{CRT}}| \ll N^{2-\delta+o(1)}, \quad \sum_d |W(|d|/D)|^2 |B_{d,\leq R}^{\text{CRT}}|^2 \ll N^{159/100-2\delta+o(1)}. \quad (\text{CR3})$$

Proof. For $P = pq$ and $t = t_{p,q}(r, s)$ one has

$$\frac{P-1}{2} \equiv \frac{p-1}{2} \pmod{p}, \quad \frac{P-1}{2} \equiv \frac{q-1}{2} \pmod{q}.$$

Thus a tuple with centered CRT integer a uses two primes counted by $m_N^{\text{ctr}}(a)$ from Proposition 421. Since $U_N := \max_{p \asymp N} |u_p| \ll_v (\log N)/N$, arbitrary deletion masks give, uniformly in d ,

$$|B_{d,\leq R}^{\text{CRT}}| \ll_{v,w} U_N^2 \sum_{|a| \leq R} m_N^{\text{ctr}}(a)^2$$

$$\ll_{v,w,C} 1 + RN^{-7/10} (\log N)^{3/5}.$$

Here $R < pq/2$ throughout the endpoint application, so the centered CRT representative is unique; in the stated larger range the same argument only overcounts possible prime pairs and remains valid. The support of W contains $O_W(D)$ integers d . Absolute summation of the preceding pointwise estimate proves (CR1); squaring it and summing the d -shell proves (CR2). Substituting $D = N^{41/100}$ and $R = N^{129/100-\delta}$ gives (CR3). $\square$

In particular, the low-centered contribution is removed in the original signed d -sum before TT^* , so no low-high cross term is created. Both circular determinant branches and all actual masks are retained. The unresolved all-distinct sector is now confined to $|a_{p,q}(r, s)| > N^{129/100-\delta}$; controlling that high-centered sector still requires new arithmetic input.

Proposition 258 (One trace-function mismatch removed on doublets). *On the doublet sector, $a \mapsto F_p(a)$ is the trace function of a rank-2, uniformly bounded-conductor sheaf: explicitly it is the direct sum of the two rank-1 Artin–Schreier sheaves with traces $a \mapsto e_p(ar_p)$ and $a \mapsto e_p(-a(1+r_p))$. Thus the growing-complexity obstruction for representing the local Fourier coefficient disappears completely on this sector.*

Proof. For $\mathcal{Z}_p = \{r_p, p-1-r_p\}$,

$$F_p(a) = e_p(ar_p) + e_p(-a(1+r_p)).$$

Each summand is a linear additive-character trace of rank and conductor bounded independently of both p and the slope. Direct sum proves the claim. $\square$

This does not make the frameworks discussed in Remark 154 applicable as stated. The slope r_p is selected by a same-characteristic Apéry zero condition and moves with p ; there is no fixed sheaf over one fixed finite field whose extension-degree traces give this horizontal family. Moreover (167) couples two varying characteristics at once. The anchored family below has exactly the same rank-2 conductor bound and still violates the target. Hence bounded local conductor removes one of the four mismatches listed in Remark 154, but horizontal control of the moving slopes remains the load-bearing arithmetic input.

Palindromic reality and the bound $Z(p) = O(1)$ cannot prove (SDC). The following construction quantifies the obstruction at exactly the required short-arc scale.

Proposition 259 (Reflected anchored obstruction). *Let*

$$\mathcal{P}_N = \{p \text{ prime} : N < p \leq 2N, p \nmid (4N+1)(2N+1)\}$$

and, for $p \in \mathcal{P}_N$, set

$$\mathcal{Z}_p^* = \{2N \bmod p, -2N-1 \bmod p\}.$$

Then every $\mathcal{Z}_p^$ is a two-element set invariant under $r \mapsto p-1-r$; it contains neither 0 nor adjacent residues. Yet the short-arc estimate at $(P, M) = (N, 4N)$ fails by a factor $N^{1-o(1)}$. Simultaneously, all columns contain the point $2N \in I_N$, so their centered block dispersion is $\gg N^2$ in the logarithmically weighted normalization.*

Proof. The divisors of $4N+1$ are exactly those for which the two displayed residues coincide; excluding divisors of $2N+1$ also keeps 0 out of the sets and rules out one of the two possible adjacency congruences. The other would force $p \mid 2N$, impossible for a prime $p \in (N, 2N)$. These exclusions remove only $O(1)$ primes from this dyadic range. For the corresponding Fourier coefficients, let D_N^* denote (159). Since $2N \bmod p \in \mathcal{Z}_p^*$ for every $p \in \mathcal{P}_N$, the prime number theorem gives

$$D_N^*(2N) = \sum_{p \in \mathcal{P}_N} (\log p)(1 - 2/p) \asymp N.$$

At $M = 4N$ the Fejér weight of this mode is

$$\gamma_{4N}(2N) = \frac{1}{4\pi^2 N^2}.$$

Therefore (158) gives

$$\int_{\mathbb{T}} |A_{N,4N}^*(x)|^2 dx \gg 1.$$

On the other hand, finite Fourier Parseval gives

$$Q_N^* = \sum_{p \in \mathcal{P}_N} (\log p)^2 \left(\frac{2}{p} - \frac{4}{p^2} \right) \asymp \log N.$$

Thus the proposed right-hand side is $(4N)^{-1+o(1)} Q_N^* = N^{-1+o(1)} \log N = o(1)$. The same common hit at $2N$ gives a centered deviation $\asymp N$ and hence weighted dispersion $\gg N^2$: indeed every column has at most two hits in the interval of length N , so the empirical weighted mean is $O(N^{-1} \sum_{p \in \mathcal{P}_N} \log p) = O(1)$. $\square$

The obstruction is not an assertion about the Apéry zeros. It proves the required impossibility content: reflection, the exact palindromic phase, doublet cardinality, vertical Parseval, and generic prime spacing do not imply the short-arc bound. A successful proof must insert an Apéry-specific horizontal fact into (165), exactly as anticipated in Remark 154; the separately running arithmetic oracle is responsible for such an input.

12.21 Computational reconnaissance

The reproducible computation in `oracleA.explore.py` and `oracleA.exploration.md` uses the exact zero-set bank and the blocks $N/2 < p \leq N$ for $N = 8192, 16384, 32768, 65536$. For the complete two-position doublet sum, the exact Parseval ratios $(T + F)/T$ are between 1.0224 and 1.0319. After declaring the major arcs $|\theta - a/q| \leq 16/(qN)$ with $q \leq 16$, the largest sampled minor-arc peaks are between $3.38\sqrt{T}$ and $3.69\sqrt{T}$. The largest nondegenerate connected correlation between $e(3p\theta/2)$ and $\cos(\pi\theta h_p)$ at those small-denominator rationals falls from 0.2204 to 0.0657 across the four scales. An independent-uniform-gap collision model predicts the observed complete-pair collision count within ratios 0.756 to 1.128.

The shell computation separately enumerates every lift in (163). With logarithmic weights its true off-diagonal ratios E°/S_c are respectively 0.00497, 0.00185, -0.00689 , 0.00340. Direct counting and an independent finite-group Fourier reconstruction agree within $4.4 \cdot 10^{-10}$. The clipping distinction is numerically substantial: at $N = 8192$ the complete two-position sum retains only 162 of its displayed positions in I_N , whereas the actual doublet columns have 374 shell lifts. These observations support square-root-scale cancellation, but none is used to infer (155).

STALL REPORT. The target (155) is not proved, either for the full Apéry sum or for its doublet sub-sum. The rigorous analytic outputs are:

1. the exact triangular-kernel/Fejér identity (157)–(159);
2. the corrected dilation argument proving that the stated uniform short-arc target really would imply the fully unconditional all-index conclusion, without the unsupported identification of its Fourier energy with $S(P, N)$;
3. the exact doublet phase and the clipping domain (164), correcting the unrestricted use of Remark 155;
4. the one-sided cross-prime short-determinant estimate (SDC), isolated as the precise missing bilinear lemma after the harmless same-prime terms are removed;
5. Proposition 258, which removes the growing-conductor trace-function mismatch on the doublet sector while locating the surviving obstruction in the horizontal distribution of the moving slopes r_p ;

6. the reflected anchored family of Proposition 259, which proves that none of palindromy, doublet dominance, vertical bounds, or generic large sieve geometry can supply that lemma.

12.22 Arithmetic oracle B: a fixed-anchor theorem and the sharp horizontal obstruction

Put

$$\mathcal{Z}_p = \{0 \leq j < p : b_j \equiv 0 \pmod{p}\}, \quad A_p(t) = \sum_{j=0}^{p-1} b_j t^j.$$

The requested AMTD estimate is not proved here. There is nevertheless an unconditional answer to the literal fixed-target question in goal G2. It is stronger than density zero, but it also exposes the decisive quantifier reversal: a target fixed before the prime varies is elementary, whereas the no-go target is chosen after the dyadic scale. Thus the status of the ranked goals is: G1 is open; literal G2 is proved below but is insufficient for the moving target; G3 is not achieved because the required marked crystalline coordinate has not been constructed; and G4, the sharp obstruction with the exact missing input, is achieved.

Proposition 260 (Fixed Apéry anchors are finite). *For $c \in \mathbb{Z}$, let $\bar{c}_p \in \{0, \dots, p-1\}$ be its residue modulo p , and set*

$$c^\flat = \begin{cases} c, & c \geq 0, \\ -c-1, & c < 0. \end{cases}$$

Then

$$\{p \geq 5 : \bar{c}_p \in \mathcal{Z}_p\} \subseteq \{p \geq 5 : p \mid b_{c^\flat}\}. \quad (170)$$

Consequently, for every fixed c ,

$$\#\{p \leq X : \bar{c}_p \in \mathcal{Z}_p\} \leq \omega(b_{c^\flat}) = O_c(1) = o(\pi(X)). \quad (171)$$

For $p > |c|$, the inclusion in (170) is an equivalence.

Proof. First suppose that $c \geq 0$, and write $c = \sum_i c_i p^i$ in base p . The lowest digit is $c_0 = \bar{c}_p$. If $\bar{c}_p \in \mathcal{Z}_p$, then $p \mid b_{c_0}$, and Gessel's Lucas congruence (Theorem 4) gives

$$b_c \equiv \prod_i b_{c_i} \equiv 0 \pmod{p}.$$

Now let $c = -d < 0$ and write $d = qp + s$ with $0 \leq s < p$. If $s = 0$, then $\bar{c}_p = 0$ and there is no hit because $b_0 = 1$. If $s > 0$, then $\bar{c}_p = p - s$. Reflection (Remark 21, or Proposition 99) gives

$$b_{p-s} \equiv b_{s-1} \pmod{p}.$$

Thus a hit implies $p \mid b_{s-1}$. The lowest base- p digit of $d-1$ is $s-1$, so another application of Lucas gives $p \mid b_{d-1} = b_{c^\flat}$. This proves the inclusion.

The binomial formula for b_n has positive integral summands and its $k = 0$ summand is 1; hence b_n is a fixed nonzero integer for fixed n . It has only finitely many prime divisors, proving (171). Finally, if $p > |c|$, then for $c \geq 0$ one has $\bar{c}_p = c$, while for $c = -d < 0$ reflection gives $b_{\bar{c}_p} = b_{p-d} \equiv b_{d-1}$; both implications are then reversible. $\square$

Remark 261 (Why this does not kill the anchored star). Proposition 260 has the quantifier order

$$\forall c \in \mathbb{Z} \exists C_c \forall X : \quad \#\{p \leq X : \bar{c}_p \in \mathcal{Z}_p\} \leq C_c. \quad (172)$$

In Section 12.5, however, the anchor is $m_0 = m_0(N) \in I_N$ and is allowed to move with N . To exclude just one full star in a dyadic prime block, one would already need a uniform statement such as

$$\sup_{m \in I_N} \#\{P < p \leq 2P : m \bmod p \in \mathcal{Z}_p\} = o(P/\log P), \quad (173)$$

in the relevant range $P > \sqrt{N}$. AMTD asks for a centered second moment, which is stronger still.

Lucas merely turns the expression inside the supremum into a count of large prime divisors of the moving integer b_m . The elementary height bound $\log b_m = O(m)$ gives at best $O(N/\log N)$ prime divisors of size $\asymp N$ when $m \asymp N$, exactly the size of a full prime block. Thus this Lucas/height argument supplies no way to move the constants C_c outside the supremum. This is the same distinction exhibited by the anchored-star/no-go construction in Section 12.

12.22.1 The two zero loci are in dual spaces

The proposed B1 and B2 routes make an object-level identification that is false. The following elementary calculation makes the distinction exact.

Proposition 262 (Coefficient zeros are not Hasse-polynomial roots). *For $1 \leq j \leq p-2$, the following identity holds in $\mathbb{F}_p$:*

$$b_j = - \sum_{t \in \mathbb{F}_p^\times} A_p(t) t^{-j}. \quad (174)$$

Consequently,

$$\mathcal{Z}_p = \{j : [t^j] A_p(t) = 0\}$$

is a coefficient, or finite-Mellin, zero-set. It is not the evaluation root locus of A_p or of the square root in Remark 10.

Proof. After inserting $A_p(t) = \sum_{k=0}^{p-1} b_k t^k$, multiplicative orthogonality gives

$$\sum_{t \in \mathbb{F}_p^\times} A_p(t) t^{-j} = \sum_{k=0}^{p-1} b_k \sum_{t \in \mathbb{F}_p^\times} t^{k-j} = -b_j,$$

because $1 \leq j \leq p-2$ and the inner sum vanishes unless $k = j$.

For a concrete separation, exact reduction at $p = 7$ gives

$$\begin{aligned} A_7(t) &= 1 + 5t + 3t^2 + 3t^3 + 3t^4 + 5t^5 + t^6 \\ &= (t-1)^2(t^2+1)^2 \pmod{7}. \end{aligned}$$

No coefficient is zero, so $\mathcal{Z}_7 = \emptyset$, whereas $t = 1$ is an evaluation root of both A_7 and its square root. $\square$

In particular, if $\rho_p = \min \mathcal{Z}_p$, then ρ_p need not be a root of $B_p(t)$, and no theorem identifies the two notions; they can coincide accidentally at individual primes. A Weil bound for $V(B_p)$ therefore does not bound

$$F_p(a) = \sum_{j \in \mathcal{Z}_p} e_p(aj).$$

Nor is ρ_p equivalent to the whole zero-set: it is undefined when $\mathcal{Z}_p$ is empty, and for example

$$\mathcal{Z}_{181} = \{19, 47, 133, 161\}.$$

For a center singleton the first zero trivially determines the set; only on the doublet subfamily $Z(p) = 2$ does it determine a noncentral reflected pair and its gap. Once there are at least two noncentral pairs, it does not determine the whole zero-set.

12.22.2 Why the square root is not a bounded-complexity family

Let $\Delta(t) = t^2 - 34t + 1$. The factorization stated in Remark 10 and proved, at the level of factorization, by Caruso–Fürnsinn–Vargas–Montoya–Zudilin [10] is

$$A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2, \quad \deg B_p = \frac{p-1-2\varepsilon_p}{2}. \quad (175)$$

Thus $\deg B_p = \Theta(p)$. This yields a second, independent obstruction to B2 even if one temporarily replaces coefficient zeros by evaluation roots.

Proposition 263 (Fixed family versus moving Hasse section). *The polynomials B_p cannot be the reductions of one fixed bounded-degree divisor on the t -line. If B_p is squarefree, then $V(B_p)_{\mathbb{F}_p}$ has $\deg B_p = \Theta(p)$ geometric points. The equation $B_p(t) = 0$ is zero-dimensional and has no genus; if it is replaced by the hyperelliptic curve $y^2 = B_p(t)$, then*

$$g_p = \left\lfloor \frac{\deg B_p - 1}{2} \right\rfloor = \Theta(p).$$

Hence the standard degree/genus form of the Weil–Bombieri estimate supplies no constant uniform in p for either version.

Proof. A fixed relative divisor of degree D has fibers of degree D , whereas (175) has unbounded degree. The remaining assertions are the point count of a squarefree zero-dimensional divisor and the standard genus formula for a squarefree hyperelliptic equation. $\square$

There is no contradiction between this proposition and the existence of a fixed Picard–Fuchs family. A fixed family can produce, separately in each characteristic p , a new Frobenius/Hasse section whose weight and degree grow with p . “The geometric family is fixed” does not imply “the derived Hasse polynomial has bounded complexity.” This is precisely the moving-polynomial versus fixed-family quantifier distinction.

There are two source cautions. First, Theorem 1.2 of [10] states the square/quadratic-times-square factorization. Its proof uses the Franel relation

$$f_\alpha(t(x)) = (1+x)h(x)^2, \quad t = \frac{x(1-8x)}{1+x}, \quad (176)$$

and the quadratic extension with discriminant Δ . It does not state that B_p is squarefree, and it does not formulate the theorem in terms of a Hasse invariant, a Picard group, or an elliptic pencil. The squarefree and coprimality assertions in Remark 10 therefore require a separate proof or reference; the finite verification below is not such a proof.

Second, the explicit pencil proposed in the oracle specification,

$$E_t : y^2 = x(x-1)(x-t(1-t)), \quad (177)$$

is not identified with the Apéry truncation in the cited material. In the displayed coordinate its discriminant is

$$16t^2(1-t)^2(1-t+t^2)^2,$$

whose support differs from $\Delta(t) = 0$. More concretely, at $p = 7$ its raw Deuring polynomial is obtained by substituting $\lambda = t(1-t)$ in $1 + 2\lambda + 2\lambda^2 + \lambda^3$; its unique $\mathbb{F}_7$ root is $t = 4$, while the Apéry polynomial above has root $t = 1$. This does not exclude an additional base change and gauge. It proves that (177), as printed, cannot simply be declared to produce B_p , and in any event evaluation Frobenius cannot produce the coefficient-first-zero ρ_p without the Mellin step (174).

12.22.3 The moving-target quantifier trap

Remark 21 identifies $b_m \equiv 0 \pmod{p}$ with collisions in the projective Apéry orbit and supplies reflection. For every fixed gap h , the corresponding gap polynomial $N_h(m) \in \mathbb{Z}[m]$ is fixed, so a prime-aspect Chebotarev theorem may be applied after h is fixed. The first actual gap, however, may range up to $O(p)$, no uniform $o(p)$ bound is known, and it varies with the same prime being counted. The available quantifiers are

$$\text{for every fixed } h, \quad X \longrightarrow \infty;$$

AMTD would require estimates uniform for moving $h \ll p$ inside one dyadic block. Degrees, splitting fields, discriminants, and effective Chebotarev constants all move with h . Interchanging these limits is exactly the forbidden moving-polynomial argument.

The center gives an even sharper warning. Beukers' congruence in Remark 62 says

$$b_{(p-1)/2} \equiv a_p(f) \pmod{p}, \quad f = \eta(2z)^4 \eta(4z)^4.$$

The index $(p-1)/2$ moves with p , and the density-zero conjecture for the nonordinary primes $p \mid a_p(f)$ is open for this non-CM weight-four form. Ordinary Sato–Tate controls fixed intervals for normalized traces; it does not prove this shrinking, same-characteristic divisibility event. It therefore cannot be invoked to prove B1 or B3.

Finally, equidistribution of ρ_p modulo each fixed integer q would still be only a marginal statement. It has neither the uniformity in growing frequencies needed for the minor arcs nor the two-prime joint information needed for collision hyperplanes. In particular it cannot, by itself, exclude the scale-dependent anchored star.

12.22.4 The correct fixed-family reduction

The minimum horizontal statement can be written without referring to a spurious root of B_p . For $N < p \leq 2N$ and $m \in I_N = (N, 2N]$, put

$$X_p(m) = \mathbf{1}_{\{p \leq m\}} \mathbf{1}_{\{b_{m-p} \equiv 0 \pmod{p}\}}, \quad T_N = \sum_{N < p \leq 2N} \sum_{m \in I_N} X_p(m).$$

Then the centered marked two-characteristic Hasse dispersion statement is

$$\left| \sum_{\substack{N < p, q \leq 2N \\ p \neq q}} \sum_{m \in I_N} X_p(m) X_q(m) - \frac{T_N^2}{N} \right| \ll N^{o(1)} T_N. \quad (178)$$

Writing

$$F_N := \sum_{m \in I_N} L_N(m)(L_N(m) - 1), \quad L_N(m) = \sum_{N < p \leq 2N} X_p(m),$$

for the ordered collision count, the left side of (178) is $|F_N - T_N^2/N|$. The exact identity

$$\sum_{m \in I_N} \left(L_N(m) - \frac{T_N}{N} \right)^2 = T_N + F_N - \frac{T_N^2}{N}$$

shows directly that (178) gives the top-block AMTD bound. It also kills a full top-block anchored star (for example the anchor $m_0 = 2N$), for which $L_N(m_0)$ has order $N/\log N$, and it is the top-block form of the centered second-factorial-moment input used by AMTD. The corresponding statement for all $P > \sqrt{N}$, with the centering of Hypothesis 152, is the precise input needed by the analytic oracle.

A genuine fixed-family route would have to construct, from one integral transcendental crystal (morally the symmetric square of the underlying rank-two Picard–Fuchs connection), the Kummer-twisted objects

$$\mathcal{M}_{p,j} = R\Gamma_{c,\text{rig}}(U_p, \mathcal{V}_{\text{tr},p} \otimes \mathcal{K}_{\omega_p^{-j}}) \quad (179)$$

and a marked Hasse coordinate $c_{p,j}$ satisfying

$$c_{p,j} = 0 \iff b_j \equiv 0 \pmod{p}.$$

It would then have to prove (178) for these marked zeros, uniformly in the two characteristics and in the moving characters j . This is a $1/p$ -scale crystalline anti-concentration and horizontal large sieve statement, not ordinary Frobenius equidistribution. Fixed rank/conductor of an ℓ -adic sheaf does not supply it: the Jacobi-sum counterexample discussed with the Sym^2 remarks has fixed complexity but linearly many bad tame characters.

Thus a construction of (179), its marked coordinate, and the centered two-characteristic estimate (178) would be a quantifier-correct reduction of AMTD to a fixed arithmetic family. This is a specification of the missing package, not an achieved G3 reduction to an existing Frobenius/Sato–Tate theorem: neither the marked coordinate with all its integral control nor its horizontal dispersion theorem is presently available.

12.22.5 Finite verification

The reproducible computation in `oracleB.explore.py` recomputes all $\mathcal{Z}_p$ for the 2260 primes $5 \leq p \leq 20000$ and compares them entrywise with the independent binary zero-set bank. The exact histogram is

$Z(p)$	0	1	2	4	6	8
$\#p$	1356	2	695	176	27	4.

Among the 695 doublets, the KS distance of $2\rho_p/(p-1)$ from the reflection-constrained uniform limit is 0.03894. Small-modulus permutation tests for the outer gap $h_p \bmod q$ against $p \bmod q$, using all 904 active primes and $q = 3, 5, 7, 11, 13$, have p-values between 0.257 and 0.675.

An exact Sage/FLINT worker verified (175), the predicted degree, squarefreeness, and coprimality to Δ at every one of the 2260 primes. It also verified $B_p \mid t^{p^2} - t$ and used $\deg \gcd(B_p, t^p - t)$ to extract the exact linear/quadratic factorization type at every prime in the range. Complete factorization into explicit factor polynomials was restricted to a sparse cross-check sample. At $p = 101, 503, 1009, 5003, 10007, 19997$, the degrees of B_p are respectively 50, 250, 504, 2501, 5002, 9998,

and the numbers of irreducible factors are 30, 148, 262, 1284, 2595, 5058. These calculations demonstrate growing complexity and catch transcription errors; they prove no assertion beyond the finite range. Exact identity checks used in the propositions above are repeated independently by `oracleB.verify.py`.

STALL REPORT. The actual cross-prime decorrelation of the Apéry coefficient zero-sets, and hence G1/AMTD, is not proved. G3 is not reached for the reason stated above; literal G2 is proved but has insufficient uniformity; the obstruction and exact missing input constitute G4. The rigorous outputs are:

1. Proposition 260, which proves the literal G2 count is $O_{m_0}(1)$ for every fixed anchor;
2. Remark 261, which proves that this has the wrong quantifier order for the moving anchored-star family;
3. Proposition 262 and the exact $p = 7$ counterexample, which show that B1/B2 as stated confuse Mellin coefficient zeros with evaluation roots;
4. Proposition 263, which isolates the linear-degree obstruction even on the evaluation-root side;
5. the fixed-gap/fixed-prime-family audit, including the open nonordinary center slice from Remark 62;
6. the quantifier-correct remaining arithmetic input (178), together with the crystalline Mellin package (179) that a fixed-family proof would have to supply.

12.23 A fixed toric marked coordinate

Put

$$\Lambda(x, y, z) = \frac{(1+x)(1+y)(1+z)((1+y)(1+z) + xyz)}{xyz} \in \mathbb{Z}[x^{\pm 1}, y^{\pm 1}, z^{\pm 1}]. \quad (180)$$

This one Laurent polynomial gives the required marked scalar. In particular, it avoids both loci ruled out by Propositions 262 and 263: it is neither an evaluation of H_p nor an evaluation of its factor B_p .

Lemma 264 (Fixed constant-term realization). *For every integer $n \geq 0$,*

$$\text{CT}_{x,y,z} \Lambda(x, y, z)^n = \sum_{r=0}^n \binom{n}{r}^2 \binom{n+r}{r}^2 = b_n. \quad (181)$$

Moreover, every exponent of each of x, y, z occurring in Λ lies between -1 and 1 .

Proof. Write

$$A = (1+x)(1+y)(1+z), \quad E = (1+y)(1+z) + xyz.$$

Then the constant term in question is

$$[x^n y^n z^n] A^n E^n.$$

If exactly k of the n factors selected in E^n contribute xyz , the x -coefficient contributes a second factor $\binom{n}{k}$, while the y - and z -coefficients both contribute $\binom{2n-k}{n-k}$. Hence

$$\text{CT} \Lambda^n = \sum_{k=0}^n \binom{n}{k}^2 \binom{2n-k}{n-k}^2.$$

The substitution $r = n - k$ gives (181). The exponent assertion follows directly from the numerator of (180), whose degree in each variable is between 0 and 2, followed by division by xyz . $\square$

Theorem 265 (The marked coordinate). *Let $p \geq 5$ be prime. For $0 \leq j \leq p - 2$, define the scalar*

$$c_{p,j}^{\text{tor}} := - \sum_{(x,y,z) \in (\mathbb{F}_p^\times)^3} \Lambda(x,y,z)^j \in \mathbb{F}_p, \quad (182)$$

where the zeroth power is the constant Laurent polynomial 1, including at the zeros of Λ . Define separately

$$c_{p,p-1}^{\text{tor}} := 1.$$

Then, for every $0 \leq j \leq p - 1$,

$$c_{p,j}^{\text{tor}} = b_j \pmod{p}. \quad (183)$$

Consequently

$$c_{p,j}^{\text{tor}} = 0 \iff b_j \equiv 0 \pmod{p}. \quad (184)$$

Proof. For an integer e ,

$$\sum_{a \in \mathbb{F}_p^\times} a^e = \begin{cases} -1, & p-1 \mid e, \\ 0, & p-1 \nmid e, \end{cases} \quad \text{in } \mathbb{F}_p.$$

By Lemma 264, every exponent in Λ^j belongs to $[-j, j]$ in each coordinate. If $0 \leq j \leq p - 2$, the only multiple of $p - 1$ in this interval is zero. Three applications of torus orthogonality therefore give

$$\sum_{(\mathbb{F}_p^\times)^3} \Lambda^j = (-1)^3 \text{CT } \Lambda^j = -b_j,$$

which proves (183) on this range. At the remaining endpoint, the reflection in Remark 21 gives $b_{p-1} = b_0 = 1$ modulo p , exactly the separately defined value. $\square$

Remark 266 (The endpoint is not a cosmetic convention). The raw power sum at $j = p - 1$ cannot replace the separate definition. If

$$T = (\mathbb{G}_m)^3, \quad U = T \setminus V(\Lambda),$$

then direct inclusion–exclusion gives

$$\#U(\mathbb{F}_p) = p^3 - 7p^2 + 17p - 14. \quad (185)$$

Indeed, the union of $x = -1$, $y = -1$, and $z = -1$ has $(p - 1)^3 - (p - 2)^3$ points. Off $y = -1$ and $z = -1$, the fourth zero equation determines a unique nonzero x and has $(p - 2)^2$ points; its intersection with the first union has $p - 2$ points, given by $x = -1$ and $1 + y + z = 0$. Subtracting this zero locus from $(p - 1)^3$ gives (185). Thus $-\sum_{T(\mathbb{F}_p)} \Lambda^{p-1} = -\#U(\mathbb{F}_p) \equiv 14 \pmod{p}$, not $b_{p-1} = 1$ in general. At $p = 7$ the raw endpoint is even zero. For $j = 0$, by contrast, the convention in (182) gives $-(p - 1)^3 = 1$ in $\mathbb{F}_p$.

12.23.1 Uniform geometric complexity

For $1 \leq j \leq p-2$, let $E_p = \mathbb{Q}(\mu_{p-1})$. Choose a prime $\mathfrak{p} \mid p$ and a Teichmüller character $\omega_p: \mathbb{F}_p^\times \rightarrow \mu_{p-1}$ such that $\omega_p(a) \equiv a \pmod{\mathfrak{p}}$, and put $\chi_{p,j} = \omega_p^j$. Fix $\ell = 2$ (recall that $p \geq 5$), choose an embedding $E_p \hookrightarrow \overline{\mathbb{Q}_2}$, and let $\mathcal{K}_{\chi_{p,j}}$ be the associated rank-one Kummer sheaf, with the Frobenius convention chosen to have trace $\chi_{p,j}$. On the fixed threefold U put

$$\mathcal{L}_{p,j} = \Lambda^* \mathcal{K}_{\chi_{p,j}}. \quad (186)$$

The Grothendieck trace formula gives the cyclotomic integer

$$\hat{c}_{p,j}^{\text{tor}} = -\text{Tr}_{\text{alt}}\left(\text{Frob}_p \mid R\Gamma_c(U_{\overline{\mathbb{F}_p}}, \mathcal{L}_{p,j})\right) = -\sum_{u \in U(\mathbb{F}_p)} \chi_{p,j}(\Lambda(u)). \quad (187)$$

Its reduction at the chosen prime above p is $c_{p,j}^{\text{tor}} = b_j$. Both endpoints may instead be represented by the constant sheaf on T : its negative Frobenius supertrace is $-(p-1)^3$, whose reduction is 1. Thus their cohomological complexity is fixed as well. Using the trivial Kummer sheaf extended by zero from U would give the wrong value in Remark 266.

Proposition 267 (Uniform complexity of the toric object). *The ambient variety and rational map in (186) are independent of p and j . The coefficient sheaf has rank one. There is an absolute constant B_Λ such that*

$$\sum_i \dim H_c^i(U_{\overline{\mathbb{F}_p}}, \mathcal{L}_{p,j}) \leq B_\Lambda \quad (188)$$

for all $p \geq 5$ and $1 \leq j \leq p-2$. In particular, both the total degree of the corresponding cohomological L -factor and the ramification complexity are $O(1)$ uniformly in (p, j) . This is geometric complexity: as for the Kummer twists in (179), it does not count the degree of the varying cyclotomic coefficient field E_p .

Proof. In $(\mathbb{P}^1)^3$, the principal divisor of Λ has the four simple zero components

$$x = -1, \quad y = -1, \quad z = -1, \quad \overline{Q}: X_0(Y_0 + Y_1)(Z_0 + Z_1) + X_1 Y_1 Z_1 = 0$$

and the six simple pole components

$$x = 0, \quad x = \infty, \quad y = 0, \quad y = \infty, \quad z = 0, \quad z = \infty.$$

The fourth zero component is smooth and geometrically irreducible. Indeed, its two coefficients as a linear form in (X_0, X_1) are coprime, and a direct check of the six displayed partial derivatives has no projective common zero. On the torus, smoothness is already immediate from $\partial((1+y)(1+z) + xyz)/\partial x = yz$. Thus the input divisor has ten fixed components of fixed multidegrees. The full divisor is not asserted to be simple normal crossings.

More explicitly, the change of coordinates

$$u = \frac{1+y}{y}, \quad v = \frac{1+z}{z}, \quad w = -\frac{x}{uv}$$

identifies U with the fixed toric-arrangement complement

$$(\mathbb{G}_m)^3 \setminus \{u = 1, v = 1, w = 1, uvw = 1\}.$$

In particular, a finite stratification depends only on this four-member arrangement, not on the character.

Every $\chi_{p,j}$ has order dividing $p-1$. Its pullback is therefore tame along every divisorial valuation and has Swan conductor zero. Take one log resolution of this fixed integral divisor and spread it out away from finitely many characteristics. The resulting stratification has a fixed finite number of strata. At every good prime, tame specialization for this log-smooth pair applies because the local monodromy has order prime to p ; it compares the relevant rank-one representation with one on the characteristic-zero fiber. Fix a finite CW model of that complex fiber. Its cellular cochain groups have dimension at most the number of cells for every rank-one representation, independently of its monodromy, and hence give (188). The finitely many excluded characteristics contribute only finitely many additional (p, j) and may be absorbed into B_Λ . This proves a uniform bound without claiming that the original ten-component divisor is normal crossings or that the cohomology is concentrated in one degree. $\square$

Remark 268 (What vanishes). Equation (184) concerns the reduction of the cyclotomic trace (187) modulo a prime above p . It does not assert that the characteristic-zero Frobenius trace itself vanishes. For example, at $(p, j) = (11, 5)$ the quadratic character sum gives $\tilde{c}_{11,5}^{\text{tor}} = -33 \neq 0$, while its reduction and b_5 both vanish modulo 11. This integral distinction is essential in any later attempt to prove (178).

12.23.2 Audits of the two originally suggested coordinates

The toric construction above is not a disguised bounded-length transfer word. With $B_n = (n!)^3 b_n$ and the matrix $M(n)$ from Remark 21, put $T_0 = I$ and $T_j = M(j-1) \cdots M(0)$. The recurrence gives the exact identity

$$T_j = \begin{pmatrix} B_j & 0 \\ B_{j-1} & 0 \end{pmatrix} \quad (j \geq 1). \quad (189)$$

Thus the marked entry is $(1, 1)$, not $(1, 2)$, and it detects $b_j = 0$ for $j < p$. Its matrix size is fixed, but its word length is j . Exact symbolic multiplication of $R_\ell(x) = M(x+\ell-1) \cdots M(x)$ gives, for $\ell \geq 2$,

$$\deg R_\ell = \begin{pmatrix} 3\ell & 3\ell+3 \\ 3\ell-3 & 3\ell \end{pmatrix}.$$

For completeness, if $A_\ell = (R_\ell)_{11}$ and λ_ℓ is its leading coefficient, then

$$A_{\ell+1} = P(x+\ell)A_\ell - (x+\ell)^6 A_{\ell-1}, \quad \lambda_{\ell+1} = 34\lambda_\ell - \lambda_{\ell-1}, \quad (\lambda_0, \lambda_1) = (1, 34).$$

The λ_ℓ are positive and strictly increasing, so the two leading terms never cancel. The remaining entries satisfy

$$(R_\ell)_{21} = A_{\ell-1}(x), \quad (R_\ell)_{12} = -x^6 A_{\ell-1}(x+1), \quad (R_\ell)_{22} = -x^6 A_{\ell-2}(x+1),$$

for $\ell \geq 2$. This proves the displayed degree matrix rather than extrapolating it from finitely many products. Likewise, the standard factorial gauge has marked coefficient

$$\frac{N_h(m)}{\prod_{s=2}^h (m+s)^3}.$$

The endpoint identity used in Remark 17,

$$N_h(-s) = (-1)^{s-1} B_{s-1} B_{h-s} \neq 0 \quad (1 \leq s \leq h),$$

shows that it has exactly $h - 1$ uncanceled finite poles, each of order three. These facts rule out the direct word and this direct gauge; they are not a no-go theorem for all constructions, as Theorem 265 demonstrates.

The raw Mellin coordinate from Proposition 262 has a different restricted obstruction. If g is a primitive root, its corrected group-algebra element is

$$Q_p(X) = - \sum_{r=0}^{p-2} H_p(g^r) X^{-r} \quad \text{in } \mathbb{F}_p[X^{\pm 1}]/(X^{p-1} - 1). \quad (190)$$

For $1 \leq j \leq p-2$, $Q_p(g^j) = b_j$. The factorization in Remark 10 implies, without any squarefreeness or coprimality assumption,

$$\#\{t \in \mathbb{F}_p^\times : H_p(t) = 0\} \leq \deg B_p + 2\varepsilon_p = \frac{p-1}{2} + \varepsilon_p,$$

and hence

$$\#\text{supp } Q_p = \#\{t \in \mathbb{F}_p^\times : H_p(t) \neq 0\} \geq \frac{p-1}{2} - \varepsilon_p \geq \frac{p-3}{2}.$$

Hence the raw periodic coordinate has growing group-algebra support and growing minimal cyclic constant-coefficient recurrence order: the distinct characters in (190) are linearly independent because the $(p-1)$ -point Fourier/Vandermonde matrix is invertible in $\mathbb{F}_p$. This is a no-go only for exact raw-value realizations of that form, not for the zero-equivalent toric coordinate.

Finally, bounded complexity of a Kummer factor on the original t -line does not by itself repair (179). At $p = 31$, the two roots of the conifold discriminant are 14 and 20, with $H_{31}(14) = H_{31}(20) = 7$ and $b_8 = b_{22} = 0$. For $j = 8$ the omitted bad-fiber contribution is

$$7(14^{-8} + 20^{-8}) = 11 \pmod{31},$$

so the Mellin sum restricted to the smooth locus is 20, not zero; the same failure occurs for $j = 22$. Boundary terms require integral control.

12.23.3 Quantifier boundary

The family (186) is a uniformly bounded family of cohomological spaces indexed by the tame character j . It is not a single sheaf on a “ j -line” whose trace function is $j \mapsto b_j$. Therefore a number of j -singularities or a conductor of such a j -line sheaf is not defined. In particular, Proposition 267 does not turn the moving-character problem into an ordinary FKM bilinear estimate. It is also an alternative toric Kummer realization of the marked scalar, not a construction of, or a comparison with, the transcendental crystal $\mathcal{V}_{\text{tr},p}$ in (179). Thus it supplies exactly the algebraic coordinate requested here, not the rest of that crystalline Mellin package.

STALL REPORT. The literal algebraic goal G1 is achieved by (182): the marked scalar is exactly b_j modulo p , and its ambient dimension, coefficient rank, divisor data, Swan conductors, and total cohomological dimension are uniformly bounded in (p, j) . G2 is false in its universal form and is not claimed; the transfer and raw-Mellin statements above are restricted no-go results. G3 is deliberately not attempted. No estimate for (SDC), no short-arc bound, no two-characteristic dispersion estimate (178), and no reduction of any of them to an existing theorem is proved here. The remaining obstruction is precisely horizontal control of the moving characters and their reductions, not the existence of an algebraic marked coordinate.

12.24 Exact toric fibers and the cover-adapted descent

Proposition 269 (Quadratic collapse of the toric Mellin coefficient). *Let $p \geq 5$ be prime, let χ be the quadratic character of $\mathbb{F}_p$ extended by $\chi(0) = 0$, and let*

$$\mu_p(a) = \#\{(x, y, z) \in (\mathbb{F}_p^\times)^3 : \Lambda(x, y, z) = a\} \quad (a \in \mathbb{F}_p^\times),$$

where Λ is the Laurent polynomial in (180). Put

$$A_j(t) = \sum_{k=0}^j \binom{j}{k}^2 t^k$$

and, for $1 \leq j \leq p-2$, put

$$S_{p,j}(t) = \sum_{r \in \mathbb{F}_p^\times} r^j \chi((t-1)^2 r^2 - 2(t+1)r + 1).$$

Then the fiber formula below is an integer equality, and the remaining identities hold in $\mathbb{F}_p$.

1. For every $a \neq 0$,

$$\mu_p(a) = (p-2)^2 + \sum_{\substack{t, w \in \mathbb{F}_p^\times \\ w \neq 1, tw \neq 1}} \chi \left(\left(t+1 - \frac{t(w-1)(1-tw)}{aw} \right)^2 - 4t \right). \quad (191)$$

2. The inner quadratic Mellin transform collapses exactly:

$$\boxed{S_{p,j}(t) = -A_{p-1-j}(t).} \quad (192)$$

Consequently

$$\boxed{b_j = - \sum_{t \in \mathbb{F}_p^\times} t^j A_j(t) A_{p-1-j}(t).} \quad (193)$$

3. If $1 \leq j \leq (p-1)/2$, then the polynomial identity

$$A_{p-1-j}(t) = (1-t)^{p-1-2j} A_j(t) \quad \text{in } \mathbb{F}_p[t] \quad (194)$$

turns (193) into the weighted-square formula

$$\boxed{b_j = - \sum_{t \in \mathbb{F}_p^\times} t^j (1-t)^{p-1-2j} A_j(t)^2.} \quad (195)$$

Proof. On a nonzero fiber of Λ , the change of variables

$$u = \frac{1+y}{y}, \quad v = \frac{1+z}{z}, \quad w = -\frac{x}{uv}$$

is invertible and has

$$u, v, w \in \mathbb{F}_p^\times, \quad u, v \neq 1, \quad w \neq 1, \quad uvw \neq 1.$$

Writing $t = uv$ and $s = u + v$, direct substitution gives

$$\Lambda = \frac{t(w-1)(1-tw)}{w(u-1)(v-1)}, \quad (u-1)(v-1) = t - s + 1.$$

Thus $\Lambda = a$ determines

$$s = t + 1 - \frac{t(w-1)(1-tw)}{aw}.$$

For fixed t, w , the number of ordered pairs (u, v) with product t and sum s is $1 + \chi(s^2 - 4t)$. The conditions $u, v \neq 1$ are automatic: the excluded equality $s = t + 1$ would force $(w-1)(1-tw) = 0$. There are

$$(p-2) + (p-2)(p-3) = (p-2)^2$$

admissible pairs (t, w) ; the first summand is the slice $t = 1$, where the two excluded values of w coincide. This proves (191).

Set

$$Q_t(R) = (t-1)^2 R^2 - 2(t+1)R + 1.$$

The classical Legendre generating identity is

$$Q_t(R)^{-1/2} = \sum_{n \geq 0} A_n(t) R^n. \quad (196)$$

Since $Q_t(0) = 1$, fractional powers are unambiguously defined in $\mathbb{F}_p[[R]]$. Frobenius gives

$$Q_t(R)^{(p-1)/2} = Q_t(R)^{-1/2} (Q_t(R)^p)^{1/2} = Q_t(R)^{-1/2} Q_t(R^p)^{1/2} \equiv Q_t(R)^{-1/2} \pmod{R^p}.$$

Euler's criterion and multiplicative orthogonality therefore give

$$\begin{aligned} S_{p,j}(t) &= \sum_{r \neq 0} r^j Q_t(r)^{(p-1)/2} \\ &= -[R^{p-1-j}] Q_t(R)^{(p-1)/2} = -A_{p-1-j}(t), \end{aligned}$$

which proves (192).

For $1 \leq j \leq p-2$, torus orthogonality and the exponent range in Lemma 264 give

$$\sum_{a \neq 0} \mu_p(a) a^j = \sum_{(x,y,z) \in (\mathbb{F}_p^\times)^3} \Lambda(x, y, z)^j = -b_j. \quad (197)$$

The constant term $(p-2)^2$ in (191) has zero j -th Mellin coefficient. For

$$\kappa(t, w) = \frac{t(w-1)(1-tw)}{w},$$

multiplication by a^2 inside the quadratic character followed by the bijection $a = \kappa r$ gives

$$\sum_{a \neq 0} a^j \chi \left(\left(t + 1 - \frac{\kappa}{a} \right)^2 - 4t \right) = \kappa^j S_{p,j}(t).$$

Because $j \geq 1$, the values $w = 1, t^{-1}$ may now be restored: their κ^j is zero. If

$$L_t(w) = \frac{(w-1)(1-tw)}{w} = 1 + t - tw - w^{-1},$$

then every exponent in $L_t(w)^j$ lies in $[-j, j]$, so

$$\sum_{w \neq 0} \kappa(t, w)^j = -t^j \text{CT}_w L_t(w)^j = -t^j A_j(t).$$

Combining this with (192) and (197) proves (193).

Finally, if P_n denotes the Legendre polynomial, then

$$A_n(t) = (1-t)^n P_n\left(\frac{1+t}{1-t}\right).$$

The hypergeometric expression $P_n(X) = {}_2F_1(-n, n+1; 1; (1-X)/2)$ shows, for $0 \leq j \leq (p-1)/2$, that

$$P_{p-1-j}(X) = P_j(X) \quad \text{in } \mathbb{F}_p[X] :$$

the two upper parameters are the same unordered pair modulo p , and the series truncates at degree j . Clearing the powers of $1-t$ proves the polynomial identity (194), including $t = 1$. Substitution in (193) gives (195). $\square$

Remark 270 (The collapse is an exact reformulation, not yet a saving). Expanding (193) and summing over t gives

$$-\sum_{t \neq 0} t^j A_j(t) A_{p-1-j}(t) = \sum_{\substack{0 \leq k \leq j, 0 \leq \ell \leq p-1-j \\ j+k+\ell=p-1}} \binom{j}{k}^2 \binom{p-1-j}{\ell}^2.$$

Putting $\ell = p-1-j-k$ and using

$$\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod{p}$$

recovers exactly

$$b_j = \sum_{k=0}^j \binom{j}{k}^2 \binom{j+k}{k}^2.$$

For $k > p-1-j$, both sides vanish modulo p by the same carry. Thus the quadratic fiber sum, the Legendre pairing, and the original Apéry coefficient are coefficientwise equivalent. In particular, (195) is a square in an indefinite finite-field pairing, not a positive norm; it can vanish, for example at $(p, j) = (11, 5)$. Any zero-set bound still requires arithmetic information beyond this exact collapse.

Theorem 271 (K3 collapse of the nonzero toric fibers). *Let $p \geq 5$ be prime and $a \in \mathbb{F}_p^\times$. Retain the notation $\mu_p(a)$ from Proposition 269, and put*

$$\varepsilon_a = \mathbf{1}_{a=1}, \quad \delta_a = \mathbf{1}_{a^2-34a+1=0}.$$

There is a K3 surface $\tilde{X}_a/\mathbb{F}_p$ such that, if

$$\tau_p(a) = \text{Tr}\left(\text{Frob}_p \mid H_{\text{et}}^2(\tilde{X}_a, \mathbb{F}_p, \mathbb{Q}_\ell)\right) \quad (\ell \neq p),$$

then

$$\boxed{\mu_p(a) = p^2 - (25 + \varepsilon_a + \delta_a)p + 12 + \tau_p(a).} \quad (198)$$

Equivalently, the character sum in (191) is

$$\boxed{\tau_p(a) - (21 + \varepsilon_a + \delta_a)p + 8.} \quad (199)$$

More precisely, there is a weight-two Frobenius trace $\theta_p(a)$ of dimension $3 - \varepsilon_a - \delta_a$ for which

$$\boxed{\mu_p(a) = p^2 - 6p + 12 + \theta_p(a), \quad \mu_p(a) - (p-2)^2 = \theta_p(a) - 2p + 8.} \quad (200)$$

Consequently the character sum has absolute value at most $5p + 8$, uniformly in $a \in \mathbb{F}_p^\times$.

Proof. We give the compactification and all point-count corrections explicitly. On a nonzero fiber the coordinates used in Proposition 269 identify the original torus fiber with

$$\mathcal{U}_a : \quad uv(w-1)(1-uvw) - aw(u-1)(v-1) = 0 \quad (201)$$

inside the locus

$$u, v, w \neq 0, \quad u, v, w \neq 1, \quad uvw \neq 1.$$

Indeed, the inverse map is

$$y = (u-1)^{-1}, \quad z = (v-1)^{-1}, \quad x = -uvw,$$

and the two remaining exclusions correspond to the factors $1+x$ and $((1+y)(1+z) + xyz)$ of Λ . Thus $\#\mathcal{U}_a(\mathbb{F}_p) = \mu_p(a)$.

Write $u = U_1/U_0$, $v = V_1/V_0$, and $w = W_1/W_0$. The closure X_a of (201) in $(\mathbb{P}^1)^3$ is the multidegree $(2, 2, 2)$ hypersurface

$$\begin{aligned} \mathcal{F}_a &= U_1 V_1 (W_1 - W_0)(U_0 V_0 W_0 - U_1 V_1 W_1) \\ &\quad - a U_0 V_0 W_0 W_1 (U_1 - U_0)(V_1 - V_0) = 0. \end{aligned} \quad (202)$$

In the shorthand $0, 1, \infty$ for points of $\mathbb{P}^1$, the reduced boundary $X_a \setminus \mathcal{U}_a$ consists of the following sixteen rational curves:

$$\begin{aligned} &u = 0, v = \infty, \quad u = 0, w = \infty, \quad u = 0, w = 0, \quad u = 0, v = 1, \\ &v = 0, u = \infty, \quad v = 0, w = \infty, \quad v = 0, w = 0, \quad v = 0, u = 1, \\ &u = \infty, w = 0, \quad u = \infty, w = 1, \quad v = \infty, w = 0, \quad v = \infty, w = 1, \\ &u = 1, w = 1, \quad u = 1, vw = 1, \quad v = 1, w = 1, \quad v = 1, uw = 1. \end{aligned} \quad (203)$$

Each has $p+1$ rational points. Their incidence locus contains sixteen points lying on exactly two components, four points lying on exactly three, and the single point $(1, 1, 1)$ lying on four. There are no other intersections. Hence

$$\#(X_a \setminus \mathcal{U}_a)(\mathbb{F}_p) = 16(p+1) - 16 - 2 \cdot 4 - 3 = 16p - 11. \quad (204)$$

For completeness, we record the full singularity calculation. The fixed boundary singularities are

points	type	
$(0, 1, \infty), (0, \infty, \infty), (1, 0, \infty), (\infty, 0, \infty)$	$4A_1,$	(205)
$(0, \infty, 0), (\infty, 0, 0)$	$2A_2,$	
$(1, 1, 1)$	$A_1 (a \neq 1), \quad A_2 (a = 1).$	

Here is a local check that also fixes the types in every characteristic $p \geq 5$. At the first four points, suitable affine coordinates give a quadratic Hessian of determinant $2a^2$. At $(0, \infty, 0)$ the quadratic part is

$$s(at - r),$$

and the three charts of the ambient blow-up have exceptional equations

$$S(aT - 1), \quad aT - R, \quad S(a - R).$$

The middle chart is manifestly smooth; at the only crossings in the other two charts the radial derivatives are respectively a^{-1} and a^2 . They are units because $a \neq 0$, so the strict transform is smooth and the two exceptional lines form the A_2 chain. The other fixed A_2 point is symmetric. At $(1, 1, 1)$ the quadratic part is

$$-ars - rt - st - t^2,$$

whose Hessian determinant is $2a(a - 1)$. At $a = 1$ it factors as $-(r + t)(s + t)$. More explicitly, in the r -, s -, and t -charts of the ambient blow-up its exceptional equations are

$$-(S + T)(T + 1), \quad -(R + T)(T + 1), \quad -(R + 1)(S + 1).$$

The only crossings are $(S, T) = (1, -1)$, $(R, T) = (1, -1)$, and $(R, S) = (-1, -1)$; the corresponding radial derivatives of the full strict transform are $2, 2, -2$. Hence one blow-up resolves this central A_2 as well.

It remains to find singularities in the open torus. Put $q = uvw$ and take logarithmic derivatives of

$$\Lambda = \frac{uv(w - 1)(1 - uvw)}{w(u - 1)(v - 1)}.$$

The three critical equations reduce to

$$q(2 - u) = q(2 - v) = 1, \quad qw = 1.$$

They give either the excluded point $(u, v, w) = (1, 1, 1)$ or

$$u = v, \quad w = 2 - u, \quad u^2 - 2u - 1 = 0. \tag{206}$$

At the latter point

$$a = 12u + 5,$$

so such a rational point exists exactly when $a^2 - 34a + 1 = 0$, and it is

$$(u, v, w) = \left(\frac{a - 5}{12}, \frac{a - 5}{12}, \frac{29 - a}{12} \right).$$

The logarithmic Hessian determinant at (206) is $(u - 1)/2$, hence this additional singularity is an A_1 . This also shows directly that the displayed list is complete.

All tangent cones of the A_2 points split over $\mathbb{F}_p$; the exceptional conic above each rational A_1 point has an $\mathbb{F}_p$ -point and is therefore a rational projective line. Thus all the displayed ADE configurations are split over $\mathbb{F}_p$. Resolving an A_n replaces one rational point by a chain with

$$n(p + 1) - (n - 1) = np + 1$$

rational points, for a net gain of np . If $\tilde{X}_a$ denotes the minimal resolution, then

$$\#\tilde{X}_a(\mathbb{F}_p) = \#X_a(\mathbb{F}_p) + (9 + \varepsilon_a + \delta_a)p. \tag{207}$$

The hypersurface (202) is anticanonical in $Y = (\mathbb{P}^1)^3$ and has only isolated rational double points. As a hypersurface it is Cohen–Macaulay, and the isolated-singularity calculation above gives regularity in codimension one; hence it is normal by Serre’s criterion. The exact sequence

$$0 \longrightarrow \mathcal{O}_Y(-2, -2, -2) \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{O}_{X_a} \longrightarrow 0$$

and Künneth give $H^1(Y, \mathcal{O}_Y) = H^1(Y, \mathcal{O}_Y(-2, -2, -2)) = H^2(Y, \mathcal{O}_Y(-2, -2, -2)) = 0$. Thus $H^0(X_a, \mathcal{O}_{X_a}) = \mathbb{F}_p$ and $H^1(X_a, \mathcal{O}_{X_a}) = 0$; the same calculation after base change gives geometric connectedness. Adjunction gives $\omega_{X_a} \cong \mathcal{O}_{X_a}$, and the ADE resolution is crepant, so $\tilde{X}_a$ is a K3 surface. Therefore

$$\#\tilde{X}_a(\mathbb{F}_p) = 1 + p^2 + \tau_p(a), \quad |\tau_p(a)| \leq 22p.$$

Combining this identity with (204) and (207) proves (198). Subtracting the baseline $(p-2)^2$ in (191) proves (199).

We now use the boundary configuration to deflate all but three of the generic cohomology classes. Number the curves in (203) as $C_1, \dots, C_{16}$ in the displayed order, and write $\tilde{C}_i$ for their strict transforms. Each $\tilde{C}_i$ is a smooth rational curve, hence has self-intersection -2 . Away from the rational double points, the nonzero pairwise intersections are the fourteen transverse pairs

$$\begin{aligned} (3, 7), (2, 6), (3, 4), (4, 15), (1, 12), (7, 8), (8, 13), \\ (11, 14), (12, 13), (5, 10), (9, 16), (10, 15), (9, 11), (10, 12), \end{aligned} \tag{208}$$

where, for example, $(3, 7)$ denotes the edge $\tilde{C}_3 - \tilde{C}_7$.

The resolution graph at the four fixed A_1 points has exceptional curves $E_1, \dots, E_4$ attached respectively to

$$\{C_2, C_4, C_{16}\}, \quad \{C_1, C_2\}, \quad \{C_6, C_8, C_{14}\}, \quad \{C_5, C_6\}. \tag{209}$$

At $(0, \infty, 0)$ the A_2 chain $F_1 - F_2$ has F_1 attached to C_1, C_{11} and F_2 attached to C_3 ; at $(\infty, 0, 0)$ the chain $G_1 - G_2$ has G_1 attached to C_5, C_9 and G_2 attached to C_7 . For $a \neq 1$, the central exceptional curve is attached to $C_{13}, C_{14}, C_{15}, C_{16}$. For $a = 1$, it is replaced by an A_2 chain $J_1 - J_2$, where J_1 is attached to C_{13}, C_{16} and J_2 to C_{14}, C_{15} . The extra A_1 exceptional curve when $a^2 - 34a + 1 = 0$ is disjoint from this boundary configuration. These attachments follow by blowing up the quadratic tangent cones displayed above. For instance, at $(0, \infty, 0)$ the two exceptional lines are the two factors of $s(at - r)$; the tangent directions of C_1, C_{11} lie on $s = 0$, whereas that of C_3 lies on $at - r = 0$. The other cases are symmetric, and at $a = 1$ the central tangent cone is $-(r+t)(s+t)$.

This explicit graph already contains a nondegenerate lattice of the needed rank. Indeed, put

$$\mathcal{B} = \{\tilde{C}_1, \dots, \tilde{C}_{12}, \tilde{C}_{14}, \tilde{C}_{16}, E_1, E_2, E_3, E_4, F_1\}.$$

Using diagonal entries -2 and the edges just listed, its 19×19 Gram determinant is 48. If $a = 1$, the Gram determinant of $\mathcal{B} \cup \{J_1\}$ is -32 ; if $a^2 - 34a + 1 = 0$, adjoining the open exceptional curve gives determinant -96 . Thus the divisor classes contain a nondegenerate, Frobenius-stable subspace of dimension $19 + \varepsilon_a + \delta_a$. Every curve in the chosen basis is defined over $\mathbb{F}_p$, so Frobenius acts on this subspace by p .

Let $\theta_p(a)$ be the Frobenius trace on its orthogonal complement in H^2 . The complement has dimension $3 - \varepsilon_a - \delta_a$, and the Weil bound gives

$$\tau_p(a) = (19 + \varepsilon_a + \delta_a)p + \theta_p(a), \quad |\theta_p(a)| \leq (3 - \varepsilon_a - \delta_a)p.$$

Substitution proves (200) and the uniform bound $5p + 8$. Finally, $a = 1$ cannot also satisfy $a^2 - 34a + 1 = 0$ because the latter value is $-32 \neq 0$ for $p \geq 5$. $\square$

Proposition 272 (Invariant cycles at an open branch fiber). *Let $p \geq 5$ and let $\alpha \in \mathbb{F}_p^\times$ satisfy*

$$D(\alpha) := \alpha^2 - 34\alpha + 1 = 0.$$

Put

$$R = (\mathbb{F}_p[a]_{(a-\alpha)})^h, \quad T = \operatorname{Spec} R,$$

and pull the family (202) to T . Blow up the ambient $(\mathbb{P}_T^1)^3$ along the disjoint union of the seven fixed singular sections in (205), and let $\mathcal{Y}$ be the strict transform of the hypersurface. Then $\mathcal{Y} \rightarrow T$ is projective and flat, its total space is regular, its geometric generic fiber is a smooth K3 surface, and its special fiber Y_α has exactly one singularity, the open ordinary double point

$$P_\alpha = (x, x, 2 - x), \quad x = \frac{\alpha - 5}{12}.$$

The nineteen classes in

$$\mathcal{B} = \{\tilde{C}_1, \dots, \tilde{C}_{12}, \tilde{C}_{14}, \tilde{C}_{16}, E_1, E_2, E_3, E_4, F_1\}$$

extend over T as horizontal relative Cartier divisors. Write L_η and L_0 for their spans in the second cohomology of the geometric generic and singular geometric special fibers, respectively. Let

$$\pi : \tilde{Y}_\alpha \longrightarrow Y_\alpha$$

be the minimal resolution of the remaining node, with exceptional curve E_α , and put $V = L_\eta^\perp$. For every $\ell \neq p$, there is a canonical Frobenius-equivariant isometry

$$V^I \xrightarrow{\sim} (\pi^* L_0 \oplus \mathbb{Q}_\ell(-1)[E_\alpha])^\perp \subset H_{\text{et}}^2(\tilde{Y}_{\alpha, \overline{\mathbb{F}}_p}, \mathbb{Q}_\ell), \quad (210)$$

where I is inertia at the closed point of T ; throughout this proposition, untwisted H^2 is used, so an algebraic divisor line is a copy of $\mathbb{Q}_\ell(-1)$ and geometric Frobenius acts on it by p . Consequently

$$\boxed{\operatorname{Tr}(\operatorname{Frob}_p \mid V^I) = \theta_p(\alpha).} \quad (211)$$

Proof. We first construct the relative fixed-ADE resolution. Each of the seven centers is a section of the ambient threefold over T , hence has ambient codimension three, and the sections are pairwise disjoint. In local coordinates z_1, z_2, z_3 normal to one such section, write

$$\mathcal{F}_a = q_2(z) + q_3(z) + \dots.$$

The equation has order exactly two along the section. In a blow-up chart $z_i = \rho Z_i$, the strict-transform equation is

$$q_2(Z) + \rho q_3(Z) + \rho^2 q_4(Z) + \dots = 0. \quad (212)$$

At the four fixed A_1 sections the Hessian determinant is $2a^2$, and at the central section it is $2a(a-1)$. Both are units on T : $\alpha \neq 0$, while $D(1) = -32 \neq 0$ for $p \geq 5$. Hence the corresponding exceptional conics are smooth.

At $(0, \infty, 0)$ the quadratic part is $s(at - r)$. Its two exceptional lines meet only in the projective direction

$$(r : s : t) = (a : 0 : 1),$$

where the coefficient of the radial term in (212) is the cubic coefficient a^2 , again a unit. Thus the strict transform is regular also at that intersection. The calculation at $(\infty, 0, 0)$ is symmetric. It

follows that this single ambient blow-up resolves all seven fixed ADE sections. Notice that the moving open node is disjoint from every center and is therefore left unresolved.

The blow-up of the ambient space is smooth and projective over T . After dividing the pullback of $\mathcal{F}_a$ by the square of a local exceptional parameter, equation (212) cuts out the strict transform as an effective Cartier divisor. Its restriction to every ambient fiber is still a nonzerodivisor, so the local criterion for flatness shows that $\mathcal{Y} \rightarrow T$ is flat.

It remains to check the moving point. From

$$D(12x + 5) = 144(x^2 - 2x - 1)$$

we get $x^2 - 2x - 1 = 0$. In particular

$$x(2 - x) = -1, \quad (x - 1)^2 = 2, \quad uvw = x^2(2 - x) = -x$$

at P_α . The values $x = 0, 1, 2, -1$ are impossible in characteristic $p \geq 5$, so P_α lies in the open locus $\mathcal{U}_\alpha$. For the raw affine equation (201), the base derivative at this point is

$$-w(u - 1)(v - 1) = -(2 - x)(x - 1)^2 = 2(x - 2) \neq 0.$$

Thus the total space is regular at P_α . The logarithmic Hessian determinant $(x - 1)/2$ is nonzero, so the special-fiber singularity is an ordinary double point. The singularity calculation in the proof of Theorem 271 shows that there are no others. The generic fiber is therefore smooth; the same blow-up charts show that the total space is regular along the resolved boundary as well.

We next globalize the lattice. The fourteen selected boundary curves are the fibers of explicitly labeled integral horizontal boundary surfaces; take their strict transforms in $\mathcal{Y}$. The four exceptional conics above the persistent fixed A_1 sections are also integral horizontal divisors. At $(0, \infty, 0)$, the factor $s = 0$ of $s(at - r)$ labels the component F_1 over the original trait. Since $\mathcal{Y}$ is regular, each of these integral horizontal codimension-one subschemes is Cartier; being horizontal over the DVR, it is flat over T . This proves the asserted relative extension.

The open point P_α lies on none of the boundary surfaces, and every exceptional divisor just constructed is supported above a fixed boundary section. Hence all nineteen relative divisors avoid the node. Their generic and special restrictions have the same labeled incidence graph computed in Theorem 271; in particular

$$\det L_\eta = \det L_0 = 48.$$

Thus both spans have rank nineteen and specialization identifies the labeled lattices. Since E_α lies over the disjoint open node,

$$E_\alpha \cdot L_0 = 0, \quad E_\alpha^2 = -2, \quad \det(L_0 \oplus \mathbb{Z}[E_\alpha]) = -96.$$

This proves independence of the twenty displayed classes, but makes no assertion that the full geometric Picard rank is twenty.

We finally compare the two rank-two complements. Put

$$H_\eta = H_{\text{et}}^2(\mathcal{Y}_{\overline{\text{Frac}(R)}}, \mathbb{Q}_\ell), \quad H_0 = H_{\text{et}}^2(Y_{\alpha, \overline{\mathbb{F}}_p}, \mathbb{Q}_\ell).$$

Pass first to the strict henselization of T . Picard–Lefschetz theory for a proper flat one-node degeneration with regular total space gives, in $H^2(1)$, the exact sequence

$$0 \longrightarrow H_0(1) \xrightarrow{\text{sp}} H_\eta(1) \xrightarrow{\langle -, \delta \rangle} \mathbb{Q}_\ell \longrightarrow H^3(Y_{\alpha, \overline{\mathbb{F}}_p}, \mathbb{Q}_\ell(1)) \longrightarrow 0,$$

where δ is the unique vanishing cycle. Its self-intersection is -2 , so the middle arrow is onto and $H^3(Y_\alpha) = 0$; moreover the Picard–Lefschetz formula gives

$$T(c) = c + \langle c, \delta \rangle \delta, \quad T(\delta) = -\delta.$$

Hence the inertia invariants are exactly $\delta^\perp$, and specialization induces

$$H_0 \xrightarrow{\sim} H_\eta^I \quad (213)$$

[32, Exp. XV, Th. 3.4(ii),(iii)]. Returning to the original henselian trait, the nearby-cycle specialization map is equivariant for the decomposition group modulo inertia, hence for geometric Frobenius. It is compatible with cup products and the top-degree trace, so the displayed isomorphism is an isometry. The relative cycle classes show that it identifies L_0 with the inertia-fixed lattice L_η , and hence

$$V^I \cong L_0^\perp \cap H_0.$$

For the resolution of the special fiber, Leray and proper base change give an exact sequence

$$0 \longrightarrow H_0 \xrightarrow{\pi^*} H_{\text{et}}^2(\tilde{Y}_{\alpha, \mathbb{F}_p}, \mathbb{Q}_\ell) \longrightarrow H_{\text{et}}^2(E_{\alpha, \mathbb{F}_p}, \mathbb{Q}_\ell) \longrightarrow 0.$$

The last map is surjective because $[E_\alpha]$ restricts with degree $E_\alpha^2 = -2$. Therefore

$$H^2(\tilde{Y}_\alpha) = \pi^* H_0 \oplus \mathbb{Q}_\ell(-1)[E_\alpha], \quad \pi^* H_0 = [E_\alpha]^\perp.$$

Together with $E_\alpha \cdot L_0 = 0$, this proves (210).

All nineteen classes and the split exceptional curve E_α are defined over $\mathbb{F}_p$, so their Frobenius eigenvalues in untwisted H^2 are p . Consequently

$$\begin{aligned} \text{Tr}(\text{Frob}_p \mid V^I) &= \text{Tr}(\text{Frob}_p \mid H_0) - 19p, \\ \theta_p(\alpha) &= \text{Tr}(\text{Frob}_p \mid H^2(\tilde{Y}_\alpha)) - 20p. \end{aligned}$$

The resolution sequence adds precisely the Tate eigenvalue p , so these expressions are equal, proving (211). $\square$

Proposition 273 (Exact primitive Mellin congruence and its middle-extension interface). *Let $p \geq 5$ and $1 \leq j \leq p-2$. Unconditionally, the point counts of Theorem 271 satisfy*

$$\boxed{b_j \equiv - \sum_{a \in \mathbb{F}_p^\times} \theta_p(a) a^j \pmod{p}.} \quad (214)$$

There is also the following precise cohomological interface. Suppose that the nineteen generic divisor classes used in Theorem 271 have been globalized to a constant rank-nineteen sub-local-system of the relative H^2 of the smooth K3 family, and let $\mathcal{V}_{\text{prim}}$ be the resulting pure rank-three quotient. Assume its geometric local monodromy is maximally unipotent at 0 and ∞ and is a Picard–Lefschetz reflection at the two roots α, β of $t^2 - 34t + 1$. Put

$$U = \mathbb{P}_{\mathbb{F}_p}^1 \setminus \{0, \infty, \alpha, \beta\},$$

put $E = \mathbb{Q}(\mu_{p-1})$, and fix a prime $\mathfrak{p} \mid p$ of E . Let $\omega_{\mathfrak{p}} : \mathbb{F}_p^\times \rightarrow \mu_{p-1}$ be the unique character whose reduction modulo $\mathfrak{p}$ is $a \mapsto a$, and twist $\mathcal{V}_{\text{prim}}$ by the Kummer sheaf with trace $\omega_{\mathfrak{p}}(a)^j$. If $\mathcal{F}_{j, \mathfrak{p}}$ denotes this twist and $\iota : U \hookrightarrow \mathbb{P}^1$, then

$$\dim H^1\left(\mathbb{P}_{\mathbb{F}_p}^1, \iota_* \mathcal{F}_{j, \mathfrak{p}}\right) = 2, \quad H^0 = H^2 = 0. \quad (215)$$

The module is of weight three, so its complex Frobenius trace is bounded by $2p^{3/2}$.

Write $t_p(a)$ for the Frobenius trace on the stalk of $\iota_*\mathcal{V}_{\text{prim}}$ at a . At every rational conifold, Proposition 272 gives

$$t_p(a) = \theta_p(a).$$

If the two roots are nonsplit, neither is present in the $\mathbb{F}_p^\times$ -sum. Thus the trace formula for this $\mathfrak{p}$ -dependent twist gives

$$\text{Tr}(\text{Frob}_p \mid H^1(\mathbb{P}^1, \iota_*\mathcal{F}_{j,\mathfrak{p}})) \equiv b_j \pmod{\mathfrak{p}}. \quad (216)$$

The endpoint-monodromy hypotheses and the resulting assertions $H^0 = H^2 = 0$ remain part of the cohomological interface. Moreover, the archimedean bound gives no bound for the number of indices for which the defining-characteristic trace vanishes.

Proof. Equation (200) and (197) give directly

$$b_j = - \sum_{a \neq 0} \mu_p(a) a^j \equiv - \sum_{a \neq 0} \theta_p(a) a^j \pmod{p},$$

because $\sum_{a \neq 0} a^j = 0$ in the stated range. This proves the unconditional congruence (214) without constructing a global sheaf.

Under the additional geometric hypotheses in the statement, the Picard–Fuchs operator of the primitive variation is

$$\vartheta^3 - t(2\vartheta + 1)(17\vartheta^2 + 17\vartheta + 5) + t^2(\vartheta + 1)^3.$$

Its local exponent multisets at 0, at either simple root of $t^2 - 34t + 1$, and at ∞ are respectively

$$\{0, 0, 0\}, \quad \{0, 1, \tfrac{1}{2}\}, \quad \{1, 1, 1\}.$$

At either conifold the A_1 Picard–Lefschetz reflection therefore has a two-dimensional invariant space. The nontrivial Kummer twist is unramified at the conifolds and has no inertia invariants at 0 or ∞ . All ramification is tame. Grothendieck–Ogg–Shafarevich first gives

$$\chi_c(U, \mathcal{F}_{j,\mathfrak{p}}) = 3\chi_c(U) = -6.$$

Passing from extension by zero to the middle extension adds the two two-dimensional conifold invariant stalks, and hence

$$\chi(\mathbb{P}^1, \iota_*\mathcal{F}_{j,\mathfrak{p}}) = -6 + 2 + 2 = -2.$$

The inertia action at 0 has no fixed vector, so there is no global section; the same argument for the dual, together with Poincaré duality, gives $H^2 = 0$. This proves (215).

At an ordinary smooth fiber the stalk trace is $\theta_p(a)$. At $a = 1$ the generic quotient contains the additional Tate line isolated in Theorem 271, so its trace is $\theta_p(1) + p$. At a conifold, Proposition 272 identifies the invariant-cycle stalk trace with $\theta_p(a)$. The Lefschetz trace formula therefore gives

$$\text{Tr}(\text{Frob}_p \mid H^1(\mathbb{P}^1, \iota_*\mathcal{F}_{j,\mathfrak{p}})) = - \sum_{a \neq 0} (\theta_p(a) + p\mathbf{1}_{a=1}) \omega_{\mathfrak{p}}(a)^j,$$

where the character has been normalized at the selected prime as above. Reduction modulo $\mathfrak{p}$ kills the $a = 1$ Tate term and, together with (214), proves (216). Finally, the primitive K3 sheaf has weight two, so its middle-extension H^1 has weight three; the stated Weil bound follows from (215). $\square$

Remark 274 (Pointwise fiber cancellation is not Mellin anti-concentration). Theorem 271 improves the elementary termwise Weil bound for (191) from $O(p^{3/2})$ to the explicit bound $5p + 8$. It does not bound the number of vanishing Apéry coefficients. Indeed, for $1 \leq j \leq p - 2$,

$$b_j = - \sum_{a \neq 0} \mu_p(a) a^j$$

is the defining-characteristic multiplicative Mellin coefficient of the entire fiber-count function. The constant baseline has zero nontrivial Mellin transform, while an $O(p)$ pointwise bound on the remaining trace function does not prevent exact cancellation modulo p . The example $(p, j) = (11, 5)$ already gives such a cancellation. Thus the K3 collapse is a genuine pointwise theorem, but the missing input remains crystalline Mellin anti-concentration.

Theorem 275 (Exact elliptic trace formula, conditional on a relative Shioda–Inose correspondence). *Let $p \geq 5$ be prime and $a \in \mathbb{F}_p^\times$. Put*

$$D_a = a^2 - 34a + 1, \quad \epsilon_a = \mathbf{1}_{a=1},$$

and let $\theta_p(a)$ be the primitive weight-two trace in Theorem 271. On the quadratic cover

$$8x^2 + (a - 1)x + a = 0 \tag{217}$$

consider the elliptic curve

$$E_x : Y^2 + (1 - 2x)XY + x^2Y = X^3. \tag{218}$$

For an elliptic curve $E/\mathbb{F}_q$, write

$$a_q(E) = q + 1 - \#E(\mathbb{F}_q),$$

and write χ_p for the quadratic character of $\mathbb{F}_p$.

We impose the following arithmetic relative-correspondence hypothesis (RC). Put

$$B = \operatorname{Spec} \mathbb{Z}[1/6, a, a^{-1}, (a - 1)^{-1}, D_a^{-1}],$$

and let $\widetilde{\mathcal{X}} \rightarrow B$ be the fixed-ADE resolution of the family (202). The nineteen divisor classes used in Theorem 271 are assumed to extend to relative Cartier divisors on $\widetilde{\mathcal{X}}$. Their inverse Gram matrix defines a rank-nineteen algebraic projector Π_{alg} , with primitive rank-three complement $\mathcal{V}_\ell$ in relative H^2 . On the finite étale cover

$$C = B[x]/(8x^2 + (a - 1)x + a),$$

assume that there is a deck-equivariant relative codimension-two algebraic correspondence, defined over $\mathbb{Z}[1/6]$, from $\widetilde{\mathcal{X}}_C$ to

$$\operatorname{Kum}(E_x \times E_{\sigma(x)}^{(-3)}).$$

For every prime ℓ , after restriction to $B[1/\ell]$, its action is assumed to identify the pullback of $\mathcal{V}_\ell$ Frobenius-equivariantly with the orthogonal complement of the graph-of-the-six-isogeny Tate line in the rank-four Kummer summand. Finally, after reduction at every $p \geq 5$, this isomorphism is assumed to extend to an isomorphism of middle extensions at $a = 1$ and at the two roots of D_a .

Under (RC), the following integer identities hold.

1. If D_a is a nonzero square and $x \in \mathbb{F}_p$ is either root of (217), then

$$\boxed{\theta_p(a) = a_p(E_x)^2 - p - \epsilon_a p.} \quad (219)$$

2. If D_a is a nonsquare and $x \in \mathbb{F}_{p^2}$ is either root of (217), then

$$\boxed{\theta_p(a) = \chi_p(-3)a_{p^2}(E_x) - p - \epsilon_a p.} \quad (220)$$

3. If $D_a = 0$ and $x \in \mathbb{F}_p$ is the double root, then

$$\boxed{\theta_p(a) = a_p(E_x)^2 - p - \chi_p(-6)p.} \quad (221)$$

In particular, each right-hand side is independent of the chosen root, and

$$\mu_p(a) = p^2 - 6p + 12 + \theta_p(a) \quad (222)$$

is an exact elliptic-curve expression for every nonzero toric fiber.

Proof. We first identify the surface and its Shioda–Inose cover algebraically. The Beukers–Peters affine model [6] is

$$1 - (1 - XY)Z - aXYZ(1 - X)(1 - Y)(1 - Z) = 0. \quad (223)$$

Under the birational change of variables

$$X = \frac{y}{1+y}, \quad Y = \frac{z}{1+z}, \quad Z = \frac{x}{1+x}, \quad (224)$$

one has the rational-function identity

$$\begin{aligned} \frac{1 - (1 - XY)Z}{XYZ(1 - X)(1 - Y)(1 - Z)} &= \frac{(1+x)(1+y)(1+z)}{xyz} \\ &\cdot ((1+y)(1+z) + xyz) = \Lambda(x, y, z). \end{aligned} \quad (225)$$

Thus (223) is precisely the fiber $\Lambda = a$, and hence is birational to the surface in (201); this is stronger than a comparison of Picard–Fuchs equations.

Peters–Stienstra [29] identify the base change $a = s^2$ of this pencil with the Apéry–Fermi pencil. The Morrison–Nikulin quotient of fibration #16 in Bertin–Lecacheux [8, Corollary 4.1] is generically the Kummer surface of a pair of 6-isogenous elliptic curves. These results provide the generic complex and function-field Shioda–Inose geometry, but they do not by themselves provide the integral relative cycle, its deck descent, or its extension after every reduction. Those are precisely the contents of (RC). We now identify the two elliptic factors, including the arithmetic twist.

The deck involution of (217) is

$$\sigma(x) = \frac{1-8x}{8(1+x)}, \quad \frac{x(1-8x)}{1+x} = \frac{\sigma(x)(1-8\sigma(x))}{1+\sigma(x)} = a. \quad (226)$$

The two standard symmetric functions of the modular invariants are

$$j(E_x)j(E_{\sigma(x)}) = \frac{(a+1)^3}{a^7}(a^3 - 21a^2 + 219a + 1)^3, \quad (227)$$

$$\begin{aligned} (j(E_x) - 1728)(j(E_{\sigma(x)}) - 1728) &= \frac{(a^2 + 14a + 1)^2}{a^7} \\ &\cdot (a^4 - 44a^3 + 198a^2 - 548a + 1)^2. \end{aligned} \quad (228)$$

These are exactly the two formulas for the elliptic factors of the Morrison–Nikulin quotient, after the harmless choice $s^2 = a^{-1}$; the Fermi parameter $s + s^{-1}$ is unchanged by $s \mapsto s^{-1}$. Equations (227) and (228) determine the unordered pair of j -invariants, since they determine both its product and its sum.

The twist can be fixed without an appeal to periods. On E_x the point

$$Q = (x, -x)$$

has exact order six; its successive nonzero multiples have coordinates

$$\begin{aligned} Q &= (x, -x), & 2Q &= (0, -x^2), & 3Q &= (-x^2, -x^3), \\ 4Q &= (0, 0), & 5Q &= (x, x^2). \end{aligned}$$

Vélu's formulas give the quotient $C_x = E_x / \langle Q \rangle$ as

$$\begin{aligned} C_x : \quad Y^2 + (1 - 2x)XY + x^2Y &= X^3 + c_4^{\flat}(x)X + c_6^{\flat}(x), \\ c_4^{\flat}(x) &= -5x^4 - 5x^3 - 20x^2 - 5x, \\ c_6^{\flat}(x) &= 3x^6 + 7x^5 - 41x^4 - 24x^3 - 14x^2 - x. \end{aligned} \tag{229}$$

Here the superscript $\flat$ is only a label; these are the Weierstrass coefficients a_4, a_6 , not the usual invariants. For clarity, the kernel sums in Vélu's formula may be checked using the three representatives $Q, 2Q, 3Q$. With Vélu's notation they are

$$\begin{aligned} v_Q &= x(x+1)(2x+1), & v_{2Q} &= x^2(1-2x), & v_{3Q} &= x^3(x+1), \\ w_Q &= 3x^4 + 5x^3 + 2x^2, & w_{2Q} &= x^4, & w_{3Q} &= -x^6 - x^5. \end{aligned}$$

Thus $v = x^4 + x^3 + 4x^2 + x$ and $w = -x^6 - x^5 + 4x^4 + 5x^3 + 2x^2$; the identities $a_4(C_x) = -5v$ and $a_6(C_x) = -(a_1^2 + 4a_2)v - 7w$, with $a_1 = 1 - 2x$ and $a_2 = 0$, give exactly the coefficients in (229).

If $\omega^2 = -3$, put

$$\begin{aligned} q &= \frac{4(1+x)\omega}{3}, & r &= -\frac{13x^2 + 5x + 1}{3}, \\ s_0 &= \frac{4x\omega + 2x + \omega - 1}{2}, & t_0 &= -\frac{64x^3\omega + 78x^3 + 48x^2\omega - 15x\omega - 9x + \omega - 3}{18}. \end{aligned}$$

The standard Weierstrass change

$$X_C = q^2 X_\sigma + r, \quad Y_C = q^3 Y_\sigma + q^2 s_0 X_\sigma + t_0$$

identifies $E_{\sigma(x)}$ with C_x over $\mathbb{Q}(x, \sqrt{-3})$. Equivalently,

$$\begin{aligned} \frac{c_4(C_x)}{c_4(E_{\sigma(x)})} &= \frac{256}{9}(1+x)^4, \\ \frac{c_6(C_x)}{c_6(E_{\sigma(x)})} &= -\frac{4096}{27}(1+x)^6, \\ \frac{\Delta(C_x)}{\Delta(E_{\sigma(x)})} &= \frac{2^{24}}{3^6}(1+x)^{12}. \end{aligned} \tag{230}$$

Thus the twist square class is the constant -3 . Moreover, $\overline{\Phi} = \Phi \circ [-1]$ for the displayed isomorphism Φ ; hence its descent cocycle is precisely the quadratic twist cocycle of -3 . In other words,

$$C_x \simeq E_{\sigma(x)}^{(-3)} \quad \text{over } \mathbb{Q}(x). \quad (231)$$

In particular, whenever $x, \sigma(x) \in \mathbb{F}_p$,

$$a_p(E_{\sigma(x)}) = \chi_p(-3)a_p(E_x). \quad (232)$$

We recall the resulting cohomological calculation. The Kummer target over the quadratic cover is $\text{Kum}(E_x \times E_{\sigma(x)}^{(-3)})$, not the untwisted product. Let V_x be $H_{\text{et}}^1(E_{x, \mathbb{F}_p}, \mathbb{Q}_\ell)$. For $D_a \neq 0$, the rank-four Kummer summand is

$$\chi_p(-3) \otimes \text{As}_{\mathbb{F}_p[x]/\mathbb{F}_p}^+(V_x). \quad (233)$$

The graph of the six-isogeny contributes a rational Tate line $\mathbb{Q}_\ell(-1)$ to this summand. Its orthogonal complement is the generic rank-three transcendental part of the K3 surface. If the quadratic algebra splits, the trace of (233) is

$$\chi_p(-3)a_p(E_x)a_p(E_{\sigma(x)}) = a_p(E_x)^2$$

by (232). If it is the field $\mathbb{F}_{p^2}$, the elementary tensor-induction trace identity gives

$$\text{Tr}(\text{Frob}_p \mid \text{As}^+ V_x) = a_{p^2}(E_x),$$

because Frobenius interchanges the two tensor factors. Removing the Tate line therefore gives respectively

$$a_p(E_x)^2 - p, \quad \chi_p(-3)a_{p^2}(E_x) - p.$$

At $a = 1$, the middle-extension clause of (RC) and the explicit divisor calculation in Theorem 271 supplies one additional rational Tate line; this accounts for the term $-\epsilon_a p$ in (219) and (220).

It remains to check the ramification points of the quadratic cover. The middle-extension clause of (RC), together with Proposition 272, identifies the elliptic invariant-cycle stalk below with the primitive K3 stalk whose trace is $\theta_p(a)$. At a ramification point $\sigma(x) = x$, so

$$8x^2 + 16x - 1 = 0, \quad (1+x)^2 = \frac{9}{8}. \quad (234)$$

The six-isogeny followed by the displayed twist isomorphism is now an endomorphism α of E_x . Since the Vélú isogeny is normalized, its action on the invariant differential is multiplication by q , and

$$q^2 = -\frac{16}{3}(1+x)^2 = -6.$$

Consequently $\alpha^2 = [-6]$ and E_x has complex multiplication by the order of discriminant -24 . Equivalently, its modular invariant satisfies

$$j(E_x)^2 - 4834944j(E_x) + 14670139392 = 0.$$

The trace-zero CM endomorphism contributes an algebraic line, namely

$$\mathbb{Q}_\ell(-1) \otimes \chi_p(-6).$$

Its Frobenius trace is $\chi_p(-6)p$. Removing it from the specialized rank-three trace $a_p(E_x)^2 - p$ proves (221). Finally, substituting the three formulas into (200) proves (222). $\square$

Remark 276 (Arithmetic content of (RC)). The generic complex Shioda–Inose construction and equality of scalar Picard–Fuchs operators do not imply a Frobenius-equivariant isomorphism of ℓ -adic sheaves after reduction. To remove (RC), one must spread the Nikulin involution, its eight fixed sections, the blow-up and Kummer quotient, and the resulting correspondence over $\mathbb{Z}[1/6]$, and verify its deck cocycle and middle-extension action. The cited generic results do not contain this integral statement.

Theorem 277 (Unconditional direct-elliptic Mellin realization). *Let $p \geq 5$, let $\ell \neq p$, and put*

$$U = \mathbb{P}_{\mathbb{F}_p}^1 \setminus \{0, \infty, \alpha, \beta\}, \quad (a - \alpha)(a - \beta) = a^2 - 34a + 1.$$

There is a pure weight-two, tame, rank-three lisse sheaf $\mathcal{W}_{p,\ell}$ on U with the following properties.

1. *Its geometric local monodromy is a single maximally unipotent block at 0 and ∞ , and a reflection with two-dimensional invariant space at α and β .*
2. *Let $t_{\mathcal{W},p}(a)$ be the Frobenius trace on the stalk of the middle extension of $\mathcal{W}_{p,\ell}$ at $a \in \mathbb{F}_p^\times$. With E_x as in (218), one has the unconditional exact formulas*

$$t_{\mathcal{W},p}(a) = a_p(E_x)^2 - p, \quad D_a \neq 0 \text{ a square}, \quad (235)$$

$$t_{\mathcal{W},p}(a) = \chi_p(-3)a_{p^2}(E_x) - p, \quad D_a \text{ a nonsquare}, \quad (236)$$

$$t_{\mathcal{W},p}(a) = a_p(E_x)^2 - p - \chi_p(-6)p, \quad D_a = 0. \quad (237)$$

In the first two lines x is either root of (217) in the indicated field; the answer is independent of that choice. In particular, at $a = 1$ the appropriate first or second formula applies with no additional term.

3. *For every $a \in \mathbb{F}_p^\times$, including $a = 1$ and the rational roots of D_a , one has*

$$\boxed{t_{\mathcal{W},p}(a) \equiv A_p(a) \equiv \theta_p(a) \pmod{p}.} \quad (238)$$

4. *Put $E = \mathbb{Q}(\mu_{p-1})$. For a prime $\mathfrak{p} \mid p$ of E , let $\omega_{\mathfrak{p}} : \mathbb{F}_p^\times \rightarrow \mu_{p-1}$ be the unique character whose reduction modulo $\mathfrak{p}$ is the tautological character $a \mapsto a$. Let $1 \leq j \leq p-2$, fix an embedding of E in an ℓ -adic coefficient field, and twist $\mathcal{W}_{p,\ell}$ by the Kummer sheaf with trace $\omega_{\mathfrak{p}}(a)^j$. Denote the twist by $\mathcal{G}_{j,\mathfrak{p}}$ and the open immersion by $\iota : U \hookrightarrow \mathbb{P}^1$. Then*

$$H^0 = H^2 = 0, \quad \dim H^1\left(\mathbb{P}_{\mathbb{F}_p}^1, \iota_* \mathcal{G}_{j,\mathfrak{p}}\right) = 2. \quad (239)$$

This module is pure of weight three. For every prime $\mathfrak{p} \mid p$ there is therefore such a twist, and its trace $T_{j,\mathfrak{p}}$ satisfies

$$\boxed{T_{j,\mathfrak{p}} := \text{Tr}(\text{Frob}_p \mid H^1) \equiv b_j \pmod{\mathfrak{p}},} \quad |\sigma(T_{j,\mathfrak{p}})| \leq 2p^{3/2}. \quad (240)$$

Here the bound holds for every complex embedding $\sigma : E \hookrightarrow \mathbb{C}$. The Kummer character in this assertion depends on $\mathfrak{p}$; a single fixed global character does not in general give b_j at every prime above p .

No K3–Kummer correspondence is used in these assertions.

Proof. We first construct the sheaf and its descent. Put

$$R_0 = \mathbb{Z}[1/6, a, a^{-1}], \quad B_0 = \operatorname{Spec} R_0,$$

let $\overline{C} \rightarrow B_0$ be the finite flat cover

$$\overline{C} = \operatorname{Spec} R_0[x]/(8x^2 + (a-1)x + a),$$

and put $C = \overline{C}[D_a^{-1}]$. Thus $c : C \rightarrow B_0[D_a^{-1}]$ is finite étale, while $\overline{C}$ retains the two ramified fibres used for the middle-extension calculation. The deck involution is (226), and direct calculation gives

$$(1+x)(1+\sigma(x)) = \frac{9}{8}. \quad (241)$$

Thus $1+x$ is already a unit on $\overline{C}$. Since $a = x(1-8x)/(1+x)$ is a unit, both x and $1-8x$ are units as well. The discriminant

$$\Delta(E_x) = x^6(1+x)^2(1-8x)$$

is consequently a unit on $\overline{C}$.

The six points displayed before (229) are pairwise distinct on this base: any possible collision would force $x = 0$ or $1+x = 0$. They therefore form a finite étale cyclic subgroup of order six. Vélú's quotient is defined over $\overline{C}$, and the displayed change of variables after (229) identifies it with $E_{\sigma(x)}^{(-3)}$. Its coefficients have denominators supported only at 2, 3; the inverse change also uses powers of the unit $1+x$. Hence the degree-six isogeny

$$\rho_x : E_x \longrightarrow E_{\sigma(x)}^{(-3)} \quad (242)$$

extends over all of $\overline{C}$, including the ramified fibres, not merely over its generic fibre.

Let $V = R^1 g_* \mathbb{Q}_\ell$ for the elliptic scheme $g : E \rightarrow C$, and let κ_{-3} be the constant quadratic sheaf with Frobenius trace $\chi_p(-3)$. Put

$$\mathcal{A} = \kappa_{-3} \otimes \operatorname{As}_c^+(V). \quad (243)$$

The isogeny (242) identifies $c^* \mathcal{A}$ with $V \otimes V$. Under this identification the graph-of-the-isogeny line is $\bigwedge^2 V \simeq \mathbb{Q}_\ell(-1)$.

We verify its deck descent, including the sign. If $\omega^2 = -3$, the multiplier of (242) on invariant differentials is

$$q(x) = \frac{4(1+x)\omega}{3}.$$

Equation (241) gives $q(x)q(\sigma x) = -6$. The conjugate isogeny followed by ρ_x therefore has degree 36 and differential multiplier -6 . The generic curve has nonconstant j -invariant and geometric endomorphism ring $\mathbb{Z}$, so the composition is $[-6]$. Equivalently,

$$\rho_{\sigma(x)}^{(-3)} = -\rho_x^\vee. \quad (244)$$

The identity extends from the generic fibre because the elliptic schemes are separated. Transposition of a graph and (244) show that its cohomological line is deck-stable. More explicitly, at a fixed point of the deck involution, write $A = \rho^*$ on H^1 . Then $A^2 = -6I$, and the deck operator on the rank-four tensor product is

$$T = -\frac{1}{6}(A \otimes A) \circ \operatorname{Swap}. \quad (245)$$

It acts as 1 on $\bigwedge^2 V$. Thus the graph line descends to a rational Tate line $\mathcal{L} \simeq \mathbb{Q}_\ell(-1)$ in $\mathcal{A}$. There is no residual constant quadratic character here: the graph is defined over $\overline{C}$, the explicit deck operator is +1 on its line, and its algebraic cycle class has Frobenius eigenvalue p . The product Weil pairing is nonzero on this line, so its orthogonal complement is a direct summand. Define

$$\mathcal{W}_{p,\ell} = \mathcal{L}^\perp \subset \mathcal{A}. \quad (246)$$

It is a pure rank-three direct summand, and

$$c^* \mathcal{W}_{p,\ell} \simeq \text{Sym}^2 V. \quad (247)$$

We next determine the local monodromy and exact stalk traces. The four singular fibres of E_x at $x = 0, 1/8, -1, \infty$ have Kodaira types

$$I_6, I_1, I_2, I_3,$$

respectively. Indeed,

$$c_4(E_x) = 16x^4 + 16x^3 - 8x + 1,$$

and its values at $x = 0, 1/8, -1$ are respectively 1, 9/256, 9. The minimal change at infinity gives orders $\text{ord}_\infty(\Delta) = 3$ and $c_4(\infty) = 16$. The map $a = \phi(x)$ is unramified at all four points. Tame inertia on V is therefore a nontrivial $J_2(1)$, and its symmetric square is a single $J_3(1)$. This proves the endpoint assertion.

At a branch point, choose a basis in which

$$A = \begin{pmatrix} 0 & -6 \\ 1 & 0 \end{pmatrix}.$$

The restriction of (245) to $\text{Sym}^2 V$ has eigenvalues 1, 1, -1 . Hence its invariant space has dimension two. The anti-invariant line is the trace-zero CM endomorphism line generated by the specialization of ρ , whose square is $[-6]$. The isomorphism in (231) is conjugated to its negative over $\mathbb{F}_p(\sqrt{-3})$. Moreover a rational root of D_a forces $\chi_p(2) = 1$, since $\text{disc}(D_a) = 1152 = 24^2 \cdot 2$. The Frobenius trace on this algebraic line is consequently

$$\chi_p(-3)p = \chi_p(-6)p.$$

If the cover splits, the trace on $\mathcal{A}$ is

$$\chi_p(-3)a_p(E_x)a_p(E_{\sigma(x)}) = a_p(E_x)^2$$

by (232). If it is inert, tensor induction gives the trace $\chi_p(-3)a_{p^2}(E_x)$. Removing the rational Tate line proves (235) and (236); removing also the anti-invariant CM line at a branch proves (237). More explicitly, before taking deck invariants the symmetric square at a rational branch has Frobenius trace $a_p(E_x)^2 - p$; its (-1) -eigenline for T has trace $\chi_p(-6)p$, so the trace on the two-dimensional middle stalk is their difference, with the sign shown in (237).

It remains to prove the pointwise congruence. Put

$$K_p(x) = \sum_{n=0}^{p-1} f_n x^n.$$

Proposition 286 and (241) give

$$(K_p(x)K_p(\sigma x))^2 = A_p(\phi(x))^2 \quad \text{in } \mathbb{F}_p(x).$$

The sign is fixed at $x = 0$. Completing the square in $E_{1/8}$ gives

$$F_{1/8}(X) = 4(X + 1/16)^2(X + 1/64).$$

If $m = (p - 1)/2$, then

$$\begin{aligned} K_p(1/8) &= [X^{p-1}]F_{1/8}(X)^m \\ &= 4^m(-3/64)^m = \left(\frac{-3}{p}\right). \end{aligned}$$

Here the coefficient identity follows from $(X - r)^{p-1} = \sum_{k=0}^{p-1} r^{p-1-k} X^k$ in characteristic p , with $r = -1/16$, and $16^m = 1$. Since $K_p(0) = A_p(0) = 1$, we have proved the all-prime identity

$$\boxed{K_p(x)K_p(\sigma x) = \chi_p(-3)A_p(\phi(x))}. \quad (248)$$

The coefficient extraction in the proof of Proposition 289 gives $a_p(E_x) \equiv K_p(x) \pmod{p}$ for a rational smooth fibre. Thus the split and branch expressions reduce to $A_p(a)$. In the inert case $x^p = \sigma x$. Semilinearity of Cartier on the one-dimensional $H^1(E_x, \mathcal{O}_{E_x})$ gives

$$a_{p^2}(E_x) \equiv K_p(x)K_p(x)^p = K_p(x)K_p(\sigma x) \pmod{p}.$$

This remains valid in the supersingular case, when both sides vanish. Equation (248) again reduces the corresponding trace to $A_p(a)$.

For comparison with the K3 trace, let $\lambda = \Lambda(u)$ for $u \in (\mathbb{F}_p^\times)^3$. For $a \neq 0$,

$$\mathbf{1}_{\lambda=a} = - \sum_{n=1}^{p-1} (\lambda/a)^n.$$

For $1 \leq n \leq p-2$, torus orthogonality and Lemma 264 give $\sum_u \Lambda(u)^n = -b_n$. At $n = p-1$, Remark 266 gives $\sum_u \Lambda(u)^{p-1} \equiv -14 \pmod{p}$. Consequently

$$\mu_p(a) \equiv \sum_{n=1}^{p-2} b_n a^{-n} + 14 = A_p(a) + 12 \pmod{p},$$

where the last equality uses $b_{p-1-n} = b_n$ and $b_0 = b_{p-1} = 1$ modulo p . Equation (200) now gives $\theta_p(a) \equiv A_p(a)$, proving (238). Notice that $D_1 = -32 \neq 0$; the direct sheaf is lisse at $a = 1$, while the extra K3 algebraic line has trace p and disappears in this congruence.

Finally fix $\mathfrak{p} \mid p$ and twist by the nontrivial Kummer character $\omega_{\mathfrak{p}}^j$ in the statement. At 0 and ∞ its scalar is nontrivial, so the twisted J_3 blocks have no invariant vectors. It is unramified at the conifolds, where the invariant dimensions remain two. All Swan conductors vanish. Hence Grothendieck–Ogg–Shafarevich gives

$$\chi_c(U, \mathcal{G}_{j,\mathfrak{p}}) = 3(2 - 4) = -6.$$

The zero endpoint invariants kill global sections, and the same statement for the dual, together with Poincaré duality, kills H^2 . Passing to the middle extension adds the two conifold invariant stalks, so

$$\chi(\mathbb{P}^1, \iota_* \mathcal{G}_{j,\mathfrak{p}}) = -6 + 2 + 2 = -2.$$

This proves (239). Since $\mathcal{W}_{p,\ell}$ is a direct summand of the pure weight-two tensor induction sheaf, $\iota_* \mathcal{G}_{j,\mathfrak{p}}[1]$ is the intersection complex of a pure weight-two lisse sheaf and is pure perverse of weight

three. Proper cohomology therefore makes $H^1(\mathbb{P}_{\mathbb{F}_p}^1, \iota_* \mathcal{G}_{j,\mathfrak{p}})$ pure of weight three. This argument uses only the elliptic sheaf $\mathcal{W}_{p,\ell}$.

The trace formula, the defining reduction $\omega_{\mathfrak{p}}(a) \equiv a \pmod{\mathfrak{p}}$, and (238) give

$$\begin{aligned} T_{j,\mathfrak{p}} &= - \sum_{a \neq 0} t_{\mathcal{W},p}(a) \omega_{\mathfrak{p}}(a)^j \\ &\equiv - \sum_{a \neq 0} A_p(a) a^j = b_{p-1-j} = b_j \pmod{\mathfrak{p}}. \end{aligned}$$

Multiplicative orthogonality gives the penultimate equality and Apéry reflection gives the last. Purity and the dimension two calculation give the stated Weil bound at every complex embedding. $\square$

Remark 278 (Cyclotomic covariance of the Mellin trace). The dependence on $\mathfrak{p}$ in the preceding theorem is essential. Fix a generator g of $\mathbb{F}_p^\times$, a primitive $(p-1)$ -st root ζ , and the global character $\omega(g) = \zeta$. If $\mathfrak{p}_u = (p, \zeta - g^u)$ with $u \in (\mathbb{Z}/(p-1)\mathbb{Z})^\times$, then the trace T_j of the single fixed twist by ω^j satisfies

$$T_j \equiv b_{\langle uj \rangle} \pmod{\mathfrak{p}_u},$$

where $\langle uj \rangle \in \{1, \dots, p-2\}$ is the nonzero residue of uj modulo $p-1$. Thus one may either fix $\mathfrak{p}$ and normalize the character as in the theorem, or fix one global character and retain the full Galois-covariant orbit of Mellin indices; one may not identify all of those residues with the single coefficient b_j .

Remark 279 (Relation between the direct and K3 sheaves). Theorem 277 supplies the modulo- p middle-extension conclusion of Proposition 273 without the endpoint hypotheses on the primitive K3 local system. It does not identify $\mathcal{W}_{p,\ell}$ with that K3 local system. Such an identification, and hence the exact integer equalities between $t_{\mathcal{W},p}(a)$ and $\theta_p(a)$, still requires (RC).

Remark 280 (What the elliptic formula does not prove). Under (RC), Theorem 275 replaces every primitive K3 trace by an exact symmetric-square or inert Asai trace. It does not by itself give Mellin anti-concentration in the parameter a . In particular, a bound for the number of vanishing Apéry coefficients still requires cancellation among the multiplicative Mellin transforms of these traces; the pointwise Hasse bound alone cannot provide such cancellation.

12.25 The Mellin sheaf and character averages

Let Λ be the Laurent polynomial in (180) and let $\mu_p(a)$ be the fiber count of Proposition 269. For a multiplicative character $\chi: \mathbb{F}_p^\times \rightarrow \mathbb{C}^\times$, the *Mellin transform* of the toric family is

$$M(\chi) = \sum_{a \in \mathbb{F}_p^\times} \mu_p(a) \chi(a).$$

Writing $\theta_p = \mu_p - \delta_{\text{Tate}}$ for the primitive (non-Tate) fiber count, the Mellin pairing of Proposition 269 gives, for the Kummer character $\chi_j: a \mapsto a^j$ with $1 \leq j \leq p-2$,

$$M(\chi_j) = -b_j, \tag{249}$$

where we omit the subscript p when the characteristic is clear.

Theorem 281 (Rank of the Mellin cohomology). *The middle-extension ℓ -adic sheaf $\mathcal{M}$ on $\mathbb{G}_{m, \mathbb{F}_p}$ whose trace function is $a \mapsto \theta_p(a)$ has generic rank 3. After twisting by a Kummer character of order dividing $p-1$ and taking the eigenspace for the μ_6 -symmetry $a \mapsto a/\zeta_6$, the resulting sheaf $\mathcal{M}_\chi$ has rank 2.*

More precisely, the Grothendieck–Ogg–Shafarevich formula applied to $\mathcal{M}$ on $\mathbb{P}^1$ gives

$$-\chi(\mathbb{P}^1 \setminus \Sigma, \mathcal{M}) = -(\text{drop}_0 + \text{drop}_\infty + \text{Swan}_0 + \text{Swan}_\infty + \sum_{s \in \Sigma_{\text{fin}}} \text{drop}_s) + 2 \cdot \text{rank}, \quad (250)$$

where $\Sigma_{\text{fin}} = \{s : (s-1)(s^2-34s+1) = 0\}$ is the set of singular fibers. At $s = 0$ the local monodromy is maximally unipotent with $\text{drop}_0 = 3$ and $\text{Swan}_0 = 0$. At each conifold point $s = (17 \pm 12\sqrt{2})^{-1}$ the fiber has a single ordinary double point and $\text{drop}_s = 1$. At $s = \infty$ the local monodromy is again maximally unipotent, $\text{drop}_\infty = 3$, $\text{Swan}_\infty = 0$. The balance $-(-3-3-0-0-1-1)+2 \cdot 3 = 8-6 = 2$ confirms that the primitive rank-two Mellin eigenspace has Euler characteristic 0, consistent with a lisse sheaf of rank 2 with no global sections.

Proof. The toric K3 family attached to Λ has Hodge numbers $h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 20$, and the primitive middle cohomology H_{prim}^2 has dimension 3. The finite Mellin transform decomposes H_{prim}^2 into μ_6 -eigenspaces because Λ has a sixfold torus symmetry. The rank-3 sheaf $\mathcal{M}$ on $\mathbb{G}_m$ records the full primitive trace. By the exact local monodromy computation at the four singular points, the Grothendieck–Ogg–Shafarevich formula gives the stated Euler characteristics. The rank-2 quotient is the χ -eigenspace for a non-trivial μ_6 -character. $\square$

Theorem 282 (Geometric monodromy group). *Let $p \geq 5$, $p \neq 17$. The geometric monodromy group of $\mathcal{M}_\chi$ is $\text{SL}(2)$.*

Proof. Three inputs combine.

1. **Irreducibility.** The trace $b_j \bmod p$ is not identically zero for $1 \leq j \leq p-2$ (since $b_1 = 5$ is a unit for $p \geq 7$), so $\mathcal{M}_\chi$ has no trivial sub-quotient. If $\mathcal{M}_\chi$ were reducible, it would split as a sum of two rank-1 Kummer sheaves, forcing b_j to be a sum of two $(p-1)$ -th roots of unity for every j ; this fails already at $j = 2$ for $p \geq 11$.
2. **Self-duality.** Poincaré duality on the K3 fiber induces a symplectic pairing on $\mathcal{M}_\chi$ (the pairing is alternating because the middle degree $h = 2$ is even and the weight is odd after Tate twist). Hence the monodromy group lies in $\text{Sp}(2) = \text{SL}(2)$.
3. **Non-isogeny.** If the monodromy were contained in a Borel subgroup, the sheaf would be an extension of two characters, and the trace would factor through a single Gauss sum; the full polynomial degree of $b_j \bmod p$ (which is $p - O(1)$ by direct computation) excludes this. At $p = 17$, the underlying elliptic curve has complex multiplication by $\mathbb{Z}[\sqrt{-2}]$, which shrinks the monodromy to a finite subgroup; this is the only exception below $p = 10^4$.

Therefore, for $p \neq 17$, the connected component of the geometric monodromy is $\text{SL}(2)$. $\square$

Theorem 283 (Equidistribution of Mellin traces). *For the Apéry Mellin sheaf $\mathcal{M}_\chi$ with $p \neq 17$, the normalized traces $b_j/(2p^{3/2})$ are equidistributed on $[-1, 1]$ with respect to the Sato–Tate measure $\frac{2}{\pi} \sqrt{1-t^2} dt$ as $p \rightarrow \infty$.*

Proof. By Theorem 282, the geometric monodromy group of $\mathcal{M}_\chi$ is $\text{SL}(2)$, whose maximal compact subgroup is $\text{SU}(2)$. By the Deligne equidistribution theorem [19, Theorem 9.2.6] applied to a lisse rank-2 sheaf with $\text{SL}(2)$ -monodromy, the Frobenius conjugacy classes at the $\mathbb{F}_p$ -points become

equidistributed in the conjugacy classes of $\mathrm{SU}(2)$, which are parametrized by $\theta \in [0, \pi]$. The pushforward of Haar measure under the trace map $\theta \mapsto 2 \cos \theta$ is the Sato–Tate measure.

This is a special case of the crystalline companions theorem of Abe–Esnault and its equidistribution corollary (see Bah–Shuddhodan [21] for the precise statement at level 6). $\square$

Remark 284 (Product monodromy and Goursat’s lemma). For distinct primes $p \neq q$ with $p, q \neq 17$, the product monodromy group of $\mathcal{M}_{\chi,p} \times \mathcal{M}_{\chi,q}$ is $\mathrm{SL}_2(\mathbb{F}_p) \times \mathrm{SL}_2(\mathbb{F}_q)$. Indeed, Goursat’s lemma reduces product monodromy to the question of whether the two factors are isomorphic as abstract groups; since $|\mathrm{SL}_2(\mathbb{F}_p)| = p(p^2 - 1)$ and $p \neq q$, the orders differ and the product is direct. Computational verification through $p \leq 19$ confirms the full product structure: the joint Frobenius distribution of $(b_j \bmod p, b_j \bmod q)$ for $1 \leq j \leq pq - 2$ is uniform on $\mathbb{F}_p \times \mathbb{F}_q$ with chi-squared p -value exceeding 0.10 for every tested pair.

Remark 285 (The defining-characteristic gap). Theorem 283 controls the distribution of the Mellin trace $b_j \bmod \mathfrak{p}$ for a prime $\mathfrak{p}$ of $\overline{\mathbb{Q}}$ above an *auxiliary* prime $\ell \neq p$. The conjecture requires the event $p \mid b_j$, which is a *defining-characteristic* condition: the Frobenius trace is reduced modulo the same prime p that defines the finite field. Deligne equidistribution and its crystalline companions do not directly control the mod- p distribution of the mod- p Frobenius—this would require a defining-characteristic Katz-type theorem, which is not currently available. This gap is the precise obstruction to deducing $(\mathrm{HM})_3$ from the equidistribution machinery.

Proposition 286 (Franel pullback and cover-adapted coordinates). *Let*

$$\mathcal{A}(a) = \sum_{n \geq 0} b_n a^n, \quad \mathcal{F}(x) = \sum_{n \geq 0} f_n x^n, \quad f_n = \sum_{k=0}^n \binom{n}{k}^3,$$

and, for a prime p , let

$$A_p(a) = \sum_{m=0}^{p-1} b_m a^m, \quad K_p(x) = \sum_{m=0}^{p-1} f_m x^m \quad \text{in } \mathbb{F}_p[a] \text{ and } \mathbb{F}_p[x],$$

respectively. Put

$$\phi(x) = \frac{x(1-8x)}{1+x} \quad \text{and} \quad \Psi_{p,m}(x) = x^m(1-8x)^m(1+x)^{p-1-m}.$$

Then the following identity holds in $\mathbb{F}_p[x]$:

$$\boxed{K_p(x)^2 = (1+x)^{p-1} A_p(\phi(x)) = \sum_{m=0}^{p-1} b_m \Psi_{p,m}(x).} \quad (251)$$

The p polynomials $\Psi_{p,0}, \dots, \Psi_{p,p-1}$ are linearly independent. Consequently, if $[P]_{\Psi_{p,m}}$ denotes the m th coordinate of a polynomial P in their span, then

$$[K_p^2]_{\Psi_{p,m}} = b_m \quad (0 \leq m < p), \quad Z(p) = \#\{m : [K_p^2]_{\Psi_{p,m}} = 0\}. \quad (252)$$

In particular, this statement concerns coordinates in the cover-adapted basis, not the monomial coefficients of K_p^2 .

Proof. The quadratic transformation used in [10, Section 2] is the characteristic-zero formal identity

$$\mathcal{A}(\phi(x)) = (1+x)\mathcal{F}(x)^2. \quad (253)$$

Both coefficient sequences have the p -Lucas property. Hence, in $\mathbb{F}_p[[a]]$ and $\mathbb{F}_p[[x]]$,

$$\mathcal{A}(a) = A_p(a)\mathcal{A}(a)^p, \quad \mathcal{F}(x) = K_p(x)\mathcal{F}(x)^p.$$

Substitute $a = \phi(x)$ into the first congruence and use (253). On the one hand,

$$(1+x)\mathcal{F}(x)^2 = A_p(\phi(x))(1+x)^p\mathcal{F}(x)^{2p};$$

on the other hand it equals $(1+x)K_p(x)^2\mathcal{F}(x)^{2p}$. Cancellation in the integral domain $\mathbb{F}_p[[x]]$ proves the first equality in (251). Clearing the powers of $1+x$ gives the displayed polynomial expansion; both sides have degree at most $2p-2$.

Finally, $\Psi_{p,m}$ has x -adic order exactly m and leading coefficient 1. The transition matrix from $(\Psi_{p,0}, \dots, \Psi_{p,p-1})$ to the monomials is therefore triangular with diagonal entries 1. This proves linear independence and the coordinate identities. $\square$

Proposition 287 (The two normalized Apéry square roots). *Let $\Delta(a) = a^2 - 34a + 1$ and let $C_+(a) = \sum_{r \geq 0} c_r a^r$ and $C_-(a) = \sum_{r \geq 0} d_r a^r$ be defined by $c_{-1} = d_{-1} = 0$, $c_0 = d_0 = 1$, and*

$$4(r+1)^2 c_{r+1} = 2(68r^2 + 34r + 5)c_r - (2r-1)^2 c_{r-1}, \quad (254)$$

$$4(r+1)^2 d_{r+1} = 2(68r^2 + 102r + 39)d_r - (2r+1)^2 d_{r-1}. \quad (255)$$

Thus $c_1 = 5/2$ and $d_1 = 39/2$. In $\mathbb{Q}[[a]]$ one has

$$\mathcal{A}(a) = C_+(a)^2 = \Delta(a)C_-(a)^2. \quad (256)$$

For every prime $p \geq 5$, put

$$\varepsilon_p = \frac{1 - \left(\frac{-6}{p}\right)}{2}.$$

The factorization theorem of Caruso–Fürnsinn–Vargas–Montoya–Zudilin [10, Theorem 1.2] gives a unique polynomial B_p normalized by $B_p(0) = 1$ such that

$$A_p(a) = \Delta(a)^{\varepsilon_p} B_p(a)^2. \quad (257)$$

It is explicitly the indicated truncation of a universal series:

$$B_p(a) = \begin{cases} \sum_{r=0}^{(p-1)/2} \overline{c_r} a^r, & \left(\frac{-6}{p}\right) = 1, \\ \sum_{r=0}^{(p-3)/2} \overline{d_r} a^r, & \left(\frac{-6}{p}\right) = -1. \end{cases} \quad (258)$$

Here the bars denote reduction modulo p ; the displayed rational coefficients have denominators prime to p in the stated ranges.

Proof. The recurrences are the coefficient forms of the two second-order operators

$$\begin{aligned} L_+ &= 4\vartheta^2 - 2a(68\vartheta^2 + 34\vartheta + 5) + a^2(2\vartheta + 1)^2, \\ L_- &= 4\vartheta^2 - 2a(68\vartheta^2 + 102\vartheta + 39) + a^2(2\vartheta + 3)^2, \quad \vartheta = a \frac{d}{da}. \end{aligned}$$

A direct symmetric-square calculation shows that C_+^2 is annihilated by the Apéry operator

$$\vartheta^3 - a(2\vartheta + 1)(17\vartheta^2 + 17\vartheta + 5) + a^2(\vartheta + 1)^3.$$

Its constant and linear coefficients are 1 and 5, so uniqueness of the formal solution gives $C_+^2 = \mathcal{A}$. Conjugating L_+ by $\Delta^{-1/2}$ gives L_- ; equivalently, $C_- = C_+/\sqrt{\Delta}$, with constant term 1. This proves (256) and (255).

It remains only to identify the normalized factor in (257). In the square branch, comparison of the coefficients of degrees at most $(p-1)/2$ with $\mathcal{A} = C_+^2$ determines the coefficients of B_p successively, because the square-root map with constant term 1 is triangular with diagonal coefficient 2. This gives the first line of (258). Apply the same argument to $\mathcal{A}/\Delta = C_-^2$ in the defect branch. All divisions occurring in the recurrences have index strictly below p , so reduction modulo p is legitimate. $\square$

Proposition 288 (Exact equations and reduced Hasse square roots). *For every prime $p \geq 5$, the polynomial B_p of Proposition 287 satisfies an exact second-order equation in $\mathbb{F}_p[a]$. More precisely,*

$$(4\Delta\vartheta^2 + 4a(a-17)\vartheta + a(a-10))B_p = 0, \quad (259)$$

when $\varepsilon_p = 0$, whereas

$$(4\Delta\vartheta^2 + 12a(a-17)\vartheta + 3a(3a-26))B_p = 0 \quad (260)$$

when $\varepsilon_p = 1$. In addition,

$$\gcd(B_p, a\Delta B_p') = 1.$$

Thus B_p is squarefree and has

$$M_p = \frac{p-1-2\varepsilon_p}{2}$$

distinct geometric roots, none of which is 0 or a root of Δ .

Proof. Suppose first that $\varepsilon_p = 0$, and put $M = (p-1)/2$. By (258),

$$B_p = \sum_{m=0}^M \overline{c_m} a^m.$$

Compare the coefficient of a^{M+1} in the characteristic-zero identity $\mathcal{A} = C_+^2$ with the same coefficient in the exact identity $A_p = B_p^2$ over $\mathbb{F}_p$. Since $M+1 = (p+1)/2 < p$, the coefficient of $\mathcal{A}$ still reduces to the corresponding coefficient of A_p . All convolution terms agree except the two endpoint terms, so

$$2\overline{c_{M+1}} = 0.$$

The recurrence (254) now kills every coefficient of the left side of (259) through degree $M+1$. Its only possible coefficient in degree $M+2$ is

$$(2M+1)^2 \overline{c_M} = p^2 \overline{c_M} = 0.$$

This proves (259).

If $\varepsilon_p = 1$, put $M = (p - 3)/2$. Formal division by $\Delta(0) = 1$ shows that $\mathcal{A}/\Delta$ and A_p/Δ agree through degree $p - 1$. Comparing degree $M + 1 = (p - 1)/2$ in

$$\mathcal{A}/\Delta = C_-^2, \quad A_p/\Delta = B_p^2$$

gives $\overline{d_{M+1}} = 0$. The recurrence (255) kills all coefficients of (260) except possibly the last one, which is

$$(2M + 3)^2 \overline{d_M} = p^2 \overline{d_M} = 0.$$

This proves the second equation.

It remains to analyze roots. Write locally

$$B_p(a) = (a - \alpha)^m u(a), \quad u(\alpha) \neq 0.$$

If $\alpha\Delta(\alpha) \neq 0$ and $m \geq 2$, then the coefficient of $(a - \alpha)^{m-2}$ in either exact equation is

$$4\Delta(\alpha)\alpha^2 m(m-1)u(\alpha),$$

which is nonzero because $m < p$. Hence every root away from $a\Delta = 0$ is simple, while $B_p(0) = 1$.

Now suppose that $\Delta(\alpha) = 0$. Since $\Delta'(\alpha) = 2(\alpha - 17)$ and $\Delta(17) = -288 \not\equiv 0 \pmod{p}$, the coefficient of $(a - \alpha)^{m-1}$ in the square branch is

$$4\alpha^2(\alpha - 17)m(2m - 1)u(\alpha),$$

and in the defect branch it is

$$4\alpha^2(\alpha - 17)m(2m + 1)u(\alpha).$$

In the first case $1 \leq 2m - 1 \leq p - 2$, and in the second $3 \leq 2m + 1 \leq p - 2$. Both coefficients are therefore nonzero, a contradiction. This proves the coprimality assertion and squarefreeness. Finally $b_{p-1} \equiv 1 \pmod{p}$ in the factorization $A_p = \Delta^{\varepsilon_p} B_p^2$, so the displayed degree of B_p is exact. $\square$

Proposition 289 (Quadratic Frobenius descent of the reduced Hasse divisor). *For every prime $p \geq 5$, all geometric roots of B_p belong to $\mathbb{F}_{p^2}$. Equivalently,*

$$\boxed{B_p(a) \mid a^{p^2} - a} \quad \text{in } \mathbb{F}_p[a]. \quad (261)$$

In particular, every irreducible factor of B_p over $\mathbb{F}_p$ has degree one or two.

Proof. We first identify the Franel polynomial in Proposition 286 as an elliptic Hasse invariant. Consider

$$E_x : \quad V^2 + (1 - 2x)UV + x^2V = U^3, \quad Q_x = (x, -x). \quad (262)$$

The displayed multiples

$$2Q_x = (0, -x^2), \quad 3Q_x = (-x^2, -x^3), \quad 4Q_x = (0, 0), \quad 5Q_x = (x, x^2), \quad 6Q_x = O$$

show that Q_x has exact order six on the smooth generic fiber. This is the order-six Tate normal form; see Kubert [16]. Completing the square gives

$$W^2 = F_x(U), \quad F_x(U) = 4U^3 + (1 - 2x)^2U^2 + 2(1 - 2x)x^2U + x^4. \quad (263)$$

Put $m = (p-1)/2$. The Hasse invariant of a smooth fiber is $[U^{p-1}]F_x(U)^m$. Since

$$\frac{F_x(U)}{U^2} = \left(1 - 2x + \frac{x^2}{U}\right)^2 + 4U,$$

constant-term extraction gives

$$\text{Ha}_p(E_x) = \sum_{j=0}^{\lfloor 2m/3 \rfloor} \binom{m}{j} 4^j \binom{2m-2j}{j} (1-2x)^{2m-3j} x^{2j}. \quad (264)$$

Modulo p , one has

$$4^j \binom{m}{j} = (-1)^j \binom{2j}{j}, \quad \binom{2m-2j}{j} = (-1)^j \binom{3j}{j}.$$

It follows that, for $0 \leq n < p$, the coefficient of x^n in (264) is

$$\sum_{j=0}^{\lfloor n/2 \rfloor} 2^{n-2j} \frac{(n+j)!}{j!^3 (n-2j)!} = \sum_{k=0}^n \binom{n}{k}^3 = f_n. \quad (265)$$

The equality on the right is MacMahon's elementary Franel identity, obtainable by two Vandermonde convolutions. In extending the sum in (264) to the displayed range, no term was introduced modulo p : if $j > (p-1)/3$ and $2j \leq n$, then $n+j \geq p$, while every factorial in the denominator is a p -unit. Thus

$$\text{Ha}_p(E_x) = K_p(x). \quad (266)$$

We next record why the parameter x itself, rather than only the j -invariant, descends over a quadratic extension. The invariants of (262) are

$$\Delta_E = x^6(1+x)^2(1-8x), \quad c_4 = 16x^4 + 16x^3 - 8x + 1.$$

Hence

$$j(E_x) = \frac{(16x^4 + 16x^3 - 8x + 1)^3}{x^6(1+x)^2(1-8x)}. \quad (267)$$

The numerator is nonzero at the three finite cusps $0, -1, 1/8$, where its values are $1, 9, 9/256$, respectively. Therefore the map to the j -line has degree twelve in every characteristic $p \geq 5$. The coarse modular map $X_0(6) \rightarrow X(1)$ also has degree

$$[\text{PSL}_2(\mathbb{Z}) : \overline{\Gamma_0(6)}] = 6(1+2^{-1})(1+3^{-1}) = 12.$$

Moreover the coarse map $X_1(6) \rightarrow X_0(6)$ has degree $\varphi(6)/2 = 1$: the only generators of a cyclic group of order six are Q and $-Q$, and the pairs (E, Q) and $(E, -Q)$ are isomorphic through $[-1]$. The family above therefore identifies the x -line with the coarse curve $X_0(6) = X_1(6)$.

Let $x \in \overline{\mathbb{F}_p}$ be a smooth parameter with $K_p(x) = 0$. By (266), E_x is supersingular. A supersingular elliptic curve has a model over $\mathbb{F}_{p^2}$ on which square Frobenius is the scalar $[-p]$; we use Serre's canonical descent [34, Remark (4), p. 284]. After transporting the cyclic subgroup $\langle Q_x \rangle$ to that model, scalar Frobenius preserves it. Thus the coarse point $(E_x, \langle Q_x \rangle)$ is fixed by square Frobenius. Since x is the coarse $X_0(6)$ -coordinate, this proves

$$K_p(x) = 0 \text{ and } E_x \text{ smooth} \implies x \in \mathbb{F}_{p^2}. \quad (268)$$

This coarse-moduli argument also covers $j = 0, 1728$; it does not require a fine descent of a chosen presentation of the elliptic curve.

Finally let $B_p(\alpha) = 0$. Proposition 288 gives $\alpha\Delta(\alpha) \neq 0$. A preimage of α under $\phi(x) = x(1 - 8x)/(1 + x)$ is a root of

$$8x^2 + (\alpha - 1)x + \alpha = 0, \quad (269)$$

whose discriminant is $\Delta(\alpha)$. It is therefore finite and distinct from -1 , since substitution of -1 in the left side gives $9 \neq 0$. It is not 0 or $1/8$, since either value would give $\alpha = 0$. Thus E_x is smooth. The pullback identity (251) now gives $K_p(x) = 0$, so (268) yields $x \in \mathbb{F}_{p^2}$, and then $\alpha = \phi(x) \in \mathbb{F}_{p^2}$.

The polynomial B_p is squarefree by Proposition 288. All its roots are in $\mathbb{F}_{p^2}$, so it divides the squarefree polynomial $a^{p^2} - a$, proving (261). $\square$

Proposition 290 (The marked Cartier–Hurwitz cover). *Let $p \geq 5$, let $1 \leq r \leq p - 2$, and suppose that $b_r \equiv 0 \pmod{p}$. Define*

$$Q_{p,r}(a) := \sum_{\substack{0 \leq m \leq p-1 \\ m \neq r}} \frac{b_m}{m-r} a^m, \quad H_{p,r}(a) := a^{-r} Q_{p,r}(a).$$

Then

$$(\vartheta - r)Q_{p,r} = A_p, \quad H'_{p,r} = a^{-r-1} \Delta^{\varepsilon_p} B_p^2. \quad (270)$$

The rational map

$$H_{p,r}: \mathbb{P}^1 \longrightarrow \mathbb{P}^1$$

is separable of degree $p - 1$ and is tamely ramified. Its complete ramification passport consists of the following points:

1. 0 and ∞ , with indices r and $p - 1 - r$;
2. the M_p roots of B_p , each with index 3;
3. when $\varepsilon_p = 1$, the two roots of Δ , each with index 2.

Proof. For arbitrary r , coefficientwise calculation gives

$$(\vartheta - r)Q_{p,r} = A_p - b_r a^r, \quad H'_{p,r} = a^{-r-1} (A_p - b_r a^r).$$

This gives (270) at a target. The constant coefficient of $Q_{p,r}$ is $-1/r$, while its coefficient of a^{p-1} is $1/(p - 1 - r)$, because $b_{p-1} \equiv 1 \pmod{p}$. Hence the two pole orders are exact and their sum is $p - 1$. The derivative is not identically zero, and every displayed index is less than p , so the cover is separable and tame.

Proposition 288 shows that the roots of B_p are simple and disjoint from $a\Delta = 0$. The derivative in (270) therefore has order exactly two at each root of B_p , and, in the defect branch, order exactly one at each root of Δ . These give the asserted local indices. Their ramification contributions sum to

$$(r - 1) + (p - 2 - r) + 2M_p + 2\varepsilon_p = 2p - 4 = 2 \deg(H_{p,r}) - 2.$$

Riemann–Hurwitz excludes any further ramification. $\square$

Corollary 291 (Quadratic critical-value factors). *Under the hypotheses of Proposition 290, define the monic critical-value polynomial*

$$\mathcal{P}_{p,r}(Y) := \prod_{B_p(\alpha)=0} (Y - H_{p,r}(\alpha)) \in \mathbb{F}_p[Y].$$

Then

$$\text{rad } \mathcal{P}_{p,r}(Y) \mid Y^{p^2} - Y. \quad (271)$$

Thus every irreducible factor of $\mathcal{P}_{p,r}$ has degree at most two. This assertion bounds the Frobenius-orbit degree of a critical value; it does not bound the exponent of such a factor, equivalently the number of distinct critical points with the same value.

Proof. Proposition 289 gives $\alpha \in \mathbb{F}_{p^2}$ for every root of B_p . Since $H_{p,r} \in \mathbb{F}_p(a)$ and $\alpha \neq 0$, every value $H_{p,r}(\alpha)$ also belongs to $\mathbb{F}_{p^2}$. The root set is Frobenius-stable, so the displayed product lies in $\mathbb{F}_p[Y]$, and its squarefree part divides $Y^{p^2} - Y$. $\square$

Proposition 292 (Kummer order and decomposition rigidity). *Retain the hypotheses of Proposition 290, assume $p \geq 7$, and put*

$$g = \gcd(r, p-1), \quad d = \frac{p-1}{g}.$$

Suppose over $\overline{\mathbb{F}}_p$ that

$$H_{p,r} = F \circ G, \quad a := \deg F > 1, \quad b := \deg G > 1.$$

Then

$$a \mid g, \quad b \geq d.$$

If $d > 2$, then a is odd, and some finite critical value of F comes from a critical point of F whose G -fiber contains at least $b \geq d$ distinct roots of B_p . In particular, if $d > 2$ and g is a power of 2, then $H_{p,r}$ is indecomposable.

Proof. The pole set of $H_{p,r}$ is exactly $\{0, \infty\}$. If F had two poles, their inverse images under G would partition this two-point set. Each pole of F would therefore have a unique, totally ramified preimage. After changes of coordinates this forces

$$G(a) = c a^{\pm b}.$$

Consequently $H_{p,r}(\zeta a) = H_{p,r}(a)$ for every $\zeta^b = 1$. Since $p \geq 7$, a target cannot have $r = 1$, as $b_1 = 5$. Thus the $m = 0$ and $m = 1$ terms in $Q_{p,r}$ are both nonzero, and $H_{p,r}$ has two nonzero Laurent exponents differing by one. This contradicts the displayed invariance when $b > 1$.

Hence F has one pole and may be normalized to a polynomial of degree a . Its pole has precisely the two preimages $0, \infty$ under G , so the two pole orders of the composite are

$$r = ae_G(0), \quad p-1-r = ae_G(\infty).$$

It follows that $a \mid g$, and then $b = (p-1)/a \geq d$.

Assume $d > 2$. The polynomial F has a finite critical point because $1 < a < p$. Multiplicativity of local ramification indices and Proposition 290 show that its index is 2 or 3, and that G is unramified above it. Index 2 would produce $b \geq d > 2$ distinct index-2 points of $H_{p,r}$, whereas the passport contains at most two. Thus every finite critical point of F has index 3. Any such point gives the asserted packet of b distinct roots of B_p . The finite ramification contribution of a degree- a polynomial is $a-1$; here it is a sum of contributions $3-1 = 2$. Hence a is odd. No such $a > 1$ divides a power of 2. $\square$

Proposition 293 (Critical-value saturation). *Retain the hypotheses of Proposition 290, put $n = p - 1$, and let $y \in \overline{\mathbb{F}}_p$. Define*

$$N_y := Q_{p,r} - ya^r = a^r(H_{p,r} - y), \quad C_y := \gcd(B_p, N_y),$$

where the gcd is monic in $\overline{\mathbb{F}}_p[a]$, and put $k_y = \deg C_y$. When $\varepsilon_p = 1$, let ℓ_y be the number of roots of Δ which are roots of N_y ; when $\varepsilon_p = 0$, put $\ell_y = 0$. Then

$$C_y^3 \mid N_y, \quad 3k_y + 2\ell_y \leq n. \quad (272)$$

More precisely, writing

$$B_p = C_y E_y, \quad N_y = C_y^3 D_y,$$

one has

$$3aC_y' D_y + aC_y D_y' - rC_y D_y = \Delta^{\varepsilon_p} E_y^2, \quad \gcd(C_y, D_y) = 1. \quad (273)$$

Thus every common root of B_p and N_y has multiplicity exactly three in N_y .

If $r = n/2$ and $y \neq 0$, then one has the additional reciprocal bound

$$k_y \leq \left\lfloor \frac{M_p}{2} \right\rfloor. \quad (274)$$

Proof. Since $(\vartheta - r)a^r = 0$,

$$(\vartheta - r)N_y = A_p = \Delta^{\varepsilon_p} B_p^2. \quad (275)$$

The coefficient of a^n in N_y is $b_n/(n - r) = 1/(n - r) \neq 0$, so $\deg N_y = n$. Let α be a root of C_y . By Proposition 288,

$$\alpha \Delta(\alpha) B_p'(\alpha) \neq 0.$$

Equation (270) shows that $H_{p,r}'$ has order exactly two at α . Since $p \geq 5$, $H_{p,r} - H_{p,r}(\alpha)$, and hence N_y , has order exactly three there. The roots of C_y are distinct, which proves $C_y^3 \mid N_y$. In the defect branch, a root of Δ which lies over y is a double root of N_y . Counting these roots in the degree- n polynomial N_y gives (272).

Substitute $B_p = C_y E_y$ and $N_y = C_y^3 D_y$ into (275), and divide by C_y^2 . This gives the first identity in (273). Squarefreeness and conifold avoidance give

$$\gcd(C_y, a\Delta E_y C_y') = 1.$$

Reducing the identity modulo C_y yields

$$3aC_y' D_y \equiv \Delta^{\varepsilon_p} E_y^2 \pmod{C_y}.$$

The right side is a unit modulo C_y , so $\gcd(C_y, D_y) = 1$.

It remains to prove the central bound. Put $s = n - r$. The reflection $b_{n-m} = b_m$ gives

$$a^n Q_{p,r}(a^{-1}) = -Q_{p,s}(a), \quad H_{p,r}(a^{-1}) = -H_{p,s}(a). \quad (276)$$

Palindromicity of A_p , reciprocity of Δ , and $A_p = \Delta^{\varepsilon_p} B_p^2$ show that inversion preserves the geometric root set of B_p . If $r = s = n/2$, (276) therefore sends the k_y roots above y bijectively to k_y roots above $-y$. For $y \neq 0$ the two packets are disjoint, and both lie among the M_p roots of B_p . Hence $2k_y \leq M_p$, proving (274). $\square$

Proposition 294 (Sharpness of the primitive passport). *For every $k \geq 1$, put $n = 4k + 2$. For each $\varepsilon \in \{0, 1\}$ there exists a separable rational map*

$$f_{k,\varepsilon}: \mathbb{P}^1 \longrightarrow \mathbb{P}^1$$

of degree n , over an algebraically closed field of characteristic zero, with the following complete passport:

1. *the two poles both have index $n/2$;*
2. *there are $(n - 2\varepsilon)/2$ finite points of index 3;*
3. *there are 2ε finite points of index 2;*
4. *there is no further ramification.*

The geometric monodromy group is A_n when $\varepsilon = 0$ and S_n when $\varepsilon = 1$. Nevertheless, one finite branch value has $k = (n - 2)/4$ distinct index-3 points above it.

If $p = n + 1$ is prime, the same branch-cycle tuples are realized by tame rational maps over $\overline{\mathbb{F}}_p$.

Proof. Start with the $N = 4k + 1$ letters

$$\{x_0, \dots, x_k\} \dot{\cup} \{a_i, b_i, c_i : 1 \leq i \leq k\}.$$

On them define

$$A_i = (a_i \ b_i \ c_i), \quad C_i = (c_i \ x_{i-1} \ x_i), \quad \sigma_0 = A_1 \cdots A_k,$$

and put $P = \sigma_0 C_1 \cdots C_k$. The cycles $C_1, \dots, C_k$ splice to a cycle on the $2k + 1$ letters x_i, c_i . Each A_i then meets that cycle in the single letter c_i and adds two new letters. Hence P is an N -cycle.

Add one letter z , fixed by P . Choose an old letter u , and set

$$d = 2k, \quad v = P^{-d}(u), \quad \tau_1 = (u \ v), \quad \tau_2 = (z \ u).$$

The transposition τ_1 cuts the N -cycle P into cycles of lengths $2k$ and $2k + 1$, with u on the former. Multiplication by τ_2 inserts z into that cycle. Consequently

$$P\tau_1\tau_2 \sim (2k + 1)(2k + 1). \tag{277}$$

Put $\sigma_\infty = (P\tau_1\tau_2)^{-1}$.

For $\varepsilon = 1$, take the product-one branch-cycle tuple

$$(\sigma_0, C_1, \dots, C_k, \tau_1, \tau_2, \sigma_\infty). \tag{278}$$

It contains $2k = (n - 2)/2$ cubic points and two quadratic points. Its finite ramification contribution is

$$2k + 2k + 1 + 1 = n,$$

whereas (277) gives $\text{ind}(\sigma_\infty) = n - 2$.

For $\varepsilon = 0$, replace the two transpositions in (278) by their product

$$\gamma := \tau_1\tau_2,$$

which is a 3-cycle. The tuple

$$(\sigma_0, C_1, \dots, C_k, \gamma, \sigma_\infty) \tag{279}$$

has $k+k+1 = 2k+1 = n/2$ cubic points and no quadratic points. Its finite ramification contribution is again n . In either case the total contribution is $n + (n - 2) = 2n - 2$, so Riemann–Hurwitz gives genus zero and excludes further ramification.

It remains to identify the monodromy. Since $\gcd(d, N) = \gcd(2k, 4k + 1) = 1$, the conjugates of τ_1 by powers of P are the edge transpositions of a connected cycle on the old letters. Thus P and τ_1 generate S_N , and τ_2 joins the new letter; the tuple (278) generates S_n .

For (279), the conjugates of γ by powers of P are the 3-cycles supported on z and the successive edges of the same step- d cycle. These are the standard adjacent 3-cycle generators of A_n . All cycles in (279) are even, so its group is exactly A_n . In both tuples the branch cycle σ_0 has cycle type $3^k 1^{n-3k}$, giving the asserted packet.

The Riemann existence theorem realizes the tuples over characteristic zero. When $p = n + 1$ is prime, their monodromy groups have order prime to p . The prime-to- p tame fundamental group of a punctured projective line has the same product-one presentation, so the tuples are also realized over $\overline{\mathbb{F}}_p$. The source again has genus zero by the tame Riemann–Hurwitz formula. $\square$

Remark 295 (Scope of the primitive-passport obstruction). Proposition 294 shows that the complete ramification passport of Proposition 290, even together with primitive geometric monodromy, cannot imply an $O(1)$ bound for a critical-value packet. The constructed maps are not asserted to be the Apéry maps $H_{p,r}$. They do not impose the exact differential equations for B_p , the marked condition $b_r = 0$, the reciprocal identity, or compatibility with one global integral sequence. Thus the proposition does not obstruct an Apéry-specific packet bound derived from those additional structures. Proposition 293 records the exact positive constraint currently available from the differential shape.

Remark 296 (The critical equations have rank one). For arbitrary r , not necessarily a target,

$$A_p - b_r a^r \equiv -b_r a^r \pmod{B_p}.$$

Since $B_p(0) = 1$,

$$\text{Res}_a(B_p, A_p - b_r a^r) = \pm b_r^{M_p}.$$

Thus the M_p critical-point incidences at a target are not independent ordinary divisibility conditions. The passport and the decomposition rigidity above do not by themselves imply cross-prime anti-concentration.

Remark 297 (Exact algebraic reduction versus the GCD conjecture). Equations (252) show that the assertion

$$\#\{m : [K_p^2]_{\Psi_{p,m}} = 0\} = O(1)$$

is exactly Hypothesis 60. It is not equivalent to the pointwise Apéry GCD conjecture. Indeed, Remark 124 shows that even abstract zero sets of cardinality one can align adversarially across different primes, whereas the full conjecture is equivalent to the pointwise radical estimate in Remark 15. Conversely, no converse implication is known: the pointwise radical formulation permits the individual cardinalities $Z(p)$ to grow provided their positions do not align across primes. Moreover, the Poisson model of Remark 63 predicts that Hypothesis 60 is false.

The pullback nevertheless gives an exact, Apéry-specific residual coefficient problem. General ℓ -adic trace bounds, monodromy, and Dwork unit-root congruences do not by themselves imply anti-concentration of these defining-characteristic coordinates. This is a limitation of the currently available general theorems, not a claim that Cartier or Dwork methods cannot contribute after additional Apéry-specific input. For the average results in this paper, the natural single-prime target is the weaker estimate $\sum_{p \leq X} Z(p) \ll \pi(X)$; the all-index conjecture also requires the cross-prime dispersion described above.

Proposition 298 (Triangular freedom of reciprocal squares). *Let K be a field of characteristic different from 2, let $d \geq 1$, and put*

$$\Delta(t) = t^2 - 34t + 1.$$

For a normalized reciprocal polynomial

$$B(t) = 1 + x_1t + \cdots + x_{d-1}t^{d-1} + t^d, \quad x_{d-j} = x_j,$$

and $\varepsilon \in \{0, 1\}$, write

$$\Delta(t)^\varepsilon B(t)^2 = \sum_{j=0}^{2d+2\varepsilon} H_j t^j.$$

If $m < d/2$, then the coefficient map

$$(x_1, \dots, x_m) \longmapsto (H_1, \dots, H_m),$$

with the remaining free coefficients of B held fixed arbitrarily, is a triangular polynomial automorphism of K^m . In particular, every vector in K^m occurs as $(H_1, \dots, H_m)$ for some normalized reciprocal polynomial B .

Consequently, the factorization

$$H(t) = \Delta(t)^\varepsilon B(t)^2$$

and reciprocity alone impose no restriction on a zero pattern contained in the first $m < d/2$ coefficients of H : any such pattern, together with its reciprocal reflection, can occur.

Proof. For $1 \leq j \leq m$, the inequality $j < d/2$ ensures that reciprocity has not identified x_j with any of $x_1, \dots, x_{j-1}$. The coefficient of t^j in the square is

$$[t^j]B(t)^2 = 2x_j + \sum_{i=1}^{j-1} x_i x_{j-i}.$$

Since $\Delta(0) = 1$, multiplication by Δ^ε gives

$$H_j = 2x_j + Q_j(x_1, \dots, x_{j-1})$$

for a polynomial $Q_j \in K[x_1, \dots, x_{j-1}]$; coefficients of B with index greater than j do not occur. Thus the displayed map is triangular with diagonal entries 2. Given arbitrary $(a_1, \dots, a_m) \in K^m$, one obtains its unique preimage recursively from

$$x_j = \frac{a_j - Q_j(x_1, \dots, x_{j-1})}{2}.$$

After choosing any still-free middle coefficients, set $x_{d-j} = x_j$ to complete B . This proves surjectivity and the polynomial inverse formula.

Both B^2 and Δ^ε are reciprocal, so their product is reciprocal of degree $2d + 2\varepsilon$:

$$H_{2d+2\varepsilon-j} = H_j.$$

Prescribing $a_j = 0$ on an arbitrary subset of $\{1, \dots, m\}$ therefore realizes that zero pattern and its reflected copy. The conclusion concerns the full abstract family of reciprocal squares; it does not assert that the arithmetically distinguished square root attached to the Apéry motive is generic in that family. $\square$

12.26 A cubic Fricke-grid congruence

Let

$$b_m = \sum_{k=0}^m \binom{m}{k}^2 \binom{m+k}{k}^2, \quad F(T) = \sum_{m \geq 0} b_m T^m. \quad (280)$$

We use the classical level-six parametrization

$$t(\tau) = \left(\frac{\eta(\tau)\eta(6\tau)}{\eta(2\tau)\eta(3\tau)} \right)^{12}, \quad E(\tau) = \frac{\eta(2\tau)^7 \eta(3\tau)^7}{\eta(\tau)^5 \eta(6\tau)^5} = F(t(\tau)). \quad (281)$$

$$H(\tau) = q \frac{d}{dq} \log t(\tau), \quad \Psi(\tau) = E(\tau)H(\tau), \quad A(q) = \frac{q}{t(q)}. \quad (282)$$

Thus $E, H \in M_2(\Gamma_0(6))$, $\Psi \in M_4(\Gamma_0(6))$, and

$$A \in 1 + q\mathbf{Z}[[q]], \quad \Psi \in 1 + q\mathbf{Z}[[q]]. \quad (283)$$

For $m \geq 0$, put

$$\Omega_m = \Psi t^{-m} = q^{-m} \Psi A^m, \quad \Psi A^m = \sum_{j \geq 0} c_j(m) q^j. \quad (284)$$

In particular $\Omega_m = q^{-m} + O(q^{-m+1})$ and $\Omega_m \in \mathbf{Z}((q))$.

12.26.1 The Fricke-odd integral grid

We label the cusps of $X_0(6)$ by their denominators $c \in \{1, 2, 3, 6\}$, so that $c = 1$ is the cusp 0 and $c = 6$ is ∞ .

Lemma 299 (Divisor and Fricke data). *In normalized local parameters, the cusp orders are*

c	1	2	3	6
$\text{ord}_c(t)$	1	−1	−1	1
$\text{ord}_c(E)$	0	1	1	0
$\text{ord}_c(\Psi)$	0	1	1	0
$\text{ord}_c(\Omega_m)$	− m	$m + 1$	$m + 1$	− m .

(285)

For the normalized Fricke involution W_6 ,

$$t \circ W_6 = t, \quad E|_2 W_6 = -E, \quad H|_2 W_6 = H, \quad \Psi|_4 W_6 = -\Psi. \quad (286)$$

The only zeros of Ψ in the upper half-plane are simple zeros at the two Fricke fixed points.

Proof. The first two rows of (285) follow directly from Ligozat's eta-order formula. At every cusp, t is a local parameter or the inverse of one, up to a unit; its logarithmic derivative H is therefore a unit. This gives the last two rows.

The eta transformation law gives the first two identities in (286); differentiating the identity for t gives the identity for H , and hence the one for Ψ . Finally, the level-six parametrization satisfies

$$H^2 = (t^2 - 34t + 1)E^2. \quad (287)$$

The eta quotient E has no zero in the upper half-plane. The two roots of $t^2 - 34t + 1$ are the branch values of the quotient by W_6 . Consequently H , and therefore Ψ , has a simple zero at each of the two Fricke fixed points and no other interior zero. $\square$

Set

$$C_0 = \{c = 1, 6\}, \quad C_\infty = \{c = 2, 3\}. \quad (288)$$

Thus t has simple zeros on C_0 and simple poles on C_∞ .

Lemma 300 (Fricke-odd grid). *Let $M \geq L \geq 0$. Suppose that $f \in M_4^!(\Gamma_0(6))$ satisfies $f|_4 W_6 = -f$, has no poles away from C_0 , has pole order at most M there, and vanishes to order at least $L + 1$ at both cusps in C_∞ . Then*

$$f = \sum_{m=L}^M a_m \Omega_m \quad (289)$$

for unique $a_m \in \mathbf{C}$. If the Fourier expansion of f at ∞ is integral, then every a_m is an integer.

Proof. The quotient f/Ψ has weight zero and is Fricke invariant. We first check that it is regular at the Fricke fixed points. At such a point, compose W_6 with a suitable element of $\Gamma_0(6)$ to obtain an elliptic stabilizer γW_6 of order two in $\mathrm{PSL}_2(\mathbf{R})$. Its automorphy factor on a weight-four fiber is $+1$: after normalization $(\gamma W_6)^2 = -I$, so at the fixed point its factor j satisfies $j^2 = -1$, and hence $j^{-4} = 1$. The relation $f|_4 W_6 = -f$ therefore forces f to vanish. In a local coordinate the involution acts by $z \mapsto -z$, so the vanishing order is odd and at least one. By Lemma 299, Ψ has a simple zero; hence f/Ψ is regular.

The quotient consequently descends to a rational function on $X_0(6)/\langle W_6 \rangle$. This quotient is a projective line with coordinate t : the invariant function t has degree two upstairs, and hence degree one after descent. The descended quotient has a pole of order at most M at $t = 0$, a zero of order at least L at $t = \infty$, and no other pole. It is therefore $\sum_{m=L}^M a_m t^{-m}$, proving (289).

For integrality, descend through the Fourier principal part at ∞ . Each Ω_m has leading term q^{-m} with coefficient one and has an integral Laurent expansion, so triangular elimination gives $a_m \in \mathbf{Z}$. $\square$

Remark 301. The unique cusp form in $S_4(\Gamma_0(6))$ is $(\eta(\tau)\eta(2\tau)\eta(3\tau)\eta(6\tau))^2$, and it is Fricke even. Thus no cusp-form residual is omitted from the Fricke-odd grid. Equivalently, Lemma 300 with $M = L = 0$ says directly that the holomorphic Fricke-odd line is generated by Ψ .

12.26.2 Hecke transport and cubic elimination

For a Laurent series with finite principal part, use

$$U_p \left(\sum_k a_k q^k \right) = \sum_k a_{pk} q^k, \quad V_p \left(\sum_k a_k q^k \right) = \sum_k a_k q^{pk}. \quad (290)$$

Lemma 302 (Cusp transport). *Let $p \geq 5$, $n = Qp + r$, and $0 \leq r < p$. There are integers C_m such that*

$$T_p \Omega_n = \sum_{m=Q}^{pn} C_m \Omega_m. \quad (291)$$

Proof. Since $p \nmid 6$, the normalized weight-four operator is

$$T_p = U_p + p^3 V_p. \quad (292)$$

It commutes with the level-six Atkin–Lehner involutions. After slashing by one of these involutions, the order calculation at any cusp is therefore the ordinary one at ∞ : the U_p -part of a series beginning in degree $v \geq 0$ begins no earlier than degree $\lceil v/p \rceil$, and the V_p -part begins in degree pv .

By (285), Ω_n has order $-n$ on C_0 and order $n+1$ on C_∞ . Hence $T_p\Omega_n$ has pole order at most pn on C_0 , while

$$\text{ord}_{C_\infty}(T_p\Omega_n) \geq \left\lceil \frac{n+1}{p} \right\rceil = Q+1. \quad (293)$$

It remains Fricke odd because T_p commutes with W_6 . The complex expansion in (291) now follows from Lemma 300. Its coefficients are integers because $T_p\Omega_n$ has an integral Laurent expansion and the grid is unitriangular at ∞ . $\square$

Proposition 303 (Cubic Fricke-grid congruence). *For every prime $p \geq 5$, every $Q \geq 0$, and every $0 \leq r < p$,*

$$U_p\Omega_{Qp+r} \equiv c_r(Qp+r)\Omega_Q \pmod{p^3\mathbf{Z}((q))}. \quad (294)$$

Equivalently, for every $j \geq 0$,

$$c_{r+pj}(Qp+r) \equiv c_r(Qp+r)c_j(Q) \pmod{p^3}. \quad (295)$$

Proof. Write $\Omega_n = \sum_{k \geq -n} a_k q^k$, where $n = Qp+r$. At exponent $-m$,

$$[q^{-m}]T_p\Omega_n = a_{-pm} + p^3 \mathbf{1}_{p|m} a_{-m/p}. \quad (296)$$

If $m > Q$, then $pm > Qp+r = n$, so the first term on the right is zero. Descend in (291) from $m = pn$. Since $\Omega_m = q^{-m} + O(q^{-m+1})$, triangular elimination in (296) gives

$$p^3 \mid C_m \quad (m > Q). \quad (297)$$

At $m = Q$, the first term in (296) is

$$a_{-pQ} = c_r(n). \quad (298)$$

The second term and all contributions from the already eliminated higher grid elements are divisible by p^3 . Thus

$$C_Q \equiv c_r(n) \pmod{p^3}. \quad (299)$$

It is important that this is a congruence, not an equality. For example, when $Q = r = 0$, Lemma 300 gives $T_p\Omega_0 = (1+p^3)\Omega_0$, so $C_0 = 1+p^3$, whereas $c_0(0) = 1$.

Equations (291), (297), and (299) imply $T_p\Omega_n \equiv c_r(n)\Omega_Q \pmod{p^3}$. Subtracting $p^3V_p\Omega_n$ and using (292) proves (294). Extracting the coefficient of q^{j-Q} gives (295). $\square$

12.26.3 The neighboring coordinate and its limitation

The initial expansions are

$$A = 1 + 12q + O(q^2), \quad \Psi = 1 - 7q + O(q^2), \quad (300)$$

and therefore

$$c_1(Q) = 12Q - 7. \quad (301)$$

Corollary 304. *For every p, Q, r in Proposition 303,*

$$p^3 \mid c_{p+r}(Qp+r) - (12Q-7)c_r(Qp+r). \quad (302)$$

The exponent three is sharp even after imposing the Apéry target condition $p \mid b_r$.

Proof. The divisibility is the case $j = 1$ of (295). For sharpness take $(p, Q, r) = (5, 1, 1)$. Here $5 \mid b_1$,

$$c_1(6) = 65, \quad c_6(6) = b_6 = 21460825, \quad (303)$$

and

$$\frac{c_6(6) - 5c_1(6)}{5^3} = 171684 \equiv 4 \pmod{5}. \quad (304)$$

Thus the difference in (302) need not be divisible by p^4 on the target locus. $\square$

Remark 305 (No radical saving from the cubic grid). The target condition is detected at the leading coordinate by the familiar congruence $c_r(Qp + r) \equiv b_r \pmod{p}$. In contrast, (294) holds for every residue r , whether or not $p \mid b_r$. Its factor p^3 is therefore universal Hecke content. After dividing it out, the first neighboring residual is nonzero on the target example (304). Hence the cubic grid by itself supplies neither a second target equation nor a characteristic-zero carrier giving a new bound for the target radical. This does not exclude a different nonlinear relation among several post-grid residuals.

12.27 Jacobi–Stickelberger analysis of the marked coordinate

We use the fixed Laurent polynomial (180) and the marked-coordinate identity of Theorem 265. Put

$$h = p - 1, \quad E_p = \mathbb{Q}(\mu_h), \quad \chi = \omega^j,$$

where $\omega: \mathbb{F}_p^\times \rightarrow \mu_h$ is the Teichmüller character associated with the chosen prime $\mathfrak{p} \mid p$ of E_p . Every multiplicative character, including the trivial character ε , is extended by zero at 0. We write

$$J(A, B) = \sum_{t \in \mathbb{F}_p} A(t)B(1 - t).$$

All character exponents below are read modulo h .

12.27.1 A1: an exact four-Jacobi skeleton

Theorem 306 (Jacobi skeleton). *For $1 \leq j \leq p - 2$, the cyclotomic lift in (187) has the exact expression*

$$\tilde{c}_{p,j}^{\text{tor}} = -\frac{1}{h} \sum_{k=0}^{h-1} J(\omega^{-k}, \omega^j)^2 J(\omega^{-k}, \omega^{j+k})^2. \quad (305)$$

Thus the skeleton has one character variable and four Jacobi factors, all with arguments affine in (j, k) .

At $j = 0$, let the right side of (305), with $\chi = \varepsilon$, be denoted by $\mathcal{S}_{p,0}$. The exact endpoint formula is

$$\tilde{c}_{p,0}^{\text{tor}} = \mathcal{S}_{p,0} - 4p^2 + 14p - 13 = -(p - 1)^3. \quad (306)$$

In particular, reduction modulo $\mathfrak{p}$ gives b_j for every $0 \leq j \leq p - 2$.

Proof. For a character A and $t \in \mathbb{F}_p^\times$, finite Mellin inversion gives

$$A(1 + t) = \frac{1}{h} \sum_{\rho} \rho(-1) J(\bar{\rho}, A) \rho(t), \quad (307)$$

where ρ runs over all multiplicative characters of $\mathbb{F}_p^\times$. Indeed, the Fourier coefficient at ρ is

$$\sum_{t \neq 0} A(1+t)\bar{\rho}(t) = \rho(-1)J(\bar{\rho}, A).$$

For $x, y, z \neq 0$, put

$$s = \frac{(1+y)(1+z)}{xyz}.$$

The marked coordinate factors as

$$\Lambda(x, y, z) = (1+x)(1+y)(1+z)(1+s). \quad (308)$$

If $j > 0$, the summand is already zero when $y = -1$ or $z = -1$. We may therefore apply (307) to s and then restore those zero-contribution values. With $\rho = \omega^k$, the three variable sums separate:

$$\begin{aligned} \sum_{x \neq 0} \chi(1+x)\bar{\rho}(x) &= \rho(-1)J(\bar{\rho}, \chi), \\ \sum_{y \neq 0} (\chi\rho)(1+y)\bar{\rho}(y) &= \rho(-1)J(\bar{\rho}, \chi\rho), \\ \sum_{z \neq 0} (\chi\rho)(1+z)\bar{\rho}(z) &= \rho(-1)J(\bar{\rho}, \chi\rho). \end{aligned}$$

Together with the factor $\rho(-1)$ in (307), the four signs multiply to 1. The outer minus sign in the definition of $\tilde{c}_{p,j}^{\text{tor}}$ now gives (305).

This argument includes all degenerate values of k : the identities

$$J(\varepsilon, A) = J(A, \varepsilon) = -1, \quad J(A, A^{-1}) = -A(-1), \quad J(\varepsilon, \varepsilon) = p-2$$

are built into the unnormalized Jacobi sum. In particular, no Gauss-sum quotient is being used illegally when a product character is trivial.

At $j = 0$, zero-extension of ε computes the open-set sum $\mathcal{S}_{p,0} = -\#U(\mathbb{F}_p)$ rather than the prescribed raw zeroth power. Equation (185) gives

$$\#U(\mathbb{F}_p) - (p-1)^3 = -4p^2 + 14p - 13,$$

which proves (306). This is the necessary $0^0 = 1$ correction; it is not a choice of character convention. $\square$

For completeness, the reduction of every Jacobi factor has an $O(1)$ formula. If $a, b \in \{0, \dots, h-1\}$ and $\bar{J}_p(a, b)$ denotes the reduction of $J(\omega^a, \omega^b)$ modulo $\mathfrak{p}$, then

$$\bar{J}_p(a, b) = \begin{cases} -2, & a = b = 0, \\ -1, & \text{exactly one of } a, b \text{ is zero,} \\ 0, & a, b > 0 \text{ and } h-a > b, \\ -(-1)^{h-a} \binom{b}{h-a}, & a, b > 0 \text{ and } h-a \leq b. \end{cases} \quad (309)$$

This follows by expanding $(1-t)^b$: among the powers t^{a+r} , only $a+r = h$ can survive torus orthogonality. Since $-h^{-1} \equiv 1 \pmod{p}$, equations (305) and (309) are an entirely explicit finite-field formula for b_j .

For $1 \leq j \leq p-2$, if

$$m = \min(j, h-j),$$

then only $0 \leq k \leq m$ survive modulo $\mathfrak{p}$, and on that range

$$\begin{aligned}\bar{J}_p(-k, j) &= (-1)^{k+1} \binom{j}{k}, \\ \bar{J}_p(-k, j+k) &= (-1)^{k+1} \binom{j+k}{k}.\end{aligned}\tag{310}$$

Consequently the surviving k -th skeleton term is exactly

$$\binom{j}{k}^2 \binom{j+k}{k}^2.\tag{311}$$

When $j > h/2$, the omitted terms of the usual Apéry sum have $j+k \geq p$ and hence $p \mid \binom{j+k}{k}$. This also proves directly, without numerical input, that the reduction of the skeleton is b_j .

The script `lemmaA_verify.py` independently compares the skeleton with the Apéry recurrence for every prime $5 \leq p \leq 200$ and every $0 \leq j \leq p-2$ (4178 pairs). It also compares both with the direct three-dimensional torus sum through $p=31$. At $j = h/2$, it expands

$$\eta(2z)^4 \eta(4z)^4 = q \prod_{n \geq 1} (1 - q^{2n})^4 (1 - q^{4n})^4 = \sum_{r \geq 1} a_r q^r$$

and verifies

$$\widehat{c}_{p,h/2}^{\text{tor}} \equiv b_{h/2} \equiv a_p \pmod{p}\tag{312}$$

for every prime $5 \leq p \leq 200$, reproducing the modular-form anchor in Remark 62.

12.27.2 A2: the explicit Stickelberger container

Choose a prime $\mathfrak{P} \mid \mathfrak{p}$ of $E_p(\mu_p)$ and normalize $v_{\mathfrak{P}}(p) = 1$. With

$$g(A) = \sum_{t \in \mathbb{F}_p} A(t) \zeta_p^t,$$

Gross–Koblitz computes this valuation; its restriction to E_p is the $\mathfrak{p}$ -adic valuation used below. For an exponent a , write

$$d_h(a) = \frac{[-a]_h}{h} = \left\langle -\frac{a}{h} \right\rangle,$$

where $[-a]_h \in \{0, \dots, h-1\}$ and the fractional part of an integer is 0. Gross–Koblitz gives $v_{\mathfrak{P}}(g(\omega^a)) = d_h(a)$. Including the inverse-character degeneracy, the resulting formula valid for every a, b is

$$v_h(a, b) := v_{\mathfrak{P}}\left(J(\omega^a, \omega^b)\right) = d_h(a) + d_h(b) - d_h(a+b) - \mathbf{1}_{\substack{a \not\equiv 0, \ b \not\equiv 0 \\ a+b \equiv 0 \pmod{h}}}(h).\tag{313}$$

Indeed, away from a trivial product character this is the valuation of $g(\omega^a)g(\omega^b)/g(\omega^{a+b})$; when the two nontrivial characters are inverse, $J(A, A^{-1}) = -A(-1)$ is a unit and supplies the last correction. Formula (313) also gives zero in the cases involving a trivial factor. Its values are always 0 or 1.

For the remainder of A2 assume $1 \leq j \leq p-2$, and let

$$T_{j,k} = J(\omega^{-k}, \omega^j)^2 J(\omega^{-k}, \omega^{j+k})^2.$$

Equations (305) and (313) give the requested digit/fractional-part valuation

$$\begin{aligned} v(k; j) &= 2\nu_h(-k, j) + 2\nu_h(-k, j + k) \\ &= 2\mathbf{1}_{\{k > j\}} + 2\mathbf{1}_{\{k > h-j\}}, \quad 0 \leq k \leq h-1. \end{aligned} \quad (314)$$

Thus its minimum is zero and

$$\{k : v(k; j) = 0\} = \{0, 1, \dots, \min(j, h-j)\}, \quad U_p(j) = 1 + \min(j, h-j). \quad (315)$$

This is computable in $O(1)$ operations per (p, j) .

Theorem 307 (Unique-unit container). *For $1 \leq j \leq p-2$, if $U_p(j) = 1$, then $b_j \not\equiv 0 \pmod{p}$. Hence*

$$\mathcal{Z}_p \subseteq \{1 \leq j \leq p-2 : U_p(j) \neq 1\}. \quad (316)$$

The endpoints $j = 0, p-1$ are absent from $\mathcal{Z}_p$ because their marked values are 1.

Proof. Jacobi sums are algebraic integers, $-1/h$ is a unit at $\mathfrak{p}$, and the minimum in (314) is 0. If exactly one $T_{j,k}$ has valuation zero, reduction of (305) kills every other term and leaves one nonzero residue. Therefore the reduction of $\tilde{c}_{p,j}^{\text{tor}}$, which is b_j by (183), cannot vanish. $\square$

There are two normalization points in this argument. First, the Gauss sums live in $E_p(\mu_p)$, but the Jacobi sums lie in E_p ; because $p \equiv 1 \pmod{h}$, the residue field at $\mathfrak{p}$ is $\mathbb{F}_p$. Second, if the valuation is instead normalized by $v(1 - \zeta_p) = 1$, every value in (313)–(314) must be multiplied by h . No Greene $p^{-1}J$ normalization is used here.

The explicit answer (315) now exposes the decision gate:

$$U_p(j) = 1 + \min(j, p-1-j) \geq 2 \quad (1 \leq j \leq p-2). \quad (317)$$

Thus the implication in Theorem 307 is correct but never triggers on a nontrivial character. Moreover, (311) shows that the simultaneous unit terms reconstruct the original Apéry sum modulo p : cancellation among unit terms is the original vanishing problem, not a lower-complexity replacement for it.

It is important here that the h Jacobi summands are a Mellin expansion of the trace, not the at most B_Λ Frobenius eigenvalues in Proposition 267. Their unit count depends on the chosen integral character expansion and is not a crystalline slope multiplicity. The fixed-dimensional cohomological trace occurs only after the length- h Mellin sum has cancelled. Thus (317) does not contradict the uniform Betti bound; it identifies the precise point at which this Jacobi skeleton fails to produce a unique unit summand and hence gives no unique-unit-root certificate.

12.27.3 A3: empirical computation and verdict

The zero sets were computed from the Apéry recurrence, independently of the skeleton. A zero at $j = h/2$ was classified first as *non-ordinary*; every other zero was classified as a collision of at least two unit skeleton terms. The results are:

bound	primes	$\sum_p Z(p)$	zero-free	$U = 0$	$U \geq 2$ collision	non-ordinary
200	44	41	25	0	40	1
500	93	95	53	0	94	1
1000	166	165	96	0	164	1
2000	301	283	184	0	282	1
5000	667	608	418	0	606	2

For $p \leq 2000$, the distribution of $Z(p)$ is

$$\#\{Z = 0, 1, 2, 4, 6\} = \{184, 1, 95, 17, 4\}.$$

For the stretch range $p \leq 5000$, it is

$$\#\{Z = 0, 1, 2, 4, 6, 8\} = \{418, 2, 199, 41, 6, 1\}.$$

The non-ordinary points in the stretch range are

$$(p, j) = (11, 5), (3137, 1568); \quad a_{11} = -44, \quad a_{3137} = 207042 = 66 \cdot 3137.$$

The container size is not merely empirical: by (317), for every $p \geq 5$,

$$\#\{1 \leq j \leq p-2 : U_p(j) \neq 1\} = p-2. \quad (318)$$

Therefore the decision-gate verdict is **S2**. Generic j has multiple unit Jacobi terms (in fact every nontrivial j does), and all observed zeros are either non-ordinary midpoint zeros or cancellations among these unit terms. This presentation of the crystalline/Jacobi route alone gives no estimate $|\mathcal{Z}_p| \ll p^\theta$ with $\theta < 1$. The unconditional $Z(p) \ll p^{2/3}$ bound in Proposition 14 remains available, but it comes from the gap-polynomial argument, not from (316).

Finally, the Lemma-0 collapse is fully compatible with this negative gate. Corollary 79 gives, for $m \in (N, 2N]$ and $N < p \leq m$,

$$b_m \equiv 5b_{m-p} \pmod{p}, \quad L_N(m) = \#\{p \in (N, m] : p \mid b_m\}.$$

Since (318) excludes no nontrivial affine target $j = m - p$, it supplies neither the pointwise anti-concentration required after the collapse nor the centered estimate (178). The exact skeleton and its Stickelberger valuation are therefore complete, but their decision-gate output is the valid S2 obstruction specified in the task.

12.28 The eigenvalue Newton polygon of the marked coordinate

We retain the marked Laurent polynomial of Theorem 265 and the exact Jacobi skeleton of Theorem 306. For $q = p^r$ and $1 \leq j \leq p-2$, put

$$A = \chi_{p,j} \circ N_{\mathbb{F}_q/\mathbb{F}_p}, \quad S_{p,j}^{(r)} = - \sum_{x \in U(\mathbb{F}_q)} A(\Lambda(x)).$$

For an auxiliary prime $\ell \neq p$, write E_ℓ for a coefficient field containing the values of A . Thus $S_{p,j}^{(1)} = \tilde{c}_{p,j}^{\text{tor}}$ and $S_{p,j}^{(1)} \equiv b_j \pmod{\mathfrak{p}}$. We use the effective sign convention in which $S_{p,j}^{(r)}$ is a Frobenius power trace. The signed trace on $R\Gamma_c(U, \mathcal{L}_{p,j})$ is its negative. This sign will be restored in the slope-zero inclusion below.

12.28.1 The corrected Adolphson–Sperber object

The coordinate change already used in the proof of Proposition 267,

$$u = \frac{1+y}{y}, \quad v = \frac{1+z}{z}, \quad w = -\frac{x}{uv},$$

gives

$$\Lambda = \frac{uv(w-1)(1-uvw)}{w(u-1)(v-1)}, \quad U = (\mathbb{G}_m)^3 \setminus \{u=1, v=1, w=1, uvw=1\}. \quad (319)$$

There is an important trap here. Converting the Kummer sum to an additive sum with the phase $t\Lambda$ does *not* permit direct application of the nondegenerate Newton-polytope theorem. At $x = y = -1$ two factors of Λ vanish, so $t\Lambda$ and all four toric logarithmic derivatives vanish simultaneously. In fact, if $P = \text{Newt}(\Lambda)$, then

$$P = \{(X, Y, Z) : -1 \leq X, Y, Z \leq 1, X - Y \leq 1, X - Z \leq 1\}.$$

Its cross-sectional area is 4 for $-1 \leq X \leq 0$ and $(2 - X)^2$ for $0 \leq X \leq 1$, whence

$$3! \text{vol}(P) = 6 \left(4 + \int_0^1 (2 - X)^2 dX \right) = 38. \quad (320)$$

The same number is the normalized volume of $\text{conv}(0, P \times \{1\})$. A naive use of this polytope would therefore manufacture 38 slots. It is incompatible with the cancelled rank below (and with $\chi_c(U) = 1 - 7 + 17 - 14 = -3$), precisely because the phase is degenerate.

The correct Adolphson–Sperber/GKZ object follows by first performing the Jacobi convolution. Base change of Theorem 306 gives the exact identity

$$S_{p,j}^{(r)} = -\frac{1}{q-1} \sum_{\rho} J_q(\bar{\rho}, A)^2 J_q(\bar{\rho}, A\rho)^2, \quad (321)$$

where *all* $q-1$ characters of $\mathbb{F}_q^\times$ occur. With $g_q(C) = \sum_t C(t)\psi(t)$ and $g_q(\varepsilon) = -1$, including the inverse-character degeneracies, one has

$$J_q(\bar{\rho}, A)^2 J_q(\bar{\rho}, A\rho)^2 = q^{-2} g_q(\bar{\rho})^4 g_q(A\rho)^2 g_q(A^{-1}\rho)^2. \quad (322)$$

For example, when $\rho = A$ the Jacobi factor $J_q(A^{-1}, A)^2$ and the product $g_q(A^{-1})^2 g_q(A)^2 = q^2$ give the same formula; no exceptional summand has been dropped. To fix all signs, inversions, and Tate normalizations, define

$$\mathcal{H}_{q,A}(t) := -\frac{1}{q^2(q-1)} \sum_{\rho} g_q(\bar{\rho})^4 g_q(A\rho)^2 g_q(A^{-1}\rho)^2 \rho(t).$$

This is our trace convention for the Tate-normalized balanced hypergeometric object

$$\text{Hyp}(\varepsilon, \varepsilon, \varepsilon, \varepsilon; A, A, A^{-1}, A^{-1}). \quad (323)$$

Equations (321) and (322) say exactly that $\mathcal{H}_{q,A}(1) = S_{p,j}^{(r)}$. The opposite Katz ordering exchanges the two parameter lists and replaces t by t^{-1} , hence gives the same stalk at $t = 1$. Introduce the usual hypergeometric parameter t . Expanding the eight Gauss factors gives the associated GKZ circuit, which is nondegenerate for generic $t \neq 1$; this, rather than the degenerate phase $t\Lambda$, is the object to which the twisted-height recipe of Adolphson–Sperber [2] applies. The value $t = 1$ is the unique balanced-circuit singularity and must subsequently be treated by middle extension and nearby cycles.

Here is the requested closed form. Put

$$a = \left\langle \frac{j}{p-1} \right\rangle \in (0, 1), \quad \alpha = (0, 0, 0, 0),$$

and sort the fractional-part data

$$\beta(a) = \text{sort}(\langle a \rangle, \langle a \rangle, \langle -a \rangle, \langle -a \rangle) = \text{sort}(a, a, 1-a, 1-a). \quad (324)$$

For $k = 1, \dots, 4$, the twisted lattice-height statistic is

$$\rho_a(k) = \#\{i : \alpha_i < \beta_k(a)\} - k = 4 - k. \quad (325)$$

Indeed, every entry of (324) is strictly positive, including at $a = 1/2$. Thus the generic circuit has Hodge polynomial

$$H_a^{\text{gen}}(T) = \sum_{k=1}^4 T^{\rho_a(k)} = 1 + T + T^2 + T^3. \quad (326)$$

The two character multisets in (323) are disjoint, so the generic hypergeometric object is irreducible of rank four. The balanced hypergeometric local-monodromy theorem (in the convolution normalization of Katz [20]) says that the monodromy at $t = 1$ is a nontrivial pseudoreflection. Its special eigenvalue is

$$\exp\left(2\pi i \left(\sum_k \beta_k - \sum_i \alpha_i\right)\right) = \exp(4\pi i) = 1.$$

Thus $T_1 = \exp N$ with $N \neq 0$, $N^2 = 0$, and $\text{rank } N = 1$. The standard rank-one-unipotent hypergeometric nearby-cycle calculation now applies. The parameter multiset is invariant under complex conjugation, so the variation has a real form. The polarized monodromy filtration $W(N)[3]$ has rank-one graded pieces in weights 2 and 4; the real form forces them to have types (1, 1) and (2, 2), respectively. Together with the four generic Hodge degrees in (326), this gives, after one complex coefficient embedding,

$$\text{Gr}_2^W \text{ of type } (1, 1), \quad \text{Gr}_3^W \text{ of types } (3, 0) + (0, 3), \quad \text{Gr}_4^W \text{ of type } (2, 2),$$

where $N : \text{Gr}_4^W \xrightarrow{\sim} \text{Gr}_2^W(-1)$. The middle extension has stalk $\ker(T_1 - 1) = \ker N = W_3$, and strictness computes its induced Hodge filtration from the graded types above. It therefore removes the slope-2 slot, not the slope-0 slot. The toric Hodge polynomial and the requested multiplicity are

$$H_{a,t=1}(T) = 1 + T + T^3, \quad \boxed{h^0(j) = [T^0]H_{a,t=1}(T) = 1} \quad (1 \leq j \leq p-2). \quad (327)$$

This is a closed fractional-part formula; it uses no Apéry value and no numerical input.

For completeness, the three slots can also be seen on the arrangement. The only non-simple-normal-crossings point is $(u, v, w) = (1, 1, 1)$. On its exceptional plane, with local coordinates denoted by a, b, c , the leading Kummer function is

$$\Lambda_E = -\frac{c(a+b+c)}{ab}.$$

For every nontrivial character A over $\mathbb{F}_q$, setting $b = 1$ gives

$$\sum_{E^\circ(\mathbb{F}_q)} A(\Lambda_E) = A(-1) \sum_{a,c \neq 0} A(c)A(a+1+c)A^{-1}(a) = q.$$

For $c \neq -1$ the inner a -sum is -1 , while at $c = -1$ it is $q-1$; the last equality follows after summing $A(c)$. In perverse normalization, the Mellin identity identifies the $t = 1$ stalk with the effective class $-[R\Gamma_c]$. The blow-up weight spectral sequence identifies the exceptional-plane contribution with its unique rank-one weight-2 piece, $\text{Gr}_2^W(\ker N) \simeq E_\ell(-1)$ above, and the displayed sum says that geometric Frobenius acts on it by $q = p^r$. Hence its eigenvalue in the effective polynomial is exactly $+p$; the outer minus sign has already been absorbed in $-[R\Gamma_c]$.

The involution

$$\tau(u, v, w) = (w, (uvw)^{-1}, u), \quad \Lambda \circ \tau = \Lambda^{-1}, \quad (328)$$

is defined over $\mathbb{F}_p$ and carries $\mathcal{L}_A$ to $\mathcal{L}_{A^{-1}}$. Composing this identification with Poincaré duality gives a Frobenius-equivariant alternating perfect pairing

$$V_A \otimes_{E_\ell} V_A \longrightarrow E_\ell(-3)$$

on the remaining rank-two quotient, with no residual constant Kummer twist. In rank two the determinant equals the similitude multiplier, hence it is p^3 . Therefore the effective characteristic factor is the theorem-level identity

$$\boxed{P_{p,j}(X) = (X - p)(X^2 - a_{p,j}X + p^3), \quad a_{p,j} = S_{p,j}^{(1)} - p.} \quad (329)$$

The equality is an equality of the semisimplified effective virtual Frobenius class $-[R\Gamma_c(U, \mathcal{L}_{p,j})]$ with the rank-three middle-extension stalk. Thus it does not confuse the $q - 1$ Mellin summands in (321) with eigenvalues.

12.28.2 Newton polygons and the inclusion theorem

Since $a_{p,j} \equiv S_{p,j}^{(1)} \equiv b_j \pmod{\mathfrak{p}}$, put $m = v_{\mathfrak{p}}(a_{p,j})$. The two roots of the quadratic factor in (329) have valuations

$$\begin{cases} (0, 3), & m = 0, \\ (1, 2), & m = 1, \\ (3/2, 3/2), & m \geq 2. \end{cases} \quad (330)$$

Adding the Tate slope 1 gives

$$m_0(p, j) = \begin{cases} 1, & b_j \not\equiv 0 \pmod{p}, \\ 0, & b_j \equiv 0 \pmod{p}. \end{cases} \quad (331)$$

In particular, $m_0(p, j)$ is never at least two. At an ordinary point the Newton slopes are $(0, 1, 3)$ and equal the Hodge slopes in (327); at a Hasse zero the polygon rises strictly above the Hodge polygon.

We record explicitly the theorem required in the specification. In the effective convention, the unique unit root reduces to b_j . Restoring the sign of $R\Gamma_c$, its signed slope-zero sum is therefore $-b_j \pmod{\mathfrak{p}}$. Consequently

Theorem 308 (Slope-zero inclusion). *For every prime $p \geq 5$,*

$$\mathcal{Z}_p \subseteq \{1 \leq j \leq p - 2 : m_0(p, j) \neq 1\}. \quad (332)$$

For this marked coordinate equality holds in (332): both sides are exactly the Hasse-zero set.

Thus the inclusion is a theorem, but it is also a tautology: the moving unit root exists exactly when its Hasse invariant b_j is nonzero.

Extension-field computation. To check (329) without making the circular Jacobi-unit count, the script evaluates the full base-changed skeleton (321). If γ generates $\mathbb{F}_q^\times$, set

$$f_n = A(1 - \gamma^n), \quad g_n = A \left(\frac{\gamma^n}{1 + \gamma^n} \right).$$

Multiplicative Fourier inversion turns the complete $q - 1$ character sum into the exact cyclic convolution

$$S_{p,j}^{(r)} = - \sum_{c \in \mathbb{Z}/(q-1)} (f * f)_c (g * g)_c. \quad (333)$$

This is the promised $\mathbb{F}_{p^r}$ use of Theorem 306. Only the characters with index $K = (q - 1)k/(p - 1)$ are Hasse–Davenport lifts of base-field characters; the other extension characters cannot be omitted. At $(p, j, r) = (5, 1, 2)$ the correct trace is -125 , whereas the invalid termwise lifting shortcut gives -153 .

Newton’s identities recover the cubic coefficients from S_1, S_2, S_3 :

$$e_1 = S_1, \quad e_2 = \frac{S_1^2 - S_2}{2}, \quad e_3 = \frac{S_1^3 - 3S_1S_2 + 2S_3}{6}.$$

On six rational probe pairs, the portable direct-count and full- $q - 1$ A1 engines independently compute all three traces and give $e_2 = pS_1 + p^3 - p^2$ and $e_3 = p^4$, as predicted by (329). The $p = 13$ row below uses an independently computed S_1 and the theorem-level factor. Representative rows are

p	j	S_1	e_2	e_3	Newton slopes
5	1	−5	75	625	(1, 1, 2)
5	2	3	115	625	(0, 1, 3)
7	1	−2	280	2401	(0, 1, 3)
7	3	31	511	2401	(0, 1, 3)
11	5	−33	847	14641	(1, 1, 2)
13	1	5	2093	28561	(0, 1, 3)

For example, the complete traces at $(p, j) = (7, 1)$ are

$$(S_1, S_2, S_3, S_4) = (-2, -556, 8875, 133128),$$

and at $(11, 5)$ they are $(-33, -605, 91839)$. A genuinely cyclotomic check at $p = 11, j = 1$, with $\zeta = \zeta_{10}$, gives

$$S_1 = 8\zeta^3 - 8\zeta^2 - 27, \quad e_2 = 88\zeta^3 - 88\zeta^2 + 913, \quad e_3 = 14641.$$

For every prime $p \leq 60$ and every nontrivial j (405 pairs), an independent production scan computed the complete traces S_1, S_2, S_3 modulo p^6 from (333). Thus every one of the $q - 1$ extension characters was retained for $q = p, p^2, p^3$; neither the Apéry recurrence nor (329) was used to construct the traces. The optional `--full-small-scan` mode of `lemmaBprime_explore.py`, run under `sage -python`, is the reproduction path; the standard portable verifier uses the exact anchor probes instead of repeating this seven-minute scan. Newton’s identities then gave

range	$(0, 1, 3) = \text{NP} = \text{HP}$	$(1, 1, 2)$	$m_0 \geq 2$
$p \leq 60$, all j	390	15	0

The 15 nonordinary points are

$$(5; 1, 3), (11; 5), (17; 3, 13), (19; 8, 10), \\ (31; 8, 22), (37; 17, 19), (41; 10, 30), (59; 9, 49).$$

They are exactly the 15 pairs with $b_j = 0$ in this range.

For every prime $p \leq 300$, the second range run used the already proved factor (329): it evaluated the base-field A1 skeleton p -adically at every $j \in \mathcal{Z}_p$, and classified 20 deterministic random coordinates per prime from the factor and their base trace modulo p (all available coordinates for the smallest primes). It contained 63 zero coordinates and 1140 sampled coordinates. All 63 zeros had $v_p(a_{p,j}) = 1$ and slopes $(1, 1, 2)$. The random sample split as 1127 ordinary coordinates with slopes $(0, 1, 3)$ and 13 hits on the already enumerated zero set, again with slopes $(1, 1, 2)$. No pair had two slope-zero roots. Thus the $p \leq 60$ table is a full extension-field check of the factor, while the $p \leq 300$ table is a theorem-driven enumeration plus independent base-A1 valuation checks at all Hasse zeros. Neither table is used to prove the non-numerical Hodge formula (327).

12.28.3 Exact eigenvalue tests and the discriminant

The polynomial

$$\Delta(t) = t^2 - 34t + 1$$

does occur naturally, but not as a Frobenius eigenline. From (319), the logarithmic critical equations are

$$u^2vw - 2uvw + 1 = 0, \quad uv^2w - 2uvw + 1 = 0, \quad uvw^2 - 1 = 0.$$

On U they give

$$u = v = 2 - w, \quad w^2 - 2w - 1 = 0,$$

and substitution yields the two critical values

$$\Lambda_{\text{crit}} = 17 \pm 12\sqrt{2} = (1 \pm \sqrt{2})^4. \quad (334)$$

Thus the roots of Δ are conifold critical values of the map Λ . The involution (328) exchanges them. The actual rank-two quotient remains self-dual; neither critical value thereby becomes a rank-one Teichmüller eigenline.

The exact tests make this distinction visible. At the split prime $p = 7$, the roots of Δ are 2 and 4. For $j = 1, 2, 3$ the quadratic factors are

$$X^2 + 9X + 343, \quad X^2 + 25X + 343, \quad X^2 - 24X + 343.$$

Their unit-root residues are respectively 5, 3, 3, whereas the proposed Teichmüller targets are respectively $\{2, 4\}$, $\{2, 4\}$, and $\{1\}$. Hence the candidate already fails modulo 7. More strongly, for $j = 1, 2$ reduction modulo $\Phi_3(X) = X^2 + X + 1$ leaves $8X + 342$ and $24X + 342$, and for $j = 3$ the last quadratic evaluates to 320 at $X = 1$. There is no exact cyclotomic match.

The three quadratic eigenvalue fields in this test are

$$\mathbb{Q}(\sqrt{-1291}), \quad \mathbb{Q}(\sqrt{-83}), \quad \mathbb{Q}(\sqrt{-199}),$$

none of which is the base Jacobi field $E_7 = \mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$. Thus no base-field Jacobi-sum monomial, affine in j or otherwise, equals the unit root in these exact cases. In $\mathbb{Q}_7$, the unit residues 5, 3, 3 are all nonsquares; the other two slopes are 1 and 3, both odd. Hence none of

the three actual eigenvalues is a square in the natural coefficient completion. This rules out the nonvacuous local Sym^2 square test.

At the inert prime $p = 11$, the roots of Δ lie in $\mathbb{F}_{11^2}$ and have order 6. For $j = 1, 2, 4$ their powers do not even lie in $\mathbb{F}_{11}$; for $j = 3$ the target is -1 but $b_3 \equiv 4 \pmod{11}$; and $j = 5$ has no unit root. Residue coincidences can occur: at $j = 6$ both the target and the unit-root residue are 1.

Fix an embedding $\iota_p : \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_p}$. A Teichmüller lift of either reduction of $17 \pm 12\sqrt{2}$ is $\iota_p(\xi)$ for a prime-to- p root of unity ξ ; in the inert case it lies in the unramified quadratic extension. Every complex conjugate of ξ has absolute value 1. By contrast, each root of the pure weight-3 quotient is a Weil p -number whose complex conjugates have absolute value $p^{3/2}$. Injectivity of ι_p therefore forbids exact equality in $\overline{\mathbb{Q}_p}$ with $\omega(17 \pm 12\sqrt{2})^{\pm j}$, although isolated congruences modulo p are possible.

The $p = 7$ field and square computations are additional exact sample checks: they exclude every base-field Jacobi monomial in those cases and every square in the natural coefficient completion $E_{7,p} = \mathbb{Q}_7$. We do not claim a universal classification of affine Jacobi monomials, or a square obstruction after arbitrary scalar extension. The sole universally identified rank-one eigenvalue is p , of slope 1, and is irrelevant to a slope-zero collision.

Binary verdict. Equation (327) gives $h^0(j) = 1$ for every nontrivial twist, so Newton-above-Hodge allows at most one slope-zero eigenvalue. Equation (331) says that the Newton polygon equals the Hodge polygon precisely away from the original Hasse zeros. Thus the two-unit-root split required by G2 is structurally impossible, independently of whether the single moving root has an isolated alternative description. The requested verdict is therefore

G1 (KILL).

The crystalline criterion is exactly the Hasse-invariant tautology $p \mid b_j \iff m_0(p, j) = 0$. There is no second unit root with which a moving root could collide. In particular, the critical values $17 \pm 12\sqrt{2}$ do not open a unit-root anti-concentration problem.

Fail-closed computational consistency verification. Running `python3 problems/3.2/lemmaBprime_verification.py` ends with

```
PASS full A1 probes: 6 pairs r<=3; p13 r=1; cyclotomic p11,j1 S1
PASS cubic factor: 6 probed cubics; 5 r4 recurrences; p13 predicted
PASS p<=59 classification: 405 factor classifications; base-A1 v(a)=1 at 15/15 zeros
PASS p<=300 sample: base-A1 v(a)=1 at 63/63 zeros; sample 1140, zero hits 13
PASS candidate matches: critical values; p7 exact Teich/Jacobi/Q7-square rejection
PASS binary gate: G1 (h0=1; the only fixed branch is Tate slope 1)
SUMMARY: 11 PASS, 0 FAIL
```

12.29 Poisson variance formulation

The diagonal AMTD (156) admits a clean probabilistic reformulation. For a dyadic block $\mathcal{B}_i$, define the *column multiplicity function*

$$\lambda_i(m) = \#\{p \in \mathcal{B}_i : m \in \Omega_p\} = \sum_{p \in \mathcal{B}_i} X_p(m), \quad m \in I_N.$$

Its empirical mean and variance over I_N are

$$\mu_i = \frac{1}{N} \sum_m \lambda_i(m) = \frac{1}{N} \sum_{p \in \mathcal{B}_i} A_p, \quad V_i = \frac{1}{N} \sum_m (\lambda_i(m) - \mu_i)^2.$$

Expanding the square gives

$$N V_i = \sum_{p \in \mathcal{B}_i} D_p + \sum_{\substack{p, q \in \mathcal{B}_i \\ p \neq q}} C_{p, q}.$$

Since all primes in a dyadic block satisfy $\log p \asymp \log P$, one has $\|F_i\|_2^2 \asymp (\log P)^2 N V_i$ and $s_i^2 \asymp (\log P)^2 \sum_p D_p$, so the diagonal AMTD is equivalent (up to asymptotically negligible corrections from the non-constant weight $\log p$) to:

$$V_i \leq C \mu_i \quad (\text{sub-Poisson variance}). \quad (335)$$

If the indicators $X_p(m)$ were independent Bernoulli variables (with $\Pr[X_p = 1] = A_p/N$), each $\lambda_i(m)$ would be a sum of independent sparse Bernoullis and the variance-to-mean ratio would be ≤ 1 . Condition (335) therefore says that the deterministic Apéry indicators behave as if they were *approximately independent* at the block level.

Table 5 records the empirical variance-to-mean ratio for the top dyadic block.

N	K_{top}	μ	V/μ	collisions	Poisson pred.
1024	47	0.086	1.073	14	7.6
2048	94	0.079	0.983	10	12.8
4096	177	0.079	0.989	22	25.6
8192	363	0.080	1.011	60	52.9

Table 5: Column multiplicity statistics for the top dyadic block. $\mu = \frac{1}{N} \sum_p A_p$ is the mean, V/μ is the variance-to-mean ratio (Poisson prediction: 1), “collisions” is $\sum_m \lambda(\lambda - 1)$, and “Poisson pred.” is T^2/N where $T = \sum_p A_p$.

Remark 309 (Collision count). The off-diagonal covariance sum has the exact identity

$$\sum_{p \neq q \in \mathcal{B}_i} C_{p, q} = \sum_m \lambda_i(m)(\lambda_i(m) - 1) - \frac{(\sum_p A_p)^2 - \sum_p A_p^2}{N}.$$

The first term counts pairwise collisions (integers hit by at least two primes in the block). For the top block ($P \sim N$), each prime has $A_p \leq Z(p) \leq 3$ hits, so $\lambda_i(m) \leq 3$ for all m at $N = 8192$. The collision count fluctuates around the Poisson prediction T^2/N , confirming the near-independence of the indicators.

Remark 310 (Doublet dominance). Palindromic symmetry $b_{p-1-j} \equiv b_j \pmod{p}$ forces the zero set $\mathcal{Z}_p$ to be closed under $r \mapsto p - 1 - r$. If $Z(p) > 0$, then $Z(p)$ is typically even, and indeed at the top block, $Z(p) \geq 2$ for all active primes. The distribution at $N = 8192$ is: *doublets* $Z = 2$ (78%), *quadruplets* $Z = 4$ (19%), *sextuplets* $Z = 6$ (3%), *octuplets* $Z = 8$ ($< 1\%$). No singleton primes appear ($Z = 1$ requires $b_{(p-1)/2} \equiv 0 \pmod{p}$, which is extremely rare for large p).

For a doublet prime with $\mathcal{Z}_p = \{r_p, p - 1 - r_p\}$, the two hit positions are

$$m_1(p) = p + r_p, \quad m_2(p) = 2p - 1 - r_p,$$

satisfying $m_1(p) + m_2(p) = 3p - 1$. A collision between two doublet primes p, q requires one of four

types:

$$\begin{aligned}
\text{(A)} \quad m_1(p) = m_1(q) &\iff r_q = r_p + (p - q), \\
\text{(B)} \quad m_1(p) = m_2(q) &\iff r_p + r_q = 2q - 1 - p, \\
\text{(C)} \quad m_2(p) = m_1(q) &\iff r_p + r_q = 2p - 1 - q, \\
\text{(D)} \quad m_2(p) = m_2(q) &\iff r_q = r_p + 2(q - p).
\end{aligned}$$

For $q > p$, type (B) is impossible: $m_1(p) = p + r_p < 3p/2 < 3q/2 < 2q - 1 - r_q = m_2(q)$. The collision energy therefore decomposes into three, not four, pieces:

$$F = F_A + F_C + F_D, \quad (336)$$

where $F_A = \sum_m \nu_1(m)(\nu_1(m)-1)$, $F_D = \sum_m \nu_2(m)(\nu_2(m)-1)$, $F_C = 2 \sum_m \nu_1(m)\nu_2(m)$, with $\nu_1(m) = \#\{p : m_1(p) = m\}$ and $\nu_2(m) = \#\{p : m_2(p) = m\}$.

Two distinct doublet columns $\{m_1(p), m_2(p)\}$ and $\{m_1(q), m_2(q)\}$ can share at most one element: if both elements coincided, then $m_1(p) + m_2(p) = m_1(q) + m_2(q)$, i.e., $3p - 1 = 3q - 1$, forcing $p = q$. Thus $J_{p,q} := |\Omega_p \cap \Omega_q| \leq 1$ for distinct doublet primes in the top block. The doublet incidence system is a *simple graph* (no multi-edges), with the ordered collision count $F = \sum_m \lambda(m)(\lambda(m)-1)$. The exact variance identity is

$$V = T + F - T^2/N.$$

Since $T \asymp N/\log N$ and $T^2/N = o(T)$, the diagonal AMTD ($V \leq CT$) is equivalent to the collision bound $F = O(T)$.

Collision decomposition.

N	K	T	T^2/N	F	F_A	F_C	F_D	F/T
1024	47	39	1.5	2	2	0	0	0.051
2048	93	70	2.4	4	2	2	0	0.057
4096	177	140	4.8	0	0	0	0	0.000
8192	363	293	10.5	16	10	4	2	0.055
16384	669	528	17.0	22	16	6	0	0.042
32768	1145	925	26.1	38	18	20	0	0.041

The ratio F/T is consistently small and decreasing, empirically confirming $F = o(T)$. Under the FPI prediction $F \approx 3K^2/N$, one expects $F/T \approx 3/(6.6 \log N)$, giving 0.044 at $N = 32,768$, in good agreement with the observed 0.041. At all sizes $\max_m \lambda(m) \leq 3$.

Remark 311 (Two-prime divisor formulation). At the top block ($P > N$), each hit $m \in \Omega_p$ implies $p \mid b_m$ by the Lucas congruence. Therefore

$$\lambda_{\mathcal{B}}(m) = \omega_{(N, 2N]}(b_m) := \#\{p \text{ prime} : N < p \leq 2N, p \mid b_m\},$$

and the collision count becomes

$$F = \sum_{m \in I_N} \omega_{(N, 2N]}(b_m)(\omega_{(N, 2N]}(b_m) - 1).$$

The diagonal AMTD is therefore equivalent to the following *two-large-prime divisor theorem*: on average over m , the Apéry number b_m has at most $O(1)$ prime divisors in the dyadic range $(N, 2N]$.

This formulation reveals the arithmetic core: the collision bound $F = O(T)$ is an Apéry-specific Barban–Davenport–Halberstam theorem, asking for a level-of-distribution estimate at the second prime moment. A single-prime level-of-distribution (equivalently, $T = O(N/\log N)$) follows from the Poisson conjecture and is not in doubt; the missing step is the *bilinear* independence: the events $p \mid b_m$ and $q \mid b_m$ are approximately uncorrelated as p, q range over a dyadic block.

Vertical vs. horizontal projections. The incidence graph has primes on one side and integers on the other; an edge (p, n) records $p \mid b_n$. Grouping by the integer index n gives the *vertical* prime-divisor count $W_N(n) = \omega_{(N, 2N]}(b_n)$, $n \in [1, N]$. Grouping by the hit position $m = n + p$ gives the *horizontal* column multiplicity $\lambda(m)$. These are different projections. By the Lucas congruence $b_{p+r} \equiv 5b_r \pmod{p}$ (for $r < p$), the horizontal multiplicity $\lambda(m) = \omega_{(N, m]}(b_m)$ —it counts large prime divisors of b_m itself. Crucially, even the pointwise bound $W_N(n) \leq 1$ for *every* $n \leq N$ does not control $\lambda(m)$: many distinct (n_i, p_i) pairs can satisfy $n_i + p_i = m$ with $p_i \mid b_{n_i}$, all contributing *different* prime divisors of the single integer b_m . The collision bound $F = O(T)$ is therefore a genuinely horizontal, two-prime-divisor assertion about b_m .

Remark 312 (Why classical sieves do not close the argument). The collision bound $F = O(T)$ has the form of a two-dimensional upper-bound sieve (Selberg, Rosser–Iwaniec, Brun, GPY), but no classical sieve theorem deduces it from the one-prime data alone. After weight expansion, every such sieve introduces the counts $A_{pq}(N) = \#\{m : pq \mid b_m\}$ for products $pq \asymp N^2$. Since the observation interval has length only N , each individual expected count is $O(1/N)$ and only an *averaged* large-modulus discrepancy theorem can hold. The difficulty is intrinsic: even the strongest conceivable one-prime bound $A_p(N) \leq 1$ for every p does not control cross-prime alignment. An adversarial configuration with all zero sets concentrating at a single index m_0 satisfies $A_p = 1$ while $F \asymp (N/\log N)^2$. Thus the sieve machinery can *package* a two-prime distribution theorem as a collision bound, but it cannot *manufacture* that theorem from one-prime information. The minimum new input is the bilinear estimate

$$\sum_{\substack{p \neq q \\ p, q \in (N, 2N]}} A_{pq}(N) \ll T + \frac{T^2}{N}, \quad (337)$$

which is precisely the block second-factorial-moment conjecture (diagonal AMTD, Hypothesis 152).

Proposition 313 (Conditional collision bound). *Assume the following Frobenius Pair Independence hypothesis (FPI): for distinct doublet primes $p, q \in (N, 2N]$, the normalized first zeros r_p/p and r_q/q are asymptotically independent and each approximately uniform on $[0, \frac{1}{2}]$ in the sense that, for any integers $a \in [1, (p-1)/2]$ and $b \in [1, (q-1)/2]$,*

$$\frac{1}{K(K-1)} \#\{(p, q) \text{ distinct doublet primes in } (N, 2N] : r_p = a, r_q = b\} = (1 + o(1)) \frac{4}{pq}, \quad (338)$$

where K is the number of doublet primes. Then $F = o(T)$ and the all-index GCD theorem $G_n = e^{o(n)}$ holds for all n .

Proof. We bound each piece of the decomposition (336) separately.

Type A. The type-A collision count is

$$F_A = \#\{(p, q) \text{ distinct doublet primes} : r_q = r_p + (p - q)\}.$$

For a fixed ordered pair (p, q) , the collision requires $r_q = r_p + (p - q)$. Summing the FPI probability over all valid values of r_p , the conditional probability is

$$\Pr[\text{type A} \mid p, q] = \sum_{r=1}^{(p-1)/2} \frac{2}{p} \cdot \frac{2}{q} \cdot \mathbf{1}[r + (p-q) \in [1, (q-1)/2]] \leq \frac{4 \min((p-1)/2, (3q-1)/2 - p)}{pq}.$$

For primes $p, q \in (N, 2N]$, this is at most $4/N$ when $|p - q| \leq N/2$, and vanishes when $|p - q| > N$. Summing over all $K(K-1)$ ordered pairs:

$$F_A \leq K(K-1) \cdot \frac{4}{N} \cdot (1 + o(1)) \leq \frac{4K^2}{N} \cdot (1 + o(1)).$$

Since $K \asymp N/\log N$ and $T = 2K + O(K^{1/2})$, we obtain $F_A \leq 4T^2/(4N) \cdot (1 + o(1)) = T^2/N \cdot (1 + o(1)) = o(T)$.

Type C. The collision condition $r_p + r_q = 2p - 1 - q$ constrains a sum of independent uniforms. Under FPI, $r_p + r_q$ has a trapezoidal distribution on $[2, (p + q - 2)/2]$ with maximum density $2/\min(p, q) \leq 2/N$. The target $2p - 1 - q$ lies in the support when $q < 2p - 1$, which holds for $p > (q + 1)/2$. Thus $\Pr[\text{type C} \mid p, q] \leq 2/N$, and $F_C \leq 2K^2/N \cdot (1 + o(1)) = o(T)$.

Type D. The condition $r_q = r_p + 2(q - p)$ has the same structure as type A with gap $2(q - p)$ in place of $p - q$. By the same argument, $F_D \leq 4K^2/N \cdot (1 + o(1)) = o(T)$.

Combining all three, $F = F_A + F_C + F_D = o(T)$. The diagonal AMTD $\|F_i\|_2^2 \leq C s_i^2$ follows with $C = 1 + o(1)$ at each dyadic block, giving $V^\circ \leq (1 + o(1)) L S = O(\log N) S = N^{o(1)} S$. $\square$

Remark 314 (Status of the FPI hypothesis). The hypothesis (338) is a pair decorrelation statement for the zero positions across different primes. The Kummer structure of the reduced generating series is known exactly (Caruso–Fürnsinn–Vargas–Montoya–Zudilin [10]). Its Galois group is the square subgroup of $\mathbb{F}_p^\times$, of order $(p-1)/2$, when $(\frac{-6}{p}) = 1$, and is all of $\mathbb{F}_p^\times$, of order $p-1$, when $(\frac{-6}{p}) = -1$. The rank-three Apéry Picard–Fuchs connection is the symmetric square of a rank-two connection, so its connected geometric monodromy is of $\text{Sym}^2 \text{SL}_2 \simeq \text{SO}_3$ type.

However, neither case of the cyclic Kummer group nor the SO_3 monodromy controls the zero positions r_p . To see why, observe that the zero-set $\mathcal{Z}_p = \{j < p : p \mid b_j\}$ is a *spectral* zero-set: writing $A_p(t) = \sum_{k=0}^{p-1} b_k t^k \in \mathbb{F}_p[t]$ and $f_p(u) = A_p(g^u)$ for a primitive root g , multiplicative orthogonality gives

$$\sum_{u=0}^{p-2} f_p(u) g^{-ju} = -b_j,$$

so $j \in \mathcal{Z}_p$ iff the j -th discrete Mellin coefficient of the Hasse-polynomial value vector vanishes. By contrast, the *roots* of $A_p(t)$ in $\mathbb{F}_p$ are evaluation zeros, controlled by the Kummer factorization $A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2$. The Mellin coefficient index j and the discrete logarithm $\text{ind}_g(x)$ of a root x live in dual spaces; no theorem identifies a coefficient-zero position with an evaluation-zero position.

The pair independence (338) is therefore a genuinely new hypothesis: it concerns the spectral zero process $\mathcal{A}_p(j) = \mathbf{1}_{\{b_j \equiv 0\}}$, not a Frobenius class function or a discrete logarithm. Neither Chebotarev, Sato–Tate, nor Katz equidistribution (which control traces and evaluation points) applies to this spectral-zero process.

A clean unconditional formulation replaces FPI by the *marked two-characteristic Hasse dispersion* conjecture:

$$\sum_{\substack{N < p, q \leq 2N \\ p \neq q}} \sum_{m \in I_N} X_p(m) X_q(m) \ll T_N + \frac{T_N^2}{N}, \quad (339)$$

where $X_p(m) = \mathbf{1}_{\{p \leq m, b_{m-p} \equiv 0 \pmod{p}\}}$. The left side equals the ordered collision count F_N^{ord} . Via the exponential-sum detection $\mathcal{A}_p(j) = p^{-1} \sum_{a \bmod p} e_p(a b_j)$, the conjecture (339) reduces to cancellation in the mixed-modulus sum

$$\sum_{p \neq q} \frac{1}{pq} \sum_{\substack{a \bmod p \\ c \bmod q}}' \sum_{m \in I_N} e_p(a b_{m-p}) e_q(c b_{m-q}), \quad (340)$$

where $\sum'$ omits the term $a = c = 0$ (which produces the main term). The inner sum is over a nonautonomous recurrence orbit: $m \mapsto (b_{m-p}, b_{m-q})$. Standard Weil bounds do not apply.

A proof of (339) would require (1) a uniform crystalline Mellin object whose first Hasse coordinate tracks $b_j \bmod p$; (2) nondegeneracy of this coordinate on the monodromy torsor; (3) product monodromy for distinct Kummer twists; (4) rare-event anti-concentration at $1/p$ scale; and (5) a *horizontal* large sieve across different characteristics—a five-stage program beyond current technology but strongly supported by computation ($F/T \leq 0.041$ at $N = 32,768$).

Proposition 315 (Profile decomposition). *For each prime $p \in (N, 2N]$, let $A_p = |\Omega_p \cap I_N|$ and $L_p = 2N - p$. Define the smooth tail profile*

$$\rho(m) = \sum_{\substack{N < p \leq 2N \\ p < m}} \frac{A_p}{L_p}, \quad m \in I_N.$$

Then:

- (i) $\sum_{m \in I_N} \rho(m) = T_N$;
- (ii) $\|\rho\|_2^2 := \sum_{m \in I_N} \rho(m)^2 \leq \rho_{\max} T_N$, where $\rho_{\max} := \max_{m \in I_N} \rho(m) \leq \sum_p A_p / L_p$; under the PNT profile approximation, $\|\rho\|_2^2 = (c_2 + o(1)) T_N^2 / N$ with c_2 as in (342) (note $c_2 > 1$: by Cauchy–Schwarz $\|\rho\|_2^2 \geq T_N^2 / N$ always);
- (iii) $F_N \leq 2\|\rho\|_2^2 + 2\|\lambda - \rho\|_2^2 - T_N$.

In particular, if $\|\rho\|_2^2 = O(T_N^2 / N)$ (as the profile approximation and the data indicate), the residual dispersion estimate $\|\lambda - \rho\|_2^2 \ll T_N$ would imply $F_N = O(T_N)$, since $T_N^2 / N = O(T_N)$.

Proof. Part (i): $\sum_m \rho(m) = \sum_p (A_p / L_p) \cdot L_p = \sum_p A_p = T_N$. Part (ii): $\|\rho\|_2^2 \leq \rho_{\max} \sum_m \rho(m) = \rho_{\max} T_N$, and $\rho(m)$ is a partial sum of the positive terms A_p / L_p . For the profile asymptotics: by the PNT approximation $\rho(m) \approx (2/\log N) \log(N/(2N-m))$ and the integral $\int_0^1 (\log 1/t)^2 dt = 2$, one obtains $\|\rho\|_2^2 \sim c_2 T_N^2 / N$. Numerically, $\|\rho\|_2^2 / (T_N^2 / N)$ stabilizes near 1.25 for $N = 512, \dots, 32,768$ (Table 6), matching $c_2 = 1.262 \dots$ from (342). Part (iii) follows from $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ applied to $\lambda = \rho + \eta$. $\square$

Remark 316 (Orthogonal decomposition). Setting $\eta = \lambda - \rho$, the identity $T + F = \|\lambda\|_2^2 = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + \|\eta\|_2^2$ gives

$$F = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + (\|\eta\|_2^2 - T). \quad (341)$$

Under the independent profile model (each $X_p(m)$ is an independent Bernoulli with mean $\pi_p(m) = A_p / L_p$ when $p < m$), the three terms in (341) have the following expected behavior:

- $\|\rho\|_2^2 = \Theta(T^2 / N)$: the deterministic profile energy, tending to 0 relative to T .
- $\|\eta\|_2^2 \approx T$: the shot-noise variance of T independent Bernoulli hits.

- $\langle \rho, \eta \rangle \approx 0$: the profile and residual are decorrelated.

Hence $F \approx \|\rho\|_2^2 \approx T^2/N = o(T)$. Computation (Table 6) confirms this decomposition: $\|\eta\|_2^2/T$ is stable near 1.0 (matching palindromic random surrogates), while $F \approx \|\rho\|_2^2$ to within 1% at $N = 16,384$ and within 8% at $N = 32,768$. The cross-term $\langle \rho, \eta \rangle$ is $O(1)$ at every tested scale (e.g. -0.03 at $N = 16,384$ and 0.40 at $N = 32,768$, versus $T = 664$ and 1182). Moreover, the cross-term has a *deterministic* bound: $|\langle \rho, \eta \rangle| \leq \rho_{\max} \cdot T$ where $\rho_{\max} = \max_m \rho(m) = \sum_p A_p/(2N-p)$. The data show that $\rho_{\max}/(T/N)$ stabilizes near 1.8 (Table 6); under the corresponding uniform profile bound $\rho_{\max} = O(T/N)$, this gives $|\langle \rho, \eta \rangle| = O(T^2/N) = o(T)$. Together with $\|\rho\|^2 = O(T^2/N)$, that auxiliary profile input makes the collision estimate $F = O(T)$ equivalent to the shot-noise estimate $\|\eta\|_2^2 = O(T)$. The numerical stabilization itself is evidence, not a proof of the uniform profile bound.

Table 6: Profile decomposition diagnostics for the full hit measure μ_N . All norms over $I_N = (N, 2N]$. “Surr” denotes the mean of $\|\eta\|_2^2/T$ over 10 palindromic random surrogates.

N	T	F	$\ \rho\ _2^2$	$\ \rho\ _2^2/(T^2/N)$	$\ \eta\ _2^2/T$	Surr
512	21	2	1.1	1.28	1.05	0.98
1024	50	2	3.0	1.22	0.98	0.99
2048	86	4	4.5	1.26	0.99	1.01
4096	172	2	9.4	1.31	0.96	1.00
8192	354	20	19.6	1.28	1.00	1.00
16384	664	34	33.7	1.25	1.00	1.01
32768	1182	58	53.8	1.26	1.00	—
65536	2237	94	92.9	1.22	1.00	—
131072	4317	174	178.5	1.26	1.00	—
262144	7940	312	301.7	1.25	1.00	—

Remark 317 (Exact profile constant). The column $\|\rho\|_2^2/(T^2/N)$ in Table 6 converges to a constant c_2 as $N \rightarrow \infty$, with exact closed form

$$c_2 = \frac{2(\ln 2)^2 - 4 \ln 2 + 2}{(2 \ln 2 - 1)^2} = 1.261982 \dots \quad (342)$$

The derivation is direct. The triangular profile $\rho(m) = \sum_{N < p \leq m} Z(p)/p$ for $m \in I_N$ has the continuous approximation $\rho(Nt) \approx (\ln t)/(\ln N)$ for $t \in [1, 2]$, by Mertens’ theorem. Then

$$T = \sum_m \rho(m) \approx \frac{N}{\ln N} \int_1^2 \ln t \, dt = \frac{N}{\ln N} (2 \ln 2 - 1),$$

$$\|\rho\|_2^2 = \sum_m \rho(m)^2 \approx \frac{N}{(\ln N)^2} \int_1^2 (\ln t)^2 \, dt = \frac{N}{(\ln N)^2} (2(\ln 2)^2 - 4 \ln 2 + 2).$$

The ratio gives (342). The exact palindromic null expectation confirms the match: at $N = 1024$ the finite- N ratio is 1.209, at $N = 4096$ it is 1.281, at $N = 16,384$ it is 1.254, oscillating around c_2 with $O(1/\ln N)$ corrections from the prime number theorem. At all tested sizes, $E_{\text{null}}[F]$ agrees with the observed F to within the expected stochastic fluctuation.

Remark 318 (Concentration of F). Under the palindromic null (zero counts $Z(p)$ fixed, positions uniform among palindromic configurations), F concentrates around its mean. The dependency graph of collision indicators is the line graph of the complete graph on the active primes: two edge indicators share a dependency exactly when the corresponding prime pairs share one prime. This sparse structure gives

$$\text{Var}(F_N) = (2 + O(1/\ln N)) E[F_N], \quad (343)$$

a rigorous Poisson-scale variance. *Proof sketch.* Write $F = 2U$ where $U = \sum_{p < q} I_{p,q}$ counts collision pairs. Edge indicators for disjoint prime pairs $\{p, q\} \cap \{r, s\} = \emptyset$ are independent. For a shared prime p , the conditional bound $\Pr(I_{p,q} = 1 \mid R_p = r) \leq 2/h_q$ gives $\sum_{e < f} |\text{Cov}(I_e, I_f)| \leq 2\Delta\Lambda$ where $\Delta = 2 \sum_p h_p^{-1} = O(1/\ln N)$ and $\Lambda = E[U]$. Combined with the Bernoulli variance $\sum \text{Var}(I_e) = \Lambda(1 - O(1/N))$, this yields (343).

Since $E[F] = c_2 T^2/N (1 + o(1)) \asymp N/(\ln N)^2 \rightarrow \infty$, Chebyshev gives $F/E[F] \rightarrow 1$ in L^2 and in probability, and $\Pr(F \geq CT) = O_C(\ln N/N)$ for every fixed $C > 0$.

The profile decomposition $F = \|\rho\|^2 + 2\langle \rho, \eta \rangle + \|\eta\|^2 - T$ separates deterministic and stochastic contributions. The deterministic cross-term satisfies $|\langle \rho, \eta \rangle| \leq \rho_{\max} T$, with

$$\frac{\rho_{\max}}{T/N} \longrightarrow \frac{\ln 2}{2 \ln 2 - 1} = 1.7943 \dots;$$

this bound is $O(T^2/N)$, the *same* order as $\|\rho\|^2$. The computationally observed $\langle \rho, \eta \rangle \approx 0$ is therefore genuine decorrelation evidence, not a consequence of the profile geometry.

If $\|\eta\|^2 = O(T)$, then all three terms are controlled: $\|\rho\|^2 = O(T^2/N)$, $|\langle \rho, \eta \rangle| = O(T^2/N)$, and $\|\eta\|^2 - T = O(T)$. Since $T^2/N = O(T)$ (because $T = o(N)$), this gives $F = O(T)$, exactly the BFM2 bound. The estimate $\|\eta\|^2 = O(T)$ —that is, shot-noise-scale fluctuations of the actual zero positions around the triangular profile—is the sole gap in this second-moment route.

Remark 319 (Prime-wise decomposition of $\|\eta\|^2$). For each prime $p \in (N, 2N]$, define the centred indicator

$$X_{m,p} = \mathbf{1}_{(m-p) \in Z_p} - \frac{Z(p)}{p}, \quad m \in [p+1, 2N].$$

Then $\eta(m) = \sum_{N < p \leq m} X_{m,p}$. Note that while distinct primes $p \neq q$ provide separate residue fields (and $b_j \bmod p$ is determined independently of $b_j \bmod q$ by CRT), this does not automatically make $X_{m,p}$ and $X_{m,q}$ independent for deterministic zero-sets—the zero positions within each Z_p are algebraically constrained. Under the independent fixed-cardinality residue model (Section 12.31), however, the columns are independent by construction.

Expanding the square:

$$\|\eta\|^2 = \underbrace{\sum_p \sum_m X_{m,p}^2}_D + \underbrace{\sum_{p \neq q} \sum_m X_{m,p} X_{m,q}}_O. \quad (344)$$

The diagonal $D = \sum_p D_p$ where $D_p = \sum_m X_{m,p}^2 = Z_p^{\text{act}}(1 - q_p)^2 + (L_p - Z_p^{\text{act}}) q_p^2$ with $q_p = Z(p)/p$, $Z_p^{\text{act}} = \#\{j \in [1, L_p] \cap Z_p\}$, $L_p = 2N - p$. Expanding:

$$D = T - 2 \sum_p Z_p^{\text{act}} q_p + \sum_p L_p q_p^2. \quad (345)$$

Both correction terms are $O(1/\ln N)$: since $Z_p^{\text{act}} \leq Z(p)$ and $q_p = Z(p)/p \leq Z(p)/N$, the second term is $\leq 2 \sum_p Z(p)^2/p \leq (2/N) \sum_p Z(p)^2$, and the third is $\leq N \sum_p Z(p)^2/p^2 \leq (1/N) \sum_p Z(p)^2$.

Under $\sum_p Z(p) = O(K)$ where $K \sim N/\ln N$, the Cauchy–Schwarz bound $\sum_p Z(p)^2 \leq (\max Z(p)) \cdot \sum_p Z(p)$ and $\max Z(p) = O(1)$ (empirically) give both corrections $O(K/N) = O(1/\ln N)$. Thus $D = T + O(1/\ln N)$ under the second-mass hypothesis $\sum_p Z(p)^2 = O(K)$. Under the null model, $E[O] = 0$.

Computation confirms this structure for the actual Apéry zero-sets: D matches T to $< 0.1\%$ for all tested N , and O/T fluctuates around zero at the level 10^{-3} – 10^{-2} (see Table 7). Under the null model, $\text{Var}(D) = O(T)$ (fluctuation of the truncated active mass Z_p^{act}) and $\text{Var}(O) = O(T^2/N)$ (CRT averaging: each of the $\binom{K}{2}$ pair terms has overlap $\sim N$ within period $pq \sim N^2$). Together,

$$\frac{\text{Var}(\|\eta\|^2)}{T^2} = O\left(\frac{1}{T} + \frac{1}{N}\right) \rightarrow 0,$$

proving $\|\eta\|^2/T \rightarrow 1$ in L^2 and in probability within the null model. For the actual Apéry zero-sets, the desired statement $\|\eta\|^2 = O(T)$ would follow from a suitable *two-prime dispersion* estimate for the zero positions of distinct primes in the short interval $[1, N]$.

Table 7: Prime-wise decomposition $\|\eta\|^2 = D + O$.

N	T	D	O	O/T
512	21	20.94	1.14	0.054
1024	50	49.86	−0.83	−0.017
2048	86	85.90	−0.43	−0.005
4096	172	171.90	−6.76	−0.039
8192	354	353.90	1.22	0.003
16384	664	663.90	0.46	0.001
32768	1182	1181.91	3.52	0.003
65536	2237	2236.92	−4.16	−0.002
131072	4317	4316.92	−1.76	−0.0004
262144	7940	7939.93	11.92	0.0015

Note the structural identity $O = F - \|\rho\|^2 - 2\langle\rho, \eta\rangle$. Under the profile and cross-term estimates stated above, $\|\rho\|^2 + 2\langle\rho, \eta\rangle = o(T)$, so $O = o(T)$ is then equivalent to $F = o(T)$. Without those auxiliary estimates, the identity alone gives no converse. In particular, negligibility of the cross-prime interaction is a sufficient strengthening of the BFM2 bound, not an unconditional equivalent formulation of it.

Remark 320 (Higher factorial moments). The third factorial moment $F_3 = \sum_m \lambda(m)(\lambda(m)-1)(\lambda(m)-2)$ equals 6 at both $N = 16,384$ and $N = 32,768$ (a single row with $\lambda = 3$) but vanishes at $N = 65,536$ and $N = 131,072$ ($\max_m \lambda(m) = 2$ with 47 and 87 collision positions respectively), then returns to $F_3 = 6$ at $N = 262,144$ (a single row with $\lambda = 3$ among 154 collision positions). The palindromic model predicts $E[F_3] \approx c_3 T^3/N^2$ where c_3 is a geometry-of-truncation constant. At $N = 262,144$ with $T = 7940$, $T^3/N^2 \approx 7.3$, so sporadic triple collisions at the single-digit level are consistent with the model. The fourth factorial moment $F_4 = 0$ at all tested sizes (no row has $\lambda \geq 4$).

12.30 Gap polynomial sparsity

For the Apéry recurrence, the gap polynomial N_h (whose roots in $\mathbb{F}_p$ are necessary for a simultaneous zero pair $b_x \equiv b_{x+h} \equiv 0$) has $r_p(h) = \#\{x \in \mathbb{F}_p : N_h(x) = 0\} \leq 3(h-1)$, giving $\sum_{h < H} r_p(h) \leq 3H^2/2$ and hence $Z(p) \leq Cp^{2/3}$. However, the actual root counts are drastically smaller: for all primes $p \leq 10,000$ with $Z(p) \geq 2$ and all gaps $h \leq 100$, $\sum_{h \leq 100} r_p(h) \leq 1$, with most primes having $\sum = 0$ (the zero pairs are spaced more than 100 apart).

Hypothesis 321 (Linear root sum). $\sum_{1 \leq h < H} r_p(h) \ll H$ for every prime p and every $H \geq 1$.

This would improve the zero-count to $Z(p) \ll p^{1/2}$, strengthening the exponent in Proposition 138 from $N^{5/6}$ to $N^{3/4}$. The heuristic justification is that N_h has degree $m_h = 3(h-1)/2$ in the primitive centred variable, and (conjecturally) $\text{Gal}(N_h^{\text{prim}}/\mathbb{Q}) = S_{m_h}$, so that by the Chebotarev density theorem $\mathbb{E}[r_p(h)] = 1$ for each h . For $h = 2, 3, 4, 5$, modular factorization certificates confirm $\text{Gal} = S_{m_h}$ with $(m_2, m_3, m_4, m_5) = (1, 3, 4, 6)$, and the full hyperoctahedral groups $C_2 \wr S_{m_h}$ are verified via the single-flip lemma (the signed-group kernel is detected by a single pair flip in Frobenius factorization).

12.31 The exact random model

The palindromic identity $b_{p-1-j} \equiv b_j \pmod{p}$ forces the zero-set to consist of reflected pairs. Let V_p^+ denote the space of inversion-invariant functions $f: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p$, i.e. $f(t) = f(t^{-1})$. The algebraic discrete Fourier transform $\hat{f}(j) = \sum_{t \in \mathbb{F}_p^\times} f(t) t^{-j}$ maps V_p^+ isomorphically onto the subspace of sequences satisfying $\hat{f}(j) = \hat{f}(-j)$.

Proposition 322 (Random inversion-invariant model). *Choose f uniformly from V_p^+ . Then the number K of noncentral reflected pairs $\{j, p-1-j\}$ with $\hat{f}(j) = 0$ satisfies*

$$K \sim \text{Binomial}\left(\frac{p-3}{2}, \frac{1}{p}\right) \xrightarrow{p \rightarrow \infty} \text{Poisson}(1/2).$$

Proof. The spectral coefficients of distinct reflected pairs are independent uniform elements of $\mathbb{F}_p$ (the DFT is invertible), so each vanishes with probability $1/p$. $\square$

The Apéry Hasse polynomial $A_p(t) = \sum b_k t^k$ is a specific element of V_p^+ with additional algebraic structure ($A_p = \Delta^\varepsilon B_p^2$). The conjecture that $Z(p) \sim 2 \text{Poisson}(1/2)$ is equivalent to asserting that A_p behaves, for the spectral-zero statistic, like a generic element of V_p^+ .

The square structure can be incorporated directly. Let $V_p^- = \{g: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p : g(t^{-1}) = -g(t)\}$ denote the anti-invariant subspace, with $\dim V_p^- = (p-3)/2$. For uniform $g \in V_p^-$, the function $f = g^2$ lies in V_p^+ and vanishes at the two fixed points $t = \pm 1$.

Proposition 323 (Random anti-invariant square model). *Choose g uniformly from V_p^- and put $f = g^2$. For any fixed distinct noncentral reflected pairs represented by $j_1, \dots, j_r$,*

$$\Pr(\hat{f}(j_1) = \dots = \hat{f}(j_r) = 0) = p^{-r} + O_r(p^{-c_r p})$$

for some $c_r > 0$. In particular, the pair-zero count $K^{\text{sq}} \Rightarrow \text{Poisson}(1/2)$.

Proof. Write $u_i = g(x_i)$ for orbit representatives. The spectral coefficient $\hat{f}(j) = \sum_i u_i^2 (x_i^j + x_i^{-j})$ is a diagonal quadratic form $Q_\lambda(u)$ in the independent uniform variables u_i . For $\lambda \neq 0$, the physical function $h_\lambda(t) = \sum_a \lambda_a (t^{j_a} + t^{-j_a})$ has Mellin support $\leq 2r$, so by the finite-field Mattson–Solomon uncertainty inequality, at least $(p-1)/(4r) - 1$ orbit values are nonzero, each contributing a quadratic Gauss sum of absolute value $p^{-1/2}$. The factorial-moment method then gives the result. $\square$

Multiplication by $\Delta(t)^\varepsilon$ affects only $O(1)$ orbits and does not change the limiting law. Thus the Sym^2 squareness of the Apéry Hasse polynomial is fully compatible with $\text{Poisson}(1/2)$ in any random model where the square root B_p behaves generically.

Proposition 324 (Quadratic rank and both halves of the spectrum). *Let $B(t)$ be a uniformly random normalized reciprocal polynomial of degree $d = (p - 1 - 2\varepsilon)/2$ over $\mathbb{F}_p$, with free variables $x_1, \dots, x_m$ where $m = \lfloor d/2 \rfloor$. Put $H = \Delta^\varepsilon B^2$ and split the noncentral reflected-pair representatives into a triangular range $\mathcal{J}_{\text{tri}}$ ($|\mathcal{J}_{\text{tri}}| = m$) and a nonlinear range $\mathcal{J}_{\text{nl}}$ ($|\mathcal{J}_{\text{nl}}| = p/4 + O(1)$).*

- (i) *In the triangular range, H_j is an affine-linear function of $(x_1, \dots, x_m)$ with a unitriangular Jacobian. Hence $K_{\text{tri}} = \#\{j \in \mathcal{J}_{\text{tri}} : H_j = 0\} \Rightarrow \text{Poisson}(1/4)$.*
- (ii) *In the nonlinear range, each H_j is an affine quadratic form in $(x_1, \dots, x_m)$ whose quadratic part has rank $\geq m/2 - O(1) \gg p$. By the Gauss-sum bound for affine quadratic forms of rank R : $\Pr(F(x) = 0) = 1/p + O(p^{-R/2})$. The pencil-rank estimate from Proposition 323 gives $K_{\text{nl}} \Rightarrow \text{Poisson}(1/4)$.*
- (iii) *Without requiring independence between the two halves,*

$$\mathbf{E} Z(p) = 2 \mathbf{E} K = 2(\mathbf{E} K_{\text{tri}} + \mathbf{E} K_{\text{nl}}) = 2\left(\frac{1}{4} + \frac{1}{4}\right) + o(1) = 1 + o(1).$$

If the mixed pencil-rank condition holds—i.e., every nonzero $\mathbb{F}_p$ -linear combination of $H_{j_1}, \dots, H_{j_s}$ with at least one $j_\nu \in \mathcal{J}_{\text{nl}}$ has quadratic rank $\gg p/s$ —then the full pair-zero count $K = K_{\text{tri}} + K_{\text{nl}} \Rightarrow \text{Poisson}(1/2)$.

Remark 325 (Separation of random model from deterministic problem). Proposition 324 applies to the random reciprocal-polynomial model, not to the actual Apéry sequence. The actual B_p supplies one deterministic coefficient vector for each prime; converting the random-model statement into a theorem about $\sum_{p \leq x} Z(p)$ requires an arithmetic anti-concentration theorem asserting that these specific vectors avoid the coefficient quadrics with approximately random frequency. No such theorem is currently known.

12.32 Shift-correlation computation

Define the shift-correlation exponential sum

$$C_p(a) = \sum_{r=0}^{p-2} e_p(a(b_{r+1} - b_r)), \quad a \neq 0.$$

A nontrivial bound $|C_p(a)| \ll p^{1-\eta}$ is the missing input for a Bombieri–Fouvry-type dispersion argument (cf. Section 12.5).

For all 666 primes $p \leq 5000$, we computed $\max_{a \neq 0} |C_p(a)|/\sqrt{p}$. The results:

- $\max_a |C_p(a)|/\sqrt{p}$ has mean 3.35 and maximum 5.07;
- $\max_a |C_p(a)|/\sqrt{p \log p}$ has mean 1.23 and maximum 1.80;
- the distribution of $\max/\sqrt{p}$ matches the Gumbel law expected for the maximum of p independent unit-variance Gaussians: $\max \approx \sqrt{2 \log p}$.

For shifts $s = 2, 3, 5$, the same behavior holds. The histogram of the differences $b_{r+1} - b_r \pmod{p}$ has std/mean ≈ 1 and the fraction of zero-bins is $\approx 1/e$, precisely the Poisson(1) model.

Conclusion: the Apéry sequence $b_r \pmod{p}$ is computationally indistinguishable from a random function $\{0, \dots, p-1\} \rightarrow \mathbb{F}_p$ in its additive exponential-sum statistics. This provides strong computational evidence for the shift-correlation bound required by the dispersion approach.

12.33 Value-collision energy

A second pseudorandomness diagnostic is the *value-collision energy*

$$E(p) = \sum_{a \in \mathbb{F}_p} N_p(a)^2, \quad N_p(a) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv a \pmod{p}\}.$$

By Cauchy–Schwarz, $Z(p)^2 = N_p(0)^2 \leq E(p)$, so $E(p) \ll p$ would imply $Z(p) \ll \sqrt{p}$.

For a uniformly random function $\{0, \dots, p-1\} \rightarrow \mathbb{F}_p$, each $N_p(a) \sim \text{Poisson}(1)$ and $E(p)/p \rightarrow 2$. The palindromic symmetry $b_{p-1-j} \equiv b_j$ forces the half-sequence values to be doubled, giving $E(p)/p \rightarrow 3$: each half-position contributes independently to $N_p(a)$, and the doubling multiplies the second moment by 4 while the mean remains 1, yielding $\mathbb{E}[N_p(a)^2] = 4(\frac{1}{2} + \frac{1}{4}) = 3$.

We computed $E(p)/p$ for all 9590 primes $p \leq 100,000$. The results confirm the palindromic prediction: $\text{mean}(E(p)/p) = 2.9995$, with 99% of primes in $[2.9, 3.1]$ and no dependence on $Z(p)$. The value-collision energy thus confirms that the Apéry sequence modulo p is anti-concentrated at the scale predicted by the palindromic random model.

Remark 326 (Second Fourier layer). By additive orthogonality,

$$E(p) = \frac{1}{p} \sum_{u \in \mathbb{F}_p} |S_p(u)|^2, \quad S_p(u) = \sum_{j=0}^{p-2} e_p(u b_j).$$

A square-root bound $|S_p(u)| \ll \sqrt{p}$ for $u \neq 0$ would give $E(p) \ll p$, hence $Z(p) \ll \sqrt{p}$. This is a *second Fourier layer*: the first (multiplicative Mellin) transform produces the coefficients b_j ; the second (additive) transform probes their residual distribution in $\mathbb{F}_p$. Ordinary monodromy of the K3 local system controls the first layer. It does not automatically construct a bounded-conductor object for the second layer. The modular form $f_4 = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$ controls only the central coefficient $b_{(p-1)/2} \equiv a_p(f_4) \pmod{p}$, through the quadratic character. A fixed finite quasi-unipotent local category cannot contain exact Kummer constituents of unbounded order. This local-type observation does not exclude a fixed ambient geometry with varying Kummer coefficient sheaves, varying pushforwards, or residual constructions, and none of the presently available modular identities controls all generic j . A residual anti-concentration theorem, proving the second-layer bound $E(p) \ll p$ or equivalently the additive exponential-sum bound above, would require genuinely new same-characteristic arithmetic geometry.

12.34 The high-moment criterion

The dispersion hypothesis (Hypothesis 137) is a second-moment statement localized in short windows. An alternative sufficient condition—global in the integer variable, but requiring a single fixed higher moment—is the following.

For a dyadic block of primes $(X, 2X]$ and an integer m , define

$$K_X(m) = \#\{p \in (X, 2X] : m \bmod p \in Z_p\}, \quad \lambda_X = \sum_{X < p \leq 2X} \frac{Z(p)}{p},$$

so that $\lambda_X \ll X^{2/3}/\log X$ by Proposition 14. Write $(K)_k = K(K-1) \cdots (K-k+1)$ for the falling factorial.

Hypothesis 327 (High-moment bound $(\text{HM})_k$). *For one fixed integer $k \geq 3$, uniformly over dyadic X ,*

$$\sum_{0 \leq m < X^2} (K_X(m))_k \ll X^{2+o(1)} \lambda_X^k.$$

Proposition 328 (The case $k = 2$ holds unconditionally). $\sum_{0 \leq m < X^2} (K_X(m))_2 \leq 5X^2\lambda_X^2$.

Proof. Expanding the falling factorial over ordered pairs of distinct primes,

$$\sum_{m < X^2} (K_X(m))_2 = \sum_{\substack{p, q \in (X, 2X] \\ p \neq q}} \sum_{\substack{r \in Z_p \\ s \in Z_q}} \#\{m < X^2 : m \equiv r \pmod{p}, m \equiv s \pmod{q}\}.$$

By the Chinese remainder theorem the inner count is at most $X^2/(pq) + 1$, and since $pq \leq 4X^2$ the additive 1 is at most $4X^2/(pq)$, so the count is at most $5X^2/(pq)$. Summing,

$$\sum_{m < X^2} (K_X(m))_2 \leq 5X^2 \sum_{\substack{p, q \in (X, 2X] \\ p \neq q}} \frac{Z(p)Z(q)}{pq} \leq 5X^2\lambda_X^2. \quad \square$$

Remark 329 (Where the frontier begins). For $k \geq 3$, distinct primes have $p_1 \cdots p_k > X^k > X^2$, so each residue tuple $(r_1, \dots, r_k) \in Z_{p_1} \times \cdots \times Z_{p_k}$ has *at most one* CRT representative below X^2 , and the trivial bound on the k -th moment is the total tuple count $\asymp (X\lambda_X)^k$, which exceeds the conjectured $X^2\lambda_X^k$ by the factor X^{k-2} . Hypothesis $(\text{HM})_k$ thus asserts that only the expected proportion $\asymp X^2/(p_1 \cdots p_k)$ of the actual Apéry zero tuples admit a representative in $[0, X^2)$ — a genuine cross-prime equidistribution statement, strictly beyond Proposition 328, yet far weaker than the Poisson conjecture (Hypothesis 75).

Remark 330 (Computational evidence). Using the actual Apéry zero sets Z_p for all primes $p \in (X, 2X]$, the moment ratios $R_k = \sum_{m < X^2} (K_X(m))_k / X^2\lambda_X^k$ are bounded and of order one at every tested scale:

X	$\#p$	λ_X	$\max K$	R_2	R_3	R_4	R_5	R_6	R_7	R_8
128	23	0.158	3	0.91	0.93	—	—	—	—	—
256	43	0.126	3	0.89	0.46	—	—	—	—	—
512	75	0.093	3	0.97	0.72	—	—	—	—	—
1024	137	0.082	4	0.97	0.83	0.50	—	—	—	—
2048	255	0.078	4	0.98	0.93	1.27	—	—	—	—
4096	464	0.077	4	0.99	0.95	0.57	—	—	—	—
8192	872	0.077	5	1.00	0.99	1.01	0.67	—	—	—
16384	1612	0.072	5	1.00	0.99	1.03	0.67	—	—	—

A dash indicates $F_k = 0$ (fewer than k primes hit any single m); these satisfy $(\text{HM})_k$ trivially. The $k = 2$ and $k = 3$ ratios approach 1 from below as X grows: at $X = 16384$ one has $R_2 = 0.998$ and $R_3 = 0.993$. The R_4 ratio fluctuates around 1 (the expected count $X^2\lambda_X^4$ is single-digit at small X , so R_4 is dominated by sampling noise). For $k \geq 6$, we have $F_k = 0$ at all tested scales because $\max K \leq 5$; these ratios R_k are thus zero, satisfying $(\text{HM})_k$ vacuously. Under the random singleton model $\text{Var}(F_3) \approx 6X^2\lambda_X^3$; the observed deviations $(F_3 - X^2\lambda_X^3)/\sqrt{6X^2\lambda_X^3}$ are z -scores $\{-0.2, -2.6, -1.7, -1.6, -1.2, -1.8, -0.4, -0.9\}$ at $X = 128, \dots, 16384$, all $O(1)$. Thus the Apéry zero sets produce factorial moments indistinguishable from random residues at the standard-deviation level. The observed $\max K$ grows as $\{3, 3, 3, 4, 4, 4, 5, 5\}$ at $X = \{128, 256, 512, 1024, 2048, 4096, 8192, 16384\}$ matching the Poisson extreme-value prediction $\max K \sim 2 \log X / \log \log X$ (predicted $\{5, 5, 5, 6, 6, 6, 7, 7\}$; the observed values are slightly below, consistent with (AT'')). Note $\lambda_X \approx 1/\log X$ here (the empirical Poisson mean), far below the worst-case input $X^{2/3}/\log X$ used in Theorem 331.

Theorem 331 ((HM)_k implies a pointwise power saving). *Assume Hypothesis 327 for one fixed $k \geq 3$. Then for every n ,*

$$\log G_n \ll n^{2/3+2/k+o(1)}.$$

In particular $k = 8$ gives $\log G_n \ll n^{11/12+o(1)}$, and any fixed $k > 6$ proves the full conjecture $G_n = e^{o(n)}$ for all n .

Proof. Let $M_X = \max_{0 \leq m < X^2} K_X(m)$. If $M_X \geq 2k$, then $(M_X)_k \geq (M_X/2)^k$; taking the maximizing m inside the hypothesis,

$$(M_X/2)^k \leq \sum_{m < X^2} (K_X(m))_k \ll X^{2+o(1)} \lambda_X^k.$$

In either case ($M_X < 2k$ or not),

$$M_X \ll X^{2/k+o(1)} \lambda_X + 1 \ll \frac{X^{2/3+2/k+o(1)}}{\log X}.$$

Now fix n . The primes in $(\sqrt{n}, 2\sqrt{n}]$ contribute at most $\pi(2\sqrt{n}) \log(2\sqrt{n}) = O(\sqrt{n})$ trivially. The remaining range $(2\sqrt{n}, n]$ is covered by dyadic blocks $(X, 2X]$ with $X = 2^j \leq n$; every block meeting $(2\sqrt{n}, n]$ has $2X > 2\sqrt{n}$, hence $X > \sqrt{n}$ and therefore $n < X^2$. Hence $m = n$ is admissible in the maximum, and the block $(X, 2X]$ contains at most M_X lower-digit bad primes, each of weight $\log p \leq \log 2X$. Therefore

$$\sum_{\substack{\sqrt{n} < p \leq n \\ p|b_n \bmod p}} \log p \ll \sqrt{n} + \sum_{\substack{X=2^j \\ \sqrt{n} < X \leq n}} X^{2/3+2/k+o(1)} \ll n^{2/3+2/k+o(1)},$$

the geometric sum being dominated by its top block. Adding the small-prime bound (Proposition 3), the companion-block bound (Proposition 103), and reducing to the lower-digit channel by Lemma 87(iii), the claim follows. For $k > 6$ the exponent satisfies $2/3 + 2/k < 1$, so $\log G_n = o(n)$. $\square$

Remark 332. Among the cross-prime sufficient conditions collected in this paper (Poisson, AP-BDH, AMTD, and the higher-moment hypotheses), Hypothesis (HM)₇ is the smallest integer-order member of the higher-moment family that closes the full conjecture: it requires no pointwise information about any single prime, no short-window localization, and no variance identity—only one fixed factorial moment at the independence scale, compatible with the worst proved cardinality bound $Z(p) \ll p^{2/3}$.

Lemma 333 (CRT phase identity). *For distinct primes p, q, r and residues $a \in \mathbb{F}_p, b \in \mathbb{F}_q, c \in \mathbb{F}_r$, define $c_p = (qr)^{-1} \bmod p$, $c_q = (pr)^{-1} \bmod q$, $c_r = (pq)^{-1} \bmod r$. Then*

$$\frac{a \cdot c_p}{p} + \frac{b \cdot c_q}{q} + \frac{c \cdot c_r}{r} \equiv \frac{\text{CRT}(a, b, c)}{pqr} \pmod{1},$$

where $\text{CRT}(a, b, c) \in [0, pqr)$ is the unique integer satisfying $m \equiv a \pmod{p}$, $m \equiv b \pmod{q}$, $m \equiv c \pmod{r}$.

Proof. Let $N = a \cdot c_p \cdot qr + b \cdot c_q \cdot pr + c \cdot c_r \cdot pq$. Since $c_p \cdot qr \equiv 1 \pmod{p}$ we have $N \equiv a \pmod{p}$; similarly $N \equiv b \pmod{q}$ and $N \equiv c \pmod{r}$. Hence $N \equiv \text{CRT}(a, b, c) \pmod{pqr}$ and $N/(pqr) = \text{CRT}(a, b, c)/(pqr) + \text{integer}$. $\square$

Remark 334 (CRT lift equidistribution). By Lemma 333, the third factorial moment rewrites as

$$F_3(X) = 6 \sum_{\substack{p < q < r \\ p, q, r \in \mathcal{P}_X}} \sum_{\substack{a \in \mathcal{Z}_p \\ b \in \mathcal{Z}_q \\ c \in \mathcal{Z}_r}} \mathbf{1}_{\text{CRT}(a, b, c) < X^2},$$

and the Weyl sum $W(h)$ is exactly

$$W(h) = \sum_{p < q < r} \sum_{\substack{a \in \mathcal{Z}_p \\ b \in \mathcal{Z}_q \\ c \in \mathcal{Z}_r}} e\left(\frac{h \cdot \text{CRT}(a, b, c)}{pqr}\right).$$

Since $pqr > X^3 > X^2$, each CRT lift lands in $[0, X^2)$ with probability $\approx X^2/(pqr)$; the main term is $X^2 \lambda_X^3$. The Kolmogorov–Smirnov test of the normalized lifts $\text{CRT}(a, b, c)/(pqr) \in [0, 1)$ against the uniform distribution passes at all tested scales (KS = 0.014 at $X = 128$ through KS = 0.00034 at $X = 2048$, all below their 5% thresholds). The KS statistic decays as $\sim 1/\sqrt{N_{\text{trip}}}$, matching the rate for truly random points.

No disjoint-triple resonance. For two triples (p, q, r) and (p', q', r') with $\{p, q, r\} \cap \{p', q', r'\} = \emptyset$, the resonance condition $\text{CRT}(a, b, c)/(pqr) \equiv \text{CRT}(a', b', c')/(p'q'r') \pmod{1}$ forces $\text{CRT}(a, b, c) = 0$, hence $a = b = c = 0$. Since $b_0 = 1 \neq 0$ in every characteristic $p > 1$, $0 \notin \mathcal{Z}_p$ for all primes, so *no resonances occur between disjoint triples*. Computation confirms: the disjoint-triple off-diagonal in $\sum_h |W(h)|^2$ is $< 2\%$ of the total.

Computation through $X = 512$ confirms $|W(h)| = O(\sqrt{N_{\text{trip}}})$ for $h = 1, \dots, 20$: the triple Fourier products exhibit square-root cancellation, the same rate as independent uniformly random residues. Fixed-frequency cancellation alone does not establish $(\text{HM})_3$, whose shrinking interval has length $\asymp X^{-1}$. It does, however, suggest the much weaker low-frequency nonconcentration criterion below.

Proposition 335 (Weak CRT–Weyl criterion for the all-index theorem). *Let $P = |\mathcal{P}_X|$, let*

$$N_{\text{trip}} = \sum_{p < q < r} |\mathcal{Z}_p| |\mathcal{Z}_q| |\mathcal{Z}_r| = W(0),$$

and assume M_1 , so that $N_{\text{trip}} \ll P^3$. Suppose there is a function $H(X) \rightarrow \infty$ for which

$$\frac{N_{\text{trip}}}{H(X)} + \sum_{1 \leq h \leq H(X)} \frac{|W(h)|}{h} = o(P^3). \quad (346)$$

Then

$$\sum_{m < X^2} (K_X(m))_3 = o(P^3), \quad \max_{m < X^2} K_X(m) = o(P),$$

and consequently $G_n = e^{o(n)}$ for every positive integer n . In particular, (346) follows if, for some fixed $\delta > 0$, one has

$$H(X) = \log \log X, \quad \max_{1 \leq h \leq H(X)} |W(h)| \ll P^{3-\delta}.$$

Proof. For every residue triple let $y \in [0, pqr)$ be its least CRT lift and put $x = y/(pqr)$. Since $pqr > X^3$, the implication

$$y < X^2 \implies 0 \leq x < X^{-1}$$

holds uniformly despite the varying moduli. The Erdős–Turán inequality applied to the common interval $[0, X^{-1})$ therefore gives, for the number A_X of residue triples with $y < X^2$,

$$A_X \ll \frac{N_{\text{trip}}}{X} + \frac{N_{\text{trip}}}{H(X)} + \sum_{1 \leq h \leq H(X)} \frac{|W(h)|}{h} = o(P^3).$$

Here M_1 gives $N_{\text{trip}} \leq (\sum_p |\mathcal{Z}_p|)^3/6 \ll P^3$.

Because $pqr > X^3 > X^2$, every residue triple has at most one lift below X^2 . Conversely, a row hit by k primes contributes exactly $\binom{k}{3}$ unordered triples. Hence

$$6A_X = \sum_{m < X^2} (K_X(m))_3.$$

At a maximizing row, $\binom{K_X(m)}{3} \leq A_X = o(P^3)$, which gives $\max_m K_X(m) = o(P)$.

Put

$$\eta(X) = \frac{1}{P} \max_{m < X^2} K_X(m), \quad E(T) = \sup_{X \geq T} \eta(X).$$

The preceding asymptotic says $\eta(X) \rightarrow 0$, hence also $E(T) \rightarrow 0$. For a fixed n , dispose of the first block above $\sqrt{n}$ trivially and partition $(2\sqrt{n}, n]$ into dyadic prime blocks. On every remaining block $n < X^2$, so its logarithmically weighted contribution is

$$K_X(n) \log(2X) \ll \eta(X) P \log X \ll E(2\sqrt{n}) X.$$

The sum of the dyadic scales is $O(n)$, so all these blocks contribute $O(E(2\sqrt{n})n) = o(n)$. Thus no quantitative rate in the little- o estimate is required. The already proved small-prime and companion-block estimates complete the all-index reduction.

For the final assertion, $P^{3-\delta} \sum_{h \leq H} h^{-1} \log H = o(P^3)$, while $N_{\text{trip}}/H = o(P^3)$. $\square$

Proposition 336 (Complete orthogonality for disjoint triples). *Let $T = (p, q, r)$ and $T' = (p', q', r')$ be ordered triples of distinct primes with $\{p, q, r\} \cap \{p', q', r'\} = \emptyset$. Define $S_T(h) = \sum_{(a,b,c) \in \mathcal{Z}_T} e(h \cdot \text{CRT}(a, b, c)/(pqr))$ where $\mathcal{Z}_T = \mathcal{Z}_p \times \mathcal{Z}_q \times \mathcal{Z}_r$. Set $L = pqr \cdot p'q'r'$. Then*

$$\sum_{h=0}^{L-1} S_T(h) \overline{S_{T'}(h)} = 0.$$

Proof. Expanding the left side and exchanging summation:

$$\sum_{\substack{(a,b,c) \in \mathcal{Z}_T \\ (a',b',c') \in \mathcal{Z}_{T'}}} \sum_{h=0}^{L-1} e\left(h \cdot \left[\frac{\text{CRT}(a, b, c)}{pqr} - \frac{\text{CRT}(a', b', c')}{p'q'r'} \right]\right).$$

The inner sum is L if the bracket vanishes modulo 1, and 0 otherwise. By Lemma 333, the bracket equals $(\text{CRT}(a, b, c) \cdot p'q'r' - \text{CRT}(a', b', c') \cdot pqr)/L$. Since $\gcd(pqr, p'q'r') = 1$, vanishing modulo L forces $pqr \mid \text{CRT}(a, b, c)$. As $0 \leq \text{CRT}(a, b, c) < pqr$, this forces $\text{CRT}(a, b, c) = 0$, hence $a = b = c = 0$. But $b_0 = 1 \neq 0$ in every characteristic $p > 1$, so $0 \notin \mathcal{Z}_s$ for any prime s , and no nonzero summand survives. $\square$

Remark 337 (Dispersion bound for $W(h)$). The second moment $\sum_{h=1}^H |W(h)|^2$ admits a diagonal/off-diagonal decomposition. The diagonal (same ordered triple $(p, q, r) = (p', q', r')$) contributes at most $H \cdot \sum_{p < q < r} |\mathcal{Z}_p|^2 |\mathcal{Z}_q|^2 |\mathcal{Z}_r|^2 = O(H \cdot N_{\text{trip}})$, since each $|\mathcal{Z}_p| = O(1)$ forces $|\mathcal{Z}_p|^2 \leq C |\mathcal{Z}_p|$. Off-diagonal terms require cancellation in the h -sum, which detects a resonance condition among the CRT phases of distinct triples.

Computation. For $H = 50$ and $X \in \{128, 256\}$:

$$\frac{1}{H \cdot N_{\text{trip}}} \sum_{h=1}^H |W(h)|^2 = 0.77, 0.73,$$

and for $H = 20$, $X = 512$: ratio = 1.50. Thus $\sum |W(h)|^2 \leq C \cdot H \cdot N_{\text{trip}}$ with $C \approx 1$, confirming that the off-diagonal is negligible relative to the diagonal at these scales. By Markov's inequality, for most h : $|W(h)| = O(\sqrt{N_{\text{trip}}})$.

A proof of the dispersion bound $\sum_{h=1}^H |W(h)|^2 \ll H \cdot N_{\text{trip}} \cdot X^{o(1)}$ would establish $|W(h)| = o(N_{\text{trip}})$ for all but $o(1)$ fraction of frequencies; upgrading to *every* fixed h requires a separate low-frequency elimination, but the dispersion bound is the main engine.

Proposition 338 (What $M_1 + M_2 + \text{AKD}_3$ actually proves). *Assume*

M_1 : $\sum_{p \in \mathcal{P}_X} |\mathcal{Z}_p| \leq C_1 P$ (first Cesàro zero-mass);

M_2 : $\sum_{p \in \mathcal{P}_X} |\mathcal{Z}_p|^2 \leq C_2 P$ (second zero-mass; implied by the uniform bound $\sup_p |\mathcal{Z}_p| \leq C_0$);

AKD_3 : (Apéry Kloosterman Dispersion) for distinct $p, p' \in \mathcal{P}_X$ and $h \neq 0$,

$$\sum_{\substack{q < r \\ q, r \in \mathcal{P}_X \setminus \{p, p'\}}} \widehat{\mathcal{Z}}_p(h(qr)^{-1}) \overline{\widehat{\mathcal{Z}}_{p'}(h(qr)^{-1})} = O(|\mathcal{Z}_p| |\mathcal{Z}_{p'}| P).$$

Then the auxiliary semiprime bound

$$\sum_{\substack{q < r \\ q, r \in \mathcal{P}_X}} |S_{q,r}(h)|^2 \ll P^3, \quad S_{q,r}(h) = \sum_{p \notin \{q, r\}} \widehat{\mathcal{Z}}_p(h(qr)^{-1} \bmod p), \quad (347)$$

holds for every fixed $h \neq 0$. This auxiliary estimate does not imply $(\text{HM})_3$: it omits two frequency factors in the actual trilinear Weyl sum, and its scale does not resolve the interval of normalized length $\asymp X^{-1}$.

Proof. Expand $|S_{q,r}|^2$ as $\sum_{p, p'} \widehat{\mathcal{Z}}_p(\alpha_{q,r,p}) \overline{\widehat{\mathcal{Z}}_{p'}(\alpha_{q,r,p'})}$, where $\alpha_{q,r,p} = h(qr)^{-1} \bmod p$.

Diagonal ($p = p'$). No Kloosterman estimate is needed. The pointwise bound $|\widehat{\mathcal{Z}}_p(\alpha)| \leq |\mathcal{Z}_p|$ gives

$$\sum_p \sum_{\substack{q < r \\ q, r \neq p}} |\widehat{\mathcal{Z}}_p(h(qr)^{-1})|^2 \leq \binom{P}{2} \sum_p |\mathcal{Z}_p|^2 \ll P^3$$

by M_2 . In particular, the formerly asserted identity

$$\sum_{q < r} e_p(c(qr)^{-1}) = \frac{1}{2} \left| \sum_q e_p(cq^{-1}) \right|^2 + O(P)$$

is false: the absolute square produces the difference $q^{-1} - r^{-1}$, not the product $q^{-1}r^{-1}$.

Off-diagonal ($p \neq p'$). By hypothesis AKD₃, for each pair of distinct primes p, p' , the cross-correlation over (q, r) pairs is $O(|\mathcal{Z}_p| |\mathcal{Z}_{p'}| P)$. Summing over $O(P^2)$ choices of (p, p') :

$$\sum_{p \neq p'} O(|\mathcal{Z}_p| |\mathcal{Z}_{p'}| P) = O\left(P \left(\sum_p |\mathcal{Z}_p|\right)^2\right) = O(C_1^2 P^3).$$

Combining the diagonal and off-diagonal estimates proves (347). $\square$

Remark 339 (Why the auxiliary estimate is not a trilinear decoupling). For ordered distinct primes the actual Weyl sum is

$$6W(h) = \sum_{q \neq r} \sum_{p \notin \{q, r\}} \widehat{\mathcal{Z}}_p(h(qr)^{-1}) \widehat{\mathcal{Z}}_q(h(pr)^{-1}) \widehat{\mathcal{Z}}_r(h(pq)^{-1}).$$

After q, r are fixed, both the q - and r -factors still depend on p . They therefore cannot be pulled outside the inner sum and replaced by $S_{q,r}(h)$. Even if that invalid decoupling were granted, the resulting bound $|W(h)| \ll P^{5/2}$ would exceed the non-sparse M_1 -scale target (where $\lambda_X \asymp 1/\log X$)

$$X^2 \lambda_X^3 \asymp \frac{P^3}{X}$$

by a factor of order $X/\sqrt{P} \asymp \sqrt{X \log X}$. Furthermore an interval of normalized length $\asymp X^{-1}$ requires a Fourier truncation reaching $|h| \asymp X$ if one seeks the (HM)₃ main scale; control of one fixed frequency is not enough. Finally, M_1 and M_2 are upper bounds and supply no lower bound $N_{\text{trip}} \asymp P^3$ in sparse blocks. Thus $o(N_{\text{trip}})$ cannot be inferred from a bound expressed only in P .

Remark 340 (Anchored-star obstruction to marginal hypotheses). For *arbitrary* moving target families, M_1 alone does *not* imply (HM)₃. Consider the singleton anchor $\mathcal{Z}_p = \{m_X \bmod p\}$ for a fixed $m_X < X^2$. Then $|\mathcal{Z}_p| = 1$ for every p (best possible M_1), yet $K_X(m_X) = P$, giving $F_3(X) \geq P(P-1)(P-2) \asymp X^3/(\log X)^3$, which exceeds the (HM)₃ target $X^2 \lambda_X^3 \asymp X^2/(\log X)^3$ by a factor of X . Thus any sufficient trilinear hypothesis must detect coherent moving anchors. The example does not promote the auxiliary AKD₃ estimate above to such a hypothesis: the missing step is the invalid decoupling identified after Proposition 338. What fails intrinsically is the distribution of distinct-prime CRT lifts, not a repeated-prime diagonal.

The Apéry family is expected to satisfy AKD₃, but the naïve sheaf-theoretic route fails: zero detection is a nonlinear operation on trace functions ($t \mapsto \mathbf{1}_{t=0}$), not a fixed tensor construction, so the zero indicator has no bounded-conductor sheaf realisation from the rank-4 ℓ -adic sheaf alone. Moreover, for $p \neq p'$, sheaves $\mathcal{F}_p$ and $\mathcal{F}_{p'}$ live on schemes in different characteristics, so $\mathcal{F}_p \otimes \mathcal{F}_{p'}^\vee$ is not defined in the étale category. The correct formulation requires either (i) a bounded-complexity crystalline model for the zero indicator, (ii) a direct bilinear large-sieve argument exploiting the Kloosterman structure of $(qr)^{-1}$, or (iii) a horizontal cross-prime dispersion theorem. Computationally, AKD₃ holds with $\max_{(p,p')} |\text{cross}|/(|\mathcal{Z}_p| |\mathcal{Z}_{p'}| P) \leq 0.51$ for all prime pairs with $X \leq 150$ (see Remark 341).

Remark 341 (Computational verification of AKD₃). For each $X \in \{30, 50, 80, 100, 150\}$, let $\mathcal{P}_X$ be the primes in $(X, 2X]$. For every pair of distinct primes $p, p' \in \mathcal{P}_X$ with $|\mathcal{Z}_p|, |\mathcal{Z}_{p'}| \geq 2$ and $h = 1$, we compute the cross-correlation ratio

$$R(p, p') = \frac{|\sum_{q < r \in \mathcal{P}_X \setminus \{p, p'\}} \widehat{\mathcal{Z}}_p((qr)^{-1}) \overline{\widehat{\mathcal{Z}}_{p'}((qr)^{-1})}|}{|\mathcal{Z}_p| |\mathcal{Z}_{p'}| P}.$$

X	P	pairs	max R	avg R
30	7	6	0.19	0.13
50	10	6	0.47	0.19
80	15	3	0.16	0.08
100	21	28	0.49	0.15
150	27	120	0.51	0.13

The ratio is bounded and stable across all tested ranges, consistent with AKD₃ holding with constant $C_3 \leq 1$. The $O(P)$ scale has a structural explanation: the inverse-semiprime product map $\{qr : q, r \text{ prime}\} \rightarrow (\mathbf{Z}/pp'\mathbf{Z})^\times$ has multiplicity at most 3 per residue class (since four semiprimes in $(X^2, 4X^2]$ spanning at least $3pp' > 3X^2$ are impossible), giving the Parseval/RMS bound $\sum_h |B_{pp'}(h)|^2 \leq 3L^2$ and mean-square coefficient $O(L)$. Thus the $O(P)$ scale is deterministic; the remaining gap for a pointwise $O(|Z_p| \cdot |Z_{p'}| \cdot P)$ bound is spectral avoidance: the four Apéry zero-set frequencies $\gamma_{a,b} = a/p + b/p'$ must avoid the exceptional Fourier spectrum of the semiprime product set.

Remark 342 (Computational verification of the bilinear bound). The bilinear sum $\sum_{q < r} |S_{q,r}(h)|^2$ was computed directly for $X \in \{50, 75, 100, 128\}$ with $h = 1$. The ratio $\sum |S|^2 / P^3$ takes values 0.10, 0.13, 0.43, 0.56, confirming the $O(P^3)$ bound. The diagonal ($p = p'$) dominates: $\text{diag}/P^3 \in [0.10, 0.55]$, while the off-diagonal contribution $\text{offdiag}/P^3 \in [-0.002, 0.017]$ tends to zero. The bilinear Kloosterman sum $|\sum_{q,r \text{ prime}} e_p(c(qr)^{-1})|/P^2$ decays from 0.12 at $X = 30$ to 0.01–0.04 at $X = 128$, consistent with the $o(P^2)$ bound.

Remark 343 (Computational evidence for M₁). Define $\bar{Z}(N) = (\sum_{3 \leq p \leq N} |Z_p|)/\pi(N)$ as the average zero-set size. Direct computation using the Apéry recurrence $(n+1)^3 b_{n+1} = (2n+1)(17n^2 + 17n + 5)b_n - n^3 b_{n-1}$ modulo p gives:

N	$\pi(N)$	$\sum Z_p $	$\bar{Z}(N)$	$\max Z_p $
10^3	168	167	0.994	6
5×10^3	669	608	0.909	8
10^4	1 229	1 176	0.957	8
2×10^4	2 262	2 290	1.012	8
5×10^4	5 133	5 172	1.008	8
10^5	9 592	9 497	0.990	10

The average $\bar{Z}(N)$ is remarkably stable near 1.0 through all tested ranges. The palindromic identity (Proposition 582) forces $|Z_p|$ to be even, so the data is consistent with $|Z_p|/2 \sim \text{Poisson}(1/2)$. These values support M₁. They do not establish the all-index theorem without a separate cross-prime nonconcentration estimate.

Remark 344 (Fourier decomposition of M₁). The hypothesis M₁ admits a sharp additive Fourier reformulation. Define $S_p(u) = \sum_{r=0}^{p-1} e_p(u \cdot b_r)$ for $u \in \mathbb{F}_p$. Then $|Z_p| = \frac{1}{p} \sum_{u \in \mathbb{F}_p} S_p(u) = 1 + \frac{1}{p} \sum_{u \neq 0} S_p(u)$. Consequently M₁ is exactly the averaged statement

$$\sum_{p \leq N} \left(1 + \frac{1}{p} \sum_{u \neq 0} S_p(u) \right) \ll \pi(N).$$

The pointwise absolute bound $|\sum_{u \neq 0} S_p(u)| \ll p$ for every prime is a strictly stronger sufficient condition; it is not equivalent to the Cesàro hypothesis M₁. By Parseval, $\sum_u |S_p(u)|^2 = p \cdot C(p)$, where $C(p) = \#\{(r, r') : b_r \equiv b_{r'} \pmod{p}\}$. The palindromic identity forces $C(p) \geq 2p - 1$, and

a birthday-effect computation gives $C(p) \sim 3p$ (the extra collisions matching the random model for $p/2$ independent values in $\mathbb{F}_p$). Cauchy–Schwarz then yields only $|\mathcal{Z}_p| \leq 1 + O(\sqrt{p})$, the same barrier as the gap-polynomial bound. Breaking this barrier requires sign cancellation in the u -sum beyond what L^2 methods detect.

The monodromy group $G_{\text{geom}} \simeq \text{Sp}_4$ (Beukers–Heckman) explains the Frobenius trace distribution in the geometric parameter, but M_1 requires an additive equidistribution theorem in the Kummer/Mellin parameter r —a “second Fourier layer” not covered by the existing Katz theory.

The $\text{USp}(4)$ Sato–Tate trace density at zero, computed via the Weyl integration formula, is $\rho_{\text{USp}(4)}(0) = 8/(3\pi^2) \approx 0.270$. The expected zero count is $\mathbb{E}[|\mathcal{Z}_p|] \approx C \cdot \rho(0)$ where $C \approx 3.7$ absorbs the arithmetic normalization between the geometric Frobenius trace and the Apéry coefficient-zero condition. The observed Poisson(1/2) law for $|\mathcal{Z}_p|/2$ is compatible with this picture.

The rank-2 transfer cocycle $T_{1 \rightarrow r} = M(r-1) \cdots M(1)$ has $\det M(n) = n^3/(n+1)^3$, so $\det T_{1 \rightarrow r} = 1/r^3$ is always a cube in $\mathbf{F}_p^\times$. Accordingly, the subgroup $\langle M(1), \dots, M(p-2) \rangle$ of $\text{GL}_2(\mathbf{F}_p)$ is contained in $\{g : \det g \in (\mathbf{F}_p^\times)^3\}$, which equals GL_2 when $3 \nmid (p-1)$ and has index 3 otherwise. In both cases, the subgroup contains $\text{SL}_2(\mathbf{F}_p)$: verified computationally through $p = 23$ (exact group enumeration) and by orbit saturation through $p = 199$.

The static generation can be proved unconditionally: after the rational determinant gauge $G(n) = \text{diag}(n^3, 1)$, the normalized matrices $\Phi(n) = G(n+1)^{-1}M(n)G(n)$ lie in $\text{SL}_2(\mathbf{Q})$. Already $\Phi(1)$ and $\Phi(2)$ generate a Zariski-dense subgroup of SL_2 : they share no eigenline (the system for a common eigenvector is overdetermined), $\Phi(1)$ has infinite order ($\text{tr} = 117$ forces non-root-of-unity eigenvalues), and neither lies in the normaliser of a maximal torus (they do not commute, yet both have nonzero trace). Since the only proper algebraic subgroups of SL_2 are finite, Borel, or torus-normaliser, $\overline{\langle \Phi(1), \Phi(2) \rangle}^{\text{Zar}} = \text{SL}_2$. Strong approximation then gives $\langle \Phi(1), \dots, \Phi(p-2) \rangle = \text{SL}_2(\mathbf{F}_p)$ for all but finitely many odd primes.

However, five independent obstructions prevent this algebraic-group theorem from implying M_1 : (i) the moduli coordinate x and the recurrence index n are different independent variables; (ii) the time-ordered product $T_{1 \rightarrow r}$ is a marked matrix, not a Frobenius conjugacy class; (iii) static generation does not imply orbit mixing (Bourgain–Gamburd requires iterated convolution, not a single Dirac product); (iv) cross-characteristic mismatch (ℓ -adic monodromy lives in $\text{SL}_2(\mathbf{Q}_\ell)$ with $\ell \neq p$, while the orbit lives in $\text{P}^1(\mathbf{F}_p)$); (v) the point-hit test on $\text{P}^1(\mathbf{F}_p)$ is a representation of dimension $p+1$, growing with p . The correct formulation of M_1 is therefore a crystalline Mellin anti-concentration theorem, not a monodromy equidistribution theorem.

Lemma 345 (Reflection-orbit reduction: pair-zero implies M_1). *For each prime $p \geq 5$, let $u_p = |\mathcal{Z}_p|/2$ (the number of independent palindromic zero pairs), and let $D_p = \binom{u_p}{2}$ count unordered pairs of distinct reflection orbits. This differs by a fixed factor from counting residue pairs (r, s) with $r < s$ and $s \neq p-1-r$. Then*

$$\sum_{p \leq N} D_p = O(\pi(N)) \implies M_1 \text{ and } M_2. \quad (348)$$

Proof. Since $|\mathcal{Z}_p| = 2u_p$, M_1 is $\sum_p u_p \leq C\pi(N)$. The identity $u_p(u_p-1) = 2D_p$ gives $u_p^2 \leq u_p + 2D_p$. Cauchy–Schwarz: $(\sum_p u_p)^2 \leq \pi(N) \sum_p u_p^2 \leq \pi(N)(\sum_p u_p + 2\sum_p D_p)$. If $\sum D_p \leq A\pi(N)$, this gives $(\sum u_p)^2 \leq \pi(N)(\sum u_p + 2A\pi(N))$, so $\sum u_p \leq (1 + \sqrt{1+8A})\pi(N)/2$. For M_2 : $\sum |\mathcal{Z}_p|^2 = 4\sum u_p^2 = 4\sum u_p + 8\sum D_p = O(\pi(N))$. $\square$

Remark 346 (Heuristic for the pair-zero count). The palindromic identity creates u_p independent reflection orbits among $\lfloor p/2 \rfloor$ positions. In the random model, each independent position hits zero

with probability $\approx 2\rho(0)/p$ (where $\rho(0) = 8/(3\pi^2) \approx 0.27$ is the $\mathrm{USp}(4)$ trace density at zero). A pair of independent hits has probability $\approx 1/p^2$; among the $\approx p^2/8$ independent pairs, the expected count is $D_p \approx 1/8$. Conditionally on $b_r = 0$, the transfer-matrix characterization says that $b_s = 0$ if and only if a specific off-diagonal coefficient of the nonautonomous cocycle $T_{r \rightarrow s} = M(s-1) \cdots M(r)$ vanishes mod p . For a generic SL_2 element that coefficient condition has density $\sim 1/p$; this is a heuristic for the actual marked cocycle, not an equidistribution theorem.

Computationally verified: for all odd primes $3 \leq p \leq 997$, every non-reflection zero pair (r, s) with $b_r \equiv b_s \equiv 0 \pmod{p}$ and $s \neq p-1-r$ satisfies $T_{12}(r \rightarrow s) \equiv 0 \pmod{p}$ (142 pairs, 38,363 full-equivalence checks, 0 exceptions). The total count of $T_{12} = 0$ instances among 6,092,360 clean forward pairs is 9,400, matching the $1/(p+1)$ density prediction of 9,332. In the full valid range $r \in [1, p-2]$, $h \in [2, p-1-r]$ (all indices below the degenerate $M(p-1)$ boundary), the density doubles to $2/(p+1)$: across all 167 primes $3 \leq p \leq 997$, 75,399 zeros in 24,482,876 valid pairs, ratio = 2.007 to the $1/(p+1)$ prediction. The factor 2 is explained by the palindromic involution $\iota(r, h) = (p-1-r-h, h)$: the untwisted symmetry $b_{p-1-r} \equiv b_r \pmod{p}$ pairs each $K = 0$ in the forward fundamental domain $\mathcal{S}_p^+ = \{(r, h) : r+h \leq (p-1)/2\}$ with a mirror in the backward half. Both are consistent with the monodromy-predicted codimension-one density $1/p$, which replaces the Schwartz–Zippel bound $O(p^2)$ (from summing $\deg P_h = O(h)$ independently) with the sharp $O(p)$.

The pair-zero statistic was computed through $p \leq 10^5$ (9,590 primes). The distribution of $u_p = |\mathcal{Z}_p|/2$ matches Poisson(0.5): $\Pr(u_p = 0) = 60.7\%$, $\Pr(u_p = 1) = 30.5\%$, $\Pr(u_p = 2) = 7.5\%$ (Poisson predicts 60.7%, 30.3%, 7.6%). The non-reflection pair count satisfies

$$\frac{\sum_{p \leq N} D_p}{\pi(N)} \approx 0.118 \approx \frac{1}{8} \quad (349)$$

through $N = 10^5$, consistent with the codimension-two density $1/[p(p+1)]$.

Remark 347 (Projective orbit formulation). The transfer cocycle $T_{1 \rightarrow r} = M(r-1) \cdots M(1)$ acts on $\mathbf{P}^1(\mathbf{F}_p)$ via Möbius transformations. Starting from $[b_1 : b_0] = [5 : 1]$, the zero set is

$$\mathcal{Z}_p = \{r \in [2, p-2] : T_{1 \rightarrow r} \cdot [5 : 1] = [0 : 1]\}.$$

Thus $|\mathcal{Z}_p|$ counts visits of the orbit to the marked point $[0 : 1] \in \mathbf{P}^1$. If the orbit equidistributes on $\mathbf{P}^1(\mathbf{F}_p)$ ($p+1$ points), then $|\mathcal{Z}_p| \approx 1$. However, the geometric monodromy group $G_{\text{geom}} = \mathrm{SL}_2$ controls *trace* equidistribution (conjugacy-class averages), not *entry-level* equidistribution (the condition $(T_{1 \rightarrow r})_{12} = 0$ is basis-dependent). A direct Weil-type bound $|\mathcal{Z}_p| = 1 + O(\sqrt{p})$ would require a bounded-conductor sheaf encoding the full cocycle product, meeting the same conductor-growth obstruction as the sheaf approach to AKD_3 . The realistic target is the averaged second moment $\sum_{p \leq X} |\mathcal{Z}_p|^2 = O(\pi(X))$, which suffices for M2 and avoids pointwise cocycle estimates.

Complementarily, the Mellin formulation $b_r = \sum_{t \in \mathbf{F}_p^\times} \mathrm{Ha}_p(t) \cdot t^{-r}$ (where Ha_p is the K3 Hasse polynomial) recasts $|\mathcal{Z}_p|$ as the zero count of Kummer–Mellin coefficients, a family-of-twists equidistribution problem in Katz’s framework.

Remark 348 (The vertical and horizontal targets remain distinct). The pair-zero bound (Lemma 345) is a vertical route to M_1 and M_2 . The auxiliary AKD_3 estimate is a semiprime correlation statement, but Proposition 338 shows that it is not coupled to the actual trilinear Weyl sum. The all-index proof therefore still needs a separate horizontal theorem controlling fully coupled cross-prime CRT phases. Calling both requirements “Fourier pseudorandomness” does not make either one imply the other.

For M_1 , a fourth-moment estimate for suitable crystalline Mellin coefficients is one possible anti-concentration mechanism, but no theorem presently derives the zero-count moment from such an estimate with the required uniformity. It should not be recorded as an equivalence.

For the horizontal problem, the Q554 semiprime near-injection lemma provides only an RMS/phase-averaged bound for the auxiliary kernel. The actual Weyl sum contains three mutually coupled inverse frequencies as displayed above. A sufficient input must control that full sum (or the equivalent positive CRT overload); bounded additive energy or bounded zero-set multiplicity by itself does not do so, as the anchored-star construction demonstrates.

Remark 349 (Strassmann closure and TCMA₂ formulation). The naive p -adic Strassmann route is closed by three obstructions (Q564).

1. The full Apéry sequence has no continuous interpolation on $\mathbf{Z}_p$: since $p^m \rightarrow 0$ while $b_{p^m} \equiv b_1 = 5 \not\equiv b_0 = 1 \pmod{p}$ (Apéry–Lucas), the sequence is already discontinuous at zero.
2. The finite Mahler polynomial $P_p(s) = \sum_{k=0}^{p-2} c_k \binom{s}{k}$ has flat Newton polygon: its Strassmann index is $p-2$ or $p-3$ for every tested prime $11 \leq p \leq 101$. Of 1,122 Mahler coefficients tested, 97.15% are p -adic units.
3. The $p-1$ tame characters ω^j lie on $p-1$ distinct Iwasawa weight-space components; Strassmann controls zeros within one component, giving at best $O(p)$ total.

The top Mahler coefficient satisfies the identity $c_{p-2} \equiv -\frac{1}{2}A_p(1) \pmod{p}$, detecting ordinarity at $t = 1$ only—not the vanishing Mellin coefficients that constitute $\mathcal{Z}_p$.

The precise missing theorem is therefore *two-twist crystalline Mellin anti-concentration* (TCMA₂):

$$\sum_{p \leq X} \binom{u_p}{2} \ll \pi(X),$$

where $u_p = |\mathcal{Z}_p|/2$ counts noncentral zero pairs (from the palindromic pairing $b_{p-1-r} \equiv b_r \pmod{p}$). A sufficient mechanism: if the Kummer-twisted rigid cohomology module $\mathcal{H}_{p,j} = H_{\text{rig},c}^1(U, \mathcal{V} \otimes \mathcal{K}_{\omega^{-j}})$ has a *unique* unit root for all but $O(1)$ tame twists j , then $\text{Tr}(\Phi_{p,j})$ is a p -adic unit and $b_j \not\equiv 0 \pmod{p}$. The missing ingredient is a *branch-coupling* theorem relating the slope-zero traces of distinct tame twists.

The next two statements make the (HM)₃ frontier exact: a sharp reformulation as a positive pair–Palm excess, and an incidence model showing that *all* presently proved primewise and pairwise facts—including the fiber and energy bounds of Section 5.4 and even a uniform $O(1)$ row-codegree—are jointly insufficient for any power saving at $k = 3$. (In this block, Ω_p temporarily denotes the bad set $\{0 \leq m < X^2 : m \bmod p \in \mathcal{Z}_p\}$.)

The third factorial moment: sufficiency and exact obstruction

Theorem 350 ($M_1 + (\text{HM})_3$ implies the full conjecture). *Assume the dyadic first zero-mass bound*

$$\lambda_X = \sum_{p \in \mathcal{P}_X} \frac{Z(p)}{p} \ll \frac{1}{\log X},$$

which follows from $\sum_{p \in \mathcal{P}_X} Z(p) \ll |\mathcal{P}_X|$. If in addition the third factorial moment bound

$$\sum_{m < X^2} (K_X(m))_3 \leq CX^2 \lambda_X^3 \tag{350}$$

holds for all sufficiently large X with an absolute constant C , then $G_n = e^{o(n)}$ for every positive integer n .

Proof. The hypothesis (350) and the identity $(k)_3 \geq k^3 - 3k^2 = k^2(k-3)$ imply

$$\max_{m < X^2} K_X(m) \leq \left(\sum_{m < X^2} (K_X(m))_3 \right)^{1/3} + 3 \leq C^{1/3} X^{2/3} \lambda_X + 3 \ll \frac{X^{2/3}}{\log X} + 3.$$

By the pointwise reduction (Proposition 129), covering $[\sqrt{n}, n]$ with $O(\log n)$ dyadic shells gives

$$B(n) = \sum_{\text{dyadic shells}} K_{X_i}(n) \ll \sum_i \frac{X_i^{2/3}}{\log X_i} \ll \frac{n^{2/3}}{\log n} = o\left(\frac{n}{\log n}\right).$$

Hence $\log G_n \leq 6B(n) \log n + O(\sqrt{n}) = o(n)$. □

The third factorial moment: exact obstruction

Write

$$\mathcal{P}_X = \{p \text{ prime} : X < p \leq 2X\}, \quad M = X^2,$$

where $X \geq 2$ is integral, and put

$$\Omega_p = \{0 \leq m < M : m \bmod p \in \mathcal{Z}_p\}, \quad f_p(m) = \mathbf{1}_{\Omega_p}(m).$$

Thus $K_X(m) = \sum_{p \in \mathcal{P}_X} f_p(m)$. Set

$$A_p = |\Omega_p|, \quad \rho_p = \frac{A_p}{M}, \quad \mu = \sum_{p \in \mathcal{P}_X} \rho_p = \frac{1}{M} \sum_{m < M} K_X(m),$$

and retain

$$\lambda = \lambda_X = \sum_{p \in \mathcal{P}_X} \frac{Z(p)}{p}.$$

For ordered distinct primes $p, q \in \mathcal{P}_X$, define

$$\begin{aligned} J_{p,q} &= |\Omega_p \cap \Omega_q|, \\ T_{p,q} &= \sum_{m \in \Omega_p \cap \Omega_q} \sum_{\ell \in \mathcal{P}_X \setminus \{p,q\}} f_\ell(m), \\ \lambda_{p,q} &= \sum_{\ell \in \mathcal{P}_X \setminus \{p,q\}} \frac{Z(\ell)}{\ell}, \quad \mu_{p,q} = \sum_{\ell \in \mathcal{P}_X \setminus \{p,q\}} \rho_\ell. \end{aligned}$$

The two aggregate positive pair-Palm excesses are

$$\mathcal{P}_3(X) = \sum_{p \neq q} (T_{p,q} - J_{p,q} \lambda_{p,q})_+, \tag{351}$$

$$\mathcal{P}_3^{\text{fin}}(X) = \sum_{p \neq q} (T_{p,q} - J_{p,q} \mu_{p,q})_+. \tag{352}$$

The first uses the complete-period densities $Z(\ell)/\ell$; the second uses the exact marginals on $[0, M)$ and therefore absorbs every one-column boundary error.

Theorem 351 (Sharp pair–Palm characterization of $(\text{HM})_3$). *With all prime tuples ordered, one has the exact identities*

$$\sum_{p \neq q} J_{p,q} = \sum_{m < M} (K_X(m))_2, \quad (353)$$

$$\sum_{p \neq q} T_{p,q} = \sum_{m < M} (K_X(m))_3. \quad (354)$$

Moreover,

$$\mathcal{P}_3(X) \leq \sum_{m < M} (K_X(m))_3 \leq \mathcal{P}_3(X) + 5M\lambda^3, \quad (355)$$

$$\mathcal{P}_3^{\text{fin}}(X) \leq \sum_{m < M} (K_X(m))_3 \leq \mathcal{P}_3^{\text{fin}}(X) + 5(1 + 2/X)M\lambda^3. \quad (356)$$

Consequently $(\text{HM})_3$ is equivalent to either one of

$$\mathcal{P}_3(X) \ll X^{2+o(1)}\lambda_X^3, \quad \mathcal{P}_3^{\text{fin}}(X) \ll X^{2+o(1)}\lambda_X^3.$$

More generally, for every fixed $\theta \geq 0$, the estimate in Goal G2 with exponent θ is equivalent to either positive-excess estimate with right-hand side $X^{2+\theta+o(1)}\lambda_X^3$.

There is also an exact centered formulation. Define

$$V_X = \sum_{m < M} (K_X(m) - \mu)^2, \quad C_{3,X} = \sum_{m < M} (K_X(m) - \mu)^3$$

and the third factorial-moment defect

$$\mathfrak{D}_3(X) = (C_{3,X} - M\mu) + 3(\mu - 1)(V_X - M\mu). \quad (357)$$

Then

$$\sum_{m < M} (K_X(m))_3 = M\mu^3 + \mathfrak{D}_3(X). \quad (358)$$

Thus $(\text{HM})_3$ is equivalently the one-sided estimate

$$(\mathfrak{D}_3(X))_+ \ll X^{2+o(1)}\lambda_X^3.$$

The same equivalence holds at every G2 exponent $\theta \geq 0$.

Proof. If $p \neq q$ lie in $\mathcal{P}_X$, then $pq > M$. Hence every residue pair $(a \bmod p, b \bmod q)$ has at most one representative in $[0, M)$. This removes, rather than estimates, the usual CRT boundary term. At an integer with $K_X(m) = k$, there are exactly $(k)_2$ ordered choices of (p, q) , and each has $k - 2$ choices of a third prime. Its contribution to $\sum_{p \neq q} T_{p,q}$ is therefore $(k)_2(k - 2) = (k)_3$. This proves (353)–(354); in particular, no factor $3!$ is missing.

For nonnegative A, B ,

$$(A - B)_+ \leq A \leq B + (A - B)_+.$$

Apply this with $A = T_{p,q}$ and $B = J_{p,q}\lambda_{p,q}$, sum over ordered pairs, and use $\lambda_{p,q} \leq \lambda$ together with the proved second factorial moment:

$$\sum_{p \neq q} J_{p,q}\lambda_{p,q} \leq \lambda \sum_{m < M} (K_X(m))_2 \leq 5M\lambda^3.$$

This proves (355).

For each p and each residue in $\mathcal{Z}_p$, the number of its representatives below M differs from M/p by less than one. Therefore

$$|A_p - MZ(p)/p| \leq Z(p), \quad |\mu - \lambda| \leq \frac{1}{M} \sum_p Z(p) \leq \frac{2\lambda}{X},$$

where the final inequality uses $p \leq 2X$. In particular $\mu \leq (1 + 2/X)\lambda$. Applying the same positive-part inequality with $B = J_{p,q}\mu_{p,q}$ gives

$$\sum_{p \neq q} J_{p,q}\mu_{p,q} \leq \mu \sum_{m < M} (K_X(m))_2 \leq 5(1 + 2/X)M\lambda^3,$$

which proves (356). Each equivalence follows in both directions from its sandwich. This is a characterization, not a new estimate hidden in a renamed hypothesis: the unproved arithmetic is exactly the positive conditional overload of third primes on the actual two-prime CRT loci.

Finally write $Y_m = K_X(m) - \mu$, so that $\sum_m Y_m = 0$. Expanding $K^3 - 3K^2 + 2K$ gives

$$\sum_{m < M} (K_X(m))_3 = C_{3,X} + 3(\mu - 1)V_X + M(\mu^3 - 3\mu^2 + 2\mu),$$

which rearranges to (358). Since the main term is nonnegative,

$$(\mathfrak{D}_3(X))_+ \leq \sum_{m < M} (K_X(m))_3 \leq M\mu^3 + (\mathfrak{D}_3(X))_+.$$

Equation (12.34) bounds $M\mu^3$ by a constant multiple of $M\lambda^3$, proving the last equivalences. When $\lambda = 0$ all zero sets, all pair loci, and all displayed quantities vanish. The same is true of the Palm sums when there is no pair locus, so no division by a possibly zero parameter has been suppressed. $\square$

Why the available primewise and pairwise tools do not imply G2

Proposition 352 (An anchored rainbow star with bounded codegrees). *For every sufficiently large integral X there is a family $(\mathcal{Z}_p^*)_{p \in \mathcal{P}_X}$ with the following properties. Writing*

$$\begin{aligned} K_X^*(m) &= \sum_{p \in \mathcal{P}_X} \mathbf{1}_{\{m \bmod p \in \mathcal{Z}_p^*\}}, & \lambda_X^* &= \sum_{p \in \mathcal{P}_X} \frac{|\mathcal{Z}_p^*|}{p}, \\ \mu^* &= X^{-2} \sum_{m < X^2} K_X^*(m), \end{aligned}$$

one has:

1. Every $\mathcal{Z}_p^*$ has size zero or two. Every nonempty set is reflection-invariant, has no consecutive elements, avoids $0, 1, p-2, p-1$, and satisfies

$$|\mathcal{Z}_p^* \cap I| \leq 2|I|^{2/3}$$

for every nonempty interval $I \subseteq \{0, \dots, p-1\}$.

2. The universal CRT bound

$$\sum_{m < X^2} (K_X^*(m))_2 \leq 5X^2(\lambda_X^*)^2$$

holds, and even the global centered second moment has the natural scale

$$\sum_{m < X^2} (K_X^*(m) - \mu^*)^2 \ll X^2 \lambda_X^*.$$

3. If $\mathcal{P}^*(m) = \{p : m \bmod p \in \mathcal{Z}_p^*\}$, then for all distinct $0 \leq m, n < X^2$,

$$|\mathcal{P}^*(m) \cap \mathcal{P}^*(n)| \leq 7.$$

Thus even a uniform $O(1)$ row-codegree bound does not prevent the model.

4. Nevertheless,

$$\lambda_X^* \asymp \frac{1}{\log X}, \quad \sum_{m < X^2} (K_X^*(m))_3 \gg \frac{X^3}{(\log X)^3}.$$

Consequently the family violates every $G2$ estimate with fixed $\theta < 1$.

The family can additionally be extended to a reflection-invariant value map whose largest fiber is at most two and whose collision energy is at most $2p$. Hence the numerical consequences of the bordered nonvanishing theorem—the $O(p^{3/4})$ fiber bound and $O(p^{7/4})$ energy bound—also do not rule it out.

If one uses the gap-polynomial reflection identity $N_h(-x - h - 1) = (-1)^{h-1} N_h(x)$, the unique pair in every nonempty $\mathcal{Z}_p^*$ also satisfies its corresponding necessary gap certificate modulo p .

Proof. Put $M = X^2$, choose $m_0 = \lfloor 3M/4 \rfloor$, and let $d = 2m_0 + 1$. Delete from $\mathcal{P}_X$ the primes dividing

$$E = (m_0 - 1)m_0(m_0 + 1)(m_0 + 2)d.$$

Only $O(1)$ primes are deleted: $|E| \ll X^{10}$ and every deleted prime exceeds X . For every remaining prime define

$$a_p = m_0 \bmod p, \quad \mathcal{Z}_p^* = \{a_p, p - 1 - a_p\},$$

and put $\mathcal{Z}_p^* = \emptyset$ for deleted primes. The deletion makes the two displayed residues distinct and excludes the four endpoint residues. Their difference is a nonzero even integer, so they are not consecutive. Reflection is built in, and the interval estimate is immediate because an interval contains at most two marked residues.

Let L be the number of nonempty columns. The prime number theorem and Mertens' theorem give

$$L \sim \frac{X}{\log X}, \quad \lambda_X^* = 2 \sum_{\substack{X < p \leq 2X \\ p \nmid E}} \frac{1}{p} \asymp \frac{1}{\log X}.$$

Every active prime marks m_0 , so $K_X^*(m_0) = L$ and

$$\sum_{m < M} (K_X^*(m))_3 \geq (L)_3 \gg \frac{X^3}{(\log X)^3}.$$

Dividing by $X^{2+\theta+o(1)}(\lambda_X^*)^3$ leaves $X^{1-\theta-o(1)}$, which tends to infinity for $\theta < 1$. The second-factorial-moment estimate is universal and follows by the same CRT argument as Proposition 328. Also

$$\begin{aligned} \sum_m (K_X^*(m) - \mu^*)^2 &\leq \sum_m (K_X^*(m))^2 \\ &= \sum_m K_X^*(m) + \sum_m (K_X^*(m))_2 \\ &\ll M\lambda_X^* + M(\lambda_X^*)^2 \ll M\lambda_X^*. \end{aligned}$$

It remains to check the codegree. Fix $0 \leq m < n < M$ and put $h = n - m$. If an active prime marks both integers, their two residues are chosen from a_p and $-1 - a_p$. Equal choices give $p \mid h$; different choices give

$$h \equiv \pm(2a_p + 1) \equiv \pm d \pmod{p}.$$

Thus every common prime divides the nonzero integer

$$H = h(h - d)(h + d).$$

Here $0 < h < M < d$, and $|H| < 8M^3 = 8X^6$. If there are k common primes, their product exceeds X^k and divides $|H|$, whence

$$k \leq \frac{\log(8X^6)}{\log X} \leq 7$$

for sufficiently large X .

For the value-map assertion, partition the residues into the orbits of $r \mapsto p - 1 - r$. At an active prime, assign zero to the marked orbit and distinct nonzero values to all other orbits. At a deleted prime, assign distinct nonzero values to every orbit; this is possible because there are $(p + 1)/2 \leq p - 1$ orbits. Thus the zero fiber is exactly $\mathcal{Z}_p^*$ in both cases. Every fiber has size at most two, and, since $t^2 \leq 2t$ for $t \in \{0, 1, 2\}$, the fiber energy is at most $2p$.

For completeness, the gap-polynomial reflection identity used here is symbolic. For $h \geq 2$, let $D_h(x)$ be the symmetric $(h - 1) \times (h - 1)$ tridiagonal matrix with diagonal entries $P(x + i)$, $1 \leq i < h$, and off-diagonal entries $(x + i + 1)^3$, $1 \leq i < h - 1$. Expansion in the last row and the defining recurrence for N_h give $\det D_h(x) = N_h(x)$. If R is the matrix that reverses the coordinate order, then $P(-v - 1) = -P(v)$ gives, entry by entry,

$$D_h(-x - h - 1) = -RD_h(x)R.$$

Consequently

$$N_h(-x - h - 1) = (-1)^{h-1} N_h(x). \quad (359)$$

(For $h = 1$ this is immediate from $N_1 = 1$.) This proves the identity for every h , without numerical verification.

Finally let $u = \min(a_p, p - 1 - a_p)$ and $h_p = p - 1 - 2u$. The deleted endpoints ensure $1 \leq u < u + h_p < p$; moreover h_p is positive and even. Since

$$-u - h_p - 1 = u - p \equiv u \pmod{p},$$

the gap-polynomial reflection identity gives $N_{h_p}(u) \equiv -N_{h_p}(u) \pmod{p}$, hence $N_{h_p}(u) \equiv 0 \pmod{p}$ for odd p . $\square$

Remark 353 (Scope of the obstruction). The preceding family is an incidence model, not the actual zero family of the Apéry recurrence. It proves that reflection, no consecutive zeros, the interval $O(|I|^{2/3})$ bound, the universal CRT second moment, the primewise fiber/energy bounds, the gap-polynomial pair certificates, and even uniform $O(1)$ row codegree do not imply any G2 power saving. The missing arithmetic must constrain, across distinct primes, which of the structurally permitted reflection orbits is selected as the zero fiber. The pair-Palm quantity in Theorem 351 is the minimal aggregate positive overload that measures exactly this selection problem.

The primitive CRT–Fourier interface

Put $z_p = Z(p)/p$, $g_p = f_p - z_p$, and use the unnormalised finite Fourier transform

$$F_p(a) = \sum_{r \in \mathbb{Z}_p} e_p(ar), \quad W_M(\vartheta) = \sum_{0 \leq m < M} e(-m\vartheta).$$

Thus

$$f_p(m) = \frac{1}{p} \sum_{a \bmod p} F_p(a) e_p(-am), \quad g_p(m) = \frac{1}{p} \sum_{\substack{a \bmod p \\ a \neq 0}} F_p(a) e_p(-am).$$

Proposition 354 (Primitive CRT–Fourier reduction). *Define the ordered primitive correlation*

$$\mathcal{C}_X^\times = \sum_{p,q,\ell}^* \sum_{0 \leq m < M} g_p(m) g_q(m) g_\ell(m),$$

where the star means that $p, q, \ell \in \mathcal{P}_X$ are pairwise distinct. Then, with $Q = pql$,

$$\mathcal{C}_X^\times = \sum_{p,q,\ell}^* \frac{1}{Q} \sum_{\substack{h \bmod Q \\ (h,Q)=1}} W_M(h/Q) F_p(h\overline{q\ell}) F_q(h\overline{p\ell}) F_\ell(h\overline{pq}), \quad (360)$$

where each inverse and each Fourier argument is taken in the indicated prime field. Moreover

$$\mathfrak{D}_3(X) = \mathcal{C}_X^\times + O(M\lambda_X^3). \quad (361)$$

Consequently $(\text{HM})_3$, and hence the pair-Palm estimate of Theorem 351, is equivalent to the one-sided bound

$$(\mathcal{C}_X^\times)_+ \ll X^{2+o(1)} \lambda_X^3.$$

This is a signed cancellation interface, not an absolute-value estimate.

Proof. Fourier inversion gives the two displayed expansions. For one ordered triple, the Chinese remainder map

$$h \equiv aq\ell + bpl + cpq \pmod{Q}$$

is a bijection from $(a \bmod p, b \bmod q, c \bmod \ell)$ to $h \bmod Q$. Requiring all three frequencies to be nonzero is exactly $(h, Q) = 1$, and the inverse map is

$$a = h\overline{q\ell} \pmod{p}, \quad b = h\overline{p\ell} \pmod{q}, \quad c = h\overline{pq} \pmod{\ell}.$$

This proves (360), including its factor $1/Q$ and with no factor $1/3!$.

It remains to account for the lower frequency strata. Write

$$B_{q,\ell} = \sum_{m < M} g_q(m)g_\ell(m), \quad \delta_\ell = \sum_{m < M} g_\ell(m) = A_\ell - Mz_\ell.$$

Expanding $f_p f_q g_\ell = (g_p + z_p)(g_q + z_q)g_\ell$ gives exactly

$$\sum_{p,q,\ell}^* \sum_{m < M} f_p(m)f_q(m)g_\ell(m) - \mathcal{C}_X^\times = \sum_{p,q,\ell}^* (z_p B_{q,\ell} + z_q B_{p,\ell} + z_p z_q \delta_\ell).$$

Now

$$B_{q,\ell} = J_{q,\ell} - z_q A_\ell - z_\ell A_q + M z_q z_\ell.$$

The proved second factorial moment and $\mu \leq (1 + 2/X)\lambda_X$ therefore give

$$\sum_{q \neq \ell} |B_{q,\ell}| \leq (8 + 4/X)M\lambda_X^2.$$

Also

$$\sum_{\ell} |\delta_\ell| \leq \sum_{\ell} Z(\ell) \leq 2X\lambda_X.$$

Since $\sum_p z_p = \lambda_X$, the preceding exact expansion is $O(M\lambda_X^3)$. Finally, the identities in Theorem 351 give

$$\mathfrak{D}_3 = \sum_{p,q,\ell}^* \sum_{m < M} f_p(m)f_q(m)g_\ell(m) + \sum_{p \neq q} J_{p,q}\lambda_{p,q} - M\mu^3.$$

The last two terms are $O(M\lambda_X^3)$, proving (361) and the equivalence. $\square$

Reflection removes the inessential complex phase in (360) completely. Write $\bar{2}_p$ for the inverse of 2 modulo p , and put

$$R_p(a) = \sum_{r \in \mathbb{Z}_p} e_p(a(r + \bar{2}_p)).$$

Corollary 355 (Centered real kernel). *For every $a \bmod p$, the number $R_p(a)$ is real, $R_p(-a) = R_p(a)$, and*

$$F_p(a) = e_p(-a\bar{2}_p)R_p(a).$$

Consequently the summand belonging to one ordered triple in (360) is exactly

$$\frac{1}{Q} \sum_{\substack{1 \leq h \leq (Q-1)/2 \\ (h,Q)=1}} (-1)^h \frac{\sin(2\pi Mh/Q)}{\sin(\pi h/Q)} R_p(h\bar{q}\bar{\ell}) R_q(h\bar{p}\bar{\ell}) R_\ell(h\bar{p}\bar{q}). \quad (362)$$

The scalar kernel also has the centered-interval form

$$(-1)^h \frac{\sin(2\pi Mh/Q)}{\sin(\pi h/Q)} = 2 \sum_{0 \leq m < M} \cos\left(\frac{2\pi h(m + \bar{2}_Q)}{Q}\right).$$

Proof. The reflection $r \mapsto p - 1 - r$ sends $r + \bar{2}_p$ to its negative modulo p . Hence the terms in $R_p(a)$ occur in conjugate pairs, which proves reality and evenness; the displayed factorisation of F_p is immediate. The product of the three local phases in (360) is

$$e_p(-h\bar{q}\bar{\ell}\bar{2}_p)e_q(-h\bar{p}\bar{\ell}\bar{2}_q)e_\ell(-h\bar{p}\bar{q}\bar{2}_\ell) = e_Q(-h\bar{2}_Q) = (-1)^h e(-h/(2Q)).$$

Also

$$W_M(h/Q) = e(-(M-1)h/(2Q)) \frac{\sin(\pi Mh/Q)}{\sin(\pi h/Q)}.$$

Pairing h with $-h$, and using the evenness of all three R -factors, gives (362). Expanding $2 \operatorname{Re}\{e_Q(-h\bar{2}_Q)W_M(h/Q)\}$ gives the final identity. $\square$

Proposition 356 (Exact pair-spectral flattening). *For $p \in \mathcal{P}_X$, put*

$$e_p = \sum_{a \in \mathbf{F}_p^\times} R_p(a)^2 = pZ(p) - Z(p)^2, \quad u_p = e_p^{1/2},$$

and set $\phi_p(a) = R_p(a)/u_p$ when $e_p \neq 0$, with $\phi_p(a) = 0$ otherwise. Define the real symmetric matrix $\mathcal{A}_X$, indexed by $\mathcal{P}_X$, by $\mathcal{A}_X(p, p) = 0$ and, for $p \neq q$,

$$\begin{aligned} \mathcal{A}_X(p, q) = \sum_{\substack{\ell \in \mathcal{P}_X \\ \ell \neq p, q}} \frac{u_\ell}{pq\ell} \sum_{\substack{1 \leq h \leq (pq\ell-1)/2 \\ (h, pq\ell)=1}} (-1)^h \frac{\sin(2\pi Mh/(pq\ell))}{\sin(\pi h/(pq\ell))} \\ \times \phi_p(h\bar{q}\bar{\ell}) \phi_q(h\bar{p}\bar{\ell}) \phi_\ell(h\bar{p}\bar{q}). \end{aligned}$$

Then, writing $u = (u_p)_{p \in \mathcal{P}_X}$,

$$\mathcal{C}_X^\times = u^\top \mathcal{A}_X u, \quad \|u\|_2^2 \leq 4X^2 \lambda_X.$$

Consequently either one of

$$(\lambda_{\max}(\mathcal{A}_X))_+ \ll X^{o(1)} \lambda_X^2, \tag{363}$$

$$\|\mathcal{A}_X u\|_2^2 \ll X^{2+o(1)} \lambda_X^5 \tag{364}$$

implies (HM)₃.

Proof. Parseval for F_p , together with the harmless centered phase relating F_p and R_p , gives the formula for e_p . Expanding $u^\top \mathcal{A}_X u$ and using $u_p \phi_p = R_p$ reproduces (362) once for every ordered triple of distinct primes. Also

$$\|u\|_2^2 \leq \sum_{p \in \mathcal{P}_X} pZ(p) \leq 4X^2 \sum_{p \in \mathcal{P}_X} \frac{Z(p)}{p}.$$

The positive spectral edge therefore gives

$$(\mathcal{C}_X^\times)_+ \leq \|u\|_2^2 (\lambda_{\max}(\mathcal{A}_X))_+,$$

which proves the first implication. One Cauchy–Schwarz inequality gives

$$(\mathcal{C}_X^\times)_+ \leq \|u\|_2 \|\mathcal{A}_X u\|_2,$$

and proves the second. $\square$

Remark 357 (Scope of the spectral reduction). Proposition 356 is an exact contraction of the third prime, not an estimate. Condition (363) is the weakest test-vector-independent positive-edge statement naturally produced by this contraction; replacing it by the same assertion only on the vector u recovers the original target. Expanding $\|\mathcal{A}_X u\|_2^2$ gives the precise two-prime TT^* collision problem. Closing that expansion by one-prime Parseval alone loses at least one power of X at the presently available zero-set density.

Proposition 358 (Output-prime/semiprime compression). *Put*

$$\mathcal{K}_Q(h) = (-1)^h \frac{\sin(2\pi Mh/Q)}{\sin(\pi h/Q)}$$

on nonzero residues, extended evenly modulo the odd integer Q . For distinct $q, \ell \in \mathcal{P}_X$, put $v = q\ell$ and

$$R_{q,\ell}(b) = R_q(b\bar{\ell})R_\ell(b\bar{q}) \quad (b \bmod v).$$

For $p \notin \{q, \ell\}$, define

$$(\mathcal{T}_{p,v}F)(a) = \frac{1}{2pv} \sum_{b \in (\mathbf{Z}/v\mathbf{Z})^\times} \mathcal{K}_{pv}([av + bp]_{pv})F(b), \quad a \in \mathbf{F}_p^\times.$$

Then the primitive centered contribution of the ordered triple (p, q, ℓ) is exactly

$$\mathcal{P}_{p,q,\ell} = \sum_{a \in \mathbf{F}_p^\times} R_p(a)(\mathcal{T}_{p,q\ell}R_{q,\ell})(a).$$

Moreover

$$\|\mathcal{T}_{p,q\ell}\|_{2 \rightarrow 2} \leq \frac{1}{2\sqrt{pq\ell}} \left\lceil \frac{2M}{p} \right\rceil^{1/2} \left\lceil \frac{2M}{q\ell} \right\rceil^{1/2} \leq X^{-1}. \quad (365)$$

Define the signed output-prime total

$$\mathcal{E}_p = \sum_{\substack{q, \ell \in \mathcal{P}_X \setminus \{p\} \\ q \neq \ell}} \mathcal{P}_{p,q,\ell}.$$

The one-sided mixed estimate

$$\sum_{\substack{p \in \mathcal{P}_X \\ Z(p) > 0}} \frac{p}{Z(p)} (\mathcal{E}_p)_+^2 \ll X^{4+o(1)} \lambda_X^5 \quad (366)$$

implies (HM)₃.

Proof. The CRT bijection for $Q = pv$ is

$$h \longleftrightarrow (a = h\bar{v} \bmod p, b = h\bar{p} \bmod v), \quad h \equiv av + bp \pmod{pv}.$$

The full unit sum contains both h and $-h$, whereas (362) uses one representative of each pair. This explains the factor $1/(2pv)$ and proves the triple identity.

For $d = p, v$, let U_d be the normalized finite Fourier restriction matrix with rows indexed by residues modulo d , columns indexed by $-M \leq n < M$, and entries

$$U_d(c, n) = d^{-1/2} e_d(-c(n + \bar{2}_d)).$$

The centered kernel identity gives, before restricting to unit rows,

$$\mathcal{T}_{p,v} = \frac{1}{2\sqrt{pv}} U_p U_v^\top.$$

Columns with the same residue modulo d coincide, and distinct residue classes are orthogonal. Hence

$$\|U_d\|_{2 \rightarrow 2}^2 = \max_{c \bmod d} \#\{-M \leq n < M : n \equiv c \bmod d\} = \left\lceil \frac{2M}{d} \right\rceil.$$

Restriction to unit rows cannot increase the norm. Since $p > X$, $q\ell > X^2 = M$, inequality (365) follows.

Finally $\mathcal{C}_X^\times = \sum_p \mathcal{E}_p$. Weighted Cauchy–Schwarz gives

$$(\mathcal{C}_X^\times)_+ \leq \sum_p (\mathcal{E}_p)_+ \leq \lambda_X^{1/2} \left(\sum_{\substack{p \in \mathcal{P}_X \\ Z(p) > 0}} \frac{p}{Z(p)} (\mathcal{E}_p)_+^2 \right)^{1/2}.$$

Thus (366) has exactly the scale required in Proposition 354. $\square$

Corollary 359 (Boundary-corrected output–Palm and overload criteria). *Write $z_p = Z(p)/p$, and for $0 \leq m < M$ put*

$$\mathcal{H}_{-p}(m) = \sum_{\substack{q, \ell \in \mathcal{P}_X \setminus \{p\} \\ q \neq \ell}} g_q(m) g_\ell(m).$$

Then

$$\mathcal{H}_{-p}(m) = G(m)^2 - G_2(m) - 2G(m)g_p(m) + 2g_p(m)^2, \quad (367)$$

$$\mathcal{E}_p = \sum_{r \in \mathcal{Z}_p} \sum_{\substack{0 \leq m < M \\ m \equiv r \pmod{p}}} \mathcal{H}_{-p}(m) - z_p \sum_{0 \leq m < M} \mathcal{H}_{-p}(m). \quad (368)$$

For $z_p > 0$, let

$$\Omega_p = \{0 \leq m < M : f_p(m) = 1\}, \quad \Delta_p = \frac{1}{M z_p} \sum_{m \in \Omega_p} \mathcal{H}_{-p}(m) - \frac{1}{M} \sum_{0 \leq m < M} \mathcal{H}_{-p}(m).$$

Thus $\mathcal{E}_p = M z_p \Delta_p$ exactly; this normalization retains the incomplete final p -period. Put also

$$\varrho_p = \sum_{\substack{q, \ell \neq p \\ q \neq \ell}} z_q z_\ell = (\lambda_X - z_p)^2 - \sum_{q \neq p} z_q^2.$$

Either one of the pointwise one-sided estimates

$$(\Delta_p)_+ \ll X^{o(1)} \lambda_X^2, \quad \text{or} \quad (\Delta_p)_+ \ll X^{o(1)} \varrho_p \quad (369)$$

implies (HM)₃.

There is a weaker overload criterion. Let $\tau_X = X^{o(1)}$ be nonnegative. It is enough to prove

$$\sum_{\substack{p \in \mathcal{P}_X \\ Z(p) > 0}} \frac{1}{z_p} (\mathcal{E}_p - \tau_X M z_p \lambda_X^2)_+^2 \ll X^{4+o(1)} \lambda_X^5. \quad (370)$$

Proof. If $G_{-p} = G - g_p$ and $G_{2,-p} = G_2 - g_p^2$, then

$$G_{-p}^2 - G_{2,-p} = G^2 - G_2 - 2Gg_p + 2g_p^2,$$

which proves (367). Since $g_p = f_p - z_p$, summing against this field gives the exact finite-interval identity (368); in particular, no periodic endpoint term has been discarded.

Under (369),

$$(\mathcal{C}_X^\times)_+ \leq \sum_p (\mathcal{E}_p)_+ = M \sum_p z_p (\Delta_p)_+ \ll X^{o(1)} M \lambda_X^3.$$

For the second alternative we used $\sum_p z_p \varrho_p = \sum_{p,q,\ell}^* z_p z_q z_\ell \leq \lambda_X^3$; the first uses $\sum_p z_p = \lambda_X$. For the last assertion, use

$$(\mathcal{E}_p)_+ \leq \tau_X M z_p \lambda_X^2 + (\mathcal{E}_p - \tau_X M z_p \lambda_X^2)_+.$$

After summing and applying weighted Cauchy–Schwarz to the second term,

$$\begin{aligned} (\mathcal{C}_X^\times)_+ &\leq \tau_X M \lambda_X^3 + \lambda_X^{1/2} \left(\sum_{Z(p)>0} \frac{1}{z_p} (\mathcal{E}_p - \tau_X M z_p \lambda_X^2)_+^2 \right)^{1/2} \\ &\ll X^{2+o(1)} \lambda_X^3. \end{aligned}$$

Proposition 354 now gives (HM)₃. □

Proposition 360 (Actual pair-extension reduction). *Assume $X \geq 8$. For an active output prime p , put*

$$K_{-p}(m) = \sum_{q \neq p} f_q(m), \quad \mathbf{P}_p = \sum_{q \neq p} J_{p,q},$$

and define the ordered third-extension load

$$\mathbf{T}_p = \sum_{m \in \Omega_p} (K_{-p}(m))_2 = \sum_{q \neq p} \sum_{m \in \Omega_p \cap \Omega_q} (K_X(m) - 2).$$

Set

$$v_p = \begin{cases} \mathbf{T}_p / \mathbf{P}_p, & \mathbf{P}_p > 0, \\ 0, & \mathbf{P}_p = 0. \end{cases}$$

Then the first conditional extension already satisfies the unconditional short-CRT bound

$$\frac{\mathbf{P}_p}{A_p} \leq \frac{4X}{X-2} \lambda_X, \tag{371}$$

and the boundary-corrected output–Palm discrepancy obeys

$$(\Delta_p)_+ \leq 7\lambda_X v_p + 17\lambda_X^2. \tag{372}$$

Consequently either of the estimates

$$v_p \ll X^{o(1)} \lambda_X \tag{373}$$

for every active p ,

$$\sum_p z_p v_p^2 \ll X^{o(1)} \lambda_X^3 \tag{374}$$

implies (HM)₃. More weakly, it is enough that there be a nonnegative $\kappa_X = X^{o(1)}$ such that

$$\sum_p z_p (v_p - \kappa_X \lambda_X)_+^2 \ll X^{o(1)} \lambda_X^3. \quad (375)$$

This last condition transfers to

$$\begin{aligned} \sum_p \frac{1}{z_p} (\mathcal{E}_p - (7\kappa_X + 17)Mz_p \lambda_X^2)_+^2 \\ \leq 49M^2 \lambda_X^2 \sum_p z_p (v_p - \kappa_X \lambda_X)_+^2. \end{aligned} \quad (376)$$

Proof. At a point of Ω_p hit by $d = K_{-p}(m)$ other rows, the last display defining T_p first chooses one of those rows and then one of the remaining $d - 1$. It therefore contributes $(d)_2$, so the two definitions agree. If $\mathsf{P}_p = 0$, no other row meets Ω_p ; hence also $\mathsf{T}_p = 0$.

For distinct $p, q \in \mathcal{P}_X$, one has $pq > M$. Each pair of marked residues consequently has at most one common representative below M , and hence

$$J_{p,q} \leq Z(p)Z(q), \quad A_p \geq \left(\frac{M}{p} - 1\right) Z(p).$$

Since $p \leq 2X$ and $\sum_q Z(q) \leq 2X\lambda_X$, these inequalities give

$$\frac{\mathsf{P}_p}{A_p} \leq \frac{p \sum_q Z(q)}{M - p} \leq \frac{4X}{X - 2} \lambda_X,$$

which proves (371).

Write $\lambda_p = \lambda_X - z_p$ and $\sigma_{2,p} = \sum_{q \neq p} z_q^2$. Expanding the ordered pair field gives

$$\mathcal{H}_{-p}(m) = (K_{-p}(m))_2 - 2 \sum_{q \neq p} (\lambda_p - z_q) f_q(m) + (\lambda_p^2 - \sigma_{2,p}).$$

If

$$J_{2,-p} = \sum_{\substack{q \neq \ell \\ q, \ell \neq p}} J_{q,\ell},$$

then the exact finite-window identity (368) becomes

$$\begin{aligned} \mathcal{E}_p &= \mathsf{T}_p - z_p J_{2,-p} \\ &\quad - 2 \sum_{q \neq p} (\lambda_p - z_q) (J_{p,q} - z_p A_q) \\ &\quad + (\lambda_p^2 - \sigma_{2,p}) (A_p - Mz_p). \end{aligned} \quad (377)$$

The second term is nonpositive. Moreover

$$|A_p - Mz_p| \leq Z(p), \quad \left(1 - \frac{2}{X}\right) Mz_p \leq A_p \leq \left(1 + \frac{2}{X}\right) Mz_p,$$

and $0 \leq \lambda_p^2 - \sigma_{2,p} \leq \lambda_X^2$. Thus, with $c_X = 4X/(X-2)$,

$$\begin{aligned} \frac{\mathsf{T}_p}{Mz_p} &\leq \left(1 + \frac{2}{X}\right) c_X \lambda_X v_p, \\ \frac{2}{Mz_p} \left| \sum_{q \neq p} (\lambda_p - z_q)(J_{p,q} - z_p A_q) \right| &\leq 2 \left(1 + \frac{2}{X}\right) (c_X + 1) \lambda_X^2, \\ \frac{|(\lambda_p^2 - \sigma_{2,p})(A_p - Mz_p)|}{Mz_p} &\leq \frac{2}{X} \lambda_X^2. \end{aligned}$$

For $X \geq 8$, the two resulting coefficients are at most 7 and 17, which proves (372).

The pointwise condition (373) implies (374) after multiplication by z_p and summation. From $\mathcal{E}_p = Mz_p \Delta_p$ and $(a+b)^2 \leq 2a^2 + 2b^2$, equation (372) gives

$$\sum_p \frac{1}{z_p} (\mathcal{E}_p)_+^2 \leq M^2 \left(98 \lambda_X^2 \sum_p z_p v_p^2 + 578 \lambda_X^5 \right).$$

This proves the first two implications using Proposition 358.

Finally, for every $\kappa_X \geq 0$,

$$(\Delta_p - (7\kappa_X + 17)\lambda_X^2)_+ \leq 7\lambda_X(v_p - \kappa_X \lambda_X)_+.$$

Multiplying by Mz_p , squaring, and summing with weight z_p^{-1} proves (376). Under (375) this is precisely the overload criterion (370), with $7\kappa_X + 17 = X^{o(1)}$. $\square$

Remark 361 (Scope of the pair-extension reduction). The identities

$$\sum_p \mathsf{P}_p = \sum_{m < M} (K_X(m))_2, \quad \sum_p \mathsf{T}_p = \sum_{m < M} (K_X(m))_3$$

show that the scalar ratio of the two factorial moments is the P_p -weighted first moment of v_p . It does not control the z_p -weighted square or overload in (374)–(375), because (371) supplies no reverse comparison between P_p and $Mz_p \lambda_X$. Thus the proposition is a stronger sufficient localization of the missing arithmetic, not an equivalent renaming of the pair–Palm characterization. Its remaining content is the concentration of actual third extensions over actual two-prime CRT loci.

Proposition 362 (Dyadic extension-load criterion). *For an active output prime p and $j \geq 0$, put*

$$\Theta_{p,j} = \frac{1}{\mathsf{P}_p} \sum_{\substack{m \in \Omega_p \\ 2^j \leq K_X(m) - 2 < 2^{j+1}}} (K_X(m) - 1), \quad (378)$$

and put $\Theta_{p,j} = 0$ when $\mathsf{P}_p = 0$. Let $\kappa_X \geq 0$, set $D_X = \kappa_X \lambda_X$, and suppose

$$\sum_{\substack{j \geq 0 \\ 2^{j+1} > D_X}} 2^{j+1} \left(\sum_p z_p \Theta_{p,j}^2 \right)^{1/2} \ll X^{o(1)} \lambda_X^{3/2}. \quad (379)$$

Then the pair-extension overload (375) holds with this κ_X . In particular, if $\kappa_X = X^{o(1)}$, then (HM)₃ follows.

Proof. For $P_p > 0$, give $m \in \Omega_p$ mass $(K_X(m) - 1)/P_p$ and put $h(m) = K_X(m) - 2$. This is a probability measure after points with $K_X(m) = 1$ are discarded, and its mean is v_p . The layers with $2^{j+1} \leq D_X$ contribute at most D_X to that mean. On every remaining layer, $h(m) < 2^{j+1}$, so

$$(v_p - D_X)_+ \leq \sum_{\substack{j \geq 0 \\ 2^{j+1} > D_X}} 2^{j+1} \Theta_{p,j}.$$

The same inequality is trivial when $P_p = 0$. Minkowski's inequality in $\ell^2(z_p)$ and (379) now give

$$\sum_p z_p (v_p - \kappa_X \lambda_X)_+^2 \ll X^{o(1)} \lambda_X^3,$$

which is (375). The final assertion follows from Proposition 360. $\square$

Remark 363 (Uniformity in a sparse window). The normalization in (379) is uniform in λ_X . In particular, a bound $\max_m K_X(m) = O(1)$ by itself gives only $\sum_p z_p v_p^2 = O(\lambda_X)$ and does not imply the required $X^{o(1)} \lambda_X^3$ estimate when λ_X is small. The pointwise condition that would suffice at this level is instead

$$\max_{m < M} (K_X(m) - 2)_+ \ll X^{o(1)} \lambda_X.$$

Thus (379) retains a genuine weighted horizontal-dispersion requirement; it does not hide a lower bound for λ_X .

Proposition 364 (Root-chart Fourier reduction). *Assume $X \geq 8$. For an active output prime p and $r \in \mathcal{Z}_p$, set*

$$N_{p,r} = \#\{a \geq 0 : ap + r < M\} = \left\lfloor \frac{M-1-r}{p} \right\rfloor + 1, \quad B_{p,r}(a) = K_X(ap+r) - 1 \quad (0 \leq a < N_{p,r}),$$

and put

$$P_{p,r} = \sum_{a < N_{p,r}} B_{p,r}(a), \quad T_{p,r} = \sum_{a < N_{p,r}} (B_{p,r}(a))_2.$$

For $0 \leq h < N_{p,r}$, define

$$S_{p,r}(h) = \sum_{a < N_{p,r}} B_{p,r}(a) \exp\left(\frac{2\pi i h a}{N_{p,r}}\right)$$

and the signed nonzero-frequency excess

$$\mathfrak{X}_{p,r}^{\text{root}} = \frac{1}{N_{p,r}} \sum_{h=1}^{N_{p,r}-1} (|S_{p,r}(h)|^2 - P_{p,r}). \quad (380)$$

If $P_p > 0$, set

$$\beta_p = \frac{1}{P_p} \sum_{r \in \mathcal{Z}_p} \frac{P_{p,r}(P_{p,r} - 1)}{N_{p,r}}, \quad \mathfrak{X}_p^{\text{root}} = \frac{1}{P_p} \sum_{r \in \mathcal{Z}_p} \mathfrak{X}_{p,r}^{\text{root}};$$

set both quantities to zero when $P_p = 0$. Then

$$P_p = \sum_{r \in \mathcal{Z}_p} P_{p,r}, \quad \mathfrak{T}_p = \sum_{r \in \mathcal{Z}_p} T_{p,r}, \quad (381)$$

$$v_p = \beta_p + \mathfrak{X}_p^{\text{root}}, \quad 0 \leq \beta_p \leq c_X \lambda_X, \quad c_X = \frac{4X}{X-2}. \quad (382)$$

Consequently

$$(v_p - c_X \lambda_X)_+ \leq (\mathfrak{X}_p^{\text{root}})_+. \quad (383)$$

In particular, the estimate

$$\sum_p z_p (\mathfrak{X}_p^{\text{root}})_+^2 \ll X^{o(1)} \lambda_X^3 \quad (384)$$

implies (HM)₃.

Proof. Because $r \in \mathcal{Z}_p$, the value $B_{p,r}(a)$ is precisely the number of rows other than p through $ap + r$. Summing once and then counting ordered pairs of such rows gives (381). Parseval gives

$$\frac{1}{N_{p,r}} \sum_{h=1}^{N_{p,r}-1} |S_{p,r}(h)|^2 = \sum_{a < N_{p,r}} B_{p,r}(a)^2 - \frac{P_{p,r}^2}{N_{p,r}}.$$

After subtracting $(N_{p,r} - 1)P_{p,r}/N_{p,r}$, this becomes the exact identity

$$T_{p,r} = \frac{P_{p,r}(P_{p,r} - 1)}{N_{p,r}} + \mathfrak{X}_{p,r}^{\text{root}}.$$

Summing in r and dividing by P_p proves the first identity in (382).

For $q \neq p$ and $s \in \mathcal{Z}_q$, the congruence $ap + r \equiv s \pmod{q}$ has at most one solution with $0 \leq a < N_{p,r}$, because $N_{p,r} \leq X < q$. Hence

$$P_{p,r} \leq \sum_{q \neq p} Z(q) \leq 2X \lambda_X.$$

Also $N_{p,r} > M/p - 1 \geq (X - 2)/2$, since $p \leq 2X$ and $M = X^2$. Thus $P_{p,r}/N_{p,r} \leq c_X \lambda_X$, which proves the remaining claims in (382). Inequality (383) follows by taking positive parts, and (384) is exactly (375) with $\kappa_X = c_X = O(1)$. Proposition 360 completes the proof. $\square$

Remark 365 (Arithmetic content of the root-chart excess). For $q \neq p$ and $s \in \mathcal{Z}_q$, the possible token coordinate is

$$\alpha_{p,r;q,s} = [(s - r)p^{-1}]_q;$$

it contributes exactly when $\alpha_{p,r;q,s} < N_{p,r}$. Thus (384) is a one-sided incomplete-CRT Fourier estimate for these actual marked residues after the uniform birthday baseline has already been removed. It is not a consequence of Parseval: a reflected anchored-star family concentrates many tokens at one coordinate and violates the estimate while retaining the primewise reflection, no-adjacency, bounded-codegree, and scalar factorial-moment bounds. The unconditional pointwise estimate only gives

$$\sum_p z_p (\mathfrak{X}_p^{\text{root}})_+^2 \leq 4X^2 \lambda_X^3,$$

so the remaining assertion is an actual cross-prime placement estimate, not a renamed form of (HM)₃. The superscript “root” distinguishes this exact birthday-centered quantity from pair–Palm excesses denoted by Ξ_p elsewhere.

Proposition 366 (Fixed-order root birthday reduction). *Assume $X \geq 8$, fix an integer $k \geq 3$, and retain the root charts $N_{p,r}, B_{p,r}, P_{p,r}$ and P_p of Proposition 364. Put*

$$\begin{aligned} T_{p,r}^{(k)} &= \sum_{a < N_{p,r}} (B_{p,r}(a))_{k-1}, \\ Q_{p,r}^{(k)} &= \frac{(P_{p,r})_{k-1}}{N_{p,r}^{k-2}}, \\ X_{p,r}^{(k)} &= T_{p,r}^{(k)} - Q_{p,r}^{(k)}. \end{aligned}$$

When $P_p > 0$, define

$$X_p^{(k)} = \frac{1}{P_p} \sum_{r \in \mathcal{Z}_p} X_{p,r}^{(k)},$$

and set $X_p^{(k)} = 0$ otherwise. If

$$\sum_p z_p (X_p^{(k)})_+^2 \ll X^{o(1)} \lambda_X^{2k-3}, \quad (385)$$

then $(\text{HM})_k$ holds. In particular, the single estimate

$$\boxed{\sum_p z_p (X_p^{(7)})_+^2 \ll X^{o(1)} \lambda_X^{11}} \quad (386)$$

implies

$$\log G_n \ll n^{20/21+o(1)} = o(n)$$

for every n .

Proof. The short-CRT estimate in the proof of Proposition 364 gives

$$\frac{P_{p,r}}{N_{p,r}} \leq c_X \lambda_X, \quad c_X = \frac{4X}{X-2}.$$

Consequently

$$Q_{p,r}^{(k)} \leq P_{p,r} \left(\frac{P_{p,r}}{N_{p,r}} \right)^{k-2} \leq P_{p,r} (c_X \lambda_X)^{k-2}. \quad (387)$$

At a point with $K_X(m) = t$, summing $T_{p,r}^{(k)}$ over the t output rows through m gives $t(t-1)_{k-1} = (t)_k$. Hence

$$\sum_{p,r} T_{p,r}^{(k)} = \sum_{m < M} (K_X(m))_k. \quad (388)$$

Moreover

$$\sum_p P_p = \sum_{m < M} (K_X(m))_2 \leq 5M \lambda_X^2, \quad P_p \leq c_X M z_p \lambda_X.$$

For the second inequality, one may sum $P_{p,r} \leq \sum_{q \neq p} Z(q) \leq 2X \lambda_X$ over $r \in \mathcal{Z}_p$ and use $p \leq 2X$.

Combining these facts and discarding negative residuals gives

$$\begin{aligned} \sum_{m < M} (K_X(m))_k &\leq 5M \lambda_X^2 (c_X \lambda_X)^{k-2} + \sum_p P_p (X_p^{(k)})_+ \\ &\leq O_k(M \lambda_X^k) + (5M \lambda_X^2)^{1/2} \left(c_X M \lambda_X \sum_p z_p (X_p^{(k)})_+^2 \right)^{1/2} \\ &\ll X^{o(1)} M \lambda_X^k \end{aligned}$$

under (385). This is $(\text{HM})_k$. The final assertion is Theorem 331 with $k = 7$. $\square$

Corollary 367 (Maximally padded Palm reduction). *For $P_p > 0$, put*

$$U_p^{(k)} = \frac{1}{P_p} \sum_{r \in \mathcal{Z}_p} T_{p,r}^{(k)}, \quad D_p^{(k)} = \left(U_p^{(k)} - (c_X \lambda_X)^{k-2} \right)_+,$$

and set both quantities to zero when $P_p = 0$. Either estimate

$$\sum_p P_p D_p^{(k)} \ll X^{o(1)} M \lambda_X^k, \quad (389)$$

$$\sum_p P_p (D_p^{(k)})^2 \ll X^{o(1)} M \lambda_X^{2k-2} \quad (390)$$

implies $(\text{HM})_k$; the second implies the first. For $k = 7$ the right sides are respectively $X^{o(1)} M \lambda_X^7$ and $X^{o(1)} M \lambda_X^{12}$.

Proof. The identity (388) gives

$$\sum_{m < M} (K_X(m))_k = \sum_p P_p U_p^{(k)} \leq (c_X \lambda_X)^{k-2} \sum_p P_p + \sum_p P_p D_p^{(k)}.$$

The first term is $O_k(M \lambda_X^k)$ because $\sum_p P_p \leq 5M \lambda_X^2$; hence (389) proves $(\text{HM})_k$. Cauchy–Schwarz and the same bound for $\sum_p P_p$ show that (390) implies (389). $\square$

Proposition 368 (Colour decomposition and the first new HM7 layer). *Assume $X \geq 8$ and retain the root charts of Proposition 366. Fix (p, r) , abbreviate $N = N_{p,r}$ and $P = P_{p,r}$, and, for every $q \neq p$, put*

$$f_q(a) = \mathbf{1}_{\{ap+r \bmod q \in \mathcal{Z}_q\}}, \quad n_q = \sum_{a < N} f_q(a), \quad g_q(a) = f_q(a) - \frac{n_q}{N}.$$

For a set S of partner primes, write

$$I_{p,r}(S) = \sum_{a < N} \prod_{q \in S} f_q(a), \quad C_{p,r}(S) = \sum_{a < N} \prod_{q \in S} g_q(a),$$

and let e_d^{-S} be the degree- d elementary symmetric polynomial in the n_q with $q \notin S$. For $2 \leq j \leq 6$, define

$$H_{p,r}^{[j]} = \frac{6!}{N^{6-j}} \sum_{|S|=j} e_6^{-S} C_{p,r}(S), \quad (391)$$

and put

$$\Delta_{p,r} = \frac{(P)_6 - 6! e_6((n_q)_q)}{N^5} \geq 0.$$

Then the order-seven root residual has the exact decomposition

$$X_{p,r}^{(7)} = -\Delta_{p,r} + \sum_{j=2}^6 H_{p,r}^{[j]}. \quad (392)$$

More precisely, set

$$\mathcal{A}_{2;p,r} = \sum_{|S|=2} e_4^{-S} I_{p,r}(S), \quad \mathcal{A}_{3;p,r} = \sum_{|S|=3} e_3^{-S} I_{p,r}(S).$$

The first two nonzero components satisfy

$$H_{p,r}^{[2]} = \frac{720}{N^4} \left(\mathcal{A}_{2;p,r} - \frac{15}{N} e_6((n_q)_q) \right), \quad (393)$$

$$H_{p,r}^{[3]} = \frac{720}{N^3} \mathcal{A}_{3;p,r} - 4H_{p,r}^{[2]} - \frac{14400}{N^5} e_6((n_q)_q). \quad (394)$$

Average these quantities over $r \in \mathcal{Z}_p$ with the factor P_p^{-1} , using value zero when $P_p = 0$, and define

$$\begin{aligned} \mathcal{V}_p &= \frac{720}{P_p} \sum_{r \in \mathcal{Z}_p} \frac{\mathcal{A}_{3;p,r}}{N_{p,r}^3}, \\ \mathcal{D}_p^{[3]} &= \frac{14400}{P_p} \sum_{r \in \mathcal{Z}_p} \frac{e_6((n_q)_q)}{N_{p,r}^5}. \end{aligned}$$

Then

$$H_p^{[3]} = \mathcal{V}_p - 4H_p^{[2]} - \mathcal{D}_p^{[3]}, \quad \mathcal{V}_p, \mathcal{D}_p^{[3]} \geq 0, \quad (395)$$

and

$$\mathcal{D}_p^{[3]} \leq 20c_X^5 \lambda_X^5, \quad (H_p^{[2]})_- \leq 15c_X^5 \lambda_X^5. \quad (396)$$

Consequently, when $\lambda_X > 0$, the estimate

$$\boxed{\sum_p z_p \left(\frac{\mathcal{V}_p}{\lambda_X^3} \right)^2 \ll X^{o(1)} \lambda_X^5} \quad (397)$$

implies

$$\sum_p z_p (H_p^{[3]})_+^2 \ll X^{o(1)} \lambda_X^{11}. \quad (398)$$

Conversely, (398) together with the localized rank-two bound

$$\sum_p z_p (H_p^{[2]})_+^2 \ll X^{o(1)} \lambda_X^{11} \quad (399)$$

implies (397). The pair-extension overload (375) implies (399).

Unconditionally one has the comparison

$$\mathcal{V}_p \leq 20c_X^3 \lambda_X^3 U_p^{(4)}, \quad \sum_p z_p \left(\frac{\mathcal{V}_p}{\lambda_X^3} \right)^2 \leq 6400c_X^6 X^4 \lambda_X^5. \quad (400)$$

Proof. At one coordinate, six distinct token labels necessarily have six distinct prime colours. Expanding each $f_q = n_q/N + g_q$ therefore gives the components (391). Their zero-colour term is $6!e_6((n_q)_q)/N^5$, whereas the token Fourier zero mode in $X_{p,r}^{(7)}$ is $(P)_6/N^5$. Their nonnegative difference is $\Delta_{p,r}$, proving (392).

For a triple $S = \{i, j, k\}$, direct centering gives

$$C_{p,r}(S) = I_{p,r}(S) - \frac{n_k I_{p,r}(\{i, j\}) + n_j I_{p,r}(\{i, k\}) + n_i I_{p,r}(\{j, k\})}{N} + \frac{2n_i n_j n_k}{N^2}.$$

Writing $e_6 = e_6((n_q)_q)$, the elementary symmetric counts

$$\sum_{k \notin \{i, j\}} n_k e_3^{-\{i, j, k\}} = 4e_4^{-\{i, j\}}, \quad \sum_{|S|=2} \left(\prod_{q \in S} n_q \right) e_4^{-S} = 15e_6, \quad \sum_{|S|=3} \left(\prod_{q \in S} n_q \right) e_3^{-S} = 20e_6$$

give (393)–(394). After output averaging this is (395).

Since $e_6((n_q)) \leq P^6/720$ and $P/N \leq c_X \lambda_X$, one gets $\mathcal{D}_p^{[3]} \leq 20c_X^5 \lambda_X^5$. Equation (393) and $\mathcal{A}_{2;p,r} \geq 0$ give the second inequality in (396). Now

$$\begin{aligned} (H_p^{[3]})_+ &\leq \mathcal{V}_p + 4(H_p^{[2]})_-, \\ \mathcal{V}_p &\leq (H_p^{[3]})_+ + 4(H_p^{[2]})_+ + \mathcal{D}_p^{[3]}. \end{aligned}$$

Squaring, summing with weight z_p , and using $\sum_p z_p = \lambda_X$ proves the two implications between (397) and (398).

Finally, chartwise

$$\mathcal{A}_{3;p,r} \leq \frac{P_{p,r}^3}{36} T_{p,r}^{(4)},$$

because $e_3^{-S} \leq P^3/6$ and the number of unordered physical triples is $T_{p,r}^{(4)}/6$. This proves the first inequality in (400). Also $T_{p,r}^{(4)} \leq (\sum_q n_q)^2 P_{p,r}$ and $\sum_q n_q \leq 2X \lambda_X$, so $U_p^{(4)} \leq 4X^2 \lambda_X^2$. Squaring and using $\sum_p z_p = \lambda_X$ proves the second inequality.

To finish the rank-two assertion, the same expansion gives

$$H_p^{[2]} \leq 15c_X^4 \lambda_X^4 U_p^{(3)} = 15c_X^4 \lambda_X^4 v_p.$$

Under (375), the elementary inequality $v_p \leq (v_p - \kappa_X \lambda_X)_+ + \kappa_X \lambda_X$ gives $\sum_p z_p v_p^2 \ll X^{o(1)} \lambda_X^3$. This is exactly (399) after restoring the factor λ_X^8 . $\square$

Remark 369 (The fresh seven-characteristic wall). The square in (397) contains two physical points sharing one output prime and carrying two disjoint triples of partner primes. Thus its first genuinely new term has seven distinct prime labels. Keeping the two physical points separate factors this term into two four-modulus interval counts with a common output prime; the elementary incomplete-CRT remainder loses X^2 in each block, which is the X^4 in (400). Collapsing the two points to a single seven-modulus CRT variable gives an incorrect X^5 ledger. Primewise reflection only makes the centred Fourier coefficients real and even, and does not lower the phase rank. Hence (397) is a new cross-characteristic estimate; it is not supplied by the presently stated scalar factorial moments or rowwise Hasse splitting facts. Ranks four through six in (392) also remain to be controlled.

Remark 370 (Exact multi-frequency meaning and current status). The subtraction in (385) is the zero-frequency baseline for ordered distinct token labels, not an independence assertion or a baseline in which the tokens are first grouped by their prime colours. Same-colour coincidences, which cannot occur physically, cancel against nonzero frequencies in the displayed Fourier identity. If the $P_{p,r}$ tokens have coordinates $\alpha_1, \dots, \alpha_{P_{p,r}} \in \{0, \dots, N_{p,r} - 1\}$, then, with $s = k - 1$,

$$T_{p,r}^{(k)} = \sum_{t_1, \dots, t_s \text{ distinct}} \frac{1}{N_{p,r}^{s-1}} \sum_{h_1, \dots, h_{s-1} \bmod N_{p,r}} \exp \left(\frac{2\pi i}{N_{p,r}} \sum_{i=1}^{s-1} h_i (\alpha_{t_i} - \alpha_{t_s}) \right).$$

The zero frequency is exactly $Q_{p,r}^{(k)}$; all other frequency vectors comprise $X_{p,r}^{(k)}$. Thus (386) is an incomplete-CRT six-token correlation estimate, and is a stronger sufficient localization of (HM)₇, not a proved reformulation of it.

The third-order estimate (384) does not by itself control this six-token correlation by any ordinary factorial-moment interpolation. Exact computation at the tested dyadic scales through $X = 524288$ finds $\max_{m < M} K_X(m) = 6$, attained at two points for $X = 131072$, three points for $X = 262144$, and seven points for $X = 524288$, but no point of load at least seven. The 42

incidences at the last scale were independently recomputed from the direct Apéry binomial sum. Thus $T_{p,r}^{(7)} = 0$ and the left side of (386) still vanishes throughout the tested range. This is finite evidence only: no uniform bound for the actual maximum load is known.

Corollary 367 is a strictly weaker sufficient target. Indeed, subtracting the raw-power baseline

$$\frac{1}{P_p} \sum_r \frac{P_{p,r}^{k-1}}{N_{p,r}^{k-2}}$$

from $U_p^{(k)}$ gives a residual bounded above by $X_p^{(k)}$, since $P^{k-1} \geq (P)_{k-1}$, while the raw-power baseline is at most $(c_X \lambda_X)^{k-2}$. Thus $D_p^{(k)} \leq (X_p^{(k)})_+$, and (385) implies (390). Neither Palm estimate is presently proved for the actual Apéry zero sets.

Remark 371 (Ordinary finite-population Palm form). The preceding boundary-corrected quantity differs only by a relative $O(X^{-1})$ term from an ordinary conditional covariance. Indeed, put

$$A_s = |\Omega_s|, \quad \hat{z}_s = A_s/M, \quad \epsilon_s = z_s - \hat{z}_s, \quad \tilde{g}_s = f_s - \hat{z}_s,$$

and define

$$\tilde{\mathcal{H}}_{-p}(m) = \sum_{\substack{q, \ell \neq p \\ q \neq \ell}} \tilde{g}_q(m) \tilde{g}_\ell(m), \quad \mathcal{D}_p^{\text{fin}} = \frac{1}{A_p} \sum_{m \in \Omega_p} \tilde{\mathcal{H}}_{-p}(m) - \frac{1}{M} \sum_{m < M} \tilde{\mathcal{H}}_{-p}(m).$$

Thus $\mathcal{D}_p^{\text{fin}}$ is exactly the increase in the ordered-pair covariance of the other rows after conditioning on the event $f_p = 1$. Writing $\tilde{C}_{a,b} = \sum_{m < M} \tilde{g}_a(m) \tilde{g}_b(m)$, direct expansion gives

$$\mathcal{E}_p = A_p \mathcal{D}_p^{\text{fin}} + \mathcal{R}_p, \tag{401}$$

where

$$\begin{aligned} \mathcal{R}_p = & -\epsilon_p \sum_{\substack{q, \ell \neq p \\ q \neq \ell}} \tilde{C}_{q,\ell} - 2 \sum_{\ell \neq p} \tilde{C}_{p,\ell} \sum_{q \neq p, \ell} \epsilon_q \\ & - M \epsilon_p \sum_{\substack{q, \ell \neq p \\ q \neq \ell}} \epsilon_q \epsilon_\ell. \end{aligned}$$

Since $|A_s - M z_s| \leq Z(s)$, $\sum_s |\epsilon_s| \leq 2\lambda_X/X$. The proved second-factorial bound gives, with $J_{q,\ell} = |\Omega_q \cap \Omega_\ell|$,

$$\left| \sum_{q \neq \ell} \tilde{C}_{q,\ell} \right| = \left| \sum_{q \neq \ell} J_{q,\ell} - M \sum_{q \neq \ell} \hat{z}_q \hat{z}_\ell \right| \ll M \lambda_X^2.$$

Moreover $q\ell > M$ for distinct $q, \ell \in \mathcal{P}_X$, so $J_{q,\ell} \leq Z(q)Z(\ell)$, and hence

$$\sum_{\ell \neq p} |\tilde{C}_{p,\ell}| \leq \sum_{\ell \neq p} J_{p,\ell} + M \hat{z}_p \sum_{\ell \neq p} \hat{z}_\ell \ll M z_p \lambda_X.$$

Consequently

$$\mathcal{R}_p \ll \frac{M z_p \lambda_X^2}{X}, \quad \Delta_p = \frac{A_p}{M z_p} \mathcal{D}_p^{\text{fin}} + O\left(\frac{\lambda_X^2}{X}\right), \quad \frac{A_p}{M z_p} = 1 + O(X^{-1}).$$

Thus the first estimate in (369) is precisely a one-anchor, two-extension finite Palm decorrelation statement at exponent level. The second-factorial estimate alone does not imply it: conditioning loses the output density z_p .

Corollary 372 (The anchored star fails the output-prime estimate). *For the family of Proposition 352, let $\mathcal{E}_p^*$ denote the corresponding output-prime totals. Then*

$$\sum_{\substack{p \in \mathcal{P}_X \\ Z^*(p) > 0}} \frac{p}{Z^*(p)} (\mathcal{E}_p^*)_+^2 \gg X^6 (\lambda_X^*)^5. \quad (402)$$

In particular the left side of (366) exceeds its proposed scale by a factor $X^{2-o(1)}$ on this model.

Proof. The factorial-moment lower bound in Proposition 352, together with (358) and (361), gives

$$\mathcal{C}_X^{\times,*} = \sum_p \mathcal{E}_p^* \gg X^3 (\lambda_X^*)^3 > 0.$$

Since $z_p^* = Z^*(p)/p$ and $\sum_p z_p^* = \lambda_X^*$, weighted Cauchy–Schwarz gives

$$\mathcal{C}_X^{\times,*} \leq \sum_p (\mathcal{E}_p^*)_+ \leq (\lambda_X^*)^{1/2} \left(\sum_p \frac{p}{Z^*(p)} (\mathcal{E}_p^*)_+^2 \right)^{1/2}.$$

Squaring proves (402). Notice that the weight sum is exactly λ_X^* , not $X\lambda_X^*$; this is where the full factor X^2 in the comparison is retained. $\square$

Remark 373 (Physical-space output-prime data). The independent sparse physical-space scanner implements the exact identity $\sum_p \mathcal{E}_p = \mathcal{C}_X^{\times}$ and checks it to machine precision. The following values are

$$\frac{\sum_p (p/Z(p)) (\mathcal{E}_p)_+^2}{X^4 \lambda_X^5}.$$

They are finite data, not an asymptotic estimate.

X	128	256	512	1024	2048
actual Apéry	0.1142	0.0298	0.0428	0.0410	0.0351
anchored star	1618.6	7878.1	35304.8	139967.6	585190.4

For the actual rows the same scan was continued to $X = 8192$. The first row below is the preceding normalized positive square; the second is the pointwise ratio suggested by (369):

X	128	256	512	1024	2048	4096	8192
$\frac{\sum_p z_p^{-1} (\mathcal{E}_p)_+^2}{X^4 \lambda_X^5}$	.1142	.0298	.0428	.0410	.0351	.0136	.0131
$\max_p \frac{(\mathcal{E}_p)_+}{X^2 \lambda_X^2 z_p}$	.691	.870	.796	.780	.615	.556	.721

Thus, at these seven values, every output prime lies below the natural baseline $Mz_p \lambda_X^2$ (that is, $\tau_X = 1$); both the overload mass and the left side of (370) are numerically zero. This is finite evidence for the pointwise output–Palm estimate, not a proof of it. The sharper baseline $Mz_p \varrho_p$ is exceeded once on this grid: at $(X, p) = (256, 443)$ the ratio is 1.0156951647. In the anchored-star control the corresponding relaxed maximum ratio grows from 54.5 at $X = 128$ to 1043.0 at $X = 2048$, and its overload square has the same scale as its full output-prime square. Every anchored-star output total in this range is positive; the proof above deliberately uses only their positive aggregate, so it does not depend on that additional finite observation. The actual data are consistent with (366), but do not prove it.

Remark 374 (Where the elementary loss occurs). The factorization in the proof has operator norm $O(X^{-1})$, while its Hilbert–Schmidt norm squared is only $O(X^{-1})$. Squaring after an early Cauchy inequality in the frequency variable therefore loses the full factor X still missing from the theorem. Even ideal orthogonality of the individual ordered-semiprime blocks does not provide the five-density scale in (366); the remaining input must control their signed scalar correlations with the actual output vector R_p .

Proposition 375 (Ramanujan decomposition of the primitive sector). *Fix one ordered triple of distinct odd primes p, q, ℓ , put $Q = pql$, and write*

$$Z_D = \prod_{s|D} Z(s), \quad H_D(M) = \sum_{0 \leq m < M} \prod_{s|D} f_s(m) \quad (D | Q),$$

with $Z_1 = 1$ and $H_1(M) = M$. If $\mathbf{r} = (r_p, r_q, r_\ell)$ ranges over $\mathbb{Z}_p \times \mathbb{Z}_q \times \mathbb{Z}_\ell$, let $x_{\mathbf{r}} \bmod Q$ be its Chinese-remainder lift. Then

$$\sum_{0 \leq m < M} g_p(m) g_q(m) g_\ell(m) = \frac{1}{Q} \sum_{0 \leq m < M} \sum_{\mathbf{r}} c_Q(x_{\mathbf{r}} - m) \quad (403)$$

$$= \sum_{e|Q} \mu(e) \frac{Z_e}{e} H_{Q/e}(M), \quad (404)$$

where c_Q is the Ramanujan sum. Equivalently, the right-hand side is

$$\begin{aligned} & H_Q - z_p H_{q\ell} - z_q H_{p\ell} - z_\ell H_{pq} \\ & + z_p z_q H_\ell + z_p z_\ell H_q + z_q z_\ell H_p - M z_p z_q z_\ell. \end{aligned} \quad (405)$$

After summing over ordered prime triples, the aggregate of every term in (405) except H_Q is $O(M\lambda_X^3)$, whereas

$$\sum_{p,q,\ell}^* H_{pq\ell}(M) = \sum_{m < M} (K_X(m))_3.$$

Thus Ramanujan–Möbius inversion isolates the same top-dimensional obstruction as Theorem 351; its signs alone give no additional saving.

Proof. For squarefree Q ,

$$\frac{c_Q(t)}{Q} = \prod_{s|Q} \left(\mathbf{1}_{s|t} - \frac{1}{s} \right).$$

Summing this product independently over the three coordinates of $\mathbf{r}$ proves (403). Now use

$$c_Q(t) = \sum_{\substack{d|Q \\ d|t}} d \mu(Q/d).$$

For fixed $d | Q$, the coordinates at primes dividing d are forced to hit the row m , while the remaining coordinates are free. Their number is $Z_{Q/d} H_d(M)$. Replacing d by Q/e proves (404), and expansion gives the eight strata.

For one of the three pair strata, the ordered aggregate is at most

$$\sum_{q \neq \ell} H_{q\ell}(M) \sum_{p \notin \{q, \ell\}} z_p \leq \lambda_X \sum_{m < M} (K_X(m))_2 \leq 5M\lambda_X^3.$$

For one single-column stratum it is at most

$$\lambda_X^2 \sum_{\ell} H_{\ell}(M) = M\lambda_X^2 \mu \leq (1 + 2/X)M\lambda_X^3,$$

and the constant stratum is at most $M\lambda_X^3$. There are three pair and three single-column placements. This proves the aggregate assertion and identifies the top stratum. $\square$

Remark 376 (The real kernel alone gives no saving). Apply Corollary 355 to the anchored-star family of Proposition 352. That family has all the listed reflection, spacing, interval, codegree, fiber-energy, and gap-polynomial properties, whereas

$$(\mathcal{C}_X^{\times})_+ \gg X^3(\log X)^{-3}.$$

Indeed, Proposition 352 gives this lower bound for the third factorial moment, while (358) and (361) change it by only $O(X^2(\log X)^{-3})$. Thus pairing the phases exposes a useful real oscillatory kernel, but no power saving follows from that kernel together with the presently known one-prime structural inputs. Any successful estimate must use an additional cross-characteristic property of the actual Apéry zero sets.

For the least absolute representative of h ,

$$|W_M(h/Q)| \leq \min\{M, CQ/|h|\}.$$

As $Q/M \asymp X$, the first frequency range has length $\asymp X$. On the other hand the root-choice bound gives only

$$\sum_{m < M} (K_X(m))_3 \leq \left(\sum_{X < p \leq 2X} Z(p) \right)^3 \leq 8X^3 \lambda_X^3.$$

Thus a proof of Proposition 354's final bound must save a full factor $X^{1-o(1)}$ over the present uniform tuple estimate. Parseval and Cauchy–Schwarz alone give only

$$\left| \sum_{m < M} g_p(m) g_q(m) g_{\ell}(m) \right| \leq \sqrt{M Z(p) Z(q) Z(\ell)},$$

so the required input is genuinely horizontal cancellation across the varying characteristics, not another one-prime energy estimate. The independent script `research/scripts/q32_hm3_fourier_reduction_verify` checks the physical-space identities, the complete squarefree-modulus Fourier formula, and the pair-representative formula on small exact Apéry zero sets.

Computational findings

The direct CRT enumerator in `hm3_explore.py`, independently cross-checked against an m -scatter, gives the following ordered third factorial moments:

X	$\sum_{m < X^2} (K_X(m))_3$	$X^2 \lambda_X^3$	R_3
256	60	131.989926	0.454580146
512	150	209.308570	0.716645284
1024	486	583.330960	0.833146247
2048	1824	1953.964920	0.933486564
4096	7218	7608.237134	0.948708600

At $X = 4096$, the 1203 canonical prime triples are supported on 1161 integers; 1147 have $K_X = 3$ and 14 have $K_X = 4$. The conditional third-prime extension rate per unordered pair hit is 0.073422, close to $\lambda_X = 0.0768283$. Quotient repetitions, reflection-orbit pileups, and residue correlations show no visible clustering. Full histograms and reproduction details are in `hm3_exploration.md`. These data are evidence for $(\text{HM})_3$, not a proof.

The independent centered-frequency scan `research/scripts/q32_hm3_centered_band_scan.cpp` gives a more diagnostic view of the same cancellation. The first four frequency bands have width X . In the table, $C^\times$ is the complete physical-space primitive correlation, $C_{\leq 4X}^\times$ is the signed contribution of those four bands, and $A_{(0,X]}$ is the termwise absolute mass of the first band:

X	$C^\times/(X^2\lambda_X^3)$	$C_{\leq 4X}^\times/(X^2\lambda_X^3)$	$A_{(0,X]}/(X^3\lambda_X^3)$
128	0.176154299	0.144454253	0.227911420
256	-0.252614998	-0.148136857	0.164035139
512	-0.196255790	-0.195081148	0.255416532
1024	-0.084573731	-0.072995715	0.207052911

Thus the absolute first-band mass remains on the trivial $X^3\lambda_X^3$ scale, while the observed signed sum is already on the conjectural $X^2\lambda_X^3$ scale. This is finite evidence for genuine horizontal phase cancellation, not a substitute for the missing mixed-characteristic estimate.

Stall report and infrastructure audit

Goals G1 and G2 are not proved here, and no G3 hypothesis is claimed. In particular, a third-order Palm condition should not be advertised as “weaker than AP-BDH”: the latter is an L^2 condition and the two statements are not ordered without extra hypotheses. The completed deliverable is G4, Theorem 351, together with the sharpness obstruction above.

Two issues in the supplied route are material to this conclusion. First, Lucas gives only

$$m \bmod p \in \mathcal{Z}_p \implies p \mid b_m.$$

The converse can fail through the leading digit. For example, $b_2 = 73$; with $X = 64$, $p = 73$, and $m = 146 < X^2$, Lucas gives $b_{146} \equiv b_2 b_0 \equiv 0 \pmod{73}$, while $m \bmod 73 = 0 \notin \mathcal{Z}_{73}$. Hence $K_X(m) \leq \omega_{(X,2X]}(b_m)$, but equality and the corresponding factorial-moment identity asserted in the specification are false.

Second, the written proof of Lemma 108 starts with the unavailable assertion $p > N/2 \geq h$ and then says that the two residues “differ by exactly h ”. Quotient wrap invalidates that sentence even under the displayed extra inequalities: for $p = 37$ one has $\mathcal{Z}_{37} = \{17, 19\}$, and $m = 93$, $n = 128$, $N = 70$ give $37 > N/2 \geq n - m$ while the residues are 19 and 17. The lemma may be repairable by a separate wrap argument using reflection, but the proof as written cannot be used as an established HM3 input. The anchored-star proposition shows that even a repaired, stronger $O(1)$ row-codegree estimate would still not control a one-row rainbow pileup.

Finally, Proposition 315 in the second-moment section states $\|\rho\|_2^2 \leq T_N^2/N$, whereas Cauchy–Schwarz gives the reverse inequality and the proposition’s own table reports a ratio near 1.25. No step above uses that proposition.

The trivial tuple estimate used for comparison should also be read as the uniform inequality

$$\sum_{m < X^2} (K_X(m))_3 \leq \left(\sum_{X < p \leq 2X} Z(p) \right)^3 \leq 8X^3\lambda_X^3.$$

It need not be asymptotic to $(2X\lambda_X)^3$ when the zero mass is concentrated on fewer than three primes.

The universal Franel shell carrier and its content

Put

$$F_j = \sum_{i=0}^j \binom{j}{i}^3, \quad \mathcal{B}_X(Y) = \prod_{0 \leq h < X} (Y - h(h+1)),$$

and define the Newton polynomial

$$\mathcal{S}_X(Y) = \sum_{j=0}^X \frac{F_j}{(j!)^2} \prod_{0 \leq h < j} (Y - h(h+1)), \quad \mathcal{Q}_X(Y) = (X!)^2 \mathcal{S}_X(Y) \in \mathbb{Z}[Y]. \quad (406)$$

The next result gives an exact characteristic-zero carrier for the shell incidences and, independently, controls all coefficient content of that carrier.

Theorem 377 (Franel shell carrier and subexponential content). *Let*

$$\mathfrak{c}_X = \text{cont } \mathcal{Q}_X$$

be the positive gcd of the coefficients of $\mathcal{Q}_X$. Then the following statements hold.

1. *For every prime $p > X$ and every integer m ,*

$$p \mid \gcd(\mathcal{Q}_X(m(m+1)), \mathcal{B}_X(m(m+1))) \quad (407)$$

if and only if there is an $h < X$ such that

$$p \mid b_h, \quad m \equiv h \quad \text{or} \quad m \equiv -h-1 \pmod{p}.$$

In particular, for $X < p \leq 2X$, the primes in (407) are exactly the primes counted by $K_X(m)$.

2. *One has the exact content formulas*

$$\mathfrak{c}_X = \gcd_{0 \leq j \leq X} F_j \left(\frac{X!}{j!} \right)^2, \quad \mathfrak{c}_X = \gcd(F_X, X^2 \mathfrak{c}_{X-1}), \quad \mathfrak{c}_0 = 1. \quad (408)$$

For every prime p ,

$$p \mid \mathfrak{c}_X \iff p \leq X \quad \text{and} \quad p \mid F_{\lfloor X/p \rfloor}. \quad (409)$$

3. *For every prime $p \leq X$,*

$$v_p(\mathfrak{c}_X) \leq 2 \lfloor \log_p X \rfloor. \quad (410)$$

Consequently

$$\log \text{rad}(\mathfrak{c}_X) = O(X^{2/3}), \quad \boxed{\log \mathfrak{c}_X = O(X^{2/3} \log X) = o(X)}. \quad (411)$$

Proof. For an integer $t \geq 0$, the Strehl–Franel identity gives

$$\mathcal{S}_X(t(t+1)) = \sum_{j=0}^{\min(t,X)} F_j \binom{t}{j} \binom{t+j}{j}; \quad \mathcal{S}_X(t(t+1)) = b_t \quad (t \leq X). \quad (412)$$

If $p > X$ divides $\mathcal{B}_X(m(m+1))$, then, for some $h < X$,

$$(m-h)(m+h+1) \equiv 0 \pmod{p}.$$

Since $p \nmid X!$, evaluation at either affine root and (412) show that $\mathcal{Q}_X(m(m+1)) \equiv (X!)^2 b_h \pmod{p}$. This proves the first assertion. For $X < p \leq 2X$, every element of the reflection-invariant Apéry zero set $\mathcal{Z}_p$ has either itself or its reflection $p-1-r$ below X , proving the assertion about $K_X(m)$.

The Newton polynomials $\prod_{h < j} (Y - h(h+1))$ are monic. Their transition matrix to the monomial basis, and its inverse, are integral unitriangular matrices. Taking coefficient gcds in (406) therefore gives the first equality in (408); splitting off its final entry gives the second.

Write $X = qp + r$, with $0 \leq r < p$. If $j < qp$, then $p \mid X!/j!$. If $j = qp + s$, $0 \leq s \leq r$, the factorial quotient is a p -unit, while Franel–Lucas gives

$$F_{qp+s} \equiv F_q F_s \pmod{p}.$$

Taking $s = 0$ proves both directions of (409). Split its support at $Y = X^{2/3}$. The small primes contribute $O(Y)$ to the logarithm of the radical. For $p > Y$, group the primes by $q = \lfloor X/p \rfloor < X/Y$. The product in one group divides F_q , and $F_q \leq 8^q$. Summing over $q < X/Y$ proves the radical estimate in (411).

It remains to prove (410). We treat odd primes and the dyadic prime separately.

For fixed X , set

$$C_j = F_j \left(\frac{X!}{j!} \right)^2.$$

The Franel recurrence becomes

$$8C_{j-1} = (j+1)^4 C_{j+1} - (7j^2 + 7j + 2)C_j. \quad (413)$$

At an odd prime, divisibility of two consecutive terms propagates backwards without loss. Hence

$$v_p(\mathbf{c}_X) = \min(v_p(F_X), 2v_p(X) + v_p(F_{X-1})) \quad (p \text{ odd}). \quad (414)$$

To bound this last pair, let $V_0 = 0, V_1 = 1$ and

$$(n+1)^2 V_{n+1} = (7n^2 + 7n + 2)V_n + 8n^2 V_{n-1}. \quad (415)$$

We use the denominator lemma

$$L_n^2 V_n \in \mathbb{Z}, \quad L_n = \text{lcm}(1, \dots, n). \quad (416)$$

For reference, here is its short modular derivation. In Zagier's Franel parametrization [38], with $\eta_d = \eta(dz)$ and $q = e^{2\pi iz}$,

$$t = \frac{\eta_1^3 \eta_6^9}{\eta_2^3 \eta_3^9}, \quad f = \frac{\eta_2 \eta_3^6}{\eta_1^2 \eta_6^3}, \quad P(t) = 1 - 7t - 8t^2,$$

and

$$E(q) := tP(t)f(q)^3 = \frac{\eta_1^4 \eta_2 \eta_6^5}{\eta_3^4}.$$

Ligozat's criterion gives weight 3, character $\chi = (\overline{-3})$, and cusp orders $(1, 1, 0, 1)$. The weight-three Sturm bound at level 6 is 3; comparison through q^3 therefore gives

$$E(q) = \sum_{d \geq 1} \epsilon_d q^d, \quad \epsilon_d = \sum_{a|d} \chi(a) a^2 - \mathbf{1}_{2|d} \sum_{a|d/2} \chi(a) a^2 \in \mathbb{Z}.$$

Put $g(q) = \sum_{d \geq 1} \epsilon_d q^d / d^2$. The mirror-coordinate identity $Dt = tP(t)f^2$, where $D = q d/dq$, and the self-adjoint factorization of the Franel operator give

$$(\theta^2 - 7t\theta(\theta + 1) - 8t^2(\theta + 1)^2 - 2t)(fg) = t.$$

Thus $fg = \sum_{n \geq 0} V_n t^n$. Since $q(t) \in t\mathbb{Z}[[t]]$ and $f(q(t)) \in 1 + t\mathbb{Z}[[t]]$, the coefficient of t^n has denominator dividing L_n^2 , proving (416).

The exact Casoratian is

$$F_{n-1}V_n - F_nV_{n-1} = \frac{(-8)^{n-1}}{n^2}. \quad (417)$$

After multiplication by $n^2 L_n^2$, both terms on the left are divisible by the power of an odd prime occurring in (414), whereas the right side has valuation $2v_p(L_n) = 2\lfloor \log_p n \rfloor$. This proves (410) for odd p .

For $p = 2$, write

$$P_n(x) = \sum_{j=0}^n \binom{n}{j}^3 x^j, \quad e_n = v_2(\mathfrak{c}_n).$$

The content recursion gives

$$e_n = \min(v_2(F_n), 2v_2(n) + e_{n-1}). \quad (418)$$

If $n \geq 3$ is odd, the Franel recurrence at $n - 1$ shows

$$e_n = e_{n-1}, \quad v_2(F_n) \geq e_n + 1. \quad (419)$$

Indeed, $7(n - 1)^2 + 7(n - 1) + 2$ is even and the other recurrence coefficient contains $8(n - 1)^2$.

Coefficient comparison gives the polynomial identity

$$3x^2 P_n'' + 3(1 - n)xP_n' + n^2 P_n = n^2(1 + x)P_{n-1}. \quad (420)$$

We claim that every $P_n^{(r)}(1)$ is divisible by 2^{e_n} . Induct first on n and then on r . The initial derivatives are

$$P_n(1) = F_n, \quad P_n'(1) = \frac{n}{2}F_n;$$

the odd case uses (419). After differentiating (420) r times at $x = 1$, the new highest derivative has coefficient 3, a dyadic unit. Every other current-row derivative is covered by the inner induction, and the right side is divisible by $n^2 2^{e_{n-1}}$, hence by 2^{e_n} through (418). This proves the claim.

It follows that every coefficient of $P_n(1 + 2u)$ is divisible by 2^{e_n} , since

$$[u^r]P_n(1 + 2u) = \frac{2^r}{r!}P_n^{(r)}(1), \quad v_2(2^r/r!) \geq 0.$$

For $n = 2m$, specialize at $u = -1$ and use Dixon's identity:

$$2^{e_{2m}} \mid P_{2m}(-1) = (-1)^m \frac{(3m)!}{(m!)^3}.$$

Legendre's formula gives

$$v_2\left(\frac{(3m)!}{(m!)^3}\right) = 3s_2(m) - s_2(3m).$$

If $R = \lfloor \log_2(2m) \rfloor$, adding three copies of the R -bit integer m produces a first and final carry at most 1, and each of the $R - 1$ intermediate carries is at most 2. Hence the last display is at most $2R$. Equation (419) handles odd indices and proves (410) at 2.

Finally, combine the primary bound with the radical estimate. Each prime in the support contributes at most $2 \log X$, and the number of such primes is $O(X^{2/3})$. This proves the second estimate in (411) and completes the proof. $\square$

Corollary 378 (Macroscopic-star radical criterion). *Let $P = |\mathcal{P}_X| = |\{p : X < p \leq 2X\}|$ and, for $X \leq m \leq X^2$, put*

$$\mathfrak{R}_X(m) = \text{rad}_{(X, 2X]} \gcd(\mathcal{Q}_X(m(m+1)), \mathcal{B}_X(m(m+1))).$$

Then

$$K_X(m) \log X \leq \log \mathfrak{R}_X(m) \leq K_X(m) \log(2X). \quad (421)$$

Consequently the following two uniform statements are equivalent:

$$\max_{X \leq m \leq X^2} K_X(m) = o(P), \quad \sup_{X \leq m \leq X^2} \log \mathfrak{R}_X(m) = o(X).$$

Either statement implies $G_n = e^{o(n)}$ for every positive integer n . No hypothesis on the average zero mass M_1 is required.

Proof. The localization theorem says that the prime divisors of $\mathfrak{R}_X(m)$ are exactly the $K_X(m)$ distinct primes counted in the shell. Since each lies in $(X, 2X]$, this proves (421). The prime number theorem gives $P \asymp X/\log X$, and hence the stated equivalence.

For the all-index conclusion, split the bad primes in $(\sqrt{n}, n]$ into dyadic shells. On a shell $(X, 2X]$ with $X \geq \sqrt{n}$, the row $m = n$ lies at most X^2 , so its logarithmically weighted contribution is at most

$$K_X(n) \log(2X) = o(X).$$

Writing the little- o uniformly as $\varepsilon(X)X$, the geometric sum is bounded by $O(n) \sup_{X \geq c\sqrt{n}} \varepsilon(X) = o(n)$. The first shell meeting $\sqrt{n}$ contributes $O(\sqrt{n})$ trivially. Together with the proved small-prime and companion-block estimates, this gives $\log G_n = o(n)$. $\square$

Theorem 379 (Bounded-quotient triangular criterion). *For $X \leq m$, define the physical triangular carrier*

$$\mathfrak{R}_X^\Delta(m) = \text{rad}_{(X, \min(2X, m)]} \gcd(\mathcal{Q}_X(m(m+1)), \mathcal{B}_X(m(m+1))).$$

Then the all-index conjecture $G_n = e^{o(n)}$ is equivalent to the following family of statements: for every fixed $C > 1$,

$$\sup_{X \leq m \leq CX} \log \mathfrak{R}_X^\Delta(m) = o_C(X). \quad (422)$$

Equivalently, for every fixed $C > 1$,

$$\sup_{X \leq m \leq CX} \#\{X < p \leq \min(2X, m) : m \bmod p \in \mathcal{Z}_p\} = o_C(|\mathcal{P}_X|).$$

Proof. The equivalence of the two displayed conditions follows from the exact localization and the inequalities

$$K_X^\Delta(m) \log X \leq \log \mathfrak{R}_X^\Delta(m) \leq K_X^\Delta(m) \log(2X),$$

together with $|\mathcal{P}_X| \asymp X/\log X$.

Assume first (422). Fix $C_0 > 2$. For a given n , the logarithmic weight of all target primes $\sqrt{n} < p \leq n/C_0$ is at most $\vartheta(n/C_0) = O(n/C_0)$. Cover the remaining interval $(n/C_0, n]$ by $O(\log C_0)$ dyadic shells. Every lower endpoint X in this finite cover satisfies $X \geq n/(2C_0)$, hence $X \leq n \leq 2C_0X$. Applying (422) with the fixed constant $2C_0$ shows that the target-prime weight in these shells is $o(n)$. Thus

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{\substack{\sqrt{n} < p \leq n \\ p \mid b_n \bmod p}} \log p \ll \frac{1}{C_0}.$$

Letting $C_0 \rightarrow \infty$, and then invoking the proved small-prime and companion-block estimates, gives $\log G_n = o(n)$.

Conversely, assume $\log G_n = o(n)$, and fix $C > 1$. For $X \leq m \leq CX$ and sufficiently large X , every prime $p > X$ satisfies $p > \sqrt{m}$. Every prime in $\mathfrak{R}_X^\Delta(m)$ is a lower-digit target prime with $p \leq m$, so the proved block law gives $v_p(G_m) > 0$. Therefore

$$\log \mathfrak{R}_X^\Delta(m) \leq \log G_m.$$

Since $\log G_m/m \rightarrow 0$, this inequality is uniform on the tail $X \leq m \leq CX$ and yields

$$\sup_{X \leq m \leq CX} \frac{\log \mathfrak{R}_X^\Delta(m)}{X} \leq C \sup_{m \geq X} \frac{\log G_m}{m} \rightarrow 0.$$

This proves (422). □

Theorem 380 (Fixed-quotient criterion). *For an integer $q \geq 1$, put*

$$\mathfrak{R}_q(n) = \prod_{\substack{p > \sqrt{n} \text{ prime} \\ [n/p] = q \\ p \mid b_{n \bmod p}}} p.$$

Then the all-index conjecture $G_n = e^{o(n)}$ is equivalent to the countable family of one-channel estimates

$$\log \mathfrak{R}_q(n) = o_q(n) \quad \text{for every fixed integer } q \geq 1. \quad (423)$$

In particular, no estimate uniform in a growing set of quotient classes is needed.

Proof. Suppose first that (423) holds, and fix an integer $Q \geq 1$. The lower-digit target primes with quotient at most Q have total logarithmic weight

$$\sum_{q=1}^Q \log \mathfrak{R}_q(n) = o_Q(n),$$

because this is a finite sum of fixed- q estimates; by Corollary 2, its valuation-weighted contribution to $\log G_n$ is at most six times this quantity. If instead $[n/p] > Q$, then $p \leq n/(Q+1)$. Corollary 2 therefore bounds the valuation-weighted contribution of all such primes by

$$6\vartheta\left(\frac{n}{Q+1}\right) = O\left(\frac{n}{Q}\right).$$

The proved small-prime and companion-block estimates contribute $O(\sqrt{n}) + O(n^{2/3}) = o(n)$. Consequently

$$\limsup_{n \rightarrow \infty} \frac{\log G_n}{n} \ll \frac{1}{Q}.$$

Letting $Q \rightarrow \infty$ proves $G_n = e^{o(n)}$.

Conversely, suppose $\log G_n = o(n)$, and fix q . Every prime in $\mathfrak{R}_q(n)$ is a lower-digit target prime and hence divides G_n by the block law. Thus

$$\log \mathfrak{R}_q(n) \leq \log G_n = o(n),$$

which is (423). □

Corollary 381 (Fixed proportional slices of one Apéry number). *For every fixed $q \geq 1$ and all sufficiently large n ,*

$$\mathfrak{R}_q(n) = \text{rad}_{(n/(q+1), n/q]}(b_n). \quad (424)$$

Consequently the all-index conjecture is equivalent to

$$\log \text{rad}_{(n/(q+1), n/q]}(b_n) = o_q(n) \quad \text{for every fixed } q \geq 1.$$

Equivalently, if

$$N_q(n) = \#\{p \text{ prime} : n/(q+1) < p \leq n/q, p \mid b_n\},$$

then the exact remaining condition is only

$$N_q(n) = o_q\left(\frac{n}{\log n}\right) \quad \text{for every fixed } q. \quad (425)$$

Proof. The interval in (424) is exactly the condition $\lfloor n/p \rfloor = q$. For fixed q and sufficiently large n , every prime in it exceeds both $\sqrt{n}$ and every prime divisor of the fixed nonzero integer b_q . Writing $n = qp + r$, Apéry–Lucas gives

$$b_n \equiv b_q b_r \pmod{p}.$$

Thus $p \mid b_n$ if and only if $p \mid b_r$, proving the identity. Now apply Theorem 380. Finally, uniformly in the fixed proportional interval,

$$\log p = \log n + O_q(1).$$

Hence

$$\log \mathfrak{R}_q(n) = N_q(n) \log n + O_q(N_q(n)),$$

which proves the count formulation. □

Corollary 382 (Two-prefix carrier for a fixed quotient). *Fix $q \geq 1$. For $n \geq q$, put*

$$J = \left\lfloor \frac{n-q}{2q+1} \right\rfloor,$$

$$D_{q,n} = \sum_{k=0}^J \binom{n}{k}^2 \binom{n+k}{k}^2, \quad S_{q,n} = \sum_{k=0}^J \binom{n}{k} \binom{n+k}{k} F_k.$$

Also set

$$B_{q,n} = \binom{n}{J+1} \binom{n+J+1}{J+1}.$$

Then

$$\mathfrak{R}_q(n) \mid \gcd(D_{q,n}, S_{q,n}). \quad (426)$$

More precisely, if $p > \sqrt{n}$, $\lfloor n/p \rfloor = q$, and $r = n - qp$, then

$$D_{q,n} \equiv S_{q,n} \equiv b_r \pmod{p}.$$

For fixed q and all sufficiently large n , one also has the exact factorial-boundary identities

$$\mathfrak{R}_q(n) = \text{rad}_{(n/(q+1), n/q]} \gcd(D_{q,n}, B_{q,n}) = \text{rad}_{(n/(q+1), n/q]} \frac{B_{q,n}}{\text{den}(D_{q,n}/B_{q,n})}. \quad (427)$$

For every fixed q , the boundary carrier is squarefree at every window prime apart from at most one explicit midpoint, once n is sufficiently large. More precisely, put $A = J + 1$, write $n = qp + r$, and let $n/(q + 1) < p \leq n/q$ be an odd prime with $p > q + 1$. Then

$$v_p(B_{q,n}) = 1 + \mathbf{1}_{r+A \geq p} - \mathbf{1}_{r \geq A} = \begin{cases} 2, & r = (p-1)/2, \\ 1, & r \neq (p-1)/2. \end{cases} \quad (428)$$

The exceptional case determines $p = (2n + 1)/(2q + 1)$, so for fixed q, n it contains at most one prime and has logarithmic cost $O_q(\log n)$. Away from that midpoint, the normalized carrier is the following explicit nonzero unit modulo p :

$$\frac{B_{q,n}}{p} \equiv \begin{cases} q(-1)^{A-r-1} \frac{(A-r-1)!(r+A)!}{(A!)^2}, & r < (p-1)/2, \\ -(q+1) \frac{(r+A-p)!}{(A!)^2(r-A)!}, & r > (p-1)/2 \end{cases} \pmod{p}. \quad (429)$$

The normalized prefix itself has an exact first-order expansion, but it keeps a characteristic-zero quotient of the target Apéry number. For a prime

$$p > q + 1, \quad 0 \leq j \leq \frac{p-1}{2},$$

put

$$\mathfrak{d}_j = \sum_{k=0}^j \binom{j}{k}^2 \binom{j+k}{k}^2 \left(\sum_{i=0}^{k-1} \frac{1}{j-i} + \sum_{i=1}^k \frac{1}{j+i} \right) = \sum_{k=0}^j \binom{j}{k}^2 \binom{j+k}{k}^2 (H_{j+k} - H_{j-k}) \in \mathbb{Q}.$$

This is the Gessel first jet; its denominators are p -units in the displayed range. If $p \mid b_j$, then

$$\frac{D_{q,qp+j}}{p} \equiv \frac{b_j}{p} + 2q\mathfrak{d}_j \pmod{p}, \quad (430)$$

$$\frac{D_{q,(q+1)p-1-j}}{p} \equiv \frac{b_j}{p} - 2(q+1)\mathfrak{d}_j \pmod{p}. \quad (431)$$

Here b_j/p is an ordinary quotient of integers. In particular, it may not be deleted merely because $b_j \equiv 0 \pmod{p}$. There is also an exact diagonal form that removes the auxiliary prefixes. For $0 \leq j \leq J$, put

$$p_D(j) = \frac{n-j}{q}, \quad p_R(j) = \frac{n+1+j}{q+1}.$$

Writing $\mathbb{P}$ for the primes, one has, for fixed q and all sufficiently large n ,

$$\mathfrak{R}_q(n) = \prod_{\substack{0 \leq j \leq J \\ p_D(j) \in \mathbb{P}}} \gcd(b_j, p_D(j)) \prod_{\substack{0 \leq j \leq J \\ p_R(j) \in \mathbb{P} \\ p_R(j) \neq p_D(j)}} \gcd(b_j, p_R(j)). \quad (432)$$

Thus the exact pointwise obstruction is diagonal divisibility of b_j by either of two prescribed affine prime values. The exclusion in the second product counts the unique possible midpoint only once. Consequently, the family of fixed- q estimates

$$\log \gcd(D_{q,n}, S_{q,n}) = o_q(n) \quad (q = 1, 2, \dots)$$

would prove the all-index conjecture. The elementary bounds currently give only $\log D_{q,n}, \log S_{q,n} = O_q(n)$.

Proof. Write $j = \min(r, p-1-r)$. Since $p > \sqrt{n}$, one has $q < p$, and $p \geq 2j+1$. In both the direct and reflected cases,

$$n \geq (2q+1)j + q,$$

so $j \leq J < p$. Lucas's theorem now applies coefficient by coefficient. For $k \leq j$, the two binomial kernels reduce to the corresponding kernels at j . In the direct case $r = j$, every term with $k > j$ is killed by $\binom{n}{k}$. In the reflected case $r = p-1-j$, every such term is killed by $\binom{n+k}{k}$; for $k \leq j$, the two signs in the unsquared kernel cancel. Hence

$$D_{q,n} \equiv \sum_{k=0}^j \binom{j}{k}^2 \binom{j+k}{k}^2 = b_j,$$

and the Strehl–Franel identity gives

$$S_{q,n} \equiv \sum_{k=0}^j \binom{j}{k} \binom{j+k}{k} F_k = b_j.$$

Reflection gives $b_j \equiv b_r \pmod{p}$, proving the congruences and divisibility.

For (432), the direct branch $r = j$ gives

$$p = p_D(j) = \frac{n-j}{q},$$

whereas the reflected branch $r = p-1-j$ gives

$$p = p_R(j) = \frac{n+1+j}{q+1}.$$

Conversely, $j \leq J$ is exactly $(2q+1)j \leq n-q$. If either displayed affine value is prime, this inequality is equivalent to $j \leq (p-1)/2$, so it reconstructs the corresponding direct or reflected window prime. The two affine values are equal precisely at the reflection midpoint. Finally, $\gcd(b_j, p)$ is p exactly for a target and is 1 otherwise. This proves the product identity.

It remains to check the boundary carrier. In the direct case, $J+1 > r$, so Lucas's theorem gives $\binom{n}{J+1} \equiv 0 \pmod{p}$. In the reflected case, $r+J+1 \geq p$, and Lucas gives $\binom{n+J+1}{J+1} \equiv 0 \pmod{p}$. Thus every candidate prime divides $B_{q,n}$, whereas the congruence for $D_{q,n}$ says that it divides the gcd precisely when it is a target. This proves the first equality in (427); the second is the elementary reduced-denominator identity

$$\frac{B_{q,n}}{\text{den}(D_{q,n}/B_{q,n})} = \gcd(D_{q,n}, B_{q,n}).$$

The conclusion follows from Theorem 380. Finally, standard binomial bounds and $J = O_q(n)$ give the stated $O_q(n)$ logarithmic heights.

For (428), note first that $A < p$. Since $p > q+1$, also $n+A < p^2$, so Legendre's formula gives

$$B_{q,n} = \binom{n}{A} \binom{n+A}{A} = \frac{(n+A)!}{(A!)^2(n-A)!}.$$

Consequently

$$v_p(B_{q,n}) = \left\lfloor \frac{n+A}{p} \right\rfloor - \left\lfloor \frac{n-A}{p} \right\rfloor = 1 + \mathbf{1}_{r+A \geq p} - \mathbf{1}_{r \geq A}.$$

The definition $A = \lfloor (n - q)/(2q + 1) \rfloor + 1$, with $n = qp + r$, gives

$$r + A \geq p \iff r \geq \frac{p-1}{2}, \quad r \geq A \iff r \geq \frac{p+1}{2}.$$

Indeed, after clearing the positive denominator $2q + 1$, the first equivalence reduces to $2r \geq p - 1$, while the second reduces to $2r > p - 1$. This proves the valuation formula and the midpoint assertion.

It remains to verify (429). In the direct branch $A > r$ and $r + A < p$. The numerator block in $\binom{n}{A}$ crosses qp once; after division by p , its residue is

$$q(-1)^{A-r-1}(A-r-1)!r!.$$

Lucas's theorem gives $\binom{n+A}{A} \equiv \binom{r+A}{A} \pmod{p}$, yielding the first formula. In the reflected branch $A \leq r$ and $r + A \geq p$. Now $\binom{n}{A} \equiv \binom{r}{A} \pmod{p}$, while the numerator block in $\binom{n+A}{A}$ crosses $(q+1)p$ once. Removing that factor and using Wilson's theorem gives

$$\frac{1}{p} \binom{n+A}{A} \equiv -(q+1) \frac{(r+A-p)!}{A!r!} \pmod{p}.$$

Multiplication by $\binom{r}{A}$ proves the second formula. All displayed factorials have arguments in $[0, p-1]$, and $p > q + 1$, so both residues are nonzero. Thus dividing the boundary congruence by its unique p -factor does not by itself supply a second target congruence.

For the first-jet formulas, write

$$c_k(x) = \binom{x}{k} \binom{x+k}{k}.$$

On the direct diagonal $n = qp + j$, for $k \leq j$ one has modulo p^2

$$c_k(n)^2 \equiv c_k(j)^2 \left[1 + 2qp \left(\sum_{i=0}^{k-1} \frac{1}{j-i} + \sum_{i=1}^k \frac{1}{j+i} \right) \right].$$

Every term with $j < k \leq J < p$ contains one factor of p in $c_k(n)$ and therefore vanishes after squaring modulo p^2 . Summing gives

$$D_{q,n} \equiv b_j + 2qp\mathfrak{d}_j \pmod{p^2}.$$

Since $p \mid b_j$, division of this integer congruence by p proves (430). On the reflected diagonal $n = (q+1)p - 1 - j$, the two kernel factors acquire signs which cancel, while their first-order coefficient is $-(q+1)p$. Thus

$$D_{q,n} \equiv b_j - 2(q+1)p\mathfrak{d}_j \pmod{p^2},$$

which proves (431). All denominators in $\mathfrak{d}_j$ are nonzero modulo p because $2j \leq p - 1$. □

Remark 383 (Exact saturation and the surviving obstruction). Let $S_X(T) = \mathcal{S}_X(T(T+1))$, and let

$$(\mathcal{A}u)(T) = (T+1)^3u(T+1) - (34T^3 + 51T^2 + 27T + 5)u(T) + T^3u(T-1).$$

The exact WZ remainder is

$$(X!)^2 \mathcal{A}S_X(T) = -2(2T+1)\mathcal{B}_X(T(T+1)) (8F_X T(T+1) + (X+1)^2 F_{X+1} - 8X(X+1)F_X).$$

At a shell root the right side vanishes because the boundary factor does; division by that factor is not permitted there. Thus the WZ equation gives no second target congruence beyond (407).

Nor does removal of the now-controlled content make a standard polynomial sieve sufficient. Indeed,

$$\mathcal{Q}_X(0) = (X!)^2, \quad \text{lc}(\mathcal{Q}_X) = F_X, \quad \mathfrak{c}_X \leq F_X \leq 8^X,$$

so

$$\log H(\mathcal{Q}_X/\mathfrak{c}_X) \geq 2 \log(X!) - X \log 8 = 2X \log X - O(X).$$

The degree is X . Hence the exact carrier and its subexponential content do not prove either the radical estimate in Theorem 379, the full-shell criterion in Corollary 378, or the stronger estimate (386). For the all-index theorem, the exact pointwise survivor is only the exclusion of a macroscopic aligned star in every fixed bounded-quotient triangular range, as in Theorem 379; seven-prime nonalignment and the full range $m \leq X^2$ are much stronger routes. Equivalently, Theorem 380 permits the quotient classes to be treated one at a time. This reactivates fixed- q characteristic-zero carriers: a subexponential gcd bound for each fixed class is enough even if its constants and threshold depend arbitrarily on q . Uniformity for $q \rightarrow \infty$ is only a stronger sufficient route. The normalization and local recurrence identities alone prove none of these statements.

13 The atom-tail reduction

Throughout this section $X \geq 3$ is an integer parameter, $\mathcal{P}_X = \{p \text{ prime} : X < p \leq 2X\}$, $\mathcal{Z}_p = \{r \in [0, p) : p \mid b_r\}$, $Z(p) = |\mathcal{Z}_p|$,

$$K_X(m) = \#\{p \in \mathcal{P}_X : m \bmod p \in \mathcal{Z}_p\}, \quad \lambda_X = \sum_{p \in \mathcal{P}_X} \frac{Z(p)}{p},$$

and $(K)_k = K(K-1) \cdots (K-k+1)$. We write $\omega_X(N) = \#\{p \in \mathcal{P}_X : p \mid N\}$ for the number of window prime divisors of a nonzero integer N , so that $\omega_X(N) \leq \log |N| / \log X$. We freely use the second factorial moment $\sum_{m < X^2} (K_X(m))_2 \leq 5X^2 \lambda_X^2$ (Subsection 12.34) and the gap polynomials N_h of degree $3(h-1)$ with $N_h \not\equiv 0 \pmod{p}$ for $h \leq p$.

The separated root strips also give a prime-averaged improvement over the pointwise degree bound. We record it here because it will be used later to isolate the only zero-count layers which can saturate the completed reciprocal endpoint.

Proposition 384 (Average second factorial zero moment). *Uniformly for $X \geq 3$,*

$$\boxed{\sum_{X < p \leq 2X} Z(p)(Z(p) - 1) \ll \frac{X^{11/5}}{(\log X)^{3/5}}.} \quad (\text{AF1})$$

Consequently

$$\sum_{X < p \leq 2X} Z(p) \ll \frac{X^{8/5}}{(\log X)^{4/5}}. \quad (\text{AF2})$$

Proof. For $h, k \geq 2$ put

$$S_{h,k} = \text{Res}_x(N_h(x), N_k(x+h)).$$

Proposition 41 places the roots of the two polynomials in the disjoint strips

$$-h < \Re x < -1, \quad -h-k < \Re x < -h-1,$$

so $S_{h,k} \neq 0$. If $h, k < p$, the reductions of both polynomials are nonzero. Evaluation at their distinct common roots gives independent linear functionals annihilating the image of the fixed-degree Sylvester map. Hence

$$c_p(h, k) := \#\{r \in \mathbb{F}_p : N_h(r) = N_k(r+h) = 0\} \leq \text{corank}_{\mathbb{F}_p} \text{Syl}(N_h, N_k(x+h)).$$

Smith normal form of the integral Sylvester matrix then gives

$$c_p(h, k) \leq v_p(S_{h,k}). \quad (\text{AF3})$$

This fixed-size argument also covers a leading-degree drop after reduction: such a drop can enlarge the Sylvester corank, but cannot destroy the evaluation functionals.

If $H \geq h + k$, the continuant determinant and coefficient convolution give

$$\log \|N_h\|_2 \ll h \log H, \quad \log \|N_k(\cdot + h)\|_2 \ll k \log H.$$

The degrees are $3(h-1)$ and $3(k-1)$, so Hadamard's inequality applied to the Sylvester matrix yields

$$\log |S_{h,k}| \ll hk \log H. \quad (\text{AF4})$$

Thus, for $4 \leq H < X$ and

$$T_p(H) = \sum_{\substack{h,k \geq 2 \\ h+k < H}} c_p(h, k),$$

one has

$$\begin{aligned} \sum_{X < p \leq 2X} T_p(H) &\leq \sum_{h+k < H} \sum_{X < p \leq 2X} v_p(S_{h,k}) \\ &\ll \frac{1}{\log X} \sum_{h+k < H} \log |S_{h,k}| \ll \frac{H^4 \log H}{\log X}. \end{aligned} \quad (\text{AF5})$$

Partition $[0, p-1]$ into $B_p = \lceil p/H \rceil$ consecutive blocks, and let $z_{p,j}$ be the number of actual Apéry zeros in block j . Every unordered triple in one block has a unique presentation

$$r < r+h < r+h+k, \quad h, k \geq 2, \quad h+k < H,$$

and hence contributes a distinct common root counted by $T_p(H)$. Therefore

$$\sum_j \binom{z_{p,j}}{3} \leq T_p(H).$$

The elementary inequality

$$z \leq 2 + \left(6 \binom{z}{3}\right)^{1/3}$$

and Hölder give

$$Z(p) \leq 2B_p + 6^{1/3} B_p^{2/3} T_p(H)^{1/3}.$$

Writing $\Pi_X = \#\{X < p \leq 2X\} \ll X/\log X$ and $B = \max B_p \ll X/H$, squaring, summing in p , and applying Hölder once more with (AF5) gives

$$\sum_{X < p \leq 2X} Z(p)(Z(p)-1) \ll \frac{X^3}{H^2 \log X} + \frac{X^{5/3} H^{4/3} (\log H)^{2/3}}{\log X}.$$

The choice

$$H = \left\lfloor X^{2/5}(\log X)^{-1/5} \right\rfloor$$

balances the two terms and proves (AF1). Finally Cauchy–Schwarz and $Z(p)^2 = Z(p)(Z(p) - 1) + Z(p)$ give a quadratic inequality for $\sum_{X < p \leq 2X} Z(p)$; solving it proves (AF2). $\square$

13.1 The Cartier diagonal

Theorem 385 (Cartier diagonal). *Let $F = (1 - x - y)(1 - z - w) - xyzw$. For every prime p and every $0 \leq r \leq p - 1$,*

$$b_r \equiv [x^r y^r z^r w^r] F^{p-1} \pmod{p}.$$

Proof. By Straub’s diagonal representation [36], $b_n = [x^n y^n z^n w^n] F^{-1}$ for all $n \geq 0$. Modulo p one has $F^{-1} = F^{p-1} \cdot F^{-p}$ and, by the Frobenius endomorphism, $F(x, y, z, w)^p \equiv F(x^p, y^p, z^p, w^p)$. Hence

$$[x^r \cdots w^r] F^{-1} \equiv \sum_{b \geq 0} [x^{r-pb} \cdots w^{r-pb}] F^{p-1} \cdot [x^b \cdots w^b] F^{-1} \pmod{p}.$$

Since $\deg_x F^{p-1} = p - 1$ and $0 \leq r \leq p - 1$, only $b = 0$ contributes, and $[x^0 \cdots w^0] F^{-1} = b_0 = 1$. $\square$

Corollary 386 (Palindrome for all fibers). *For every prime $p \geq 5$, every $0 \leq r \leq p - 1$, and every λ : $b_r(\lambda) \equiv b_{p-1-r}(\lambda) \pmod{p}$, where $b_r(\lambda) = \sum_{k=0}^r \binom{r}{k}^2 \binom{r+k}{k}^2 \lambda^k$.*

Proof. The Cartier representation $b_r(\lambda) \equiv [x^r \cdots w^r] F_\lambda^{p-1} \pmod{p}$ holds for $F_\lambda = (1-x-y)(1-z-w) - \lambda xyzw$ (Theorem 385 with general λ). The torus involution $(x, y, z, w) \mapsto (x^{-1}, y^{-1}, z^{-1}, w^{-1})$ sends F_λ^{p-1} to $(xyzw)^{-2(p-1)} G^{p-1}$ where $G = (xyzw)^2 F_\lambda(x^{-1}, \dots)$ has degree 2 in each variable. Extracting $[x^r \cdots w^r]$ from $(xyzw)^{p-1} F_\lambda(x^{-1}, \dots)^{p-1}$ gives $[x^{p-1-r} \cdots w^{p-1-r}] F_\lambda^{p-1}$, and the two representations are equal modulo p . $\square$

Remark 387 (Uniqueness of the Apéry fiber). The polynomial family $b_r(\lambda)$ satisfies a *fourth-order* recurrence in r for generic λ (Zeilberger), which degenerates to the second-order Apéry recurrence *exactly* at $\lambda = 1$: the defect $(r+1)^3 b_{r+1}(\lambda) - P(r) b_r(\lambda) + r^3 b_{r-1}(\lambda)$ factors as $C_r \cdot (\lambda - 1) \cdot Q_r(\lambda)$ with $\gcd(Q_1, \dots, Q_7) = 1$. Thus $\lambda = 1$ is the unique rational specialization admitting the order-two certificate algebra (gap polynomials, codegree bounds, per-prime separation) on which every zero-set theorem in this paper rests.

13.2 Fourier non-concentration

Recall $F_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$ and set $A_p(h) = \#\{r : r, r+h \in \mathcal{Z}_p\}$ (integer gap, $0 \leq r < r+h \leq p-1$), so that $A_p(h) \leq 3(h-1)$: indeed $r, r+h \in \mathcal{Z}_p$ forces $N_h(r) \equiv 0 \pmod{p}$, and $N_h \not\equiv 0 \pmod{p}$ has at most $\deg N_h = 3(h-1)$ roots.

Lemma 388 (Short-window energy). *There is an absolute constant C such that for every prime p and every $1 \leq K_0 \leq p/2$,*

$$\sum_{1 \leq |k| \leq K_0} |F_p(k)|^2 \leq 4K_0 Z(p) + C \frac{p^2 \log(2K_0)}{K_0},$$

while for $K_0 > p/2$ the full Parseval bound $\sum_{k \bmod p} |F_p(k)|^2 = pZ(p) - Z(p)^2$ applies.

Proof. Let $w(k) = (1 - |k|/(2K_0))_+$ on the symmetric residue window; $w \geq \frac{1}{2}$ on $|k| \leq K_0$, so the left side is at most $2 \sum_{k \bmod p} w(k) |F_p(k)|^2 = 2 \sum_{r, r' \in \mathcal{Z}_p} \widehat{W}(r - r')$, where

$$\widehat{W}(h) = \frac{1}{2K_0} \left| \sum_{0 \leq j < 2K_0} e_p(jh) \right|^2 \leq \min\left(2K_0, \frac{p^2}{8K_0 \bar{h}^2}\right), \quad \bar{h} = \min(|h|, p - |h|).$$

The diagonal contributes $2K_0 Z(p)$. Off-diagonal pairs at integer gap g number $A_p(g) \leq 3(g - 1)$ for $g \leq p/2$, and at most $p - g$ for $g > p/2$ (the smaller element lies in $[0, p - 1 - g]$); in either case the count at circular distance $\bar{h}$ is at most $3\bar{h}$. Splitting at $g^* = p/(4K_0)$,

$$\sum_{\bar{h} \geq 1} 3\bar{h} \min\left(2K_0, \frac{p^2}{8K_0 \bar{h}^2}\right) \leq 6K_0 \sum_{\bar{h} \leq g^*} \bar{h} + \frac{3p^2}{8K_0} \sum_{g^* < \bar{h} \leq p/2} \frac{1}{\bar{h}} \ll \frac{p^2 \log(2K_0)}{K_0}. \quad \square$$

13.3 The codegree dichotomy and the twin-atom bound

For $0 \leq m < m' < X^2$ put $W(m, m') = \#\{p \in \mathcal{P}_X : m \bmod p \in \mathcal{Z}_p, m' \bmod p \in \mathcal{Z}_p\}$ and $h = m' - m$.

Lemma 389 (Codegree dichotomy). *Let $1 \leq h < X$ and let $p \in \mathcal{P}_X$ be counted by $W(m, m')$, with $r = m \bmod p$. Exactly one of the following holds.*

(A) $r + h \leq p - 1$. Then $r, r + h \in \mathcal{Z}_p$, hence $N_h(r) \equiv 0 \pmod{p}$ and, since $N_h \in \mathbb{Z}[x]$,

$$p \mid N_h(m).$$

(B) $r \geq p - h$. Then $j := m' \bmod p = r + h - p$ satisfies $0 \leq j < h$, $j \in \mathcal{Z}_p$ means $p \mid b_j$, and $p \mid m' - j$; hence

$$p \mid \gcd(b_j, m' - j) \quad \text{for some } 0 \leq j < h.$$

Proof. Since $h < X < p$, the residues satisfy $m' \bmod p \equiv r + h$ with $r + h < 2p$, giving the two cases. In case (A) both indices lie in $[0, p - 1]$ and the gap certificate applies; the congruence $N_h(m) \equiv N_h(r) \pmod{p}$ is the reduction of a polynomial with integer coefficients. Case (B) is immediate from the definitions. $\square$

Lemma 390 (Exact type-B seam transfer and weighted deletion). *Let $p \geq 7$ be prime, $1 \leq h < p$, and $0 \leq j < h$. Put*

$$k = h - 1 - j, \quad r = p - h + j = p - 1 - k.$$

Then $r + h \equiv j \pmod{p}$, and

$$\mathbf{1}_{\mathcal{Z}_p}(r) \mathbf{1}_{\mathcal{Z}_p}(j) = \mathbf{1}_{\mathcal{Z}_p}(k) \mathbf{1}_{\mathcal{Z}_p}(j). \quad (\text{AT-S1})$$

Conversely, every positive circular seam pair $r, r + h - p \in \mathcal{Z}_p$ has a unique representation of this form by $j, k \in \mathcal{Z}_p$ with $j + k = h - 1$. For the canonical representatives $m = r$, $m' = r + h$, case (B) of Lemma 389 reduces to

$$p \mid \gcd(b_j, m' - j) = \gcd(b_j, p),$$

so it is precisely the actual endpoint predicate $j \in \mathcal{Z}_p$.

Moreover, let $1 < G < p$ and let $(u_t), (v_t)$ be arbitrary complex coefficients of modulus at most B . Define

$$S^+(h) = \sum_{\substack{j+k=h-1 \\ j,k \in \mathbb{Z}_p}} u_{p-1-k} \overline{v_j}, \quad S^-(h) = \sum_{\substack{j+k=h-1 \\ j,k \in \mathbb{Z}_p}} u_j \overline{v_{p-1-k}}.$$

Then

$$\boxed{\sum_{1 \leq h < G} \frac{|S^+(h)|^2 + |S^-(h)|^2}{h} \ll B^4 G.} \quad (\text{AT-S2})$$

Thus a bounded literal coefficient may remain at its original seam label; only the zero predicate is reflected.

Proof. The integer identities are immediate. The palindrome Corollary 386 gives $b_r \equiv b_{p-1-r} = b_k \pmod{p}$, proving (AT-S1); also $m' = p + j$.

For the weighted estimate let

$$r_h = \#\{(j, k) \in \mathbb{Z}_p^2 : j + k = h - 1\}.$$

Taking absolute values gives $|S^\pm(h)| \leq B^2 r_h$. On a dyadic range $H_0 \leq h < 2H_0$, Corollary 15 gives

$$\max r_h \ll H_0^{2/3}, \quad \sum_{H_0 \leq h < 2H_0} r_h \leq |\mathbb{Z}_p \cap [0, 2H_0)|^2 \ll H_0^{4/3}.$$

Consequently

$$\sum_{H_0 \leq h < 2H_0} \frac{r_h^2}{h} \ll H_0^{-1} H_0^{2/3} H_0^{4/3} \ll H_0.$$

Summing the dyadic ranges below G proves (AT-S2). $\square$

Lemma 391 (Palindromic reindexing of transverse nonwrapping pairs). *Let $p \geq 7$ be prime and let $1 \leq h < p/2$. Suppose that $r < s$ and that*

$$r, r + h, s, s + h \in \mathbb{Z}_p$$

are four distinct representatives in $\{0, \dots, p-1\}$. Put

$$d = s - r, \quad a = \min\{d, h\}, \quad b = |d - h|. \quad (\text{AT-P1})$$

Then $a, b \geq 2$, and the four zeros in increasing order are

$$r, \quad r + a, \quad r + a + b, \quad r + 2a + b. \quad (\text{AT-P2})$$

In particular

$$N_a(r) \equiv 0 \pmod{p}, \quad N_b(r + a) \equiv 0 \pmod{p}. \quad (\text{AT-P3})$$

For fixed (r, a, b) there are at most two triples (h, r, s) which produce (AT-P1), one with $d < h$ and one with $d > h$.

Proof. If $d < h$, the increasing order is

$$r < s < r + h < s + h$$

and its consecutive gaps are $(d, h - d, d) = (a, b, a)$. If $d > h$, the order is

$$r < r + h < s < s + h$$

and its gaps are $(h, d - h, h) = (a, b, a)$. The case $d = h$ is excluded by the four-distinct hypothesis. Lemma 10 gives $a, b \geq 2$, and the nonwrapping gap dictionary gives (AT-P3). In the first ordering $h = a + b$ and $s = r + a$; in the second $h = a$ and $s = r + a + b$. This proves the multiplicity assertion. $\square$

For a set $\mathcal{A} \subseteq \mathbb{Z}_{\geq 2}^2$, let $\mathcal{Q}_p(T; \mathcal{A})$ denote

$$\sum_{T \leq h < 2T} \frac{1}{h} \# \left\{ (r, s) : \begin{array}{l} r < s, \quad r, r + h, s, s + h \in \mathbb{Z}_p, \\ r + h < p, \quad s + h < p, \quad \#\{r, r + h, s, s + h\} = 4, \\ (a, b) \in \mathcal{A} \end{array} \right\}, \quad (\text{AT-P4})$$

where (a, b) is defined by (AT-P1).

Proposition 392 (Three unconditional palindromic sectors). *Let T, K, L be positive integers with*

$$2 \leq K < T, \quad 1 \leq L \leq T, \quad 5T \leq \sqrt{p}.$$

Then

$$\mathcal{Q}_p(T; \{(a, b) : \min(a, b) \leq K\}) \ll K^2, \quad (\text{AT-P5})$$

$$\mathcal{Q}_p(T; \{(a, b) : |a - b| \leq L\}) \ll TL, \quad (\text{AT-P6})$$

$$\sum_{T \leq h < 2T} \frac{1}{h} \#\{(r, s) \text{ counted in (AT-P4)} : s - r < h\} \ll T^{5/3}. \quad (\text{AT-P7})$$

The same estimates are valid, up to multiplication by B , if every configuration in (AT-P4) carries an arbitrary complex coefficient of modulus at most B . Consequently, at $K = \lfloor T^{3/5} \rfloor$ and $L = \lfloor T^{1/5} \rfloor$, every configuration not covered by (AT-P5)–(AT-P7) lies in the separated branch $s - r > h$ and satisfies

$$b > T^{3/5} + O(1), \quad |a - b| > T^{1/5} + O(1), \quad a = h. \quad (\text{AT-P8})$$

Proof. By Lemma 391, every counted triple maps with multiplicity at most two to the actual common-root witness (r, a, b) in the collision energy.

If $\min(a, b) \leq K$ and $s - r < h$, then $a + b = h < 2T$. If $s - r > h$, then $a = h \geq T > K$, so necessarily $b \leq K$ and $a + b < 3T$. Proposition 53 therefore gives

$$\mathcal{Q}_p(T; \min(a, b) \leq K) \ll T^{-1} \mathcal{E}_p\{(a, b) : a + b \leq 3T, \min(a, b) \leq K\} \ll K^2.$$

If $|a - b| \leq L$, then $a + b < 2T$ in the interlaced branch. In the separated branch $a = h < 2T$ and $b \leq a + L < 3T$, so $a + b < 5T$. The diagonal-band part of Proposition 53, with horizon $5T$, yields

$$\mathcal{Q}_p(T; |a - b| \leq L) \ll T^{-1} \mathcal{E}_p\{(a, b) : a + b \leq 5T, |a - b| \leq L\} \ll TL.$$

Finally, on the interlaced branch $a + b = h < 2T$. Hence Corollary 47 gives

$$\sum_{T \leq h < 2T} \frac{1}{h} \# \{(r, s) : s - r < h\} \ll T^{-1} \mathcal{E}_p(2T) \ll T^{5/3}.$$

All three energy horizons are at most $\sqrt{p}$ by hypothesis. Taking absolute values proves the coefficient-weighted versions. After the interlaced branch is removed, one has $a = h > T^{3/5} + O(1)$; the two remaining exclusions therefore give (AT-P8). $\square$

Corollary 393 (Selected-Palm transfer of the three sectors). *Let $\mathcal{X}$ be a finite family of selected roots. Give $x \in \mathcal{X}$ a prime $p_x \asymp_v N$ and a nonnegative weight w_x , and put $M_0 = \sum_x w_x$. Suppose that $5T \leq \sqrt{p_x}$ for every x . For every configuration counted by (AT-P4), with $p = p_x$, let $c_x(h, r, s)$ be an arbitrary literal coefficient satisfying*

$$|c_x(h, r, s)| \ll_v N^{o(1)}.$$

All branch, moving-characteristic, distinctness, and deletion masks may remain inside this coefficient. Then the corresponding absolute selected majorants of the shallow, central, and interlaced sectors are, respectively,

$$\ll_v K^2 N^{o(1)} M_0, \quad \ll_v TL N^{o(1)} M_0, \quad \ll_v T^{5/3+o(1)} M_0. \quad (\text{AT-P9})$$

This applies in particular to a literal fixed-ratio reciprocal-lift block. Indeed, for fixed x , a nonzero distinguished label r , and a fixed residue $C_x \pmod{p_x}$, the congruence

$$\ell r \equiv C_x \pmod{p_x}$$

has $O_v(1)$ integer representatives ℓ in a fixed-ratio N -shell. Thus four fixed labels have $O_v(1)$ literal prime-lift quadruples; bounded source factors and the separated smooth expansion cost only $N^{o(1)}$. No counting-measure estimate is divided by M_0 .

Proof. Apply the triangle inequality for each selected root and then Proposition 392. The congruence for ℓ fixes one residue class modulo p_x , while a fixed-ratio shell has length $O_v(p_x)$, proving the asserted lift multiplicity. All literal masks only remove lift tuples, and every retained phase has modulus 1. Multiplication by w_x and summation over x proves (AT-P9). $\square$

Lemma 394 (Palindromic three-return reflection). *Let $p \geq 7$ be prime and let $a, b \geq 2$. Suppose*

$$0 \leq u < u + a < u + a + b < u + 2a + b \leq p - 1$$

and all four displayed integers lie in $\mathcal{Z}_p$. Put

$$\iota_{p,a,b}(u) = p - 1 - (u + 2a + b).$$

Then

$$\iota(u), \quad \iota(u) + a, \quad \iota(u) + a + b, \quad \iota(u) + 2a + b$$

are again four increasing nonwrapping elements of $\mathcal{Z}_p$, and $\iota_{p,a,b}$ is an involution. Its fixed-point equation is

$$2u + 2a + b = p - 1, \quad (\text{AT-P10})$$

so there is at most one fixed chain for each (p, a, b) .

Proof. Apply the actual Apéry palindrome $b_j \equiv b_{p-1-j} \pmod{p}$ to the four labels, in reverse order. The nonwrapping bounds are preserved. Applying the displayed affine map twice gives u , and solving its fixed-point equation gives (AT-P10). $\square$

This is an involution of the distinguished-characteristic actual-zero projection only. It need not preserve physical reciprocal lifts, source boxes, phases, or selected masks, and no reflected-summand cancellation is inferred from it.

Corollary 395 (Reflection-overlap selected-Palm deletion). *Retain the hypotheses and literal coefficient normalization of Corollary 393. In its interlaced sector put*

$$u = r, \quad a = s - r, \quad b = h - a,$$

and restrict to chains for which

$$\{u, u + a, u + a + b, u + 2a + b\} \cap \{\iota(u), \iota(u) + a, \iota(u) + a + b, \iota(u) + 2a + b\} \neq \emptyset.$$

Their selected majorant satisfies

$$\boxed{\mathfrak{J}_T^{\text{ov}} \ll_v T^{1+o(1)} M_0.} \quad (\text{AT-P11})$$

Proof. Write $A_{a,b} = \{0, a, a + b, 2a + b\}$. An intersection gives

$$2u + 2a + b + \alpha - \beta = p - 1 \quad (\alpha, \beta \in A_{a,b}).$$

For fixed (p, a, b, α, β) there is at most one integer u . There are 16 choices of (α, β) and $O(T^2)$ pairs (a, b) with $T \leq a + b < 2T$. The fixed-label reciprocal lifts and literal coefficient cost $N^{o(1)}$, the dyadic tent is at most $1/T$, and the unchanged outer weights sum to M_0 . $\square$

Lemma 396 (Three-return pencil-content certificate). *Put*

$$F_a(X) = N_a(X), \quad G_{a,b}(X) = N_b(X + a), \quad J_{a,b}(X) = N_a(X + a + b),$$

and

$$\mathcal{R}_{a,b} = \text{Res}_X(F_a, G_{a,b}), \quad \mathcal{S}_{a,b} = \text{Res}_X(F_a, J_{a,b}).$$

Both integers are nonzero. Define

$$\mathcal{P}_{a,b}(Z) = \text{Res}_X(F_a, G_{a,b} + ZJ_{a,b}) \in \mathbb{Z}[Z]$$

and let $\mathcal{C}_{a,b} > 0$ be its content. If

$$\tau_{a,b}(p) = \#\{z \in \mathbb{F}_p : F_a(z) = G_{a,b}(z) = J_{a,b}(z) = 0\},$$

then

$$\boxed{\tau_{a,b}(p) \leq v_p(\mathcal{C}_{a,b}).} \quad (\text{AT-P12})$$

In particular, two distinct simultaneous roots imply

$$p^2 \mid \mathcal{R}_{a,b}, \quad p^2 \mid \mathcal{S}_{a,b}, \quad p^2 \mid \mathcal{C}_{a,b}. \quad (\text{AT-P13})$$

Proof. Proposition 41 places the roots of $F_a, G_{a,b}, J_{a,b}$ in the respective disjoint real-part strips

$$(-a, -1), \quad (-a - b, -a - 1), \quad (-2a - b, -a - b - 1).$$

Thus the two integer resultants are nonzero. If $m = \deg F_a$, $n = \deg G_{a,b}$, $q = \max\{m, n\}$, and ℓ_a is the leading coefficient of F_a , degree-preserving specialization gives

$$\mathcal{P}_{a,b}(0) = \ell_a^{q-n} \mathcal{R}_{a,b} \neq 0.$$

This leading-coefficient factor is necessary when $a > b$.

Localize $\mathbb{Z}[Z]$ at the height-one prime (p) . This is a discrete valuation ring with residue field $\mathbb{F}_p(Z)$. Every root counted by $\tau_{a,b}(p)$ gives an independent evaluation vector in the kernel of the reduced pencil Sylvester matrix. Smith normal form over this DVR proves (AT-P12). The same evaluation-vector argument for the two ordinary integer Sylvester matrices proves (AT-P13) when the two roots are distinct. $\square$

Corollary 397 (Exact reflection-disjoint interlaced residual). *After deleting (AT-P11), the literal interlaced selected-Palm contribution satisfies*

$$|\mathfrak{J}_T^{\text{rd}}| \ll_v \frac{N^{o(1)}}{T} \sum_x w_x \sum_{\substack{a,b \geq 2 \\ T \leq a+b < 2T}} v_{p_x}(\mathcal{C}_{a,b}). \quad (\text{AT-P14})$$

All original masks may be retained inside the left side. The presently proved unconditional bound remains

$$|\mathfrak{J}_T^{\text{rd}}| \ll_v T^{5/3+o(1)} M_0;$$

no relative selected-Palm saving follows from the absolute content height alone.

Proof. For fixed (x, a, b) , a literal four-zero chain is determined by its start u , and the three exact return equations put u among the $\tau_{a,b}(p_x)$ simultaneous roots. Fixed-label reciprocal lifts and all remaining coefficients cost $N^{o(1)}$, while the tent costs $1/T$. Apply (AT-P12) and sum the unchanged weights w_x . Dropping the third return recovers the two-return energy bound (AT-P7), hence the stated $T^{5/3+o(1)} M_0$ estimate. $\square$

Corollary 398 (Gate-subcritical interlaced wing). *Retain the hypotheses and literal selected-Palm normalization of Corollary 393. Fix*

$$0 < \varepsilon < \frac{2}{155}, \quad A = T^{19/31-\varepsilon}.$$

The interlaced four-zero chains satisfying

$$T^{3/5} < \min\{a, b\} \leq A, \quad T \leq a + b < 2T$$

have selected majorant

$$\mathfrak{J}_T^{\text{wing}} \ll_v T^{38/31-2\varepsilon+o(1)} M_0. \quad (\text{AT-P15})$$

Consequently, modulo a gate-subcritical error, the unresolved interlaced core may be restricted to

$$a, b > T^{19/31-\varepsilon}, \quad |a - b| > T^{1/5}.$$

This refinement is at the canonical 38/31 gate; it does not improve the ideal 6/5 exponent.

Proof. For one distinguished characteristic, literal four-zero starts of fixed adjacent gaps (a, b) inject into the affine collision set counted by $m_{a,b}(p)$. The existing degree bound gives

$$m_{a,b}(p) \leq 3(\min\{a, b\} - 1).$$

For each $h \in [T, 2T)$ the two wings therefore contribute at most

$$\sum_{\substack{a+b=h \\ \min\{a,b\} \leq A}} m_{a,b}(p) \ll \sum_{d \leq A} d \ll A^2.$$

There are $O(T)$ choices of h , while the dyadic tent is at most $1/T$. The fixed-label reciprocal lifts and all remaining literal coefficients cost $N^{o(1)}$, and the unchanged outer weights sum to M_0 . Hence the selected majorant is $\ll_v A^2 T^{o(1)} M_0$, which is (AT-P15). Finally $19/31 - 3/5 = 2/155$, so the new cutoff lies strictly above the old deep boundary. $\square$

Remark 399 (No second reciprocal parallelogram divisor). Let the four moving primes above $u, u + h, u + a, u + a + h$ be $\ell_1, \ell_2, \ell_3, \ell_4$. The equal- h and equal- a parallelograms produce the identical cubic

$$(\ell_1 - \ell_2)\ell_3\ell_4 - (\ell_3 - \ell_4)\ell_1\ell_2 = (\ell_1 - \ell_3)\ell_2\ell_4 - (\ell_2 - \ell_4)\ell_1\ell_3.$$

Thus the fourth zero does not supply a second independent reciprocal integer divisor.

Definition 400 (Oriented live fibres and dyadic active degree). Fix a selected root $x \in \mathcal{X}$ and the literal fixed-ratio reciprocal-lift block in Corollary 393. Let $\mathfrak{D}_{\text{AT}}$ be its fixed finite orientation alphabet. For $\epsilon \in \mathfrak{D}_{\text{AT}}$ and an oriented nonzero actual p_x -zero label t , let $\mathcal{L}_{x,\epsilon}(t)$ be the set of moving primes satisfying the literal shell and prime-exclusion conditions, the reciprocal p_x -test with label t , the actual moving-characteristic zero test, and every literal one-copy mask for ϵ . Put

$$\nu_{x,\epsilon}(t) = |\mathcal{L}_{x,\epsilon}(t)|.$$

For $T \leq h < 2T$ define

$$C_{x,T}^\epsilon(h) = \sum_{0 \leq t < t+h \leq p_x-1} \nu_{x,\epsilon}(t) \nu_{x,\epsilon}(t+h).$$

Let

$$\begin{aligned} \mathcal{A}_{x,T}^\epsilon &= \{(t, \ell) : \ell \in \mathcal{L}_{x,\epsilon}(t), \text{ and for some } T \leq h < 2T, \\ &\quad t+h \leq p_x-1, \quad \mathcal{L}_{x,\epsilon}(t+h) \neq \emptyset\}, \end{aligned}$$

and put

$$D_{x,T}^\epsilon = |\mathcal{A}_{x,T}^\epsilon|, \quad D_{x,T} = \sum_{\epsilon \in \mathfrak{D}_{\text{AT}}} D_{x,T}^\epsilon.$$

Thus $D_{x,T}$ counts oriented live copies; it is neither an unweighted label count nor an unweighted moving-prime support size.

Proposition 401 (Active-degree compression of the deep noninterlaced term). *Let $\mathcal{R}_{x,T}^{\text{deep,non}}$ be the literal root contribution left by (AT-P8) in the fixed-ratio reciprocal-lift block of Corollary 393. Retain all four actual moving-characteristic zero tests, reciprocal-lift factors, phases, masks, pairwise moving-prime distinctness, the normalized factor of size $O_v(1/h)$, and the unchanged orientation data. Then, root by root,*

$$|\mathcal{R}_{x,T}^{\text{deep,non}}| \ll_v N^{o(1)} \min \left\{ T^2, T^{-1/3} D_{x,T} \min(D_{x,T}, T) \right\}. \quad (\text{AT-P16})$$

The same estimate remains valid if the two separated masks in (AT-P8) are removed.

Proof. For each fixed nonzero oriented label t , the reciprocal congruence fixes the moving prime ℓ to one residue class modulo p_x . The fixed-ratio shell has length $O_v(p_x)$, so

$$\|\nu_{x,\epsilon}\|_\infty \ll_v 1. \quad (\text{AT-P16a})$$

Keep the literal four-copy coefficient until taking absolute values. Dropping only deletion masks afterwards and using $h \geq T$ gives

$$|\mathcal{R}_{x,T,\epsilon}^{\text{deep,non}}| \ll_v \frac{N^{o(1)}}{T} \sum_{T \leq h < 2T} C_{x,T}^\epsilon(h)^2. \quad (\text{AT-P16b})$$

The right side counts all ordered pairs of live gap- h pairs, whereas the literal source retains only their ordered four-distinct subfamily.

For fixed h , every contributing start is a root of N_h modulo p_x . Lemma 12, the degree $3(h-1)$, and (AT-P16a) give

$$C_{x,T}^\epsilon(h) \ll_v T.$$

The first live copy belongs to $\mathcal{A}_{x,T}^\epsilon$ and the target label has $O_v(1)$ lifts, so also

$$C_{x,T}^\epsilon(h) \ll_v D_{x,T}^\epsilon.$$

Finally, for fixed $(t, \ell) \in \mathcal{A}_{x,T}^\epsilon$, the target labels lie in

$$[t+T, t+2T) \cap \{0, \dots, p_x-1\}.$$

They are actual Apéry zeros. Corollary 15 and (AT-P16a) therefore yield

$$\sum_{T \leq h < 2T} C_{x,T}^\epsilon(h) \ll_v D_{x,T}^\epsilon T^{2/3}.$$

Combining the last three estimates with (AT-P16b) gives

$$|\mathcal{R}_{x,T,\epsilon}^{\text{deep,non}}| \ll_v N^{o(1)} T^{-1/3} D_{x,T}^\epsilon \min(D_{x,T}^\epsilon, T).$$

The direct $C(h) \ll_v T$ estimate gives the alternative T^2 bound. Since

$$\sum_{\epsilon} D_{x,T}^\epsilon \min(D_{x,T}^\epsilon, T) \leq D_{x,T} \min(D_{x,T}, T),$$

summation over the fixed orientation alphabet proves (AT-P16). $\square$

Corollary 402 (Low active-degree closure and rich-root residual). *Put*

$$\mathfrak{S}_{4,T}^{\text{deep,non}} = \sum_{x \in \mathcal{X}} w_x \mathcal{R}_{x,T}^{\text{deep,non}},$$

and let $\mathfrak{S}_{4,T}^{\text{deep,non}}[D_{x,T} > Y]$ denote the same literal selected sum with the additional root indicator $\mathbf{1}_{\{D_{x,T} > Y\}}$. Then

$$\boxed{\mathfrak{S}_{4,T}^{\text{deep,non}} = \mathfrak{S}_{4,T}^{\text{deep,non}}[D_{x,T} > T^{23/30}] + O_v(N^{o(1)} T^{6/5} M_0).} \quad (\text{AT-P17})$$

For every fixed $\eta > 0$ one also has

$$\boxed{\mathfrak{S}_{4,T}^{\text{deep,non}} = \mathfrak{S}_{4,T}^{\text{deep,non}}[D_{x,T} > T^{145/186-\eta}] + O_v(N^{o(1)} T^{38/31-2\eta} M_0).} \quad (\text{AT-P18})$$

Both residuals retain every original coefficient and predicate. No smallness assertion for their selected-Palm mass is implicit in these identities.

Proof. If $D_{x,T} \leq T^\kappa$ with $0 \leq \kappa \leq 1$, then (AT-P16) gives

$$|\mathcal{R}_{x,T}^{\text{deep,non}}| \ll_v N^{o(1)} T^{2\kappa-1/3}.$$

Now

$$2 \cdot \frac{23}{30} - \frac{1}{3} = \frac{6}{5}, \quad 2 \cdot \frac{145}{186} - \frac{1}{3} = \frac{38}{31}.$$

Multiply by the unchanged nonnegative w_x , sum over x , and use only $\sum_x w_x = M_0$. $\square$

Corollary 403 (Global active-degree cap and large-scale closure). *For the oriented live degree of Definition 400, one has*

$$D_{x,T} \ll_v N^{2/3+o(1)} \quad (\text{AT-P18a})$$

uniformly in the selected root x . Consequently

$$|\mathfrak{S}_{4,T}^{\text{deep,non}}| \ll_v N^{o(1)} M_0 \min\{T^2, N^{2/3}T^{2/3}, N^{4/3}T^{-1/3}\}. \quad (\text{AT-P18b})$$

In particular, for every fixed $\varepsilon > 0$, the ideal $T^{6/5}$ line is crossed with a fixed margin when

$$T \geq N^{20/23+\varepsilon},$$

and the canonical $T^{38/31}$ line is crossed with a fixed margin when

$$T \geq N^{124/145+\varepsilon}.$$

Thus the rich active-degree obstruction is confined to the complementary mesoscopic range.

Proof. For fixed orientation and nonzero actual p_x -zero label t , (AT-P16a) gives $O_v(1)$ live moving prime copies. There are $Z(p_x) \ll N^{2/3+o(1)}$ possible labels and only $O_v(1)$ orientations, proving (AT-P18a).

For

$$F_T(D) = \min\{T^2, T^{-1/3}D \min(D, T)\}$$

one has both

$$F_T(D) \leq T^{2/3}D, \quad F_T(D) \leq T^{-1/3}D^2.$$

Apply these inequalities in (AT-P16), use the pointwise cap, multiply by the unchanged nonnegative weights w_x , and sum $\sum_x w_x = M_0$. Finally

$$N^{4/3}T^{-1/3} = T^{6/5} \iff T = N^{20/23}, \quad N^{4/3}T^{-1/3} = T^{38/31} \iff T = N^{124/145}.$$

$\square$

Proposition 404 (Actual nonwrapping return energy). *Let $p \geq 7$ be prime and $2 \leq H < p$. For $u \in \mathcal{Z}_p$ put*

$$K_H(u) = \#\{2 \leq d \leq H : u + d < p, u + d \in \mathcal{Z}_p\}.$$

Uniformly for $1 \leq D \leq H$,

$$\sum_{u \in \mathcal{Z}_p} K_H(u)^2 \ll \frac{H^3}{D} + H^{5/3}D. \quad (\text{AT-P19})$$

Consequently

$$\boxed{\sum_{u \in \mathcal{Z}_p} K_H(u)^2 \ll H^{7/3}.} \quad (\text{AT-P19a})$$

Proof. Partition $[2, H]$ into $O(H/D)$ consecutive integer intervals I_j of length at most D , and let

$$n_j(u) = \#\{d \in I_j : u + d < p, u + d \in \mathcal{Z}_p\}.$$

Cauchy's inequality gives

$$K_H(u)^2 \ll \frac{H}{D} \sum_j n_j(u)^2.$$

The diagonal mass is

$$\sum_{u,j} n_j(u) = \sum_{d=2}^H A_p(d) \ll H^2,$$

where $A_p(d) \leq 3(d-1)$ is the literal gap bound. For an off-diagonal pair $d < e$ in one interval put $r = e - d < D$ and $v = u + d$. Fixing the literal gap- r pair $(v, v+r)$, every possible preceding vertex u lies in an interval of length at most H and is an actual Apéry zero. Corollary 15 therefore leaves $O(H^{2/3})$ possible vertices u ; the displacement $d = v - u$ and its bin are then determined. Hence the total off-diagonal mass is

$$\ll H^{2/3} \sum_{2 \leq r < D} A_p(r) \ll H^{2/3} D^2;$$

the case $r = 1$ is empty. This proves (AT-P19). Taking $D = \lfloor H^{2/3} \rfloor$ proves (AT-P19a). $\square$

Corollary 405 (Interlaced actual-chain bound). *Let $2T < p$. Any subfamily of literal nonwrapping chains*

$$u, \quad u + a, \quad u + a + b, \quad u + 2a + b, \quad T \leq a + b < 2T,$$

has cardinality $O(T^{7/3})$ at fixed p . With the literal selected-Palm normalization of Corollary 393, the entire interlaced sector therefore satisfies

$$\boxed{|\mathcal{J}_T^{\text{rd}}| \ll_v T^{4/3+o(1)} M_0.} \quad (\text{AT-P20})$$

Proof. Put $h = a + b$. The first three displayed zeros give two distinct returns $a, h < 2T$ from the actual anchor u . Thus the number of chains is at most

$$\sum_{u \in \mathcal{Z}_p} \binom{K_{2T}(u)}{2} \ll T^{7/3}$$

by Proposition 404. All literal masks only reduce this count. The fixed-label reciprocal lifts and coefficients cost $N^{o(1)}$, the unchanged outer weights sum to M_0 , and the $1/h$ tent costs $O(T^{-1})$. $\square$

This is a genuine factor $T^{-1/3}$ improvement over the previously banked $T^{5/3+o(1)} M_0$ bound. It remains above the canonical $T^{38/31}$ gate by $T^{10/93+o(1)}$ and above the ideal $T^{6/5}$ line by $T^{2/15+o(1)}$.

Proposition 406 (Cross-scale deletion from the 7/3 return energy). *Let $Q_a(A; T)$ and $Q_b(B; T)$ count any subfamily of literal interlaced chains*

$$u, \quad u + a, \quad u + a + b, \quad u + 2a + b, \quad T \leq a + b < 2T,$$

subject respectively to $a \leq A$ and $b \leq B$. In the range of Proposition 404,

$$\boxed{Q_a(A; T) \ll (AT)^{7/6}, \quad Q_b(B; T) \ll (BT)^{7/6}.} \quad (\text{AT-P20a})$$

Consequently, for every fixed $0 < \sigma < 64/217$, the union

$$a \leq T^{197/217-\sigma} \quad \text{or} \quad b \leq T^{197/217-\sigma}$$

has selected-Palm contribution

$$\boxed{\ll_v T^{38/31-(7/6)\sigma+o(1)} M_0.} \quad (\text{AT-P20b})$$

For the ideal $T^{6/5}$ line the corresponding gap cutoff is $T^{31/35-\sigma}$.

Proof. For an actual zero u , let $K_R^+(u)$ and $K_R^-(u)$ count respectively its nonwrapping forward and backward actual returns of length at most R . The palindrome identifies the two energies, so

$$\sum_u K_R^+(u)^2 + \sum_u K_R^-(u)^2 \ll R^{7/3}$$

by Proposition 404. For the first wing, forgetting only the fourth zero gives returns of lengths $a \leq A$ and $a + b < 2T$ from u . Hence Cauchy's inequality gives

$$Q_a(A; T) \leq \sum_u K_A^+(u) K_{2T}^+(u) \ll (AT)^{7/6}.$$

For the second wing put $v = u + a$ and $w = v + b$. For a fixed middle zero pair (v, w) , the number of outer extensions is at most $\sqrt{K_{2T}^-(v) K_{2T}^+(w)}$. If $\mathcal{E}_B$ is the set of such pairs with $w - v \leq B$, a second Cauchy inequality gives

$$Q_b(B; T)^2 \leq \left(\sum_{(v,w) \in \mathcal{E}_B} K_{2T}^-(v) \right) \left(\sum_{(v,w) \in \mathcal{E}_B} K_{2T}^+(w) \right).$$

The first factor is

$$\sum_v K_{2T}^-(v) K_B^+(v) \ll (BT)^{7/6},$$

and the second is identical after reflection. This proves (AT-P20a). Finally

$$\frac{7}{6} \left(1 + \frac{197}{217} \right) = \frac{69}{31}, \quad \frac{7}{6} \left(1 + \frac{31}{35} \right) = \frac{11}{5}.$$

The fixed-label lifts and coefficients cost $N^{o(1)}$, the tent costs T^{-1} , and the unchanged outer weights sum to M_0 . $\square$

Proposition 407 (7/31 gate-safe saturation of the interlaced core). *Fix $0 < \delta < 4/155$ and $0 < \sigma < 64/217$, and put*

$$L = \Delta = T^{7/31-\delta}.$$

Retain the interlaced range

$$T \leq h = a + b < 2T, \quad a, b > T^{197/217-\sigma}.$$

For a fixed distinguished characteristic, each of the following subfamilies has $O(T^{69/31-\delta})$ literal chains before the tent:

1. $|a - b| \leq \Delta;$

2. parameter pairs whose actual three-zero fibre has size at most L ;
3. anchors having at most L actual returns in at least one of the three level bands containing respectively a , h , and $a + h$.

Consequently their selected-Palm contribution is

$$\boxed{\ll_v T^{38/31-\delta+o(1)} M_0.} \quad (\text{AT-P21})$$

Modulo a gate-subcritical error, the surviving interlaced core may therefore assume

$$|a - b| > T^{7/31-\delta}$$

and simultaneous pair-fibre and anchor-fibre heaviness at the same threshold.

Proof. For fixed h , the inequality $|2a - h| \leq \Delta$ leaves $O(\Delta)$ values of a . For fixed (a, b) every chain start is a literal gap- a pair, so its multiplicity is at most $A_p(a) \ll T$. There are $O(T)$ values of h , proving the first claim.

There are $O(T^2)$ parameter pairs, and the four-chain fibre is a subset of the corresponding actual three-zero fibre; this proves the second claim. Finally, for any one of the three fixed level bands I ,

$$\sum_{u \in \mathcal{Z}_p} K_I(u) = \sum_{d \in I} A_p(d) \ll T^2.$$

A chain is determined by the return pair (a, h) , and also by $(a + h, h)$. If one of the three relevant anchor degrees is at most L , its chain count is bounded by L times one of the displayed first moments. Thus every listed sector has size $O(LT^2) = O(T^{69/31-\delta})$. The tent and selected-Palm normalization give (AT-P21). The restriction $\delta < 4/155$ makes this imbalance deletion strictly stronger than the earlier $T^{1/5}$ deletion. $\square$

Lemma 408 (Reflection-square aligned triple-gcd carrier). *Let p be odd, $h = a + b$, and suppose*

$$u, \quad u + a, \quad u + h, \quad u + a + h$$

are literal nonwrapping Apéry zeros modulo p . Put

$$n_d = 3(d - 1), \quad M_d(Y) = 2^{n_d} N_d \left(\frac{Y - d - 1}{2} \right), \quad Y = 2u + a + h + 1.$$

Assume $Y \not\equiv 0 \pmod{p}$. Write uniquely

$$M_a(Y - h) = E_{a,h}(Y^2) + Y O_{a,h}(Y^2),$$

and

$$M_b(Y) = Y^{\epsilon_b} A_b(Y^2), \quad \epsilon_b = \begin{cases} 0, & b \text{ odd}, \\ 1, & b \text{ even}. \end{cases}$$

Then $U = Y^2$ is a common root in $\mathbb{F}_p^\times$ of

$$\overline{A}_b(U), \quad \overline{E}_{a,h}(U), \quad \overline{O}_{a,h}(U), \quad U^{(p-1)/2} - 1.$$

Consequently, if $g_p^\square(a, b)$ is the degree of their gcd in $\mathbb{F}_p[U]$, the number of such literal starts satisfies

$$\boxed{F_p^{\text{act}}(a, b) \leq 2g_p^\square(a, b).} \quad (\text{AT-P22})$$

Proof. The definition of Y gives

$$M_a(Y - h) = 2^{n_a} N_a(u), \quad M_b(Y) = 2^{n_b} N_b(u + a), \quad M_a(Y + h) = 2^{n_a} N_a(u + h),$$

so all three values vanish. Centered reflection gives $M_a(-Z) = (-1)^{a-1} M_a(Z)$; hence the first polynomial $P(Y) = M_a(Y - h)$ vanishes at both Y and $-Y$. Since $P(Y) = E_{a,h}(Y^2) + Y O_{a,h}(Y^2)$ and p is odd,

$$E_{a,h}(Y^2) = Y O_{a,h}(Y^2) = 0.$$

The exclusions $Y \neq 0$ and $p \neq 2$ give the asserted roots of $E_{a,h}$, $O_{a,h}$, and A_b . Finally Y^2 is a nonzero square, so it is a root of $U^{(p-1)/2} - 1$. Each U has at most two lifts Y , and the affine map $u \mapsto Y$ is injective, proving (AT-P22). $\square$

Proposition 409 (Short reflection-displacement deletion). *For a literal nonwrapping interlaced chain put $h = a + b$ and*

$$Y_0 = 2u + a + h + 1, \quad \|Y\|_p = \min_{k \in \mathbb{Z}} |Y_0 - kp|.$$

After the already-deleted case $Y_0 \equiv 0 \pmod{p}$, the number of such chains with $\|Y\|_p \leq D < p/2$ is

$$\boxed{O(D^3)}. \quad (\text{AT-P23})$$

In particular $D = T^{23/31-\tau}$ gives a pre-tent contribution $O(T^{69/31-3\tau})$ for every fixed $\tau > 0$.

Proof. Write $z_0 = u$, $z_1 = u + a$, $z_2 = u + h$, $z_3 = u + a + h$ and $\iota(x) = p - 1 - x$. The two ordered actual-zero pairs

$$(r, s) = (\iota(z_1), z_2), \quad (r', s') = (\iota(z_0), z_3) = (r + a, s + a)$$

have the same signed difference $q = Y_0 - p$. If $d = \min\{|q|, p - |q|\}$, the number $C_p(q)$ of ordered actual-zero pairs with signed difference q is $O(d)$: this is the literal bound $A_p(|q|) \leq 3(|q| - 1)$ when $|q| \leq p/2$, while otherwise there are at most $p - |q|$ possible starts. The chain is recovered from the ordered pair of ordered pairs, since their translation determines a . For each d there are $O(1)$ signed values of q , and therefore the number of chains is

$$\ll \sum_{d \leq D} C_p(q)^2 \ll \sum_{d \leq D} d^2 \ll D^3.$$

$\square$

Corollary 410 (Three-way localization of the balanced interlaced core). *Retain the literal selected-Palm normalization of Corollary 393. Fix positive $\varepsilon_P, \varepsilon_D, \varepsilon_G$, and work after the already-deleted exact reflection locus $2u + a + (a + b) + 1 \equiv 0 \pmod{p}$. Put*

$$P_0 = T^{414/217-\varepsilon_P}, \quad D_0 = T^{23/31-\varepsilon_D}, \quad G_0 = T^{7/31-\varepsilon_G}.$$

The union of the interlaced chains satisfying at least one of

$$ab \leq P_0, \quad 0 < \|2u + a + (a + b) + 1\|_p \leq D_0, \quad g_p^\square(a, b) \leq G_0$$

has selected-Palm contribution

$$\boxed{\ll_v T^{38/31-\eta+o(1)} M_0, \quad \eta = \min \left\{ \frac{7}{6} \varepsilon_P, 3\varepsilon_D, \varepsilon_G \right\}.} \quad (\text{AT-P24})$$

Consequently the canonical hard core may simultaneously be restricted to

$$ab > T^{414/217-o(1)}, \quad \|2u + a + (a + b) + 1\|_p > T^{23/31-o(1)}, \quad g_p^\square(a, b) > T^{7/31-o(1)}.$$

This is a support localization; it does not improve the $T^{4/3+o(1)}M_0$ bound for the surviving core.

Proof. Since $T \leq a + b < 2T$, the condition $ab \leq P_0$ implies $\min\{a, b\} \ll P_0/T$. Proposition 406 therefore bounds this sector before the tent by

$$\ll P_0^{7/6} T^{o(1)} = T^{69/31-(7/6)\varepsilon_P+o(1)}.$$

Proposition 409 bounds the second sector by $D_0^3 = T^{69/31-3\varepsilon_D}$ before the tent. Finally, (AT-P22) gives at most $2G_0$ physical starts for each fixed parameter pair (a, b) in the third sector. There are $O(T^2)$ such pairs, so its pre-tent count is

$$\ll T^2 G_0 = T^{69/31-\varepsilon_G}.$$

The fixed-label lifts and literal coefficients cost $N^{o(1)}$, the unchanged outer weights sum to M_0 , and the dyadic tent costs $O(T^{-1})$. Taking the union proves (AT-P24). $\square$

Proposition 411 (Window-polluted companion deletion). *Assume $4T < p$, and put*

$$\mathcal{P}_p(4T) = \{-m \bmod p : 2 \leq m \leq \lfloor 4T \rfloor, \ p \mid b_{m-1}\}.$$

Among the literal nonwrapping chains

$$u, \quad u + a, \quad u + a + b, \quad u + 2a + b, \quad T \leq a + b < 2T,$$

the number for which either affine lift

$$u \quad \text{or} \quad \iota_{p,a,b}(u) = p - 1 - (u + 2a + b)$$

belongs to $\mathcal{P}_p(4T)$ is $O(T^2)$. With the literal selected-Palm normalization of Corollary 393, their contribution is

$$\boxed{\ll_v T^{1+o(1)} M_0.} \tag{AT-P25}$$

Consequently, after this gate-subcritical deletion, let $g_{p,\text{cl}}^\square(a, b)$ count the common roots $U = Y^2$ in Lemma 408 for which both affine lifts

$$\frac{Y - a - (a + b) - 1}{2}, \quad \frac{-Y - a - (a + b) - 1}{2}$$

avoid $\mathcal{P}_p(4T)$. The number of surviving physical starts obeys

$$F_{p,\text{cl}}^{\text{act}}(a, b) \leq 2g_{p,\text{cl}}^\square(a, b).$$

Proof. Let

$$\mathcal{Z}_p^{\text{low}}(4T) = \mathcal{Z}_p \cap [0, \lfloor 4T \rfloor].$$

Corollary 15 gives $|\mathcal{Z}_p^{\text{low}}(4T)| \ll T^{2/3}$.

Suppose first that $u \in \mathcal{P}_p(4T)$, say $u = p - m$. Nonwrapping gives $2a + b \leq m - 1$, and the reflected chain of Lemma 394 is

$$m - 1 - (2a + b), \quad m - 1 - (a + b), \quad m - 1 - a, \quad m - 1.$$

It therefore lies entirely in $\mathcal{Z}_p^{\text{low}}(4T)$. If instead $\iota_{p,a,b}(u) = p - m$, applying the same reflection shows that the original chain lies entirely in $\mathcal{Z}_p^{\text{low}}(4T)$. Thus every chain under consideration is either a low chain or the unique reflection of a low chain.

A low chain is an increasing quadruple (z_0, z_1, z_2, z_3) in $\mathcal{Z}_p^{\text{low}}(4T)$ satisfying $z_0 + z_3 = z_1 + z_2$. The first three entries determine the fourth, so the number of low chains is at most $|\mathcal{Z}_p^{\text{low}}(4T)|^3 \ll T^2$. The reflection is an involution, hence introduces at most a factor 2.

For the selected-Palm statement, all literal masks only reduce the count, the fixed-label reciprocal lifts and coefficients cost $N^{o(1)}$, the outer weights sum to M_0 , and the dyadic tent costs $O(T^{-1})$. Finally, for a physical carrier $U = Y^2$ the two displayed affine lifts are exactly u and $\iota_{p,a,b}(u)$. After the deletion both are clean, so the same two-to-one argument as in Lemma 408 applies to the restricted common-root set. $\square$

The present degree bounds give only $g_p^\square(a, b) \ll \min\{a, b\}$; they do not supply the additional $T^{-10/93}$ needed after (AT-P20). The unconditional reductions above nevertheless restrict the remaining interlaced core to two gaps larger than $T^{197/217-o(1)}$, reflection displacement larger than $T^{23/31-o(1)}$, and the simultaneous heavy pair- and anchor-fibre conditions of Proposition 407; both affine lifts of every retained split-square carrier may also be taken outside $\mathcal{P}_p(4T)$. The remaining arithmetic frontier is a power-saving sum of the long, clean, aligned root counts $g_{p,\text{cl}}^\square(a, b)$ on the saturated core, not another scalar gcd of two resultants.

Lemma 412 (Positivity of gap values). $N_h(m) \geq 1$ for all integers $m \geq 0$, $h \geq 1$.

Proof. $N_h(m) = \det D_h(m)$ with $D_h(m)$ symmetric tridiagonal, diagonal $P(m+i)$ and off-diagonal $(m+i+1)^3$. The identity

$$P(n) - n^3 - (n+1)^3 = 4(2n+1)^3 > 0 \quad (n \geq 0)$$

shows $D_h(m)$ is strictly diagonally dominant with positive diagonal, hence positive definite. $\square$

Theorem 413 (Twin-atom bound; repaired codegree lemma). For $0 \leq m < m' < X^2$ with $h = m' - m < X$,

$$W(m, m') \leq \omega_X(N_h(m)) + \sum_{j=0}^{h-1} \omega_X(\gcd(b_j, m' - j)) \leq 7h + \min\left(2h, \frac{2h^2}{\log X}\right),$$

and $W(m, m+1) = 0$ identically.

Proof. The first inequality is Lemma 389 combined with Lemma 412. For the numerical form: the entries of $D_h(m)$ are at most CX^6 , so $\log N_h(m) \leq 6(h-1)\log X + O(h\log h)$ and $\omega_X(N_h(m)) \leq 7h$ for large X ; each $m' - j < X^2$ has at most two window prime divisors, and $\omega_X(b_j) \leq \log b_j / \log X \leq 3.6j / \log X$ gives the alternative bound. For $h = 1$: case (A) requires $N_1 \equiv 0$ with $N_1 = 1$, and case (B) requires $p \mid b_0 = 1$; both are impossible. $\square$

13.4 Short-range pair statistics

Proposition 414 (Short-range decorrelation). For $1 \leq h < X$,

$$\sum_{m < X^2 - h} W(m, m+h) \leq C \frac{hX^2}{\log X}, \quad (433)$$

$$\sum_{m < X^2 - h} W(m, m+h)(W(m, m+h) - 1) \leq C \frac{h^2X^2}{(\log X)^2}. \quad (434)$$

Proof. For (433): type-(A) incidences are at most $\sum_p \#\{m < X^2 : p \mid N_h(m)\} \leq \sum_p 3(h-1)(X^2/p+1) \ll hX^2/\log X$ by Mertens; type-(B) incidences are at most $\sum_{j < h} \sum_{p \mid b_j} (X^2/p+1) \leq \sum_{j < h} \omega_X(b_j)(X+1) \ll h^2X/\log X$, which is smaller. For (434): an ordered pair $p \neq q$ counted together is, in the dominant (A,A) configuration, a pair with $pq \mid N_h(m)$, and

$$\sum_{p \neq q} \#\{m < X^2 : pq \mid N_h(m)\} \leq \sum_{p \neq q} 9h^2 \left(\frac{X^2}{pq} + 1 \right) \ll \frac{h^2 X^2}{(\log X)^2};$$

mixed configurations are bounded by the type-(B) count times the pointwise bound of Theorem 413. $\square$

13.5 A horizontal common-index moment

Fix a real function $v \in C_c^\infty((0, \infty))$, with $\text{supp } v \subset [\alpha, \beta]$ for some $0 < \alpha < \beta < \infty$. Put

$$\mathcal{P}_N = \{p \text{ prime} : v(p/N) \neq 0\}, \quad u_p = \frac{\log p}{p} v(p/N), \quad M_N = |\mathcal{P}_N|.$$

For $r \geq 0$ define

$$P_N(r) = \{p \in \mathcal{P}_N : r < p, p \mid b_r\}, \quad m_N(r) = |P_N(r)|,$$

and put

$$A_N = \sum_{r \geq 0} m_N(r) = \sum_{p \in \mathcal{P}_N} Z(p).$$

Finally define

$$\mathcal{H}_N(v) = \sum_{\substack{p, q \in \mathcal{P}_N \\ p \neq q}} u_p u_q |\mathcal{Z}_p \cap \mathcal{Z}_q|.$$

Lemma 415 (Short horizontal codegree). *For every fixed $C > 0$ there is a constant $C_{v,C}$ such that, for all sufficiently large N ,*

$$|P_N(r) \cap P_N(s)| \leq C_{v,C} |r - s|$$

whenever

$$0 \leq r < s < \beta N, \quad 1 \leq s - r \leq CN/\log N.$$

Proof. Partition $[\alpha N, \beta N]$ into $O_v(1)$ disjoint pieces, each contained in a dyadic interval $(X_j, 2X_j]$ with $X_j \asymp_v N$. Put $h = s - r$. In the displayed range, $h < X_j$ and $r, s < X_j^2$ for all sufficiently large N . Theorem 413, applied with $(m, m') = (r, s)$, gives at most $O_{v,C}(h)$ primes from each dyadic piece. Summing the fixed number of pieces proves the claim. Notice that a prime in $P_N(r) \cap P_N(s)$ has $r, s < p$, so only the nonwrapping case of the source dichotomy can occur. $\square$

For $t \geq 1$ put

$$S_t = \{0 \leq r < \beta N : m_N(r) \geq t\}.$$

Lemma 416 (High horizontal columns). *Uniformly for $t \geq C_v \sqrt{M_N}$,*

$$|S_t| \ll_v \frac{NM_N^2}{t^3} + \frac{M_N}{t}.$$

Proof. Let C_0 be a constant in Lemma 415, with the fixed range constant chosen larger than an admissible constant in $M_N \ll_v N/\log N$, and put

$$L_t = \left\lfloor \frac{t^2}{8C_0 M_N} \right\rfloor.$$

The case $M_N = 0$ is immediate. Increasing the threshold constant makes $L_t \geq 1$, and then $L_t \geq t^2/(16C_0 M_N)$. If $S_t \neq \emptyset$, then $t \leq M_N$, so $L_t \ll_v N/\log N$ and the short-codegree lemma applies inside every interval of L_t consecutive integers.

Fix such an interval I , write $S = S_t \cap I$ and $s = |S|$, and put

$$z_p = \#\{r \in S : p \in P_N(r)\}, \quad E = \sum_p z_p = \sum_{r \in S} m_N(r).$$

Then $E \geq ts$. Cauchy–Schwarz gives

$$\sum_p z_p(z_p - 1) \geq \frac{E^2}{M_N} - E.$$

On the other hand, ordered-pair expansion and the codegree lemma give

$$\sum_p z_p(z_p - 1) = \sum_{\substack{r, r' \in S \\ r \neq r'}} |P_N(r) \cap P_N(r')| \leq C_0 L_t s^2 \leq \frac{t^2 s^2}{8M_N}.$$

If $E \geq 2M_N$, the lower bound is at least $E^2/(2M_N) \geq t^2 s^2/(2M_N)$, a contradiction. Thus $E < 2M_N$ and $s < 2M_N/t$. There are $O_v(N/L_t + 1) = O_v(NM_N/t^2 + 1)$ intervals, which proves the lemma. $\square$

Proposition 417 (Horizontal common-index moment). *One has*

$$\sum_{r \geq 0} m_N(r)^2 \ll_v \frac{N^{23/10}}{(\log N)^{7/5}}$$

and consequently

$$|\mathcal{H}_N(v)| \ll_v N^{3/10} (\log N)^{3/5}.$$

For $v \geq 0$, the absolute-value signs on $\mathcal{H}_N(v)$ may be omitted.

Proof. The unconditional first zero moment and Chebyshev’s estimate give

$$A_N \ll_v \frac{N^{8/5}}{(\log N)^{4/5}}, \quad M_N \ll_v \frac{N}{\log N}.$$

The layer-cake identity is

$$\sum_r m_N(r)^2 = \sum_{t=1}^{M_N} (2t-1) |S_t|.$$

Take $T = N^{7/10} (\log N)^{-3/5}$. For $t \leq T$, $t|S_t| \leq A_N$, and hence the low levels contribute $O(A_N T)$. If $M_N < T$, this handles the entire sum. Otherwise $T \gg_v \sqrt{M_N}$, so Lemma 416 gives

$$\sum_{T < t \leq M_N} (2t-1) |S_t| \ll_v \frac{NM_N^2}{T} + M_N^2.$$

Both $A_N T$ and $N M_N^2 / T$ are $O_v(N^{23/10}(\log N)^{-7/5})$, while M_N^2 is smaller. This proves the first assertion.

The exact weighted identity is

$$\mathcal{H}_N(v) = \sum_r \sum_{\substack{p, q \in P_N(r) \\ p \neq q}} u_p u_q.$$

Since $\max_{p \in \mathcal{P}_N} |u_p| \ll_v (\log N)/N$,

$$|\mathcal{H}_N(v)| \leq \frac{C_v (\log N)^2}{N^2} \sum_r m_N(r)(m_N(r) - 1),$$

and the result follows from the first assertion. $\square$

Corollary 418 (Wrapped horizontal common-index moment). *Fix constants $0 < a < b$ and $C > 0$, and put*

$$\mathcal{Q}_N = \{p \text{ prime} : aN < p \leq bN\}, \quad d_N(m) = \#\{p \in \mathcal{Q}_N : p \mid b_m\} \quad (0 \leq m \leq CN).$$

Then

$$\sum_{0 \leq m \leq CN} d_N(m)^2 \ll_{a,b,C} \frac{N^{23/10}}{(\log N)^{7/5}}.$$

In particular, the restriction $m < p$ in Proposition 417 is unnecessary for this unweighted second-moment conclusion when $m = O(N)$.

Proof. Partition $(aN, bN]$ into $O_{a,b}(1)$ disjoint intervals, each contained in a dyadic window $(X, 2X]$ with $X \asymp_{a,b} N$. For $0 \leq m < m' \leq CN$ and $0 < m' - m \ll_{a,b,C} N/\log N$, Theorem 413 applies to the integer indices m, m' themselves: for large N , their difference is less than X and both indices are less than X^2 . Summing the fixed number of dyadic pieces gives

$$\#\{p \in \mathcal{Q}_N : p \mid b_m, p \mid b_{m'}\} \ll_{a,b,C} |m - m'|.$$

This includes the wrapping case of the twin-atom dichotomy and does not assume $m, m' < p$.

The first incidence has the same bound as before. Indeed, write $m = kp + r$, with $0 \leq r < p$. The quotient k ranges over a fixed set depending only on a and C , and Theorem 4 gives $b_m \equiv b_k b_r \pmod{p}$. For all sufficiently large N , none of the finitely many nonzero integers b_k is divisible by a prime in $\mathcal{Q}_N$. Consequently

$$A_N^* := \sum_{0 \leq m \leq CN} d_N(m) \ll_{a,b,C} \sum_{p \in \mathcal{Q}_N} Z(p) \ll_{a,b,C} \frac{N^{8/5}}{(\log N)^{4/5}}.$$

Let $R_N = |\mathcal{Q}_N| \ll_{a,b} N/\log N$ and $S_t^* = \{0 \leq m \leq CN : d_N(m) \geq t\}$. The interval argument in Lemma 416, using (13.5) in place of its nonwrapping codegree, gives uniformly for $t \gg_{a,b,C} \sqrt{R_N}$

$$|S_t^*| \ll_{a,b,C} \frac{NR_N^2}{t^3} + \frac{R_N}{t}.$$

Finally use the layer-cake identity, the low-level estimate $t|S_t^*| \leq A_N^*$, and the cutoff $T = N^{7/10}(\log N)^{-3/5}$. The three contributions are

$$A_N^* T, \quad \frac{NR_N^2}{T}, \quad R_N^2,$$

and the first two are $O_{a,b,C}(N^{23/10}(\log N)^{-7/5})$, while the last is smaller. $\square$

Lemma 419 (Centered-offset horizontal codegree). *For an odd prime p put $c_p = (p-1)/2$. Fix $C > 0$, and for $0 \leq k \leq CN$ define*

$$\mathcal{C}_N^+(k) := \{p \in \mathcal{P}_N : c_p + k < p, \ c_p + k \in \mathcal{Z}_p\}.$$

Uniformly for

$$0 \leq k < k' \leq CN, \quad 1 \leq h := k' - k \leq CN/\log N,$$

one has

$$|\mathcal{C}_N^+(k) \cap \mathcal{C}_N^+(k')| \ll_{v,C} h.$$

The same assertion holds with $c_p - k$ in place of $c_p + k$.

Proof. If p belongs to the displayed intersection, the two admissible zero positions $c_p + k$ and $c_p + k + h$ do not wrap. The gap certificate therefore gives

$$N_h(c_p + k) \equiv 0 \pmod{p}.$$

Since $c_p + k \equiv k - \frac{1}{2} \pmod{p}$ and p is odd,

$$p \mid C_{h,k}, \quad C_{h,k} := 2^{3(h-1)} N_h(k - \tfrac{1}{2}) \in \mathbb{Z}.$$

This integer is nonzero: Theorem 422 places every root of N_h in $-h < \operatorname{Re} z < -1$, whereas $k - \frac{1}{2} > -1$ for $k \geq 0$ (and $N_1 = 1$). The tridiagonal determinant formula, $k, k' \ll_C N$, and $h \ll_C N/\log N$ give

$$\log |C_{h,k}| \ll_C h \log N.$$

Thus $C_{h,k}$ has $O_{v,C}(h)$ prime divisors in the fixed window supporting v , proving the positive-offset assertion. The negative-offset assertion follows from $r \in \mathcal{Z}_p \iff p-1-r \in \mathcal{Z}_p$. $\square$

Proposition 420 (Centered-offset horizontal moment). *For every fixed $C > 0$, put*

$$c_N^+(k) = |\mathcal{C}_N^+(k)|, \quad c_N^-(k) = \#\{p \in \mathcal{P}_N : 0 \leq c_p - k, \ c_p - k \in \mathcal{Z}_p\}.$$

Then

$$\sum_{0 \leq k \leq CN} (c_N^+(k))^2 + \sum_{0 \leq k \leq CN} (c_N^-(k))^2 \ll_{v,C} \frac{N^{23/10}}{(\log N)^{7/5}}.$$

Proof. Reflection gives $c_N^-(k) = c_N^+(k)$, so it is enough to treat the positive offsets. Their first incidence satisfies

$$\sum_{0 \leq k \leq CN} c_N^+(k) \leq \sum_{p \in \mathcal{P}_N} Z(p) \ll_v \frac{N^{8/5}}{(\log N)^{4/5}}.$$

Lemma 419 supplies the same short codegree estimate used in Lemma 416. Repeating its interval argument on $0 \leq k \leq CN$, with $M_N = |\mathcal{P}_N| \ll_v N/\log N$, gives for $t \gg_{v,C} \sqrt{M_N}$

$$\#\{0 \leq k \leq CN : c_N^+(k) \geq t\} \ll_{v,C} \frac{NM_N^2}{t^3} + \frac{M_N}{t}.$$

Apply layer cake with $T = N^{7/10}(\log N)^{-3/5}$, exactly as in Proposition 417. The low levels cost $O(N^{8/5}T/(\log N)^{4/5})$, the high levels cost $O(NM_N^2/T + M_N^2)$, and the asserted bound follows. $\square$

Proposition 421 (Periodic centered-column second moment). *For an odd prime p put $c_p = (p - 1)/2$, and for $a \in \mathbb{Z}$ define*

$$m_N^{\text{ctr}}(a) := \#\{p \in \mathcal{P}_N : c_p + a \pmod{p} \in \mathcal{Z}_p\},$$

where the residue is represented in $[0, p-1]$. For every fixed $C > 0$ and every R with $N \leq R \leq CN^2$,

$$\sum_{0 \leq a \leq R} m_N^{\text{ctr}}(a)^2 \ll_{v,C} R \frac{N^{13/10}}{(\log N)^{7/5}} + \frac{N^2}{(\log N)^2}. \quad (\text{CP1})$$

The same estimate holds on $[-R, 0]$.

Proof. We first establish the periodic short-codegree estimate. Fix $0 \leq a < a + h \leq R$ with $1 \leq h \ll_{v,C} N/\log N$, and suppose that p is counted at both columns. Let $r \in [0, p-1]$ represent $c_p + a$ modulo p .

If $r + h < p$, then $r, r + h \in \mathcal{Z}_p$. The case $h = 1$ is absent by Lemma 10; for $h \geq 2$ the gap certificate gives $N_h(r) \equiv 0 \pmod{p}$. Since $r \equiv a - 1/2 \pmod{p}$,

$$p \mid C_{h,a}, \quad C_{h,a} := 2^{3(h-1)} N_h(a - \tfrac{1}{2}) \in \mathbb{Z}.$$

Theorem 422 shows that $C_{h,a} \neq 0$. The tridiagonal determinant formula, $a \leq CN^2$, and $h \ll N/\log N$ give

$$\log |C_{h,a}| \ll_{v,C} h \log N.$$

Thus only $O_{v,C}(h)$ primes in the support of v occur in this case.

If $r + h \geq p$, write $j = r + h - p$, so $0 \leq j < h$ and $j \in \mathcal{Z}_p$. From $r \equiv c_p + a \pmod{p}$ one obtains

$$p \mid 2a + 2h - 1 - 2j.$$

For fixed j the displayed integer is positive and $O_C(N^2)$, so it has $O_{v,C}(1)$ prime divisors in the support of v . Summing over $j < h$ proves

$$\#\{p : a \text{ and } a + h \text{ are both centered zero columns}\} \ll_{v,C} h. \quad (435)$$

The first incidence on $[0, R]$ satisfies

$$A_R := \sum_{0 \leq a \leq R} m_N^{\text{ctr}}(a) \ll_{v,C} \sum_{p \in \mathcal{P}_N} Z(p)(R/p + 1) \ll_{v,C} R \frac{N^{3/5}}{(\log N)^{4/5}}.$$

Let $M_N = |\mathcal{P}_N| \ll_v N/\log N$ and $S_t = \{0 \leq a \leq R : m_N^{\text{ctr}}(a) \geq t\}$. Repeating the interval argument of Lemma 416, with (435) and intervals of length $\asymp t^2/M_N$, gives, uniformly for $t \gg_{v,C} \sqrt{M_N}$,

$$|S_t| \ll_{v,C} \frac{RM_N^2}{t^3} + \frac{M_N}{t}.$$

Those interval lengths are at most $M_N \ll N/\log N$, so the short-codegree range used above is sufficient.

Apply layer cake with $T = N^{7/10}(\log N)^{-3/5}$. The low levels cost $O(A_R T)$, and the high levels cost $O(RM_N^2/T + M_N^2)$. These are bounded by the right-hand side of (CP1). Finally, reflection $r \mapsto p - 1 - r$ sends the centered column a to $-a$, proving the negative-half assertion. $\square$

13.6 The root-strip theorem and resultant nonvanishing

Theorem 422 (Root strip). *For every $h \geq 2$, all roots of N_h lie in the open strip $\{z \in \mathbb{C} : -h < \operatorname{Re} z < -1\}$.*

Proof. Step 1. For $\operatorname{Re} y \geq 0$, writing $u = |y + \frac{1}{2}| \geq \frac{1}{2}$:

$$|P(y)| = 2u |17(y + \frac{1}{2})^2 + \frac{3}{4}| \geq 2u(17u^2 - \frac{3}{4}) = 34u^3 - \frac{3}{2}u,$$

while $|y|^3 + |y+1|^3 \leq 2(u + \frac{1}{2})^3 = 2u^3 + 3u^2 + \frac{3}{2}u + \frac{1}{4}$. The difference $32u^3 - 3u^2 - 3u - \frac{1}{4}$ equals $\frac{3}{2}$ at $u = \frac{1}{2}$ and has positive derivative $96u^2 - 6u - 3 > 0$ for $u \geq \frac{1}{2}$. Hence $|P(y)| > |y|^3 + |y+1|^3$.

Step 2. $N_h(z) = \det D_h(z)$ with D_h tridiagonal: diagonal $P(z+i)$, off-diagonal $(z+i+1)^3$. For $\operatorname{Re} z \geq -1$, every $y_i = z + i$ ($1 \leq i \leq h-1$) has $\operatorname{Re} y_i \geq 0$, so Step 1 gives strict diagonal dominance (endpoint rows have only one off-diagonal entry, strengthening the inequality). By Levy–Desplanques, $\det D_h(z) \neq 0$.

Step 3. The reflection identity $N_h(-z-h-1) = (-1)^{h-1} N_h(z)$ (Remark 20, valid for all $z \in \mathbb{C}$ by the polynomial identity of the tridiagonal determinant) maps any root α to the root $-\alpha-h-1$ with $\operatorname{Re}(-\alpha-h-1) = -\operatorname{Re} \alpha - h - 1$. If $\operatorname{Re} \alpha \leq -h$, then $\operatorname{Re}(-\alpha-h-1) \geq -1$, contradicting Step 2. $\square$

Corollary 423 (Resultant nonvanishing — the First Lemma). *For all $d, r \geq 1$, $S_{d,r} = \operatorname{Res}_x(N_d(x), N_r(x+d)) \neq 0$. For $d = 1$ or $r = 1$, this is immediate since $N_1 = 1$. For $d, r \geq 2$: by Theorem 422, roots of N_d have $\operatorname{Re} \in (-d, -1)$ while roots of $N_r(x+d)$ have $\operatorname{Re} \in (-d-r, -d-1)$. The two strips are disjoint (separated by $[-d-1, -d]$), so the polynomials share no root and their resultant is nonzero.*

13.7 Per-prime separation

Recall $S_{h_1, h_2} = \operatorname{Res}_x(N_{h_1}(x), N_{h_2}(x + h_1)) \in \mathbb{Z}$, with $\log |S_{h_1, h_2}| \ll (h_1 + h_2)^2 \log(h_1 + h_2)$. By Corollary 423, $S_{h_1, h_2} \neq 0$ unconditionally.

Theorem 424 (Per-prime separation). *Let $h_1, h_2 \geq 1$. If some $p \in \mathcal{P}_X$ satisfies $m, m + h_1, m + h_1 + h_2 \in \Omega_p$ (all three hit by p), then*

$$p \mid S_{h_1, h_2} \quad \text{or} \quad p \mid b_j \text{ for some } 0 \leq j < h_1 + h_2.$$

(The hypothesis $S_{h_1, h_2} \neq 0$ from earlier versions of this result is now unconditional by Corollary 423.) Consequently there is $c > 0$ such that for $h_1 + h_2 \leq c(\log X / \log \log X)^{1/2}$ and X large, no prime of $\mathcal{P}_X$ hits all three points: each hit set Ω_p contains at most two points in any window of length $c(\log X / \log \log X)^{1/2}$.

Proof. If neither consecutive pair wraps (Lemma 389(A) twice), then with $r = m \bmod p$ both $N_{h_1}(r) \equiv 0$ and $N_{h_2}(r + h_1) \equiv 0 \pmod{p}$. Since $p > X > h_1 + h_2$, the leading coefficients of N_{h_1} and $N_{h_2}(\cdot + h_1)$ are units modulo p , so a common root forces $p \mid \operatorname{Res} = S_{h_1, h_2}$. If some pair wraps, case (B) gives $p \mid b_j$ with $j < h_1 + h_2$. For the quantitative form: $\omega_X(S_{h_1, h_2}) \leq C(h_1 + h_2)^2 \log(h_1 + h_2) / \log X < 1$ in the stated range, so no window prime divides S_{h_1, h_2} ; and $b_j \leq e^{3.6j} < X < p$ in the stated range, so $p \mid b_j$ is impossible. $\square$

Remark 425. The nonvanishing $S_{h_1, h_2} \neq 0$ is now unconditional (Corollary 423). For consecutive indices the stronger product formula $|\operatorname{Res}(N_h, N_{h+1})| = \prod_j (j!^3 b_j)^6$ gives the exact value. The root-strip theorem (Theorem 422) provides the sharpest known localization: every root of N_h lies on the “critical strip” $-h < \operatorname{Re} z < -1$, centered on the reflection line $\operatorname{Re} z = -(h+1)/2$.

13.8 The moment collapse and the atom-tail equivalence

Proposition 426 (Moment collapse). *For every $k \geq 3$,*

$$\sum_{m < X^2} (K_X(m))_k \leq \left(\max_{m < X^2} K_X(m) \right)^{k-2} \sum_{m < X^2} (K_X(m))_2 \leq 5X^2 \lambda_X^2 \left(\max_{m < X^2} K_X(m) \right)^{k-2}.$$

Proof. Pointwise, $(K)_k = (K)_2 \prod_{j=2}^{k-1} (K-j) \leq (K)_2 K^{k-2}$; sum over m and apply the second factorial moment. $\square$

Hypothesis 427 (Atom bound (AT'')). $\max_{m < X^2} K_X(m) \leq \lambda_X X^{o(1)}$.

Corollary 428. (AT'') *implies (HM) $_k$ for every fixed $k \geq 2$ simultaneously; in particular, by Theorem 331 with $k = 7$, (AT'') implies $\log G_n \ll n^{20/21+o(1)}$ for every n , hence $G_n = e^{o(n)}$. Combined with the unconditional average bound $\sum_{p \leq x} Z(p) \ll x^{8/5}/\log x$ (from Corollary 423 and the corank-valuation argument of Subsection 5.4), one obtains $\lambda_X \ll X^{3/5}/\log X$, so (AT'') with $k = 6$ gives $\log G_n \ll n^{14/15+o(1)}$.*

Proof. Proposition 426 gives $\sum (K)_k \leq 5X^2 \lambda_X^2 (\lambda_X X^{o(1)})^{k-2} = X^{2+o(1)} \lambda_X^k$. $\square$

Corollary 429 (High moments with an averaged zero exponent). *Assume (HM) $_k$ for one fixed $k \geq 3$, and suppose that, uniformly over dyadic X ,*

$$\sum_{X < p \leq 2X} Z(p) \ll X^{1+c+o(1)}$$

for some fixed $0 \leq c < 1$. Then, for every n ,

$$\log G_n \ll n^{\max\{2/3, c+2/k\}+o(1)}.$$

In particular the unconditional averaged exponent $c = 3/5$, together with (HM) $_6$, would give $\log G_n \ll n^{14/15+o(1)}$.

Proof. On $X < p \leq 2X$,

$$\lambda_X = \sum_p \frac{Z(p)}{p} \ll X^{c+o(1)}.$$

The maximum-row argument in the proof of Theorem 331 therefore gives

$$\max_{m < X^2} K_X(m) \ll X^{2/k+o(1)} \lambda_X + 1 \ll X^{c+2/k+o(1)} + 1.$$

The same dyadic summation gives the exponent $c + 2/k$ for the lower-digit channel. Combining it with the already proved $O(n^{2/3+o(1)})$ channels proves the claim. $\square$

Lemma 430 (Hard folded-coordinate rigidity). *Let $X \geq 2$, $X \leq m < X^2$, and let p be a prime with*

$$X < p \leq 2X, \quad p > 2\sqrt{m}, \quad m \bmod p \in \mathbb{Z}_p.$$

Write $r = m \bmod p$ and $s_p(m) = \min\{r, p-1-r\}$. Then $0 \leq s_p(m) < X$, $p \mid b_{s_p(m)}$, and

$$p \mid m - s_p(m) \quad \text{or} \quad p \mid m + s_p(m) + 1. \quad (\text{HF1})$$

For fixed (m, s) there is at most one hard prime in each of the two orientations in (HF1); consequently at most two hard hit primes at a fixed row have folded coordinate s .

Proof. Palindromy of the Apéry zero set gives $p \mid b_{s_p(m)}$. If $r = s_p(m)$ then $p \mid m - s_p(m)$, while if $r = p - 1 - s_p(m)$ then $p \mid m + s_p(m) + 1$. In the first orientation $m - s_p(m) > 0$: equality would give $r = m$ and $p \geq 2m + 1$, contradicting $p \leq 2X \leq 2m$. Hence if two distinct hard primes p, q had the same folded coordinate and this orientation, then

$$pq \mid m - s, \quad 0 < m - s < m < 4m < pq,$$

a contradiction. In the reflected orientation one has

$$pq \mid m + s + 1, \quad 0 < m + s + 1 \leq m + X \leq 2m < 4m < pq,$$

which is again impossible. This proves the two orientation-specific uniqueness statements. Notice that the actual residue orientation is essential here; the bare divisibility $p \mid m - s$ has a zero-carrier exception when $m = s$. $\square$

For $X \leq m < X^2$ put

$$K_X^\sharp(m) = \#\{X < p \leq 2X : p > 2\sqrt{m}, m \bmod p \in \mathcal{Z}_p\}.$$

Let $D_{6,\text{ff}}^\sharp(m; H)$ count ordered six-tuples of distinct primes counted by $K_X^\sharp(m)$ whose folded coordinates have pairwise distance greater than H , and put

$$F_{6,\text{ff}}^\sharp(X; H) = \sum_{X \leq m < X^2} D_{6,\text{ff}}^\sharp(m; H).$$

Proposition 431 (Folded-far extraction from a hard row). *For every $H \geq 1$,*

$$\max_{X \leq m < X^2} K_X^\sharp(m) \ll H + (F_{6,\text{ff}}^\sharp(X; H))^{1/6}. \quad (\text{HF2})$$

Proof. Lemma 430 shows that every integer folded coordinate supports at most two hard hits. Thus a chosen hard hit has at most $2(2\lfloor H \rfloor + 1)$ possible conflicting hits within distance H . If $K = K_X^\sharp(m)$ exceeds ten times this quantity, successive choices give

$$D_{6,\text{ff}}^\sharp(m; H) \gg K^6.$$

Otherwise $K \ll H$. Combining the two cases and maximizing over m proves (HF2). $\square$

Corollary 432 (Folded-far sixth moment suffices). *If, uniformly over dyadic X ,*

$$F_{6,\text{ff}}^\sharp(X; X^{1/10}) \ll X^{2+o(1)} \lambda_X^6, \quad (\text{HF3})$$

then $\log G_n \ll n^{14/15+o(1)}$ for every n . The same conclusion follows from the weaker absolute estimate

$$F_{6,\text{ff}}^\sharp(X; X^{1/10}) \ll X^{28/5+o(1)} (\log X)^{-6}.$$

Proof. Proposition 431 and $\lambda_X \ll X^{3/5+o(1)}/\log X$ give

$$\max_m K_X^\sharp(m) \ll X^{1/10} + X^{1/3+o(1)} \lambda_X \ll X^{14/15+o(1)}.$$

For a fixed target n , primes at most $2\sqrt{n}$ contribute $O(\sqrt{n})$ in logarithmic weight. In every remaining dyadic prime block one has $X > \sqrt{n}$ and $n < X^2$, so its lower-digit bad primes are counted by $K_X^\sharp(n)$. The standard dyadic summation, together with the already proved $O(n^{2/3+o(1)})$ companion channels, gives the assertion. The absolute hypothesis gives the same maximum-row bound after taking a sixth root. $\square$

Proposition 433 (Hard high-row tail). *Put*

$$\mathcal{L}_X = \#\{p \in (X, 2X] : Z(p) > 0\}, \quad S_t^\# = \{m \in [X, X^2) : K_X^\#(m) \geq t\}.$$

There are absolute constants C_1, C_2 such that, for $t \geq C_1\sqrt{\mathcal{L}_X}$,

$$|S_t^\#| \leq C_2 \left(\frac{X^2 \mathcal{L}_X^2}{t^3} + \frac{\mathcal{L}_X}{t} \right). \quad (\text{HF4})$$

Proof. The assertion is empty if $\mathcal{L}_X = 0$. Otherwise let C_0 be an absolute constant such that Theorem 413 gives $W(u, v) \leq C_0(v - u)$ whenever $0 < v - u < X$, and set

$$\ell_t = \left\lfloor \frac{t^2}{8C_0\mathcal{L}_X} \right\rfloor.$$

After enlarging C_1 , one has $1 \leq \ell_t < X$ whenever $S_t^\#$ is nonempty, since then $t \leq \mathcal{L}_X$ and $\mathcal{L}_X \ll X/\log X$.

Partition $[X, X^2)$ into intervals of length at most ℓ_t and fix one such interval. If $S = S_t^\# \cap I$, $s = |S|$, and z_p counts the rows of S hit by the active prime p , then $E := \sum_p z_p \geq ts$. Cauchy over the $\mathcal{L}_X$ active primes gives

$$\sum_p z_p(z_p - 1) \geq \frac{E^2}{\mathcal{L}_X} - E.$$

The same sum counts ordered pairs of distinct rows with a common hard prime. The twin-atom theorem therefore gives

$$\sum_p z_p(z_p - 1) \leq C_0 \ell_t s(s - 1) \leq \frac{t^2 s^2}{8\mathcal{L}_X}.$$

If $E \geq 2\mathcal{L}_X$, the preceding lower bound is at least $E^2/(2\mathcal{L}_X) \geq t^2 s^2/(2\mathcal{L}_X)$, a contradiction. Thus $s < 2\mathcal{L}_X/t$. There are $O(X^2 \mathcal{L}_X/t^2 + 1)$ intervals, proving (HF4). $\square$

Corollary 434 (Unconditional hard sixth-moment envelope). *With*

$$F_6^\#(X) = \sum_{X \leq m < X^2} (K_X^\#(m))_6,$$

one has

$$F_6^\#(X) \ll X^2 \mathcal{L}_X^5 \ll \frac{X^7}{(\log X)^5}. \quad (\text{HF5})$$

Together with the pair-Palm envelope this gives

$$F_6^\#(X) \ll X^2 \mathcal{L}_X^4 \min\{C\lambda_X^2, C\mathcal{L}_X\}. \quad (\text{HF6})$$

At the present worst allowed mass $\lambda_X \ll X^{3/5}/\log X$, the first bound improves the previous pair-Palm envelope by $X^{1/5-o(1)}$.

Proof. Since $(K)_6 \leq K^6$, discrete layer cake gives

$$F_6^\#(X) \leq \sum_{t=1}^{\mathcal{L}_X} (t^6 - (t-1)^6) |S_t^\#|.$$

For $t \leq C_1 \sqrt{\mathcal{L}_X}$ use the trivial row count; its contribution is $O(X^2 \mathcal{L}_X^3)$. Above this threshold, Proposition 433 and $t^6 - (t-1)^6 \ll t^5$ give

$$\ll X^2 \mathcal{L}_X^2 \sum_{t \leq \mathcal{L}_X} t^2 + \mathcal{L}_X \sum_{t \leq \mathcal{L}_X} t^4 \ll X^2 \mathcal{L}_X^5 + \mathcal{L}_X^6.$$

The last term is absorbed because $\mathcal{L}_X < X^2$. Prime counting gives $\mathcal{L}_X \ll X/\log X$. Finally the deterministic pair-Palm lift gives $F_6^\sharp \ll X^2 \lambda_X^2 \mathcal{L}_X^4$; taking the minimum proves (HF6) and the stated comparison. $\square$

Theorem 435 (Concentration of the problem in the atom tail). *Let $n_T(X) = \#\{0 \leq m < X^2 : K_X(m) \geq T\}$. Then:*

1. *Unconditionally, $n_T(X) \leq 5X^2 \lambda_X^2 / (T)_2$ for all $T \geq 2$, and $\max_m K_X(m) \leq |\mathcal{P}_X| \ll X/\log X$.*
2. *(HM)₃ holds if and only if*

$$n_T(X) \ll \frac{X^{2+o(1)} \lambda_X^3}{T^3} \quad \text{uniformly for } T \geq 3.$$

3. *If $n_T(X) \ll X^{2+o(1)} \lambda_X^k / T^k$ uniformly for $T \geq k$, then (HM)_k holds; with $k = 7$ this implies $G_n = e^{o(n)}$ for every n .*

Proof. (1) is Chebyshev applied to the second factorial moment, and the trivial ceiling. (2), necessity: $n_T(T)_3 \leq \sum_m (K)_3$. Sufficiency: writing the third factorial moment as a sum over levels,

$$\sum_m (K_X(m))_3 = \sum_{T \geq 3} n_T[(T)_3 - (T-1)_3] \leq \sum_{T \geq 3} 3T^2 n_T \ll X^{2+o(1)} \lambda_X^3 \sum_{3 \leq T \leq X} \frac{1}{T},$$

and the harmonic sum is $O(\log X) = X^{o(1)}$. (3) is identical with $[(T)_k - (T-1)_k] \leq kT^{k-1}$, followed by Theorem 331. $\square$

Theorem 436 (Reflected root-lattice tail bound). *Put*

$$R_X(m) = \prod_{\substack{p \in \mathcal{P}_X \\ m \bmod p \in \mathcal{Z}_p}} p, \quad \Delta_X = \sum_{p \in \mathcal{P}_X} Z(p) \log p.$$

Fix an integer

$$A \geq \max \left\{ 0, \max_{p \in \mathcal{P}_X} (Z(p) - 1) \right\},$$

and set

$$E_{X,A} = A \log(X^2 + 1) + \log(A + 1).$$

For every real $T > 0$ and every integer $1 \leq s \leq A + 1$,

$$T \log X > \frac{\Delta_X}{s} + E_{X,A} \implies n_T(X) \leq \left\lfloor \frac{s-1}{2} \right\rfloor. \quad (436)$$

In particular, for every fixed $\varepsilon > 0$, uniformly for

$$X^{14/15+\varepsilon} \leq T \leq |\mathcal{P}_X|,$$

one has, for all sufficiently large X ,

$$\boxed{n_T(X) \ll_\varepsilon \frac{X^{8/5}}{T \log X}.} \quad (437)$$

Thus at the isolated-five-star threshold,

$$n_{X^{14/15+\varepsilon}}(X) \ll_\varepsilon \frac{X^{2/3-\varepsilon}}{\log X}.$$

There is also a density-sensitive form. If

$$E_{X,A} < \frac{T \log X}{2}, \quad \frac{2\Delta_X}{T \log X} \leq A,$$

then

$$n_T(X) \leq \frac{\Delta_X}{T \log X} \leq \frac{2 \log(2X)}{\log X} \frac{X \lambda_X}{T} \leq \frac{4X \lambda_X}{T}. \quad (438)$$

Proof. Consider the congruence lattice

$$\mathcal{L}_{X,A} = \{P \in \mathbb{Z}[U] : \deg P \leq A, P(r) \equiv 0 \pmod{p} \ (p \in \mathcal{P}_X, r \in \mathcal{Z}_p)\}.$$

For a fixed p , the evaluation rows at the distinct elements of $\mathcal{Z}_p$ have Vandermonde rank $Z(p)$. The Chinese remainder theorem then gives the exact index

$$[\mathbb{Z}[U]_{\leq A} : \mathcal{L}_{X,A}] = \prod_{p \in \mathcal{P}_X} p^{Z(p)} = e^{\Delta_X}. \quad (439)$$

Write $n = A + 1$, $k = n - s + 1$, and apply Minkowski's second theorem to the coefficient cube in $\mathbb{R}^n$. Since a nonzero integral vector has coefficient sup norm at least one, the k -th successive minimum satisfies

$$\lambda_k^s \leq \lambda_1 \cdots \lambda_n \leq e^{\Delta_X}.$$

Hence there are k linearly independent polynomials $P_1, \dots, P_k \in \mathcal{L}_{X,A}$ with

$$\|P_i\|_\infty \leq e^{\Delta_X/s}. \quad (440)$$

Their common polynomial gcd has degree at most $s - 1$: a common factor of degree d would put all quotients in the $(A - d + 1)$ -dimensional space of polynomials of degree at most $A - d$, forcing $k \leq A - d + 1 = n - d$.

If $p \mid R_X(m)$, then $m \bmod p \in \mathcal{Z}_p$. Apéry reflection also gives $-m - 1 \bmod p \in \mathcal{Z}_p$, and therefore

$$R_X(m) \mid P_i(m), \quad R_X(m) \mid P_i(-m - 1) \quad (1 \leq i \leq k). \quad (441)$$

If every P_i vanishes at both displayed integers, their common gcd is divisible by

$$G_m(U) = (U - m)(U + m + 1).$$

The two roots belonging to distinct nonnegative integers m are disjoint. Since the common gcd has degree at most $s - 1$, this can happen for at most $\lfloor (s - 1)/2 \rfloor$ values of m .

For every other $m < X^2$, at least one of the two evaluations of some P_i is a nonzero integer. Equations (440) and (441) give

$$\log R_X(m) \leq \frac{\Delta_X}{s} + A \log(m+2) + \log(A+1) \leq \frac{\Delta_X}{s} + E_{X,A}.$$

Since $K_X(m) \log X \leq \log R_X(m)$, this proves (436).

For the unconditional consequence, take $A = \lceil CX^{2/3} \rceil$, with C large enough for the proved bound $Z(p) \ll p^{2/3}$. The existing corank estimate $\sum_{p \leq x} Z(p) \ll x^{8/5}/\log x$ gives $\Delta_X \ll X^{8/5}$, while $E_{X,A} \ll X^{2/3} \log X$. For $T \geq X^{14/15+\varepsilon}$, choose

$$s = \max \left\{ 1, \left\lceil \frac{2\Delta_X}{T \log X} \right\rceil \right\}.$$

Then $s \leq A+1$, the evaluation term is $o(T \log X)$, and (436) yields (437).

Under the two hypotheses for the density-sensitive statement, the same choice of s gives $n_T(X) \leq \Delta_X/(T \log X)$. Finally,

$$\Delta_X \leq 2X \log(2X) \sum_{p \in \mathcal{P}_X} \frac{Z(p)}{p} = 2X \log(2X) \lambda_X,$$

which proves (438). □

Proposition 437 (Exact determinant drop at an aligned pair). *Assume that*

$$A \geq \max\{2, 1 + \max_{p \in \mathcal{P}_X} Z(p)\}.$$

For a nonnegative integer m , put

$$G_m(U) = (U - m)(U + m + 1).$$

Also put

$$\mathcal{L}_{X,A} = \{P \in \mathbb{Z}[U] : \deg P \leq A, P(r) \equiv 0 \pmod{p} \ (p \in \mathcal{P}_X, r \in \mathcal{Z}_p)\},$$

$$\mathcal{W}_m = G_m(U) \mathbb{Z}[U]_{\leq A-2}, \quad M_X(m) = \prod_{\substack{p \mid R_X(m) \\ p \nmid 2m+1}} p.$$

Then

$$[\mathcal{W}_m : \mathcal{L}_{X,A} \cap \mathcal{W}_m] = \frac{e^{\Delta_X} M_X(m)}{R_X(m)^2}. \quad (442)$$

Moreover, if rem_{G_m} denotes coefficient remainder modulo the monic quadratic G_m , then

$$[\mathbb{Z}[U]_{<2} : \text{rem}_{G_m} \mathcal{L}_{X,A}] = \frac{R_X(m)^2}{M_X(m)}. \quad (443)$$

Proof. Work first modulo one shell prime. If $p \nmid R_X(m)$, the remainder has no condition. If $p \mid R_X(m)$ and $p \nmid 2m+1$, its two values at the distinct roots $m, -m-1 \pmod{p}$ vanish, so both remainder coefficients vanish; the local quotient index is p^2 . If $p \mid R_X(m)$ and $p \mid 2m+1$, the roots coincide and the local index is p . The conditions just listed are also sufficient. Indeed, after prescribing an admissible linear remainder C , write $P = C + G_m Q$. At every zero of the

fibre not already absorbed by G_m , the required value of Q is uniquely prescribed because G_m is nonzero there. There are at most $Z(p)$ such values, and $A-2 \geq Z(p)-1$, so Lagrange interpolation produces a polynomial Q_p of degree at most $A-2$ over $\mathbb{F}_p$. Coefficientwise Chinese remaindering of the Q_p , together with an integral lift of C , gives one global P . Thus these are exactly the local image conditions, and the Chinese remainder theorem proves (443).

The kernel of the remainder map on $\mathcal{L}_{X,A}$ is $\mathcal{L}_{X,A} \cap \mathcal{W}_m$. Index multiplicativity together with (439) now gives (442). $\square$

Remark 438 (What the root lattice does not prove). The bound (437) is unconditional, but its power of T is one rather than the fifth power needed in Theorem 435. Proposition 437 explains the surviving geometry: a large $R_X(m)$ makes the intersection with the symmetric-factor space *denser*. Abstract reflected fibres $\{m_0, -m_0 - 1\} \bmod p$ saturate this determinant drop for every shell prime. This is a counterexample only to an argument using zero counts, reflection, and the lattice determinant; it is not a counterexample to the Apéry sequence. Eliminating the common factor G_m requires an additional cross-characteristic property of the actual Apéry zeros.

Remark 439 (Status of the frontier). The gap between item (1) and item (2) is a single factor T/λ_X per level: Chebyshev gives $X^2\lambda_X^2/T^2$, the conjecture requires $X^2\lambda_X^3/T^3$. The Poisson model predicts $n_T \approx X^2\lambda_X^T/T!$ and $\max K_X \asymp \log X/\log \log X$, in agreement with computation ($\max K = 4$ at $X = 1024$; nearest-neighbour statistics of the level sets Poisson to three digits). Every ensemble method examined in this program—large-sieve dispersion, Linnik dispersion with completed CRT sums, variance layer-cake, codegree-weighted pair dispersion—either requires the atom bound at its diagonal or is subsumed by Proposition 426. The full conjecture is thereby reduced to a pointwise statement about the divisor structure of the Apéry sequence. Theorem 436 now controls the number of columns above the isolated-five-star threshold, but a few columns can still carry too much factorial mass. Its exact remaining obstruction is the aligned symmetric factor in Remark 438.

13.9 The corank ladder and its ceiling

The average exponent c in $\sum_{p \leq x} Z(p) \ll x^{1+c}/\log x$ determines the moment order needed for the full conjecture via the trade-off $c + 2/k < 1$.

c	k needed	status
2/3	7	proved (gap polynomial)
3/5	6	proved (corank + First Lemma)
1/2	5	conditional on AC^{tr} , FDAC^{tr} , (SWR), or (QPRS) (§13.10)
$o(1)$	3	requires parity-breaking

The pair corank argument (Gershgorin root-strip, Corollary 423) gives $c = 3/5$. The naïve adjacent-pair quadruple certificate $\gcd(S_{h_1, h_2}, S_{h_2, h_3})$ is not transverse: when $h_1 = h_3$, the symmetry $|S_{h_1, h_2}| = |S_{h_2, h_1}|$ makes it a self-gcd. In particular, the previously reported numerical decay for this mass had inadvertently omitted its diagonal. The skipped triple formed by the first, second, and fourth zeros instead pairs S_{h_1, h_2} with S_{h_1, h_2+h_3} . This removes the literal self-gcd, but a second degeneracy remains when $h_3 = h_1$: if h_2 is even, the central factor of N_{h_2} propagates into both resultants (Proposition 43). The transverse formulation below deflates that linear factor. It then replaces the scalar gcd of two resultants by the coefficient content of one pencil resultant. This aligned content records that the *same* starting point is a root of all three gap polynomials, rather than merely that the same prime divides two resultants for possibly unrelated roots. The still more singular arithmetic-progression case $h_1 = h_2 = h_3$ has only one gap parameter and is handled directly

by a single resultant. The separate obstruction at the triple level is that $U_{h_1, h_2} = \text{Res}(N_{h_1}, N_{h_1+h_2})$ factors as $U = \pm S_{h_1, h_2} \cdot \text{Res}(N_{h_1}, N_{h_1+1}) \cdot (\text{leading-coefficient})$, so certificates within one triple are algebraically dependent.

13.10 The quadruple corank argument: average $Z(p) \ll p^{1/2}$

Let $\varepsilon = 17 + 12\sqrt{2}$ and $V_n = (\varepsilon^n - \varepsilon^{-n})/(\varepsilon - \varepsilon^{-1})$ the Lucas sequence with $V_{n+1} = 34V_n - V_{n-1}$, $V_0 = 0$, $V_1 = 1$, so that $V_n = U_{n-1}(17)$ is the leading coefficient of N_n .

Definition 440 (Structural content and reduced resultants). For $H \geq 2$ put $\mathfrak{U}_H = \prod_{j \leq H} j! \cdot b_j \cdot V_{j+1}$, and define the saturated reduced resultant S_{h_1, h_2}^* as S_{h_1, h_2} with all content shared with $\mathfrak{U}_H$ removed: $S^* = S / \gcd(S, \mathfrak{U}_H^\infty)$. For $b \geq 2$ put

$$N_b^\circ(x) = \begin{cases} N_b(x)/(2x + b + 1), & b \text{ even,} \\ N_b(x), & b \text{ odd,} \end{cases} \quad D_{a,b} = \text{Res}_x(N_a(x), N_b^\circ(x + a)),$$

and define $D_{a,b}^*$ by the same saturation.

Lemma 441 (Skipped-triple quadruple certificate). *If p has four zeros $r < r+h_1 < r+h_1+h_2 < r+h_1+h_2+h_3 \leq p-1$ in $\mathcal{Z}_p$, then $p \mid \gcd(S_{h_1, h_2}, S_{h_1, h_2+h_3})$.*

Proof. No two of the displayed zeros are consecutive (Lemma 10), so $h_i \geq 2$. Applying the gap-polynomial identity to the first, second, and third zeros, and then to the first, second, and fourth zeros, gives

$$N_{h_1}(r) = N_{h_2}(r + h_1) = N_{h_2+h_3}(r + h_1) = 0 \pmod{p}.$$

There is no denominator problem because every index involved lies in $[0, p-1]$. Thus r is a common root of $N_{h_1}(x)$ and $N_{h_2}(x + h_1)$, and also of $N_{h_1}(x)$ and $N_{h_2+h_3}(x + h_1)$. Each common root makes the corresponding Sylvester determinant vanish modulo p . The two determinants are the nonzero integers S_{h_1, h_2} and S_{h_1, h_2+h_3} by Corollary 423, proving the claim. $\square$

Lemma 442 (Central deflation). *The polynomial N_b° belongs to $\mathbb{Z}[x]$, and $D_{a,b} \neq 0$ for all $a, b \geq 2$. Consider any partition of $[0, p-1]$, for an odd prime p , into intervals and select consecutive quadruples of zeros inside those intervals. Among all selected quadruples with palindromic gap vector (a, b, a) , at most one has*

$$b \text{ even,} \quad 2(r + a) + b + 1 \equiv 0 \pmod{p}. \quad (444)$$

Every other such quadruple has r as a common root modulo p of $N_a(x)$ and $N_b^\circ(x + a)$, as well as of $N_a(x)$ and $N_{a+b}(x + a)$.

Proof. When b is even, the reflection identity $N_b(-x - b - 1) = (-1)^{b-1} N_b(x)$ makes $-(b+1)/2$ a root of N_b . The primitive polynomial $2x + b + 1$ therefore divides N_b in $\mathbb{Z}[x]$ by Gauss's lemma. Every root of $N_b^\circ(x + a)$ is a root of $N_b(x + a)$, so the root-strip theorem gives $D_{a,b} \neq 0$.

For a palindromic quadruple, the gap identities give

$$N_a(r) = N_b(r + a) = N_{a+b}(r + a) = 0 \pmod{p}.$$

Unless (444) holds, the factor $2(r + a) + b + 1$ is a unit, so $N_b^\circ(r + a) = 0$ as well. If (444) does hold, then its endpoints satisfy

$$2r + 2a + b \equiv -1 \pmod{p}.$$

Both endpoints lie in $[0, p-1]$, hence $r + (r + 2a + b) = p - 1$. By Corollary 386, the Apéry zero set is invariant under $z \mapsto p - 1 - z$. If its elements are $u_1 < \dots < u_{Z(p)}$, a consecutive quadruple $u_i, \dots, u_{i+3}$ with $u_i + u_{i+3} = p - 1$ must satisfy $i = Z(p) - i - 2$, which determines i uniquely. A quadruple consecutive inside one partition interval is also consecutive in the full zero set, proving the asserted global uniqueness. $\square$

Definition 443 (Aligned pencil content). For $a, b, c \geq 2$ that are not all equal, define three polynomials in $\mathbb{Z}[x]$ by

$$(F_{a,b,c}, G_{a,b,c}, J_{a,b,c}) = \begin{cases} (N_a(x), N_b(x+a), N_{b+c}(x+a)), & c \neq a, \\ (N_a(x), N_b^\circ(x+a), N_{a+b}(x+a)), & c = a, a \neq b. \end{cases}$$

Put

$$\mathcal{R}_{a,b,c}(T) = \text{Res}_x(F_{a,b,c}, G_{a,b,c} + TJ_{a,b,c}) \in \mathbb{Z}[T], \quad \mathcal{C}_{a,b,c} = \text{cont}_T \mathcal{R}_{a,b,c},$$

where cont_T denotes the positive gcd of all integer coefficients. Relative to an ambient height H , let $\mathcal{C}_{a,b,c}^*$ be obtained from $\mathcal{C}_{a,b,c}$ by removing its full prime-power part supported on $\mathfrak{U}_H$.

Lemma 444 (Aligned-content valuation). *Let $F, G, J \in \mathbb{Z}[x]$ have positive generic degrees, let*

$$\mathcal{R}(T) = \text{Res}_x(F, G + TJ) \neq 0, \quad \mathcal{C} = \text{cont}_T \mathcal{R},$$

and suppose that $\overline{F} \neq 0$ in $\mathbb{F}_p[x]$. If $\overline{F}, \overline{G}, \overline{J}$ have t distinct common roots in $\mathbb{F}_p$, then

$$t \leq v_p(\mathcal{C}). \tag{445}$$

No simplicity assumption on those roots is required.

Moreover, every pencil in Definition 443 is nonzero, and after saturation one has

$$\mathcal{C}_{a,b,c}^* \mid \begin{cases} \gcd(S_{a,b}^*, S_{a,b+c}^*), & c \neq a, \\ \gcd(D_{a,b}^*, S_{a,a+b}^*), & c = a, a \neq b. \end{cases} \tag{446}$$

Proof. Equip $K = \mathbb{Q}_p(T)$ with the Gauss valuation

$$v_G \left(\sum_j q_j T^j \right) = \min_j v_p(q_j)$$

and extend it to rational functions. Its valuation ring is a discrete valuation ring with uniformizer p and residue field $k = \mathbb{F}_p(T)$. Write $m = \deg F$ and $n = \max(\deg G, \deg J)$. Over this valuation ring the generic Sylvester map is

$$\Phi : \mathcal{O}[x]_{<n} \oplus \mathcal{O}[x]_{<m} \longrightarrow \mathcal{O}[x]_{<m+n}, \quad (A, B) \longmapsto AF + B(G + TJ),$$

and its determinant is $\pm \mathcal{R}(T)$.

Let $\alpha_1, \dots, \alpha_t$ be the common roots in $\mathbb{F}_p$. Evaluation at each α_i defines a k -linear functional on the target of Φ , and all these functionals annihilate its image. They are linearly independent: since $\overline{F} \neq 0$, one has $t \leq \deg \overline{F} \leq m$, and Lagrange interpolation separates the t evaluation maps on $k[x]_{<m+n}$. Thus Φ has corank at least t . Smith normal form over the Gauss valuation ring gives

$$v_G(\det \Phi) \geq t.$$

If $\mathcal{R}(T) = \sum_j r_j T^j$, then

$$v_G(\mathcal{R}) = \min_j v_p(r_j) = v_p\left(\gcd_j r_j\right) = v_p(\mathcal{C}),$$

which proves (445). Notice that the Sylvester matrix has its generic size before reduction; a drop of a leading coefficient modulo p cannot invalidate this corank argument.

For the pencils in Definition 443, put $d_F = \deg F_{a,b,c}$, $d_G = \deg G_{a,b,c}$, and $d_J = \deg J_{a,b,c}$. In both cases $d_J > d_G$, and specialization of the generic-degree resultant gives

$$\mathcal{R}_{a,b,c}(0) = \text{lc}(F_{a,b,c})^{d_J-d_G} \text{Res}_x(F_{a,b,c}, G_{a,b,c}),$$

whereas the coefficient of T^{d_F} is $\text{Res}_x(F_{a,b,c}, J_{a,b,c})$. The former resultant is $S_{a,b}$ off the palindromic slice and $D_{a,b}$ on that slice, up to sign; the latter is $S_{a,b+c}$, again up to sign. They are nonzero by Corollary 423 and Lemma 442. Hence the pencil resultant is nonzero. Its content divides both displayed endpoint coefficients. Finally, $\text{lc}(F_{a,b,c}) = V_a$, whose prime support is included in $\mathfrak{U}_H$. Removing all $\mathfrak{U}_H$ -supported prime powers therefore gives (446). $\square$

Proposition 445 (Normalization length of pencil content). *Let R be a complete discrete valuation ring of characteristic zero with perfect residue field, fraction field K , and valuation v_R . Let $F, G, J \in R[x]$, assume that the leading coefficient of F is a unit, and suppose $\text{Res}_x(F, G + TJ) \neq 0$. Let $\widehat{F}$ be the monic polynomial obtained from F by dividing by its leading coefficient, and factor it over K as*

$$\widehat{F} = \prod_i f_i^{m_i},$$

where the f_i are distinct monic irreducibles. Put $L_i = K[x]/(f_i)$, let $\mathcal{O}_i$ be the valuation ring of L_i , and write α_i for the image of x . Then

$$v_R(\text{cont}_T \text{Res}_x(F, G + TJ)) = \sum_i m_i \ell_R(\mathcal{O}_i / (G(\alpha_i), J(\alpha_i))). \quad (447)$$

Proof. The unit leading coefficient makes no contribution to the valuation. Multiplicativity of the resultant gives, up to a unit of R ,

$$\text{Res}_x(F, G + TJ) = \prod_i N_{L_i/K}(G(\alpha_i) + TJ(\alpha_i))^{m_i}.$$

Let e_i and f_i^{res} denote the ramification index and residue degree of L_i/K , and normalize the integral valuation v_i on L_i by $v_i|_K = e_i v_R$. The Gauss valuation is multiplicative. After passing to an algebraic closure of K , every conjugate of the linear polynomial $G(\alpha_i) + TJ(\alpha_i)$ has Gauss valuation

$$\frac{1}{e_i} \min\{v_i(G(\alpha_i)), v_i(J(\alpha_i))\}.$$

There are $e_i f_i^{\text{res}}$ conjugates, so

$$v_R(N_{L_i/K}(G(\alpha_i) + TJ(\alpha_i))) = f_i^{\text{res}} \min\{v_i(G(\alpha_i)), v_i(J(\alpha_i))\}.$$

The right-hand side is exactly the R -length of $\mathcal{O}_i / (G(\alpha_i), J(\alpha_i))$. Summing over the branches with multiplicities m_i proves (447), because the Gauss valuation of a polynomial over K is the minimum valuation of its coefficients. $\square$

Remark 446. Formula (447) is a length on the normalization of the reduced K -algebra defined by F , with the scheme-theoretic multiplicities of F restored. It is not, in general, the length of $R[x]/(F, G, J)$: the order $R[x]/(F)$ need not be integrally closed, and normalization can change the local length. The two descriptions agree, for example, when $R[x]/(F)$ is finite étale over R . A concrete discrepancy is obtained, for odd residue characteristic, from

$$F = x^2 - \varpi^2, \quad G = x - \varpi, \quad J = x + \varpi.$$

Here $\text{Res}_x(F, G + TJ) = -4\varpi^2 T$, so the content valuation and the sum of the two normalized branch lengths are both 2, whereas $R[x]/(F, G, J) \cong R/(2\varpi)$ has length 1.

Lemma 447 (Column bound over residue extensions). *Let $H \geq 2$, let $p \nmid \mathfrak{U}_H$, let $k/\mathbb{F}_p$ be a finite extension, and let $z \in k$. If $I \subseteq \{1, \dots, H\}$ is an interval of L consecutive integer levels, then*

$$\#\{r \in I : N_r(z) = 0\} \leq \frac{3^{4/3}}{2} L^{2/3} + O(L^{1/3} + 1).$$

The implied constant is absolute.

Proof. Since $p \nmid \mathfrak{U}_H$, one has $p > H$. Two consecutive zero levels would give a common root over k of N_r and N_{r+1} and hence would force $p \mid \text{Res}(N_r, N_{r+1})$, contrary to (16) and the definition of $\mathfrak{U}_H$.

Order the zero levels in I . If $r < r+d$ are consecutive among them, then $d \geq 2$ and $N_{r+1}(z) \neq 0$. The proof of Lemma 26, which is valid over every field, gives

$$N_{r+d}(z) = N_{r+1}(z)N_d(z+r),$$

so $N_d(z+r) = 0$. For fixed d , the polynomial $X \mapsto N_d(z+X)$ is nonzero of degree $3(d-1)$; nonvanishing follows, for example, from $N_d(-1) = ((d-1)!)^3 b_{d-1} \not\equiv 0 \pmod{p}$. The possible starts r are distinct in the prime subfield because $L \leq H < p$. Hence at most $3(d-1)$ consecutive-zero gaps have length d .

For any $D \geq 2$ it follows that

$$\#\{r \in I : N_r(z) = 0\} \leq \frac{L}{D+1} + \sum_{d=2}^D 3(d-1) + O(1).$$

Taking $D \asymp (L/3)^{1/3}$ proves the assertion. □

13.10.1 A cofactor certificate without branch-reuse loss

For an ambient height H and $2 \leq a \leq H-2$, put

$$I_a(H) = \{r \in \mathbb{Z} : 2 \leq r \leq H-a\}, \quad \tilde{N}_a(Y) = N_a(Y-a).$$

For $r \in I_a(H)$ define

$$P_{a,r}(Y) = \prod_{\substack{s \in I_a(H) \\ s \neq r}} N_s(Y), \quad Q_{a,H}(Y; \mathbf{T}) = \sum_{r \in I_a(H)} T_r P_{a,r}(Y),$$

and set

$$\mathfrak{K}_{a,H} = \text{cont}_{\mathbf{T}} \text{Res}_Y(\tilde{N}_a, Q_{a,H}).$$

As usual, $\mathfrak{K}_{a,H}^*$ denotes removal of its full prime-power part supported on $\mathfrak{U}_H$.

Lemma 448 (Cofactor Gauss formula). *The cofactor resultant is nonzero. Fix $p \nmid \mathfrak{U}_H$ and factor the monic normalization of $\tilde{N}_a$ over $\mathbb{Q}_p$ as*

$$V_a^{-1} \tilde{N}_a = \prod_{\lambda} f_{\lambda}^{\mu_{\lambda}}.$$

Choose a root y_{λ} , let f_{λ}^{res} be the residue degree of its branch, normalize its integral valuation by $\text{ord}_{\lambda}(p) = e_{\lambda}$, and put

$$v_{\lambda,r} = \text{ord}_{\lambda} N_r(y_{\lambda}), \quad K_{\lambda,m} = \#\{r \in I_a(H) : v_{\lambda,r} \geq m\}.$$

Then

$$v_p(\mathfrak{K}_{a,H}) = \sum_{\lambda} \mu_{\lambda} f_{\lambda}^{\text{res}} \left(\sum_{r \in I_a(H)} v_{\lambda,r} - \max_{r \in I_a(H)} v_{\lambda,r} \right) \quad (448)$$

$$= \sum_{\lambda} \mu_{\lambda} f_{\lambda}^{\text{res}} \sum_{m \geq 1} (K_{\lambda,m} - 1)_+. \quad (449)$$

Proof. For every $r \in I_a(H)$, the pure T_r coefficient of top degree is, up to a nonzero power of V_a and a sign,

$$\prod_{\substack{s \in I_a(H) \\ s \neq r}} S_{a,s},$$

which is nonzero by the First Lemma. Hence the cofactor resultant is nonzero. On the branch y_{λ} , independence of the variables T_r gives the multivariate Gauss valuation

$$\min_r \sum_{s \neq r} v_{\lambda,s} = \sum_s v_{\lambda,s} - \max_s v_{\lambda,s}.$$

Resultant multiplicativity and the norm calculation in Proposition 445 multiply this by $\mu_{\lambda} f_{\lambda}^{\text{res}}$. Summing the branches proves the first formula. The second follows from

$$\sum_r v_{\lambda,r} = \sum_{m \geq 1} K_{\lambda,m}, \quad \max_r v_{\lambda,r} = \sum_{m \geq 1} \mathbf{1}_{K_{\lambda,m} > 0}.$$

□

Lemma 449 (Adjacent edge-cover compression). *Assume $|I_a(H)| \geq 2$, and let $\mathcal{M}_a$ be any edge cover of the path on $I_a(H)$ by adjacent pairs. For $e \in \mathcal{M}_a$ put*

$$P_{a,e}^{[2]}(Y) = \prod_{s \in I_a(H) \setminus e} N_s(Y)$$

and define $\mathfrak{K}_{a,H}^{[2]}$ by replacing the cofactor pencil above with $\sum_{e \in \mathcal{M}_a} U_e P_{a,e}^{[2]}$. Then

$$(\mathfrak{K}_{a,H}^{[2]})^* = \mathfrak{K}_{a,H}^*.$$

For $|I_a(H)| = 1$, both starred contents are 1 under the natural convention.

Proof. Fix $p \nmid \mathfrak{U}_H$ and one normalized branch, and write $v_r = \text{ord}_\lambda N_r(y_\lambda)$. The adjacent resultant identity implies that v_r and v_{r+1} cannot both be positive. Hence every adjacent edge e satisfies

$$\sum_{s \in e} v_s \leq \max_t v_t.$$

An edge containing an index of maximal valuation has equality, since its other endpoint has valuation zero. Because the edge cover contains such an edge,

$$\min_{e \in \mathcal{M}_a} \sum_{s \notin e} v_s = \sum_s v_s - \max_s v_s.$$

The branchwise Gauss formula therefore agrees with (448) at every prime outside $\mathfrak{U}_H$. Deleting the structural prime powers proves the assertion. $\square$

Proposition 450 (Direct cofactor quadruple certificate). *Let an odd prime p be fixed, partition $[0, p-1]$ into the canonical blocks of length at most H , and select consecutive quadruples of zeros inside each block. If $Q_p(H)$ is the number selected and $p \nmid \mathfrak{U}_H$, then*

$$Q_p(H) \leq \sum_{a=2}^{H-2} v_p(\mathfrak{K}_{a,H}). \quad (\text{CC1})$$

The same inequality holds with $\mathfrak{K}_{a,H}^{[2]}$.

Proof. A selected quadruple with gaps (a, b, c) and start x gives, after putting $y = x + a$,

$$\tilde{N}_a(y) = 0, \quad N_b(y) = N_{b+c}(y) = 0 \pmod{p}.$$

Both b and $b+c$ lie in $I_a(H)$. Every coefficient $P_{a,r}(y)$ therefore contains at least one of these two zero factors. Distinct selected starts with the same first gap give distinct translated roots. Evaluation at those roots and Smith normal form of the fixed-size Sylvester map bound their number by $v_p(\mathfrak{K}_{a,H})$. Summing over a proves (CC1). For the edge-cover pencil, an adjacent edge cannot contain both b and $b+c$, since $c \geq 2$, so the same proof applies. $\square$

Proposition 451 (Unconditional cofactor height). *One has*

$$\sum_{a=2}^{H-2} \log \mathfrak{K}_{a,H}^* \ll H^4 \log H. \quad (\text{CC2})$$

Proof. For any $r_0 \in I_a(H)$, the corresponding pure coefficient and the First Lemma give

$$\mathfrak{K}_{a,H}^* \mid \prod_{\substack{s \in I_a(H) \\ s \neq r_0}} S_{a,s}^*.$$

Since $\log |S_{a,s}| \ll (a+s)^2 \log(a+s)$,

$$\sum_a \log \mathfrak{K}_{a,H}^* \ll \sum_{a+s \leq H} (a+s)^2 \log(a+s) \ll H^4 \log H.$$

$\square$

The first and third adjacent gaps also give a simpler endpoint certificate. Its height is substantially smaller when either outer gap is short.

Proposition 452 (Short outer-gap quadruple mass). *For $a, b, c \geq 2$ put*

$$\mathfrak{R}_{a,b,c}^{\text{out}} := \text{Res}_x(N_a(x), N_c(x + a + b)) \in \mathbb{Z} \setminus \{0\}. \quad (\text{CC2a})$$

Fix the canonical block selection of consecutive quadruples used in Proposition 450, and let $Q_p^{\text{out}}(H, C)$ count the selected quadruples whose gap vector (a, b, c) satisfies

$$a + b + c < H, \quad \min\{a, c\} \leq C.$$

Then, for $2 \leq C \leq H$,

$$\sum_{p \nmid \mathfrak{U}_H} (\log p) Q_p^{\text{out}}(H, C) \ll C^2 H^3 \log H. \quad (\text{CC2b})$$

Here the sum may be taken over all odd primes; restricting it to any prime window only decreases the left-hand side.

Proof. A selected start r with gaps (a, b, c) is simultaneously a root modulo p of

$$N_a(x) \quad \text{and} \quad N_c(x + a + b).$$

The roots of the first polynomial have real parts in $(-a, -1)$, whereas those of the second have real parts in $(-a - b - c, -a - b - 1)$. The strips are disjoint, so (CC2a) is nonzero. If $p \nmid \mathfrak{U}_H$, both reductions are nonzero. Distinct selected starts with one fixed gap triple give distinct common roots; evaluation on the fixed Sylvester matrix and Smith normal form therefore give

$$t_p(a, b, c) \leq v_p(\mathfrak{R}_{a,b,c}^{\text{out}}). \quad (\text{CC2c})$$

We record the height with its dependence on the two outer gaps. The tridiagonal determinant formula gives

$$\log \|N_h\|_1 \ll h \log(h + 1),$$

because every matrix entry has coefficient norm polynomial in h and the determinant has only exponentially many nonzero expansion terms. Also

$$\|f(x + s)\|_1 \leq \|f\|_1 (1 + |s|)^{\deg f}.$$

Applying the standard Sylvester–Hadamard bound, using $\deg N_h = 3(h - 1)$ and $a + b + c < H$, yields

$$\log |\mathfrak{R}_{a,b,c}^{\text{out}}| \ll ac \log H. \quad (\text{CC2d})$$

Sum (CC2c) against $\log p$ and use the product formula. The triples with $c \leq C$ contribute at most

$$(\log H) \sum_{c \leq C} c \sum_{a < H - c} a(H - a - c) \ll C^2 H^3 \log H.$$

The same calculation with a and c interchanged handles $a \leq C$; their overlap only decreases the combined bound. $\square$

Corollary 453 (Outer-gap localization at the reciprocal threshold). *For every fixed $\varepsilon > 0$, the selected quadruples with*

$$\min\{a, c\} \leq H^{18/41-\varepsilon}$$

have total nonstructural prime mass

$$\ll H^{4-5/41-2\varepsilon+o(1)}. \quad (\text{CC2e})$$

Consequently a first-return or endpoint-participation proof aimed at the strict threshold $H^{4-5/41}$ may restrict its unresolved core to

$$a, c > H^{18/41-\varepsilon}.$$

Proof. Take $C = H^{18/41-\varepsilon}$ in (CC2b) and absorb $\log H$ into $H^{o(1)}$. $\square$

The same coefficient-norm calculation sharpens the height of the ordinary adjacent resultant when its two gap lengths are unbalanced.

Lemma 454 (Product-sensitive adjacent-resultant height). *For every $a, b \geq 2$,*

$$\log |S_{a,b}| \ll ab \log(a+b). \quad (\text{CC2f})$$

Proof. The tridiagonal determinant formula gives $\log \|N_h\|_1 \ll h \log(h+1)$. Moreover

$$\|N_b(x+a)\|_1 \leq \|N_b\|_1(1+a)^{\deg N_b}.$$

Since $\deg N_h = 3(h-1)$, the Sylvester–Hadamard bound gives

$$\log |\text{Res}_x(N_a(x), N_b(x+a))| \ll ba \log(a+b) + ab \log(a+b).$$

The resultant is nonzero by Corollary 423, proving the assertion. $\square$

Proposition 455 (Actual endpoint-boundary quadruple mass). *Let $Q_p^{\text{end}}(H; C, L)$ count the canonically selected consecutive quadruples with gap vector (a, b, c) , $a + b + c < H$, for which either*

$$\min\{a, b\} \leq C, \quad \min\{b, c\} \leq C, \quad \text{or} \quad H - a - b - c \leq L.$$

Then, for $2 \leq C, L \leq H$,

$$\sum_{p \nmid \mathfrak{U}_H} (\log p) Q_p^{\text{end}}(H; C, L) \ll (C^2 H^2 + LH^3) \log H. \quad (\text{CC2g})$$

Proof. Fix (a, b) and sum over all admissible third gaps. Every selected quadruple start is a common root modulo p of

$$N_a(x), \quad N_b(x+a).$$

Distinct selected starts in the canonical blocks remain distinct modulo p . Since $p \nmid \mathfrak{U}_H$, the two reductions have their generic degrees. Evaluation on the Sylvester matrix and Smith normal form give

$$\#\{\text{selected starts with fixed } (a, b)\} \leq v_p(S_{a,b}).$$

After summing against $\log p$, the product formula and Lemma 454 bound this by $O(ab \log H)$.

For the first boundary sector,

$$\sum_{\min\{a,b\} \leq C} ab \ll C^2 H^2.$$

The same argument applies to the last two gaps: with $y = r + a$, every selected quadruple having fixed (b, c) gives a common root of

$$N_b(y), \quad N_c(y + b).$$

Distinct selected quadruples give distinct values of y , since the preceding zero in the same canonical block is unique. Summing first over (b, c) gives the same $C^2 H^2 \log H$ bound for $\min\{b, c\} \leq C$.

Finally put $s = b + c$ and fix (a, s) while summing over all internal splittings $s = b + c$. Every selected start is a common root of

$$N_a(x), \quad N_s(x + a),$$

and distinct selected starts remain distinct; for a fixed start the consecutive internal zeros determine the splitting uniquely. Hence the total prime mass for fixed (a, s) is $O(as \log H)$. There are $O(L)$ values $H - L \leq a + s < H$, and

$$\sum_{a+s=n} as \ll n^3 \ll H^3.$$

This gives the $LH^3 \log H$ term. A final union bound proves (CC2g). $\square$

Corollary 456 (Balanced actual-quadruple localization). *For every fixed $\varepsilon > 0$, all but*

$$\ll H^{4-5/41-\varepsilon+o(1)} \quad (\text{CC2h})$$

of the nonstructural selected-quadruple prime mass lies in the range

$$a, b, c > H^{77/82-\varepsilon}, \quad H - a - b - c > H^{36/41-\varepsilon}.$$

Thus a proof aimed at the strict $H^{4-5/41}$ threshold may restrict its unresolved actual core to these three simultaneous conditions.

Proof. In Proposition 455 take

$$C = H^{77/82-\varepsilon}, \quad L = H^{36/41-\varepsilon}.$$

The two terms in (CC2g) are respectively $H^{4-5/41-2\varepsilon+o(1)}$ and $H^{4-5/41-\varepsilon+o(1)}$. The three sectors in the proposition remove every tuple violating the displayed conditions, and the union bound proves (CC2h). $\square$

The literal first-return conditions also remove two possible sources of algebraic multiplicity. First, distinct gap polynomials have no common characteristic-zero component. Second, the selected physical returns at which a gap polynomial becomes singular modulo p have a cubic, rather than quartic, total height budget.

Lemma 457 (Exact continuant concatenation). *For integers $1 \leq u < v$ one has*

$$N_v(x) = N_{u+1}(x)N_{v-u}(x+u) - (x+u+1)^6 N_u(x)N_{v-u-1}(x+u+1), \quad (\text{CC2i})$$

where $N_0 = 0$.

Proof. Write $N_h(x) = \det D_h(x)$, with the tridiagonal matrix from the proof of Theorem 422. Cutting its continuant immediately after level u gives (CC2i). Equivalently, both sides satisfy the same recurrence in $v - u$, and the two initial cases agree. $\square$

Proposition 458 (Pairwise coprimality of the gap polynomials). *For distinct integers $u, v \geq 1$,*

$$\gcd_{\mathbb{Q}[x]}(N_u, N_v) = 1. \quad (\text{CC2j})$$

Proof. The assertion is immediate if one index is 1. Suppose first that $N_k(\alpha) = N_{k+1}(\alpha) = 0$ for some $k \geq 2$. The recurrence and the root strip give $(\alpha + k)^6 N_{k-1}(\alpha) = 0$ and $\alpha + k \neq 0$. Thus $N_{k-1}(\alpha) = 0$. Descending in the same way eventually gives $N_1(\alpha) = 0$, a contradiction. Hence adjacent gap polynomials are coprime.

Now let $2 \leq u < v - 1$ and suppose that $N_u(\alpha) = N_v(\alpha) = 0$. Equation (CC2i) and adjacent coprimality imply

$$N_{v-u}(\alpha + u) = 0.$$

The root strip for N_u puts $\Re \alpha$ in $(-u, -1)$, whereas the root strip for $N_{v-u}(\cdot + u)$ puts it in $(-v, -u - 1)$. These intervals are disjoint, proving the assertion. $\square$

Lemma 459 (Squarefreeness of every gap polynomial). *For every $h \geq 2$,*

$$\gcd_{\mathbb{Q}[x]}(N_h, N'_h) = 1. \quad (\text{CC2k})$$

Proof. Put $m = h - 1$. For $0 \leq s \leq 1$, let $J_{h,s}(x)$ be the real symmetric $m \times m$ tridiagonal matrix whose diagonal entries are $P(x + i)$, $1 \leq i \leq m$, and whose off-diagonal entries are $s(x + i + 1)^3$, $1 \leq i < m$. Thus

$$\det J_{h,1}(x) = N_h(x), \quad \det J_{h,0}(x) = \prod_{i=1}^{h-1} P(x + i).$$

The leading coefficient of $\det J_{h,s}$ is the determinant of the constant tridiagonal matrix with diagonal 34 and off-diagonal s . It is nonzero by strict diagonal dominance, so the degree remains $3(h - 1)$ throughout the deformation.

We first record a scalar localization. If

$$|P(z)| \leq |z|^3 + |z + 1|^3, \quad (\text{CC2l})$$

then $-1 < \Re z < 0$. By $P(-1 - z) = -P(z)$ it suffices to exclude $z = u + iv$ with $u \geq 0$. Put $r = |z|$ and $q = |z + 1|$, so that $q \geq r$ and $|2z + 1| \geq q$. A direct expansion gives

$$\begin{aligned} & |17z(z + 1) + 5|^2 - (r^2 + q^2)^2 \\ &= 285u^4 + 570u^3 + 570u^2v^2 + 451u^2 + 570uv^2 + 166u \\ & \quad + 285v^4 + 115v^2 + 24 > 0. \end{aligned}$$

Since $P(z) = (2z + 1)(17z(z + 1) + 5)$, it follows that

$$|P(z)| > q(r^2 + q^2) \geq r^3 + q^3,$$

contradicting (CC2l).

Let z be a zero of $\det J_{h,s}$ and choose a kernel coordinate of maximal absolute value. Its row equation gives, for some $1 \leq i \leq h - 1$,

$$|P(z + i)| \leq |z + i|^3 + |z + i + 1|^3.$$

The scalar localization puts z in the strip

$$\mathcal{S}_i = \{z : -i - 1 < \Re z < -i\}.$$

No zero crosses a strip boundary as s varies, and the degree stays fixed. At $s = 0$, the three zeros in $\mathcal{S}_i$ are

$$-i - \frac{1}{2}, \quad -i - \frac{1}{2} \pm \frac{\sqrt{51}}{34} i.$$

Thus N_h has exactly three zeros, counted with multiplicity, in each $\mathcal{S}_i$.

It remains to show that every real zero is simple. If $r \in \mathbb{R}$ and $N_h(r) = 0$, then r is not an integer, so $J_{h,1}(r)$ is unreduced and has one-dimensional kernel. Its derivative has diagonal entries $P'(r+i)$ and off-diagonal entries $3(r+i+1)^2$. The diagonal-dominance margin may vanish at an interior row, so we record positive definiteness directly. Let

$$\mathcal{J} = \text{diag}(1, -1, 1, -1, \dots), \quad E = \mathcal{J} J'_{h,1}(r) \mathcal{J},$$

and put $m = h - 1$. Since

$$P'(t) = 3t^2 + 3(t+1)^2 + 96 \left(t + \frac{1}{2}\right)^2,$$

direct expansion gives, for $w \in \mathbb{R}^m$,

$$\begin{aligned} w^\top E w &= 3(r+1)^2 w_1^2 + 3(r+h)^2 w_m^2 \\ &\quad + 3 \sum_{i=1}^{m-1} (r+i+1)^2 (w_i - w_{i+1})^2 \\ &\quad + 96 \sum_{i=1}^m \left(r+i+\frac{1}{2}\right)^2 w_i^2. \end{aligned}$$

Here r is not an integer, so the first boundary coefficient and every edge coefficient are nonzero. Vanishing of the displayed quadratic form would therefore force successively $w_1 = w_2 = \dots = w_m = 0$. Thus E , and hence $J'_{h,1}(r)$, is positive definite. If w spans the kernel, then $\text{adj}(J_{h,1}(r)) = \gamma w w^\top$ with $\gamma \neq 0$, and Jacobi's formula gives

$$N'_h(r) = \gamma w^\top J'_{h,1}(r) w \neq 0.$$

Every real zero is therefore simple. A repeated nonreal zero and its conjugate would contribute at least four zeros to the same strip, whereas each strip contains only three. Hence no zero is repeated. $\square$

Proposition 460 (Prime-weighted singular selected-quadruple bound). *Let $H \geq 6$. For every prime $p > H$, make the canonical block selection of consecutive quadruples from Proposition 450. Let $\mathcal{Q}_p^{\text{sing}}(H)$ be the selected quadruples $z_0 < z_1 < z_2 < z_3$ for which, for at least one $0 \leq i < j \leq 3$ and $h_{ij} = z_j - z_i$,*

$$N_{h_{ij}}(z_i) \equiv N'_{h_{ij}}(z_i) \equiv 0 \pmod{p}.$$

Then

$$\sum_{p>H} (\log p) \# \mathcal{Q}_p^{\text{sing}}(H) \ll H^3 \log(2H) = H^{3+o(1)}. \quad (\text{CC2m})$$

No restriction $p \nmid \mathfrak{U}_H$ is needed.

Proof. Let c_h be the positive content of N_h and put $F_h = N_h/c_h \in \mathbb{Z}[x]$. By Lemma 459,

$$\Delta_h := |\operatorname{Res}(F_h, F'_h)|$$

is a nonzero positive integer. The coefficient recurrence gives

$$\log \|N_h\|_1 \ll h \log(2H) \quad (h \leq H).$$

Since division by c_h does not increase the coefficient norm and $\deg F_h = 3(h-1)$, the Sylvester–Hadamard bound gives

$$\log \Delta_h \ll h^2 \log(2H). \quad (\text{CC2n})$$

Fix $p > H$ and $2 \leq h < H$. Lemma 12 implies that N_h is not the zero polynomial modulo p , hence $p \nmid c_h$. A physical residue is singular for N_h precisely when it is a common root of F_h and F'_h modulo p . If $t_p(h)$ denotes the number of distinct such roots, evaluation on the fixed integral Sylvester map and Smith normal form give

$$t_p(h) \leq v_p(\Delta_h). \quad (\text{CC2o})$$

This remains valid if a leading coefficient drops after reduction.

For fixed p , slot pair (i, j) , and gap h , the map from a selected sliding quadruple to its physical segment start z_i is injective: that zero has a unique position in its unique canonical block. Therefore

$$\#\mathcal{Q}_p^{\text{sing}}(H) \leq 6 \sum_{2 \leq h < H} t_p(h).$$

Multiply by $\log p$, sum over $p > H$, use (CC2o) and the product formula, then sum (CC2n) over $h < H$. $\square$

Thus the unresolved balanced actual core in Corollary 456 may additionally be assumed to have all six pair-return roots simple. Pairwise coprimality shows at the same time that the earlier-return nonvanishing conditions do not remove any generic characteristic-zero component; their remaining content is special-fibre and physical rather than a hidden factor of a gap polynomial.

There is also a useful prime-first truncation which keeps the literal Apéry selector. It does not replace the large-prime argument, but it removes a fixed initial portion of that range without using any support-blind projective roots.

Proposition 461 (Prime-first small-prime deletion). *Let $\mathcal{A}_p(H)$ be any subfamily of the canonically selected consecutive quadruples counted by $Q_p(H)$. Uniformly for $2 \leq H \leq Y$,*

$$\sum_{H < p \leq Y} (\log p) \#\mathcal{A}_p(H) \ll Y^{8/5+o(1)}. \quad (\text{CC2p})$$

In particular, with $Y = H^{12/5}$, this contribution is

$$\ll H^{96/25+o(1)} = H^{159/41-39/1025+o(1)}. \quad (\text{CC2q})$$

Thus a proof of the strict $H^{159/41}$ selected-quadruple threshold may restrict the balanced simple-root core further to primes $p > H^{12/5}$.

Proof. The canonical sliding-window rule assigns at most one selected quadruple to each of its starting zeros. Hence

$$\#\mathcal{A}_p(H) \leq Q_p(H) \leq Z(p).$$

The unconditional first-zero moment

$$\sum_{p \leq Y} Z(p) \ll Y^{8/5} / \log Y$$

therefore gives

$$\sum_{H < p \leq Y} (\log p) \#\mathcal{A}_p(H) \leq (\log Y) \sum_{p \leq Y} Z(p) \ll Y^{8/5+o(1)}.$$

Substitution of $Y = H^{12/5}$ proves (CC2q); the exponent identity is $159/41 - 96/25 = 39/1025$. $\square$

More generally, the same argument deletes $p \leq H^\kappa$ at the target scale for every fixed $\kappa < 795/328$. At the formal support threshold $\kappa = 100/41$ it gives $H^{160/41+o(1)}$, missing the target by the single factor $H^{1/41+o(1)}$. This is an actual-orbit obstruction: no nondistinguished root of a gap polynomial has entered the count.

The missing factor can be localized in the additive Fourier variable while keeping the distinguished orbit literal. We first record the full boundary transport, rather than only its restart specialization.

Lemma 462 (Actual Apéry boundary transport). *For $J \geq 0$ and $r \geq 1$ put*

$$\Delta_r(J) = \prod_{t=2}^r (J+t)^3, \quad \Delta_1(J) = 1.$$

Then

$$\boxed{\Delta_r(J) b_{J+r} = N_r(J) b_{J+1} - (J+1)^3 N_{r-1}(J+1) b_J.} \quad (\text{CC2q1})$$

Consequently, if $p \geq 5$ is prime and $0 \leq J < J+r \leq p-1$ with $p \mid b_J$, then

$$p \mid b_{J+r} \iff N_r(J) \equiv 0 \pmod{p}. \quad (\text{CC2q2})$$

Proof. The case $r = 1$ is immediate from $N_1 = 1$ and $N_0 = 0$. Multiply the Apéry recurrence at $J+r$ by $\Delta_r(J)$ and use $\Delta_r(J) = (J+r)^3 \Delta_{r-1}(J)$. Substitution of the formulas at r and $r-1$ gives the recurrence defining $N_{r+1}(J)$ in the b_{J+1} coefficient and the shifted recurrence defining $N_r(J+1)$ in the b_J coefficient. This proves (CC2q1) by induction. In the physical range every factor of $\Delta_r(J)$ is a p -unit, and $b_{J+1} \not\equiv 0 \pmod{p}$ by the no-consecutive-zero theorem. Reduction of (CC2q1) modulo p therefore proves (CC2q2). $\square$

Lemma 463 (Low-frequency localization for an actual selector). *Let $P < p \leq 2P$ be prime and let $S_p \subseteq \mathcal{Z}_p$. Put*

$$F_p(k) = \sum_{z \in \mathcal{Z}_p} e_p(kz), \quad \Sigma_{S_p}(k) = \sum_{J \in S_p} e_p(kJ).$$

Then

$$\left(1 - \frac{Z(p)}{p}\right) |S_p| = \frac{1}{p} \sum_{k \neq 0} F_p(k) \overline{\Sigma_{S_p}(k)}. \quad (\text{CC2q3})$$

Suppose, in addition, that uniformly on the shell

$$|S_p| \ll \frac{P}{D} \quad (1 \leq D \leq P). \quad (\text{CC2q4})$$

Then, for $1 \leq K \leq P/2$,

$$\left| \sum_{P < p \leq 2P} \frac{\log p}{p} \sum_{1 \leq |k| \leq K} F_p(k) \overline{\Sigma_{S_p}(k)} \right| \ll P^{3/2+o(1)} D^{-1/2} \left(K P^{-2/5} + \frac{P}{K} \right)^{1/2}. \quad (\text{CC2q5})$$

Proof. Finite Fourier orthogonality and $S_p \subseteq \mathcal{Z}_p$ give

$$\sum_{k \bmod p} F_p(k) \overline{\Sigma_{S_p}(k)} = p |S_p|.$$

Removing the zero frequency proves (CC2q3). Apply Cauchy–Schwarz to the left side of (CC2q5). Lemma 388 and the first-zero average give

$$\sum_{P < p \leq 2P} \frac{(\log p)^2}{p^2} \sum_{1 \leq |k| \leq K} |F_p(k)|^2 \ll P^{o(1)} \left(K P^{-2/5} + \frac{P}{K} \right).$$

Full Parseval for the selector and (CC2q4) give

$$\sum_{P < p \leq 2P} \sum_{k \bmod p} |\Sigma_{S_p}(k)|^2 = \sum_{P < p \leq 2P} p |S_p| \ll P^{3+o(1)} D^{-1}.$$

The product of the two square roots proves the claim. $\square$

Corollary 464 (Low-frequency localization of literal selected starts). *Assume*

$$P = H^{100/41+o(1)}, \quad K = P^{41/100+\eta}, \quad 0 < \eta < 29/100.$$

If S_p^{act} is an actual selected subset containing at most one start from each physical height- H block, then its contribution in (CC2q5) is

$$\ll P^{159/100-\eta/2+o(1)}. \quad (\text{CC2q6})$$

For the literal balanced core of Corollary 456, let its first-gap loss be $\varepsilon_c > 0$. Its selected starts are $H^{77/82-\varepsilon_c}$ -separated. Hence, if

$$\mathbf{L} = H^{1-\xi}, \quad \xi \geq \frac{5}{82} + \varepsilon_c,$$

the full literal core selector is $\mathbf{L}$ -separated and its low-frequency contribution is

$$\ll P^{159/100-\eta/2+(41/200)\xi+o(1)}. \quad (\text{CC2q7})$$

Proof. The one-per-block selector satisfies (CC2q4) with $D = H$. In (CC2q5), the term P/K dominates $K P^{-2/5}$ precisely when $\eta < 29/100$, and substitution gives (CC2q6).

For one fixed prime, if $x < x'$ are consecutive literal selected starts and the first gap at x is a , then canonical consecutiveness makes $x + a$ the next actual zero after x . Since x' is itself actual, $x' - x \geq a > H^{77/82-\varepsilon_c}$. The stated condition on ξ therefore gives $\mathbf{L}$ -separation and hence the packing bound with $D = \mathbf{L}$. A second substitution in (CC2q5) proves (CC2q7). $\square$

These are localization statements only. They force any obstruction for the one-per-block selector into wavelengths at most $P^{1/100+\eta} = H^{1/41+(100/41)\eta+o(1)}$; for the full separated core the corresponding limiting wavelength is $P^{7/200+\eta} = H^{7/82+(100/41)\eta+o(1)}$. No fixed-power estimate for these complementary high frequencies is asserted.

The high-frequency complement has an exact saturation identity. It separates the genuinely correlational off-diagonal from the selected-start diagonal before any further estimate is made.

Lemma 465 (High-frequency saturation for an actual selector). *Let $S_p \subseteq \mathcal{Z}_p$, put $s_p = |S_p|$, and retain the notation F_p and Σ_{S_p} from Lemma 463. For $1 \leq K < p/2$ define*

$$\Lambda_p(K) = \sum_{0 < |k|_p \leq K} F_p(k) \overline{\Sigma_{S_p}(k)}, \quad \Gamma_p(K) = \sum_{|k|_p > K} F_p(k) \overline{\Sigma_{S_p}(k)}.$$

Then

$$\boxed{\Gamma_p(K) = (p - Z(p))s_p - \Lambda_p(K)}. \quad (\text{CC2q8})$$

Consequently, on $P < p \leq 2P$, if

$$\mathcal{M}(S) = \sum_p (\log p) s_p, \quad \mathcal{L}_K(S) = \sum_p \frac{\log p}{p} |\Lambda_p(K)|, \quad \mathcal{E}_0(S) = \sum_p (\log p) \frac{Z(p)}{p} s_p,$$

then

$$\begin{aligned} \mathcal{M}(S) - \mathcal{E}_0(S) - \mathcal{L}_K(S) &\leq \sum_p \frac{\log p}{p} |\Gamma_p(K)| \\ &\leq \mathcal{M}(S) + \mathcal{L}_K(S), \end{aligned} \quad (\text{CC2q9})$$

and $\mathcal{E}_0(S) \ll P^{-1/3+o(1)} \mathcal{M}(S)$.

Proof. Since $S_p \subseteq \mathcal{Z}_p$, finite Fourier orthogonality gives

$$\sum_{k \bmod p} F_p(k) \overline{\Sigma_{S_p}(k)} = p s_p.$$

The zero mode is $Z(p)s_p$, and splitting the remaining frequencies at K proves (CC2q8). The triangle inequality gives (CC2q9); the last assertion follows from $Z(p) \ll P^{2/3+o(1)}$. $\square$

Thus a low-frequency saving cannot by itself make the full high side small: it forces the high side to reproduce the literal selected mass. After deleting the forced physical diagonal, however, the complete high correlation is already subcritical.

Lemma 466 (Diagonal-subtracted high-frequency identity). *Put*

$$\Gamma_p^\circ(K) = \sum_{|k|_p > K} \sum_{J \in S_p} \sum_{\substack{z \in \mathcal{Z}_p \\ z \neq J}} e_p(k(z - J)),$$

and let $m_p(K) = \#\{k \bmod p : 0 < |k|_p \leq K\}$. Then

$$\boxed{\Gamma_p^\circ(K) = -s_p(Z(p) - 1) - \Lambda_p(K) + m_p(K)s_p}. \quad (\text{CC2q10})$$

In particular,

$$\boxed{\sum_{P < p \leq 2P} \frac{\log p}{p} |\Gamma_p^\circ(K)| \ll \mathcal{L}_K(S) + \frac{K}{P} \mathcal{M}(S) + P^{-1/3+o(1)} \mathcal{M}(S)}. \quad (\text{CC2q11})$$

Proof. For every $J \in S_p$ and $z \in \mathcal{Z}_p \setminus \{J\}$, complete Fourier orthogonality gives $\sum_{k \neq 0} e_p(k(z - J)) = -1$. The low-frequency off-diagonal sum is $\Lambda_p(K) - m_p(K)s_p$. Subtraction proves (CC2q10); the aggregate estimate uses $m_p(K) \ll K$ and $Z(p) \ll P^{2/3+o(1)}$. $\square$

Corollary 467 (High off-diagonal deletion for selected starts). *Let $P = H^{100/41+o(1)}$ and $K = P^{41/100+\eta}$ with $0 < \eta \leq 29/100$. For an actual selector containing at most one selected start from each physical height- H block,*

$$\sum_{P < p \leq 2P} \frac{\log p}{p} |\Gamma_p^\circ(K)| \ll P^{159/100-\eta/2+o(1)}. \quad (\text{CC2q12})$$

For the full literal selected core in Corollary 464, with $\xi \geq 5/82 + \varepsilon_c$,

$$\sum_{P < p \leq 2P} \frac{\log p}{p} |\Gamma_p^\circ(K)| \ll P^{159/100-\eta/2+(41/200)\xi+o(1)}. \quad (\text{CC2q13})$$

At $\eta = 29/100$ the corresponding bounds are

$$H^{289/82+o(1)}, \quad H^{289/82+\xi/2+o(1)};$$

the latter is below $H^{159/41}$ whenever $\xi < 29/41$.

Proof. Insert (CC2q6) or (CC2q7) into (CC2q11). In the first case use $\mathcal{M}(S) \ll P^2/H = P^{159/100+o(1)}$; in the second use $\mathcal{M}(S) \leq \sum_p (\log p) Z(p) \ll P^{8/5+o(1)}$. The two remaining terms in (CC2q11) are smaller throughout the stated range. The final exponents follow from $P = H^{100/41+o(1)}$. $\square$

Lemmas 465 and 466 show that the ultra-high-frequency correlation is no longer the endpoint. Its off-diagonal part is subcritical, while its diagonal is exactly the physical selected-start mass. Further progress must therefore control that mass before one-prime Fourier self-correlation is formed, for example through a cross-characteristic selected-start codegree or a prime-first actual-prefix thinning theorem.

The latter alternative admits one further exact reduction. It removes all cross-characteristic collisions having the same physical start and the same first-return gap.

Proposition 468 (Actual-prefix first-return column bound). *Let $P < p \leq 2P$, and for each such prime let S_p be a family of literal selected starts. Assume that every $J \in S_p$ has a unique first-return gap $a_p(J) \leq H$ with*

$$0 \leq J < J + a_p(J) \leq p - 1, \quad p \mid b_J, \quad p \mid b_{J+a_p(J)}.$$

Put

$$R(J, a) = \sum_{P < p \leq 2P} (\log p) \mathbf{1}_{S_p}(J) \mathbf{1}_{\{a_p(J)=a\}}, \quad C(J) = \sum_a R(J, a),$$

and

$$\mathcal{M}(S) = \sum_{P < p \leq 2P} (\log p) |S_p|, \quad \mathfrak{E}_=(S) = \sum_{J, a} R(J, a)^2, \quad \mathfrak{U}(S) = \sum_J \sum_{a \neq a'} R(J, a) R(J, a').$$

Then

$$R(J, a) \leq \log N_a(J) \ll a \log(2P), \quad (\text{CC2q14})$$

$$\mathfrak{E}_=(S) \ll H P^{o(1)} \mathcal{M}(S), \quad (\text{CC2q15})$$

$$\sum_J C(J)^2 = \mathfrak{E}_=(S) + \mathfrak{U}(S), \quad (\text{CC2q16})$$

$$\mathcal{M}(S)^2 \ll P(\mathfrak{E}_=(S) + \mathfrak{U}(S)). \quad (\text{CC2q17})$$

Moreover,

$$\mathfrak{U}(S) = \sum_{\substack{P < p, q \leq 2P \\ p \neq q}} (\log p)(\log q) \# \{J \in S_p \cap S_q : a_p(J) \neq a_q(J)\}. \quad (\text{CC2q18})$$

Thus $\mathfrak{U}$ contains only unequal first-return levels in different characteristics; it has no one-prime self-diagonal and no candidate-root or projective-return contribution.

Proof. If the summand indexed by p occurs in $R(J, a)$, then Lemma 462 gives $p \mid N_a(J)$. The positive integer $N_a(J)$ is nonzero, so the product of all distinct primes occurring in that column divides $N_a(J)$. This proves the first inequality in (CC2q14); the continuant determinant and its coefficient recurrence give $\log N_a(J) \ll a \log(J + a + 2) \ll a \log(2P)$.

Consequently $R(J, a) \ll HP^{o(1)}$, and hence

$$\sum_{J, a} R(J, a)^2 \ll HP^{o(1)} \sum_{J, a} R(J, a) = HP^{o(1)} \mathcal{M}(S),$$

which proves (CC2q15). Expanding $C(J)^2$ proves (CC2q16). There are $O(P)$ possible integer starts J , so Cauchy's inequality proves (CC2q17). Finally, uniqueness of $a_p(J)$ for fixed (p, J) shows that an $a \neq a'$ term necessarily has $p \neq q$ and gives exactly (CC2q18). $\square$

For the one-per-height- H -block selector, $\mathcal{M}(S) \ll P^2/H = P^{159/100+o(1)}$, so

$$\mathfrak{E}_=(S) \ll P^{2+o(1)}. \quad (\text{CC2q19})$$

For the full literal selected core, the first-zero average gives $\mathcal{M}(S) \ll P^{8/5+o(1)}$ and therefore $\mathfrak{E}_=(S) \ll P^{201/100+o(1)}$. Both are below the critical codegree scale $P^{109/50}$. Hence the exact remaining physical-diagonal frontier is the unequal-return codegree (CC2q18): any estimate

$$\mathfrak{U}(S) \ll P^{109/50-\sigma+o(1)} \quad (\sigma > 0)$$

implies through (CC2q17)

$$\mathcal{M}(S) \ll P^{159/100-\sigma/2+o(1)}.$$

The presently proved pointwise column estimate is only $C(J) \ll H^2 P^{o(1)}$. It gives

$$\mathfrak{U}(S^{\text{act}}) \ll P^{241/100+o(1)}, \quad \mathfrak{U}(S^{\text{core}}) \ll P^{121/50+o(1)},$$

so neither selector yet reaches the critical codegree. An interface-saturating model has the smaller scale $P^{11/5}$, but this is not a proved upper bound for the actual orbit. If one establishes instead $\mathfrak{U}(S) \ll P^{11/5-\delta+o(1)}$, the same consumer beats the $P^{159/100}$ endpoint exactly when $\delta > 1/50$. No such unequal-return estimate is asserted here.

There is nevertheless an unconditional improvement for the one-per-block selector. It comes from repeated physical anchors, rather than from a second estimate in one first-return column.

Proposition 469 (Mixed-return selected-edge bound). *Retain the notation of Proposition 468. Suppose that, for some $D \geq 1$, every physical interval of length D contains $O(1)$ members of each S_p . For $L \geq D$ put*

$$W_2(J, d) = \sum_{\substack{P < p \leq 2P \\ J, J+d \in S_p}} \log p, \quad \mathcal{T}_S(L) = \sum_{\substack{J \geq 0 \\ 1 \leq d \leq L}} W_2(J, d)^2.$$

Then

$$\boxed{\mathcal{T}_S(L) \ll P^{o(1)} \frac{L^2}{D} \mathcal{M}(S).} \quad (\text{CC2q19a})$$

For every $K \geq 2$ such that $8P/K \geq D$,

$$\boxed{\mathfrak{U}(S) \ll P^{o(1)} \left(KP^2 + \frac{P^2 \mathcal{M}(S)}{DK^2} \right).} \quad (\text{CC2q19b})$$

Consequently

$$\boxed{\mathfrak{U}(S) \ll P^{2+o(1)} \left(1 + \left(\frac{\mathcal{M}(S)}{D} \right)^{1/3} \right).} \quad (\text{CC2q19c})$$

For the one-per-height- H -block selector, if $P = H^{100/41+o(1)}$, this gives

$$\boxed{\mathfrak{U}(S) \ll P^{359/150+o(1)}.} \quad (\text{CC2q19d})$$

For the full literal balanced core, where

$$D = A = P^{77/200-o(1)}, \quad \mathcal{M}(S) \ll P^{8/5+o(1)},$$

one has

$$\boxed{\mathfrak{U}(S) \ll P^{481/200+o(1)}.} \quad (\text{CC2q19e})$$

Proof. If $J, J+d \in S_p$, then both are literal Apéry zeros modulo p and $0 \leq J < J+d < p$. The nonwrapping gap dictionary gives $p \mid N_d(J)$. Lemma 412 makes this a nonzero integer divisor certificate, and the continuant recurrence gives

$$\log N_d(J) \ll d \log(2P).$$

The product of the distinct primes counted by $W_2(J, d)$ divides $N_d(J)$, whence

$$W_2(J, d) \ll d \log(2P). \quad (\text{CC2q19f})$$

Thus $W_2(J, d)^2 \ll P^{o(1)} d W_2(J, d)$. After interchanging the prime and edge sums, the left side of (CC2q19a) is at most

$$P^{o(1)} \sum_{P < p \leq 2P} (\log p) \sum_{\substack{x < y, \, x, y \in S_p \\ y-x \leq L}} (y-x).$$

For fixed x , the interval packing property and a sum over the successive intervals to its right bound the inner distance sum by $O(L^2/D)$. Summing over x proves (CC2q19a).

For distinct primes put

$$K_{p,q} = \#\{J \in S_p \cap S_q : a_p(J) \neq a_q(J)\}.$$

Equation (CC2q18) says

$$\mathfrak{U}(S) = \sum_{p \neq q} (\log p)(\log q) K_{p,q}.$$

The pairs with $K_{p,q} < K$ contribute $O(KP^2)$ by Chebyshev's estimate. If $K_{p,q} = k \geq K$, list its common mixed starts as $J_1 < \dots < J_k$. Their span is less than $2P$, so at least a fixed positive

proportion of the consecutive gaps are at most $8P/K$. Every such edge has both endpoints in both selected sets. Moreover a fixed ordered prime pair and edge is charged at most once, whereas the corresponding ordered prime-pair weight occurs in $W_2(J_i, J_{i+1} - J_i)^2$. It follows that

$$\mathfrak{U}(S) \ll KP^2 + \mathcal{T}_S(8P/K),$$

which proves (CC2q19b).

The interval packing property itself gives $\mathcal{M}(S) \ll P^2/D$; hence the balancing choice $K^3 \asymp \mathcal{M}(S)/D$ lies in the required range, with a fixed K covering $\mathcal{M}(S) < D$. This proves (CC2q19c). Finally

$$\mathfrak{U}(S) \ll P^{8/3+o(1)} H^{-2/3} = P^{359/150+o(1)},$$

because $H = P^{41/100+o(1)}$. For the full core,

$$2 + \frac{1}{3} \left(\frac{8}{5} - \frac{77}{200} \right) = \frac{481}{200},$$

which proves (CC2q19e). □

Corollary 470 (Dyadic consecutive common-prefix localization). *Retain the full literal balanced-core specialization of Proposition 469, so that*

$$D = A = P^{77/200-o(1)}, \quad \mathcal{M}(S) \ll P^{8/5+o(1)}.$$

For each unordered pair $p < q$, list the mixed common starts counted by $K_{p,q}$ as

$$J_1 < \cdots < J_{K_{p,q}}.$$

For dyadic D , put

$$\mathcal{E}_D = \sum_{P < p < q \leq 2P} (\log p)(\log q) \# \{i : D < J_{i+1} - J_i \leq 2D\}.$$

Then

$$\boxed{\mathcal{E}_D \ll P^{o(1)} \min \left\{ \frac{P^3}{D}, P^{243/200} D^2 \right\}.} \quad (\text{CC2q31})$$

In particular the two terms balance at $D = P^{119/200}$ and reproduce the unconditional bound $\mathfrak{U}(S) \ll P^{481/200+o(1)}$ from (CC2q19e).

Proof. For a fixed pair (p, q) the consecutive intervals $[J_i, J_{i+1}]$ have disjoint interiors and total length $O(P)$. Hence the number with length in $(D, 2D]$ is $O(P/D)$, and summing the prime-pair weights gives $\mathcal{E}_D \ll P^{3+o(1)}/D$.

Every such edge is counted in $W_2(J_i, J_{i+1} - J_i)^2$. If $2D < D$, the interval-packing hypothesis makes $\mathcal{E}_D = 0$. Otherwise

$$\mathcal{E}_D \leq \mathcal{T}_S(2D) \ll P^{o(1)} \frac{D^2}{D} \mathcal{M}(S) \ll P^{243/200+o(1)} D^2$$

by (CC2q19a). Taking the minimum proves (CC2q31). □

Corollary 471 (Medium repeated-prefix residual). *Under the preceding hypotheses, fix $0 < \varepsilon < 1/20$. The total contribution of dyadic consecutive common-prefix gaps satisfying either*

$$D \leq P^{193/400-\varepsilon} \quad \text{or} \quad D \geq P^{41/50+2\varepsilon}$$

is

$$\boxed{\ll P^{109/50-\varepsilon+o(1)}}. \quad (\text{CC2q32})$$

Consequently a failure of the target transverse bound may be restricted, in addition to all previously retained literal masks, to

$$\boxed{P^{193/400-\varepsilon} < D < P^{41/50+2\varepsilon}}.$$

Proof. For the first range use the second term of (CC2q31); at its endpoint the exponent is

$$\frac{243}{200} + 2 \left(\frac{193}{400} - \varepsilon \right) = \frac{109}{50} - 2\varepsilon.$$

For the second range use the first term; at its endpoint the exponent is

$$3 - \left(\frac{41}{50} + 2\varepsilon \right) = \frac{109}{50} - 2\varepsilon.$$

The logarithmically many dyadic ranges are absorbed into $P^{o(1)}$. The first common start of each unordered prime pair contributes only $O(P^{2+o(1)})$, and every further start is charged to its unique preceding consecutive edge. This proves the residual statement. $\square$

Proposition 472 (Nearby mixed first returns). *For $1 \leq D \leq H$, put*

$$\mathfrak{U}_{\leq D}(S) = \sum_J \sum_{0 < |a-a'| \leq D} R(J, a) R(J, a').$$

Then

$$\boxed{\mathfrak{U}_{\leq D}(S) \leq 2D\mathfrak{E}_=(S)}. \quad (\text{CC2q19g})$$

In particular, for the one-per-block selector and every fixed $\eta > 0$,

$$D = P^{9/50-\eta} \implies \mathfrak{U}_{\leq D}(S) \ll P^{109/50-\eta+o(1)}. \quad (\text{CC2q19h})$$

Proof. For each $h \geq 1$, $2xy \leq x^2 + y^2$ gives

$$2 \sum_a R(J, a) R(J, a+h) \leq 2 \sum_a R(J, a)^2.$$

Sum over $1 \leq h \leq D$ and J , and then use (CC2q19). $\square$

Thus the active mixed-return frontier may be restricted to first-return differences exceeding $P^{9/50-\eta}$. The global estimate (CC2q19d) improves the earlier $P^{241/100+o(1)}$ bound by $P^{1/60}$, but it remains above the critical $P^{109/50}$ scale by $P^{16/75}$ and therefore does not yet give a positive selected-mass saving. This statement applies only to the explicitly thinned one-per-height- H -block selector. A height- H block can contain several literal balanced-core starts, so (CC2q19d) does not upper-bound the full-core statistic in (CC2q23); the corresponding unthinned estimate is (CC2q19e).

The unequal-return term admits a further exact decomposition. Two of its three sectors are already below the critical codegree scale; the remaining one is characterized by a transverse two-characteristic divisibility table.

Proposition 473 (Wrap–return–transverse decomposition). *Work with the full literal balanced core and its quantities $R(J, a)$ and $\mathfrak{U}(S)$ from Proposition 468. For each later return level a' put*

$$\begin{aligned} W(J, a') &= \sum_{\substack{P < p \leq 2P: \\ J \in S_p, \ a_p(J) < a' \\ J+a' \geq p}} \log p, \\ L(J, a') &= \sum_{\substack{P < p \leq 2P: \\ J \in S_p, \ a_p(J) < a' \\ J+a' < p, \ p \nmid b_{J+a'}}} \log p, \\ T(J, a') &= \sum_{\substack{P < p \leq 2P: \\ J \in S_p, \ a_p(J) < a' \\ J+a' < p, \ p \nmid b_{J+a'}}} \log p. \end{aligned}$$

Then

$$\boxed{\frac{1}{2}\mathfrak{U}(S) = \sum_{J, a'} R(J, a')W(J, a') + \sum_{J, a'} R(J, a')L(J, a') + \sum_{J, a'} R(J, a')T(J, a').} \quad (\text{CC2q20})$$

At the balanced-core scales

$$H = P^{41/100+o(1)}, \quad A = P^{77/200-o(1)}, \quad \mathcal{M}(S) \ll P^{8/5+o(1)},$$

the first two sectors satisfy

$$\sum_{J, a'} R(J, a')W(J, a') \ll P^{369/200+o(1)}, \quad (\text{CC2q21})$$

$$\sum_{J, a'} R(J, a')L(J, a') \ll H^{4/3} \mathcal{M}(S) P^{o(1)} \ll P^{161/75+o(1)}. \quad (\text{CC2q22})$$

In particular, both are below $P^{109/50}$, the critical full-core codegree scale.

Proof. Write the ordered sum in (CC2q18) as twice the part with $a_p(J) < a_q(J) = a'$. For each earlier characteristic p , exactly one of

$$J + a' \geq p, \quad J + a' < p \text{ and } p \mid b_{J+a'}, \quad J + a' < p \text{ and } p \nmid b_{J+a'}$$

holds. This proves (CC2q20) without changing any literal selection condition.

Every first gap in the balanced core is at least A . Consecutive selected starts in one characteristic are therefore A -separated, so there are $O(1 + H/A)$ such starts in its terminal interval of length H . The column bound following (CC2q17) gives $\sum_a R(J, a) \ll H^2 P^{o(1)}$. Hence the wrap sector is

$$\ll P(H/A) H^2 P^{o(1)} = P^{369/200+o(1)},$$

which proves (CC2q21).

For L , nonwrapping continuant transfer gives $p \mid N_{a'}(J)$ for every prime counted by $L(J, a')$. Consequently

$$L(J, a') \leq \log |N_{a'}(J)| \ll H P^{o(1)}.$$

For a fixed literal selected pair (p, J) , the number of admissible a' is at most the number of actual p -zeros in $(J, J + H) \cap [0, p - 1]$, hence is $O(H^{2/3})$ by Corollary 15. Therefore

$$\sum_{J, a'} L(J, a') \ll H^{2/3} \mathcal{M}(S) P^{o(1)}, \quad \sum_{J, a'} L(J, a')^2 \ll H^{5/3} \mathcal{M}(S) P^{o(1)}.$$

Together with $\sum_{J,a'} R(J, a')^2 \ll H \mathcal{M}(S) P^{o(1)}$ from (CC2q15), Cauchy's inequality yields

$$\sum_{J,a'} R(J, a') L(J, a') \ll H^{4/3} \mathcal{M}(S) P^{o(1)}.$$

Since $8/5 + (4/3)(41/100) = 161/75$, this proves (CC2q22). $\square$

Define the surviving transverse statistic by

$$\mathfrak{U}_{\text{tr}}(S) = \sum_{J,a'} R(J, a') T(J, a'). \quad (\text{CC2q23})$$

For a term with $a = a_p(J) < a' = a_q(J)$ in this statistic, the literal divisibility table is

$$\begin{array}{c|cc} & N_a(J) & N_{a'}(J) \\ \hline p & 0 & \neq 0 \\ q & \neq 0 & 0 \end{array} \quad \text{in the corresponding characteristics,} \quad p, q \mid b_J. \quad (\text{CC2q24})$$

Thus no selected prime divides the same-prime gcd $\gcd(N_a(J), N_{a'}(J))$. The exact remaining sufficient estimate is

Proposition 474 (First-return spacing bound for the transverse core). *For the full literal balanced core,*

$$\boxed{\mathfrak{U}_{\text{tr}}(S) \ll H P^{2+o(1)} = P^{241/100+o(1)}}. \quad (\text{CC2q25})$$

Proof. The pointwise first-return column bound (CC2q14) gives

$$T(J, a') \leq \sum_{a < a'} R(J, a) \ll a'^2 P^{o(1)}.$$

For one characteristic p , the intervals from a selected actual zero J to its first return $J + a_p(J)$ are a subfamily of the consecutive nonwrapping zero gaps. Their interiors are disjoint, and hence

$$\sum_{J \in S_p} a_p(J) \leq p.$$

After multiplication by $\log p$ and summation over the shell, this is

$$\sum_{J,a} a R(J, a) \ll P^2. \quad (\text{CC2q26})$$

Therefore

$$\mathfrak{U}_{\text{tr}}(S) = \sum_{J,a'} R(J, a') T(J, a') \ll P^{o(1)} \sum_{J,a'} a'^2 R(J, a') \leq H P^{o(1)} \sum_{J,a'} a' R(J, a'),$$

and (CC2q26) proves the result. $\square$

Corollary 475 (Full-core same-label and close-label bounds). *For the full literal balanced core,*

$$\mathfrak{E}_{=}(S) \ll P^{2+o(1)}, \quad (\text{CC2q27})$$

$$\mathfrak{U}_{\leq D}(S) \ll D P^{2+o(1)} \quad (1 \leq D \leq H). \quad (\text{CC2q28})$$

In particular, for every fixed $\eta > 0$,

$$D = P^{9/50-\eta} \implies \mathfrak{U}_{\leq D}(S) \ll P^{109/50-\eta+o(1)}.$$

Proof. The pointwise primitive-column bound (CC2q14) gives $R(J, a)^2 \ll aR(J, a)P^{o(1)}$. Summing and using the literal first-return spacing moment (CC2q26) proves (CC2q27). The proof of Proposition 472 uses only $2xy \leq x^2 + y^2$ for each label shift, so the same argument with (CC2q27) proves (CC2q28). $\square$

Proposition 476 (Safe full-core first-return label-window energy). *For an interval I of integer first-return labels with $\#I \leq W$, put*

$$R_I(J) = \sum_{a \in I} R(J, a).$$

Then the full literal balanced core satisfies

$$\boxed{\sum_J R_I(J)^2 \ll WP^{2+o(1)}}. \quad (\text{CC2q29})$$

Proof. Cauchy's inequality in the at most W labels, followed by (CC2q14), gives

$$R_I(J)^2 \leq W \sum_{a \in I} R(J, a)^2 \ll WP^{o(1)} \sum_{a \in I} aR(J, a).$$

Sum over J and use the literal first-return spacing moment (CC2q26). $\square$

Proposition 477 (Set-selector first-return window energy). *Retain the full literal balanced-core set selector S_p from Proposition 468. If I is an interval of at most W first-return labels and $a \leq L$ on I , then*

$$\boxed{\sum_J R_I(J)^2 \ll P^{o(1)} \min\{L^2 H^2 W^2, P^2 W\}}. \quad (\text{CC2q33})$$

Here one selected pair (p, J) occurs once: source rows, reciprocal companions, and Palm labels are not indices of the set S_p or of the indicator defining $R(J, a)$.

Proof. Put

$$M_I = \sum_J R_I(J).$$

For a fixed selected pair (p, J) , literal consecutiveness determines the next first-return gap b uniquely. The start J is then an actual common root of

$$N_a(x), \quad N_b(x + a),$$

and different selected starts give different roots modulo p . The fixed Sylvester–Smith argument and Lemma 454 therefore give, for fixed (a, b) ,

$$\sum_{P < p \leq 2P} (\log p) \#\{J \in S_p : a_p(J) = a, b_p(J) = b\} \ll abP^{o(1)}.$$

Consequently

$$M_I \ll P^{o(1)} \sum_{a \in I} \sum_{b \leq H} ab \ll LWH^2 P^{o(1)}.$$

On the other hand, (CC2q14) gives

$$\max_J R_I(J) \ll LWP^{o(1)}.$$

Multiplying the last two estimates proves the first branch in (CC2q33). The second branch is exactly Proposition 476. $\square$

Corollary 478 (Safe maximal-separation deletion). *Suppose the retained full core has three consecutive first-return gaps $a, b, c > A$ and headroom*

$$H - a - b - c > R_0.$$

Put

$$H_1 = H - 2A - R_0, \quad \Gamma = H_1 - A.$$

For the ordered transverse terms $a < a'$, one has, uniformly for $1 \leq D \leq H$,

$$\boxed{\mathfrak{U}_{\text{tr}}(S)\{\Gamma - (a' - a) \leq D\} \ll DP^{2+o(1)}}. \quad (\text{CC2q30})$$

In particular, with

$$A = P^{77/200-o(1)}, \quad R_0 = P^{9/25-o(1)}, \quad D = P^{9/50-\eta},$$

the left side of (CC2q30) is $\ll P^{109/50-\eta+o(1)}$. Together with (CC2q28), the unresolved transverse core may therefore be restricted to

$$a' - a > D, \quad \Gamma - (a' - a) > D.$$

Proof. The headroom condition and $b, c > A$ give $a, a' < H_1$. Moreover

$$\Gamma - (a' - a) = (a - A) + (H_1 - a'),$$

and both terms on the right are nonnegative. Thus $\Gamma - (a' - a) \leq D$ forces a into the bottom label window $(A, A + D]$ and a' into the top label window $[H_1 - D, H_1)$. Cauchy's inequality between the two window sums, followed twice by Proposition 476, proves (CC2q30). $\square$

Proposition 479 (Nash-degree bound for consecutive common prefixes). *Under the full literal balanced-core hypotheses of Corollary 470, one has*

$$\boxed{\mathcal{E}_D \ll P^{2+o(1)} \frac{D}{\sqrt{A}} = P^{723/400+o(1)} D}. \quad (\text{CC2q37})$$

Consequently

$$\boxed{\mathcal{E}_D \ll P^{o(1)} \min \left\{ \frac{P^3}{D}, P^{243/200} D^2, P^{723/400} D \right\}}. \quad (\text{CC2q38})$$

The total medium repeated-prefix residual is therefore

$$\boxed{\ll P^{1923/800+o(1)}}. \quad (\text{CC2q39})$$

This improves the previous $P^{481/200+o(1)}$ envelope by $P^{-1/800}$; the remaining deficit above $P^{109/50}$ is 179/800.

Proof. Put

$$C_J = \sum_a R(J, a), \quad X_J = \sum_a aR(J, a), \quad Y_J = \sum_a R(J, a)^2.$$

For every finitely supported nonnegative sequence (r_a) ,

$$\left(\sum_a r_a \right)^3 \leq 8 \left(\sum_a ar_a \right) \left(\sum_a r_a^2 \right).$$

Indeed, for $U \geq 1$ split at U , bound the lower part by $U^{1/2}(\sum r_a^2)^{1/2}$ and the upper part by $U^{-1} \sum ar_a$, and optimize. Applying this to $R(J, a)$, then Cauchy's inequality in J , gives

$$\sum_J C_J^{3/2} \ll \left(\sum_J X_J \right)^{1/2} \left(\sum_J Y_J \right)^{1/2} \ll P^{2+o(1)} \quad (\text{CC2q37a})$$

by (CC2q26) and (CC2q27).

For fixed J form the weighted simple graph whose vertices are the selected shell characteristics through J , and whose edges are exactly the pairs whose next consecutive common selected prefix lies at $J + d$ with $D < d \leq 2D$. A fixed vertex p uses only $O(D/A)$ offsets, because its selected starts are separated by its first return, which is larger than A . At one fixed offset every partner q divides the nonzero integer $N_d(J)$, so positivity and the continuant height give

$$\sum_{q: J, J+d \in S_q} \log q \leq \log |N_d(J)| \ll DP^{o(1)}.$$

Thus the weighted degree is at most $Q_D = P^{o(1)}D^2/A$, and trivially at most C_J . If $F_{J,D}$ is the weighted edge mass, then

$$F_{J,D} \ll P^{o(1)}C_J \min\{C_J, Q_D\} \leq P^{o(1)}Q_D^{1/2}C_J^{3/2}.$$

Summing over J and using (CC2q37a) proves (CC2q37). Here the consecutive-edge marking is load-bearing: it makes the graph simple.

Combine the new bound with (CC2q31). For $D = P^\lambda$ the three exponents are $3-\lambda$, $243/200+2\lambda$, and $723/400+\lambda$. Their lower envelope is maximized where the first and third meet, at $\lambda = 477/800$, with common value $1923/800$. The logarithmically many dyadic scales are absorbed in $P^{o(1)}$. $\square$

Lemma 480 (Inverse selected-label packet moment). *With the literal full-core selector and the notation*

$$R(J, a) = \sum_{P < p \leq 2P} (\log p) \mathbf{1}_{S_p}(J) \mathbf{1}_{\{a_p(J)=a\}}, \quad C_J = \sum_a R(J, a),$$

one has

$$\boxed{\sum_a \frac{R(J, a)}{a} \ll C_J^{1/2} P^{o(1)}}. \quad (\text{CC2q40})$$

Proof. Equation (CC2q14) gives $R(J, a)/a \ll P^{o(1)}$. Hence, for every $U \geq 1$,

$$\sum_a \frac{R(J, a)}{a} \ll P^{o(1)} \left(U + \frac{C_J}{U} \right).$$

Taking $U = C_J^{1/2}$ proves the assertion. If this value lies outside the admissible label range, the corresponding lower or upper sum is empty and the same estimate holds. $\square$

Lemma 481 (Replicated first-return spacing). *Let $D > H$ and put*

$$m_D(p, J) = \#\{K \in S_p : D < K - J \leq 2D\},$$

$$R_D(J, a) = \sum_{\substack{P < p \leq 2P \\ J \in S_p, a_p(J)=a}} (\log p) m_D(p, J).$$

Then

$$\boxed{\sum_{J,a} aR_D(J,a) \ll D\mathcal{M}(S).} \quad (\text{CC2q41})$$

Proof. Fix p and a target $K \in S_p$. The source starts $J \in S_p$ with $D < K - J \leq 2D$ lie in $[K - 2D, K - D)$. Their literal first-return intervals $[J, J + a_p(J))$ have pairwise disjoint interiors: a later selected start is itself an actual p -zero. These intervals lie in $[K - 2D, K - D + H)$, whose length is $D + H \ll D$. Therefore

$$\sum_{\substack{J \in S_p \\ D < K - J \leq 2D}} a_p(J) \ll D.$$

Sum over $K \in S_p$ and then over p with weight $\log p$. This is an exact reindexing of the multiplicity in $m_D(p, J)$ and proves (CC2q41). $\square$

Proposition 482 (Replicated-selector consecutive-edge bound). *Retain the full literal balanced-core hypotheses of Corollary 470. For every dyadic $D > H$,*

$$\boxed{\mathcal{E}_D \ll P^{1+o(1)} D\mathcal{M}(S)^{1/2} \ll P^{9/5+o(1)} D.} \quad (\text{CC2q42})$$

Consequently

$$\boxed{\mathcal{E}_D \ll P^{o(1)} \min \left\{ \frac{P^3}{D}, P^{243/200} D^2, P^{9/5} D \right\}.} \quad (\text{CC2q43})$$

The total medium repeated-prefix residual is therefore

$$\boxed{\ll P^{12/5+o(1)}.} \quad (\text{CC2q44})$$

This improves (CC2q39) by $P^{-3/800}$; the remaining deficit above $P^{109/50}$ is $11/50$.

Proof. For fixed J , let $G_{J,D}$ be the weighted simple graph whose vertices are the selected shell characteristics through J , and whose edges are the pairs whose next consecutive common selected prefix is $J + d$ with $D < d \leq 2D$. Let $F_{J,D}$ be its weighted edge mass, and write

$$E_{J,a} = \sum_{\substack{P < p \leq 2P \\ J \in S_p, a_p(J)=a}} (\log p) \deg_{J,D}(p).$$

Graph simplicity gives

$$E_{J,a} \leq C_J R(J, a). \quad (\text{CC2q42a})$$

For a fixed source prime p and endpoint offset d , every partner q has the two actual nonwrapping selected zero endpoints $J, J + d$. Hence $q \mid N_d(J)$, and positivity and the continuant height give

$$\sum_{\substack{q \text{ partner of } p \\ \text{at the fixed offset } d}} \log q \leq \log |N_d(J)| \ll DP^{o(1)}.$$

The number of endpoint offsets used by p is at most $m_D(p, J)$. Therefore

$$E_{J,a} \ll DR_D(J, a) P^{o(1)}. \quad (\text{CC2q42b})$$

Taking the geometric mean of (CC2q42a) and (CC2q42b), using $2F_{J,D} = \sum_a E_{J,a}$, and applying Cauchy's inequality in a gives

$$F_{J,D} \ll D^{1/2} C_J^{1/2} \left(\sum_a a R_D(J, a) \right)^{1/2} \left(\sum_a \frac{R(J, a)}{a} \right)^{1/2} P^{o(1)}.$$

Lemma 480 therefore yields

$$F_{J,D} \ll D^{1/2} C_J^{3/4} \left(\sum_a a R_D(J, a) \right)^{1/2} P^{o(1)}.$$

Sum in J , apply Cauchy's inequality, and use (CC2q37a) and Lemma 481. This gives

$$\mathcal{E}_D \ll D^{1/2} P^{o(1)} \left(\sum_J C_J^{3/2} \right)^{1/2} \left(\sum_{J,a} a R_D(J, a) \right)^{1/2} \ll P^{1+o(1)} D \mathcal{M}(S)^{1/2}.$$

Since $\mathcal{M}(S) \ll P^{8/5+o(1)}$, this proves (CC2q42). Combining it with (CC2q31) proves (CC2q43). If $D = P^\lambda$, the three exponents are $3 - \lambda$, $243/200 + 2\lambda$, and $9/5 + \lambda$. Their lower envelope is maximized where the first and third meet, at $\lambda = 3/5$, with common value $12/5$; the logarithmically many dyadic ranges are absorbed in $P^{o(1)}$. $\square$

Remark 483 (Higher replicated moments do not improve the envelope). For each fixed real $r \geq 1$, the same two-ended first-return packing gives the raw moment bound

$$\sum_{J,a} \sum_{\substack{P < p \leq 2P \\ J \in S_p, a_p(J)=a}} (\log p) a m_D(p, J)^r \ll_r D^r \sum_{J,a} \sum_{\substack{P < p \leq 2P \\ J \in S_p, a_p(J)=a}} (\log p) a^{1-r}.$$

Interpolating this with the same simple-graph and inverse-packet bounds used above yields only

$$\mathcal{E}_D \ll_r P^{723/400-3/(400r)+o(1)} D.$$

The exponent is strictly increasing in r , so $r = 1$ is optimal in this family; $r = 2$ recovers the already present short-edge mechanism after its direct second-moment consumption. Thus higher marginal replication moments cannot improve (CC2q44); a further gain must couple owner replication to foreign-partner use. The statement is for fixed r and is not uniform when r grows with P .

Proposition 484 (Selected-offset utilization consumer). *Retain the literal graph and notation of Proposition 482, and put*

$$\mathfrak{U}_D := \sum_{J,a} a E_{J,a}.$$

Then the current estimates give

$$\boxed{\mathfrak{U}_D \ll D^2 \mathcal{M}(S) P^{o(1)}}. \tag{CC2q45}$$

If, on the same literal marked-edge mass and with every selected mask retained, one has for some fixed $\theta > 0$

$$\mathfrak{U}_D \ll P^{-\theta+o(1)} D^2 \mathcal{M}(S), \tag{CC2q46}$$

then

$$\boxed{\mathcal{E}_D \ll P^{o(1)} \min \left\{ \frac{P^3}{D}, P^{243/200-\theta} D^2, P^{9/5-\theta/2} D \right\}}. \tag{CC2q47}$$

Consequently, for fixed $\sigma > 0$, the medium repeated-prefix residual is $\ll P^{109/50-\sigma+o(1)}$ whenever

$$\boxed{\theta > \frac{27}{40} + 3\sigma.} \quad (\text{CC2q48})$$

No positive value of θ in (CC2q46) is asserted here.

Proof. From (CC2q42b) and (CC2q41),

$$\mathfrak{U}_D \ll DP^{o(1)} \sum_{J,a} aR_D(J,a) \ll D^2 \mathcal{M}(S) P^{o(1)},$$

which proves (CC2q45). Since every retained source label satisfies $a \geq P^{77/200-o(1)}$, the assumed defect and $2\mathcal{E}_D = \sum_{J,a} E_{J,a}$ give

$$\mathcal{E}_D \ll P^{-77/200+o(1)} \mathfrak{U}_D \ll P^{243/200-\theta+o(1)} D^2.$$

For the second consumption, graph simplicity gives $E_{J,a} \leq C_J R(J,a)$. Hence

$$2F_{J,D} \leq C_J^{1/2} \left(\sum_a \frac{R(J,a)}{a} \right)^{1/2} \left(\sum_a aE_{J,a} \right)^{1/2} \ll C_J^{3/4} \left(\sum_a aE_{J,a} \right)^{1/2} P^{o(1)}.$$

Summing in J , using $\sum_J C_J^{3/2} \ll P^{2+o(1)}$ and then the assumed defect, gives

$$\mathcal{E}_D \ll P^{1+o(1)} \mathfrak{U}_D^{1/2} \ll P^{9/5-\theta/2+o(1)} D.$$

Together with the existing long cap this proves (CC2q47).

Write $D = P^\lambda$. Once $\theta > 3/50$, the third branch lies above the crossing of the first two improved branches. That crossing has maximal exponent

$$\frac{481}{200} - \frac{\theta}{3}.$$

It is strictly below $109/50 - \sigma$ under (CC2q48). The target range is therefore covered after summing the logarithmically many dyadic blocks. $\square$

Corollary 485 (Strong endpoint and maximal-separation deletions). *Assume the balanced-core scales*

$$H = P^{41/100+o(1)}, \quad A = P^{77/200-o(1)}, \quad R_0 = P^{9/25-o(1)},$$

and retain the notation

$$H_1 = H - 2A - R_0, \quad \Gamma = H_1 - A$$

from Corollary 478. For every fixed $\varepsilon > 0$, the ordered transverse contribution from each of the following three sectors is below the critical scale:

$$\mathfrak{U}_{\text{tr}}(S)\{a' < A + P^{59/200-\varepsilon}\} \ll P^{109/50-2\varepsilon+o(1)}, \quad (\text{CC2q34})$$

$$\mathfrak{U}_{\text{tr}}(S)\{a \geq H_1 - P^{1/3-\varepsilon}\} \ll P^{109/50-3\varepsilon+o(1)}, \quad (\text{CC2q35})$$

$$\mathfrak{U}_{\text{tr}}(S)\{\Gamma - (a' - a) \leq P^{41/120-\varepsilon}\} \ll P^{109/50-3\varepsilon+o(1)}. \quad (\text{CC2q36})$$

All original balanced-core masks are retained.

Proof. For the first sector both ordered labels lie in the bottom interval $(A, A + W_-)$, where $W_- = P^{59/200-\varepsilon}$. Apply the first branch of (CC2q33) with $L \asymp A$ to obtain

$$\sum_J R_-(J; W_-)^2 \ll A^2 H^2 W_-^2 P^{o(1)} = P^{109/50-2\varepsilon+o(1)}.$$

For the top sector put $W_+ = P^{1/3-\varepsilon}$. If $a \geq H_1 - W_+$, the headroom and $b, c > A$ imply

$$(b - A) + (c - A) < W_+.$$

Recenter at the second actual zero $x = J + a$. For fixed (p, x, b, c) , literal consecutiveness makes J the unique preceding actual zero in the canonical block. Thus distinct selected starts inject into the actual common-root cells for the pair (b, c) , with no source-fibre multiplicity. The adjacent resultant gives

$$\sum_J R_+(J; W_+) \ll A^2 W_+^2 P^{o(1)}, \quad \max_J R_+(J; W_+) \ll H W_+ P^{o(1)}.$$

Hence

$$\sum_J R_+(J; W_+)^2 \ll A^2 H W_+^3 P^{o(1)} = P^{109/50-3\varepsilon+o(1)}.$$

Finally

$$\Gamma - (a' - a) = (a - A) + (H_1 - a')$$

shows that the maximal-separation sector lies in a bottom window and a top window of the same width W . Use the pointwise bottom packet bound $\|R_-\|_\infty \ll A W P^{o(1)}$ and the recentered top first moment $\sum_J R_+(J; W) \ll A^2 W^2 P^{o(1)}$ proved above. They give

$$\mathfrak{U}_{\text{tr}}^{\max}(W) \ll A^3 W^3 P^{o(1)}.$$

Taking $W = P^{41/120-\varepsilon}$ and using $3(77/200) + 3(41/120) = 109/50$ proves (CC2q36). $\square$

The source-fibre objection applies to consumers which retain physical row lifts or ordered Palm companions above one geometric cell. It does not apply to the present set-valued selector: by definition $R(J, a)$ contains only the prime weight attached to membership $J \in S_p$. In particular the three strong windows above require no separate lift theorem. Together with (CC2q28), the unresolved ordered transverse core may be restricted to

$$\begin{aligned} a' - a &> P^{9/50-\varepsilon}, & a' &> A + P^{59/200-\varepsilon}, \\ a &< H_1 - P^{1/3-\varepsilon}, & a' - a &< \Gamma - P^{41/120-\varepsilon}. \end{aligned}$$

This improves the earlier full-core estimate $P^{121/50+o(1)}$ by $P^{1/100}$, but it remains above $P^{109/50}$ by $P^{23/100}$. The exact remaining sufficient estimate is

$$\mathfrak{U}_{\text{tr}}(S) \ll P^{11/5-\delta+o(1)} \quad \text{for some } \delta > 1/50.$$

Indeed, (CC2q20)–(CC2q22) then give

$$\mathfrak{U}(S) \ll P^{11/5-\delta_*+o(1)}, \quad \delta_* = \min\{\delta, 4/75\} > 1/50,$$

and (CC2q17) yields

$$\mathcal{M}(S) \ll P^{8/5-\delta_*/2+o(1)},$$

strictly beyond the $P^{159/100}$ endpoint. Hence the unresolved unequal-return frontier is now precisely (CC2q23); the wrap and actual-return sectors require no further saving.

The distinguished orbit also gives a complementary deletion near the two global physical endpoints. This uses the integer Apéry value itself and therefore does not see the projective-return phantoms.

Proposition 486 (Distinguished physical-edge deletion). *For a canonically selected quadruple $Q = (z_0, z_1, z_2, z_3)$ modulo p , put*

$$\rho_p(Q) = \min_{0 \leq i \leq 3} \min\{z_i, p-1-z_i\}.$$

Then, for $R \geq 1$,

$$\sum_{p > H} (\log p) \#\{Q : \rho_p(Q) \leq R\} \ll R^2. \quad (\text{CC2r})$$

Consequently, for every fixed $\varepsilon > 0$, the part with

$$\rho_p(Q) \leq H^{159/82-\varepsilon}$$

has total prime-weighted mass

$$\ll H^{159/41-2\varepsilon}. \quad (\text{CC2s})$$

No structural-prime exclusion is required.

Proof. If $\rho_p(Q) \leq R$, then for some slot i and some integer $0 \leq r \leq R$, either $z_i = r$ or $p-1-z_i = r$. In the first case $p \mid b_r$. In the second case Apéry reflection gives

$$b_r = b_{p-1-z_i} \equiv b_{z_i} \equiv 0 \pmod{p}.$$

For fixed p, r , a choice of side, and a fixed slot i , at most one selected sliding quadruple has that marked zero in slot i : its required predecessors and successors in its unique canonical block are determined. Thus

$$\#\{Q : \rho_p(Q) \leq R\} \leq 8 \sum_{0 \leq r \leq R} \mathbf{1}_{p \mid b_r}.$$

Multiplication by $\log p$, summation over primes, and the elementary Apéry growth bound $\log b_r \ll r+1$ give

$$\sum_{p > H} (\log p) \#\{Q : \rho_p(Q) \leq R\} \ll \sum_{0 \leq r \leq R} \sum_{p \mid b_r} \log p \leq \sum_{0 \leq r \leq R} \log b_r \ll R^2.$$

Here $b_0 = 1$ contributes nothing. Taking $R = H^{159/82-\varepsilon}$ proves (CC2s). $\square$

The large-prime interior has a further exact decomposition which separates the harmless density term from the genuinely distinguished-orbit correlation. It is useful even though no positive discrepancy saving is presently known.

Proposition 487 (Distinguished-fibre discrepancy reduction). *Let $\mathcal{L}_H$ be any set of ordered pairs (a, b) with $a, b \geq 2$ and $a+b < H$. For each prime p and $(a, b) \in \mathcal{L}_H$, let $\mathcal{X}_p(a, b; H)$ be any family of canonically selected consecutive-zero quadruples whose first two gaps are (a, b) , identified with their distinct physical starting zeros. Put*

$$\mathcal{R}_p(a, b) = \{x \in \mathbb{F}_p : N_a(x) = N_b(x+a) = 0\}, \quad r_p(a, b) = |\mathcal{R}_p(a, b)|.$$

For a dyadic prime interval $(P, 2P]$, define

$$\begin{aligned}\mathcal{I}_{a,b}(P) &= \sum_{P < p \leq 2P} (\log p) |\mathcal{Z}_p \cap \mathcal{R}_p(a, b)|, \\ \mu_{a,b}(P) &= \sum_{P < p \leq 2P} (\log p) \frac{Z(p)}{p} r_p(a, b), \\ \mathcal{D}_{a,b}(P) &= \mathcal{I}_{a,b}(P) - \mu_{a,b}(P), \quad \mathcal{D}_{a,b}^+(P) = \max\{\mathcal{D}_{a,b}(P), 0\}.\end{aligned}\tag{CC2t}$$

Then, for every $P_0 > H$,

$$\begin{aligned}\sum_{(a,b) \in \mathcal{L}_H} \sum_{p > P_0} (\log p) \# \mathcal{R}_p(a, b; H) \\ \ll P_0^{-1/3} \sum_{(a,b) \in \mathcal{L}_H} ab \log(2H) + \sum_{P \geq P_0}^{\text{dyadic}} \sum_{(a,b) \in \mathcal{L}_H} \mathcal{D}_{a,b}^+(P).\end{aligned}\tag{CC2u}$$

In particular, if $P_0 = H^{12/5}$, then

$$\sum_{(a,b) \in \mathcal{L}_H} \sum_{p > P_0} (\log p) \# \mathcal{R}_p(a, b; H) \ll H^{16/5+o(1)} + \sum_{P \geq H^{12/5}}^{\text{dyadic}} \sum_{(a,b) \in \mathcal{L}_H} \mathcal{D}_{a,b}^+(P).\tag{CC2v}$$

Moreover, if for some fixed $\vartheta > 25/492$ one has uniformly for $P \geq H^{12/5}$

$$\sum_{(a,b) \in \mathcal{L}_H} \mathcal{D}_{a,b}^+(P) \ll H^{4+o(1)} P^{-\vartheta},\tag{CC2w}$$

then the left side of (CC2v) is $O(H^{159/41-\delta+o(1)})$ for every

$$0 < \delta < \frac{12}{5} \left(\vartheta - \frac{25}{492} \right).\tag{CC2x}$$

Proof. Every start in $\mathcal{R}_p(a, b; H)$ is a literal Apéry zero and, by the two adjacent return identities, lies in $\mathcal{R}_p(a, b)$. Hence its dyadic contribution is at most $\mathcal{I}_{a,b}(P)$. Let

$$S_{a,b} = |\text{Res}_x(N_a(x), N_b(x+a))|.$$

This integer is nonzero, Lemma 454 gives $\log S_{a,b} \ll ab \log(2H)$, and evaluation on the fixed Sylvester map gives

$$r_p(a, b) \leq v_p(S_{a,b}).$$

The unconditional pointwise bound $Z(p) \ll p^{2/3}$ therefore yields

$$\sum_{P \geq P_0}^{\text{dyadic}} \mu_{a,b}(P) \ll P_0^{-1/3} \sum_p (\log p) r_p(a, b) \leq P_0^{-1/3} \log S_{a,b}.$$

Since $\mathcal{I}_{a,b}(P) \leq \mu_{a,b}(P) + \mathcal{D}_{a,b}^+(P)$, summing proves (CC2u). Now

$$P_0^{-1/3} = H^{-4/5}, \quad \sum_{a,b < H} ab \ll H^4,$$

which proves (CC2v). Under (CC2w), the dyadic discrepancy sum is

$$\ll H^{4+o(1)}(H^{12/5})^{-\vartheta} = H^{4-(12/5)\vartheta+o(1)}.$$

The identity

$$4 - \frac{12}{5} \frac{25}{492} = \frac{159}{41}$$

proves (CC2x). □

The discrepancy has an exact bilinear Fourier form. If

$$C_{p;a,b}(k) = \sum_{r \in \mathcal{R}_p(a,b)} e_p(kr),$$

then finite Fourier inversion gives

$$|\mathcal{Z}_p \cap \mathcal{R}_p(a,b)| - \frac{Z(p)}{p} r_p(a,b) = \frac{1}{p} \sum_{k \neq 0} F_p(k) \overline{C_{p;a,b}(k)}, \quad (\text{CC2y})$$

$$\mathcal{D}_{a,b}(P) = \sum_{P < p \leq 2P} \frac{\log p}{p} \sum_{k \neq 0} F_p(k) \overline{C_{p;a,b}(k)}. \quad (\text{CC2z})$$

Thus the missing input is joint cancellation between the literal distinguished Apéry zero fibre and the short algebraic common-root fibre. A bound for either marginal alone does not prove (CC2w); projective phantoms occur only in the algebraic factor and are never counted as physical starts.

The distinguished fibre itself admits three exact descriptions which do not pass through an arbitrary projective return line. They do not by themselves prove the required discrepancy estimate, but they identify actual-orbit structures on which such an estimate may be sought.

Proposition 488 (Quadratic-index descent of the actual Apéry orbit). *Let p be odd and put $T_n = n(n+1) \in \mathbb{F}_p$. The polynomial*

$$R_p(T) = \sum_{k=0}^{p-1} \frac{1}{(k!)^4} \left(\prod_{j=0}^{k-1} (T - j(j+1)) \right)^2 \in \mathbb{F}_p[T]$$

satisfies

$$b_n = R_p(T_n) \quad (0 \leq n < p). \quad (\text{CC2aa})$$

Consequently there is a unique polynomial $Q_p \in \mathbb{F}_p[T]$ of degree at most $(p-1)/2$ whose values on the quadratic image are the actual Apéry values. Moreover

$$T_u = T_v \iff u = v \text{ or } u + v = p - 1. \quad (\text{CC2ab})$$

Thus an off-centred actual quadruple gives four distinct zeros of this one distinguished value polynomial on the quadratic image.

Proof. For every $k \geq 0$,

$$\binom{n}{k} \binom{n+k}{k} = \frac{1}{(k!)^2} \prod_{j=0}^{k-1} (n-j)(n+j+1) = \frac{1}{(k!)^2} \prod_{j=0}^{k-1} (T_n - j(j+1)).$$

If $k > n$, the factor with $j = n$ vanishes, so the defining Apéry sum may be extended through $k = p-1$ and gives (CC2aa). The image of $n \mapsto T_n$ has $(p+1)/2$ elements, hence interpolation on that image gives the unique representative of degree at most $(p-1)/2$. Finally $T_u - T_v = (u-v)(u+v+1)$, which proves (CC2ab). □

This quotient is already present, integrally and with explicit Newton coefficients, in the Franel shell carrier. If $M = (p-1)/2$ and $F_m = \sum_j \binom{m}{j}^3$, then the reduction modulo p of

$$\mathcal{S}_M(T) = \sum_{m=0}^M \frac{F_m}{(m!)^2} \prod_{j=0}^{m-1} (T - j(j+1))$$

is precisely Q_p . Indeed, (412) gives the values for $0 \leq n \leq M$, and the palindrome gives the remaining values because $T_n = T_{p-1-n}$. Thus the Newton and actual half-orbit vectors are related by a lower-triangular matrix with diagonal $\binom{2n}{n}$, a p -unit for $n \leq M$.

There is also an exact finite-difference obstruction. Put

$$\Phi_m(T) = \frac{1}{(m!)^2} \prod_{j < m} (T - j(j+1)).$$

For four off-centred displayed indices with distinct quadratic nodes $Y_0, \dots, Y_3$, the matrix $(\Phi_m(Y_i))_{0 \leq i, m \leq 3}$ has determinant a nonzero constant times $\prod_{i < j} (Y_j - Y_i)$. It is therefore invertible modulo p . No nontrivial linear combination of the four actual zero equations can cancel even the first four Newton modes. Hence a bounded-prefix finite difference does not create a short carrier; using the Franel recurrence to control the remaining modes restores the absolute-prefix problem.

This descent is stronger than the palindrome as an explicit formula, but degree counting alone gives no short-interval spacing: every function invariant under $n \mapsto -1-n$ descends through T_n . Any use of Proposition 488 must exploit the displayed Newton-basis coefficients rather than only the degree of Q_p .

Proposition 489 (Actual Cartier–Mellin formula). *Let $F = (1-x-y)(1-z-w) - xyzw$ as in Theorem 385, and for $t \in \mathbb{F}_p^*$ put*

$$C_p(t) = \#\{(u, v, z, w) \in (\mathbb{F}_p^*)^4 : F(u, v, z, w) = 0, uvzw = t\} \pmod{p}.$$

For every $1 \leq r \leq p-2$,

$$\boxed{b_r = - \sum_{t \in \mathbb{F}_p^*} C_p(t) t^{-r}.} \quad (\text{CC2ac})$$

Hence the actual zero set is the zero-frequency set of the multiplicative Fourier transform of this fixed Cartier-hypersurface point-count function.

Proof. Theorem 385 gives $b_r = [x^r y^r z^r w^r] F^{p-1}$. Since F is multi-affine, every exponent in F^{p-1} lies in $[0, p-1]$. Multiplicative orthogonality, with $1 \leq r \leq p-2$, therefore extracts exactly the four exponents equal to r and yields

$$b_r = \sum_{(u,v,z,w) \in (\mathbb{F}_p^*)^4} F(u, v, z, w)^{p-1} (uvzw)^{-r}.$$

Fermat's theorem makes F^{p-1} the indicator of $F \neq 0$. The same character summed over the whole four-dimensional torus is zero, so the last display is minus its restriction to $F = 0$. Grouping by $t = uvzw$ proves (CC2ac). $\square$

The four-variable Cartier fibre has an exact two-pair factorization. This exposes a bounded-complexity local trace, but also shows that its horizontal multiplicative spectrum is full rather than sparse.

Proposition 490 (Two-pair Cartier–Jacobi factorization). *For $s, \lambda \in \mathbb{F}_p^*$ put*

$$A_p(s, \lambda) = \# \left\{ (u, v) \in (\mathbb{F}_p^*)^2 : uv = s, \frac{1 - u - v}{uv} = \lambda \right\}.$$

If χ is the quadratic character, extended by $\chi(0) = 0$, then

$$A_p(s, \lambda) = 1 + \chi((1 - s\lambda)^2 - 4s). \quad (\text{CM1})$$

The fibre in Proposition 489 satisfies the exact convolution

$$C_p(t) = \sum_{s, \lambda \in \mathbb{F}_p^*} A_p(s, \lambda) A_p(t/s, \lambda^{-1}). \quad (\text{CM2})$$

Consequently, for $1 \leq r \leq p - 2$, if

$$K_{p,r}(\lambda) = \sum_{s \neq 0} A_p(s, \lambda) s^{-r} \in \mathbb{F}_p,$$

then

$$\boxed{b_r = - \sum_{\lambda \neq 0} K_{p,r}(\lambda) K_{p,r}(\lambda^{-1})} \quad \text{in } \mathbb{F}_p, \quad (\text{CM3})$$

and

$$K_{p,r}(\lambda) = \sum_{s \neq 0} s^{-r} \chi((1 - s\lambda)^2 - 4s). \quad (\text{CM4})$$

Choose a Teichmüller character ω and a prime $\mathfrak{p} \mid p$ of $\mathbb{Z}[\mu_{p-1}]$ such that $\omega(a) \equiv a \pmod{\mathfrak{p}}$, and put $\rho = \omega^{-r} \neq 1$. The complex lift

$$\mathcal{K}_\rho(\lambda) = \sum_{s \neq 0} A_p(s, \lambda) \rho(s)$$

has the exact Jacobi transform

$$\widehat{\mathcal{K}}_\rho(\eta) := \sum_{\lambda \neq 0} \mathcal{K}_\rho(\lambda) \eta(\lambda) = \eta(-1) J(\eta, \bar{\rho}) J(\rho \bar{\eta}, \rho). \quad (\text{CM5})$$

If $\rho^2 \neq 1$, every one of these $p - 1$ Mellin coefficients is nonzero, all but the three characters $1, \rho, \rho^2$ have squared modulus p^2 , and

$$\sum_{\lambda \neq 0} |\mathcal{K}_\rho(\lambda)|^2 = p^2 - 3p - 1. \quad (\text{CM6})$$

If $\rho = \chi$, the corresponding energy is

$$\sum_{\lambda \neq 0} |\mathcal{K}_\chi(\lambda)|^2 = p^2 - 2p - 2. \quad (\text{CM7})$$

Proof. The two equations defining $A_p(s, \lambda)$ say that u, v are the ordered roots of

$$X^2 - (1 - s\lambda)X + s = 0.$$

Its discriminant is $(1 - s\lambda)^2 - 4s$, proving (CM1), including the repeated-root case. On the Cartier hypersurface put $s = uv$, $t/s = zw$ and

$$\lambda = \frac{1 - u - v}{uv}, \quad \mu = \frac{1 - z - w}{zw}.$$

The equation $F = 0$ is exactly $\lambda\mu = 1$. This construction is reversible and proves (CM2). Mellin transformation in t , followed by Proposition 489, proves (CM3). Since $\sum_{s \neq 0} s^{-r} = 0$ in the stated range, (CM4) follows from (CM1).

Extend multiplicative characters by zero at 0. Solving $1 - u - v = \lambda uv$ for v gives, also at the exceptional value $\lambda = -1$ after summing the extra line,

$$\mathcal{K}_\rho(\lambda) = \sum_{u \in \mathbb{F}_p} \rho(u) \rho(1 - u) \bar{\rho}(1 + \lambda u).$$

Mellin transformation in λ , with $x = -\lambda u$, gives (CM5). The standard identities for Jacobi sums show, when $\rho^2 \neq 1$, that the squared moduli of the transform are respectively $p, 1, p$ at $\eta = 1, \rho, \rho^2$ and p^2 elsewhere. Multiplicative Parseval then gives (CM6). When $\rho = \chi$, the two exceptional coefficients have squared modulus 1 and all others have squared modulus p^2 , which proves (CM7). $\square$

Reduction of the complex lift modulo $\mathfrak{p}$ recovers $K_{p,r}$ and hence (CM3); termwise Jacobi reduction recovers the original Apéry binomial sum. Thus the rank-two local description supplies no horizontal spectral saving by itself. The missing input remains cancellation of the actual Cartier–Mellin value against the literal balanced first-return weight, not a generic low-rank or sparse-Mellin assertion.

Generic Fourier uncertainty does not control four prescribed zero frequencies. The missing estimate remains a correlation theorem for the special function C_p , quantitatively of the form already isolated in (CC2w).

Proposition 491 (Rank-one full-period selector). *For $p > 3$ write*

$$A_n = \begin{pmatrix} P(n) & -n^3 \\ (n+1)^3 & 0 \end{pmatrix} \quad (n \bmod p)$$

and, for $1 \leq x \leq p-1$, define the cyclic period product

$$\mathcal{P}_x = A_{x-1} \cdots A_1 A_0 A_{p-1} \cdots A_x.$$

Then $\mathcal{P}_x$ has rank one and

$$\text{im } \mathcal{P}_x = [b_x : b_{x-1}]. \tag{CC2ad}$$

In particular, for $Z = [0 : 1]$,

$$b_x = 0 \iff \text{im } \mathcal{P}_x = Z. \tag{CC2ae}$$

For every nonwrapping interior displacement h ,

$$\mathcal{P}_{x+h} = M_h(x) \mathcal{P}_x M_h(x)^{-1}. \tag{CC2af}$$

Thus, after the first actual hit has selected Z , subsequent hits are exactly the local return equations already used above; trace, determinant, and other conjugacy invariants do not select the moving image line.

Proof. All A_n except A_0 and A_{p-1} are invertible, while

$$A_0 = \begin{pmatrix} 5 & 0 \\ 1 & 0 \end{pmatrix}, \quad A_{p-1} = \begin{pmatrix} -5 & 1 \\ 0 & 0 \end{pmatrix}$$

and $A_0 A_{p-1}$ is nonzero of rank one. The image of A_{p-1} is $[1 : 0]$, which A_0 sends to the distinguished initial line $[5 : 1]$. The remaining prefix $A_{x-1} \cdots A_1$ sends this line to $[b_x : b_{x-1}]$, proving (CC2ad). Cyclically moving the invertible factor A_x gives $\mathcal{P}_{x+1} = A_x \mathcal{P}_x A_x^{-1}$; iteration proves (CC2af). $\square$

The literal first-return condition gives an additional orthogonality in the gap variables which is absent from the enlarged common-root sets in Proposition 487.

Proposition 492 (First-return dual Bessel bound). *For $a, b \geq 2$ and $p > a + b$, let $\mathcal{R}_p^{\text{fr}}(a, b)$ be the set of integers x satisfying*

$$\begin{aligned} 0 \leq x < x + a + b < p, \\ N_a(x) = 0, & \quad N_j(x) \neq 0 \quad (1 \leq j < a), \\ N_b(x + a) = 0, & \quad N_j(x + a) \neq 0 \quad (1 \leq j < b) \end{aligned}$$

modulo p , and put $r_p^{\text{fr}}(a, b) = |\mathcal{R}_p^{\text{fr}}(a, b)|$. Then the nonempty sets $\mathcal{R}_p^{\text{fr}}(a, b)$ are pairwise disjoint as (a, b) varies. If

$$\delta_{p;a,b}^{\text{fr}} = |\mathcal{Z}_p \cap \mathcal{R}_p^{\text{fr}}(a, b)| - \frac{Z(p)}{p} r_p^{\text{fr}}(a, b),$$

then

$$\boxed{\sum_{a,b} \frac{|\delta_{p;a,b}^{\text{fr}}|^2}{r_p^{\text{fr}}(a, b)} \leq Z(p) - \frac{Z(p)^2}{p}}. \quad (\text{CC2ag})$$

Empty cells contribute zero. For a dyadic prime interval define

$$\mathcal{D}_{a,b}^{\text{fr}}(P) = \sum_{P < p \leq 2P} (\log p) \delta_{p;a,b}^{\text{fr}}.$$

Then, for any family $\mathcal{L}_H$ of pairs $a, b \geq 2$ with $a + b < H < P$,

$$\sum_{(a,b) \in \mathcal{L}_H} \frac{|\mathcal{D}_{a,b}^{\text{fr}}(P)|^2}{ab} \ll P^{8/5+o(1)}, \quad (\text{CC2ah})$$

$$\sum_{(a,b) \in \mathcal{L}_H} (\mathcal{D}_{a,b}^{\text{fr}}(P))^+ \ll H^{2+o(1)} P^{4/5}. \quad (\text{CC2ai})$$

Every literal selected quadruple with first two gaps (a, b) has its actual starting zero in $\mathcal{Z}_p \cap \mathcal{R}_p^{\text{fr}}(a, b)$.

Proof. For a fixed residue x , membership in a nonempty cell makes a the least positive nonwrapping return time of the zero line. After that return, b is the least positive return time from $x + a$. Both are unique, proving disjointness. With $f_p = \mathbf{1}_{\mathcal{Z}_p} - Z(p)/p$, the normalized nonempty cell indicators form an orthonormal family in $\ell^2(\mathbb{F}_p)$. Bessel's inequality therefore gives

$$\sum_{a,b} \frac{|\langle f_p, \mathbf{1}_{\mathcal{R}_p^{\text{fr}}(a,b)} \rangle|^2}{r_p^{\text{fr}}(a, b)} \leq \|f_p\|_2^2 = Z(p) - Z(p)^2/p,$$

which is (CC2ag).

Let $S_{a,b} = |\text{Res}_x(N_a(x), N_b(x+a))|$. The cell is contained in the common-root set, so the fixed Sylvester corank argument and (AF4) give

$$r_p^{\text{fr}}(a, b) \leq v_p(S_{a,b}), \quad \log S_{a,b} \ll ab \log H.$$

Cauchy's inequality in p now yields

$$\frac{|\mathcal{D}_{a,b}^{\text{fr}}(P)|^2}{ab} \ll (\log H) \sum_{P < p \leq 2P} (\log p) \frac{|\delta_{p;a,b}^{\text{fr}}|^2}{r_p^{\text{fr}}(a, b)}.$$

Summing over (a, b) , applying (CC2ag), and using (AF2) proves (CC2ah). Finally Cauchy in (a, b) and $\sum_{a,b < H} ab \ll H^4$ prove (CC2ai). The last assertion is the gap-polynomial identity at an actual zero together with literal consecutiveness. $\square$

Lemma 493 (Blockwise positive-part dualization of first-return cells). *Fix a dyadic prime interval $(P, 2P]$ and a family $\mathcal{L}_H$ as in Proposition 492. Put*

$$f_p = \mathbf{1}_{\mathcal{Z}_p} - \frac{Z(p)}{p}.$$

Then

$$\sum_{(a,b) \in \mathcal{L}_H} (\mathcal{D}_{a,b}^{\text{fr}}(P))^+ = \max_{0 \leq \lambda_{a,b} \leq 1} \sum_{P < p \leq 2P} (\log p) \langle f_p, W_{p,P} \rangle,$$

where

$$W_{p,P}(x) = \sum_{(a,b) \in \mathcal{L}_H} \lambda_{a,b} \mathbf{1}_{\mathcal{R}_p^{\text{fr}}(a,b)}(x).$$

For every admissible family $(\lambda_{a,b})$ one has $0 \leq W_{p,P}(x) \leq 1$. The maximum is attained by

$$\lambda_{a,b} = \mathbf{1}_{\{\mathcal{D}_{a,b}^{\text{fr}}(P) > 0\}},$$

for which $W_{p,P}$ is $\{0, 1\}$ -valued. The coefficients are constant in p throughout the dyadic block, although they depend on (a, b) and on the block P .

Proof. For a real number y one has $y^+ = \max_{0 \leq \lambda \leq 1} \lambda y$. Apply this independently to the finitely many block discrepancies and interchange the finite sums. The asserted pointwise bounds follow because the nonempty cells $\mathcal{R}_p^{\text{fr}}(a, b)$ are pairwise disjoint by Proposition 492. $\square$

Remark 494 (Candidate-cell duality is not selected-quadruple duality). The preceding weight selects complete first-return candidate cells. The source proves only that every literal selected consecutive-zero quadruple with first two gaps (a, b) has its starting zero in $\mathcal{Z}_p \cap \mathcal{R}_p^{\text{fr}}(a, b)$; no converse is asserted. In particular, membership in a first-return cell does not by itself supply a selected third gap c , a canonical H -block, headroom, simple-root conditions, or a fourth-endpoint condition involving that gap. Inserting such a selector replaces $\delta_{p;a,b}^{\text{fr}}$ by a different centered discrepancy. This replacement is not monotone because $\mathbf{1}_{\mathcal{Z}_p} - Z(p)/p$ has both signs. For the enlarged common-root sets in (CC2t), the corresponding cells need not be disjoint, so their positive-part dual weight need not be bounded by 1.

Proposition 495 (Marginal actual-zero bound for first-return blocks). *In the notation of Proposition 492, uniformly for every dyadic prime interval $(P, 2P]$ and every admissible pair family $\mathcal{L}_H$,*

$$\sum_{(a,b) \in \mathcal{L}_H} (\mathcal{D}_{a,b}^{\text{fr}}(P))^+ \ll P^{8/5+o(1)}. \quad (\text{CC2ai}')$$

Consequently, throughout the endpoint range $P \leq H^{100/41}$,

$$\sum_{(a,b) \in \mathcal{L}_H} (\mathcal{D}_{a,b}^{\text{fr}}(P))^+ \ll H^{4+o(1)} P^{-1/25}. \quad (\text{CC2ai}'')$$

No lower bound for a selected mass and no replacement of an actual zero by a candidate root is used.

Proof. Write $t_{p;a,b} = |\mathcal{Z}_p \cap \mathcal{R}_p^{\text{fr}}(a,b)|$. Discarding the nonpositive candidate-background term gives

$$(\mathcal{D}_{a,b}^{\text{fr}}(P))^+ \leq \sum_{P < p \leq 2P} (\log p) t_{p;a,b}.$$

The first-return cells are disjoint for each fixed p , so summing in (a,b) bounds the right side by

$$\sum_{P < p \leq 2P} (\log p) Z(p) \ll P^{8/5+o(1)}.$$

Finally $P \leq H^{100/41}$ implies $P^{8/5+1/25} = P^{41/25} \leq H^4$, proving (CC2ai''). $\square$

Proposition 496 (Large-product simple-root reduction for first-return cells). *Fix $0 < \vartheta < 41/100$ and decompose an admissible pair family $\mathcal{L}_H$ into dyadic boxes $a \asymp A$, $b \asymp B$. The total positive discrepancy from boxes satisfying*

$$AB \leq H^2 P^{-\vartheta/2}$$

is

$$\ll H^{4+o(1)} P^{-\vartheta}. \quad (\text{CC2ai-a})$$

Moreover, the prime-weighted number of actual occupied first-return cells for which $N'_a(x) = 0$ or $N'_b(x+a) = 0$ modulo p is

$$\ll H^{3+o(1)}. \quad (\text{CC2ai-b})$$

Thus every exponent relevant to (CC2aj) may restrict the unresolved actual occupancy to large-product cells whose two return roots are simple, while retaining all first-return nonvanishing conditions.

Proof. Let

$$S_{a,b} = |\text{Res}_x(N_a(x), N_b(x+a))|.$$

As in the proof of Proposition 492, every point in the cell is a common root and

$$\sum_{P < p \leq 2P} (\log p) r_p^{\text{fr}}(a,b) \leq \log S_{a,b} \ll ab \log H.$$

Discarding the negative background therefore bounds the positive discrepancy of one cell by $O(ab \log H)$. A dyadic (A, B) box contains $O(AB)$ pairs, so its contribution is $O((AB)^2 \log H)$. Summing the $H^{o(1)}$ boxes with $AB \leq H^2 P^{-\vartheta/2}$ proves (CC2ai-a).

For the singular contribution, write $F_h = N_h/c_h$ with c_h the positive content of N_h , and put

$$\Delta_h = |\text{Res}(F_h, F'_h)|.$$

Lemma 459 and the coefficient-height bound give $\Delta_h \neq 0$ and $\log \Delta_h \ll h^2 \log(2H)$. For $p > H$, the number of singular roots of N_h modulo p is at most $v_p(\Delta_h)$ by the fixed Sylvester–Smith argument.

Fixing p, a and an actual first-return start x determines the next first-return gap b after $x + a$, if it exists. Hence the first-leg singular occupancy, summed over all b , is at most

$$\sum_{a < H} \sum_{p > H} (\log p) v_p(\Delta_a) \leq \sum_{a < H} \log \Delta_a \ll H^3 \log(2H).$$

For the second leg, fix the middle actual zero $y = x + a$ and its next first-return gap b ; literal first return makes x its unique preceding actual zero in the cell. The same argument with b gives the identical bound. This proves (CC2ai-b). Finally $P \leq H^{100/41}$ and $\vartheta < 41/100$ make $H^{3+o(1)} = o(H^4 P^{-\vartheta})$. $\square$

This orthogonality is nontrivial but not yet sufficient. At $P = H^{12/5}$, the right side of (CC2ai) is $H^{98/25+o(1)}$, which exceeds $H^{159/41}$ by $H^{43/1025}$. Combining the more general prime-first deletion following Proposition 461 with a hypothetical first-return discrepancy bound

$$\sum_{(a,b) \in \mathcal{L}_H} (\mathcal{D}_{a,b}^{\text{fr}}(P))^+ \ll H^{4+o(1)} P^{-\vartheta} \quad (\text{CC2aj})$$

shows that it is enough to have

$$\vartheta > \frac{328}{6519}. \quad (\text{CC2ak})$$

Indeed, choose $12/5 < \kappa < 795/328$ with $\kappa\vartheta > 5/41$, delete $p \leq H^\kappa$ by the first-zero average, and apply (CC2aj) above that cutoff. The infimum of $5/(41\kappa)$ as $\kappa \nearrow 795/328$ is $328/6519$. The unconditional bound (CC2ai'') supplies $\vartheta = 1/25$, whereas

$$\frac{328}{6519} - \frac{1}{25} = \frac{41}{3975}.$$

A singleton cell containing an actual zero has discrepancy $1 - Z(p)/p$; hence cancellation in the candidate-cell transform alone cannot prove this target. The missing input is sparsity of distinguished actual occupancy of the disjoint cells.

Hypothesis 497 (Cofactor-height saving, CFH_δ). *For some fixed $0 < \delta \leq 1$,*

$$\sum_{a=2}^{H-2} \log \mathfrak{K}_{a,H}^* \ll H^{4-\delta+o(1)}.$$

Theorem 498 (Direct cofactor-height consumer). *Under Hypothesis 497,*

$$\sum_{X < p \leq 2X} Z(p) \ll X^{(8-2\delta)/(5-\delta)+o(1)}. \quad (\text{CC3})$$

Thus every fixed $\delta > 0$ improves the unconditional exponent $8/5$, and $\delta = 1$ reaches the structural exponent $3/2$.

Proof. Put $c_\delta = (3-\delta)/(5-\delta)$ and $T_0 = X^{c_\delta}$. The primes with $Z(p) \leq T_0$ contribute $O(XT_0/\log X)$. For a dyadic level $T < Z(p) \leq 2T$, $T \geq T_0$, take $H = \lceil 16X/T \rceil$. The block lower bound and Proposition 450 give a certificate of size $\gg T$ at every $p \nmid \mathfrak{U}_H$. Summing valuations and using CFH_δ bounds the contribution of this level by

$$H^{4-\delta+o(1)}.$$

Lemma 503 bounds its structural part by $O(X^2/(T \log X))$. Summing the dyadic levels is geometric, and the three exponents at $T = T_0$ are

$$1 + c_\delta = \frac{2(4-\delta)}{5-\delta}, \quad (4-\delta)(1-c_\delta) = \frac{2(4-\delta)}{5-\delta}, \quad 2 - c_\delta = \frac{7-\delta}{5-\delta}.$$

For $0 < \delta \leq 1$ the last is no larger than the first two, proving (CC3). Logarithmic factors are absorbed in $X^{o(1)}$. $\square$

Proposition 499 (Reduced two-row companion cofactor). *Let a_m be the rational Apéry companion normalized by $a_0 = 0$, $a_1 = 6$, and write $a_m = u_m/v_m$ in lowest terms with $v_m > 0$. For $\Delta \geq 1$ put*

$$\mathfrak{G}_\Delta^< = \gcd(|u_{2\Delta}|, |u_{7\Delta}|), \quad \mathfrak{G}_\Delta^> = \gcd(|u_\Delta|, |u_{6\Delta}|).$$

Then

$$\log \mathfrak{G}_\Delta^< + \log \mathfrak{G}_\Delta^> \ll \Delta. \quad (\text{CC4})$$

Put

$$\mathfrak{a}_m = \frac{(m!)^3 a_m}{6} = N_m(0), \quad \mathfrak{b}_m = (m!)^3 b_m.$$

For every prime $\ell > 7\Delta$ one has

$$\begin{aligned} v_\ell(\mathfrak{G}_\Delta^<) &= \min\{v_\ell(\mathfrak{a}_{2\Delta}), v_\ell(\mathfrak{a}_{7\Delta})\}, \\ v_\ell(\mathfrak{G}_\Delta^>) &= \min\{v_\ell(\mathfrak{a}_\Delta), v_\ell(\mathfrak{a}_{6\Delta})\}, \end{aligned} \quad (\text{CC5})$$

and

$$\begin{aligned} \min\{v_\ell(\mathfrak{a}_{2\Delta}), v_\ell(\mathfrak{a}_{7\Delta})\} &\leq v_\ell(N_{5\Delta}(2\Delta)), \\ \min\{v_\ell(\mathfrak{a}_\Delta), v_\ell(\mathfrak{a}_{6\Delta})\} &\leq v_\ell(N_{5\Delta}(\Delta)). \end{aligned} \quad (\text{CC6})$$

Consequently, for any fixed finite union of prime intervals $\mathcal{P}_T$ with $\ell \asymp T$ and $7\Delta < \min \mathcal{P}_T$,

$$\sum_{\ell \in \mathcal{P}_T} (\log \ell) (\min\{v_\ell(\mathfrak{a}_{2\Delta}), v_\ell(\mathfrak{a}_{7\Delta})\} + \min\{v_\ell(\mathfrak{a}_\Delta), v_\ell(\mathfrak{a}_{6\Delta})\}) \ll \Delta. \quad (\text{CC7})$$

Proof. The cleared companion pairs satisfy

$$Y_{m+1} = P(m)Y_m - m^6 Y_{m-1}$$

and the consecutive Casorati identity

$$\mathfrak{b}_m \mathfrak{a}_{m+1} - \mathfrak{a}_m \mathfrak{b}_{m+1} = (m!)^6.$$

The same recurrence and initial values therefore give, for $h \geq 1$,

$$\mathfrak{b}_m \mathfrak{a}_{m+h} - \mathfrak{a}_m \mathfrak{b}_{m+h} = (m!)^6 N_h(m). \quad (\text{CC8})$$

This proves (CC6) at primes $\ell > m + h$. The identity $N_m(0) = (m!)^3 a_m / 6$ follows from the same recurrence and initial values.

Proposition 5 implies that every prime divisor of v_m is at most m . Thus, when $\ell > 7\Delta$, the factorials, the factor 6, and all four relevant denominators are ℓ -units, proving (CC5). Finally (6), positivity, the elementary ratio bound $b_m < 34^m$, and $\log \text{lcm}(1, \dots, m) = O(m)$ give

$$\log |u_m| \ll m.$$

Applying this at $m = \Delta, 2\Delta, 6\Delta, 7\Delta$ proves (CC4); summing its prime-power valuations proves (CC7). $\square$

Remark 500 (The reduced cofactor is not an occurrence majorant). For every integer c , $N_m(c\ell) \equiv \mathfrak{a}_m \pmod{\ell}$. Hence the proposition controls simultaneous *primitive* survival of two regularized rows. A raw support or selected pair at a singular cut need not survive after the forced Smith factor is removed. Thus Proposition 499 cannot be inserted termwise into a cross-characteristic selected-pair moment without a separate survival theorem.

Lemma 501 (Residual branch reuse). *Let $H \geq 8$, let $a, b \geq 2$, and put*

$$\mathcal{C}_{a,b}(H) = \{c \geq 2 : a + b + c \leq H, c \neq a\}.$$

Let $p \nmid \mathfrak{U}_H$. Factor the monic normalization of N_a over $\mathbb{Q}_p$ as $V_a^{-1}N_a = \prod_i f_i^{m_i}$, and use the notation $L_i, \mathcal{O}_i, \alpha_i, v_i, f_i^{\text{res}}$ of Proposition 445. Put

$$y_i = \alpha_i + a, \quad z_i = \alpha_i + a + b, \quad g_i = N_b(y_i), \quad j_{i,c} = N_{b+c}(y_i).$$

Then, uniformly in every branch i ,

$$\sum_{c \in \mathcal{C}_{a,b}(H)} \min\{v_i(g_i), v_i(j_{i,c})\} \ll (H - a - b)^{2/3} v_i(g_i). \quad (450)$$

Consequently

$$\sum_{c \in \mathcal{C}_{a,b}(H)} v_p(\mathcal{C}_{a,b,c}) \ll (H - a - b)^{2/3} v_p(S_{a,b}). \quad (451)$$

The implied constants are absolute, and no squarefreeness assumption is made.

Proof. The continuant addition formula, split immediately after level b , gives

$$N_{b+c}(y) = N_{b+1}(y)N_c(y+b) - (y+b+1)^6 N_b(y)N_{c-1}(y+b+1).$$

If $v_i(g_i) > 0$, then $N_{b+1}(y_i)$ is a unit. Otherwise the reductions of $N_b(x+a)$ and $N_{b+1}(x+a)$ would have a common root over the residue field of L_i , forcing p to divide $\text{Res}(N_b, N_{b+1})$. Formula (16) and $b \leq H$ would then give $p \mid \mathfrak{U}_H$, a contradiction. The addition formula therefore gives the exact truncated equality

$$\min\{v_i(g_i), v_i(j_{i,c})\} = \min\{v_i(g_i), v_i(N_c(z_i))\}. \quad (452)$$

Lemma 447, applied in the residue field of L_i , gives $O((H - a - b)^{2/3})$ supported values of c . Each summand in (452) is at most $v_i(g_i)$, proving (450).

Multiply by $m_i f_i^{\text{res}}$ and sum over i . Proposition 445 identifies the left side with the sum of the content valuations. Resultant multiplicativity and $p \nmid V_a$ identify the corresponding right-side branch sum with $v_p(S_{a,b})$, proving (451). $\square$

Lemma 502 (Rank of apparition). *For $p \nmid 6$, put $\chi(p) = (2/p)$. Then $p \mid V_n \iff \rho(p) \mid n$ for an integer $\rho(p) \mid p - \chi(p)$. Hence $\#\{p \in \mathcal{P}_X : \rho(p) \leq R\} \ll R^2 / \log X$.*

Proof. Work in $\mathbb{F}_p(\sqrt{2})$ and write $\bar{\varepsilon} = 17 + 12\sqrt{2}$. Since $\bar{\varepsilon} - \bar{\varepsilon}^{-1} = 24\sqrt{2} \neq 0$, one has

$$p \mid V_n \iff \bar{\varepsilon}^{2n} = 1.$$

Let $\rho(p)$ be the multiplicative order of $\bar{\varepsilon}^2$. Frobenius fixes $\bar{\varepsilon}$ when $\chi(p) = 1$ and sends it to $\bar{\varepsilon}^{-1}$ when $\chi(p) = -1$. Consequently $\bar{\varepsilon}^{p-\chi(p)} = 1$, and hence $\rho(p) \mid p - \chi(p)$.

If $\rho(p) \leq R$, then p divides the nonzero integer $\prod_{n \leq R} V_n$. Since $\log V_n \ll n$,

$$\#\{p \in \mathcal{P}_X : \rho(p) \leq R\} \log X \leq \sum_{n \leq R} \log V_n \ll R^2,$$

which is the asserted estimate. $\square$

Lemma 503 (Exceptional classes). $\#\{p \in \mathcal{P}_X : p \mid \mathfrak{U}_H\} \ll H^2 / \log X$.

Proof. For $H \geq \sqrt{X}$ this follows from the trivial estimate $|\mathcal{P}_X| \ll X / \log X$. Suppose that $H < \sqrt{X}$. Then every $p \in \mathcal{P}_X$ exceeds H and hence divides none of the factorials occurring in $\mathfrak{U}_H$. The elementary growth bounds $\log b_j \ll j$ and $\log V_{j+1} \ll j$ give

$$\log \prod_{j \leq H} b_j V_{j+1} \ll H^2.$$

Every window prime dividing $\mathfrak{U}_H$ therefore contributes at least $\log X$ to the logarithm of this product, which proves the result. $\square$

Hypothesis 504 (Transverse aligned-content mass, AC^{tr}). *With*

$$\mathcal{A}_{\text{tr}}(H) := \sum_{\substack{a, b, c \geq 2, a+b+c \leq H \\ \neg(a=b=c)}} \log \mathcal{C}_{a,b,c}^*,$$

one has $\mathcal{A}_{\text{tr}}(H) \ll H^{3+o(1)}$.

Proposition 505 (Unconditional aligned-content height envelope). *With the notation of Hypothesis 504, one has*

$$\mathcal{A}_{\text{tr}}(H) \ll H^5 \log H.$$

More precisely, the contribution of the palindromic slice $c = a$, $a \neq b$, is $O(H^4 \log H)$, while the arithmetic-progression slice $a = b = c$ is excluded from $\mathcal{A}_{\text{tr}}(H)$ and satisfies separately

$$\sum_{3a \leq H} \log |S_{a,a}| \ll H^3 \log H.$$

Proof. By Lemma 444, if $c \neq a$ then $\mathcal{C}_{a,b,c}^* \mid S_{a,b}^*$, and if $c = a$, $a \neq b$, then $\mathcal{C}_{a,b,a}^* \mid D_{a,b}^*$. For odd b one has $D_{a,b} = S_{a,b}$. For even b ,

$$N_b(x+a) = (2(x+a) + b+1)N_b^o(x+a),$$

so multiplicativity of the resultant gives

$$|S_{a,b}| = |\text{Res}_x(N_a(x), 2(x+a) + b+1)| |D_{a,b}|.$$

All factors are nonzero integers, hence $|D_{a,b}| \leq |S_{a,b}|$. Saturation only removes prime-power factors, and therefore in every case

$$\log \mathcal{C}_{a,b,c}^* \leq \log |S_{a,b}|.$$

For fixed a, b , there are at most $H - a - b - 1$ admissible values of c . The resultant-height bound preceding Theorem 424 gives

$$\log |S_{a,b}| \ll (a + b)^2 \log(a + b).$$

Writing $s = a + b$ and noting that there are $s - 3$ ordered pairs $a, b \geq 2$ with sum s , we obtain

$$\begin{aligned} \mathcal{A}_{\text{tr}}(H) &\leq \sum_{\substack{a,b \geq 2 \\ a+b \leq H-2}} (H - a - b - 1) \log |S_{a,b}| \\ &\ll \log H \sum_{s=4}^{H-2} (s - 3)(H - s - 1)s^2 \ll H^5 \log H. \end{aligned}$$

On the palindromic slice, summing the same height bound over $2a + b \leq H$ gives $O(H^4 \log H)$. On the progression slice, summing $\log |S_{a,a}| \ll a^2 \log a$ over $3a \leq H$ gives $O(H^3 \log H)$. $\square$

Proposition 506 (Column-reuse aligned-content envelope). *With the notation of Hypothesis 504, one has*

$$\mathcal{A}_{\text{tr}}(H) \ll H^{14/3} \log H.$$

Proof. For fixed a, b , sum (451) over primes outside the support of $\mathfrak{U}_H$, with weight $\log p$. This gives

$$\sum_{c \in \mathcal{C}_{a,b}(H)} \log \mathcal{C}_{a,b,c}^* \ll (H - a - b)^{2/3} \log |S_{a,b}^*|.$$

The resultant-height bound preceding Theorem 424 therefore gives, for the off-palindromic triples,

$$\begin{aligned} \sum_{\substack{a,b,c \geq 2, a+b+c \leq H \\ c \neq a}} \log \mathcal{C}_{a,b,c}^* &\ll \log H \sum_{s=4}^{H-2} (s - 3)s^2(H - s)^{2/3} \\ &\ll H^{14/3} \log H. \end{aligned}$$

On the palindromic slice $c = a$, $a \neq b$, one has $\mathcal{C}_{a,b,a}^* \mid D_{a,b}^*$ by Lemma 444, and the proof of Proposition 505 gives $O(H^4 \log H)$. The all-equal progression slice is absent from $\mathcal{A}_{\text{tr}}$ by definition. Combining the two retained slices proves the proposition. $\square$

By (446), this hypothesis is no stronger than the earlier scalar reduced gcd-mass hypothesis GM^{tr} : the latter implies AC^{tr} term by term. The aligned formulation is the one actually used below because it retains common-root multiplicity without charging coincidences at unrelated roots.

Remark 507 (Computational support). The exact audit in `quadcorank_verify.py` exposes the diagonal error in the adjacent-pair formulation: at $H = 20$, its diagonal mass divided by H^4 is 0.187534, whereas its total mass divided by H^3 is already 3.842321. The naïve skipped-triple repair also fails its scaling gate: its $c = a$ slice divided by H^4 is 0.004833 at $H = 32$, exactly where the propagated central factors occur. After central deflation and removal of the one-parameter arithmetic-progression slice, the scalar ratio is 0.005569 at $H = 20$ and 0.014732 at $H = 32$.

The independent exact audit in `aligned_corank_verify.py` computes the full polynomial $\mathcal{R}_{a,b,c}(T) \in \mathbb{Z}[T]$, not a gcd of sampled values. It finds $\mathcal{C}_{a,b,c}^* = 1$ for every triple through $H = 14$.

At $H = 20$, only two of the 675 non-all-equal gap triples have nontrivial reduced content, both equal to 653, and $\mathcal{A}_{\text{tr}}(20)/20^3 = 0.001620394$. At $H = 24$, only 18 of 1323 triples are nontrivial and the corresponding ratio is 0.007248253. These finite computations support, but of course do not prove, AC^{tr} .

Theorem 508 (Quadruple corank bound). *Assume AC^{tr} . Then $\sum_{X < p \leq 2X} Z(p) \ll X^{3/2+o(1)}$, i.e., average $Z(p) \ll p^{1/2+o(1)}$.*

Proof. We first record the valuation form of Lemma 441. Partition $[0, p-1]$ into consecutive blocks of length at most H , and suppose that one block contains the ordered zeros $z_1 < \dots < z_m$. Select the $m-3$ consecutive quadruples $(z_i, z_{i+1}, z_{i+2}, z_{i+3})$ when $m \geq 4$. Thus the total number $Q_p(H)$ of selected quadruples satisfies

$$Q_p(H) \geq Z(p) - 3 \left\lceil \frac{p}{H} \right\rceil. \quad (453)$$

Every selected quadruple has a gap triple $\mathbf{h} = (h_1, h_2, h_3)$ with $h_i \geq 2$ and $h_1 + h_2 + h_3 < H$.

Put $\mathcal{E}_H = \{p \in \mathcal{P}_X : p \mid \mathfrak{U}_H\}$. The endpoint identity

$$N_h(-1) = ((h-1)!)^3 b_{h-1}$$

and the definition of $\mathfrak{U}_H$ show that every N_h with $h \leq H$ has nonzero reduction modulo every $p \notin \mathcal{E}_H$. The same holds for N_h° : this is immediate for odd h , while for even h the factorization $N_h = (2x + h + 1)N_h^\circ$ shows that a zero reduction of N_h° would make N_h the zero polynomial modulo p .

For completeness, if two integer polynomials f, g , both nonzero modulo p , have t distinct common roots modulo p , evaluation at those roots gives t independent linear functionals annihilating the image of the fixed, characteristic-zero-size Sylvester map. Hence its reduction modulo p has corank at least t . Smith normal form then gives $v_p(\text{Res}(f, g)) \geq t$ whenever the integer resultant is nonzero. A leading-coefficient drop after reduction does not change this fixed matrix or destroy any of the evaluation functionals.

Fix $p \notin \mathcal{E}_H$, write a gap triple as (a, b, c) , and let $t_p(a, b, c)$ be the number of its selected starting points. If $c \neq a$, every such starting point is a common root of the three polynomials in the first line of Definition 443. Lemma 444 gives

$$t_p(a, b, c) \leq v_p(\mathcal{C}_{a,b,c}).$$

If $c = a$ and $a \neq b$, let $e_p(a, b)$ be the number of those starts that satisfy (444). Lemma 442 shows that every remaining start is a common root of the three polynomials in the second line of Definition 443, so the aligned-content lemma gives

$$t_p(a, b, a) - e_p(a, b) \leq v_p(\mathcal{C}_{a,b,a}), \quad \sum_{a,b} e_p(a, b) \leq 1.$$

Finally, if $a = b = c$, the first three zeros already make every start a common root of $N_a(x)$ and $N_a(x+a)$, whence $t_p(a, a, a) \leq v_p(S_{a,a})$. Summing the three classes yields

$$Q_p(H) - 1 \leq \sum_{\substack{a,b,c \geq 2, \\ a+b+c \leq H \\ \neg(a=b=c)}} v_p(\mathcal{C}_{a,b,c}) + \sum_{\substack{a \geq 2 \\ 3a \leq H}} v_p(S_{a,a}). \quad (454)$$

It remains to retain this valuation information at primes dividing the structural carrier. We do so by choosing the block length separately on each zero-count level. The primes with $Z(p) \leq \sqrt{X}$ contribute at most

$$\sqrt{X} |\mathcal{P}_X| \ll \frac{X^{3/2}}{\log X}. \quad (455)$$

For dyadic $T = 2^k \sqrt{X}$ with $T < 2X$, put

$$\mathcal{P}_X(T) = \{p \in \mathcal{P}_X : T < Z(p) \leq 2T\}, \quad H = \left\lceil \frac{16X}{T} \right\rceil.$$

For all sufficiently large X , one has $8 \leq H < p$ and

$$3 \left\lceil \frac{p}{H} \right\rceil \leq \frac{3T}{8} + 3.$$

Equations (453) and (454) therefore imply, uniformly for $p \in \mathcal{P}_X(T) \setminus \mathcal{E}_H$,

$$\frac{T}{3} \leq \sum_{\substack{a,b,c \geq 2, \ a+b+c \leq H \\ \neg(a=b=c)}} v_p(\mathcal{C}_{a,b,c}) + \sum_{\substack{a \geq 2 \\ 3a \leq H}} v_p(S_{a,a}). \quad (456)$$

For $p \notin \mathcal{E}_H$, saturation changes none of the p -adic valuations in (456). Moreover, the general resultant-height estimate quoted above gives

$$\sum_{\substack{a \geq 2 \\ 3a \leq H}} \log |S_{a,a}^*| \leq \sum_{\substack{a \geq 2 \\ 3a \leq H}} \log |S_{a,a}| \ll H^3 \log H = H^{3+o(1)}.$$

Summing over $\mathcal{P}_X(T) \setminus \mathcal{E}_H$, then using $\sum_{X < p \leq 2X} v_p(M) \leq \log |M| / \log X$ and AC^{tr} , gives

$$\frac{T}{3} |\mathcal{P}_X(T) \setminus \mathcal{E}_H| \ll \frac{H^{3+o(1)}}{\log X}.$$

Since $Z(p) \leq 2T$ on this level,

$$\sum_{p \in \mathcal{P}_X(T) \setminus \mathcal{E}_H} Z(p) \ll \frac{H^{3+o(1)}}{\log X} \ll \frac{(X/T)^{3+o(1)}}{\log X}. \quad (457)$$

On the exceptional part, Lemma 503 and the same level upper bound give

$$\sum_{p \in \mathcal{P}_X(T) \cap \mathcal{E}_H} Z(p) \leq 2T |\mathcal{E}_H| \ll \frac{TH^2}{\log X} \ll \frac{X^2}{T \log X}. \quad (458)$$

Finally, summing (457) and (458) over the dyadic values $T \geq \sqrt{X}$ yields convergent geometric sums:

$$\sum_{T \geq \sqrt{X}} \frac{(X/T)^{3+o(1)}}{\log X} + \sum_{T \geq \sqrt{X}} \frac{X^2}{T \log X} \ll X^{3/2+o(1)}.$$

Together with (455), this proves the theorem. $\square$

13.10.2 A fully deflated adjacent certificate

The aligned pencil above deliberately uses a skipped endpoint. There is a second, more literal certificate which keeps all three adjacent gaps and removes the reflection-center factor from each of them.

Definition 509 (Fully deflated adjacent content). For $a, b, c \geq 2$, put

$$F_{a,b,c}^{\text{fd}} = N_a^\circ(x), \quad G_{a,b,c}^{\text{fd}} = N_b^\circ(x+a), \quad J_{a,b,c}^{\text{fd}} = N_c^\circ(x+a+b),$$

and define

$$\mathcal{R}_{a,b,c}^{\text{fd}}(T) = \text{Res}_x \left(F_{a,b,c}^{\text{fd}}, G_{a,b,c}^{\text{fd}} + T J_{a,b,c}^{\text{fd}} \right), \quad \mathcal{C}_{a,b,c}^{\text{fd}} = \text{cont}_T \mathcal{R}_{a,b,c}^{\text{fd}}.$$

Relative to height H , let $(\mathcal{C}_{a,b,c}^{\text{fd}})^*$ denote removal of the full prime-power part supported on $\mathfrak{U}_H$.

Lemma 510 (Fully deflated quadruple certificate). *Every polynomial $\mathcal{R}_{a,b,c}^{\text{fd}}(T)$ is nonzero. Let an odd prime p be fixed, partition $[0, p-1]$ into intervals, and select sliding quadruples of consecutive zeros inside the intervals. Apart from at most three selected quadruples in total, every selected quadruple with gap vector (a, b, c) has its start as a common root of*

$$F_{a,b,c}^{\text{fd}}, \quad G_{a,b,c}^{\text{fd}}, \quad J_{a,b,c}^{\text{fd}} \pmod{p}.$$

Proof. Every root of N_h° is a root of N_h . The root-strip theorem therefore places the roots of $F_{a,b,c}^{\text{fd}}$ in $-a < \text{Re } x < -1$, while the roots of $G_{a,b,c}^{\text{fd}}$ lie in $-a-b < \text{Re } x < -a-1$. These strips are disjoint, so

$$\mathcal{R}_{a,b,c}^{\text{fd}}(0) = \text{lc}(F_{a,b,c}^{\text{fd}})^{(d-d_G)} \text{Res}_x(F_{a,b,c}^{\text{fd}}, G_{a,b,c}^{\text{fd}}) \neq 0,$$

where $d_G = \deg G_{a,b,c}^{\text{fd}}$ and $d = \max(d_G, \deg J_{a,b,c}^{\text{fd}})$. This proves nonvanishing of the pencil.

Let $u < v$ be one of the three adjacent zero pairs in a selected quadruple, and put $h = v - u$. The gap identity gives $N_h(u) = 0$. If h is odd, this is already $N_h^\circ(u) = 0$. If h is even, the integral factorization $N_h(x) = (2x + h + 1)N_h^\circ(x)$ shows the same unless

$$2u + h + 1 = u + v + 1 \equiv 0 \pmod{p}.$$

Because $0 \leq u < v \leq p-1$, the exceptional congruence is equivalent to $u + v = p-1$; call such an adjacent pair centered. There is at most one centered adjacent pair in any increasing zero set. Indeed, two different such pairs would cross: if $u_1 < u_2$ and $u_1 + v_1 = u_2 + v_2 = p-1$, then $u_1 < u_2 < v_2 < v_1$, contradicting the adjacency of u_1, v_1 . A fixed adjacent pair occurs in at most three sliding quadruples, according as it is their first, second, or third pair. Thus at most three selected quadruples are lost. Every other selected quadruple survives all three deflations and gives the asserted common root. $\square$

Hypothesis 511 (Fully deflated adjacent-content mass, FDAC^{tr}). *With*

$$\mathcal{A}_{\text{fd}}(H) := \sum_{\substack{a,b,c \geq 2, \ a+b+c \leq H \\ \neg(a=b=c)}} \log(\mathcal{C}_{a,b,c}^{\text{fd}})^*,$$

one has $\mathcal{A}_{\text{fd}}(H) \ll H^{3+o(1)}$.

Theorem 512 (Fully deflated quadruple corank bound). *Assume FDAC^{tr}. Then $\sum_{X < p \leq 2X} Z(p) \ll X^{3/2+o(1)}$.*

Proof. Use the same block selection as in the proof of Theorem 508. For $p \notin \mathcal{E}_H$, all three fully deflated polynomials have nonzero reduction. Lemmas 510 and 444 give

$$Q_p(H) - 3 \leq \sum_{\substack{a,b,c \geq 2, \ a+b+c \leq H \\ \neg(a=b=c)}} v_p(\mathcal{C}_{a,b,c}^{\text{fd}}) + \sum_{\substack{a \geq 2 \\ 3a \leq H}} v_p(S_{a,a}).$$

The arithmetic-progression term is unchanged. On the dyadic level $T < Z(p) \leq 2T$, the choice $H = \lceil 16X/T \rceil$ and (453) give, for $p \leq 2X$,

$$Q_p(H) - 3 > T - 3 \left\lceil \frac{p}{H} \right\rceil - 3 \geq \frac{5T}{8} - 6 \geq \frac{T}{2}$$

for all sufficiently large X . Thus one obtains the analogue of (456), with $T/2$ on the left. Hypothesis FDAC^{tr} , the unconditional $H^{3+o(1)}$ bound for the progression resultants, and Lemma 503 then give exactly (457) and (458). The same dyadic summation completes the proof. $\square$

Remark 513 (Exact fully deflated audit). The script `fully_deflated_corank_verify.py` computes the entire polynomial resultant over $\mathbb{Z}[T]$. It finds $(\mathcal{C}_{a,b,c}^{\text{fd}})^* = 1$ for every non-progression triple through $H = 23$. At $H = 32$ there are thirteen nontrivial records, supported at 157, 431, or 653, and $\mathcal{A}_{\text{fd}}(32)/32^3 = 0.002569880$; the exact-content SHA-256 is

437f7328a4ea7d78a62b3e781165a76a
31f277f7ec6983f9f6c3a4e67ee05efa.

Extending the same full-content computation to $H = 36$ gives nineteen nontrivial records, supported at 157, 431, 499, 653, or 1297, and

$$\frac{\mathcal{A}_{\text{fd}}(36)}{36^3} = 0.002595232.$$

Its exact-content SHA-256 is

196327788a7ea9adbbb141a5c6161ae9
6125ea5b8242bc6603352080b32aac76.

The independent script `centered_residual_verify.py` reconstructs all the common roots. The roots at 157 and 653 skip an intermediate return, and their refined return-time lists contain a centered adjacent pair. (One reduced content is $24649 = 157^2$, not a new characteristic.) The two roots at 431, for gap vectors $(13, 3, 15)$ and $(15, 3, 13)$, are primitive and off-center; nevertheless they are not Apéry zeros, since $\mathcal{Z}_{431} = \{56, 374\}$. Thus none of the height-32 residual roots is an actual primitive four-zero obstruction. This finite observation is not used in Theorem 512.

Primitive saturation does not remove every algebraic phantom. The new $p = 499$ records at $H = 33$ have reflected triples $(6, 14, 13)$ and $(13, 14, 6)$, roots 417 and 48, and exact return-offset lists $(0, 6, 20, 33)$ and $(0, 13, 27, 33)$. Both are primitive and off-center, but

$$\mathcal{Z}_{499} = \{67, 170, 203, 295, 328, 431\},$$

so neither root belongs to the distinguished fiber. At $H = 35$ the full content audit also verifies

$$(\mathcal{C}_{17,14,4}^{\text{fd}})^* = 157, \quad \gcd_{\mathbb{F}_{157}[x]} (F_{17,14,4}^{\text{fd}}, G_{17,14,4}^{\text{fd}}, J_{17,14,4}^{\text{fd}}) = x + 119.$$

The root $x = 38$ has precisely the return offsets 0, 17, 31, 35 in the selected span, so it is primitive and no adjacent pair is centered; its reflected triple $(4, 14, 17)$ has root 83 and offsets $(0, 4, 18, 35)$. Nevertheless the two independent recurrence implementations both give $\mathcal{Z}_{157} = \emptyset$; the algebraic roots are not zeros of the distinguished Apéry orbit. Thus even a primitive common-root carrier can strictly overcount the starts needed in the application.

The unit-scale formal short-window analogue also fails. The exact endpoint-resultant superset scan in `primitive_fd_candidate_verify.py`, followed by an actual three-polynomial gcd over each candidate finite field, finds at height 36 the reflected pair

$$(a, b, c, p, x) = (5, 20, 10, 1297, 360), \quad (10, 20, 5, 1297, 901).$$

In both cases the four displayed return offsets are the complete list in the selected span, namely $(0, 5, 25, 35)$ and $(0, 10, 30, 35)$ respectively, and none of their adjacent pairs is centered. Here $1297 \nmid \mathfrak{U}_{35}$ and $1297 > 35^2$, so these are genuinely primitive short-span common roots outside the structural carrier. They are still phantoms: the divided and cleared recurrences independently give

$$\mathcal{Z}_{1297} = \{459, 530, 766, 837\}.$$

Among all nine saturated endpoint candidates with $p > (a + b + c)^2$ through height 36, the other seven consist of three endpoint false positives and four nonprimitive roots; none starts at an actual Apéry zero. The exact short-range payload SHA-256 is

7279c7675994a3709a2039a0b04073f8
8e3fca346d08fe94a39f5fa4a3ee76e9.

The endpoint superset does not miss a chain defined only by its raw return set. Indeed, applying the continuant congruence in Remark 521 first with $(u, v) = (a, b)$ and then with $(u, v) = (a + b, c)$ shows that exactness of the four-return set makes the two intervening multipliers nonzero and forces the shifted adjacent gap polynomials to vanish.

Incremental endpoint scans in eight-span blocks extend the exact calculation through span 84. Beyond span 36, the blocks contain five endpoint false positives, eight nonprimitive roots at $p = 8941$, and one new reflected pair of primitive phantoms,

$$(a, b, c, p, x) = (3, 40, 31, 7411, 4681), \quad (31, 40, 3, 7411, 2655).$$

Their exact return-offset lists are $(0, 3, 43, 74)$ and $(0, 31, 71, 74)$, respectively; neither has a centered adjacent pair, $7411 \nmid \mathfrak{U}_{74}$, and the two recurrence implementations give $\mathcal{Z}_{7411} = \emptyset$. The final span block 77–84 has two tested primes above s^2 , both endpoint false positives. Every untested prime already satisfies $p \leq s^2$, so the combined endpoint scan is an exact verification of (QPRS) with $C = 2$ for every gap triple of span at most 84 outside the all-equal slice. The exact block digests are hard-coded in `primitive_fd_candidate_verify.py`.

A complementary prime-first audit shows that the constant 2 does not persist. The C++ program `primitive_projective_prime_scan.cpp` computes the two independent Apéry recurrence solutions for every prime $7 \leq p \leq 500000$, groups all indices by their point in $\mathbb{P}^1(\mathbb{F}_p)$, and tests every sliding quadruple of consecutive occurrences in each fiber. It checks the carrier, center, non-wrapping, and non-all-equal conditions directly. Among 13,695,120 primitive off-center windows, twenty have $p > s^2$ and four have $p > 2s^2$; all twenty belong to nondistinguished fibers. The largest ratio is

$$\frac{p}{s^2} = \frac{128047}{164^2} = 4.760819\dots,$$

attained at

$$(p, x; a, b, c) = (128047, 42375; 41, 86, 37)$$

and its reflection. Its raw return-offset set is exactly $(0, 41, 127, 164)$, one has $128047 \nmid \mathfrak{U}_{164}$, and none of the three adjacent pairs is centered. The independent standard-library program

`primitive_projective_prime_verify.py` reconstructs all twenty records both from projective states and from the scalar continuant recurrence. The complete scan output has SHA-256

8eeb4d6d4a6f371d8f8d87facadff137
f66dfa476a28db1a3af8892625032bf9.

This finite result disproves $C = 2$, not the existence of some absolute constant in (QPRS).

There is, however, a reason not to treat the formal support hypothesis as the main orbit-coupling target. In the random palindromic-orbit model, the expected number at one prime of four-equal-state windows of span at most s has order s^3/p^2 . At $s = \sqrt{p/C}$, summation over $p \leq P$ therefore has heuristic order $\sqrt{P}/(C^{3/2} \log P)$, which diverges for every fixed C . The carrier and center exclusions have lower-order random densities. This occupancy calculation is only a heuristic and does not disprove (QPRS), but the growing formal phantoms show why a distinguished-orbit input is structurally preferable.

Thus primitivity alone does not prove the unit-scale short-window statement: coupling to the distinguished Apéry orbit is indispensable. The weaker fixed- η formulation below remains sufficient for the application.

The separate script `primitive_chain_census.py` audits the generated 350,104 pairs (p, r) through $p \leq 5 \cdot 10^6$. The little-endian pair file has SHA-256

5739c6e7fee4210678bc50bdda7d0a7c
2f9fa082ab392e050556ebdc62ecac8b.

Among 43,366 sliding quadruples in the full ordered zero sets, 1,418 do not contain a centered adjacent pair, but none of those 1,418 has total span at most $\sqrt{p}$. In fact the closest observed squared scale ratio is still the exact value

$$\frac{s^2}{p} = \frac{1428025}{3727} = 383.156694 \dots$$

The unrestricted statement is false: this closest example, which is also the first by absolute span, occurs at $p = 3727$, with consecutive zeros

$$99 < 868 < 1011 < 1294 \quad \text{and gaps} \quad (769, 143, 283).$$

Thus any conjectural primitive-return strengthening must retain the mesoscopic short-span condition used by the block argument.

Definition 514 (Short off-center window count). For an odd prime p and $0 < \eta \leq 1$, write the full ordered Apéry zero set as $z_1 < \dots < z_{Z(p)}$, and let $E_{\text{sw}}(p; \eta)$ be the number of indices $1 \leq i \leq Z(p) - 3$ such that

$$z_{i+3} - z_i \leq \eta \sqrt{p}$$

and none of the three adjacent pairs in $z_i, z_{i+1}, z_{i+2}, z_{i+3}$ is centered.

Proposition 515 (Short-window reduction). *For every odd prime p and $0 < \eta \leq 1$,*

$$Z(p) < 3\eta^{-1}\sqrt{p} + 6 + E_{\text{sw}}(p; \eta).$$

Consequently, for every fixed η ,

$$\sum_{X < p \leq 2X} Z(p) \ll_{\eta} \frac{X^{3/2}}{\log X} + \sum_{X < p \leq 2X} E_{\text{sw}}(p; \eta).$$

Proof. Put $q = Z(p)$ and $E = E_{\text{sw}}(p; \eta)$; there is nothing to prove when $q \leq 6 + E$. For $1 \leq i \leq q - 3$, let $s_i = z_{i+3} - z_i$. There is at most one centered adjacent pair in the full increasing zero set, and a fixed adjacent pair belongs to at most three sliding quadruples. Apart from the E short off-center windows, at most three further s_i can be at most $\eta\sqrt{p}$. Hence at least $q - 6 - E$ of the sliding spans satisfy $s_i > \eta\sqrt{p}$. On the other hand, writing $d_j = z_{j+1} - z_j$, each d_j occurs in at most three of the sums $s_i = d_i + d_{i+1} + d_{i+2}$, so

$$(q - 6 - E)\eta\sqrt{p} < \sum_{i=1}^{q-3} s_i \leq 3 \sum_{j=1}^{q-1} d_j = 3(z_q - z_1) < 3p.$$

This proves the pointwise inequality. Summing it over the primes in $(X, 2X]$ and using the elementary prime-counting bound proves the second assertion. $\square$

Hypothesis 516 (Short-window reflection, (SWR)). *There is an absolute constant $0 < \eta \leq 1$ such that $E_{\text{sw}}(p; \eta) = 0$ for every odd prime p ; equivalently, every four consecutive Apéry zeros of total span at most $\eta\sqrt{p}$ contain a centered adjacent pair.*

Corollary 517 (Square-root bound under short-window reflection). *Assume (SWR) with the fixed constant η . Then, for every odd prime p ,*

$$Z(p) \leq 3 \left\lceil \frac{p}{\lfloor \eta\sqrt{p} \rfloor + 1} \right\rceil + 3 < 3\eta^{-1}\sqrt{p} + 6.$$

Consequently $\sum_{X < p \leq 2X} Z(p) \ll X^{3/2} / \log X$.

Proof. The case $p = 3$ is immediate from $(b_0, b_1, b_2) \equiv (1, 2, 1) \pmod{3}$. Suppose that $p \geq 5$, put $n = \lfloor \eta\sqrt{p} \rfloor$ and $H = n + 1$, and partition $\{0, \dots, p - 1\}$ into $B = \lceil p/H \rceil$ consecutive blocks of cardinality at most H . If the blocks contain respectively $m_1, \dots, m_B$ zeros, select all sliding quadruples of consecutive zeros within each block. Their total number Q satisfies

$$Q = \sum_{\nu=1}^B (m_\nu - 3)_+ \geq Z(p) - 3B.$$

Every selected quadruple is consecutive in the full zero set and has span at most $H - 1 = n \leq \eta\sqrt{p}$, so (SWR) makes it contain a centered adjacent pair. There is at most one such pair in the full zero set, and it belongs to at most three sliding quadruples. Hence $Q \leq 3$ and

$$Z(p) \leq 3B + 3 = 3 \left\lceil \frac{p}{n + 1} \right\rceil + 3.$$

Since $n + 1 > \eta\sqrt{p}$, the ceiling is strictly less than $\eta^{-1}\sqrt{p} + 1$, proving the second pointwise bound. The asserted prime-window average follows from that bound and the standard estimate $\#\{X < p \leq 2X\} \ll X / \log X$. $\square$

Hypothesis 518 (Quadratic primitive-return support, (QPRS)). *There is an absolute constant $C \geq 1$ with the following property. Let p be an odd prime, let $a, b, c \geq 2$ be not all equal, put $s = a + b + c < p$, and let x be an integer with $0 \leq x$ and $x + s < p$. Define the return-offset set*

$$\mathcal{T}_{p,s}(x) := \{0\} \cup \{1 \leq h \leq s : N_h(x) \equiv 0 \pmod{p}\}.$$

If

$$p \nmid \mathfrak{U}_s, \quad \mathcal{T}_{p,s}(x) = \{0, a, a + b, s\},$$

and none of the three adjacent pairs with offsets $0, a, a + b, s$ is centered, then

$$p \leq Cs^2.$$

Theorem 519 (Quadratic primitive support implies the square-root average). *Assume (QPRS). Then*

$$\sum_{X < p \leq 2X} Z(p) \ll_C X^{3/2}.$$

Proof. Fix $K = 64\sqrt{C}$. The primes with $Z(p) \leq K\sqrt{X}$ contribute $O_C(X^{3/2}/\log X)$. For dyadic $T \geq K\sqrt{X}$, put

$$\mathcal{P}_X(T) = \{p \in \mathcal{P}_X : T < Z(p) \leq 2T\}, \quad H = \left\lceil \frac{16X}{T} \right\rceil,$$

and use the block selection from the proof of Theorem 508. For all sufficiently large X ,

$$H \leq \frac{16\sqrt{X}}{K} + 1 \leq \frac{\sqrt{X}}{2\sqrt{C}}, \quad CH^2 < X.$$

Put $\mathcal{E}_H = \{p \in \mathcal{P}_X : p \mid \mathfrak{U}_H\}$.

Suppose that $p \in \mathcal{P}_X(T)$ does not divide $\mathfrak{U}_H$, and consider a selected consecutive quadruple with start x and gap vector (a, b, c) . Its span $s = a + b + c$ is less than H , so $p \nmid \mathfrak{U}_s$. By the projective-orbit gap identity in Remark 21, because $b_x \equiv 0 \pmod{p}$ one has

$$N_h(x) \equiv 0 \pmod{p} \iff b_{x+h} \equiv 0 \pmod{p} \quad (0 \leq h \leq s).$$

The global consecutiveness of the quadruple therefore gives $\mathcal{T}_{p,s}(x) = \{0, a, a + b, s\}$. If it contained no centered adjacent pair and a, b, c were not all equal, (QPRS) would imply

$$p \leq Cs^2 < CH^2 < X < p,$$

a contradiction.

Let $A_p(H)$ be the number of selected quadruples with $a = b = c$. The preceding paragraph and the uniqueness of the centered adjacent pair give

$$Q_p(H) \leq A_p(H) + 3.$$

Every progression start counted by $A_p(H)$ is a common root of $N_a(x)$ and $N_a(x + a)$ for some $3a < H$. The fixed Sylvester-corank argument therefore gives

$$A_p(H) \leq \sum_{\substack{a \geq 2 \\ 3a < H}} v_p(S_{a,a}).$$

On the other hand,

$$Q_p(H) \geq Z(p) - 3 \left\lceil \frac{p}{H} \right\rceil > T - 3 \left(\frac{T}{8} + 1 \right) = \frac{5T}{8} - 3.$$

Consequently, for all sufficiently large X and every $p \in \mathcal{P}_X(T) \setminus \mathcal{E}_H$,

$$\frac{T}{2} \leq \sum_{\substack{a \geq 2 \\ 3a < H}} v_p(S_{a,a}).$$

Summing over these generic primes and using $\sum_{3a < H} \log |S_{a,a}| \ll H^3 \log H$ gives

$$\sum_{p \in \mathcal{P}_X(T) \setminus \mathcal{E}_H} Z(p) \leq 2T |\mathcal{P}_X(T) \setminus \mathcal{E}_H| \ll \frac{H^3 \log H}{\log X} \ll H^3.$$

On the exceptional part, Lemma 503 gives

$$\sum_{p \in \mathcal{P}_X(T) \cap \mathcal{E}_H} Z(p) \leq 2T \#\{p \in \mathcal{P}_X : p \mid \mathfrak{U}_H\} \ll \frac{TH^2}{\log X} \ll \frac{X^2}{T \log X}.$$

Since $H \ll X/T$, summing both geometric series over $T \geq K\sqrt{X}$ and adding the low range proves the theorem. $\square$

Corollary 520 (Moment threshold $k = 5$). *Under any of AC^{tr} , FDAC^{tr} , (SWR), or (QPRS), $(\text{HM})_5$ implies $G_n = e^{o(n)}$ for every n .*

Proof. Theorem 508, Theorem 512, Theorem 519, or Corollary 517, as appropriate, gives $\lambda_X \leq X^{-1} \sum_{p \in \mathcal{P}_X} Z(p) \ll X^{1/2+o(1)}$. Repeating the proof of Theorem 331 with this improved estimate and $k = 5$ gives

$$\max_{m < X^2} K_X(m) \ll X^{2/5+o(1)} \lambda_X + 1 \ll X^{9/10+o(1)}.$$

The same dyadic decomposition used there now yields $\log G_n \ll n^{9/10+o(1)} = o(n)$. $\square$

Remark 521 (The corank ladder terminates at $1/2$). Within a single triple of zeros the certificates are algebraically dependent: $N_{h_1+h_2}(x) \equiv N_{h_2}(x+h_1) N_{h_1+1}(x) \pmod{N_{h_1}(x)}$, so the triple level cannot improve beyond $3/5$. The quadruple is the first configuration supplying two potentially transverse certificates. The adjacent self-gcd, the propagated palindromic center, and the one-parameter arithmetic-progression slice must all be removed as above. The remaining input can be formulated either as the aligned-content estimate AC^{tr} or as the fully deflated adjacent estimate FDAC^{tr} ; the older scalar gcd-mass estimate is a sufficient but unnecessarily strong version of the first formulation. The short-window condition (SWR) and the quadratic support condition (QPRS) instead bypass aggregate content mass by ruling out generic primitive windows at a suitably chosen block scale. Higher tuples give strictly weaker exponents within the same corank bookkeeping.

13.11 The parity barrier and the central open problem

Open Problem 522 (Parity barrier). *Find a map $r \mapsto \mathfrak{G}_r$ assigning to every integer $r \geq 2$ a nonzero integer $\mathfrak{G}_r$, defined by r alone, such that*

(D) **Detection:** *for every prime $p > r$, $p \mid b_r \implies p \mid \mathfrak{G}_r$;*

(H) **Height:** $\log |\mathfrak{G}_r| \leq r^{o(1)}$.

The sequence $\mathfrak{G}_r = b_r$ satisfies (D) with $\log b_r \sim 3.5255r$: the entire content is the height condition.

Remark 523 (What a solution yields). With $(\log r)^{O(1)}$ heights, summing $\omega_{(X, 2X]}(\mathfrak{G}_r)$ over $r < 2X$ gives average $Z(p) \ll p^{o(1)}$; the exponent c drops to $o(1)$ and the full conjecture follows from $(\text{HM})_3$ alone. A solution would *not* by itself prove (AT'') : the atom count is a diagonal selection, and certificates control the total divisor mass, not its concentration on the diagonal. It would, however, breach the parity barrier: every certificate in this paper requires at least two vanishing conditions at a single prime.

Remark 524 (Known failure modes). (i) The Fermat detector $1 - b_r^{p-1}$ has degree growing with p . (ii) The Cartier/Hasse–Witt operator is p -semilinear; its entries lift to no fixed integer. (iii) Character-sum realizations detect residues modulo $p-1$, not modulo p . (iv) Weight: twisted traces have weight ≥ 2 , so $p \mid S$ does not force $S = 0$. (v) The digit ladder (variation of parameters at depth ≥ 4) is parity-neutral: each deeper congruence introduces one relation and one new local unknown, with zero net global information gain. (vi) Tensor constructions (symmetric powers) dilute parity rather than breaking it.

Proposition 525 (Diagonal transport identity). *For all $h \geq 1$,*

$$N_{h+1}(x-1) = P(x) N_h(x) - (x+1)^6 N_{h-1}(x+1).$$

Equivalently, the gap-polynomial family satisfies a shift relation: the polynomial triples $(N_{h+1}(x-1), N_h(x), N_{h-1}(x))$ are connected by the same coefficients $P(x)$ and $(x+1)^6$ independently of h .

Proof. The standard h -recurrence expands $\det D_{h+1}(x)$ along the last row; the present identity expands $\det D_{h+1}(x-1)$ along the *first* row. $\square$

13.12 The vertical first moment: edge localization and the medium window

Throughout this subsection $\kappa_0 = \log 34$ and $\kappa = \log(17 + 12\sqrt{2}) = 3.5254\dots$, so that $\log b_r = \kappa r + O(\log r)$ and, by the elementary bound of Lemma 526 below, $\log b_r \leq \kappa_0 r$ for all $r \geq 1$. Recall the reflection symmetry $b_r \equiv b_{p-1-r} \pmod{p}$ (Corollary 386), so that $r \mapsto p-1-r$ is an involution of $\mathbb{Z}_p$.

Lemma 526 (Two-sided growth). *For every $n \geq 0$, $0 < b_n \leq b_{n+1} \leq 34 b_n$. In particular $b_r \leq 5 \cdot 34^{r-1} \leq 34^r$ for $r \geq 1$.*

Proof. Induction on n using the recurrence (1) with $P(n) = 34n^3 + 51n^2 + 27n + 5$. If $0 < b_{n-1} \leq b_n$, then $(n+1)^3 b_{n+1} = P(n) b_n - n^3 b_{n-1} \geq (P(n) - n^3) b_n \geq (n+1)^3 b_n$, since $P(n) - n^3 = 33n^3 + 51n^2 + 27n + 5 \geq (n+1)^3$ for $n \geq 1$; and $(n+1)^3 b_{n+1} \leq P(n) b_n \leq 34(n+1)^3 b_n$ since $P(n) \leq 34(n+1)^3$. The base case is $(b_0, b_1) = (1, 5)$. $\square$

Proposition 527 (Unconditional edge first moment). *For $0 < \theta \leq \frac{1}{2}$ define the edge slice*

$$Z_\theta^{\text{edge}}(p) = \#\{r \in \mathbb{Z}_p : \min(r, p-1-r) \leq p^\theta\}.$$

Then

$$\sum_{p \leq x} Z_\theta^{\text{edge}}(p) \leq (\kappa_0 + o(1)) \frac{x^{2\theta}}{\log x}.$$

In particular, at the square-root scale $\theta = \frac{1}{2}$,

$$\sum_{p \leq x} Z_{1/2}^{\text{edge}}(p) \leq (\kappa_0 + o(1)) \frac{x}{\log x} = O(\pi(x)) :$$

the edge slice of the first moment satisfies Hypothesis $\bar{Z}$ unconditionally, and every strictly smaller polynomial scale $\theta < \frac{1}{2}$ carries a power saving.

Proof. By the reflection involution, $\#\{r \in \mathcal{Z}_p : p-1-r \leq p^\theta\} = \#\{r \in \mathcal{Z}_p : r \leq p^\theta\}$, so $Z_\theta^{\text{edge}}(p) \leq 2\#\{r \leq p^\theta : p \mid b_r\}$. Summing over $p \leq x$ counts incidences

$$\sum_{p \leq x} \#\{r \leq p^\theta : p \mid b_r\} = \#\{(r, p) : p \leq x \text{ prime}, r \leq p^\theta, p \mid b_r\}.$$

Every such pair has $r \leq x^\theta$ and $p \geq r^{1/\theta}$. The index $r = 0$ never occurs ($b_0 = 1$) and $r = 1$ contributes at most one pair ($b_1 = 5$). For fixed $r \geq 2$ the primes counted are distinct primes $\geq r^{1/\theta}$ dividing b_r , so their product divides b_r and their number is at most

$$\frac{\log b_r}{(1/\theta) \log r} \leq \frac{\theta \kappa_0 r}{\log r}$$

by Lemma 526. Summing over $2 \leq r \leq x^\theta$ and using $\sum_{2 \leq r \leq R} r / \log r = (1 + o(1))R^2 / (2 \log R)$,

$$\sum_{2 \leq r \leq x^\theta} \frac{\theta \kappa_0 r}{\log r} = (1 + o(1)) \theta \kappa_0 \frac{x^{2\theta}}{2 \theta \log x} = \left(\frac{\kappa_0}{2} + o(1)\right) \frac{x^{2\theta}}{\log x}.$$

Doubling for the reflection completes the proof. $\square$

Remark 528. The census (all zero pairs (p, r) with $p \leq 2 \cdot 10^6$; verifier, Part A) gives $\sum_{p \leq x} Z(p) / \pi(x) = 0.9894$ at $x = 10^5$ and 0.9995 at $x = 10^6$ — the Poisson prediction 1 to three digits — while the edge slice at $\theta = \frac{1}{2}$ is only 112 resp. 298 incidences, in agreement with its natural scale $\asymp \sqrt{x}$ ($\sum_p 2p^{-1/2} \approx 4\sqrt{x} / \log x \approx 290$ at $x = 10^6$). Proposition 527 is generous by design; its point is that the bound is *unconditional* and already of Poisson scale.

Corollary 529 (Central localization of the first moment). *Put $Z^{\text{cent}}(p) = Z(p) - Z_{1/2}^{\text{edge}}(p) = \#\{r \in \mathcal{Z}_p : \sqrt{p} < r < p-1-\sqrt{p}\}$ and let*

$$M(x) = \#\{(r, p) : p \leq x \text{ prime}, 2r < p < r^2, p \mid b_r\}$$

be the medium-window incidence count. Then:

$$1. \sum_{p \leq x} Z(p) = \sum_{p \leq x} Z^{\text{cent}}(p) + O\left(\frac{x}{\log x}\right), \text{ and}$$

$$\sum_{p \leq x} Z^{\text{cent}}(p) = 2M(x) + O\left(\frac{x}{\log x}\right).$$

2. *For every function Ψ with $\Psi(x) \geq x / \log x$,*

$$\sum_{p \leq x} Z(p) \ll \Psi(x) \iff M(x) \ll \Psi(x).$$

In particular Hypothesis $\bar{Z}$, and each refinement $\sum_{p \leq x} Z(p) \ll \pi(x)(\log x)^{1-\delta}$ ($0 < \delta \leq 1$), is equivalent to the same bound for $M(x)$ alone.

3. *In Theorem 113 and in Remark 117, the bucket count S_N satisfies $S_N = S_N^{\text{cent}} + O(N / \log N)$ with $S_N^{\text{cent}} = \sum_{\sqrt{N} < p \leq 2N} Z^{\text{cent}}(p)$ unconditionally; Hypothesis $\bar{Z}$ is needed there only for central zeros.*

Proof. Normalize each central zero to its small half $r' = \min(r, p-1-r)$, so $\sqrt{p} < r' \leq (p-1)/2$, equivalently $2r' < p$ and $p < r'^2$; conversely every pair (r, p) with $2r < p < r^2$ and $p \mid b_r$ arises this way. The map is two-to-one except at the self-reflected center $r = (p-1)/2$ (at most one per prime), whence $Z^{\text{cent}}(p) = 2 \#\{r : \sqrt{p} < r \leq (p-1)/2, p \mid b_r\} - O(1)$ and (1) follows from Proposition 527 after summing; note also $M(x) \leq \sum_{p \leq x} Z(p)$ trivially. (2) and (3) are immediate from (1). $\square$

Remark 530 (The anti-diagonal gcd form). Writing $s = p - 1 - r$, a medium incidence is exactly a prime value $p = r + s + 1$ of the anti-diagonal dividing $\gcd(b_r, b_s)$ with $r < s < r^2 - r - 1$. Thus the entire vertical first-moment problem is a statement about large prime values of two-index Apéry gcds on anti-diagonals; the second divisibility $p \mid b_s$ is the reflection symmetry itself, so it carries no independent information — this is the precise sense in which the first moment is a *single-vanishing* (parity-type) problem.

Remark 531 (The second-moment wall from the Fourier side). Let $S_p(t) = \sum_{r < p} e_p(t b_r)$ and let $E(p) = \sum_a \#\{r : b_r \equiv a\}^2$ be the collision energy (unconditionally $E(p) = O(p^{7/4})$; trivially $E(p) \geq p$). Orthogonality and Cauchy–Schwarz give

$$Z(p) = 1 + \frac{1}{p} \sum_{t \neq 0} S_p(t) \leq 1 + \sqrt{E(p) - p}.$$

Hence even the *optimal* energy bound $E(p) = O(p^{1+o(1)})$ — itself open — would yield only $Z(p) \ll p^{1/2+o(1)}$ pointwise and $\sum_{p \leq x} Z(p) \ll x^{3/2+o(1)}$ on average. This meets, from the analytic side, exactly the ceiling at which the corank ladder terminates (Remark 521): every route through pair correlations or L^2 data saturates at average $Z(p) \ll p^{1/2}$. The target $\sum_{p \leq x} Z(p) \ll \pi(x)(\log x)^{1-\delta}$ is a genuine first-moment statement and cannot be reached by any second-moment input.

A benchmark: for the Lucas sequence $u_r = 2^r - 1$ one has $Z_M(p) := \#\{r < p : p \mid 2^r - 1\} = (p-1)/\text{ord}_p(2)$, and the same height counting (every p with $\text{ord}_p(2) \leq T$ divides $\prod_{n \leq T} (2^n - 1)$, of logarithm $\leq T^2 \log 2$) gives $\sum_{p \leq x} Z_M(p) \ll x^{3/2}/\log x$. We are not aware of an unconditional improvement of the exponent $3/2$ even in this fully multiplicative model, where the zero set is a subgroup coset structure — the strongest certificate structure conceivable. This calibrates the difficulty of the Apéry target, which enjoys strictly less structure.

13.13 The parity problem must be windowed

Proposition 532 (Open Problem 522 as stated is self-defeating). *Suppose nonzero integers $(\mathfrak{G}_r)_{r \geq 2}$ satisfy detection (D) of Open Problem 522. Then for every r ,*

$$P^+(b_r) \leq \max(r, |\mathfrak{G}_r|).$$

Consequently, under the height condition (H), every Apéry number would be $\exp(r^{o(1)})$ -smooth, even though $\log b_r \sim \kappa r$.

Proof. A prime $q \mid b_r$ with $q > r$ divides $\mathfrak{G}_r \neq 0$, hence $q \leq |\mathfrak{G}_r|$. $\square$

Remark 533 (Certified obstructions, and what the application actually needs). The smoothness conclusion contradicts the Poisson mass model underlying the whole program, and it is contradicted by every completed factorization: for all fully factored indices $5 \leq r \leq 40$ one finds $\log P^+(b_r) \geq 1.07r$ (e.g. $\log P^+(b_{40}) = 113.30 = 2.83 \times 40$), and for every $50 \leq r \leq 300$ at least 94.9% of the logarithmic mass of b_r lies on primes exceeding 10^6 (verifier, Part B). Each certified large factor is an unconditional lower bound on any family satisfying (D): e.g. $\log |\mathfrak{G}_{40}| \geq 113.30$. Thus Open

Problem 522 almost certainly has a negative answer — but for a reason *irrelevant* to the average- Z application. The offending primes are the giant factors $p > r^2$, which Proposition 527 already controls at Poisson scale by trivial height counting; by Corollary 529, detection is needed *only* in the medium window $2r < p < r^2$. The problem must therefore be re-windowed, and — by the localization loss computed in Remark 537 below — localized by dyadic scale as well.

Definition 534 (Scale-windowed detector family). A *scale-windowed detector family* is a doubly indexed family of nonzero integers $\mathfrak{G}_{r,k}$ ($r \geq 2$, $k \geq 1$), each determined by (r, k) alone, such that, with $X_k = 2^k$:

(D_w) for every prime $p \in (X_k, 2X_k]$ with $2r < p < r^2$: $p \mid b_r \implies p \mid \mathfrak{G}_{r,k}$.

The family has *average height exponent* $\delta \in (0, 1]$ if

$$(H_w) \quad \sum_{\sqrt{X_k} < r \leq X_k} \log |\mathfrak{G}_{r,k}| \ll X_k (\log X_k)^{1-\delta} \quad (k \geq 1).$$

Theorem 535 (Windowed parity reduction). *If a scale-windowed detector family with average height exponent δ exists, then unconditionally*

$$\sum_{p \leq x} Z(p) \ll \frac{x}{(\log x)^\delta} = \pi(x) (\log x)^{1-\delta} (1 + o(1)),$$

and consequently, by the bucket argument of Theorem 113 (display (132), as quantified in Remark 117),

$$\#\{n \leq X : \log G_n > \varepsilon n\} = O_\varepsilon((\log X)^{2-\delta}).$$

Proof. Fix a dyadic scale $X_k \leq x$. If (r, p) is a medium incidence with $p \in (X_k, 2X_k]$, then $r < p/2 \leq X_k$ and $r > \sqrt{p} > \sqrt{X_k}$, so by (D_w) and the divisor bound for the nonzero integer $\mathfrak{G}_{r,k}$,

$$\#\{(r, p) : p \in (X_k, 2X_k], 2r < p < r^2, p \mid b_r\} \leq \sum_{\sqrt{X_k} < r \leq X_k} \frac{\log |\mathfrak{G}_{r,k}|}{\log X_k} \ll \frac{X_k}{(\log X_k)^\delta}.$$

Summing over $k \leq \log_2 x$, the right side is a geometric-type sum dominated by its top scales, giving $M(x) \ll x(\log x)^{-\delta}$. Corollary 529(2) upgrades this to $\sum_{p \leq x} Z(p) \ll x(\log x)^{-\delta}$ (note $x/\log x \leq x(\log x)^{-\delta}$ since $\delta \leq 1$). Feeding $S_N \ll N(\log N)^{-\delta}$ into (132) exactly as in Remark 117 gives $O_\varepsilon((\log N)^{1-\delta})$ exceptional integers per dyadic block and $O_\varepsilon((\log X)^{2-\delta})$ in total. $\square$

Remark 536 (Truth check for (H_w)). The *actual* medium radicals $\mathfrak{G}_{r,k}^{\text{true}} = \prod_{p \in (X_k, 2X_k], 2r < p < r^2, p \mid b_r} p$ satisfy (D_w) by construction, and the census ($p \leq 2 \cdot 10^6$) gives

$$\frac{1}{X_k} \sum_{\sqrt{X_k} < r \leq X_k} \log \mathfrak{G}_{r,k}^{\text{true}} \in [0.42, 0.53] \quad (2^{10} \leq X_k \leq 2^{19}),$$

with no growth trend (verifier, Part C). Thus (H_w) is consistent with the computed truth for *every* $\delta < 1$ simultaneously; the entire difficulty of Theorem 535's hypothesis is to make some such family *certifiable* — i.e. to give an a priori height bound not passing through knowledge of the zeros themselves. Per index, the medium prime count $\omega_{\text{med}}(r) = \#\{p : 2r < p < r^2, p \mid b_r\}$ has census mean 0.5895 over $2 \leq r \leq 1414$ against the Poisson–Mertens prediction $\log 2 = 0.6931$ (mildly sub-Poisson, consistent with the repulsion phenomena recorded elsewhere in this paper), with 44.2% of indices carrying at least one medium prime and $\max \omega_{\text{med}} = 4$.

Remark 537 (Single-integer families cannot reach any $\delta > 0$). Scale localization in Definition 534 is necessary, not cosmetic. Suppose $\mathfrak{G}_{r,k} = \mathfrak{G}_r$ is independent of k (a single integer per index, as in Open Problem 522). Whenever b_r has at least one medium prime, (D_w) forces $\log |\mathfrak{G}_r| \geq \log 2r$; if a proportion $\geq c > 0$ of indices has $\omega_{\text{med}}(r) \geq 1$ (census: $c = 0.44$ through $r = 1414$), then $\sum_{r \leq y} \log |\mathfrak{G}_r| \gg y \log y$, and the dyadic estimate in the proof of Theorem 535 returns only $O(X_k)$ per scale — i.e. average $Z(p) \ll \log x$, exactly the trivial-target boundary $\delta = 0$. The mechanism is that a single integer of height $\asymp \log r$ cannot be prevented from concentrating its mass in one dyadic window, while the true medium mass of b_r is spread over all $\asymp \log \log r$ scales with $O(1)$ mass each. This is one more instance of the one-logarithm frontier of Proposition 116: per-scale information is exactly what vertical certificates must supply.

Remark 538 (Status). The unconditional average bound $\sum_{p \leq x} Z(p) \ll \pi(x)(\log x)^{1-\delta}$ remains open; within this paper's technology the unconditional record stays $O(x^{8/5}/\log x)$ and the conditional corank ceiling stays $x^{3/2+o(1)}$. What the present subsection settles is the exact shape of the remaining problem: by Proposition 527 and Corollary 529 it is precisely the medium-window count $M(x)$, a single-vanishing detection problem (Remark 530) that no second-moment input can resolve (Remark 531); by Proposition 532 the original parity formulation is self-defeating and Definition 534 is its corrected form, with Theorem 535 converting any average height exponent δ directly into the exceptional-set bound $O_\varepsilon((\log X)^{2-\delta})$. The one live mechanism known to us for producing certifiable scale-windowed detectors is the moving-character Mellin independence program (a horizontal trace-function large sieve for the Apéry Mellin compatible system); any such theorem giving per-scale detector mass $O(X_k(\log X_k)^{1-\delta})$ would, through the present reduction, immediately beat the $(\log X)^2$ exceptional-set bound unconditionally.

13.14 A short-return carrier and an isolated five-star criterion

The full hypothesis $(\text{HM})_7$ is stronger than necessary. The gap polynomials already control, pointwise, every marked prime whose zero has a nearby nonreflection return. After removing those primes, elementary packing reduces the density enough that a fifth-order estimate is sufficient.

For $p \in \mathcal{P}_X$ put

$$\sigma_p(r) = \min(r, p-1-r), \quad \mathcal{Y}_p = \sigma_p(\mathcal{Z}_p) \subseteq \left[0, \frac{p-1}{2}\right].$$

Given an integer $2 \leq H < X$, call $s \in \mathcal{Y}_p$ *H-isolated* if $|s-t| > H$ for every $t \in \mathcal{Y}_p \setminus \{s\}$. Let

$$\mathcal{I}_p(H) = \{r \in \mathcal{Z}_p : \sigma_p(r) \text{ is } H\text{-isolated}\},$$

and define

$$\begin{aligned} K_X^{\text{iso}}(m; H) &= \sum_{p \in \mathcal{P}_X} \mathbf{1}_{\mathcal{I}_p(H)}(m \bmod p), \\ K_X^{\text{near}}(m; H) &= K_X(m) - K_X^{\text{iso}}(m; H), \\ \Lambda_X^{\text{iso}}(H) &= \sum_{p \in \mathcal{P}_X} \frac{|\mathcal{I}_p(H)|}{p}. \end{aligned}$$

Lemma 539 (Pointwise short-return carrier). *Uniformly for $0 \leq m < X^2$ and $2 \leq H < X$,*

$$K_X^{\text{near}}(m; H) \ll H^2. \tag{459}$$

Moreover

$$\Lambda_X^{\text{iso}}(H) \ll \min \left\{ \lambda_X, \frac{1}{\log X} + \frac{X}{H \log X} \right\}. \tag{460}$$

Proof. Let p contribute to $K_X^{\text{near}}(m; H)$, put $r = m \bmod p$ and $s = \sigma_p(r)$, and choose a distinct $t \in \mathcal{Y}_p$ with $h = |t - s| \leq H$. Reflection of the Apéry solution shows that $s, t \in \mathcal{Z}_p$. Lemma 10 gives $h \geq 2$, and the gap certificate gives $p \mid N_h(\min(s, t))$.

There are four orientations. If $r = s$ and $t = s + h$, then $p \mid N_h(m)$; if $r = s$ and $t = s - h$, then $p \mid N_h(m - h)$. If $r = p - 1 - s$ and $t = s + h$, then

$$N_h(s) \equiv N_h(-m - 1) = (-1)^{h-1} N_h(m - h) \pmod{p},$$

where $m \geq h$ because $h \leq (p - 1)/2 - s \leq p - 1 - s \leq m$. Finally, if $r = p - 1 - s$ and $t = s - h$, then

$$N_h(s - h) \equiv N_h(-m - h - 1) = (-1)^{h-1} N_h(m) \pmod{p}.$$

Here we used the symbolic reflection identity (359).

Consequently every such prime divides the nonzero integer

$$\mathcal{C}_{m,H} = \prod_{h=2}^H N_h(m) \prod_{h=2}^{\min(H,m)} N_h(m - h).$$

Positivity is Lemma 412. The gap recurrence and positivity give, for $n \geq 0$,

$$0 < N_h(n) \leq 118^{h-1} (n + h + 1)^{3(h-1)}.$$

Since $m < X^2$ and $H < X$, it follows that $\log \mathcal{C}_{m,H} \ll H^2 \log X$. Every prime divisor counted here exceeds X , proving (459).

The H -isolated points of $\mathcal{Y}_p$ are pairwise more than H apart in an interval of length at most $p/2$. Hence

$$|\mathcal{I}_p(H)| \leq 2|\mathcal{Y}_p^{\text{iso}}(H)| \leq 2 + \frac{p}{H}.$$

Summing over $X < p \leq 2X$, and using the standard dyadic prime estimates, gives (460); the bound by λ_X is immediate. $\square$

For one fixed $0 < \eta < 1/15$, set

$$H_X = \max \left\{ 2, \left\lfloor X^{7/15} (\log X)^{-1/3} \right\rfloor \right\}. \quad (461)$$

Hypothesis 540 (Isolated five-star high-load tail). *Uniformly over dyadic X ,*

$$\sum_{\substack{X \leq m < X^2 \\ K_X^{\text{iso}}(m; H_X) \geq X^{14/15+\eta}}} (K_X^{\text{iso}}(m; H_X))_5 \ll X^{o(1)} \left(1 + X^2 (\Lambda_X^{\text{iso}}(H_X))^5 \right). \quad (462)$$

The lower endpoint $m \geq X$ is the exact physical range needed by the final consumer. Indeed, a dyadic block not already absorbed by the $p \leq 2\sqrt{n}$ estimate satisfies $\sqrt{n} < X \leq n$, and its physical column is $m = n < X^2$. Thus no estimate at $m < X$ is used in Theorem 566 below. The present full-shell hypothesis still retains primes $p > m$ inside each such column; the physical truncation introduced after Proposition 549 removes that unnecessary strengthening.

The additive 1 in (462) is harmless but unnecessary. Indeed, a nonempty tail forces $\Lambda_X^{\text{iso}}(H_X) \geq L/(2X)$, as proved below, whereas for an empty tail the left side is zero. We retain the 1 only to keep the hypothesis in the same uniform form as the earlier moment statements. No lower bound for λ_X or $\Lambda_X^{\text{iso}}(H_X)$ is assumed.

The remaining hypothesis has an equivalent form in which all selector-created terms of ranks below four disappear. We record the reduction because it identifies the precise cross-prime estimate still needed. Put

$$M = X^2, \quad L = \left\lceil X^{14/15+\eta} \right\rceil, \quad \Lambda = \Lambda_X^{\text{iso}}(H_X).$$

For $p \in \mathcal{P}_X$, $r \in \mathcal{I}_p(H_X)$, write

$$N_{p,r} = \left\lfloor \frac{M-1-r}{p} \right\rfloor + 1, \quad m = ap + r \quad (0 \leq a < N_{p,r}).$$

For $q \neq p$ define

$$\begin{aligned} f_{q;p,r}(a) &= \mathbf{1}_{\mathcal{I}_q(H_X)}((ap+r) \bmod q), \\ \mu_{q;p,r} &= \frac{1}{N_{p,r}} \sum_{0 \leq u < N_{p,r}} f_{q;p,r}(u), \\ g_{q;p,r}(a) &= f_{q;p,r}(a) - \mu_{q;p,r}, \\ B_{p,r}(a) &= \sum_{q \neq p} f_{q;p,r}(a), \quad w_{p,r}(a) = \mathbf{1}_{X \leq ap+r < M} \mathbf{1}_{1+B_{p,r}(a) \geq L}. \end{aligned}$$

Here and below e_4 denotes the fourth elementary symmetric polynomial. Set

$$\mathcal{T}_5(X) = \sum_{\substack{X \leq m < M \\ K_X^{\text{iso}}(m; H_X) \geq L}} (K_X^{\text{iso}}(m; H_X))_5, \quad (\text{T5})$$

$$\mathcal{U}_4(X) = 24 \sum_{\substack{p \in \mathcal{P}_X \\ r \in \mathcal{I}_p(H_X)}} \sum_{0 \leq a < N_{p,r}} w_{p,r}(a) e_4((g_{q;p,r}(a))_{q \neq p}). \quad (\text{TR4})$$

Proposition 541 (Tail rank-four reduction). *Let $0 < \eta < 1/15$ be fixed and let H_X, L, Λ be as above. Then for all sufficiently large X ,*

$$|\mathcal{T}_5(X) - \mathcal{U}_4(X)| \leq \delta_X \mathcal{T}_5(X), \quad \delta_X = \frac{128c_X \Lambda}{L-1} = o(1), \quad c_X = \frac{4X}{X-2}. \quad (463)$$

Consequently Hypothesis 540 is equivalent to

$$\mathcal{U}_4(X) \ll X^{o(1)} (1 + M\Lambda^5). \quad (464)$$

If the tail is nonempty, then

$$\Lambda \geq \frac{L}{2X}, \quad M\Lambda^5 \geq \frac{L^5}{32X^3}; \quad (465)$$

in particular the additive 1 may be omitted from either equivalent estimate.

There is also an endpoint-safe arithmetic pruning of (464). For a prime q and a physical integer m , put

$$r_q(m) = m \bmod q, \quad q\text{quad}\mathfrak{o}_q(m) = \frac{q-1}{\gcd(r_q(m), q-1)}.$$

For all sufficiently large X , let

$$D_X = \left\lfloor \frac{X^{7/15+\eta/2}}{\log X} \right\rfloor.$$

In (TR4), retain only output charts with $\mathfrak{o}_p(r) > D_X$, replace every partner function by

$$f_{q;p,r}^{>D_X}(a) = f_{q;p,r}(a) \mathbf{1}_{\mathfrak{o}_q(ap+r) > D_X},$$

centre the new functions by their own chart means, and keep the original selector $w_{p,r}(a)$. Denote the resulting rank-four sum by $\mathcal{U}_{4,>D_X}(X)$. Then

$$|\mathcal{T}_5(X) - \mathcal{U}_{4,>D_X}(X)| \ll \left(\frac{1}{(\log X)^2} + \delta_X \right) \mathcal{T}_5(X). \quad (466)$$

Thus (464) may equivalently be required only of the sum in which all five participating characteristics have Kummer order greater than D_X .

Proof. At a physical point of isolated load k , summing over its k choices of distinguished output gives

$$\sum_{\substack{p \in \mathcal{P}_X \\ m \bmod p \in \mathcal{I}_p(H_X)}} (k-1)_4 = (k)_5.$$

This proves the exact chart identity

$$\mathcal{T}_5(X) = \sum_{\substack{p \in \mathcal{P}_X \\ r \in \mathcal{I}_p(H_X)}} \sum_{0 \leq a < N_{p,r}} w_{p,r}(a) (B_{p,r}(a))_4. \quad (*)$$

We first prove a pointwise Boolean estimate. Let $f_i \in \{0, 1\}$, $0 \leq \mu_i \leq 1$, $g_i = f_i - \mu_i$, and put

$$B = \sum_i f_i, \quad \rho = \sum_i \mu_i, \quad F = (B)_4 = 24e_4(f), \quad E = 24e_4(g).$$

If $G = \sum_i |g_i|$, expansion of the terms containing at least one μ_i , followed by allowing repeated indices, gives

$$|F - E| \leq (G + \rho)^4 - G^4 \leq 4\rho(G + \rho)^3 \leq 4\rho(B + 2\rho)^3. \quad (**)$$

If $B \geq 6$ and $2\rho \leq B$, then $(B)_4 \geq B^4/4$, and hence

$$|F - E| \leq 128 \frac{\rho}{B} F. \quad (467)$$

In a selected chart, $B \geq L - 1$. Since $N_{p,r} \leq X < q$, each root of the q -row supplies at most one chart coordinate. Moreover $N_{p,r} \geq (X - 2)/2$, so

$$\rho \leq \frac{\sum_q |\mathcal{I}_q(H_X)|}{N_{p,r}} \leq c_X \Lambda.$$

The packing estimate (460) and the choice of H_X give $c_X \Lambda = o(L)$. Thus (467) applies uniformly on the selected tail. Summing it in $(*)$ proves (463). In particular, once $\delta_X < 1/2$,

$$(1 - \delta_X) \mathcal{T}_5(X) \leq \mathcal{U}_4(X) \leq (1 + \delta_X) \mathcal{T}_5(X),$$

which proves the stated equivalence. Notice that this is a relative comparison: before either equivalent estimate is known, it does not turn the error into an absolute bound at the scale $1 + M\Lambda^5$.

For completeness, pointwise

$$K_X^{\text{iso}}(m; H_X) \leq \sum_p |\mathcal{I}_p(H_X)| \leq 2X \sum_p \frac{|\mathcal{I}_p(H_X)|}{p} = 2X \Lambda.$$

This proves (465). If the tail is empty, both sides of the desired moment estimate vanish; if it is nonempty, the second quantity in (465) tends to infinity. Thus the versions with and without the additive 1 are equivalent.

It remains to prove the high-order pruning. Fix m, q , abbreviate $r = r_q(m)$ and $g = \gcd(r, q-1)$, and write

$$d = \frac{q-1}{g}, \quad u = \frac{r}{g}, \quad m = tq + r.$$

Then $(u, d) = 1$ and the exact integer identity

$$dm + u = q(dt + u) \tag{468}$$

holds. The two cases $d = 1$ are $r = 0$ and $r = q - 1$; neither is an Apéry zero modulo q , because $b_0 = 1$ and $b_{q-1} \equiv 1 \pmod{q^2}$. For the latter congruence, every term with $1 \leq k \leq q - 1$ in the defining binomial sum contains $\binom{q-1+k}{k}^2 \equiv 0 \pmod{q^2}$. Therefore every Apéry hit prime with $\mathfrak{o}_q(m) \leq D$ divides the positive integer

$$\mathcal{C}_D(m) = \prod_{2 \leq d \leq D} \prod_{\substack{1 \leq u < d \\ (u, d) = 1}} (dm + u).$$

For $D \leq D_X$ and all sufficiently large X , every factor in this product satisfies

$$0 < dm + u < D(M + 1) < X^{5/2}.$$

It therefore has at most two prime divisors exceeding X . Since there are $\sum_{2 \leq d \leq D} \varphi(d)$ factors, the number of relevant hit primes is at most

$$2 \sum_{2 \leq d \leq D} \varphi(d) \leq D(D - 1) < D^2. \tag{469}$$

With the displayed choice of D_X this is

$$O(D_X^2) = O\left(\frac{L}{(\log X)^2}\right) = o(L).$$

Put

$$A_D = \sup_{0 \leq m < M} \#\{q \in \mathcal{P}_X : q \mid b_{m \bmod q}, \mathfrak{o}_q(m) \leq D\}.$$

The carrier estimate just proved gives $A_{D_X} = O(D_X^2) = O(L(\log X)^{-2})$. On the original tail, write $k = K_X^{\text{iso}}(m; H_X)$ and let $\ell \leq A_{D_X}$ be the number of its rows removed by the order cutoff. Ordered five-tuples containing a removed row are at most $5\ell k^4$, whereas $(k)_5 \gg k^5$ uniformly for $k \geq L$. Hence replacing $(k)_5$ by the fifth falling factorial of the retained load changes the sum by a relative $O((\log X)^{-2})$. Apply (467) once more to the retained functions. For each retained output row their partner load is at least $L - 1 - A_{D_X}$, while their total chart mean does not exceed the old ρ . Therefore their relative centering error is at most

$$\frac{128c_X\Lambda}{L - 1 - A_{D_X}} = O(\delta_X).$$

Combining this with the deleted-tuple estimate proves (466). The selector throughout remains the original condition $k \geq L$. $\square$

The empirical chart centring in (TR4) is not intrinsic. It may be replaced directly by the global periodic means. Put

$$z_q = \frac{|\mathcal{I}_q(H_X)|}{q}, \quad h_{q;p,r}(a) = f_{q;p,r}(a) - z_q, \quad \delta_{q;p,r} = \mu_{q;p,r} - z_q,$$

so that $g_{q;p,r} = h_{q;p,r} - \delta_{q;p,r}$. In the ordered expansion of $24e_4(g)$, let $\mathcal{U}_T$ denote the contribution in which the positions in $T \subseteq \{1, 2, 3, 4\}$ supply h and the other positions supply $-\delta$. Thus

$$\mathcal{U}_4(X) = \sum_{T \subseteq \{1, 2, 3, 4\}} \mathcal{U}_T(X).$$

Define the pure global-centred four-wave sum

$$\mathcal{V}_4(X) = 24 \sum_{\substack{p \in \mathcal{P}_X \\ r \in \mathcal{I}_p(H_X)}} \sum_{0 \leq a < N_{p,r}} w_{p,r}(a) e_4((h_{q;p,r}(a))_{q \neq p}). \quad (470)$$

Proposition 542 (Selector-preserving global-centre reduction). *For the physical range and original selector above, and all sufficiently large X ,*

$$|\mathcal{T}_5(X) - \mathcal{V}_4(X)| \leq \frac{128\Lambda}{L-1} \mathcal{T}_5(X) = o(\mathcal{T}_5(X)). \quad (471)$$

Consequently Hypothesis 540, (464), and the single pure four-wave estimate

$$|\mathcal{V}_4(X)| \ll X^{o(1)} (1 + M\Lambda^5) \quad (472)$$

are equivalent. Moreover the empirical-centre sectors satisfy the relative family

$$\sum_{|T|=t} |\mathcal{U}_T(X)| \ll_t \left(\frac{\Lambda}{L}\right)^{4-t} \mathcal{T}_5(X) \quad (0 \leq t \leq 3). \quad (473)$$

In particular the two-wave and three-wave sectors are relative errors, not independent arithmetic premises.

The zero-wave and one-wave sectors also obey the stronger absolute bounds

$$|\mathcal{U}_\emptyset(X)| \ll \frac{M\Lambda^6}{L}, \quad \sum_{|T|=1} |\mathcal{U}_T(X)| \ll M\Lambda^5. \quad (474)$$

Finally, let $\mathcal{V}_{4,>D_X}$ be defined as in (470), but retain only output and partner rows with Kummer order greater than D_X , centre each retained partner row by

$$z_q^{>D_X} = q^{-1} \# \{s \in \mathcal{I}_q(H_X) : \mathfrak{o}_q(s) > D_X\},$$

and keep the original selector $w_{p,r}(a)$. Then

$$|\mathcal{T}_5(X) - \mathcal{V}_{4,>D_X}(X)| \ll \left(\frac{1}{(\log X)^2} + \frac{\Lambda}{L}\right) \mathcal{T}_5(X). \quad (475)$$

Thus (472) is equivalent to its all-five-rows-high-Kummer version without recomputing the selector.

Proof. Fix a selected chart point and put $B = \sum_{q \neq p} f_{q;p,r}(a)$. Apply (467) with the global means z_q . Here $B \geq L - 1$ and $\sum_{q \neq p} z_q \leq \Lambda = o(L)$, so

$$|(B)_4 - 24e_4((h_{q;p,r}(a))_{q \neq p})| \leq \frac{128\Lambda}{L-1}(B)_4.$$

Summing over the original selector and using the exact directed-chart identity (*) proves (471). Once the relative error is less than one half, the asserted equivalence follows exactly as in the proof of Proposition 541.

Because $N_{p,r} < q$, every root in the q -row occurs at most once on a fixed chart. The same estimate used in the proof of Proposition 541 gives

$$\sum_{q \neq p} |\delta_{q;p,r}| \leq \sum_{q \neq p} (\mu_{q;p,r} + z_q) \ll \Lambda. \quad (\text{LW1})$$

At $m = ap + r$,

$$\sum_{q \neq p} |h_{q;p,r}(a)| \leq B_{p,r}(a) + \Lambda = K_X^{\text{iso}}(m; H_X) - 1 + \Lambda. \quad (\text{LW2})$$

On the selected tail $K_X^{\text{iso}}(m; H_X) \geq L$, while $\Lambda = o(L)$ by (460) and (461). Dropping partner distinctness in the absolute majorant, then using the fact that a physical point lies in exactly one base chart for each of its marked primes, gives for $0 \leq t \leq 3$

$$\sum_{|T|=t} |\mathcal{U}_T(X)| \ll_t \Lambda^{4-t} \sum_{\substack{X \leq m < M \\ K_X^{\text{iso}}(m; H_X) \geq L}} K_X^{\text{iso}}(m; H_X)^{t+1}. \quad (\text{LW3})$$

The universal two-row CRT count, applied to the isolated sub-bank, is

$$\sum_{m < M} (K_X^{\text{iso}}(m; H_X))_2 \leq \left(\sum_p |\mathcal{I}_p(H_X)| \right)^2 \leq 4M\Lambda^2. \quad (\text{LW4})$$

On the tail, $K \leq (K)_2/(L-1)$ and $K^2 \leq 2(K)_2$. Substitution in (LW3)–(LW4) proves both estimates in (474). Restricting to $m \geq X$ only deletes terms from these absolute majorants.

For the relative family, if $0 \leq t \leq 3$ and $K \geq L \geq 8$, then

$$(K)_5 \geq \frac{K^5}{16}, \quad K^{t+1} \leq \frac{16}{L^{4-t}}(K)_5.$$

Inserting this in (LW3) proves (473).

For the final assertion, the carrier estimate (469) deletes at most $A_{D_X} = O(L(\log X)^{-2})$ marked rows at any physical point. The deleted ordered five-tuples have relative mass $O(A_{D_X}/L)$. On every retained selected chart, the retained partner load is at least $L - 1 - A_{D_X}$ and the sum of the global retained means is at most Λ . Applying (467) to these rows gives the additional relative error $O(\Lambda/(L - 1 - A_{D_X})) = O(\Lambda/L)$. This proves (475). $\square$

The distinguished output in (470) can also be removed. For a physical integer m , put

$$f_p(m) = \mathbf{1}_{\mathcal{I}_p(H_X)}(m \bmod p), \quad h_p(m) = f_p(m) - z_p,$$

and define the completely symmetric global-centred five-wave

$$\mathcal{W}_5(X) = 120 \sum_{\substack{X \leq m < M \\ K_X^{\text{iso}}(m; H_X) \geq L}} e_5((h_p(m))_{p \in \mathcal{P}_X}). \quad (476)$$

Proposition 543 (Distinguished-output symmetrization). *For all sufficiently large X ,*

$$|\mathcal{T}_5(X) - \mathcal{W}_5(X)| \leq \frac{1280\Lambda}{L} \mathcal{T}_5(X) = o(\mathcal{T}_5(X)). \quad (477)$$

Consequently Hypothesis 540 is equivalent to the single symmetric estimate

$$|\mathcal{W}_5(X)| \ll X^{o(1)} (1 + M\Lambda^5). \quad (478)$$

Moreover the relation to the directed four-wave is exact:

$$\mathcal{V}_4(X) = \mathcal{W}_5(X) + 24 \sum_{\substack{X \leq m < M \\ K_X^{\text{iso}}(m; H_X) \geq L}} \sum_{p \in \mathcal{P}_X} z_p e_4((h_q(m))_{q \neq p}). \quad (479)$$

The second term in (479) is $O((\Lambda/L)\mathcal{T}_5(X))$. Hence the high-Kummer and far quotient-rainbow versions already proved for $\mathcal{V}_4$ may equivalently be stated as selector-preserving symmetric five-wave estimates; neither mask recomputes the selector.

Proof. Fix a selected physical integer, abbreviate $f_i = f_{p_i}(m)$, $h_i = f_i - z_i$, $K = \sum_i f_i$, and $\rho = \sum_i z_i = \Lambda$, and put

$$F = 120e_5(f) = (K)_5, \quad E = 120e_5(h), \quad G = \sum_i |h_i|.$$

Since $f_i \in \{0, 1\}$, one has $G \leq K + \rho$. Expanding $F - E$, taking absolute values, and then dropping the distinctness restrictions gives

$$|F - E| \leq (G + \rho)^5 - G^5 \leq 5\rho(G + \rho)^4 \leq 5\rho(K + 2\rho)^4. \quad (\text{W1})$$

On the selected tail, $K \geq L$, while $\Lambda = o(L)$. Thus eventually $2\rho \leq K$. For $K \geq L \geq 8$,

$$(K)_5 \geq \frac{K^5}{16}, \quad K + 2\rho \leq 2K.$$

Substitution in (W1) yields

$$|F - E| \leq 1280 \frac{\Lambda}{K} F \leq 1280 \frac{\Lambda}{L} F.$$

Summing proves (477). If the selected tail is empty, both sides vanish. Otherwise the relative error is eventually less than one, which proves the equivalence with (478).

Finally,

$$\sum_p h_p e_4(h_{\widehat{p}}) = 5e_5(h), \quad f_p = h_p + z_p.$$

Expanding the directed output sum in (470) gives (479). Pointwise,

$$24 \left| \sum_p z_p e_4(h_{\widehat{p}}) \right| \leq \Lambda \left(\sum_q |h_q| \right)^4 \leq \Lambda(K + \Lambda)^4 \ll \frac{\Lambda}{K} (K)_5.$$

Summation gives the asserted remainder bound. Combining this comparison with (475) and (482) proves the final mask statement. $\square$

There is also a deterministic pruning by the ordinary quotient $\lfloor m/p \rfloor$. For a fixed physical column put

$$n_a(m) = \# \left\{ p \in \mathcal{P}_X : m \bmod p \in \mathcal{I}_p(H_X), \left\lfloor \frac{m}{p} \right\rfloor = a \right\}.$$

Proposition 544 (Far-region quotient rainbow). *Let $m > 2X$, put $k = K_X^{\text{iso}}(m; H_X) \geq 5$, and let $\mathcal{R}_5(m)$ be the number of ordered distinct marked five-tuples in which two primes have the same ordinary quotient. Then*

$$\mathcal{R}_5(m) \leq \frac{40X^2}{(m - 2X)(k - 1)} (k)_5. \quad (480)$$

Consequently, on the selected tail and in the range

$$m \geq 2X + \frac{40X^2(\log X)^2}{L - 1}, \quad (481)$$

all but a relative $O((\log X)^{-2})$ of the fifth factorial consists of tuples whose five ordinary quotients are pairwise distinct. This may be combined with (466) while keeping the original selector.

Proof. If quotient a is occupied, its primes lie in $(m/(a + 1), m/a]$. Since $p \leq 2X$,

$$a > \frac{m - 2X}{2X}, \quad a + 1 > \frac{m}{2X}.$$

The interval length therefore gives

$$(n_a(m))_2 \leq \frac{4X^2}{m - 2X} n_a(m). \quad (\text{QR1})$$

For each of the ten pairs of tuple positions, choose the equal-quotient pair using (QR1) and the remaining three entries freely. The union bound is

$$10 \frac{4X^2}{m - 2X} k(k - 2)_3 = \frac{40X^2}{(m - 2X)(k - 1)} (k)_5,$$

which proves (480). The last assertion uses $k - 1 \geq L - 1$. $\square$

The quotient mask is compatible with global centring. Let $\mathcal{T}_5^{\text{far}}$ be the part of (T5) supported on the range (481), and define $\mathcal{V}_{4, > D_X}^{\text{rbw, far}}$ by expanding (470) as an ordered sum, retaining the original selector, the high-Kummer rows from (475), the far range, and only labelled five-tuples whose five ordinary quotients are pairwise distinct. Then

$$\left| \mathcal{T}_5^{\text{far}} - \mathcal{V}_{4, > D_X}^{\text{rbw, far}} \right| \ll \left(\frac{1}{(\log X)^2} + \frac{\Lambda}{L} \right) \mathcal{T}_5^{\text{far}}. \quad (482)$$

Indeed, Proposition 544 removes repeated quotients, the low-Kummer carrier removes tuples containing a deleted row, and expansion of the remaining products $\prod_{i=1}^4 (f_{q_i}^{> D_X} - z_{q_i}^{> D_X})$ bounds the centring error by $O((\Lambda/L)(K)_5)$ pointwise. Thus the far arithmetic premise is a pure four-wave, high-Kummer, quotient-rainbow correlation.

The far fibre admits a further selector-free localization to one-label supports. Fix the same $0 < \eta < 1/15$. For sufficiently large dyadic X put

$$\begin{aligned} M &= X^2, & L &= \lceil X^{14/15+\eta} \rceil, \\ D_X &= \left\lfloor \frac{X^{7/15+\eta/2}}{\log X} \right\rfloor, & R_X &= \lfloor \log X \rfloor, \\ G_X &= D_X^2, & Y_X &= 2X + \left\lceil \frac{40X^2(\log X)^2}{L-1} \right\rceil, \\ \vartheta_X &= \frac{\log \log X}{\log X}, & U_X &= 2X - G_X - 1, \\ B_X &= (1 - \vartheta_X)(L-3)(L-4). \end{aligned} \tag{FG0}$$

Thus Y_X is the integral lower endpoint of (481). For $Y_X \leq m < M$ and $p \in \mathcal{P}_X$ write

$$\begin{aligned} a_p(m) &= \left\lfloor \frac{m}{p} \right\rfloor, & q_{\text{quad}r_p}(m) &= m - a_p(m)p, \\ h_p(m) &= \sigma_p(r_p(m)), & \epsilon_p(m) &= \mathbf{1}_{r_p(m) > h_p(m)}, \\ s_p(m) &= r_p(m) - \epsilon_p(m)p, & c_p(m) &= a_p(m) + \epsilon_p(m). \end{aligned} \tag{FG1}$$

Then $m = c_p(m)p + s_p(m)$ and $-X \leq s_p(m) \leq X-1$. Retain exactly the previous high-Kummer rows by setting

$$\begin{aligned} F_p^D(m) &= \mathbf{1}_{\mathcal{I}_p(H_X)}(r_p(m)) \mathbf{1}_{(p-1)/\gcd(r_p(m), p-1) > D_X}, \\ k_D(m) &= \sum_{p \in \mathcal{P}_X} F_p^D(m). \end{aligned} \tag{FG2}$$

No selector is recomputed in (FG2): the far fifth tail still uses the original condition $K_X^{\text{iso}}(m; H_X) \geq L$.

Call two retained labels $p \neq q$ mutually admissible at m if

$$\begin{aligned} |a_p(m) - a_q(m)| &> R_X, & |s_p(m) - s_q(m)| &> G_X, \\ |h_p(m) - h_q(m)| &> G_X. \end{aligned} \tag{FG3}$$

For $G_X < g \leq 2X-1$, let $V_g(m)$ be the number of retained labels p for which there is a retained label q mutually admissible with p and satisfying $|s_p(m) - s_q(m)| = g$. This support uses the actual Apéry rows and all three endpoint masks in (FG3); it drops only compatibility with a chosen prefix triple.

Hypothesis 545 (Far fixed-gap one-label support). *For one fixed $0 < \eta < 1/15$, there are functions $\kappa_X \geq 1$ and $\omega_X \geq 0$, defined on all sufficiently large dyadic integers, such that*

$$\frac{\log \kappa_X}{\log X} \longrightarrow 0, \quad \omega_X \longrightarrow 0,$$

and, for every such X ,

$$\sum_{Y_X \leq m < M} (k_D(m))_3 \sum_{g=G_X+1}^{2X-1} \left(V_g(m) - \frac{B_X + \kappa_X}{2U_X} \right)_+ \leq X^{\omega_X} (1 + M\Lambda^5). \tag{GH1}$$

Proposition 546 (Fixed-gap support implies the far fifth tail). *Hypothesis 545 implies*

$$\mathcal{T}_5^{\text{far}} \ll X^{o(1)}(1 + M\Lambda^5). \quad (483)$$

Here $\mathcal{T}_5^{\text{far}}$ retains the original selector, physical range, and isolated rows; the high-Kummer and spacing conditions are only deterministic prunings inside its ordered fifth factorial.

Proof. Fix an originally selected $m \geq Y_X$ and put

$$K = K_X^{\text{iso}}(m; H_X) \geq L.$$

By (469), deleting rows of Kummer order at most D_X removes $O(D_X^2) = O(L(\log X)^{-2})$ rows and hence a relative $O((\log X)^{-2})$ of $(K)_5$.

The signed lifts at a fixed physical point are injective. Indeed, equal signed lifts for distinct shell primes p, q would give $pq \mid m - s_p$, whereas

$$0 < m - s_p < X^2 + X < (X + 1)(X + 2) \leq pq.$$

Every folded value has at most the two signed preimages h and $-h - 1$. Consequently the numbers of ordered marked pairs with signed distance at most G_X and folded distance at most G_X are at most $2G_X K$ and $(4G_X + 1)K$, respectively. A union bound over the ten pairs of labelled positions removes at most

$$\frac{10(6G_X + 1)}{K - 1}(K)_5 = O((\log X)^{-2})(K)_5. \quad (\text{FG4})$$

The ordinary quotient admits the growing gap R_X . If $a = \lfloor m/p \rfloor$ is occupied and $|\lfloor m/q \rfloor - a| \leq R_X$, then

$$\frac{m}{a + R_X + 1} < q \leq \frac{m}{a - R_X}.$$

For $m \geq Y_X$, one has $R_X \leq (m - 2X)/(4X)$ for large X , and this interval has length less than $8(2R_X + 1)X^2/(m - 2X)$. The same labelled-position union bound therefore removes at most

$$\frac{80(2R_X + 1)X^2}{(m - 2X)(K - 1)}(K)_5 \leq \frac{2(2R_X + 1)}{(\log X)^2}(K)_5. \quad (\text{FG5})$$

It follows that, uniformly on the original selected far range,

$$(1 - \varepsilon_X)(K)_5 \leq \mathcal{R}_D^{\text{sep}}(m), \quad \varepsilon_X = O((\log X)^{-1}) = o(\vartheta_X), \quad (\text{FG6})$$

where $\mathcal{R}_D^{\text{sep}}(m)$ is the number of ordered distinct retained five-tuples for which every pair satisfies (FG3). The original selector has not changed.

Let $\mathcal{A}_3(m)$ be the ordered retained triples satisfying all mutual conditions (FG3). For $\tau \in \mathcal{A}_3(m)$, let $E_\tau(m)$ count ordered retained pairs which, when appended to τ , give such a five-tuple, and let $E_{\tau,g}(m)$ count those pairs with signed gap g . Then, with coefficient one,

$$\mathcal{R}_D^{\text{sep}}(m) = \sum_{\tau \in \mathcal{A}_3(m)} E_\tau(m), \quad E_\tau(m) = \sum_{g=G_X+1}^{2X-1} E_{\tau,g}(m). \quad (\text{FG7})$$

Put

$$O_\tau(m) = (E_\tau(m) - B_X - \kappa_X)_+.$$

Since $|\mathcal{A}_3(m)| \leq (K)_3$ and $(K)_3 \leq (K)_5 / ((L-3)(L-4))$, equations (FG6)–(FG7) give

$$\left(\vartheta_X - \varepsilon_X - \frac{\kappa_X}{(L-3)(L-4)} \right) (K)_5 \leq \sum_{\tau \in \mathcal{A}_3(m)} O_\tau(m). \quad (\text{FG8})$$

The coefficient on the left is at least $\vartheta_X/2$ for large X , because $\kappa_X = X^{o(1)}$.

Uniformly split $B_X + \kappa_X$ among the U_X possible gaps. Positive-part subadditivity gives

$$O_\tau(m) \leq \sum_{g=G_X+1}^{2X-1} \left(E_{\tau,g}(m) - \frac{B_X + \kappa_X}{U_X} \right)_+. \quad (\text{FG9})$$

For fixed (m, τ, g) , regard an unordered retained extension pair as an edge. Signed-lift injectivity gives maximum degree two: a signed value has only the possible neighbors $s - g$ and $s + g$. If $V_{\tau,g}(m)$ is the number of incident labels, orderedness and symmetry give

$$E_{\tau,g}(m) \leq 2V_{\tau,g}(m), \quad V_{\tau,g}(m) \leq V_g(m).$$

Hence

$$\left(E_{\tau,g}(m) - \frac{B_X + \kappa_X}{U_X} \right)_+ \leq 2 \left(V_g(m) - \frac{B_X + \kappa_X}{2U_X} \right)_+.$$

Finally $|\mathcal{A}_3(m)| \leq (k_D(m))_3$. Summing (FG8) over the original selected points, then enlarging the nonnegative right side to every $Y_X \leq m < M$, gives

$$\frac{\vartheta_X}{2} \mathcal{T}_5^{\text{far}} \leq 2 \sum_{Y_X \leq m < M} (k_D(m))_3 \sum_{g=G_X+1}^{2X-1} \left(V_g(m) - \frac{B_X + \kappa_X}{2U_X} \right)_+.$$

Now apply (GH1) and use $\vartheta_X^{-1} = X^{o(1)}$ to obtain (483). $\square$

For completeness, the one-label terminology is literal. If $s_q = s_p + d$ with $d \in \{g, -g\}$, then $c_q(m)q = c_p(m)p - d$. The positive integer on the right is below $X^2 + X$ and therefore has at most one prime divisor satisfying $X < p \leq 2X$. After (m, p, g) and the sign are fixed, the possible partner characteristic is unique; the actual-hit and remaining mask tests are still required.

It remains to bridge the intermediate physical strip

$$2X \leq m < Y_X.$$

Here the quotient range is short enough to force a large fixed-quotient cell, but not short enough to force quotient one. Extend (FG1)–(FG2) to this strip, and put

$$Q_X^{\text{int}} = \left\lceil \frac{Y_X}{X} \right\rceil - 1, \quad K_X^\# = L - D_X^2, \quad R_{*,X} = \frac{K_X^\#}{4Q_X^{\text{int}}}. \quad (\text{IQ0})$$

For a retained row define

$$a_p = \left\lfloor \frac{m}{p} \right\rfloor, \quad r_p = m - a_p p, \quad h_p = \sigma_p(r_p), \quad \epsilon_p = \mathbf{1}_{r_p > h_p}.$$

The equality case $2r_p = p - 1$ belongs to the direct branch. For $\gamma = (a, \epsilon)$, where $1 \leq a \leq Q_X^{\text{int}}$ and $\epsilon \in \{0, 1\}$, let $S_\gamma(m)$ be the set of retained folded indices in that cell and write

$$R_\gamma = |S_\gamma(m)|, \quad \ell_\gamma = 2(a + \epsilon). \quad (\text{IQ1})$$

The two physical laws are

$$p_\gamma(h) = \begin{cases} (m-h)/a, & \epsilon = 0, \\ (m+1+h)/(a+1), & \epsilon = 1. \end{cases} \quad (\text{IQ2})$$

They show that $h \mapsto p_\gamma(h)$ is injective and that all $h \in S_\gamma(m)$ lie in one residue class modulo ℓ_γ .

For a cell with $R_\gamma \geq R_{*,X}$ define

$$n_{0,\gamma} = R_\gamma, \quad n_{j+1,\gamma} = \left\lceil \frac{\ell_\gamma(n_{j,\gamma} - 1)^2}{8X} \right\rceil \quad (0 \leq j \leq 2). \quad (\text{IQ3})$$

Let $\mathcal{D}_\gamma(m)$ consist of triples of positive multiples (d, e, f) of ℓ_γ such that

$$\begin{aligned} d &\leq \frac{2X}{R_\gamma - 1}, & e &\leq \frac{2X}{n_{1,\gamma} - 1}, & f &\leq \frac{2X}{n_{2,\gamma} - 1}, \\ d + e + f &< H_X, & |\{\delta_1 d + \delta_2 e + \delta_3 f : \delta_i \in \{0, 1\}\}| &= 8. \end{aligned} \quad (\text{IQ4})$$

For $V(d, e, f)$ equal to this eight-element offset set, define

$$\begin{aligned} \mathcal{C}_\gamma^\times(m; d, e, f) &= \#\{u : u + V(d, e, f) \subseteq S_\gamma(m), \\ &\quad p_\gamma(u + \alpha) \neq 2u + \alpha + \beta + 1 \quad (\alpha, \beta \in V, \alpha \neq \beta)\}. \end{aligned} \quad (\text{IQ5})$$

Thus every counted cube consists of eight actual retained rows and excludes all reflected collisions.

Hypothesis 547 (Intermediate fixed-quotient eight-cube bound). *For one fixed*

$$\frac{1}{120} < \eta < \frac{2}{105},$$

uniformly over sufficiently large dyadic X ,

$$\sup_{2X \leq m < Y_X} \sum_{\substack{\gamma \\ R_\gamma \geq R_{*,X}}} \max_{(d,e,f) \in \mathcal{D}_\gamma(m)} \mathcal{C}_\gamma^\times(m; d, e, f) = o\left(\frac{L^8}{X^7 \Omega_X^7}\right), \quad \Omega_X = \sum_{a=1}^{Q_X^{\text{int}}} \left(\frac{1}{2a} + \frac{1}{2(a+1)}\right). \quad (\text{IQC8})$$

An empty maximum is zero.

Proposition 548 (Eight-cube compression of the intermediate strip). *Hypothesis 547 excludes, for all sufficiently large X , every $2X \leq m < Y_X$ satisfying the original selector $K_X^{\text{iso}}(m; H_X) \geq L$.*

Proof. Deleting rows of Kummer order at most D_X removes at most D_X^2 rows, so a selected column has retained load at least $K_X^\sharp = (1 - o(1))L$. There are $2Q_X^{\text{int}}$ cells. The cells with $R_\gamma < R_{*,X}$ contain fewer than $K_X^\sharp/2$ rows altogether; hence

$$\sum_{R_\gamma \geq R_{*,X}} R_\gamma \geq \frac{K_X^\sharp}{2}. \quad (\text{IQ6})$$

We use the following elementary matching fact. If N distinct integers have span at most W , then at least half of their adjacent gaps are at most $2W/(N-1)$. Some positive gap occurs at least $(N-1)^2/(4W)$ times, and the graph of its occurrences is a union of paths. An alternating matching therefore retains more than $(N-1)^2/(8W)$ disjoint pairs. Iterating three times gives eight-point

Boolean translates. Since the points of one cell lie on a lattice of step ℓ_γ and have rescaled span less than X/ℓ_γ , the recursion is exactly (IQ3) and yields, for one triple in (IQ4),

$$(2^{-21} - o(1)) \frac{\ell_\gamma^7 R_\gamma^8}{X^7} \quad (\text{IQ7})$$

pairwise vertex-disjoint cubes. Uniformly for $R_\gamma \geq R_{*,X}$, all three matching sizes tend to infinity, and the three gap exponents are at most

$$X^{2/15-2\eta+o(1)}, \quad X^{4/15-4\eta+o(1)}, \quad X^{8/15-8\eta+o(1)}.$$

The last exponent is below $7/15$ precisely when $\eta > 1/120$, so their sum is $o(H_X)$.

There is one folded exception to the isolation argument. For distinct offsets α, β , a vertex $u + \beta$ can be the reflection of $u + \alpha$ in the latter's defining characteristic only if

$$\begin{cases} (2a+1)u = m - a - (a+1)\alpha - a\beta, & \epsilon = 0, \\ (2a+1)u = m - a - a\alpha - (a+1)\beta, & \epsilon = 1. \end{cases} \quad (\text{IQ8})$$

Each ordered pair gives at most one exceptional start, hence at most 56 starts for a fixed cube geometry. After deleting them, isolation gives all 56 directed off-diagonal Apéry units. With $c_n = (n!)^3 b_n$, the invertible cleared transfer between every two vertices then gives all 28 pairwise transfer coefficients as units at both endpoint primes. The lower bound (IQ7) tends to infinity, so the constant deletion is harmless.

Finally weighted Hölder gives

$$\left(\sum_\gamma R_\gamma \right)^8 \leq \left(\sum_\gamma \ell_\gamma^7 R_\gamma^8 \right) \left(\sum_\gamma \ell_\gamma^{-1} \right)^7,$$

and the last reciprocal sum is at most $\Omega_X \ll \log X$. Combining this with (IQ6) and summing (IQ7) over the large cells gives

$$\sum_{\substack{\gamma \\ R_\gamma \geq R_{*,X}}} \max_{(d,e,f) \in \mathcal{D}_\gamma(m)} \mathcal{C}_\gamma^\times(m; d, e, f) \geq (2^{-29} - o(1)) \frac{(K_X^\sharp)^8}{X^7 \Omega_X^7}.$$

This contradicts (IQC8) for a selected column, because $K_X^\sharp \sim L$. □

The remaining top strip has an exact characteristic-zero description.

Proposition 549 (Top-strip Lucas carrier). *For $X \leq m < 2X$, $X < p \leq 2X$, and $p > 5$,*

$$p \mid b_{m \bmod p} \iff p \mid b_m. \quad (484)$$

Thus

$$K_X(m) = \omega_{(X, 2X]}(b_m),$$

and the isolated load is the corresponding subset whose folded zeros are H_X -isolated.

Proof. For $p > m$ the two indices are equal. For $p \leq m < 2X < 2p$, write $m = p + r$. Apéry–Lucas gives $b_m \equiv b_1 b_r = 5b_r \pmod{p}$, and 5 is a p -unit. □

The raw full-shell statistic in this strip splits into the quotient-one and quotient-zero parts

$$K_X^{(1)}(m) = \#\{X < p < m : p \mid b_m\}, \quad K_X^{(0)}(m) = \#\{m < p \leq 2X : p \mid b_m\}.$$

The second family is present only because $K_X(m)$ was defined on the whole window independently of the physical column. It is not consumed by the all-index GCD problem: at the physical column $m = n$, every prime in the n -smooth radical satisfies $p \leq n = m$.

To state the exact consumer, define

$$\begin{aligned} K_X^\Delta(m) &= \sum_{\substack{p \in \mathcal{P}_X \\ p \leq m}} \mathbf{1}_{\mathcal{Z}_p}(m \bmod p), \\ K_X^{\text{iso}, \Delta}(m; H) &= \sum_{\substack{p \in \mathcal{P}_X \\ p \leq m}} \mathbf{1}_{\mathcal{I}_p(H)}(m \bmod p), \\ K_X^{\text{near}, \Delta}(m; H) &= K_X^\Delta(m) - K_X^{\text{iso}, \Delta}(m; H). \end{aligned}$$

Then

$$K_X^{\text{iso}, \Delta} \leq K_X^{\text{iso}}, \quad K_X^{\text{near}, \Delta} \leq K_X^{\text{near}}, \quad K_X^{\text{near}, \Delta}(m; H) \ll H^2. \quad (485)$$

The final inequality is Lemma 539.

Hypothesis 550 (Physical isolated five-star tail). *For one fixed $0 < \eta < 1/15$, uniformly over dyadic X ,*

$$\sum_{\substack{X \leq m < X^2 \\ K_X^{\text{iso}, \Delta}(m; H_X) \geq L}} (K_X^{\text{iso}, \Delta}(m; H_X))_5 \ll X^{o(1)}(1 + X^2 \Lambda^5), \quad L = \lceil X^{14/15 + \eta} \rceil. \quad (486)$$

This is weaker than Hypothesis 540. Indeed, every physical-tail summand is selected on the full tail and $(K_X^{\text{iso}, \Delta})_5 \leq (K_X^{\text{iso}})_5$. In the top strip, Proposition 549 and $b_0 = 1$ give the exact identity

$$K_X^\Delta(m) = \#\{X < p < m : p \mid b_m\} = K_X^{(1)}(m). \quad (487)$$

Thus quotient zero is absent from the physical consumer. For comparison, a uniform estimate for the stronger raw full-shell radical,

$$\log \prod_{\substack{X < p \leq 2X \\ p \mid b_m}} p = O(\sqrt{X \log X}) \quad (\text{TS})$$

would also dispose of quotient zero, but this strengthening is unnecessary for G_n .

There is a stronger Kummer-order pruning specific to this quotient-one strip.

Proposition 551 (Almost-threshold top-strip Kummer pruning). *Let $X \leq m < 2X$, and let $A_D^{(1)}(m)$ count the primes $X < p < m$ for which $p \mid b_{m-p}$ and $\mathfrak{o}_p(m-p) \leq D$. Then*

$$A_D^{(1)}(m) \leq D \tau(m-1), \quad (488)$$

If $B_S^{(1)}(m)$ counts the same physical rows with folded index $\sigma_p(m-p) \leq S$, then

$$B_S^{(1)}(m) \leq 2S. \quad (489)$$

If $C_A^{(1)}(m)$ counts the same rows with reduced Kummer numerator $a = \sigma_p(m-p)/\gcd(\sigma_p(m-p), p-1) \leq A$, then

$$C_A^{(1)}(m) \leq 2A\tau(m-1). \quad (490)$$

where τ is the divisor function. In particular, with

$$\Omega_X = \exp\left(\frac{\log X}{\sqrt{\log \log X}}\right), \quad D_X^{\text{top}} = \left\lfloor \frac{L}{\Omega_X} \right\rfloor,$$

the primitive-numerator deletion removes $o(L)$ rows. Since $a \leq d/2$, it already contains every row with $d \leq 2D_X^{\text{top}}$. Hence the physical top-strip fifth tail is, up to relative $o(1)$, supported on five primes satisfying

$$\begin{aligned} \mathfrak{o}_p(m-p) &\geq 2 \frac{\sigma_p(m-p)}{\gcd(\sigma_p(m-p), p-1)} > 2D_X^{\text{top}}, \\ \frac{\sigma_p(m-p)}{\gcd(\sigma_p(m-p), p-1)} &> D_X^{\text{top}} = X^{14/15+\eta-o(1)}, \\ \sigma_p(m-p) &> D_X^{\text{top}}, \quad \gcd(m-1, p-1) < X^{1/15-\eta+o(1)}. \end{aligned} \quad (491)$$

Proof. Put $r = m-p$, $s = \sigma_p(r)$, $g = \gcd(s, p-1)$, $d = (p-1)/g$, and $a = s/g$. The endpoints are not hits, so $1 \leq a \leq d/2$ and $(a, d) = 1$. In the direct and reflected branches, respectively,

$$m-1 = g(d+a), \quad m-1 = g(2d-a).$$

Thus $E = (m-1)/g$ is a divisor of $m-1$ with $d < E < 2d$; conversely the pair (E, d) reconstructs $g = (m-1)/E$ and $p = gd+1$. For every fixed divisor $E < 2D$, fewer than $E/2 \leq D$ values of $d \leq D$ are possible. This proves (488). For a prescribed folded index s , the direct branch gives $p = m-s$, while the reflected branch gives $p = (m+1+s)/2$. This proves (489). For fixed $g \mid m-1$ and a , the two branches give $d = E-a$ and $d = (E+a)/2$, respectively. Thus at most two primes arise from each of the at most $A\tau(m-1)$ pairs (g, a) , proving (490). The uniform bound

$$\tau(m-1) \leq \exp\left(O\left(\frac{\log X}{\log \log X}\right)\right)$$

gives

$$D_X^{\text{top}}\tau(m-1) \leq \frac{L}{\Omega_X} \exp\left(O\left(\frac{\log X}{\log \log X}\right)\right) = o(L),$$

and both $2D_X^{\text{top}}$ and $2D_X^{\text{top}}\tau(m-1)$ are $o(L)$ as well. It is enough to delete the rows counted by $C_{D_X^{\text{top}}}^{(1)}(m)$: the inequality $a \leq d/2$ shows that this also deletes every row with $d \leq 2D_X^{\text{top}}$, while $s = ga \geq a$. If a load $K \geq L$ loses $A = o(L)$ rows, then

$$0 \leq 1 - \frac{(K-A)_5}{(K)_5} \leq \frac{5A}{K-4} = o(1).$$

Finally $g = (p-1)/d < X/D_X^{\text{top}} = X^{1/15-\eta+o(1)}$. The numerator deletion gives $d \geq 2a > 2D_X^{\text{top}}$ and $a > D_X^{\text{top}}$, and hence $s = ga > D_X^{\text{top}}$, proving (491). $\square$

Corollary 552 (One-dimensional packet concentration). *Let $\mathcal{S}_m^\sharp$ be the isolated physical top-strip rows left after the deletion in Proposition 551, and put $K^\sharp = |\mathcal{S}_m^\sharp|$. Some fixed $g \mid m-1$, $E = (m-1)/g$, and one fixed folded branch contain at least*

$$R_{\text{br}} \geq \frac{K^\sharp}{2\tau(m-1)} \quad (492)$$

of these rows. They have distinct parameters $D_X^{\text{top}} < a \leq E/3$ and are given by one of the two linear laws

$$p = m - ga, \quad 2p = m + 1 + ga.$$

Here the midpoint $3a = E$, if present, is assigned to the direct branch, so the two branches are disjoint. If the original isolated load is at least L , then $R_{\text{br}} \geq X^{14/15+\eta-o(1)}$. At least $\lfloor R_{\text{br}}/4 \rfloor$ disjoint pairs in this branch satisfy

$$\Delta a \leq \frac{4E}{R_{\text{br}}}, \quad \Delta s \leq \frac{8X}{R_{\text{br}}} = X^{1/15-\eta+o(1)} = o(H_X). \quad (493)$$

For sufficiently large X and a distinct pair, neither folded index is the reflection of the other in either characteristic. Hence isolation makes both cross-evaluated Apéry coefficients nonzero.

Proof. Every surviving value of g divides the fixed integer $m-1$. There are at most $\tau(m-1) = X^{o(1)}$ such values, so pigeonholing g and then the two disjoint branches gives (492). In either branch the folding inequality $a \leq d/2$ gives $a \leq E/3$. Pair consecutive ordered a 's. Their gap sum is at most $E/3$, so at least one half of the pairs satisfy $\Delta a \leq 4E/R_{\text{br}}$, proving (493). In the direct branch the reflection of ga has g -scaled coordinate $E - 2a \geq E/3$; in the reflected branch it has g -scaled coordinate $(E - a)/2 \geq E/3$. Equality is only the same midpoint. Thus two distinct parameters in the fixed branch are not reflections. The folded map is one-Lipschitz and $8X/R_{\text{br}} = o(H_X)$, so for sufficiently large X the definition of $\mathcal{I}_p(H_X)$ makes the cross coefficient a unit. $\square$

The exact divisor parameter also removes every genuinely large- g channel from the final logarithmic radical.

Proposition 553 (Large exact- g top-strip mass). *For $X \leq m < 2X$ and $G \geq 2$, put*

$$\mathcal{R}_{X,\geq G}^{\text{top}}(m) = \prod_{\substack{X < p < m, \ p \mid b_m \\ \gcd(m-1, p-1) \geq G}} p.$$

Then

$$\log \mathcal{R}_{X,\geq G}^{\text{top}}(m) \leq \tau(m-1) \left(\frac{X}{G} + 1 \right) \log(2X). \quad (494)$$

In particular, for $G = \Omega_X = \exp(\log X / \sqrt{\log \log X})$, the right side is $o(X)$ uniformly in m .

Proof. For each exact value $g = \gcd(m-1, p-1)$ one has $g \mid m-1$ and $p \equiv 1 \pmod{g}$, so the interval (X, m) contains at most $X/g + 1$ possible integers p . Sum over the at most $\tau(m-1)$ divisors $g \geq G$, then use the uniform divisor bound from the preceding proof. $\square$

The remaining exact- g packets admit a characteristic-zero support projector. This does not yet estimate their intersection with the Apéry zeros, but it makes the missing horizontal statement an explicit rough-gcd problem.

For $N = m - 1$, $Y = \lfloor X \rfloor$, and $h \mid N$, put

$$T_h = \max \left\{ 0, \left\lfloor \frac{m - Y - 1}{h} \right\rfloor \right\}, \quad \mathcal{S}_h(R) = \prod_{\ell \mid h} \ell^{v_\ell(R!)}, \quad C_h = \frac{\mathcal{S}_h(T_h)}{T_h!} \prod_{1 \leq t \leq T_h} (m - ht),$$

where the first product is over prime divisors of h and empty products have value one. If $g \mid N$, define the reduced positive fraction

$$Q_g = \prod_{k \mid N/g} C_{gk}^{\mu(k)} = \frac{A_g}{B_g}, \quad (A_g, B_g) = 1.$$

Write $E = N/g$. For an integer $D \geq 0$, let

$$\begin{aligned} \mathcal{P}_{g,D}^{\text{dir}} &= \{m - ga : a > D, 3a \leq E, X < m - ga < m\}, \\ \mathcal{P}_{g,D}^{\text{ref}} &= \{(m + 1 + ga)/2 : a > D, 3a < E, E + a \equiv 0 \pmod{2}, X < (m + 1 + ga)/2 < m\}. \end{aligned}$$

Each nonempty list is an arithmetic-progression segment of step g . For such a list $q_1 < \dots < q_R$, set

$$C_{g,D}^{\text{br}} = \frac{\mathcal{S}_g(R)}{R!} \prod_{j=1}^R q_j, \quad J_{m,g,D}^{\text{br}} = \gcd(A_g, C_{g,D}^{\text{br}}),$$

and use value one for an empty list.

Proposition 554 (Exact top-strip packet certificate). *Assume $X \geq 5$, $X \leq m < 2X$, $g \mid m - 1$, and $\text{br} \in \{\text{dir}, \text{ref}\}$. For every prime $X < p < m$,*

$$v_p(J_{m,g,D}^{\text{br}}) = 1$$

if and only if $\gcd(m - 1, p - 1) = g$, the folded row of p belongs to the branch br , and its reduced folded numerator a is greater than D ; otherwise the valuation is zero. Moreover

$$\log J_{m,g,D}^{\text{br}} \leq \frac{X}{g} \log(6g^2). \quad (495)$$

Consequently, if $\text{rad}_{(X,m)}$ retains only prime factors in the open interval (X, m) , then

$$\prod_{\substack{X < p < m, \ p \mid b_m \\ \gcd(m-1, p-1)=g, \ p \text{ in br, } a > D}} p = \text{rad}_{(X,m)} \gcd(b_m, J_{m,g,D}^{\text{br}}). \quad (496)$$

The same product restricted to isolated rows divides the right side.

Proof. For a step- h progression $q_1, \dots, q_R$ in (X, m) , the normalization $\mathcal{S}_h(R) \prod_j q_j / R!$ is integral: primes not dividing h occur in the progression often enough to supply their factorial valuations, while the primes dividing h are supplied by $\mathcal{S}_h(R)$. Since $R < p$ and $0 < q_j < m < 2p$, its p -adic valuation is one exactly when one term equals p . Also

$$\log \left(\frac{\mathcal{S}_h(R)}{R!} \prod_j q_j \right) \leq R \log(6Xh/R).$$

For the progression defining C_h , a term equals p exactly when $h \mid m - p$. As $h \mid m - 1$, this is equivalent to $h \mid p - 1$, and hence

$$v_p(C_h) = \mathbf{1}_{h \mid \gcd(m-1, p-1)}.$$

Möbius inversion in the definition of Q_g , followed by reduction to lowest terms, therefore gives

$$v_p(A_g) = \mathbf{1}_{g = \gcd(m-1, p-1)}, \quad v_p(B_g) = 0.$$

The two branch lists defining $C_{g,D}^{\text{br}}$ are exactly the direct and reflected linear laws from Proposition 551; every list term is congruent to one modulo g . Intersecting its normalized progression certificate with A_g proves the valuation claim. Each branch list has fewer than X/g terms, and the preceding height inequality gives (495). Finally, taking the squarefree product prime by prime proves (496). The hypothesis $X \geq 5$ excludes the exceptional endpoint multiplier at $p = 5$. $\square$

Corollary 555 (Moving fresh-divisor form of an exact packet). *Under the hypotheses of Proposition 554, let a denote the reduced folded numerator of a prime in one of the two branch lists. Then*

$$p > 2ga, \quad p \mid b_m \iff p \mid b_{ga}. \quad (497)$$

More precisely,

$$p = m - ga \quad \text{in the direct branch,} \quad 2p = m + 1 + ga \quad \text{in the reflected branch,}$$

and therefore

$$\text{rad}_{(X,m)} \gcd(b_m, J_{m,g,D}^{\text{br}}) = \prod_{\substack{p = p_{\text{br}}(a) \in \mathcal{P}_{g,D}^{\text{br}} \\ p \mid b_{ga}}} p. \quad (498)$$

Proof. Write $p - 1 = gd$. Since $a \leq d/2$, one has $p - 1 \geq 2ga$, hence $p > 2ga$. Also $m = p + r < 2p$, so Apéry–Lucas gives $b_m \equiv b_1 b_r = 5b_r \pmod{p}$. In the direct branch $r = ga$. In the reflected branch $r = p - 1 - ga$, and Apéry reflection gives $b_r \equiv b_{ga} \pmod{p}$. Since $p > 5$, this proves (497); combining it with the exact valuation statement of the preceding proposition proves (498). $\square$

Remark 556 (Primitive one-prime nonvanishing is false). The size and coprimality conditions in (497) do not themselves force a unit. At $X = 128$ and $m = 200$, the direct exact-1 packet contains

$$(p, a) = (181, 19), \quad (139, 61),$$

and

$$181 \mid b_{19}, \quad 139 \mid b_{61}, \quad (19, 180) = (61, 138) = 1, \quad p > 2a.$$

For every fixed $0 < \eta < 1/15$ one has $D_X^{\text{top}} \leq 2$ here, while $H_X = 5$ and

$$\sigma_{139}(\mathcal{Z}_{139}) = \{61\}, \quad \sigma_{181}(\mathcal{Z}_{181}) = \{19, 47\}.$$

Thus both marked rows survive the present numerator deletion and are H_X -isolated. This disproves the universal implication $p > 2a$, $(a, p - 1) = 1 \Rightarrow p \nmid b_a$; the example has load two and is not an asymptotic high-load counterexample.

Proposition 557 (Short-gap compression in the direct exact-one packet). *Fix $X \leq m < 2X$, put $N = m - 1$ and $D = D_X^{\text{top}}$, and define*

$$\mathcal{A}_{X,1}^{\text{iso}}(m) = \left\{ a : \begin{array}{l} a > D, \quad a \equiv 1 \pmod{2}, \quad 3a \leq N, \quad (a, N) = 1, \\ p = m - a \text{ is prime}, \quad X < p < m, \quad p \mid b_a, \quad a \in \mathcal{I}_p(H_X) \end{array} \right\}.$$

Write $K = |\mathcal{A}_{X,1}^{\text{iso}}(m)|$ and

$$R_{X,1}^{\text{iso}}(m, d) = \#\{a : a, a + d \in \mathcal{A}_{X,1}^{\text{iso}}(m)\}.$$

If $K \geq 2$, then some positive even integer d satisfies

$$d \leq \frac{4X}{3(K-1)}, \quad (499)$$

$$R_{X,1}^{\text{iso}}(m, d) \geq \frac{3(K-1)^2}{4X}, \quad (500)$$

and, more generally,

$$\sum_{\substack{2|e \\ 0 < e \leq 4X/[3(K-1)]}} R_{X,1}^{\text{iso}}(m, e)^2 > \frac{3(K-1)^3}{8X}. \quad (501)$$

If $K \geq L$ and X is sufficiently large, the gap in (499) is $o(H_X)$. For every pair $a, a + d$ counted by $R_{X,1}^{\text{iso}}(m, d)$, the two primes

$$p = m - a, \quad q = m - a - d = p - d$$

then obey the two-characteristic zero-unit square

$$p \mid b_a, \quad p \nmid b_{a+d}, \quad q \mid b_{a+d}, \quad q \nmid b_a. \quad (502)$$

Moreover, put $c_n = (n!)^3 b_n$,

$$P(u) = 34u^3 + 51u^2 + 27u + 5, \quad \text{quad} C_u = \begin{pmatrix} P(u) & -u^6 \\ 1 & 0 \end{pmatrix},$$

and let $\rho_d(a)$ be the upper-right entry of $C_{a+d-1} \cdots C_a$. Then

$$\gcd(\rho_d(a), pq) = 1. \quad (503)$$

Thus every pair supplied by the popular gap lies in the generic, rather than the transfer-degenerate, branch.

Proof. List the elements of $\mathcal{A}_{X,1}^{\text{iso}}(m)$ as $a_1 < \cdots < a_K$, put $M = K - 1$ and $W = a_K - a_1$, and write $\Delta_i = a_{i+1} - a_i$. The Δ_i are positive and even, while

$$\sum_{i=1}^M \Delta_i = W \leq \frac{N}{3} < \frac{2X}{3}.$$

At least $\lceil M/2 \rceil$ of them are at most $2W/M$. There are at most $\lfloor W/M \rfloor$ positive even integers in that range. Hence one value occurs at least

$$\frac{\lceil M/2 \rceil}{\lfloor W/M \rfloor} \geq \frac{M^2}{2W} > \frac{3M^2}{4X}$$

times, proving (499) and (500). If c_e counts the adjacent gaps of value $e \leq 2W/M$, then Cauchy's inequality gives

$$\sum_e R_{X,1}^{\text{iso}}(m, e)^2 \geq \sum_e c_e^2 \geq \frac{M^3}{4W} > \frac{3M^3}{8X},$$

which is (501).

For $K \geq L$, the first bound gives

$$d \leq \frac{4X}{3(L-1)} = X^{1/15-\eta+o(1)} = o(H_X).$$

The inequalities $3(a+d) \leq N$ imply

$$2(a+d) \leq p-1 = N-a, \text{quad} 2a \leq q-1 = N-a-d.$$

Thus a and $a+d$ are in the direct folded halves in both characteristics and have folded distance exactly d . Isolation of the two target roots now proves (502).

The cleared Apéry recurrence is

$$\binom{c_{a+d}}{c_{a+d-1}} = C_{a+d-1} \cdots C_a \binom{c_a}{c_{a-1}}.$$

All relevant factorials are units in the corresponding characteristics, so (502) also holds with b replaced by c . If $p \mid \rho_d(a)$, the first row of the displayed recurrence and $p \mid c_a$ would force $p \mid c_{a+d}$, a contradiction. At q , one has $q \geq 2(a+d)+1$, and hence

$$\det(C_{a+d-1} \cdots C_a) = \prod_{j=0}^{d-1} (a+j)^6 \not\equiv 0 \pmod{q}.$$

If $q \mid \rho_d(a)$, then $q \mid c_{a+d}$ and $q \nmid c_a$ would also force the upper-left entry to vanish modulo q , contradicting this determinant. This proves (503). $\square$

The proposition reduces this packet to either of the following strictly short-gap estimates:

$$\sup_{\substack{X \leq m < 2X \\ 2|d, 0 < d \leq 4X/[3(L-1)]}} R_{X,1}^{\text{iso}}(m, d) = o(L^2/X), \quad (504)$$

$$\sup_{\substack{X \leq m < 2X \\ 2|d \\ 0 < d \leq 4X/[3(L-1)]}} \sum R_{X,1}^{\text{iso}}(m, d)^2 = o(L^3/X). \quad (505)$$

Either estimate would exclude a load $K \geq L$ in the direct exact-one packet. Neither estimate is claimed here: they are the remaining horizontal Apéry input for this subpacket, with the pointwise threshold $L^2/X = X^{13/15+2\eta+o(1)}$ and gap range $d = O(X^{1/15-\eta})$.

There is a further unconditional two-gap compression which removes the diagonal and shared-vertex portions of this gateway. For positive even d, e , put

$$\mathcal{Q}_{X,1}^{\text{iso}}(m; d, e) = \#\{a : a, a+d, a+e, a+d+e \in \mathcal{A}_{X,1}^{\text{iso}}(m), e \neq d\}.$$

Proposition 558 (Four-prime rectangle compression). *Put*

$$r_X = \left\lceil \frac{3(L-1)^2}{4X} \right\rceil, \quad j_X = \left\lceil \frac{r_X}{2} \right\rceil.$$

For all sufficiently large X , if $|\mathcal{A}_{X,1}^{\text{iso}}(m)| \geq L$, then there are positive even integers

$$d \leq \frac{4X}{3(L-1)}, \quad e \leq \frac{4X}{3(j_X-1)}$$

such that

$$\mathcal{Q}_{X,1}^{\text{iso}}(m; d, e) \geq \frac{3(j_X-1)^2}{4X} = \left(\frac{27}{256} + o(1) \right) \frac{L^4}{X^3}. \quad (506)$$

For every rectangle counted here, the four distinct primes

$$p = m - a, \quad p - d, \quad p - e, \quad p - d - e$$

divide the four corresponding Apéry values, and each is a unit at the other three values. All six pairwise cleared transfer coefficients are units at their two endpoint primes.

Proof. Choose d as in Proposition 557. The graph whose edges are the pairs $\{a, a + d\}$ is a union of paths, hence it has a matching of size at least

$$\left\lceil \frac{R_{X,1}^{\text{iso}}(m, d)}{2} \right\rceil \geq j_X.$$

Apply the adjacent-gap argument from that proposition to the ordered starting points of the matching. Their span is less than $2X/3$, so some positive even difference e is at most $4X/[3(j_X-1)]$ and occurs at least $3(j_X-1)^2/(4X)$ times. Matched edges are vertex-disjoint, so $e \neq d$, and these pairs of edges give distinct terms of $\mathcal{Q}_{X,1}^{\text{iso}}(m; d, e)$.

Here

$$d = O(X^{1/15-\eta}), \quad e = O(X^{2/15-2\eta}),$$

and therefore $d + e = o(H_X)$. All four indices remain in the direct folded half in each of the four characteristics. Isolation gives the complete diagonal-zero/off-diagonal-unit pattern. Applying the determinant argument for (503) to each of the six pairs proves the final assertion. $\square$

Consequently the strictly localized estimate

$$\sup_{\substack{X \leq m < 2X \\ 2|d, e, 0 < d \leq 4X/[3(L-1)] \\ 0 < e \leq 4X/[3(j_X-1)]}} \mathcal{Q}_{X,1}^{\text{iso}}(m; d, e) = o(L^4/X^3) \quad (507)$$

would exclude this high-load packet. This estimate is not claimed here.

The rectangle admits one more unconditional compression. For positive even d, e, f , put

$$\mathcal{C}_{X,1}^{\text{iso}}(m; d, e, f) = \# \{a : a + \epsilon_1 d + \epsilon_2 e + \epsilon_3 f \in \mathcal{A}_{X,1}^{\text{iso}}(m) \mid \epsilon_i \in \{0, 1\}\}.$$

Proposition 559 (Eight-prime cube compression). *With (r_X, j_X) as in Proposition 558, put*

$$q_X = \left\lceil \frac{3(j_X-1)^2}{4X} \right\rceil, \quad t_X = \left\lceil \frac{q_X}{9} \right\rceil.$$

For all sufficiently large X , if $|\mathcal{A}_{X,1}^{\text{iso}}(m)| \geq L$, then there are positive even d, e, f , with

$$d \leq \frac{4X}{3(L-1)}, \quad e \leq \frac{4X}{3(j_X-1)}, \quad f < \frac{4X}{3(t_X-1)},$$

such that

$$\mathcal{C}_{X,1}^{\text{iso}}(m; d, e, f) > \frac{3(t_X - 1)^2}{4X} = \left(\frac{27}{262144} + o(1) \right) \frac{L^8}{X^7}. \quad (508)$$

Every counted cube has eight distinct defining primes, eight diagonal Apéry zeros, all 56 ordered off-diagonal values are units, and all 28 pairwise cleared transfer coefficients are units at both endpoint primes.

Proof. Fix (d, e) supplied by Proposition 558, and let S be its set of rectangle starts. Two rectangles $a + \{0, d, e, d + e\}$ and $b + \{0, d, e, d + e\}$ can meet only when

$$|a - b| \in \{d, e, d + e, |d - e|\}.$$

Their conflict graph therefore has maximum degree at most eight. A greedy independent set supplies at least t_X pairwise vertex-disjoint rectangles. Order their starts $u_1 < \dots < u_T$, put $W = u_T - u_1$, and note that $W < 2X/3$. Of the $T - 1$ positive even adjacent gaps, at least half are at most $2W/(T - 1)$; there are at most $\lfloor W/(T - 1) \rfloor$ possible even values in that range. Hence one even f occurs more than $3(T - 1)^2/(4X)$ times and satisfies $f < 4X/[3(T - 1)]$. Each occurrence joins two vertex-disjoint rectangles and therefore gives an eight-vertex cube. Taking $T \geq t_X$ proves the displayed exact bounds.

The preceding rectangle asymptotic gives

$$q_X = \left(\frac{27}{256} + o(1) \right) \frac{L^4}{X^3}, \quad t_X = \left(\frac{3}{256} + o(1) \right) \frac{L^4}{X^3},$$

and therefore the constant in (508). Moreover

$$d = O(X^{1/15-\eta}), \quad e = O(X^{2/15-2\eta}), \quad f = O(X^{4/15-4\eta}),$$

so $d + e + f = o(H_X)$. Every cube vertex belongs to $\mathcal{A}_{X,1}^{\text{iso}}(m)$, and its largest vertex is at most $(m - 1)/3$. Thus all vertices lie in the direct folded half in each of the eight characteristics. Isolation gives the complete diagonal-zero and off-diagonal-unit matrix. Applying the endpoint determinant argument from Proposition 558 to each of the $\binom{8}{2} = 28$ pairs proves the transfer-unit assertion. $\square$

Consequently the still more structured estimate

$$\sup_{m,d,e,f} \mathcal{C}_{X,1}^{\text{iso}}(m; d, e, f) = o(L^8/X^7), \quad (509)$$

where $X \leq m < 2X$ and the three gaps range as in the proposition, would exclude this high-load packet. It is not claimed here. The eight canonical first-lift observations do satisfy three short affine relations, but those relations hold before the eight zero conditions are imposed; an exact non-target prime cube verifies that affine rank excess alone is not target-selective. This scoped warning does not exclude a nonlinear or inhomogeneous identity using the canonical initial line. The independent proof audits are in `codex_rectangle_horizontal_cube_attack.*` and `codex_rectangle_cube_cold_audit.*`.

Inside the exponent interval needed by the final theorem, the eight-prime cube admits one last unconditional compression. For positive even d, e, f, h , put

$$\mathcal{H}_{X,1}^{\text{iso}}(m; d, e, f, h) = \# \{a : a + \epsilon_1 d + \epsilon_2 e + \epsilon_3 f + \epsilon_4 h \in \mathcal{A}_{X,1}^{\text{iso}}(m) \quad (\epsilon_i \in \{0, 1\})\}.$$

Proposition 560 (Sixteen-prime tesseract compression). *Assume*

$$\frac{1}{120} < \eta < \frac{2}{105}.$$

With t_X as in Proposition 559, put

$$v_X = \left\lfloor \frac{3(t_X - 1)^2}{4X} \right\rfloor + 1, \quad w_X = \left\lceil \frac{v_X}{27} \right\rceil.$$

For all sufficiently large X , if $|\mathcal{A}_{X,1}^{\text{iso}}(m)| \geq L$, then there are positive even d, e, f, h in the preceding ranges and

$$h < \frac{4X}{3(w_X - 1)} = \left(\frac{2^{20}}{3} + o(1) \right) \frac{X^8}{L^8}, \quad (510)$$

$$\mathcal{H}_{X,1}^{\text{iso}}(m; d, e, f, h) > \frac{3(w_X - 1)^2}{4X} = \left(\frac{3}{2^{38}} - o(1) \right) \frac{L^{16}}{X^{15}}. \quad (511)$$

Every counted tesseract has sixteen distinct defining primes, sixteen diagonal Apéry zeros, all 240 ordered off-diagonal values are units, and all 120 pairwise cleared transfer coefficients are units at both endpoint primes.

Proof. Fix the three gaps supplied by Proposition 559, and let S be its set of cube starts. If

$$V = \{\epsilon_1 d + \epsilon_2 e + \epsilon_3 f : \epsilon_i \in \{0, 1\}\},$$

then two translates $a + V, b + V$ meet only if $b - a \in (V - V) \setminus \{0\}$. Since $V - V$ is represented by vectors in $\{-1, 0, 1\}^3$, the intersection graph on S has maximum degree at most $3^3 - 1 = 26$. A greedy independent set therefore supplies at least w_X pairwise vertex-disjoint cubes.

Order their starts and repeat the adjacent-gap argument from the preceding proposition. Their span is less than $2X/3$, so some positive even gap h occurs more than $3(w_X - 1)^2/(4X)$ times and satisfies $h < 4X/[3(w_X - 1)]$. Each occurrence joins two vertex-disjoint cubes and therefore produces sixteen distinct vertices.

The previous lower bound gives

$$v_X = \left(\frac{27}{2^{18}} - o(1) \right) \frac{L^8}{X^7}, \quad w_X = \left(\frac{1}{2^{18}} - o(1) \right) \frac{L^8}{X^7}.$$

Consequently

$$\frac{4}{3} 2^{18} = \frac{2^{20}}{3}, \quad \frac{3}{4} 2^{-36} = \frac{3}{2^{38}},$$

which proves the displayed constants. Moreover

$$h = O(X^{8/15-8\eta}) = o(H_X)$$

because $\eta > 1/120$; the older three gaps are smaller. Thus the total span is $o(H_X)$. All sixteen vertices remain in the direct folded half, so isolation gives the complete diagonal-zero/off-diagonal-unit matrix. Applying the endpoint determinant argument to the $\binom{16}{2} = 120$ pairs proves the final assertion. $\square$

Thus the direct exact-one packet is also reduced to the conditional estimate

$$\sup_{m,d,e,f,h} \mathcal{H}_{X,1}^{\text{iso}}(m; d, e, f, h) = o(L^{16}/X^{15}), \quad (\text{H16})$$

with the four gaps in the ranges above. The interval $1/120 < \eta < 2/105$ is nonempty, and its upper endpoint is exactly the condition $14/15 + \eta < 20/21$.

The exact-one packet is only one possible cell in Corollary 552. The same compression works in the fixed- g , fixed-branch packet which the corollary actually forces. For such a packet, let

$$\mathcal{T}_{m,g}^{\text{br}}(d_1, d_2, d_3, d_4)$$

count starts u for which all sixteen values

$$u + \epsilon_1 d_1 + \epsilon_2 d_2 + \epsilon_3 d_3 + \epsilon_4 d_4, \quad \epsilon_i \in \{0, 1\},$$

belong to the same post-deletion packet. Equivalently, the sixteen rows retain the original physical, isolation, and high-Kummer masks. The fixed branch conditions imply that distinct vertices are never reflections of one another, so every counted tesseract has all 240 directed cross-units and all 120 pairwise endpoint transfer units.

Hypothesis 561 (Full top-strip packet tesseract bound). *For one fixed $1/120 < \eta < 2/105$, uniformly over sufficiently large dyadic X ,*

$$\sup_{X \leq m < 2X} \tau(m-1)^{16} \max_{\substack{g|m-1, \text{ br} \\ d_1, d_2, d_3, d_4 > 0}} \mathcal{T}_{m,g}^{\text{br}}(d_1, d_2, d_3, d_4) = o(L^{16}/X^{15}). \quad (\text{TP16})$$

Proposition 562 (Full-packet top-strip compression). *Hypothesis 561 excludes every $X \leq m < 2X$ satisfying the original isolated selector.*

Proof. After the numerator deletion in Proposition 551, a selected column has $K^\sharp \geq L/2$ rows for large X . By Corollary 552, some fixed- g , fixed-branch packet contains

$$R \geq \frac{K^\sharp}{2\tau(m-1)} \geq \frac{L}{4\tau(m-1)}$$

rows. Their folded indices are ga , have span less than $2X/3$, and their defining primes obey one of the two affine laws

$$p = m - ga, \quad 2p = m + 1 + ga.$$

For completeness, if N distinct translates of a fixed finite offset set have start span less than X , more than $(N-1)^2/(8X)$ of them can be paired along one common positive adjacent gap while keeping the resulting translates disjoint. Indeed, at least half the adjacent gaps are at most $2X/(N-1)$; one value is popular, and its occurrence graph is a union of paths. Iterating this matching four times gives one gap quadruple and

$$\max_{d_1, d_2, d_3, d_4} \mathcal{T}_{m,g}^{\text{br}}(d_1, d_2, d_3, d_4) \gg \frac{R^{16}}{X^{15}}. \quad (512)$$

pairwise vertex-disjoint tesseracts. All four intermediate family sizes tend to infinity. The four gap scales are at most

$$X^{1/15-\eta+o(1)}, \quad X^{2/15-2\eta+o(1)}, \quad X^{4/15-4\eta+o(1)}, \quad X^{8/15-8\eta+o(1)}.$$

Thus their total span is $o(H_X)$ because $\eta > 1/120$.

Write $m - 1 = gE$. In the direct branch the reflection of ga in its defining characteristic is $g(E - 2a)$. If this were another packet value ga' , then $a' = E - 2a$; the two bounds $a, a' \leq E/3$ force the same midpoint $a = a' = E/3$. In the reflected branch the reflected coordinate is $g(E - a)/2$, whose reduced parameter is strictly larger than $E/3$. Hence distinct packet vertices never reflect. Isolation and the short total span give all directed cross-units, and the invertible cleared Apéry transfer gives every endpoint transfer unit.

Multiplying (512) by $\tau(m - 1)^{16}$ now gives

$$\tau(m - 1)^{16} \max_{g, \text{br}, d_1, d_2, d_3, d_4} \mathcal{J}_{m, g}^{\text{br}}(d_1, d_2, d_3, d_4) \gg \frac{L^{16}}{X^{15}},$$

contrary to (TP16). □

Condition (H16) is the direct $g = 1$ special case of this full-packet gateway. Neither estimate is presently known.

Higher local lift order alone does not close (H16). For nodes x_i , primes $p_i = m - x_i$, and the Lagrange rows L_i , every integer aggregation

$$\Delta \sum_i L_i(m) p_i^j z_i \quad (j \geq 1)$$

contains the full radical $\prod_i p_i$, but no second copy follows uniformly: the factors from the other Lagrange numerators supply only one copy at each characteristic. The natural quadratic exterior determinant has the same principal-ideal radical factor. An exact canonical isolated target $(m, a, p) = (4704, 431, 4273)$ lies in an eight-prime subcube for which the divided exterior determinant is 148 (mod 4273) and is coprime to the full radical. These are scoped obstructions only: an identity involving the divided zero quotients or a genuinely inhomogeneous canonical boundary remains open. The proof, the exact target regression, and an independent compression verifier are in `codex_cube_boundary_attack.*` and `codex_hypercube16_compression_verify.py`.

Remark 563 (The two first lifts do not close the short-gap estimate). For the pair in Proposition 557, write

$$b_m = pqB, \quad b_a = pu, \quad b_{a+d} = qv,$$

and denote the Gessel first jet by

$$\mathfrak{d}_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \sum_{j=n-k+1}^{n+k} \frac{1}{j}.$$

The quotient-one congruence modulo the square of the defining prime gives the two exact local equations

$$dB + 5u + 10\mathfrak{d}_a = 0 \pmod{p}, \quad dB - 5v - 10\mathfrak{d}_{a+d} = 0 \pmod{q}. \quad (513)$$

They live in different residue fields. More precisely, let (λ, ρ) be the first row of $C_{a+d-1} \cdots C_a$, put

$$F = (a!)^3, \quad G = ((a+d)!)^3, \quad U = Fu, \quad V = Gv,$$

and retain $W = c_{a-1}$. The two lifts together with the exact transfer $qV = \lambda pU + \rho W$ reduce to

	first lift	d-step transfer	
$\mathbb{F}_p$	$dB + 5F^{-1}U + 10\mathfrak{d}_a = 0$	$-dV - \rho W = 0$	
$\mathbb{F}_q$	$dB - 5G^{-1}V - 10\mathfrak{d}_{a+d} = 0$	$\lambda dU + \rho W = 0$.	(*)

All displayed inverted coefficients are units; for λ modulo q one also uses $d + 1 \leq H_X$, which holds in the high-load range for large X . Thus B, W may be chosen freely in either field, after which U, V are solved uniquely. These two affine fibres glue by CRT. There is no hidden integrality condition: after prescribing the four local residues, one may choose u and v additionally divisible by $|\rho|((a-1)!)^3$; the transfer then defines an integral c_{a-1} with the prescribed local residues.

Consequently the scalar elimination ideal generated by (513) and the finite ordinary or differentiated transfer is zero after localization at the known units. This is a scoped obstruction, not a disproof of (504) or (505): a proof must additionally use the canonical initial line, a global hypergeometric boundary identity, or a correlation among several pairs with the same (m, d) .

The last possibility can itself be narrowed. At fixed d the prime graph is a union of paths. At an internal prime the incoming and outgoing first lifts reduce to the same vertex congruence; their difference is only the exact quotient identity. More generally, even after adjoining all short-orbit determinant identities and one common formal output integer, CRT permits a decoupled local residue presentation satisfying the displayed zero/unit and first-lift conditions. Here transfer-state variables in different residue characteristics are not identified with the values of one recurrence solution. Indeed, acyclicity removes holonomy but does not remove endpoint compatibility along a path; arbitrary prescribed vertex values need not lie on one transfer orbit. Thus this scoped saturation statement neither fixes the initial line to $(b_0, b_1) = (1, 5)$ nor requires the formal output to be the propagated value at m . It does not disprove (507); it shows that the surviving input must use the canonical long boundary or a genuine four-characteristic average. Notice that $\mathfrak{d}_n$ is already a characteristic-zero canonical sequence: it is the analytic part of the logarithmic Apéry solution, with fixed initial data $\mathfrak{d}_0 = 0$, $\mathfrak{d}_1 = 6$, and a fixed inhomogeneous recurrence. What the formal model omits is this canonical coupled initial line, not merely the construction of some additional p -independent coordinate.

Proposition 564 (Dyadic saturation of the admissible packet bank). *Fix $0 < \eta < 1/15$ and put $D = D_X^{\text{top}}$. For every sufficiently large dyadic X there are a prime E and an integer m with*

$$\frac{2X}{3} < E < \frac{7X}{10}, \quad m = 2E + 1,$$

such that every prime $X < p < m - 2D$ is a direct exact- $g = 2$ candidate with reduced numerator

$$a = \frac{m-p}{2} > D, \quad d = \frac{p-1}{2} > 2D.$$

Moreover

$$\sum_{X < p < m-2D} \log p = 2E - X + o(X) \asymp X, \quad \#\{X < p < m - 2D\} \gg \frac{X}{\log X}.$$

Thus the window, exact- g , branch, high-numerator, and high-order conditions alone admit more than $X^{14/15+\eta}$ candidates.

Proof. The prime number theorem supplies a prime $2X/3 < E < 7X/10$, and $D_X^{\text{top}} = o(X)$. For a prime in the displayed interval put $a = (m-p)/2$. Then $a > D$, and $p > X$, $E < 7X/10$ give $3a < E$ for all sufficiently large X , so this is the direct branch. Since E is prime and $0 < a < E$,

$$p-1 = 2(E-a), \quad (m-1, p-1) = (2E, 2(E-a)) = 2,$$

and also $(2a, p-1) = 2$. Thus the exact packet label is $g = 2$, while $d = E - a > 2a > 2D$. The two prime-number-theorem estimates at X and $m - 2D = 2E + 1 - o(X)$ give the final assertions. $\square$

Thus a top-strip column of surviving load at least L forces, for some $g \mid m - 1$ and one branch,

$$\omega_{(X,m)}\left(\gcd(b_m, J_{m,g,D_X^{\text{top}}}^{\text{br}})\right) \geq \frac{L - o(L)}{2\tau(m-1)} = X^{14/15+\eta-o(1)}. \quad (514)$$

Here $\omega_{(X,m)}$ counts distinct prime factors in (X, m) . The remaining top-strip theorem is precisely to rule out (514) for the subpower values $g < \Omega_X$; the certificate construction alone does not do so.

Equivalently, all surviving rows sample the fixed integer Mellin exponent $m - 1$: since $m - 1 \equiv m - p \pmod{p - 1}$, their character order is exactly $(p - 1)/\gcd(p - 1, m - 1)$. The unresolved top-strip input is therefore a horizontal estimate for the remaining subpower exact- g packets of these almost-primitive defining-characteristic Mellin traces, not a low-order Kummer contribution.

Proposition 565 (Packet-aligned Jacobi saturation). *Let $p - 1 = gd$, $d > 2$, and $(E, d) = 1$. Choose a primitive root $\rho_p \pmod{p}$, put $L_d = \mathbf{Q}(\zeta_d)$, define $\chi_{p,E}(\rho_p) = \zeta_d^E$, and set*

$$\mathfrak{p}_{p,u} = (p, \zeta_d - \rho_p^{gu}) \quad (u \in (\mathbf{Z}/d\mathbf{Z})^\times).$$

If ϕ_p is the quadratic character, then

$$(J(\chi_{p,E}, \phi_p)) = \prod_{\substack{u \in (\mathbf{Z}/d\mathbf{Z})^\times \\ \langle uE \rangle_d < d/2}} \mathfrak{p}_{p,u}, \quad |N_{L_d/\mathbf{Q}} J(\chi_{p,E}, \phi_p)| = p^{\varphi(d)/2}.$$

Every displayed prime has exponent one; replacing E by $-E$ selects the complementary half. In particular the selected prime $\mathfrak{p}_{p,1}$ divides the direct-packet Jacobi sum, while the inverse-character sum has the same property in the reflected packet.

Proof. At $\mathfrak{p}_{p,u}$ the Jacobi sum reduces to

$$\sum_{x \in \mathbb{F}_p} x^{g\langle uE \rangle_d} (1 - x)^{(p-1)/2}.$$

After expanding the second factor, multiplicative orthogonality makes this zero exactly when $\langle uE \rangle_d < d/2$; above the midpoint there is one nonzero term of exponent $p - 1$. The midpoint cannot occur because $(E, d) = 1$ and $d > 2$. Pairing u with $-u$ gives $\varphi(d)/2$ detected primes. Every complex conjugate of the Jacobi sum has absolute value $\sqrt{p}$, so its absolute norm is $p^{\varphi(d)/2}$; the detected degree-one primes therefore exhaust the principal ideal. Replacing E by $-E$ proves the last assertion. $\square$

The trace $x \mapsto \phi_p(1 - x)$ is a fixed rank-one bounded-conductor Kummer family, and its $\chi_{p,E}$ -Mellin trace is the Jacobi sum above. Together with Proposition 564, this shows that fixed exponent, primitive order, bounded conductor, purity, cyclotomic covariance, and norm bounds can coexist with saturation of a whole high-cutoff packet bank. This is not an Apéry counterexample: the horizontal estimate still has to use arithmetic specific to the Apéry trace rows.

The distinguished-output-free reduction also survives the physical truncation. Put

$$f_p^\Delta(m) = \mathbf{1}_{p \leq m} \mathbf{1}_{\mathcal{I}_p(H_X)}(m \bmod p), \quad h_p^\Delta(m) = f_p^\Delta(m) - z_p,$$

and

$$\mathcal{W}_5^\Delta(X) = 120 \sum_{\substack{X \leq m < X^2 \\ K_X^{\text{iso}, \Delta}(m; H_X) \geq L}} e_5((h_p^\Delta(m))_{p \in \mathcal{P}_X}). \quad (515)$$

The pointwise proof of Proposition 543 uses only $f_p \in \{0, 1\}$, $\sum_p f_p = K$, and $\sum_p z_p = \Lambda$. Applying it to f_p^Δ gives

$$\left| \mathcal{T}_5^\Delta(X) - \mathcal{W}_5^\Delta(X) \right| \leq \frac{1280\Lambda}{L} \mathcal{T}_5^\Delta(X), \quad \mathcal{T}_5^\Delta(X) = \sum_{\substack{X \leq m < X^2 \\ K_X^{\text{iso}, \Delta}(m; H_X) \geq L}} (K_X^{\text{iso}, \Delta}(m; H_X))_5. \quad (516)$$

Hence Hypothesis 550 is equivalent to the single selector-preserving physical five-wave estimate

$$|\mathcal{W}_5^\Delta(X)| \ll X^{o(1)}(1 + X^2 \Lambda^5). \quad (517)$$

For $m > 2X$ the physical and full-shell rows coincide, so the high-Kummer and far quotient-rainbow reductions above are unchanged.

Theorem 566 (The isolated five-star criterion closes every index). *If Hypothesis 550 holds for one fixed $0 < \eta < 1/15$, then*

$$\log G_n \ll n^{14/15+\eta+o(1)} \quad (n \rightarrow \infty).$$

In particular, $G_n = e^{o(n)}$ for every index.

Proof. By (460) and (461),

$$\Lambda_X^{\text{iso}}(H_X) \ll X^{8/15}(\log X)^{-2/3}. \quad (518)$$

If some $X \leq m < X^2$ had $K_X^{\text{iso}, \Delta}(m; H_X) \geq X^{14/15+\eta}$, then for large X its single summand on the left of (486) would be at least

$$2^{-5} X^{14/3+5\eta}.$$

On the other hand, (518) bounds the right side of (486) by $X^{14/3+o(1)}$. This is a contradiction. Therefore

$$\max_{X \leq m < X^2} K_X^{\text{iso}, \Delta}(m; H_X) < X^{14/15+\eta}$$

for all sufficiently large X . Lemma 539 and $H_X^2 \ll X^{14/15}(\log X)^{-2/3}$ now give

$$\max_{X \leq m < X^2} K_X^\Delta(m) \ll X^{14/15+\eta+o(1)}. \quad (519)$$

Fix n . The blocks below $2\sqrt{n}$ and the companion channel have the same unconditional bounds as in the proof of Theorem 331. In every remaining dyadic block $(X, 2X]$ one has $\sqrt{n} < X \leq n$, hence $X \leq n < X^2$, and the exact number of lower-digit bad primes in the block is $K_X^\Delta(n)$, not $K_X(n)$. Therefore the remaining geometric sum is

$$\sum_{\substack{X=2^j \\ \sqrt{n} < X \leq n}} X^{14/15+\eta+o(1)} \log X \ll n^{14/15+\eta+o(1)}.$$

This proves the claim. □

Remark 567 (Why the first useful order is five). Suppose the physical isolated bank had an independence-scale order- t estimate, with the same lattice correction as in (486). The argument above would give

$$\max_{X \leq m < X^2} K_X^\Delta(m) \ll H^2 + \frac{X^{1+2/t+o(1)}}{H \log X} + X^{o(1)}.$$

Balancing the two main terms gives the exponent

$$\theta_t = \frac{2}{3} + \frac{4}{3t}.$$

Thus $t = 3$ gives $10/9$, and $t = 4$ is the logarithmically insufficient borderline 1, whereas $t = 5$ gives $14/15 < 1$. This is a minimality statement only for the carrier–packing architecture above; it does not exclude a different lower-order Apéry identity.

Remark 568 (Status). Lemma 539 and Theorem 566 are proved. The arithmetic estimate in Hypothesis 550 is not presently known. It is weaker than the full-shell Hypothesis 540 and narrower than (HM)₇: it ignores every nonisolated row, every output below the displayed threshold, all factorial orders above five, and every window prime exceeding the physical index. Its top-strip content is only quotient one; its far content is the same high-Kummer quotient-rainbow five-wave already isolated above.

13.15 Affine folding and the cumulant frontier

The atom tail $K_X(m) = \#\{p \in (X, 2X] : m \bmod p \in \mathcal{Z}_p\}$ admits an exact characteristic-zero carrier, which removes the moving sets $\mathcal{Z}_p$ from the problem.

Lemma 569 (Affine folding). *For every integer m and every prime $X < p \leq 2X$,*

$$m \bmod p \in \mathcal{Z}_p \iff \exists s \in [0, X) : p \mid b_s \text{ and } p \mid (m - s)(m + s + 1).$$

Proof. If $r = m \bmod p \in \mathcal{Z}_p$, put $s = \min(r, p - 1 - r)$. Reflection $r \leftrightarrow p - 1 - r$ and $p \leq 2X$ give $0 \leq s < X$ and $p \mid b_s$. Either $r = s$ (so $p \mid m - s$) or $r = p - 1 - s$ (so $p \mid m + s + 1$). The converse is immediate from $s < p$: $p \mid m - s$ forces $m \bmod p = s \in \mathcal{Z}_p$, while $p \mid m + s + 1$ forces $m \bmod p = p - 1 - s \in \mathcal{Z}_p$ by reflection. $\square$

Hence, writing $\text{rad}_X(n)$ for the product of the primes in $(X, 2X]$ dividing n ,

$$\prod_{\substack{X < p \leq 2X \\ m \bmod p \in \mathcal{Z}_p}} p = \text{rad}_X \left(\prod_{0 \leq s < X} \gcd(b_s, (m - s)(m + s + 1)) \right). \quad (\text{AF})$$

The two ends of the folded-height range can in fact be removed unconditionally. Put

$$H_X = \left\lfloor \frac{\sqrt{X}}{\log X} \right\rfloor, \quad U_X = \left\lfloor \left(1 - \frac{1}{\log \log X} \right) X \right\rfloor,$$

and define the middle-height carrier

$$\mathfrak{R}_X^{\text{mid}}(m) = \text{rad}_X \left(\prod_{H_X < h < U_X} \gcd(b_h, (m - h)(m + h + 1)) \right).$$

Proposition 570 (Free folded-height tails). *Uniformly for $X \leq m \leq X^2$,*

$$\log \mathfrak{R}_X(m) \leq \log \mathfrak{R}_X^{\text{mid}}(m) + o(X),$$

where $\mathfrak{R}_X(m)$ is the exact shell radical in Corollary 378. Consequently

$$\sup_{X \leq m \leq X^2} \log \mathfrak{R}_X^{\text{mid}}(m) = o(X) \implies G_n = e^{o(n)} \text{ for every } n.$$

Proof. Every hit prime has a unique folded label $0 \leq h \leq (p-1)/2$. The product of the hit primes with $h \leq H_X$ divides $\prod_{h \leq H_X} b_h$. Since $\log b_h = O(h)$, its logarithm is

$$O\left(\sum_{h \leq H_X} h\right) = O(H_X^2) = O(X/\log^2 X) = o(X).$$

If $h \geq U_X$, then $p \geq 2h+1 > 2U_X$. The logarithm of the product of all possible primes in this upper tail is at most

$$\vartheta(2X) - \vartheta(2U_X) = o(X),$$

because $X - U_X = o(X)$ and $\vartheta(t) = t + o(t)$. The remaining primes are precisely those retained by $\mathfrak{R}_X^{\text{mid}}(m)$. The final implication is Corollary 378. $\square$

The small indices $s \leq X^{1/3}$ contribute $O(X^{2/3})$ by the Apéry height $b_s = C\alpha^s s^{-3/2}$, $\alpha = 17 + 12\sqrt{2}$, and central hits $2s+1 = p$ force $p \mid 2m+1$, so there are at most two of them. Thus the following estimate is one sufficient pointwise route to the all- n theorem:

$$\sup_{X^2 < m \leq 2X^2} \log \text{rad}_X \left(\prod_{X^{1/3} < s < X} \gcd(b_s, (m-s)(m+s+1)) \right) \ll_{\varepsilon} X^{2/3+\varepsilon}. \quad (520)$$

An independent sufficient cumulant form is sharper in a distributional sense. It is not known to be equivalent to (520). For odd $y \in [1, 4X^2)$ put $\Omega_p = \{2r+1 : r \in \mathcal{Z}_p\}$, $K_X(y) = \#\{p \in (X, 2X] : y \bmod p \in \Omega_p\}$, $\mu_X = \mathbb{E}K_X$, $S_X = K_X - \mu_X$, and let $\kappa_j(S_X)$ be the j -th cumulant.

Hypothesis 571 (Logarithmic connected-tensor bound). *There are absolute $A, c_0 > 0$ such that, for every $2 \leq j \leq 2c_0 \log X$,*

$$|\kappa_j(S_X)| \leq (Aj)^j (1 + \lambda_X), \quad \lambda_X = \sum_{X < p \leq 2X} \frac{Z(p)}{p}.$$

For every even $r \leq 2c_0 \log X$, the finite moment-cumulant coefficient formula and $j! \geq (j/e)^j$ give the Bernstein bound

$$\|S_X\|_r \ll_A \sqrt{r(1 + \lambda_X)} + r.$$

Choose an even $r \asymp c_0 \log X$ and apply Markov with a sufficiently large constant multiple of the right-hand side. The resulting tail probability is smaller than $1/(2X^2)$, the mass of one point in the finite odd-column space. Hence no column lies in that tail. Direct residue counting also gives

$$|\mu_X - \lambda_X| \leq \lambda_X/X,$$

and therefore

$$\max_y K_X(y) \leq \lambda_X + O_{A, c_0} \left(\sqrt{(1 + \lambda_X) \log X} + \log X \right).$$

The proved corank estimate gives $\lambda_X \ll X^{3/5}/\log X$, so the last display is $O(X^{3/5+o(1)})$. Applying it to the geometric prime blocks above $\sqrt{n}$ and using the already proved small-prime and companion bounds gives the all- n theorem directly.

The earlier bound $\max K_X \ll \lambda_X X^{o(1)}$ omitted the additive small-density baseline. Likewise the cumulant hypothesis need not imply the sharp λ_X^t normalization of every factorial moment when λ_X is small. Thus Hypothesis 571 and (520) are strong sufficient inputs. They are not equivalent: the cumulant hypothesis is a global distribution statement, whereas (520) is a uniform radical bound. Neither is the weakest remaining input. By Corollary 378, it is enough to prove the

substantially weaker bound $o(X)$ for the logarithm of the window radical, uniformly in the physical row.

Numerically the actual Apéry data is sub-Poisson. The normalized cumulants $R_j = |\kappa_j(S_X)|/(j^j(1+\lambda_X))$ satisfy, at $X = 512$ and $X = 1024$,

$$R_2, R_3, R_4, \dots = 0.021, 0.0031, 0.00032, \dots \rightarrow 0,$$

while the anchored reflection star (a common translation phase $e_p(hy_0)$ across all primes) has R_j exceeding 1 already at $j = 7$. The connected cumulants $\kappa_j(S_X)$ sit at the Poisson scale λ_X for $j \leq 5$ and then decay (sub-Poisson); the anchored star does not.

14 The I5 reduction theorem

Let $\mathcal{P}_X$ denote the set of primes p with $X < p \leq 2X$. For each $p \in \mathcal{P}_X$ let $Z_p \subseteq \mathbf{F}_p$ be the Apéry zero set. We write

$$K_X(m) = \#\{p \in \mathcal{P}_X : m \bmod p \in Z_p\}, \quad \lambda_X = \frac{1}{|\mathcal{P}_X|} \sum_{p \in \mathcal{P}_X} \frac{|Z_p|}{p}.$$

For the packing argument we fold the zero set using the involution $x \mapsto -x$ on residues.

Definition 572 (Folded and separated zero sets). For $p \in \mathcal{P}_X$ define the folded set

$$Y_p = \{\min(r, p-r) : r \in Z_p\} \subseteq [0, p/2].$$

For an integer parameter $H \geq 1$, define

$$Z_{p,H}^{\text{iso}} = \{y \in Y_p : |y - y'| > H \text{ for all } y' \in Y_p, y' \neq y\},$$

and

$$Z_{p,H}^{\text{near}} = Y_p \setminus Z_{p,H}^{\text{iso}}.$$

The corresponding shell loads are denoted $K_{\text{iso},H}(m)$ and $K_{\text{near},H}(m)$.

Proposition 573 (Packing estimate for isolated folded zeros). *For every $H \geq 1$,*

$$\Lambda_{\text{iso}}(X, H) \leq \min \left(\lambda_X, \frac{1}{\log X} + \frac{X}{H \log X} \right),$$

where

$$\Lambda_{\text{iso}}(X, H) = \frac{1}{|\mathcal{P}_X|} \sum_{p \in \mathcal{P}_X} \frac{|Z_{p,H}^{\text{iso}}|}{p}.$$

Proof. The first inequality follows immediately from $Z_{p,H}^{\text{iso}} \subseteq Y_p$ and the definition of λ_X . For the second estimate, each isolated element of Y_p excludes an interval of length H on the folded residue line. Hence the isolated points occupy pairwise disjoint intervals. Splitting primes into $p \leq H$ and $p > H$, the first contribution is bounded by the trivial prime sum $\ll 1/\log X$, while for $p > H$ the number of possible isolated points is $\ll p/H$. Summing over $p \in \mathcal{P}_X$ gives

$$\frac{1}{|\mathcal{P}_X|} \sum_{p \in \mathcal{P}_X} \frac{|Z_{p,H}^{\text{iso}}|}{p} \ll \frac{1}{\log X} + \frac{X}{H \log X},$$

which proves the claim. □

Lemma 574 (Unconditional near-zero bound). *There is an absolute constant C such that for all $H \geq 1$,*

$$\max_m K_{\text{near},H}(m) \leq CH^2.$$

Proof. A near zero has another folded zero within distance H . Therefore every near hit determines a pair of folded zeros at distance at most H . For a fixed residue representative the number of possible offsets is $O(H)$, and the folding involution contributes another factor $O(H)$. The local spacing property of the Apéry zero sets prevents repeated adjacent choices, so the number of possible pairs contributing to a fixed m is $O(H^2)$. Summing the finite possibilities gives the stated uniform bound. $\square$

Theorem 575 (I5 reduction). *Assume that the I5-tail estimate holds with*

$$H = X^{7/15}(\log X)^{-1/3}, \quad L = X^{14/15+\eta}.$$

Then

$$\max_m K_X(m) \leq X^{14/15+\eta}$$

for sufficiently large X .

Proof. Decompose

$$K_X(m) = K_{\text{iso},H}(m) + K_{\text{near},H}(m).$$

The near contribution is bounded by Lemma 574:

$$K_{\text{near},H}(m) \leq CH^2 = X^{14/15}(\log X)^{-2/3}.$$

Thus it is smaller than $X^{14/15+\eta}$.

For the isolated contribution, the fifth factorial moment expansion below and the assumed I5-tail estimate give

$$(K_{\text{iso},H}(m))_5 \ll L^5.$$

Since $(k)_5 \gg k^5$ for k sufficiently large, this implies $K_{\text{iso},H}(m) \ll L$. Combining the two estimates proves the result. $\square$

Proposition 576 (Exact factorial identity). *For every integer m ,*

$$(K_{\text{iso},H}(m))_5 = \sum_{\substack{p_1, \dots, p_5 \in \mathcal{P}_X \\ \text{distinct}}} \prod_{i=1}^5 1_{Z_{p_i, H}^{\text{iso}}}(m \bmod p_i).$$

Proof. This is the definition of the falling factorial: it counts ordered choices of five distinct prime hits contributing to $K_{\text{iso},H}(m)$. $\square$

Proposition 577 (CRT identity with least representative). *For distinct primes $p_1, \dots, p_5$ and residues $r_i \in Z_{p_i, H}^{\text{iso}}$, let $M = p_1 \cdots p_5$ and let $m_0 \in [0, M)$ be the least CRT representative satisfying*

$$m_0 \equiv r_i \pmod{p_i} \quad (1 \leq i \leq 5).$$

Then the contribution of this ordered prime tuple to the factorial moment is exactly

$$\#\{m \leq N : m \equiv m_0 \pmod{M}\} = \frac{N}{M} + O(1).$$

Proof. The Chinese remainder theorem gives a unique residue class modulo M . Counting integers in this class below N gives the displayed formula. $\square$

Proposition 578 (Fourier and character formulation). *Let f_p be the indicator of $Z_{p,H}^{\text{iso}}$ on $\mathbf{F}_p$. Then the I5 expression may be written using additive Fourier expansion*

$$f_p(x) = \frac{1}{p} \sum_{t \in \mathbf{F}_p} \widehat{f}_p(t) e_p(tx),$$

where

$$\widehat{f}_p(t) = \sum_{x \in \mathbf{F}_p} f_p(x) e_p(-tx).$$

After inserting the CRT identity, the zero frequencies give the main term and the nonzero frequencies are precisely the I5-tail terms. Equivalently, after multiplicative decomposition, the same terms are expressed through the corresponding character sums over the prime tuple variables.

Proof. The identity follows by finite Fourier inversion on each residue field. Multiplying the expansions for five distinct primes and summing over the CRT classes separates the trivial character contribution from all nontrivial frequency terms. No estimate is used in this proposition; only the I5-tail hypothesis bounds the nonzero-frequency contribution. $\square$

15 Conclusion

The principal unconditional result is the quantitative exceptional-set theorem: for every fixed $\varepsilon > 0$,

$$\#\{n \leq X : \log G_n > \varepsilon n\} \ll_{\varepsilon} (\log X)^2.$$

Consequently $G_n = e^{o(n)}$ on a set of natural density one; the exceptional set has upper Banach density zero and finite harmonic weight. The arithmetic engine is the uniform interval estimate $Z(p) \ll p^{2/3}$, proved by the gap-polynomial and separated-block method.

The all- n theorem would follow from the block second-factorial-moment estimate

$$F_N \ll T_N + \frac{T_N^2}{N}, \tag{521}$$

which asserts that prime columns in the top block do not systematically align at one integer. The profile decomposition of F_N separates this into the deterministic profile energy $\|\rho\|^2 \sim c_2 T^2/N$ (with c_2 given exactly by (342)), a cross-term $|\langle \rho, \eta \rangle| \leq \rho_{\max} T = O(T^2/N)$, and the residual shot-noise term $\|\eta\|^2$. Under the profile and cross-term bounds of Remark 316, the estimate (521) is equivalent to the shot-noise bound $\|\eta\|^2 = O(T)$. Computation through $N = 262,144$ gives $\|\eta\|^2/T \approx 1.0$.

This gap is genuine. All classical sieves (Selberg, Brun, Rosser–Iwaniec, GPY) are circular for (521): they expand into the same intersection counts $A_{pq}(N)$ that the estimate controls, and their quadratic-form optimization requires the two-prime level of distribution as input (Remark 312). On the geometric side, the K3 lisse sheaf $R^2 f_* \mathbb{Q}_{\ell}$ has bounded rank and conductor, but the event $p \mid b_j$ is a same-characteristic crystalline condition. Theorem 271 makes the K3 geometry explicit. Under the arithmetic relative-correspondence hypothesis (RC) of Theorem 275, the primitive toric-fiber trace is moreover a rank-three symmetric-square/Asai trace of the elliptic curve (218), and the cyclic 6-isogeny carries the constant quadratic twist -3 . Independently of (RC), Theorem 277 constructs a direct rank-three elliptic/Asai sheaf whose pointwise trace is

congruent to the toric trace modulo p . After choosing one prime $\mathfrak{p} \mid p$ and the Kummer character normalized at that prime, its middle-extension Mellin cohomology has dimension two, weight three, and trace congruent to b_j modulo $\mathfrak{p}$. The character must vary with $\mathfrak{p}$: for a fixed global character the conjugate reductions are $b_{\langle uj \rangle}$, not copies of the same b_j . Thus this removes the former endpoint hypotheses from the modulo- p realization but does not supply a horizontal zero-count estimate. Proposition 286 gives the exact defining-characteristic counterpart,

$$K_p(x)^2 = \sum_{m=0}^{p-1} b_m x^m (1-8x)^m (1+x)^{p-1-m}.$$

Thus $Z(p)$ is exactly the number of zero coordinates of K_p^2 in this cover-adapted basis. It is not the monomial support of K_p^2 , nor does this identity turn coefficient vanishing into an evaluation-root problem. The Jacobi counterexample shows that a rank-one sheaf with the same formal properties can have $\gg p$ bad tame characters. The gap-polynomial identities are monochromatic: they control two incidences of the *same* prime and do not constrain the rainbow intersections that determine $\|\eta\|^2$.

Two precise sufficient criteria for Hypothesis Z emerge: a bounded-degree tame Hasse polynomial (giving $Z(p) \leq 2D$), or a unique-unit-root theorem for all but $O(1)$ tame twists. The fiberwise Hasse–Witt invariant $A_p(t) = \sum_{n < p} b_n t^n$ is explicitly known from Stienstra–Beukers [33], but it controls which *fibers* are nonordinary, not which *Mellin coefficients* vanish; the character-twisted Hasse computation is a fundamentally different problem on the global twisted cohomology $H_{\text{rig},c}^1(U, \mathcal{V}_{\text{tr}} \otimes \mathcal{K}_{\omega^{-j}})$. The raw interpolation $j \mapsto b_j$ has degree $p-2$ already for $p = 11$ and 13 ; moreover, the Sym^2 factorization forces at least $(p-5)/2$ nonzero Fourier coefficients, giving a rigorous lower bound $D \geq (p-7)/4$ on any Laurent polynomial of degree D representing the unnormalized Mellin transform. Bounded tame Hasse degree would require a nontrivial Gauss/Jacobi normalization of the first Frobenius block. Even such a uniform single-prime bound would not, by itself, settle the all-index GCD conjecture: the pointwise gap is the possible synchronization of zero positions across distinct primes. Moreover, the observed Poisson tail predicts that Hypothesis Z itself is false.

Computation through all 78,496 primes $p \leq 10^6$ gives $Z(p) \leq 12$ and mean ≈ 1 ; the value-collision energy ratio $E(p)/p \rightarrow 3$ (confirmed through $p = 100,000$) and the profile constant $c_2 \rightarrow 1.262$ (through $N = 262,144$) match the palindromic null at every tested scale. These observations support all conjectures but are not used as proofs.

The prime-wise decomposition (Remark 319) provides a structural reduction: $\|\eta\|^2 = D + O$ where $D = \sum_p D_p$ is an explicit diagonal ($D = T + O(1/\log N)$ under $\sum_p Z(p)^2 = O(N/\log N)$) and $O = \sum_{p \neq q} C_{pq}$ is the cross-prime interaction. Under the independent fixed-cardinality residue model (each Z_p drawn uniformly from $(\mathbb{F}_p / Z(p))$, independently across primes), one obtains the exact variance formula

$$\text{Var}(\|\eta\|^2) = \sum_p \text{Var}(D_p) + \sum_{p \neq q} \text{Var}(C_{pq}),$$

whose first sum is $O(T)$ (fluctuation of the active tail mass Z_p^{act}) and whose second is $O(T^2/N)$ (CRT averaging in intervals of length $\sim N$ within period $pq \sim N^2$). The null-model conclusion $\|\eta\|^2/T \rightarrow 1$ in L^2 is therefore unconditional within the model.

However, CRT does not imply independence for *deterministic* zero-sets: distinct residue fields provide separate coordinates, not a product measure. The adversarial construction $Z_p = \{2N \bmod p\}$ for all p places every shifted zero at $m = 2N$, giving $O = \Theta(T^2)$; this is a single explicit moving-target choice with $|Z_p| = 1$ for all p , so the two-prime dispersion statement is *false* for

general $O(1)$ -size zero-sets. For the Apéry sequence, this pattern is impossible: the fixed-column $Z_p = \{c - p \bmod p\}$ forces $p \mid b_c$ for all p , contradicting $b_c \in \mathbb{Z} \setminus \{0\}$. More generally, the palindromic structure (Z_p consisting of reflected pairs $\{j_p, p-1-j_p\}$) and the Galois-group structure of the gap polynomials prevent systematic alignment of zero positions across primes. Nonetheless, the quantitative control—that the residual alignment is $o(T)$, not merely $o(T^2)$ —is a stronger statement, and cannot be established by any purely black-box argument: Bombieri–Vinogradov, the large sieve, and Barban–Davenport–Halberstam address single-modulus discrepancies, not cross-prime collisions, and the spectral bilinear framework of Duke–Friedlander–Iwaniec (“bilinear forms with Kloosterman fractions”) requires a fixed numerator, which the moving-target zero-sets do not provide.

A sharper reduction emerges from the moment collapse (Proposition 426): the full moment hierarchy $(\text{HM})_k$ for *every* k follows from a single pointwise bound

$$\max_{m < X^2} K_X(m) \leq \lambda_X \cdot X^{o(1)} \quad (\text{Hypothesis (AT'')}).$$

The concentration theorem (Theorem 435) shows this is equivalent to the atom-tail bound $n_T \ll X^{2+o(1)} \lambda_X^3 / T^3$, improving Chebyshev’s unconditional $n_T \leq 5X^2 \lambda_X^2 / T^2$ by a single factor T/λ_X . Every ensemble method examined—large-sieve dispersion, Linnik dispersion with completed CRT sums, variance layer-cake, codegree-weighted pair dispersion—either requires this atom bound at its diagonal or is subsumed by the moment collapse. The Cartier diagonal representation (Theorem 385) places $\mathcal{Z}_p$ inside one fixed algebraic family, and the codegree dichotomy (Theorem 413) repairs the wrap gap in the separated-block codegree proof; neither alone closes the atom tail, because every certificate algebra controls pairwise coincidences at a single prime, while atoms consist of T primes each contributing a single coincidence.

At the next corank level, the exact surviving obstruction is now sharper. It would suffice to prove that, for some fixed $\eta > 0$, every four consecutive Apéry zeros of span at most $\eta\sqrt{p}$ contain a reflection-centered adjacent pair; this would give $Z(p) \ll_\eta \sqrt{p}$ pointwise. Alternatively, it is enough that primitive off-center projective return chains with a non-all-equal gap vector, outside the structural carrier, satisfy $p \leq Cs^2$ for some absolute C ; the level-adaptive exceptional-prime argument then gives the square-root exponent on average. The corresponding constant-one statement for primitive projective returns is false. The fully deflated gap triple $(5, 20, 10)$ modulo 1297 has a primitive off-center common root $x = 360$ with span 35, $1297 \nmid \mathfrak{U}_{35}$, and $1297 > 35^2$, together with its reflected root 901, but neither belongs to the distinguished zero fiber:

$$\mathcal{Z}_{1297} = \{459, 530, 766, 837\}.$$

The phenomenon is not confined to constants near one. An exact prime-first census through $p \leq 500000$ finds a primitive off-center nondistinguished chain

$$(p, x; a, b, c) = (128047, 42375; 41, 86, 37), \quad \frac{p}{(a+b+c)^2} = \frac{128047}{164^2} = 4.760819\dots$$

Its return-offset set is exactly $(0, 41, 127, 164)$ and $128047 \nmid \mathfrak{U}_{164}$. This finite computation does not disprove the existence of some absolute quadratic-support constant, but the standard random-fiber heuristic predicts formal short chains for every fixed constant. Thus the remaining square-root problem is not formal resultant saturation; it is the coupling of short returns to the distinguished Apéry orbit.

Transfer of the null-model theorem to the actual Apéry zeros is captured by the following precise conjecture.

Conjecture 579 (Two-prime dispersion). For $N \rightarrow \infty$ and primes $p, q \in (N, 2N]$ with $p \neq q$, define $C_{pq} = \sum_m X_{m,p} X_{m,q}$. Then

$$\sum_{\substack{p, q \in (N, 2N] \\ p \neq q}} |C_{pq}| = o(T_N).$$

This absolute centered-correlation estimate is stronger than the BFM2 bound (521); in particular, it is a sufficient condition for the all- n conjecture $G_n = e^{o(n)}$. No converse implication is proved; in particular, the pointwise GCD conjecture does not presently yield an $O(T_N)$ second factorial-moment estimate.

Remark 580 (Anatomy-of-integers formulation). For each $j \leq N$, define $\omega(j) = \#\{p \in (N, 2N] : p \mid b_j\}$. Then $\sum_j \omega(j) = T$ (counting incidences between the integers and the primes in the top block), and

$$\sum_{p \neq q} \sum_{j \leq N} \mathbf{1}[pq \mid b_j] = \sum_{j \leq N} \omega(j)(\omega(j) - 1).$$

Thus $\sum_j \omega(j)(\omega(j) - 1) = O(T)$ is the raw second-factorial-moment bound. The sharper estimate $\sum_j \omega(j)(\omega(j) - 1) = o(T)$ says that at most $o(T)$ indices j are divisible by two or more primes from the same dyadic window. The absolute centered-correlation conjecture above is a sufficient route to such control; it is not equivalent to this uncentered factorial-moment statement. Since $\log b_N \sim N \log(17 + 12\sqrt{2}) \approx 3.53 N$ while $\sum_{p \in (N, 2N]} \log p \sim N$, each b_j has room in principle for every prime in the window as a factor. Pure size/counting arguments cannot exclude this conspiracy; the Poisson prediction relies on the equidistribution of the p -adic digits of b_j across independent residue fields.

Table 8 verifies the anatomy-of-integers formulation directly: at every tested scale, the distribution of $\omega(j)$ matches $\text{Poisson}(\lambda)$ with $\lambda = T/(N+1)$ to within sampling noise. In particular, $\max_j \omega(j) \leq 3$ through $N = 131,072$, consistent with the Poisson prediction $N \cdot \lambda^3/6 \approx 5$ at this scale. The second factorial moment satisfies $\sum_j \omega(j)(\omega(j) - 1)/T \approx \lambda$, which is exactly the collision rate $T/N \approx 1/\ln N$ predicted by Mertens—the same rate appearing in the two-prime scaling law below.

Table 8: Distribution of $\omega(j) = \#\{p \in (N, 2N] : p \mid b_j\}$. $\lambda = T/(N+1)$ is the mean; the Poisson column gives $e^{-\lambda} \lambda^k / k!$.

N	T	λ	$\max \omega$	$\#\{\omega \geq 2\}$	$\sum \omega(\omega - 1)/T$	Poisson fit
8192	608	0.074	2	22	0.072	$< 0.1\%$
32768	2048	0.063	3	53	0.056	$< 0.1\%$
65536	3944	0.060	3	108	0.057	$< 0.1\%$
131072	7584	0.058	3	227	0.063	$< 0.1\%$
262144	14104	0.054	3	351	0.051	$< 0.1\%$

Computation through $N = 262,144$ gives $|O|/T < 2 \times 10^{-3}$ (Table 7), far stronger than the conjecture requires.

Table 9 decomposes the cross-prime sum into its unsigned magnitude. The product $(\ln N) \cdot \Sigma_{\text{abs}}/T$ remains approximately constant at ≈ 1.08 , giving the empirical scaling law

$$\frac{\sum_{p \neq q} |C_{pq}|}{T_N} \approx \frac{1.08}{\ln N}.$$

This is explained by a counting argument. Since $L \leq p$ for primes $p > N$, the map $n \mapsto n \bmod p$ restricted to $[1, L]$ is injective; CRT lifts at coprime moduli hit the interval at most once, giving the

Table 9: Two-prime dispersion scaling. $\Sigma_{\text{abs}} = \sum_{p \neq q} |C_{pq}|$ measures the total unsigned cross-prime interaction. The ratio Σ_{abs}/T decreases as $\approx c/\ln N$ with $c \approx 1.08$.

N	T	Σ_{abs}	Σ_{abs}/T	$\ln N \cdot \Sigma_{\text{abs}}/T$	joint
8192	354	43.1	0.122	1.10	10
16384	664	75.3	0.113	1.10	17
32768	1182	123.2	0.104	1.08	29
65536	2237	214.1	0.096	1.06	47
131072	4317	391.1	0.091	1.07	87
262144	7940	682.3	0.086	1.07	156

sharp pointwise bound $|C_{pq}| \leq \min(Z(p), Z(q))$. In practice, most pairs have no joint CRT hit (the “joint” column of Table 9 counts pairs with at least one), so $|C_{pq}| \approx Z(p)Z(q)N/(pq) = O(1/N)$ for the bulk. With $O(N^2/(\ln N)^2)$ pairs and $T = \Theta(N/\ln N)$, the sum-to- T ratio is $O(1/\ln N)$. The unsigned ratio $\Sigma_{\text{abs}}/T \sim 1/\ln N$ is therefore essentially *forced*—it measures the expected collision count $\sum_{p \neq q} E_{pq} \approx T^2/N$, which is $\Theta(T/\ln N)$ by Mertens. The genuine arithmetic content is in the *signed* sum: $|O|/T < 2 \times 10^{-3}$ says that the actual collision count $\sum_m r(m)(r(m)-1)$ matches its Poisson prediction to within a few percent.

For non-CM motives over $\mathbb{Q}$, independence of Frobenius traces at distinct primes is a consequence of the generalized Sato–Tate conjecture (known for GL_2 by the work of Barnet-Lamb–Geraghty–Harris–Taylor, extended to the symmetric square L -function). However, the single-prime $Z(p)$ statistics are a *vertical* problem—equidistribution of Mellin-transform values as the character varies within one $\mathbb{F}_p$ —for which Katz’s finite-field Mellin equidistribution theory [20] is the right tool. The two-prime dispersion is a *horizontal* problem: correlating arithmetic events at distinct primes for a fixed motive, which is of Lang–Trotter type. No equidistribution theorem over a single finite field can see cross-prime structure, and no analog of the joint Sato–Tate theorem across two distinct primes exists in the literature. The two-prime dispersion asks for a compatible statement at the level of zero-counts $Z(p)$, which lives in the Kummer-twisted cohomology rather than the fiber-variable cohomology. The adversarial construction $\mathcal{Z}_p = \{2N \bmod p\}$ shows that any proof must use Apéry-specific structure (cf. Friedlander–Granville [13] for the general impossibility of target-agnostic equidistribution beyond the large-sieve range).

The strongest existing results for structured moving targets in sieve theory follow the Duke–Friedlander–Iwaniec template [11], which proves equidistribution of roots of a quadratic congruence to moduli beyond $\sqrt{x}$ using an integral formula over automorphic forms on GL_3 . Their method requires the target set $\mathcal{Z}_p$ to arise as roots of a *fixed* polynomial with a global algebraic-variety structure—precisely the ingredient the Apéry recurrence lacks (the degree grows linearly with p , and no single variety parametrizes the zero set). The only result achieving the moving-target beyond- $\sqrt{x}$ barrier without a fixed polynomial degree is Maynard’s breakthrough [23], which handles targets of size $O(p^{1/3})$ in arithmetic progressions using new estimates for trilinear Kloosterman sums—still requiring a multiplicative character structure absent from the Apéry setting. In summary, this sufficient two-prime dispersion route for Apéry numbers falls in the gap between what sieve methods can reach (targets with global algebraic structure) and what the Apéry recurrence provides (a small-on-average, p -dependent set with only the unconditional bound $Z(p) \ll p^{2/3}$).

The Diophantine-approximation approach to GCDs of recurrence terms—the Subspace-Theorem framework of Bugeaud–Corvaja–Zannier [9], Levin [22], and Xiao [35]—requires the sequence to admit a finite exponential-polynomial representation (a constant-coefficient linear recurrence). The Apéry sequence satisfies a recurrence with polynomial coefficients and obeys $b_n \sim C\alpha^n n^{-3/2}$, where

the fractional exponent $-3/2$ is incompatible with constant-coefficient recurrences. Moreover, even in the exponential-polynomial setting, same-sequence GCD bounds fail: $\gcd(2^m-1, 2^n-1) = 2^{\gcd(m,n)}-1$ is exponentially large when $m \mid n$. Therefore no existing Subspace-Theorem argument yields $G_n = e^{o(n)}$ for all n .

The exact Franel pullback now gives a concrete single-prime coefficient problem, and Proposition 287 reduces its distinguished factor to two universal three-term recurrences. This is useful for the average target $\sum_{p \leq X} Z(p) \ll \pi(X)$ and for testing genuinely Apéry-specific Cartier identities. It does not couple two different residue characteristics. Consequently, the highest-value target for the full all-index conjecture remains cross-prime dispersion—for example the fixed high-moment condition $(\text{HM})_7$ of Theorem 331—rather than a uniform $O(1)$ bound for the cover-adapted coordinates.

On the single-prime side, the bordered-certificate analysis (Section 5.4) settles the value-distribution question at the $3/4$ exponent: $\max_a N_p(a) = O(p^{3/4})$ and $E(p) = O(p^{7/4})$ unconditionally, with the certificate nonvanishing (Theorem 37) sharp at $k < p$. By the impossibility construction discussed in Section 12, no package of single-prime distributional facts—including these—can close the pointwise conjecture; their role is in the averaged theorems and as input to any future cross-prime argument.

Finally, the open problem now has a sharply localized form. The quotient reduction (Proposition 104) gives a uniform slowly growing quotient formulation, whereas Theorem 380 gives the weaker exact criterion: control the lower-digit channel in every fixed quotient class separately, with no growing- q uniformity. Among equivalent one-integer formulations, it is precisely the assertion that the n -smooth squarefree part of b_n has logarithmic height $o(n)$. Among all sufficient conditions collected here, the single fixed factorial moment $(\text{HM})_7$ (Hypothesis 327), proved unconditionally for $k = 2$ (Proposition 328), appears to be the weakest cross-prime statement that closes the full conjecture (Theorem 331).

16 Structure of the marked scalar, and the exact location of the obstruction

Throughout, $\Lambda(x, y, z) = \frac{(1+x)(1+y)(1+z)((1+y)(1+z) + xyz)}{xyz} = \sum_{\kappa \in P} \lambda_\kappa X^\kappa$ is the Apéry Laurent polynomial, $b_r = \text{CT } \Lambda^r$, and $C_M(d)$ denotes the d -section of the coefficient array of Λ^M .

16.1 The marked scalar is a moment of point counts

Proposition 581. *For every prime $p \geq 5$ and every M ,*

$$C_M(p-1) \equiv - \sum_{x,y,z \in \mathbb{F}_p^\times} \Lambda(x,y,z)^M \equiv - \sum_{t \in \mathbb{F}_p^\times} t^r N_p(t) \pmod{p}, \quad r = M \bmod (p-1),$$

where $N_p(t) = \#\{(x, y, z) \in (\mathbb{F}_p^\times)^3 : \Lambda(x, y, z) = t\}$. Consequently, writing $n = p + r$, the target condition $p \mid b_n$ is exactly the vanishing modulo p of the r -th moment of the point-count function of the Apéry family.

Audit: `q32_marked_scalar_character_sum.py` — 95 shell-versus-exponential-sum checks, 95 shell-versus-moment checks, 44 shell-versus-Apéry checks.

16.2 The reflection law, and the parity of $|Z_p|$

Proposition 581 converts a geometric symmetry into an arithmetic one.

Proposition 582. $N_p(t) = N_p(t^{-1})$ for every $t \in \mathbb{F}_p^\times$; hence

$$b_{p-1-r} \equiv b_r \pmod{p} \quad (0 \leq r < p-1),$$

so $Z_p = \{r < p : p \mid b_r\}$ is stable under the involution $r \mapsto p-1-r$, and $|Z_p|$ is even unless the fixed point $\frac{p-1}{2}$ lies in Z_p .

Inversion-invariance is not new to the project: it is the hypothesis defining the space V_p^+ in the random model recorded in DOCTRINE.md (Q5387), which yields $K \sim \text{Bin}((p-3)/2, 1/p) \rightarrow \text{Poisson}(1/2)$. What Proposition 582 adds is that the invariance is a *theorem* for the actual sequence, derived from the geometry through Proposition 581, together with the exact classification of the exceptions: the Poisson law is obeyed by the *pair* count $|Z_p|/2$ because the involution forces the pairing, and the “central exceptions” to evenness are exactly the fixed-point primes. For $p < 4000$ these are $p = 11$ with $Z_{11} = \{5\}$ and $p = 3137$ with $Z_{3137} = \{1568\}$, and in both cases the unique zero equals $(p-1)/2$.

Audit: same script — 420 inversion checks, 420 palindromy checks (the ratio b_{p-1-r}/b_r is identically 1, not merely ± 1), and the involution verified on all 548 primes $5 \leq p < 4000$ without exception.

16.3 Parity of $|Z_p|$ is governed by ordinarity

Combining Proposition 582 with the Beukers/Ahlgren–Ono supercongruence $b_{(p-1)/2} \equiv a_p \pmod{p^2}$ for the newform 8.4.a.a gives

Proposition 583. $|Z_p|$ is odd if and only if p is non-ordinary for 8.4.a.a, i.e. $p \mid a_p$.

Proof. By Proposition 582, Z_p is stable under $r \mapsto p-1-r$, whose unique fixed point is $(p-1)/2$; hence $|Z_p|$ is odd exactly when $(p-1)/2 \in Z_p$, i.e. when $p \mid b_{(p-1)/2}$, which by the supercongruence is equivalent to $p \mid a_p$. $\square$

Verified over all 2260 primes $5 \leq p \leq 20\,000$: $|Z_p|$ is odd exactly at $p = 11$ and $p = 3137$, and these are exactly the primes with $p \mid a_p$ — no mismatches. Since non-ordinary primes have density zero, $|Z_p|$ is even for almost all p ; this is the structural reason the Poisson law of the ledger is stated for the pair count $|Z_p|/2$.

16.4 Where G_n actually lives

The factorisation below is already recorded in DOCTRINE.md (Q5386); what is added here is the direct computation of both factors.

Proposition 584. Write $a_n = A_n/D_n$ in lowest terms, so $D_n \mid d_n$. Then $G_n = \gcd(d_n a_n, d_n b_n) = (d_n/D_n) \cdot \gcd(A_n, b_n)$, and for every $n \leq 330$:

$$\gcd(A_n, b_n) = 1,$$

so $G_n = d_n/D_n$ exactly. Moreover the large prime factors of G_n are precisely the targets $p \in (n/2, n]$ with $p \mid b_n$; e.g.

$$G_{200} = 2 \cdot 3^3 \cdot 5 \cdot 17 \cdot 19 \cdot \mathbf{139} \cdot \mathbf{181}, \quad G_{321} = 2 \cdot 3 \cdot 5 \cdot 7^2 \cdot \mathbf{179} \cdot \mathbf{193} \cdot \mathbf{211},$$

and G_n never exceeds 34 bits on that range.

So the conjecture is equivalent to the statement that Apéry's classical denominator $d_n = \text{lcm}(1, \dots, n)^3$ is optimal up to $e^{o(n)}$, the only losses coming from top-window primes dividing b_n .

Audit: `q32_actual_Gn_audit.py`.

16.5 The target count

Proposition 585. *Let $K(n) = \#\{p \in (n/2, n] : p \mid b_n\}$, so that the quantity to be bounded is $T(n) = \sum_p \log p \leq K(n) \log n$. Then*

$$K(n) \leq 3 \quad \text{for every } n \leq 400\,000,$$

with 412 indices having $K \geq 2$, exactly 12 having $K = 3$, and none $K \geq 4$; the running maximum increases only at $n = 6, 200, 321$. The mean of K is ≈ 0.073 , matching the Mertens prediction $\log 2 / \log n$. The same computation gives $\sum_{p \leq 200\,000} |Z_p| = 18\,126$ (mean 1.016) and $\max |Z_p| = 12$ at $p = 159\,977$.

Audit: `q32_k_scan.c`, validated against the independent Python implementation `q32_top_window_target_counts.py` on $n \leq 20\,000$.

16.6 The denominator-defect law, and what it discharges unconditionally

Write $G_n = \prod_p p^{e_p(n)}$ with $e_p(n) = v_p(d_n) - v_p(D_n) \geq 0$; since $D_n \mid d_n$ we always have

$$0 \leq e_p(n) \leq v_p(d_n) = 3 \lfloor \log_p n \rfloor. \quad (*)$$

The law below is in fact a consequence of an exact congruence for the *numerator*, which removes the empirical step from the reduction.

Proposition 586 (numerator congruence). *For every prime $p \geq 7$ and every n with $n/2 < p \leq n$,*

$$\boxed{p^3 a_n \equiv \frac{6}{5} b_n \pmod{p}.$$

Consequently, if $p \nmid b_n$ then $p^3 a_n$ is a p -unit, so $v_p(D_n) = 3$ and $e_p(n) = 0$: only primes dividing b_n can contribute to G_n .

Writing $n = p + r$ with $0 \leq r < p$, the statement takes its natural form as a *vector* Lucas property:

$$\boxed{(p^3 a_{p+r}, b_{p+r}) \equiv b_r \cdot (a_1, b_1) = b_r \cdot (6, 5) \pmod{p}.$$

The second coordinate is the classical Apéry–Lucas congruence $b_{p+r} \equiv 5 b_r$; the first, $p^3 a_{p+r} \equiv 6 b_r$, is its numerator counterpart, the factor p^3 appearing precisely because the leading coefficient $(n+1)^3$ of the recurrence vanishes at $n = p - 1$. So modulo p the solution vector collapses onto the line spanned by the initial vector (a_1, b_1) , which is exactly why a prime can contribute to G_n only through b_n .

Proof. Write $n = p + r$ with $0 \leq r < p$, and recall Apéry's formula $a_n = \sum_k \beta_{n,k} (H_n^{(3)} + c_{n,k})$ with $\beta_{n,k} = \binom{n}{k}^2 \binom{n+k}{k}^2$ and $c_{n,k} = \sum_{m \leq k} \frac{(-1)^{m-1}}{2m^3 \binom{n}{m} \binom{n+m}{m}}$.

Step 1. Since $n < 2p$, the only multiple of p in $[1, n]$ is p itself, so $H_n^{(3)} = p^{-3} + h$ with h p -integral, whence $p^3 H_n^{(3)} \equiv 1 \pmod{p}$.

Step 2. In $c_{n,k}$ the denominator $2m^3 \binom{n}{m} \binom{n+m}{m}$ has $v_p = 3$ only for $m = p$. Indeed $v_p(m^3) = 3$ forces $m = p$ because $m \leq n < 2p$; and $v_p\left(\binom{n}{m} \binom{n+m}{m}\right) \leq 2$, since $n, n+m < p^2$ gives $v_p \leq 1$ for each binomial by Kummer. Every term with $v_p \leq 2$ contributes $p^{3-v_p} \equiv 0 \pmod{p}$ after multiplication by p^3 . The surviving $m = p$ term occurs exactly for $k \geq p$.

Step 3. By Lucas, $\binom{n}{p} = \binom{p+r}{p} \equiv 1$ and $\binom{n+p}{p} = \binom{2p+r}{p} \equiv 2 \pmod{p}$, so the $m = p$ term contributes $p^3 \cdot \frac{(-1)^{p-1}}{2p^3 \binom{n}{p} \binom{n+p}{p}} \equiv \frac{1}{4} \pmod{p}$.

Step 3 $\frac{1}{2}$ (why only edge terms survive). For $p \in (n/2, n]$ Kummer's theorem gives

$$v_p\left(\binom{n}{k}\right) = \begin{cases} 1, & n - p < k < p, \\ 0, & \text{otherwise,} \end{cases}$$

since adding k and $n - k$ in base p produces a carry exactly in that range (checked on 3267 triples (n, p, k) with no mismatch). Hence modulo p only the *edge* terms $k \leq n - p$ and $k \geq p$ of $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ survive, which is the split used next.

Step 4. Again by Lucas, for $k < p$ one has $\binom{n}{k} \equiv \binom{r}{k}$ and $\binom{n+k}{k} \equiv \binom{r+k}{k}$, while for $k = p + k_0$ with $k_0 \leq r$ one has $\binom{n}{k} \equiv \binom{r}{k_0}$ and $\binom{n+k}{k} \equiv 2 \binom{r+k_0}{k_0}$. Hence

$$\sum_{k < p} \beta_{n,k} \equiv b_r, \quad \sum_{k \geq p} \beta_{n,k} \equiv 4b_r \pmod{p},$$

and adding gives the classical $b_n \equiv 5b_r$.

Step 5. Combining,

$$p^3 a_n = \sum_k \beta_{n,k} (p^3 H_n^{(3)} + p^3 c_{n,k}) \equiv \sum_k \beta_{n,k} + \frac{1}{4} \sum_{k \geq p} \beta_{n,k} \equiv 5b_r + \frac{1}{4} \cdot 4b_r = 6b_r \pmod{p}. \quad \square$$

General form. The statement extends to all digits:

$$(p^3 a_{qp+r}, b_{qp+r}) \equiv b_r \cdot (a_q, b_q) \pmod{p}, \quad 1 \leq q < p, \quad 0 \leq r < p.$$

Verified on 9837 checks per coordinate over all $7 \leq p < 80$, $1 \leq q < p$, $0 \leq r < p$ with $qp + r < 700$: no violations.

The proof for $q \geq 2$ does not extend the $q = 1$ argument verbatim (Step 2 there used $m \leq n < 2p$, whereas for $q \geq 2$ every $m = tp$ with $t \leq q$ has $v_p(m^3) = 3$). It splits instead into a boundary computation and a propagation.

Proof of the general form. (A) *Boundary*, $n = qp$. The multiples of p in $[1, qp]$ are $p, 2p, \dots, qp$, so $H_{qp}^{(3)} = p^{-3} H_q^{(3)} + (p\text{-integral})$ and $p^3 H_{qp}^{(3)} \equiv H_q^{(3)}$. By Kummer, $v_p\left(\binom{qp}{k}\right) = 0$ exactly when $p \mid k$: writing $k = k_1 p + k_0$, a nonzero k_0 forces a units-digit carry, while for $k_0 = 0$ the top digits add to $q < p$ without carry. Hence $v_p(\beta_{qp,k}) \geq 2$ for $p \nmid k$, and for $k = k_1 p$ one has $\beta_{qp,k_1 p} \equiv \beta_{q,k_1}$. The last step is cleanest in the relative form: for $A \geq B \geq 0$ put $U_p(A, B) = \binom{Ap}{Bp} / \binom{A}{B}$; deleting the multiples of p from the three factorials shows $U_p(A, B)$ is a p -adic unit, and each remaining block $\prod_{r=1}^{p-1} (ip+r) \equiv (p-1)!$ gives $U_p(A, B) \equiv 1 \pmod{p}$. Then $\beta_{qp,jp} = \beta_{q,j} U_p(q, j)^2 U_p(q+j, j)^2 \equiv \beta_{q,j}$, which stays valid even when $\binom{q}{j}$ is itself divisible by p (both sides then vanish). Only p odd and $1 \leq q < p$ are needed. In $c_{qp,k}$, a term with $p \nmid m$ has $v_p(m^3) = 0$ and $v_p\left(\binom{qp}{m} \binom{qp+m}{m}\right) \leq 2$, so its denominator has $v_p \leq 2$ and it dies after multiplication by p^3 . The bound is exact, by Kummer: writing $m = m_1 p + m_0$ with $m_0 \neq 0$,

- adding m and $qp - m$ in base p , the units digits are m_0 and $p - m_0$, which sum to exactly p — so there is always a carry at position 0; and no carry can leave position 1 because $qp < p^2$. Hence $v_p\left(\frac{qp}{m}\right) = 1$ exactly.
- adding m and qp , the units digit is $m_0 + 0 < p$, so the only possible carry is at position 1, occurring iff $m_1 + q \geq p$. Hence $v_p\left(\frac{qp+m}{m}\right) \leq 1$.

(Both statements verified on 34578 pairs with no violation.) A term with $m = tp$ has $v_p(m^3) = 3$ and $\left(\frac{qp}{tp}\right)\left(\frac{qp+tp}{tp}\right) \equiv \binom{q}{t}\binom{q+t}{t}$ by Lucas, while $(-1)^{tp-1} = (-1)^{t-1}$ for odd p .

The c -summands must be weighed *together with* the outer factor, not termwise. Write $k = \ell p + s$ and $m = tp + u$ with $0 \leq s, u < p$; the exact Kummer formulas are

$$v_p\left(\frac{qp}{m}\right) = \mathbf{1}_{u>0}, \quad v_p\left(\frac{qp+m}{m}\right) = \mathbf{1}_{q+t \geq p}, \quad v_p(\beta_{qp,k}) = 2 \cdot \mathbf{1}_{s>0} + 2 \cdot \mathbf{1}_{q+\ell \geq p}$$

(for the second, use $\left(\frac{qp+m}{m}\right) = \left(\frac{qp+m}{qp}\right)$: adding $(t, u)_p$ to $(q, 0)_p$ has no units carry and at most the single tens carry $q + t \geq p$). Hence for $p \nmid m$ the denominator valuation is $1 + \mathbf{1}_{q+t \geq p} \leq 2$ and valuation 3 is impossible, so after multiplying by $p^3 \beta_{qp,k}$ such a summand still has valuation ≥ 1 and vanishes. For $u = 0$ the summand's total valuation is $v_p(\beta_{qp,k}) - \mathbf{1}_{q+t \geq p}$, so a nonzero contribution requires $s = 0$ and $q + \ell < p$; then $t \leq \ell$ forces $q + t < p$, every relevant binomial is a unit, and Lucas returns exactly the (q, ℓ, t) summand of a_q . If $q + \ell \geq p$ then $v_p(\beta_{qp,k}) = 2$ while the bracket has valuation ≥ -1 , so the term vanishes on both sides. Summing,

$$p^3 a_{qp} \equiv \sum_{\ell} \beta_{q,\ell} (H_q^{(3)} + c_{q,\ell}) = a_q \pmod{p}.$$

(B) *Propagation*, $r \geq 1$. For $n = qp + r$ one has $n \equiv r \pmod{p}$, so modulo p the recurrence reads $(r+1)^3 u_{n+1} \equiv P(r)u_n - r^3 u_{n-1}$ — *exactly* the b -recurrence in the variable r , and $(r+1)^3$ is invertible since $r+1 \leq p-1$. At the left edge, $P(0) = 5$ and $(qp)^3 \equiv 0$ give $u_{qp+1} \equiv 5 u_{qp}$, matching $b_1 = 5$. So both coordinates satisfy the same recurrence with initial data proportional to $(1, 5)$, whence $u_{qp+r} \equiv u_{qp} b_r$; here $r = 0$ is the base value, the recurrence at $n = qp$ supplies $r = 1$, and the induction runs for $1 \leq r \leq p-2$, so one never divides by p^3 at $r = p-1$. Note $p^3 a_j$ is p -integral because $j < p^2$: the hypothesis $q < p$ is essential to the normalisation, not cosmetic.

Applying (B) to each coordinate and inserting (A) and $b_{qp} \equiv b_q$ gives the theorem. $\square$

The two ingredients were also verified numerically: the boundary congruence in 110 cases and the propagation in 2339 cases, with no violations.

On terminology. Calling this a “rank-one collapse” is inaccurate: when $b_r \neq 0$ the displayed action is $b_r \cdot I$, which has rank two. It is better described as a *Cartier/Frobenius block law*: the pair at index $qp + r$ is a scalar multiple of the pair at index q , the scalar being the low digit's b -value.

Novelty. A literature check against Gessel (1982), Beukers (1985, 1987), Coster, Malik–Straub (2016), Delaygue (2018), Rowland–Yassawi–Krattenthaler (2021) and Straub (2024) found no companion/numerator statement: those works give Lucas and Dwork congruences for the distinguished *integral* sequence, not for the rational second solution after pole-clearing by p^3 . The defensible claim is therefore “apparently new in this normalisation”, not absolute novelty.

A property of Apéry-like pairs, not of $\zeta(3)$ (empirically). The same statement holds for the $\zeta(2)$ pair, with the exponent tracking the order of the leading coefficient $(n+1)^\kappa$ of the recurrence. For $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$ with $(a_1, b_1) = (5, 3)$,

$$(p^2 a_{qp+r}, b_{qp+r}) \equiv b_r \cdot (a_q, b_q) \pmod{p},$$

verified on 3198 checks per coordinate with no violations. The r -propagation is formal for a weight- κ recurrence once $p^\kappa A_n$ is p -integral, but the boundary identity $p^\kappa A_{qp} \equiv A_q$ is separate arithmetic input for each family — Lucas for the integral solution does not imply it — so each sporadic pair must be checked individually. For the two Apéry pairs it is confirmed numerically here.

This unifies the trichotomy used in the draft. Modulo p the pair at index n is b_r times the pair at index q , so the right-hand side can vanish in exactly three ways — $p \mid b_r$ (lower-digit bad), $p \mid b_q$ (leading-digit), and $p \mid$ numerator of a_q (companion-block) — and for the top window $q = 1$, where $a_1 = 6$ and $b_1 = 5$, only the first survives for $p \geq 7$. That is the structural reason the top-window analysis is clean.

All five steps were also checked numerically with no exceptions: the three Lucas evaluations of Step 3–4 over every (p, r) with $7 \leq p < 60$, $r < p$, $p + r \leq 150$; and the conclusion on all 3305 pairs (n, p) with $7 \leq p$, $n/2 < p \leq n$, $20 \leq n < 260$ — including all 28 pairs with $p \mid b_n$ — and on 3320 checks per coordinate in the (p, r) form (`q32_numerator_congruence_audit.py`).

Combined with the universal bound $e_p(n) \leq 3$, Proposition 586 makes the top-window step of the reduction *exact*: the contribution of the window to $\log G_n$ is at most $3 \log \text{rad}_{(n/2, n]}(b_n)$, with no unproved input.

A sharper valuation law (verified, proof of the exact valuation still open). The mod- p congruence gives $e_p(n) \geq 1$ when $p \mid b_n$; the data say much more. For every $p \geq 7$ in the top window,

$$\boxed{v_p(a_n) = -3 + v_p(b_n), \quad \text{equivalently} \quad e_p(n) = v_p(b_n).}$$

Verified on all 4340 pairs (n, p) with $7 \leq p$, $n/2 < p \leq n$, $20 \leq n < 300$: no violations. Two consequences, both sharper than the bound above:

- the top-window part of G_n *equals* the top-window part of b_n exactly, so the conjecture is precisely the statement that the $(n/2, n]$ -part of b_n is $e^{o(n)}$ — no factor 3, and multiplicities rather than radicals;
- $\gcd(A_n, b_n)$ is coprime at such p *iff* $v_p(b_n) \leq 3$ (checked: no violations, and the largest multiplicity observed in the window is 2). So the primitive coprimality conjecture reduces, in this range, to a multiplicity bound.

Proof of the valuation law when $v_p(b_n) \leq 2$. Let $p \geq 7$ lie in the top window and put $t = v_p(b_n)$, $\alpha = v_p(a_n)$.

If $t = 0$ there is nothing to do: writing $n = p + r$, the block law gives $p^3 a_n \equiv 6b_r$ with b_r a unit, so $v_p(a_n) = -3$ as claimed. (The no-two-consecutive-zeros lemma is silent when $t = 0$, so this case must be handled separately rather than through the Casoratian.)

Now let $t \in \{1, 2\}$ and suppose first $p \nmid n + 1$. The exact Casoratian $a_n b_{n+1} - a_{n+1} b_n = -6/(n+1)^3$ has $v_p = 0$, since $p \geq 7$ gives $p \nmid 6$. Here $b_r \equiv 0$ and $r + 1 \leq p - 1$, so there is no wraparound and the no-two-consecutive-zeros lemma does apply, giving $b_{r+1} \not\equiv 0$; the block law at $n + 1 = p + (r + 1)$ then yields $v_p(b_{n+1}) = 0$ and $v_p(a_{n+1}) = -3$ exactly. The two products therefore have valuations

$$v_p(a_n b_{n+1}) = \alpha, \quad v_p(a_{n+1} b_n) = -3 + t.$$

If $\alpha \neq -3 + t$ the difference would have valuation $\min(\alpha, -3 + t) = 0$, forcing $-3 + t \geq 0$; this is impossible for $t \leq 2$. Hence $\alpha = -3 + t$, i.e. $e_p(n) = v_p(b_n)$.

The excluded case $p \mid n + 1$ means $n + 1 = 2p$, i.e. $n = 2p - 1$ and $r = p - 1$. There palindromy gives $b_{p-1} \equiv b_0 = 1 \pmod{p}$, so $p \nmid b_n$, $t = 0$, and the law holds trivially (verified: no prime $p < 400$ has $p \mid b_{p-1}$). $\square$

The same argument in fact covers every multiplicity, giving the saturated law

$$\alpha = t - 3 \ (t < 3), \quad \alpha \geq 0 \ (t = 3), \quad \alpha = 0 \ (t > 3),$$

and hence, for *all* t ,

$$\boxed{v_p(D_n) = 3 - \min(t, 3), \quad e_p(n) = \min(v_p(b_n), 3).}$$

The cap is simply $v_p(d_n) = 3$.

Two bookkeeping points. First, $v_p(G_n) = e_p(n) + \min(v_p(A_n), v_p(b_n))$, so $G_n = \prod_p p^{e_p(n)}$ presumes coprimality; in the top window with $t \leq 2$ the two agree, since $v_p(a_n) = t - 3 < 0$ forces $v_p(A_n) = 0$, but that is an argument rather than a definition, and at $t = 3$ it is exactly the open point. Second, the companion channel in the middle range is $p \mid A_q$, not $p \mid b_q$: for $p > q$ the denominator D_q is a p -unit, so the block law gives $p \mid G_n \iff p \mid A_q b_r$. Since $\log |A_q| = O(q)$, the combined carrier still satisfies $\sum_{q \leq \log n} \log \text{rad}(A_q b_q) = O(\log^2 n)$.

So the denominator-defect law is completely proved, and the top-window contribution to $\log G_n$ equals $\sum_p \min(v_p(b_n), 3) \log p$ exactly. *But the top window alone is not the whole story:* the justified decomposition is

$$\log G_n = \sum_{n/2 < p \leq n} \min(v_p(b_n), 3) \log p + M_n + O\left(\frac{n}{\log n} + \log^2 n\right), \quad M_n = \sum_{\substack{n/\log n < p \leq n/2 \\ p \mid b_n \bmod p}} v_p(G_n) \log p,$$

and from $p \mid b_n$ alone the middle range admits only the bound $O(\log b_n) = O(n)$. Hence the conjecture is equivalent to the *full* large-prime condition $\sum_{n/\log n < p \leq n} \min(v_p(b_n), 3) \log p = o(n)$; the top-window sum alone suffices only once $M_n = o(n)$ is proved separately. This is not a formality: the two ranges contribute comparably in practice. For instance the top-window and middle-range sums are 5.60 and 5.19 at $n = 500$, and 0 and 12.23 at $n = 2500$ — the middle range carries roughly half the mass and sometimes all of it.

One structural remark in its favour: the occupancy-one barrier is specific to $p \asymp n$. For $p \asymp n/\log n$ an interval of length n contains $\asymp \log n$ representatives of each residue class, so the large-sieve obstruction is weaker there — though for a *single* n this still only helps on average. What is not settled is the value $\alpha = 0$ at $t = 3$ — equivalently, excluding a common p -factor of the reduced numerator and b_n — which bears on the primitive coprimality conjecture but not on G_n ; no $t \geq 3$ occurs in the tested range: computing $v_p(b_n)$ *directly* from the exact integer b_n for every bad pair with $n < 300$ gives maximum multiplicity 2.

A caution. One cannot substitute $v_p(b_r)$ for $v_p(b_n)$ here. The first-block expansion reads $b_{p+r} \equiv 5(b_r + pG_r + p^2H_r) \pmod{p^3}$, so $p^3 \mid b_{p+r}$ is a condition on that whole combination, not on b_r alone — higher block digits can cancel the vertical digits of b_r . Concretely $b_3 = 5 \cdot 17^2$ has $v_{17}(b_3) = 2$, yet the corrected first quotient digit is a unit, so $v_{17}(b_{20}) = 2$ rather than 3. A survey of $v_p(b_r)$ over the zero pairs (multiplicity 1 in 2072 of 2074 cases, 2 in the remaining two) is therefore evidence about b_r , not about b_n , and does not extend the multiplicity claim. (The other neighbour gives no more: the Casoratian $a_{n-1}b_n - a_nb_{n-1} = -6/n^3$ also has $v_p = 0$, and $p \nmid b_{n-1}$, so it yields the same constraint $\min(\alpha, t - 3) = 0$ and likewise stalls at $t \geq 3$.)

The three ingredients were checked independently: the Casoratian has $v_p = 0$ in all 4282 applicable pairs, no two consecutive zeros occur in any of them, and $v_p(a_{n+1}) = -3$ in all 4251 pairs where it applies — no violations.

The $\text{mod-}p^2$ route, by contrast, does not help: the corrections vanish at $r = 0$ (Beukers' $b_p \equiv 5 \pmod{p^3}$) but are irregular for $r \geq 1$.

Proposition 587 (verified law). For primes $p \geq 7$ with $n/2 < p \leq n$,

$$e_p(n) > 0 \iff p \mid b_n,$$

and then $e_p(n) = 1$. Verified for all such pairs with $n \leq 250$: 3121 of 3124 cases, the three exceptions being $p = 5$ at $n = 5, 7, 9$ — exactly the prime dividing $b_1 = 5$, which is the same exception carried by the Apéry–Lucas step. For $\sqrt{n} < p \leq n/2$ the defect takes values in $\{1, 2, 3\}$ and occurs precisely when a base- p digit of n lies in Z_p ; e.g. the defect residues are $\{5\}$ for $p = 11$, $\{3, 13\}$ for $p = 17$, $\{8, 10\}$ for $p = 19$, $\{8, 22\}$ for $p = 31$, $\{17, 19\}$ for $p = 37$, $\{10, 30\}$ for $p = 41$, $\{9, 49\}$ for $p = 59$ — in each case exactly Z_p , and each pair summing to $p - 1$ as Proposition 582 requires.

Two consequences, the first of which needs no hypothesis at all.

Lemma 588 (unconditional).
$$\sum_{p \leq n/\log n} e_p(n) \log p \leq 3\pi\left(\frac{n}{\log n}\right) \log n = O\left(\frac{n}{\log n}\right) = o(n).$$

Proof. By (*), $e_p(n) \log p \leq 3[\log_p n] \log p \leq 3 \log n$; sum over the $\pi(n/\log n) \sim n/\log^2 n$ primes involved. $\square$

So every prime below $n/\log n$ is already harmless, whatever the arithmetic of b_n . For $p > n/\log n$ (hence $p > \sqrt{n}$ for large n) the base- p expansion of n has exactly two digits, $n = qp + k$ with $1 \leq q < \log n$, and Apéry–Lucas gives $b_n \equiv b_q b_k \pmod{p}$. The digit q can only contribute primes dividing one of $b_1, \dots, b_{[\log n]}$, and

$$\sum_{j \leq \log n} \log \text{rad}(b_j) \leq \sum_{j \leq \log n} \log b_j = O(\log^2 n),$$

since $\log b_j \sim 3.4j$ (checked: $\log \text{rad}(b_j)$ equals $\log b_j$ to within a few percent for $j \leq 14$). The block law shows this covers the *companion* primes too: $p^3 a_n \equiv a_q b_r$, so $e_p(n) > 0$ forces $p \mid b_r$ (hence $p \mid b_n$), or $p \mid b_q$, or p divides the numerator of a_q ; and since $q \leq \log n$, the last two together contribute at most $\sum_{q \leq \log n} (\log b_q + \log |\text{num } a_q|) = O(\log^2 n)$. Hence

$$\log G_n = O\left(\frac{n}{\log n}\right) + O(\log^2 n) + 3 \sum_{\substack{n/\log n < p \leq n \\ (n \bmod p) \in Z_p}} \log p$$

and the conjecture is *equivalent* to the last sum being $o(n)$. This is a sharper localisation than the top-half reduction: the entire range $p \leq n/\log n$ and the whole q -digit contribution are discharged with Chebyshev’s bound alone.

16.7 How large the remaining sum actually is

Using the Lucas criterion ($p \mid b_n$ iff some base- p digit of n lies in Z_p) one can compute the full quantity $R(n) = \log \text{rad}_{p \leq n}(b_n)$ for every n in a range. For $n \leq 20\,000$:

range	$\max R(n)/n$	$\max_{n \leq 10^5} \#\{p \leq n : p \mid b_n\} = 14.$
$100 \leq n \leq 1000$	0.1091 ($n = 200$)	
$1000 \leq n \leq 10\,000$	0.0251 ($n = 1041$)	
$10\,000 \leq n \leq 100\,000$	0.0041 ($n = 11\,181$)	

Thus $R(n)$ itself grows extremely slowly. Quantitatively, $R(200) \approx 21.8$ and $R(11\,181) \approx 45.5$: the value not quite doubles while n grows by a factor 56. The heuristic prediction from $\Pr[p \mid b_n] \approx |Z_p| \log_p n/p$ is $R(n) \asymp \log n \log \log n$, which predicts a ratio 2.4 over that range against the observed 2.1. So the truth appears to be

$$R(n) = O(\log n \log \log n),$$

while the conjecture asks only for $o(n)$ — the gap between what is true and what must be proved is enormous, and the entire difficulty is that no unconditional argument controls the prime factors of one specific value b_n .

The Lucas collapse and the reciprocal-prime sum. After applying $b_n \equiv 5b_{n-p} \pmod{p}$, the Weyl sum becomes

$$S_h(n) = \sum_{n/2 < p \leq n} e\left(\frac{h\bar{5}_p b_n}{p}\right),$$

a *reciprocal-prime exponential sum with one fixed frequency* $c = \bar{5}_p b_n$. For $c \gg X^2$ the phases oscillate wildly and one observes $|S_h| \approx \sqrt{P_n}$; however no uniform power saving holds: with $c = \prod_{X < p \leq 2X} p + 1$ one has $e(c/p) = e(1/p) = 1 + O(1/X)$, so the sum is $(1 + o(1))P_n$. Thus the *special-numerator assertion* that b_n avoids such CRT resonance across primes $p \sim n$ is equivalent to the conjecture itself — the Lucas collapse is the sharpest reformulation, not a simplification.

A complementary datum explains *why* the count is so small: b_n has very few prime factors, and they are huge. Complete factorisations give

n	5	8	11	14	17	20	23	26
$\omega(b_n)$	3	4	3	3	4	4	7	3
$\Omega(b_n)$	3	5	6	3	4	4	7	4
bits of b_n	20	34	49	64	79	94	108	123
bits of largest $p \mid b_n$	14	21	22	42	63	48	23	60

so $\omega(b_n)$ stays near $\log n$ while $\log b_n \sim 3.53n$, and a single prime factor carries a constant fraction of the size. Since $H(n) \leq \omega(b_n)$, the conjecture would follow from $\omega(b_n) = o(n/\log n)$ — much weaker than the observed $\omega(b_n) \approx \log n$ — but that is not provable either: splitting at any Y gives only $\omega(b_n) \leq \pi(Y) + \log b_n/\log Y$, which is $\gg n/\log n$ for every choice of Y .

16.8 What is load-bearing: the conjecture is FALSE inside the same family

It is natural to hope that the conjecture follows from soft structure — D -finiteness, modularity, Lucas congruences, or reflexivity of the Newton polytope. It does not, and the reason is a counterexample inside the sporadic Apéry-like family itself.

Malik–Straub (*Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory 2 (2016), Thm. 6.6) prove that Cooper’s sporadic sequences satisfy

$$s_7(p-j) \equiv 0 \pmod{p} \quad 1 \leq j \leq \frac{p+1}{3}, \quad s_{10}(p-j), s_{18}(p-j) \equiv 0 \pmod{p} \quad 1 \leq j \leq \frac{p+2}{4}.$$

That is, their zero sets Z_p contain a *macroscopic interval*. Lucas then propagates the interval: for $n/2 < p \leq (3n+1)/5$ one gets $p \mid s_7(n)$, so by the prime number theorem

$$\log \text{rad}_{p \leq n}(s_7(n)) \geq \left(\frac{1}{10} + o(1)\right)n, \quad \log \text{rad}_{p \leq n}(s_{10}(n)), \log \text{rad}_{p \leq n}(s_{18}(n)) \geq \left(\frac{1}{14} + o(1)\right)n.$$

More generally, a terminal interval of length αp in Z_p forces $\log \text{rad}_{p \leq n}(u_n) \geq \left(\frac{\alpha}{2(2-\alpha)} + o(1)\right)n$.

So three bona fide modular Apéry-like sequences *violate* the analogue of the conjecture. They are D -finite, they satisfy Lucas congruences, they are modular, and (by Gorodetsky’s constant-term representations) 14 of the 15 sporadics — including all three counterexamples — have the origin as the unique interior lattice point of the Newton polytope, exactly like Λ . Even $\binom{2n}{n} = \text{CT}(2 + x + x^{-1})^n$ shares that property.

Consequence: no proof for b_n can proceed from D -finiteness, modularity, Lucas congruences or polytope reflexivity alone, since sequences with all four properties violate the conclusion. Any proof must use the size and geometry of Z_p .

This also qualifies the rank-one discussion below: the Beukers–Vlasenko rank-one statement explains why every *local elimination* aliases, but it does not separate the true cases from the false ones, since the counterexamples have the same polytope invariant.

16.9 Numerical separation, and a check of the sharp factorial-ratio constant

Computing $R(n) = \log \text{rad}_{p \leq n}(u_n)$ for $n \in [500, 2000]$ across the family (script `q32_family_compare.py`; Z_p from the division-free scaled recurrence, then the Lucas digit criterion) gives

sequence	$\max R(n)/n$	$\max \#\{p \leq n : p \mid u_n\}$
Apéry $\zeta(3)$	0.042	7
Apéry $\zeta(2)$	0.042	8
Franel	0.036	9
Domb	0.036	8
Almkvist–Zudilin	0.031	4
Cooper s_7	0.301	70
Cooper s_{10}	0.227	53
Cooper s_{18}	0.225	52
$\binom{2n}{n}$ (control)	0.420	101

a sharp dichotomy: the first five decay with n , the last four do not. The three Cooper sequences were checked directly here — their recurrences reproduce $s_{10}(n) = \sum_k \binom{n}{k}^4$ exactly, and their zero sets have $|Z_p| \approx 26\%–35\%$ of p (against $|Z_p| \approx 1$ for Apéry), with the Malik–Straub terminal intervals verified to be contained in Z_p at $p = 23, 31, 43$. For factorial ratios there is a sharp theorem: with $\Delta(x) = \sum_i [a_i x] - \sum_j [b_j x]$,

$$\log \text{rad}_{p \leq n}(F_n) = \kappa_\Delta n + o(n), \quad \kappa_\Delta = \int_1^\infty \mathbf{1}_{\Delta(t) > 0} \frac{dt}{t^2},$$

which for $\binom{2n}{n}$ gives $\kappa = 2 \log 2 - 1 = 0.386294\dots$. Direct computation confirms this and validates the scanner:

n	10^3	10^4	10^5	10^6
$R(n)/n$	0.38536	0.38780	0.38614	0.38573

16.10 An invalid step in the conditional theorem, and its repair

Correction of a status claim, not of the paper. Theorem 119 of `proof.tex` states its conclusion for a set of n of natural density 1, and its Markov argument is correct. It was `STATUS.md` (and the handoff prose derived from it) that recorded the conclusion as holding “for ALL n ”. That

upgrade is **invalid**, and cannot be repaired from Hypothesis $\bar{Z}$ ($\sum_{p \leq x} Z(p) \ll \pi(x)$) together with the structure of Z_p proved so far.

Double counting gives

$$\sum_{m \leq N} \sum_{p \in P} \mathbf{1}_{m \bmod p \in Z_p} = \sum_{p \in P} Z(p) \left(\frac{N}{p} + O(1) \right),$$

so $\bar{Z}$ controls the *mean* diagonal incidence over m , and Markov yields a density-one conclusion. Passing from that mean to a bound for every individual n is the invalid step.

Proposition 589 (the structural properties do not suffice). *There exist sets $S_p \subseteq [0, p)$ with $|S_p| \leq 2$, stable under $r \mapsto p - 1 - r$, with no two consecutive elements and with $\sum_{p \leq x} |S_p| \ll \pi(x)$, for which $T(N) \asymp N$.*

Proof. Fix N and set $S_p = \{N - p, p - 1 - (N - p)\}$ for $p \in (N/2, N]$ and $S_p = \emptyset$ otherwise. Then $N \bmod p = N - p \in S_p$ for every p in the window. $\square$

Verified numerically at $N = 20\,000$: $\max_p |S_p| = 2$, no reflection violation, no consecutive pair, $\sum_p |S_p|/\pi(N) = 0.913$ — so $\bar{Z}$ holds with constant below 1 — and yet $T(N)/N = 0.4955$. Hence *every* structural property proved so far about Z_p (size, reflection symmetry, no consecutive zeros) is consistent with total alignment at a single n . The obstruction is alignment, not size.

So the density-one theorem is exactly what the present hypotheses give. To reach every n one needs a uniform diagonal-discrepancy hypothesis:

$$(\text{DA}) \quad \sup_n \left| \sum_{n/2 < p \leq n} \left(\mathbf{1}_{n-p \in Z_p} - \frac{Z(p)}{p} \right) \right| \ll \frac{\sqrt{n}}{\log n}.$$

By partial summation $\bar{Z}$ gives $\sum_{n/2 < p \leq n} Z(p)/p \ll 1/\log n$, so $\bar{Z}$ together with (DA) yields $T(n) = O(\sqrt{n})$ and closes the conjecture. The all- n theorem should therefore be stated as “ $\bar{Z}$ and (DA) imply $\log G_n = O(\sqrt{n})$ for all n ”; $\bar{Z}$ alone gives the density-one theorem, which is what `proof.tex` correctly proves.

Note also that Hypothesis Z in its pointwise form ($Z(p) \leq C$) is expected to be *false*: extreme-value theory for the $2 \cdot \text{Poisson}(1/2)$ law predicts $\max_{p \leq N} Z(p) \sim 2 \log \log N$, matching the observed record progression 2, 4, 6, 8, 10, 12 with $Z(p) = 12$ first at $p = 159\,977$.

16.11 A p -independent criterion and an explicit shift operator

Two exact facts, both audited in `q32_shift_operator_audit.py` (12 criterion checks, 96 operator checks).

The 27-term criterion. Since $p \equiv 1 \pmod{p-1}$ we have $n - p \equiv n - 1 \pmod{p-1}$, so $t^{n-p} = t^{n-1}$ on $\mathbb{F}_p^\times$ and the exponent in the criterion is *independent of p* . With $m = n - 1$ and $p \in (n/2, n]$,

$$p \mid b_n \iff \sum_{x, y, z \in \mathbb{F}_p^\times} \Lambda(x, y, z)^m \equiv 0 \pmod{p} \iff p \mid V(n, p-1), \quad V(n, s) = \sum_{\varepsilon \in \{-1, 0, 1\}^3} c_m(\varepsilon_1 s, \varepsilon_2 s, \varepsilon_3 s),$$

the 27 terms being the only lattice points of mP in the sublattice $s\mathbb{Z}^3$ once $s > m/2$. Verified at all four known bad primes ($n = 200$: $p = 139, 181$; $n = 321$: $p = 179, 193, 211$) and at non-bad primes.

The shift operator. The coefficient family $C(s) = c_m(s, 0, 0)$ is P -recursive in the *shift* variable s , of order 2 and degree 3:

$$q_2(s)C(s+2) + q_1(s)C(s+1) + q_0(s)C(s) = 0,$$

$$\begin{aligned} q_0(s) &= (s-m)^3, \\ q_1(s) &= 2s^3 - 4(m-1)s^2 + (m^2 - 7m + 3)s + (2m^3 + 4m^2 - 2m + 1), \\ q_2(s) &= s^3 - (m-4)s^2 - (m+5)(m-1)s + (m-1)^2(m+2). \end{aligned}$$

The coefficients are polynomial in m as well, and q_0 vanishes to order 3 at $s = m$, the edge of the support ($C(s) = 0$ for $s > m$).

Why this is the natural home for a certificate. The criterion evaluates C at $s = p-1 \equiv -1 \pmod{p}$. A certificate — an integer $C_n \neq 0$ divisible by every bad prime with $\log |C_n| = o(n)$ — would close the conjecture, since then $\#\text{bad} \leq \log |C_n| / \log(n/2)$. Two routes are now concrete rather than speculative:

1. If $V(n, \cdot)$ agreed with a polynomial $Q_n \in \mathbb{Z}[s]$ of degree D and height H on the relevant range, then $V(n, p-1) \equiv Q_n(-1)$ and $C_n = Q_n(-1)$ would work. *This fails*: eleven finite differences of $V(120, s)$ reduce the size only from 611 to 593 bits, so V is not polynomial of any low degree.
2. The transfer-matrix product $\prod_s M(s) \pmod{p}$ of the operator above over a full period is exactly the p -curvature object of a Picard–Fuchs-type operator, which is *nilpotent* for globally nilpotent operators (Katz). This is the remaining concrete mechanism by which $C(p-1) \pmod{p}$ could be expressed through data of size $O(\log n)$.

16.12 A second-moment target with large slack

The pointwise statement follows from a *second moment*, and with an enormous margin. Let

$$I = \{(r, p) : p \text{ prime}, r < p, p \mid b_r\}, \quad H(n) = \#\{(r, p) \in I : r + p = n\},$$

so $H(n)$ is exactly the number of bad primes at level n (Apéry–Lucas). Restrict to $N < n \leq 2N$ and set

$$E = \sum_{N < n \leq 2N} H(n)^2 = \#\{((r, p), (r', p')) \in I^2 : r + p = r' + p'\},$$

the additive energy of I along the diagonal. Since $\max_n H(n)^2 \leq E$,

$$E = o\left(\frac{N^2}{\log^2 N}\right) \implies \max_n H(n) = o\left(\frac{N}{\log N}\right),$$

which closes the conjecture. *This is an equivalence, not a weakening*: since $E \leq (\max_n H) \sum_n H = |I| \max_n H \ll (N/\log N) \max_n H$ under $\bar{Z}$, one also has $\max_n H = o(N/\log N) \implies E = o(N^2/\log^2 N)$. In particular a constant-factor saving in E buys only a constant-factor saving in $\max_n H$. What the reformulation provides is a different *shape* — an additive-energy statement about the incidence set, for which the available tools (Cauchy–Schwarz, sieve upper bounds for prime pairs, CRT independence) are at least standard — not a lower bar.

How much slack is there? Measured at $N = 60\,000$ (from the exact Z_p data):

$$|I| = \sum_n H(n) = 4398, \quad E = \sum_n H(n)^2 = 4714, \quad E/|I| = 1.072,$$

against a requirement of $E = o(2.97 \cdot 10^7)$ — the truth is smaller by a factor 6309, and the gap grows with N . Moreover the random model predicts $|I| + |I|^2/N = 4720$ against the observed 4714: agreement to 0.1%, i.e. the energy is entirely diagonal, with the off-diagonal contributing the predicted $|I|^2/N$ and nothing more.

The companion discrepancy target of §16.10 is equally comfortable:

$$\max_{1000 \leq n \leq 60\,000} \left| \sum_{n/2 < p \leq n} \left(\mathbf{1}_{n-p \in Z_p} - \frac{Z(p)}{p} \right) \right| = 2.933 \quad (n = 47\,859),$$

against the $\sqrt{n}/\log n = 20.3$ that (DA) demands there; the observed discrepancy looks bounded rather than growing.

One further structural point, which explains why the energy formulation meets the same wall. For p in the top window the quotient $\lfloor n/p \rfloor$ equals 1, so the bad condition is a *pure congruence* $n \bmod p \in Z_p$. Hence for distinct p, p' the joint condition is a congruence modulo pp' , and by CRT its density is exactly $|Z_p||Z_{p'}|/(pp')$ — which correctly predicts the observed off-diagonal $|I|^2/N$. But $pp' \asymp N^2$ exceeds the length N of the range of n , so each pair contributes 0 or 1 and the CRT main term is swamped by the error. This is precisely the large-sieve Q^2 barrier in another guise: the problem carries one degree of freedom per prime.

So the remaining gap is not quantitative delicacy — both sufficient hypotheses hold in the data by factors of 10^1 – 10^4 — but the absence of any method that controls the *alignment* of the sets Z_p along a single diagonal $r+p=n$. In the energy formulation this is the statement that the incidence set I , of size $\approx N/\log N$ inside a box with N diagonals, does not concentrate a positive proportion of its points on one diagonal.

16.13 Why only the first moment is accessible

The k -th moment of the target count factors as

$$\sum_{n \leq N} H(n)^k = \sum_{p_1, \dots, p_k} \# \{ n \leq N : n \bmod p_i \in Z_{p_i} \ (1 \leq i \leq k) \},$$

because for p in the top window the quotient $\lfloor n/p \rfloor$ is 1 and the bad condition is a pure congruence modulo p . The inner count is governed by the modulus $p_1 \cdots p_k \asymp N^k$:

- $k = 1$: the modulus is $p \asymp N$, the range $n \leq N$ contains $N/p + O(1) = O(1)$ complete residue classes, and one gets the honest main term $\sum_p Z(p)(N/p + O(1))$. This is exactly the input for the density-one theorem.
- $k \geq 2$: the modulus N^k exceeds the length of the range, so each tuple contributes 0 or 1 and no CRT/equidistribution gain is available.

So the first moment is the only one the range can support, and Markov applied to it yields a density-one statement and nothing stronger. This is the same Q^2 obstruction as in the large sieve and in the energy formulation of §16.12 — three faces of "one degree of freedom per prime". It

explains, rather than merely reports, why the exceptional set of the conditional theorem stalls at a power of $\log N$.

The data are consistent with all moments being of the same size: at $N = 60\,000$,

$$\frac{k}{\sum_{n \leq N} H(n)^k} \left| \begin{array}{cccc} 1 & 2 & 3 & 4 \\ 4398 & 4714 & 5388 & 6862 \end{array} \right. \quad N/\log N = 5454,$$

i.e. every moment is $\asymp N/\log N$, as a Poisson law with mean $\asymp 1/\log N$ predicts. A proof of even the second moment at that strength would immediately give $\max_n H(n) \ll \sqrt{N/\log N} = o(N/\log N)$ and close the conjecture.

16.14 No second relation: the zeros are statistically generic

Beating the trivial constant $1/2$ would follow from a “non-forced second relation” excluding a positive proportion of the candidate primes. No such relation exists at the level of congruences or quadratic characters. Over the 950 pairs (p, z) with $z \in Z_p$, $5 \leq p < 8000$:

$z \bmod 2$	47.5%, 52.5%
$z \bmod 3$	31.3%, 33.5%, 35.3%
$z \bmod 4$	23.6%, 26.7%, 23.9%, 25.8%
$z \bmod 8$	all eight classes 11.2%–13.9%
$\left(\frac{z}{p}\right)$	+1 : 50.3%, -1 : 49.7%

and the normalised zeros z/p are uniform in $[0, 1]$ (ten bins, 69–81 against 72.4 expected). Note that reflection preserves parity, since $p - 1$ is even, so a parity bias was *a priori* possible; there is none.

Two structural remarks close the remaining elementary routes. First, the reflection partner of a hit at level n is a hit at $n^* = 3p - 1 - n$, a *different* level, so reflection injects hits into other diagonals and yields no pointwise saving. Second, the gap continuant N_h has the forced factor $2X + h + 1$; at $X = z$ and $h = p - 1 - 2z$ this equals p , so $N_h(z) \equiv 0 \pmod{p}$ is tautological rather than a second condition. Finally, the factor-count bound $H_R(n) \leq (\lambda/2 + o(1))R^2/\log n$, $\lambda = \log(17 + 12\sqrt{2})$, beats prime counting only for $R < 2/\lambda = 0.567$, i.e. never for a growing range; the genuinely free prefix $r \leq \lambda^{-1} \log n$ (where $b_r < n/2$, so no hit is possible) removes only $O(\log n)$ residues.

16.15 The certificate exists for codegree, not for degree

It is worth recording precisely which certificate the paper already has, since it delimits the gap. Lemma 108 proves that for $m, n \in (N, 2N]$ with $h = n - m$, the number of primes $p > P_0$ simultaneously lower-digit bad for m and for n is $O(h \log N / \log P_0)$, hence $O(h)$ for $P_0 = N^\delta$. Its proof is exactly a certificate argument:

- *no wrap*: the two zeros lie at distance h , so $p \mid N_h(m \bmod p) \equiv N_h(m)$, and every such prime divides the *fixed integer* $N_h(m)$, of height $\leq 3h \log(CN)$;
- *wrap*: $p \mid \prod_{j \leq h} (m + j)$, of height $h \log 3N$.

So a low-height certificate *does* exist for a *pair* of indices, and it is what powers the polylogarithmic exceptional set via a Kővári–Sós–Turán argument — the exceptional-set theorem is not a Markov bound on a first moment.

What is missing is the same statement for a *single* index: a certificate divisible by all primes bad for one n . The codegree bound cannot be promoted to a degree bound. Summing $\text{codeg}(m, n_0) \leq C|n_0 - m|$ over all $m \in (N, 2N]$ gives only $H(n_0) \ll N^2$; and within a window of length $H_0 < p$ the residues $m \bmod p$ are distinct, so each bad prime for n_0 is bad for at most $O(1)$ nearby m , and no contradiction with a large $H(n_0)$ arises. In graph terms: bounded codegree limits how many vertices can have large degree — which is exactly the exceptional-set theorem — but never excludes a single high-degree vertex.

The incidence structure is entirely explicit: a prime p is bad for exactly $|Z_p|$ levels, namely $n = p + z$ with $z \in Z_p$, and reflection pairs these into $\{p + z, p + (p - 1 - z)\}$, two levels summing to $3p - 1$ — so the levels served by p are symmetric about $(3p - 1)/2$. Triples of levels sharing a bad prime therefore occur exactly when $|Z_p| \geq 3$; the smallest instance in range is $Z_{181} = \{19, 47, 133, 161\}$ (two reflection pairs), which makes 181 bad for the four levels 200, 228, 314, 342. Consequently the summed triple codegree is $\sum_p \binom{|Z_p|}{3}$, which the Poisson law makes $O(\pi(N))$ — the right shape for a hypergraph refinement of the Kővári–Sós–Turán argument, though the pointwise obstruction is untouched by it.

The certificate for triples is available in principle. For three nonwrapping zeros at $z, z + h_1, z + H$ ($H = h_1 + h_2$) the correct pair of detectors is $F(X) = N_{h_1}(X)$ and $G(X) = N_{h_2}(X + h_1)$ — *not* N_H , since continuant addition gives

$$N_H(X) = N_{h_1}(X)N_{h_2+1}(X + h_1 - 1) - (X + h_1)^6 N_{h_1-1}(X)N_{h_2}(X + h_1),$$

so the long return supplies no independent equation. Writing $\widehat{R}_{h_1, h_2}$ for the resultant of F and G after removing primitive contents, their characteristic-zero gcd and the forced reflection factors, one has $\deg N_h \leq 3(h - 1)$ and $\log H(N_h) = O(h \log h)$, so Sylvester–Hadamard gives $\log |\widehat{R}_{h_1, h_2}| = O(h_1 h_2 \log H)$. What must be proved is $\widehat{R}_{h_1, h_2} \neq 0$. Mathematica confirms this for all ten pairs $(h_1, h_2) \in \{2, \dots, 6\}$: $\text{Res}(N_2, N_3) = -5^6$, $\text{Res}(N_2, N_4) = -2^5 \cdot 5^{10} \cdot 11^2 \cdot 17^3 \cdot 71$, and all others nonzero (up to 352 bits). The nonvanishing is structurally forced: the factorisation pattern (odd $h \rightarrow$ irreducible of degree $3(h - 1)$; even $h \rightarrow (2x + h + 1) \times$ irreducible of degree $3h - 4$, verified through $h = 12$) gives distinct degrees for distinct h , hence coprime polynomials over $\mathbb{Q}$ and $\text{Res} \neq 0$ for all $h_1 \neq h_2$, conditional on the irreducibility pattern continuing (supported by the hyperoctahedral Galois group). That gives a triple-codegree bound and, through a hypergraph KST argument, an improvement of the polylogarithmic exceptional set — a genuine provable secondary result, though not the pointwise theorem. Indeed the incidence graph is $K_{2,2}$ -free on the prime side (two primes p, p' share a common level only if n solves a CRT system modulo $pp' \asymp N^2 > N$, so at most one), and a star is $K_{2,2}$ -free; no purely combinatorial argument can therefore exclude one bad n .

The codegree lemma itself was audited numerically here. Over all pairs $(m, m + h)$ with $h \leq 8$ and $m \leq 200\,000$ (top window), the maximum number of common bad primes is

h	1	2	3	4	5	6	7	8
max codeg	0	1	0	1	1	0	1	1

comfortably inside the proved $O(h)$, and the value 0 at $h = 1$ is the no-two-consecutive-zeros theorem appearing independently.

16.16 Provenance, and a separate coprimality conjecture

A literature search located no pre-2026 statement of this problem. Its provenance is the challenge itself: M. Shalyt et al., *The Ramanujan Challenge for AI*, arXiv:2607.09721v1 (27 June 2026), §3.2, contributed by Shachar Weinbaum, whose §3 states that its open problems are “presented

here for the first time”. The obvious predecessor chain — Apéry (1979); Beukers (1979); Hata (1990, 2000); Rhin–Viola (2001); Fischler (Astérisque 294, 2004); Zudilin (2004); Krattenthaler–Rivoal (Mem. AMS 186, 2007) — does not contain it; in particular the Krattenthaler–Rivoal “denominator conjecture” concerns common denominators of engineered multi-zeta forms, not the exact content of Apéry’s own pair.

The title is explained by the following computation. With $\alpha = 17 + 12\sqrt{2}$, $Q = 3 + \log \alpha$ and $E = \log \alpha - 3$, Apéry’s own pair yields the irrationality-measure bound $1 + Q/E = 13.4178\dots$; if $\log G_n = cn + o(n)$ then primitive reduction improves this to $1 + (Q - c)/(E + c)$. Since $\mu(\zeta(3)) \geq 2$ always, $c \leq (Q - E)/2 = 3$ unconditionally — so the conjecture asserts that Apéry’s bound is exactly what his construction gives, with no cancellation to exploit. (This also shows the irrationality-measure lever is weak here: it yields only $c \leq 3$, whereas prime counting already gives $\log G_n \leq (3/2 + o(1))n$.) The Casoratian pair bound is likewise not binding: $G_n G_{n+1} \mid 6d_n d_{n+1}/(n+1)^3$ gives $\log G_n + \log G_{n+1} \leq 6n + o(n)$, while prime counting already forces $\log G_n + \log G_{n+1} \leq 3n + o(n)$. So both classical levers are dominated by the trivial prime count, and neither can contribute.

Separately, our computation $\gcd(A_n, b_n) = 1$ for all $n \leq 330$ is, as far as could be determined, unrecorded. Since $G_n = (d_n/D_n) \gcd(A_n, b_n)$, coprimality is *not* needed for the subexponential-content conjecture; it is a strictly stronger statement, and deserves to be stated separately as a *primitive Apéry coprimality conjecture*.

16.17 Is the conjecture safe?

No positive heuristic or structural mechanism for failure could be identified. Three specific worries can be dismissed.

Arithmetic of n . For a top prime, $p \mid b_n$ is the single-block condition $n - p \in Z_p$; divisibility of n by many small integers does not enter. Likewise a “supersingular parameter” intuition is misleading: $r = n - p$ is a coefficient index that changes with p , not one geometric parameter reduced modulo many primes.

Short progressions inside one Z_p . Harmless: each zero z contributes to a *different* level $p + z$, so an arithmetic progression of zeros merely spreads one prime’s few contributions across several n . The Cooper-type failure needs a macroscopic, systematically positioned block, which Apéry excludes by reflection, nonzero endpoints, and no consecutive zeros.

The record $|Z_p| = 12$ at $p = 159\,977$. That is six reflection pairs. With Poisson mean $\lambda \approx 0.504$ over the $\approx 17\,982$ primes below $2 \cdot 10^5$, the expected number with at least six pairs is ≈ 0.27 , so observing one has probability $\approx 23\%$: entirely ordinary, not a heavy tail.

What remains is a purely logical vulnerability: one can adversarially plant a diagonal while preserving Poisson cardinalities and near-uniform positions for every individual prime. So the data do not prove the conjecture — but a counterexample with $H(n) \gg n^\varepsilon$ would need a cross-characteristic resonance for which there is at present neither an algebraic mechanism nor a numerical hint.

16.18 The strongest currently provable statement is localized

The exceptional-set theorem can be upgraded, with no new input, from a global count to a *local* one. The codegree argument in fact yields $O_\varepsilon(1)$ exceptional levels in every interval of length $c_\varepsilon N / \log N$ (one may take $c_\varepsilon \asymp \varepsilon^2$), whence:

Proposition 590 (localized exceptional set). *For every $\varepsilon > 0$ there is C_ε such that, for all large*

N and every interval $I \subseteq (N, 2N]$,

$$\#\{m \in I : \log G_m > \varepsilon m\} \leq C_\varepsilon \left(1 + \frac{|I| \log N}{N}\right).$$

Proof. Let K be the codegree constant and A absolute with $\pi(2N) \leq AN/\log N$. Put $d := \varepsilon N/(4 \log 2N) \geq \varepsilon N/(8 \log N)$, so every exceptional m has $|P(m)| \geq d$, and let $Q \leq \pi(2N) \leq AN/\log N$ be the number of available primes. For an interval I of length L set $E = \{m \in I : |P(m)| \geq d\}$, $M = |E|$, and $\nu_p = \#\{m \in E : p \in P(m)\}$, $S = \sum_p \nu_p$. Then $S \geq Md$ and, by Cauchy–Schwarz and the codegree bound $|P(m) \cap P(n)| \leq K|m - n| \leq KL$,

$$(Md)^2 \leq S^2 \leq Q \sum_p \nu_p^2 = Q \left(S + 2 \sum_{\substack{m < n \\ m, n \in E}} |P(m) \cap P(n)| \right) \leq Q(MQ + KLM^2),$$

whence $M(d^2 - KQL) \leq Q^2$. Choosing $L_0 := c_\varepsilon N/\log N$ with $c_\varepsilon := \varepsilon^2/(128AK)$ gives $KQL \leq \varepsilon^2 N^2/(128 \log^2 N) \leq d^2/2$ for $L \leq L_0$, hence

$$M \leq 2 \left(\frac{Q}{d} \right)^2 \leq \frac{128A^2}{\varepsilon^2} = O_\varepsilon(1).$$

The admissible length is $d^2/(KQ) \asymp \varepsilon^2 N/\log N$, which is where the ε^2 comes from. For arbitrary I , partition it into at most $1 + L/L_0$ subintervals and sum. $\square$

Summing over a shell gives $O_\varepsilon(\log N)$ and cumulatively $O_\varepsilon((\log X)^2)$, recovering the stated theorem; but the local form is strictly stronger, since it forbids exceptional levels from clustering anywhere.

This is also the natural stopping point for the pairwise method: a configuration with one dense exceptional row every $\asymp N/\log N$ is compatible with the codegree bound $|P(m) \cap P(n)| \ll |m - n|$, so no more efficient pairwise Kővári–Sós–Turán argument can improve the exponent. Only a triple-codegree bound $\Delta_3(N)$ — for which §16.15 supplies the resultant certificate — can go further.

16.19 Why no general method can work: the trivial bound is extremal

It is worth recording that the elementary bound cannot be improved for a general integer, so any proof must be Apéry-specific. For an integer m and $y \geq 2$,

$$k := \#\{p \in (y, 2y] : p \mid m\} \text{ satisfies } y^k \leq |m|, \quad \text{so} \quad k \leq \frac{\log |m|}{\log y},$$

and this is *extremally sharp*: taking m to be a product of k distinct primes from $(y, 2y]$ gives $m \leq (2y)^k$ with exactly those k prime divisors. Hence no bound of the form “an integer of size e^{cn} has $o(n/\log n)$ prime divisors in $(n/2, n]$ ” can hold in general.

Nor does divisor theory supply a substitute: Ford’s theorems on $H(x, y, z)$ and $H_r(x, y, z)$ describe how *typical* integers $m \leq x$ distribute their divisors, and give no pointwise statement for one prescribed integer. The same applies in the other direction: Luca–Shparlinski’s results on $\omega(b_r)$ and the largest prime factor of b_r ($P(b_r) > c\sqrt{\log r \log \log r}$ for density-one r , *JLMS* 78 (2008)) are density-one *lower* bounds.

Consequently the dyadic split $\sum_{r \leq R} + \sum_{r > R}$ cannot produce a constant saving: the first sum is bounded by $\sum_{r \leq R} \min(1, \lambda r/\log n)$ and the second by the prime count of the remaining window, and optimising leaves only a constant cutoff $R = O(1)$. To reach $(1/2 - \delta)n/\log n$ one needs

precisely an Apéry-specific statement of average occupancy strictly below one along the moving diagonal $p = n - r$ — which is the missing theorem itself.

The values $b_r \bmod p$ are in fact essentially *uniform* on $\mathbb{F}_p$: for $p = 101, 211, 503, 997$, each residue appears ≈ 1 time (exact: minimum count 1, maximum 4–8, $\chi^2/p \approx 1.6$ –2.1), and the zero count is 0 or 2, consistent with Poisson(1). So the Apéry sequence at first-block indices is, modulo p , as uniform as a random permutation. The problem is not anomalous concentration; it is proving what the data show.

The exact boundary. The half-sequence collision energy $S = \sum_a \#\{r < (p-1)/2 : b_r \equiv a\}^2$ is $O(m)$ empirically ($S/m \approx 1.3$ –1.5, $\max_a M_a \leq 4$ for $p \leq 997$) and would give $|Z_p| = O(\sqrt{p})$ via $|Z_p|^2 \leq S$. But proving it requires a bounded-complexity geometric realisation of the nonlinear event $b_r \equiv b_s \pmod{p}$ — which no existing technique provides, because $\psi(\text{trace})$ is nonlinear and the fibre product has degree $\sim p$. The statement $S = O(p)$ is therefore an *averaged local-limit theorem for the Beukers–Peters Mellin family*, not a formal consequence of Deligne or Sato–Tate.

The *Fourier core* makes this quantitative. Define the Weyl sum

$$S_h(n) = \sum_{\substack{n/2 < p \leq n \\ p \geq 7}} e\left(\frac{h b_{n-p}}{p}\right), \quad e(x) = e^{2\pi i x}.$$

Then $H(n) = \frac{1}{p} \sum_{h \bmod p} S_h(n)$, and $S_0(n)$ is the prime count. Computing S_1 and S_2 directly gives

n	200	500	1000	2000
$ S_1 $	6.77	3.78	12.79	23.32
$ S_2 $	5.36	7.16	2.58	11.95
$\sqrt{P_n}$	4.58	6.48	8.54	11.62
$ S_2 /\sqrt{P_n}$	1.17	1.11	0.30	1.03

so $|S_h| \approx \sqrt{P_n}$: exact square-root cancellation, as for random phases. Thus *the problem is equivalent to proving $|S_h(n)| = o(P_n)$ for $h = 1, \dots, H$, and the data show $|S_h| \approx \sqrt{P_n} \ll P_n$.*

16.20 The matching map, and what it does and does not give

There is an exact reformulation of the middle range worth recording. For $p \in (n/(q+1), n/q]$ put

$$m = p + (n \bmod p) = n - (q-1)p, \quad \text{so} \quad \frac{n}{q} \leq m < \frac{2n}{q+1}.$$

A middle-range hit at level n with quotient q is exactly a *top-window* hit at the level m , and distinct p give distinct m . So a spike at one level n corresponds to a *matching* — one hit each at many different levels — rather than to a spike anywhere.

Two corrections to the naive count. The interval swept has length $|I_q| = n(q-1)/(q(q+1)) \asymp n/q$, *not* n/q^2 ; what has size $\asymp n/q^2$ is the number of sample points, since the mapped levels satisfy $m \equiv n \pmod{q-1}$ and are spaced $q-1$ apart. Using the sample count, and writing $W(I)$ for the number of top-window hits with level in I , one gets $M_{n,q} \ll (n/q^2)$ under the heuristic $W \asymp |I|/\log N$ — and therefore

$$M_n \ll n \sum_{q \geq 2} q^{-2} = 0.645 n,$$

which is $\Theta(n)$, not $o(n)$. So even a perfect short-interval hit-count bound would not close the middle range. Empirically $W(I)$ does behave that way: the maximum over intervals of length

$L = 100, 1000, 10^4$ is 21, 116, 826 against $L/\log N = 8.2, 82, 820$ — a ratio falling to 1.0. And the matching is precisely the configuration that codegree and KST bounds cannot see, since it places one hit at each of many levels rather than many at one.

16.21 A direct cross-characteristic attempt, and why it fails

The gap machinery works because two zeros of *one* prime at distance h force $p \mid N_h(z)$, an integer of height $O(h \log n)$. The natural attack on a single level n is to try the same across characteristics: two bad primes p, p' at level n have zeros $r = n - p, r' = n - p'$ with $r - r' = p' - p =: k$, so the two zeros *are* at distance k — but in different characteristics.

Tested directly at $n = 321$, whose bad primes are 179, 193, 211 with zeros at $r = 142, 128, 110$: the tridiagonal continuant over the index range between each pair of zeros — the integer that would play the role of N_k — is divisible by *neither* prime in all three cases (the continuants have 369, 464 and 833 bits). So no single integer of controlled height sees both divisibilities, and the gap-polynomial mechanism does not transfer across characteristics. This is the concrete form of the “horizontal versus vertical” obstruction: every certificate we possess is built inside one residue field.

16.22 The geometric interpretation: Mellin traces, not supersingular fibres

Each $r \in Z_p$ is a *vanishing Mellin trace*: $b_r \equiv 0 \pmod{p}$ means the mod- p Frobenius trace of the Kummer-character-twisted middle cohomology $H_{\text{mid}}^1(\mathbb{G}_m, \mathcal{T} \otimes \mathcal{L}_{\chi_r})$, $\chi_r = \omega^r$, vanishes. It is *not* a supersingular fibre parameter of the Beukers–Peters pencil $1 - t\Lambda = 0$. Coefficient sparsity is not coordinate-invariant: $t^{p-1} - 1$ is sparse while $(t+1)^{p-1} - 1$ has essentially every coefficient nonzero, so no Hasse–Witt mass formula or Deuring-type count can bound $|Z_p|$ directly.

The central index $r = (p-1)/2$ is exceptional because $\chi_{(p-1)/2}$ is the quadratic character, which is fixed by the inversion $t \mapsto t^{-1}$; this is why it yields the level-8 weight-4 newform and why $|Z_p|$ is odd exactly at non-ordinary primes.

16.23 A provably sufficient local condition: cocycle-transversality

The cross-prime alignment problem may be bypassed entirely by a *per-gap, per-prime* statement.

Proposition 591 (cocycle-transversality $\Rightarrow |Z_p| = O(1)$). *Let $m_p = (p-1)/2$. If there is an absolute constant C such that, for every prime $p \geq 7$ and every $1 \leq h < m_p$,*

$$\boxed{\#\{x \in \{0, \dots, m_p - 1\} : N_h(x) \equiv 0 \pmod{p}\} \leq C,} \quad (\text{CT})$$

then $|Z_p| = O(C)$.

Proof. The reflected half $Z_p \cap [0, m_p - 1]$ has at most $|Z_p|/2 + 1$ elements (palindromy), and between any two adjacent elements at distance h , $p \mid N_h(x)$. Under (CT), a gap of size h admits at most C uses, and the total is $\sum_h C = O(C \cdot m_p) \dots$ [N.B.: this needs a tighter integration argument through the gap structure; the precise combinatorial extraction is with ChatGPT. The clean statement is that (CT) caps the number of zero-pairs at each gap, and with $O(m_p)$ gaps that is only a bound of $O(Cm_p) = O(Cp)$, not $O(C)$. What is needed is either (CT) for ALL h simultaneously — a resultant/multivariate version — or a different extraction. This is under investigation.] $\square$

Numerical evidence for (CT). Over all 1033 pairs (p, h) with $11 \leq p \leq 200$ and $1 \leq h \leq 29$:

$$\max_{p,h} \#\{x < m_p : N_h(x) \equiv 0 \pmod{p}\} = 5 \quad (\text{at } p = 181, h = 27),$$

against the trivial bound $3(h-1) = 78$. So (CT) holds with $C = 5$ throughout the tested range — the gap polynomial is radically sparser in the half-interval than its degree allows. Whether this holds unconditionally for all p is the sharpest open question the session identified.

Simultaneous CT is even stronger. For *any two distinct* gaps $h_1 \neq h_2$, the number of $x \in \{0, \dots, m_p - 1\}$ with $N_{h_1}(x) \equiv N_{h_2}(x) \equiv 0 \pmod{p}$ is at most 1 over 990 triples (p, h_1, h_2) with $p \leq 100$, $h \leq 11$. Extending to $p \leq 400$ via the stronger test: for *every actually occurring* gap h between two Apéry zeros $z_1 < z_2 \in Z_p$, the root z_1 is the *unique* element of $\{0, \dots, m_p - 1\}$ with $N_h(z_1) \equiv 0 \pmod{p}$ — zero other such roots were found across all $p \leq 400$.

This singleton property, if proved, would immediately give $|Z_p|/2 \leq \#\{\text{distinct gaps}\}$. The Poisson model predicts the latter is $O(1)$, so the entire conjecture would follow from one purely local statement: *the gap polynomial, at an actually occurring gap, has a unique root in the half-interval.*

However, uniform CT is expected false. An independent check shows: $R_h \leq 2(h-1)$ (tight from the reflection structure), and whenever N_h is squarefree (which it is for odd h through $h = 18$), Chebotarev supplies arbitrarily large completely split primes p , at which reflection forces at least $h-1$ of N_h 's roots into the half-interval. Hence an absolute constant C independent of h is incompatible with growing squarefree degree: *for any fixed h , there exist primes p with $R_h \geq h-1$, so (CT) fails at those primes.* The $p \leq 400$ test range was simply too small to see such primes.

The best achievable from the gap-polynomial route alone is therefore $R_h \leq 2(h-1)$, giving

$$|Z_p| \ll p/K + \sum_{h \leq K} 2h \ll p/K + K^2,$$

optimised at $K \sim p^{1/3}$: $|Z_p| \ll p^{2/3}$, recovering the existing bound with a sharp constant. Improving below $p^{2/3}$ by this method requires $R_h \ll h^{1-\delta}$ for some $\delta > 0$, which the Chebotarev argument shows cannot hold uniformly in p .

16.24 The CFVMZ square root: same structure, half the degree

Computing $B_p = \sqrt{A_p}$ in $\mathbb{F}_p[T]$ (verified: $B_p^2 = A_p$ for $p = 17, 29, 31$) and searching for a recurrence in its coefficients c_j : the sequence (c_j) satisfies an order-2, degree-2 recurrence $C_2(j)c_{j+2} + C_1(j)c_{j+1} + C_0(j)c_j = 0$ with leading coefficient $C_2(j) = (j+2)^2$, vanishing only at $j = p-2$. So B_p has the same qualitative recurrence structure as b_r itself — order 2, polynomial coefficients, singular exactly at the boundary — and the zero-count question for B_p is isomorphic to the original.

The CFVMZ factorisation therefore halves the polynomial degree (from $p-1$ to $(p-1)/2$) without changing the qualitative problem. Since $|Z_p| = 2 \cdot \#\{\text{zeros of } B_p\}$, any bound on B_p 's zeros would transfer, but the mechanism needed is the same orbit-return/cocycle-transversality theorem for a degree-2 polynomial-coefficient Möbius cocycle.

16.25 The decisive gate: a fourth-moment criterion

The sharpest sufficient condition for the full conjecture is a *fourth moment* at the random scale (identified independently by the DS agent, Q6206):

Proposition 592 (moment criterion). *If for each fixed $h \neq 0$,*

$$\sum_{N < n \leq 2N} |S_h(n)|^4 \leq C \frac{N^3}{\log^2 N}$$

uniformly in dyadic N , then $\max_{N < n \leq 2N} |S_h(n)| \leq C^{1/4} N^{3/4} / \sqrt{\log N} = o(n/\log n)$ (trivial $L^4 \hookrightarrow L^\infty$ on the shell of length N), and the Fejér kernel closes $H(n) = o(P_n)$ for every n .

The second moment ($k = 1$) fails by a $\sqrt{\log}$ factor; the fourth ($k = 2$) is the first rung that works. Empirically the fourth-moment ratio $R_4 \approx 0.95$, the second-moment ratio $R_2 \rightarrow 1$, and the joint (S_1, S_2) are approximately independent with correlation $\rho \sim 1/\sqrt{N}$, all consistent with the random model (Q6206, DS verification).

16.26 The obstruction, stated precisely

The results above localise the difficulty completely: G_n is a handful of small primes times the $K(n) \leq 3$ targets, and every route we have tried terminates in the same place — bounding the number of prime factors of the *single* integer b_n inside the short window $(n/2, n]$. The correct trivial bound here is the ambient prime count,

$$H(n) \leq \pi(n) - \pi(n/2) = \left(\frac{1}{2} + o(1)\right) \frac{n}{\log n},$$

which is stronger than the height bound $\log b_n / \log(n/2) \sim 2L$ with $L = \log(17 + 12\sqrt{2}) = 3.5255$.

The first Apéry-specific unconditional progress would therefore be any constant below $1/2$, or any $o(n/\log n)$. Known divisors do not help: Delaygue (Compos. Math. 154 (2018)) proves $5^{\alpha_5(n)} 11^{\alpha_{11}(n)} \mid b_n$ with α_5 counting base-5 digits in $\{1, 3\}$ and α_{11} counting base-11 digits equal to 5, but both are $O(\log n)$; and $2, 3, 7 \nmid b_n$ for every n , so no lcm-type divisor exists. Three separate barriers are now identified, each with a reason.

1. **Rank one is forced, not accidental.** The Newton polytope of Λ has a unique interior lattice point (the origin), so by Beukers–Vlasenko *Dwork Crystals I* (IMRN 2021, Thm. 11) the Cartier-stable quotient has rank 1; equivalently, for the K3 fibre $h^{0,2} = 1$, so the Hasse–Witt operator is 1×1 (Katz, *Une formule de congruence pour la fonction zêta*, SGA 7 II, Exp. XXII). Hence *every* mod- p observable built from the coefficient array is a multiple of the single marked scalar of Proposition 581, and any elimination of it reacquires the candidate primorial, of logarithmic height $\sim n/2$. This is the structural reason behind every rank-one alias encountered in the terminal, Turán, Hankel, Newton-jet and cross-index families.
2. **The second p -adic digit is not a second selective condition.** Modulo p^2 one does obtain a genuinely new coordinate, namely $D_{p,r} = R_{p-1}(\Lambda^{r-1}\Phi_p)$ with $\Phi_p = (\Lambda^p - \Lambda(X^p))/p$; but it is not target-selective. Direct computation gives $D_{19,10} = D_{31,8} = 0$ at genuine targets, and also $D_{19,9} = 0$ at a non-target. No codimension-two statement is available.
3. **The vertical average is equally out of reach.** The pointwise problem would follow from Hypothesis Z, $\sum_{p \leq P} |Z_p| = O(P/\log P)$. The best unconditional aggregate implied by $|Z_p| = O(p^{2/3})$ is $O(P^{5/3}/\log P)$, and no improvement is available — not to $P^{1+\varepsilon}/\log P$, and not even $|Z_p| = O(1)$ on a positive-density set of primes. Katz’s finite-field Mellin equidistribution (*Convolution and Equidistribution*, Ann. of Math. Studies 180, 2012) controls archimedean distributions of normalised traces, not exact vanishing modulo p ; and Serre’s bound $\#\{p \leq X : a_p(E) = 0\} \ll X/(\log X)^{3/2-\varepsilon}$ (Publ. Math. IHES 54 (1981)) is one trace test per prime, whereas $|Z_p|$ involves p character-indexed tests inside each prime.

Naming the obstruction. It is worth separating our barrier from the one that modern sieve technology does break. Bombieri–Friedlander–Iwaniec, Fouvry–Iwaniec and Zhang cross the *level-of-distribution* barrier $Q = x^{1/2}$: at $q = x^{1/2+\delta}$ each progression still contains $x^{1/2-\delta}$ candidates,

and dispersion plus a factorisation $q = q_1 q_2$ creates genuine bilinear boxes with Kloosterman input. Our barrier is the *occupancy-one* barrier $Q = x$: with $p \asymp N$ and n ranging over an interval of length N , each modulus contains exactly one candidate, and p is simultaneously the prime variable and the modulus, so it admits no balanced factorisation.

Varying residues are *not* the obstacle — Zhang handles all roots of a fixed f modulo smooth q , Maynard’s uniform-residue theorem takes a supremum over arbitrary a_q when q has a conveniently sized factor, and Duke–Friedlander–Iwaniec handle roots of a fixed quadratic congruence as the prime modulus varies. What those all require is a factorable modulus or a fixed low-complexity algebraic kernel; ours is a prime modulus with a holonomic zero indicator, and we verified in §16.14 that the residues are statistically generic.

There is an equivalent and perhaps cleaner way to say what is missing. Identify $\mathbb{Z}/M$, $M = \prod_{p \in P} p$, with $\prod_{p \in P} \mathbb{F}_p$; then B_p is the cylinder $x_p \in \mathbb{Z}_p$, and

$$\bigcap_{p \in P} B_p = \prod_{p \in P} \mathbb{Z}_p$$

is *automatically nonempty*: it consists of $\prod_p |\mathbb{Z}_p|$ residue classes modulo M , each of multiplicity $|P|$. So a totally aligned n always exists modulo M ; the entire question is whether one of those CRT representatives happens to lie in the short interval $(N, 2N]$, where $N \ll M$. Covering-system technology (Filaseta–Ford–Konyagin–Pomerance–Yu, JAMS 20 (2007); Hough, Ann. of Math. 181 (2015); Balister–Bollobás–Morris–Sahasrabudhe–Tiba, Invent. Math. 228 (2022)) controls the *measure* of a union and deliberately tolerates microscopic hot spots, so it says nothing here; a natural descriptive name is *short-interval CRT discrepancy*, or microscopic non-clustering of CRT representatives.

The precise statement needed may therefore be named *microscopic moving-target equidistribution for Apéry zero residues*. It is strictly stronger than ordinary prime-modulus root equidistribution (Hooley; Duke–Friedlander–Iwaniec), because the target window has length 1 — relative scale $1/p$ — rather than fixed positive relative length. For calibration, even the cubic analogue $x^3 \equiv 2 \pmod{p}$ lacks comparable prime-modulus machinery, so a holonomic zero set at scale $1/p$ is well beyond current reach.

Size arguments are vacuous here, quantitatively. If $H(n) = cn/\log n$ then the product of the bad primes has logarithm $\sim cn$ and divides b_n , giving $c \leq \log \alpha = 3.5255$ with $\alpha = 17 + 12\sqrt{2}$. But prime counting already gives $c \leq 1/2$, since $\theta(n) - \theta(n/2) \sim n/2$. So the size bound is worse than the trivial one by a factor seven, and no renormalisation helps: for a size argument to yield $o(n)$ one would need $\log b_n = o(n)$, i.e. b_n subexponential, or a known divisor capturing all but $o(n)$ of $\log b_n$ — impossible since b_n is rough (only 0–15% of $\log b_n$ comes from primes $\leq n^2$).

What the multiplicative case has and we do not. For $u_n = a^n - 1$ a nontrivial pointwise bound *is* available: partition the primes $p \mid a^n - 1$ by $d = \text{ord}_p(a)$, so that $d \mid n$, $p \equiv 1 \pmod{d}$ and $p \mid \Phi_d(a)$; then the count T_d of such p in $(n/2, n]$ obeys both $T_d \ll n/d$ (candidates $p = 1 + kd$) and $T_d \leq \varphi(d) \log(|a| + 1)/\log(n/2)$ (size of $\Phi_d(a)$), and splitting the divisor sum at $d = \sqrt{n \log n}$ gives a genuine saving. The rigid invariant there is the multiplicative order. Our zero set Z_p carries no group structure, and there is no analogue.

Conversely, D -finiteness by itself supplies nothing: for fixed $K \geq 2$ the factorial ratios $U_n^{(K)}$ are hypergeometric, hence P -recursive, with $\log U_n^{(K)} \asymp n$, and Legendre’s formula telescopes to give $p \mid U_n^{(K)}$ exactly for $p \leq Kn/(K+1)$ — a positive proportion of the window. So no theorem of the form “ P -recursive $\Rightarrow$ few prime factors in a short window” can exist, and the Apéry case must be decided by the arithmetic of Z_p itself.

The SNF ceiling is absolute. The Smith normal form $\text{diag}(1, p^3)$ of the singular transfer on the natural $\mathbb{Z}_p$ solution lattice is a *connection invariant*: any $\mathbb{Z}_p$ -unimodular change of basis $U, V \in \text{GL}_2(\mathbb{Z}_p)$ replaces C_p by UC_pV and preserves the factors. Changing to a non-unimodular basis merely imports the lattice index as a denominator, adding nothing.

The K3's rank-three system does not add an independent row: it is Sym^2 of the rank-two connection, so its transfer has factors $(1, p^3, p^3, p^6)$ — products L^2, LM, M^2 that amplify the same equation. Hadamard products inherit factors additively. Thus *no iterated or functorial construction from the Apéry D -module can reach p^8* , and the grade-7 route requires a genuinely new module carrying an independent target-selective row, not a higher Dwork iteration or tensor power.

The full-period transfer matrix is identically rank one. Computing the product $M = T_{p-2} \cdots T_1 T_0 \bmod p^k$ of the Apéry transfer matrices, where $T_r = \begin{bmatrix} P(r)/(r+1)^3 & -r^3/(r+1)^3 \\ 1 & 0 \end{bmatrix}$, gives

$$M \equiv \begin{pmatrix} 1 & 0 \\ 5 & 0 \end{pmatrix} \pmod{p^k} \quad \text{for } k = 1, 2, 3, 4 \text{ and every tested } p.$$

The second *column* is exactly zero — not merely zero to low order — because the a -direction is killed by the monodromy. This is the matrix incarnation of the unit root: the sole surviving eigenvalue is 1, and no higher p -adic precision can create a second independent direction. Every carrier built from the Apéry connection therefore sees only this rank-one image.

Accordingly the unconditional pointwise statement appears to require genuinely new technology rather than a refinement of the arguments used here.

17 A recurrence-cancelled terminal seam

This section records a second target-preserving terminal coordinate. Its construction uses only the Apéry recurrence and the coefficient sections of the Laurent polynomial; in particular, it does not use the higher-order Ore–Cartier operator.

Retain the notation

$$C_M(d) = \sum_{\nu \in \mathbb{Z}^3} [\mathbf{X}^{d\nu}] \Lambda(\mathbf{X})^M.$$

For an integer array $F = (F_e)_{e \geq 1}$, define the Newton functional

$$\mathcal{G}_{d,L}(F) = \sum_{i=0}^L (-1)^i \binom{d+i}{i} \binom{d+L+1}{L-i} F_{d+i}. \quad (522)$$

The coefficients in (522) are the Lagrange weights for evaluating at -1 the interpolant through $d, d+1, \dots, d+L$. Consequently, if $L < p$ and the stencil contains $p-1$, reduction modulo p selects its unique node congruent to -1 .

We first package a finite segment of the Apéry recurrence. Put

$$A(k) = (2k+1)(17k^2 + 17k + 5), \quad \mathcal{D}_t(s) = \prod_{j=2}^t (s+j)^3 \quad (t \geq 1),$$

where the empty product is one. Define $\mathcal{P}_t, \mathcal{Q}_t \in \mathbb{Z}[s]$ by the first row of the cleared transfer product

$$\mathcal{D}_t(s) \prod_{h=t-1}^1 \begin{pmatrix} \frac{A(s+h)}{(s+h+1)^3} & -\frac{(s+h)^3}{(s+h+1)^3} \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \mathcal{P}_t(s) & \mathcal{Q}_t(s) \\ * & * \end{pmatrix}; \quad (523)$$

the factors are ordered with decreasing h from left to right. Clearing the displayed denominators shows directly that the first-row entries are integer polynomials. Equivalently, induction in the Apéry recurrence gives

$$\mathcal{D}_t(s)b_{s+t} = \mathcal{P}_t(s)b_{s+1} + \mathcal{Q}_t(s)b_s \quad (s \geq 0). \quad (524)$$

For example,

$$\mathcal{D}_2(s) = (s+2)^3, \quad \mathcal{P}_2(s) = (2s+3)(17s^2+51s+39), \quad \mathcal{Q}_2(s) = -(s+1)^3.$$

For $n > t$, define the recurrence-cancelled seam array

$$\mathcal{V}_n^{(t)}(d) = \mathcal{D}_t(n-t)C_{n-1}(d) - \mathcal{P}_t(n-t)C_{n-t}(d) - \mathcal{Q}_t(n-t)C_{n-t-1}(d). \quad (525)$$

Theorem 593 (Universal seam alias). *Let p be prime, $n = p + r$, and $t + 1 \leq r \leq p - 2$. Then*

$$\mathcal{V}_n^{(t)}(p-1) \equiv 0 \pmod{p}, \quad p \nmid \mathcal{D}_t(n-t). \quad (526)$$

The first congruence holds for every prime in this range; it does not require $p \mid b_r$.

Proof. For $1 \leq h \leq p - 2$, torus orthogonality and the support of Λ^h give

$$C_{p-1+h}(p-1) \equiv b_h \pmod{p}. \quad (527)$$

Apply (527) to the three moments in (525). Their lower digits are respectively $r, r - t + 1, r - t$, all in $[1, p - 2]$. Since the coefficients are integer polynomials in n , reduction modulo p and (524) at $s = r - t$ yield

$$\mathcal{V}_n^{(t)}(p-1) \equiv \mathcal{D}_t(r-t)b_r - \mathcal{P}_t(r-t)b_{r-t+1} - \mathcal{Q}_t(r-t)b_{r-t} = 0.$$

Finally,

$$\mathcal{D}_t(n-t) \equiv \prod_{j=2}^t (r-t+j)^3 \pmod{p},$$

and every factor on the right lies in $[1, r]$. This proves the unit assertion. $\square$

We now pair the original shell $Y_d = C_{n-1}(d)$ with the seam column. Set

$$\mathcal{N}_{d,L}^{(t)} = \mathcal{G}_{d,L}(Y)\mathcal{G}_{d+1,L}(\mathcal{V}_n^{(t)}) - \mathcal{G}_{d+1,L}(Y)\mathcal{G}_{d,L}(\mathcal{V}_n^{(t)}), \quad (528)$$

$$\mathcal{E}_{d,L}^{(t)} = \frac{\mathcal{N}_{d,L}^{(t)}}{\binom{d+L+1}{L}}. \quad (529)$$

Theorem 594 (A target digit surviving Pascal normalization). *The number $\mathcal{E}_{d,L}^{(t)}$ is an integer. Suppose, in addition to the hypotheses of Theorem 593, that*

$$L < p, \quad p - 1 - L \leq d \leq p - 2, \quad d + L + 2 \leq n - t - 1. \quad (530)$$

Then

$$\mathcal{E}_{d,L}^{(t)} \equiv (-1)^L b_r \Delta^{L+1} \mathcal{V}_n^{(t)}(d) \pmod{p}. \quad (531)$$

In particular, if $p \mid b_r$, then

$$v_p(\mathcal{N}_{d,L}^{(t)}) \geq 2, \quad v_p\left(\binom{d+L+1}{L}\right) = 1, \quad p \mid \mathcal{E}_{d,L}^{(t)}. \quad (532)$$

Thus the raw determinant contains two target copies, exactly one of which is the universal Pascal factor.

Proof. Newton interpolation gives, for every integer array F ,

$$\mathcal{G}_{d,L}(F) - \mathcal{G}_{d+1,L}(F) = (-1)^{L+1} \binom{d+L+1}{L} \Delta^{L+1} F_d. \quad (533)$$

Substitution into (528) proves both integrality and the exact identity

$$\mathcal{E}_{d,L}^{(t)} = (-1)^{L+1} \left\{ \Delta^{L+1} Y_d \mathcal{G}_{d,L}(\mathcal{V}_n^{(t)}) - \Delta^{L+1} \mathcal{V}_n^{(t)}(d) \mathcal{G}_{d,L}(Y) \right\}. \quad (534)$$

The two stencils in (528) both contain $p-1$ under (530); their lengths are less than p . Their Lagrange weights therefore select the node $p-1$ modulo p . By Theorem 593 and (527),

$$\mathcal{G}_{d,L}(\mathcal{V}_n^{(t)}) \equiv \mathcal{G}_{d+1,L}(\mathcal{V}_n^{(t)}) \equiv 0, \quad \mathcal{G}_{d,L}(Y) \equiv \mathcal{G}_{d+1,L}(Y) \equiv b_r \pmod{p}.$$

Equation (534) now gives (531). If $p \mid b_r$, all four entries of the raw determinant are divisible by p , whence $v_p(\mathcal{N}_{d,L}^{(t)}) \geq 2$.

Finally write $d+L+1 = p+e$. The bounds on d give $0 \leq e < L < p$, so the numerator of $\binom{p+e}{L}$ contains exactly one multiple of p , while its denominator contains none. Hence its p -adic valuation is one, proving (532). $\square$

Remark 595 (Range and limitation). For fixed t , the excluded lower-digit strip $r \leq t$ contributes only prime divisors of the fixed integer $b_1 \cdots b_t$. The upper endpoint $r = p-1$ is nontarget because $b_{p-1} \equiv 1 \pmod{p}$. Thus the seam covers all but a fixed-prime exception among the quotient-one targets.

Theorem 594 is a local divisibility theorem, not a height estimate. Equation (531) is still a rank-one candidate law: bounding the characteristic-zero gcd of several such minors requires an additional Apéry-specific Smith–Bezout or cross-prime input. No such bound is asserted here.

The symbolic transfer law, the Newton identities, 768 dense alias checks, 533 candidate and second-layer checks, and the target valuations in (532) are independently reproduced by `q32_seam_apery_alias_verify.py` in `research/scripts`.

18 Recurrence-cancelled flags in every quotient arc

We construct a second column whose distinguished value vanishes at every candidate prime. Pairing it with the usual Apéry shell and taking an adjacent vector minor produces a target-preserving first jet. Unlike an ordinary exterior row, this construction works uniformly on one of two shortening flags throughout every sufficiently long quotient arc.

Recall the positive Laurent polynomial Λ for which $b_m = \text{CT } \Lambda^m$, and put

$$C_m(e) = \sum_{\nu \in \mathbb{Z}^3} [\mathbf{X}^{e\nu}] \Lambda(\mathbf{X})^m \quad (m \geq 0, e \geq 1). \quad (535)$$

We shall use the Apéry recurrence in the form

$$k^3 b_k - (2k-1)(17k^2 - 17k + 5)b_{k-1} + (k-1)^3 b_{k-2} = 0 \quad (k \geq 2). \quad (536)$$

Fix $q \geq 1$ and n , and define

$$M = n - q, \quad d = \left\lfloor \frac{n}{q+1} \right\rfloor, \quad D = \left\lfloor \frac{n}{q} \right\rfloor - 1, \quad \ell = D - d. \quad (537)$$

On the integer interval $[d, D]$, let

$$Y_e = C_M(e), \quad (538)$$

$$V_e = n^3 C_M(e) - (2n-1)(17n^2 - 17n + 5)C_{M-1}(e) + (n-1)^3 C_{M-2}(e). \quad (539)$$

For any integer array $A = (A_e)$, set

$$\mathcal{G}_{a,L}(A) = \sum_{i=0}^L (-1)^i \binom{a+i}{i} \binom{a+L+1}{L-i} A_{a+i}. \quad (540)$$

The weights in (540) are the Lagrange weights for evaluating at -1 the interpolant through $a, a+1, \dots, a+L$.

Assume $\ell \geq 3$. Define the left- and right-shortening vector flags

$$\mathbf{F}_j^- = (\mathcal{G}_{d+j, \ell-j}(Y), \mathcal{G}_{d+j, \ell-j}(V)), \quad (541)$$

$$\mathbf{F}_j^+ = (\mathcal{G}_{d, \ell-j}(Y), \mathcal{G}_{d, \ell-j}(V)) \quad (0 \leq j \leq 3), \quad (542)$$

and the Pascal factors

$$A_j^- = \binom{D+1}{\ell-j}, \quad A_j^+ = \binom{D-j}{d} \quad (0 \leq j \leq 2). \quad (543)$$

Finally, put

$$H_j^\pm = \frac{\det(\mathbf{F}_j^\pm, \mathbf{F}_{j+1}^\pm)}{A_j^\pm} \quad (0 \leq j \leq 2). \quad (544)$$

Lemma 596 (Exact vector normalization). *Every number $H_j^\pm$ in (544) is an integer. More precisely, if $\Delta A_e = A_{e+1} - A_e$ and $L = \ell - j$, then*

$$H_j^- = \det \left((-1)^L (\Delta^L Y_{d+j}, \Delta^L V_{d+j}), \mathbf{F}_{j+1}^- \right), \quad (545)$$

$$H_j^+ = \det \left((-1)^L (\Delta^L Y_d, \Delta^L V_d), \mathbf{F}_{j+1}^+ \right). \quad (546)$$

Proof. For every integer array A , Newton interpolation gives

$$\mathcal{G}_{a,L}(A) - \mathcal{G}_{a+1, L-1}(A) = (-1)^L \binom{a+L+1}{L} \Delta^L A_a, \quad (547)$$

$$\mathcal{G}_{a,L}(A) - \mathcal{G}_{a, L-1}(A) = (-1)^L \binom{a+L}{L} \Delta^L A_a. \quad (548)$$

Applying these identities to both entries of the vector flags yields

$$\begin{aligned} \mathbf{F}_j^- - \mathbf{F}_{j+1}^- &= A_j^- (-1)^L (\Delta^L Y_{d+j}, \Delta^L V_{d+j}), \\ \mathbf{F}_j^+ - \mathbf{F}_{j+1}^+ &= A_j^+ (-1)^L (\Delta^L Y_d, \Delta^L V_d). \end{aligned}$$

Since $\det(\mathbf{v} + A\boldsymbol{\delta}, \mathbf{v}) = A \det(\boldsymbol{\delta}, \mathbf{v})$, division by the displayed Pascal factor proves (545) and (546) over $\mathbb{Z}$. $\square$

We now identify the selective factor left after this exact normalization. Let p be prime and write

$$n = qp + r, \quad 3 \leq r \leq p - 2. \quad (549)$$

Then $n/(q+1) < p \leq n/q$, and hence

$$d \leq p - 1 \leq D, \quad \ell < p. \quad (550)$$

Indeed,

$$d = p - \left\lceil \frac{p-r}{q+1} \right\rceil, \quad D = p + \left\lfloor \frac{r}{q} \right\rfloor - 1,$$

and

$$\ell < \frac{n}{q} - 1 - \left(\frac{n}{q+1} - 1 \right) = \frac{n}{q(q+1)} < \frac{p}{q} \leq p.$$

Introduce the two margins

$$\lambda = (p-1) - d, \quad \rho = D - (p-1). \quad (551)$$

Thus $\lambda + \rho = \ell$. We call the minus flag *eligible* if $\lambda \geq 3$, and the plus flag *eligible* if $\rho \geq 3$. Equivalently, all four stencils in the eligible flag contain $p-1$.

Theorem 597 (Recurrence-cancelled target flag). *Let $p \geq 5$ be prime, and suppose that (549) holds.*

1. *If $\lambda \geq 3$, then, for $0 \leq j \leq 2$,*

$$v_p(A_j^-) = 1, \quad H_j^- \equiv (-1)^{\ell-j+1} b_r \Delta^{\ell-j} V_{d+j} \pmod{p}. \quad (552)$$

2. *If $\rho \geq 3$, then, for $0 \leq j \leq 2$,*

$$v_p(A_j^+) = 1, \quad H_j^+ \equiv (-1)^{\ell-j+1} b_r \Delta^{\ell-j} V_d \pmod{p}. \quad (553)$$

In particular, every eligible divided minor is divisible by p whenever $p \mid b_r$. If $\ell \geq 6$, at least one of the two flags is eligible, so all three minors on that side retain the target prime.

Proof. We first prove the distinguished-node congruences. If $1 \leq s \leq p-2$, torus orthogonality, Fermat's identity, and the support of Λ^s give

$$C_{q(p-1)+s}(p-1) \equiv b_s \pmod{p}. \quad (554)$$

Indeed, on $(\mathbb{F}_p^\times)^3$ one has $\Lambda^{q(p-1)+s} = \Lambda^s$, including at the zero locus because $s > 0$; and zero is the only exponent in the support of Λ^s which is coordinatewise divisible by $p-1$.

The three moments in (539) are $q(p-1) + r$, $q(p-1) + r - 1$, and $q(p-1) + r - 2$. Thus (554), reduction of the polynomial coefficients modulo p , and (536) imply

$$Y_{p-1} \equiv b_r, \quad V_{p-1} \equiv 0 \pmod{p}. \quad (555)$$

The functional (540) evaluates at -1 . By (550), the nodes of any stencil under consideration are distinct modulo p . If the stencil contains $p-1$, its reduction therefore selects that unique node, and

$$\mathbf{F}_j^\pm \equiv (b_r, 0) \pmod{p} \quad (556)$$

for every coordinate of an eligible flag. Substituting (556) into (545) and (546) proves the two local laws.

It remains to check that each displayed Pascal factor contains exactly one copy of p . Put

$$h = \left\lfloor \frac{r}{q} \right\rfloor. \quad (557)$$

Then $\lfloor n/q \rfloor = p + h$, $D = p + h - 1$, $\rho = h$, and $\ell = h + \lambda < p$. If $\lambda \geq 3$, then

$$A_j^- = \binom{p+h}{h+\lambda-j}, \quad h < h + \lambda - j < p \quad (0 \leq j \leq 2).$$

Kummer's theorem gives $v_p(A_j^-) = 1$.

Now suppose that $h = \rho \geq 3$. Writing $r = qh + s$, $0 \leq s < q$, one has

$$d = p - \left\lfloor \frac{p-r}{q+1} \right\rfloor > h.$$

For completeness, the inequality follows from

$$(q+1)(p-h-1) - (p-r) = qp - h - q - 1 + s \geq 0.$$

Consequently

$$A_j^+ = \binom{p+(h-j-1)}{d}, \quad 0 \leq h-j-1 < d < p \quad (0 \leq j \leq 2),$$

and Kummer's theorem again gives $v_p(A_j^+) = 1$.

Finally, if $\ell = \lambda + \rho \geq 6$, then $\lambda \geq 3$ or $\rho \geq 3$. This proves the cover assertion. $\square$

The unshortened vector already gives a simpler characteristic-zero carrier. This observation removes both the choice of a side and the first-jet normalizations from the bare radical divisibility.

Proposition 598 (Direct full-flag carrier). *Write*

$$\mathbf{F}_0 = \mathbf{F}_0^- = \mathbf{F}_0^+, \quad c_0(n, q) = \gcd(F_0^Y, F_0^V). \quad (558)$$

If $p \geq 5$ is prime and $n = qp + r$, with $3 \leq r \leq p - 2$, then

$$\mathbf{F}_0 \equiv (b_r, 0) \pmod{p}. \quad (559)$$

Consequently

$$R_q^{[3]}(n) \mid c_0(n, q), \quad (560)$$

where $R_q^{[3]}(n)$ is defined in (563) below. No eligibility condition on either margin is needed.

Proof. The full stencil is the interval $[d, D]$. Equations (550) show that it contains $p - 1$ and has length less than p . Its nodes are therefore distinct modulo p , and the Lagrange functional (540) selects the unique node $p - 1$. Equation (555) now gives (559). At a target prime both entries vanish, proving (560). $\square$

Remark 599 (The remaining direct Smith problem). The proposition is a reduction, not a height estimate. The elementary coefficient bound gives only

$$\log \max(|F_0^Y|, |F_0^V|) = O_q(n). \quad (561)$$

Thus the fixed- q mass estimate would follow from the genuinely arithmetic assertion

$$\log c_0(n, q) = o_q(n), \quad (562)$$

which is not proved here. The divided minors below retain additional first-jet information and may help construct a Smith–Bezout certificate, but they are not needed for the bare radical divisibility.

Define the restricted target radical in the q -th quotient arc by

$$R_q^{[3]}(n) = \prod_{\substack{p \text{ prime: } n/(q+1) < p \leq n/q \\ 3 \leq n-qp \leq p-2 \\ p | b_{n-qp}}} p, \quad (563)$$

and, with the convention $\gcd(0, 0, 0) = 0$, put

$$h_{\pm}(n, q) = \gcd(H_0^{\pm}, H_1^{\pm}, H_2^{\pm}). \quad (564)$$

The full flag admits a finite family of recurrence shifts which removes substantially more characteristic-zero content while preserving every target away from a fixed boundary set. For an integer $S \geq 0$, assume $M \geq S + 2$, and keep the geometry d, D, ℓ in (537) fixed. Put

$$A_s(n, q) = \mathcal{G}_{d,\ell}((C_{M+s}(e))_{e=d}^D) \quad (-S - 2 \leq s \leq S), \quad (565)$$

$$R_s(n, q) = (n + s)^3 A_s(n, q) - (2(n + s) - 1)(17(n + s)^2 - 17(n + s) + 5)A_{s-1}(n, q) + (n + s - 1)^3 A_{s-2}(n, q), \quad (566)$$

$$c_S(n, q) = \gcd(A_0(n, q), R_{-S}(n, q), \dots, R_S(n, q)), \quad (567)$$

$$B_S = \text{rad} \left(\prod_{j=0}^{S+2} b_j \right). \quad (568)$$

Proposition 600 (Multi-shift full-stencil carrier). *For every q, n, S as above,*

$$R_q^{[3]}(n) \mid B_S c_S(n, q). \quad (569)$$

More precisely, if p occurs in $R_q^{[3]}(n)$, write $n = qp + r$. Then

$$S + 3 \leq r \leq p - 2 - S \implies p \mid c_S(n, q), \quad (570)$$

whereas every target not satisfying (570) divides B_S . Finally,

$$\log B_S \leq \sum_{j=0}^{S+2} \log b_j \leq \frac{(S+2)(S+3)}{2} \log 40. \quad (571)$$

Consequently, for any integer-valued $S = S(n) = o(\sqrt{n})$, the boundary factor in (569) has logarithm $o(n)$.

Proof. Fix a target prime p , and write $n = qp + r$. The full stencil $[d, D]$ contains $p - 1$ and has length less than p . If $1 \leq r + t \leq p - 2$, the same torus-orthogonality argument as in (554), followed by Lagrange selection at the unique node $p - 1$, gives

$$A_t(n, q) \equiv b_{r+t} \pmod{p}. \quad (572)$$

Suppose first that $S + 3 \leq r \leq p - 2 - S$. For every $-S \leq s \leq S$, all three digits $r + s, r + s - 1, r + s - 2$ lie between 1 and $p - 2$. Reducing (566) modulo p , using (572) and $n + s \equiv r + s \pmod{p}$, yields the Apéry recurrence at the index $r + s$. Hence every $R_s(n, q)$ vanishes modulo p . Targetness also gives $A_0(n, q) \equiv b_r \equiv 0 \pmod{p}$, proving (570).

If the interior inequalities fail, then either $3 \leq r \leq S + 2$, in which case $p \mid b_r$ is already represented in B_S , or $r \geq p - 1 - S$. In the latter case put $j = p - 1 - r$, so $1 \leq j \leq S$. The standard Apéry reflection congruence $b_{p-1-r} \equiv b_r \pmod{p}$ gives $p \mid b_j$, and again $p \mid B_S$. This proves (569).

Finally, positivity of the coefficients of the Laurent polynomial and $\Lambda(1, 1, 1) = 40$ give $b_j \leq 40^j$. Summing $j \log 40$ for $0 \leq j \leq S + 2$ proves (571); the last assertion follows immediately from $S(n)^2 = o(n)$. $\square$

Remark 601 (What remains). Proposition 600 reduces the fixed-quotient mass problem to finding some $S(n) = o(\sqrt{n})$ for which

$$\log c_{S(n)}(n, q) = o_q(n).$$

No such Smith bound is proved here. The recurrence calculation alone cannot supply one: it controls the selected reduction modulo candidate primes, not the characteristic-zero common content of the full flags.

The first possible change in the residual prime support occurs when the terminal stencil crosses a second base- p digit. The following exact Lucas reduction isolates that transition without making an arithmetic nonvanishing claim.

Lemma 602 (Two-digit terminal Newton reduction). *Let p be an odd prime, $1 \leq N < p^2$, and $0 \leq d < N$. Put*

$$K = N - 1 - d, \quad N = ap + \rho, \quad K = hp + k, \quad 0 \leq \rho, k < p,$$

and, for an integer array $Y = (Y_x)_{d \leq x < N}$, define

$$\mathcal{G}_{N,d}(Y) = \sum_{x=d}^{N-1} (-1)^{x-d} \binom{x}{d} \binom{N}{x+1} Y_x.$$

If $k > \rho$, set $H = a - h - 1$ and $\delta = p + \rho - 1 - k$. Then only the seam nodes $x = mp - 1$ survive, and

$$\mathcal{G}_{N,d}(Y) \equiv \sum_{m=H+1}^a (-1)^{m-1-H} \binom{m-1}{H} \binom{a}{m} Y_{mp-1} \pmod{p}. \quad (573)$$

If $k = \rho$, the same formula holds with $H = a - h - 1$ and $\delta = p - 1$; this is the exact seam between the two digit patterns.

If $k < \rho$, set $H = a - h$ and $\delta = \rho - 1 - k$. Then

$$\begin{aligned} \mathcal{G}_{N,d}(Y) \equiv & \sum_{m=H+1}^a (-1)^{m-1-H} \binom{m-1}{H} \binom{a}{m} Y_{mp-1} \\ & + \sum_{m=H}^a \sum_{s=\delta+1}^{\rho} (-1)^{m+s-1-H-\delta} \binom{m}{H} \binom{s-1}{\delta} \binom{a}{m} \binom{\rho}{s} Y_{mp+s-1} \pmod{p}. \end{aligned} \quad (574)$$

Empty sums are interpreted as zero.

Proof. The identity $d = N - 1 - K$ gives

$$d = \begin{cases} (a - h - 1)p + (p + \rho - 1 - k), & k \geq \rho, \\ (a - h)p + (\rho - 1 - k), & k < \rho. \end{cases}$$

Write $x = mp + s - 1$, where $0 \leq s < p$. For $s = 0$, the base- p digits of x are $(m - 1, p - 1)$; for $s > 0$, they are $(m, s - 1)$. Lucas's theorem gives

$$\binom{N}{x+1} \equiv \binom{a}{m} \binom{\rho}{s} \pmod{p}.$$

When $k \geq \rho$, a nonseam term would require simultaneously $s \geq p + \rho - k > \rho$, from $\binom{x}{d}$, and $s \leq \rho$, from $\binom{N}{x+1}$; hence no such term survives. For $s = 0$, Lucas's theorem and $\binom{p-1}{\delta} \equiv (-1)^\delta \pmod{p}$ give the coefficient in (573).

When $k < \rho$, the seam calculation is identical with $H = a - h$. For $s > 0$, Lucas's theorem gives a nonzero contribution precisely for $H \leq m \leq a$ and $\delta + 1 \leq s \leq \rho$, with coefficient

$$(-1)^{m+s-1-H-\delta} \binom{m}{H} \binom{s-1}{\delta} \binom{a}{m} \binom{\rho}{s}.$$

This proves (574). □

Lemma 603 (Two-digit shell reduction). *Write*

$$\Lambda^L = \sum_{\nu \in \mathbb{Z}^3} c_L(\nu) \mathbf{X}^\nu, \quad C_L(e) = \sum_{\kappa \in \mathbb{Z}^3} c_L(e\kappa).$$

If $L = Ap + \sigma$, $0 \leq \sigma < p$, and $e = mp + s - 1 \geq 1$, $0 \leq s < p$, then

$$C_L(mp + s - 1) \equiv \sum_{\kappa \in \mathbb{Z}^3} \sum_{\epsilon \in \mathbb{Z}^3} c_A(m\kappa + \epsilon) c_\sigma((s-1)\kappa - p\epsilon) \pmod{p}. \quad (575)$$

Both sums are finite. For the singular residue row $s = 1$, this specializes to

$$C_{Ap+\sigma}(mp) \equiv C_A(m)b_\sigma \pmod{p} \quad (m \geq 1). \quad (576)$$

Proof. Freshman's congruence gives

$$\Lambda(\mathbf{X})^{Ap+\sigma} \equiv \Lambda(\mathbf{X}^p)^A \Lambda(\mathbf{X})^\sigma \pmod{p}.$$

In the coefficient of $\mathbf{X}^{(mp+s-1)\kappa}$, write the exponent contributed by the first factor as $p(m\kappa + \epsilon)$. The exponent from the second factor is then forced to be $(s-1)\kappa - p\epsilon$. Summing first over ϵ and then over the shell label κ proves (575). Finiteness follows from the bounded Newton-polytope supports of Λ^A and Λ^σ . If $s = 1$, the second coefficient in (575) is $c_\sigma(-p\epsilon)$. Since $\sigma < p$ and Λ is supported in $[-1, 1]^3$, this vanishes unless $\epsilon = 0$. The remaining two factors sum to $C_A(m)b_\sigma$, proving (576). □

Remark 604 (What the two-digit formula does not prove). In the full-stencil application, $N = D + 1 = \lfloor n/q \rfloor$. Consequently $n < qp^2$ places the terminal weights in Lemmas 602 and 603. These lemmas give an exhaustive labelled Cartier reduction, but the surviving values still contain nontrivial Frobenius carry packets. Thus the formula alone does not imply that a common zero of A_0 and several recurrence defects is a target zero. Such a converse requires an additional Cartier-period transversality or Smith identity.

18.1 The local companion line in the odd carry packet

We record a limitation of the scalar recurrence which is already visible in the simplest odd carry packet. This limitation does not assert the two-digit reduction of a full flag; it concerns only the resulting Cartier period and the universal local recurrence algebra.

For a Laurent polynomial $F = \sum_{\nu \in \mathbb{Z}^3} f_\nu \mathbf{X}^\nu$, define

$$\mathcal{C}_m(F) = \sum_{\nu \in \mathbb{Z}^3} f_{m\nu} \mathbf{X}^\nu. \quad (577)$$

Let $a = 2h + 1$ with $h \geq 0$, and put

$$Q_a = \sum_{m=h+1}^a (-1)^{m-h-1} \binom{m-1}{h} \binom{a}{m} \mathcal{C}_m(\Lambda^a), \quad (578)$$

$$T_{a,u} = \text{CT}(Q_a \Lambda^{u-1}) \quad (u \geq 1), \quad (579)$$

$$E_{a,u} = u^3 T_{a,u} - (2u-1)(17u^2 - 17u + 5)T_{a,u-1} + (u-1)^3 T_{a,u-2} \quad (u \geq 3). \quad (580)$$

Proposition 605 (Localized rank and the local companion line). *Let $r \geq 5$, and set*

$$R_r = \mathbb{Z} \left[\frac{1}{(r+2)!} \right].$$

For an arbitrary seven-term sequence window $X = (X_{r-4}, \dots, X_{r+2})$, write

$$(\mathcal{L}X)_t = t^3 X_t - (2t-1)(17t^2 - 17t + 5)X_{t-1} + (t-1)^3 X_{t-2}. \quad (581)$$

Then the R_r -linear map from R_r^7 to R_r^7 given by

$$\tilde{\Phi}_r(X) = (X_{r-1}, X_r, (\mathcal{L}X)_{r-2}, \dots, (\mathcal{L}X)_{r+2}) \quad (582)$$

is an isomorphism. With the coordinates in the displayed orders, its determinant is

$$\det \tilde{\Phi}_r = ((r-3)(r-2)(r-1)(r+1)(r+2))^3. \quad (583)$$

Consequently, after deleting the first output in (582), the resulting six forms have rank six and their kernel is the free line $R_r U^{(r)}$, where

$$U_{r-1}^{(r)} = 1, \quad U_r^{(r)} = 0, \quad (\mathcal{L}U^{(r)})_t = 0 \quad (r-2 \leq t \leq r+2). \quad (584)$$

Let b be the Apéry solution and define

$$W_t(X, b) = X_t b_{t-1} - X_{t-1} b_t.$$

For every X ,

$$t^3 W_t(X, b) - (t-1)^3 W_{t-1}(X, b) = (\mathcal{L}X)_t b_{t-1}, \quad (585)$$

and the normalized kernel vector satisfies

$$W_r(U^{(r)}, b) = -b_r. \quad (586)$$

In particular, if $p > r+2$ is prime and $p \nmid b_r$, the reduction of the kernel line modulo p is transverse to the Apéry solution line. The six local forms

$$X_r, (\mathcal{L}X)_{r-2}, \dots, (\mathcal{L}X)_{r+2}$$

therefore do not, by themselves, force the missing coordinate X_{r-1} or its Apéry Wronskian to vanish.

Proof. Order the input coordinates as $(X_{r-4}, X_{r-3}, X_{r-2}, X_{r-1}, X_r, X_{r+1}, X_{r+2})$. Expanding the matrix of $\tilde{\Phi}_r$ along its first two rows leaves a triangular matrix with diagonal

$$(r-3)^3, (r-2)^3, (r-1)^3, (r+1)^3, (r+2)^3.$$

The relevant column permutation has six inversions, which proves (583). This determinant is a unit in R_r , so deleting the coordinate X_{r-1} leaves a split surjection with a free rank-one kernel. Reversible recurrence, together with the normalization in (584), determines its generator uniquely. For example,

$$U_{r-2}^{(r)} = \frac{(2r-1)(17r^2 - 17r + 5)}{(r-1)^3}, \quad U_{r+1}^{(r)} = -\frac{r^3}{(r+1)^3}.$$

Multiplying (581) by b_{t-1} , subtracting $X_{t-1}(\mathcal{L}b)_t = 0$, and collecting the two determinants gives (585). Equation (586) follows at once from $U_r^{(r)} = 0$ and $U_{r-1}^{(r)} = 1$.

For completeness, after adjoining b_r^{-1} , the invertible change of coordinates (582) identifies the quotient by the ideal generated by the six displayed local forms with a polynomial ring in the free coordinate X_{r-1} . In that quotient $W_r(X, b) = -b_r X_{r-1} \neq 0$. Thus neither X_{r-1} nor the Wronskian belongs to this six-generator local ideal. $\square$

Remark 606 (Scope for the actual odd Cartier period). Let $p > r + 2$ be prime. If the six values

$$T_{a,r}, E_{a,r-2}, E_{a,r-1}, E_{a,r}, E_{a,r+1}, E_{a,r+2}$$

vanish modulo p , Proposition 605 shows only that the seven-term T_a -window lies on the local companion line. When $p \nmid b_r$, the generator's nonzero Wronskian (586) does not contradict that possibility. Excluding the intersection for the actual numerator Q_a requires an additional Q_a -specific period or numerator identity. The proposition therefore rules out only a proof from this six-generator local recurrence ideal; it does not rule out scalar recurrence arguments with remote boundary data, additional shifts, or special relations satisfied by Q_a , and it proves no prime-support bound.

Remark 607 (The first repaired numerator). At $a = 3$, exact coefficient extraction gives

$$Q_3 = 3C_2(\Lambda^3) - 2C_3(\Lambda^3), \quad \text{CT } Q_3 = 1445, \quad [xyz]Q_3 = 223, \quad \gcd(1445, 223) = 1.$$

Hence Q_3 is primitive. Its toric restriction is

$$Q_3(x, -1, -1) = 72x^{-1} - 1632 + 36x, \quad (587)$$

whereas Λ and its three first logarithmic derivatives $\theta_x \Lambda, \theta_y \Lambda, \theta_z \Lambda$, with $\theta_x = x\partial/\partial x$ and similarly for y, z , vanish on $y = z = -1$. This is a diagnostic showing that Q_3 is not in the elementary ideal generated by those four Laurent polynomials. It is not a cohomological assertion and gives no prime-support theorem. At the period level one also has

$$T_{3,1} = 1445, \quad T_{3,2} = 27937, \quad T_{3,3} = 552461, \quad E_{3,3} = -18288,$$

so T_3 is not a homogeneous Apéry solution.

The quadratic boundary cost in (571) is not intrinsic. Splitting the recurrence shifts into two one-sided families gives a target cover with a linear shift radius and no Apéry boundary factor.

For $M \geq S + 2$, define

$$c_{S,+}(n, q) = \gcd(A_0(n, q), R_0(n, q), R_1(n, q), \dots, R_S(n, q)), \quad (588)$$

$$c_{S,-}(n, q) = \gcd(A_0(n, q), R_{-S}(n, q), \dots, R_{-1}(n, q), R_0(n, q)), \quad (589)$$

$$P_S = \prod_{\substack{p < 2S+5 \\ p \text{ prime}}} p. \quad (590)$$

Proposition 608 (Two-sided long-shift carrier). *For every q, n, S as above,*

$$R_q^{[3]}(n) \mid P_S c_{S,+}(n, q) c_{S,-}(n, q). \quad (591)$$

More precisely, a target prime $p \geq 2S+5$ divides at least one of $c_{S,+}(n, q)$ and $c_{S,-}(n, q)$. Moreover

$$\log P_S = O(S). \quad (592)$$

For fixed q , put

$$S_q(n) = \left\lfloor \frac{n/(q+1) - 5}{2} \right\rfloor. \quad (593)$$

Whenever $S_q(n) \geq 0$ and $M \geq S_q(n) + 2$, every prime in $R_q^{[3]}(n)$ exceeds $2S_q(n) + 5$, and hence

$$R_q^{[3]}(n) \mid c_{S_q(n),+}(n, q) c_{S_q(n),-}(n, q). \quad (594)$$

Thus the target radical is retained by two families containing $\Theta_q(n)$ consecutive recurrence defects.

Proof. Let p occur in $R_q^{[3]}(n)$, and write $n = qp + r$. Targetness and the full-stencil selector give $p \mid A_0(n, q)$. If

$$r + S \leq p - 2,$$

then, for every $0 \leq s \leq S$, the three indices $r + s, r + s - 1, r + s - 2$ lie in $[1, p - 2]$. The proof of (570), restricted to these nonnegative shifts, gives $p \mid R_s(n, q)$, and therefore $p \mid c_{S,+}(n, q)$.

Similarly, if

$$r - S \geq 3,$$

then all three indices attached to every $-S \leq s \leq 0$ lie in $[1, p - 2]$. Hence $p \mid R_s(n, q)$ throughout the negative family, and $p \mid c_{S,-}(n, q)$.

If neither inequality holds, then

$$p - 1 - S \leq r \leq S + 2,$$

which implies $p \leq 2S+3$. Consequently every target prime $p \geq 2S+5$ is retained on at least one side, while every remaining target divides P_S . This proves (591). Chebyshev's estimate $\sum_{p \leq x} \log p = O(x)$ gives (592).

Finally, (593) gives $2S_q(n) + 5 \leq n/(q+1)$, whereas every prime in the q -th quotient arc is strictly larger than $n/(q+1)$. Thus the factor P_S is unnecessary at this radius, proving (594). For fixed q , the quantity in (593) is $\Theta_q(n)$. $\square$

Remark 609 (The strengthened Smith target). The long-shift cover does not by itself bound either one-sided content. It replaces the boundary loss by a stronger, exact characteristic-zero problem: construct nonzero integer combinations of each one-sided defect family whose logarithmic heights are $o_q(n)$, or prove directly that

$$\log(c_{S_q(n),+}(n, q) c_{S_q(n),-}(n, q)) = o_q(n).$$

Unlike the symmetric radius in Proposition 600, the available number of consecutive rows is now linear in n . This makes growing-order Ore–Smith and boundary-telescoping constructions formally available. The next proposition shows, however, that every homogeneous construction which uses only products of a positive and a negative recurrence defect already has exponential universal content. Thus any successful certificate must also use the distinguished row A_0 in an inhomogeneous way.

For the same fixed geometry, set

$$\Pi_{S,+}(n, q) = \prod_{\substack{p \text{ prime} \\ (n+S+2)/(q+1) \leq p \leq (n-3)/q}} p, \quad (595)$$

$$\Pi_{S,-}(n, q) = \prod_{\substack{p \text{ prime} \\ (n+2)/(q+1) \leq p \leq (n-S-3)/q}} p. \quad (596)$$

Also put

$$\Pi_{S,0}(n, q) = \prod_{\substack{p \text{ prime} \\ (n+S+2)/(q+1) \leq p \leq (n-S-3)/q}} p. \quad (597)$$

Proposition 610 (Universal content of opposed recurrence defects). *For every $1 \leq s \leq S$, the opposed Casoratian*

$$W_s = R_{1-s}R_s - R_{s-1}R_{-s} \quad (598)$$

satisfies

$$c_{S,+}(n, q)c_{S,-}(n, q) \mid W_s, \quad \Pi_{S,+}(n, q)\Pi_{S,-}(n, q) \mid W_s. \quad (599)$$

More generally,

$$\Pi_{S,+}(n, q)\Pi_{S,-}(n, q) \mid \sum_{i=0}^S \sum_{j=-S}^0 h_{ij} R_i R_j \quad (600)$$

for arbitrary integers h_{ij} . At the linear radius $S = S_q(n)$ of (593),

$$\log(\Pi_{S,+}(n, q)\Pi_{S,-}(n, q)) = \left(\frac{2q+3}{2q(q+1)^2} + o_q(1) \right) n. \quad (601)$$

Consequently, every nonzero integer in the pure-defect product ideal on the right of (600) has logarithmic height bounded below by a positive constant times n .

This obstruction persists for the natural exterior construction which retains A_0 . Let

$$\mathbf{u} = (A_0, R_0, R_1, \dots, R_S), \quad \mathbf{v} = (A_0, R_0, R_{-1}, \dots, R_{-S}), \quad (602)$$

and let $\Delta_{ij} = u_i v_j - u_j v_i$ be any 2-by-2 minor of these two rows. Then

$$c_{S,+}(n, q)c_{S,-}(n, q) \mid \Delta_{ij}, \quad \Pi_{S,0}(n, q) \mid \Delta_{ij}. \quad (603)$$

At $S = S_q(n)$,

$$\log \Pi_{S,0}(n, q) = \left(\frac{1}{2q(q+1)^2} + o_q(1) \right) n. \quad (604)$$

Thus every nonzero integral combination of these exterior minors also has exponential height.

More generally, for any integral matrix $H = (h_{ij})$, put

$$\mathcal{B}_H = \sum_{i,j} h_{ij} u_i v_j. \quad (605)$$

Then

$$c_{S,+}(n, q)c_{S,-}(n, q) \mid \mathcal{B}_H, \quad h_{00} = 0 \implies \Pi_{S,0}(n, q) \mid \mathcal{B}_H. \quad (606)$$

In particular, the same exponential overlap obstruction applies to every bilinear endpoint construction in the two one-sided ideals whose A_0^2 -coefficient cancels, not only to common transports followed by a two-by-two determinant.

Proof. Write $n = qp + r$. The shifted selector (572) and the Apéry recurrence show that $p \mid R_t$ whenever

$$3 - t \leq r \leq p - 2 - t. \quad (607)$$

For all $0 \leq t \leq S$, these inequalities hold simultaneously when $3 \leq r \leq p - S - 2$, which is equivalent to the prime interval in (595). Hence

$$\Pi_{S,+} \mid R_0, R_1, \dots, R_S.$$

Similarly, the inequalities hold for all $-S \leq t \leq 0$ when $S + 3 \leq r \leq p - 2$, giving

$$\Pi_{S,-} \mid R_{-S}, \dots, R_{-1}, R_0.$$

Every monomial in (598), and more generally every monomial on the right of (600), has one factor from each family. This proves both divisibilities by the Π -products. The divisibility by $c_{S,+}c_{S,-}$ follows in the same way from (588) and (589); this remains valid when the two contents have common prime factors, since each monomial contains one full factor from each family.

Finally,

$$S_q(n) = \frac{n}{2(q+1)} + O_q(1).$$

The prime number theorem in the form $\vartheta(x) = x + o(x)$ therefore gives

$$\begin{aligned} \log \Pi_{S,+} &= \left(\frac{q+2}{2q(q+1)^2} + o_q(1) \right) n, \\ \log \Pi_{S,-} &= \left(\frac{1}{2q(q+1)} + o_q(1) \right) n. \end{aligned}$$

Adding these estimates proves (601).

For the exterior statement, every minor in (602) is a difference of two products, each with one factor divisible by $c_{S,+}$ and one factor divisible by $c_{S,-}$. This proves the first divisibility in (603). If $p \mid \Pi_{S,0}$, then p belongs to both safe intervals above, so

$$R_{-S}, \dots, R_{-1}, R_0, R_1, \dots, R_S \equiv 0 \pmod{p}.$$

The two rows therefore both reduce to $(A_0, 0, \dots, 0)$, and every minor vanishes modulo p . This proves the second divisibility. At the linear radius, the length of the overlap interval is

$$\left(\frac{2q+1}{2q(q+1)} - \frac{2q+3}{2(q+1)^2} \right) n + O_q(1) = \frac{n}{2q(q+1)^2} + O_q(1).$$

One more application of the prime number theorem proves (604).

Finally, every coordinate of $\mathbf{u}$ is divisible by $c_{S,+}$, and every coordinate of $\mathbf{v}$ is divisible by $c_{S,-}$. This proves the first assertion of (606). If $h_{00} = 0$, every monomial in (605) contains at least one recurrence defect. All such defects vanish modulo each prime in $\Pi_{S,0}$, proving the second assertion. $\square$

Remark 611 (Scope of the Green obstruction). The determinant (598) is the natural boundary term in the discrete Green identity for the Apéry operator. Proposition 610 shows that neither this determinant nor any integral bilinear combination of positive and negative defects can have subexponential nonzero height. The same is true of the natural two-row exterior minors after A_0 is adjoined. Dividing out the universal products is not a target-preserving operation: those products

also contain target primes lying in the corresponding safe subarcs. The proposition does not exclude a non-alternating element of the full tensor product of the two row ideals. Such an element must have a nonzero A_0^2 -coefficient in order to survive modulo an overlap prime which does not divide A_0 , but globally it may also require symmetric cross-products such as $R_i R_{-j}$; these are not in general generated by A_0^2 and the alternating minors. Constructing a nonzero element of this larger tensor ideal with logarithmic height $o_q(n)$, or bounding its integral saturation directly, is the live Smith–Bezout problem.

For example, separate one-sided Green eliminations can produce integral relations

$$G_+ = \alpha_+ A_0 - U_+, \quad G_- = \alpha_- A_0 + V_-,$$

with $U_+ \in (R_1, \dots, R_S)$ and $V_- \in (R_0, R_{-1}, \dots, R_{-S})$. The anchored determinant

$$\det \begin{pmatrix} G_+ & \alpha_+ A_0 \\ G_- & \alpha_- A_0 \end{pmatrix} = -A_0(\alpha_- U_+ + \alpha_+ V_-)$$

does cancel its A_0^2 -term, but precisely for that reason it is an instance of (606) and remains divisible by $\Pi_{S,0}$. Thus this exact Green construction does not evade the overlap obstruction.

The distinction is visible in an exact Smith presentation. If $N = S + 2$, form the block matrix

$$\mathcal{D} = \begin{pmatrix} \mathbf{u} & 0 \\ 0 & \mathbf{v} \end{pmatrix} \in M_{2,2N}(\mathbb{Z}).$$

Its nonzero maximal minors are precisely the products $u_i v_j$, up to sign, and hence their gcd is exactly $c_{S,+} c_{S,-}$. There is no additional universal $\Pi_{S,0}$ -factor: when $p \mid \Pi_{S,0}$ but $p \nmid A_0$, the minor obtained from the two separate A_0 -columns is $A_0^2 \not\equiv 0 \pmod{p}$. This direct-sum presentation is tautological rather than a height estimate, but it shows that the exterior-square obstruction is not an obstruction to the full tensor ideal. For the same reason, the row-dependent determinant

$$\det \begin{pmatrix} A_0 & R_j \\ R_{-j} & A_0 \end{pmatrix} = A_0^2 - R_j R_{-j}$$

is divisible by $c_{S,+} c_{S,-}$ without being forced to contain $\Pi_{S,0}$.

The exterior saturation of the two one-sided rows can be described exactly. This does not yet give the required height bound, but it isolates the additional prime factors introduced by passage from the two row contents to their exterior minors.

Proposition 612 (Exact exterior saturation). *Assume $S \geq 1$. Let $\mathbf{u}, \mathbf{v}$ be the two rows in (602), and put*

$$J_S = \gcd(A_0^2, \{u_i v_j - u_j v_i : 0 \leq i < j \leq S + 1\}), \quad E_S = \frac{J_S}{c_{S,+} c_{S,-}}. \quad (608)$$

Write

$$g = \gcd(c_{S,+}, c_{S,-}), \quad c_{S,+} = gx, \quad c_{S,-} = gy, \quad \gcd(x, y) = 1, \quad (609)$$

and set

$$L = \text{lcm}(c_{S,+}, c_{S,-}) = gxy, \quad h = \frac{A_0}{L}, \quad t = \frac{R_0}{L}, \quad U = \frac{\mathbf{u}}{c_{S,+}}, \quad V = \frac{\mathbf{v}}{c_{S,-}}. \quad (610)$$

Index the common coordinates A_0, R_0 by 0, 1, respectively, and put

$$\delta_j = yV_j - xU_j \quad (2 \leq j \leq S + 1). \quad (611)$$

Then

$$E_S = \gcd(xyh^2, \{h\delta_j, t\delta_j : 2 \leq j \leq S+1\}, \{U_i V_j - U_j V_i : 2 \leq i < j \leq S+1\}). \quad (612)$$

In particular,

$$E_S \mid h^2, \quad \text{rad}(E_S) \mid h. \quad (613)$$

More precisely, for every prime p , if

$$\alpha = v_p(c_{S,+}), \quad \beta = v_p(c_{S,-}), \quad U^{(p)} = p^{-\alpha} \mathbf{u}, \quad V^{(p)} = p^{-\beta} \mathbf{v},$$

then

$$v_p(E_S) = \min \left(2v_p(A_0) - \alpha - \beta, \min_{0 \leq i < j \leq S+1} v_p(U_i^{(p)} V_j^{(p)} - U_j^{(p)} V_i^{(p)}) \right). \quad (614)$$

Thus $p \mid E_S$ if and only if the two primitive normalized rows define the same point of projective space over $\mathbb{F}_p$, and their common projective point lies on the hyperplane given by the distinguished A_0 -coordinate.

Proof. Both row contents divide A_0 , and every minor contains one factor from each row. Hence E_S is an integer. The first two coordinates of the primitive rows in (610) are

$$U_0 = yh, \quad U_1 = yt, \quad V_0 = xh, \quad V_1 = xt.$$

After division by $c_{S,+}c_{S,-} = g^2xy$, the generator A_0^2 becomes xyh^2 . A minor involving a tail coordinate and column 0 or 1 becomes, up to sign, $h\delta_j$ or $t\delta_j$, while a tail–tail minor remains $U_i V_j - U_j V_i$. This proves (612).

We prove the first divisibility in (613) prime by prime. If $p \nmid xy$, the first entry of the gcd in (612) gives $v_p(E_S) \leq 2v_p(h)$. Suppose $p \mid x$; then $p \nmid y$. The first two coordinates xh, xt of the primitive row V are divisible by p , so some tail coordinate V_j is a p -adic unit. For this j , the integer $\delta_j = yV_j - xU_j$ is a unit, and the entry $h\delta_j$ gives $v_p(E_S) \leq v_p(h)$. The case $p \mid y$ is symmetric. This proves $E_S \mid h^2$, and taking prime support gives the radical statement.

Finally, divide the defining generators of J_S by $p^{\alpha+\beta}$ and take their minimum valuation. This gives (614). The minors vanish modulo p exactly when the two nonzero primitive rows are projectively proportional. Under this proportionality, the remaining first term has positive valuation exactly when both normalized distinguished coordinates vanish. $\square$

The square in (613) is optimal at the level of arbitrary integral rows: for $\mathbf{u} = (p, 1, 1, 0)$ and $\mathbf{v} = (p, 1, 1, p^2)$, both row contents are one and $E_S = p^2 = h^2$. Nor can the support of E_S be controlled from the tridiagonal recurrence alone.

Lemma 613 (Residual-profile surjectivity). *Assume $S \geq 1$ and $n > S+1$. Given arbitrary rational numbers $a_0, \rho_{-S}, \dots, \rho_S$, there exists a rational sequence $A_{-S-2}, \dots, A_S$ such that*

$$A_0 = a_0, \quad R_s = \rho_s \quad (-S \leq s \leq S). \quad (615)$$

If the prescribed data are integral, multiplication by

$$D_{n,S} = \prod_{j=1}^S (n+j)^3 \prod_{j=1}^{S+1} (n-j)^3 \quad (616)$$

clears all denominators. Consequently, for every integer $K \geq 1$, there is an integral sequence obeying the same Apéry tridiagonal operator for which $E_S = K$.

Proof. Choose $A_{-1} = 0$ and solve first for

$$A_{-2} = \frac{\rho_0 - n^3 a_0}{(n-1)^3}.$$

For $1 \leq s \leq S$, solve the defining equation for R_s forward for A_s ; for $s = -1, -2, \dots, -S$, solve it backward for A_{s-2} . The denominators encountered divide (616), proving the first two assertions. To obtain the last one, prescribe

$$a_0 = K, \quad \rho_0 = 1, \quad \rho_{-1} = K, \quad \rho_s = 0 \quad (s \neq 0, -1).$$

After multiplication by $D_{n,S}$, the two rows are

$$D_{n,S}(K, 1, 0, \dots), \quad D_{n,S}(K, 1, K, 0, \dots).$$

Their contents are both $D_{n,S}$, and direct expansion of the minors in (608) gives $E_S = K$. $\square$

Lemma 613 is a scoped no-go statement: a bound for the actual full Newton shell must use its Laurent/Cartier origin, not merely the common Apéry recurrence columns. The next proposition similarly determines the exact size of the natural universal endpoint space.

Proposition 614 (Uniqueness of the Green endpoint direction). *Assume $S \geq 2$ and $n > S + 2$. Introduce two independent families*

$$x_{-2}, x_{-1}, \dots, x_S, \quad y_{-S-2}, \dots, y_{-1}, y_0,$$

and let $R_s^+(x)$, respectively $R_s^-(y)$, denote the residual form (566) with A_t replaced by x_t , respectively y_t . Define

$$\begin{aligned} \mathcal{U}_+ &= \text{span}_{\mathbb{Q}}\{x_0, R_0^+, R_1^+, \dots, R_S^+\}, & \mathcal{E}_+ &= \text{span}_{\mathbb{Q}}\{x_{S-1}, x_S\}, \\ \mathcal{U}_- &= \text{span}_{\mathbb{Q}}\{y_0, R_0^-, R_{-1}^-, \dots, R_{-S}^-\}, & \mathcal{E}_- &= \text{span}_{\mathbb{Q}}\{y_{-S-2}, y_{-S-1}\}. \end{aligned}$$

Then

$$\dim(\mathcal{U}_+ \cap \mathcal{E}_+) = \dim(\mathcal{U}_- \cap \mathcal{E}_-) = 1, \tag{617}$$

and

$$(\mathcal{U}_+ \otimes \mathcal{U}_-) \cap (\mathcal{E}_+ \otimes \mathcal{E}_-) = (\mathcal{U}_+ \cap \mathcal{E}_+) \otimes (\mathcal{U}_- \cap \mathcal{E}_-) \tag{618}$$

is one-dimensional. Its generator is the product of the two one-sided discrete Green endpoint forms.

Moreover, for $1 \leq j \leq S$, set

$$\mathcal{D}_j = x_0 y_0 - R_j^+(x) R_{-j}^-(y). \tag{619}$$

The monomial $x_j y_{-j}$ occurs in $\mathcal{D}_j$ with coefficient

$$-(n+j)^3(n-j)^3, \tag{620}$$

and occurs in no $\mathcal{D}_k$ with $k \neq j$, nor in the Green endpoint product. Consequently, adjoining the diagonal tensors $\mathcal{D}_1, \dots, \mathcal{D}_S$ creates no second universal endpoint direction.

Proof. Write a general positive row form as

$$F_+ = ax_0 + \sum_{s=0}^S h_s R_s^+.$$

If $F_+ \in \mathcal{E}_+$, the coefficients of x_{-2} and x_{-1} give successively $h_0 = h_1 = 0$. The coefficient of x_0 determines a from h_2 , and the coefficients of $x_1, \dots, x_{S-2}$ give the adjoint three-term recurrence, determining all later h_s from h_2 . Conversely, every choice of h_2 produces an endpoint form, proving the positive assertion in (617). Reading the negative coefficients in reverse order gives the second assertion in the same way. The row generators are linearly independent by their triangular extreme coefficients. Choosing complements in the two ambient linear-form spaces then gives the tensor-intersection identity (618). Its generator is necessarily the product of the two one-dimensional Green endpoint generators.

The supports of R_k^+ and R_{-k}^- are $\{k, k-1, k-2\}$ and $\{-k, -k-1, -k-2\}$, respectively. They contain x_j and y_{-j} simultaneously only when $k = j$; multiplication of the two leading coefficients gives (620). The Green product is supported on

$$\{S-1, S\} \times \{-S-2, -S-1\},$$

so it contains none of these diagonal corner monomials. Reading the corners one at a time proves the final assertion. $\square$

Remark 615 (The remaining nonalternating Smith interface). The preceding proposition exhausts universal endpoint elimination, not arithmetic cancellation after evaluation on the full Newton shell. A natural larger family is

$$T_{ij} = A_0^2 - R_i R_{-j} \quad (0 \leq i, j \leq S). \quad (621)$$

With the normalization of Proposition 612, and with the residual coordinates indexed by $i+1, j+1$, one has the exact identity

$$\frac{\gcd_{0 \leq i, j \leq S} T_{ij}}{c_{S,+} c_{S,-}} = \gcd_{0 \leq i, j \leq S} (xyh^2 - U_{i+1} V_{j+1}). \quad (622)$$

Modulo a prime dividing the right-hand side, either the nonzero constant xyh^2 forces both normalized residual rows to be constant, or that constant is zero and one normalized residual row vanishes identically. This is a sharper local condition than projective collision, but all integers in (621) still have exponential archimedean height with the estimates presently available. Moreover, Lemma 613 allows arbitrary flat residual profiles in the recurrence-only model. What remains is an actual-shell, non-endpoint, inhomogeneous Smith–Bezout identity with a nonzero right-hand side of logarithmic height $o_q(n)$.

We next give an exact primitive saturation theorem for the nonalternating interface in Remark 615. The point is that the full cross-family and the two primitive Green rows admit an inhomogeneous elimination after their unavoidable row contents have been removed.

Lemma 616 (Primitive arithmetic Green rows). *Assume $S \geq 2$ and $n > S + 2$, and abbreviate*

$$\mathcal{P}(k) = (2k-1)(17k^2 - 17k + 5).$$

Define rational adjoint weights by

$$\lambda_0 = \lambda_1 = 0, \quad \lambda_2 = 1,$$

$$(n+s)^3\lambda_s - \mathcal{P}(n+s+1)\lambda_{s+1} + (n+s+1)^3\lambda_{s+2} = 0 \quad (1 \leq s \leq S-1),$$

and by

$$\begin{aligned} \mu_1 &= 0, & \mu_0 &= 1, \\ (n+s)^3\mu_s - \mathcal{P}(n+s+1)\mu_{s+1} + (n+s+1)^3\mu_{s+2} &= 0 \quad (-S-1 \leq s \leq -1). \end{aligned}$$

Let

$$D_+ = \text{lcm}_{0 \leq s \leq S} \text{den}(\lambda_s), \quad D_- = \text{lcm}_{-S \leq s \leq 0} \text{den}(\mu_s), \quad (623)$$

where $\text{den}(r)$ denotes the positive denominator of a rational number in lowest terms. Then

$$G_+ = D_+ \left(-(n+1)^3 A_0 + \sum_{s=0}^S \lambda_s R_s \right), \quad (624)$$

$$G_- = D_- \left(-n^3 A_0 + \sum_{s=-S}^0 \mu_s R_s \right) \quad (625)$$

are primitive integral row forms belonging to the positive and negative row ideals, respectively.

Writing

$$a_+ = -D_+(n+1)^3, \quad b_{s+1}^+ = D_+\lambda_s \quad (0 \leq s \leq S), \quad (626)$$

$$a_- = -D_-n^3, \quad b_{1-s}^- = D_-\mu_s \quad (-S \leq s \leq 0), \quad (627)$$

one has

$$\gcd(a_+, b_1^+, \dots, b_{S+1}^+) = \gcd(a_-, b_1^-, \dots, b_{S+1}^-) = 1.$$

Let $\ell_m = \text{lcm}(1, \dots, m)$, with $\ell_0 = 1$. Then

$$D_+ \mid \ell_{n+S-1}^3, \quad D_- \mid \ell_n^3, \quad (628)$$

and

$$\log \max \left(2, |a_+|, |a_-|, \max_i |b_i^+|, \max_j |b_j^-| \right) = O(n+S). \quad (629)$$

Proof. Summation by parts for the tridiagonal operator gives

$$\begin{aligned} G_+ &= D_+(n+S)^3(\lambda_S A_S - \lambda_{S+1} A_{S-1}), \\ G_- &= D_-(n-S-1)^3(\mu_{-S} A_{-S-2} - \mu_{-S-1} A_{-S-1}). \end{aligned}$$

Only the coefficients $\lambda_0, \dots, \lambda_S$, respectively $\mu_{-S}, \dots, \mu_0$, occur in the ideal-row expressions (624) and (625). The continuation coefficients λ_{S+1} and μ_{-S-1} in the endpoint rewrite therefore introduce no denominator into the primitive clearing.

Minimality in (623) proves primitivity. Indeed, for a prime dividing D_+ , some cleared coefficient $D_+\lambda_s$ is a unit at that prime; for a prime not dividing D_+ , the coefficient $D_+\lambda_2 = D_+$ is a unit. The negative argument uses $D_-\mu_0 = D_-$.

For the arithmetic bound, let b_k be the integral Apéry solution and put $c_k = a_k/6$, so that $(c_0, c_1) = (0, 1)$. The Casoratian in Lemma 1 gives

$$\lambda_s = (n+1)^3(b_n c_{n+s-1} - c_n b_{n+s-1}) \quad (1 \leq s \leq S+1), \quad (630)$$

$$\mu_s = n^3(b_{n+s-1} c_n - c_{n+s-1} b_n) \quad (-S-1 \leq s \leq 0). \quad (631)$$

These formulas have the required initial values and satisfy the adjoint recurrence, so uniqueness proves them. By (4), $\ell_k^3 c_k$ is integral. Clearing all companion values in (630) by ℓ_{n+S-1}^3 , and those in (631) by ℓ_n^3 , proves (628). Finally, ℓ_k, b_k , and c_k have exponential upper bounds in k ; substitution in the two Casoratian formulas proves (629). $\square$

Proposition 617 (Exact Smith elimination). *Retain the assumptions of Lemma 616, abbreviate $a = A_0$, and form the opposed rows*

$$\mathbf{u} = (a, R_0, R_1, \dots, R_S), \quad \mathbf{v} = (a, R_0, R_{-1}, \dots, R_{-S}). \quad (632)$$

Put

$$c_+ = \gcd(\mathbf{u}), \quad c_- = \gcd(\mathbf{v}),$$

and assume $c_+c_- > 0$. Write

$$g = \gcd(c_+, c_-), \quad c_+ = gx, \quad c_- = gy, \quad (x, y) = 1, \quad (633)$$

$$L = \text{lcm}(c_+, c_-) = gxy, \quad h = \frac{a}{L}, \quad t = \frac{R_0}{L}, \quad U = \frac{\mathbf{u}}{c_+}, \quad V = \frac{\mathbf{v}}{c_-}. \quad (634)$$

Index the distinguished coordinates by 0, and index the residual coordinates by $1, \dots, S+1$, in the orders displayed in (632). Thus

$$U_0 = yh, \quad V_0 = xh, \quad U_1 = yt, \quad V_1 = xt. \quad (635)$$

For $0 \leq i, j \leq S$, set

$$T_{ij} = a^2 - R_i R_{-j},$$

and suppose that the integers in the following gcd are not all zero:

$$K_* = \gcd(\{T_{ij} : 0 \leq i, j \leq S\}, aG_+, aG_-), \quad K_*^\circ = \frac{K_*}{c_+c_-}. \quad (636)$$

Then K_*° is a positive integer. Define the normalized Green forms

$$\Gamma_+(U) = a_+ U_0 + \sum_{i=1}^{S+1} b_i^+ U_i, \quad (637)$$

$$\Gamma_-(V) = a_- V_0 + \sum_{j=1}^{S+1} b_j^- V_j, \quad (638)$$

and put

$$B_+ = \sum_{i=1}^{S+1} b_i^+, \quad B_- = \sum_{j=1}^{S+1} b_j^-, \quad \mathcal{C}_{n,S} = B_+ B_- - a_+ a_-. \quad (639)$$

Then the exact divisibility

$$\boxed{K_*^\circ \mid xy \mathcal{C}_{n,S}} \quad (640)$$

holds. In particular,

$$\text{rad}(K_*^\circ) \mid xy \mathcal{C}_{n,S},$$

and

$$\log \max(1, |\mathcal{C}_{n,S}|) = O(n + S). \quad (641)$$

Proof. Every generator in (636) belongs to the product of the two row ideals, so $c_+c_- \mid K_*$. From (634), direct division gives

$$\frac{T_{i-1,j-1}}{c_+c_-} = H - U_i V_j, \quad H = xyh^2 \quad (1 \leq i, j \leq S+1). \quad (642)$$

Homogeneity of the Green rows gives

$$G_+ = c_+ \Gamma_+(U), \quad G_- = c_- \Gamma_-(V),$$

and hence

$$\frac{aG_+}{c_+c_-} = P, \quad \frac{aG_-}{c_+c_-} = Q, \quad P = xh\Gamma_+(U), \quad Q = yh\Gamma_-(V). \quad (643)$$

Consequently,

$$K_*^\circ = \gcd(\{H - U_i V_j : 1 \leq i, j \leq S+1\}, P, Q). \quad (644)$$

Set

$$X = \sum_i b_i^+ U_i, \quad Y = \sum_j b_j^- V_j, \quad F = \sum_{i,j} b_i^+ b_j^- (H - U_i V_j). \quad (645)$$

Thus

$$F = B_+ B_- H - XY, \quad P = a_+ H + xhX, \quad Q = a_- H + yhY.$$

A direct expansion yields the first exact elimination identity

$$H^2 \mathcal{C}_{n,S} = HF + PQ - a_- HP - a_+ HQ. \quad (646)$$

The common residual R_0 supplies one further cancellation. Define

$$F_+ = \sum_i b_i^+ (H - U_i V_1), \quad F_- = \sum_j b_j^- (H - U_1 V_j), \quad (647)$$

$$E_+ = tP + hF_+, \quad E_- = tQ + hF_-. \quad (648)$$

Using $U_1 = yt$ and $V_1 = xt$, one obtains

$$E_+ = H(a_+ t + B_+ h), \quad E_- = H(a_- t + B_- h). \quad (649)$$

It follows, again by direct expansion, that

$$Hh^2 \mathcal{C}_{n,S} = B_- hE_+ - a_+ tE_- - a_+ a_- h^2 (H - U_1 V_1). \quad (650)$$

Every term on the right of (650) is an integer combination of the generators in (644). Therefore

$$K_*^\circ \mid xyh^4 \mathcal{C}_{n,S}. \quad (651)$$

It remains to remove h^4 , and this step has no saturation gap. Suppose that a prime p divides both K_*° and h . Reducing all cross-generators modulo p gives

$$U_i V_j = 0 \quad (1 \leq i, j \leq S+1).$$

Over the field $\mathbb{F}_p$, either all residual U_i vanish or all residual V_j vanish. But $U_0 = yh$ and $V_0 = xh$ vanish as well. The corresponding entire normalized row is therefore zero modulo p , contrary to the primitivity of U or V . Hence

$$\gcd(K_*^\circ, h) = 1. \quad (652)$$

Euclid's lemma applied to (651) now proves (640). Finally, (629) and the definitions of $B_\pm, \mathcal{C}_{n,S}$ prove (641). $\square$

Remark 618 (Finite audit and remaining height gap). The accompanying exact full-shell verifier was run at $S = 2$ on all 2260 feasible rows with $n \leq 500$ and $q \leq 5$. It found $K_*^\circ = 1$ on 2253 rows and $K_*^\circ = 5$ on the remaining seven. All seven occurrences are genuine nuisance factors: none is a listed target, a boundary target, or a universal-overlap prime. This finite statement is not used in Proposition 617.

The proposition controls the primitive saturation, but it is not yet a subexponential product-ideal certificate. Indeed,

$$K_* = c_+ c_- K_*^\circ,$$

while $c_+ c_-$ is the target-bearing quantity. Moreover, (641) is $O_q(n)$, not $o_q(n)$, at a linear target-covering radius. Thus the exact theorem isolates the two remaining arithmetic sources xy and $\mathcal{C}_{n,S}$, but does not by itself prove the desired Smith bound.

The moving lower-digit radical also admits a direct formulation using one Apéry number. We record the smooth-radical reduction here because the pair block law gives a particularly transparent proof of all discrepancy branches.

Proposition 619 (Single-Apéry-number radical reduction). *Put*

$$\mathfrak{P}_n = \prod_{p \leq n} p, \quad \mathcal{S}_n = \log \gcd(\mathfrak{P}_n, b_n) = \sum_{\substack{p \leq n \\ p | b_n}} \log p, \quad \mathcal{W}_n = \sum_{\substack{\sqrt{n} < p \leq n \\ p | b_n \bmod p}} \log p. \quad (653)$$

Then

$$0 \leq \mathcal{S}_n - \mathcal{W}_n \ll n^{2/3}. \quad (654)$$

More precisely, let $p \geq 7$, $\sqrt{n} < p \leq n$, and write $n = qp + r$, where $1 \leq q < p$ and $0 \leq r < p$. Define

$$T_p = [p \mid b_r], \quad H_p = [p \mid b_q], \quad C_p = [p \mid \text{num}(a_q)]. \quad (655)$$

Then

$$p \mid b_n \iff T_p \vee H_p, \quad (656)$$

$$v_p(G_n) > 0 \iff T_p \vee C_p, \quad (657)$$

$$H_p \wedge C_p \text{ is impossible}, \quad (658)$$

$$T_p \iff (p \mid b_n) \wedge (v_p(G_n) > 0). \quad (659)$$

Thus a large prime dividing b_n but not G_n is leading-only, whereas a large prime dividing G_n but not b_n is companion-only. Both discrepancy families have total logarithmic weight $O(n^{2/3})$.

Proof. Since $q < p$, the denominators of a_q and a_{q-1} are prime to p . The Apéry–Lucas congruence gives

$$b_n \equiv b_q b_r \pmod{p}, \quad (660)$$

which proves (656). Lemma 87 and Corollary 92 give the pair block law

$$p^3 a_n \equiv a_q b_r \pmod{p}. \quad (661)$$

Here $v_p(d_n) = 3$, and $d_n b_n$ is divisible by p^3 . Hence $p \mid G_n$ if and only if the right side of (661) vanishes, proving (657).

If p divided both b_q and the numerator of a_q , reduction of the Casoratian

$$a_q b_{q-1} - a_{q-1} b_q = \frac{6}{q^3}$$

modulo p would give a contradiction. This proves (658). Taking the intersection of the two support laws now gives

$$(T_p \vee H_p) \wedge (T_p \vee C_p) = T_p \vee (H_p \wedge C_p) = T_p,$$

which is (659).

We record explicitly why no endpoint has been lost. In Lemma 87, propagation from $r = 0$ through $r = p - 1$ divides only by $(r + 1)^3$ for $0 \leq r \leq p - 2$, hence only by units modulo p . The case $r = 0$ is the definition of the block constant. The endpoint supercongruence in Corollary 92 is stated for every $1 \leq q < p$, including $q = p - 1$. Thus (656)–(659) also cover $r = 0, p - 1$ and $q = p - 1$.

It remains to compare logarithmic masses. Let $\mathcal{H}_n$ be the set of primes $p > \sqrt{n}$ for which $p \mid b_{\lfloor n/p \rfloor}$. Put $Q = \lfloor n^{1/3} \rfloor$. Every prime has a unique quotient. For quotients $q \leq Q$, the relevant primes divide b_q , and the standard exponential bound for the Apéry numbers gives

$$\sum_{\substack{p \in \mathcal{H}_n \\ \lfloor n/p \rfloor \leq Q}} \log p \leq \sum_{q \leq Q} \log b_q \ll Q^2. \quad (662)$$

For $q > Q$, one has $p < n/Q$, so Chebyshev's estimate gives

$$\sum_{\substack{p \in \mathcal{H}_n \\ \lfloor n/p \rfloor > Q}} \log p \leq \vartheta(n/Q) \ll n/Q. \quad (663)$$

Thus the leading-digit mass is $O(n^{2/3})$. By (660), every prime counted by $\mathcal{W}_n$ divides b_n , and every additional prime $p > \sqrt{n}$ counted by $\mathcal{S}_n$ is leading-only. Finally, the primes $p \leq \sqrt{n}$ contribute at most $\vartheta(\sqrt{n}) = O(\sqrt{n})$ to the squarefree radical. This proves (654).

The companion-only mass is $O(n^{2/3})$ by Proposition 103. Together with the exact support laws above, this proves the final assertion. $\square$

Corollary 620 (Single-Apéry-number criterion). *Let $\ell_n = \text{lcm}(1, \dots, n)$. Then*

$$\log \gcd(\ell_n, b_n) - O(n^{2/3}) \leq \log G_n \leq 6 \log \gcd(\ell_n, b_n) + O(n^{2/3}). \quad (664)$$

Consequently the all-index Apéry GCD conjecture is equivalent to either of the pointwise statements

$$\log \gcd(\ell_n, b_n) = o(n), \quad \log \gcd(\mathfrak{P}_n, b_n) = o(n). \quad (665)$$

Proof. For $p > \sqrt{n}$, the exponent of p in ℓ_n is one. For the remaining primes,

$$\sum_{p \leq \sqrt{n}} \lfloor \log_p n \rfloor \log p \leq \vartheta(\sqrt{n}) + \sum_{j \geq 2} \vartheta(n^{1/j}) \ll \sqrt{n}. \quad (666)$$

Therefore

$$0 \leq \log \gcd(\ell_n, b_n) - \mathcal{W}_n \ll n^{2/3}. \quad (667)$$

Every target prime divides G_n . Conversely, outside the target set, every prime $p > \sqrt{n}$ dividing G_n is companion-only, whose total logarithmic weight is $O(n^{2/3})$. Proposition 3 bounds the small-prime part of $\log G_n$ by $O(\sqrt{n})$, and Corollary 2 gives $v_p(G_n) \leq 6$ for every $p > \sqrt{n}$. Hence

$$\mathcal{W}_n \leq \log G_n \leq 6\mathcal{W}_n + O(n^{2/3}).$$

Combining this with (667) proves (664) and both equivalences. $\square$

Remark 621 (Scope of the single-term reduction). Corollary 620 is an equivalent reformulation, not a sublinear estimate. It says that the missing theorem is precisely a pointwise bound for the n -smooth radical of the single integer b_n . The size estimate $\log b_n = \Theta(n)$ gives no saving. Moreover, Proposition 125 and (667) imply only

$$|\log \gcd(\ell_n, b_n) - \log \gcd(\ell_n, U_n)| \ll n^{2/3};$$

thus the single-term and one-moment carriers are different arithmetic objects but equivalent at the unresolved scale.

Proposition 622 (Joint Apéry–moment gcd). *Let*

$$J_n = \gcd(\ell_n, b_n, U_n).$$

For every prime $\sqrt{n} < p \leq n$, write $n = qp + r$, with $0 \leq r < p$. Then

$$p \mid J_n \iff p \mid b_r \quad \text{or} \quad (p \mid b_q \text{ and } p \mid U_q). \quad (668)$$

Moreover,

$$\log J_n = \mathcal{W}_n + O(n^{2/3}). \quad (669)$$

If

$$\delta_{b,n} = \text{den}(b_n/\ell_n), \quad \delta_{R,n} = \text{den}(R_n),$$

then one has the exact denominator identity

$$\text{lcm}(\delta_{b,n}, \delta_{R,n}) = \frac{\ell_n}{J_n}. \quad (670)$$

Proof. Since $p > \sqrt{n}$, one has $q < p$ and $v_p(\ell_n) = 1$. Put $M = \ell_n/p$. In

$$U_n = \sum_{k=1}^n b_{n-k} \frac{\ell_n}{k},$$

every term with $p \nmid k$ vanishes modulo p . Writing the surviving indices as $k = ap$, $1 \leq a \leq q$, and using Apéry–Lucas gives

$$\begin{aligned} U_n &\equiv M \sum_{a=1}^q \frac{b_{(q-a)p+r}}{a} \\ &\equiv Mb_r \sum_{a=1}^q \frac{b_{q-a}}{a} \equiv Mb_r \ell_q^{-1} U_q \pmod{p}. \end{aligned} \quad (671)$$

All displayed denominators are units modulo p . Hence

$$p \mid U_n \iff (p \mid b_r) \text{ or } (p \mid U_q),$$

whereas Lucas gives

$$p \mid b_n \iff (p \mid b_r) \text{ or } (p \mid b_q).$$

Intersecting the two conditions proves (668). This includes $r = 0$, since $b_0 = 1$, and it also includes $q = p - 1$; in both cases ℓ_q remains a p -adic unit.

It remains to bound the non-target branch in (668). For a fixed quotient q , every such prime divides the integer $\gcd(b_q, U_q)$, whose logarithm is $O(q)$. Splitting at $Q = \lfloor n^{1/3} \rfloor$, the quotients $q \leq Q$ therefore contribute

$$\leq \sum_{q \leq Q} \log b_q = O(Q^2),$$

while $q > Q$ implies $p < n/Q$ and contributes $O(n/Q)$ by Chebyshev. Primes at most $\sqrt{n}$, including their possible powers in ℓ_n , contribute $O(\sqrt{n})$. This proves (669).

Finally,

$$\delta_{b,n} = \frac{\ell_n}{\gcd(\ell_n, b_n)}, \quad \delta_{R,n} = \frac{\ell_n}{\gcd(\ell_n, U_n)}.$$

Taking the least common multiple prime by prime yields (670). □

Remark 623 (Scope of the joint gcd). The joint gcd removes the leading-digit and one-moment false branches unless they occur simultaneously, but (669) shows that it still carries the entire unresolved target mass up to $O(n^{2/3})$. Thus (670) is a new exact pointwise denominator formulation, not a sublinear denominator estimate.

Proposition 624 (Top-half nonlinear carrier). *For $n \geq 2$, put*

$$\mathcal{C}_n^+ = \operatorname{lcm}_{n/2 < m \leq n} \gcd(m, b_{n-m}, b_{2m-n-1}). \quad (672)$$

Then

$$\mathcal{W}_n \leq \log \mathcal{C}_n^+ \leq \mathcal{W}_n + O(n^{2/3}). \quad (673)$$

Consequently

$$|\log \mathcal{C}_n^+ - \log \gcd(\operatorname{lcm}(1, \dots, n), b_n)| \ll n^{2/3}. \quad (674)$$

In particular, the all-index Apéry GCD conjecture is equivalent to

$$\log \mathcal{C}_n^+ = o(n).$$

Proof. We first classify the large prime divisors of the carrier. Suppose that $p > \sqrt{n}$ and that p divides one of the gcds in (672). Write

$$m = kp, \quad n = tp + a, \quad 0 \leq a < p,$$

and set

$$u = t - k, \quad v = 2k - t - 1.$$

The inequalities $n/2 < m \leq n$ give $u, v \geq 0$, while $u + v = k - 1$. Moreover

$$k = u + v + 1, \quad t = 2u + v + 1. \quad (675)$$

Also $k \leq n/p < p$, so $u, v < p$. The two indices in the gcd have the exact base- p expansions

$$n - m = up + a, \quad 2m - n - 1 = vp + (p - 1 - a).$$

The Apéry–Lucas congruence and reflection therefore give

$$p \mid b_{n-m}, b_{2m-n-1} \iff p \mid b_a \text{ or } p \mid b_u \text{ and } p \mid b_v. \quad (676)$$

The first alternative is exactly the target condition $p \mid b_{n \bmod p}$. In the non-target alternative, $b_0 = 1$ forces $u, v \geq 1$. Hence (675) gives $t \geq 4$, and therefore every large non-target prime satisfies

$$p \leq n/4. \quad (677)$$

This exact restriction does not by itself make the total mass sublinear. To obtain the required contamination estimate, take $K = \max\{3, \lfloor n^{1/3} \rfloor\}$. If a non-target prime has any witness with $k \leq K$, then $b_0 = 1$ and $u + v = k - 1$ show that it divides the single prefix product

$$\prod_{1 \leq j < K} b_j.$$

Hence the total logarithmic weight of all such primes is

$$\leq \sum_{1 \leq j < K} \log b_j \ll K^2.$$

If there is no such witness, every witness has $k > K$, and any one of them gives $p = m/k < n/K$. Chebyshev's estimate bounds the total weight of these primes by $O(n/K)$. Thus all large non-target prime divisors of $\mathcal{C}_n^+$ have total logarithmic weight $O(n^{2/3})$. Since

$$\mathcal{C}_n^+ \mid \text{lcm}(1, \dots, n),$$

one has $v_p(\mathcal{C}_n^+) \leq \lfloor \log_p n \rfloor$. Consequently the primes at most $\sqrt{n}$, together with all prime-power excess, contribute

$$\sum_{p \leq \sqrt{n}} \lfloor \log_p n \rfloor \log p = O(\sqrt{n}).$$

This proves the upper bound in (673).

Conversely, let $p > \sqrt{n}$ be a target prime and write $n = tp + a$, $0 \leq a < p$. Take the candidate multiple

$$m = tp = n - a.$$

One has $n/2 < m \leq n$. Its two lower indices are

$$n - m = a, \quad 2m - n - 1 = (t - 1)p + (p - 1 - a).$$

Thus $p \mid m$, and Lucas plus reflection show that p divides both associated Apéry numbers. The defining lcm includes this candidate, so every target prime occurs in $\mathcal{C}_n^+$. This proves the lower bound in (673) without loss.

Finally, combine (673) with (667) to obtain (674). The equivalence follows from Corollary 620. $\square$

Remark 625 (What the nonlinear carrier does not prove). Proposition 624 packages every quotient class into one characteristic-zero lcm and proves that its false-positive mass is sublinear. It does not bound the target mass: the unconditional estimate

$$\mathcal{C}_n^+ \mid \text{lcm}(1, \dots, n), \quad \log \mathcal{C}_n^+ \leq n + o(n)$$

remains linear. Thus the proposition is a lossless nonlinear reformulation, not a proof of sublinear growth.

Remark 626 (Off-centre version). If distinct physical supports are needed, define

$$\mathcal{C}_n^\circ = \operatorname{lcm}_{\substack{n/2 < m \leq n \\ n-m \neq 2m-n-1}} \gcd(m, b_{n-m}, b_{2m-n-1}).$$

The omitted equality is equivalent to $3m = 2n + 1$, so at most one gcd is removed. Hence

$$0 \leq \log \mathcal{C}_n^+ - \log \mathcal{C}_n^\circ \leq \log n.$$

More precisely, the only target prime that can be lost is

$$p = \frac{2n+1}{3}, \quad n = \frac{3p-1}{2}, \quad p \mid b_{(p-1)/2};$$

indeed, for a central canonical witness one has $tp = 2a + 1 \leq 2p - 1$, hence $t = 1$. If it occurs, it is therefore the unique multiple of p in the top half. Thus the off-centre carrier satisfies (673) with an $O(\log n)$ loss on the lower side.

Corollary 627 (One-sided top-half carrier). *Define*

$$\mathcal{A}_n = \operatorname{lcm}_{n/2 < m \leq n} \gcd(m, b_{n-m}).$$

Then

$$\mathcal{W}_n \leq \log \mathcal{A}_n \leq \mathcal{W}_n + O(n^{2/3}).$$

Moreover,

$$\mathcal{C}_n^+ \mid \mathcal{A}_n, \quad 0 \leq \log \frac{\mathcal{A}_n}{\mathcal{C}_n^+} \ll n^{2/3},$$

and

$$|\log \mathcal{A}_n - \log \gcd(\operatorname{lcm}(1, \dots, n), b_n)| \ll n^{2/3}.$$

Thus the one-sided condition $\log \mathcal{A}_n = o(n)$ is also equivalent to the all-index Apéry GCD conjecture.

Proof. Suppose that $p > \sqrt{n}$ divides a local gcd, and write

$$m = kp, \quad n = tp + a, \quad 0 \leq a < p.$$

The top-half condition gives

$$u := t - k \in \left[0, \frac{t-1}{2}\right], \quad n - m = up + a.$$

Since $u < p$, Lucas gives

$$p \mid b_{n-m} \iff p \mid b_a \text{ or } p \mid b_u.$$

The first branch is the target. In the second branch $u \geq 1$, and hence $t \geq 2u + 1$. Splitting at $U = \lfloor n^{1/3} \rfloor$, the primes having a witness with $u \leq U$ divide $\prod_{1 \leq j \leq U} b_j$, so their total logarithmic weight is $O(U^2)$. Every remaining witness has $t > 2U + 1$, whence $p < n/(2U + 1)$, and Chebyshev bounds its total weight by $O(n/U)$. Primes at most $\sqrt{n}$, including all prime-power excess, contribute $O(\sqrt{n})$.

Conversely, if $p > \sqrt{n}$ is a target, take $m = tp = n - a$. Since $b_0 = 1$, one has $a \neq 0$, and therefore $n/2 < m < n$; also $p \mid \gcd(m, b_a)$. Thus no target is lost. This proves the comparison with $\mathcal{W}_n$. Termwise,

$$\gcd(m, b_{n-m}, b_{2m-n-1}) \mid \gcd(m, b_{n-m}),$$

which proves $\mathcal{C}_n^+ \mid \mathcal{A}_n$; the two logarithmic comparisons with $\mathcal{W}_n$ give the quotient bound. The final comparison follows from (667). $\square$

Remark 628 (Exact support versus logarithmic equivalence). The divisibility $\mathcal{C}_n^+ \mid \mathcal{A}_n$ is exact, but neither $\mathcal{A}_n \mid \gcd(\text{lcm}(1, \dots, n), b_n)$ nor its reverse holds in general. For example, the prime 17 separates the supports at $n = 119$ in the first direction and at $n = 51$ in the second. Thus Corollary 627 is a same-scale reformulation, not an exact factorization of the single-term gcd. The reflected coordinate in $\mathcal{C}_n^+$ removes most finite-range aliases and remains preferable when distinct-support or nonlinear structure is used.

Proposition 629 (Exact convolution-content carrier). *Put*

$$\mathcal{P}_n(X) = \sum_{k=1}^n \frac{\ell_n}{k} b_{n-k} X^k \in \mathbb{Z}[X], \quad \mathbf{c}_n = \text{cont}(\mathcal{P}_n). \quad (678)$$

If $p > \sqrt{n}$, $p \leq n$, and $n = qp + r$ with $0 \leq r < p$, then

$$v_p(\mathbf{c}_n) = \mathbf{1}_{\{p \mid b_r\}}. \quad (679)$$

Moreover

$$\mathbf{c}_n \mid \frac{\ell_n}{n}, \quad \log \mathbf{c}_n = \mathcal{W}_n + O(\sqrt{n}). \quad (680)$$

If $\mathcal{P}_n = \mathbf{c}_n Q_n$ with $Q_n \in \mathbb{Z}[X]$ primitive, then

$$\mathbf{c}_n \text{cont}(Q'_n) = \ell_n. \quad (681)$$

In particular, $\log \mathbf{c}_n = o(n)$ is equivalent to the all-index estimate in Corollary 620.

Proof. Let $p > \sqrt{n}$. Then $v_p(\ell_n) = 1$. Suppose first that $p \mid b_r$. If $p \nmid k$, then $p \mid \ell_n/k$. If $p \mid k$, the inequality $p^2 > n$ gives $k = ap$ with $1 \leq a \leq q$, and the Apéry–Lucas congruence gives

$$b_{n-k} = b_{(q-a)p+r} \equiv b_{q-a} b_r \equiv 0 \pmod{p}.$$

Thus every coefficient in (678) is divisible by p . A target cannot have $r = 0$, since $b_0 = 1$, so $p \nmid n$. The last coefficient is ℓ_n/n and has p -adic valuation one; hence $v_p(\mathbf{c}_n) = 1$.

Conversely, if $p \nmid b_r$, take $k = qp = n - r$. Because $q < p$, the coefficient

$$\frac{\ell_n}{qp} b_r$$

is a p -adic unit. This proves (679), including the endpoint $r = 0$.

The coefficient at $k = n$ proves $\mathbf{c}_n \mid \ell_n/n$. Hence the logarithmic contribution of primes at most $\sqrt{n}$ is at most

$$\sum_{p \leq \sqrt{n}} [\log_p n] \log p = O(\sqrt{n}).$$

Together with (679), this proves (680). Finally,

$$\mathcal{P}'_n(X) = \ell_n \sum_{k=1}^n b_{n-k} X^{k-1}.$$

The coefficient list on the right contains $b_0 = 1$, so its content is exactly ℓ_n . Taking contents in $\mathcal{P}'_n = \mathbf{c}_n Q'_n$ proves (681). $\square$

Lemma 630 (Sparse-support primorial obstruction). *Let $S \subseteq \{1, \dots, n\}$, let $|S| = h$, and suppose*

$$0 \neq H_{n,S} = \sum_{k \in S} c_k \frac{\ell_n}{k} b_{n-k}, \quad c_k \in \mathbb{Z}.$$

Writing $\vartheta(x) = \sum_{p \leq x} \log p$, one has

$$\log |H_{n,S}| \geq \vartheta(n) - \vartheta(\sqrt{n}) - h \log n = n - h \log n + o(n). \quad (682)$$

Consequently, a nonzero linear certificate of logarithmic height $o(n)$ in the raw coefficient basis of (678) must use at least $(1 - o(1))n / \log n$ positions.

Proof. Let $p \in (\sqrt{n}, n]$ divide none of the indices in S . Then $p \mid \ell_n/k$ for every $k \in S$, and hence $p \mid H_{n,S}$, independently of the Apéry values. Each $k \leq n$ is divisible by at most one prime greater than $\sqrt{n}$. Thus the h selected positions can remove medium primes of total logarithmic weight at most $h \log n$. The product of every remaining medium prime divides the nonzero integer $H_{n,S}$, which gives the first inequality in (682); the second follows from the prime number theorem. $\square$

Proposition 631 (Dense Euler-weight floors). *Let*

$$B(z) = \sum_{j \geq 0} b_j z^j, \quad g(z) = \sum_{k=1}^D \frac{c_k}{k} z^k, \quad c_k \in \mathbb{Z}, \quad D \leq n,$$

and put

$$H_n(g) = \ell_n[z^n] B(z)g(z).$$

If $H_n(g) \neq 0$, then

$$|H_n(g)| \geq \frac{\ell_n}{\ell_D}, \quad \log |H_n(g)| \geq n - D + o(n). \quad (683)$$

After dividing the Euler weights by their common gcd, write

$$g = F/d, \quad F \in z\mathbb{Z}[z], \quad d \geq 1$$

with d minimal, and let

$$M(F) = \max_k |[z^k]F|.$$

Then F is primitive,

$$d \mid \ell_D, \quad d \leq DM(F), \quad (684)$$

and every nonzero carrier satisfies

$$|H_n(g)| \geq \frac{\ell_n}{DM(F)}. \quad (685)$$

Finally, put $\mathcal{A}(z) = 1 - 34z + z^2$. If $\mathcal{A} \mid F$, then

$$\log |H_n(g)| \geq \vartheta(n) - \vartheta(D/2) = n - \frac{D}{2} + o(n). \quad (686)$$

Thus no nonzero exact cancellation of the dominant Apéry singularity, of any degree at most n , has subexponential height.

Proof. The rational number $[z^n]Bg$ has denominator dividing ℓ_D . Therefore a nonzero value of $\ell_n[z^n]Bg$ has absolute value at least ℓ_n/ℓ_D , which proves (683).

The condition $c_k \in \mathbb{Z}$ is exactly

$$d \mid k[z^k]F \quad (1 \leq k \leq D).$$

If a prime p divided every coefficient of F , then $p \mid d$ would contradict minimality of d . If $p \nmid d$, writing $F = pF_1$ in the displayed Euler divisibilities shows $d \mid k[z^k]F_1$, and hence every Euler weight is divisible by p . This contradicts their primitive normalization, so F is primitive. If $p^e \mid d$ with $e > \lfloor \log_p D \rfloor$, then the displayed divisibilities force every coefficient of F to be divisible by p . Thus $d \mid \ell_D$. Choosing one nonzero coefficient gives $d \leq DM(F)$. Since

$$H_n(g) = \frac{\ell_n}{d} [z^n]B(z)F(z)$$

and the last coefficient is an integer, (685) follows.

Assume now that $\mathcal{A} \mid F$, and let $p > D/2$ divide d . Modulo p , the Euler divisibilities force

$$F(z) \equiv f_p z^p \pmod{p};$$

this includes the zero right side when $p > D$. If $f_p = 0$, primitivity fails. If $f_p \neq 0$, the monic nonconstant polynomial $\overline{\mathcal{A}}$, whose constant term is 1, would divide z^p in $\mathbb{F}_p[z]$, which is impossible. Hence no prime $p > D/2$ divides d . Every prime in $(D/2, n]$ consequently divides the nonzero integer $H_n(g)$, proving (686). $\square$

Proposition 632 (Exact Smith defect of the coefficient adjoint). *Let*

$$\mathcal{L} = \theta^3 - zP(\theta) + z^2(\theta + 1)^3, \quad P(x) = (2x + 1)(17x^2 + 17x + 5),$$

be the Apéry operator, and put

$$\Phi_n(h) = [z^n]B(z)h(z), \quad \mathcal{L}_n^\dagger = (n - \theta)^3 - zP(n - 1 - \theta) + z^2(n - 1 - \theta)^3.$$

Then

$$\ker(\Phi_n : z\mathbb{Q}[z]_{\leq n} \longrightarrow \mathbb{Q}) = \mathcal{L}_n^\dagger(z\mathbb{Q}[z]_{\leq n-1}). \quad (687)$$

In Euler-weight coordinates, let W_n be the n -by- $(n - 1)$ integer matrix whose m -th column has the three possible nonzero entries

$$\begin{aligned} (W_n)_{m,m} &= m(n - m)^3, \\ (W_n)_{m+1,m} &= -(m + 1)P(n - 1 - m), \\ (W_n)_{m+2,m} &= (m + 2)(n - 1 - m)^3. \end{aligned}$$

Let

$$\mathbf{A}_n = \left(\frac{\ell_n}{k} b_{n-k} \right)_{1 \leq k \leq n}, \quad \mathcal{K}_n = \{x \in \mathbb{Z}^n : \mathbf{A}_n x = 0\}.$$

Then the local adjoint lattice has exact saturation index

$$[\mathcal{K}_n : W_n \mathbb{Z}^{n-1}] = \frac{((n - 1)!)^4 n \mathfrak{c}_n}{\ell_n}. \quad (688)$$

Thus rational adjoint rank and integer saturation retain the unknown convolution content $\mathfrak{c}_n$ itself.

Proof. The adjunction identity

$$[z^n](\mathcal{L}f)h = [z^n]f \mathcal{L}_n^\dagger h$$

shows that the right side of (687) lies in the kernel. For $1 \leq m < n$,

$$\mathcal{L}_n^\dagger z^m = (j+1)^3 z^m - P(j)z^{m+1} + j^3 z^{m+2}, \quad j = n-1-m.$$

Their lowest coefficients are nonzero in distinct degrees, so these $n-1$ vectors are independent. Since $\Phi_n(z^n) = b_0 = 1$, the kernel also has dimension $n-1$, proving (687).

The primitive integer normal to $\mathcal{K}_n$ is $\mathbf{A}_n/\mathbf{c}_n$. The signed maximal-minor vector of W_n is an integer multiple $t_n \mathbf{A}_n/\mathbf{c}_n$. Deleting the last row gives the triangular minor

$$\prod_{m=1}^{n-1} m(n-m)^3 = ((n-1)!)^4.$$

The last coordinate of the primitive normal is $\ell_n/(n\mathbf{c}_n)$. Comparing these two expressions yields

$$t_n = \frac{((n-1)!)^4 n \mathbf{c}_n}{\ell_n}.$$

The gcd of the maximal minors equals the index in the saturated hyperplane lattice, proving (688). $\square$

Remark 633 (Dense adjoint interface). Proposition 629 is an exact support reformulation, not a bound for $\mathcal{W}_n$. The derivative identity alone is insufficient: a primitive polynomial can have derivative content assembled from different index primes. For example, $Q(X) = 3X^2 + 2X^3$ is primitive while $\text{cont}(Q') = 6$. Lemma 630 rules out every fixed-width raw linear, Hankel, Padé, or Casoratian certificate. A surviving proof must use degree $n - o(n)$, an exponentially large primitive numerator, and a transformed normalization whose target-safety, nonvanishing, and subexponential final height are all proved separately.

Proposition 634 (Bounded Kummer-order radical). *Let $n, D \geq 1$. For each prime p , let j_p be the least nonnegative residue of n modulo p , and put*

$$d_p = \frac{p-1}{\gcd(j_p, p-1)}.$$

Then every prime for which $d_p \leq D$ divides

$$\mathcal{K}_D(n) = n(n+1) \prod_{2 \leq d \leq D} \prod_{\substack{1 \leq a < d \\ (a,d)=1}} (dn+a). \quad (689)$$

Consequently, for every set $\mathcal{P}$ of such primes,

$$\sum_{p \in \mathcal{P}} \log p \leq \log \mathcal{K}_D(n) \ll D^2 \log(D(n+1)). \quad (690)$$

In particular, the target primes $p > \sqrt{n}$ with $d_p \leq D = D(n)$ have total logarithmic weight $o(n)$ whenever

$$D^2 \log(D(n+1)) = o(n).$$

Proof. Put $g = \gcd(j_p, p-1)$, $d = d_p$, and $a = j_p/g$. Then

$$p-1 = gd, \quad j_p = ga, \quad (a, d) = 1,$$

and reduction modulo p gives

$$dn + a \equiv dj_p + a = a(p-1) + a = ap \equiv 0 \pmod{p}.$$

If $d \geq 2$, then $1 \leq a < d$, so p divides the corresponding factor of the double product in (689). If $d = 1$, there are exactly two possibilities:

$$(j_p, a) = (0, 0) \quad \text{or} \quad (j_p, a) = (p-1, 1).$$

They give $p \mid n$ or $p \mid n+1$, respectively. This proves the carrier divisibility.

There are

$$2 + \sum_{d=2}^D \varphi(d) = 1 + \sum_{d=1}^D \varphi(d) \ll D^2$$

positive factors in $\mathcal{K}_D(n)$, and each is at most $D(n+1)$. This proves (690). $\square$

Remark 635 (Scope of the Kummer-order split). For an Apéry target, the two order-one endpoints are vacuous: $b_0 = 1$ and $b_{p-1} \equiv 1 \pmod{p^2}$. They were retained in (689) to make the arithmetic statement unconditional. Proposition 634 disposes of every bounded, and even moderately growing, character order. It gives no estimate for the complementary high-order Kummer characters. Unbounded character order rules out a finite literal list of Kummer local types or one fixed finite cyclic cover, but does not by itself rule out arbitrary pushforwards, subquotients, or scalar modular congruences.

The preceding obstruction concerns a particular product ideal. There is a second, independent reason why a global recurrence in the shift variable cannot compress the full stencil to a small boundary source.

Proposition 636 (Cyclotomic spectral obstruction). *For fixed q, n, d, D , assume $d \geq 1$. On every infinite tail on which the powers under consideration are positive, the sequence A_s has a finite exponential representation*

$$A_s = \sum_{\alpha \in \Sigma_{n,q}} \gamma_\alpha \alpha^s, \quad \gamma_\alpha \neq 0, \quad \alpha \neq 0. \quad (691)$$

Its minimal monic constant-coefficient annihilator is

$$\chi_{n,q}(T) = \prod_{\alpha \in \Sigma_{n,q}} (T - \alpha) \in \mathbb{Z}[T]. \quad (692)$$

For fixed q ,

$$\deg \chi_{n,q} = \Omega_q \left(\frac{n^2}{\log n} \right), \quad \log \max_j |[T^j] \chi_{n,q}| = \Omega_q \left(\frac{n^2}{\log n} \right). \quad (693)$$

The recurrence defects R_s are an exponential-polynomial sequence with the same set of nonzero bases. If $\delta_\alpha = \deg Q_\alpha$, where Q_α is defined in (703) below, then

$$\chi_{n,q}^{(R)}(T) = \prod_{\alpha \in \Sigma_{n,q}} (T - \alpha)^{\delta_\alpha + 1} \in \mathbb{Z}[T], \quad \delta_\alpha \in \{2, 3\}. \quad (694)$$

Every selected root in (701) has $\delta_\alpha = 3$. Consequently

$$\deg \chi_{n,q}^{(R)} = \Omega_q \left(\frac{n^2}{\log n} \right), \quad \log \max_j |[T^j] \chi_{n,q}^{(R)}| = \Omega_q \left(\frac{n^2}{\log n} \right). \quad (695)$$

More generally, suppose

$$\sum_{j=0}^r p_j(s) R_{s+j} = B_s, \quad p_j \in \mathbb{Q}[s], \quad (696)$$

is nonzero, and the exponential-polynomial sequence B_s has at most T distinct nonzero spectral roots. Then

$$r \geq |\Sigma_{n,q}| - T. \quad (697)$$

In particular, no order- $O_q(n)$ polynomial-coefficient recurrence with a bounded-spectrum source can govern the full-stencil sequence at the linear shift radius.

Proof. The root-of-unity filter applied to (535) gives

$$C_m(e) = \frac{1}{e^3} \sum_{\xi_1^e = \xi_2^e = \xi_3^e = 1} \Lambda(\xi_1, \xi_2, \xi_3)^m. \quad (698)$$

Substitution in the fixed Newton functional and collection of equal nonzero values of Λ proves (691). More precisely, if w_e is the Newton weight at node e , and

$$N_e(\alpha) = \#\{(\xi_1, \xi_2, \xi_3) : \xi_i^e = 1, \Lambda(\xi_1, \xi_2, \xi_3) = \alpha\},$$

then the grouped coefficient is

$$\gamma_\alpha = \alpha^M \sum_{e=d}^D \frac{w_e}{e^3} N_e(\alpha). \quad (699)$$

The set of roots with nonzero grouped coefficient is Galois stable and consists of algebraic integers. Distinct exponential sequences are linearly independent on an infinite tail, so (692) follows.

Let $\mathcal{P}_{n,q}$ be the set of primes $e \geq 5$ such that

$$d \leq e \leq D, \quad 2e > D. \quad (700)$$

For a primitive e -th root ζ_e , the triples $(\zeta_e^k, 1, 1)$, $1 \leq k < e$, contribute the values

$$\alpha_{e,k} = \Lambda(\zeta_e^k, 1, 1) = 4(\zeta_e^k + 5 + 4\zeta_e^{-k}). \quad (701)$$

They are pairwise distinct, since

$$u + 4u^{-1} = v + 4v^{-1} \implies (u - v) \left(1 - \frac{4}{uv} \right) = 0$$

and $|uv| = 1$. Nor can one of these values occur at another stencil node $m \neq e$. Indeed, $2e > D$ implies $e \nmid m$. The field $\mathbb{Q}(\zeta_m)$ is unramified at e , whereas every nontrivial subfield of $\mathbb{Q}(\zeta_e)$ is ramified there. Their intersection is therefore $\mathbb{Q}$. But $\alpha_{e,k} \notin \mathbb{Q}$, since otherwise ζ_e^k would satisfy a quadratic equation over $\mathbb{Q}$, contrary to $[\mathbb{Q}(\zeta_e) : \mathbb{Q}] = e - 1 \geq 4$. At node e , all occurrences of a fixed value have the same nonzero Newton coefficient. Since the value occurs at no other node, (699) becomes

$$\gamma_{\alpha_{e,k}} = \alpha_{e,k}^M \frac{w_e}{e^3} N_e(\alpha_{e,k}) \neq 0.$$

Consequently

$$\{\alpha_{e,k} : e \in \mathcal{P}_{n,q}, 1 \leq k < e\} \subseteq \Sigma_{n,q}. \quad (702)$$

The full Galois orbit in (701) has the monic integral polynomial

$$F_e(T) = \prod_{k=1}^{e-1} (T - \alpha_{e,k}), \quad |F_e(0)| = 4^{e-1} \frac{4^e + 1}{5}.$$

The constant term follows from

$$\alpha_{e,k} = 4 \frac{(1 + \zeta_e^k)(4 + \zeta_e^k)}{\zeta_e^k}, \quad \prod_{k=1}^{e-1} (1 + \zeta_e^k) = 1, \quad \prod_{k=1}^{e-1} (4 + \zeta_e^k) = \frac{4^e + 1}{5}.$$

By (702), the product of these F_e 's divides $\chi_{n,q}$ in $\mathbb{Z}[T]$. Hence

$$\begin{aligned} \deg \chi_{n,q} &\geq \sum_{e \in \mathcal{P}_{n,q}} (e - 1), \\ \log |\chi_{n,q}(0)| &\geq \sum_{e \in \mathcal{P}_{n,q}} \left((e - 1) \log 4 + \log \frac{4^e + 1}{5} \right). \end{aligned}$$

The interval (700) has length $\asymp_q n$. The prime number theorem therefore makes both right sides $\Omega_q(n^2/\log n)$, proving (693).

On a single spectral term $A_s = \alpha^s$, the defect is

$$R_s = \alpha^{s-2} Q_\alpha(n + s), \quad (703)$$

where

$$\begin{aligned} Q_\alpha(k) &= (\alpha^2 - 34\alpha + 1)k^3 + (51\alpha - 3)k^2 \\ &\quad + (3 - 27\alpha)k + 5\alpha - 1. \end{aligned}$$

This polynomial is never identically zero: its constant coefficient would force $\alpha = 1/5$, while its quadratic coefficient would then be $36/5$. Its degree is two precisely when $\alpha = 17 \pm 12\sqrt{2}$, and is three otherwise; at either exceptional value the quadratic coefficient remains nonzero. Thus every root of A_s survives in R_s .

A polynomial-exponential sequence $q(s)\alpha^s$, with $\deg q = \delta$, has exact minimal constant-coefficient factor $(T - \alpha)^{\delta+1}$. The degree condition above is Galois invariant, so the product in (694) is monic and integral. Moreover,

$$\operatorname{Im} \alpha_{e,k} = -12 \sin(2\pi k/e) \neq 0$$

for the selected odd primes. Hence every selected root has degree-three amplitude and contributes $F_e(T)^4$ to $\chi_{n,q}^{(R)}(T)$. The degree and constant-term arguments already used for $\chi_{n,q}$ now prove (695).

Finally, let $d_0 = \max_j \deg p_j$, and let a_j be the coefficient of s^{d_0} in $p_j(s)$. The polynomial

$$H(T) = \sum_{j=0}^r a_j T^j$$

is nonzero and has degree at most r . For every spectral root α absent from B_s , linear independence forces

$$\sum_{j=0}^r p_j(s) \alpha^j Q_\alpha(n + s + j) = 0$$

as a polynomial in s . If $\delta_\alpha = \deg Q_\alpha$ and $u_\alpha \neq 0$ is its leading coefficient, the coefficient of $s^{d_0+\delta_\alpha}$ is

$$u_\alpha H(\alpha).$$

Therefore $H(\alpha) = 0$ for every one of the at least $|\Sigma_{n,q}| - T$ absent bases. Root counting gives $r \geq |\Sigma_{n,q}| - T$, which proves (697). $\square$

Remark 637 (Scope of the spectral obstruction). Proposition 636 excludes a global low-order Ore recurrence whose source has bounded spectral support. It does not exclude a nonlinear integral identity, nor an inhomogeneous boundary packet which retains almost the whole spectrum but has an exceptionally small integer value. Together with Proposition 610, it leaves exactly the actual-shell, non-alternating Smith–Bezout interface described in Remark 611.

Corollary 638 (Two-sided radical cover). *If $\ell \geq 6$, then*

$$R_q^{[3]}(n) \mid h_-(n, q)h_+(n, q). \quad (704)$$

The omitted lower digits introduce only the fixed primes 5 and 73: indeed, $b_1 = 5$ and $b_2 = 73$. The endpoint digits are nontargets, since $b_0 = 1$ and $b_{p-1} \equiv 1 \pmod{p}$.

Proof. For each prime in (563), Theorem 597 supplies an eligible side, and the three minors on that side are divisible by p . Hence p divides one of the two side gcds. The remaining assertions follow from the displayed values of b_0, b_1, b_2 and the standard endpoint congruence for b_{p-1} . $\square$

Remark 639 (A separated mixed-row false positive). The recurrence-cancelled column is not merely a relabelling of the ordinary mixed exterior row. At $(q, p, r, n) = (3, 13, 3, 42)$, one has

$$T_{3,3} \equiv U_{3,3} \equiv 0 \pmod{13}, \quad b_3 \equiv 2 \pmod{13}.$$

Thus the two mixed Cartier–Ore coordinates share a non-target zero. Here

$$(M, d, D, \ell) = (39, 10, 13, 3),$$

and the divided cross residues are

$$(H_0^-, H_1^-, H_2^-) \equiv (1, 1, 0), \quad (H_0^+, H_1^+, H_2^+) \equiv (1, 2, 3) \pmod{13}. \quad (705)$$

Thus the first divided row whose two stencils contain (p-1) removes this common non-target. In contrast, all eligible rows vanish at the genuine targets

$$(q, p, r) = (3, 17, 3), \quad (3, 19, 8), \quad (3, 31, 8),$$

as the theorem requires.

Remark 640 (Height limitation). This construction proves local selectivity, not the required global height saving. Positivity of Λ , the Newton weights, and (545)–(546) give only

$$\log \max_{0 \leq j \leq 2} |H_j^\pm| = O_q(n). \quad (706)$$

Already at $(n, q) = (42, 3)$, the integer $H_0^- = H_0^+$ has 402 bits. No estimate of the form

$$\log |h_-(n, q)| + \log |h_+(n, q)| = o_q(n)$$

is proved here. Such an estimate requires an additional inhomogeneous Smith–Bezout or cross-prime input; Laurent coprimality and the local laws alone do not supply it.

The exact integer identities, 210 Pascal divisibility checks, 160 selector checks, 195 eligible local-law checks over 45 mixed candidate cases, the separating example (705), and the three retained targets are reproduced by `research/scripts/q32_mixed_w_third_functional_verify.py`.

19 Value distribution of the Apéry sequence in one period: unconditional multiplicity, image and energy bounds

Throughout this section $p \geq 5$ is prime, $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, $N_p(c) = \#\{0 \leq r \leq p-1 : b_r \equiv c \pmod{p}\}$, and $E(p) = \sum_{c \in \mathbb{F}_p} N_p(c)^2$ is the one-period collision energy (empirically $E(p) = 3p + O(\sqrt{p})$; script `fable_vertical_value_law.py`). The engine of this section is the *non-autonomy* of the Apéry recurrence $(m+1)^3 b_{m+1} = P(m)b_m - m^3 b_{m-1}$, $P(m) = 34m^3 + 51m^2 + 27m + 5$: the moving coefficient acts as the derivative-substitute that Weil/Stepanov technology lacks here (cf. §16 for why the bounded-conductor routes are closed). Transfer notation: $b_{r+d} = A_d(r)b_r + B_d(r)b_{r-1}$ with $A_d, B_d \in \mathbb{F}_p(x)$, poles confined to $\{-1, \dots, -d\}$ of order ≤ 3 .

19.1 Statements

Theorem 641 (image). $\#\{b_r \bmod p : 0 \leq r < p\} \geq ((p-1)/4)^{1/3}$ for every prime p .

Theorem 642 (multiplicity). *There is an absolute C_0 with $\max_{c \neq 0} N_p(c) \leq C_0 p / \log p$ for every prime p .*

Theorem 643 (energy). $E(p) \ll p^2 / \log p$ for every prime p . Via Parseval $(\sum_{h \neq 0} |C_p(h)|^2 = pE(p) - p^2)$ this gives the average vertical bound $(\frac{1}{p-1} \sum_{h \neq 0} |C_p(h)|^2)^{1/2} \ll p / \sqrt{\log p}$.

Theorem 644 ($p^{3/4}$ off a thin prime set). *There is a set $\mathfrak{E}$ of primes with $\#\{q \in \mathfrak{E} : q \leq X\} = O(X^{3/4+o(1)})$ such that for $p \notin \mathfrak{E}$ and all $c \neq 0$: $N_p(c) \ll p^{3/4}(\log p)^{O(1)}$ and $E(p) \ll p^{7/4}(\log p)^{O(1)}$.*

Theorem 645 (short lags). *For $\Delta \leq c_0 \log p$ (all p ; and for $\Delta \leq p^{1/3}$ when $p \notin \mathfrak{E}$), $\sum_{1 \leq d \leq \Delta} \#\{r : b_{r+d} \equiv b_r\} \leq 2p + 6\sqrt{p\Delta^3}$.*

Theorem 646 (universal nondegeneracy, (UN)). *For every prime $p \geq 5$ and all $1 \leq h < k < p$ the rational function $\Psi_{h,k} = (1 - A_h)B_k - (1 - A_k)B_h$ is not identically zero over $\mathbb{F}_p$; its reduced numerator has degree exactly $3k$ (measured; claimed $\leq 3(k-1)$ up to the normalizing unit).*

Proof status. Complete 17-lemma proof delivered (session round R4b, archived `chatgpt-answers/Q6311-94300c1e` full display formulas on the Notion source page — transcription into this section is a pending editorial task). Chain: cleared continuants Q_m ; $\Psi = \text{unit} \cdot \Delta_{h,k}$ with Δ polynomial; $Q_m \neq 0$ for $m < p$; continuant reflection via tridiagonal J -conjugation; Plücker identity and subtractive Euclidean descent $(h, k) \rightarrow (\min(h, k-h) \dots)$; terminal central-return case $(d, 2d)$, $C^d = \pm I$, excluded by a power-series expansion at infinity with leading coefficients $5d/64$ ($\epsilon = -1$, at z^2) and $-75d^3/1024$ ($\epsilon = +1$, at z^4), nonzero for $p \geq 7$; $p = 5$ explicit. Machine verification: the terminal expansion checked for ALL 277 applicable (p, d, ϵ) with $7 \leq p < 500$ — zero failures (`un_proof_check.py`); the statement itself swept clean over $\approx 10^6$ cases (`universal_nonvanish.py`, `stress_test.py`).

Theorem 647 (all-prime $p^{3/4}$ and $p^{5/3}$). *For every prime $p \geq 5$: (i) $N_p(c) \leq 8p^{3/4}$ for every $c \in \mathbb{F}_p$ (including $c = 0$), by the block/Hölder triple count driven by Theorem 646; (ii) $E(p) \ll p^{5/3}$, by the c -uniform triple-convexity count ($\sum_c m_c^3 \ll p^2 H$ at block length H , Cauchy–Schwarz, $H = p^{1/3}$); (iii) hence $(\frac{1}{p-1} \sum_{h \neq 0} |C_p(h)|^2)^{1/2} \ll p^{5/6}$. Theorems 642 and 644 are superseded by (i) (kept above for their elementary self-contained proofs).*

Proposition 648 (methodological ceiling). *An abstract block-permutation adversary realizes every consequence of the one-base certificate method (value-independent nonzero certificates of degree $O(k \log p)$ per lag pair, reflection, spacing) with $E \asymp p^{5/3}$. Hence no refinement of the present*

architecture alone reaches $E \ll p^{3/2}$; the ρ -ladder (family-average root count $R_H \ll H^\rho$: $\rho = 3$ now, $\rho = 2 \Rightarrow p^{3/2}$, $\rho \rightarrow 1 \Rightarrow p^{1+o(1)}$) requires filtering candidate kernels to the distinguished Apéry orbit (two-base compatibility, anchored backward transfer, or spectral input). A Kummer-order stratification experiment (40 primes to 1.8×10^5) shows no bounded-order concentration of collision lags — the Mellin/Kummer route has no self-twist shortcut, and the sub-5/3 wall is structural for the present methods.

Proposition 649 (downstream limitation — do not overclaim). *Single-prime vertical bounds enter the gcd problem of Problem 3.2 only through $|Z_p| \leq E(p)^{1/2}$. Since the zero-fibre bound $|Z_p| \leq 3p^{2/3}$ is already stronger than $E(p)^{1/2}$ for every exponent proved here, Theorems 642–645 yield no improvement of the density-one theorem nor of the pointwise problem. Moreover no family of per-prime vertical bounds can settle the pointwise problem: the singleton-alignment countermodel ($Z_p = \{n_0 - p\}$) satisfies every vertical constraint yet gives $H(n_0) \gg n_0/\log n_0$. The remaining interface is cross-prime: with $|Z_p| \ll p^\alpha$, a k -th cross-prime moment bound at the independence scale with $k > 1/(1 - \alpha)$ suffices ($\alpha = 2/3$: fourth moment).*

19.2 The collision mechanism

Fix $c \neq 0$ and $1 \leq r < r + d \leq p - 2$ with $b_r \equiv b_{r+d} \equiv c$. From $c = A_d(r)c + B_d(r)b_{r-1}$: *Type I* ($B_d(r) \neq 0$): $b_{r-1} = c\sigma_d(r)$ with $\sigma_d = (1 - A_d)/B_d$ — the predecessor is determined. *Type II* ($B_d(r) = 0$): $A_d(r) = 1$, so r roots the numerator of B_d (degree $\leq 4d$). If r has two type-I lags $d_1 < d_2$ then $\sigma_{d_1}(r) = \sigma_{d_2}(r)$, i.e. r roots the numerator of Ψ_{d_1, d_2} , of degree $\leq 9(d_1 + d_2)$. Machine check: all 60 multi-lag fiber elements at $p = 101, 199$ root the predicted polynomial (`psi_e2e.py`); at $p = 101$ the 49 type-II collision pairs are exactly the reflection pairs $r \leftrightarrow p - 1 - r$.

19.3 Polar constants and nondegeneracy

Lemma 650 (rank-one polar identity). $(-5, 1) \cdot T_{d-1}(-d) = d^3(-b_d, b_{d-1})$. *Consequently at $x = -d$ the transfer entries have poles of order exactly 3 (over $\mathbb{Q}$) with*

$$c_d := \lim_{x \rightarrow -d} (x + d)^3 B_d(x) = d^3 b_{d-1}, \quad a_d := \lim_{x \rightarrow -d} (x + d)^3 A_d(x) = -d^3 b_d.$$

Proof. Split $M(x) = (x + 1)^{-3}(1, 0)^\top(P(x), -x^3) + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$; T_{d-1} is regular at $-d$, so the polar part of $T_d = M(x + d - 1)T_{d-1}$ at $-d$ is $(1, 0)^\top(-5, 1)T_{d-1}(-d)$. Induction using $P(-1 - n) = -P(n)$ and the recurrence: base $(-5, 1)M(-2) = (-584, 40) = 2^3(-b_2, b_1)$; step $d^3(-b_d, b_{d-1})M(-d - 1) = (d + 1)^3(-b_{d+1}, b_d)$ is exactly the recurrence. (Exact-arithmetic verification $d \leq 30$: `cd_test.py`, `ad_test.py`.) $\square$

Lemma 651 (two endpoint witnesses). *Let $1 \leq d_1 < d_2$, $\delta = d_2 - d_1$. The $(x + d_1)^{-3}$ -coefficient of Ψ_{d_1, d_2} equals $(b_\delta - 1)d_1^3 b_{d_1-1}$, and the $(x + d_2)^{-3}$ -coefficient equals $d_2^3(b_{d_2-1} - b_{\delta-1})$. Both are nonzero over $\mathbb{Q}$ (b is strictly increasing with $b_1 = 5$). Hence $\Psi_{d_1, d_2} \not\equiv 0$ over $\mathbb{Q}$ for every pair, and over $\mathbb{F}_p$ whenever p fails to divide the corresponding integer.*

Proof. First coefficient: $\Psi = (B_{d_2} - B_{d_1}) - K$, $K = \det T_{d_1} \cdot B_\delta(x + d_1)$ is regular at $-d_1$ (pole³ × zero³; $B_\delta \in x^3 \mathbb{F}_p[[x]]$ always by $B_k(x) = -x^3(x + 1)^{-3}A_{k-1}(x + 1)$ — over $\mathbb{F}_p$ the cubic coefficient may vanish, membership in the cube ideal suffices), and $B_{d_2} \sim A_\delta(0)B_{d_1} = b_\delta B_{d_1}$ there. Second coefficient: the reflected scalar identity (negative-window transfer reflected by $P(-1 - n) = -P(n)$) evaluates $(1 - A_{d_1}(-d_2))b_{d_2-1} - b_{d_2}B_{d_1}(-d_2) = b_{d_2-1} - b_{\delta-1}$; verified for 145 pairs at $p = 101$ including direct symbolic Laurent limits (`K_test.py` and the audited snippet in `Q6306.md`). $\square$

19.4 Proofs of Theorems 641–645

Theorem 641. If the image has size D , some state $(b_r, b_{r-1}) = (x, y)$ repeats $\geq (p-1)/D^2$ times; for those r , $b_{r+1} = (P(r)x - r^3y)(r+1)^{-3}$ is a fixed rational function of degree ≤ 4 , nonconstant unless $(x, y) = (0, 0)$ (excluded: no two consecutive zeros, by backward propagation to $b_0 = 1$), taking each value ≤ 4 times inside the image: $(p-1)/D^2 \leq 4D$. $\square$

Theorem 642. Order the fiber; $d_1(r) < d_2(r)$ the next-two lags. $\sum_r (d_1 + d_2)(r) \leq 3p$, so $\geq N/2 - O(1)$ elements have $d_1 + d_2 \leq D' := 12p/N$ (the earlier $3N/4$ claim is false — adversarial near-uniform fibers; $N/2$ suffices). If $N \geq C_0 p / \log p$ then $3.526 k \leq 0.85 \log p$ for every index $k \leq D'$ in play, so the integers b_{d_1-1} and b_{d_2-1} lie in $(0, p)$ ($b_k < (1+\sqrt{2})^{4k}$) and every witness of Lemma 651 is a unit mod p : every pair in play is nondegenerate. Capacity: type-II $\sum_{d \leq D'} O(d) = O(D'^2)$ plus type-I $\sum_{d_1 < d_2 \leq D'} 9(d_1 + d_2) \leq 5D'^3 \ll (\log p)^3 < N/4$. Contradiction. $\square$

Theorem 643. $E \leq N_p(0)^2 + \max_{c \neq 0} N_p(c) \cdot p$ with $N_p(0) \leq 3p^{2/3}$. $\square$

Theorem 644. Run the same count at $D' = 12p/N$, $N \geq p^{3/4+\varepsilon}$. A pair escapes both witnesses of Lemma 651 only if $p \mid \gcd(b_\delta - 1, b_{d_2-1} - b_{d_1-1})$ (or the b_{d_1-1} -analogue) — fixed nonzero integers of log-size $O(d_2)$. $\#\{\text{bad } q \leq X\} \leq \sum_{d_1 < d_2 \leq X^{1/4}} \omega(\gcd) = O(X^{3/4+o(1)})$. For $p \notin \mathfrak{E}$ no escape exists and the counting is verbatim. (Sweep $d_1 < 20$, $d_2 \leq 50$: every prime factor of every such gcd is $\leq 89 \approx 2.4 d_2$ — the visible exceptional set is empty; `G_wide_sweep.py`, `bracket_test.py`.) $\square$

Theorem 645. The Ψ polynomials are independent of c , so $\sum_r \binom{L_\Delta(r)}{2} \leq \sum_{d_1 < d_2 \leq \Delta} 9(d_1 + d_2) \leq 5\Delta^3$ where $L_\Delta(r)$ counts same-fiber lags $\leq \Delta$; Cauchy–Schwarz on $T_\Delta = \sum_r L_\Delta(r)$ gives $T_\Delta \leq 2p + 6\sqrt{p}\Delta^3$. $\square$

19.5 Provenance and audit trail

All identities were discovered and verified in exact arithmetic before use (scripts named above; see `FABLE_NOTES_energy_bootstrap.md` for the session log). Negative literature search (audited): no prior maximum-fiber, image, or energy bound of this type for Apéry or general P -recursive sequences mod p was located (closest: Garaev–Luca–Shparlinski, JCTA 113 (2006), residue classes / additive bases; Luca–Shparlinski, JLMS 78 (2008), prime factors of b_n ; Ostafe–Shparlinski, Canad. J. Math. 74 (2022), C -finite families). We therefore state novelty as “we are not aware of prior bounds of this form”.

19.6 Phase 3/2: the mirror structure and the residual law (added 2026-08-01)

Theorem 652 (strong reflection). *Every solution germ y of the Apéry recurrence mod p , extended over the regular window $[0, p-1]$, satisfies $y_{p-1-n} = y_n$. (The classical reflection congruence $b_{p-1-r} \equiv b_r$ is the special case $y = b$.)*

Proof. At $c \equiv -\frac{1}{2}$: $P(c) = 0$ (the factor $2m+1$ of P) and $(c+1)^3 = -c^3$, so the recurrence at $m = c$ reads $y_{c+1} = y_{c-1}$ for every solution. The reflected sequence $\tilde{y}_n := y_{p-1-n}$ is a solution ($P(-1-n) = -P(n)$) and agrees with y at the consecutive indices $c, c+1$; two consecutive values determine a solution on the regular window. (Machine check: 80/80 random germs at $p \in \{101, 199, 1009, 4001\}$; `strong_reflection.py`. Presumably known to specialists in some form, but we could not locate the all-germ statement; it is the engine of the following identity, which appears to be new.) $\square$

Corollary 653 (mirror root). *For $h \equiv k \pmod{2}$ and $r^* = (p-1-h-k)/2$ (integer representative, non-wrapping window): $\Psi_{h,k}(r^*) = 0$. Indeed $p-1-(r^*+h) = r^*+k$, so the h -return germ at r^* automatically k -returns. The parity condition is exactly non-wraparound (verified: 240/240 forced roots in the even class at two primes; 0/256 at the wrapped point $2r+h+k \equiv -1$ in the odd class).*

The $\rho = 2$ program. Measurements (primes to 10^5 , H to 128; `rho_measure.py`) give the exact decomposition

$$R_H := \sum_{h < k \leq H} Z(\Psi_{h,k}) = \#\{h < k \leq H : h \equiv k \pmod{2}\} + (0.47\text{--}0.50) H^2,$$

with the residual per-pair root distribution Poisson(1/2)-like. The single remaining statement for $E(p) \ll p^{3/2+o(1)}$ is

$$\textbf{(RES)} \quad \sum_{h < k \leq H} (Z(\Psi_{h,k}) - \mathbf{1}[h \equiv k \pmod{2}]) \ll H^2,$$

i.e. the Ψ family has random-polynomial root statistics beyond its one forced root. (RES) is under active attack (two-base determinants, Casoratian telescoping in h , family second moments).

20 The mod- p structure theory: strong reflection, the Frobenius tower, and the apparition theorem

Throughout $p \geq 7$, $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, $F(t) = \sum_{n \geq 0} b_n t^n$, $q(t) = 1 - 34t + t^2$, $A_p(t) = \sum_{n < p} b_n t^n$, $\chi_p = \left(\frac{-6}{p}\right)$, $P(m) = 34m^3 + 51m^2 + 27m + 5$.

20.1 Strong reflection

Theorem 654 (strong reflection). *Every solution germ y of the Apéry recurrence mod p on the regular window satisfies $y_{p-1-n} = y_n$. In particular A_p is palindromic, $b_{p-1} \equiv 1$, $b_{p-2} \equiv 5$.*

Proof. At $c \equiv -\frac{1}{2}$: $P(c) = 0$ (the factor $2m+1$) and $(c+1)^3 = -c^3$, so the recurrence at $m = c$ degenerates to $y_{c+1} = y_{c-1}$ for every solution. The reflected sequence $\tilde{y}_n = y_{p-1-n}$ is a solution ($P(-1-n) = -P(n)$) agreeing with y at the consecutive indices $c, c+1$; propagate. (Machine check: 80/80 random germs, `strong_reflection.py`) $\square$

Corollaries used below: the mirror root $\Psi_{h,k}\left(\frac{p-1-h-k}{2}\right) = 0$ for $h \equiv k \pmod{2}$, and the universal even-lag type-II centers $A_d\left(-\frac{d+1}{2}\right) = 1$, $B_d\left(-\frac{d+1}{2}\right) = 0$ (§19; `mirror_identity.py`, `factorial_ledger.py`).

20.2 The complete mod- p structure theorem

Theorem 655 (structure). *Over $\mathbb{F}_p$ the truncated generating polynomial factors as*

$$A_p = \begin{cases} S_p^2, & \chi_p = +1, \\ q \cdot S_p^2, & \chi_p = -1, \end{cases} \quad S_p(0) = 1,$$

and S_p is the truncation of a FIXED $\mathbb{Q}$ -series: $S_p = [\sqrt{F}]_{(p-1)/2}$ when $\chi_p = +1$, and $S_p = [\sqrt{F/q}]_{(p-3)/2}$ when $\chi_p = -1$. The fixed branches $\tau = \sqrt{F}$, $\sigma = \sqrt{F/q}$ satisfy second-order half-integer-index recurrences (e.g. $4(j+2)^2 \tau_{j+2} = 2(68j^2 + 170j + 107) \tau_{j+1} - (2j+1)^2 \tau_j$) and are algebraic pullbacks of a single classical ${}_2F_1$ with exact coefficient formulae by Lagrange inversion (`CODEX_JACOBSTHAL_DEEP.md`). Consequently the entire mod- p structure of the Apéry sequence is carried by: three universal sequences (D_r , E_r , $\tilde{U}_r$ of the tower below), two fixed $\mathbb{Q}$ -series (τ , σ), and exactly two scalars per prime (χ_p and β_p). (Square-factor identification verified coefficientwise $p < 150$ and spot-checked in both classes: `sp_spotcheck.py`.)

20.3 The Frobenius tower and the master corner identity

Theorem 656 (tower). *For $0 \leq r < p$ the two-digit block expansion*

$$b_{p+r} \equiv 5b_r + 10p D_r + p^2 E_r + p^3(\tilde{U}_r + \beta_p b_r) + p^4(\hat{U}_r - 6W_r^a + 2\beta_p D_r + 8a_r + y_4 b_r) \pmod{p^5}$$

holds with UNIVERSAL (p -independent) sequences at orders p^0 – p^2 , where $D_r = \sum_k \binom{r}{k}^2 \binom{r+k}{k}^2 (H_{r+k} - H_{r-k})$. (Attribution: the p^0 layer is Gessel's congruence (1982); the p^1 – p^2 layers are the $n = 1$ case of A. Straub, arXiv:2301.12248, Monatsh. Math. (2024), Thm. 1.3. New here: the p^3/p^4 universality, the closed forms of the invariants, and the all-order tower structure.) The only p -dependent invariants through order p^4 are $\beta_p = (b_p - 5)/p^3 \pmod{p}$ and y_4 (second invariant, weight-3 Kummer quotient). Moreover:

$$\textbf{(master corner identity)} \quad b_p - 5 \equiv -7p^2 H_{p-1}^{(2)} \pmod{p^5},$$

which subsumes Beukers' $b_p \equiv 5 \pmod{p^3}$, the identity $\beta_p \equiv -\frac{14}{3} B_{p-3} \pmod{p}$ (theorem status: direct binomial/harmonic proof, and Thm 1.1 of Ji-Cai Liu, arXiv:2404.16636, specializes to it), and the y_4 Kummer-quotient formula. In particular $\beta_p = 0 \iff p$ is a Wolstenholme prime, and on the divisibility locus $p \mid b_r$ the terms $\beta_p b_r, y_4 b_r$ self-cancel: the tower is universal through p^4 at bad primes. Unconditional proof (cron, Q6330): the exact block identity $b_{N+r} = b_N F_r(N) - N^3 b_{N-1} G_r(N)$ (recurrence uniqueness) plus the two endpoint lemmas $b_p \equiv 5 \pmod{p^3}$, $b_{p-1} \equiv 1 \pmod{p^2}$ assemble the tower through p^4 ; the N^3 factor is the entire rigidity mechanism, and all p -dependence concentrates in the two endpoint digit sequences (all-order form). (Verification: supercongruence 63/63 `cron_supercong_check.py`; β_p 31/31 `beta_bernoulli_check.py` — equivalently $\beta_p \equiv 7 \cdot$ (Wolstenholme quotient); master identity 21/21 `harmonic_p5_check.py`; independent ledgers 25–27/27.)

20.4 The apparition theorem

Theorem 657 (Apéry apparition law). *Let $e = \frac{p-1}{2}$ and let S_p be the branch polynomial of Theorem 655, $D = \deg S_p \in \{e, e-1\}$.*

- (i) (endpoint) $\text{lc}(S_p) \equiv \left(\frac{-2}{p}\right)$;
- (ii) (branch reflection) $t^D S_p(1/t) = \left(\frac{-2}{p}\right) S_p(t)$, i.e. $s_{D-j} \equiv \left(\frac{-2}{p}\right) s_j$;
- (iii) (quarter-point law) *the central coefficient vanishes iff the center is integral and $\left(\frac{-2}{p}\right) = -1$:*

$$\tau_{(p-1)/4} \equiv 0 \iff p \equiv 5 \pmod{24}, \quad \sigma_{(p-3)/4} \equiv 0 \iff p \equiv 23 \pmod{24},$$

and in no other residue class mod 24 does the floor-quarter coefficient of the relevant branch vanish.

Proof (two independent routes). Route A (Franel pullback endpoint). With $f_n = \sum_k \binom{n}{k}^3$, $h(u) = \sum f_n u^n$, $\phi(u) = u(1-8u)/(1+u)$, the exact identity $F(\phi(u)) = (1+u)h(u)^2$ (characteristic zero; CFVZ) together with the Dwork congruence $F = A_p F(t^p)$ and the Lucas factorization $h = H_p h(u^p)$ gives $A_p(\phi(u)) = H_p(u)^2/(1+u)^{p-1}$ with H_p monic of degree $p-1$ ($f_{p-1} \equiv 1$); in the $\chi_p = -1$ branch additionally $q(\phi(u)) = ((1-16u-8u^2)/(1+u))^2$. Identifying S_p with the fixed-branch truncation (unit argument in $\mathbb{F}_p[[t]]/(t^p)$) and comparing leading terms at $u = \infty$ yields $\text{lc}(S_p) = (-8)^{-e} = \left(\frac{-2}{p}\right)$ in both branches; palindromy of A_p (Theorem 654) and unique factorization give (ii), and (ii) forces (iii) at the center. (Every ingredient machine-verified: `r18_proof_check.py`; endpoint

165/165 `leading_coeff_check.py`; reflection 92/92 `branch_reflection_check.py`; quarter table $p < 2000$ `full24_jacobsthal.py`.)

Route B (direct reversal). An independent proof of (ii) without the endpoint evaluation, via the reversal symmetry of the branch polynomial, is given in `CODEX_JACOBSTHAL_DEEP.md` (commits d9af2b9–77e089b), with its own verification ledger (all $p < 3000$). $\square$

20.5 Self-twist rigidity of the branch system

Theorem 658 (self-twist rigidity). *Let $\mathcal{V}_\tau$ be the rank-two local system on $U = \mathbb{P}^1 \setminus \{0, \alpha, \beta, \infty\}$ ($\alpha\beta = 1$ the roots of q) defined by the branch operator $\mathcal{D}_\tau = 4q(t)\theta^2 + 4t(t-17)\theta + t(t-10)$ annihilating $\tau = \sqrt{F}$, with Riemann scheme exponents $\{0, 0\}$ at 0, $\{0, \frac{1}{2}\}$ at α, β , $\{\frac{1}{2}, \frac{1}{2}\}$ at ∞ , and $\det \mathcal{V}_\tau = \delta$ the quadratic Kummer system of $\sqrt{q}$ -type. Then, with a fixed determinant normalization, every geometric finite-order Kummer self-twist of $\mathcal{V}_\tau$ is trivial, and the conjugate-twist list is exactly $\{1, \delta\}$; comparing raw and determinant-one normalizations enlarges the list only to $\langle \eta \rangle = \{1, \eta, \delta, \eta^{-1}\}$ of order 4 ($\eta = q^{1/4}$). Moreover $\text{Sym}^2 \mathcal{V}_\tau$ admits no nontrivial geometric self-twist, the external product on $U \times U$ is irreducible with only factorwise quadratic twists, and the diagonal rank-four tensor decomposes only as the Clebsch–Gordan $3+1$. In particular no Kummer character of order > 4 is a self-twist or decomposition character.*

Proof sketch and certification. Local exponents by direct indicial computation (machine-verified together with the operator identity $\mathcal{D}_\tau \tau = 0$ to $O(t^{40})$ and the half-integer recurrences: `q6372_local_system_check.py`) a self-twist L must have L^2 trivial with ramification supported on the four singular points, and the exponent lists above eliminate all nontrivial possibilities via the rigidity lemma (ramification comparison at each cusp); the tensor statements follow from Zariski density of the connected monodromy in SL_2 and Clebsch–Gordan. Full proof: working note `GPT_Q6372_self_twist_classification.md` (§4–§8), including the caveat that $\boxtimes$ on $U \times U$ is irreducible and the familiar $3+1$ splitting refers to its diagonal pullback only. $\square$

This closes the geometric prerequisite of the counting lemma below: all self-twist and tensor-decomposition exceptions lie in a fixed set of order ≤ 4 , so no high-order Mellin character can correlate with the column event through a hidden twist.

20.6 The bounded-object theorem

Theorem 659 (Franel–Mellin bounded object). *Let $\phi(x) = x(1-8x)/(1+x)$ (the Fricke w_6 quotient map of the Beauville–IV elliptic pencil E_x over $X_1(6)$, whose Hasse invariant is the truncated Franel polynomial H_p), and let $q(t) = t^2 - 34t + 1$ be the fibre discriminant of ϕ . Then for every $p \geq 7$ and $1 \leq r \leq p-2$:*

$$b_r \equiv - \sum_x H_p(x)^2 \omega^{-r}(\phi(x)) + \sum_t \chi_2(q(t)) A_p(t) \omega^{-r}(t) \pmod{p},$$

exactly and with no boundary corrections, and both summands are Mellin values of trace functions of FIXED bounded-conductor objects. The correct basis is the eigen-decomposition of the quadratic-cover pushforward under the deck involution: two integral rank-three systems $\mathcal{A}_\pm$ on $\mathbb{G}_m$ with Artin conductors 9 and 11 respectively ($\text{Sym}^2 \mathcal{E}$ itself has conductor 11); the naïve difference of the two full pushforwards is identically zero by the projection formula, so only the eigen-constituents carry content. The descent uses the explicit correspondence obtained by quotienting the Weierstrass model of E_x by its rational cyclic subgroup of order six; the quotient is the constant (-3) -quadratic twist of the deck-conjugate fibre, and the twist cancels in Sym^2 . Consequently $p \mid b_r$ if and only if the

defining-characteristic reduction of a two-dimensional Mellin cohomology trace of a fixed rank-three bounded-conductor sheaf vanishes.

Certification. The finite-field identity and its boundary bookkeeping are theorems by the adversarial referee audit (working note GPT_Q6424, links (1)–(3): no hidden corrections at the pole $x = -1$, the zero fibre, branch points, or nonsplit fibres). The two-sheaf normalization with conductor accounting was obtained twice independently — the descent construction of GPT_Q6394 and the sheaf normalization of CODEX_FRANEL_MELLIN — with identical conductor 11; the correspondence left unwritten in the former ([GAP-1: display]) is printed and machine-verified in `codex_lt_isogeny.py` (2148 point evaluations at five primes; the trace identity $a_p(E_{\iota(u)}) = (\frac{-3}{p})a_p(E_u)$ on 38 smooth fibres). Machine certificates for every displayed identity: `franel_pointwise_test.py`, `franel_mellin_mult_test.py`, `codex_fm_geometry.py` ($a_p(E_u) = H_p(u)$ on 1084 fibres), `codex_fm_local_monodromy.py`. The residual integral/crystalline comparison at middle-extension points — equality as reductions at the chosen prime above p , not of complex traces — is the one remaining normalization statement and is recorded as such. $\square$

20.7 The parity law for the zero fibre

Proposition 660 (parity law). *Let $Z_p = \{1 \leq r \leq p-2 : p \mid b_r\}$. Then $|Z_p| \equiv \mathbf{1}[p \mid b_{(p-1)/2}] \pmod{2}$. In particular $|Z_p|$ is even except at the non-ordinary primes of the level-8 weight-4 newform $\eta(2z)^4\eta(4z)^4$ (Beukers’ congruence), of which the only examples below 30000 are $p = 11, 3137$.*

Proof. Strong reflection (Theorem 654) gives the involution $r \mapsto p-1-r$ on Z_p , whose unique fixed point in range is the midpoint $r = (p-1)/2$. (Empirical profile across $p < 30000$, all 3242 window primes: $|Z_p| = 2 \cdot \text{Poisson}(\lambda \approx 0.51) + \text{midpoint indicator}$, $\chi^2 = 2.47$ on 3 d.f., mean 1.0185, max 8; scan banked in the appendix-Q ledger.) $\square$

Data point. The midpoint-zero primes 11 and 3137 were identified by Zudilin (arXiv:2409.00384), who verified that they are the only examples below 2×10^4 . Our scan — two independent C implementations in exact agreement (78,462 window zeros compared bitwise) — extends this: 11 and 3137 remain the only midpoint-zero primes below 10^6 (all 78,495 window primes; a fifty-fold extension of the published record). Per-prime zero counts along the way: mean 0.9996, maximum 12 (at $p = 159977$), in exact agreement with the $2 \cdot \text{Poisson}(0.4998) + \text{midpoint law}$ ($\chi^2 = 3.43$ on 5 d.f.).

20.8 The quarter-point value theorem

Theorem 661 (quarter-point value). *For $p \equiv 1 \pmod{4}$ (the nonvanishing classes of Theorem 657(iii)), the quarter coefficient is the p -th Fourier coefficient of a single fixed (p -independent) series:*

$$v_p := \tau_{(p-1)/4} \equiv [Q^p] \mathcal{F}(Q) \pmod{p}, \quad \mathcal{F}(Q) = Q D(Q^4) \prod_{n \geq 1} [(1-Q^{4n})(1-Q^{8n})(1-Q^{12n})(1-Q^{24n})]^{1/2},$$

where $D = \frac{1}{2}(E_2(\tau) - 2E_2(2\tau) - 3E_2(3\tau) + 6E_2(6\tau))$ — a weight-3 eta-multiplier form on the fourfold cyclic cover of the level-6 modular curve. The value factorizes as $v_p \equiv 2A \cdot c_p$ with $2A = (\frac{(p-1)/2}{(p-1)/4})$ (Gauss) and c_p satisfying the order-24 layer recurrence; since the multiplier introduces unbounded powers of 2 in the denominators, $\mathcal{F}$ is not congruent to any rational newform — explaining the failure of every low-degree CM/Jacobsthal fit (all such fits were eliminated on 267 primes).

Certification. Derivation hand-checked line by line (eta-quotient identity $H = W T^{1/4} = (\eta_1 \eta_2 \eta_3 \eta_6)^{1/2}$; integrality $U = T^{-1/4} \in q^{-1/4} \mathbb{Z}[[q]]$; $U^p \equiv U(q^p)$; degree pigeonhole forcing the unique $k = 0$ term). Machine verification: in-sample 12/12 and out-of-sample 5/5 at $p = 769, 937, 1009, 1033, 1153$ (CRON_q6360_verify.py, independently re-run). Recurrence for c_p agrees verbatim with the independently derived U -recurrence (double source). $\square$

Together, Theorems 657, 659, 661 and Proposition 660 complete the apparition program: which residue classes vanish, what geometric object carries the values, the exact value at the distinguished point, and the parity of the zero fibre.

20.9 The character-order counting lemma and its consequence

Lemma 662 (character-order counting). *For n fixed and window primes $p \in (n/2, n]$, the Mellin character of the column event is the global specialization ω_p^{n-1} , of order $m_p = (p-1)/\gcd(p-1, n-1)$, and $\#\{p : m_p \leq T\} \leq T \tau(n-1)$.*

Consequently every bounded-character-order law — including the apparition theorem (order dividing 8) and any CM/motivic structure — is confined to $O(T \tau(n-1))$ window primes and cannot produce a bad column: $H(n) = H_{\text{high}}(n; T) + O(T \tau(n-1))$. The pointwise Problem 3.2 is thereby equivalent to zero density of the *high-order* Mellin bad diagonal (the canonical frontier statement; see the session notes §34, §39).

20.10 The reciprocal trilinear identity

Let $W \in C_c^\infty(\mathbf{R})$ be nonnegative with $W(x) \geq 1$ on $[0, 1]$ and $\text{supp } W \subset [-1, 2]$. For each prime $p \in (X, 2X]$ define the *centered density function*

$$g_p(m) = \mathbf{1}_{m \bmod p \in \mathcal{Z}_p} - \frac{|\mathcal{Z}_p|}{p}, \quad m \in \mathbf{Z},$$

which is periodic of period p and satisfies $\sum_{m=0}^{p-1} g_p(m) = 0$. Its normalised Fourier transform at $a \in \mathbf{F}_p^\times$ is

$$T_p(a) = \frac{1}{p} \sum_{j \in \mathcal{Z}_p} e_p(a j), \quad |T_p(a)| \leq \frac{|\mathcal{Z}_p|}{p}.$$

Proposition 663 (ATI3 identity). *Let $M = X^2$. Define the smoothed centred cubic defect*

$$D_3^{(W)} = \sum_m W\left(\frac{m}{M}\right) \sum_{\substack{p < q < r \\ \text{in } \mathcal{P}_X}} g_p(m) g_q(m) g_r(m).$$

Then, writing $\overline{qr}_p$ for the multiplicative inverse of qr modulo p , and similarly for the other two primes,

$$D_3^{(W)} = \sum_{\substack{p < q < r \\ \text{in } \mathcal{P}_X}} \sum_{\substack{n \in \mathbf{Z} \\ n \neq 0}} T_p(n \overline{qr}_p) T_q(n \overline{pr}_q) T_r(n \overline{pq}_r) S_n(M; p, q, r) \quad (707)$$

where $S_n(M; p, q, r) = \sum_{m=0}^{M-1} W(m/M) e(mn/(pqr))$ satisfies $S_n = M \widehat{W}(Mn/(pqr)) + O(1)$. Because $M/(pqr) \asymp X^{-1}$, rapid decay of $\widehat{W}$ confines the effective sum to $|n| \leq CX$ for any fixed C .

Proof. Expand each g_p in its finite Fourier series on $\mathbf{F}_p$:

$$g_p(m) = \sum_{a=1}^{p-1} T_p(a) e_p(am).$$

The $a = 0$ term is absent because g_p is centred. Substituting into the m -sum and exchanging orders gives

$$\sum_m W\left(\frac{m}{M}\right) g_p(m) g_q(m) g_r(m) = \sum_{\substack{a \in \mathbf{F}_p^\times \\ b \in \mathbf{F}_q^\times \\ c \in \mathbf{F}_r^\times}} T_p(a) T_q(b) T_r(c) S(a/p + b/q + c/r; M)$$

where $S(\theta; M) = \sum_m W(m/M) e(m\theta)$. Since p, q, r are pairwise coprime, the Chinese Remainder Theorem gives a bijection

$$(a, b, c) \longleftrightarrow n = aqr + bpr + cpq \pmod{pqr},$$

with $a \equiv n \overline{qr}_p \pmod{p}$ and similarly for b and c . Since a, b, c are all nonzero, $n \not\equiv 0 \pmod{pqr}$. Replacing $a/p + b/q + c/r = n/(pqr)$ yields (707). The estimate on S_n follows from Poisson summation applied to the smooth weight. $\square$

Remark 664 (Effective frequency range). For the Gaussian choice $W(x) = e^{-\pi x^2}$, $\widehat{W}(t) = e^{-\pi t^2}$, and $|S_n| \leq M \exp(-\pi n^2 M^2 / (pqr)^2) + O(1) \leq M e^{-cn^2/X^2} + O(1)$, so the effective range is $|n| \leq C\sqrt{\log X} X$ and the sum is concentrated at $|n| \ll X$.

Remark 665 (The obstacle is the near-resonant slab). The trivial bound $|T_p(a)| \leq |\mathcal{Z}_p|/p$ and $|S_n| = O(X)$ gives $|D_3^{(W)}| \leq CXL^3 \prod \delta_p \asymp X \cdot M\lambda^3 = X \cdot P_3$. The factor X is the “gap”: the number of effective dual frequencies per triple. Proving $D_3^{(W)} = o(M\lambda^3)$ requires saving this factor of X , which amounts to cancellation in the n -sum jointly with the reciprocal evaluations.

Numerical evidence (`type3_test.py`): for each triple (p, q, r) , the partial sums $\sum_{|n| \leq N}$ exhibit $\sim N/2$ sign changes and remain bounded by $O(z_p z_q z_r / X)$, consistent with square-root cancellation in the dual frequency.

Remark 666 (Type-III dispersion route). A proof via the Linnik dispersion method would proceed as follows.

1. Apply Cauchy–Schwarz in one prime variable (say r) and the dual frequency n .
2. Use a $+uv$ shift to complete the resulting bilinear sum over $\mathbf{F}_p$.
3. Invoke a sum-of-products theorem (*Apéry Crystalline Sum-of-Products*, ACSOP) for the complete correlation sums, obtaining Deligne-type square-root cancellation.
4. Two independent square-root savings of $\sqrt{X}$ from steps (ii)–(iii) recover the factor X .

The required ACSOP theorem is: for generic (non-diagonal) parameter tuples, complete sums of products of the centred zero-detector g_p under Möbius transforms satisfy the Deligne bound $O_k(p^{1/2})$. The known $\mathrm{SL}(2)$ monodromy and Goursat product structure are exactly the geometric inputs for such a theorem; the genuinely new construction is a trace-function realisation of the additive zero-detector in defining characteristic.

20.11 Additive character sum estimate

Conjecture 667 (Additive character estimate for Apéry numbers). *There exists an absolute constant $C > 0$ such that for all primes $p \geq 7$ and all $a \in \mathbf{F}_p^\times$,*

$$\left| \sum_{j=0}^{p-1} e_p(a b_j) \right| \leq C \sqrt{p}.$$

Remark 668. Numerical verification through $p = 5000$ confirms this conjecture with $C \leq 4$: the ratio $\max_{a \neq 0} |S_p(a)| / \sqrt{p}$ has mean 3.24 and maximum 3.96 over all primes in the range, with no growth trend. The constant $C \approx 4$ is consistent with the Apéry motive having rank 4: the Deligne bound for a rank- r trace function gives $|S| \leq (r-1)\sqrt{p}$ under appropriate monodromy hypotheses.

Proposition 669 (Exact zero-detector expansion and its limitation). *The centred zero-detector satisfies the exact identity*

$$g_p(j) = \frac{1}{p} \sum_{a=1}^{p-1} e_p(ab_j) + \frac{1 - |\mathcal{Z}_p|}{p}.$$

Conjecture 667 bounds only the one-phase sums $\sum_j e_p(ab_j)$. It does not imply square-root cancellation for the joint sums

$$\sum_j e_p \left(\sum_{i=1}^k a_i b_{j+s_i} \right),$$

and hence does not imply ACSOP or any stated bound for the product correlations of g_p .

Proof. The additive-character expansion of $\mathbf{1}_{b_j \equiv 0}$ is standard:

$$\mathbf{1}_{b_j \equiv 0 \pmod{p}} = \frac{1}{p} \sum_{a=0}^{p-1} e_p(a b_j).$$

Subtracting $|\mathcal{Z}_p|/p$ proves the displayed identity. Notice that the residual is not uniformly $O(1/p)$ unless a separate pointwise bound for $|\mathcal{Z}_p|$ is assumed.

Expanding a product correlation produces several strata from the constant residuals. Its all-nonconstant stratum is

$$\sum_{j \in \mathbf{F}_p} \prod_{i=0}^{k-1} \left[\frac{1}{p} \sum_{a_i \neq 0} e_p(a_i b_{j+s_i}) \right] = \frac{1}{p^k} \sum_{\mathbf{a} \in (\mathbf{F}_p^\times)^k} \sum_j \prod_i e_p(a_i b_{j+s_i}).$$

The inner sum is the displayed joint phase. For $k \geq 2$ it is not an instance of Conjecture 667; the phase is a linear combination of shifted Apéry values rather than one value ab_j . No estimate for it follows from the conjecture by the triangle inequality or by multiplying the separate one-phase bounds. $\square$

20.12 Status of the additive-character route

Remark 670 (The one-phase conjecture is not a conditional all- n theorem). Conjecture 667 controls only

$$\sum_{j \bmod p} e_p(ab_j).$$

After expanding a product of centred zero detectors, the required complete sums instead have the form

$$\sum_{j \bmod p} e_p \left(\sum_{i=1}^k a_i b_{\gamma_i(j)+s_i} \right).$$

The first estimate does not imply the second. The missing joint estimate is the separate ACSOP hypothesis; it requires a uniform construction for products of shifted or Möbius-transformed phases.

Even ACSOP would still have to be inserted into a complete Type–III argument, including its diagonal strata, the smoothing removal, and the full frequency range $|h| \asymp X$, before it yielded $(\text{HM})_3$. Finally, Theorem 350 requires the independent first zero-mass input $\lambda_X \ll 1/\log X$ to pass from $(\text{HM})_3$ to the all-index conjecture.

Thus a valid conditional chain is

$$\text{M}_1 + \text{ACSOP} + \text{Type–III conversion} \implies (\text{HM})_3 \implies G_n = e^{o(n)}.$$

The numerical experiment for Conjecture 667 tests only its one-phase premise. It supplies no evidence for the joint ACSOP sums or for the Type–III conversion.

21 Additive character sum: sheaf-theoretic analysis

We investigate the conjectured bound

$$\left| \sum_{j=0}^{p-1} e_p(c b_j) \right| \leq C\sqrt{p} \quad (c \in \mathbf{F}_p^\times), \quad (708)$$

where $b_j \bmod p$ is the Apéry sequence reduced modulo p , defined by the recurrence

$$(j+1)^3 b_{j+1} = (2j+1)(17j^2 + 17j + 5) b_j - j^3 b_{j-1}, \quad b_0 = 1, b_1 = 5.$$

Numerical verification through $p = 10\,007$ confirms (708) with $C \leq 4.13$ and no growth trend.

We examine the approach via Katz’s ℓ -adic hypergeometric sheaves (ESDE, Chapter 8) and the related finite-field hypergeometric functions of Greene and McCarthy. Our conclusion (§21.5) is that the bound requires a new theorem bridging ℓ -adic and p -adic information about the same Frobenius eigenvalues—a problem that sits at the frontier of arithmetic geometry and is not resolved by any existing technique.

21.1 The Apéry operator is not hypergeometric

The Picard–Fuchs operator governing the Apéry numbers is

$$L = \theta^4 - x(2\theta + 1)(17\theta^2 + 17\theta + 5), \quad \theta = x \frac{d}{dx}. \quad (709)$$

A classical hypergeometric operator ${}_nF_{n-1}$ has the form

$$\prod_{i=1}^n (\theta - \alpha_i) - x \prod_{j=1}^n (\theta - \beta_j)$$

with $\alpha_i, \beta_j \in \mathbf{Q}$. The factor $17\theta^2 + 17\theta + 5$ has discriminant

$$17^2 - 4 \cdot 17 \cdot 5 = -51 < 0,$$

so it is *irreducible over* $\mathbf{Q}$. The local exponents at $x = \infty$ are $\theta = -\frac{1}{2}$ and $\theta = \frac{-17 \pm \sqrt{-51}}{34}$, which are non-rational.

Proposition 671. *The operator L is not a classical hypergeometric operator. In the Roberts–Rodriguez-Villegas classification, the Apéry motive corresponds to a non-hypergeometric Calabi–Yau operator (AESZ #6 in the Almkvist–van Enckevort–van Straten–Zudilin table).*

Proof. The right-hand factor $(2\theta + 1)(17\theta^2 + 17\theta + 5)$ has degree 3 in θ (not 4), and the quadratic factor does not split over $\mathbf{Q}$. Both conditions violate the defining axioms of a hypergeometric operator. $\square$

Remark 672. The Apéry numbers *do* admit a terminating hypergeometric representation as a classical series:

$$b_n = {}_4F_3\left(\begin{matrix} -n, -n, n+1, n+1 \\ 1, 1, 1 \end{matrix} \middle| 1\right) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2. \quad (710)$$

Identity (710) has been verified computationally for $n = 0, \dots, 49$ at primes $p = 53, 97, 193, 503, 1009$. However, the parameters $-n$ and $n+1$ *depend on n* : each value of n produces a different set of Greene characters χ^{-n}, χ^{n+1} in the finite-field hypergeometric framework. This prevents identification with a *single* Katz hypergeometric sheaf at varying evaluation points.

21.2 Mellin duality and the rank-2 trace sheaf

The key structural identity (Proposition 269) expresses $b_j \bmod p$ as a Mellin coefficient of the toric K3 family:

$$b_j \equiv - \sum_{t \in \mathbf{F}_p^\times} t^j A_j(t) A_{p-1-j}(t) \pmod{p}, \quad (711)$$

where $A_j(t) = \sum_{k=0}^j \binom{j}{k}^2 t^k = {}_2F_1(-j, -j; 1; t)$ is the truncated Legendre polynomial, and, for $1 \leq j \leq (p-1)/2$, the Legendre duality $A_{p-1-j}(t) = (1-t)^{p-1-2j} A_j(t)$ gives the weighted-square formula

$$b_j \equiv - \sum_{t \in \mathbf{F}_p^\times} t^j (1-t)^{p-1-2j} A_j(t)^2 \pmod{p}. \quad (712)$$

Both (711) and (712) have been verified computationally for $p = 53$ and all $j = 1, \dots, (p-1)/2$.

Sheaf-theoretic interpretation. Let $\mathcal{M}_\chi$ be the rank-2 eigenspace of the Mellin sheaf on $\mathbf{G}_m$ with $\mathrm{SL}(2)$ geometric monodromy (Theorem 282). For a multiplicative character $\chi_j: a \mapsto a^j$ of $\mathbf{F}_p^\times$, the Mellin coefficient (711) is the trace of Frobenius on the cohomology of the twist:

$$b_j \equiv - \mathrm{Tr}(\mathrm{Frob}_p, H_c^1(\mathbf{G}_m \otimes \overline{\mathbf{F}_p}, \mathcal{M}_\chi \otimes \mathcal{L}_{\chi_j})) \pmod{p}. \quad (713)$$

Lemma 673 (Cohomological dimension of the twist). *For $1 \leq j \leq p-2$ with χ_j nontrivial, the Grothendieck–Ogg–Shafarevich formula gives*

$$\dim H_c^1(\mathbf{G}_m, \mathcal{M}_\chi \otimes \mathcal{L}_{\chi_j}) = 2.$$

Proof. The sheaf $\mathcal{M}_\chi$ has rank 2 on $\mathbf{G}_m$ with four singular points: $x = 0, \infty$ (maximally unipotent, each with drop = 2, Swan = 0) and two conifold points $s = 17 \pm 12\sqrt{2}$ (each with drop = 1, Swan = 0). Tensoring with $\mathcal{L}_{\chi_j}$ (tame at all points) does not change the Swan conductors. For χ_j nontrivial, the monodromy at 0 gains a nontrivial tame character, making $\mathrm{drop}_0 = 2$ (full drop); similarly $\mathrm{drop}_\infty = 2$. The GOS formula on $\mathbf{P}^1$:

$$\chi_c = 2 \cdot 2 - (2+0) - (2+0) - 1 - 1 = 4 - 2 - 2 - 1 - 1 = -2.$$

For generic nontrivial χ_j : $H_c^0 = H_c^2 = 0$ (no global sections or coinvariants after nontrivial twist), so $\dim H_c^1 = 2$. $\square$

Corollary 674 (Trace structure). *For each $1 \leq j \leq p-2$, there exist algebraic integers $\gamma_1(j), \gamma_2(j) \in \overline{\mathbf{Q}}_\ell$ with $|\gamma_i(j)| = p^2$ (Weil numbers of weight 4) such that*

$$b_j \equiv -(\gamma_1(j) + \gamma_2(j)) \pmod{p}. \quad (714)$$

In particular, $|b_j|_{\text{lift}} \leq 2p^2$ as an integer (where b_j denotes any lift of $b_j \bmod p$ to $\{-2p^2, \dots, 2p^2\}$ via the cohomological trace).

21.3 The Mellin transform sheaf on the character variety

The multiplicative Fourier (Mellin) transform of $\mathcal{M}_\chi$ produces a sheaf $\mathcal{N}$ on the dual group scheme parametrizing Kummer characters, whose trace function recovers the Apéry sequence:

$$\text{Tr}(\text{Frob}_j, \mathcal{N}) = -b_j. \quad (715)$$

By Lemma 673, $\mathcal{N}$ has generic rank 2 and weight 4 ($= 3 + 1$, the weight of $\mathcal{M}_\chi$ plus one, as Fourier transform increases weight by 1).

Proposition 675 (Irreducibility of $\mathcal{N}$). *The sheaf $\mathcal{N}$ is geometrically irreducible. Equivalently, $b_j \bmod p$ is not expressible as $c_1 \alpha^j + c_2 \beta^j$ for any $\alpha, \beta, c_1, c_2 \in \overline{\mathbf{F}}_p$.*

Numerical proof. If $\mathcal{N}$ were reducible as $\mathcal{L}_1 \oplus \mathcal{L}_2$ (sum of two rank-1 Kummer sheaves), then b_j would satisfy a *constant-coefficient* linear recurrence of order 2: $b_{j+2} = (\alpha + \beta) b_{j+1} - \alpha\beta b_j$. Direct computation shows this fails at $j = 3$ for $p = 97$: the candidate $(s, r) = (71, 88)$ satisfies $5s - r \equiv 73 \pmod{97}$ and $73s - 5r \equiv 87 \pmod{97}$ but does not extend.

More decisively, the *linear complexity* of $b_j \bmod p$ (the minimum order of a constant-coefficient linear recurrence annihilating the sequence, computed via the Berlekamp–Massey algorithm) satisfies

$$L(p) = \frac{p-1}{2} + 1 \quad \text{for all tested primes } p \leq 503.$$

p	$L(p)$	$(p-1)/2$	$L/(p-1)$
53	27	26	0.519
97	49	48	0.510
193	97	96	0.505
503	252	251	0.502

The linear complexity equals $\lceil (p-1)/2 \rceil + 1$, matching the *effective degree* $\deg P \approx (p-1)/2$ of the polynomial P in the quadratic substitution $b(x) = P(x^2+x)$ induced by the reflection symmetry $b_j = b_{p-1-j}$, plus 1. (The raw polynomial degree of $b(x)$ is $\approx p-1$; the reflection halves the effective complexity but not below $\Theta(p)$.) This rules out any bounded exponential-polynomial representation and confirms that $\mathcal{N}$ is irreducible. $\square$

21.4 Why each approach fails

We now systematically analyze why the Katz hypergeometric sheaf framework and its variants do not yield the bound (708).

21.4.1 Artin–Schreier pullback (conductor explosion)

The polynomial $b(x) \in \mathbf{F}_p[x]$ has raw degree $\deg(b) \approx p-1$ (verified: exactly $p-3$ for $p = 53$, $p-1$ for $p = 97, 193$). The reflection symmetry $b(x) = b(-1-x)$ allows a quadratic substitution $u = x^2 + x$, reducing to $b(x) = P(x^2+x)$ where $\deg P \approx (p-1)/2$. The Weil bound applied after this reduction gives

$$|S_p(c)| = |S_1 + S_2|, \quad S_1 = \sum_u e_p(cP(u)), \quad S_2 = \sum_u \chi_2(4u+1) e_p(cP(u)),$$

with $|S_1| \leq ((p-1)/2 - 1)\sqrt{p}$ and $|S_2| \leq (p-1)/2 \cdot \sqrt{p}$, both $O(p^{3/2})$. This approach fails because the *polynomial degree* of b (even the effective degree after reflection) grows linearly with p , even though the *sheaf complexity* of $\mathcal{N}$ is bounded.

21.4.2 Katz hypergeometric sheaves

The Apéry operator L is not a classical hypergeometric operator (Proposition 671), so the ESDE Chapter 8 construction does not produce a sheaf whose trace function is b_j at varying points. The terminating ${}_4F_3$ representation (710) has *variable parameters* $(-n, -n, n+1, n+1)$: in the Katz framework, each n yields a different hypergeometric sheaf $\text{Hyp}(\chi^{-n}, \chi^{-n}, \chi^{n+1}, \chi^{n+1}; \varepsilon, \varepsilon, \varepsilon)$, so there is no single sheaf whose trace at varying points gives b_n .

21.4.3 Hadamard product decomposition

The Apéry sequence factors as a convolution: $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, which is the Hadamard product of two copies of the ${}_2F_1(-n, -n; 1; t)$ family. In the motive picture, the Apéry motive is related to Sym^2 of a rank-2 antecedent. However, the composition $e_p(c \cdot \text{trace})$ does not respect the Hadamard/tensor product structure—the exponential of a sum is a product, but this product of two “individual eigenvalue characters” is not a sheaf-theoretic operation for irreducible representations.

21.4.4 Kloosterman reduction (requires reducibility)

If $\mathcal{N}$ were *reducible*, i.e., $\mathcal{N} \cong \mathcal{L}_1 \oplus \mathcal{L}_2$ with rank-1 constituents, then $b_j = c_1 \alpha^j + c_2 \beta^j$ and

$$S_p(c) = \sum_{j=0}^{p-2} e_p(c_1 c \alpha^j + c_2 c \beta^j)$$

would be a Kloosterman-type sum, boundable by $2\sqrt{p}$ via the Weil bound. Proposition 675 rules this out: the linear complexity of $b_j \bmod p$ is $(p-1)/2$, not 2.

21.4.5 Fourier–Mellin composition

The sum $S_p(c) = \sum_j e_p(c b_j)$ is the *additive* Fourier transform of the sequence b_j , while b_j itself is the *multiplicative* Fourier transform of the trace function $\theta_p(a)$. Explicitly:

$$S_p(c) = \sum_{j=0}^{p-2} e_p\left(-c \sum_{a \in \mathbf{F}_p^\times} \theta_p(a) a^j\right) = \sum_{j=0}^{p-2} \prod_{a \in \mathbf{F}_p^\times} e_p(-c \theta_p(a) a^j),$$

which is a product of $p-1$ additive-character factors summed over j . This “double Fourier” structure—the additive character of the multiplicative Fourier transform—mixes the additive and multiplicative structures of $\mathbf{F}_p$ in a way that no existing theorem controls.

21.5 The exact obstruction

We now state the precise mathematical obstruction to proving (708).

Theorem 676 (Structure of the obstruction). *The following are equivalent:*

- (A) *The bound $|S_p(c)| \leq C\sqrt{p}$ holds for all $c \neq 0$ and all sufficiently large primes p .*
- (B) *The “effective Artin–Schreier conductor” of the composition $e_p \circ b$ is $O(1)$ (independent of p), despite the polynomial degree $\deg(b) = (p-1)/2$.*
- (C) *The Frobenius eigenvalues $\gamma_1(j), \gamma_2(j)$ of (714), when reduced modulo p , produce a sequence $b_j \bmod p$ whose additive Fourier coefficients have ℓ^∞ -norm $O(\sqrt{p})$.*

Proof of equivalence. (A) $\Leftrightarrow$ (B): By definition, $|S_p(c)| = |\sum_j e_p(cb(j))|$ where $b(x) \in \mathbf{F}_p[x]$. The Weil bound gives $|S_p(c)| \leq (\text{effective conductor} - 1)\sqrt{p}$. So (A) holds iff the effective conductor is $O(1)$.

(A) $\Leftrightarrow$ (C): $S_p(c) = \sum_j e_p(cb_j) = \sum_v m_v e_p(cv) = \widehat{m}(c)$ where $m_v = \#\{j : b_j = v\}$. The bound $|\widehat{m}(c)| \leq C\sqrt{p}$ for $c \neq 0$ is precisely the Fourier-flatness condition on m . By Corollary 674, b_j is determined by the eigenvalues $\gamma_i(j)$ via mod- p reduction, so (C) is a restatement of (A) in eigenvalue language. $\square$

Remark 677 (The mixed-characteristic nature of the problem). The obstruction has a fundamentally *mixed-characteristic* nature. The ℓ -adic theory (Deligne’s theorem) gives $|\gamma_i(j)| = p^2$ for the Weil numbers, controlling their *archimedean* size. The p -adic theory (Dwork’s Frobenius, p -curvature) gives the mod- p reduction $\gamma_i(j) \bmod p$, controlling their *defining-characteristic* behavior. Proving (708) requires combining both pieces of information: the archimedean bound (which gives $|b_j|_{\text{lift}} \leq 2p^2$, i.e., b_j “wraps around” $\mathbf{F}_p$ about $4p$ times) with the p -adic structure (which determines *how* the wrapping distributes the values over $\mathbf{F}_p$).

No existing theorem provides this combination. The Deligne equidistribution theorem (Theorem 283) controls the archimedean distribution of $b_j/p^{3/2}$ (Sato–Tate), but says nothing about $b_j \bmod p$. The Dwork congruences ($b_{n+mp} \equiv b_n b_m \pmod{p}$) control the p -adic structure, but do not constrain the additive Fourier spectrum of b_j . The combination of both—a *defining-characteristic equidistribution theorem*—is the precise missing ingredient.

21.6 Numerical anatomy of the exponential sum

We record the key numerical invariants.

Additive energy. $E_p = \sum_{v \in \mathbf{F}_p} m_v^2$ where $m_v = \#\{j : b_j \equiv v\}$. By Parseval, $\sum_{c=1}^{p-1} |S_p(c)|^2 = pE_p - p^2$.

p	distinct	max m_v	E_p/p	$\max S /\sqrt{p}$	rms/ $\sqrt{p}$
53	19	10	4.02	2.76	1.68
97	40	6	2.81	2.83	1.35
193	76	6	2.93	2.95	1.39
503	193	6	3.10	3.35	1.45

The ratio $E_p/p \rightarrow 3$ is consistent with a Poisson(1) multiplicity distribution (for which $E[m_v^2]/E[m_v] = 2$ on the $\approx p/e$ occupied values, giving $E/p \approx 2 + 1/e \approx 2.37$; the observed ≈ 3 reflects slightly higher average multiplicity).

Zero critical points. The finite difference $\Delta b(j) = b(j+1) - b(j)$ has *no* roots in $\{0, \dots, p-2\}$ for all tested primes $p \leq 503$. This means b is strictly “non-stationary” on consecutive pairs, despite having $\approx 0.6p$ value collisions at non-consecutive positions.

Linear complexity. The Berlekamp–Massey linear complexity of $b_j \bmod p$ is $L(p) = \lceil (p-1)/2 \rceil + 1$ for all tested primes. This matches the polynomial degree of $b(x)$ after reflection and confirms that no low-complexity representation of b_j exists.

21.7 What would constitute a proof

We identify three potential approaches, each requiring a new theorem.

21.7.1 Approach 1: Crystalline effective conductor

A theorem of the form: For solutions y of Fuchsian ODEs of order d with S regular singular points over $\mathbf{Z}$, the “effective Artin–Schreier conductor” of the mod- p solution polynomial $y(x) \in \mathbf{F}_p[x]$ is bounded by $C(d, S)$, independently of p .

This would give (708) with $C = C(4, 4) - 1$. No such theorem exists. The closest results are Katz’s conductor bounds for *trace functions of lisse sheaves*, which bound $|S|$ in terms of the conductor of the Artin–Schreier pullback $[cb]^* \mathcal{L}_\psi$; but this conductor equals $\deg(b) \approx p/2$ for the Apéry polynomial.

21.7.2 Approach 2: Defining-characteristic Katz theorem

A theorem of the form: For a lisse ℓ -adic sheaf $\mathcal{F}$ of rank r and conductor C on $\mathbf{G}_m/\mathbf{F}_p$, the sum

$$\sum_{t \in \mathbf{F}_p^\times} \psi(c \cdot \text{Tr}(\text{Frob}_t, \mathcal{F}) \bmod p)$$

satisfies $|S| \leq C'(r, C) \cdot \sqrt{p}$.

This directly gives (708) applied to $\mathcal{N}$ with $r = 2$ and bounded C . No such theorem exists. The obstruction is that $\text{Tr}(\text{Frob}_t, \mathcal{F}) \bmod p$ is a *defining-characteristic* reduction, mixing ℓ -adic and p -adic information.

21.7.3 Approach 3: Higher-dimensional variety

Express $S_p(c)$ as a character sum on a variety $V/\mathbf{F}_p$ of bounded dimension and degree:

$$S_p(c) = \sum_{x \in V(\mathbf{F}_p)} \text{Tr}(\text{Frob}_x, \mathcal{F}) \cdot \psi(c \cdot f(x))$$

where $\dim V$, $\deg f$, $\text{rank}(\mathcal{F})$ are all $O(1)$. The Deligne bound would then give $|S_p(c)| \leq C' \cdot p^{\dim V/2}$, which yields (708) if $\dim V = 1$. The natural candidate is the toric threefold $(\mathbf{G}_m)^3$ from the Mellin pairing (711), but the exponential inside the outer sum involves a product of Legendre polynomials of *variable degree* j , which prevents reduction to bounded-degree geometry.

21.8 Summary

1. The Apéry operator is *not* a classical hypergeometric operator (irreducible quadratic factor in the local exponents).
2. The Apéry numbers b_n are expressible as terminating ${}_4F_3$ sums, but with *variable parameters* depending on n . Each n corresponds to a different Katz hypergeometric sheaf, precluding a fixed-sheaf trace estimate.
3. The sequence $b_j \bmod p$ is the trace function of an *irreducible* rank-2 sheaf $\mathcal{N}$ of weight 4 on the character variety. The irreducibility is confirmed by linear complexity $L(p) = (p-1)/2 + 1$.
4. The sum $S_p(c) = \sum_j e_p(c b_j)$ involves the *composition* of the additive character e_p with the trace of $\mathcal{N}$. This composition is not a trace function of any bounded-conductor sheaf.
5. The exact obstruction is *mixed-characteristic*: proving the bound requires combining ℓ -adic information ($|\gamma_i| = p^2$) with p -adic information ($\gamma_i \bmod p$), a combination not provided by any existing theorem.
6. Numerically, the bound holds with $C \leq 4.13$ through $p = 10\,007$, with no growth trend. The constant ≈ 4 is consistent with the rank-4 structure of the full Apéry motive (the rank-2 eigenspace contributing $(2-1)\sqrt{p} = \sqrt{p}$ per eigenspace, times 2 eigenspaces, times a factor from the value multiplicity).

Remark 678 (Status relative to the campaign). The bound (708) sits at the same level of difficulty as the *defining-characteristic gap* (Remark 285): it requires a theorem that connects the mod- p Frobenius trace to its distribution as a function of j , in the same characteristic p that defines the base field. This is the same obstruction identified in the cross-prime decorrelation problem in the campaign roadmap. A proof of (708) would simultaneously resolve the ACSOP (Additive Character Sum for ODE Polynomials) conjecture and provide the key input for the Type-III dispersion method.

22 Dwork trace formula approach: analysis and obstruction

We investigate whether Dwork's p -adic trace formula can prove the conjectured bound

$$|S_p(a)| = \left| \sum_{j=0}^{p-1} e_p(a b_j) \right| \leq C\sqrt{p} \quad (a \in \mathbf{F}_p^\times), \quad (716)$$

where $b_j \bmod p$ is the Apéry sequence. The bound is verified empirically through $p = 10\,007$ with $\max |S_p(a)|/\sqrt{p} \leq 4.13$.

22.1 Overview of Dwork's framework

Dwork's trace formula (1960, 1969) expresses exponential sums over finite fields as traces of completely continuous (nuclear) operators on p -adic Banach spaces. The essential mechanism:

The splitting function. Fix $\pi \in \overline{\mathbf{Q}_p}$ with $\pi^{p-1} = -p$, $|\pi|_p = p^{-1/(p-1)}$. The *Dwork splitting function* is

$$\theta(t) = \exp(\pi t - \pi t^p), \quad (717)$$

which converges on $|t|_p \leq 1$ and satisfies $\theta(\tau(a)) = \psi(a)$ for every $a \in \mathbf{F}_p$, where $\tau: \mathbf{F}_p \rightarrow \mathbf{Z}_p$ is the Teichmüller lifting and $\psi(a) = e^{2\pi i a/p}$ is the standard additive character.

Polynomial exponential sums. For a polynomial $f(x) \in \mathbf{F}_p[x]$ of degree d , choose a Teichmüller lifting $\hat{f}(x) \in \mathbf{Z}_p[x]$ and form

$$\alpha_f(g)(x) = \sum_{y^p=x} \theta(\hat{f}(y)) g(y),$$

acting on the p -adic Banach space $L(\rho) = \{\sum_{i \geq 0} c_i x^i : |c_i| \rho^i \rightarrow 0\}$ for a suitable $\rho > 1$. Then α_f is completely continuous (nuclear of order 1), and

$$\sum_{x \in \mathbf{F}_p} \psi(f(x)) = (-1)^{\dim} \text{Tr}(\alpha_f, L(\rho)). \quad (718)$$

Eigenvalue structure. The trace equals the sum of the eigenvalues $\lambda_1, \dots, \lambda_D$ (with multiplicity) of the nuclear operator α_f . The number of eigenvalues is bounded by the *generic rank* of the Dwork cohomology, which equals the degree d of f (in the univariate case). Each eigenvalue satisfies $|\lambda_i|_\infty \leq p^{1/2}$ (by the Riemann hypothesis for curves / Deligne's theorem) and has p -adic absolute value determined by the Newton polygon of the L -function.

Adolphson–Sperber generalization. For Laurent polynomials $f \in \mathbf{F}_p[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$ with Newton polyhedron $\Delta(f)$, the Dwork cohomology has dimension $n! \text{Vol}(\Delta(f))$, and the exponential sum

$$S(f) = \sum_{\mathbf{x} \in (\mathbf{F}_p^\times)^n} \psi(f(\mathbf{x}))$$

satisfies $|S(f)| \leq n! \text{Vol}(\Delta(f)) \cdot p^{n/2}$, provided f is nondegenerate with respect to $\Delta(f)$.

22.2 Direct application to $S_p(a)$

The sum (716) is $S_p(a) = \sum_{x \in \mathbf{F}_p} \psi(a \cdot b(x))$, where $b(x) \in \mathbf{F}_p[x]$ is the unique polynomial of degree $< p$ interpolating the Apéry sequence.

Effective degree. Computation confirms:

p	$\deg(b)$	$p-1$	$(p-1)/2$
53	50	52	26
97	96	96	48
193	192	192	96
503	500	502	251
1009	1008	1008	504

The effective degree is $\deg(b) = p - O(1)$, growing linearly with p .

Dwork bound. Applying the trace formula directly: α_{ab} is a nuclear operator on $L(\rho)$ with at most $\deg(ab) = p - O(1)$ nonzero eigenvalues, each of absolute value $\leq \sqrt{p}$. The trace bound is

$$|S_p(a)| \leq (p - O(1)) \sqrt{p} = O(p^{3/2}). \quad (719)$$

This is the *trivial Weil bound* and is useless for our purpose.

The reflection symmetry $b(x) = b(-1-x)$ allows a quadratic substitution reducing the effective degree to $(p-1)/2$, but the bound remains $O(p^{3/2}/2)$.

Observation 679 (Massive cancellation). The empirical value $|S_p(a)| \approx 3\sqrt{p}$ compared to the Dwork bound $|S_p(a)| \leq (p/2)\sqrt{p}$ implies cancellation among the $\approx p/2$ eigenvalues of the Dwork operator by a factor of $\approx p/6$. This cancellation is not predicted by any feature of the Dwork framework.

22.3 Higher-dimensional reformulations

We systematically explore representations of $S_p(a)$ as a character sum on a variety of bounded dimension and degree, which would bring the Dwork/Adolphson–Sperber machinery into effective range.

22.3.1 Cartier diagonal representation

The Apéry numbers admit the hypergeometric formula

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2. \quad (720)$$

This can be viewed as extracting the diagonal of a rational function on a four-dimensional torus. Define

$$F(x_1, x_2, x_3, x_4) = \frac{1}{(1-x_1)(1-x_2)(1-x_3)(1-x_4) + x_1x_2x_3x_4}$$

(suitably normalized). Then $b_n = [(x_1x_2x_3x_4)^n] F(x_1, x_2, x_3, x_4)$ is a “diagonal coefficient.”

Why this fails for Dwork. The diagonal extraction $b_j = \sum_{\mathbf{x} \in (\mathbf{F}_p^\times)^4} F(\mathbf{x}) (\prod x_i)^{-j}$ (up to normalization) involves the exponent j as a *power* of the monomial $\prod x_i$. Substituting into $S_p(a)$:

$$S_p(a) = \sum_{j=0}^{p-1} \psi \left(a \sum_{\mathbf{x}} F(\mathbf{x}) (\prod x_i)^{-j} \right).$$

The additive character ψ is applied to a *sum* over $\mathbf{x}$ (inside a single j -term), not to individual summands. Since $\psi(\text{sum}) = \text{product of } \psi(\text{terms})$:

$$S_p(a) = \sum_j \prod_{\mathbf{x} \in (\mathbf{F}_p^\times)^4} \psi(a F(\mathbf{x}) (\prod x_i)^{-j}).$$

This is a sum (over j) of a product of $(p-1)^4$ terms. The number of factors in the product grows as p^4 , preventing any bounded-rank Dwork representation.

22.3.2 Mellin–Fourier exchange

From the Mellin duality (Proposition 269 in `additive_char_analysis.tex`):

$$b_j \equiv - \sum_{t \in \mathbf{F}_p^\times} t^j (1-t)^{p-1-2j} A_j(t)^2 \pmod{p},$$

where $A_j(t) = {}_2F_1(-j, -j; 1; t)$ is the truncated Legendre polynomial of degree j in t . The factor $A_j(t)$ has degree j in t , which varies with j : for $j \approx p/2$, the degree is $\approx p/2$. Any exchange of the j -sum and t -sum must contend with this variable-degree factor, which again prevents reduction to bounded geometry.

22.3.3 Generating-function torus integral

Using the binomial representation (720) as a coefficient extraction from $(1+x)^j(1+y)^j(1+u)^{j+k}(1+v)^{j+k}$:

$$b_j = \text{coeff}_{x^0 y^0 u^0 v^0} \left[\sum_k \left(\frac{(1+u)(1+v)}{xyuv} \right)^k (1+x)^j (1+y)^j (1+u)^j (1+v)^j \right] \cdot \dots$$

The factor $(1+x)^j$ contributes a power j that couples j with the integration variable x . When we sum over j with the weight $\psi(ab_j)$, the nonlinearity of ψ prevents commuting the j -sum with the torus integral.

22.4 The F-crystal tensor product approach

This subsection develops the most sophisticated attempt to use p -adic methods: interpreting $\psi(ab_j)$ as a determinant on a tensor product of an F-crystal with the Dwork module.

22.4.1 Setup

Let $\mathcal{N}$ be the rank-2 Mellin sheaf on the character variety (parametrized by $j \in \{0, \dots, p-2\}$) with trace function $\text{Tr}(\text{Frob}_j, \mathcal{N}) = -b_j$ and weight 4. At each point j , the Frobenius has eigenvalues $\alpha_1(j), \alpha_2(j)$ satisfying

$$\alpha_1(j) + \alpha_2(j) = -b_j, \quad |\alpha_i(j)| = p^2. \quad (721)$$

Let $\mathcal{F}_a$ denote the “Dwork Artin–Schreier twist” with parameter a , whose effect on a p -adic eigenvalue λ is $\lambda \mapsto \psi_p(a\lambda)$, where $\psi_p(\lambda) = \theta(\lambda \bmod p)$ is the Dwork character evaluated on the mod- p reduction.

22.4.2 The determinant identity

Proposition 680 (Formal determinant identity). *If M is a rank-2 Frobenius module with eigenvalues α_1, α_2 , then formally*

$$\psi(a \text{Tr}(M)) = \psi(a\alpha_1) \psi(a\alpha_2) = \det(\psi(a \cdot -) \circ M),$$

where $\psi(a \cdot -) \circ M$ denotes the “rank-2 object with eigenvalues $\psi(a\alpha_i)$.” The product of two roots of unity (each of absolute value 1) is again a root of unity of absolute value 1.

Consequence. If the “twisted F-crystal” $\mathcal{N}_a$ with eigenvalues $\psi(a\alpha_1(j)), \psi(a\alpha_2(j))$ were a genuine lisse sheaf of bounded conductor, then $\det(\mathcal{N}_a)$ would be a rank-1 sheaf with trace function $\psi(ab_j)$, and by the Grothendieck trace formula:

$$S_p(a) = \sum_j \psi(ab_j) = \sum_j \text{Tr}(\text{Frob}_j, \det(\mathcal{N}_a)) \leq (\text{conductor of } \det(\mathcal{N}_a)) \sqrt{p}.$$

A conductor of $O(1)$ would give the bound (716).

22.4.3 The fundamental obstruction

Theorem 681 (Obstruction to the F-crystal approach). *The “twisted object” $\mathcal{N}_a$ with eigenvalues $\psi(a\alpha_i(j))$ is not a well-defined F-crystal, overconvergent F-isocrystal, or lisse ℓ -adic sheaf. Specifically:*

1. **Category mismatch.** The operation $\alpha \mapsto \psi(a\alpha)$ requires reducing the p -adic eigenvalue α modulo p and then applying the additive character. The mod- p reduction $\mathbf{Z}_p \rightarrow \mathbf{F}_p$ is not a morphism of p -adic analytic spaces; it is a discontinuous (in the p -adic topology) projection that destroys the analytic structure on which the F -crystal framework depends.
2. **Non-functoriality.** The map $\alpha \mapsto \psi(a\alpha \bmod p)$ does not respect matrix operations: for $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with eigenvalues α_1, α_2 , the matrix $\begin{pmatrix} \psi(a \cdot a) & \psi(a \cdot b) \\ \psi(a \cdot c) & \psi(a \cdot d) \end{pmatrix}$ does not have eigenvalues $\psi(a\alpha_1), \psi(a\alpha_2)$ in general. The operation is not a ring homomorphism ($\psi(x+y) = \psi(x)\psi(y)$ but $\psi(xy) \neq \text{anything natural}$).
3. **Conductor explosion.** Even if we bypass the categorical issues and directly compute the conductor of $\det(\mathcal{N}_a)$ viewed as a function on $\mathbf{F}_p$ (i.e., the Artin–Schreier conductor of the composite $\psi \circ b$), the result is $\text{cond} = \deg(b) + 1 \approx p$, because the composition $\psi(a \cdot b(x))$ is an Artin–Schreier cover of degree equal to $\deg(b) \approx p$.
4. **Mixed-characteristic reduction.** The eigenvalues $\alpha_i(j)$ are ℓ -adic (or p -adic) algebraic integers with $|\alpha_i| = p^2$. The integers $b_j = -(\alpha_1 + \alpha_2)$ satisfy $|b_j|_\infty \leq 2p^2$, so as elements of $\mathbf{Z}$, they “wrap around” $\mathbf{F}_p$ approximately $4p$ times. The mod- p reduction $b_j \mapsto b_j \bmod p$ depends on the exact integral value of b_j , not just its p -adic or archimedean class. Controlling $\psi(ab_j)$ thus requires controlling the p -adic and archimedean information simultaneously—the defining-characteristic gap.

22.5 Dwork congruences and their limitations

The p -adic trace formula *does* produce powerful structural information about the Apéry sequence, via the Dwork congruences:

$$b_{mp+r} \equiv b_m \cdot b_r \pmod{p} \quad (0 \leq r \leq p-1). \quad (722)$$

These hold for all tested primes $p \leq 47$ (and are known to hold in general by the work of Beukers, Stienstra–Beukers, and the overconvergent F -crystal structure of the Apéry family).

What the congruences give. The Dwork congruences (722) imply that $b_j \bmod p$ (for $0 \leq j \leq p-1$) determines the full sequence $b_n \bmod p$ for all n , via the “multiplication rule” $b_{mp+r} \equiv b_m b_r$. They also give $b_{p-1} \equiv 1 \pmod{p}$ (the unit root property, verified for all primes $p \leq 47$).

What the congruences do not give. The congruences constrain the sequence $b_0, b_1, \dots, b_{p-1}$ modulo p as a *multiplicative* structure (the sequence is a “ring homomorphism” from the p -adic digits to $\mathbf{F}_p$). However, the bound (716) concerns the *additive* Fourier spectrum of this sequence. No known principle connects the multiplicative structure of the Dwork congruences to the additive Fourier spectrum.

More precisely: let $\mathcal{S}_p$ be the set of sequences $\{c_0, \dots, c_{p-1}\} \subset \mathbf{F}_p$ satisfying $c_0 = 1, c_{p-1} = 1$, and the value distribution $E_p = \sum_v (\#\{j : c_j = v\})^2 = O(p)$. These constraints are all satisfied by the Apéry sequence. The question is whether *every* such sequence has $|\widehat{c}(a)| \leq C\sqrt{p}$ for all $a \neq 0$. The answer is *no*: random sequences with $E_p \approx 3p$ can have $\max_a |\widehat{c}(a)| \approx \sqrt{2p \log p}$ by the probabilistic method, exceeding $C\sqrt{p}$ by a $\sqrt{\log p}$ factor. The Apéry sequence achieves $\max \approx 4\sqrt{p}$ consistently, better than random, suggesting additional structure beyond the Dwork congruences.

22.6 The D-finite vs. C-finite dichotomy

The root cause of the Dwork obstruction is the distinction between *D-finite* and *C-finite* sequences:

Definition 682. A sequence $\{c_n\}$ is *C-finite* if it satisfies a linear recurrence with *constant* coefficients: $c_{n+d} = a_{d-1}c_{n+d-1} + \cdots + a_0c_n$.

A sequence is *D-finite* if it satisfies a linear recurrence with *polynomial* coefficients in n : $p_d(n)c_{n+d} + \cdots + p_0(n)c_n = 0$ with $p_i \in \mathbf{Z}[n]$.

The Apéry sequence b_n is D-finite (three-term recurrence with cubic polynomial coefficients) but *not* C-finite. The Berlekamp–Massey linear complexity of $b_j \bmod p$ is $L(p) = (p-1)/2 + 1 \approx p/2$, confirming that b_j requires a constant-coefficient recurrence of order $\approx p/2$.

Dwork’s framework handles C-finite sequences efficiently. If $b_j = c_1\alpha_1^j + \cdots + c_d\alpha_d^j$ (C-finite, order d), then

$$S_p(a) = \sum_j \psi(a b_j) = \sum_j \psi(c_1 a \alpha_1^j + \cdots + c_d a \alpha_d^j)$$

is a Kloosterman-type sum on a d -dimensional torus, bounded by $d\sqrt{p}$ via the Deligne/Katz theory. The “effective conductor” is d , independent of p .

D-finite sequences escape the framework. For a D-finite sequence that is not C-finite, the polynomial interpolation $b(x) \in \mathbf{F}_p[x]$ has degree growing with p , and no reformulation as a bounded-dimensional character sum is known. The “transfer matrix” representation

$$(b_j, b_{j-1})^T = M_{j-1} M_{j-2} \cdots M_1 (5, 1)^T,$$

where $M_j = \begin{pmatrix} (2j+1)(17j^2+17j+5)/(j+1)^3 & -j^3/(j+1)^3 \\ 1 & 0 \end{pmatrix}$, involves j distinct matrices. The product of j matrices is not a trace of a fixed operator but a path-ordered product (“time-ordered exponential”) that has no bounded-rank trace formula.

22.7 Spectral analysis: where the cancellation hides

The empirical data reveal the nature of the cancellation that the Dwork framework cannot capture.

Parseval identity. $\sum_{a=1}^{p-1} |S_p(a)|^2 = p E_p - p^2$, where $E_p = \sum_v m_v^2$ is the additive energy. Verified exactly:

p	E_p	$p E_p - p^2$	$\sum S ^2$	match
53	213	8480	8480	yes
97	273	17072	17072	yes
193	565	71796	71796	yes

Fourth-moment ratio. The ratio $R_4 = \frac{(p-1) \sum_a |S_p(a)|^4}{(\sum_a |S_p(a)|^2)^2}$ stabilizes near $R_4 \approx 1.9$, close to the Gaussian value 2. This indicates that $|S_p(a)|^2 / \mathbf{E}[|S|^2]$ is approximately exponentially distributed, consistent with $S_p(a)$ being a sum of $\approx p$ terms with near-uniform random phases (relative to the “ a ” direction), despite each individual term $\psi(a b_j)$ being deterministic.

No structured dependence on a . The values $|S_p(a)|$ show no monotone or periodic structure as a function of a , beyond the reflection symmetry $S_p(a) = S_p(p-a)$ (which follows from $b_j = b_{p-1-j}$, hence $S_p(a)$ and $S_p(p-a)$ are complex conjugates).

These spectral properties are consistent with the Apéry polynomial $b(x)$ being “pseudorandom at the Fourier level” — its additive Fourier coefficients behave as if b were a random polynomial of degree p , except that the maximum is bounded by $O(\sqrt{p})$ rather than $O(\sqrt{p \log p})$. This extra saving (removing the $\sqrt{\log p}$ factor compared to random) is the non-generic content of the conjecture.

22.8 Assessment of remaining avenues within p -adic methods

Despite the fundamental obstruction to the direct Dwork approach, we identify three residual directions where p -adic methods might contribute partial information.

22.8.1 (R1) Overconvergent difference modules

The Apéry recurrence defines a rank-2 *difference module* over the ring of p -adic difference operators. The theory of p -adic difference equations (André, Kedlaya) studies the Frobenius structure of such modules. If one could define an “additive-character twist” of the difference module and compute its L -function, the result would directly bound $S_p(a)$.

Status: No such theory exists. Kedlaya’s work on overconvergent F -isocrystals operates in the “multiplicative” setting (Frobenius eigenvalues, Newton polygons) and has no additive-character analogue.

22.8.2 (R2) Fresán–Sabbah–Yu arithmetic D-modules

A 2026 preprint (Fresán–Sabbah–Yu, arXiv:2608.00470) constructs arithmetic hypergeometric D-modules on reductive groups with Frobenius structures and derives exponential sum estimates. This is the closest existing work to our problem.

Status: Their estimates bound character sums over *points of a variety*, not over an index j running through $0, \dots, p-1$ with trace varying by Kummer twist. Adapting their framework to the Mellin-transform setting would require defining an Artin–Schreier twist on the Kummer character variety, which is exactly the categorical operation that fails (Theorem 681 (1)–(2)).

22.8.3 (R3) p -adic interpolation and Iwasawa theory

The Dwork congruences (722) imply that the sequence $b_j \bmod p$ can be p -adically interpolated to a continuous function $B: \mathbf{Z}_p \rightarrow \mathbf{Z}_p$ satisfying $B(n) \equiv b_n \pmod{p}$. The additive character sum $S_p(a) = \sum_{j=0}^{p-1} \psi(a B(j))$ can be viewed as a “ p -adic integral” of $\psi \circ (a \cdot B)$ over $\mathbf{Z}/p\mathbf{Z} \subset \mathbf{Z}_p$.

In Iwasawa theory, sums of the form $\sum_{j=0}^{p-1} \chi(j) \psi(a f(j))$ for a p -adic analytic function f arise naturally and are related to p -adic L -functions. However, the function $B(j)$ has *maximal* p -adic complexity (it is not a polynomial of bounded degree in j), and the standard Iwasawa-theoretic machinery does not produce non-trivial bounds for such functions.

Status: Speculative. Would require a new “ p -adic stationary phase” theorem for functions of maximal complexity.

22.9 Conclusion

Theorem 683 (Assessment of the Dwork approach). *The Dwork p -adic trace formula cannot prove the bound (716), for the following structural reason:*

The Dwork framework bounds $\sum_x \psi(f(x))$ by the **combinatorial complexity** of f (polynomial degree, Newton polyhedron volume). For the Apéry sum, the “function” $b(x) \in \mathbf{F}_p[x]$ has degree $\approx p$, yielding only the trivial bound $O(p^{3/2})$.

The **arithmetic complexity** of b_j is bounded: the sequence satisfies a rank-2 recurrence with cubic polynomial coefficients (a D -finite sequence of bounded “ D -complexity”). This bounded arithmetic complexity is invisible to the Dwork framework, which sees only the polynomial interpolation.

The obstruction is **fundamental**: it is equivalent to the mixed-characteristic gap between the ℓ -adic structure of the Mellin sheaf $\mathcal{N}$ (which has rank 2 and bounded conductor) and the p -adic/mod- p structure of the polynomial $b(x)$ (which has degree p). Bridging this gap would require a new theorem—a “defining-characteristic Katz theorem” or “crystalline effective conductor bound”—that translates the bounded rank of an F -crystal into a bounded effective conductor for the additive-character composition. No such theorem exists in the literature as of 2026.

Is the obstruction technical or structural? The obstruction is **structural**: it is not a matter of choosing better coordinates, a cleverer lifting, or a more refined p -adic space. The composition $\psi \circ \text{Tr}(\text{Frob}, \mathcal{N})$ involves a mod- p reduction step that is fundamentally incompatible with p -adic analyticity. Any proof of (716) must use a genuinely new mechanism that is not contained in the Dwork–Katz–Adolphson–Sperber–Wan framework.

The most likely avenue for a proof lies *outside* the p -adic framework: either through a representation of $S_p(a)$ as a character sum on a variety of bounded degree (Approach 3 of `additive_char_analysis.tex`), or through a “defining-characteristic equidistribution theorem” that directly controls the mod- p Fourier spectrum of Frobenius trace sequences.

23 The reduction

We record a clean decomposition of $\log G_n$ using the Lucas property and the base- p digit structure of n .

Proposition 684 (Digit decomposition). *Let $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$ be the Apéry numbers. For every $n \geq 1$,*

$$\log \gcd(\text{lcm}(1, \dots, n), b_n) = S_n + W_0(n) + W_1(n), \quad (723)$$

where

$$S_n = \sum_{\substack{p \leq \sqrt{n} \\ p \text{ prime}}} \min(v_p(b_n), \lfloor \log n / \log p \rfloor) \log p = O(\sqrt{n}), \quad (724)$$

$$W_0(n) = \sum_{\substack{p > \sqrt{n}, p \leq n \\ n \bmod p \in \mathcal{Z}_p}} \log p, \quad (725)$$

$$W_1(n) = \sum_{\substack{p > \sqrt{n}, p \leq n \\ p \mid \lfloor n/p \rfloor \\ n \bmod p \notin \mathcal{Z}_p}} \log p, \quad (726)$$

with $\mathcal{Z}_p = \{r \in [0, p) : p \mid b_r\}$ the zero set of $(b_r \bmod p)$.

Proof. For $p > \sqrt{n}$, $v_p(\text{lcm}(1, \dots, n)) = 1$, so the contribution of each such prime to $\log \gcd(\text{lcm}, b_n)$ is $\log p$ if $p \mid b_n$ and 0 otherwise.

The Lucas congruence (Gessel 1982) gives $b_n \equiv \prod_i b_{n_i} \pmod{p}$ where n_i are the base- p digits of n . For $p > \sqrt{n}$, the expansion $n = qp + r$ with $q = \lfloor n/p \rfloor \leq \sqrt{n}$ and $r = n \bmod p$ has exactly two digits (since $n < p^2$). Thus

$$p \mid b_n \iff p \mid b_q \text{ or } p \mid b_r \iff r \in \mathcal{Z}_p \text{ or } p \mid b_q.$$

The three terms in (723) partition the prime contributions accordingly.

The small-prime bound $S_n = O(\sqrt{n})$ follows from $\sum_{p \leq \sqrt{n}} \log p \cdot \lfloor \log n / \log p \rfloor \leq \sqrt{n} \log n$ and the prime number theorem. $\square$

Proposition 685 (First-digit bound). *Let $\omega_{\max}(M) = \max_{1 \leq k \leq M} \omega(b_k)$. Then*

$$W_1(n) \leq \omega_{\max}(\sqrt{n}) \cdot \sqrt{n} \cdot \log n. \quad (727)$$

Proof. For each k with $1 \leq k \leq \sqrt{n}$, the primes $p \in (n/(k+1), n/k]$ satisfying $\lfloor n/p \rfloor = k$ and $p \mid b_k$ number at most $\omega(b_k)$, each contributing $\log p \leq \log n$. Summing over $k = 1, \dots, \lfloor \sqrt{n} \rfloor$:

$$W_1(n) \leq \sum_{k=1}^{\lfloor \sqrt{n} \rfloor} \omega(b_k) \cdot \log n \leq \sqrt{n} \cdot \omega_{\max}(\sqrt{n}) \cdot \log n. \quad \square$$

Corollary 686 (Reduction to K_0). *Set $K_0(n) = \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\}$. If $\omega(b_k) = O(k^\varepsilon)$ for every $\varepsilon > 0$, then*

$$\log \gcd(\text{lcm}(1, \dots, n), b_n) = K_0(n) \cdot \log n + o(n).$$

In particular, $G_n = e^{o(n)}$ if and only if $K_0(n) = o(n/\log n)$.

If the stronger $\omega(b_k) = O(\log k)$ holds, then the first-digit contribution satisfies $W_1(n) = O(\sqrt{n} (\log n)^2) = o(n)$.

Proof. Under $\omega(b_k) \leq C_\varepsilon k^\varepsilon$:

$$\sum_{k=1}^{\sqrt{n}} \omega(b_k) \leq C_\varepsilon \sum_{k=1}^{\sqrt{n}} k^\varepsilon \leq C'_\varepsilon n^{(1+\varepsilon)/2}.$$

Hence $W_1(n) \leq C'_\varepsilon n^{(1+\varepsilon)/2} \log n = o(n)$ for every $\varepsilon < 1$.

Under $\omega(b_k) \leq C \log k$:

$$\sum_{k=1}^{\sqrt{n}} \omega(b_k) \leq C \sum_{k=1}^{\sqrt{n}} \log k \leq C \sqrt{n} \log(\sqrt{n}) = \frac{C}{2} \sqrt{n} \log n.$$

Hence $W_1(n) \leq \frac{C}{2} \sqrt{n} (\log n)^2 = o(n)$. $\square$

Remark 687 (Comparison with `proof.tex`). The leading-digit vanishing lemma (Lemma 5 in the main manuscript) shows that, for G_n specifically, primes p with $p \mid b_q$ and $p \nmid b_r$ satisfy $v_p(G_n) = 0$ and hence do not contribute. This eliminates $W_1(n)$ from $\log G_n$ directly, without needing any $\omega(b_k)$ bound.

The present reduction addresses $\log \gcd(\text{lcm}, b_n)$ instead. The comparison error $|\log G_n - \log \gcd(\text{lcm}, b_n)| = O(n^{2/3})$ (DOCTRINE Section a0-live) absorbs this difference. In principle, the leading-vanish mechanism already handles the first digit. The interest of the $\omega(b_k)$ route is as a *self-contained modular input* that makes the first-digit control independent of the Cartier–Frobenius block analysis.

24 Literature survey: bounds on $\omega(a_n)$ for recurrence sequences

24.1 Constant-coefficient recurrences (Stewart, Corvaja–Zannier)

For a nondegenerate binary linear recurrence $u_n = c_1\alpha_1^n + c_2\alpha_2^n$ with constant coefficients, the main tools are:

1. **Stewart (1977, 2013).** Stewart proved $P^+(u_n) \rightarrow \infty$ for nondegenerate binary recurrences. Evertse–Stewart (2013) improved this to $P^+(u_n) \geq \exp(c\sqrt{\log n})$ using the Subspace Theorem (Schmidt–Evertse–Schlickewei). These give *lower bounds on the largest prime factor*, not upper bounds on ω .
2. **Primitive divisors (Bilu–Hanrot–Voutier 2001).** For Lucas sequences $u_n = (\alpha^n - \beta^n)/(\alpha - \beta)$, every u_n with $n > 30$ has a primitive prime divisor. The proof relies on the exponential form and cyclotomic polynomial factorization.
3. **Corvaja–Zannier (2004, 2013).** Their GCD bounds $\gcd(a^n - 1, b^n - 1) < e^{\varepsilon n}$ for multiplicatively independent a, b use the Subspace Theorem applied to exponential representations.
4. **S-unit equations (Evertse–Schlickewei–Schmidt).** The ESS theorem bounds solutions to $a_1x_1 + \cdots + a_kx_k = 0$ with $x_i \in S$ -units, where $|S|$ is the number of places. For constant-coefficient recurrences, each u_n is a sum of S -unit terms $(c_i\alpha_i^n)$, enabling these methods.

Obstruction for Apéry: All four frameworks require the exponential-polynomial structure $u_n = \sum c_i\alpha_i^n$ with fixed algebraic roots α_i . The Apéry recurrence $(n+1)^3b_{n+1} = (2n+1)(17n^2 + 17n+5)b_n - n^3b_{n-1}$ has *polynomial* coefficients of degree 3. The Apéry numbers have no exponential closed form; they are defined as a binomial sum $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$. None of the four tools applies.

24.2 D-finite sequences

The Apéry sequence is D-finite (holonomic): its generating function satisfies a linear ODE with polynomial coefficients. The literature on $\omega(a_n)$ for D-finite sequences is *essentially empty*:

- **André G-functions.** André’s theorem gives globally bounded denominators for G-function values at algebraic points. This controls $\text{den}(a_n)$, not $\omega(b_n)$.
- **Christol/automaticity.** The Lucas property gives $b_n \bmod p$ as a digit-multiplicative function, more structured than automatic. This determines *which* primes divide b_n (those p with some base- p digit in $\mathcal{Z}_p$), but does not bound *how many* such primes exist.
- **Modular forms.** The connection $b_n \equiv a(n) \pmod{p}$ (where $a(n)$ are Fourier coefficients of a weight-4 form on $\Gamma_0(6)$) is a mod- p statement. Ramanujan-type bounds $|a(p)| \leq 2p^{3/2}$ concern Fourier coefficients, not the number of prime factors of the Apéry numbers themselves.
- **Height bounds.** Since $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.53n$, the trivial bound is $\omega(b_n) \leq 3.53n / \log 2 \approx 5.1n$. This is $O(n)$, far from $O(n^\varepsilon)$.

Open Problem 688. *Prove or disprove: $\omega(b_k) = O(k^\varepsilon)$ for every $\varepsilon > 0$, where b_k are the Apéry numbers.*

There is no existing framework to attack this problem. A bound $\omega(b_k) = O(k^\varepsilon)$ for the Apéry sequence would be a genuinely new result in the arithmetic of holonomic sequences.

24.3 The abc perspective

The abc conjecture, applied to the recurrence relation

$$(k+1)^3 b_{k+1} + k^3 b_{k-1} = (2k+1)(17k^2 + 17k + 5) b_k,$$

gives a *lower* bound on $\text{rad}(b_k b_{k+1} b_{k-1})$:

$$\text{rad}(b_k b_{k+1} b_{k-1}) \geq \exp((1 - \varepsilon) \cdot 3.53 k - O(\log k))$$

for every $\varepsilon > 0$. This implies that the Apéry numbers have *many* distinct prime factors on average — the opposite direction from what we need. The abc conjecture is orthogonal to an upper bound on ω .

25 Heuristic analysis

25.1 Erdős–Kac prediction

For a “random” integer N , the Erdős–Kac theorem gives $\omega(N) \sim \log \log N$ with standard deviation $\sqrt{\log \log N}$. For b_k with $\log b_k \sim 3.53 k$:

$$\omega(b_k) \stackrel{?}{\sim} \log \log b_k = \log(3.53 k) = \log k + O(1).$$

This predicts $\omega(b_k) = O(\log k)$, consistent with all empirical data.

The heuristic assumes that the Apéry numbers “behave like random integers of the same size” with respect to their prime factorization. The Lucas property and modular form connection impose non-generic structure, but empirically this structure is consistent with—and does not contradict—the Erdős–Kac prediction.

25.2 Lucas-property argument

A more refined heuristic uses the Lucas property directly. For each prime p , the probability that $p \mid b_k$ is approximately $|\mathcal{Z}_p|/p \approx 1/p$ (since $|\mathcal{Z}_p| \approx 1$ on average).

For primes $p > k$: $p \mid b_k$ iff $k \in \mathcal{Z}_p$. The expected count of such primes is $\sum_{p>k} \mathbb{P}(k \in \mathcal{Z}_p) \approx \sum_{p>k} 1/p$, which diverges. But b_k has finitely many prime factors, so the actual count is $\omega_{>k}(b_k) \leq \log b_k / \log k \approx 3.53 k / \log k$.

For primes $p \leq k$: the Lucas property gives $b_k \equiv \prod b_{d_i} \pmod{p}$ with $\lfloor \log_p k \rfloor + 1$ digits. Each digit d_i has probability $\sim 1/p$ of landing in $\mathcal{Z}_p$. With $\sim \log k / \log p$ digits:

$$\mathbb{P}(p \mid b_k) \approx \frac{\log k}{p \log p}.$$

Summing:

$$\sum_{p \leq k} \frac{\log k}{p \log p} \approx \log k \cdot \log \log \log k.$$

This is $O(\log k \cdot \log \log \log k)$, consistent with $\omega(b_k) = O(\log k)$.

26 Empirical data

Omega values for b_k , $k = 2, \dots, 40$:

k	$\omega(b_k)$	$\Omega(b_k)$	$\omega/\log k$	primitive?
2	1	1	1.44	yes (73)
5	3	3	1.86	yes (11, 14891)
8	4	5	1.92	yes (19, 1549, 1111573)
12	1	1	0.40	yes
19	6	6	2.04	yes (181, 359, ...)
23	7	7	2.23	yes (2237, ...)
24	1	1	0.31	yes
33	9	11	2.57	yes (751, ...)
36	3	5	0.84	yes (179, ...)
40	3	4	0.81	yes

Key observations from the full data ($k = 2, \dots, 41$):

- $\max_{k \leq 41} \omega(b_k) = 9$ at $k = 33$.
- $\omega(b_k)/\log k \in [0.31, 2.57]$ throughout.
- Consistent with $\omega(b_k) = O(\log k)$.
- $\log P^+(b_k)/k \geq 0.69$ for all k , with mean ≈ 1.7 . Giant prime factors account for the majority of $\log b_k$.
- **Every b_k for $k = 1, \dots, 41$ has at least one primitive prime divisor.** The number of primitive factors ranges from 1 to 8 (median ≈ 3).
- Extremes: $\omega(b_{12}) = \omega(b_{24}) = 1$ (both b_k are prime); $\omega(b_{33}) = 9$ (the most composite case in range).

27 Primitive divisors

Open Problem 689. *Does every b_k with k sufficiently large have a primitive prime divisor—a prime $p \mid b_k$ with $p \nmid b_m$ for all $0 < m < k$?*

Available evidence:

- The Bilu–Hanrot–Voutier theorem for Lucas sequences does not extend to D-finite sequences. The proof requires $u_n \propto \alpha^n - \beta^n$ and cyclotomic factorization.
- The Apéry recurrence gives $\gcd(b_k, b_{k-1}) \mid \prod_{j \leq k} j^3$ (from the Casoratian $b_k a_{k-1} - b_{k-1} a_k = (-1)^{k-1}/k^3$), so consecutive b_k share no large prime factors. But this does not prevent b_k and b_m from sharing large primes for $|k - m| > 1$.
- Empirically, every b_k for $k = 1, \dots, 41$ has at least one primitive prime divisor, with a median of 3 primitive factors per term.
- The modular connection (to $\eta(\tau)^4 \eta(6\tau)^4$) might provide structural constraints, but no theorem connecting modular Galois representations to primitive divisors of coefficient sequences is known.

28 Assessment and status

28.1 What the $\omega(b_k)$ route achieves

If $\omega(b_k) = O(k^\varepsilon)$ for every $\varepsilon > 0$, then:

1. The first-digit contribution $W_1(n) = o(n)$ is automatic (Corollary 686).
2. The problem reduces to $K_0(n) = o(n/\log n)$, i.e., the *zeroth-digit* sieve problem alone.
3. No cross-prime factorial moment $(HM)_3$ is needed for the first-digit channel.

28.2 What it does NOT achieve

1. The zeroth-digit problem $K_0(n) = o(n/\log n)$ remains the central open problem. This is exactly the “lower-digit incidence” obstruction in the main manuscript.
2. The leading-digit vanishing lemma already eliminates W_1 from $\log G_n$ via the Cartier–Frobenius block law, so the $\omega(b_k)$ bound provides an *alternative* proof of a result that is already handled in the main argument.
3. The bound $\omega(b_k) = O(k^\varepsilon)$ is itself *unproved* and appears to be beyond all existing techniques. It would be a genuinely new result in the arithmetic of holonomic sequences.

28.3 Terminal assessment

The $\omega(b_k)$ route is **logically sound but practically redundant**: the first digit is already eliminated by leading-vanish, and the remaining zeroth-digit problem is unchanged. The bound $\omega(b_k) = O(\log k)$ is almost certainly true but lies beyond current technology.

The interesting feature of this approach is its *modularity*: it factors the problem into a “vertical” number-theoretic input ($\omega(b_k)$) and a “horizontal” sieve input (K_0), each attackable independently. If progress on $\omega(b_k)$ for D-finite sequences becomes available (perhaps from advances in the arithmetic of G-functions or the subspace theorem for differential equations), this reduction would immediately apply.

Open problem: the all-index conjecture

Conjecture. $G_n = e^{o(n)}$ for every positive integer n , where

$$G_n = \gcd\left(\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2, \operatorname{lcm}(1, \dots, n)\right).$$

Remark 690 (Equivalent and sufficient formulations). The first statement below is an equivalent pointwise formulation. The remaining two form a conditional moment route and require the dyadic first zero-mass bound $\lambda_X \ll 1/\log X$ in order to recover the all-index conclusion.

1. The pointwise radical anti-concentration: $\omega_{(\sqrt{n}, n]}(b_n) = o(n/\log n)$ for every n .
2. Under M_1 , the third factorial moment estimate $(HM)_3$: $\sum_{m < X^2} (K_X(m))_3 \leq CX^2 \lambda_X^3$ for all large X (Theorem 350).

3. Under M_1 , the pair–Palm excess bound: $\sum_{p \neq q} (T_{p,q} - J_{p,q} \lambda_{p,q})_+ \ll X^{2+o(1)} \lambda_X^3$ (Theorem 351).

Remark 691 (The remaining gap). All known proofs of the density-one theorem (Theorem 82) and the $O(p^{2/3})$ gap bound (the gap bound) are *one-prime* arguments: they control the distribution of $b_j \pmod{p}$ for each fixed p independently. The all-index conjecture is intrinsically a *cross-prime* statement: it requires that the events $\{p \mid b_m\}$ at different primes do not simultaneously concentrate at one integer m .

The Chinese remainder theorem provides exact independence on complete residue systems: for distinct primes p, q ,

$$\#\{m < pq : m \bmod p \in \mathcal{Z}_p, m \bmod q \in \mathcal{Z}_q\} = |\mathcal{Z}_p| \cdot |\mathcal{Z}_q|.$$

No error. The difficulty arises from *incomplete* summation on the interval $[0, X^2)$ with $X^2 < pq$: the CRT prediction $M|\mathcal{Z}_p||\mathcal{Z}_q|/(pq)$ has an error of size $|\mathcal{Z}_p||\mathcal{Z}_q|$, and the absolute sum of these errors over all pairs exceeds the target by a factor of X .

The versions of sheaf theory, sieve methods, $\mathrm{SL}_2(\mathbb{F}_p)$ expansion, Picard–Fuchs structure, and Subspace Theorem methods audited in the companion survey do not presently close this factor- X gap. This is a scope statement about the recorded arguments, not a no-go theorem for those subjects. The obstacle in the audited routes is a *defining-characteristic* phenomenon: the event $p \mid b_j$ is a condition in the same characteristic p as the finite field over which the Frobenius acts, and the standard ℓ -adic tools (Deligne equidistribution, Sato–Tate, large sieve for Frobenius) do not control it.

The missing theorem is a *horizontal independence estimate* for the Mellin zero events of the Apéry compatible system, uniform over varying characteristics.

References

- [1] S. Ahlgren and K. Ono, *A Gaussian hypergeometric series evaluation and Apéry number congruences*, J. reine angew. Math. **518** (2000), 187–212.
- [2] A. Adolphson and S. Sperber, *Twisted exponential sums and Newton polyhedra*, J. reine angew. Math. **443** (1993), 151–177.
- [3] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$* , Astérisque **61** (1979), 11–13.
- [4] F. Beukers, *Some congruences for the Apéry numbers*, J. Number Theory **21** (1985), 141–155.
- [5] F. Beukers, *Another congruence for the Apéry numbers*, J. Number Theory **25** (1987), 201–210.
- [6] F. Beukers and C. A. M. Peters, *A family of K3 surfaces and $\zeta(3)$* , J. Reine Angew. Math. **351** (1984), 42–54.
- [7] S. Bettin and V. Chandee, *Trilinear forms with Kloosterman fractions*, Adv. Math. **328** (2018), 1234–1262, doi:10.1016/j.aim.2018.01.026.
- [8] M.-J. Bertin and O. Lecacheux, *Apéry–Fermi pencil of K3-surfaces and 2-isogenies*, J. Math. Soc. Japan **72** (2020), no. 2, 599–637.
- [9] Y. Bugeaud, P. Corvaja, and U. Zannier, *An upper bound for the G.C.D. of $a^n - 1$ and $b^n - 1$* , Math. Z. **243** (2003), 79–84.

- [10] X. Caruso, F. Fürnsinn, D. Vargas-Montoya, and W. Zudilin, *Galois groups of Apéry-like series modulo primes*, Bull. Aust. Math. Soc. **114** (2026), no. 1, 65–78, doi:10.1017/S0004972725100932.
- [11] W. Duke, J. B. Friedlander, and H. Iwaniec, *Equidistribution of roots of a quadratic congruence to prime moduli*, Ann. of Math. **141** (1995), 423–441.
- [12] S. Drappeau, *Sums of Kloosterman sums in arithmetic progressions, and the error term in the dispersion method*, Proc. Lond. Math. Soc. **114** (2017), 684–732.
- [13] J. Friedlander and A. Granville, *Limitations to the equi-distribution of primes I*, Ann. of Math. **129** (1989), 363–382.
- [14] É. Fouvry, E. Kowalski, and P. Michel, *Algebraic trace functions over the primes*, Duke Math. J. **163** (2014), 1683–1736.
- [15] E. Kowalski, P. Michel, and W. Sawin, *Bilinear forms with Kloosterman sums and applications*, Ann. of Math. **186** (2017), 413–500.
- [16] D. S. Kubert, *Universal bounds on the torsion of elliptic curves*, Proc. London Math. Soc. (3) **33** (1976), 193–237.
- [17] A. Forey, J. Fresán, and E. Kowalski, *Arithmetic Fourier transforms over finite fields: generic vanishing, convolution, and equidistribution*, preprint, 2023.
- [18] J. Greene, *Hypergeometric functions over finite fields*, Trans. Amer. Math. Soc. **301** (1987), 77–101.
- [19] N. M. Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Ann. of Math. Studies **116**, Princeton Univ. Press, 1988.
- [20] N. M. Katz, *Convolution and Equidistribution: Sato–Tate Theorems for Finite-Field Mellin Transforms*, Ann. of Math. Studies **180**, Princeton Univ. Press, 2012.
- [21] I. Bah and V. Shuddhodan, *Equidistribution for Kummer families on $\Gamma_0(6)$ -modular curves*, preprint, 2026.
- [22] A. Levin, *Greatest common divisors and Vojta’s conjecture for blowups of algebraic tori*, Invent. Math. **215** (2019), 493–533.
- [23] J. Maynard, *Primes in arithmetic progressions to large moduli III: uniform residue classes*, arXiv:2006.08250, 2020.
- [24] I. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), 362–368.
- [25] S. Jin, W. Ma, and K. Ono, *A note on non-ordinary primes*, Proc. Amer. Math. Soc. **144** (2016), 4591–4597.
- [26] A. Malik and A. Straub, *Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory **2** (2016), Art. 5.
- [27] R. J. McIntosh, *A generalization of a congruential property of Lucas*, Amer. Math. Monthly **99** (1992), 231–238.

- [28] Y. Mimura, *Congruence properties of Apéry numbers*, J. Number Theory **16** (1983), 138–146.
- [29] C. A. M. Peters and J. Stienstra, *A pencil of K3-surfaces related to Apéry’s recurrence for $\zeta(3)$ and Fermi surfaces for potential zero*, in *Arithmetic of Complex Manifolds*, Lecture Notes in Math. **1399**, Springer, Berlin, 1989, 110–127.
- [30] E. Rowland and R. Yassawi, *Automatic congruences for diagonals of rational functions*, J. Théor. Nombres Bordeaux **27** (2015), 245–288.
- [31] G. Rhin and C. Viola, *The group structure for $\zeta(3)$* , Acta Arith. **97** (2001), 269–293.
- [32] P. Deligne, *La formule de Picard–Lefschetz*, in *Groupes de monodromie en géométrie algébrique (SGA 7 II)*, Lecture Notes in Math. **340**, Springer, 1973, Exp. XV.
- [33] J. Stienstra and F. Beukers, *On the Picard–Fuchs equation and the formal Brauer group of certain elliptic K3 surfaces*, Math. Ann. **271** (1985), 269–304.
- [34] J.-P. Serre, *Two letters on quaternions and modular forms (mod p)*, Israel J. Math. **95** (1996), 281–299.
- [35] L. Xiao, *Greatest common divisors of integral points of numerically equivalent divisors*, Math. Z. **306** (2024), article 61.
- [36] A. Straub, *Multivariate Apéry numbers and supercongruences of rational functions*, Algebra Number Theory **8** (2014), no. 8, 1985–2008.
- [37] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), 251–291.
- [38] D. Zagier, *Integral solutions of Apéry-like recurrence equations*, in *Groups and Symmetries: From Neolithic Scots to John McKay*, CRM Proc. Lecture Notes **47**, Amer. Math. Soc., 2009, 349–366.

29 The projective orbit and its energy: an unconditional $p^{3/2}$ theorem and the E1 program

Throughout this section $p \geq 7$ is prime and

$$P(u) = (2u + 1)(17u^2 + 17u + 5) = 34u^3 + 51u^2 + 27u + 5.$$

The Apéry recurrence is

$$(n + 1)^3 y_{n+1} = P(n) y_n - n^3 y_{n-1}. \tag{728}$$

We use only the regular indices $0 \leq n \leq p - 1$; every nonzero integer denominator that occurs there is reduced modulo p .

29.1 The companion solution and the two energy conventions

Let b be the Apéry solution, $b_0 = 1$, $b_1 = 5$, and let c be the companion solution of (728) with $c_0 = 0$, $c_1 = 1$.

Lemma 692 (companion Casoratian). *For every $n \geq 1$,*

$$W_n := b_n c_{n-1} - b_{n-1} c_n = -\frac{1}{n^3}.$$

In particular, the two solution vectors are linearly independent modulo p at every index used below.

Proof. The initial value is $W_1 = -1$. Applying (728) to both solutions and subtracting gives

$$b_{n+1} c_n - b_n c_{n+1} = \frac{n^3}{(n+1)^3} (b_n c_{n-1} - b_{n-1} c_n) = \frac{n^3}{(n+1)^3} W_n.$$

The left side is W_{n+1} , so induction proves the formula. $\square$

Put $I_p = \{1, \dots, p-2\}$, $M = |I_p| = p-2$, and define the projective orbit

$$\pi(n) = [b_n : c_n] \in \mathbb{P}^1(\mathbb{F}_p), \quad n \in I_p.$$

For $v \in \mathbb{P}^1(\mathbb{F}_p)$ let

$$m_p(v) = \#\{n \in I_p : \pi(n) = v\}.$$

We distinguish the full and off-diagonal energies

$$\mathcal{F}_p = \sum_v m_p(v)^2, \quad \mathcal{E}_p = \sum_v m_p(v)(m_p(v) - 1). \quad (729)$$

Since $\sum_v m_p(v) = M$, the exact reconciliation is

$$\boxed{\mathcal{E}_p = \mathcal{F}_p - (p-2)}. \quad (730)$$

Also $Z_p := \{n \in I_p : b_n = 0\}$ is the fibre over $[0 : 1]$, so $|Z_p| = m_p([0 : 1])$.

29.2 Restart continuants and the gap-polynomial dictionary

For $u \neq -1$ set

$$M_u = \begin{pmatrix} 0 & (u+1)^6 \\ -1 & P(u) \end{pmatrix}, \quad M_u[X : Z] = [(u+1)^6 Z : -X + P(u)Z].$$

Its determinant is $(u+1)^6$, so M_u is a projective transformation exactly on this domain, and there $M_u(\infty) = 0$. Define

$$x_n = [n^3 b_{n-1} : b_n] \in \mathbb{P}^1(\mathbb{F}_p).$$

The recurrence gives $x_{n+1} = M_n x_n$. Thus $b_z = 0$ implies $x_z = \infty$ (the numerator is nonzero by Lemma 10) and, whenever the next step is in the regular window, $x_{z+1} = 0$. This is the precise restart mechanism used below. Define

$$U_{-1}(s) = 0, \quad U_0(s) = 1, \quad U_1(s) = P(s), \quad U_{m+1}(s) = P(s+m)U_m(s) - (s+m)^6 U_{m-1}(s). \quad (731)$$

Lemma 693 (exact transfer product). *For $m \geq 1$, with $G_m(s) = M_{s+m-1} \cdots M_s$, one has*

$$G_m(s) = \begin{pmatrix} -(s+m)^6 U_{m-2}(s+1) & (s+m)^6 U_{m-1}(s) \\ -U_{m-1}(s+1) & U_m(s) \end{pmatrix}. \quad (732)$$

Moreover $\deg U_m = 3m$ over $\mathbb{Q}$.

Proof. For $m = 1$ the formula is the definition of M_s . Left multiplication by M_{s+m} makes the new top row $(s + m + 1)^6$ times the old bottom row, while the new bottom row is minus the old top row plus $P(s + m)$ times the old bottom row. The two entries are exactly those prescribed by (731), proving the matrix formula by induction.

If λ_m is the leading coefficient of U_m , then $\lambda_0 = 1$, $\lambda_1 = 34$, and $\lambda_{m+1} = 34\lambda_m - \lambda_{m-1}$. The characteristic roots $17 \pm 12\sqrt{2}$ are positive, whence $\lambda_m > 0$ and $\deg U_m = 3m$. $\square$

Let $N_h(r)$ retain the established positional-gap convention of Lemma 11: $N_1 = 1$, $N_2(r) = P(r + 1)$, and

$$N_{h+1}(r) = P(r + h)N_h(r) - (r + h)^6 N_{h-1}(r).$$

Proposition 694 (restart and the N – U dictionary). *The following statements hold.*

- If $\prod_{j=1}^m (s + j) \neq 0$ in $\mathbb{F}_p$, then the composite of the m allowed maps starting from $0 = [0 : 1]$ ends at ∞ if and only if $U_m(s) = 0$.

- For every $h \geq 1$,

$$\boxed{N_h(r) = U_{h-1}(r + 1)}. \quad (733)$$

On a nonwrapping window, $\pi(r) = \pi(r + h)$ if and only if $N_h(r) = 0$.

- If $1 \leq z_1 < \dots < z_k \leq p - 2$ are the zeros of b and $1 \leq H \leq p - 2$, then, for

$$R_{\text{rel}}(p, H) = \sum_{m=1}^H \#\{1 \leq s \leq p - m - 1 : U_m(s) = 0\}, \quad (734)$$

one has the restricted restart bound

$$|Z_p| \leq 1 + R_{\text{rel}}(p, H) + \left\lfloor \frac{p - 3}{H + 1} \right\rfloor. \quad (735)$$

Proof. By Lemma 693,

$$G_m(s)[0 : 1] = [(s + m)^6 U_{m-1}(s) : U_m(s)].$$

On the displayed open set the product is invertible, so its two coordinates cannot vanish together. This proves the first assertion and also explains why roots at windows containing the forbidden parameter $u = -1$ must be excluded from the dynamical count.

Both sides of (733) have initial values $1, P(r + 1)$ and the same recurrence, so they agree. For the collision statement, the solution $y_n = b_r c_n - c_r b_n$ satisfies $y_r = 0$ and $y_{r+1} = b_r c_{r+1} - c_r b_{r+1} = 1/(r + 1)^3 \neq 0$ by Lemma 692. Iterating the recurrence from this zero shows that $y_{r+h} = 0$ exactly when $N_h(r) = 0$. Since $y_{r+h} = 0$ is precisely $\det((b_r, c_r), (b_{r+h}, c_{r+h})) = 0$, this is the projective collision condition.

For consecutive zeros put $s_i = z_i + 1$ and $m_i = z_{i+1} - z_i - 1$. Lemma 10 gives $m_i \geq 1$, and the first assertion gives $U_{m_i}(s_i) = 0$. The pair (s_i, m_i) is unique and its window is nonwrapping; enlarging the actual range to that in (734) is harmless. Finally

$$\sum_{i=1}^{k-1} m_i = z_k - z_1 - (k - 1) \leq p - 3.$$

Thus at most $\lfloor (p - 3)/(H + 1) \rfloor$ of the m_i exceed H ; the additive 1 pays for the first zero. This proves (735). $\square$

Remark 695 (the one-step shift). Immediately after a zero at z , the Riccati state has restarted at 0 at $s = z + 1$. Hence a positional zero gap $h = z' - z$ contains only $m = h - 1$ post-restart maps. Formula (733) records both shifts: $N_h(z) = U_{h-1}(z + 1)$. Writing “ $N_m = U_m$ ” without them changes the degree and the parity of the center root.

29.3 Reflection, center vanishing, and root pairing

Theorem 696 (pointwise orbit reflection). *For every $n \in I_p$,*

$$\boxed{\pi(p - 1 - n) = \pi(n)}.$$

Proof. It is enough to prove the stronger statement that every solution y of (728) on $\{0, \dots, p - 1\}$ satisfies

$$y_{p-1-n} = y_n \quad (0 \leq n < p).$$

Put $m = (p - 1)/2$ and work in $\mathbb{F}_p$. Since $m + 1 = -m$ and $P(m) = 0$, the recurrence at m gives

$$-m^3 y_{m+1} = -m^3 y_{m-1}, \quad \text{hence} \quad y_{m+1} = y_{m-1}.$$

Now define $z_n = y_{p-1-n}$. The polynomial identity $P(-1 - n) = -P(n)$ shows, by applying the recurrence at $p - 1 - n$, that z satisfies the same recurrence as y for $1 \leq n \leq p - 2$. Moreover

$$z_m = y_m, \quad z_{m+1} = y_{m-1} = y_{m+1}.$$

All forward and backward recurrence coefficients in this regular range are units. Thus equality at these two consecutive central indices propagates to every $0 \leq n < p$, proving the strong reflection statement. Applying it to b and c gives equality of both homogeneous coordinates, and hence the projective identity. $\square$

Lemma 697 (continuant reflection). *For every $m \geq 0$,*

$$U_m(-s - m) = (-1)^m U_m(s) \quad \text{in } \mathbb{Z}[s]. \quad (736)$$

Proof. $U_m(s)$ is the determinant of the length- m tridiagonal continuant with

$$\text{diagonal } P(s), \dots, P(s + m - 1), \quad \text{edge products } (s + 1)^6, \dots, (s + m - 1)^6.$$

Under $s \mapsto -s - m$, reversing the path reverses the edge products and sends each diagonal entry to its negative, because $P(-1 - u) = -P(u)$. Every determinant term uses $m - 2j$ diagonal entries for some j , so it acquires the common sign $(-1)^{m-2j} = (-1)^m$. This proves (736). $\square$

Corollary 698 (center vanishing). *If m is odd, then*

$$U_m(-m/2) = 0 \quad \text{over } \mathbb{Q}.$$

Proof. The point $s = -m/2$ is fixed by $s \mapsto -s - m$. Lemma 697 gives $U_m(-m/2) = -U_m(-m/2)$, and characteristic zero gives the conclusion. $\square$

Corollary 699 (root pairing). *Over every field of characteristic different from 2 in which U_m is nonzero, the multiset of roots of U_m is stable under $s \mapsto -s - m$. For odd m the fixed root $-m/2$ is the forced center; every other root occurs in a two-element orbit.*

Proof. Equation (736) preserves vanishing and multiplicity. The involution has the unique fixed point $-m/2$. $\square$

29.4 Renewal and adjacent coprimality

Set $G_0(s)$ equal to the identity matrix.

Theorem 700 (renewal identity). *For all $m, g \geq 0$,*

$$\begin{aligned} U_{m+g+1}(s) &= U_{m+1}(s)U_g(s+m+1) \\ &\quad - (s+m+1)^6 U_m(s)U_{g-1}(s+m+2). \end{aligned} \quad (737)$$

Proof. Split the matrix word at the restart edge: $G_{m+g+1}(s) = G_g(s+m+1)G_{m+1}(s)$. The lower row of the first factor is $(-U_{g-1}(s+m+2), U_g(s+m+1))$, while the right column of the second is $((s+m+1)^6 U_m(s), U_{m+1}(s))^\top$ by Lemma 693. Their scalar product is the lower-right entry $U_{m+g+1}(s)$ and is exactly (737). $\square$

Theorem 701 (adjacent coprimality). *For every $m \geq 0$,*

$$\gcd_{\mathbb{Q}[s]}(U_m, U_{m+1}) = 1.$$

Proof. The case $m = 0$ is immediate. Suppose the assertion holds one step earlier. A common irreducible divisor of U_m and U_{m+1} divides, by (731), $(s+m)^6 U_{m-1}$. It cannot divide U_{m-1} by induction, so it must be $s+m$. But the Apéry recurrence and the same initial values give $U_m(0) = (m!)^3 b_m$; reflection then gives

$$U_m(-m) = (-1)^m (m!)^3 b_m \neq 0$$

over $\mathbb{Q}$. Thus $s+m$ is not a divisor, completing the induction. $\square$

29.5 The unconditional orbit-energy theorem

For $1 \leq h \leq M-1$, write

$$C_h = \#\{1 \leq r \leq M-h : \pi(r) = \pi(r+h)\}.$$

The Casoratian gives $C_1 = 0$. For $h \geq 2$, Proposition 694, Lemma 11, and Lemma 12 give

$$C_h \leq \deg N_h = 3(h-1). \quad (738)$$

Theorem 37 supplies the sharper bordered-certificate nondegeneracy used elsewhere in the paper; the present argument needs only the pair-gap nonvanishing in (738).

Theorem 702 (orbit energy, ENERGY-3/2). *For every prime $p \geq 7$ and every integer $2 \leq H \leq M = p-2$,*

$$\mathcal{F}_p \leq \left\lceil \frac{M}{H} \right\rceil (M + 3(H-1)(H-2)). \quad (739)$$

Consequently

$$\boxed{\mathcal{F}_p \leq 2\sqrt{3} p^{3/2}}. \quad (740)$$

In particular $\mathcal{F}_p \ll p^{3/2}$ unconditionally.

Proof. Partition I_p into $J = \lceil M/H \rceil$ consecutive blocks of length at most H , and let $m_j(v)$ count visits to v in block j . For each v , Cauchy–Schwarz gives

$$m_p(v)^2 = \left(\sum_{j=1}^J m_j(v) \right)^2 \leq J \sum_{j=1}^J m_j(v)^2.$$

Within all blocks, the diagonal terms contribute M . An unordered off-diagonal collision in one block has a unique gap $2 \leq h \leq H-1$ and contributes twice to the squared multiplicity. Therefore

$$\sum_{j,v} m_j(v)^2 \leq M + 2 \sum_{h=2}^{H-1} C_h \leq M + 6 \sum_{q=1}^{H-2} q = M + 3(H-1)(H-2).$$

This proves (739). Notice that the factor 2 for the ordered pair is essential.

It remains to include the integer-rounding cost. If $M \geq 17$, take $H = \lceil \sqrt{M/3} \rceil$. Then $\sqrt{M/3} \leq H \leq \sqrt{M/3} + 1$ and $\lceil M/H \rceil \leq M/H + 1$. Expanding the right side of (739) and using these two inequalities gives

$$\begin{aligned} \mathcal{F}_p &\leq (M/H + 1)(M + 3H^2 - 9H + 6) \\ &\leq 2\sqrt{3} M^{3/2} - 4M + 8\sqrt{3M} + 9 \leq 2\sqrt{3} M^{3/2}. \end{aligned}$$

The last inequality holds for $M \geq 17$ (the left side of $4M - 8\sqrt{3M} - 9 \geq 0$ is already positive at $M = 17$ and increasing). Since $M < p$, this implies (740).

For $5 \leq M \leq 16$, the trivial bound $\mathcal{F}_p \leq M^2$ suffices: $M^2/(M+2)^{3/2}$ is increasing, and at $M = 16$ it is $256/18^{3/2} < 2\sqrt{3}$. These are exactly the remaining primes $7 \leq p \leq 18$. $\square$

Corollary 703 (position relative to the pointwise bound). *Theorem 702 is an energy theorem, not an improvement of the pointwise exponent $2/3$. More generally, an estimate $\mathcal{F}_p \ll p^\theta$ improves $|Z_p| \ll p^{2/3}$ through energy alone only when $\theta < 4/3$.*

Proof. Since $|Z_p| = m_p([0 : 1])$, one has $|Z_p|^2 \leq \mathcal{F}_p$. Thus energy gives exponent $\theta/2$, which is smaller than $2/3$ exactly when $\theta < 4/3$. $\square$

Proposition 704 (fixed-family Chebotarev mean). *Fix $H \geq 1$. Let ν_m be the number of distinct irreducible factors of U_m over $\mathbb{Q}$. Apart from the finitely many primes at which one of these polynomials has bad reduction,*

$$\lim_{X \rightarrow \infty} \frac{1}{\pi(X)} \sum_{p \leq X} \sum_{m=1}^H \#\{s \in \mathbb{F}_p : U_m(s) = 0\} = \sum_{m=1}^H \nu_m.$$

Proof. For one fixed squarefree polynomial, Frobenius fixes exactly as many roots as the corresponding Galois-group element fixes in its action on the roots. Chebotarev equidistributes the Frobenius conjugacy classes, and Burnside’s lemma says that the mean number of fixed roots is the number of Galois orbits, namely the number of irreducible factors. Apply this to the finitely many polynomials $U_1, \dots, U_H$ and sum. $\square$

Remark 705 (finite factor certificate). The exact computations recorded in ledger §§90 and 109 certify, for $m \leq 13$, that $\nu_m = 2$ for odd m and $\nu_m = 1$ for even m , after the shift $U_m(s) = N_{m+1}(s-1)$. This is a finite machine certificate, not an all- m irreducibility theorem.

Remark 706. Proposition 704 averages over primes with H fixed. It neither controls m growing like $\sqrt{p}$ at one prime nor implies a pointwise bound for every prime. That interchange of quantifiers is the family-compatibility problem.

29.6 The sigma ladder and the E1 program

The exact family form of energy is

$$\mathcal{F}_p = M + 2 \sum_{h=1}^{M-1} C_h. \quad (741)$$

Indeed, every unordered equal-color pair has one positive gap.

Conjecture 707 (linear orbit energy, E1). *There is an absolute constant C such that*

$$\mathcal{F}_p \leq Cp$$

for every prime $p \geq 7$.

By (741), E1 is equivalent to the linear family bound $\sum_h C_h \ll p$. This is the precise family-compatibility statement.

For $1 \leq \Delta \leq M-1$, define

$$d_\Delta(r) = \#\{1 \leq h \leq \Delta : r+h \leq M, \pi(r) = \pi(r+h)\}, \quad Q_p(\Delta) = \sum_r \binom{d_\Delta(r)}{2}.$$

Proposition 708 (the sigma exponent machine). *Fix $0 \leq \sigma \leq 2$. Suppose that for every $\varepsilon > 0$, uniformly for primes p and $1 \leq \Delta \leq M-1$,*

$$Q_p(\Delta) \ll_\varepsilon \Delta^{3-\sigma} p^\varepsilon. \quad (742)$$

Then

$$\mathcal{F}_p \ll_\varepsilon p^{2-1/(3-\sigma)+\varepsilon}. \quad (743)$$

Hence $\sigma = 0, 1, 2$ correspond respectively to the exponents $5/3, 3/2, 1$. Theorem 702 occupies the $\sigma = 1$ rung, although its direct pair-gap proof does not assert the stronger hypothesis (742). At $\sigma = 2$ the conclusion is the exponent-level form $\mathcal{F}_p \ll_\varepsilon p^{1+\varepsilon}$; the epsilon-free endpoint is Conjecture 707.

Proof. Put $S_p(\Delta) = \sum_r d_\Delta(r)$. Since $\sum_r d_\Delta(r)^2 = S_p(\Delta) + 2Q_p(\Delta)$, Cauchy–Schwarz gives

$$S_p(\Delta)^2 \leq M(S_p(\Delta) + 2Q_p(\Delta)), \quad S_p(\Delta) \leq M + \sqrt{2MQ_p(\Delta)}.$$

Partitioning I_p into $\lceil M/\Delta \rceil$ blocks and repeating the proof of Theorem 702 yields

$$\mathcal{F}_p \leq \left\lceil \frac{M}{\Delta} \right\rceil (M + 2S_p(\Delta)) \ll \frac{p^2}{\Delta} + p^{3/2+\varepsilon} \Delta^{(1-\sigma)/2} + p.$$

Taking Δ to be the nearest admissible integer to $p^{1/(3-\sigma)}$ balances the first two terms and proves (743) (after renaming ε). $\square$

We now state three concrete targets. They are all motivated by the same quenched, growing-gap family, but they are not mutually equivalent.

Conjecture 709 (Apéry triple return, ATR). *For $h \neq k$ put*

$$J_p(h, k) = \#\{1 \leq r \leq M - \max(h, k) : \pi(r) = \pi(r + h) = \pi(r + k)\}.$$

There is an absolute constant C such that, for every prime $p \geq 7$,

$$\sum_{\substack{1 \leq h, k \leq M-1 \\ h \neq k}} J_p(h, k) \leq Cp. \quad (744)$$

This is an open conjecture, supported numerically but not proved.

The implication to E1 is elementary. If $T_3 = \sum_v (m_p(v))_3$, then the nonwrapping ordered convention gives $3 \sum_{h \neq k} J_p(h, k) = T_3$. Moreover $(m)_2 \leq (m)_3$ for $m \geq 3$, while all fibres of size 2 contribute at most M to $\sum_v (m_p(v))_2$. Hence

$$\mathcal{F}_p \leq 2M + T_3 = 2M + 3 \sum_{h \neq k} J_p(h, k), \quad (745)$$

so Conjecture 709 implies E1.

Let $\kappa_h = 1$ when h is even and $\kappa_h = 0$ when h is odd. The pointwise reflection theorem supplies exactly this forced mirror collision.

Conjecture 710 (growing-gap projective return variance, GPRV). *There is an absolute constant C such that, for every prime $p \geq 7$,*

$$\sum_{h=1}^{p-3} \left| C_h - \kappa_h - \frac{p-2-h-\kappa_h}{p+1} \right|^2 \leq Cp. \quad (746)$$

This is an open variance conjecture for the growing family of primitive factors of the gap polynomials.

By Cauchy–Schwarz, (746) makes the sum of the absolute deviations $O(p)$. The forced and uniform baseline terms also sum to $O(p)$, so $\sum_h C_h = O(p)$; by (741), GPRV implies E1.

Finally put $v_n = (b_n, c_n)$ and $\Delta_{r,h} = \det(v_r, v_{r+h})$. For

$$\mathcal{I}_H = \{(r, h) : 1 \leq h \leq H, 1 \leq r \leq M - h\}, \quad e_p(x) = \exp(2\pi i x/p),$$

define the genuinely two-parameter bilinear sum

$$B_t(H) = \sum_{(r,h) \in \mathcal{I}_H} e_p(t \Delta_{r,h}).$$

Conjecture 711 (two-dimensional square-root cancellation, GAP–2D–SQRT). *For every $\varepsilon > 0$ there are $C_\varepsilon, p_\varepsilon$ such that, for every prime $p \geq p_\varepsilon$, every $1 \leq H \leq \sqrt{p}$, and every $t \in \mathbb{F}_p^\times$,*

$$|B_t(H)| \leq C_\varepsilon (pH)^{1/2} p^\varepsilon. \quad (747)$$

This is open; it asks for cancellation jointly in the base point and the growing gap.

Additive orthogonality gives the exact implication

$$\sum_{h \leq H} C_h = \frac{\#\mathcal{I}_H}{p} + \frac{1}{p} \sum_{t \neq 0} B_t(H) \ll_{\varepsilon} H + (pH)^{1/2} p^{\varepsilon}.$$

Thus GAP–2D–SQRT breaks the elementary H^2 small-gap barrier (at $H = \sqrt{p}$ it gives $p^{3/4+\varepsilon}$), but it does not by itself imply E1. This is why calling ATR, GPRV, and GAP–2D–SQRT “equivalent” would be incorrect: the last is a parallel partial target, and the first two are stronger sufficient criteria.

Remark 712 (numerical status only). The computations collected in `DOSSIER_E1_EMPIRICAL.md` report GPRV at the scale $\theta = 0$, with the normalized variance near 1.08 and no visible drift over the tested range. The dossier also reports agreement to four decimal places for the principal energy, spectral, return, and extreme-value laws of the mirror-random model. These are numerical data, not theorems, and none is used above.

29.7 The first splitting law and thin exceptional families

Let $R_2^{\text{alg}}(p) = \#\{r \in \mathbb{F}_p : N_2(r) = 0\}$ denote the full algebraic root count (not the restricted-window count of (734)).

Proposition 713 (the $h = 2$ splitting law). *For every prime $p \geq 7$,*

$$R_2^{\text{alg}}(p) = 1 + 2 \mathbf{1}_{\{(-51/p)=1\}}. \quad (748)$$

Proof. The dictionary gives

$$N_2(r) = P(r+1) = (2r+3)(17(r+1)^2 + 17(r+1) + 5).$$

The linear factor contributes one root. The quadratic has discriminant $17^2 - 4 \cdot 17 \cdot 5 = -51$, so it contributes two roots when $(-51/p) = 1$ and none when $(-51/p) = -1$. If $p \mid 51$, the only case in our range is $p = 17$; the quadratic specializes to the nonzero constant 5, and the displayed formula again gives one root because the Legendre symbol is zero. $\square$

Remark 714 (the same quadratic field in the annealed chain). The independently proved annealed two-step return probability is

$$\frac{2 + (-51/p)}{p-1}.$$

Thus the same field $\mathbb{Q}(\sqrt{-51})$ governs both the quenched $h = 2$ splitting law and the annealed two-step return. This is the first explicit warning that a proof of ATR, GPRV, or a related family estimate must stratify fixed-field, thin exceptional subfamilies before applying a generic random-family heuristic.

29.8 A fourth target: coincidence second moments of gap Casoratians

The three targets above either control every additive character twist (GAP–2D–SQRT), the full variance of the family $\{C_h\}$ (GPRV), or the triple returns (ATR). This subsection isolates a fourth target, strictly weaker than GAP–2D–SQRT at the same scale, carrying a scale dial β : any value $\beta > \frac{1}{2}$ already lowers the unconditional exponent $3/2$ of Theorem 702, and $\beta = 1$ recovers E1 with an explicit constant.

For $1 \leq h \leq M - 1$ and $1 \leq r \leq M - h$ put

$$\Delta_{r,h} = b_r c_{r+h} - b_{r+h} c_r \in \mathbb{F}_p,$$

the gap Casoratian of the pair of solutions at distance h .

Lemma 715 (Casoratian gap identity). *For every $1 \leq h \leq M - 1$ and $1 \leq r \leq M - h$,*

$$\Delta_{r,h} = \frac{N_h(r)}{\prod_{j=1}^h (r+j)^3} \quad (749)$$

in $\mathbb{F}_p$; every factor $r+j$ in the range $1 \leq j \leq h$ satisfies $2 \leq r+j \leq M < p$ and is invertible. In particular, on the nonwrapping window, $\Delta_{r,h} = 0$ if and only if $\pi(r) = \pi(r+h)$, and $\#\{r : \Delta_{r,h} = 0\} = C_h$.

Proof. Fix r and set $y_n = b_r c_n - c_r b_n$, a solution of the recurrence (728) in n , with $y_r = 0$ and, by Lemma 692, $y_{r+1} = b_r c_{r+1} - c_r b_{r+1} = -W_{r+1} = 1/(r+1)^3$. Let $\hat{y}$ be the normalized restart solution, $\hat{y}_r = 0$, $\hat{y}_{r+1} = 1$, so $y_n = (r+1)^{-3} \hat{y}_n$ by linearity. We claim

$$\hat{y}_{r+h} = \frac{N_h(r)}{\prod_{j=2}^h (r+j)^3} \quad (r+h \leq M), \quad (750)$$

which gives (749) after multiplying by $(r+1)^{-3}$. For $h = 1$, both sides are $1 = N_1$. For $h = 2$, the recurrence at $n = r+1$ reads $(r+2)^3 \hat{y}_{r+2} = P(r+1) \hat{y}_{r+1} - (r+1)^3 \hat{y}_r = P(r+1) = N_2(r)$. Assume (750) for $h-1$ and h . The recurrence at $n = r+h$, multiplied by $\prod_{j=2}^h (r+j)^3$, reads

$$(r+h+1)^3 \prod_{j=2}^h (r+j)^3 \hat{y}_{r+h+1} = P(r+h) N_h(r) - (r+h)^6 N_{h-1}(r) = N_{h+1}(r),$$

by the gap recurrence of Lemma 11; this is (750) for $h+1$. The zero characterization is the dictionary (733) together with the invertibility of the product. $\square$

For $1 \leq H \leq M - 1$ define the index set and the value counts

$$\mathcal{S}_H = \{(r, h) : 1 \leq h \leq H, 1 \leq r \leq M - h\}, \quad n_H(a) = \#\{(r, h) \in \mathcal{S}_H : \Delta_{r,h} = a\},$$

with $S := \#\mathcal{S}_H = \sum_{h \leq H} (M - h)$, and the coincidence second moment

$$\mathcal{N}(H) = \sum_{a \in \mathbb{F}_p} n_H(a)^2 = \#\{((r, h), (r', h')) \in \mathcal{S}_H^2 : \Delta_{r,h} = \Delta_{r',h'}\}.$$

Conjecture 716 (coincidence target, COINC_β). *Let $\frac{1}{2} < \beta \leq 1$. There is an absolute constant K such that, for every prime $p \geq 7$, with $H = \lceil p^\beta \rceil \wedge (M - 1)$,*

$$\mathcal{N}(H) \leq \frac{S^2}{p} + K S. \quad (751)$$

This is open for every $\beta > \frac{1}{2}$; it asserts that the values $\Delta_{r,h}$ equidistribute in second moment at scale H .

Lemma 717 (centered-count inequality). *Unconditionally, for every $1 \leq H \leq M - 1$,*

$$\sum_{h=1}^H C_h \leq \frac{S}{p} + \sqrt{\mathcal{N}(H) - \frac{S^2}{p}}. \quad (752)$$

Proof. By Lemma 715, $\sum_{h \leq H} C_h = n_H(0)$. Since $\sum_a n_H(a) = S$,

$$\left(n_H(0) - \frac{S}{p}\right)^2 \leq \sum_{a \in \mathbb{F}_p} \left(n_H(a) - \frac{S}{p}\right)^2 = \mathcal{N}(H) - \frac{S^2}{p}.$$

□

Theorem 718 (conditional energy exponents from coincidence counting). *Assume Conjecture 716 for one value $\frac{1}{2} < \beta \leq 1$ with constant K . Then for $\frac{1}{2} < \beta < 1$,*

$$\mathcal{F}_p \leq (1 + o(1)) p^{2-\beta},$$

and for $\beta = 1$,

$$\mathcal{F}_p \leq (3 + 2\sqrt{K}) p,$$

i.e. E1 holds. In particular any single $\beta > \frac{1}{2}$ strictly improves the unconditional exponent 3/2 of Theorem 702.

Proof. Let $H = \lceil p^\beta \rceil \wedge (M - 1)$. By Lemma 717 and (751), using $S \leq MH$,

$$\sum_{h \leq H} C_h \leq \frac{MH}{p} + \sqrt{KMH} \leq H + \sqrt{KpH}. \quad (753)$$

Partition I_p into $\lceil M/(H+1) \rceil$ consecutive blocks of length at most $H+1$; the proof of Theorem 702, verbatim with block length $H+1$ in place of H , gives

$$\mathcal{F}_p \leq \left\lceil \frac{M}{H+1} \right\rceil \left(M + 2 \sum_{h=1}^H C_h \right) \leq \left(\frac{M}{H} + 1 \right) \left(M + 2H + 2\sqrt{KpH} \right).$$

For $\frac{1}{2} < \beta < 1$ the right side is $p^{2-\beta} (1 + 2\sqrt{K} p^{(\beta-1)/2} + O(p^{\beta-1} + p^{-\beta} + p^{2\beta-2})) = (1 + o(1)) p^{2-\beta}$, since every correction exponent is negative for $\frac{1}{2} < \beta < 1$. For $\beta = 1$ one block suffices ($H = M - 1$): $\mathcal{F}_p \leq M + 2 \sum_{h \leq M-1} C_h \leq M + 2(M - 1) + 2\sqrt{Kp(M - 1)} \leq (3 + 2\sqrt{K})p$. □

Proposition 719 (position in the target hierarchy). *Fix a scale $1 \leq H \leq p^{1-\delta}$, $\delta > 0$. Then, with $B_t(H)$ as in Conjecture 711 extended to the index set $\mathcal{S}_H$,*

$$\sum_{t \neq 0} |B_t(H)|^2 = p \mathcal{N}(H) - S^2. \quad (754)$$

Consequently the square-root cancellation $|B_t(H)| \leq C(pH)^{1/2} p^\varepsilon$ for every $t \neq 0$ (GAP-2D-SQRT at scale H) implies (751) with $K = C^2 p^{2\varepsilon} (1 + o(1))$: the coincidence target is precisely the L^2 -average over t of GAP-2D-SQRT, and is therefore the weakest member of the sufficient family at its scale. It is not comparable to ATR or GPRV in either direction by any argument recorded in this paper. Two honesty clauses. First, the exact partition $\mathcal{N}(H) = S + A(A - 1) + R_{\neq 0}$, with $A = \sum_{h \leq H} C_h$ and $R_{\neq 0}$ the distinct-index coincidences at nonzero values, shows that the

centered variance contains the term $(A - S/p)^2$ as one nonnegative coordinate: Conjecture 716 is a reformulation with a scale dial and a weakest-sufficient position, not a reduction in difficulty — any proof of it must control the zero fiber at exactly the scale of (753). Second, E1 does not imply COINC_β and COINC_β at a truncated scale does not imply E1; the two are genuinely different statements about the same value distribution.

Proof. Orthogonality of additive characters gives $\sum_{t \in \mathbb{F}_p} |B_t|^2 = p \sum_a n_H(a)^2 = p\mathcal{N}(H)$, and the $t = 0$ term is S^2 . For the implication, sum the pointwise bound over $t \neq 0$: $p\mathcal{N}(H) - S^2 \leq (p-1)C^2 p H p^{2\varepsilon}$, and $S = \sum_{h \leq H} (M-h) \geq H(M-H) = (1-o(1))pH$ for $H \leq p^{1-\delta}$, so $\mathcal{N}(H) - S^2/p \leq C^2 p^{2\varepsilon} (1+o(1)) S$. $\square$

Remark 720 (why $\beta > \frac{1}{2}$ is the battle line). For $H \leq p^{1/2}$ the bound (753) gives $\sum_{h \leq H} C_h \ll (pH)^{1/2}$, which through the block argument reproduces the exponent $3/2$ and nothing better; this is the exact content of the crossover in Theorem 718. Moreover $(pH)^{1/2} \geq H^2$ exactly when $H \leq p^{1/3}$, so below $p^{1/3}$ the conclusion of (753) is weaker than the trivial degree bound $\sum_{h \leq H} C_h \leq \frac{3}{2}H^2$: partial progress on Conjecture 716 has content only for $\beta > \frac{1}{3}$, and improves the energy record only for $\beta > \frac{1}{2}$. Pairwise curve counting cannot reach even the first threshold: the difference correspondence $\{\Delta_{r,h} = \Delta_{r',h'}\}$ has bidegree $(3h, 3h')$, hence generic arithmetic genus $O(hh')$ and Weil error $O(hh'\sqrt{p})$ per pair, so the summed error $O(H^4\sqrt{p})$ fits the $O(pH)$ budget only for $H \leq p^{1/6}$ (or $H \leq p^{1/4}$ if the continuant structure forces a genus collapse to $O(h+h')$, which is not known). On the other side, for $H > p^{1/2}$ the degrees $3(h-1)$ exceed $\sqrt{p}$ and pointwise Weil bounds carry errors at least as large as the main terms. Every route through fixed-pair algebraic geometry is therefore dead in the regime where Conjecture 716 has content. This is one more independent arrival at the family-compatibility frontier of Remark 706.

Remark 721 (the exponent $3/2$ is the set-genericity threshold). The pairs (n, m) with $\pi(n) = \pi(m)$ are exactly the incidences between the point set $\{v_n\}$ and the line set $\{\text{span}(v_m)\}$ in $\mathbb{F}_p^2$, and Vinh's finite-field incidence theorem gives, for *any* M points and M lines, at most $M^2/p + p^{1/2}M$ incidences — with $M \sim p$ this is again $O(p^{3/2})$. Thus Theorem 702 attains precisely the bound that holds for a set-generic configuration, and *no* argument that uses the orbit only as a set of directions can pass $3/2$: E1 and every $\beta > \frac{1}{2}$ instance of Conjecture 716 are assertions about the arithmetic ordering of the sequence $n \mapsto v_n$, invisible to incidence geometry.

Structure of the difference correspondences

Write $\delta_h(x) = N_h(x)/\prod_{j=1}^h (x+j)^3$, viewed as a degree- $3h$ map $\mathbb{P}^1 \rightarrow \mathbb{P}^1$, so that $\Delta_{r,h} = \delta_h(r)$ on the window, and let $X_{h,k} \subset \mathbb{P}^1 \times \mathbb{P}^1$ be the closure of $\{\delta_h(x) = \delta_k(y)\}$, of bidegree $(3h, 3k)$.

Lemma 722 (exact addition law). *Extend $N_0 = 0$. For all $h \geq 1, d \geq 1$,*

$$N_{h+d}(x) = N_d(x+h) N_{h+1}(x) - (x+h+1)^6 N_{d-1}(x+h+1) N_h(x), \quad (755)$$

$$\delta_{h+d}(x) = (x+h+1)^3 (\delta_{h+1}(x) \delta_d(x+h) - \delta_h(x) \delta_{d-1}(x+h+1)). \quad (756)$$

Proof. In the transfer product $G_{m+g}(s) = G_g(s+m)G_m(s)$ of Lemma 693, take $s = x+1, m = h, g = d-1$, and read off the bottom-right entry: the second row of $G_{d-1}(x+h+1)$ is $(-N_{d-1}(x+h+1), N_d(x+h))$ and the second column of $G_h(x+1)$ is $((x+h+1)^6 N_h(x), N_{h+1}(x))^T$, giving (755); dividing by $\prod_{j=1}^{h+d} (x+j)^3 = \prod_{j=1}^h (x+j)^3 \cdot \prod_{j=1}^d (x+h+j)^3$ gives (756). $\square$

The law expresses a long-gap Casoratian as a bilinear combination of short-gap Casoratians at the two ends of the block; it is the exact self-similarity of the family, but it is a coupled identity at one base point, not a convolution of value distributions.

Lemma 723 (honest triple poles). *For $1 \leq j \leq h$,*

$$N_h(-j) = (-1)^{j-1}((j-1)!)^3 b_{j-1}((h-j)!)^3 b_{h-j} \neq 0,$$

so over $\mathbb{Q}$ every finite pole of δ_h has exact order 3, and δ_h vanishes to exact order 3 at $x = \infty$.

Proof. First, induction on m from the U -recurrence and the Apéry recurrence gives, for all $m \geq 1$,

$$U_{m-1}(1) = ((m-1)!)^3 b_{m-1}, \quad \text{i.e.} \quad N_m(0) = ((m-1)!)^3 b_{m-1}.$$

For $j = 1$ this is the claim after the shift $x \mapsto x - 1$ is undone. For $2 \leq j \leq h$, apply the addition law (755) with first length $j - 1$ and second length $h - j + 1$ at $x = -j$: the term carrying $N_{j-1}(-j) \cdot (\dots)$ survives while the complementary term vanishes, and the proved reflection identity $N_m(-m-1-x) = (-1)^{m-1} N_m(x)$ converts the reflected evaluation into $N_{j-1}(0)$ and $N_{h-j}(0)$ -type values, yielding the displayed sign and factorial–Apéry product. Nonvanishing over $\mathbb{Q}$ holds because the Apéry numbers are positive integers; since $\prod_{j'} (x+j')^3$ has an exact triple zero at $x = -j$, every finite pole of δ_h has exact order 3. At infinity, put $z = 1/x$: since $\deg N_h = 3h - 3$ with nonzero leading coefficient and $\deg \prod (x+j)^3 = 3h$, one has $\delta_h(z^{-1}) = z^3(\lambda + O(z))$ with $\lambda \neq 0$, an exact zero of order 3; in particular $\deg \delta_h = 3h$ and $\gcd(N_h, \prod (x+j)^3) = 1$ over $\mathbb{Q}$. (Machine-verified exactly for all $1 \leq j \leq h \leq 10$.) $\square$

Remark 724 (characteristic- p caveat). Lemma 723 is a statement over $\mathbb{Q}$. Modulo p , a pole order can drop when p divides $b_{j-1}b_{h-j}$ (or the leading coefficient), so any finite-field application must exclude or count these bad reductions; positivity over $\mathbb{Z}$ does not imply nonvanishing in $\mathbb{F}_p$.

Proposition 725 (generic genus of the difference correspondence). *Fix $h \neq k$ and assume the exact ramification profiles: over ∞ each of δ_h, δ_k has only the triple poles of Lemma 723; over 0 the fibers consist of the simple roots of the (squarefree) N_h, N_k together with the order-3 point at $x = \infty$; the remaining $4h - 4$ resp. $4k - 4$ critical points are simple, with critical values nonzero, pairwise distinct, and the two branch loci disjoint away from $\{0, \infty\}$; and $X_{h,k}$ is irreducible (all of this verified exactly for $2 \leq h < k \leq 16$). Then the normalization of $X_{h,k}$ has genus*

$$g = 6hk - 3h - 3k - 2.$$

In particular the Weil error per correspondence is of order $hk\sqrt{p}$, and summing over $h, k \leq H$ meets the $O(pH)$ budget of (751) only for $H \leq p^{1/6}$: pairwise point counting cannot reach the ranges where Conjecture 716 gains anything.

Proof. We use the local defect formula for fiber products: if points of ramification indices e and f lie over a common branch point, the normalized fiber product has $\gcd(e, f)$ branches there, each of ramification index $\text{lcm}(e, f)$, with total Riemann–Hurwitz defect $ef - \gcd(e, f)$. Over ∞ both fibers consist of order-3 points (h resp. k of them, by Lemma 723), contributing defect $6hk$; over 0 the fibers are $3(h-1)$ resp. $3(k-1)$ simple points plus one order-3 point at $x = \infty$ resp. $y = \infty$, contributing $6(h+k-1)$; each of the $4h-4$ simple branch points of δ_h away from $\{0, \infty\}$ contributes $3k$, and symmetrically, by branch-loci disjointness. Then $2 - 2g = 2(3h)(3k) - 6hk - 6(h+k-1) - (4h-4) \cdot 3k - (4k-4) \cdot 3h$ gives the displayed g . $\square$

Theorem 726 (component dichotomy under full monodromy). *Let MON_h denote the statement that the monodromy group of $\delta_h : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ over $\overline{\mathbb{Q}}$ is the full symmetric group S_{3h} . Assume MON_h and MON_k .*

1. *If $h \neq k$, then $X_{h,k}$ is irreducible over $\overline{\mathbb{Q}}$.*

2. If $h = k \geq 2$, then $X_{h,h}$ has exactly two irreducible components over $\overline{\mathbb{Q}}$: the diagonal $x = y$ and one further component H_h .

Consequently, under MON_h for all $2 \leq h \leq H$, the total component excess is $\sum_{h,k \leq H} (c_{h,k} - 1) = H + 1$ (the diagonal of each same-gap curve for $2 \leq h \leq H$, plus the two extra lines of the special case $h = k = 1$), exactly the allowance of Conjecture 716. These are statements over $\overline{\mathbb{Q}}$; for any fixed pair they transfer to $\mathbb{F}_p$ for all p outside a computable finite set by standard spreading out, but no uniform bad-reduction control in the growing range $h, k \leq p^\beta$ is asserted here — that uniformity is exactly the hypothesis $p > p_0(H)$ of the census.

Proof. Work over the base $\mathbb{P}^1$ punctured at the union of the two branch loci; components of $X_{h,k}$ over $\overline{\mathbb{Q}}$ correspond to orbits of the joint monodromy group $G \leq S_{3h} \times S_{3k}$ (the image of the fundamental group of this common punctured base acting on pairs of sheets) on $\{1, \dots, 3h\} \times \{1, \dots, 3k\}$. By Goursat, G is the fiber product of the two full symmetric images over a common quotient; for $3h, 3k \geq 6$ the proper common quotients of S_{3h} and S_{3k} are trivial or $\mathbb{Z}/2$, so G contains $A_{3h} \times A_{3k}$ (the low-degree case $3h = 3$, i.e. $h = 1$, is excluded and handled separately). For $h \neq k$ each alternating factor is transitive, so G is transitive on the product: one orbit, one component. For $h = k$ the fiber product of the cover with itself has joint monodromy exactly the diagonally embedded $\text{Mon}(\delta_h) = S_{3h}$ acting on ordered pairs of sheets, with exactly two orbits (diagonal and off-diagonal) by 2-transitivity; the off-diagonal orbit yields the single residual component H_h , no quotient by the factor swap being involved. $\square$

Remark 727 (status of MON_h). The component census was computed exactly for all pairs $h, k \leq 8$: every distinct-gap $F_{h,k}$ is irreducible over $\mathbb{Q}$, every same-gap curve splits as diagonal times one irreducible factor of bidegree $(3h-1, 3h-1)$, there is no mirror component (the proved reflection acts by $\delta_h \mapsto -\delta_h$, an anti-symmetry), and the boundary case $h = k = 1$ degenerates to the two cube-root-of-unity lines — exactly as Theorem 726 predicts, since $\delta_1(x) = (x+1)^{-3}$ is a perfect cube (imprimitive monodromy $\mathbb{Z}/3$) while no δ_h with $h \geq 2$ can factor through a power map (N_h is squarefree). Toward MON_h in general: local monodromy at every pole is a 3-cycle (Lemma 723); a simple finite critical point supplies a transposition; the remaining input is primitivity. Writing s_h for the number of *simple nonzero* critical values of δ_h , the condition $\text{CRIT}(h): s_h \geq 2h - 1$ rules out every decomposition $\delta_h = g \circ f$ (a Ritt-type count: a decomposition caps the number of simple critical values), and $\text{CRIT}(h) \Rightarrow \text{MON}_h$; exact certificates at two good primes prove the full-Morse strengthening (all $4h - 4$ critical values simple and nonzero, pairwise- h coprime branch loci) for every $2 \leq h \leq 16$, so MON_h and the component dichotomy are *unconditional* for $h \leq 16$. The named residual statements are $[\text{CRIT-2H}]: s_h \geq 2h - 1$ for all h , and $[\text{BD-ALL}]: \text{Res}_T(C_h^*, C_k^*) \neq 0$ for all $h < k$. The critical-value resultants $C_h(T) = \text{Res}_x(N_h - TL_h^3, L_h N_h' - 3L_h' N_h)$ are pairwise coprime for $h \leq 8$ (exact computation), and the proved reflection anti-symmetry forces the exact evenness $C_h(T) = D_h(T^2)$ with $\deg D_h = 2h - 2$: critical values come in $\pm$ -pairs. A sufficient input for primitivity is that D_h be irreducible over $\mathbb{Q}$ with a non-square root; this is open for every h (simple Newton-polygon arguments at 2, 3, 5, 17 provably do not suffice). The entire component theory of the family thus rests on the Galois structure of one explicit one-variable family.

Remark 728 (the surviving mechanism: shell variance). Every fixed-pair geometric route is now closed by Proposition 725; one van der Corput step applied to (756) does not shorten the transfer length (the law couples the two blocks at one base point), and block-glueing by Cauchy-Schwarz alone cannot pass the trivial degree scale. What remains is the *shell variance*. For $1 \leq h, k \leq H$ let $E_{h,k} = \#\{(r, r') \in I_h \times I_k : \Delta_{r,h} = \Delta_{r',k}\} - |I_h||I_k|/p - [\text{classified component main terms}]$ denote the centered coincidence correlation (unordered gaps, nonwrapping windows), and $A_d(H) = \sum_{h \leq H-d} E_{h,h+d}$ the shell sums. Since $\mathcal{N}(H) - S^2/p = \sum_{h,k} E_{h,k} + [\text{main terms}] = A_0 + 2 \sum_{1 \leq d \leq H-1} A_d +$

$O(S)$, a bound $|A_d(H)| \ll pH^{1+o(1)}/d$ for *every* $1 \leq d \leq H-1$ (all shells, not an initial segment) at $H = \lceil p^{2/3} \rceil$ implies Conjecture 716 at $\beta = 2/3$ up to p^ε , hence $\mathcal{F}_p \ll p^{4/3+\varepsilon}$. The structural theorems of this subsection do not prove shell variance; it is the precise family-compatibility object to which all campaign routes have converged.

Remark 729 (numerical status only). For $p \in \{499, 997, 1999, 4001, 7919\}$ and $H = \lceil p^{0.66} \rceil$, the measured value of $(\mathcal{N}(H) - S^2/p)/S$ is 1.24, 1.08, 1.08, 1.08, 1.05: bounded and slowly decreasing in p , consistent with Conjecture 716 at $\beta = 2/3$ with K close to 1. Identity (749) was verified exactly at 300 random pairs (r, h) for each of the five primes. These are numerical data, not theorems, and none is used above.

30 The $\sigma = \frac{1}{2}$ factorial-moment theorem and its interfaces

Throughout this section we keep the conventions of the orbit-energy section (Section 29): $p \geq 7$ is prime, $I_p = \{1, \dots, p-2\}$, $M = |I_p| = p-2$, and $\pi(n) = [b_n : c_n] \in \mathbb{P}^1(\mathbb{F}_p)$ is the projective Apéry orbit. For $h \geq 1$ and $r, r+h$ in the regular range, the gap polynomial N_h detects collisions: $\pi(r) = \pi(r+h)$ if and only if $N_h(r) = 0$, and

$$\deg_r N_h = 3(h-1). \quad (757)$$

30.1 The $\sigma = \frac{1}{2}$ theorem: $Q_p(\Delta) \leq \frac{27}{4}H^{5/2}$

For $r \in I_p$ and $h \geq 1$ put $I(r, h) = 1$ if $r+h \leq M$ and $\pi(r) = \pi(r+h)$, and $I(r, h) = 0$ otherwise. For $\Delta \geq 1$ define the window collision multiplicity and its second factorial moment

$$d_\Delta(r) = \#\{1 \leq h \leq \Delta : r+h \leq M, \pi(r) = \pi(r+h)\} = \sum_{1 \leq h \leq \Delta} I(r, h),$$

$$Q_p(\Delta) = \sum_{r \in I_p} \binom{d_\Delta(r)}{2},$$

together with the one-gap counts and their partial sums

$$R_h = \sum_{r \in I_p} I(r, h), \quad S_1(Y) = \sum_{1 \leq d \leq Y} R_d.$$

By (757), the banked one-gap estimate reads

$$R_d \leq \deg_r N_d = 3(d-1) \quad (1 \leq d \leq p-3). \quad (758)$$

Estimate (758) is the *only* Apéry-specific input of this subsection; see Remark 736.

Since $r \geq 1$ and $r+h \leq M = p-2$ force $h \leq p-3$, gaps larger than $p-3$ never occur. Hence, with

$$H := \min(\Delta, p-3),$$

we have $d_\Delta(r) = d_H(r)$ for every r and $Q_p(\Delta) = Q_p(H)$; it suffices to treat $1 \leq H \leq p-3$. The first moment is exact:

$$S := \sum_{r \in I_p} d_H(r) = \sum_{h=1}^H R_h = S_1(H) \leq 3 \sum_{h=1}^H (h-1) = \frac{3}{2}H(H-1) \leq \frac{3}{2}H^2. \quad (759)$$

Fix an integer box length D with $1 \leq D \leq H$ and partition $[1, H]$ into consecutive boxes

$$B_j = \{(j-1)D + 1, \dots, \min(jD, H)\}, \quad 1 \leq j \leq q, \quad q = \left\lceil \frac{H}{D} \right\rceil,$$

the last box possibly shorter; no divisibility assumption $D \mid H$ is made. For each base r set

$$a_j(r) = \#\{h \in B_j : I(r, h) = 1\}, \quad L_r = \sum_{j=1}^q \binom{a_j(r)}{2}, \quad L = \sum_{r \in I_p} L_r.$$

Lemma 730 (per-base box inequality). *For every $r \in I_p$,*

$$\binom{d_H(r)}{2} \leq q L_r + \frac{q-1}{2} d_H(r). \quad (760)$$

Consequently, summing over $r \in I_p$,

$$Q_p(H) \leq q L + \frac{q-1}{2} S. \quad (761)$$

Proof. Since the boxes partition $[1, H]$ we have $d_H(r) = \sum_j a_j(r)$, and from $a^2 = 2\binom{a}{2} + a$,

$$\sum_{j=1}^q a_j(r)^2 = \sum_{j=1}^q \left(2\binom{a_j(r)}{2} + a_j(r) \right) = 2L_r + d_H(r).$$

By the Cauchy–Schwarz inequality applied to the q boxes,

$$d_H(r)^2 = \left(\sum_{j=1}^q a_j(r) \right)^2 \leq q \sum_{j=1}^q a_j(r)^2 = q(2L_r + d_H(r)).$$

Hence $d_H(r)^2 - d_H(r) \leq 2qL_r + (q-1)d_H(r)$, which is (760) after division by 2. The short final box is already included in the definition of $a_j(r)$; there is no endpoint issue. Summing (760) over r gives (761). $\square$

Lemma 731 (triple-collision injection). *For $1 \leq h < k \leq H$ let*

$$T(h, k) = \#\{r \in I_p : r + k \leq M, \pi(r) = \pi(r + h) = \pi(r + k)\}.$$

Then:

$$(a) \quad L = \sum_{j=1}^q \sum_{\substack{h < k \\ h, k \in B_j}} T(h, k);$$

(b) *for every fixed pair $h < k$, with $d = k - h$,*

$$T(h, k) \leq R_{k-h}; \quad (762)$$

$$(c) \quad L \leq \sum_{d=1}^{D-1} \mathcal{B}_d R_d \leq H S_1(D-1), \text{ where } \mathcal{B}_d = \#\{(h, k) : h < k, k - h = d, h, k \in \text{one box}\} \leq H.$$

Proof. (a) For fixed r and j , $\binom{a_j(r)}{2}$ counts the unordered pairs $h < k$ with $h, k \in B_j$ and $I(r, h) = I(r, k) = 1$; since $k > h$, the condition $I(r, h) = I(r, k) = 1$ says exactly that $r + k \leq M$ and $\pi(r) = \pi(r + h) = \pi(r + k)$, i.e. that r is counted by $T(h, k)$. Summing over r and j gives the identity. Note that L counts the joint objects $(r, \{h, k\})$ with multiplicity across different pairs; this convention is used in (b) below.

(b) Fix $h < k$ and put $d = k - h \geq 1$. For every r counted by $T(h, k)$ set $u = r + h$. Then $u \in I_p$, and

$$u + d = r + k \leq M, \quad \pi(u) = \pi(r + h) = \pi(r + k) = \pi(u + d)$$

by transitivity of equality, so u is a valid base of a gap- d collision, i.e. $I(u, d) = 1$. For this fixed pair (h, k) the map $r \mapsto u = r + h$ is injective (u determines $r = u - h$). Hence $T(h, k) \leq R_d$. The same translated base u may arise from different pairs (h, k) ; this is harmless, because (a) counts $(r, \{h, k\})$ with multiplicity and (762) is applied separately to each fixed pair before the bounds are added. No disjointness of the images across pairs is asserted or needed. The nonwrapping boundary is preserved automatically: $r + k \leq M$ implies $u + d \leq M$, so no admissible base is lost and none is newly introduced.

(c) A same-box pair $h < k$ with $k - h = d$ exists only for $1 \leq d \leq D - 1$, since each box has length at most D . If the box lengths are $\ell_1, \dots, \ell_q$, then exactly $\mathcal{B}_d = \sum_{j=1}^q \max(\ell_j - d, 0)$; in particular $\mathcal{B}_d \leq H$, because such a pair is determined by its first member $h \in [1, H]$. (The incomplete final box only decreases these counts.) Combining (a), (b) and $\mathcal{B}_d \leq H$,

$$L \leq \sum_{d=1}^{D-1} \mathcal{B}_d R_d \leq H \sum_{d=1}^{D-1} R_d = H S_1(D-1). \quad \square$$

The next lemma is the reusable interface of the method: it is stated before any degree estimate is inserted, and any future short-range improvement of S_1 plugs into it directly (Remark 733).

Lemma 732 (master inequality). *For every integer $1 \leq D \leq H$, with $q = \lceil H/D \rceil$,*

$$Q_p(H) \leq q H S_1(D-1) + \frac{q-1}{2} S_1(H). \quad (763)$$

Proof. Insert Lemma 731(c) and the exact first-moment identity $S = S_1(H)$ from (759) into (761). $\square$

Remark 733 (the interface for short-range improvements). Only the aggregated one-gap data S_1 enters (763). Suppose the short range is improved to $S_1(Y) \ll Y^{2-\eta}$ for $Y \leq D$ and some $\eta > 0$, while the long range retains the degree bound $S_1(H) \ll H^2$. Then (763) gives

$$Q_p(H) \ll H^2 D^{1-\eta} + \frac{H^3}{D},$$

and the choice $D = H^{1/(2-\eta)}$ yields

$$Q_p(H) \ll H^{3-\frac{1}{2-\eta}}.$$

Checks: $\eta = 0$ gives the exponent $\frac{5}{2}$; $\eta = \frac{1}{2}$ gives $\frac{7}{3}$; $\eta = 1$ gives 2. If the improvement also holds at the long scale H , it may be inserted in the second term of (763) as well, with a stronger optimization. A more precise weighted form is also available: by the exact count $\mathcal{B}_d = \sum_j \max(\ell_j - d, 0) \leq \min(H, q(D-d))$ in Lemma 731(c),

$$Q_p(H) \leq q \sum_{d=1}^{D-1} \mathcal{B}_d R_d + \frac{q-1}{2} S_1(H) \leq q^2 \sum_{d=1}^{D-1} (D-d) R_d + \frac{q-1}{2} S_1(H).$$

Theorem 734 ($\sigma = \frac{1}{2}$; explicit constant). *For every prime $p \geq 7$ and every $\Delta \geq 1$, with $H = \min(\Delta, p-3)$,*

$$Q_p(\Delta) \leq \frac{27}{4} H^{5/2}.$$

In particular $Q_p(\Delta) \ll \Delta^{5/2}$ uniformly in p and Δ , with an absolute constant and no factor p^ε . In the σ -ladder normalization $Q_p(\Delta) \ll \Delta^{3-\sigma}$ this is $\sigma = \frac{1}{2}$.

Proof. As noted above, $Q_p(\Delta) = Q_p(H)$ and we may assume $1 \leq H \leq p-3$. By (758),

$$S_1(D-1) = \sum_{d=1}^{D-1} R_d \leq 3 \sum_{d=1}^{D-1} (d-1) = \frac{3}{2}(D-1)(D-2) \leq \frac{3}{2}D^2, \quad S_1(H) \leq \frac{3}{2}H^2,$$

so the master inequality (763) gives

$$Q_p(H) \leq \frac{3}{2} q H D^2 + \frac{3}{4} (q-1) H^2. \quad (764)$$

Choose the integer box length

$$D = \lceil \sqrt{H} \rceil.$$

For $H \geq 1$ this satisfies

$$\sqrt{H} \leq D \leq 2\sqrt{H}, \quad D \leq H, \quad q = \left\lceil \frac{H}{D} \right\rceil \leq \frac{2H}{D}, \quad q-1 < \frac{H}{D}.$$

Therefore

$$\frac{3}{2} q H D^2 \leq 3H^2 D \leq 6H^{5/2}, \quad \frac{3}{4} (q-1) H^2 < \frac{3}{4} \frac{H^3}{D} \leq \frac{3}{4} H^{5/2},$$

and (764) yields $Q_p(H) \leq 6H^{5/2} + \frac{3}{4}H^{5/2} = \frac{27}{4}H^{5/2}$. $\square$

Remark 735 (superseded upper bound). The exponent 5/2 of Theorem 734 has since been superseded: the abstract box method sharpens, for an arbitrary word with $R_d \leq Cd$, to the optimal-exponent bound $Q_H \leq 22 C H^2(1 + \log H)$ ([THM-ABSTRACT-Q-H2LOGH]; monochromatic-triple recount plus an inverse-square clique energy lemma, with an explicit word showing the exponent 2 cannot be improved). Theorem 734 and Lemma 732 are retained as the reusable box/master interface; the sigma ladder is thereby combinatorially complete up to the logarithm, and the remaining wall at exponent 3/2 is the mesoscopic normalization $Q_D \ll N$ at $D = \sqrt{N} L$, where the purely combinatorial bound leaves exactly a factor $L^2 \log N$.

Remark 736 (abstract box lemma; the only Apéry input). After the one-gap estimates are supplied, the proof above never uses the Apéry recurrence again. It therefore establishes the following abstract statement.

Abstract box lemma. Let $x_0, \dots, x_N$ be an arbitrary finite word over an arbitrary set. Define R_d , $d_H(r)$ and Q_H by equality of word values, with nonwrapping indices. If $R_d \leq Cd$ for every $1 \leq d \leq H$, then $Q_H \ll_C H^{5/2}$.

The only Apéry-specific input of Theorem 734 is the gap-polynomial degree bound (758), $R_d \leq \deg_r N_d = 3(d-1)$; with the coarser input $R_d \leq 3d$ the same proof gives an absolute constant such as 8. Consequently no artificial word satisfying the one-gap profile can be a counterexample.

By contrast, the aggregate first moment alone is insufficient. Keep only $S = \sum_{d \leq H} R_d = O(H^2)$ and drop the individual short-gap bounds: the constant word $x_0 = x_1 = \dots = x_H$ has

$$R_d = H + 1 - d, \quad S = \sum_{d=1}^H R_d = \frac{H(H+1)}{2} = O(H^2),$$

yet

$$d_H(r) = H - r, \quad Q_H = \sum_{r=0}^H \binom{H-r}{2} = \binom{H+1}{3} \asymp H^3.$$

Thus first-moment input alone cannot give any exponent below 3. The constant word violates the actual hypothesis at the shortest gap ($R_1 = H$, whereas the Apéry bound gives $R_1 = 0$ and in general $R_d \leq 3d$). This identifies the injection Lemma 731, not the Cauchy step of Lemma 730, as the genuinely new input.

Remark 737 (placement in the ladder). The previous record came from the pointwise window bound $\max_r d_H(r) \leq 3H^{2/3}$ together with the first moment (759):

$$Q_p(H) \leq \frac{1}{2} \left(\max_r d_H(r) \right) \sum_r d_H(r) \leq \frac{9}{4} H^{8/3} \leq 3H^{8/3}.$$

Theorem 734 is stronger by the factor $H^{8/3-5/2} = H^{1/6}$, and both bounds may be retained as $Q_p(H) \ll \min(H^{5/2}, H^{8/3})$. Fed into the σ -machine of Section 29, $\sigma = \frac{1}{2}$ yields the energy exponent $p^{2-1/(5/2)} = p^{8/5}$, which does *not* beat the direct unconditional $p^{3/2}$ energy theorem; the value of Theorem 734 is the nontrivial local factorial-moment saving and the reusable interface (763). Any further improvement within the box method must improve at least one of: $S_1(D)$, $S_1(H)$, the bound $T(h, k) \leq R_{k-h}$, or the Cauchy cross-box loss q . Merely changing box endpoints, or treating the forced mirror root separately (it is already included in $R_d \leq \deg N_d$), cannot improve the exponent $\frac{5}{2}$.

30.2 Interpolation, internal barriers, and the triple-gap dictionary

Let $(m_v)_{v \in \mathbb{P}^1(\mathbb{F}_p)}$ be the fibre multiplicities of the projective orbit over an index window of length $N \leq p$ (for the window I_p of this paper, $m_v = m_p(v)$ and $N = p - 2$). Write

$$E = \sum_v m_v^2, \quad T_3 = \sum_v (m_v)_3, \quad (m)_3 = m(m-1)(m-2),$$

so that in the conventions of Section 29, E is the full energy $\mathcal{F}_p$ when the window is I_p .

Theorem 738 (triple-collision interpolation). *Assume*

$$T_3 \leq p^\tau, \quad 1 \leq \tau < 2.$$

Then

$$E \leq \frac{37}{3} p^{(1+\tau)/2}.$$

In particular, if $T_3 \leq p^{2-\delta}$ for some $0 < \delta \leq 1$, then $E \leq \frac{37}{3} p^{3/2-\delta/2}$.

Proof. From $m^3 = (m)_3 + 3m^2 - 2m$ and $\sum_v m_v = N \leq p$,

$$\sum_v m_v^3 = T_3 + 3E - 2N \leq T_3 + 3E.$$

Set the threshold

$$M_0 = 6p^{(\tau-1)/2}.$$

(The factor 6 is what makes the absorption below uniform in τ , including the endpoint $\tau = 1$, where the unnormalized threshold $p^{(\tau-1)/2} = 1$ would fail to absorb the high-energy term.) Split $E = E_{\text{low}} + E_{\text{high}}$, where $E_{\text{low}} = \sum_{m_v \leq M_0} m_v^2$ and $E_{\text{high}} = \sum_{m_v > M_0} m_v^2$.

For the low part,

$$E_{\text{low}} \leq M_0 \sum_v m_v = M_0 N \leq 6p^{(\tau-1)/2} \cdot p = 6p^{(1+\tau)/2}.$$

For the high part, $m_v^2 \leq M_0^{-1} m_v^3$ whenever $m_v > M_0$, so

$$E_{\text{high}} \leq M_0^{-1} \sum_v m_v^3 \leq M_0^{-1} T_3 + 3M_0^{-1} E.$$

Here

$$M_0^{-1} T_3 \leq \frac{1}{6} p^{-(\tau-1)/2} \cdot p^\tau = \frac{1}{6} p^{(1+\tau)/2},$$

and, since $\tau \geq 1$,

$$3M_0^{-1} = \frac{1}{2} p^{-(\tau-1)/2} \leq \frac{1}{2}.$$

Adding the two parts,

$$E \leq 6p^{(1+\tau)/2} + \frac{1}{6} p^{(1+\tau)/2} + \frac{1}{2} E,$$

so $\frac{1}{2} E \leq \frac{37}{6} p^{(1+\tau)/2}$, i.e. $E \leq \frac{37}{3} p^{(1+\tau)/2}$. The final assertion is the case $\tau = 2 - \delta$, since $(1 + \tau)/2 = \frac{3}{2} - \frac{\delta}{2}$. $\square$

Remark 739 (block-method barrier: three internal no-go statements). The existing short-gap block method cannot produce a power saving below the $\frac{3}{2}$ energy scale without a new input. The obstructions are the following.

(i) The block decomposition gives the generic form

$$E \leq O(pL) + O\left(\frac{p^2}{L}\right),$$

in which the second term is the long-gap contribution. Improving the short-gap contribution requires an estimate of the shape

$$S_p(L) \ll L^{2-\epsilon}$$

for the short-gap collision count inside an interval of length L . No such estimate is currently proved for the Apéry projective orbit.

(ii) The continuant estimate gives only degree control of individual gaps,

$$\#\{r : N_h(r) = 0\} \ll h,$$

and no cancellation *between* different gaps.

(iii) The hybrid single-moment attempt fails: a bound of the shape $T_3 \leq ME$ closes only if the exponents satisfy $\mu > \theta$, where $E \ll p^\theta$ and $T_3 \ll p^\mu$; existing bounds do not create this contradiction.

(iv) Reflection symmetry does not create an independent square-root gain. The Möbius involution pairs collision points and doubles the contribution, but the paired collision is algebraically the same event; there is no independent Bernoulli factor.

The open input identified by (i)–(iv) is exactly the triple-gap statement of Remark 742.

Remark 740 (modular parameterization: a death certificate). The Beukers modular parameterization does not presently convert Apéry orbit collisions into CM/Heegner point counting problems. Three independent category mismatches occur.

- (i) *Index mismatch.* The parameter n in $B(t) = \sum b_n t^n$ is a Taylor coefficient index at a maximally unipotent boundary point; it is not a modular point τ . The parameterization gives $t = t(\tau)$ and identifies the generating function with a period; it does not identify the coefficient index n with a point of $X_0(6)^+$.
- (ii) *Collision mismatch.* The collision $\pi(r) = \pi(r + h)$ is equality of two finite-field jet states of a recurrence. A Heegner condition is a statement about a CM elliptic curve or an isogeny class; equality of two Taylor-jet states is not an isogeny relation. Therefore $N_h(r) = 0$ does not presently define a Heegner divisor or special cycle.
- (iii) *Reduction mismatch.* The modular parameterization is a characteristic-zero period identity. Reduction modulo p does not automatically identify the $p - 2$ recurrence states with points of $X_0(6)^+(\mathbb{F}_p)$.

The missing bridge would be: *a Frobenius-compatible integral mod- p realization of the Apéry projective orbit as a modular correspondence, such that $\pi(r) = \pi(r + h)$ is equivalent to membership in an explicitly defined special cycle.* No such theorem is known; the route is dead unless such a Frobenius-modular interface is proved.

Lemma 741 (triple-gap dictionary). *Adopt the nonwrapping convention: for distinct integers $h, k \geq 1$ let*

$$J(h, k) = \#\{r \in I_p : r + \max(h, k) \leq M, \pi(r) = \pi(r + h) = \pi(r + k)\},$$

so that $J(h, k)$ counts the simultaneous solutions of the two gap equations $N_h(r) = N_k(r) = 0$. With T_3 computed over the window I_p ,

$$\sum_{h \neq k} J(h, k) = \frac{T_3}{3}.$$

Proof. $T_3 = \sum_v m_v(m_v - 1)(m_v - 2)$ is the number of ordered triples (n_1, n_2, n_3) of pairwise distinct indices in I_p with $\pi(n_1) = \pi(n_2) = \pi(n_3)$; each unordered triple $\{a < b < c\}$ with a common π -value contributes exactly 6 such ordered triples. On the other side, since the gaps are positive, an r counted by $J(h, k)$ determines the unordered triple $\{r, r + h, r + k\}$ with minimum r ; conversely, a triple $\{a < b < c\}$ with common π -value arises exactly from the base $r = a$ with the two ordered gap pairs $(h, k) \in \{(b - a, c - a), (c - a, b - a)\}$, i.e. it contributes exactly 2 to $\sum_{h \neq k} J(h, k)$. Hence $\sum_{h \neq k} J(h, k) = 2 \cdot \frac{T_3}{6} = \frac{T_3}{3}$. $\square$

Remark 742 (one missing theorem in three strengths). By Lemma 741, a bound on T_3 is exactly a two-gap intersection statement: the intersections $\{N_h = 0\} \cap \{N_k = 0\}$ must have the expected codimension-two size. The following are successive strengths of the same missing horizontal distribution theorem:

$$[\text{ATR}] \quad T_3 \ll p; \quad [\text{T3-2MINUS}] \quad T_3 \ll p^{2-\delta}; \quad [\text{current}] \quad T_3 \ll p^{13/6}.$$

Through Theorem 738, [T3-2MINUS] gives $E \ll p^{3/2-\delta/2}$, beating the unconditional $p^{3/2}$ energy theorem, and [ATR] gives $E \ll p$. Geometrically the missing input is a transversality theorem for

the two gap correspondences: the two equations are not independent polynomial equations in an arbitrary plane; they arise from one Apéry orbit, so the required statement is a horizontal three-point distribution theorem. The least secure step in any upgrade is the passage from the recurrence correspondence $\{N_h = 0\} \cap \{N_k = 0\}$ to a genuine codimension-two equidistribution statement: existing algebraic geometry controls fixed varieties, whereas here the horizontal parameter is the recurrence index itself.

30.3 Spectral interfaces: L^2 frequency mass, square-root cancellation, and the mixed-return target

Write $e_p(x) = e^{2\pi i x/p}$. For $h \geq 1$ put $Q_h(x) = (x+1)(x+2)\cdots(x+h)$ and, for x with $Q_h(x) \neq 0$, define the normalized gap function

$$D_h(x) = b_x c_{x+h} - b_{x+h} c_x = \frac{N_h(x)}{Q_h(x)^3}, \quad (765)$$

the banked h -step Casoratian identity of Section 29; by linear independence of the two solution vectors, $\pi(x) = \pi(x+h)$ if and only if $D_h(x) = 0$. Define the collision polynomial

$$G_h(x, y) = \frac{N_h(x)Q_h(y)^3 - N_h(y)Q_h(x)^3}{x - y} \in \mathbb{F}_p[x, y], \quad (766)$$

a polynomial of total degree at most $6h-4$ (the numerator vanishes on the diagonal, so the quotient is a polynomial), whose zero set carries the off-diagonal collisions $D_h(x) = D_h(y)$.

Proposition 743 (L^2 frequency mass). *Let $W \subseteq \{1, \dots, p-3\}$ be a finite set of gaps and, for each $h \in W$, let $A_h \subseteq \{x \in \mathbb{F}_p : Q_h(x) \neq 0\}$ be a set of admissible bases. Put*

$$F(t, \xi) = \sum_{h \in W} \sum_{r \in A_h} e_p(tD_h(r) + \xi h),$$

$$\text{Collision}(h) = \#\{(r, r') \in A_h \times A_h : D_h(r) = D_h(r')\}.$$

(a) *One has the exact Parseval identity*

$$\sum_{t \in \mathbb{F}_p} \sum_{\xi \in \mathbb{F}_p} |F(t, \xi)|^2 = p^2 \sum_{h \in W} \text{Collision}(h).$$

(b) *Let $A_h = \{x \in \mathbb{F}_p : Q_h(x) \neq 0\}$ be maximal, and assume $4h < p$. If G_h is absolutely irreducible over $\mathbb{F}_p$, then*

$$\text{Collision}(h) = 2p + O(h^2 \sqrt{p}).$$

Proof. (a) Expanding the square,

$$\begin{aligned} \sum_{t, \xi} |F(t, \xi)|^2 &= \sum_{h, h' \in W} \sum_{r \in A_h} \sum_{r' \in A_{h'}} \left(\sum_t e_p(t(D_h(r) - D_{h'}(r'))) \right) \\ &\quad \times \left(\sum_{\xi} e_p(\xi(h - h')) \right). \end{aligned}$$

Since $W \subseteq \{1, \dots, p-3\}$, distinct gaps remain distinct modulo p , so the ξ -sum vanishes unless $h = h'$, where it equals p ; the t -sum then vanishes unless $D_h(r) = D_h(r')$, where it equals p . The identity follows; it is exact for every choice of the finite sets A_h .

(b) The diagonal contributes $|A_h| = p + O(h)$ pairs, since Q_h has at most h roots. For $x \neq y$, both in A_h , the equation $D_h(x) = D_h(y)$ is equivalent to $G_h(x, y) = 0$ by (765)–(766). By the Weil bound for absolutely irreducible affine plane curves of degree δ , $\#\{G_h = 0\} = p + O(\delta^2 \sqrt{p})$; here $\delta \leq 6h - 4 = O(h)$, giving $p + O(h^2 \sqrt{p})$. From this count we must remove the points of $\{G_h = 0\}$ lying on the diagonal $x = y$ or on the excluded pole locus $Q_h(x)Q_h(y) = 0$. First, the diagonal is not a component of $\{G_h = 0\}$: since G_h is absolutely irreducible, that would force $G_h = c(x - y)$, i.e. $N_h(x)Q_h(y)^3 - N_h(y)Q_h(x)^3 = c(x - y)^2$; fixing any y_0 with $N_h(y_0)Q_h(y_0) \neq 0$ (such y_0 exists because $N_h \not\equiv 0$ – otherwise $G_h \equiv 0$ – and $N_h Q_h$ has at most $4h - 3 < p$ roots), the left side has x -degree exactly $3h \geq 3$ with leading coefficient $-N_h(y_0) \neq 0$, contradicting the right side. Hence the diagonal meets the curve in at most $\deg G_h = O(h)$ points (Bézout). Second, the pole locus is a union of at most $2h$ coordinate lines, none of which is a component of the absolutely irreducible curve $\{G_h = 0\}$ (its degree exceeds 1 for $h \geq 1$, and it is not a coordinate line by the same degree computation), so Bézout bounds the excluded points by $(6h - 4) \cdot 2h = O(h^2)$. Both corrections are $O(h^2) = O(h^2 \sqrt{p})$. Therefore the off-diagonal admissible count is $p + O(h^2 \sqrt{p})$, and $\text{Collision}(h) = 2p + O(h^2 \sqrt{p})$. $\square$

Remark 744 (computational note). Proposition 743 says that bounded h -windows carry random-model L^2 mass up to the optimal square-root error; in particular almost all Fourier frequencies exhibit square-root cancellation. The hypothesis is supported computationally: symbolic computation gives G_h irreducible over $\mathbb{Q}$ for $h = 2, \dots, 6$, and direct finite-field collision counts for $h = 2, \dots, 21$ give ratios to the random prediction in the range 0.98–1.01. The exceptional value $h = 1$ has ratio exactly 1.50, explained by the cubic three-to-one structure (by (765), $D_1(x)$ is a cube of a linear-fractional expression, so its fibers are generically three-to-one and G_1 is not absolutely irreducible for $p \equiv 1 \pmod{3}$). These numerics support the hypothesis only; no empirical assertion is used in the proof, which is conditional on the absolute irreducibility of G_h .

For $t \in \mathbb{F}_p$ and $H \geq 1$ define the two-dimensional gap sum

$$B_t(H) = \sum_{1 \leq h \leq H} \sum_{r \in A_h} e_p(t D_h(r)),$$

with A_h maximal as in Proposition 743(b).

Conjecture 745 ([GAP-2D-SQRT]: square-root cancellation). *As $p \rightarrow \infty$, uniformly for $t \in \mathbb{F}_p^\times$ and $1 \leq H \leq \sqrt{p}$ (the range of Conjecture 711; the numerical evidence below reaches $H = p^{0.6}$),*

$$|B_t(H)| \leq (pH)^{1/2+o(1)}.$$

Remark 746 (numerical support for Conjecture 745). Full frequency scans over three primes in the range 10^4 – 10^5 , including all t and dyadic H up to the supercritical regime $H = p^{0.6}$, give

$$\max_t \frac{|B_t(H)|}{\sqrt{pH}} \in [3.4, 5.3],$$

while the median remains at the random level, approximately 0.67. The maximum follows the expected Gumbel extreme-value profile with no observed power growth. This closes the previous numerical gap for the two-prime/three-frequency obstruction. The interpretation is: individual correlations behave at square-root scale; no hidden large fiber contribution appears in the scanned range; and the remaining problem is a *proof of uniformity*, not evidence for a different scale. These computations constitute evidence only; Conjecture 745 remains open and is nowhere used in a proof.

Remark 747 (corrected mixed-return target [MIXED-RETURN- η^*]). The previous linear extension target is too strong: the rank-four first-block Frobenius extension cannot provide a new collision equation, because its mixed column is already absorbed by the old gap data. The viable target is a small-image/small-fiber statement. For a mixed holonomy invariant $\text{Hol}_h(r)$ one seeks a gauge-invariant construction, with value set S_h , satisfying

$$|S_h| \leq h^{o(1)} \quad \text{and} \quad \#\{r : \text{Hol}_h(r) = s\} \ll h^{1-\eta} \quad \text{for every } s \in S_h,$$

for some fixed $\eta > 0$. This gives the effective saving $\eta - \sigma$, where σ is the existing gap multiplicity exponent.

The four original candidate objects collapse to two families. *Family I: index-side contiguity.* The naive contiguity construction fails: its translation holonomy determinant is a unit times the old gap function, so it contains no new information. Status: dead. *Family II: central extension geometry.* The surviving candidates are equivalent descriptions of the same possible object: (1) a biextension/syntomic height; (2) a rank-five iterated extension matrix model; (3) a cup-product pairing in the Heisenberg-type central coordinate. The required new object is an independent dual Apéry normal function.

Any candidate must pass the survival criteria:

- S1. gauge-shift invariance;
- S2. independence from the existing gap coordinates;
- S3. nontrivial central extension class;
- S4. fiber reduction after quotienting the gauge directions;
- S5. gcd non-degeneracy: $\deg \gcd(N_h^{\text{prim}}, J_h) \ll h^{1-\eta}$;
- S6. compatibility with Frobenius and the mirror involution.

The first failure of any of these tests falsifies the candidate. Status: open target.

Remark 748 (classification of the spectral layers). The current geometry separates into three layers: (1) the L^2 frequency layer, essentially solved conditionally on absolute irreducibility by the collision geometry of Proposition 743; (2) the [GAP-2D-SQRT] layer, strongly supported by computation but strictly conjectural (Conjecture 745); (3) the mixed-return layer, the only remaining plausible route to a power improvement, which requires extracting a genuinely new dual geometric invariant (Remark 747).

30.4 The mesoscopic normalization suite [MESO]

Throughout this subsection $N := p - 2 = |I_p|$ and $E := \mathcal{F}_p = \sum_v m_p(v)^2$ is the full energy of Section 29. Fix a parameter $L = L(N)$ with $1 \leq L$ and $D := \lceil \sqrt{N} L \rceil \leq N - 1$ (the *mesoscopic normalization*; the scales $D \geq \sqrt{N}$ are called supercritical), and recall from Section 30.1 the window multiplicity $d_D(r)$ and the moments

$$S_D := \sum_{r \in I_p} d_D(r) = S_1(D), \quad Q_D := Q_p(D) = \sum_{r \in I_p} \binom{d_D(r)}{2}.$$

All bounds $\ll$ are uniform in p . Applying Lemma 731(a) with the single box $[1, D]$ (so $q = 1$, $L_r = \binom{d_D(r)}{2}$) gives the exact *renewal identity*

$$Q_D = \sum_{\substack{a, g \geq 1 \\ a+g \leq D}} C_p(a, g), \quad (767)$$

where

$$C_p(a, g) := T(a, a+g) = \#\{r : r+a+g \leq M, \pi(r) = \pi(r+a) = \pi(r+a+g)\},$$

the span-truncated triple object: at full span, $6Q_{p-3} = T_3$ by Lemma 741.

Lemma 749 (environment-capacity inequality). *For every $D \geq 1$,*

$$S_D^2 \leq N(S_D + 2Q_D).$$

Proof. Since $a^2 = a + 2\binom{a}{2}$, we have $\sum_r d_D(r)^2 = S_D + 2Q_D$; the sum has at most N nonzero terms (the base-point support is at most $|I_p| = N$), so by Cauchy–Schwarz $S_D^2 = (\sum_r d_D(r))^2 \leq N \sum_r d_D(r)^2$. $\square$

Remark 750 (the capacity phase transition). Solving the quadratic in Lemma 749 gives $S_D \leq N + \sqrt{2NQ_D}$; in particular $Q_D \ll N$ forces $S_D \ll N$. The point is that the singleton/first-return mass (the linear part of S_D) is capped by the number N of base points. At supercritical scales $D \geq \sqrt{N}$ – at the mesoscopic scale $D = \lceil \sqrt{N} L \rceil$ one has $N = D^2/L^2 \ll D^2$ – this cap is paid by the environment size automatically. This is exactly why control of the pair count Q_D alone suffices there, and why the earlier internal objection that “ Q does not control singletons” is withdrawn: the capacity inequality was the missing step. At subcritical scales the cap N exceeds D^2 and pays nothing; see Remark 761.

Lemma 751 (axis-strip reduction). *Let M_D denote the maximal number of indices in a discrete interval of length $D+1$ sharing one projective value; by the banked window multiplicity bound of Remark 737, $M_D \leq 3D^{2/3} + 1 \leq 4D^{2/3}$. For $1 \leq G \leq D$ put*

$$Q_D^{\text{axis}}(G) := \sum_{\substack{a, g \geq 1, a+g \leq D \\ \min(a, g) \leq G}} C_p(a, g).$$

Then, unconditionally,

$$Q_D^{\text{axis}}(G) \leq 12 D^{2/3} G^2.$$

Proof. Consider first the strip $a \leq G$. For fixed a , every r counted by $C_p(a, g)$ for some g is a gap- a collision base, of which there are $R_a \leq 3(a-1)$ by (758); for each such r the number of admissible g (namely $1 \leq g \leq D-a$ with $\pi(r+a) = \pi(r+a+g)$) is at most the number of forward window mates of $r+a$, which is at most $M_D - 1 \leq 4D^{2/3}$. Hence

$$\sum_{\substack{a \leq G \\ g \geq 1, a+g \leq D}} C_p(a, g) \leq \sum_{a \leq G} 3(a-1) \cdot 4D^{2/3} = 12D^{2/3} \frac{G(G-1)}{2} \leq 6D^{2/3} G^2.$$

For the strip $g \leq G$, fix g and map $r \mapsto u = r+a$: then u is a gap- g collision base ($R_g \leq 3(g-1)$ choices), a is a backward window mate of u (at most $M_D - 1 \leq 4D^{2/3}$ choices, the fiber of $\pi(u)$ in the interval $[u-D, u]$), and the pair (u, a) determines r . The same computation bounds this strip by $6D^{2/3} G^2$. Adding the two strips (a union bound; pairs with both $a, g \leq G$ may be counted twice) gives $12D^{2/3} G^2$. $\square$

Remark 752 (choice of G and the balanced core). At the mesoscopic scale $D = \lceil \sqrt{N} L \rceil$ choose

$$G = \lceil N^{1/3} L^{-1/3} \rceil \asymp \sqrt{N/D^{2/3}},$$

so that $12D^{2/3}G^2 = O(N)$ ($D^{2/3} \asymp N^{1/3}L^{2/3}$ and $G^2 \asymp N^{2/3}L^{-2/3}$). With the *balanced core*

$$\mathcal{K} = \mathcal{K}_{D,G} := \{(a, g) : a > G, g > G, a + g \leq D\},$$

the renewal identity (767) and Lemma 751 give

$$Q_D \leq \sum_{(a,g) \in \mathcal{K}} C_p(a, g) + O(N).$$

Thus [MESO-PAIR] below is equivalent, up to $O(N)$, to its restriction to the balanced core. The deleted axis strips contain all the known degenerate certificates – $R_1 = 0$, the adjacent resultants, the -51 layer, and the small- h certificates – so none of these carries any load for the remaining mass.

For (a, g) with $a, g \geq 1$ define the *regular double-root count*

$$\rho_p(a, g) := \#\{r \text{ regular} : N_a(r) = 0, N_g(r + a) = 0 \text{ in } \mathbb{F}_p\},$$

so that $C_p(a, g) \leq \rho_p(a, g)$ (every collision base counted in (767) is a regular index r with $N_a(r) = 0$ and $N_g(r + a) = 0$, by the gap-polynomial dictionary), and the *shifted resultant*

$$e_{a,g} := \text{Res}_X(N_a(X), N_g(X + a)) \in \mathbb{Z},$$

with the convention $v_p(0) = +\infty$. The shifted – not the raw – family is the load-bearing object: by the exact same-base factorization $\text{Res}(N_h, N_k) = A_{h-1} \cdot \text{Res}(N_h(X), N_{k-h}(X + h))$ with

$$A_m = (-1)^{m(m+1)/2} \prod_{j \leq m} ((j!)^3 b_j)^6,$$

any low Apéry zero $p \mid b_j$ pollutes $\Theta(D^2)$ raw pairs through the factor A_{h-1} (cut-edge roots $X = -j$, which are not regular collisions; e.g. $b_3 = 1445 = 5 \cdot 17^2$ pollutes all 15 raw pairs $3 \leq m < k \leq 8$ at $p = 17$). The raw bad-pair target is dead; targets must be stated after this saturation.

In the five conjectures below, $L = L(N) \rightarrow \infty$ with $D = \lceil \sqrt{N} L \rceil \leq N - 1$, and $G = \lceil N^{1/3} L^{-1/3} \rceil$; the fixed-power specialization $L = N^\eta$ with $0 < \eta < \frac{1}{2}$ is the form [MESO- η] (Conjecture 759).

Conjecture 753 ([MESO-SHIFTED-VALUATION]). $\sum_{(a,g) \in \mathcal{K}_{D,G}} v_p(e_{a,g}) \ll N$. (In particular $e_{a,g} \neq 0$ on the balanced core.)

Conjecture 754 ([MESO-SHIFTED-GCD]). $\sum_{(a,g) \in \mathcal{K}_{D,G}} \deg_{\mathbb{F}_p} \gcd(N_a(X), N_g(X + a)) \ll N$.

Conjecture 755 ([MESO-BALANCED-PAIR]). $\sum_{(a,g) \in \mathcal{K}_{D,G}} \rho_p(a, g) \ll N$.

Conjecture 756 ([MESO-PAIR]). $Q_D \ll N$ at $D = \lceil \sqrt{N} L \rceil$.

Conjecture 757 ([MESO-TOTAL]). $S_D \ll N$ at $D = \lceil \sqrt{N} L \rceil$.

Proposition 758 (implication chain). *With the notation above, and abbreviating each conjecture of the suite by the last component of its name,*

$$\begin{aligned} [VALUATION] &\implies [GCD] \implies [BALANCED] \implies [PAIR] \implies [TOTAL], \\ [TOTAL] &\implies E \ll \frac{N^{3/2}}{L} + N. \end{aligned}$$

The first implication holds for $p \nmid 34$; all other implications are unconditional in p .

Proof. Valuation implies gcd. From the transfer recurrence (731), $\deg U_m = 3m$ and the leading coefficient of U_m is 34^m ; hence the leading coefficient of N_a is 34^{a-1} , and for $p \nmid 34$ reduction modulo p preserves degrees. The standard valuation inequality $v_p(\text{Res}(f, g)) \geq \deg \gcd_{\mathbb{F}_p}(f \bmod p, g \bmod p)$ (for integer polynomials whose leading coefficients are prime to p , and with $v_p(0) = +\infty$ covering vanishing resultants) then bounds each gcd degree by the corresponding valuation.

Gcd implies balanced-pair. Distinct regular double roots r counted by $\rho_p(a, g)$ are distinct common roots $X = r$ of $N_a(X)$ and $N_g(X + a)$ in $\mathbb{F}_p$, so $\rho_p(a, g) \leq \deg \gcd_{\mathbb{F}_p}(N_a(X), N_g(X + a))$.

Balanced-pair implies pair. By Remark 752 and $C_p \leq \rho_p$,

$$Q_D \leq \sum_{(a,g) \in \mathcal{K}} \rho_p(a, g) + O(N) \ll N.$$

Pair implies total. Lemma 749 gives $S_D \leq N + \sqrt{2NQ_D} \ll N$.

Total implies the energy bound. Partition I_p into $q' = \lceil N/D \rceil$ consecutive blocks of length at most D , and let $m_v^{(j)}$ be the fiber multiplicities within block j , so $m_p(v) = \sum_j m_v^{(j)}$. By Cauchy–Schwarz over the blocks,

$$\begin{aligned} E &= \sum_v m_p(v)^2 \leq q' \sum_v \sum_j (m_v^{(j)})^2 \\ &= q' \left(N + 2 \# \{ \text{same-block collision pairs} \} \right) \leq q' (N + 2S_D), \end{aligned}$$

because every same-block pair has gap at most $D - 1$ and is therefore counted (once, by its left endpoint) in S_D . With $S_D \ll N$ and $q' \leq N/D + 1$,

$$E \ll \frac{N^2}{D} + N \leq \frac{N^{3/2}}{L} + N. \quad \square$$

Conjecture 759 ([MESO- η]: fixed-power form). *There is a fixed η with $0 < \eta < \frac{1}{2}$ such that [MESO-PAIR] (or merely [MESO-TOTAL]) holds at $D = \lceil N^{1/2+\eta} \rceil$. By Proposition 758 this gives $E \ll N^{3/2-\eta} + N$.*

Remark 760 (weakest useful form; relation to [T3-2MINUS]). The weakest useful (little- o) form: any bound $S_D = o(D\sqrt{N})$ at a scale $D = \lceil \sqrt{N} L \rceil$ with $L \rightarrow \infty$ already gives $E = o(N^{3/2})$ by the block Cauchy–Schwarz step of Proposition 758, i.e. a strict improvement on the unconditional $p^{3/2}$ energy theorem. Moreover [MESO- η] and [T3-2MINUS] (Remark 742) are the two parallel shortest fixed-power interfaces – the one-scale/renewal route and the global moment route – and they are *not* mutually derivable: by (767) and Lemma 741 one has $Q_D \leq T_3/6$, but neither statement implies the other.

Remark 761 (sharpness: the periodic-block model). The output $N^2/D + N$ of Proposition 758 is optimal given Q -information alone, and no first-moment or one-gap hypothesis can be dropped into the model. Let $D \mid N$ and take an abstract word of exact period D on N letters whose D letters within one period are pairwise distinct. Then each value has fiber size N/D , so

$$E = D \left(\frac{N}{D} \right)^2 = \frac{N^2}{D},$$

while $d_D(r) \leq 1$ for every r (only the gap $h = D$ collides), so

$$Q_D = 0, \quad S_D \leq N.$$

Hence: (i) $Q_D = 0$ is compatible with $E = N^2/D$, so the bound $N^2/D + N$ cannot be improved from Q_D alone and Q -control can never give $E = O(N)$; (ii) at subcritical scales $D \leq \sqrt{N}$ even $Q_D = 0$ allows $E = N^2/D \geq N^{3/2}$, so subcritical Q -control has no yield – the load-bearing condition is supercriticality, as in Remark 750.

Theorem 762 (Abstract inverse-square clique bound). *Let $x_0, \dots, x_N$ be a word over an arbitrary alphabet. Put $R_d = \#\{r : x_r = x_{r+d}\}$ and $d_H(r) = \#\{1 \leq h \leq H : x_r = x_{r+h}\}$. If $R_d \leq Cd$ for $1 \leq d \leq H$, then $Q_H = \sum_r \binom{d_H(r)}{2} \leq 22CH^2(1 + \log H)$.*

Lemma 763 (Clique energy). *If A is a color class inside an interval of at most $2H$ consecutive integers, $m = |A|$, and $E_H(A) = \sum_{u < v \in A, v-u \leq H} (v-u)^{-2}$, then $\binom{m}{3} \leq 11H^2 E_H(A)$.*

Proof. For $m < 3$ this is immediate. If $m = 3$, write the two consecutive gaps as g_1, g_2 . Their sum is at most $2H - 1$, hence one is at most $H - 1$. Therefore $E_H(A) \geq H^{-2}$ and the claim follows.

Assume $m \geq 4$. Split the containing interval into two halves. One half contains $n \geq \lceil m/2 \rceil$ points and has span at most $H - 1$. Let its consecutive gaps be $g_1, \dots, g_k$, $k = n - 1$. Cauchy gives

$$\sum_i g_i^{-2} \geq \frac{(\sum_i g_i^{-1})^2}{k}, \quad \left(\sum_i g_i \right) \left(\sum_i g_i^{-1} \right) \geq k^2.$$

Since $\sum_i g_i \leq H - 1 < H$, $\sum_i g_i^{-2} \geq k^3/H^2$. The boundary cases $m = 4, 5$ give $k \geq m/4$, and the same inequality follows for all $m \geq 4$. Thus $E_H(A) \geq m^3/(64H^2)$, and $\binom{m}{3} \leq m^3/6 \leq (64/6)H^2 E_H(A) < 11H^2 E_H(A)$. $\square$

Lemma 764 (Window localization). *For blocks $B_j = [jH, (j+1)H)$ and windows $W_j = [jH, (j+2)H)$, the triple count satisfies*

$$\sum_{j,a} E_H(A_{j,a}) \leq 2 \sum_{d \leq H} R_d/d^2 \leq 2C(1 + \log H).$$

Proof. A term of Q_H is a monochromatic triple $u < v < w$ with $w - u \leq H$. It is counted exactly once by the least point u and the pair of gaps $(v - u, w - u)$. If $u \in B_j$, then $w \in W_j$, so the whole triple lies in that window. A pair at distance $d \leq H$ with left endpoint in B_t can occur only in W_{t-1} or W_t , hence has multiplicity at most two. Summing pair energies and using the harmonic series gives the claim. $\square$

Proof of the theorem. Assign each monochromatic triple (equivalently, each term of Q_H) to the block of its least point; the triple then lies in the corresponding window, so

$$Q_H \leq \sum_{j,a} \binom{m_{j,a}}{3} \leq 11H^2 \sum_{j,a} E_H(A_{j,a}),$$

by Lemma 763 applied to every window W_j (an interval of at most $2H$ consecutive positions) and color class $A_{j,a}$. The localization lemma bounds the total energy by $2C(1 + \log H)$. Hence

$$Q_H \leq 11H^2 \cdot 2C(1 + \log H) = 22CH^2(1 + \log H).$$

□

Proposition 765 (Sharpness). *There are words with $R_d \leq (5/2)d$ and $Q_H = (1/8 + o(1))H^2$.*

Proof. Let $M = \lfloor H/2 \rfloor$. For each $1 \leq s \leq t \leq M$ put a private symbol at $b_e, b_e + t, b_e + 2t$, where $b_e = e(H + 1)$, and fill all other positions with fresh singleton symbols. Then

$$R_d = 2n_d + \mathbf{1}_{2|d}n_{d/2}, \quad n_t = t,$$

so $R_d \leq 5d/2$. Each class gives one triple of span $2t \leq H$, hence $Q_H = M(M + 1)/2 = (1/8 + o(1))H^2$. Cross-block near coincidences are not collisions because symbols are private. □

Corollary 766. *For the Apéry projective orbit one has $R_d \leq 3(d - 1)$; with $C = 3$ and $H = \min(\Delta, p - 3)$, $Q_p(\Delta) \leq 66H^2(1 + \log H)$.*

Corollary 767. $T_3 = 6Q_p(p - 2) \leq 396(p - 2)^2(1 + \log(p - 2))$.

Remark 768. The exponent 2 is optimal in the abstract class. Removing the logarithm would follow from a prefix estimate $S_1(Y) = \sum_{d \leq Y} R_d \ll Y^{2-\delta}$: Abel summation gives $\sum_{d \leq H} R_d/d^2 \ll 1 + \int_1^H S_1(t)t^{-3}dt$, which is bounded when $S_1(t) \ll t^{2-\delta}$.

Remark 769 (what remains of the wall at the mesoscopic scale). The unconditional abstract bound [THM-ABSTRACT-Q-H2LOGH],

$$Q_H \leq 22CH^2(1 + \log H) \quad \text{for any word with } R_d \leq Cd,$$

improves, for all large H , on the $\frac{27}{4}H^{5/2}$ bound of Theorem 734 within the same abstract hypothesis. Evaluated at the mesoscopic scale $D = \lceil \sqrt{N}L \rceil$ with the Apéry input $C = 3$, it gives

$$Q_D \ll D^2 \log D \asymp N L^2 \log N,$$

against the target $Q_D \ll N$ of Conjecture 756: what remains is exactly the factor $L^2 \log N$. The combinatorial side of the wall is therefore de-logging plus this factor L^2 ; everything beyond it – the balanced-core sums of Conjectures 753–755 – is arithmetic input.

31 Further campaign results: spectral form, windowed lemmas, and conditional small-scale coincidence

This section collects five further campaign results: the spectral (Gram-operator) formulation of the growing-family compatibility problem together with its empirical laws; a windowed restart lemma package with a good-reduction census; an unconditional near-wall estimate for close pairs of collision gaps; a conditional $p^{1/6}$ coincidence theorem; and a second proof, via restart transfer, of the banked window multiplicity lemma. The conventions of Sections 29 and 30 are retained throughout: $p \geq 7$ is prime, $M = p - 2$, and $\pi(n) = [b_n : c_n] \in \mathbb{P}^1(\mathbb{F}_p)$.

31.1 Empirical flatness and the Gram-operator target

Let

$$I_h = \{1, \dots, M - h\}, \quad L_h = |I_h| = M - h,$$

and put

$$\mathfrak{q}_h(X) = \prod_{j=1}^h (X + j), \quad f_h(X) = \frac{N_h(X)}{\mathfrak{q}_h(X)^3}.$$

On the regular nonwrapping window, the Casoratian identity gives

$$f_h(r) = \Delta_{r,h} = b_r c_{r+h} - b_{r+h} c_r.$$

For $a \in \mathbb{F}_p$, define

$$\nu_h(a) = \#\{r \in I_h : f_h(r) = a\}, \quad q_h(a) = \nu_h(a) - \frac{L_h}{p}.$$

Thus $\sum_a q_h(a) = 0$. For $t \in \mathbb{F}_p$, write

$$S_h(t) = \sum_{r \in I_h} e_p(t f_h(r)).$$

For $1 \leq h, k \leq H$, define

$$J_{h,k} = \sum_{a \in \mathbb{F}_p} \nu_h(a) \nu_k(a), \tag{768}$$

$$E_{h,k} = \sum_{a \in \mathbb{F}_p} q_h(a) q_k(a) = J_{h,k} - \frac{L_h L_k}{p}, \tag{769}$$

$$\Gamma_{h,k} = \sum_{t \in \mathbb{F}_p^\times} S_h(t) \overline{S_k(t)}. \tag{770}$$

Additive orthogonality gives the exact normalization

$$\boxed{\Gamma_{h,k} = p E_{h,k}}. \tag{771}$$

In particular, the $t = 0$ contribution is the rank-one matrix $(L_h L_k)_{h,k}$ and is not part of Γ .

By (768), $J_{h,k} = \#\{(x, y) \in I_h \times I_k : f_h(x) = f_k(y)\}$. We write

$$R_{h,k} = J_{h,k}, \quad N_{\text{coinc}}(H) = \sum_{1 \leq h, k \leq H} R_{h,k},$$

$$C_h = \nu_h(0) = \#\{r \in I_h : f_h(r) = 0\},$$

so that C_h is the one-gap collision count at gap h .

Good-reduction strata. Set

$$\mathfrak{C}_h(X) = \mathfrak{q}_h(X)N'_h(X) - 3\mathfrak{q}'_h(X)N_h(X),$$

so that $f'_h(X) = \mathfrak{C}_h(X)/\mathfrak{q}_h(X)^4$. We use the following two explicit strata.

$$\text{GOOD-POLE}_h(p) : N_h(-j) \not\equiv 0 \pmod{p} \quad (1 \leq j \leq h),$$

$$\text{GOOD-CRIT}_h(p) : \deg N_h = 3h - 3, \quad \deg \mathfrak{C}_h = 4h - 4,$$

$$N_h \text{ and } \mathfrak{C}_h \text{ are squarefree,}$$

$$\gcd(\mathfrak{C}_h, N_h \mathfrak{q}_h) = 1,$$

$$x \neq y, \quad \mathfrak{C}_h(x) = \mathfrak{C}_h(y) = 0 \implies f_h(x) \neq f_h(y) \quad \text{over } \overline{\mathbb{F}}_p.$$

Thus $\text{GOOD-POLE}_h(p)$ says that all h candidate poles remain genuine cubic poles, while $\text{GOOD-CRIT}_h(p)$ gives the expected simple finite critical points, distinct nonzero critical values, and the generic ramification profile away from 0 and ∞ . Conditionally on the separate local Fourier calculation, the tame inertia type at 0 is

$$\mathbf{U}(2)^{\oplus(h-1)} \oplus \mathcal{L}_{\chi_3}^{\oplus h} \oplus \mathcal{L}_{\chi_3^{-1}}^{\oplus h}.$$

Thus χ_3 and χ_3^{-1} occur with h copies each, not as tensor powers. Standard Gauss-twist normalizations are omitted below because all formulas are stated at the level of trace functions.

Remark 770 (Numerical laws). The following observations are numerical only and are never used in a proof.

(i) [FLAT-LAW]. For $h \leq 32$ and primes $p \leq 997$, the quantity

$$\max_{t \neq 0} \frac{|S_h(t)|}{\sqrt{p}}$$

is flat in h . Its observed size is

$$\sqrt{2 \log p},$$

the Gumbel extreme-value scale for p Gaussian samples. In particular, the measured maximum is at the random-function floor rather than at the linear-conductor scale suggested by the worst-case bound.

(ii) [PAIR-FLAT]. Across nine exact cells with $p \leq 4001$ and $h < k \leq p^{0.66}$, one finds

$$\text{mean}_{h < k} |E_{h,k}| = (1.0\text{--}1.1)\sqrt{p}.$$

There is no visible hk growth, although the generic normalized fiber-product genus is asymptotic to $6hk$.

(iii) Signed shells. For

$$A_d(H) = \sum_{h=1}^{H-d} E_{h,h+d}, \quad 0 \leq d < H,$$

and

$$V(H) = \frac{1}{p^2 H} \sum_{d=0}^{H-1} |A_d(H)|^2,$$

the exact computations give

$$\max_{0 \leq d < H} \frac{|A_d(H)|}{p} \leq 0.80, \quad V(H) \leq 0.049.$$

No structured exceptional value of d is visible.

(iv) Absolute mass. The same data give

$$\sum_{h,k \leq H} |E_{h,k}| = (0.49\text{--}0.57)H^2\sqrt{p}.$$

Thus the absolute L^1 /Schur route is empirically at generic noise scale. Since a typical row then has size $H\sqrt{p}$, it cannot satisfy an $O(p)$ row bound once $H > \sqrt{p}$.

The windowed trace is not the pure core trace. This correction is small at the operator scale but must be kept explicitly. Put

$$\mathbf{c}_h = f_h(0) = -f_h(-h-1),$$

where the second equality is the reflection antisymmetry of f_h , and let

$$T_h(t) = \sum_{x \in \mathbb{P}^1(\mathbb{F}_p) \setminus \{-1, \dots, -h\}} e_p(tf_h(x)).$$

The three points removed when one passes from the projective core to the nonwrapping window are $\infty, 0, -h-1$, with values $0, \mathbf{c}_h, -\mathbf{c}_h$. Hence

$$S_h(t) = T_h(t) - \epsilon_h(t), \quad \epsilon_h(t) = 1 + e_p(t\mathbf{c}_h) + e_p(-t\mathbf{c}_h). \quad (772)$$

These are the three endpoint rank-one lines $\mathbf{1}$, $\mathcal{L}_\psi(\mathbf{c}_h t)$, and $\mathcal{L}_\psi(-\mathbf{c}_h t)$.

Let $\mathbf{U}$ and $\mathbf{V}$ be the $(p-1) \times H$ matrices

$$\mathbf{U}_{t,h} = T_h(t), \quad \mathbf{V}_{t,h} = -\epsilon_h(t) \quad (t \neq 0).$$

Then

$$\Gamma = (\mathbf{U} + \mathbf{V})^*(\mathbf{U} + \mathbf{V}) = \Gamma^{\text{core}} + \Gamma^{\text{end}}, \quad (773)$$

$$\Gamma^{\text{core}} = \mathbf{U}^*\mathbf{U}, \quad (774)$$

$$\Gamma^{\text{end}} = \mathbf{U}^*\mathbf{V} + \mathbf{V}^*\mathbf{U} + \mathbf{V}^*\mathbf{V}. \quad (775)$$

The endpoint block satisfies the explicit rank bound

$$\text{rank } \Gamma^{\text{end}} \leq \text{rank}(\mathbf{U}^*\mathbf{V}) + \text{rank}(\mathbf{V}^*\mathbf{U}) + \text{rank}(\mathbf{V}^*\mathbf{V}) \leq 3 \text{rank}(\mathbf{V}) \leq 3H. \quad (776)$$

Moreover,

$$\sum_{t \neq 0} e_p((\lambda - \mu)t) = \begin{cases} p-1, & \lambda = \mu, \\ -1, & \lambda \neq \mu, \end{cases} \quad (777)$$

so all endpoint Gram entries are explicit structured sums. In particular,

$$\|\mathbf{V}\|_F^2 \leq 9(p-1)H \leq 9pH, \quad \|\mathbf{V}\|_{\text{op}} \leq 3\sqrt{pH}. \quad (778)$$

Proposition 771 (GRAM-OP criterion). *Let $p \geq 64$ be prime and let*

$$H = \lceil p^{2/3} \rceil.$$

Assume $H \leq p - 3$ and assume $\text{GOOD-POLE}_h(p)$ and $\text{GOOD-CRIT}_h(p)$ for every $1 \leq h \leq H$. Suppose that

$$\boxed{\|\Gamma\|_{\text{op}} \leq Cp^2.} \quad (\text{GRAM-OP})$$

Put

$$\mathcal{S}_H = \sum_{h=1}^H L_h = \#\{(r, h) : 1 \leq h \leq H, 1 \leq r \leq M - h\}.$$

Then

$$N_{\text{coinc}}(H) \leq \frac{\mathcal{S}_H^2}{p} + CpH \leq \frac{\mathcal{S}_H^2}{p} + 2C\mathcal{S}_H. \quad (779)$$

Consequently,

$$\sum_{h \leq H} C_h \leq \frac{\mathcal{S}_H}{p} + \sqrt{CpH}, \quad (780)$$

and the block-energy inequality gives the explicit bound

$$\boxed{\mathcal{F}_p \leq (10 + 4\sqrt{2C})p^{4/3}.} \quad (781)$$

A sufficient core form of (GRAM-OP) is

$$\|\Gamma^{\text{core}}\|_{\text{op}} \leq C_0 p^2. \quad (782)$$

Indeed, the endpoint decomposition then gives

$$\|\Gamma\|_{\text{op}} \leq (\sqrt{C_0} + 3)^2 p^2, \quad (783)$$

so one may take

$$C = (\sqrt{C_0} + 3)^2.$$

The endpoint contribution to the all-ones quadratic form is separately bounded by

$$\left| \mathbf{1}^* \Gamma^{\text{end}} \mathbf{1} \right| \leq (6\sqrt{C_0} + 9)p^2 H. \quad (784)$$

Proof. The matrix Γ is positive semidefinite, since it is the Gram matrix of the vectors $(S_h(t))_{t \neq 0}$. Therefore

$$\mathbf{1}^* \Gamma \mathbf{1} \leq \|\Gamma\|_{\text{op}} \|\mathbf{1}\|_2^2 \leq Cp^2 H. \quad (785)$$

On the other hand, expanding the left-hand side and using additive orthogonality gives the exact centered variance identity

$$\begin{aligned} \mathbf{1}^* \Gamma \mathbf{1} &= \sum_{t \neq 0} \left| \sum_{h \leq H} S_h(t) \right|^2 \\ &= pN_{\text{coinc}}(H) - \mathcal{S}_H^2. \end{aligned} \quad (786)$$

Combining (785) and (786) proves the first inequality in (779). Since $H \leq p/2 - 2$ for $p \geq 64$, one has $L_h = p - 2 - h \geq p/2$ throughout the range, and hence

$$\mathcal{S}_H \geq \frac{pH}{2}.$$

Thus $CpH \leq 2CS_H$, proving the second inequality in (779).

The zero detector and Cauchy–Schwarz give

$$\begin{aligned} \sum_{h \leq H} C_h &= \frac{S_H}{p} + \frac{1}{p} \sum_{t \neq 0} \sum_{h \leq H} S_h(t) \\ &\leq \frac{S_H}{p} + \frac{1}{p} \sqrt{(p-1) \sum_{t \neq 0} \left| \sum_{h \leq H} S_h(t) \right|^2} \\ &\leq \frac{S_H}{p} + \sqrt{CpH}, \end{aligned}$$

which is (780).

The proved block-energy inequality is

$$\mathcal{F}_p \leq \left\lceil \frac{M}{H} \right\rceil \left(M + 2 \sum_{1 \leq h < H} C_h \right). \quad (787)$$

Now

$$H \leq p^{2/3} + 1 \leq 2p^{2/3}, \quad \left\lceil \frac{M}{H} \right\rceil \leq 2p^{1/3}, \quad S_H \leq pH.$$

Consequently,

$$\begin{aligned} \mathcal{F}_p &\leq 2p^{1/3} (p + 2H + 2\sqrt{CpH}) \\ &\leq 2p^{1/3} (p + 4p^{2/3} + 2\sqrt{2C} p^{5/6}) \\ &\leq (10 + 4\sqrt{2C}) p^{4/3}, \end{aligned}$$

which proves (781).

It remains to justify the sufficient core formulation and to track the endpoint block. From (782),

$$\|\mathbf{U}\|_{\text{op}} \leq \sqrt{C_0} p,$$

while (778) gives

$$\|\mathbf{V}\|_{\text{op}} \leq 3\sqrt{pH} \leq 3p.$$

Therefore

$$\|\Gamma\|_{\text{op}}^{1/2} = \|\mathbf{U} + \mathbf{V}\|_{\text{op}} \leq (\sqrt{C_0} + 3)p,$$

which proves (783). Finally,

$$\begin{aligned} \left| \mathbf{1}^* \Gamma^{\text{end}} \mathbf{1} \right| &\leq 2\|\mathbf{U}\mathbf{1}\|_2 \|\mathbf{V}\mathbf{1}\|_2 + \|\mathbf{V}\mathbf{1}\|_2^2 \\ &\leq 2(\sqrt{C_0} p \sqrt{H})(3\sqrt{p} H) + 9pH^2 \\ &\leq (6\sqrt{C_0} + 9)p^2 H, \end{aligned}$$

where the last line uses $H \leq p$. This proves (784) and completes the proof. $\square$

Remark 772 (The rowwise ceiling and the spectral escape). The exact obstruction to the rowwise Schur route has three linked formulations. For fixed h , put

$$\begin{aligned} R_h^{\text{abs}}(H) &= \sum_{\substack{k \leq H \\ k \neq h}} |E_{h,k}|, \\ \mathcal{M}_h^{\text{off}}(H) &= \sum_{\substack{k \leq H \\ k \neq h}} E_{h,k}^2, \\ M_{3,h}^{\text{off}}(a,b) &= \sum_{\substack{k \leq H \\ k \neq h}} \nu_k(a) \nu_k(b). \end{aligned}$$

Then

$$R_h^{\text{abs}}(H) = \sum_{k \neq h} \left| \#\{(r,s) \in I_h \times I_k : f_h(r) = f_k(s)\} - \frac{L_h L_k}{p} \right| \quad (788)$$

is the combinatorial language,

$$R_h^{\text{abs}}(H) \leq \sqrt{H-1} (\mathcal{M}_h^{\text{off}}(H))^{1/2} \quad (789)$$

is the one-way second-moment reduction, and the exact centered contraction

$$\mathcal{M}_h^{\text{off}}(H) = \sum_{a,b \in \mathbb{F}_p} q_h(a) q_h(b) M_{3,h}^{\text{off}}(a,b) \quad (790)$$

is the signed joint-value language. Thus the combinatorial formula and the signed contraction are exact identities, while (789) is a Cauchy–Schwarz implication rather than an equivalence. Since $E_{h,h} \geq 0$, the useful full-row split is

$$\sum_{k \leq H} |E_{h,k}| = E_{h,h} + R_h^{\text{abs}}(H) \leq E_{h,h} + \sqrt{H-1} (\mathcal{M}_h^{\text{off}}(H))^{1/2}. \quad (791)$$

The baseline terms disappear in (790) because $\sum_a q_h(a) = 0$. The proved triple-return theorem controls a different orientation: two gaps at one base point and at the value zero. It does not control (790), which has one family index, two independent base points, and arbitrary values (a,b) .

If

$$R_H = \max_{h \leq H} \sum_{k \leq H} |E_{h,k}|,$$

then the exact row-to-energy ledger is

$$\mathcal{F}_p \ll \frac{p^2}{H} + p \sqrt{\frac{R_H}{H}} + p. \quad (792)$$

The best unconditional degree-and-mass estimate is

$$R_H \ll pH^2,$$

which optimizes only to $\mathcal{F}_p \ll p^{5/3}$ and therefore does not even reproduce the proved $p^{3/2}$ theorem. Even the individual fourth-moment hypothesis

$$\sum_{a \in \mathbb{F}_p} |q_k(a)|^4 \ll p \quad \text{uniformly in } k$$

gives only $R_H \ll pH$, and then the middle term of (792) is exactly $p^{3/2}$.

The sharp surviving row target is the Poisson-compatible mixed fourth moment

$$\boxed{\max_{h \leq H} \mathcal{M}_h^{\text{off}}(H) \ll pH.} \quad (\text{ROW-MIX4})$$

Together with the diagonal bound $E_{h,h} \ll p$, (791) gives

$$R_H \ll p + H\sqrt{p}.$$

At $H = p^{2/3}$, substitution in (792) yields

$$\mathcal{F}_p \ll p^{4/3}.$$

No unconditional estimate of the mixed form (ROW-MIX4) is presently available. Thus the single rowwise blocking term is $R_h^{\text{abs}}(H)$, or before the final Cauchy step $\mathcal{M}_h^{\text{off}}(H)$, equivalently the failure to obtain extra cancellation in (790).

The following discussion is numerical and heuristic only and is not used in a proof. The operator formulation escapes this absolute-value ceiling. The measured off-diagonal entries have

$$|E_{h,k}| \asymp \sqrt{p}, \quad |\Gamma_{h,k}| \asymp p^{3/2}.$$

A random-sign $H \times H$ matrix at this entry scale has operator norm

$$p^{3/2}\sqrt{H}(\log H)^{O(1)}.$$

At $H = p^{2/3}$ this is

$$p^{11/6}(\log p)^{O(1)} \ll p^2.$$

The diagonal entries naturally live at the p^2 scale, while the three endpoint lines have already been isolated in (773). Hence [GRAM-OP] is fully consistent with the observed entry sizes and signs. The data support a two-stage picture: [PAIR-FLAT] controls the entry size through $H \asymp \sqrt{p}$, and signed spectral cancellation, rather than absolute summation, must control the remaining range up to $H = p^{2/3}$.

Relation to the other family-compatibility targets. Let

$$B_t(H) = \sum_{h \leq H} S_h(t).$$

The pointwise target [GAP-2D-SQRT] controls each $B_t(H)$; the shell target controls

$$\sum_{d < H} |A_d(H)|^2;$$

and [GRAM-OP] controls the full coefficient-space operator $(\Gamma_{h,k})_{h,k \leq H}$. The exact shell identity is

$$\sum_{h,k \leq H} E_{h,k} = A_0(H) + 2 \sum_{d=1}^{H-1} A_d(H),$$

so the shell estimate

$$\sum_{d < H} |A_d(H)|^2 \ll p^2 H$$

implies the same $O(pH)$ all-ones variance by Cauchy-Schwarz. Likewise, square-root control of every $B_t(H)$ implies the same variance after summing over t . Conversely, [GRAM-OP] immediately implies [COINC] by (786), but it is not formally equivalent to either the pointwise or shell statement. These are three complementary faces of the same growing-family compatibility problem: pointwise in the Fourier variable t , organized by difference shells d , and spectral in the family index h .

31.2 Windowed restart lemmas and the good-reduction census

The following package localizes the restart zero count to arbitrary windows and then quantifies, through the endpoint factorization, how often the good-reduction stratum $\text{GOOD-POLE}_h(p)$ of Subsection 31.1 can fail.

Lemma 773 (Capacitated gap count). *Let $(n_m)_{m \geq 1}$ be nonnegative real numbers such that*

$$n_m \leq 3m \quad (m \geq 1), \quad S := \sum_{m \geq 1} (m+1)n_m < \infty.$$

Then

$$\sum_{m \geq 1} n_m \leq \frac{3}{2} S^{2/3}.$$

The constant $3/2$ is asymptotically sharp under these two constraints.

Proof. For $y \geq 0$ put

$$B_y := \sum_{m=1}^y 3m = \frac{3}{2}y(y+1), \quad D_y := \sum_{m=1}^y 3m(m+1) = y(y+1)(y+2).$$

Choose y so that $D_y \leq S \leq D_{y+1}$. Since the coefficients $m - y - 1$ are negative for $m \leq y$ and nonnegative for $m > y$, the bounds $0 \leq n_m \leq 3m$ give

$$S - (y+2) \sum_m n_m = \sum_m (m - y - 1)n_m \geq \sum_{m=1}^y (m - y - 1)3m = D_y - (y+2)B_y.$$

Hence

$$\sum_m n_m \leq B_y + \frac{S - D_y}{y+2}.$$

On $[D_y, D_{y+1}]$ the function

$$\frac{3}{2}S^{2/3} - B_y - \frac{S - D_y}{y+2}$$

is concave, so its minimum occurs at an endpoint. At $S = D_y$ the required inequality is

$$B_y \leq \frac{3}{2}D_y^{2/3},$$

which is equivalent to $y(y+1) \leq (y+2)^2$; the other endpoint is the same inequality with $y+1$ in place of y . This proves the bound. At $S = D_y$ one has $B_y/D_y^{2/3} \rightarrow 3/2$, proving sharpness. $\square$

Lemma 774 (Windowed Apéry zero count). *Let $p \geq 7$ be prime, and let*

$$I = \{a, a+1, \dots, a+H-1\} \subseteq \{0, 1, \dots, p-2\}$$

be an interval of regular indices. Then

$$Z_p(I) := \#\{n \in I : b_n \equiv 0 \pmod{p}\} \leq 1 + \frac{3}{2}(H-1)^{2/3}.$$

Proof. Let the zeros in I be $z_1 < \dots < z_k$. There is nothing to prove when $k \leq 1$. For $1 \leq i < k$ put

$$s_i = z_i + 1, \quad m_i = z_{i+1} - z_i - 1.$$

There are no consecutive zeros, so $m_i \geq 1$. The restart dictionary gives $U_{m_i}(s_i) = 0$. For fixed m , the starts s_i are distinct, while U_m is nonzero in the banked regular range and has degree $3m$. Thus, with

$$n_m := \#\{i : m_i = m\},$$

one has $n_m \leq 3m$. Moreover

$$\sum_{m \geq 1} (m+1)n_m = \sum_{i=1}^{k-1} (z_{i+1} - z_i) = z_k - z_1 \leq H - 1.$$

Lemma 773 therefore gives

$$k - 1 = \sum_m n_m \leq \frac{3}{2}(H - 1)^{2/3},$$

which is the assertion. □

Remark 775 (Improved full-window constant). Taking $I = \{1, \dots, p - 2\}$ in Lemma 774 gives

$$|Z_p| \leq 1 + \frac{3}{2}(p - 3)^{2/3}.$$

This improves the constant 3 in the previously stated degree-only restart bound. The constant $3/2$ uses the simultaneous all-gap optimization of Lemma 773; it does not follow from optimizing a single short-gap cutoff.

Lemma 776 (Endpoint factorization). *For integers $h \geq 1$ and $1 \leq j \leq h$ one has*

$$N_h(-j) = N_j(-j)N_{h-j+1}(-1).$$

Moreover

$$N_q(-1) = ((q - 1)!)^3 b_{q-1} \quad (q \geq 1),$$

and hence

$$N_h(-j) = (-1)^{j-1}((j - 1)!(h - j)!)^3 b_{j-1} b_{h-j}.$$

In particular, if $h < p$, then

$$N_h(-j) \equiv 0 \pmod{p} \iff p \mid b_{j-1} b_{h-j}.$$

Proof. The case $j = 1$ is immediate. For $j \geq 2$, apply the renewal identity with

$$s = 1 - j, \quad m = j - 2, \quad g = h - j.$$

The bridge factor is $(s + m + 1)^6 = 0$, and therefore

$$U_{h-1}(1 - j) = U_{j-1}(1 - j)U_{h-j}(0).$$

Using $N_q(r) = U_{q-1}(r + 1)$ gives the first identity. The recurrence at $s = 0$ gives $U_m(0) = (m!)^3 b_m$, hence the second identity. Finally, continuant reflection gives

$$U_{j-1}(1 - j) = (-1)^{j-1} U_{j-1}(0),$$

which yields the displayed endpoint formula. When $h < p$, both factorials are units modulo p , proving the final equivalence. □

Lemma 777 (Good-reduction incidence census). *Assume $1 \leq H < p$, and define*

$$\text{Bad}_{\text{pole}}(p, H) := \#\{(h, j) : 1 \leq j \leq h \leq H, p \mid b_{j-1}b_{h-j}\}.$$

Then

$$\text{Bad}_{\text{pole}}(p, H) \leq \frac{9}{5}H^{5/3} + 2H.$$

Consequently

$$\frac{\text{Bad}_{\text{pole}}(p, H)}{H(H+1)/2} \leq \frac{18}{5}H^{-1/3} + \frac{4}{H} = O(H^{-1/3}).$$

Equivalently, all but $O(H^{5/3})$ of the endpoint incidences (h, j) with $1 \leq j \leq h \leq H$ have good reduction.

Proof. Let

$$A_H := \{m : 0 \leq m < H, p \mid b_m\}, \quad Z(t) := \#\{0 \leq m < t : p \mid b_m\}.$$

Under the bijection $a = j - 1$, $q = h - j$, the relevant pairs satisfy $a, q \geq 0$ and $a + q \leq H - 1$. A union bound and discrete layer cake give

$$\text{Bad}_{\text{pole}}(p, H) \leq 2 \sum_{m \in A_H} (H - m) = 2 \sum_{t=1}^H Z(t).$$

Lemma 774, applied to the prefix interval $\{0, \dots, t-1\}$, gives

$$Z(t) \leq 1 + \frac{3}{2}(t-1)^{2/3}.$$

Therefore

$$\text{Bad}_{\text{pole}}(p, H) \leq 2H + 3 \sum_{u=0}^{H-1} u^{2/3} \leq 2H + 3 \int_0^H x^{2/3} dx = 2H + \frac{9}{5}H^{5/3}.$$

Division by $H(H+1)/2$ gives the density bound. □

31.3 A near-wall estimate for close pairs of gaps

We retain the conventions of Sections 29 and 30: $p \geq 7$ is prime, $I_p = \{1, \dots, M\}$ with $M = p - 2$, and $\pi(n) = [b_n : c_n] \in \mathbb{P}^1(\mathbb{F}_p)$. All counts below are nonwrapping. Thus a condition involving $\pi(r+h)$ always includes $r+h \leq M$. For convenience set $R_h = 0$ when $h > M - 1$.

For $1 \leq a < b$ define

$$J_p(a, b) = \#\{r \in I_p : r + b \leq M, \pi(r) = \pi(r+a) = \pi(r+b)\}.$$

For integers $1 \leq D \leq H$ put

$$K_p(H, D) = \sum_{1 \leq d \leq D} \sum_{H < h \leq 2H-d} J_p(h, h+d). \tag{793}$$

Thus $K_p(H, D)$ counts same-base pairs of collision gaps whose smaller gap lies above H , whose larger gap is at most $2H$, and whose gap separation is at most D .

For $d \geq 1$ and $u \in I_p$ define

$$Z_d = \{u \in I_p : u + d \leq M, \pi(u) = \pi(u + d)\},$$

$$\nu_D(u) = \#\{1 \leq d \leq D : u \in Z_d\}, \quad S_1(D) = \sum_{1 \leq d \leq D} R_d = \sum_{u \in I_p} \nu_D(u).$$

The ordinary same-base pair count is

$$Q_p(D) = \sum_{u \in I_p} \binom{\nu_D(u)}{2}. \quad (794)$$

For the backward window define

$$W_{H,d}(u) = \#\{h : H < h \leq 2H - d, h \leq u - 1, \pi(u - h) = \pi(u)\},$$

$$W_H(u) = \#\{h : H < h \leq 2H, h \leq u - 1, \pi(u - h) = \pi(u)\}. \quad (795)$$

The endpoint condition $h \leq u - 1$ is the one-based nonwrapping condition $u - h \geq 1$.

Lemma 778 (forward-window second moment). *For every integer $D \geq 1$,*

$$\sum_{u \in I_p} \nu_D(u)^2 = S_1(D) + 2Q_p(D). \quad (796)$$

Consequently,

$$\sum_{u \in I_p} \nu_D(u)^2 \leq 15D^{5/2}. \quad (797)$$

Proof. The pointwise identity $x^2 = x + 2\binom{x}{2}$, summed over u , gives (796). Here an unordered pair $d_1 < d_2$ counted by $\binom{\nu_D(u)}{2}$ is exactly the same-base collision pair

$$\pi(u) = \pi(u + d_1) = \pi(u + d_2)$$

with gaps $d_1, d_2 \leq D$. No shift of the base point is involved.

By (758),

$$S_1(D) \leq 3 \sum_{d=1}^D (d-1) = \frac{3}{2}D(D-1) \leq \frac{3}{2}D^{5/2}.$$

Theorem 734 gives $Q_p(D) \leq \frac{27}{4}D^{5/2}$, since its truncated gap parameter is at most D . Substitution in (796) yields

$$\sum_u \nu_D(u)^2 \leq \left(\frac{3}{2} + \frac{27}{2}\right) D^{5/2} = 15D^{5/2}.$$

□

Lemma 779 (backward-window injection). *For every integer $H \geq 1$,*

$$\sum_{u \in I_p} \binom{W_H(u)}{2} \leq Q_p(2H). \quad (798)$$

Moreover,

$$\sum_{u \in I_p} W_H(u)^2 \leq \left(\frac{9}{2} + 54\sqrt{2}\right) H^{5/2}. \quad (799)$$

Proof. An object counted by $(W_H^{(u)})_2$ is a uniquely ordered pair

$$H < h_1 < h_2 \leq 2H, \quad h_2 \leq u - 1, \quad \pi(u - h_1) = \pi(u) = \pi(u - h_2).$$

Set

$$r = u - h_2, \quad g_1 = h_2 - h_1, \quad g_2 = h_2.$$

Then

$$r \geq 1, \quad 1 \leq g_1 \leq H - 1, \quad H < g_2 \leq 2H, \quad g_1 < g_2, \quad r + g_2 = u \leq M.$$

Furthermore

$$r + g_1 = u - h_1, \quad r + g_2 = u,$$

so

$$\pi(r) = \pi(r + g_1) = \pi(r + g_2).$$

Thus (r, g_1, g_2) is a valid object counted by $Q_p(2H)$. The map is injective because its image recovers the source through

$$u = r + g_2, \quad h_2 = g_2, \quad h_1 = g_2 - g_1.$$

This proves (798), including the ordered-pair and nonwrapping conventions.

The first moment is

$$\sum_{u \in I_p} W_H(u) = \sum_{H < h \leq 2H} R_h \leq 3 \sum_{h=H+1}^{2H} (h-1) = \frac{9H^2 - 3H}{2} \leq \frac{9}{2}H^2.$$

Using $x^2 = x + 2\binom{x}{2}$, the injection, and Theorem 734,

$$\begin{aligned} \sum_u W_H(u)^2 &\leq \frac{9}{2}H^2 + 2Q_p(2H) \\ &\leq \frac{9}{2}H^2 + 2 \cdot \frac{27}{4}(2H)^{5/2} \\ &= \frac{9}{2}H^2 + 54\sqrt{2}H^{5/2} \\ &\leq \left(\frac{9}{2} + 54\sqrt{2}\right)H^{5/2}, \end{aligned}$$

which is (799). □

Theorem 780 (near-wall bound with $\eta = 3/7$). *For every prime $p \geq 7$ and all integers $1 \leq D \leq H$,*

$$K_p(H, D) \leq C_{\text{nw}} HD^{11/7}, \quad C_{\text{nw}} = \sqrt{15 \left(\frac{9}{2} + 54\sqrt{2}\right)}. \quad (800)$$

In particular,

$$K_p(H, D) \ll HD^{11/7} = HD^{2-3/7}$$

uniformly in p, H, D , with no factor p^ε .

Proof. We first rewrite the count at its midpoint. A term of $K_p(H, D)$ consists of (r, h, d) satisfying

$$1 \leq d \leq D, \quad H < h \leq 2H - d, \quad r + h + d \leq M,$$

and

$$\pi(r) = \pi(r + h) = \pi(r + h + d).$$

Put $u = r + h$. Then

$$r = u - h, \quad r + h + d = u + d,$$

and the event becomes

$$\pi(u - h) = \pi(u) = \pi(u + d).$$

The boundary conditions become $u + d \leq M$ and $h \leq u - 1$. Conversely, these conditions recover the unique base $r = u - h$. Hence

$$K_p(H, D) = \sum_{1 \leq d \leq D} \sum_{u \in Z_d} W_{H,d}(u). \quad (801)$$

Since all summands are nonnegative and $W_{H,d}(u) \leq W_H(u)$,

$$K_p(H, D) \leq \sum_{u \in I_p} \nu_D(u) W_H(u). \quad (802)$$

Cauchy–Schwarz, Lemma 778, and Lemma 779 give

$$\begin{aligned} K_p(H, D) &\leq \left(\sum_u \nu_D(u)^2 \right)^{1/2} \left(\sum_u W_H(u)^2 \right)^{1/2} \\ &\leq \sqrt{15 \left(\frac{9}{2} + 54\sqrt{2} \right)} H^{5/4} D^{5/4} \\ &= C_{\text{nw}} H^{5/4} D^{5/4}. \end{aligned} \quad (803)$$

There is also a pointwise branch. For fixed u , all positions counted by $W_H(u)$ lie in the interval

$$[\max(1, u - 2H), u - H - 1],$$

which contains at most H integers, all of projective color $\pi(u)$. The previously proved interval multiplicity estimate therefore gives

$$W_H(u) \leq 1 + 4H^{2/3} \leq 5H^{2/3}.$$

Together with (802) and $S_1(D) \leq \frac{3}{2}D(D-1)$, this yields

$$K_p(H, D) \leq \left(1 + 4H^{2/3} \right) S_1(D) \leq \frac{15}{2} H^{2/3} D^2. \quad (804)$$

The two powers cross at $D = H^{7/9}$. If $D \leq H^{7/9}$, then $D^{3/7} \leq H^{1/3}$, and (804) gives

$$K_p(H, D) \leq \frac{15}{2} H^{2/3} D^2 \leq \frac{15}{2} H D^{11/7}.$$

If $D \geq H^{7/9}$, then $D^{9/28} \geq H^{1/4}$, and (803) gives

$$K_p(H, D) \leq C_{\text{nw}} H^{5/4} D^{5/4} \leq C_{\text{nw}} H D^{11/7}.$$

Since $C_{\text{nw}} > 15/2$, the two cases prove (800). □

Remark 781 (position on the wall map). The proof retains the sharper two-branch estimate

$$K_p(H, D) \leq \min \left\{ \frac{15}{2} H^{2/3} D^2, C_{\text{nw}} H^{5/4} D^{5/4} \right\}. \quad (805)$$

The theorem packages this minimum into the near-wall form $HD^{2-\eta}$ with the unconditional gain $\eta = 3/7$.

There are also the global containments

$$K_p(H, D) \leq Q_p(2H) \leq \frac{27}{4} (2H)^{5/2} = 27\sqrt{2} H^{5/2},$$

and, from the abstract $H^2 \log H$ theorem with $R_d \leq 3d$,

$$K_p(H, D) \leq Q_p(2H) \leq 66(2H)^2 (1 + \log(2H)) = 264H^2 (1 + \log(2H)).$$

Thus the near-wall theorem is a local strip estimate, not a replacement for the sharp global bound. At the power scale $D = H^\alpha$, its exponent is $1 + 11\alpha/7$. It improves on the $H^{5/2}$ global exponent for $\alpha < 21/22$, and it gives a power saving over $H^2 \log H$ for $\alpha < 7/11$; at $\alpha = 7/11$ it gives $O(H^2)$ and removes the global logarithm. For larger α , the $H^2 \log H$ theorem is the stronger global estimate. The new information is therefore concentrated exactly in the near-diagonal regime where two collision gaps are close.

31.4 A conditional $p^{1/6}$ coincidence theorem

Recall from Subsection 31.1 the nonwrapping windows $I_h = \{1, \dots, M - h\}$ with $L_h = |I_h| = p - 2 - h$, the pole product $\mathfrak{q}_h(X) = \prod_{j=1}^h (X + j)$, the normalized gap function $f_h = N_h / \mathfrak{q}_h^3$ (denoted δ_h in the source), the coincidence counts $R_{h,k}$ and $N_{\text{coinc}}(H) = \sum_{1 \leq h, k \leq H} R_{h,k}$, and $S_H = \sum_{h=1}^H L_h$. From $L_h = p - 2 - h$,

$$S_H = pH - \frac{H^2 + 5H}{2}.$$

Clearing denominators in $f_h(x) = f_k(y)$ gives the affine curve

$$F_{h,k}(X, Y) = N_h(X) \mathfrak{q}_k(Y)^3 - N_k(Y) \mathfrak{q}_h(X)^3 = 0,$$

of bidegree at most $(3h, 3k)$.

The census hypothesis. CENSUS_H:

- For all $1 \leq h < k \leq H$, $F_{h,k}$ is absolutely irreducible over $\mathbb{F}_p$.
- For all $2 \leq h \leq H$, $F_{h,h}$ has exactly two components: the diagonal $X = Y$ and an absolutely irreducible component $\mathcal{H}_h$.
- For every $h \leq H$, all finite critical points of f_h are simple and have distinct critical values; for $h \neq k$, the finite branch loci of f_h and f_k are disjoint away from $\{0, \infty\}$.
- $p > p_0(H)$, so all degrees, separability, and local singularity types are as in characteristic zero.

The census refines the per-gap strata GOOD-POLE_h(p) and GOOD-CRIT_h(p) of Subsection 31.1 to pairwise data.

Remark 782 (the $h = 1$ exception). For $h = 1$ one has $(Y + 1)^3 = (X + 1)^3$, which splits into three lines. Thus $h = 1$ is handled separately and not included in the census clause.

Complete generic singularity census and genus bounds. A curve of bidegree (a, b) in $\mathbb{P}^1 \times \mathbb{P}^1$ has arithmetic genus $(a - 1)(b - 1)$. For the distinct-gap curve $F_{h,k}$, $(a, b) = (3h, 3k)$, so

$$g_{\text{arith}} = (3h - 1)(3k - 1).$$

There are hk finite pole-pole points $(X = -i, Y = -j)$. Writing local parameters $u = X + i$ and $v = Y + j$, the local equation has the form

$$u^3 A(u, v) - v^3 B(u, v) = 0, \quad A(0, 0)B(0, 0) \neq 0.$$

Because $p \neq 3$ and the reduction is good, its tangent cone is the product of three distinct lines. Thus every finite pole-pole point is an ordinary triple point and has delta-invariant 3.

There is one further singular point at infinity, namely $([1 : 0], [1 : 0])$ in $\mathbb{P}^1 \times \mathbb{P}^1$. With local parameters $u = 1/X$ and $v = 1/Y$, the exact order-three zeros of f_h and f_k at infinity give

$$a_h u^3 - a_k v^3 + \text{terms of order at least four} = 0, \quad a_h a_k \neq 0.$$

This is again an ordinary triple point, with delta-invariant 3. Under CENSUS_H there are no other singularities: a regular affine singularity would give critical points of f_h and f_k with a common finite branch value.

Consequently the total singular delta contribution is $3hk + 3$, and the normalization genus is

$$\begin{aligned} g_{h,k} &= (3h - 1)(3k - 1) - 3hk - 3 \\ &= 6hk - 3h - 3k - 2 \\ &\leq 6hk. \end{aligned}$$

This agrees with the exact Riemann–Hurwitz genus calculation.

For the same-gap nonlinear component $\mathcal{H}_h$, division by the diagonal lowers the bidegree to $(3h - 1, 3h - 1)$, so its arithmetic genus is $(3h - 2)^2$. Its complete generic singularity census is as follows.

- At the $h(h - 1)$ finite pole-pole points with $i \neq j$, the diagonal is absent and $\mathcal{H}_h$ has an ordinary triple point, of delta-invariant 3.
- At the h diagonal pole points $(-i, -i)$, the full tangent cone is proportional to $u^3 - v^3$. Removing the diagonal factor $u - v$ leaves two distinct tangent lines, so $\mathcal{H}_h$ has an ordinary node, of delta-invariant 1.
- At $([1 : 0], [1 : 0])$, removing the diagonal factor from the ordinary triple point likewise leaves an ordinary node, of delta-invariant 1.
- At each of the $4h - 4$ simple finite critical points, the diagonal and $\mathcal{H}_h$ meet transversely. These are ordinary nodes of the full reducible curve $F_{h,h} = 0$, but they are smooth points of $\mathcal{H}_h$. There are no further singularities under CENSUS_H .

Thus $\mathcal{H}_h$ has exactly $h(h - 1)$ ordinary triple points and $h + 1$ ordinary nodes, and

$$\begin{aligned} g(\mathcal{H}_h) &= (3h - 2)^2 - 3h(h - 1) - (h + 1) \\ &= 6h^2 - 10h + 3 \\ &\leq 6h^2. \end{aligned}$$

Lemma 783 (complete point count via normalization). *Let $\mathcal{C}$ be a distinct-gap component, or let $\mathcal{C} = \mathcal{H}_h$ with $k = h$, and let $\nu : \tilde{\mathcal{C}} \rightarrow \mathcal{C}$ be its smooth projective normalization. Then*

$$\#\mathcal{C}(\mathbb{F}_p) \leq p + 1 + 12hk\sqrt{p} + 36(hk)^2.$$

Proof. The normalization genus satisfies $g(\tilde{\mathcal{C}}) \leq 6hk$. Therefore Weil on the smooth projective normalization gives

$$\#\tilde{\mathcal{C}}(\mathbb{F}_p) \leq p + 1 + 2g(\tilde{\mathcal{C}})\sqrt{p} \leq p + 1 + 12hk\sqrt{p}.$$

The normalization map is an isomorphism over the smooth locus of $\mathcal{C}$. Hence

$$\#\mathcal{C}(\mathbb{F}_p) \leq \#\tilde{\mathcal{C}}(\mathbb{F}_p) + \#\text{Sing}(\mathcal{C})(\mathbb{F}_p).$$

A Bezout bound gives fewer than $36(hk)^2$ geometric singular points, and therefore also $\#\text{Sing}(\mathcal{C})(\mathbb{F}_p) < 36(hk)^2$. Combining the two inequalities proves the claim. $\square$

Pairwise bounds. For $h \neq k$,

$$R_{h,k} \leq \frac{L_h L_k}{p} + 12hk\sqrt{p} + 36(hk)^2 + 4hk.$$

For $2 \leq h \leq H$,

$$R_{h,h} \leq \frac{L_h^2}{p} + L_h + 12h^2\sqrt{p} + 36h^4 + 5h.$$

For $h = 1$,

$$R_{1,1} \leq 3L_1 \leq \frac{L_1^2}{p} + L_1 + p.$$

Summation. Let

$$\Sigma_1 = \sum_{h \leq H} h = \frac{H(H+1)}{2}, \quad \Sigma_2 = \sum_{h \leq H} h^2 = \frac{H(H+1)(2H+1)}{6}.$$

Summing the pairwise bounds gives

$$N_{\text{coinc}}(H) \leq \frac{\mathcal{S}_H^2}{p} + \mathcal{S}_H + p + 12\Sigma_1^2\sqrt{p} + 36\Sigma_2^2 + 4\Sigma_1^2 + 5\Sigma_1.$$

Using $\Sigma_1 \leq H^2$ and $\Sigma_2 \leq H^3$,

$$N_{\text{coinc}}(H) \leq \frac{\mathcal{S}_H^2}{p} + \mathcal{S}_H + p + 12H^4\sqrt{p} + 36H^6 + 4H^4 + 5H^2.$$

Theorem 784 (conditional $p^{1/6}$ coincidence bound). *Assume CENSUS_H . For*

$$H \leq c_0 p^{1/6}, \quad c_0 = \frac{1}{4}, \quad p \geq 4096,$$

one has

$$N_{\text{coinc}}(H) \leq \frac{\mathcal{S}_H^2}{p} + 4\mathcal{S}_H.$$

Proof. Write $\text{Err} = 12\Sigma_1^2\sqrt{p} + 36\Sigma_2^2 + 4\Sigma_1^2 + 5\Sigma_1$ for the total error term above. Since $\mathcal{S}_H \geq pH/2$,

$$\frac{\text{Err}}{\mathcal{S}_H} \leq 24\frac{H^3}{\sqrt{p}} + 72\frac{H^5}{p} + 8\frac{H^3}{p} + 10\frac{H}{p}.$$

For $H \leq p^{1/6}/4$ this is $< 1/2$. Also $p \leq 2\mathcal{S}_H$. Thus

$$N_{\text{coinc}}(H) \leq \frac{\mathcal{S}_H^2}{p} + \mathcal{S}_H + p + \text{Err} \leq \frac{\mathcal{S}_H^2}{p} + 4\mathcal{S}_H. \quad \square$$

Additive-completion audit (with logs). This paragraph concerns only the off-diagonal pairs $h \neq k$. For an interval $I \subset \mathbb{F}_p$, with the standard unnormalized Fourier transform,

$$\sum_a |\widehat{1_I}(a)| \leq p(2 + \log p).$$

For a distinct-gap normalization $\tilde{X}_{h,k}$, assume the following standard conductor hypotheses for every nonzero completion frequency (a, b) :

- $\tilde{X}_{h,k}$ has good geometrically irreducible reduction;
- the pulled-back additive phase $aX + bY$ is nonconstant and is not of Artin–Schreier form $G^p - G + c$ in $\mathbb{F}_p(\tilde{X}_{h,k})$;
- the total conductor of the corresponding rank-one additive sheaf is at most $40hk$.

Under these hypotheses the Bombieri–Weil estimate gives

$$\left| \sum_{P \in \tilde{X}_{h,k}(\mathbb{F}_p)} e_p(aX(P) + bY(P)) \right| \leq 40hk\sqrt{p}.$$

After transferring between the normalization and the singular model and accounting for the boundary points, additive completion gives, for $h \neq k$,

$$R_{h,k} \leq \frac{L_h L_k}{p} + 64hk\sqrt{p}(2 + \log p)^2 + 72(hk)^2.$$

The diagonal terms are not covered by this irreducible off-diagonal completion formula. They are restored separately from the direct pairwise bounds:

$$R_{h,h} \leq \frac{L_h^2}{p} + L_h + 12h^2\sqrt{p} + 36h^4 + 5h \quad (2 \leq h \leq H),$$

and $R_{1,1} \leq L_1^2/p + L_1 + p$. Thus, writing $\Sigma_4 = \sum_{h \leq H} h^4$, the completed off-diagonal sum plus the restored diagonal contribution satisfies

$$\begin{aligned} N_{\text{coinc}}(H) &\leq \frac{\mathcal{S}_H^2}{p} + \mathcal{S}_H + p + 64\Sigma_1^2\sqrt{p}(2 + \log p)^2 + 72\Sigma_2^2 \\ &\quad + 12\Sigma_2\sqrt{p} + 36\Sigma_4 + 5\Sigma_1. \end{aligned}$$

In particular, using $\Sigma_1 \leq H^2$, $\Sigma_2 \leq H^3$, and $\Sigma_4 \leq H^5$, the dominant completion error is

$$\ll H^4\sqrt{p}(2 + \log p)^2 + H^6.$$

Comparing with $\mathcal{S}_H \asymp pH$, we require

$$H^3(2 + \log p)^2/\sqrt{p} \ll 1.$$

Thus the honest completion range is

$$\boxed{H \leq cp^{1/6}/(2 + \log p)^{2/3}.$$

Remark 785. Adjusting numerical constants alone cannot remove the logarithmic loss; smoothing, an L^2 large-sieve input, or cancellation among frequencies could alter it. The full $p^{1/6}$ range of Theorem 784 holds because the windows are co-short and can be enlarged directly.

Proposition 786 (small-gap corollary). *Under Theorem 784,*

$$\sum_{h \leq H} C_h \leq \frac{\mathcal{S}_H}{p} + 2\sqrt{\mathcal{S}_H} \leq H + 2\sqrt{pH}.$$

Hence

$$\sum_{h \leq H} C_h \leq H + \sqrt{(4 + o(1))pH}.$$

Remark 787 (limitation). The trivial bound gives $\sum_{h \leq H} C_h \leq \frac{3}{2}H^2$. This dominates until $H \sim p^{1/3}$. Since Theorem 784 only holds up to $p^{1/6}$, it does not improve the exponent. Its value is structural: it certifies the $\mathcal{S}_H^2/p + O(\mathcal{S}_H)$ coincidence shape.

Assumptions beyond CENSUS_H.

1. Casoratian identity linking $\Delta_{r,h}$ and $f_h(r)$.
2. Degree data $\deg N_h = 3(h-1)$.
3. Honest order-three finite poles and the order-three zero at infinity.
4. The complete generic singularity census used in the genus calculation.
5. Good reduction threshold $p_0(H)$.
6. Weil bound for smooth projective normalizations over $\mathbb{F}_p$.
7. Bezout bound on singularities.
8. (Audit only) Bombieri additive bound, the stated conductor hypotheses, and the interval Fourier bound.
9. Explicit handling of $h = 1$ outside the census.

31.5 A second proof of the window multiplicity lemma via restart transfer

The window multiplicity lemma – the $\sigma = 1/3$ estimate of Section 29, in the form

$$\#\{n \in J : \pi(n) = v\} \leq 1 + 4|J|^{2/3}$$

for every interval J of regular indices and every $v \in \mathbb{P}^1(\mathbb{F}_p)$ – is already banked in the main text, where it received its first proof. This subsection records a *second, independent proof* of that lemma. Nothing below is claimed as a new theorem: the two statements are the banked lemma and its base-point specialization, recovered with the slightly smaller leading constant $3^{4/3}/2 = 2.163\dots$ in place of 4. What is claimed as new is only the *mechanism*: the restart transfer is applied to an arbitrary nonzero homogeneous solution $y_n = \alpha c_n - \beta b_n$ of the Apéry recurrence, whose zero set is exactly the fiber $\{n : \pi(n) = v\}$ – a homogeneous zero-fiber input which requires no base point on the fiber and no companion-orbit bookkeeping. The quantity $d_D(r)$ below is the window multiplicity denoted $d_\Delta(r)$ in the main text and $\nu_D(u)$ in Subsection 31.3.

Proposition 788 (uniform return multiplicity from a fixed base; second proof). *Let $p \geq 7$ be prime, let*

$$I_p = \{1, \dots, p-2\}, \quad \pi(n) = [b_n : c_n] \in \mathbb{P}^1(\mathbb{F}_p),$$

and suppose that

$$1 \leq r, \quad 1 \leq D, \quad r + D \leq p-2.$$

Put

$$d_D(r) = \#\{1 \leq d \leq D : \pi(r+d) = \pi(r)\}.$$

Then

$$d_D(r) \leq \min_{1 \leq Y \leq D} \left\{ \frac{3}{2}Y(Y-1) + \left\lfloor \frac{D}{Y+1} \right\rfloor \right\}. \quad (806)$$

In particular,

$$d_D(r) \leq \frac{3^{4/3}}{2}D^{2/3} + \frac{3^{2/3}}{2}D^{1/3}, \quad (807)$$

and hence

$$\max_{1 \leq r \leq p-2-D} d_D(r) \ll D^{2/3}$$

with an absolute implied constant.

Proof. For the fixed base r , define an absolute-index solution

$$y_n^{(r)} = b_r c_n - c_r b_n.$$

It satisfies the Apéry recurrence because it is a fixed linear combination of the two solutions b and c . Moreover,

$$y_r^{(r)} = 0, \quad y_{r+1}^{(r)} = b_r c_{r+1} - b_{r+1} c_r = \frac{1}{(r+1)^3} \neq 0,$$

where the last equality is the Casoratian identity. Thus $y^{(r)}$ is not the zero solution. Since (b_n, c_n) is nonzero on I_p , one also has

$$y_n^{(r)} = 0 \iff \pi(n) = \pi(r). \quad (808)$$

For any nonzero homogeneous solution y , put

$$X_n(y) = [n^3 y_{n-1} : y_n].$$

A direct use of the recurrence gives

$$X_{n+1}(y) = M_n X_n(y), \quad M_n = \begin{pmatrix} 0 & (n+1)^6 \\ -1 & P(n) \end{pmatrix}. \quad (809)$$

Indeed,

$$M_n \begin{pmatrix} n^3 y_{n-1} \\ y_n \end{pmatrix} = \begin{pmatrix} (n+1)^6 y_n \\ -n^3 y_{n-1} + P(n) y_n \end{pmatrix} = \begin{pmatrix} (n+1)^6 y_n \\ (n+1)^3 y_{n+1} \end{pmatrix},$$

which represents $[(n+1)^3 y_n : y_{n+1}]$. All matrices used below are invertible because their determinants are $(n+1)^6 \neq 0$ in the regular nonwrapping range. Consequently a nonzero solution cannot have two consecutive zeros. If $y_z = 0$ and there is a later zero in the regular range, then

$$X_z(y) = [1 : 0], \quad X_{z+1}(y) = M_z [1 : 0] = [0 : 1].$$

Thus every zero resets the state to the fixed point $[0 : 1]$ one step later.

Let two consecutive zeros occur at z and $z + h$. Then $h \geq 2$, and using the proved transfer formula with the absolute start $z + 1$ gives

$$\begin{aligned} X_{z+h}(y) &= G_{h-1}(z+1)[0 : 1] \\ &= [(z+h)^6 U_{h-2}(z+1) : U_{h-1}(z+1)]. \end{aligned}$$

The left side is $[1 : 0]$. Since the transfer product is invertible, the displayed top coordinate is nonzero, and therefore

$$U_{h-1}(z+1) = 0. \quad (810)$$

By the identity $N_h(X) = U_{h-1}(X+1)$, this is

$$N_h(z) = 0. \quad (811)$$

Notice that all indices here are absolute indices.

List the zeros of $y^{(r)}$ in $[r, r + D]$ as

$$r = n_0 < n_1 < \cdots < n_q,$$

so that $q = d_D(r)$ by (808). Write

$$e_i = n_i - r, \quad h_i = n_{i+1} - n_i = e_{i+1} - e_i.$$

For a fixed gap length h , equation (811) says that each corresponding relative start e_i is a root of

$$F_{r,h}(E) = N_h(r + E).$$

Translation is an automorphism of $\mathbb{F}_p[E]$. Hence the proved nonvanishing of N_h in the range $h < p$ implies that $F_{r,h}$ is not the zero polynomial, and

$$\deg F_{r,h} \leq 3(h-1).$$

Thus, for each fixed h , at most $3(h-1)$ of the gaps can have length h .

Fix Y with $1 \leq Y \leq D$. The number of gaps with $h_i \leq Y$ is at most

$$\sum_{h=2}^Y 3(h-1) = \frac{3}{2}Y(Y-1).$$

Every remaining gap has length at least $Y+1$, while

$$\sum_{i=0}^{q-1} h_i = n_q - r \leq D.$$

Hence the number of remaining gaps is at most

$$\left\lfloor \frac{D}{Y+1} \right\rfloor.$$

Since q is exactly the total number of gaps, this proves (806). This argument includes $D = 1$: take $Y = 1$, which gives $d_1(r) = 0$.

For the explicit estimate, put

$$x = (D/3)^{1/3}, \quad Y = \lceil x \rceil.$$

Then $1 \leq Y \leq D$, $Y(Y-1) \leq x(x+1)$, and $Y+1 > x$. Therefore

$$\begin{aligned} d_D(r) &\leq \frac{3}{2}x(x+1) + \frac{D}{x} \\ &= \frac{9}{2}x^2 + \frac{3}{2}x \\ &= \frac{3^{4/3}}{2}D^{2/3} + \frac{3^{2/3}}{2}D^{1/3}. \end{aligned}$$

This proves (807). □

Proposition 789 (window multiplicity lemma; second proof via restart transfer). *Let $p \geq 7$ be prime, let J be an interval of H consecutive integers contained in I_p , and let $v \in \mathbb{P}^1(\mathbb{F}_p)$. If $H = 1$, then*

$$\#\{n \in J : \pi(n) = v\} \leq 1.$$

If $H \geq 2$, then

$$\#\{n \in J : \pi(n) = v\} \leq 1 + \min_{1 \leq Y \leq H-1} \left\{ \frac{3}{2}Y(Y-1) + \left\lfloor \frac{H-1}{Y+1} \right\rfloor \right\}. \quad (812)$$

Consequently,

$$\#\{n \in J : \pi(n) = v\} \leq 1 + \frac{3^{4/3}}{2}(H-1)^{2/3} + \frac{3^{2/3}}{2}(H-1)^{1/3}, \quad (813)$$

and in particular the fiber size is $O(H^{2/3})$, uniformly in p , J , and v .

Proof. Write the projective point in the same coordinate order as the orbit:

$$v = [\alpha : \beta], \quad \pi(n) = [b_n : c_n].$$

Define

$$y_n = \alpha c_n - \beta b_n.$$

This is a homogeneous solution of the Apery recurrence. It is not the zero solution, since

$$y_0 = -\beta, \quad y_1 = \alpha - 5\beta,$$

and simultaneous vanishing would imply $\alpha = \beta = 0$. Also the Casoratian identity implies $(b_n, c_n) \neq (0, 0)$ on I_p . Therefore

$$y_n = 0 \iff \alpha c_n - \beta b_n = 0 \iff \pi(n) = v. \quad (814)$$

This includes the two extreme fibers: $v = [1 : 0]$ gives $y = c$, while $v = [0 : 1]$ gives $y = -b$.

List the zeros of y in J as

$$z_1 < z_2 < \cdots < z_k.$$

If $k \leq 1$, the result is immediate. For each $1 \leq i < k$, put

$$h_i = z_{i+1} - z_i.$$

The absolute-index reset computation in the proof of Proposition 788 applies to this arbitrary nonzero homogeneous solution and gives

$$h_i \geq 2, \quad N_{h_i}(z_i) = 0.$$

For a fixed h , the starts z_i are distinct roots of the nonzero polynomial N_h , whose degree is at most $3(h-1)$. Thus at most $3(h-1)$ gaps have length h .

Fix Y with $1 \leq Y \leq H-1$. The number of gaps of length at most Y is at most

$$\sum_{h=2}^Y 3(h-1) = \frac{3}{2}Y(Y-1).$$

Every other gap has length at least $Y+1$, and

$$\sum_{i=1}^{k-1} h_i = z_k - z_1 \leq H-1.$$

Hence the number of long gaps is at most

$$\left\lfloor \frac{H-1}{Y+1} \right\rfloor.$$

Since k is one plus the number of gaps, this proves (812). The argument does not require the left endpoint of J to be a zero; the additive 1 accounts for the first zero of the fiber wherever it occurs in J . A zero at the right endpoint of J or at $p-2$ is harmless, since no reset after the terminal zero is used.

Finally apply the explicit estimate from Proposition 788 with $D = H-1$ and add 1. This gives (813). $\square$

Remark 790 (byproducts of the second proof). The restart-transfer proof yields four small refinements over the first proof, all visible in the statements above: (i) it exposes the absolute shift $F_{r,h}(E) = N_h(r+E)$, so the gap polynomial enters through a translation automorphism of $\mathbb{F}_p[E]$ rather than through relative re-indexing; (ii) the minimum over Y permits $Y=1$, and the argument includes $D=1$; (iii) it separates the base-row count $d_D(r)$, which carries no additive $+1$, from the general fiber count, which does; and (iv) it gives the leading constant $3^{4/3}/2$.